

The form of Z_w^{+z} , as given by (9-16a) through (9-16c), as a function of f_c/f , and where f_c is the cutoff frequency of that mode, is the same as the Z_w^{+z} for the TE_z modes of a rectangular waveguide, as given by (8-20a) through (8-20c). A plot of (9-16a) through (9-16c) for any one TE_{mn}^z mode as a function of f_c/f is shown in Figure 8-2.

Example 9-1

A circular waveguide of radius $a = 3$ cm that is filled with polystyrene ($\epsilon_r = 2.56$) is used at a frequency of 2 GHz. For the dominant TE_{11}^z mode, determine the following:

- Cutoff frequency.
- Guide wavelength (in cm). Compare it to the infinite medium wavelength λ .
- Phase constant β_z (in rad/cm).
- Wave impedance.
- Bandwidth over single-mode operation (assuming only TE_z modes).

Solution:

- The dominant mode is the TE_{11} mode whose cutoff frequency is, according to (9-12a),

$$(f_c)_{11}^{TE} = \frac{1.8412}{2\pi a \sqrt{\mu\epsilon}} = \frac{1.8412(30 \times 10^9)}{2\pi(3)\sqrt{2.56}} = 1.8315 \text{ GHz}$$

- Since the frequency of operation is 2 GHz, which is greater than the cutoff frequency of 1.8315 GHz, the guide wavelength of (9-14a) for the TE_{11} mode is

$$\lambda_g = \frac{\lambda}{\sqrt{1 - \left(\frac{f_c}{f}\right)^2}}$$

where

$$\lambda = \frac{\lambda_0}{\sqrt{\epsilon_r}} = \frac{30 \times 10^9}{2 \times 10^9 \sqrt{2.56}} = 9.375 \text{ cm}$$

$$\sqrt{1 - \left(\frac{f_c}{f}\right)^2} = \sqrt{1 - \left(\frac{1.8315}{2}\right)^2} = 0.4017$$

Thus,

$$\lambda_g = \frac{9.375}{0.4017} = 23.34 \text{ cm where } \lambda = 9.375 \text{ cm}$$

- The phase constant β_z of the TE_{11} mode is found using (9-13a), or

$$\beta_z = \beta \sqrt{1 - \left(\frac{f_c}{f}\right)^2} = \frac{2\pi}{\lambda} \sqrt{1 - \left(\frac{f_c}{f}\right)^2} = \frac{2\pi}{9.375} (0.4017) = 0.2692 \text{ rad/cm}$$

which can also be obtained using

$$\beta_z = \frac{2\pi}{\lambda_g} = \frac{2\pi}{23.34} = 0.2692 \text{ rad/cm}$$

- According to (9-16a), the wave impedance of the TE_{11} mode is equal to

$$Z_{11}^{TE} = \frac{\eta}{\sqrt{1 - \left(\frac{f_c}{f}\right)^2}} = \frac{120\pi / \sqrt{2.56}}{0.4017} = 586.56 \text{ ohms}$$

- Since the next higher-order TE_{mn} mode is the TE_{21} , the bandwidth of single TE_{11} mode operation is

$$BW = 3.0542 / 1.8412 : 1 = 1.6588 : 1$$

9.2.2 Transverse Magnetic (TM^z) Modes

The transverse magnetic to z (TM^z) modes can be derived in a similar manner as the TE^z modes of Section 9.2.1 by letting

$$\mathbf{A} = \hat{\mathbf{a}}_z A_z(\rho, \phi, z) \quad (9-17a)$$

$$\mathbf{F} = 0 \quad (9-17b)$$

The vector potential $\mathbf{A}$ must satisfy the vector wave equation of (3-48), which reduces for the $\mathbf{A}$ of (9-17a) to

$$\nabla^2 A_z(\rho, \phi, z) + \beta^2 A_z(\rho, \phi, z) = 0 \quad (9-18)$$

The solution of (9-18) is obtained in a manner similar to that of (9-2), as given by (9-4), and it can be written as

$$\begin{aligned} A_z(\rho, \phi, z) = & [A_1 J_m(\beta_\rho \rho) + B_1 Y_m(\beta_\rho \rho)] \\ & \times [C_2 \cos(m\phi) + D_2 \sin(m\phi)] [A_3 e^{-j\beta_z z} + B_3 e^{+j\beta_z z}] \end{aligned} \quad (9-19)$$

with the constraint (dispersion) equation expressed as

$$\beta_\rho^2 + \beta_z^2 = \beta^2 \quad (9-19a)$$

The constants A_1 , B_1 , C_2 , D_2 , A_3 , B_3 , m , β_ρ , and β_z can be found using the following boundary conditions

$$E_\phi(\rho = a, \phi, z) = 0 \quad (9-20a)$$

or

$$E_z(\rho = a, \phi, z) = 0 \quad (9-20b)$$

$$\text{The fields must be finite everywhere} \quad (9-20c)$$

$$\text{The fields must repeat every } 2\pi \text{ radians in } \phi \quad (9-20d)$$

According to (9-20c), $B_1 = 0$ since $Y_m(\rho = 0) = \infty$. In addition, according to (9-20d),

$$m = 0, 1, 2, 3, \dots \quad (9-21)$$

Considering waves that propagate only in the $+z$ direction, (9-19) reduces to

$$A_z^+(\rho, \phi, z) = B_{mn} J_m(\beta_\rho \rho) [C_2 \cos(m\phi) + D_2 \sin(m\phi)] e^{-j\beta_z z} \quad (9-22)$$

The eigenvalues of β_ρ can be obtained by applying either (9-20a) or (9-20b). Use of (6-70) and (9-22) allows us to write the electric field component E_z^+ as

$$\begin{aligned} E_z^+ &= -j \frac{1}{\omega \mu \epsilon} \left(\frac{\partial^2}{\partial z^2} + \beta^2 \right) A_z^+ \\ &= -j B_{mn} \frac{\beta_\rho^2}{\omega \mu \epsilon} J_m(\beta_\rho \rho) [C_2 \cos(m\phi) + D_2 \sin(m\phi)] e^{-j\beta_z z} \end{aligned} \quad (9-23)$$

Application of the boundary condition (9-20b) and use of (9-23) gives

$$E_z^+(\rho = a, \phi, z) = -j B_{mn} \frac{\beta_\rho^2}{\omega \mu \epsilon} J_m(\beta_\rho a) [C_2 \cos(m\phi) + D_2 \sin(m\phi)] e^{-j\beta_z z} = 0 \quad (9-24)$$

which is only satisfied provided that

$$J_m(\beta_\rho a) = 0 \Rightarrow \beta_\rho a = \chi_{mn} \Rightarrow \beta_\rho = \frac{\chi_{mn}}{a} \quad (9-25)$$

TABLE 9-2 Zeroes χ_{mn} of $J_m(\chi_{mn}) = 0$ ($n = 1, 2, 3, \dots$) of Bessel function $J_m(x)$

	$m = 0$	$m = 1$	$m = 2$	$m = 3$	$m = 4$	$m = 5$	$m = 6$	$m = 7$	$m = 8$	$m = 9$	$m = 10$	$m = 11$
$n = 1$	2.4049	3.8318	5.1357	6.3802	7.5884	8.7715	9.9361	11.0864	12.2251	13.3543	14.4755	15.5898
$n = 2$	5.5201	7.0156	8.4173	9.7610	11.0647	12.3386	13.5893	14.8213	16.0378	17.2412	18.4335	19.6160
$n = 3$	8.6537	10.1735	11.6199	13.0152	14.3726	15.7002	17.0038	18.2876	19.5545	20.8071	22.0470	23.2759
$n = 4$	11.7915	13.3237	14.7960	16.2235	17.6160	18.9801	20.3208	21.6415	22.9452	24.2339	25.5095	26.7733
$n = 5$	14.9309	16.4706	17.9598	19.4094	20.8269	22.2178	23.5861	24.9349	26.2668	27.5838	28.8874	30.1791

In (9-25), χ_{mn} represents the n th zero ($n = 1, 2, 3, \dots$) of the Bessel function J_m of the first kind and of order m ($m = 0, 1, 2, 3, \dots$). An abbreviated list of the zeroes χ_{mn} of the Bessel function J_m is found in Table 9-2. The smallest value of χ_{mn} is 2.4049 ($m = 0, n = 1$), followed by 3.8318 ($m = 1, n = 1$), 5.1357 ($m = 2, n = 1$), etc.

By using (9-19a) and (9-25), β_z can be written as

$$(\beta_z)_{mn} = \begin{cases} \sqrt{\beta^2 - \beta_\rho^2} = \sqrt{\beta^2 - \left(\frac{\chi_{mn}}{a}\right)^2} & \text{when } \beta > \beta_\rho = \frac{\chi_{mn}}{a} & (9-26a) \\ 0 & \text{when } \beta = \beta_c = \beta_\rho = \frac{\chi_{mn}}{a} & (9-26b) \\ -j\sqrt{\beta_\rho^2 - \beta^2} = -j\sqrt{\left(\frac{\chi_{mn}}{a}\right)^2 - \beta^2} & \text{when } \beta < \beta_\rho = \frac{\chi_{mn}}{a} & (9-26c) \end{cases}$$

By following the same procedure as for the TE^z modes, we can write the expressions for the cutoff frequencies $(f_c)_{mn}$, propagation constant $(\beta_z)_{mn}$, and guide wavelength $(\lambda_g)_{mn}$ as

$$(f_c)_{mn} = \frac{\chi_{mn}}{2\pi a \sqrt{\mu\epsilon}} \quad (9-27)$$

$$(\beta_z)_{mn} = \begin{cases} \sqrt{\beta^2 - \beta_\rho^2} = \beta \sqrt{1 - \left(\frac{\beta_\rho}{\beta}\right)^2} = \beta \sqrt{1 - \left(\frac{\beta_c}{\beta}\right)^2} \\ = \beta \sqrt{1 - \left(\frac{\chi_{mn}}{\beta a}\right)^2} = \beta \sqrt{1 - \left(\frac{f_c}{f}\right)^2} & \text{when } f > f_c = (f_c)_{mn} & (9-28a) \\ 0 & \text{when } f = f_c = (f_c)_{mn} & (9-28b) \\ -j\sqrt{\beta_\rho^2 - \beta^2} = -j\beta \sqrt{\left(\frac{\beta_\rho}{\beta}\right)^2 - 1} = -j\beta \sqrt{\left(\frac{\beta_c}{\beta}\right)^2 - 1} \\ = -j\beta \sqrt{\left(\frac{\chi_{mn}}{\beta a}\right)^2 - 1} = -j\beta \sqrt{\left(\frac{f_c}{f}\right)^2 - 1} & \text{when } f < f_c = (f_c)_{mn} & (9-28c) \end{cases}$$

$$(\lambda_g)_{mn} = \begin{cases} \frac{2\pi}{\beta \sqrt{1 - \left(\frac{f_c}{f}\right)^2}} = \frac{\lambda}{\sqrt{1 - \left(\frac{f_c}{f}\right)^2}} & \text{when } f > f_c = (f_c)_{mn} & (9-29a) \\ \infty & \text{when } f = f_c = (f_c)_{mn} & (9-29b) \end{cases}$$

According to (9-27) and the values of χ_{mn} of Table 9-2, the order (lower to higher cutoff frequencies) in which the TE^z modes occur is TM_{01} , TM_{11} , TM_{21} , and so forth. The bandwidth of the first single-mode TM_{01}^z operation is $3.8318/2.4049 = 1.5933:1$, which is also less than $2:1$. Comparing the cutoff frequencies of the TE^z and TM^z modes, as given by (9-12a) and (9-27) along with the data of Tables 9-1 and 9-2, the order of the TE_{mn}^z and TM_{mn}^z modes is that of TE_{11} ($\chi'_{11} = 1.8412$), TM_{01} ($\chi_{01} = 2.4049$), TE_{21} ($\chi'_{21} = 3.0542$), TE_{01} ($\chi'_{01} = 3.8318$) = TM_{11} ($\chi_{11} = 3.8318$), TE_{31} ($\chi'_{31} = 4.2012$), and so forth. The dominant mode is TE_{11} and its bandwidth of single-mode operation is $2.4049/1.8412 = 1.3062:1$, which is much smaller than $2:1$. Plots of the field configurations over a cross section of the waveguide, both E and H , for the first 30 TE_{mn}^z and/or TM_{mn}^z modes are shown in Figure 9-2 [1].

It is apparent that the cutoff frequencies of the TE_{0n} and TM_{1n} modes are identical; therefore, they are referred to here also as degenerate modes. This is because the zeroes of the derivative of the Bessel function J_0 are identical to the zeroes of the Bessel function J_1 . To demonstrate this, let us examine the derivative of $J_0(\beta_\rho \rho)$ evaluated at $\rho = a$. Using (IV-19) we can write that

$$\frac{d}{d(\beta_\rho \rho)} J_0(\beta_\rho \rho) \big|_{\rho=a} = J'_0(\beta_\rho a) = -J_1(\beta_\rho \rho) \big|_{\rho=a} = -J_1(\beta_\rho a) \quad (9-30)$$

which vanishes when

$$J'_0(\beta_\rho a) = 0 \Rightarrow \beta_\rho a = \chi'_{0n}, \quad n = 1, 2, 3, \dots \quad (9-30a)$$

or

$$J_1(\beta_\rho a) = 0 \Rightarrow \beta_\rho a = \chi_{1n}, \quad n = 1, 2, 3, \dots \quad (9-30b)$$

The electric and magnetic field components can be written, using (6-70) and (9-22), as

$$E_\rho^+ = -j \frac{1}{\omega \mu \varepsilon} \frac{\partial^2 A_z^+}{\partial \rho \partial z} = -B_{mn} \frac{\beta_\rho \beta_z}{\omega \mu \varepsilon} J'_m(\beta_\rho \rho) [C_2 \cos(m\phi) + D_2 \sin(m\phi)] e^{-j\beta_z z} \quad (9-31a)$$

$$E_\phi^+ = -j \frac{1}{\omega \mu \varepsilon} \frac{1}{\rho} \frac{\partial^2 A_z^+}{\partial \phi \partial z} = -B_{mn} \frac{m \beta_z}{\omega \mu \varepsilon \rho} J_m(\beta_\rho \rho) [-C_2 \sin(m\phi) + D_2 \cos(m\phi)] e^{-j\beta_z z} \quad (9-31b)$$

$$\begin{aligned} E_z^+ &= -j \frac{1}{\omega \mu \varepsilon} \left(\frac{\partial^2}{\partial z^2} + \beta^2 \right) A_z^+ \\ &= -j B_{mn} \frac{\beta_\rho^2}{\omega \mu \varepsilon} J_m(\beta_\rho \rho) [C_2 \cos(m\phi) + D_2 \sin(m\phi)] e^{-j\beta_z z} \end{aligned} \quad (9-31c)$$

$$H_\rho^+ = \frac{1}{\mu} \frac{1}{\rho} \frac{\partial A_z^+}{\partial \phi} = B_{mn} \frac{m}{\mu} \frac{1}{\rho} J_m(\beta_\rho \rho) [-C_2 \sin(m\phi) + D_2 \cos(m\phi)] e^{-j\beta_z z} \quad (9-31d)$$

$$H_\phi^+ = -\frac{1}{\mu} \frac{\partial A_z^+}{\partial \rho} = -B_{mn} \frac{\beta_\rho}{\mu} J'_m(\beta_\rho \rho) [C_2 \cos(m\phi) + D_2 \sin(m\phi)] e^{-j\beta_z z} \quad (9-31e)$$

$$H_z^+ = 0 \quad (9-31f)$$

where

$$' \equiv \frac{\partial}{\partial(\beta_\rho \rho)} \quad (9-31g)$$

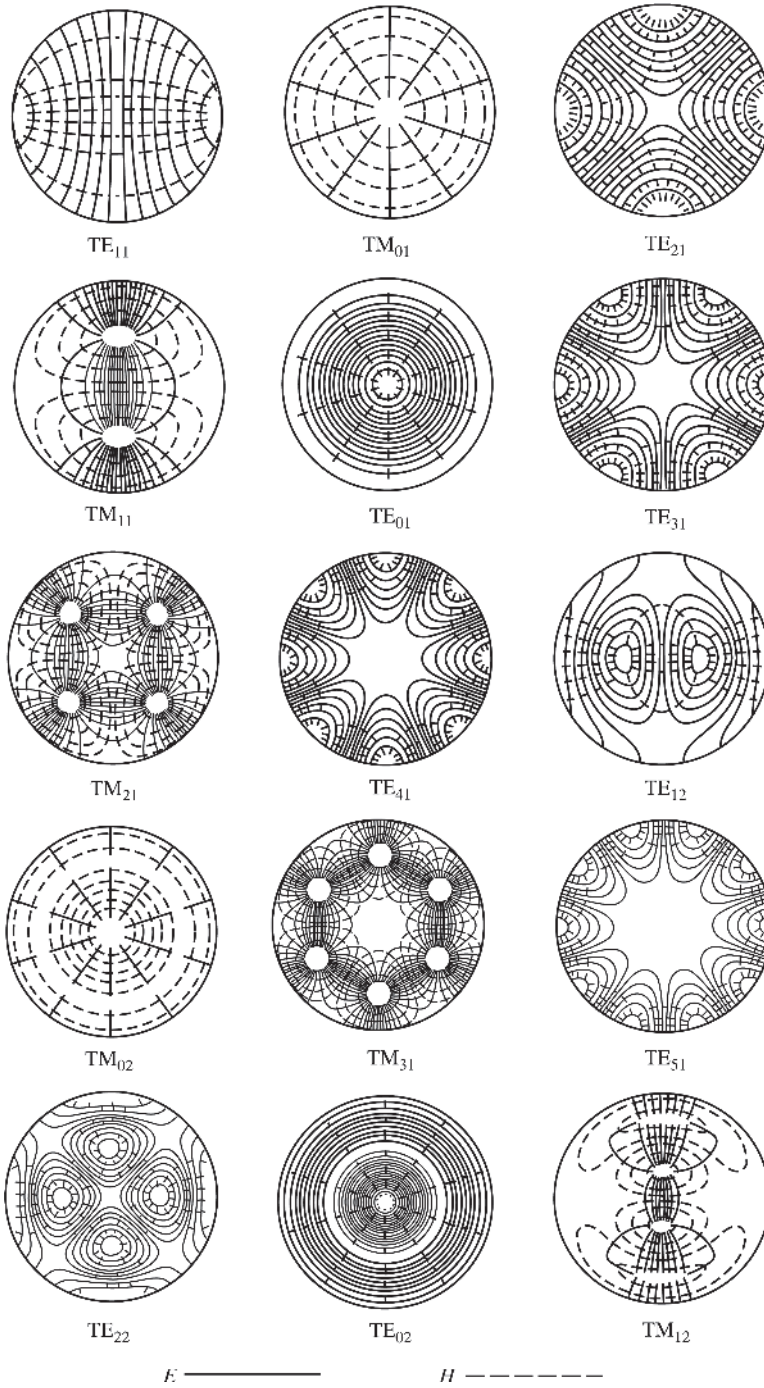


Figure 9-2 Field configurations of the first 30 TE^z and/or TM^z modes in a circular waveguide. (Source: C. S. Lee, S. W. Lee, and S. L. Chuang, "Plot of modal field distribution in rectangular and circular waveguides," *IEEE Trans. Microwave Theory Tech.*, © 1966, IEEE.)

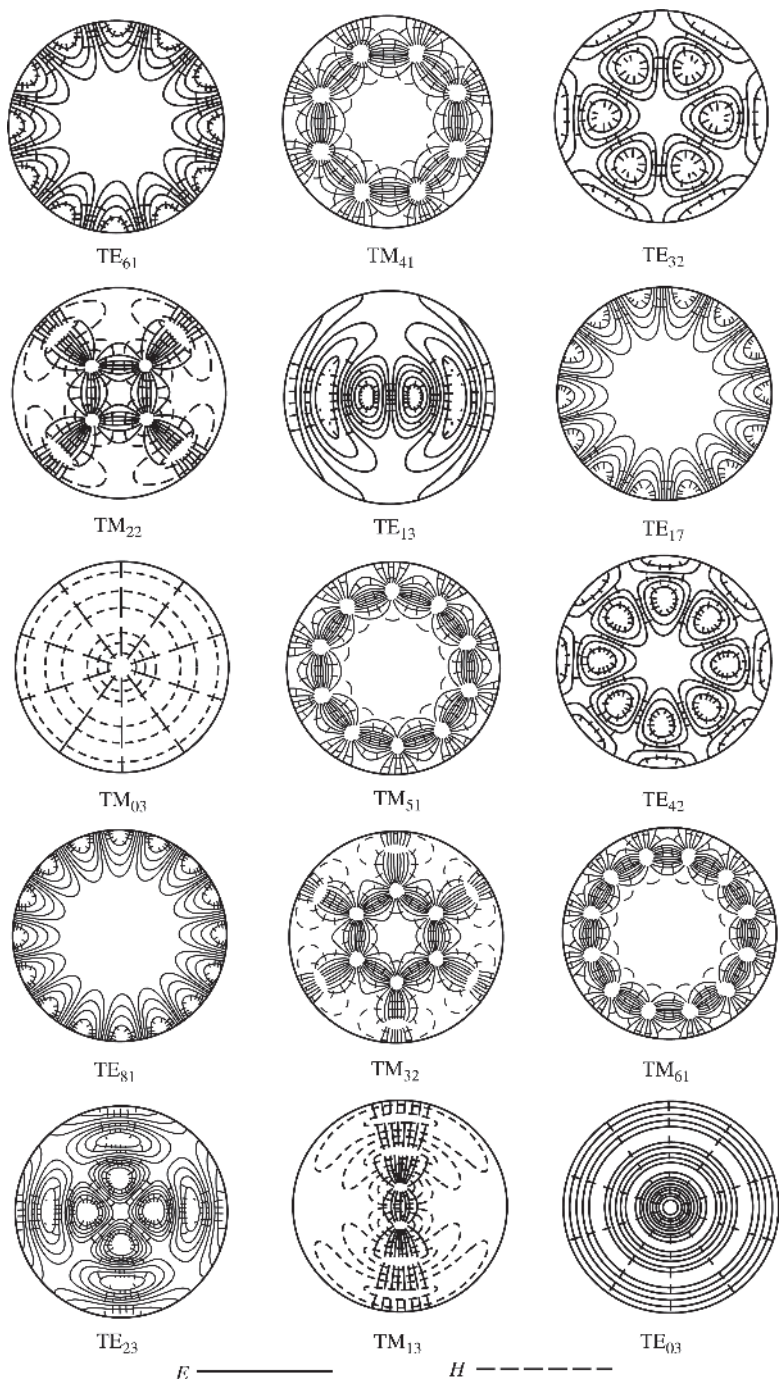


Figure 9-2 (Continued)

By using (9-31a) through (9-31f), the wave impedance in the $+z$ direction can be written as

$$(Z_w^{+z})_{mn}^{\text{TM}} = \frac{E_\rho^+}{H_\phi^+} = -\frac{E_\phi^+}{H_\rho^+} = \frac{(\beta_z)_{mn}}{\omega\epsilon} \quad (9-32)$$

With the aid of (9-28a) through (9-28c), the wave impedance of (9-32) reduces to

$$(Z_w^{+z})_{mn}^{\text{TM}} = \begin{cases} \frac{\beta \sqrt{1 - \left(\frac{f_c}{f}\right)^2}}{\omega\epsilon} = \sqrt{\frac{\mu}{\epsilon}} \sqrt{1 - \left(\frac{f_c}{f}\right)^2} = \eta \sqrt{1 - \left(\frac{f_c}{f}\right)^2} & \text{when } f > f_c = (f_c)_{mn} \\ 0 & \text{when } f = f_c = (f_c)_{mn} \\ \frac{-j\beta \sqrt{\left(\frac{f_c}{f}\right)^2 - 1}}{\omega\epsilon} = -j\sqrt{\frac{\mu}{\epsilon}} \sqrt{\left(\frac{f_c}{f}\right)^2 - 1} = -j\eta \sqrt{\left(\frac{f_c}{f}\right)^2 - 1} & \text{when } f < f_c = (f_c)_{mn} \end{cases} \quad (9-32a)$$

$$(Z_w^{+z})_{mn}^{\text{TM}} = \begin{cases} 0 & \text{when } f = f_c = (f_c)_{mn} \end{cases} \quad (9-32b)$$

$$(Z_w^{+z})_{mn}^{\text{TM}} = \begin{cases} \frac{-j\beta \sqrt{\left(\frac{f_c}{f}\right)^2 - 1}}{\omega\epsilon} = -j\sqrt{\frac{\mu}{\epsilon}} \sqrt{\left(\frac{f_c}{f}\right)^2 - 1} = -j\eta \sqrt{\left(\frac{f_c}{f}\right)^2 - 1} & \text{when } f < f_c = (f_c)_{mn} \end{cases} \quad (9-32c)$$

Examining (9-32a) through (9-32c), we can make the following statements about the wave impedance for the TM^z modes.

1. Above cutoff it is real and smaller than the intrinsic impedance of the medium inside the waveguide.
2. At cutoff it is zero.
3. Below cutoff it is imaginary and capacitive. This indicates that the waveguide below cutoff behaves as a capacitor that is an energy storage element.

The form of $(Z_w^{+z})_{mn}^{\text{TM}}$, as given by (9-32a) through (9-32c), and as a function of f_c/f , where f_c is the cutoff frequency of that mode, is the same as the $(Z_w^{+z})_{mn}^{\text{TM}}$ for the TM_{mn}^z modes of a rectangular waveguide, as given by (8-29a) through (8-29c). A plot of (9-32a) through (9-32c) for any one TM_{mn}^z mode as a function of f_c/f , is shown in Figure 8-2.

Example 9-2

Design a circular waveguide filled with a lossless dielectric medium of dielectric constant 4. The waveguide must operate in a single dominant mode over a bandwidth of 1 GHz.

1. Find its radius (in cm).
2. Determine the lower, center, and upper frequencies of the bandwidth.

Solution:

- a. The dominant mode is the TE_{11} mode whose cutoff frequency according to (9-12a) is

$$(f_c)_{11}^{\text{TE}} = \frac{\chi'_{11}}{2\pi a \sqrt{\mu\epsilon}} = \frac{1.8412(30 \times 10^9)}{2\pi(a)\sqrt{4}}$$

The next higher-order mode is the TM_{01} mode whose cutoff frequency according to (9-27) is

$$(f_c)_{01}^{\text{TM}^z} = \frac{\chi_{01}}{2\pi a \sqrt{\mu\epsilon}} = \frac{2.4049(30 \times 10^9)}{2\pi(a)\sqrt{4}}$$

The difference between the two must be 1 GHz. To accomplish this, the radius of the waveguide must be equal to

$$\frac{(2.4049 - 1.8412)30 \times 10^9}{2\pi(a)\sqrt{4}} = 1 \times 10^9 \Rightarrow a = 1.3457 \text{ cm}$$

b. The lower, upper, and center frequencies of the bandwidth are equal to

$$f_\ell = (f_c)_{11}^{\text{TE}^z} = \frac{1.8412(30 \times 10^9)}{2\pi(1.3457)2} = 3.2664 \times 10^9 = 3.2664 \text{ GHz}$$

$$f_u = (f_c)_{01}^{\text{TM}^z} = \frac{2.4049(30 \times 10^9)}{2\pi(1.3457)2} = 4.2664 \times 10^9 = 4.2664 \text{ GHz}$$

$$f_0 = f_\ell + 0.5 \times 10^9 = f_u - 0.5 \times 10^9 = 3.7664 \times 10^9 = 3.7664 \text{ GHz}$$

Whenever a given mode is desired, it is necessary to design the proper feed to excite the fields within the waveguide and detect the energy associated with such a mode. Maximization of the energy exchange or transfer is accomplished by designing the feed, which is usually a probe or antenna, so that its field pattern matches that of the field configuration of the desired mode. Usually the probe is placed near the maximum of the field pattern of the desired mode; however, that position may be varied somewhat in order to achieve some desired matching in the excitation and detection systems. Shown in Figure 9-3 are suggested designs to excite and/or detect the TE_{11} and TM_{01} modes in a circular waveguide, to transition between the TE_{10} of a rectangular waveguide and the TE_{11} mode of a circular waveguide, and to couple between the TE_{10} of a rectangular waveguide and TM_{01} mode of a circular waveguide.

9.2.3 Attenuation

The attenuation in a circular waveguide can be obtained by using techniques similar to those for the rectangular waveguide, as outlined and applied in Section 8.2.5. The basic equation is (8-64a), or

$$(\alpha_c)_{mn} = \frac{P_c \ell}{2P_{mn}} = \frac{P_\ell}{2P_{mn}} \quad (9-33)$$

which is based on the configuration of Figure 8-9.

It has been shown that the attenuation coefficients of the TE_{0n} ($n = 1, 2, \dots$) modes in a circular waveguide monotonically decrease as a function of frequency [2, 3]. This is a very desirable characteristic, and because of this the excitation, propagation, and detection of TE_{0n} modes in a circular waveguide have received considerable attention. It can be shown that the attenuation coefficients for the TE_{mn}^z and TM_{mn}^z modes inside a circular waveguide are given, respectively, by

$$\text{TE}_{mn}^z \quad (\alpha_c)_{mn}^{\text{TE}^z} = \frac{R_s}{a\eta \sqrt{1 - \left(\frac{f_c}{f}\right)^2}} \left[\left(\frac{f_c}{f}\right)^2 + \frac{m^2}{(\chi'_{mn})^2 - m^2} \right] \text{ Np/m} \quad (9-34a)$$

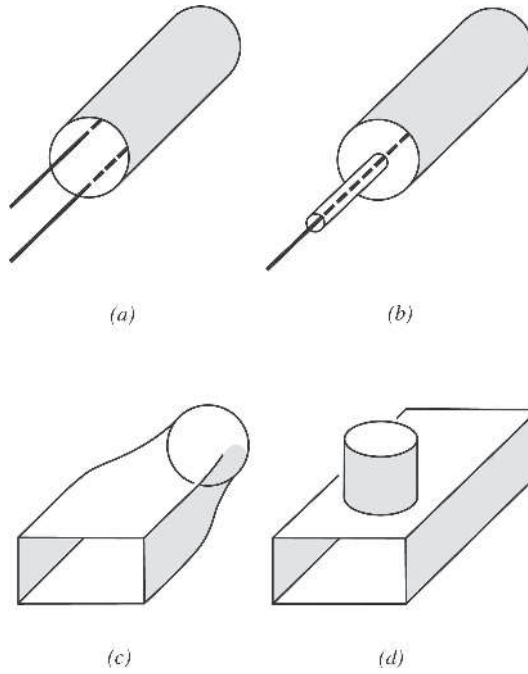


Figure 9-3 Excitation of TE_{mn} and TM_{mn} modes in a circular waveguide. (a) TE_{11} mode. (b) TM_{01} mode. (c) TE_{10} (rectangular)- TE_{11} (circular). (d) TE_{10} (rectangular)- TM_{01} (circular).

$$TM_{mn}^z$$

$$(\alpha_c)_{mn}^{TM^z} = \frac{R_s}{a\eta} \frac{1}{\sqrt{1 - \left(\frac{f_c}{f}\right)^2}} \text{ Np/m} \quad (9-34b)$$

Plots of the attenuation coefficient versus the normalized frequency f/f_c , where f_c is the cutoff frequency of the dominant TE_{11} mode, are shown for six modes in Figures 9-4a and 9-4b for waveguide radii of 1.5 and 3 cm, respectively. Within the waveguide is free space and its walls are made of copper ($\sigma = 5.7 \times 10^7 \text{ S/m}$).

Example 9-3

Derive the attenuation coefficient for the TE_{01} mode inside a circular waveguide of radius a .

Solution: According to (9-15a) through (9-15g), the electric and magnetic field components for the TE_{01} ($m = 0$, $n = 1$) mode reduce to

$$E_\rho^+ = E_z^+ = H_\phi^+ = 0$$

$$E_\phi^+ = \beta_\rho \frac{A_{01}}{\epsilon} J'_0(\beta_\rho \rho) e^{-j\beta_z z}$$

$$H_\rho^+ = -A_{01} \frac{\beta_\rho \beta_z}{\omega \mu \epsilon} J'_0(\beta_\rho \rho) e^{-j\beta_z z}$$

$$H_z^+ = -jA_{01} \frac{\beta_\rho^2}{\omega \mu \epsilon} J_0(\beta_\rho \rho) e^{-j\beta_z z}$$

where

$$\beta_\rho = \frac{\chi'_{01}}{a} = \frac{3.8318}{a}$$

Using these equations, the power through a cross section of the waveguide is equal to

$$\begin{aligned} P_{01} &= \frac{1}{2} \iint_{A_0} \operatorname{Re} [(\mathbf{E} \times \mathbf{H}^*) \cdot d\mathbf{s}] \\ &= \frac{1}{2} \iint_{A_0} \operatorname{Re} \left[\hat{\mathbf{a}}_\phi E_\phi \times (\hat{\mathbf{a}}_\rho H_\rho + \hat{\mathbf{a}}_z H_z)^* \right] \cdot \hat{\mathbf{a}}_z \, ds \\ P_{01} &= -\frac{1}{2} \operatorname{Re} \int_0^{2\pi} \int_0^a (E_\phi H_\rho^*) \rho d\rho d\phi = |A_{01}|^2 \frac{\pi \beta_z \beta_\rho^2}{\omega \mu \varepsilon^2} \int_0^a \left[J'_0 \left(\frac{\chi'_{01}}{a} \rho \right) \right]^2 \rho d\rho \end{aligned}$$

Since

$$\frac{dJ_p(cx)}{d(cx)} = -J_{p+1}(cx) + \frac{p}{cx} J_p(cx)$$

then

$$J'_0 \left(\frac{\chi'_{01}}{a} \rho \right) = \frac{d}{d(\chi'_{01} \rho / a)} J_0 \left(\frac{\chi'_{01}}{a} \rho \right) = -J_1 \left(\frac{\chi'_{01}}{a} \rho \right)$$

Thus

$$P_{01} = |A_{01}|^2 \frac{\pi \beta_z}{\omega \mu \varepsilon^2} \left(\frac{\chi'_{01}}{a} \right)^2 \int_0^a J_1^2 \left(\frac{\chi'_{01}}{a} \rho \right) \rho d\rho$$

Since

$$\int_b^c x J_p^2(cx) dx = \frac{x^2}{2} \left[J_p^2(cx) - J_{p-1}(cx) J_{p+1}(cx) \right]_b^c$$

then

$$\begin{aligned} \int_0^a \rho J_1^2 \left(\frac{\chi'_{01}}{a} \rho \right) d\rho &= \frac{a^2}{2} \left[J_1^2(\chi'_{01}) - J_0(\chi'_{01}) J_2(\chi'_{01}) \right] \\ &= -\frac{a^2}{2} J_0(\chi'_{01}) J_2(\chi'_{01}) = \frac{a^2}{2} J_0^2(\chi'_{01}) \end{aligned}$$

because

$$\begin{aligned} J_1^2(\chi'_{01}) &= J_1^2(3.8318) = 0 \\ J_2(\chi'_{01}) &= -J_0(\chi'_{01}) \end{aligned}$$

Therefore, the power of the TE₀₁ can be written as

$$P_{01} = |A_{01}|^2 \frac{\pi \beta_z}{2 \omega \mu \varepsilon^2} (\chi'_{01})^2 J_0^2(\chi'_{01})$$

The power dissipated on the walls of the waveguide is obtained using

$$P_c = \frac{R_s}{2} \iint_{S_w} (\mathbf{J}_s \cdot \mathbf{J}_s^*)_{\rho=a} ds = \frac{R_s}{2} \int_0^\ell \int_0^{2\pi} |\mathbf{J}_s|_{\rho=a}^2 a d\phi dz$$

where

$$\mathbf{J}_s|_{\rho=a} = \hat{\mathbf{n}} \times \mathbf{H}^+|_{\rho=a} = \hat{\mathbf{a}}_\phi H_z^+|_{\rho=a} = -\hat{\mathbf{a}}_\phi j \frac{\beta_\rho^2}{\omega \mu \varepsilon} A_{01} J_0(\beta_\rho a) e^{-j\beta_z z}$$

Thus,

$$P_c = |A_{01}|^2 \frac{R_s}{2} \left(\frac{\beta_p^2}{\omega \mu \varepsilon} \right)^2 a J_0^2(\chi'_{01}) \int_0^\ell \int_0^{2\pi} d\phi dz$$

or

$$\frac{P_c}{\ell} = P_\ell = |A_{01}|^2 \frac{\pi R_s}{a^3} \left[\frac{(\chi'_{01})^2}{\omega \mu \varepsilon} \right]^2 J_0^2(\chi'_{01})$$

Therefore, the attenuation coefficient of (9-33) for the TE_{01} mode can now be written as

$$\alpha_{01}(TE^z) = \frac{R_s}{a\eta} \frac{\left(\frac{f_c}{f} \right)^2}{\sqrt{1 - \left(\frac{f_c}{f} \right)^2}} \text{ Np/m}$$

It is evident from the results of the preceding example that as f_c/f becomes smaller, the attenuation coefficient decreases monotonically (as shown in Figure 9-4), which is a desirable characteristic. It should be noted that similar monotonically decreasing variations in the attenuation coefficient are evident in all TE_{0n} modes ($n = 1, 2, 3, \dots$). According to (9-15a) through (9-15f), the only tangential magnetic field components to the conducting surface of the waveguide for all these TE_{0n} ($m = 0$) modes is the H_z component, while the electric field lines are circular. Therefore, these modes are usually referred to as circular electric modes. For a constant power in the wave, the H_z component decreases as the frequency increases and approaches zero at infinite frequency. Simultaneously, the current density and conductor losses on the waveguide walls also decrease and approach zero. Because of this attractive feature, these modes have received considerable attention for long distance propagation of energy, especially at millimeter wave frequencies. Typically, attenuations as low as 1.25 dB/km (2 dB/mi) have been attained [2]. This is to be compared with attenuations of 120 dB/km for WR90 copper rectangular waveguides, and 3 dB/km at $0.85 \mu\text{m}$, and less than 0.5 dB/km at $1.3 \mu\text{m}$ for fiber optics cables.

Although the TE_{0n} modes are very attractive from the attenuation point of view, there are a number of problems associated with their excitation and retention. One of the problems is that the TE_{01} mode, which is the first of the TE_{0n} modes, is not the dominant mode. Therefore, in order for this mode to be above its cutoff frequency and propagate in the waveguide, a number of other modes (such as the TE_{11} , TM_{01} , TE_{21} , and TM_{11}) with lower cutoff frequencies can also exist. Additional modes can also be present if the operating frequency is chosen well above the cutoff frequency of the TE_{01} mode in order to provide a margin of safety from being too close to its cutoff frequency.

To support the TE_{01} mode, the waveguide must be oversized and it can support a number of other modes. One of the problems faced with such a guide is how to excite the desired TE_{01} mode with sufficient purity and suppress the others. Another problem is how to prevent coupling between the TE_{01} mode and undesired modes that can exist since the guide is oversized. The presence of the undesired modes causes not only higher losses but dispersion and attenuation distortion to the signal since each exhibits different phase velocities and attenuation. Irregularities in the inner geometry, surface, and direction (such as bends, nonuniform cross sections, etc.) of the waveguide are the main contributors to the coupling to the undesired modes. However, for the guide to be of any practical use, it must be able to sustain and propagate the desired TE_{01} and other TE_{0n} modes efficiently over bends of reasonable curvature. One technique that has been implemented to

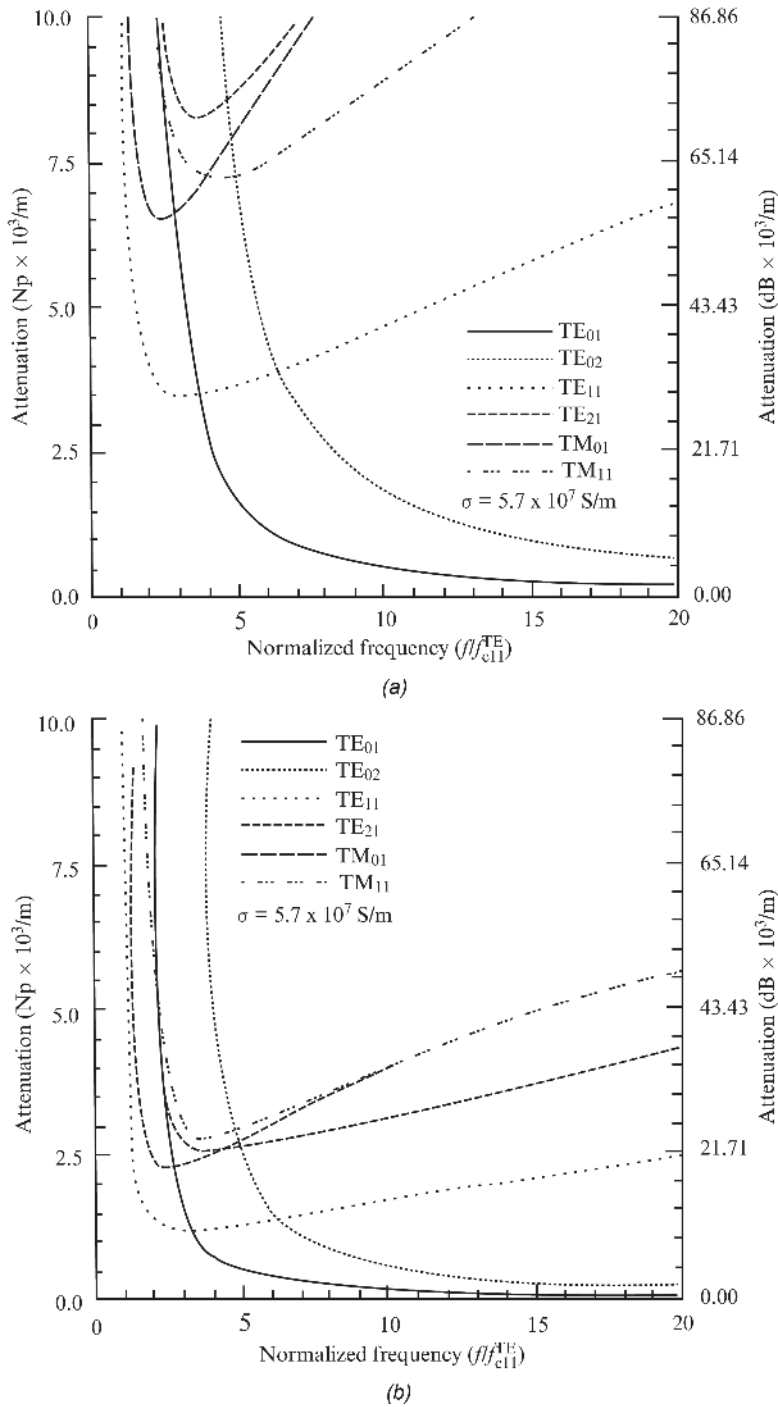


Figure 9-4 Attenuation for TE_{mn}^z and TM_{mn}^z modes in a circular waveguide. (a) $a = 1.5 \text{ cm}$. (b) $a = 3 \text{ cm}$.

achieve this is to use mode conversion before entering the corner and another conversion when exiting to convert back to the desired TE_{0n} mode(s).

Another method that has been used to discriminate against undesired modes and avoid coupling to them is to introduce filters inside the guide that cause negligible attenuation to the desired TE_{0n} mode(s). The basic principle of these filters is to introduce cuts that are perpendicular to the current paths of the undesired modes and parallel to the current direction of the desired mode(s). Since the current path of the undesired modes is along the axis (z direction) of the guide and the path of the desired TE_{0n} modes is along the circumference (ϕ direction), a helical wound wire placed on the inside surface of the guide can serve as a filter that discourages any mode that requires an axial component of current flow but propagates the desired TE_{0n} modes [3, 4].

Another filter that can be used to suppress undesired modes is to introduce within the guide very thin baffles of lossy material that will act as attenuating sheets. The surfaces of the baffles are placed in the radial direction of the guide so that they are parallel to the E_ρ and E_z components of the undesired modes (which will be damped) and normal to the E_ϕ component of the TE_{0n} modes that will remain unaffected. Typically, two baffles are used and are placed in a crossed pattern over the cross section of the guide.

A summary of the pertinent characteristics of the TE_{nm}^z and TM_{nm}^z modes of a circular waveguide are found listed in Table 9-3.

9.3 CIRCULAR CAVITY

As in rectangular waveguides, a circular cavity is formed by closing the two ends of the waveguide with plates, as shown in Figure 9-5. Coupling in and out of the cavity is done using either irises (holes) or probes (antennas), some of which were illustrated in Figure 9-3. Since the boundary conditions along the circumferential surface of the waveguide are the same as those of the cavity, the analysis can begin by modifying only the traveling waves of the z variations of the waveguide in order to obtain the standing waves of the cavity. The radial (ρ) and circumferential (ϕ) variations in both cases will be the same. For the circular cavity, both TE^z and TM^z modes can exist and will be examined here.

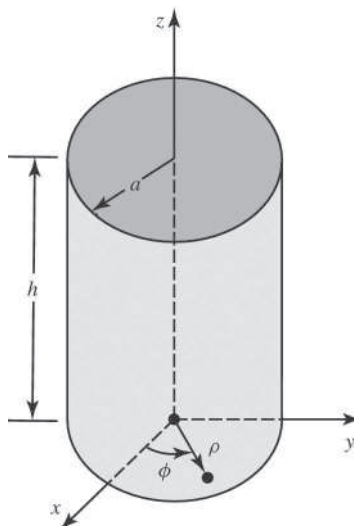


Figure 9-5 Geometry for circular cavity.

TABLE 9-3 Summary of TE_{mn}^z and TM_{mn}^z mode characteristics of circular waveguide

	$\text{TE}_{mn}^z \begin{pmatrix} m=0,1,2,\dots, \\ n=1,2,3,\dots \end{pmatrix}$	$\text{TM}_{mn}^z \begin{pmatrix} m=0,1,2,\dots, \\ n=1,2,3,\dots \end{pmatrix}$
E_ρ^+	$-A_{mn} \frac{m}{\varepsilon \rho} J_m(\beta_\rho \rho) [-C_2 \sin(m\phi) + D_2 \cos(m\phi)] e^{-j\beta_z z}$	$-B_{mn} \frac{\beta_\rho \beta_z}{\omega \mu \varepsilon} J'_m(\beta_\rho \rho) [C_2 \cos(m\phi) + D_2 \sin(m\phi)] e^{-j\beta_z z}$
E_ϕ^+	$A_{mn} \frac{\beta_\rho}{\varepsilon} J'_m(\beta_\rho \rho) [C_2 \cos(m\phi) + D_2 \sin(m\phi)] e^{-j\beta_z z}$	$-B_{mn} \frac{m \beta_z}{\omega \mu \varepsilon} \frac{1}{\rho} J_m(\beta_\rho \rho) [-C_2 \sin(m\phi) + D_2 \cos(m\phi)] e^{-j\beta_z z}$
E_z^+	0	$-j B_{mn} \frac{\beta_\rho^2}{\omega \mu \varepsilon} J_m(\beta_\rho \rho) [C_2 \cos(m\phi) + D_2 \sin(m\phi)] e^{-j\beta_z z}$
H_ρ^+	$-A_{mn} \frac{\beta_\rho \beta_z}{\omega \mu \varepsilon} J'_m(\beta_\rho \rho) [C_2 \cos(m\phi) + D_2 \sin(m\phi)] e^{-j\beta_z z}$	$B_{mn} \frac{m}{\mu} \frac{1}{\rho} J_m(\beta_\rho \rho) [-C_2 \sin(m\phi) + D_2 \cos(m\phi)] e^{-j\beta_z z}$
H_ϕ^+	$-A_{mn} \frac{m \beta_z}{\omega \mu \varepsilon} \frac{1}{\rho} J_m(\beta_\rho \rho) [-C_2 \sin(m\phi) + D_2 \cos(m\phi)] e^{-j\beta_z z}$	$-B_{mn} \frac{\beta_\rho}{\mu} J'_m(\beta_\rho \rho) [C_2 \cos(m\phi) + D_2 \sin(m\phi)] e^{-j\beta_z z}$
H_z^+	$-j A_{mn} \frac{\beta_\rho^2}{\omega \mu \varepsilon} J_m(\beta_\rho \rho) [C_2 \cos(m\phi) + D_2 \sin(m\phi)] e^{-j\beta_z z}$	0
ι		$\frac{\partial}{\partial(\beta_\rho \rho)}$
$\beta_c = \beta_\rho$	$\frac{\chi'_{mn}}{a}$	$\frac{\chi_{mn}}{a}$
f_c	$\frac{\chi'_{mn}}{2\pi a \sqrt{\mu \varepsilon}}$	$\frac{\chi_{mn}}{2\pi a \sqrt{\mu \varepsilon}}$
λ_c	$\frac{2\pi a}{\chi_{mn}}$	$\frac{2\pi a}{\chi_{mn}}$

(continued overleaf)

TABLE 9-3 (Continued)

	$\text{TE}_{mn}^z \begin{pmatrix} m=0,1,2,\dots, \\ n=1,2,3,\dots \end{pmatrix}$	$\text{TM}_{mn}^z \begin{pmatrix} m=0,1,2,\dots, \\ n=1,2,3,\dots \end{pmatrix}$
${}^c f_c \geq f_c$		$\beta \sqrt{1 - \left(\frac{f_c}{f}\right)^2} = \beta \sqrt{1 - \left(\frac{\lambda}{\lambda_c}\right)^2}$
$\lambda_g(f) \geq f_c$		$\frac{\lambda}{\sqrt{1 - \left(\frac{f_c}{f}\right)^2}} = \frac{\lambda}{\sqrt{1 - \left(\frac{\lambda}{\lambda_c}\right)^2}}$
${}^c f \geq f^d \nu$		$\frac{\nu}{\sqrt{1 - \left(\frac{f_c}{f}\right)^2}} = \frac{\nu}{\sqrt{1 - \left(\frac{\lambda}{\lambda_c}\right)^2}}$
$Z_w(f) \geq f$	$\frac{\eta}{\sqrt{1 - \left(\frac{f_c}{f}\right)^2}} = \frac{\eta}{\sqrt{1 - \left(\frac{\lambda}{\lambda_c}\right)^2}}$	$\sqrt{\left(\frac{f_c}{f}\right)^2 - 1} \sqrt{\nu} = \sqrt{\left(\frac{\lambda}{\lambda_c}\right)^2 - 1} \sqrt{u}$
$Z_w(f) \leq f$	$j \frac{\eta}{\sqrt{\left(\frac{f_c}{f}\right)^2 - 1}} = j \frac{\eta}{\sqrt{\left(\frac{\lambda}{\lambda_c}\right)^2 - 1}}$	$-j\eta \sqrt{\left(\frac{f_c}{f}\right)^2 - 1} = -j\eta \sqrt{\left(\frac{\lambda}{\lambda_c}\right)^2 - 1}$
α_c	$\frac{R_s}{m} \left[\left(\frac{f_c}{f}\right)^2 - 1 \right] \sqrt{\left(\frac{f_c}{f}\right)^2 - m^2} + \frac{m^2}{(\chi_{nm}')^2 - m^2}$	$\frac{R_s}{a\eta} \sqrt{\left(\frac{f_c}{f}\right)^2 - 1}$

9.3.1 Transverse Electric (TE^z) Modes

We begin the analysis of the TE^z modes by assuming the vector potential F_z is that of (9-7) modified so that the z variations are standing waves instead of traveling waves. Thus, using (9-7), we can write that

$$F_z(\rho, \phi, z) = A_{mn} J_m(\beta_\rho \rho) [C_2 \cos(m\phi) + D_2 \sin(m\phi)] \times [C_3 \cos(\beta_z z) + D_3 \sin(\beta_z z)] \quad (9-35)$$

where

$$\beta_\rho = \frac{\chi'_{mn}}{a} \quad (9-35a)$$

$$m = 0, 1, 2, 3, \dots \quad (9-35b)$$

To determine the permissible values of β_z , we must apply the additional boundary conditions introduced by the presence of the end plates. These additional boundary conditions are

$$E_\rho(0 \leq \rho \leq a, 0 \leq \phi \leq 2\pi, z = 0) = E_\rho(0 \leq \rho \leq a, 0 \leq \phi \leq 2\pi, z = h) = 0 \quad (9-36a)$$

$$E_\phi(0 \leq \rho \leq a, 0 \leq \phi \leq 2\pi, z = 0) = E_\phi(0 \leq \rho \leq a, 0 \leq \phi \leq 2\pi, z = h) = 0 \quad (9-36b)$$

Since both boundary conditions are not independent, using either of the two leads to the same results.

Using (6-80) and (9-35), we can write the E_ϕ component as

$$E_\phi = \frac{1}{\varepsilon} \frac{\partial F_z}{\partial \rho} = \beta_\rho \frac{A_{mn}}{\varepsilon} J'_m(\beta_\rho \rho) [C_2 \cos(m\phi) + D_2 \sin(m\phi)] [C_3 \cos(\beta_z z) + D_3 \sin(\beta_z z)] \quad (9-37)$$

where

$$' \equiv \frac{\partial}{\partial(\beta_\rho \rho)} \quad (9-37a)$$

Applying (9-36b) leads to

$$\begin{aligned} E_\phi(0 \leq \rho \leq a, 0 \leq \phi \leq 2\pi, z = 0) \\ = \beta_\rho \frac{A_{mn}}{\varepsilon} J'_m(\beta_\rho \rho) [C_2 \cos(m\phi) + D_2 \sin(m\phi)] [C_3(1) + D_3(0)] = 0 \\ \Rightarrow C_3 = 0 \end{aligned} \quad (9-38a)$$

$$\begin{aligned} E_\phi(0 \leq \rho \leq a, 0 \leq \phi \leq 2\pi, z = h) \\ = \beta_\rho \frac{A_{mn}}{\varepsilon} J'_m(\beta_\rho \rho) [C_2 \cos(m\phi) + D_2 \sin(m\phi)] D_3 \sin(\beta_z h) = 0 \\ \sin(\beta_z h) = 0 \Rightarrow \beta_z h = \sin^{-1}(0) = p\pi \\ \beta_z = \frac{p\pi}{h}, \quad p = 1, 2, 3, \dots \end{aligned} \quad (9-38b)$$

Thus, the resonant frequency is obtained using

$$\beta_\rho^2 + \beta_z^2 = \left(\frac{\chi'_{mn}}{a} \right)^2 + \left(\frac{p\pi}{h} \right)^2 = \beta_r^2 = \omega_r^2 \mu \varepsilon \quad (9-39)$$

or

$$(f_r)_{mnp}^{\text{TE}^z} = \frac{1}{2\pi\sqrt{\mu\varepsilon}} \sqrt{\left(\frac{\chi'_{mn}}{a} \right)^2 + \left(\frac{p\pi}{h} \right)^2} \quad \begin{array}{l} m = 0, 1, 2, 3, \dots \\ n = 1, 2, 3, \dots \\ p = 1, 2, 3, \dots \end{array} \quad (9-39a)$$

The values of χ'_{mn} are found listed in Table 9-1. The final form of F_z of (9-35) is

$$F_z(\rho, \phi, z) = A_{mnp} J_m(\beta_\rho \rho) [C_2 \cos(m\phi) + D_2 \sin(m\phi)] \sin(\beta_z z) \quad (9-40)$$

9.3.2 Transverse Magnetic (TM^z) Modes

The analysis for the TM^z modes in a circular cavity proceeds in the same manner as for the TE^z modes of the previous section. Using (9-22), we can write that

$$A_z(\rho, \phi, z) = B_{mn} J_m(\beta_\rho \rho) [C_2 \cos(m\phi) + D_2 \sin(m\phi)] \times [C_3 \cos(\beta_z z) + D_3 \sin(\beta_z z)] \quad (9-41)$$

where

$$\beta_\rho = \frac{\chi_{mn}}{a} \quad (9-41a)$$

$$m = 0, 1, 2, \dots \quad (9-41b)$$

Using (6-70) and (9-41), we can write the E_ϕ component as

$$\begin{aligned} E_\phi(\rho, \phi, z) &= -j \frac{1}{\omega \mu \varepsilon} \frac{1}{\rho} \frac{\partial^2 A_z}{\partial \phi \partial z} \\ &= -j B_{mn} \frac{m \beta_z}{\omega \mu \varepsilon} \frac{1}{\rho} J_m(\beta_\rho \rho) [-C_2 \sin(m\phi) + D_2 \cos(m\phi)] \\ &\quad \times [-C_3 \sin(\beta_z z) + D_3 \cos(\beta_z z)] \end{aligned} \quad (9-42)$$

Applying the boundary conditions of (9-36b) leads to

$$\begin{aligned} E_\phi(0 \leq \rho \leq a, 0 \leq \phi \leq 2\pi, z = 0) \\ = -j B_{mn} \frac{m \beta_z}{\omega \mu \varepsilon} \frac{1}{\rho} J_m(\beta_\rho \rho) [-C_2 \sin(m\phi) + D_2 \cos(m\phi)] \\ \times [-C_3(0) + D_3(1)] = 0 \Rightarrow D_3 = 0 \end{aligned} \quad (9-43a)$$

$$\begin{aligned} E_\phi(0 \leq \rho \leq a, 0 \leq \phi \leq 2\pi, z = h) \\ = j B_{mn} \frac{m \beta_z}{\omega \mu \varepsilon} \frac{1}{\rho} J_m(\beta_\rho \rho) [-C_2 \sin(m\phi) + D_2 \cos(m\phi)] [C_3 \sin(\beta_z h)] = 0 \\ \sin(\beta_z h) = 0 \Rightarrow \beta_z h = \sin^{-1}(0) = p\pi \\ \beta_z = \frac{p\pi}{h} \quad p = 0, 1, 2, 3, \dots \end{aligned} \quad (9-43b)$$

Thus, the resonant frequency is obtained using

$$\beta_\rho^2 + \beta_z^2 = \left(\frac{\chi_{mn}}{a}\right)^2 + \left(\frac{p\pi}{h}\right)^2 = \beta_r^2 = \omega_r^2 \mu \varepsilon \quad (9-44)$$

or

$$(f_r)_{mnp}^{\text{TM}^\varepsilon} = \frac{1}{2\pi\sqrt{\mu\varepsilon}} \sqrt{\left(\frac{\chi_{mn}}{a}\right)^2 + \left(\frac{p\pi}{h}\right)^2} \quad \begin{array}{l} m = 0, 1, 2, 3, \dots \\ n = 1, 2, 3, \dots \\ p = 0, 1, 2, 3, \dots \end{array} \quad (9-45)$$

The values of χ_{mn} are found listed in Table 9-2. The final form of A_z of (9-41) is

$$A_z(\rho, \phi, z) = B_{mnp} J_m(\beta_\rho \rho) [C_2 \cos(m\phi) + D_2 \sin(m\phi)] \cos(\beta_z z) \quad (9-46)$$

The resonant frequencies of the TE_{mnp}^z and TM_{mnp}^z modes, as given respectively by (9-39a) and (9-45), are functions of the h/a ratio and they are listed in Table 9-4.

The TE_{mnp}^z mode with the smallest resonant frequency is the TE_{111} , and its cutoff frequency is given by

$$(f_r)_{111}^{\text{TE}^\varepsilon} = \frac{1}{2\pi\sqrt{\mu\varepsilon}} \sqrt{\left(\frac{1.8412}{a}\right)^2 + \left(\frac{\pi}{h}\right)^2} \quad (9-47a)$$

Similarly, the TM_{mnp}^z mode with the smallest resonant frequency is the TM_{010} , and its cutoff frequency is given by

$$(f_r)_{010}^{\text{TM}^\varepsilon} = \frac{1}{2\pi\sqrt{\mu\varepsilon}} \sqrt{\left(\frac{2.4049}{a}\right)^2} \quad (9-47b)$$

Equating (9-47a) to (9-47b) indicates that the two are identical (degenerate modes) when

$$\frac{h}{a} = 2.03 \simeq 2 \quad (9-48)$$

When $h/a < 2.03$ the dominant mode is the TM_{010} , whereas for $h/a > 2.03$ the dominant mode is the TE_{111} mode.

9.3.3 Quality Factor Q

One of the most important parameters of a cavity is its quality factor, better known as the Q , which is defined by (8-84). The Q of the TM_{010} mode, which is the dominant mode when $h/a < 2.03$, is of particular interest and it will be derived here.

For the TM_{010} mode, the potential function of (9-46) reduces to

$$A_z = B_{010} J_0(\beta_\rho \rho) \quad (9-49)$$

where

$$\beta_\rho = \frac{\chi_{01}}{a} = \frac{2.4049}{a} \quad (9-49a)$$

TABLE 9-4 Resonant frequencies for the TE_{mnp} and TM_{mnp} modes of a circular cavity

$$R_{dom}^{mnp} = \frac{(f_r)_{mnp}}{(f_r)_{dom}}$$

$\frac{h}{a}$	R_{dom}^{mnp}									
0	$\frac{TM_{010}}{TM_{010}}$	$\frac{TM_{110}}{TM_{010}}$	$\frac{TM_{210}}{TM_{010}}$	$\frac{TM_{020}}{TM_{010}}$	$\frac{TM_{310}}{TM_{010}}$	$\frac{TM_{120}}{TM_{010}}$	$\frac{TM_{410}}{TM_{010}}$	$\frac{TM_{220}}{TM_{010}}$	$\frac{TM_{030}}{TM_{010}}$	$\frac{TM_{510}}{TM_{010}}$
	1.000	1.593	2.136	2.295	2.653	2.917	3.155	3.500	3.598	3.647
0.5	$\frac{TM_{010}}{TM_{010}}$	$\frac{TM_{110}}{TM_{010}}$	$\frac{TM_{210}}{TM_{010}}$	$\frac{TM_{020}}{TM_{010}}$	$\frac{TM_{310}}{TM_{010}}$	$\frac{TE_{111}}{TM_{010}}$	$\frac{TM_{011}}{TM_{010}}$	$\frac{TE_{211}}{TM_{010}}$	$\frac{TM_{120}}{TM_{010}}$	$\frac{TE_{011}}{TM_{010}}$
	1.000	1.593	2.136	2.295	2.653	2.722	2.797	2.905	2.917	3.060
1.00	$\frac{TM_{010}}{TM_{010}}$	$\frac{TE_{111}}{TM_{010}}$	$\frac{TM_{110}}{TM_{010}}$	$\frac{TM_{011}}{TM_{010}}$	$\frac{TE_{211}}{TM_{010}}$	$\frac{TM_{111}}{TM_{010}}$	$\frac{TE_{011}}{TM_{010}}$	$\frac{TM_{210}}{TM_{010}}$	$\frac{TE_{311}}{TM_{010}}$	$\frac{TM_{020}}{TM_{010}}$
	1.000	1.514	1.593	1.645	1.822	2.060	2.060	2.136	2.181	2.295
2.03	$\frac{TM_{010}}{TM_{010}, TE_{111}}$	$\frac{TE_{111}}{TM_{010}, TE_{111}}$	$\frac{TM_{011}}{TM_{010}, TE_{111}}$	$\frac{TE_{211}}{TM_{010}, TE_{111}}$	$\frac{TE_{212}}{TM_{010}, TE_{111}}$	$\frac{TM_{110}}{TM_{010}, TE_{111}}$	$\frac{TM_{012}}{TM_{010}, TE_{111}}$	$\frac{TE_{011}}{TM_{010}, TE_{111}}$	$\frac{TM_{111}}{TM_{010}, TE_{111}}$	$\frac{TE_{212}}{TM_{010}, TE_{111}}$
	1.000	1.000	1.189	1.424	1.497	1.593	1.630	1.718	1.718	1.808
3.0	$\frac{TE_{111}}{TE_{111}}$	$\frac{TM_{010}}{TE_{111}}$	$\frac{TM_{011}}{TE_{111}}$	$\frac{TE_{112}}{TE_{111}}$	$\frac{TM_{012}}{TE_{111}}$	$\frac{TE_{211}}{TE_{111}}$	$\frac{TE_{113}}{TE_{111}}$	$\frac{TE_{212}}{TE_{111}}$	$\frac{TM_{110}}{TE_{111}}$	$\frac{TM_{013}}{TE_{111}}$
	1.000	1.136	1.238	1.317	1.506	1.524	1.719	1.748	1.809	1.868
4.0	$\frac{TE_{111}}{TE_{111}}$	$\frac{TM_{010}}{TE_{111}}$	$\frac{TE_{112}}{TE_{111}}$	$\frac{TM_{011}}{TE_{111}}$	$\frac{TM_{012}}{TE_{111}}$	$\frac{TE_{113}}{TE_{111}}$	$\frac{TE_{211}}{TE_{111}}$	$\frac{TM_{013}}{TE_{111}}$	$\frac{TE_{212}}{TE_{111}}$	$\frac{TE_{114}}{TE_{111}}$
	1.000	1.202	1.209	1.264	1.435	1.494	1.575	1.682	1.717	1.819
5.0	$\frac{TE_{111}}{TE_{111}}$	$\frac{TE_{112}}{TE_{111}}$	$\frac{TM_{010}}{TE_{111}}$	$\frac{TM_{011}}{TE_{111}}$	$\frac{TE_{113}}{TE_{111}}$	$\frac{TM_{012}}{TE_{111}}$	$\frac{TM_{013}}{TE_{111}}$	$\frac{TE_{114}}{TE_{111}}$	$\frac{TE_{211}}{TE_{111}}$	$\frac{TE_{212}}{TE_{111}}$
	1.000	1.146	1.236	1.278	1.354	1.395	1.571	1.602	1.603	1.698
10.0	$\frac{TE_{111}}{TE_{111}}$	$\frac{TE_{112}}{TE_{111}}$	$\frac{TE_{113}}{TE_{111}}$	$\frac{TE_{114}}{TE_{111}}$	$\frac{TM_{010}}{TE_{111}}$	$\frac{TE_{115}}{TE_{111}}$	$\frac{TM_{011}}{TE_{111}}$	$\frac{TM_{012}}{TE_{111}}$	$\frac{TM_{013}}{TE_{111}}$	$\frac{TE_{116}}{TE_{111}}$
	1.000	1.042	1.107	1.194	1.288	1.296	1.299	1.331	1.383	1.411
∞	$\frac{TE_{11p}}{TE_{111}}$	$\frac{TM_{01p}}{TE_{111}}$	$\frac{TE_{21p}}{TE_{111}}$	$\frac{TE_{01p}}{TE_{111}}$	$\frac{TM_{11p}}{TE_{111}}$	$\frac{TE_{31p}}{TE_{111}}$	$\frac{TM_{21p}}{TE_{111}}$	$\frac{TE_{41p}}{TE_{111}}$	$\frac{TE_{12p}}{TE_{111}}$	$\frac{TM_{02p}}{TE_{111}}$
	1.000	1.306	1.659	2.081	2.081	2.282	2.790	2.888	2.896	2.998

These corresponding electric and magnetic fields are obtained using (6-70), or

$$\begin{aligned}
 E_\rho &= -j \frac{1}{\omega_r \mu \varepsilon} \frac{\partial^2 A_z}{\partial \rho \partial z} = 0 \\
 E_\phi &= -j \frac{1}{\omega_r \mu \varepsilon} \frac{1}{\rho} \frac{\partial^2 A_z}{\partial \phi \partial z} = 0 \\
 E_z &= -j \frac{1}{\omega_r \mu \varepsilon} \left(\frac{\partial^2}{\partial z^2} + \beta_r^2 \right) A_z = -j \frac{\beta_r^2}{\omega_r \mu \varepsilon} B_{010} J_0 \left(\frac{\chi_{01}}{a} \rho \right) \\
 H_\rho &= \frac{1}{\mu} \frac{1}{\rho} \frac{\partial A_z}{\partial \phi} = 0 \\
 H_\phi &= -\frac{1}{\mu} \frac{\partial A_z}{\partial \rho} = -\frac{\chi_{01}}{a} \frac{B_{010}}{\mu} J'_0 \left(\frac{\chi_{01}}{a} \rho \right) \\
 H_z &= 0
 \end{aligned} \tag{9-50}$$

The total energy stored in the cavity is given by

$$\begin{aligned}
 W &= 2W_e = \frac{\varepsilon}{2} \iiint_V |\mathbf{E}|^2 dv = |B_{010}|^2 \frac{\varepsilon}{2} \left(\frac{\beta_r^2}{\omega_r \mu \varepsilon} \right)^2 \int_0^h \int_0^{2\pi} \int_0^a J_0^2 \left(\frac{\chi_{01}}{a} \rho \right) \rho d\rho d\phi dz \\
 W &= |B_{010}|^2 \pi h \varepsilon \left(\frac{\beta_r^2}{\omega_r \mu \varepsilon} \right)^2 \int_0^a J_0^2 \left(\frac{\chi_{01}}{a} \rho \right) \rho d\rho
 \end{aligned} \tag{9-51}$$

Since [5],

$$\int_0^a \rho J_0^2 \left(\frac{\chi_{01}}{a} \rho \right) d\rho = \frac{a^2}{2} J_1^2(\chi_{01}) \tag{9-52}$$

then (9-51) reduces to

$$W = |B_{010}|^2 \frac{\pi h \varepsilon}{2} \left(\frac{a \beta_r^2}{\omega_r \mu \varepsilon} \right)^2 J_1^2(\chi_{01}) \tag{9-53}$$

Because the medium within the cavity is assumed to be lossless, the total power is dissipated on the conducting walls of the cavity. Thus, we can write that

$$\begin{aligned}
 P_d &= \frac{R_s}{2} \iint_A |\mathbf{H}|^2 ds = \frac{R_s}{2} \left\{ \int_0^{2\pi} \int_0^h |\mathbf{H}|_{\rho=a}^2 a d\phi dz + 2 \int_0^{2\pi} \int_0^a |\mathbf{H}|_{z=0}^2 \rho d\rho d\phi \right\} \\
 &= |B_{010}|^2 \frac{R_s}{2\mu^2} \left(\frac{\chi_{01}}{a} \right)^2 \left\{ \int_0^{2\pi} \int_0^h [J'_0(\chi_{01})]^2 a dz d\phi + 2 \int_0^{2\pi} \int_0^a \left[J'_0 \left(\frac{\chi_{01}}{a} \rho \right) \right]^2 \rho d\rho d\phi \right\} \\
 P_d &= |B_{010}|^2 \frac{\pi R_s}{\mu^2} \left(\frac{\chi_{01}}{a} \right)^2 \left\{ ah [J'_0(\chi_{01})]^2 + 2 \int_0^a \left[J'_0 \left(\frac{\chi_{01}}{a} \rho \right) \right]^2 \rho d\rho \right\}
 \end{aligned} \tag{9-54}$$

Because

$$J'_0 \left(\frac{\chi_{01}}{a} \rho \right) = \frac{d}{d(\chi_{01} \rho / a)} \left[J_0 \left(\frac{\chi_{01}}{a} \rho \right) \right] = -J_1 \left(\frac{\chi_{01}}{a} \rho \right) \tag{9-55a}$$

and at $\rho = a$

$$J'_0(\chi_{01}) = -J_1(\chi_{01}) \tag{9-55b}$$

(9-54) reduces to

$$\begin{aligned}
 P_d &= |B_{010}|^2 \frac{\pi R_s}{\mu^2} \left(\frac{\chi_{01}}{a} \right)^2 \left\{ ahJ_1^2(\chi_{01}) + 2 \int_0^a J_1^2 \left(\frac{\chi_{01}}{a} \rho \right) \rho d\rho \right\} \\
 &= |B_{010}|^2 \frac{\pi R_s}{\mu^2} \left(\frac{\chi_{01}}{a} \right)^2 \left\{ ahJ_1^2(\chi_{01}) + 2 \left[\frac{a^2}{2} J_1^2(\chi_{01}) \right] \right\} \\
 P_d &= |B_{010}|^2 \frac{\pi R_s}{\mu^2} \left(\frac{\chi_{01}}{a} \right)^2 a(h+a)J_1^2(\chi_{01})
 \end{aligned} \tag{9-56}$$

Using the Q definition of (8-84) along with (9-53) and (9-56), we can write that

$$Q = \frac{\omega_r W}{P_d} = \frac{\beta_r^4 h a^3}{2\omega_r \varepsilon (h+a) R_s \chi_{01}^2} = \frac{\chi_{01} \sqrt{\frac{\mu}{\varepsilon}}}{2 \left(1 + \frac{a}{h} \right) R_s} = \frac{1.2025\eta}{R_s \left(1 + \frac{a}{h} \right)} \tag{9-57}$$

since for the TM_{010} mode ($m = 0$, $n = 1$, and $p = 0$)

$$\beta_\rho^2 + \beta_z^2 = \left(\frac{\chi_{01}}{a} \right)^2 = \beta_r^2 \Rightarrow \chi_{01} = \beta_r a \tag{9-57a}$$

Example 9-4

Compare the Q values of a circular cavity operating in the TM_{010} mode to those of a square-based rectangular cavity. The dimensions of each are such that the circular cavity is circumscribed by the square-based rectangular cavity.

Solution: According to (9-57) the Q of a circular cavity of radius a (or diameter d) and height h is given by

$$Q = \frac{1.2025\eta}{R_s \left(1 + \frac{a}{h} \right)} = \frac{1.2025\eta}{R_s \left(1 + \frac{d/2}{h} \right)}$$

For a square-based rectangular cavity to circumscribe a circular cavity, one of the sides of its base must be equal to the diameter and their heights must be equal. Therefore, with a base of $a = c = d$ on each of its sides and a height $b = h$, according to (8-88a) its Q is equal to

$$Q = \frac{1.1107\eta}{R_s \left[\frac{1}{1 + \frac{a/2}{b}} \right]} = \frac{1.1107\eta}{R_s \left[\frac{1}{1 + \frac{d/2}{h}} \right]}$$

Compare these two expressions and it is evident that the Q of the circular cavity is greater than that of the square-based cavity by

$$\left(\frac{1.2025 - 1.1107}{1.1107} \right) \times 100 = 8.26\%$$

This is expected since the circular cavity does not possess as many sharp corners and edges as the square-based cavity whose volume and surface area are not as well utilized by the interior fields. It should be remembered that the Q of a cavity is proportional to volume and inversely proportional to area.

9.4 RADIAL WAVEGUIDES

For a circular waveguide the waves travel in the $\pm z$ directions and their z variations are represented by the factor $e^{\pm j\beta_z z}$. Their constant phase planes (equiphases) are planes that are parallel to each other and perpendicular to the z direction. If the waves were traveling in the $\pm\phi$ direction, their variations in that direction would be represented by $e^{\pm jm\phi}$. Such waves are usually referred to as *circulating waves*, and their equiphase surfaces are constant ϕ planes. For waves that travel in the $\pm\rho$ (radial) direction, their variations in that direction would be represented by either $H_m^{(2)}(\beta_\rho\rho)$ or $H_m^{(1)}(\beta_\rho\rho)$. Such waves are usually referred to as *radial waves*, and their equiphases are constant ρ (radius) planes. The structures that support radial waves are referred to as radial waveguides, and they will be examined here. Examples are parallel plates, wedged plates (representing horn antennas), and others.

9.4.1 Parallel Plates

When two infinite long parallel plates are excited by a line source placed between them at the center, as shown in Figure 9-6, the excited waves travel in the radial direction and form radial waves. We shall examine here both the TE^z and TM^z modes in the region between the plates.

A. Transverse Electric (TE^z) Modes For the TE^z modes of Figure 9-6, the potential function F_z can be written according to (3-67a) through (3-69b) as

$$F_z(\rho, \phi, z) = [C_1 H_m^{(1)}(\beta_\rho \rho) + D_1 H_m^{(2)}(\beta_\rho \rho)] [C_2 \cos(m\phi) + D_2 \sin(m\phi)] \times [C_3 \cos(\beta_z z) + D_3 \sin(\beta_z z)] \quad (9-58)$$

where

$$\beta_\rho^2 + \beta_z^2 = \beta^2 \quad (9-58a)$$

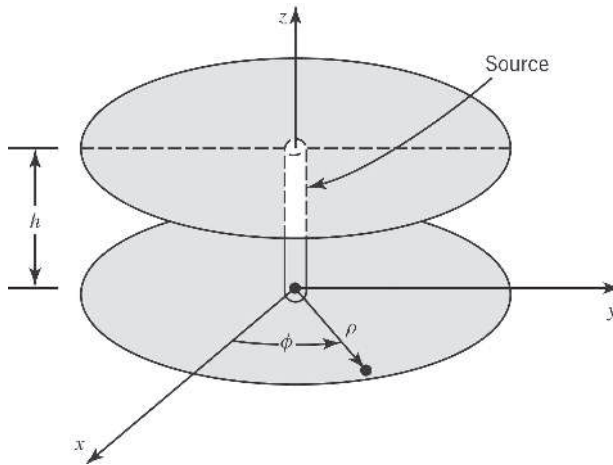


Figure 9-6 Geometry for radial waveguide.

The boundary conditions are

$$E_\rho(0 \leq \rho \leq \infty, 0 \leq \phi \leq 2\pi, z = 0) = E_\rho(0 \leq \rho \leq \infty, 0 \leq \phi \leq 2\pi, z = h) = 0 \quad (9-59a)$$

$$E_\phi(0 \leq \rho \leq \infty, 0 \leq \phi \leq 2\pi, z = 0) = E_\phi(0 \leq \rho \leq \infty, 0 \leq \phi \leq 2\pi, z = h) = 0 \quad (9-59b)$$

Since both of the preceding boundary conditions are not independent, either of the two leads to the same results.

Using (6-80) and (9-58), the E_ϕ component can be written as

$$\begin{aligned} E_\phi(\rho, \phi, z) &= \frac{1}{\varepsilon} \frac{\partial F_z}{\partial \rho} \\ &= \frac{\beta_\rho}{\varepsilon} [C_1 H_m^{(1)'}(\beta_\rho \rho) + D_1 H_m^{(2)'}(\beta_\rho \rho)] [C_2 \cos(m\phi) + D_2 \sin(m\phi)] \\ &\quad \times [C_3 \cos(\beta_z z) + D_3 \sin(\beta_z z)] \end{aligned} \quad (9-60)$$

Applying (9-59b) leads to

$$\begin{aligned} E_\phi(0 \leq \rho \leq \infty, 0 \leq \phi \leq 2\pi, z = 0) &= \frac{\beta_\rho}{\varepsilon} [C_1 H_m^{(1)'}(\beta_\rho \rho) + D_1 H_m^{(2)'}(\beta_\rho \rho)] \\ &\quad \times [C_2 \cos(m\phi) + D_2 \sin(m\phi)] [C_3(1) + D_3(0)] = 0 \\ &\Rightarrow C_3 = 0 \end{aligned} \quad (9-61a)$$

$$\begin{aligned} E_\phi(0 \leq \rho \leq a, 0 \leq \phi \leq 2\pi, z = h) &= \frac{\beta_\rho}{\varepsilon} [C_1 H_m^{(1)'}(\beta_\rho \rho) + D_1 H_m^{(2)'}(\beta_\rho \rho)] \\ &\quad \times [C_2 \cos(m\phi) + D_2 \sin(m\phi)] D_3 \sin(\beta_z h) = 0 \\ &\Rightarrow \sin(\beta_z h) = 0 \Rightarrow \beta_z h \sin^{-1}(0) = n\pi \\ \beta_z &= \frac{n\pi}{h}, \quad n = 1, 2, 3, \dots \end{aligned} \quad (9-61b)$$

Thus, F_z of (9-58) reduces to

$$\begin{aligned} F_z(\rho, \phi, z) &= [C_1 H_m^{(1)}(\beta_\rho \rho) + D_1 H_m^{(2)}(\beta_\rho \rho)] \\ &\quad \times [C_2 \cos(m\phi) + D_2 \sin(m\phi)] D_3 \sin(\beta_z z) \end{aligned} \quad (9-62)$$

where

$$\beta_\rho^2 + \beta_z^2 = \beta^2 \Rightarrow \beta_\rho = \pm \sqrt{\beta^2 - \beta_z^2} = \pm \sqrt{\beta^2 - \left(\frac{n\pi}{h}\right)^2} \quad (9-62a)$$

$$\beta_z = \frac{n\pi}{h}, \quad n = 1, 2, 3, \dots \quad (9-62b)$$

$$m = 0, 1, 2, \dots \quad (\text{because of periodicity of the fields in } \phi) \quad (9-62c)$$

Cutoff is defined when $\beta_\rho = 0$. Thus, using (9-62a)

$$\beta_\rho = \pm \sqrt{\beta^2 - \left(\frac{n\pi}{h}\right)^2} \Big|_{\substack{f=f_c \\ \beta=\beta_c}} = 0 \Rightarrow \beta_c = \frac{n\pi}{h} \quad (9-63)$$

or

$$\boxed{(f_c)_n^{\text{TE}} = \frac{n}{2h\sqrt{\mu\varepsilon}}, \quad n = 1, 2, 3, \dots} \quad (9-63a)$$

Therefore, using (9-63a), β_ρ of (9-62a) takes the following forms above, at, and below cutoff.

$$\beta_\rho = \begin{cases} \sqrt{\beta^2 - \left(\frac{n\pi}{h}\right)^2} = \beta \sqrt{1 - \left(\frac{f_c}{f}\right)^2} & f > f_c \\ 0 & f = f_c \\ -j\sqrt{\left(\frac{n\pi}{h}\right)^2 - \beta^2} = -j\beta\sqrt{\left(\frac{f_c}{f}\right)^2 - 1} = -j\alpha & f < f_c \end{cases} \quad \begin{matrix} (9-64a) \\ (9-64b) \\ (9-64c) \end{matrix}$$

Let us now examine the outward (+ ρ) traveling waves represented by $H_m^{(2)}(\beta_\rho \rho)$ and those represented simultaneously by the $\cos(m\phi)$ variations of (9-62). In that case, (9-62) reduces to

$$F_z^+(\rho, \phi, z) = A_{mn} H_m^{(2)}(\beta_\rho \rho) \cos(m\phi) \sin\left(\frac{n\pi}{h} z\right) \quad (9-65)$$

The corresponding electric and magnetic fields can be written using (6-80) as

$$E_\rho^+ = -\frac{1}{\varepsilon \rho} \frac{\partial F_z^+}{\partial \phi} = A_{mn} \frac{m}{\varepsilon \rho} H_m^{(2)}(\beta_\rho \rho) \sin(m\phi) \sin\left(\frac{n\pi}{h} z\right) \quad (9-65a)$$

$$E_\phi^+ = \frac{1}{\varepsilon} \frac{\partial F_z^+}{\partial \rho} = \beta_\rho \frac{A_{mn}}{\varepsilon} H_m^{(2)'}(\beta_\rho \rho) \cos(m\phi) \sin\left(\frac{n\pi}{h} z\right) \quad (9-65b)$$

$$E_z^+ = 0 \quad (9-65c)$$

$$H_\rho^+ = -j \frac{1}{\omega \mu \varepsilon} \frac{\partial^2 F_z^+}{\partial \rho \partial z} = -j A_{mn} \beta_\rho \frac{n\pi/h}{\omega \mu \varepsilon} H_m^{(2)'}(\beta_\rho \rho) \cos(m\phi) \cos\left(\frac{n\pi}{h} z\right) \quad (9-65d)$$

$$H_\phi^+ = -j \frac{1}{\omega \mu \varepsilon} \frac{1}{\rho} \frac{\partial^2 F_z^+}{\partial \phi \partial z} = j A_{mn} \frac{mn\pi/h}{\omega \mu \varepsilon} H_m^{(2)}(\beta_\rho \rho) \sin(m\phi) \cos\left(\frac{n\pi}{h} z\right) \quad (9-65e)$$

$$H_z^+ = -j \frac{1}{\omega \mu \varepsilon} \left(\frac{\partial^2}{\partial z^2} + \beta^2 \right) F_z^+ = -j A_{mn} \frac{\beta_\rho^2}{\omega \mu \varepsilon} H_m^{(2)}(\beta_\rho \rho) \cos(m\phi) \sin\left(\frac{n\pi}{h} z\right) \quad (9-65f)$$

$$' \equiv \frac{\partial}{\partial(\beta_\rho \rho)} \quad (9-65g)$$

The impedance of the wave in the + ρ direction is defined and given by

$$Z_w^{+\rho}(\text{TE}_n^z) = \frac{E_\phi}{H_z} = j \frac{\omega \mu}{\beta_\rho} \frac{H_m^{(2)'}(\beta_\rho \rho)}{H_m^{(2)}(\beta_\rho \rho)} \quad (9-66)$$

Since

$$H_m^{(2)'}(\beta_\rho \rho) = \frac{\partial}{\partial(\beta_\rho \rho)} [H_m^{(2)}(\beta_\rho \rho)] \quad (9-67)$$

(9-66) can be written as

$$Z_w^{+\rho}(\text{TE}_n^z) = j \frac{\omega \mu}{\beta_\rho} \frac{\frac{\partial}{\partial(\beta_\rho \rho)} [H_m^{(2)}(\beta_\rho \rho)]}{H_m^{(2)}(\beta_\rho \rho)} \quad (9-68)$$

Below cutoff ($f < f_c$) β_ρ is imaginary, and it is given by (9-64c). Therefore, for $f < f_c$

$$H_m^{(2)}(\beta_\rho \rho) = H_m^{(2)}(-j\alpha\rho) \quad (9-69a)$$

$$\frac{d}{d(\beta_\rho \rho)}[H_m^{(2)}(\beta_\rho \rho)] = \frac{d}{d(-j\alpha\rho)}[H_m^{(2)}(-j\alpha\rho)] = j \frac{d}{d(\alpha\rho)}[H_m^{(2)}(-j\alpha\rho)] \quad (9-69b)$$

For complex arguments, the Hankel function $H_m^{(2)}$ of the second kind is related to the modified Bessel function K_m of the second kind by

$$H_m^{(2)}(-j\alpha\rho) = \frac{2}{\pi} j^{m+1} K_m(\alpha\rho) \quad (9-70a)$$

$$\frac{d}{d(\alpha\rho)}[H_m^{(2)}(-j\alpha\rho)] = \frac{2}{\pi} j^{m+1} \frac{d}{d(\alpha\rho)}[K_m(\alpha\rho)] \quad (9-70b)$$

Thus, below cutoff ($f < f_c$), the wave impedance reduces to

$$\begin{aligned} Z_w^{+\rho}(\text{TE}_n^z)|_{f < f_c} &= j \frac{\omega\mu}{-j\alpha} \frac{j \frac{d}{d(\alpha\rho)}[K_m(\alpha\rho)]}{K_m(\alpha\rho)} \\ &= -j \frac{\omega\mu}{\alpha} \frac{\frac{d}{d(\alpha\rho)}[K_m(\alpha\rho)]}{K_m(\alpha\rho)} \end{aligned} \quad (9-71)$$

which is always inductive (for $f < f_c$) since $K_m(\alpha\rho) > 0$ and $d/d(\alpha\rho)[K_m(\alpha\rho)] < 0$. Therefore, below cutoff the modes are nonpropagating (evanescent) since the waveguide is behaving as an inductive storage element.

B. Transverse Magnetic (TM^z) Modes The TM^z modes of the radial waveguide structure of Figure 9-6 with the source at the center are derived in a similar manner. Using such a procedure leads to the following results:

$$\begin{aligned} A_z(\rho, \phi, z) &= [C_1' H_m^{(1)}(\beta_\rho \rho) + D_1' H_m^{(2)}(\beta_\rho \rho)] \\ &\quad \times [C_2' \cos(m\phi) + D_2' \sin(m\phi)] C_3' \cos(\beta_z z) \end{aligned} \quad (9-72)$$

$$\beta_\rho^2 + \beta_z^2 = \beta^2 \Rightarrow \beta_\rho = \pm \sqrt{\beta^2 - \beta_z^2} = \pm \sqrt{\beta^2 - \left(\frac{n\pi}{h}\right)^2} \quad (9-72a)$$

$$\beta_z = \frac{n\pi}{h}, \quad n = 0, 1, 2, \dots \quad (9-72b)$$

$$m = 0, 1, 2, \dots \quad (\text{because of periodicity of the fields in } \phi) \quad (9-72c)$$

$$(f_c)_n^{\text{TM}^z} = \frac{n}{2h\sqrt{\mu\varepsilon}}, \quad n = 0, 1, 2, \dots \quad (9-73)$$

$$\beta_\rho = \begin{cases} \sqrt{\beta^2 - \left(\frac{n\pi}{h}\right)^2} = \beta \sqrt{1 - \left(\frac{f_c}{f}\right)^2} & f > f_c \\ 0 & f = f_c \\ -j \sqrt{\left(\frac{n\pi}{h}\right)^2 - \beta^2} = -j\beta \sqrt{\left(\frac{f_c}{f}\right)^2 - 1} = -j\alpha & f < f_c \end{cases} \quad \begin{matrix} (9-74a) \\ (9-74b) \\ (9-74c) \end{matrix}$$

For outward (+ ρ) traveling waves and only $\cos(m\phi)$ variations

$$A_z^+(\rho, \phi, z) = B_{mn} H_m^{(2)}(\beta_\rho \rho) \cos(m\phi) \cos\left(\frac{n\pi}{h} z\right) \quad (9-75)$$

$$E_\rho^+ = -j \frac{1}{\omega \mu \varepsilon} \frac{\partial^2 A_z^+}{\partial \rho \partial z} = j B_{mn} \beta_\rho \frac{n\pi / h}{\omega \mu \varepsilon} H_m^{(2)'}(\beta_\rho \rho) \cos(m\phi) \sin\left(\frac{n\pi}{h} z\right) \quad (9-75a)$$

$$E_\phi^+ = -j \frac{1}{\omega \mu \varepsilon} \frac{1}{\rho} \frac{\partial^2 A_z^+}{\partial \phi \partial z} = -j B_{mn} \frac{mn\pi / h}{\omega \mu \varepsilon} H_m^{(2)}(\beta_\rho \rho) \sin(m\phi) \sin\left(\frac{n\pi}{h} z\right) \quad (9-75b)$$

$$E_z^+ = -j \frac{1}{\omega \mu \varepsilon} \left(\frac{\partial^2}{\partial z^2} + \beta^2 \right) A_z^+ = -j B_{mn} \frac{\beta_\rho^2}{\omega \mu \varepsilon} H_m^{(2)}(\beta_\rho \rho) \cos(m\phi) \cos\left(\frac{n\pi}{h} z\right) \quad (9-75c)$$

$$H_\rho^+ = \frac{1}{\mu} \frac{1}{\rho} \frac{\partial A_z^+}{\partial \phi} = -B_{mn} \frac{m}{\mu} \frac{1}{\rho} H_m^{(2)}(\beta_\rho \rho) \sin(m\phi) \cos\left(\frac{n\pi}{h} z\right) \quad (9-75d)$$

$$H_\phi^+ = -\frac{1}{\mu} \frac{\partial A_z^+}{\partial \rho} = -\beta_\rho \frac{B_{mn}}{\mu} H_m^{(2)'}(\beta_\rho \rho) \cos(m\phi) \cos\left(\frac{n\pi}{h} z\right) \quad (9-75e)$$

$$H_z = 0 \quad (9-75f)$$

$$' \equiv \frac{\partial}{\partial(\beta_\rho \rho)} \quad (9-75g)$$

$$Z_w^{+\rho}(\text{TM}_n^z) = \frac{E_z^+}{-H_\phi^+} = -j \frac{\beta_\rho}{\omega \varepsilon} \frac{H_m^{(2)}(\beta_\rho \rho)}{H_m^{(2)'}(\beta_\rho \rho)} \quad (9-76)$$

$$Z_w^{+\rho}(\text{TM}_n^z)|_{f < f_c} = j \frac{\alpha}{\omega \varepsilon} \frac{K_m(\alpha \rho)}{\frac{d}{d(\alpha \rho)}[K_m(\alpha \rho)]} \quad (9-76a)$$

which is always capacitive (for $f < f_c$) since $K_m(\alpha \rho) > 0$ and $d/d(\alpha \rho)[K_m(\alpha \rho)] < 0$. Therefore, below cutoff the modes are nonpropagating (evanescent) since the waveguide behaves as a capacitive storage element.

9.4.2 Wedged Plates

Another radial type of waveguide structure is the wedged-plate geometry of Figure 9-7 with plates along $z = 0, h$, and $\phi = 0, \phi_0$. This type of a configuration resembles and can be used to represent the structures of E - and H -plane sectoral horns [6]. In fact, the fields within the horns are found using the procedure outlined here. In general, both TE^z and TM^z modes can exist in the space

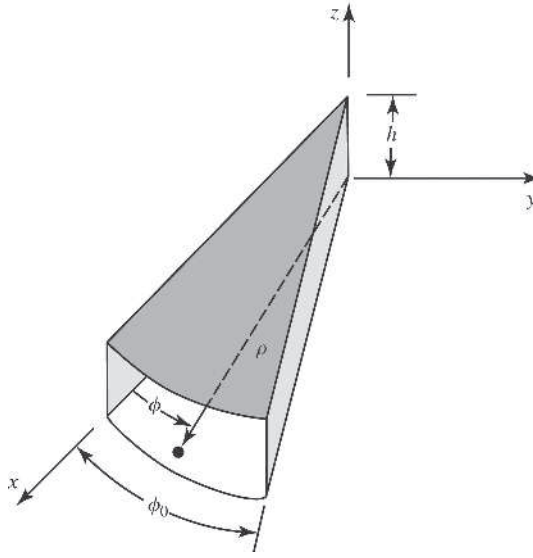


Figure 9-7 Geometry of wedged plate radial waveguide.

between the plates. The independent sets of boundary conditions of this structure that can be used to solve for the TE_{pn}^z and TM_{pn}^z modes are

$$E_\rho(0 \leq \rho \leq \infty, 0 \leq \phi \leq \phi_0, z = 0) = E_\rho(0 \leq \rho \leq \infty, 0 \leq \phi \leq \phi_0, z = h) = 0 \quad (9-77a)$$

$$E_\rho(0 \leq \rho \leq \infty, \phi = 0, 0 \leq z \leq h) = E_\rho(0 \leq \rho \leq \infty, \phi = \phi_0, 0 \leq z \leq h) = 0 \quad (9-77b)$$

or

$$E_\phi(0 \leq \rho \leq \infty, 0 \leq \phi \leq \phi_0, z = 0) = E_\phi(0 \leq \rho \leq \infty, 0 \leq \phi \leq \phi_0, z = h) = 0 \quad (9-78a)$$

$$E_z(0 \leq \rho \leq \infty, \phi = 0, 0 \leq z \leq h) = E_z(0 \leq \rho \leq \infty, \phi = \phi_0, 0 \leq z \leq h) = 0 \quad (9-78b)$$

or appropriate combinations of these. Whichever combination of independent boundary conditions is used, it leads to the same results.

A. Transverse Electric (TE^z) Modes Since the procedure used to derive this set of TE^z fields is the same as any other TE^z procedure used previously, the results of this set of TE_{pn}^z modes will be summarized here; the details are left as an end-of-chapter exercise for the reader. Only the outward radial ($+\rho$) parts will be included here.

$$F_z^+(\rho, \phi, z) = A_{pn} H_m^{(2)}(\beta_\rho \rho) \cos(m\phi) \sin(\beta_z z) \quad (9-79)$$

$\beta_\rho^2 + \beta_z^2 = \beta^2$	(9-79a)
$\beta_z = \frac{n\pi}{h} \quad n = 1, 2, 3, \dots$	(9-79b)
$m = \frac{p\pi}{\phi_0} \quad p = 0, 1, 2, \dots$	(9-79c)

$$E_\rho^+ = -\frac{1}{\varepsilon} \frac{1}{\rho} \frac{\partial F_z^+}{\partial \phi} = A_{pn} \frac{p\pi / \phi_0}{\varepsilon \rho} H_m^{(2)}(\beta_\rho \rho) \sin\left(\frac{p\pi}{\phi_0} \phi\right) \sin\left(\frac{n\pi}{h} z\right) \quad (9-79d)$$

$$E_\phi^+ = \frac{1}{\varepsilon} \frac{\partial F_z^+}{\partial \rho} = \beta_\rho \frac{A_{pn}}{\varepsilon} H_m^{(2)'}(\beta_\rho \rho) \cos\left(\frac{p\pi}{\phi_0} \phi\right) \sin\left(\frac{n\pi}{h} z\right) \quad (9-79e)$$

$$E_z^+ = 0 \quad (9-79f)$$

$$H_\rho^+ = -j \frac{1}{\omega \mu \varepsilon} \frac{\partial^2 F_z^+}{\partial \rho \partial z} = -j A_{pn} \frac{\beta_\rho \beta_z}{\omega \mu \varepsilon} H_m^{(2)'}(\beta_\rho \rho) \cos\left(\frac{p\pi}{\phi_0} \phi\right) \cos\left(\frac{n\pi}{h} z\right) \quad (9-79g)$$

$$H_\phi^+ = -j \frac{1}{\omega \mu \varepsilon} \frac{1}{\rho} \frac{\partial^2 F_z^+}{\partial \phi \partial z} = j A_{pn} \frac{\beta_z p\pi / \phi_0}{\omega \mu \varepsilon} \frac{1}{\rho} H_m^{(2)}(\beta_\rho \rho) \sin\left(\frac{p\pi}{\phi_0} \phi\right) \cos\left(\frac{n\pi}{h} z\right) \quad (9-79h)$$

$$H_z^+ = -j \frac{1}{\omega \mu \varepsilon} \left(\frac{\partial^2}{\partial z^2} + \beta^2 \right) F_z^+ = -j A_{pn} \frac{\beta_\rho^2}{\omega \mu \varepsilon} H_m^{(2)}(\beta_\rho \rho) \cos\left(\frac{p\pi}{\phi_0} \phi\right) \sin\left(\frac{n\pi}{h} z\right) \quad (9-79i)$$

$$Z_w^{+\rho}(\text{TE}_{pn}^z) = \frac{E_\phi^+}{H_z^+} = j \frac{\omega \mu}{\beta_\rho} \frac{H_m^{(2)'}(\beta_\rho \rho)}{H_m^{(2)}(\beta_\rho \rho)}; \quad ' \equiv \frac{\partial}{\partial(\beta_\rho \rho)} \quad (9-79j)$$

B. Transverse Magnetic (TM^z) Modes As for the TE_{pn}^z modes, the procedure for deriving the TM_{pn}^z for the wedged plate radial waveguide is the same as that used for TM^z modes of other waveguide configurations. Therefore, the results will be summarized here, and the details left as an end-of-chapter exercise for the reader. Only the outward radial (+ρ) parts will be included here.

$$A_z^+(\rho, \phi, z) = B_{pn} H_m^{(2)}(\beta_\rho \rho) \sin(m\phi) \cos(\beta_z z) \quad (9-80)$$

$$\beta_\rho^2 + \beta_z^2 = \beta^2 \quad (9-80a)$$

$$\beta_z = \frac{n\pi}{h}, \quad n = 0, 1, 2, \dots \quad (9-80b)$$

$$m = \frac{p\pi}{\phi_0}, \quad p = 1, 2, 3, \dots \quad (9-80c)$$

$$E_\rho^+ = -j \frac{1}{\omega \mu \varepsilon} \frac{\partial^2 A_z^+}{\partial \rho \partial z} = j B_{pn} \frac{\beta_z \beta_\rho}{\omega \mu \varepsilon} H_m^{(2)'}(\beta_\rho \rho) \sin\left(\frac{p\pi}{\phi_0} \phi\right) \sin\left(\frac{n\pi}{h} z\right) \quad (9-80d)$$

$$E_\phi^+ = -j \frac{1}{\omega \mu \varepsilon} \frac{1}{\rho} \frac{\partial^2 A_z^+}{\partial \phi \partial z} = j B_{pn} \frac{\beta_z p\pi / \phi_0}{\omega \mu \varepsilon} \frac{1}{\rho} H_m^{(2)}(\beta_\rho \rho) \cos\left(\frac{p\pi}{\phi_0} \phi\right) \sin\left(\frac{n\pi}{h} z\right) \quad (9-80e)$$

$$E_z^+ = -j \frac{1}{\omega \mu \varepsilon} \left(\frac{\partial^2}{\partial z^2} + \beta^2 \right) A_z^+ = -j B_{pn} \frac{\beta_\rho^2}{\omega \mu \varepsilon} H_m^{(2)}(\beta_\rho \rho) \sin\left(\frac{p\pi}{\phi_0} \phi\right) \cos\left(\frac{n\pi}{h} z\right) \quad (9-80f)$$

$$H_\rho^+ = \frac{1}{\mu} \frac{1}{\rho} \frac{\partial A_z^+}{\partial \phi} = B_{pn} \frac{p\pi / \phi_0}{\mu \rho} H_m^{(2)}(\beta_\rho \rho) \cos\left(\frac{p\pi}{\phi_0} \phi\right) \cos\left(\frac{n\pi}{h} z\right) \quad (9-80g)$$

$$H_\phi^+ = -\frac{1}{\mu} \frac{\partial A_z^+}{\partial \rho} = -\beta_\rho \frac{B_{pn}}{\mu} H_m^{(2)'}(\beta_\rho \rho) \sin\left(\frac{p\pi}{\phi_0} \phi\right) \cos\left(\frac{n\pi}{h} z\right) \quad (9-80h)$$

$$H_z^+ = 0 \quad (9-80i)$$

$$Z_w^{+\rho}(\text{TM}_{pn}^z) = \frac{E_z^+}{-H_\phi^+} = -j \frac{\beta_\rho}{\omega \varepsilon} \frac{H_m^{(2)}(\beta_\rho \rho)}{H_m^{(2)'}(\beta_\rho \rho)}; \quad ' \equiv \frac{\partial}{\partial(\beta_\rho \rho)} \quad (9-80j)$$

9.5 DIELECTRIC WAVEGUIDES AND RESONATORS

Guided electromagnetic propagation by dielectric media has been studied since as early as the 1920s by well-known people such as Rayleigh, Sommerfeld, and Debye. Dielectric slabs, strips, and rods have been used as waveguides, resonators, and antennas. Since the 1960s a most well-known dielectric waveguide, the fiber optic cable [7–19], has received attention and has played a key role in the general area of communication. Although the subject is very lengthy and involved, we will consider here simplified theories that give the propagation characteristics of the cylindrical dielectric rod waveguide, fiber optic cable, and the cylindrical dielectric resonator. Extensive material on each of these topics and others can be found in the literature.

9.5.1 Circular Dielectric Waveguide

The cylindrical dielectric waveguide that will be examined here is that of circular cross section, as shown in Figure 9-8. It usually consists of a high permittivity (ε_d) central core dielectric of radius a surrounded by a lower dielectric cladding (which is usually air). For simplicity, we usually assume that both are perfect dielectrics with permeabilities equal to that of free space. Such a structure can support an infinite number of modes. However, for a given set of permittivities and radius a , only a finite number of unattenuated waveguide modes exist with their fields localized in the central dielectric core. Generally, the fields within a dielectric waveguide will be TE and/or TM, as was demonstrated in Section 8.7 for the dielectric slab waveguide. However, for the cylindrical dielectric rod of Figure 9-8 pure TE(H) or TM(E) modes exist only when the field configurations are symmetrical and independent of ϕ . Modes that exhibit angular ϕ variations cannot be pure TE or TM modes. Instead field configurations that are combinations of TE (or H) and TM (or E) modes can be nonsymmetrical and possess angular ϕ variations. Such modes are usually referred to as *hybrid modes*, and are usually designated by IEEE (formerly IRE) Standards [20] as HEM_{mn} . In general, mode nomenclature for circular dielectric waveguides and resonators is not

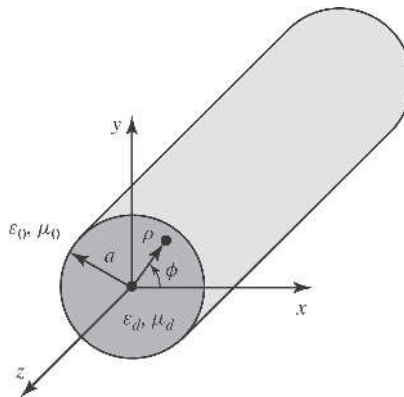


Figure 9-8 Geometry of circular dielectric waveguide.

well defined, and it is quite confusing. Another designation of the hybrid modes is to denote them as HE [when the TE(H) modes predominate] or EH [when the TM(E) modes predominate]. Pure TE or TM, or hybrid HEM (HE or EH) modes exhibit cutoff frequencies, below which unattenuating modes cannot propagate. The cutoff frequency is determined by the minimum electrical radius (a/λ) of the dielectric rod; for small values of a/λ the modes cannot propagate unattenuated within the rod. There is, however, one hybrid mode, namely the HEM_{11} (HE_{11}), which does not have a cutoff frequency. Because of its zero cutoff frequency, it is referred to as the *dominant mode* and is most widely used in dielectric rod waveguides and end-fire antennas. This HE_{11} mode is also popularly referred to as the *dipole mode*. An excellent reference on dielectric waveguides and resonators is that of [21].

The TE_{nm}^z and TM_{nm}^z electric and magnetic field components in a cylindrical waveguide are given, respectively, by (9-15a) through (9-15g) and (9-31a) through (9-31g). These expressions include both the $\cos(m\phi)$ and $\sin(m\phi)$ angular variations which, in general, both exist. The HEM modes that have only $\cos(m\phi)$ symmetry are combinations of the TE^z and TM^z modes which also exhibit $\cos(m\phi)$ symmetry. Use of the expressions (9-15a) through (9-15g) and (9-31a) through (9-31g), and selection of only the terms that possess simultaneously $\cos(m\phi)$ variations in the E_z component and $\sin(m\phi)$ variations in the H_z component, allows us to write that

HEM Modes ($\rho \leq a$)

$$E_\rho^d = -j \frac{1}{(\beta_\rho^d)^2} \left[m\omega\mu_d A_m \frac{1}{\rho} J_m(\beta_\rho^d \rho) + \beta_z \beta_\rho^d B_m J'_m(\beta_\rho^d \rho) \right] \cos(m\phi) e^{-j\beta_z z} \quad (9-81a)$$

$$E_\phi^d = j \frac{1}{(\beta_\rho^d)^2} \left[\omega\mu_d \beta_\rho^d A_m J'_m(\beta_\rho^d \rho) + m\beta_z B_m \frac{1}{\rho} J_m(\beta_\rho^d \rho) \right] \sin(m\phi) e^{-j\beta_z z} \quad (9-81b)$$

$$E_z^d = B_m J_m(\beta_\rho^d \rho) \cos(m\phi) e^{-j\beta_z z} \quad (9-81c)$$

$$H_\rho^d = -j \frac{1}{(\beta_\rho^d)^2} \left[\beta_z \beta_\rho^d A_m J'_m(\beta_\rho^d \rho) + m\omega\varepsilon_d B_m \frac{1}{\rho} J_m(\beta_\rho^d \rho) \right] \sin(m\phi) e^{-j\beta_z z} \quad (9-81d)$$

$$H_\phi^d = -j \frac{1}{(\beta_\rho^d)^2} \left[m\beta_z A_m \frac{1}{\rho} J_m(\beta_\rho^d \rho) + \omega\varepsilon_d \beta_\rho^d B_m J'_m(\beta_\rho^d \rho) \right] \cos(m\phi) e^{-j\beta_z z} \quad (9-81e)$$

$$H_z^d = A_m J_m(\beta_\rho^d \rho) \sin(m\phi) e^{-j\beta_z z} \quad (9-81f)$$

where

$$(\beta_\rho^d)^2 + \beta_z^2 = \beta_d^2 = \omega^2 \mu_d \varepsilon_d = \omega^2 \mu_0 \varepsilon_0 \varepsilon_r \mu_r \quad (9-81g)$$

$$A_m = -j \frac{(\beta_\rho^d)^2}{\omega\mu_d \varepsilon_d} A_{nm} D_2 \quad (9-81h)$$

$$B_m = -j \frac{(\beta_\rho^d)^2}{\omega\mu_d \varepsilon_d} B_{nm} C_2 \quad (9-81i)$$

$$' \equiv \frac{\partial}{\partial(\beta_\rho^d \rho)} \quad (9-81j)$$

The coefficients A_m and B_m are not independent of each other, and their relationship can be found by applying the appropriate boundary conditions.

For the dielectric rod to act as a waveguide, the fields outside the rod ($\rho \geq a$) must be of the evanescent type that exhibit a decay in the radial direction. The rate of attenuation is a function of the diameter of the rod. As the diameter of the rod decreases, the following changes occur.

1. The attenuation lessens.
2. The distance to which the fields outside the rod can extend is greater.
3. The propagation constant β_z is only slightly greater than β_0 .

As the diameter of the rod increases, the following changes occur.

1. The rate of attenuation also increases.
2. The fields are confined closer to the rod.
3. The propagation constant β_z approaches β_d .

Since in all cases β_z is greater than β_0 , the phase velocity is smaller than the velocity of light in free space. For small-diameter rods, the surface waves are said to be *loosely bound* to the dielectric surface, whereas for the larger diameters, it is said to be *tightly bound* to the dielectric surface. Therefore, the cylindrical functions that are chosen to represent the radial variations of the fields outside the rod must be cylindrical decaying functions. These functions can be either Hankel functions of order m of the first kind ($H_m^{(1)}$) or second kind ($H_m^{(2)}$) and of imaginary argument, or modified Bessel functions K_m of the second kind of order m . We choose here to use the modified Bessel functions K_m of the second kind, which are related to the Hankel functions of the first and second kind by

$$K_m(\alpha) = \begin{cases} j^{m+1} \frac{\pi}{2} H_m^{(1)}(j\alpha) & (9-82a) \\ -j^{m+1} \frac{\pi}{2} H_m^{(2)}(-j\alpha) & (9-82b) \end{cases}$$

With (9-81a) through (9-81j), and (9-82a) and (9-82b) as a guide, we can represent the corresponding electric and magnetic field components for the HEM modes outside the dielectric rod ($\rho \geq a$) by

HEM Modes ($\rho \geq a$)

$$E_\rho^0 = j \frac{1}{(\alpha_\rho^0)^2} \left[m\omega\mu_0 C_m \frac{1}{\rho} K_m(\alpha_\rho^0 \rho) + \beta_z \alpha_\rho^0 D_m K_m'(\alpha_\rho^0 \rho) \right] \cos(m\phi) e^{-j\beta_z z} \quad (9-83a)$$

$$E_\phi^0 = -j \frac{1}{(\alpha_\rho^0)^2} \left[\omega\mu_0 \alpha_\rho^0 C_m K_m'(\alpha_\rho^0 \rho) + m\beta_z D_m \frac{1}{\rho} K_m(\alpha_\rho^0 \rho) \right] \sin(m\phi) e^{-j\beta_z z} \quad (9-83b)$$

$$E_z^0 = D_m K_m(\alpha_\rho^0 \rho) \cos(m\phi) e^{-j\beta_z z} \quad (9-83c)$$

$$H_\rho^0 = j \frac{1}{(\alpha_\rho^0)^2} \left[\beta_z \alpha_\rho^0 C_m K_m'(\alpha_\rho^0 \rho) + m\omega\varepsilon_0 D_m \frac{1}{\rho} K_m(\alpha_\rho^0 \rho) \right] \sin(m\phi) e^{-j\beta_z z} \quad (9-83d)$$

$$H_\phi^0 = j \frac{1}{(\alpha_\rho^0)^2} \left[m\beta_z C_m \frac{1}{\rho} K_m(\alpha_\rho^0 \rho) + \omega\varepsilon_0 \alpha_\rho^0 D_m K_m'(\alpha_\rho^0 \rho) \right] \cos(m\phi) e^{-j\beta_z z} \quad (9-83e)$$

$$H_z^0 = C_m K_m(\alpha_\rho^0 \rho) \sin(m\phi) e^{-j\beta_z z} \quad (9-83f)$$

where

$$(j\alpha_\rho^0)^2 + \beta_z^2 = -(\alpha_\rho^0)^2 + \beta_z^2 = \beta_0^2 = \omega^2 \mu_0 \varepsilon_0 \quad (9-83g)$$

$$' \equiv \frac{\partial}{\partial(\alpha_\rho^0 \rho)} \quad (9-83h)$$

The coefficients C_m and D_m are not independent of each other or from A_m and B_m , and their relations can be found by applying the appropriate boundary conditions.

The relations between the constants A_m , B_m , C_m , and D_m and equation 9-91, which is referred to as the *eigenvalue equation*, can be used to determine the modes that can be supported by the dielectric rod waveguide. These are obtained by applying the following boundary conditions

$$E_\phi^d(\rho = a, 0 \leq \phi \leq 2\pi, z) = E_\phi^0(\rho = a, 0 \leq \phi \leq 2\pi, z) \quad (9-84a)$$

$$E_z^d(\rho = a, 0 \leq \phi \leq 2\pi, z) = E_z^0(\rho = a, 0 \leq \phi \leq 2\pi, z) \quad (9-84b)$$

$$H_\phi^d(\rho = a, 0 \leq \phi \leq 2\pi, z) = H_\phi^0(\rho = a, 0 \leq \phi \leq 2\pi, z) \quad (9-84c)$$

$$H_z^d(\rho = a, 0 \leq \phi \leq 2\pi, z) = H_z^0(\rho = a, 0 \leq \phi \leq 2\pi, z) \quad (9-84d)$$

Doing this leads to

$$\begin{aligned} & \frac{1}{(\beta_\rho^d)^2} \left[\omega \mu_d \beta_\rho^d A_m J_m'(\beta_\rho^d a) + m \beta_z B_m \frac{1}{a} J_m(\beta_\rho^d a) \right] \\ &= -\frac{1}{(\alpha_\rho^0)^2} \left[\omega \mu_0 \alpha_\rho^0 C_m K_m'(\alpha_\rho^0 a) + m \beta_z D_m \frac{1}{a} K_m(\alpha_\rho^0 a) \right] \end{aligned} \quad (9-85a)$$

$$B_m J_m(\beta_\rho^d a) = D_m K_m(\alpha_\rho^0 a) \quad (9-85b)$$

$$\begin{aligned} & -\frac{1}{(\beta_\rho^d)^2} \left[m \beta_z A_m \frac{1}{a} J_m(\beta_\rho^d a) + \omega \varepsilon_d \beta_\rho^d B_m J_m'(\beta_\rho^d a) \right] \\ &= \frac{1}{(\alpha_\rho^0)^2} \left[m \beta_z C_m \frac{1}{a} K_m(\alpha_\rho^0 a) + \omega \varepsilon_0 \alpha_\rho^0 D_m K_m'(\alpha_\rho^0 a) \right] \end{aligned} \quad (9-85c)$$

$$A_m J_m(\beta_\rho^d a) = C_m K_m(\alpha_\rho^0 a) \quad (9-85d)$$

where according to (9-81g) and (9-83g)

$$(\beta_\rho^d)^2 + \beta_z^2 = \beta_d^2 \Rightarrow (\beta_\rho^d a)^2 + (\beta_z a)^2 = (\beta_d a)^2 \Rightarrow \beta_z a = \sqrt{(\beta_d a)^2 - (\beta_\rho^d a)^2} \quad (9-85e)$$

$$-(\alpha_\rho^0)^2 + \beta_z^2 = \beta_0^2 \Rightarrow -(\alpha_\rho^0 a)^2 + (\beta_z a)^2 = (\beta_0 a)^2 \quad (9-85f)$$

Subtracting (9-85f) from (9-85e) we get that

$$\begin{aligned} (\beta_\rho^d a)^2 + (\alpha_\rho^0 a)^2 &= (\beta_d a)^2 - (\beta_0 a)^2 \Rightarrow \alpha_\rho^0 a = \sqrt{(\beta_d a)^2 - (\beta_0 a)^2 - (\beta_\rho^d a)^2} \\ &= \sqrt{(\beta_0 a)^2 (\varepsilon_r \mu_r - 1) - (\beta_\rho^d a)^2} \end{aligned} \quad (9-86)$$

Using the abbreviated notation of

$$\chi = \beta_\rho^d a \quad (9-87a)$$

$$\xi = \alpha_\rho^0 a \quad (9-87b)$$

$$\zeta = \beta_z a \quad (9-87c)$$

$$\beta_d = \beta_0 \sqrt{\varepsilon_r \mu_r} \quad (9-87d)$$

we can rewrite (9-85e) and (9-86) as

$$\zeta = \sqrt{(\beta_0 a)^2 \varepsilon_r \mu_r - \chi^2} \quad (9-88a)$$

$$\xi = \sqrt{(\beta_0 a)^2 (\varepsilon_r \mu_r - 1) - \chi^2} \quad (9-88b)$$

With the preceding abbreviated notation and with $\mu_d = \mu_0$ (9-85a) through (9-85d) can be written in matrix form as

$$Fg = 0 \quad (9-89)$$

where F is a 4×4 matrix and g is a column matrix. Each is given by

$$F = \begin{bmatrix} \frac{\omega \mu_0 a}{\chi} J'_m(\chi) & \frac{m\zeta}{\chi^2} J_m(\chi) & \frac{\omega \mu_0 a}{\xi} K'_m(\xi) & \frac{m\zeta}{\xi^2} K_m(\xi) \\ 0 & J_m(\chi) & 0 & -K_m(\xi) \\ \frac{m\zeta}{\chi^2} J_m(\chi) & \frac{\omega \varepsilon_d a}{\chi} J'_m(\chi) & \frac{m\zeta}{\xi^2} K_m(\xi) & \frac{\omega \varepsilon_0 a}{\xi} K'_m(\xi) \\ J_m(\chi) & 0 & -K_m(\xi) & 0 \end{bmatrix} \quad (9-89a)$$

$$g = \begin{bmatrix} A_m \\ B_m \\ C_m \\ D_m \end{bmatrix} \quad (9-89b)$$

Equation 9-89 has a nontrivial solution provided that the determinant of F of (9-89a) is equal to zero [i.e., $\det(F) = 0$]. Applying this to (9-89a), it can be shown that it leads to [21]

$$\begin{aligned} |F| = & \frac{\omega^2 \mu_0 \varepsilon_d a^2}{\chi^2} [J'_m(\chi)]^2 [K_m(\xi)]^2 + \frac{\omega^2 \mu_0 \varepsilon_0 a^2}{\chi \xi} J_m(\chi) J'_m(\chi) K_m(\xi) K'_m(\xi) \\ & - \frac{(m\zeta)^2}{\chi^4} [J_m(\chi)]^2 [K_m(\xi)]^2 - \frac{(m\zeta)^2}{\chi^2 \xi^2} [J_m(\chi)]^2 [K_m(\xi)]^2 \\ & + \frac{\omega^2 \mu_0 \varepsilon_d a^2}{\chi \xi} J_m(\chi) J'_m(\chi) K_m(\xi) K'_m(\xi) + \frac{\omega^2 \mu_0 \varepsilon_0 a^2}{\xi^2} [J_m(\chi)]^2 [K'_m(\xi)]^2 \\ & - \frac{(m\zeta)^2}{\chi^2 \xi^2} [J_m(\chi)]^2 [K_m(\xi)]^2 - \frac{(m\zeta)^2}{\xi^4} [J_m(\chi)]^2 [K_m(\xi)]^2 = 0 \end{aligned} \quad (9-90)$$

Dividing all the terms of (9-90) by $\omega^2 \mu_0 \varepsilon_d a^2 [K_m(\xi)]^2$ and regrouping, it can be shown that (9-90) can be placed in the form of [21]

$$G_1(\chi) G_2(\chi) - G_3^2(\chi) = 0 \quad (9-91)$$

where

$$G_1(\chi) = \frac{J'_m(\chi)}{\chi} + \frac{K'_m(\xi)J_m(\chi)}{\varepsilon_r \xi K_m(\xi)} \quad (9-91a)$$

$$G_2(\chi) = \frac{J'_m(\chi)}{\chi} + \frac{K'_m(\xi)J_m(\chi)}{\xi K_m(\xi)} \quad (9-91b)$$

$$G_3(\chi) = \frac{m\zeta}{\beta_0 a \sqrt{\varepsilon_r}} J_m(\chi) \left(\frac{1}{\chi^2} + \frac{1}{\xi^2} \right) \quad (9-91c)$$

$$\zeta = \sqrt{(\beta_0 a)^2 \varepsilon_r - \chi^2} \quad (9-91d)$$

$$\xi = \sqrt{(\beta_0 a)^2 (\varepsilon_r - 1) - \chi^2} \quad (9-91e)$$

Equation 9-91 is referred to as the *eigenvalue equation* for the dielectric rod waveguide. The values of χ that are solutions to (9-91) are referred to as the *eigenvalues* for the dielectric rod waveguide.

In order for $\xi = \alpha_\rho^0 a$ to remain real and represent decaying fields outside the dielectric rod waveguide, the values of $\chi = \beta_\rho^d a$ should not exceed a certain maximum value. From (9-88b) this maximum value $\chi_{\max}$ is equal to

$$\chi_{\max} = \left(\beta_\rho^d a \right)_{\max} = \beta_0 a \sqrt{\varepsilon_r - 1} = \omega a \sqrt{\mu_0 \varepsilon_0 (\varepsilon_r - 1)} \quad (9-92)$$

For values of $\beta_\rho^d a$ greater than $\chi_{\max}$ (see above), the values of $\xi = \alpha_\rho^0 a$ become imaginary and according to (9-82b), the modified Bessel function of the second kind is reduced to a Hankel function of the second kind that represents unattenuated outwardly traveling waves. In this case, the dielectric rod is acting as a cylindrical antenna because of energy loss from its side. Therefore, for a given value of m , there are a finite number n of χ_{mn} 's (eigenvalues) for which the dielectric rod acts as a waveguide. Each combination of allowable values of m, n that determine a given eigenvalue χ_{mn} represent the hybrid mode HEM_{mn} . HEM modes with odd values of the second subscript correspond to HE modes, whereas HEM modes with even values of the second subscript correspond to EH modes. Thus, $\text{HEM}_{m,2n-1}$ ($n=1, 2, 3, \dots$) correspond to HE_{mn} modes and $\text{HEM}_{m,2n}$ ($n=1, 2, 3, \dots$) correspond to EH_{mn} modes. According to (9-88a), if the values of χ exceed $\beta_0 a \sqrt{\varepsilon_r}$ (i.e., $\chi > \beta_0 a \sqrt{\varepsilon_r}$), then $\zeta = \beta_z a$ becomes imaginary and the waves in the dielectric rod become decaying (evanescent) along the axis (z direction) of the rod.

The allowable modes in a dielectric rod waveguide are determined by finding the values of χ , denoted by χ_{mn} , that are solutions to the transcendental equation 9-91. For each value of m there are a finite number of values of n ($n=1, 2, 3, \dots$). Examining (9-91), it is evident that for $m=0$, the left side of (9-91) vanishes when

$$\begin{aligned} & \text{TM}_{0n} \\ G_1(\chi_{0n}) &= \frac{J'_0(\chi_{0n})}{\chi_{0n}} + \frac{K'_0(\xi_{0n})J_0(\chi_{0n})}{\varepsilon_r \xi_{0n} K_0(\xi_{0n})} = -\frac{J_1(\chi_{0n})}{\chi_{0n}} - \frac{K_1(\xi_{0n})J_0(\chi_{0n})}{\varepsilon_r \xi_{0n} K_0(\xi_{0n})} = 0 \end{aligned} \quad (9-93a)$$

$$\begin{aligned} & \text{TE}_{0n} \\ G_2(\chi_{0n}) &= \frac{J'_0(\chi_{0n})}{\chi_{0n}} + \frac{K'_0(\xi_{0n})J_0(\chi_{0n})}{\xi_{0n} K_0(\xi_{0n})} = -\frac{J_1(\chi_{0n})}{\chi_{0n}} - \frac{K_1(\xi_{0n})J_0(\chi_{0n})}{\xi_{0n} K_0(\xi_{0n})} = 0 \end{aligned} \quad (9-93b)$$

since $G_3(\chi_{0n}) = 0$, $J'_0(\chi_{0n}) = -J_1(\chi_{0n})$, and $K'_0(\xi_{0n}) = -K_1(\xi_{0n})$. Equation 9-93a is valid for TM_{0n} modes and (9-93b) is applicable for TE_{0n} modes.

The nonlinear equations 9-91 through 9-91e can be solved for the values of χ_{mn} iteratively by assuming values of χ_{mn} examining the sign changes of (9-91). It should be noted that in the allowed range of χ_{mn} 's, G_1 , G_2 , and G_3 are nonsingular. Computed values of χ_{mn} as a function of $\beta_0 a$ for different HEM_{mn} modes are shown in Figure 9-9 for $\varepsilon_r = 20$ and in Figure 9-10 for $\varepsilon_r = 38$ [21]. It should be noted that for a given mn mode the values of χ_{mn} are nonconstant and vary as a function of the electrical radius of the rod. This is in contrast to the circular waveguide with conducting walls, whose χ_{mn} or χ'_{mn} values (in Tables 9-1 and 9-2) are constant for a given mode.

Once the values of χ_{mn} for a given mode have been found, the corresponding values of $\zeta = \beta_z a$ can be computed using (9-91d). When this is done for a dielectric rod waveguide of polystyrene ($\varepsilon_r = 2.56$), the values of β_z / β_0 for the HE_{11} mode as a function of the radius of the rod are shown in Figure 9-11 [10]. It is apparent that the HE_{11} mode does not possess a cutoff. In the same figure, the values of β_z / β_0 for the axially symmetric TE_{01} and TM_{01} surface wave modes, which possess a finite cutoff, are also displayed. Although in principle the HE_{11} mode has zero cutoff, the rate of attenuation exhibited by the fields outside the slab decreases as the radius of the rod becomes smaller and the wavenumber β_z approaches β_0 . Thus, for small radii, the fields outside the rod extend to large distances and are said to be loosely bound to the surface. Practically, then a minimum radius rod is usually utilized, which results in a small but finite cutoff [22]. For the larger dielectric constant material, the fields outside the dielectric waveguide are more tightly bound to the surface since larger values of β_z translate to larger values of the attenuation coefficient α_p^0 through (9-83g).

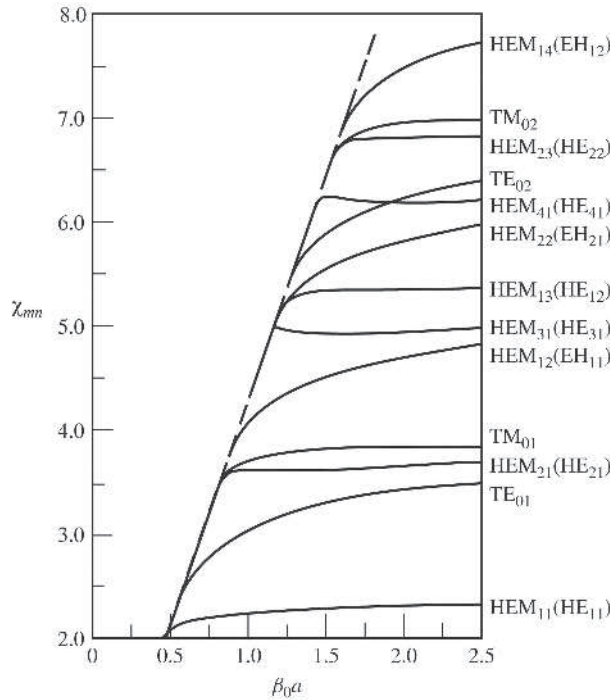


Figure 9-9 The first 13 eigenvalues of the dielectric rod waveguide ($\varepsilon_r = 20$). (Source: D. Kajfez and P. Guillon (Eds.), *Dielectric Resonators*, 1986, Artech House, Inc.)

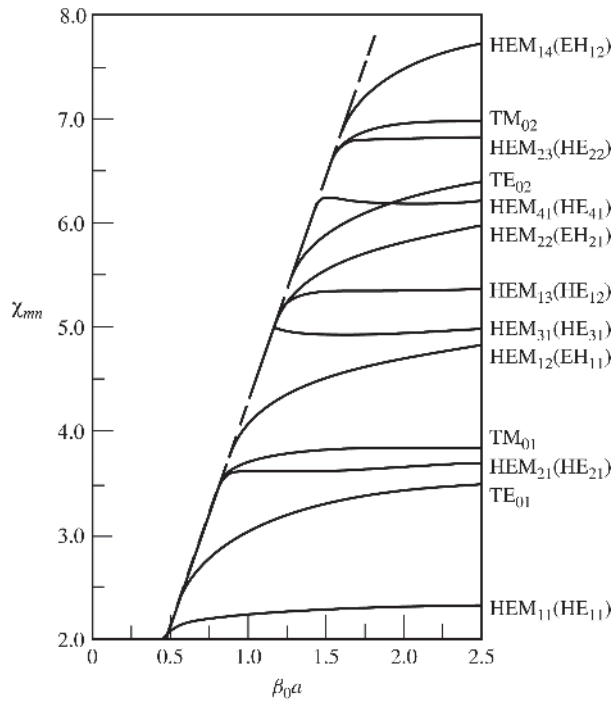


Figure 9-10 The first 13 eigenvalues of the dielectric rod waveguide ($\epsilon_r = 38$). (Source: D. Kajfez and P. Guillon (Eds.), *Dielectric Resonators*, 1986, Artech House, Inc.)

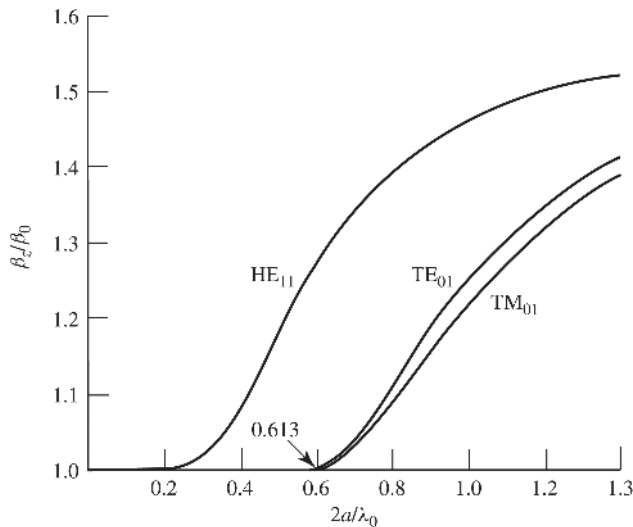


Figure 9-11 Ratio of β_z/β_0 , for first three surface-wave modes on a polystyrene rod ($\epsilon_r = 2.56$). (Source: R. E. Collin, *Field Theory of Guided Waves*, 1960, McGraw-Hill Book Co.)

Typical field patterns in the central core for the HE₁₁, TE₀₁, TM₀₁, HE₂₁, EH₁₁, and HE₃₁ modes are shown in Figure 9-12 [23]. Both E - and H -field lines are displayed over the cross section of the central core and over a cutaway a distance $\lambda_g/2$ along its length. For all the plots, the ratio $\tau = \chi^2/(\chi^2 + \xi^2) = (\beta_\rho^d)^2/[(\beta_\rho^d)^2 + (\alpha_\rho^0)^2] = 0.1$.

For all modes the sum of $\chi^2 + \xi^2$ should be a constant, which will be a function of the radius of the dielectric rod and its dielectric constant. Thus, according to (9-88b),

$$\chi^2 + \xi^2 = (\beta_0 a)^2 (\varepsilon_r - 1) = \text{constant} \quad (9-94)$$

For all modes, excluding the dominant HE_{11} , this constant should always be equal to or greater than $(2.4049)^2 = 5.7835$; that is, excluding the HE_{11} mode,

$$V^2 = \chi^2 + \xi^2 = (\beta_0 a)^2 (\varepsilon_r - 1) \geq (2.4049)^2 \quad (9-94a)$$

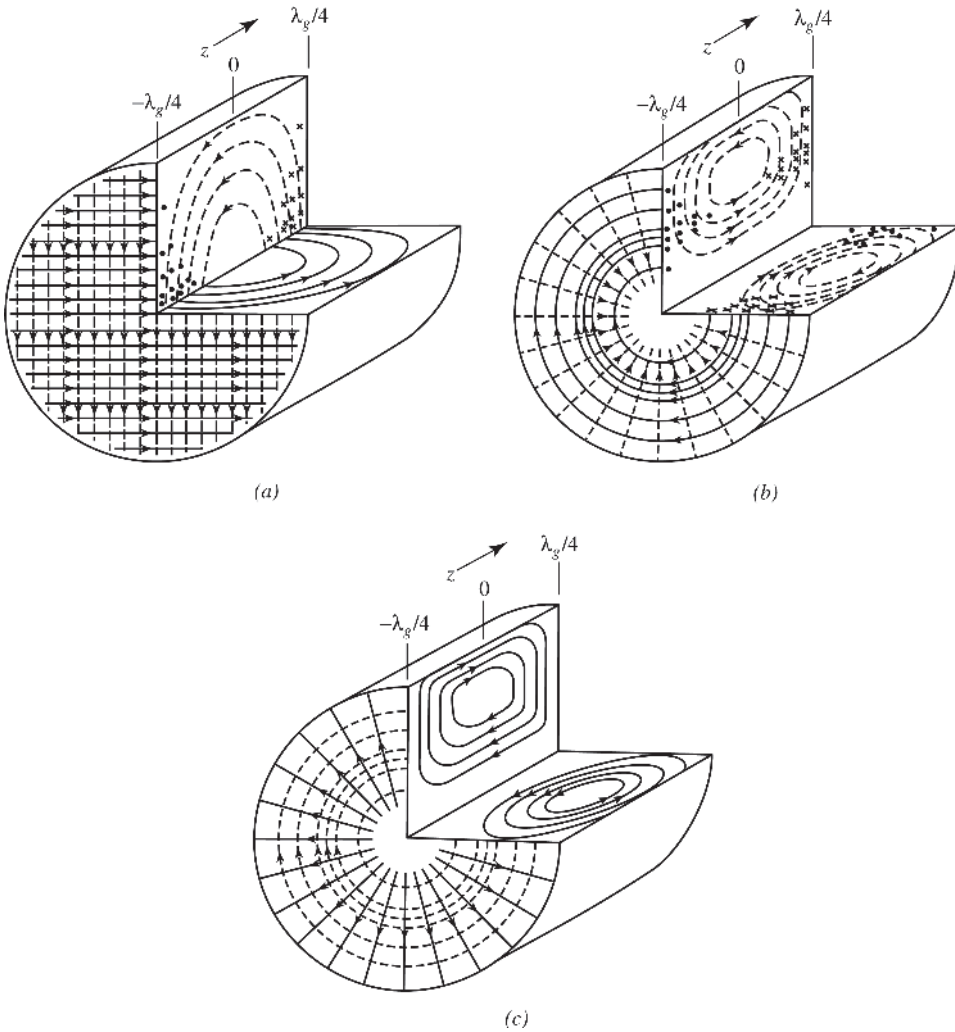


Figure 9-12 Field patterns in the central core of a dielectric rod waveguide (in all cases $\tau = 0.1$; **E**: —, **H**: ----). (a) HEM_{11} (HE_{11}) mode. (b) TE_{01} mode. (c) TM_{01} mode. (Source: T. Okoshi, *Optical Fibers*, 1982, Academic Press.)

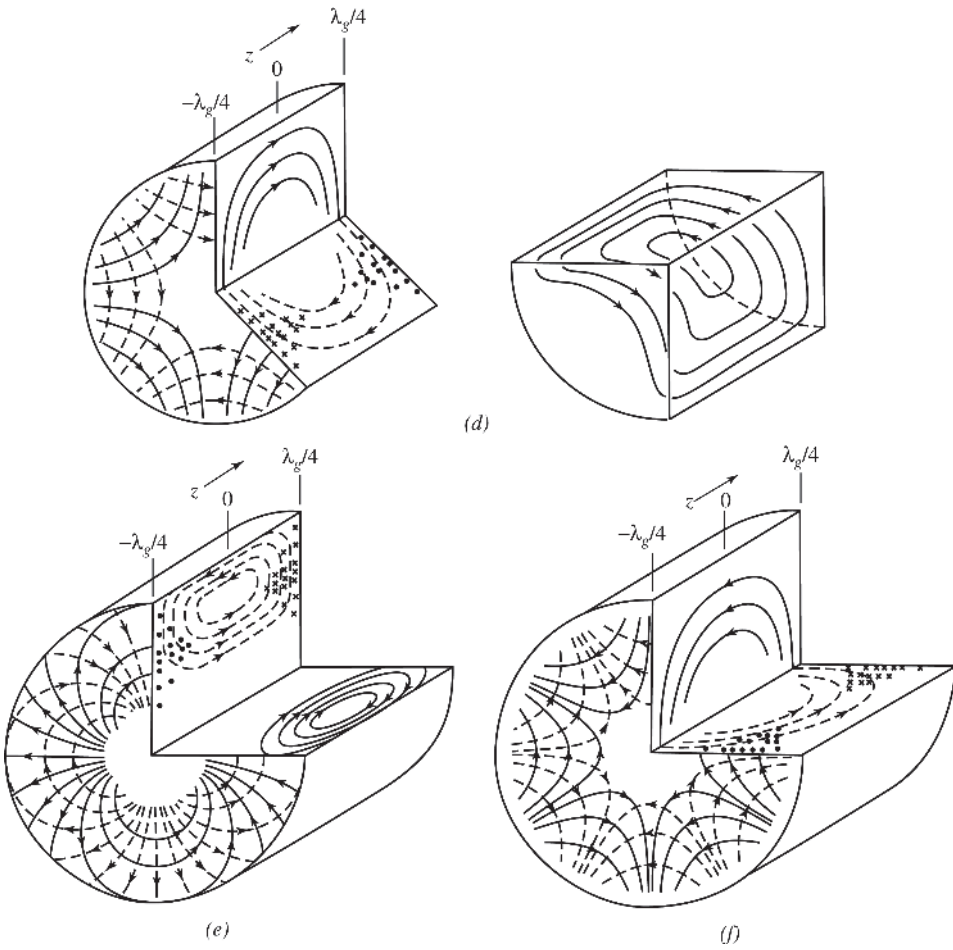


Figure 9-12 (d) HEM_{21} (HE_{21}) mode. (e) HEM_{12} (EH_{11}) mode. (f) HEM_{31} (HE_{31}) mode.

or

$$\frac{2a}{\lambda_0} \geq \frac{1}{\pi} \frac{2.4049}{\sqrt{\epsilon_r - 1}} \quad (9-94b)$$

The value of 2.4049 is used because one of the next higher-order modes is the TM_{01} mode which, according to Table 9-2, is $\chi_{nm} = \chi_{01} = 2.4049$. For values of $2a/\lambda_0$ smaller than that of (9-94b) only the dominant HE_{11} dipole mode exists as shown in Figure 9-11.

For a dielectric rod waveguide, the first 12 modes (actually first 20 modes if the twofold degeneracy for the HE_{mn} or EH_{mn} is counted), in order of ascending cutoff frequency, along with the vanishing Bessel function and its argument at cutoff, are given:

Mode(s)	$J_m(\chi_{mn}) = 0$	χ_{mn} at cutoff	Total number of propagating modes
HE ₁₁ (HEM ₁₁)	$J_1(\chi_{10}) = 0$	$\chi_{10} = 0$	2
TE ₀₁ , TM ₀₁ , HE ₂₁ (HEM ₂₁)	$J_0(\chi_{01}) = 0$	$\chi_{01} = 2.4049$	6
HE ₁₂ (HEM ₁₃), EH ₁₁ (HEM ₁₂), HE ₃₁ (HEM ₃₁)	$J_1(\chi_{11}) = 0$	$\chi_{11} = 3.8318$	12
EH ₂₁ (HEM ₂₂), HE ₄₁ (HEM ₄₁)	$J_2(\chi_{21}) = 0$	$\chi_{21} = 5.1357$	16
TE ₀₂ , TM ₀₂ , HE ₂₂ (HEM ₂₃)	$J_0(\chi_{02}) = 0$	$\chi_{02} = 5.5201$	20

According to (9-94a), a single dominant HE₁₁ mode can be maintained within the rod provided the normalized central core radius $V < 2.4049$. This can be accomplished by making the radius a of the central core small and/or choosing, between the central core and the cladding, a small dielectric constant ϵ_r . However, the smaller the size of the central core, the smaller the rate of attenuation and the less tightly the field outside it is attached to its surface. The normalized diameter over which the $e^{-1} = 0.3679 = 36.79\%$ field point outside the central core extends is shown plotted in Figure 9-13 [16]. Although the fundamental HE₁₁ mode does not cutoff as the core diameter shrinks, the fields spread out beyond the physical core and become loosely bound.

9.5.2 Circular Dielectric Resonator

Dielectric resonators are unmetallized dielectric objects (spheres, disks, parallelepipeds, etc.) of high dielectric constant (usually ceramic) and high quality factor Q that can function as energy storage devices. Dielectric resonators were first introduced in 1939 by Richtmyer [24], but for

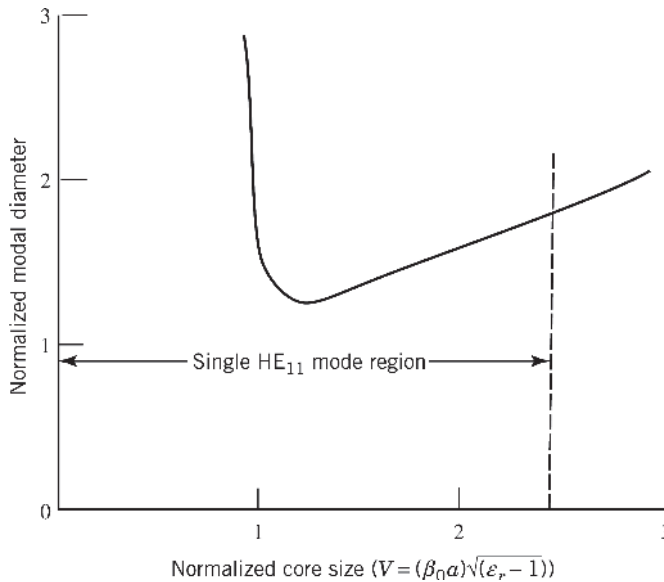


Figure 9-13 Normalized modal diameter as a function of normalized core size for dielectric rod waveguide. (Source: T. G. Giallorenzi, “Optical communications research and technology: Fiber optics,” *Proc. IEEE*, © 1978, IEEE.)

almost 25 years his theoretical work failed to generate a continuous and prolonged interest. The introduction in the 1960s of material, such as rutile, of high dielectric constant (around 100) renewed the interest in dielectric resonators [25–30]. However, the poor temperature stability of rutile resulted in large resonant frequency changes and prevented the development of practical microwave components. In the 1970s, low-loss and temperature-stable ceramics, such as barium tetratitanate and (Zr-Sn)TiO₄ were introduced and were used for the design of high-performance microwave components such as filters and oscillators. Because dielectric resonators are small, lightweight, temperature stable, high Q , and low cost, they are ideal for design and fabrication of monolithic microwave integrated circuits (MMICs) and general semiconductor devices. Such technology usually requires high Q miniature elements to design and fabricate highly stable frequency oscillators and high-performance narrowband filters. Thus, dielectric resonators have replaced traditional waveguide resonators, especially in MIC applications, and implementations as high as 94 GHz have been reported. The development of higher dielectric constant material (80 or higher) with stable temperature and low-loss characteristics will have a significant impact on MIC design using dielectric resonators.

In order for the dielectric resonator to function as a resonant cavity, the dielectric constant of the material must be large (usually 30 or greater). Under those conditions, the dielectric–air interface acts almost as an open circuit which causes internal reflections and results in the confinement of energy in the dielectric material, thus creating a resonant structure. The plane wave reflection coefficient at the dielectric–air interface is equal to

$$\Gamma = \frac{\eta_0 - \eta}{\eta_0 + \eta} = \frac{\sqrt{\frac{\mu_0}{\varepsilon_0}} - \sqrt{\frac{\mu_0}{\varepsilon}}}{\sqrt{\frac{\mu_0}{\varepsilon_0}} + \sqrt{\frac{\mu_0}{\varepsilon}}} = \frac{\sqrt{\frac{\varepsilon}{\varepsilon_0}} - 1}{\sqrt{\frac{\varepsilon}{\varepsilon_0}} + 1} = \frac{\sqrt{\varepsilon_r} - 1}{\sqrt{\varepsilon_r} + 1} \stackrel{\varepsilon_r \rightarrow \text{large}}{\simeq} +1 \quad (9-95)$$

and it approaches the value of +1 as the dielectric constant becomes very large. Under these conditions, the dielectric–air interface can be approximated by a hypothetical perfect magnetic conductor (PMC), which requires that the tangential components of the magnetic field (or normal components of the electric field) vanish (in contrast to the perfect electric conductor, PEC, which requires that the tangential electric field components, or normal components of the magnetic field, vanish). This, of course, is a well known and widely used technique in solving boundary-value electromagnetic problems. It is, however, a first-order approximation, although it usually leads to reasonable results. The magnetic wall model can be used to analyze both the dielectric waveguide and dielectric resonant cavity. Improvements to the magnetic wall approximation have been introduced and resulted in improved data [31, 32].

Although the magnetic wall modeling may not lead to the most accurate data, it will be utilized here because it is simple and instructive not only as a first-order approximation to this problem but also to other problems including antennas (e.g., microstrip antenna). The geometry of the dielectric resonator is that of Figure 9-14a, whose surface is modeled with the PMC walls of Figure 9-14b, which are represented by the independent boundary conditions

$$H_\phi(\rho = a, 0 \leq \phi \leq 2\pi, 0 \leq z \leq h) = 0 \quad (9-96a)$$

$$H_\phi(0 \leq \rho \leq a, 0 \leq \phi \leq 2\pi, z = 0) = H_\phi(0 \leq \rho \leq a, 0 \leq \phi \leq 2\pi, z = h) = 0 \quad (9-96b)$$

or

$$H_z(\rho = a, 0 \leq \phi \leq 2\pi, 0 \leq z \leq h) = 0 \quad (9-97a)$$

$$H_\rho(0 \leq \rho \leq a, 0 \leq \phi \leq 2\pi, z = 0) = H_\rho(0 \leq \rho \leq a, 0 \leq \phi \leq 2\pi, z = h) = 0 \quad (9-97b)$$

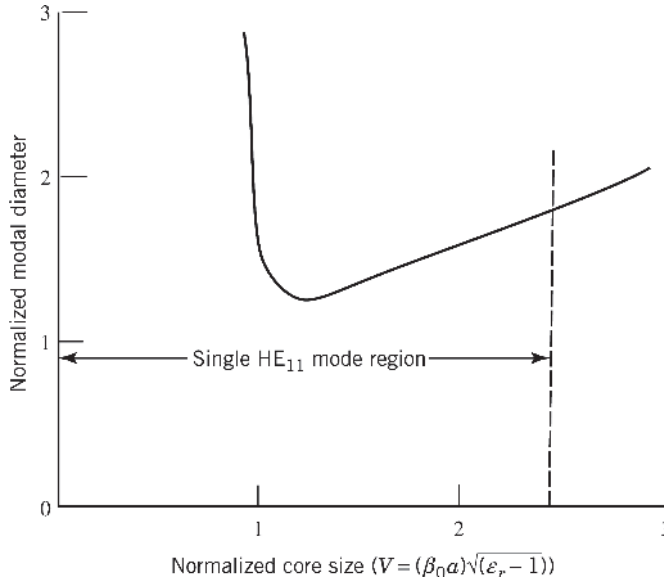


Figure 9-14 Circular dielectric resonator and its PMC modeling. (a) Dielectric resonator. (b) PMC modeling.

Either of the preceding sets leads to the same results. Since all the boundary conditions of the first set involve only the H_ϕ component, they will be applied here.

A. TE^z Modes The TE^z modes can be constructed using the vector potential F_z component of (9-35), or

$$F_z(\rho, \phi, z) = A_{mn} J_m(\beta_\rho^d \rho) [C_2 \cos(m\phi) + D_2 \sin(m\phi)] [C_3 \cos(\beta_z z) + D_3 \sin(\beta_z z)] \quad (9-98)$$

$$\text{where } m = 0, 1, 2, \dots \quad (9-98a)$$

The H_ϕ component is obtained using (6-80), or

$$H_\phi = -j \frac{1}{\omega_r \mu_d \varepsilon_d} \frac{1}{\rho} \frac{\partial^2 F_z}{\partial \phi \partial z} = -j A_{mn} \frac{m \beta_z}{\omega_r \mu_d \varepsilon_d} \frac{1}{\rho} J_m(\beta_\rho^d \rho) [-C_2 \sin(m\phi) + D_2 \cos(m\phi)] \times [-C_3 \sin(\beta_z z) + D_3 \cos(\beta_z z)] \quad (9-99)$$

Applying (9-96a) leads to

$$\begin{aligned} H_\phi(\rho = a, 0 \leq \phi \leq 2\pi, 0 \leq z \leq h) \\ = -j A_{mn} \frac{m \beta_z}{\omega_r \mu_d \varepsilon_d} \frac{1}{a} J_m(\beta_\rho^d a) [-C_2 \sin(m\phi) + D_2 \cos(m\phi)] \\ \times [-C_3 \sin(\beta_z z) + D_3 \cos(\beta_z z)] = 0 \Rightarrow J_m(\beta_\rho^d a) = 0 \Rightarrow \beta_\rho^d a = \chi_{mn} \\ \beta_\rho^d = \left(\frac{\chi_{mn}}{a} \right) \end{aligned} \quad (9-100)$$

where χ_{mn} represents the zeroes of the Bessel function of order m , many of which are found in Table 9-2. In a similar manner, the first boundary condition of (9-96b) leads to

$$\begin{aligned} H_\phi(0 \leq \rho \leq a, 0 \leq \phi \leq 2\pi, z = 0) \\ = -jA_{mn} \frac{m\beta_z}{\omega_r \mu_d \varepsilon_d} \frac{1}{\rho} J_m(\beta_\rho^d \rho) [-C_2 \sin(m\phi) + D_2 \cos(m\phi)] \\ \times [-C_3(0) + D_3(1)] = 0 \Rightarrow D_3 = 0 \end{aligned} \quad (9-100a)$$

while the second boundary condition of (9-96b) leads to

$$\begin{aligned} H_\phi(0 \leq \rho \leq a, 0 \leq \phi \leq 2\pi, z = h) \\ = -jA_{mn} \frac{m\beta_z}{\omega_r \mu_d \varepsilon_d} \frac{1}{\rho} J_m(\beta_\rho^d \rho) \\ \times [-C_2 \sin(m\phi) + D_2 \cos(m\phi)] [-C_3 \sin(\beta_z h)] = 0 \\ \Rightarrow \sin(\beta_z h) = 0 \Rightarrow \beta_z h = p\pi \\ \beta_z = \frac{p\pi}{h}, \quad p = 0, 1, 2, \dots \end{aligned} \quad (9-100b)$$

Using (9-100) and (9-100b), the resonant frequency is obtained by applying (9-4a) at resonance, that is,

$$\beta_r = \omega_r \sqrt{\mu_d \varepsilon_d} = 2\pi f_r \sqrt{\mu_d \varepsilon_d} = \sqrt{\left(\frac{\chi_{mn}}{a}\right)^2 + \left(\frac{p\pi}{h}\right)^2}$$

or

$$\boxed{(f_r)_{mnp}^{\text{TE}^z} = \frac{1}{2\pi \sqrt{\mu_d \varepsilon_d}} \sqrt{\left(\frac{\chi_{mn}}{a}\right)^2 + \left(\frac{p\pi}{h}\right)^2}} \quad \begin{aligned} m &= 0, 1, 2, \dots \\ n &= 1, 2, 3, \dots \\ p &= 0, 1, 2, \dots \end{aligned} \quad (9-101)$$

The dominant TE_{mnp}^z mode is the TE_{010}^z whose resonant frequency is equal to

$$(f_r)_{010}^{\text{TE}^z} = \frac{\chi_{01}}{2\pi a \sqrt{\mu_d \varepsilon_d}} = \frac{2.4049}{2\pi a \sqrt{\mu_d \varepsilon_d}} \quad (9-101a)$$

B. TM^z Modes The TM^z modes are obtained using a similar procedure as for the TE^z modes, but starting with the vector potential of (9-41). Doing this leads to the resonant frequency of

$$\boxed{(f_r)_{mnp}^{\text{TM}^z} = \frac{1}{2\pi \sqrt{\mu_d \varepsilon_d}} \sqrt{\left(\frac{\chi'_{mn}}{a}\right)^2 + \left(\frac{p\pi}{h}\right)^2}} \quad \begin{aligned} m &= 0, 1, 2, \dots \\ n &= 1, 2, 3, \dots \\ p &= 1, 2, 3, \dots \end{aligned} \quad (9-102)$$

where χ'_{nm} are the zeroes of the derivative of the Bessel function of order m , a partial list of which is found on Table 9-1. The dominant TM_{mnp}^z mode is the TM_{111}^z mode whose resonant frequency is equal to

$$(f_r)_{111}^{\text{TM}^z} = \frac{1}{2\pi\sqrt{\mu_d\epsilon_d}} \sqrt{\left(\frac{\chi'_{11}}{a}\right)^2 + \left(\frac{\pi}{h}\right)^2} = \frac{1}{2\pi\sqrt{\mu_d\epsilon_d}} \sqrt{\left(\frac{1.8412}{a}\right)^2 + \left(\frac{\pi}{h}\right)^2} \quad (9-102a)$$

A comparison of the resonant frequencies of (9-101) and (9-102) of the circular dielectric resonator modeled by the PMC surface with those of (9-39a) and (9-45) for the circular waveguide resonator with PEC surface shows that the TE_{mnp}^z of one are the TM_{mnp}^z of the other, and vice versa.

C. $\text{TE}_{01\delta}$ Mode Although the dominant mode of the circular dielectric resonator as predicted by the PMC modeling is the TE_{010} mode (provided $h/a < 2.03$), in practice the mode most often used is the $\text{TE}_{01\delta}$ where δ is a noninteger value less than 1. This mode can be modeled using Figure 9-15. For resonators with PEC or PMC walls, the third subscript is always an integer (including zero), and it represents the number of half-wavelength variations the field undergoes in the z direction. Since δ is a noninteger less than unity, the dielectric resonator field of a $\text{TE}_{01\delta}$ mode undergoes less than one half-wavelength variation along its length h . More accurate modelings of the dielectric resonator [31, 33, 34] indicate that $\beta_z h$ of (9-100b) is

$$\beta_z h \simeq \frac{\psi_1}{2} + \frac{\psi_2}{2} + q\pi, \quad q = 0, 1, 2, \dots \quad (9-103)$$

where

$$\frac{\psi_1}{2} = \tan^{-1} \left[\frac{\alpha_1}{\beta_z} \coth(\alpha_1 h_1) \right] \quad (9-103a)$$

$$\frac{\psi_2}{2} = \tan^{-1} \left[\frac{\alpha_2}{\beta_z} \coth(\alpha_2 h_2) \right] \quad (9-103b)$$

$$\alpha_1 = \sqrt{\left(\frac{\chi_{01}}{a}\right)^2 - \beta_0^2 \epsilon_{r1}} \quad (9-103c)$$

$$\alpha_2 = \sqrt{\left(\frac{\chi_{01}}{a}\right)^2 - \beta_0^2 \epsilon_{r2}} \quad (9-103d)$$

When $q = 0$ in (9-103), the mode is the $\text{TE}_{01\delta}$ where δ is a noninteger less than unity given by

$$\delta = \frac{1}{\pi} \left(\frac{\psi_1}{2} + \frac{\psi_2}{2} \right) \quad (9-104)$$

and it signifies the variations of the field between the ends of the dielectric resonator at $z = 0$ and $z = h$. The preceding equations have been derived by modeling the dielectric resonator as shown in Figure 9-15 where PEC plates have been placed at $z = -h_2$ and $z = h + h_1$ [21]. The medium in regions 1–5 is usually taken to be air. Thus, the dielectric resonator has been sandwiched between two PEC plates, each placed at distances h_1 and h_2 from each of its ends. The distances h_1 and h_2 can be chosen to be $0 \leq h_1, h_2 \leq \infty$.

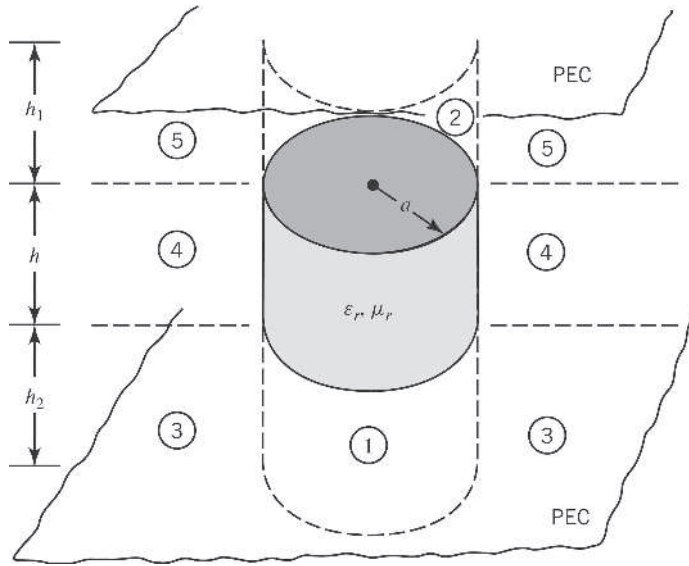


Figure 9-15 Modeling of circular dielectric rod resonator. (Source: D. Kajfez and P. Guillon (Eds.), *Dielectric Resonators*, 1986, Artech House, Inc.)

Example 9-5

Find the resonant $TE_{01\delta}$ mode and its resonant frequency when the distances h_1 and h_2 of Figure 9-15 are both zero ($h_1 = h_2 = 0$); that is, the dielectric resonator has been sandwiched between two PEC plates, each plate touching each of the ends of the dielectric resonator. Assume the resonator has radius of 5.25 mm, height of 4.6 mm, and dielectric constant of $\epsilon_r = 38$.

Solution: Since $h_1 = h_2 = 0$, then according to (9-103a) and (9-103b)

$$\frac{\psi_1}{2} = \frac{\psi_2}{2} = \tan^{-1}(\infty) = \frac{\pi}{2}$$

Thus, δ of (9-104) is equal to

$$\delta = \frac{1}{\pi} \left(\frac{\pi}{2} + \frac{\pi}{2} \right) = 1$$

and the resonant $TE_{01\delta}$ mode is the TE_{011} . For this mode $\beta_z h$ is, according to (9-103) with $p = 0$, equal to

$$\beta_z h = \pi \Rightarrow \beta_z = \frac{\pi}{h}$$

Its resonant frequency is identical to (9-101) of the PMC modeling, which reduces to

$$\begin{aligned} (f_r)_{011}^{TE^z} &= \frac{1}{2\pi\sqrt{\mu_d\epsilon_d}} \sqrt{\left(\frac{2.4049}{a}\right)^2 + \left(\frac{\pi}{h}\right)^2} \\ &= \frac{3 \times 10^{11}}{2\pi\sqrt{38}} \sqrt{\left(\frac{2.4049}{5.25}\right)^2 + \left(\frac{\pi}{4.6}\right)^2} = 6.37 \text{ GHz} \end{aligned}$$

When compared to the exact value of 4.82 GHz for a dielectric resonator immersed in free space, the preceding value has an error of +32%. A more accurate result of 4.60 GHz (error of -4.8%) can be obtained using (9-103a)–(9-104) by placing the two PECs at infinity ($h_1 = h_2 = \infty$); then the resonator is isolated in free space.

9.5.3 Optical Fiber Cable

Optical communications, which initially started as a speculative research activity, has evolved into a very practical technique that has brought new dimensions to miniaturization, data handling capabilities, and signal processing methods. This success has been primarily attributed to the development of fiber optics cables, which in 1970 exhibited attenuations of 20 dB/km and in the early 1980s reduced to less than 1 dB/km. The development of suitable solid state diode sources and detectors has certainly eliminated any remaining insurmountable technological barriers and paved the way for widespread implementation of optical communication techniques for commercial and military applications. This technology has spread to the development of integrated optics which provide even greater miniaturization of optical systems, rigid alignment of optical components, and reduced space and weight.

A fiber optics cable is a dielectric waveguide, usually of circular cross section, that guides the electromagnetic wave in discrete modes through internal reflections whose incidence angle at the interface is equal to or greater than the critical angle. The confinement of the energy within the dielectric structure is described analytically by Maxwell's equations and the boundary conditions at dielectric–dielectric boundaries, in contrast to the traditional metallic–dielectric boundaries in metallic waveguides. Such analyses have already been discussed in Section 8.7 for planar structures and Section 9.5.1 for circular geometries.

The most common geometries of fiber cables are those shown in Figure 9-16. They are classified as *step-index multimode*, *graded-index multimode*, and *single-mode step-index* [14, 16]. Typical dimensions, index-of-refraction distributions, optical ray paths, and pulse spreading by each type of cable are also illustrated in Figure 9-16. Therefore, fiber optics cables can be classified into two cases: *single-mode* and *multimode* fibers.

Single-mode fibers permit wave propagation at a single resolvable angle, whereas multimode cables transmit waves that travel at many resolvable angles all within the central core of the fiber. Both single- and multimode fibers are fabricated with a central core with a high index of refraction (dielectric constant) surrounded by a cladding with a lower index of refraction. The wave is guided in the core by total internal reflection at the core–cladding interface. Single-mode cables usually have a step-index where the diameter of the center core is very small (typically 2–16 μm) and not much larger than the wavelength of the wave it carries. Structurally, they are the simplest and exhibit abrupt index-of-refraction discontinuities along the core-cladding interface. Usually, the index of refraction of the central core is about 1.471 and that of the cladding is 1.457.

Step-index multimode cables also exhibit a well-defined central core with a constant index of refraction surrounded by a cladding with a lower index of refraction. In contrast to the single-mode step-index, the multimode step-index possesses a central core whose diameter is about 25–150 μm . When the wavelength of light is of the order of 10 μm (near the infrared region) and the central core diameter is 100 μm , such a cable can support as many as 25,000 modes (usually there are about 200 modes). However, as the central core diameter approaches 10 μm (one wavelength), the number of modes is reduced to one and the cable becomes a single-mode fiber. Because of its multimode field structure, such a cable is very dispersive and provides severe signal distortion to waveforms with broad spectral frequency content. Typically, the index of refraction of the central core is about 1.527 and that of the cladding is about 1.517.

Graded-index multimode cables also possess a relatively large central core (typical diameters of 20–150 μm) whose index of refraction continuously decreases from the core center toward the core–cladding interface, at which point the core and cladding indexes are identical. For this cable, the waves are still contained within the central core and they are continuously refocused toward the central axis of the core by its continuous lensing action. The number of modes supported by such a cable is usually about 2,000, and such a multimode structure provides waveform distortion especially to transient signals. All cables possess an outside cladding that adds mechanical strength

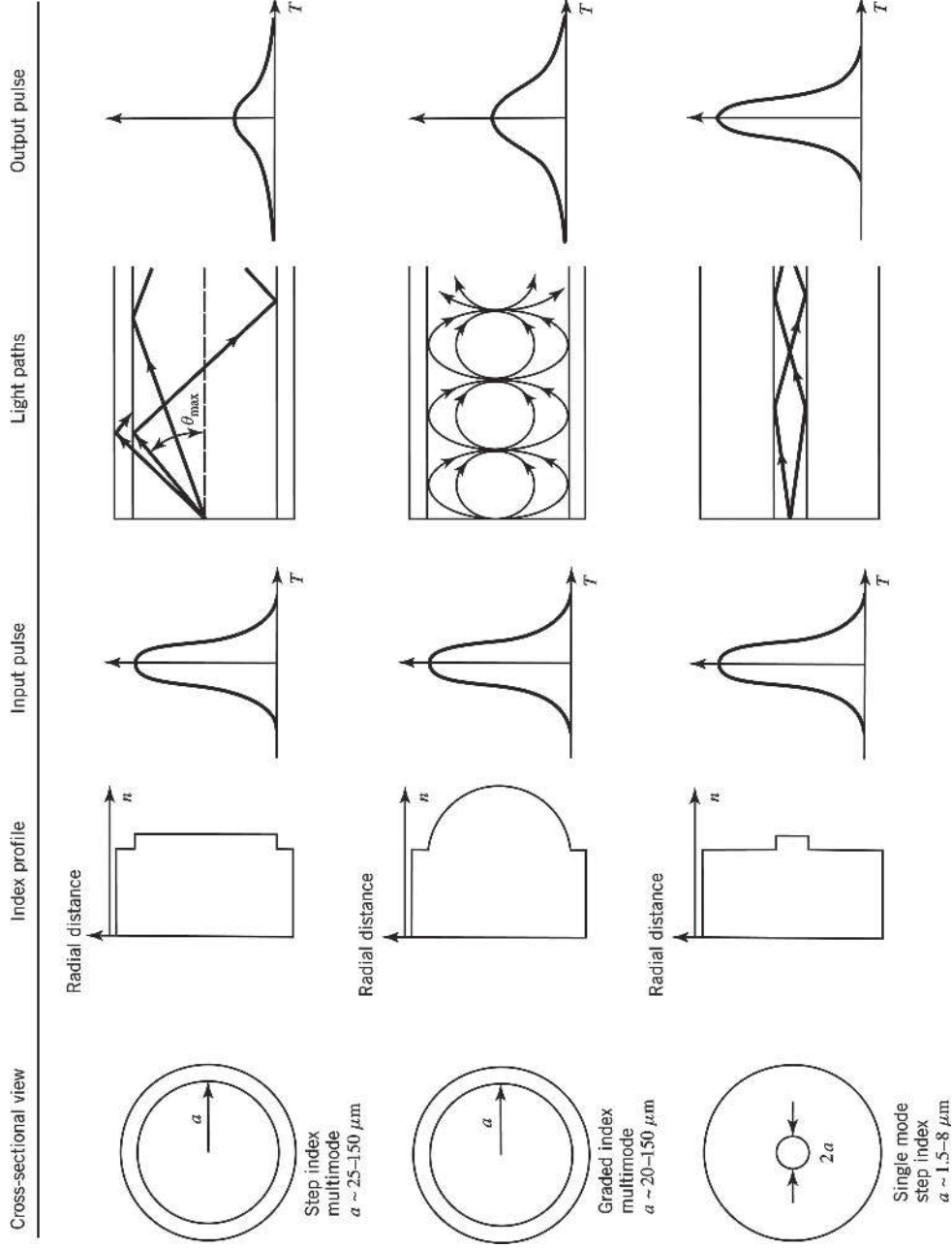


Figure 9-16 Common types and geometries of fiber cables. (Source: T. G. Giallorenzi, “Optical communications research and technology: Fiber optics,” *Proc. IEEE*, © 1978, IEEE.)

to the fiber, reduces scattering loss that is due to dielectric discontinuities, and protects the guiding central core from absorbing surface contaminants that the cable may come in contact with.

For the graded-index multimode fiber, the index of refraction $n(\rho) = \sqrt{\varepsilon_r(\rho)}$ of the central core exhibits a nearly parabolic variation in the radial direction from the center of the core toward the cladding. This variation can be represented by [16]

$$n_c(\rho) = n \left[1 + \Delta \left(\frac{a - \rho}{a} \right)^\alpha \right], \quad 0 \leq \rho \leq a \quad (9-105)$$

where $n_c(\rho)$ = index of refraction of the central core (usually $\simeq 1.562$ on axis)

n = index of refraction of the cladding (usually $\simeq 1.540$)

a = radius of the central core

Δ = parameter usually much less than unity (usually $0.01 < \Delta < 0.02$)

α = parameter whose value is close to 2 for maximum fiber bandwidth

It has been shown by Example 5-6 that for the dielectric–dielectric interface of the fiber cable to internally reflect the waves of all incidence angles, the ratio of the index of refraction of the central core at the interface to that of the cladding must be equal to or greater than $\sqrt{2}$, that is, $n_c(a)/n \geq \sqrt{2}$. However, this is not necessary if the angles of incidence of the waves are not small.

The modes that can be supported by the step-index cable (either single mode or multimode) can be found using techniques outlined in Section 9.5.1 for the dielectric rod waveguide. These modes are, in general, HEM (HE or EH) hybrid modes. The applicable equations are (9-81a) through (9-94b), where ε_r should represent the square of the ratio of the index of refraction of the central core to that of the cladding [i.e., $\varepsilon_r = (n_c/n)^2$]. The field configurations of the graded-index cable can be analyzed in terms of Hermite-Gaussian functions [35]. Because of the complexity, the analysis will not be presented here. The interested reader is referred to the literature [11, 16].

9.5.4 Dielectric-Covered Conducting Rod

Let us consider the field analysis of a dielectric-covered conducting circular rod, as shown in Figure 9-17 [36–38]. The radius of the conducting rod is a while that of the dielectric cover is b . The thickness of the dielectric sleeving is denoted by t ($t = b - a$). When the radius of the conducting rod is small, the rod will be representative of a dielectric-covered wire. In this section we will consider the TM^z and TE^z modes, which, in general, must co-exist to satisfy the boundary conditions. We will conduct an extended discussion of the modes with azimuthal symmetry—no ϕ variations—will be conducted.

A. TM^z Modes The fields in the radial direction inside the dielectric sleeving must be represented by standing wave functions while those outside the dielectric cover must be decaying in order for the rod to act as a waveguide. In general, for the TM_{nm}^z modes, the potential function A_z can be written as

$$A_z^d = \left[A_1^d J_m(\beta_\rho^d \rho) + B_1^d Y_m(\beta_\rho^d \rho) \right] \left[C_2^d \cos(m\phi) + D_2^d \sin(m\phi) \right] \\ \times \left[A_3^d e^{-j\beta_z z} + B_3^d e^{+j\beta_z z} \right], \quad \text{for } a \leq \rho \leq b \quad (9-106a)$$

$$A_z^0 = A_1^0 K_m(\alpha_\rho^0 \rho) \left[C_2^0 \cos(m\phi) + D_2^0 \sin(m\phi) \right] \\ \times \left[A_3^0 e^{-j\beta_z z} + B_3^0 e^{+j\beta_z z} \right], \quad \text{for } b \leq \rho \leq \infty \quad (9-106b)$$

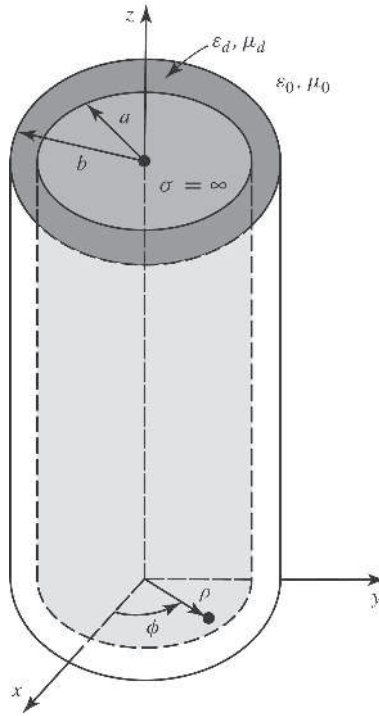


Figure 9-17 Geometry of dielectric-covered conducting rod.

However, the geometry of Figure 9-17 can only support modes with azimuthal symmetry, ($m = 0$). Therefore, for the positive traveling waves of the lowest-order mode ($m = 0$), the vector potentials reduce to

$$A_z^d = \left[A_0^d J_0(\beta_\rho^d \rho) + B_0^d Y_0(\beta_\rho^d \rho) \right] e^{-j\beta_z z}, \quad \text{for } a \leq \rho \leq b \quad (9-107)$$

where

$$(\beta_\rho^d)^2 + \beta_z^2 = \beta_d^2 = \mu_r \epsilon_r \beta_0^2 \quad (9-107a)$$

and

$$A_z^0 = A_0^0 K_0(\alpha_\rho^0 \rho) e^{-j\beta_z z}, \quad \text{for } b \leq \rho \leq \infty \quad (9-108)$$

where

$$-(\alpha_\rho^0)^2 + \beta_z^2 = \beta_0^2 \quad (9-108a)$$

The corresponding electric and magnetic fields can be written as

For $a \leq \rho \leq b$,

$$E_\rho^d = -j \frac{1}{\omega \mu_d \epsilon_d} \frac{\partial^2 A_z^d}{\partial \rho \partial z} = -\frac{\beta_z \beta_\rho^d}{\omega \mu_d \epsilon_d} \left[A_0^d J_0'(\beta_\rho^d \rho) + B_0^d Y_0'(\beta_\rho^d \rho) \right] e^{-j\beta_z z} \quad (9-109a)$$

$$E_\phi^d = -j \frac{1}{\omega \mu_d \epsilon_d} \frac{\partial^2 A_z^d}{\partial \phi \partial z} = 0 \quad (9-109b)$$

$$E_z^d = -j \frac{1}{\omega \mu_d \varepsilon_d} \left(\frac{\partial^2}{\partial z^2} + \beta_d^2 \right) A_z^d = -j \frac{(\beta_\rho^d)^2}{\omega \mu_d \varepsilon_d} \left[A_0^d J_0(\beta_\rho^d \rho) + B_0^d Y_0(\beta_\rho^d \rho) \right] e^{-j\beta_z z} \quad (9-109c)$$

$$H_\rho^d = \frac{1}{\mu_d} \frac{1}{\rho} \frac{\partial A_z^d}{\partial \phi} = 0 \quad (9-109d)$$

$$H_\phi^d = -\frac{1}{\mu_d} \frac{\partial A_z^d}{\partial \rho} = -\frac{\beta_\rho^d}{\mu_d} \left[A_0^d J_0'(\beta_\rho^d \rho) + B_0^d Y_0'(\beta_\rho^d \rho) \right] e^{-j\beta_z z} \quad (9-109e)$$

$$H_z^d = 0 \quad (9-109f)$$

$$' \equiv \frac{\partial}{\partial(\beta_\rho^d \rho)} \quad (9-109g)$$

And for $b \leq \rho \leq \infty$,

$$E_\rho^0 = -j \frac{1}{\omega \mu_0 \varepsilon_0} \frac{\partial^2 A_z^0}{\partial \rho \partial z} = \frac{\alpha_\rho^0 \beta_z}{\omega \mu_0 \varepsilon_0} A_0^0 K_0'(\alpha_\rho^0 \rho) e^{-j\beta_z z} \quad (9-110a)$$

$$E_\phi^0 = -j \frac{1}{\omega \mu_0 \varepsilon_0} \frac{1}{\rho} \frac{\partial^2 A_z^0}{\partial \phi \partial z} = 0 \quad (9-110b)$$

$$E_z^0 = -j \frac{1}{\omega \mu_0 \varepsilon_0} \left(\frac{\partial^2}{\partial z^2} + \beta_0^2 \right) A_z^0 = +j \frac{(\alpha_\rho^0)^2}{\omega \mu_0 \varepsilon_0} A_0^0 K_0(\alpha_\rho^0 \rho) e^{-j\beta_z z} \quad (9-110c)$$

$$H_\rho^0 = \frac{1}{\mu_0} \frac{1}{\rho} \frac{\partial A_z^0}{\partial \phi} = 0 \quad (9-110d)$$

$$H_\phi^0 = -\frac{1}{\mu_0} \frac{\partial A_z^0}{\partial \rho} = -\frac{\alpha_\rho^0}{\mu_0} A_0^0 K_0'(\alpha_\rho^0 \rho) e^{-j\beta_z z} \quad (9-110e)$$

$$H_z^0 = 0 \quad (9-110f)$$

$$' \equiv \frac{\partial}{\partial(\alpha_\rho^0 \rho)} \quad (9-110g)$$

The vanishing of the tangential electric fields at $\rho = a$ and the continuity of the tangential components of the electric and magnetic fields at $\rho = b$ requires that

$$E_z^d(\rho = a, 0 \leq \phi \leq 2\pi, z) = 0 \quad (9-111a)$$

$$E_z^d(\rho = b, 0 \leq \phi \leq 2\pi, z) = E_z^0(\rho = b, 0 \leq \phi \leq 2\pi, z) \quad (9-111b)$$

$$H_\phi^d(\rho = b, 0 \leq \phi \leq 2\pi, z) = H_\phi^0(\rho = b, 0 \leq \phi \leq 2\pi, z) \quad (9-111c)$$

Applying (9-111a) leads to

$$A_0^d J_0(\beta_\rho^d a) + B_0^d Y_0(\beta_\rho^d a) = 0 \quad (9-112)$$

or

$$B_0^d = -A_0^d \frac{J_0(\beta_\rho^d a)}{Y_0(\beta_\rho^d a)} \quad (9-112a)$$

whereas (9-111b) leads to

$$-\frac{(\beta_\rho^d)^2}{\omega\mu_d\varepsilon_d}\left[A_0^d J_0(\beta_\rho^d b) + B_0^d Y_0(\beta_\rho^d b)\right] = \frac{(\alpha_\rho^0)^2}{\omega\mu_0\varepsilon_0} A_0^0 K_0(\alpha_\rho^0 b) \quad (9-113)$$

which by using (9-112a) can be written as

$$A_0^0 K_0(\alpha_\rho^0 b) = -A_0^d \frac{\mu_0}{\mu_d} \frac{\varepsilon_0}{\varepsilon_d} \left(\frac{\beta_\rho^d}{\alpha_\rho^0}\right)^2 \left[J_0(\beta_\rho^d b) - \frac{J_0(\beta_\rho^d a) Y_0(\beta_\rho^d b)}{Y_0(\beta_\rho^d a)} \right] \quad (9-113a)$$

The continuity of the tangential magnetic field at $\rho = b$, as stated by (9-111c), leads to

$$-\frac{\beta_\rho^d}{\mu_d} \left[A_0^d J_0'(\beta_\rho^d b) + B_0^d Y_0'(\beta_\rho^d b) \right] = -\frac{\alpha_\rho^0}{\mu_0} A_0^0 K_0'(\alpha_\rho^0 b) \quad (9-114)$$

which by using (9-112a) can be written as

$$A_0^0 K_0'(\alpha_\rho^0 b) = A_0^d \frac{\mu_0}{\mu_d} \frac{\beta_\rho^d}{\alpha_\rho^0} \left[J_0'(\beta_\rho^d b) - \frac{J_0(\beta_\rho^d a) Y_0'(\beta_\rho^d b)}{Y_0(\beta_\rho^d a)} \right] \quad (9-114a)$$

Dividing (9-114a) by (9-113a) leads to

$$\frac{K_0'(\alpha_\rho^0 b)}{K_0(\alpha_\rho^0 b)} = -\frac{\varepsilon_d}{\varepsilon_0} \left(\frac{\alpha_\rho^0}{\beta_\rho^d}\right) \frac{\left[J_0'(\beta_\rho^d b) Y_0(\beta_\rho^d a) - J_0(\beta_\rho^d a) Y_0'(\beta_\rho^d b) \right]}{\left[J_0(\beta_\rho^d b) Y_0(\beta_\rho^d a) - J_0(\beta_\rho^d a) Y_0(\beta_\rho^d b) \right]} \quad (9-115a)$$

Subtracting (9-108a) from (9-107a) leads to

$$\left(\beta_\rho^d\right)^2 + \left(\alpha_\rho^0\right)^2 = \beta_0^2 (\mu_r \varepsilon_r - 1) \quad (9-115b)$$

which, along with the transcendental Equation 9-115a, can be used to solve for β_ρ^d and α_ρ^0 . This can be accomplished using graphical or numerical techniques similar to the ones utilized in Section 8.7 for planar structures.

The technique outlined and implemented in Section 8.7 to solve the wave numbers for planar dielectric waveguides is straightforward but complicated when utilized to solve for the wave numbers β_ρ^d and α_ρ^0 of (9-115a) and (9-115b). An approximate solution can be used to solve (9-115a) and (9-115b) simultaneously when the thickness of the dielectric cladding $t = b - a$ is small. Under these conditions the Bessel functions $J_0(\beta_\rho^d b)$ and $Y_0(\beta_\rho^d b)$ can be expanded in a

Taylor series about the point $\beta_\rho^d a$, that is,

$$\begin{aligned} J_0(\beta_\rho^d b) &\simeq J_0(\beta_\rho^d a) + \left. \frac{dJ_0(\beta_\rho^d b)}{d(\beta_\rho^d b)} \right|_{\beta_\rho^d b = \beta_\rho^d a} \beta_\rho^d (b-a) \\ &\simeq J_0(\beta_\rho^d a) - \beta_\rho^d (b-a) J_1(\beta_\rho^d a) \end{aligned} \quad (9-116a)$$

$$\begin{aligned} Y_0(\beta_\rho^d b) &\simeq Y_0(\beta_\rho^d a) + \left. \frac{dY_0(\beta_\rho^d b)}{d(\beta_\rho^d b)} \right|_{\beta_\rho^d b = \beta_\rho^d a} \beta_\rho^d (b-a) \\ &\simeq Y_0(\beta_\rho^d a) - \beta_\rho^d (b-a) Y_1(\beta_\rho^d a) \end{aligned} \quad (9-116b)$$

$$\begin{aligned} J'_0(\beta_\rho^d b) &= \left. \frac{dJ_0(\beta_\rho^d \rho)}{d(\beta_\rho^d \rho)} \right|_{\rho=b} \\ &= -J_1(\beta_\rho^d b) \simeq - \left[J_1(\beta_\rho^d a) + \left. \frac{dJ_1(\beta_\rho^d b)}{d(\beta_\rho^d b)} \right|_{\beta_\rho^d b = \beta_\rho^d a} \beta_\rho^d (b-a) \right] \\ &\simeq - \left\{ J_1(\beta_\rho^d a) + \beta_\rho^d (b-a) \left[J_0(\beta_\rho^d a) - \frac{1}{\beta_\rho^d a} J_1(\beta_\rho^d a) \right] \right\} \\ J'_0(\beta_\rho^d b) &\simeq - \left[J_1(\beta_\rho^d a) \left(1 - \frac{b-a}{a} \right) + \beta_\rho^d (b-a) J_0(\beta_\rho^d a) \right] \end{aligned} \quad (9-116c)$$

$$Y'_0(\beta_\rho^d b) \simeq - \left[Y_1(\beta_\rho^d a) \left(1 - \frac{b-a}{a} \right) + \beta_\rho^d (b-a) Y_0(\beta_\rho^d a) \right] \quad (9-116d)$$

Therefore, the numerator and denominator of (9-115a) can be written, respectively, as

$$\begin{aligned} &J'_0(\beta_\rho^d b) Y_0(\beta_\rho^d a) - J_0(\beta_\rho^d a) Y'_0(\beta_\rho^d b) \\ &\simeq -Y_0(\beta_\rho^d a) \left[J_1(\beta_\rho^d a) \left(1 - \frac{b-a}{a} \right) + \beta_\rho^d (b-a) J_0(\beta_\rho^d a) \right] \\ &\quad + J_0(\beta_\rho^d a) \left[Y_1(\beta_\rho^d a) \left(1 - \frac{b-a}{a} \right) + \beta_\rho^d (b-a) Y_0(\beta_\rho^d a) \right] \\ &\simeq \left(1 - \frac{b-a}{a} \right) [J_0(\beta_\rho^d a) Y_1(\beta_\rho^d a) - Y_0(\beta_\rho^d a) J_1(\beta_\rho^d a)] \end{aligned} \quad (9-117a)$$

$$\begin{aligned} &J_0(\beta_\rho^d b) Y_0(\beta_\rho^d a) - J_0(\beta_\rho^d a) Y_0(\beta_\rho^d b) \\ &\simeq Y_0(\beta_\rho^d a) [J_0(\beta_\rho^d a) - \beta_\rho^d (b-a) J_1(\beta_\rho^d a)] \\ &\quad - J_0(\beta_\rho^d a) [Y_0(\beta_\rho^d a) - \beta_\rho^d (b-a) Y_1(\beta_\rho^d a)] \\ &\simeq \beta_\rho^d (b-a) [J_0(\beta_\rho^d a) Y_1(\beta_\rho^d a) - Y_0(\beta_\rho^d a) J_1(\beta_\rho^d a)] \end{aligned} \quad (9-117b)$$

Substituting (9-117a) and (9-117b) into (9-115a) leads to

$$\frac{K'_0(\alpha_\rho^0 b)}{K_0(\alpha_\rho^0 b)} \simeq -\frac{\varepsilon_d}{\varepsilon_0} \left(\frac{\alpha_\rho^0}{\beta_\rho^d} \right) \left(\frac{1 - \frac{b-a}{a}}{\beta_\rho^d (b-a)} \right) = -\frac{\varepsilon_d}{\varepsilon_0} \frac{\alpha_\rho^0}{(\beta_\rho^d)^2} \left(\frac{1 - \frac{b-a}{a}}{b-a} \right) \quad (9-118)$$

For small values of a (i.e., $a \ll \lambda_0$) the attenuation constant α_ρ^0 is of the same order of magnitude as β_0 and the wave is loosely bound to the surface so that $\alpha_\rho^0 b$ is small for small values of $b - a$. Under these conditions

$$K'_0(\alpha_\rho^0 b) = \frac{dK_0(\alpha_\rho^0 \rho)}{d(\alpha_\rho^0 \rho)} \bigg|_{\rho=b} = -K_1(\alpha_\rho^0 b) \simeq -\frac{1}{\alpha_\rho^0 b} = -\frac{1}{b\alpha_\rho^0} \quad (9-119a)$$

$$K_0(\alpha_\rho^0 b) \simeq -\ln(0.89\alpha_\rho^0 b) \quad (9-119b)$$

so that (9-118) reduces for $\varepsilon_d = \varepsilon_r \varepsilon_0$ to

$$\begin{aligned} \frac{1}{b \ln(0.89\alpha_\rho^0 b)} &\simeq -\varepsilon_r \left(\frac{\alpha_\rho^0}{\beta_\rho^d} \right)^2 \left(\frac{1 - \frac{b-a}{a}}{b-a} \right) \\ (\beta_\rho^d)^2 (b-a) &\simeq -\varepsilon_r b \left(1 - \frac{b-a}{a} \right) (\alpha_\rho^0)^2 \ln(0.89\alpha_\rho^0 b) \end{aligned} \quad (9-120)$$

Substituting (9-115b) into (9-120) for $(\beta_\rho^d)^2$ leads for $t = b - a \ll a$ to

$$\left[\beta_0^2 (\varepsilon_r - 1) - (\alpha_\rho^0)^2 \right] (b-a) \simeq -\varepsilon_r b \left(1 - \frac{b-a}{a} \right) (\alpha_\rho^0)^2 \ln(0.89\alpha_\rho^0 b) \quad (9-121)$$

or

$$\boxed{\varepsilon_r b (\alpha_\rho^0)^2 \ln(0.89\alpha_\rho^0 b) \simeq \left[-\beta_0^2 (\varepsilon_r - 1) + (\alpha_\rho^0)^2 \right] (b-a)} \quad (9-121a)$$

Example 9-6

A perfectly conducting wire of radius $a = 0.09$ cm is covered with a dielectric sleeving of polystyrene ($\varepsilon_r = 2.56$) of radius $b = 0.10$ cm. At a frequency of 9.55 GHz, determine the attenuation constants α_ρ^0 , β_z , and β_ρ^d and the relative field strength at $\rho = 2\lambda_0$ compared to that at the outside surface of the dielectric sleeving ($\rho = b$).

Solution: At $f = 9.55$ GHz,

$$\begin{aligned} \lambda_0 &= \frac{30 \times 10^9}{9.55 \times 10^9} = 3.1414 \text{ cm} & \Rightarrow \beta_0 &= \frac{2\pi}{\lambda_0} = 2 \text{ rad/cm} \\ \lambda_d &= \frac{\lambda_0}{\sqrt{\varepsilon_r}} = \frac{3.1414}{\sqrt{2.56}} = 1.9634 \text{ cm} & \Rightarrow \beta_d &= \frac{2\pi}{\lambda} = \frac{2\pi}{1.9634} = 3.2 \text{ rad/cm} \end{aligned}$$

Since the thickness t of the dielectric sleeving is much smaller than the wavelength,

$$t = b - a = 0.10 - 0.09 = 0.01 \text{ cm} < \lambda_d = 1.9634 \text{ cm} < \lambda_0 = 3.1414 \text{ cm}$$

then the approximate relation of (9-121a) is applicable. Using an iterative procedure, it can be shown that

$$\alpha_\rho^0 = 0.252 \text{ Np/cm}$$

is a solution to (9-121a). Using (9-108a),

$$\beta_z = \sqrt{\beta_0^2 + (\alpha_\rho^0)^2} = \sqrt{(2)^2 + (0.252)^2} = \sqrt{4.0635} = 2.0158 \text{ rad/cm}$$

and β_ρ^d is found using (9-107a) as

$$\beta_\rho^d = \sqrt{\beta_d^2 - \beta_z^2} = \sqrt{(3.2)^2 - (2.0158)^2} = \sqrt{6.1765} = 2.485 \text{ rad/cm}$$

The field outside the sleeving is of decaying form represented by the modified Bessel function $K_0(\alpha_\rho^0 \rho)$ of (9-108). Thus,

$$\begin{aligned} \frac{E(\rho = 2\lambda_0)}{E(\rho = b)} &= \frac{E(\rho = 2\pi \text{ cm})}{E(\rho = 0.10 \text{ cm})} = \frac{K_0(\rho = 2\pi \text{ cm})}{K_0(\rho = 0.10 \text{ cm})} \\ &= \frac{\left. \frac{C_0 e^{-\alpha_\rho^0 \rho}}{\sqrt{\alpha_\rho^0 \rho}} \right|_{\rho=2\pi \text{ cm}}}{\left. \frac{C_0 e^{-\alpha_\rho^0 \rho}}{\sqrt{\alpha_\rho^0 \rho}} \right|_{\rho=0.10 \text{ cm}}} = \frac{0.1631}{6.1426} = 0.0266 = 2.66\% \end{aligned}$$

Therefore, at a distance of $\rho = 2\lambda_0$ the relative field has been reduced to a very small value; at points further away, the field intensity is even smaller.

For the dielectric-covered wire and the dielectric rod waveguide, we can define an *effective radius* as the radial distance at which point and beyond the relative field intensity is of very low value. If we use the 2.66% value of Example 9-6 as the field value with which we can define the effective radius, then the effective radius for the wire of Example 9-6 is $a_e = 2\lambda_0 = 2\pi \text{ cm}$.

The rate of attenuation α_ρ^0 can be increased and the wave can be made more tightly bound to the surface by increasing the dielectric constant and/or thickness of the dielectric sleeving. This, however, results in greater attenuation and larger losses of the wave along the direction of wave travel (axis of the wire) because of the greater field concentration near the conducting boundary.

B. TE^z Modes Following a procedure similar to that used for the TM^z_{mn} modes, it can be shown that for the dielectric-covered conducting rod of Figure 9-17, the TE^z_{mn} positive traveling waves of the lowest-order mode ($m = 0$) also possess azimuthal symmetry (no ϕ variations, $m = 0$) and can exist individually. Therefore, for the $m = 0$ mode, the vector potentials can be written as

$$F_z^d = \left[A_0^d J_0(\beta_\rho^d \rho) + B_0^d Y_0(\beta_\rho^d \rho) \right] e^{-j\beta_z z}, \quad \text{for } a \leq \rho \leq b \quad (9-122)$$

where

$$(\beta_\rho^d)^2 + \beta_z^2 = \beta_d^2 = \mu_r \epsilon_r \beta_0^2 \quad (9-122a)$$

$$F_z^0 = A_0^0 K_0(\alpha_\rho^0 \rho) e^{-j\beta_z z}, \quad \text{for } b \leq \rho \leq \infty \quad (9-123)$$

where

$$-(\alpha_\rho^0)^2 + \beta_z^2 = \beta_0^2 \quad (9-123a)$$

Leaving the details for the reader as an end-of-chapter exercise, the equations for the TE_{0n}^z modes corresponding to (9-113a), (9-114a), (9-115a), (9-115b), and (9-121a) for the TM_{0n}^z modes, are given by

$$A_0^0 K_0'(\alpha_\rho^0 b) = A_0^d \frac{\varepsilon_0}{\varepsilon_d} \frac{\beta_\rho^d}{\alpha_\rho^0} \left[J_0'(\beta_\rho^d b) - \frac{J_0'(\beta_\rho^d a) Y_0'(\beta_\rho^d b)}{Y_0'(\beta_\rho^d a)} \right] \quad (9-124a)$$

$$A_0 K_0(\alpha_\rho^0 b) = -A_0^d \frac{\mu_0}{\mu_d} \frac{\varepsilon_0}{\varepsilon_d} \left(\frac{\beta_\rho^d}{\alpha_\rho^0} \right)^2 \left[J_0(\beta_\rho^d b) - \frac{J_0'(\beta_\rho^d a) Y_0(\beta_\rho^d b)}{Y_0'(\beta_\rho^d a)} \right] \quad (9-124b)$$

$$\frac{K_0'(\alpha_\rho^0 b)}{K_0(\alpha_\rho^0 b)} = -\frac{\mu_d}{\mu_0} \left(\frac{\alpha_\rho^0}{\beta_\rho^d} \right) \left[\frac{J_0'(\beta_\rho^d b) Y_0'(\beta_\rho^d a) - J_0'(\beta_\rho^d a) Y_0'(\beta_\rho^d b)}{J_0(\beta_\rho^d b) Y_0'(\beta_\rho^d a) - J_0'(\beta_\rho^d a) Y_0(\beta_\rho^d b)} \right] \quad (9-124c)$$

$$(\beta_\rho^d)^2 + (\alpha_\rho^0)^2 = \beta_0^2 (\mu_r \varepsilon_r - 1) \quad (9-124d)$$

$$\mu_r b(b-a)(\alpha_\rho^0)^2 \ln(0.89 \alpha_\rho^0 b) \simeq 1 \quad (9-124e)$$

9.6 MULTIMEDIA

On the website that accompanies this book, the following multimedia resources are included for the review, understanding, and presentation of the material of this chapter.

- **MATLAB** computer programs:
 - a. **Cyl_Waveguide**: Computes the propagation characteristics of a cylindrical waveguide.
 - b. **Cyl_Resonator**: Computes the resonant characteristics of a cylindrical resonator.
 - c. **CircDielGuide**: Computes the propagation characteristics of a circular dielectric waveguide based on the solution of the eigenvalue equation 9-91.
- **PowerPoint (PPT)** viewgraphs, in multicolor.

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PROBLEMS

- 9.1. Design a circular waveguide filled with a lossless dielectric medium whose relative permeability is unity. The waveguide must operate in a single dominant mode over a bandwidth of 1.5 GHz. Assume that the radius of the guide is 1.12 cm. Find the
 - (a) Dielectric constant of the medium that must fill the cavity to meet the desired design specifications.
 - (b) Lower and upper frequencies of operation.
- 9.2. A dielectric waveguide, with a dielectric constant of 2.56, is inserted inside a section of a circular waveguide of radius a and length L . This section of the circular waveguide is inserted and designed to be used as a phase shifter. Assuming:
 - Dominant mode propagation
 - Radius $a = 2$ cm
 - Length $L = 5$ cm
 - Frequency = 6 GHz

determine the additional phase shift (in degrees), from what it would have been if the waveguide was filled with air, provided by the presence of the dielectric material in this section of the circular waveguide.

- 9.3. Design a circular waveguide, with radius of 2 cm and filled with a dielectric material with dielectric constant of 2.25, to be used as a delay line. What should the length (in meters) of the waveguide be so that the total delay it presents by its insertion at 3.5 GHz is 2 microseconds?
- 9.4. Design a waveguide phase shifter using a circular waveguide with a 2 cm radius, operating at 6 GHz in its dominant mode, and completely filled with a dielectric material with a dielectric constant of 2.25. Determine the length (in cm)

of the waveguide section so that the total phase shift introduced by its insertion is 180° .

- 9.5. An air-filled circular waveguide of radius a has a conducting baffle placed along its length at $\phi = 0$ extending from $\rho = 0$ to $\rho = a$, as shown in Figure P9-5. For TE^z modes, derive simplified expressions for the vector potential component, the electric and magnetic fields, and the cutoff frequencies, eigenvalues, phase constant along the axis of the guide, guide wavelength, and wave impedance. Also determine the following.
 - (a) The cutoff frequencies of the three lowest-order propagating modes in order of ascending cutoff frequency when the radius of the cylinder is 1 cm.
 - (b) The wave impedance and guide wavelength (in cm) for the lowest-order mode at $f = 1.5f_c$ where f_c is the cutoff frequency of the lowest-order mode.

Hint:

$$J'_{1/2}(x) = 0 \quad \text{for } x = 1.1655, 4.6042$$

$$J'_{3/2}(x) = 0 \quad \text{for } x = 2.4605, 6.0293$$

$$J'_{5/2}(x) = 0 \quad \text{for } x = 3.6328$$

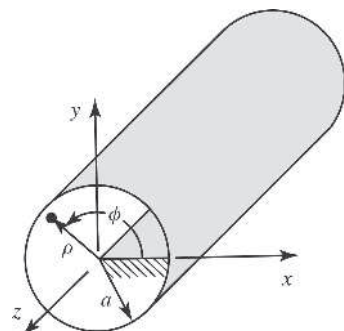


Figure P9-5

9.6. Repeat Problem 9.5 for TM^z modes. *Hint:*

$$J_{1/2}(x) = 0 \quad \text{for } x = 3.1416, 6.2832$$

$$J_{3/2}(x) = 0 \quad \text{for } x = 4.4934$$

$$J_{5/2}(x) = 0 \quad \text{for } x = 5.7635$$

9.7. The cross section of a cylindrical waveguide is a half circle, as shown in Figure P9-7. Derive simplified expressions for the vector potential component, electric and magnetic fields, eigenvalues, and cutoff frequencies for TE^z modes and TM^z modes.

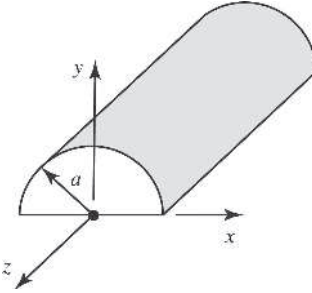


Figure P9-7

9.8. Repeat Problem 9.7 for the waveguide cross section of Figure P9-8.

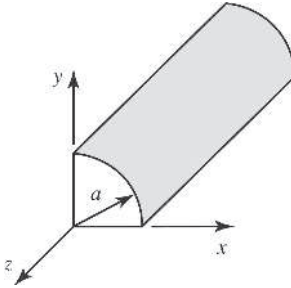


Figure P9-8

9.9. Repeat Problem 9.7 when the waveguide cross section is an angular sector as shown in Figure P9-9. Show that the zeroes of $\beta_\rho a$ are obtained using

$$(a) \text{TE}^z : J'_m(\beta_\rho a) = 0, \quad \beta_\rho = \frac{\chi'_{mn}}{a}$$

$$m = p \left(\frac{\pi}{\phi_0} \right), \quad p = 0, 1, 2, \dots$$

$$(b) \text{TM}^z : J_m(\beta_\rho a) = 0, \quad \beta_\rho = \frac{\chi_{mn}}{a}$$

$$m = p \left(\frac{\pi}{\phi_0} \right), \quad p = 1, 2, 3, \dots$$

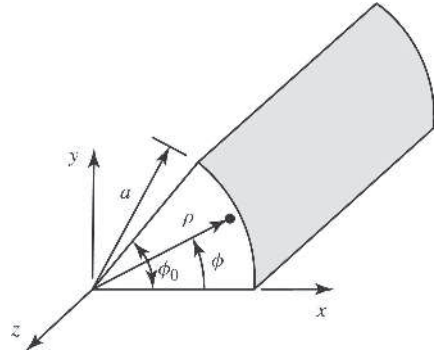


Figure P9-9

9.10. The cross section of a cylindrical waveguide is that of a coaxial line with inner radius a and outer radius b , as shown in Figure P9-10. Assume TE^z modes within the waveguide.

- Derive simplified expressions for the vector potential component, and the electric and magnetic fields.
- Show that the eigenvalues are obtained as solutions to

$$J'_m(\beta_\rho a)Y'_m(\beta_\rho b) - Y'_m(\beta_\rho a)J'_m(\beta_\rho b) = 0$$

where $m = 0, 1, 2, \dots$

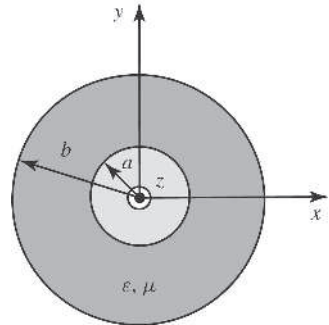


Figure P9-10

9.11. Repeat Problem 9.10 for TM^z and show that the eigenvalues are obtained as solutions to

$$J_m(\beta_\rho a)Y_m(\beta_\rho b) - Y_m(\beta_\rho a)J_m(\beta_\rho b) = 0$$

where $m = 0, 1, 2, \dots$

9.12. The cross section of a cylindrical waveguide is an annular sector with inner and outer radii of a and b , as shown in Figure 9-23. Assume TE^z modes within the waveguide.

- Derive simplified expressions for the vector potential component, and the electric and magnetic fields.

- (b) Show that the eigenvalues are determined by solving

$$J'_m(\beta_\rho b)Y'_m(\beta_\rho a) - J'_m(\beta_\rho a)Y'_m(\beta_\rho b) = 0$$

where $m = p(\pi / \phi_0)$, $p = 0, 1, 2, \dots$

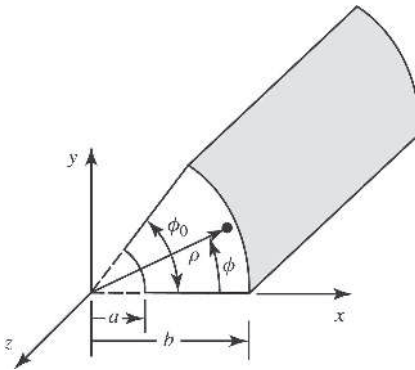


Figure P9-12

- 9.13. Repeat Problem 9.12 for TM^z modes. The eigenvalues are determined by solving

$$J_m(\beta_\rho b)Y_m(\beta_\rho a) - J_m(\beta_\rho a)Y_m(\beta_\rho b) = 0$$

where $m = p(\pi / \phi_0)$, $p = 1, 2, 3, \dots$

- 9.14. A circular waveguide with radius of 3 cm is made of copper ($\sigma = 5.76 \times 10^7$ S/m). For the dominant TE_{11} and low-loss TE_{01} modes, determine their corresponding cutoff frequencies and attenuation constants (in Np/m and dB/m) at a frequency of 7 GHz. Assume that the waveguide is filled with air.

- 9.15. Derive the attenuation coefficient α_c for the conduction losses of a circular waveguide of radius a for the TM_{01}^z . Show that α_c can be expressed as

$$\alpha_c(TM_{01}) = \frac{R_s}{\eta a \sqrt{1 - \left(\frac{f_c}{f}\right)^2}}$$

where R_s is the surface resistance of the waveguide metal and η is the intrinsic impedance of the medium within the waveguide.

- 9.16. A circular cavity, as shown in Figure 9-5, has a radius of 6 cm and a height of 10 cm. It is filled with a lossless dielectric with $\epsilon_r = 4$. Find the:

- (a) First four TE^z and/or TM^z modes according to their resonant frequency (in order of ascending values).
(b) Q of the cavity (assuming dominant mode operation). The walls of the cavity are copper ($\sigma = 5.76 \times 10^7$ S/m).

- 9.17. Design a circular cavity of radius a and height h such that the resonant frequency of the next higher-order mode is 1.5 times greater than the resonant frequency of the dominant mode. Assume that the radius is 4 cm. Find the:

- (a) Height of the cavity (in cm).
(b) Resonant frequency of the dominant mode (assume free space within the cavity).
(c) Dielectric constant of the dielectric that must be inserted inside the cavity to reduce the resonant frequency by a factor of 1.5.

- 9.18. A circular cavity of radius a and height h is completely filled with a dielectric material of dielectric constant ϵ_r . The height-to-radius ratio is $h/a = 1.9$ where $a = 2$ cm.

- (a) Identify the mode with the lowest resonant frequency.
(b) Determine the dielectric constant of the material so that the difference between the resonant frequencies of the lowest to the next lowest order modes is 50 MHz.

- 9.19. For the radial waveguide of Figure 9-6 determine the maximum spacing h (in m) to insure operation of a single lowest-order mode between the plates up to 300 MHz. Do this individually for TE_n^z modes and TM_n^z modes.

- 9.20. For the wedged-plate radial waveguide of Figure 9-7, derive for the TE^z modes the expressions for the vector potential component of (9-79), the electric and magnetic fields of (9-79d) through (9-79i), the eigenvalues of (9-79a) through (9-79c), and the wave impedance of (9-79j).

- 9.21. Given the wedged-plate geometry of Figure 9-7, assume TM^z modes and derive the expressions for the vector potential of (9-80), the electric and magnetic fields of (9-80d) through (9-80i), the eigenvalues of (9-80a) through (9-80c), and the wave impedance of (9-80j).

- 9.22.** If the dielectric constant of the materials that make up the circular dielectric waveguide of Figure 9-8 is very large (usually 30 or greater), the dielectric–air interface along the surface acts almost as an open circuit (see Section 9.5.2). Under these conditions the surface of the dielectric can be approximated by a perfect magnetic conductor (PMC). Assume that the surface of the dielectric rod can be modeled as a PMC and derive, for the TE^z modes, simplified expressions for the vector potential component, electric and magnetic fields, and cutoff frequencies. Assume that the dielectric material is rutile ($\epsilon_r \simeq 130$). Determine the cutoff frequencies of the lowest two modes when the radius of the rod is 3 cm. Verify with the MATLAB program **CircDielGuide**.
- 9.23.** Repeat Problem 9.22 for the TM^z modes.
- 9.24.** Determine the cutoff frequencies of the first four lowest-order modes (HE, EH, TE, and TM) for a dielectric rod waveguide with radius of 3 cm when the dielectric constant of the material is $\epsilon_r = 20, 38$, and 130. Verify with the MATLAB program **CircDielGuide**.
- 9.25.** Design a circular dielectric rod waveguide (find its radius in cm) so that the cutoff frequency of the TE_{01} , TM_{01} , and HE_{21} modes is 3 GHz when the dielectric constant of the material is $\epsilon_r = 2.56, 4, 9$, and 16.
- 9.26.** It is desired to operate a dielectric rod waveguide in the dominant HE_{11} mode over a frequency range of 5 GHz. Design the dielectric rod (find its dielectric constant) to accomplish this when the radius of the rod is $a = 1.315$ and 1.838 cm.
- 9.27.** A dielectric waveguide of circular cross section, radius a , and dielectric constant of 4, has at 3 GHz an effective dielectric constant of $\epsilon_{\text{reff}} = 2.78$. For the dominant mode, determine at 3 GHz the:
- Phase constant along the z and ρ directions inside the waveguide (in rad/cm).
 - Attenuation coefficient along the ρ direction outside the waveguide (in Np/cm).
- 9.28.** Design a cylindrical dielectric waveguide of circular cross section so that its dominant single mode operation is 4 GHz. The dielectric constant of the material is 4. Determine, in cm, the radius of the waveguide to accomplish this. Use an exact solution.

- 9.29.** Design a homogeneous dielectric cable to be operated at a dominant, single mode with a center frequency of 50 GHz. The radius of the cable is 1 mm. Determine the:
- Minimum value of the dielectric constant of the cable.
 - Frequency range of the dominant, single-mode operation.
- 9.30.** A lossless dielectric waveguide (no PEC walls) of radius a , with $\epsilon_r \gg 1, \mu_r = 1$, has a semi-circular cross section, as shown in Figure P9-30. For the TE^z modes, determine:
- An expression for the cutoff frequency. State the correct eigenvalues and specific allowable indices.
 - Cutoff frequency (in GHz) for the lowest order (dominant) mode assuming the radius is $a = 1.5$ cm, $\epsilon_r = 81, \mu_r = 1$.
- Show the steps, or explain, as to how you arrive at the specific formulas and corresponding allowable indices you are using.

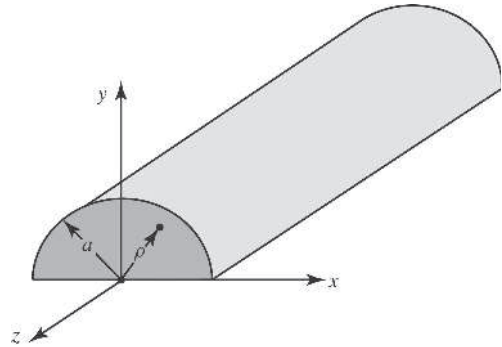


Figure P9-30

- 9.31.** Repeat Problem 9.30 for TM^z modes.
- 9.32.** A cylindrical dielectric waveguide of infinite length is totally composed of a lossless dielectric medium with a dielectric constant $\epsilon_r \gg 1, \mu_r = 1$. The cross section of the waveguide is circular. A very thin PEC baffle is placed along its length at $\phi = 0$ and extends between $0 \leq \rho \leq a$, as shown in Figure P9-5. Assume the wave is traveling in the $+z$ direction. For TM^z modes, determine:
- A reduced/simplified expression for the appropriate vector potential.
 - Expressions for the wavenumbers β_ρ^d and m (representing the ϕ variations) both within the waveguide ($0 \leq \rho \leq a$). Be specific.

- (c) An expression for the cutoff frequencies of the allowable TM^z modes.
 (d) Cutoff frequency (in GHz) of the dominant mode when $\epsilon_r = 81$, $a = 0.45$ in. (1.143 cm).

9.33. Repeat ALL parts of Problem 9.32 for TE^z modes.

9.34. For the dielectric resonator of Figure 9-14a modeled by PMC walls, as shown in Figure 9-14b, derive simplified expressions for the vector potential component, and the electric and magnetic fields when the modes are TE^z and TM^z .

9.35. Assume that the dielectric resonator of Figure 9-14a is modeled by PMC walls as shown in Figure 9-14b.

- (a) Determine the lowest TE^z or TM^z mode.
 (b) Derive an expression for the Q of the cavity for the lowest-order mode. The only losses associated with the resonator are dielectric losses within the dielectric itself.
 (c) Find the resonant frequency.
 (d) Compute the Q of the cavity.

The resonator material is rutile ($\epsilon_r \simeq 130$, $\tan \delta_e \simeq 4 \times 10^{-4}$). The radius of the disk is 0.1148 cm and its height is 0.01148 cm.

9.36. A cylindrical dielectric resonator used in microwave integrated circuit (MIC) design is comprised of a PEC ground plane covered with a section of length l of a dielectric rod of semi-circular cross section of radius a (Figure P9-36). Assuming the dielectric constant of the material is much greater than unity ($\epsilon_r \gg 1$), for TE^z modes, derive the following:

- (a) Approximate simplified expression of the appropriate vector potential component.
 (b) Determine the allowable nontrivial eigenvalues (be specific).

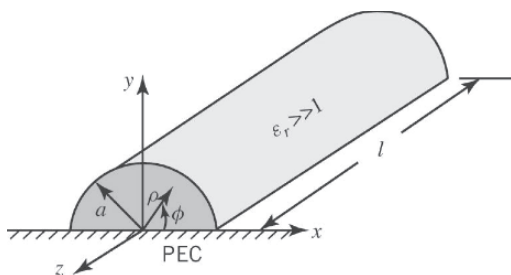


Figure P9-36

- (c) Compute the resonant frequency of the dominant mode when the radius of the rod section is 2 cm, its length is 4 cm, and the dielectric constant is 81.

9.37. Repeat Problem 9.36 for TM^z modes.

9.38. A lossless dielectric cylindrical resonator antenna (DRA) consists of a dielectric cylinder (no PEC walls), with $\epsilon_r \gg 1$, $\mu_r = 1$, of circular cross section and height h whose bottom side is resting on a PEC ground plane, as shown in Figure P9-38. Neglecting the presence of the feed and assuming a vector potential $F_z(\rho, \phi, z)$:

$$F_z(\rho, \phi, z) = A_{mm} J_m(\beta_\rho \rho) \times [C_2 \cos(m\phi) + D_2 \sin(m\phi)] \times [C_3 \cos(\beta_z z) + D_3 \sin(\beta_z z)]$$

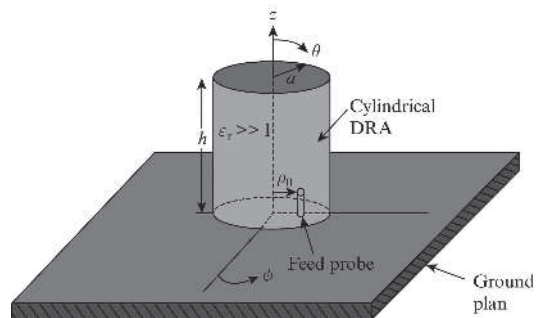


Figure P9-38

- (a) Derive an expression for the resonant frequency of the TE_{mnp}^z modes. It is important to indicate the correct allowable indices for m , n , p . You do not have to start the derivation from the very beginning if part of the derivation is already available; indicate the source/expression/equation number from where you start the derivation.

* Assume (start the count) that the lowest value of the indices, m and p , is zero, unless zero is not allowable because it leads to trivial solution.

** Note: The correct indices m , n , and p are a major part of this problem.

- (b) An expression for the resonant frequency of the dominant mode. Identify the mode and its expression and the indices m , n , and p . Be specific.

* Note: The correct indices m , n , and p are a major part of this problem.

- (c) The resonant frequency (in GHz) of the dominant mode for $\epsilon_r = 8.9$, $a = 0.5$ cm, and $h = 0.3$ cm.

9.39. Repeat Problem 9.38 for TM^z_{mnp} . Use a vector potential $A_z(\rho, \phi, z)$:

$$A_z(\rho, \phi, z) = B_{mn} J_m(\beta_\rho^d \rho) \times [C_2 \cos(m\phi) + D_2 \sin(m\phi)] \times [C_3 \cos(\beta_z z) + D_3 \sin(\beta_z z)]$$

9.40. A dielectric resonator is made of an angular sector (pie) of radius a , height h , and with a subtended angle of $\phi = \phi_0$. Assuming the dielectric constant of the material is very large ($\epsilon_r \gg 1$), for TE^z modes:

- (a) Derive an approximate simplified expression for the appropriate vector potential.
(b) Determine all the allowable eigenvalues for β_ρ, β_z and m .

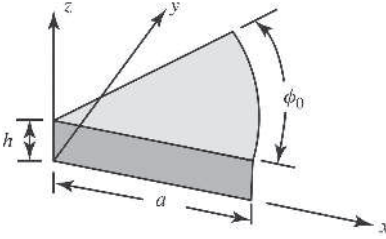


Figure P9-40

9.41. A hybrid dielectric resonator, with a general geometry as shown in Figure 9-14, can be constructed with PEC plates at $z = 0$ and $z = h$ and with open sides. If the dielectric constant of the dielectric is very large, the open dielectric surface can be modeled as a PMC surface. Using such a model for the resonator of Figure 9-14, derive expressions for the resonant frequencies assuming TE^z modes and TM^z modes. This model can also be used as an approximate representation for a circular patch (microstrip) antenna.

9.42. Repeat Problem 9.41 if the top and bottom plates of the resonator are angular sectors each with a subtended angle of ϕ_0 , as shown in Figure P9-42. In addition to the resonant frequency, show that the eigenvalues are obtained as solutions to

$$(a) \text{TE}^z : J_m(\beta_\rho a) = 0, \quad \beta_\rho = \frac{\chi_{mn}}{a} \\ m = p \left(\frac{\pi}{\phi_0} \right), \quad p = 1, 2, 3, \dots$$

$$(b) \text{TM}^z : J'_m(\beta_\rho a) = 0, \quad \beta_\rho = \frac{\chi'_{mn}}{a} \\ m = p \left(\frac{\pi}{\phi_0} \right), \quad p = 0, 1, 2, \dots$$

This model can also be used as an approximate representation for an angular patch (microstrip) antenna.

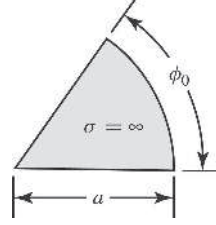


Figure P9-42

9.43. Repeat Problem 9.41 if the top and bottom plates of an annular resonator are annular patches each with inner radius a and outer radius b , as shown in Figure P9-43. In addition to finding the resonant frequency, show that the eigenvalues β_ρ are obtained as solutions to

$$(a) \text{TE}^z : J_m(\beta_\rho a) Y_m(\beta_\rho b) - Y_m(\beta_\rho a) J_m(\beta_\rho b) = 0 \\ m = 0, 1, 2, \dots$$

(b) $\text{TM}^z :$

$$J'_m(\beta_\rho a) Y'_m(\beta_\rho b) - Y'_m(\beta_\rho a) J'_m(\beta_\rho b) = 0 \\ m = 0, 1, 2, \dots$$

This model can also be used as an approximate representation for an annular patch (microstrip) antenna.

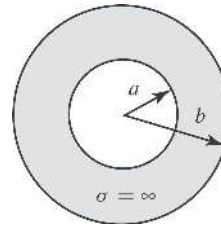


Figure P9-43

9.44. Repeat Problem 9.41 if the top and bottom plates of the resonator are annular sectors each with inner radius a and outer radius b , and subtended angle ϕ_0 , as shown in Figure P9-44. In addition to finding the resonant frequency, show that the eigenvalues β_ρ are obtained as solutions to

(a) TE^z :

$$J_m(\beta_\rho a)Y_m(\beta_\rho b) - Y_m(\beta_\rho a)J_m(\beta_\rho b) = 0$$

$$m = p \left(\frac{\pi}{\phi_0} \right), \quad p = 1, 2, 3, \dots$$

(b) TM^z :

$$J'_m(\beta_\rho a)Y'_m(\beta_\rho b) - Y'_m(\beta_\rho a)J'_m(\beta_\rho b) = 0$$

$$m = p \left(\frac{\pi}{\phi_0} \right), \quad p = 0, 1, 2, \dots$$

This model can also be used as an approximate representation for an annular patch (microstrip) antenna.

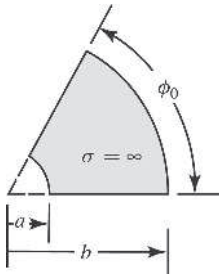


Figure P9-44

- 9.45. A dielectric resonator of height h has a geometry of an annular sector, with an inner radius

a and outer radius b . The material of the resonator has a dielectric constant of $\epsilon_r \gg 1$. For TE^z modes, derive the characteristic equation that should be used to determine the eigenvalues of the resonator. State any other known eigenvalues.

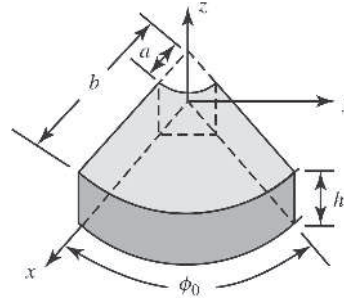


Figure P9-45

- 9.46. Repeat Problem 9.45 for TM^z modes.
- 9.47. For the dielectric covered conducting rod of Figure 9-17, assume TE_{nm}^z modes. For within and outside the dielectric sleeving, derive expressions for the vector potential components and the electric and magnetic fields. Also verify (9-124a) through (9-124e).

CHAPTER 10



Spherical Transmission Lines and Cavities

10.1 INTRODUCTION

Problems involving spherical geometries constitute an important class of electromagnetic boundary-value problems that are used to design transmission lines, cavities, antennas, and scatterers. Some of these may be constructed of metallic walls and others may be of dielectric material. In either case, the field configurations that can be supported by the structure can be obtained by analyzing the structure as a boundary-value problem. We will concern ourselves here with spherical transmission lines and cavities. Scattering by spherical structures will be examined in Chapter 11.

10.2 CONSTRUCTION OF SOLUTIONS

In Chapter 3, Section 3.4.3, we examined the solution of the scalar wave equation in spherical coordinates. It was found that the solution is that of (3-76) where the r , θ , and ϕ variations take the following forms.

1. Radial (r) variations of $f(r)$ can be represented by either:
 - a. spherical Bessel functions of the first [$j_n(\beta r)$] and second [$y_n(\beta r)$] kind, as given by (3-87a) [these functions are used to represent *standing* waves and are related to the regular Bessel functions by (3-90a) and (3-90b)],
 - b. or spherical Hankel functions of the first [$h_n^{(1)}(\beta r)$] and second [$h_n^{(2)}(\beta r)$] kind, as given by (3-87b) [these are related to the regular Hankel functions by (3-91a) and (3-91b)].
2. θ variations of $g(\theta)$ can be represented by associated Legendre functions of the first $P_n^m(\cos \theta)$ or second $Q_n^m(\cos \theta)$ kind, as given by either (3-88a) or (3-88b).
3. ϕ variations of $h(\phi)$ can be represented by either complex exponentials or cosinusoids as given, respectively, by (3-89a) and (3-89b).

It must be remembered, however, that the vector wave equation in spherical coordinates, as given by (3-72), does not reduce to three uncoupled scalar Helmholtz wave equations as stated by

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(3-73a) through (3-73c). Therefore, using the basic approach outlined in Chapter 6, Section 6.5, we cannot construct field solutions that are TE' and/or TM' . Hence, we must look for other approaches for finding field configurations that are supported by spherical structures.

One approach that can be used to find field configurations that are TE^z and/or TM^z will be to represent the potential functions by [1]

$$\underline{\text{TE}^z}$$

$$\mathbf{A} = 0 \quad (10-1a)$$

$$\mathbf{F} = \hat{\mathbf{a}}_z F_z = (\hat{\mathbf{a}}_r \cos \theta - \hat{\mathbf{a}}_\theta \sin \theta) F_z \quad (10-1b)$$

$$\underline{\text{TM}^z}$$

$$\mathbf{A} = \hat{\mathbf{a}}_z A_z = (\hat{\mathbf{a}}_r \cos \theta - \hat{\mathbf{a}}_\theta \sin \theta) A_z \quad (10-2a)$$

$$\mathbf{F} = 0 \quad (10-2b)$$

where $F_z(r, \theta, \phi)$ and $A_z(r, \theta, \phi)$ are solutions to the scalar Helmholtz equation in the spherical coordinate system. Other field configurations can be formed as superpositions of TE^z and TM^z modes. Although this is a valid approach to the problem, it will not be pursued here; it is assigned to the reader as an end-of-chapter exercise. Instead, an alternate procedure will be outlined for construction of TE' and TM' modes.

The procedure outlined in Chapter 6, Sections 6.5.2 and 6.5.3, for the construction, respectively, of TM and TE field configurations was based on the vector potentials $\mathbf{A}$ and $\mathbf{F}$, as derived in Sections 6.2 and 6.3, respectively, and summarized in Section 6.4. The final forms, which are summarized in Section 6.4, were based on the selection of the *Lorenz conditions* of (6-15) and (6-27), or

$$\psi_e = -\frac{1}{j\omega\mu\epsilon} \nabla \cdot \mathbf{A} \quad (10-3a)$$

$$\psi_m = -\frac{1}{j\omega\mu\epsilon} \nabla \cdot \mathbf{F} \quad (10-3b)$$

to represent, respectively, the scalar potential functions ψ_e and ψ_m . If that choice was not made, then the relations of the $\mathbf{E}$ and $\mathbf{H}$ fields to the potentials $\mathbf{A}$ and $\mathbf{F}$ would take a slightly different form. These forms will be outlined here.

10.2.1 The Vector Potential $\mathbf{F}$ ($\mathbf{J} = 0$, $\mathbf{M} \neq 0$)

According to (6-19), the electric field is related to the potential $\mathbf{F}$ by

$$\mathbf{E}_F = -\frac{1}{\epsilon} \nabla \times \mathbf{F} \quad (10-4)$$

Away from the source $\mathbf{M}$, the electric and magnetic fields are related by Maxwell's equation

$$\nabla \times \mathbf{E}_F = -j\omega\mu\mathbf{H}_F \quad (10-5)$$

or

$$\mathbf{H}_F = -\frac{1}{j\omega\mu} \nabla \times \mathbf{E}_F = \frac{1}{j\omega\mu\epsilon} \nabla \times \nabla \times \mathbf{F} \quad (10-5a)$$

Therefore, if the potential $\mathbf{F}$ can be related to the source ($\mathbf{M}$), then $\mathbf{E}_F$ and $\mathbf{H}_F$ can be found using (10-4) and (10-5a). We will attempt to do this next.

Taking the curl of both sides of (10-4) leads to

$$\nabla \times \mathbf{E}_F = -\frac{1}{\varepsilon} \nabla \times \nabla \times \mathbf{F} \quad (10-6)$$

Using Maxwell's equation 6-24,

$$\nabla \times \mathbf{E}_F = -\mathbf{M} - j\omega\mu\mathbf{H}_F \quad (10-7)$$

and equating it to (10-6) leads to

$$\nabla \times \nabla \times \mathbf{F} = \varepsilon\mathbf{M} + j\omega\mu\varepsilon\mathbf{H}_F \quad (10-8)$$

The electric and magnetic fields are also related by Maxwell's equation 6-20,

$$\nabla \times \mathbf{H}_F = j\omega\varepsilon\mathbf{E}_F \quad (10-9)$$

Substituting (10-4) into (10-9) and regrouping reduces to

$$\nabla \times \mathbf{H}_F = j\omega\varepsilon \left(-\frac{1}{\varepsilon} \nabla \times \mathbf{F} \right) = -j\omega \nabla \times \mathbf{F} \quad (10-10)$$

or

$$\nabla \times (\mathbf{H}_F + j\omega\mathbf{F}) = 0 \quad (10-10a)$$

Using the vector identity

$$\nabla \times (-\nabla\psi_m) = 0 \quad (10-11)$$

where ψ_m represents an arbitrary scalar potential, and equating it to (10-10a), we can write that

$$\mathbf{H}_F + j\omega\mathbf{F} = -\nabla\psi_m \quad (10-12)$$

or

$$\mathbf{H}_F = -j\omega\mathbf{F} - \nabla\psi_m \quad (10-12a)$$

Substituting (10-12a) into (10-8) leads to

$$\nabla \times \nabla \times \mathbf{F} = \varepsilon\mathbf{M} + \omega^2\mu\varepsilon\mathbf{F} - j\omega\mu\varepsilon\nabla\psi_m \quad (10-13)$$

or

$$\boxed{\nabla \times \nabla \times \mathbf{F} - \omega^2\mu\varepsilon\mathbf{F} = \varepsilon\mathbf{M} - j\omega\mu\varepsilon\nabla\psi_m} \quad (10-13a)$$

This is the desired expression, which relates the vector potential $\mathbf{F}$ to the source $\mathbf{M}$ and the associated scalar potential ψ_m . In a source-free ($\mathbf{M} = 0$) region, (10-13a) reduces to

$$\boxed{\nabla \times \nabla \times \mathbf{F} - \omega^2\mu\varepsilon\mathbf{F} = -j\omega\mu\varepsilon\nabla\psi_m} \quad (10-14)$$

In a source-free region, the procedure is to solve (10-14) for $\mathbf{F}$, and then use (10-4) and (10-5a) to find, respectively, $\mathbf{E}_F$ and $\mathbf{H}_F$.

10.2.2 The Vector Potential $\mathbf{A}$ ($\mathbf{J} \neq 0$, $\mathbf{M} = 0$)

Following a procedure similar to the one outlined in Section 10.2.1 for the vector potential $\mathbf{F}$, it can be shown, by referring also to Section 6.2, that the equations for the vector potential $\mathbf{A}$ analogous to (10-4), (10-5a), and (10-13a), are

$$\boxed{\mathbf{H}_A = \frac{1}{\mu} \nabla \times \mathbf{A}} \quad (10-15a)$$

$$\boxed{\mathbf{E}_A = \frac{1}{j\omega\epsilon} \nabla \times \mathbf{H}_A = \frac{1}{j\omega\mu\epsilon} \nabla \times \nabla \times \mathbf{A}} \quad (10-15b)$$

$$\boxed{\nabla \times \nabla \times \mathbf{A} - \omega^2 \mu\epsilon \mathbf{A} = \mu \mathbf{J} - j\omega\mu\epsilon \nabla \psi_e} \quad (10-15c)$$

In a source-free region, (10-15c) reduces to

$$\boxed{\nabla \times \nabla \times \mathbf{A} - \omega^2 \mu\epsilon \mathbf{A} = -j\omega\mu\epsilon \nabla \psi_e} \quad (10-15d)$$

The details are left as end-of-chapter exercises for the reader.

In a source-free region, the procedure is to solve (10-15d) for $\mathbf{A}$, and then use (10-15a) and (10-15b) to find, respectively, $\mathbf{H}_A$ and $\mathbf{E}_A$.

10.2.3 The Vector Potentials $\mathbf{F}$ and $\mathbf{A}$

The total fields that are due to both potentials $\mathbf{F}$ and $\mathbf{A}$ are found as superpositions of the fields of Sections 10.2.1 and 10.2.2. Doing this, we have that the total fields are obtained using

$$\boxed{\mathbf{E} = \mathbf{E}_F + \mathbf{E}_A = -\frac{1}{\epsilon} \nabla \times \mathbf{F} + \frac{1}{j\omega\epsilon} \nabla \times \mathbf{H}_A = -\frac{1}{\epsilon} \nabla \times \mathbf{F} + \frac{1}{j\omega\mu\epsilon} \nabla \times \nabla \times \mathbf{A}} \quad (10-16a)$$

$$\boxed{\mathbf{H} = \mathbf{H}_F + \mathbf{H}_A = -\frac{1}{j\omega\mu} \nabla \times \mathbf{E}_F + \frac{1}{\mu} \nabla \times \mathbf{A} = \frac{1}{j\omega\mu\epsilon} \nabla \times \nabla \times \mathbf{F} + \frac{1}{\mu} \nabla \times \mathbf{A}} \quad (10-16b)$$

where $\mathbf{F}$ and $\mathbf{A}$ are, respectively, solutions to

$$\boxed{\nabla \times \nabla \times \mathbf{F} - \omega^2 \mu\epsilon \mathbf{F} = \epsilon \mathbf{M} - j\omega\mu\epsilon \nabla \psi_m} \quad (10-16c)$$

$$\boxed{\nabla \times \nabla \times \mathbf{A} - \omega^2 \mu\epsilon \mathbf{A} = \mu \mathbf{J} - j\omega\mu\epsilon \nabla \psi_e} \quad (10-16d)$$

which for a source-free region ($\mathbf{M} = \mathbf{J} = 0$) reduce to

$$\boxed{\nabla \times \nabla \times \mathbf{F} - \omega^2 \mu\epsilon \mathbf{F} = -j\omega\mu\epsilon \nabla \psi_m} \quad (10-16e)$$

$$\boxed{\nabla \times \nabla \times \mathbf{A} - \omega^2 \mu\epsilon \mathbf{A} = -j\omega\mu\epsilon \nabla \psi_e} \quad (10-16f)$$

We will attempt now to form TE' and TM' mode field solutions using (10-16a) through (10-16f).

10.2.4 Transverse Electric (TE) Modes: Source-Free Region

It was stated previously in Section 6.5.3 that TE modes to any direction in any coordinate system can be obtained by selecting the vector potential $\mathbf{F}$ to have only a nonvanishing component in that direction while simultaneously letting $\mathbf{A} = 0$. The nonvanishing component of $\mathbf{F}$ was obtained as a solution to the scalar wave equation 6-31. The same procedure will be used here except that instead of the nonvanishing component of $\mathbf{F}$ being a solution to (6-31), which does not reduce in spherical coordinates to three scalar noncoupled wave equations, it will be a solution to (10-16e). The nonvanishing component of $\mathbf{F}$ will be the one that coincides with the direction along which the TE modes are desired. Let us construct solutions that are TE^r in a spherical coordinate system.

TE^r field configurations are constructed by letting the vector potentials $\mathbf{F}$ and $\mathbf{A}$ be equal to

$$\boxed{\mathbf{F} = \hat{\mathbf{a}}_r F_r(r, \theta, \phi)} \quad (10-17a)$$

$$\boxed{\mathbf{A} = 0} \quad (10-17b)$$

Since F_r is not a solution to the scalar Helmholtz equation

$$\nabla^2 \mathbf{F} = \nabla^2 (\hat{\mathbf{a}}_r F_r) \neq \hat{\mathbf{a}}_r \nabla^2 F_r \quad (10-18)$$

we will resort, for a source-free region, to (10-16e).

Expanding (10-16e) using (10-17a) leads to

$$\nabla \times \mathbf{F} = \nabla \times (\hat{\mathbf{a}}_r F_r) = \hat{\mathbf{a}}_\theta \frac{1}{r \sin \theta} \frac{\partial F_r}{\partial \phi} - \hat{\mathbf{a}}_\phi \frac{1}{r} \frac{\partial F_r}{\partial \theta} \quad (10-19a)$$

$$\begin{aligned} \nabla \times \nabla \times \mathbf{F} = \hat{\mathbf{a}}_r & \left\{ \frac{1}{r \sin \theta} \left[\frac{\partial}{\partial \theta} \left(-\frac{\sin \theta}{r} \frac{\partial F_r}{\partial \theta} \right) - \frac{\partial}{\partial \phi} \left(\frac{1}{r \sin \theta} \frac{\partial F_r}{\partial \phi} \right) \right] \right\} \\ & + \hat{\mathbf{a}}_\theta \left[\frac{1}{r} \left(\frac{\partial^2 F_r}{\partial r \partial \theta} \right) \right] + \hat{\mathbf{a}}_\phi \left(\frac{1}{r \sin \theta} \frac{\partial^2 F_r}{\partial r \partial \phi} \right) \end{aligned} \quad (10-19b)$$

$$\nabla \psi_m = \hat{\mathbf{a}}_r \frac{\partial \psi_m}{\partial r} + \hat{\mathbf{a}}_\theta \frac{1}{r} \frac{\partial \psi_m}{\partial \theta} + \hat{\mathbf{a}}_\phi \frac{1}{r \sin \theta} \frac{\partial \psi_m}{\partial \phi} \quad (10-19c)$$

Thus for the r , θ , and ϕ components, (10-16e) reduces to

$$\boxed{\frac{1}{r \sin \theta} \left[-\frac{\partial}{\partial \theta} \left(\frac{\sin \theta}{r} \frac{\partial F_r}{\partial \theta} \right) - \frac{\partial}{\partial \phi} \left(\frac{1}{r \sin \theta} \frac{\partial F_r}{\partial \phi} \right) \right] - \beta^2 F_r = -j\omega\mu\epsilon \frac{\partial \psi_m}{\partial r}} \quad (10-20a)$$

$$\boxed{\frac{1}{r} \frac{\partial^2 F_r}{\partial r \partial \theta} = -j \frac{\omega\mu\epsilon}{r} \frac{\partial \psi_m}{\partial \theta} \Rightarrow \frac{\partial^2 F_r}{\partial r \partial \theta} = \frac{\partial}{\partial \theta} \left(\frac{\partial F_r}{\partial r} \right) = \frac{\partial}{\partial \theta} (-j\omega\mu\epsilon \psi_m)} \quad (10-20b)$$

$$\boxed{\frac{1}{r \sin \theta} \frac{\partial^2 F_r}{\partial r \partial \phi} = -j \frac{\omega\mu\epsilon}{r \sin \theta} \frac{\partial \psi_m}{\partial \phi} \Rightarrow \frac{\partial^2 F_r}{\partial r \partial \phi} = \frac{\partial}{\partial \phi} \left(\frac{\partial F_r}{\partial r} \right) = \frac{\partial}{\partial \phi} (-j\omega\mu\epsilon \psi_m)} \quad (10-20c)$$

where $\beta^2 = \omega^2 \mu \epsilon$. The last two equations, (10-20b) and (10-20c), are satisfied simultaneously if

$$\boxed{\frac{\partial F_r}{\partial r} = -j\omega\mu\epsilon \psi_m \Rightarrow \psi_m = -\frac{1}{j\omega\mu\epsilon} \frac{\partial F_r}{\partial r}} \quad (10-21)$$

With the preceding relation for the scalar potential ψ_m , we need to find an *uncoupled* differential equation for F_r . To do this, we substitute (10-21) into (10-20a), which leads to

$$-\frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial F_r}{\partial \theta} \right) - \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2 F_r}{\partial \phi^2} - \beta^2 F_r = \frac{\partial^2 F_r}{\partial r^2} \quad (10-22)$$

or

$$\frac{\partial^2 F_r}{\partial r^2} + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial F_r}{\partial \theta} \right) + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2 F_r}{\partial \phi^2} + \beta^2 F_r = 0 \quad (10-22a)$$

which can also be written in succinct form as

$$\boxed{\left(\nabla^2 + \beta^2 \right) \frac{F_r}{r} = 0} \quad (10-22b)$$

Therefore, using this procedure, the ratio F_r/r satisfies the scalar Helmholtz wave equation but not F_r itself. A solution of F_r using (10-22b) allows us to find $\mathbf{E}_F$ and $\mathbf{H}_F$ using, respectively, (10-4) and (10-5a).

The solution of (10-22b) will be pursued in Section 10.2.6. In the meantime, the electric and magnetic field components can be written in terms of F_r by expanding (10-4) and (10-5a):

$$\underline{\text{TE}}'(\mathbf{F} = \hat{\mathbf{a}}_r F_r, \mathbf{A} = 0)$$

$$\boxed{\mathbf{E}_F = -\frac{1}{\varepsilon} \nabla \times \mathbf{F}} \quad (10-23)$$

or

$$E_r = 0 \quad (10-23a)$$

$$E_\theta = -\frac{1}{\varepsilon} \frac{1}{r \sin \theta} \frac{\partial F_r}{\partial \phi} \quad (10-23b)$$

$$E_\phi = \frac{1}{\varepsilon} \frac{1}{r} \frac{\partial F_r}{\partial \theta} \quad (10-23c)$$

$$\boxed{\mathbf{H}_F = \frac{1}{j\omega\mu\varepsilon} \nabla \times \nabla \times \mathbf{F}} \quad (10-24)$$

or

$$H_r = \frac{1}{j\omega\mu\varepsilon} \left(\frac{\partial^2}{\partial r^2} + \beta^2 \right) F_r \quad (10-24a)$$

$$H_\theta = \frac{1}{j\omega\mu\varepsilon} \frac{1}{r} \frac{\partial^2 F_r}{\partial r \partial \theta} \quad (10-24b)$$

$$H_\phi = \frac{1}{j\omega\mu\varepsilon} \frac{1}{r \sin \theta} \frac{\partial^2 F_r}{\partial r \partial \phi} \quad (10-24c)$$

where F_r/r is a solution to (10-22b).

10.2.5 Transverse Magnetic (TM) Modes: Source-Free Region

Following a procedure similar to the one outlined in the previous section for the TE^r modes, it can be shown that the TM^r fields in spherical coordinates can be constructed by letting the vector potentials $\mathbf{F}$ and $\mathbf{A}$ be equal to

$$\boxed{\mathbf{F} = 0} \quad (10-25a)$$

$$\boxed{\mathbf{A} = \hat{\mathbf{a}}_r A_r(r, \theta, \phi)} \quad (10-25b)$$

where the ratio A_r/r , and not A_r , is a solution to the scalar Helmholtz wave equation

$$\boxed{(\nabla^2 + \beta^2) \frac{A_r}{r} = 0} \quad (10-26)$$

The solution of (10-26) will be pursued in Section 10.2.6. In the meantime, the electric and magnetic field components can be written in terms of A_r , by expanding (10-15b) and (10-15a), as

$$\underline{TM^r(\mathbf{F} = 0, \mathbf{A} = \hat{\mathbf{a}}_r A_r)}$$

$$\boxed{\mathbf{E}_A = \frac{1}{j\omega\mu\epsilon} \nabla \times \nabla \times \mathbf{A}} \quad (10-27)$$

or

$$E_r = \frac{1}{j\omega\mu\epsilon} \left(\frac{\partial^2}{\partial r^2} + \beta^2 \right) A_r \quad (10-27a)$$

$$E_\theta = \frac{1}{j\omega\mu\epsilon} \frac{1}{r} \frac{\partial^2 A_r}{\partial r \partial \theta} \quad (10-27b)$$

$$E_\phi = \frac{1}{j\omega\mu\epsilon} \frac{1}{r \sin \theta} \frac{\partial^2 A_r}{\partial r \partial \phi} \quad (10-27c)$$

$$\boxed{\mathbf{H}_A = \frac{1}{\mu} \nabla \times \mathbf{A}} \quad (10-28)$$

or

$$H_r = 0 \quad (10-28a)$$

$$H_\theta = \frac{1}{\mu} \frac{1}{r \sin \theta} \frac{\partial A_r}{\partial \phi} \quad (10-28b)$$

$$H_\phi = -\frac{1}{\mu} \frac{1}{r} \frac{\partial A_r}{\partial \theta} \quad (10-28c)$$

where A_r/r is a solution to (10-26).

10.2.6 Solution of the Scalar Helmholtz Wave Equation

To find the TE^r and/or TM^r field of Sections 10.2.4 and 10.2.5 as given, respectively, by (10-23) through (10-24c) and (10-27) through (10-28c), solutions to the scalar Helmholtz wave equations 10-22b and 10-26 must be obtained for F_r/r and A_r/r (and thus, F_r and A_r). Both solutions are of the same form

$$(\nabla^2 + \beta^2)\psi = 0 \quad (10-29)$$

where

$$\psi = \begin{cases} \frac{F_r}{r} & \text{for } TE^r \text{ modes} \\ \frac{A_r}{r} & \text{for } TM^r \text{ modes} \end{cases} \quad (10-29a)$$

$$(10-29b)$$

Since the solution of ψ from (10-29) must be multiplied by r to obtain solutions for F_r or A_r , then appropriate solutions for F_r and A_r must be equal to the product of $r\psi$. The solution for F_r or A_r of (10-29) through (10-29b) must take the separable form of

$$\left. \begin{aligned} F_r(r, \theta, \phi) \\ A_r(r, \theta, \phi) \end{aligned} \right\} = f(r)g(\theta)h(\phi) \quad (10-30)$$

where $f(r)$, $g(\theta)$, and $h(\phi)$ must be represented by appropriate wave functions that satisfy the wave equation in spherical coordinates. According to (3-88a) or (3-88b), $g(\theta)$ can be represented by associated Legendre functions of the first kind $P_n^m(\cos \theta)$, or second kind $Q_n^m(\cos \theta)$, whereas $h(\phi)$ can be represented by either complex exponentials or cosinusoids as given, respectively, by (3-89a) and (3-89b).

Since the solution of ψ as given by (10-29) must be multiplied by r in order to obtain solutions to F_r and A_r as given by (10-30), it is most convenient to represent $f(r)$ *not* by spherical Bessel [$j_n(\beta r)$, $y_n(\beta r)$] or Hankel [$h_n^{(1)}(\beta r)$, $h_n^{(2)}(\beta r)$] functions, *but by another form* of spherical Bessel and Hankel functions denoted by $\hat{B}_n(\beta r)$ [for either $\hat{J}_n(\beta r)$, $\hat{Y}_n(\beta r)$, $\hat{H}_n^{(1)}(\beta r)$, or $\hat{H}_n^{(2)}(\beta r)$]. These are related to the regular spherical Bessel and Hankel functions denoted by $b_n(\beta r)$ [for either $j_n(\beta r)$, $y_n(\beta r)$, $h_n^{(1)}(\beta r)$, or $h_n^{(2)}(\beta r)$] by

$$\hat{B}_n(\beta r) = \beta r b_n(\beta r) = \beta r \sqrt{\frac{\pi}{2\beta r}} B_{n+1/2}(\beta r) = \sqrt{\frac{\pi\beta r}{2}} B_{n+1/2}(\beta r) \quad (10-31)$$

where $B_{n+1/2}(\beta r)$ is used to represent the regular cylindrical Bessel or Hankel functions of $J_{n+1/2}(\beta r)$, $Y_{n+1/2}(\beta r)$, $H_{n+1/2}^{(1)}(\beta r)$, and $H_{n+1/2}^{(2)}(\beta r)$. These new spherical Bessel and Hankel functions were introduced by Schelkunoff [2] and satisfy the differential equation

$$\left[\frac{d^2}{dr^2} + \beta^2 - \frac{n(n+1)}{r^2} \right] \hat{B}_n = 0 \quad (10-32)$$

which is obtained by substituting $b_n(\beta r) = \hat{B}_n(\beta r)/\beta r$ in

$$\frac{d}{dr} \left(r^2 \frac{db_n}{dr} \right) + [(\beta r)^2 - n(n+1)]b_n = 0 \quad (10-33)$$

Therefore, the solutions for $f(r)$ of (10-30) are of the new form of spherical Bessel or Hankel functions denoted by

$$f_1(r) = A_1 \hat{J}_n(\beta r) + B_1 \hat{Y}_n(\beta r) \quad (10-34a)$$

or

$$f_2(r) = C_1 \hat{H}_n^{(1)}(\beta r) + D_1 \hat{H}_n^{(2)}(\beta r) \quad (10-34b)$$

which are related to the regular Bessel and Hankel functions by

$$\hat{J}_n(\beta r) = \sqrt{\frac{\pi\beta r}{2}} J_{n+1/2}(\beta r) \quad (10-35a)$$

$$\hat{Y}_n(\beta r) = \sqrt{\frac{\pi\beta r}{2}} Y_{n+1/2}(\beta r) \quad (10-35b)$$

$$\hat{H}_n^{(1)}(\beta r) = \sqrt{\frac{\pi\beta r}{2}} H_{n+1/2}^{(1)}(\beta r) \quad (10-35c)$$

$$\hat{H}_n^{(2)}(\beta r) = \sqrt{\frac{\pi\beta r}{2}} H_{n+1/2}^{(2)}(\beta r) \quad (10-35d)$$

The total solution for F_r or A_r of (10-30) will be the product of the appropriate spherical wave functions representing $f(r)$, $g(\theta)$, and $h(\phi)$.

10.3 BICONICAL TRANSMISSION LINE

One form of a transmission line whose geometry conforms to the spherical orthogonal coordinate system is the biconical structure of Figure 10-1. Typically, this configuration is also representative of the biconical antenna [3–9] which exhibits very broad band frequency characteristics. Sets of fields that can be supported by such a structure can be either TE^r , TM^r , or TEM^r . Solutions to these will be examined here.

10.3.1 Transverse Electric (TE^r) Modes

According to the procedure established in Section 10.2.4, transverse (to the radial direction) electric modes (TE^r) can be constructed by choosing the potentials $\mathbf{F}$ and $\mathbf{A}$ according to (10-17a)

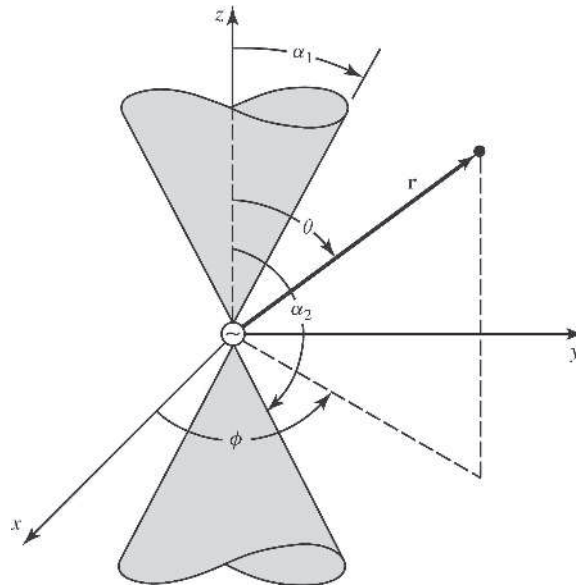


Figure 10-1 Geometry of biconical transmission line. (Source: C. A. Balanis, *Antenna Theory: Analysis and Design*. Third Edition. Copyright © 2005, John Wiley & Sons, Inc. Reprinted by permission of John Wiley & Sons, Inc.)

and (10-17b). The scalar component F_r of the vector potential must satisfy the vector wave equation (10-22b), whose solution takes the form of (3-85), or

$$F_r(r, \theta, \phi) = f(r)g(\theta)h(\phi) \quad (10-36)$$

where

$f(r)$ = a solution to (10-32) as given by either (10-34a) or (10-34b) [the form (10-34b) is chosen here]

$g(\theta)$ = a solution of (3-86b) as given by either (3-88a) or (3-88b) [the form (3-88a) is chosen here]

$h(\phi)$ = is a solution to (3-86c) as given by (3-89a) or (3-89b) [the form (3-89b) is chosen here to represent standing waves]

Therefore, F_r of (10-36) can be written, assuming the source is placed at the apex and is generating outwardly traveling waves [$C_1 = 0$ in (10-34b)], as

$$\begin{aligned} [F_r(r, \theta, \phi)]_{mm} &= D_1 \hat{H}_n^{(2)}(\beta r) [A_2 P_n^m(\cos \theta) + B_2 P_n^m(-\cos \theta)] \\ &\quad \times [C_3 \cos(m\phi) + D_3 \sin(m\phi)] \end{aligned} \quad (10-37)$$

where m = nonnegative integer ($m = 0, 1, 2, \dots$).

The corresponding electric and magnetic fields can be found using (10-23) through (10-24c) and the eigenvalues of n can be determined by applying the boundary conditions

$$E_\phi(0 \leq r \leq \infty, \theta = \alpha_1, 0 \leq \phi \leq 2\pi) = E_\phi(0 \leq r \leq \infty, \theta = \alpha_2, 0 \leq \phi \leq 2\pi) = 0 \quad (10-38)$$

According to (10-23c),

$$\begin{aligned} E_\phi &= \frac{1}{\varepsilon} \frac{1}{r} \frac{\partial F_r}{\partial \theta} = \frac{D_1}{\varepsilon} \frac{1}{r} \hat{H}_n^{(2)}(\beta r) \left[A_2 \frac{dP_n^m(\cos \theta)}{d\theta} + B_2 \frac{dP_n^m(-\cos \theta)}{d\theta} \right] \\ &\quad \times [C_3 \cos(m\phi) + D_3 \sin(m\phi)] \end{aligned} \quad (10-39)$$

Applying the first boundary condition of (10-38) leads to

$$\begin{aligned} E_\phi(0 \leq r \leq \infty, \theta = \alpha_1, 0 \leq \phi \leq 2\pi) &= \frac{D_1}{\varepsilon} \frac{1}{r} \hat{H}_n^{(2)}(\beta r) \left[A_2 \frac{dP_n^m(\cos \theta)}{d\theta} + B_2 \frac{dP_n^m(-\cos \theta)}{d\theta} \right]_{\theta=\alpha_1} \\ &\quad \times [C_3 \cos(m\phi) + D_3 \sin(m\phi)] = 0 \\ E_\phi(0 \leq r \leq \infty, \theta = \alpha_1, 0 \leq \phi \leq 2\pi) &= \frac{D_1}{\varepsilon} \frac{1}{r} \hat{H}_n^{(2)}(\beta r) \left[A_2 \frac{dP_n^m(\cos \alpha_1)}{d\alpha_1} + B_2 \frac{dP_n^m(-\cos \alpha_1)}{d\alpha_1} \right] \\ &\quad \times [C_3 \cos(m\phi) + D_3 \sin(m\phi)] = 0 \end{aligned} \quad (10-40a)$$

and the second boundary condition of (10-38) leads to

$$\begin{aligned} E_\phi(0 \leq r \leq \infty, \theta = \alpha_2, 0 \leq \phi \leq 2\pi) &= \frac{D_1}{\varepsilon} \frac{1}{r} \hat{H}_n^{(2)}(\beta r) \left[A_2 \frac{dP_n^m(\cos \alpha_2)}{d\alpha_2} + B_2 \frac{dP_n^m(-\cos \alpha_2)}{d\alpha_2} \right] \\ &\quad \times [C_3 \cos(m\phi) + D_3 \sin(m\phi)] = 0 \end{aligned} \quad (10-40b)$$

Equations 10-40a and 10-40b reduce to

$$A_2 \frac{dP_n^m(\cos \alpha_1)}{d\alpha_1} + B_2 \frac{dP_n^m(-\cos \alpha_1)}{d\alpha_1} = 0 \quad (10-41a)$$

$$A_2 \frac{dP_n^m(\cos \alpha_2)}{d\alpha_2} + B_2 \frac{dP_n^m(-\cos \alpha_2)}{d\alpha_2} = 0 \quad (10-41b)$$

which are satisfied provided the determinant of (10-41a) and (10-41b) vanishes, that is,

$$\boxed{\frac{dP_n^m(\cos \alpha_1)}{d\alpha_1} \frac{dP_n^m(-\cos \alpha_2)}{d\alpha_2} - \frac{dP_n^m(-\cos \alpha_1)}{d\alpha_1} \frac{dP_n^m(\cos \alpha_2)}{d\alpha_2} = 0} \quad (10-42)$$

Therefore, the eigenvalues of n are found as solutions to (10-42), which usually is not necessarily a very easy task.

10.3.2 Transverse Magnetic (TM') Modes

Following a procedure similar to that of the previous section and using the formulations of Section 10.2.5, it can be shown that for TM' modes the potential component A_r of (10-26) reduces to

$$\begin{aligned} [A_r(r, \theta, \phi)]_{mm} = & D_1 \hat{H}_n^{(2)}(\beta r) [A_2 P_n^m(\cos \theta) + B_2 P_n^m(-\cos \theta)] \\ & \times [C_3 \cos(m\phi) + D_3 \sin(m\phi)] \end{aligned} \quad (10-43)$$

where $m = \text{integer } (m = 0, 1, 2, \dots)$. The values of n are determined by applying the boundary conditions.

The corresponding electric and magnetic fields are obtained using (10-27) through (10-28c). By applying the boundary conditions

$$E_r(0 \leq r \leq \infty, \theta = \alpha_1, 0 \leq \phi \leq 2\pi) = E_r(0 \leq r \leq \infty, \theta = \alpha_2, 0 \leq \phi \leq 2\pi) = 0 \quad (10-44a)$$

or

$$E_\phi(0 \leq r \leq \infty, \theta = \alpha_1, 0 \leq \phi \leq 2\pi) = E_\phi(0 \leq r \leq \infty, \theta = \alpha_2, 0 \leq \phi \leq 2\pi) = 0 \quad (10-44b)$$

it can be shown that the eigenvalues of n are obtained as solutions to

$$\boxed{P_n^m(\cos \alpha_1) P_n^m(-\cos \alpha_2) - P_n^m(-\cos \alpha_1) P_n^m(\cos \alpha_2) = 0} \quad (10-45)$$

This usually is not necessarily a very easy task.

10.3.3 Transverse Electromagnetic (TEM') Modes

The lowest-order (dominant) mode of the biconical transmission line is the one for which $m = 0$ and $n = 0$. For this mode both (10-42) and (10-45) are satisfied and the potential components of (10-37) and (10-43) vanish. However, for $m = n = 0$, (10-43) could be redefined as the limit as $n \rightarrow 0$. Instead, it is usually more convenient to alternately represent the TEM mode by the TM₀₀ which is defined, using (3-88b) to represent $g(\theta)$, by [1]

$$(A_r)_{00} = B_{00} \hat{H}_0^{(2)}(\beta r) Q_0(\cos \theta) \quad (10-46)$$

since $P_0^0(\cos \theta) = P_0(\cos \theta) = 1$. The Legendre polynomial $Q_0(\cos \theta)$ can also be represented by

$$Q_0(\cos \theta) = \ln \left[\cot \left(\frac{\theta}{2} \right) \right] \quad (10-47a)$$

and the spherical Hankel function $\hat{H}_0^{(2)}(\beta r)$ can be replaced by its asymptotic form for large arguments of

$$\hat{H}_0^{(2)}(\beta r) \stackrel{\beta r \rightarrow \text{large}}{\simeq} j e^{-j\beta r} \quad (10-47b)$$

Using (10-47a) and (10-47b) reduces (10-46), for large observational distances ($\beta r \rightarrow \text{large}$), to

$$(A_r)_{00} \simeq j B_{00} \ln \left[\cot \left(\frac{\theta}{2} \right) \right] e^{-j\beta r} \quad (10-48)$$

The corresponding electric and magnetic field components are given, according to (10-27) through (10-28c), by [3]

$$E_r = \frac{1}{j\omega\mu\varepsilon} \left(\frac{\partial^2}{\partial r^2} + \beta^2 \right) A_r \simeq 0 \quad (10-49a)$$

$$E_\theta = \frac{1}{j\omega\mu\varepsilon} \frac{1}{r} \frac{\partial^2 A_r}{\partial r \partial \theta} = j B_{00} \frac{\beta}{\omega\mu\varepsilon} \frac{1}{r} \frac{1}{\sin \theta} e^{-j\beta r} \quad (10-49b)$$

$$E_\phi = \frac{1}{j\omega\mu\varepsilon} \frac{1}{r \sin \theta} \frac{\partial^2 A_r}{\partial r \partial \phi} = 0 \quad (10-49c)$$

$$H_r = 0 \quad (10-49d)$$

$$H_\theta = \frac{1}{\mu} \frac{1}{r \sin \theta} \frac{\partial A_r}{\partial \phi} = 0 \quad (10-49e)$$

$$H_\phi = -\frac{1}{\mu} \frac{1}{r} \frac{\partial A_r}{\partial \theta} = j B_{00} \frac{1}{\mu r \sin \theta} e^{-j\beta r} \quad (10-49f)$$

Using these equations, we can write the wave impedance in the radial direction as

$$Z_w^{+r} = \frac{E_\theta}{H_\phi} = \frac{\beta}{\omega\varepsilon} = \sqrt{\frac{\mu}{\varepsilon}} = \eta \quad (10-50)$$

which is the same as the intrinsic impedance of the medium.

An impedance of greater interest is the characteristic impedance that is defined in terms of voltages and currents. The voltage between two corresponding points on the cones, a distance r from the origin, is found by

$$\begin{aligned} V(r) &= \int_{\alpha_1}^{\alpha_2} \mathbf{E} \cdot d\ell = \int_{\alpha_1}^{\alpha_2} (\hat{\mathbf{a}}_\theta E_\theta) \cdot (\hat{\mathbf{a}}_\theta r d\theta) \\ &= \int_{\alpha_1}^{\alpha_2} E_\theta r d\theta = j B_{00} \frac{\beta e^{-j\beta r}}{\omega\mu\varepsilon} \int_{\alpha_1}^{\alpha_2} \frac{d\theta}{\sin(\theta)} \\ V(r) &= j B_{00} \frac{\beta e^{-j\beta r}}{\omega\mu\varepsilon} \ln \left[\frac{\cot \left(\frac{\alpha_1}{2} \right)}{\cot \left(\frac{\alpha_2}{2} \right)} \right] \end{aligned} \quad (10-51a)$$

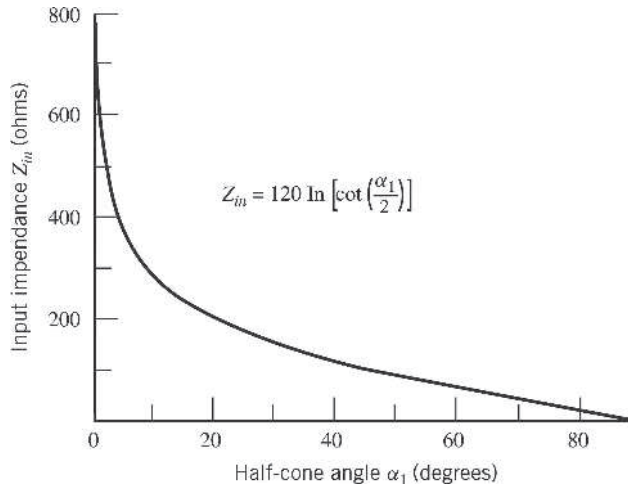


Figure 10-2 Input impedance of biconical transmission line. (Source: C. A. Balanis, *Antenna Theory: Analysis and Design*. Third Edition. Copyright © 2005, John Wiley & Sons, Inc. Reprinted by permission of John Wiley & Sons, Inc.)

The current on the surface of the cones, a distance r from the origin, is found by using (10-49f) as

$$I(r) = \oint_C \mathbf{H} \cdot d\ell = \int_0^{2\pi} (\hat{\mathbf{a}}_\phi H_\phi) \cdot (\hat{\mathbf{a}}_\phi r \sin \theta d\phi) = \int_0^{2\pi} H_\phi r \sin \theta d\phi = jB_{00} \frac{2\pi e^{-j\beta r}}{\mu} \quad (10-51b)$$

Taking the ratio of (10-51a) to (10-51b), we can define and write the characteristic impedance as

$$Z_c \equiv \frac{V(r)}{I(r)} = \frac{\beta}{2\pi\omega\epsilon} \ln \left[\frac{\cot \left(\frac{\alpha_1}{2} \right)}{\cot \left(\frac{\alpha_2}{2} \right)} \right] = \frac{\sqrt{\mu}}{2\pi} \ln \left[\frac{\cot \left(\frac{\alpha_1}{2} \right)}{\cot \left(\frac{\alpha_2}{2} \right)} \right] \equiv Z_{in} \quad (10-52)$$

Since the characteristic impedance is not a function of the radial distance r , it also represents the input impedance of the antenna at the feed terminals. For a symmetrical structure ($\alpha_2 = \pi - \alpha_1$), (10-52) reduces to

$$Z_c = \frac{\sqrt{\mu}}{2\pi} \ln \left[\cot \left(\frac{\alpha_1}{2} \right) \right]^2 = \frac{\sqrt{\mu}}{\pi} \ln \left[\cot \left(\frac{\alpha_1}{2} \right) \right] = \frac{\eta}{\pi} \ln \left[\cot \left(\frac{\alpha_1}{2} \right) \right] = Z_{in} \quad (10-52a)$$

It is apparent that the transmission line, or alternately the antenna of Figure 10-1, is a very broad band structure since its characteristic or input impedance is only a function of the included angle of the cone. A plot of (10-52a) as a function of α_1 is shown in Figure 10-2.

There are numerous other transmission lines whose geometry can be represented by the spherical orthogonal coordinate systems. They will not be discussed here but some will be assigned to the reader as end-of-chapter exercises.

10.4 THE SPHERICAL CAVITY

The metallic spherical cavity of Figure 10-3 represents a popular and classic geometry to design resonators. The field configurations that can be supported by such a structure can be TE^r and/or

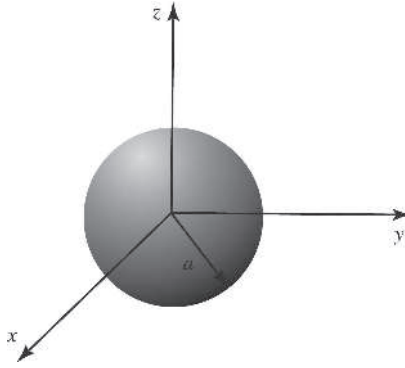


Figure 10-3 Geometry of spherical cavity.

TM' ; both will be examined here. In addition to the field expressions, the resonant frequencies and the quality factors will be the quantities of interest.

10.4.1 Transverse Electric (TE') Modes

The TE' modes in the cavity can be formed by letting the vector potentials $\mathbf{F}$ and $\mathbf{A}$ be equal to (10-17a) and (10-17b), respectively. The most appropriate form for the vector potential component F_r is

$$F_r(r, \theta, \phi) = [A_1 \hat{J}_n(\beta r) + B_1 \hat{Y}_n(\beta r)][C_2 P_n^m(\cos \theta) + D_2 Q_n^m(\cos \theta)] \times [C_3 \cos(m\phi) + D_3 \sin(m\phi)] \quad (10-53)$$

where m and n are integers. The fields must be finite at $r = 0$; thus, $B_1 = 0$ since $\hat{Y}_n(\beta r)$ possesses a singularity at $r = 0$. Additionally, the fields must also be finite at $\theta = 0, \pi$. Therefore, $D_2 = 0$ since $Q_n^m(\cos \theta)$ possesses a singularity at $\theta = 0, \pi$. Also, because the Legendre polynomial $P_0(w) = 1$, and $P_0^m(w)$ is related to $P_0(w)$ by [10-15]

$$P_0^m(w) = (-1)^m (1-w^2)^{m/2} \frac{d^m P_0(w)}{dw^m} = 0, \quad m = 1, 2, \dots \quad (10-54)$$

then, for nontrivial (nonzero) solutions, $n = 1, 2, 3, \dots$. Therefore, (10-53) reduces to

$$(F_r)_{mnp} = A_{mnp} \hat{J}_n(\beta r) P_n^m(\cos \theta) [C_3 \cos(m\phi) + D_3 \sin(m\phi)] \quad (10-55)$$

It should also be stated that $P_n^m(\cos \theta) = 0$ if $m > n$.

The corresponding electric and magnetic fields are found using (10-23) through (10-24c). Thus, we can write the electric field components of (10-23) through (10-23c), using (10-55), as

$$E_r = 0 \quad (10-56a)$$

$$E_\theta = -\frac{1}{\epsilon} \frac{1}{r \sin \theta} \frac{\partial F_r}{\partial \phi} = -A_{mnp} \frac{m}{\epsilon} \frac{1}{r \sin \theta} \hat{J}_n(\beta r) P_n^m(\cos \theta) [-C_3 \sin(m\phi) + D_3 \cos(m\phi)] \quad (10-56b)$$

$$E_\phi = \frac{1}{\epsilon} \frac{1}{r} \frac{\partial F_r}{\partial \theta} = A_{mnp} \frac{1}{\epsilon} \frac{1}{r} \hat{J}_n(\beta r) P_n^{m'}(\cos \theta) [C_3 \cos(m\phi) + D_3 \sin(m\phi)] \quad (10-56c)$$

where

$$\prime \equiv \frac{\partial}{\partial \theta} \quad (10-56d)$$

The boundary conditions that must be satisfied are

$$E_{\theta}(r = a, 0 \leq \theta \leq \pi, 0 \leq \phi \leq 2\pi) = 0 \quad (10-57a)$$

$$E_{\phi}(r = a, 0 \leq \theta \leq \pi, 0 \leq \phi \leq 2\pi) = 0 \quad (10-57b)$$

Either condition yields the same eigenfunction and corresponding eigenvalues.

Applying (10-57a) to (10-56b) leads to

$$\begin{aligned} E_{\theta}(r = a, 0 \leq \theta \leq \pi, 0 \leq \phi \leq 2\pi) &= -A_{mnp} \frac{m}{\varepsilon} \frac{1}{a \sin \theta} \hat{J}_n(\beta a) P_n^m(\cos \theta) \\ &\times [-C_3 \sin(m\phi) + D_3 \cos(m\phi)] = 0 \Rightarrow \hat{J}_n(\beta a)|_{\beta=\beta_r} = 0 \Rightarrow \beta_r a = \zeta_{np} \\ \beta_r &= \frac{\zeta_{np}}{a} \quad \begin{array}{l} n = 1, 2, 3, \dots \\ p = 1, 2, 3, \dots \end{array} \end{aligned} \quad (10-58)$$

where ζ_{np} represents the p zeroes of the spherical Bessel function $\hat{J}_n$ of order n . A listing of a limited, but for most applications sufficient, number of ζ_{np} 's is found in Table 10-1.

The resonant frequencies are found using (10-58) and can be written as

$$\beta_r = \omega_r \sqrt{\mu \varepsilon} = 2\pi f_r \sqrt{\mu \varepsilon} = \frac{\zeta_{np}}{a} \quad (10-59)$$

or

$$(f_r)_{mnp}^{\text{TE}^r} = \frac{\zeta_{np}}{2\pi a \sqrt{\mu \varepsilon}} \quad \begin{array}{l} m = 0, 1, 2, \dots \leq n \\ n = 1, 2, 3, \dots \\ p = 1, 2, 3, \dots \end{array} \quad (10-59a)$$

Since the resonant frequencies of (10-59a) obtained using the ζ_{np} 's of Table 10-1 are independent of the values of m , there are numerous degeneracies (same resonant frequencies) among the modes; for a given n and p , there are many m 's that have the same resonant frequency. To determine how many m modes exist for each set of n and p , remember that $P_n^m = 0$ if $m > n$. Therefore, for P_n^m to be nonzero, $m \leq n$. Thus, the order of degeneracy is equal to $m = n$.

TABLE 10-1 Zeroes ζ_{np} of spherical Bessel function $\hat{J}_n(\zeta_{np}) = 0$

	$n = 1$	$n = 2$	$n = 3$	$n = 4$	$n = 5$	$n = 6$	$n = 7$	$n = 8$
$p = 1$	4.493	5.763	6.988	8.183	9.356	10.513	11.657	12.791
$p = 2$	7.725	9.095	10.417	11.705	12.967	14.207	15.431	16.641
$p = 3$	10.904	12.323	13.698	15.040	16.355	17.648	18.923	20.182
$p = 4$	14.066	15.515	16.924	18.301	19.653	20.983	22.295	
$p = 5$	17.221	18.689	20.122	21.525	22.905			
$p = 6$	20.371	21.854						

According to the values of Table 10-1, the lowest ζ_{np} zeroes in ascending order, along with the number of degenerate and total modes, are

n, p	ζ_{np}	Degenerate modes	Total number of modes
$n = 1, p = 1$	$\zeta_{11} = 4.493$	$m = 0, 1$ (even, odd)	3
$n = 2, p = 1$	$\zeta_{21} = 5.763$	$m = 0, 1, 2$ (even, odd)	8
$n = 3, p = 1$	$\zeta_{31} = 6.988$	$m = 0, 1, 2, 3$ (even, odd)	15
$n = 1, p = 2$	$\zeta_{12} = 7.725$	$m = 0, 1$ (even, odd)	18
$n = 4, p = 1$	$\zeta_{41} = 8.183$	$m = 0, 1, 2, 3, 4$ (even, odd)	27

The even, odd is used to represent either the $\cos(m\phi)$ or $\sin(m\phi)$ variations of (10-55). For example, for $n = 1, p = 1$, (10-55) has a three-fold degeneracy and can be written to represent the following three modes:

$$(F_r)_{011} \text{ (even)} = A_{011} C_3 \hat{J}_1(\beta_r r) P_1^0(\cos\theta) = A_{011} C_3 \hat{J}_1\left(4.493 \frac{r}{a}\right) \cos\theta \quad (10-60a)$$

$$(F_r)_{111} \text{ (even)} = A_{111} C_3 \hat{J}_1(\beta_r r) P_1^1(\cos\theta) \cos\phi = -A_{111} C_3 \hat{J}_1\left(4.493 \frac{r}{a}\right) \sin\theta \cos\phi \quad (10-60b)$$

$$(F_r)_{111} \text{ (odd)} = A_{111} D_3 \hat{J}_1(\beta_r r) P_1^1(\cos\theta) \sin\phi = -A_{111} D_3 \hat{J}_1\left(4.493 \frac{r}{a}\right) \sin\theta \sin\phi \quad (10-60c)$$

since

$$P_1^0(\cos\theta) = P_1(\cos\theta) = \cos\theta \quad (10-60d)$$

$$P_1^1(\cos\theta) = -(1 - \cos^2\theta)^{1/2} = -\sin\theta \quad (10-60e)$$

The modes represented by (10-60b) and (10-60c) are the same except that they are rotated 90° , in the ϕ direction, from each other. The same is true between (10-60a) and (10-60b) or (10-60c) except that the rotation is in the θ and ϕ directions.

10.4.2 Transverse Magnetic (TM') Modes

Following a procedure and justification similar to that for the TE^r modes, it can be shown that the appropriate vector potential component A_r of (10-25b) takes the form

$$(A_r)_{mnp} = B_{mnp} \hat{J}_n(\beta r) P_n^m(\cos\theta) [C_3 \cos(m\phi) + D_3 \sin(m\phi)] \quad (10-61)$$

The corresponding electric and magnetic fields are found using (10-27) through (10-28c). The boundary conditions are the same as for the TE^r , as given by (10-57a) and (10-57b).

Expanding (10-27b) using (10-61) we can write that

$$E_\theta = \frac{1}{j\omega\mu\epsilon r} \frac{1}{r} \frac{\partial^2 A_r}{\partial r \partial \theta} = B_{mnp} \frac{\beta}{j\omega\mu\epsilon r} \hat{J}_n'(\beta r) P_n^{m'}(\cos\theta) [C_3 \cos(m\phi) + D_3 \sin(m\phi)] \quad (10-62)$$

Applying (10-57a) on (10-62) leads to

$$\begin{aligned} E_\theta(r=a, 0 \leq \theta \leq \pi, 0 \leq \phi \leq 2\pi) &= B_{mnp} \frac{\beta}{j\omega\mu\epsilon a} \hat{J}_n'(\beta a) P_n^{m'}(\cos\theta) \\ &\times [C_3 \cos(m\phi) + D_3 \sin(m\phi)] = 0 \Rightarrow \hat{J}_n'(\beta a)|_{\beta=\beta_r} = 0 \Rightarrow \beta_r a = \zeta'_{np} \\ \beta_r &= \frac{\zeta'_{np}}{a} \quad \begin{matrix} n = 1, 2, 3, \dots \\ p = 1, 2, 3, \dots \end{matrix} \end{aligned} \quad (10-63)$$

TABLE 10-2 Zeroes ζ'_{np} of derivative of spherical Bessel function $\hat{J}'_n(\zeta'_{np}) = 0$

	$n = 1$	$n = 2$	$n = 3$	$n = 4$	$n = 5$	$n = 6$	$n = 7$	$n = 8$
$p = 1$	2.744	3.870	4.973	6.062	7.140	8.211	9.275	10.335
$p = 2$	6.117	7.443	8.722	9.968	11.189	12.391	13.579	14.753
$p = 3$	9.317	10.713	12.064	13.380	14.670	15.939	17.190	18.425
$p = 4$	12.486	13.921	15.314	16.674	18.009	19.321	20.615	21.894
$p = 5$	15.644	17.103	18.524	19.915	21.281	22.626		
$p = 6$	18.796	20.272	21.714	23.128				
$p = 7$	21.946							

where ζ'_{np} represents the p zeroes of the derivative of the spherical Bessel function $\hat{J}'_n$ of order n . A listing of a limited, but for most applications sufficient, number of ζ'_{np} 's is found in Table 10-2.

The resonant frequencies are found using (10-63) and can be written as

$$(f_r)_{mnp}^{\text{TM}^r} = \frac{\zeta'_{np}}{2\pi a \sqrt{\mu\epsilon}} \quad \begin{array}{l} m = 0, 1, 2, \dots \leq n \\ n = 1, 2, 3, \dots \\ p = 1, 2, 3, \dots \end{array} \quad (10-64)$$

As with the TE^r modes, there are numerous degeneracies among the modes since the resonant frequencies determined by (10-64) are independent of m . For a given n , the order of degeneracy is $m = n$.

According to the values of Table 10-2, the lowest ζ'_{np} zeroes in ascending order, along with the number of degenerate and total modes, are

n, p	ζ'_{np}	Degenerate modes	Total number of modes
$n = 1, p = 1$	$\zeta'_{11} = 2.744$	$m = 0, 1$ (even, odd)	3
$n = 2, p = 1$	$\zeta'_{21} = 3.870$	$m = 0, 1, 2$ (even, odd)	8
$n = 3, p = 1$	$\zeta'_{31} = 4.973$	$m = 0, 1, 2, 3$ (even, odd)	15
$n = 4, p = 1$	$\zeta'_{41} = 6.062$	$m = 0, 1, 2, 3, 4$ (even, odd)	24
$n = 1, p = 2$	$\zeta'_{12} = 6.117$	$m = 0, 1$ (even, odd)	27

The lowest-order mode is the one found using $n = 1, p = 1$, and it has a three-fold degeneracy [$m = 0$ (even), $m = 1$ (even), and $m = 1$ (odd)]. For these, (10-61) reduces to

$$(A_r)_{011} \text{ (even)} = B_{011} C_3 \hat{J}_1(\beta_r r) P_1^0(\cos \theta) = -B_{011} C_3 \hat{J}_1 \left(2.744 \frac{r}{a} \right) \cos \theta \quad (10-65a)$$

$$(A_r)_{111} \text{ (even)} = B_{111} C_3 \hat{J}_1(\beta_r r) P_1^1(\cos \theta) \cos \phi = -B_{111} C_3 \hat{J}_1 \left(2.744 \frac{r}{a} \right) \sin \theta \cos \phi \quad (10-65b)$$

$$(A_r)_{111} \text{ (odd)} = B_{111} D_3 \hat{J}_1(\beta_r r) P_1^1(\cos \theta) \sin \phi = -B_{111} D_3 \hat{J}_1 \left(2.744 \frac{r}{a} \right) \sin \theta \sin \phi \quad (10-65c)$$

Example 10-1

For a spherical cavity of a 3-cm radius and filled with air, determine the resonant frequencies (in ascending order) of the first 11 modes (including degenerate modes).

Solution: According to (10-59a) and (10-64), using the values of ζ_{np} and ζ'_{np} from Tables 10-1 and 10-2, and taking into account the degeneracy of the modes in m as well as the even and odd forms in ϕ , we can write the resonant frequencies of the first 11 modes as

1, 2, 3:

$$\begin{aligned}(f_r)_{011}^{\text{TM}} (\text{even}) &= (f_r)_{111}^{\text{TM}} (\text{even}) = (f_r)_{111}^{\text{TM}} (\text{odd}) \\ &= \frac{2.744(30 \times 10^9)}{2\pi(3)} = 4.367 \times 10^9 \text{ Hz}\end{aligned}$$

4, 5, 6, 7, 8:

$$\begin{aligned}(f_r)_{021}^{\text{TM}} (\text{even}) &= (f_r)_{121}^{\text{TM}} (\text{even}) = (f_r)_{121}^{\text{TM}} (\text{odd}) = (f_r)_{221}^{\text{TM}} (\text{even}) \\ &= (f_r)_{221}^{\text{TM}} (\text{odd}) = \frac{3.870(30 \times 10^9)}{2\pi(3)} = 6.1593 \times 10^9 \text{ Hz}\end{aligned}$$

9, 10, 11:

$$\begin{aligned}(f_r)_{011}^{\text{TE}} (\text{even}) &= (f_r)_{111}^{\text{TE}} (\text{even}) = (f_r)_{111}^{\text{TE}} (\text{odd}) \\ &= \frac{4.493(30 \times 10^9)}{2\pi(3)} = 7.1508 \times 10^9 \text{ Hz}\end{aligned}$$

10.4.3 Quality Factor Q

As has already been pointed, the Q of the cavity is probably one of its most important parameters, and it is defined by (8-84). To derive the equation for the Q of any mode of a spherical cavity is a most difficult task. However, it is instructive to consider that of the lowest (dominant) mode, which here is any one of the three-fold degenerate modes TM_{011} (even), TM_{111} (even), or TM_{111} (odd). Let us consider the TM_{011} (even) mode.

For the TM_{011} (even) mode the potential function of (10-61) reduces to that of (10-65a), which can be written as

$$(A_r)_{011} = B'_{011} \hat{J}_1 \left(2.744 \frac{r}{a} \right) \cos \theta \quad (10-66)$$

Since the Q of the cavity is defined by (8-84), it is most convenient to find the stored energy and dissipated power by using the magnetic field, since it has only one nonzero component (the electric field has two).

The magnetic field components of the TM_{011} mode can be written using (10-28a) through (10-28c) and (10-66) as

$$H_r = 0 \quad (10-67a)$$

$$H_\theta = \frac{1}{\mu} \frac{1}{r \sin \theta} \frac{\partial A_r}{\partial \phi} = 0 \quad (10-67b)$$

$$H_\phi = -\frac{1}{\mu} \frac{1}{r} \frac{\partial A_r}{\partial \theta} = B'_{011} \frac{1}{\mu} \frac{1}{r} \hat{J}_1 \left(2.744 \frac{r}{a} \right) \sin \theta \quad (10-67c)$$

Therefore, at resonance, the total stored energy can be found using

$$W = 2W_e = 2W_m = 2 \left[\frac{\mu}{4} \iiint_V |\mathbf{H}|^2 dv \right] = \frac{\mu}{2} \int_0^{2\pi} \int_0^\pi \int_0^a |H_\phi|^2 r^2 \sin \theta dr d\theta d\phi \quad (10-68)$$

which, by substituting (10-67c), reduces to

$$\begin{aligned} W &= \frac{|B'_{011}|^2}{2\mu} \int_0^{2\pi} \int_0^\pi \int_0^a \hat{J}_1^2 \left(2.744 \frac{r}{a} \right) \sin^3 \theta \, dr \, d\theta \, d\phi \\ &= \frac{|B'_{011}|^2}{2\mu} (2\pi) \frac{4}{3} \int_0^a \hat{J}_1^2 \left(2.744 \frac{r}{a} \right) dr \end{aligned} \quad (10-68a)$$

The integral can be evaluated using the formula

$$\int_0^a \hat{J}_1^2 \left(2.744 \frac{r}{a} \right) dr = \frac{a}{2} \left[\hat{J}_1^2(2.744) - \hat{J}_0(2.744) \hat{J}_2(2.744) \right] \quad (10-69)$$

where according to (3-94) or (10-31)

$$\hat{J}_1(2.744) = 2.744 j_1(2.744) = 2.744(0.3878) = 1.0640 \quad (10-69a)$$

$$\hat{J}_0(2.744) = 2.744 j_0(2.744) = 2.744(0.1428) = 0.3919 \quad (10-69b)$$

$$\hat{J}_2(2.744) = 2.744 j_2(2.744) = 2.744(0.2820) = 0.7738 \quad (10-69c)$$

Thus, (10-69) reduces, using that $\beta_r = 2.744/a$, to

$$\begin{aligned} \int_0^a \hat{J}_1^2 \left(2.744 \frac{r}{a} \right) dr &= \frac{a}{2} [(1.0640)^2 + 0.3919(0.7738)] = \frac{a}{2} (0.8288) \\ &= \frac{a}{2.744} \frac{(2.744)(0.8288)}{2} = \frac{1.137}{\beta_r} \end{aligned}$$

and (10-68a) to

$$W = \frac{|B'_{011}|^2}{2\mu} (2\pi) \left(\frac{4}{3} \right) \frac{1.137}{\beta_r} \quad (10-70)$$

The power dissipated on the walls of the cavity can be found using

$$P_d = \frac{R_s}{2} \oint_S \mathbf{J}_s \cdot \mathbf{J}_s^* ds \quad (10-71)$$

where

$$\mathbf{J}_s = \hat{\mathbf{n}} \times \mathbf{H} |_{r=a} = -\hat{\mathbf{a}}_r \times \hat{\mathbf{a}}_\phi H_\phi |_{r=a} = \hat{\mathbf{a}}_\theta H_\phi (r=a) = \hat{\mathbf{a}}_\theta B'_{011} \frac{1}{\mu a} \hat{J}_1(2.744) \sin \theta \quad (10-71a)$$

Thus, (10-71) can be written as

$$\begin{aligned} P_d &= \frac{R_s}{2} \int_0^{2\pi} \int_0^\pi |H_\phi(r=a)|^2 a^2 \sin \theta \, d\theta \, d\phi \\ &= \frac{R_s}{2\mu^2} |B'_{011}|^2 \hat{J}_1^2(2.744) \int_0^{2\pi} \int_0^\pi \sin^3 \theta \, d\theta \, d\phi \\ &= |B'_{011}|^2 \frac{R_s}{2\mu^2} (2\pi) \left(\frac{4}{3} \right) \hat{J}_1^2(2.744) = |B'_{011}|^2 \frac{R_s}{2\mu^2} (2\pi) \left(\frac{4}{3} \right) (1.0640)^2 \\ P_d &= \frac{|B'_{011}|^2}{2\mu^2} 1.132 (2\pi) \left(\frac{4}{3} \right) R_s \end{aligned} \quad (10-72)$$

Using (10-70) and (10-72), the Q of the cavity for the TM_{011}^r mode reduces to

$$Q = \omega_r \frac{W}{P_d} = \omega_r \frac{\frac{|B'_{011}|^2}{2\mu} (2\pi) \left(\frac{4}{3}\right) \frac{1.137}{\beta_r}}{\frac{|B'_{011}|^2}{2\mu^2} (2\pi) \left(\frac{4}{3}\right) (1.132) R_s} = \frac{1.137 \omega_r \mu}{1.132 \beta_r R_s} = 1.004 \frac{\omega_r \mu}{\omega_r \sqrt{\mu \epsilon} R_s}$$

$$Q = 1.004 \sqrt{\frac{\mu}{\epsilon}} = 1.004 \left(\frac{\eta}{R_s} \right) \quad (10-73)$$

Example 10-2

Compare the Q values of a spherical cavity operating in the dominant TM_{011} (even) mode with those of a circular cylinder and cubical cavities. The dimensions of each are such that the cylindrical and spherical cavities are circumscribed by the cubical cavity.

Solution: According to (10-73), the Q of a spherical cavity of radius a operating in the dominant TM_{011} (even) mode is given by

$$Q = 1.004 \left(\frac{\eta}{R_s} \right)$$

while that of a circular cavity of diameter d and height h operating in the dominant TM_{010} mode (for $h = d$) is given by (9-57), which reduces to

$$Q = 1.2025 \frac{\eta}{R_s} \frac{1}{\left(1 + \frac{d/2}{h}\right)} = 0.8017 \left(\frac{\eta}{R_s} \right)$$

For a rectangular cavity operating in the dominant TE_{101} mode the Q is given by (8-88), which for a cubical geometry ($a = b = c$) reduces, according to (8-88a), to

$$Q = 1.1107 \frac{\eta}{R_s} \frac{1}{\left(1 + \frac{a/2}{b}\right)} = \frac{1.1107}{1.5} \left(\frac{\eta}{R_s} \right) = 0.7405 \left(\frac{\eta}{R_s} \right)$$

Comparing these three expressions, it is evident that the Q of the spherical cavity is greater than that of the circular cavity with $h = d$ by

$$\frac{1.004 - 0.8017}{0.8017} \times 100\% = 25.23\%$$

and greater than that of the cubical cavity by

$$\frac{1.004 - 0.7405}{0.7405} \times 100\% = 35.58\%$$

This is expected since the spherical cavity does not possess any sharp corners and edges, which are evident in the circular cavity and even more in the cubical cavity. Thus, the volume and surface area of the spherical cavity are better utilized by the interior fields. It should be remembered that the Q of a cavity is proportional to its volume and inversely proportional to its area.

10.5 MULTIMEDIA

On the website that accompanies this book, the following multimedia resources are included for the review, understanding, and presentation of the material of this chapter.

- **MATLAB** computer program **Sphere_Resonator**: Computes the resonant characteristics of a spherical resonator.
- **PowerPoint (PPT)** viewgraphs, in multicolor.

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PROBLEMS

- 10.1.** In Section 11.7.1 it is shown that the magnetic vector potential for an infinitesimal electric dipole of Figure 11-23a and Example 6-3 is given by (11-209)

$$A_z^{(1)} = -\hat{\mathbf{a}}_z j \frac{\mu \beta I_e \Delta \ell}{4\pi} h_0^{(2)}(\beta r)$$

where $h_0^{(2)}(\beta r)$ is the spherical Hankel function of order zero. Assume that two such dipoles

of equal amplitude, but 180° out of phase, are displaced along the x axis a distance s apart, as shown in Figure P10-1. The total magnetic potential can be written following a procedure outlined in [1] as

$$A_z' = A_z^{(1)} \left(x - \frac{s}{2}, y, z \right) - A_z^{(1)} \left(x + \frac{s}{2}, y, z \right) \stackrel{s \rightarrow 0}{\simeq} -s \frac{\partial A_z^{(1)}}{\partial x}$$

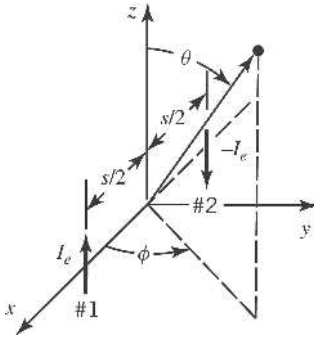


Figure P10-1

Show that the total magnetic potential A_z^t can be reduced to

$$A_z^t \stackrel{s \rightarrow 0}{\simeq} j \frac{\mu \beta^2 s I_e \Delta \ell}{4\pi} h_0^{(2)'}(\beta r) \sin \theta \cos \phi$$

$$\simeq +j \frac{\mu \beta^2 s I_e \Delta \ell}{4\pi} h_1^{(2)}(\beta r) P_1^1(\cos \theta) \cos \phi$$

where $' \equiv \partial/\partial(\beta r)$.

- 10.2.** Following the procedure of Problem 10.1, show that when the infinitesimal dipoles are displaced along the y axis, as shown in Figure P10-2, the total magnetic potential can be written as

$$A_z^t \stackrel{s \rightarrow 0}{\simeq} j \frac{\mu \beta^2 s I_e \Delta \ell}{4\pi} h_1^{(2)}(\beta r) P_1(\cos \theta) \sin \phi$$

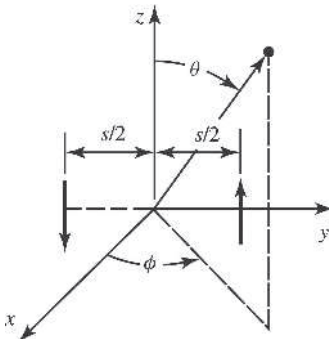


Figure P10-2

- 10.3.** Following the procedure of Problem 10.1, show that when the infinitesimal dipoles are displaced along the z axis, as shown in Figure P10-3, the total magnetic potential can be written as

$$A_z^t \stackrel{s \rightarrow 0}{\simeq} j \frac{\mu \beta^2 s I_e \Delta \ell}{4\pi} h_1^{(2)}(\beta r) P_1(\cos \theta)$$

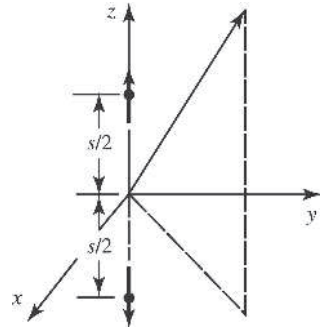


Figure P10-3

- 10.4.** Derive expressions for the electric and magnetic field components, in terms of F_z of (10-1b), that are TE^z . F_z should represent a solution to the scalar Helmholtz equation in spherical coordinates.
- 10.5.** Derive expressions for the electric and magnetic field components, in terms of A_z of (10-2a), that are TM^z . A_z should represent a solution to the scalar Helmholtz equation in spherical coordinates.
- 10.6.** For problems with sources of $\mathbf{J} \neq 0$ and $\mathbf{M} = 0$, show that the electric and magnetic fields and the vector potential $\mathbf{A}$ should satisfy (10-15a) through (10-15d).
- 10.7.** Show that using (10-25a) and (10-25b) for TM^r modes in spherical coordinates reduces (10-16f) to (10-26).
- 10.8.** By applying the boundary conditions of (10-44a) or (10-44b) on the electric field components of (10-27a) or (10-27c), where the vector potential A_r is given by (10-43), show that the eigenvalues of n are obtained as solutions to (10-45).
- 10.9.** Use Maxwell's equations

$$\nabla \times \mathbf{E} = -j\omega\mu\mathbf{H}$$

$$\nabla \times \mathbf{H} = j\omega\varepsilon\mathbf{E}$$

and assume TEM modes for the biconical antenna of Figure 10-1, these two equations reduce to only E_θ and H_ϕ components, each independent of ϕ . Then show that the H_ϕ component must satisfy the partial differential equation

$$\frac{\partial^2}{\partial r^2}(rH_\phi) = -\beta^2(rH_\phi)$$

whose solution must take the form

$$H_\phi = \frac{H_0}{\sin \theta} \frac{e^{-j\beta r}}{r}$$

whereas that of E_θ must then be written as

$$E_\theta = \eta H_\phi = \eta \frac{H_0}{\sin \theta} \frac{e^{-j\beta r}}{r}$$

- 10.10.** Show that by using the electric and magnetic field components of Problem 10.9, the power radiated by the biconical antenna of Figure 10-1 reduces to

$$P_{\text{rad}} = \iint \mathbf{S}_{\text{av}} \cdot d\mathbf{s} = 2\pi\eta |H_0|^2 \ln \left[\cot \left(\frac{\alpha}{2} \right) \right]$$

where $\mathbf{S}_{\text{av}}$ represents the average power density and $\alpha = \alpha_1 = \pi - \alpha_2$.

- 10.11.** By using the magnetic field from Problem 10.9, show that the current on the surface of the cone a distance r from the origin is equal to

$$I(r) = 2\pi H_0 e^{-j\beta r}$$

Evaluating the current at the origin $I(r=0)$, and using the definition for the radiation resistance in terms of the radiated power from Problem 10.10 and the current at the origin, show that the radiation resistance reduces to

$$R_r = \frac{2P_{\text{rad}}}{|I(r=0)|^2} = \frac{\eta}{\pi} \ln \left[\cot \left(\frac{\alpha}{2} \right) \right]$$

where $\alpha = \alpha_1 = \pi - \alpha_2$. This is the same as the characteristic impedance of (10-52a) for a symmetrical biconical transmission line.

- 10.12.** Calculate the included angle $\alpha = \alpha_1 = \pi - \alpha_2$ of a symmetrical biconical transmission line so that its characteristic impedance is:

- (a) 300 ohms.
(b) 50 ohms.

- 10.13.** For inside ($0 \leq \alpha \leq \pi/2$, $0 \leq \theta \leq \alpha$) or outside ($\pi/2 \leq \alpha \leq \pi$, $0 \leq \theta \leq \alpha$) cones shown, respectively, in Figures P10-13a and b for TE^r modes:

- (a) Show that the electric vector potential reduces to

$$(F_r)_{\text{mnp}} = A_2 P_n^m(\cos \theta) [C_1 \hat{H}_n^{(1)}(\beta r) + D_1 \hat{H}_n^{(2)}(\beta r)] \times [C_3 \cos(m\phi) + D_3 \sin(m\phi)], \quad m = 0, 1, 2, \dots$$

- (b) Show that the eigenvalues for n are obtained as solutions to

$$\left. \frac{dP_n^m(\cos \theta)}{d\theta} \right|_{\theta=\alpha} = 0$$

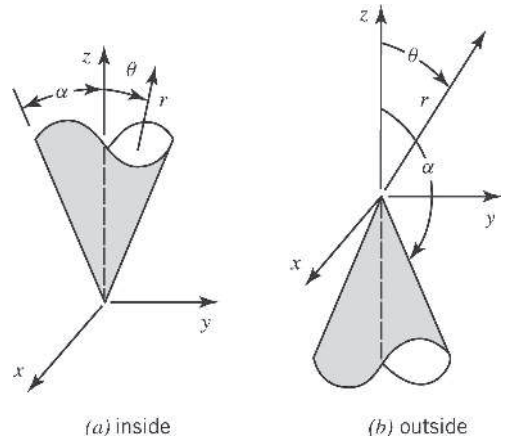


Figure P10-13

- 10.14.** Repeat Problem 10.13 for TM^r modes.

- (a) Show that the magnetic vector potential reduces to

$$(A_r)_{\text{mnp}} = A_2 P_n^m(\cos \theta) [C_1 \hat{H}_n^{(1)}(\beta r) + D_1 \hat{H}_n^{(2)}(\beta r)] \times [C_3 \cos(m\phi) + D_3 \sin(m\phi)], \quad m = 0, 1, 2, \dots$$

- (b) Show that the eigenvalues for n are obtained as solutions to

$$P_n^m(\cos \theta)|_{\theta=\alpha} = 0$$

- 10.15.** For the inside ($0 \leq \phi \leq \alpha$) or outside ($\alpha \leq \phi \leq 2\pi$) infinite-dimensions wedge of Figure P10-15, derive the reduced vector potential and allowable eigenvalues for TE^r modes.

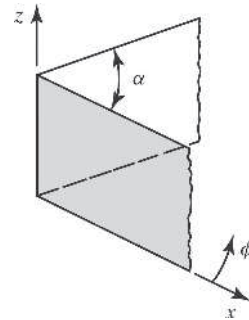


Figure P10-15

- 10.16.** Repeat Problem 10.15 for TM^r modes.

- 10.17.** A lossless dielectric sphere of radius a and dielectric constant ϵ_r is immersed into a

free-space medium with an established static electric field of

$$\mathbf{E}^i = \hat{\mathbf{a}}_z E_0$$

where E_0 is a constant. Using the spherical geometry and coordinates of Figures 10-3 and 11-28, show that the internal (*in*) and external (*ex*) electric field components to the sphere are given, respectively, by

$$E_x^{in} = E_y^{in} = 0; \quad E_z^{in} = E_0 \left(\frac{3}{\varepsilon_r + 2} \right)$$

$$E_x^{ex} = E_0 3 \left(\frac{\varepsilon_r - 1}{\varepsilon_r + 2} \right) \frac{a^3}{r^5} xz$$

$$E_y^{ex} = E_0 3 \left(\frac{\varepsilon_r - 1}{\varepsilon_r + 2} \right) \frac{a^3}{r^5} yz$$

$$E_z^{ex} = E_0 + E_0 \frac{a^3}{r^5} (2z^2 - x^2 - y^2)$$

- 10.18.** Repeat Problem 10.17 when a lossless magnetic sphere of radius a and relative permeability μ_r is immersed into a free-space medium with an established static magnetic field of

$$\mathbf{H}^i = \hat{\mathbf{a}}_z H_0$$

where H_0 is a constant. Derive the magnetic field components internal and external to the sphere in a form similar to those of the electric field of Problem 10.17.

- 10.19.** Design a spherical cavity (find its radius in cm) so that the resonant frequency of the dominant mode is 1 GHz and that of the next higher-order mode is approximately 1.41 GHz. The lossless medium within the sphere has electric constitutive parameters of $\varepsilon_r = 2.56$ and $\mu_r = 1$.
- 10.20.** Assume a spherical cavity with 2-cm radius and filled with air.
- Determine the resonant frequency of the dominant degenerate modes.
 - Find the bandwidth over which the dominant degenerate modes operate before the next higher-order degenerate modes.
 - Determine the dielectric constant that must be used to fill the sphere to reduce the resonant frequency of the dominant degenerate modes by a factor of 2.
- 10.21.** Design a spherical cavity totally filled with a lossless dielectric material so that its Q at 10 GHz, while operating in its dominant

mode, is 10,000. The surface of the cavity is made of copper with a conductivity of $\sigma = 5.7 \times 10^7$ S/m. Determine the dielectric constant of the medium that must be used to fill the cavity.

- 10.22.** A spherical cavity, because of its geometrical symmetry, is used to measure the dielectric properties of material samples, which match its geometry. To accomplish this, it is desired that the cavity is operating in the dominant mode and has a very high quality factor. The cavity is constructed of copper, ($\sigma = 5.76 \times 10^7$ S/m).
- Assuming that initially the cavity is filled with air and the desired quality factor is 10,000, determine the:
 - Resonant frequency (in GHz).
 - Radius (in cm) of the cavity.
 - While maintaining the same dimensions as in part (a), it is desired to completely fill the cavity with a lossless dielectric material in order to reduce the quality factor of the cavity of part (a) by a factor of 3. Determine the:
 - Dielectric constant of the material that must be used to accomplish this.
 - New resonant frequency (in GHz).
- 10.23.** Derive an expression for the Q of a spherical cavity when the fields within it are those of the dominant TM_{111} (even) mode. Compare the expression with (10-73).
- 10.24.** Derive an expression for the Q of a spherical cavity when the fields within it are those of the dominant TM_{111} (odd) mode. Compare the expression with (10-73).
- 10.25.** Determine the Q of the dominant mode of a spherical cavity of 2-cm radius when the medium within the cavity is:
- Air.
 - Polystyrene with a dielectric constant of 2.56.
- The cavity is made of copper whose conductivity is 5.76×10^7 S/m. Verify with the MATLAB program **Sphere_Resonator**.
- 10.26.** It is desired to design a spherical cavity whose Q at the resonant frequency of the dominant mode is 10,000. Assume that the cavity is filled with air and it is made of copper ($\sigma = 5.76 \times 10^7$ S/m). Then determine the resonant frequency of the dominant mode and the radius (in cm) of the cavity. Also find the dielectric constant of the medium that

must be used to fill the cavity to reduce its Q by a factor of 3. Verify with the MATLAB program **Sphere_Resonator**.

10.27. A spherical cavity of radius a , as shown in Figure 10-3, is filled with air. Its surface is made of a very thin layer of perfect magnetic conductor (PMC). Determine the:

- Dominant TE' modes. Identify them properly by indicating the appropriate indices (mnp). Must make a statement to justify the answer(s).
- Resonant frequency of the dominant TE' mode of the cavity.

10.28. Repeat Problem 10.27 for dominant TM' modes.

10.29. A dielectric spherical cavity, of a geometry shown in Figure 10-3, is made of a dielectric material with $\epsilon_r \gg 1$.

- Identify the approximate dominant TE'_{mnp} .
- Compute the resonant frequency for $a = 3$ cm and $\epsilon_r = 81$.

10.30. Repeat Problem 10.29 for TM'_{mnp} .

10.31. Assume a hemispherical cavity of Figure P10-31 with radius a .

- Determine the dominant mode and the expression for its resonant frequency.
- Show that the Q of the dominant mode is

$$Q = 0.574 \left(\frac{\eta}{R_s} \right)$$

- Compare the Q of part (b) with that of the spherical cavity, and those of cylindrical and square-based rectangular cavities with the same height-to-diameter ratios.

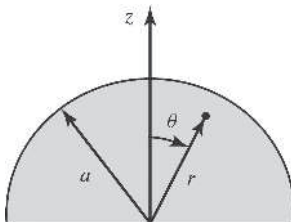


Figure P10-31

10.32. For a hemispherical PEC, air-filled cavity of radius a , as shown in Figure P10-32, determine the:

- Dominant TM'_{mnp} mode(s); identify them properly.
- Reduced/simplified vector potential. Justify it.
- Lowest resonant frequency (in GHz) when $a = 3$ cm.

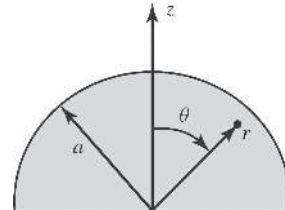


Figure P10-32

10.33. Repeat Problem 10.32 for TE'_{mnp} modes.

10.34. A hemispherical dielectric resonator, used in microwave integrated circuit (MIC) design, is comprised of a PEC ground plane covered with a dielectric half sphere of radius a , as shown in Figure P10-34. Assuming the dielectric constant of the material is much greater than unity ($\epsilon_r \gg 1$):

- Identify all the approximate TE'_{mnp} mode(s) with the lowest resonant frequency.
- Compute the resonant frequencies (in GHz) when $a = 3$ cm and $\epsilon_r = 81$.

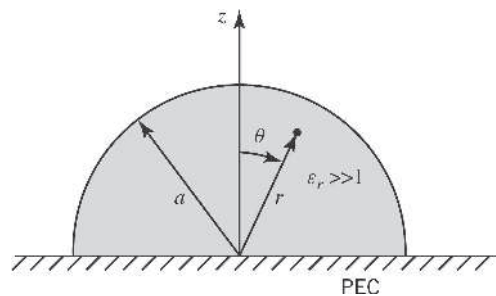


Figure P10-34

10.35. Repeat Problem 10.34 for TM'_{mnp} .

10.36. A hemispherical (half of a sphere) dielectric resonator (only in upper half sphere) is totally composed of a lossless dielectric medium with a dielectric constant of $\epsilon_r \gg 1$, $\mu_r = 1$. For TM'_{mnp} modes (even and/or odd), determine the:

- Expression for the resonant frequencies of the allowable TM'_{mnp} (even and/or odd)

modes. Be specific about all the allowable eigenvalues m, n, p for the even and/or odd modes. Be specific and justify.

- (b) A reduced/simplified expression for the appropriate vector potential. Be specific and justify it.
- (c) Resonant frequency (in GHz) of the dominant TM_{mnp}^r modes (even and/or odd, including degenerates; specify values of m, n , and p) when $a = 1.5$ cm, $\epsilon_r = 81$. Be specific to identify whether even and/or odd mode(s).

10.37. Repeat all parts of Problem 10.36 for TE_{mnp}^r modes.

10.38. A hemispherical PMC cavity, of radius a shown in Figure P10-38, is filled with air. Determine the:

- (a) Dominant TE_{mnp}^r mode(s); identify them properly by indicating the appropriate indices (m, n, p).

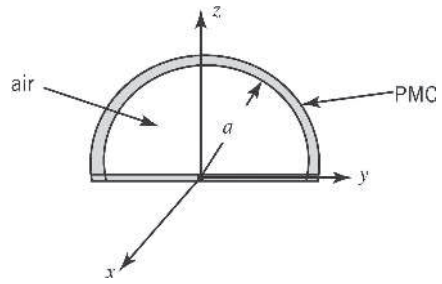


Figure P10-38

- (b) Reduced/simplified vector potential expression for the dominant mode(s). Justify it.
- (c) Lowest resonant frequency (in GHz) when $a = 3$ cm.

10.39. Repeat Problem 10.38 for TM_{mnp}^r mode(s).

CHAPTER 11



Scattering

11.1 INTRODUCTION

Previously, we have considered wave propagation in unbounded media, semi-infinite media forming planar interfaces, and conducting, dielectric, and surface waveguides. Although wave propagation in unbounded media is somewhat idealistic, it serves as a basic model for examining wave behavior while minimizing mathematical complexities. In general, however, wave propagation must be analyzed when it accounts for the presence of other structures (scatterers), especially when they are in proximity to the wave source and/or receiver.

In this chapter we want to examine wave propagation in the presence of scatterers of various geometries (planar, cylindrical, spherical). This is accomplished by introducing to the total field an additional component, referred to here as the *scattered field*, due to the presence of scatterers. The scattered field ($\mathbf{E}^s, \mathbf{H}^s$) must be such that when it is added, through superposition, to the *incident (direct) field* ($\mathbf{E}^i, \mathbf{H}^i$), the sum represents the total ($\mathbf{E}^t, \mathbf{H}^t$) field, that is,

$$\mathbf{E}^t = \mathbf{E}^i + \mathbf{E}^s \quad (11-1a)$$

$$\mathbf{H}^t = \mathbf{H}^i + \mathbf{H}^s \quad (11-1b)$$

The incident (direct) field $\mathbf{E}^i, \mathbf{H}^i$ will represent the total field produced by the sources *in the absence of any scatterers*.

The direct, scattered, and total fields will be obtained using various techniques. In general, geometrical optics (GO), physical optics (PO), modal techniques (MT), integral equations (IE), and diffraction theory [such as the geometrical theory of diffraction (GTD) and physical theory of diffraction (PTD)] can be used to analyze such problems. Typically, some of the problems are more conveniently analyzed using particular method(s). The fundamentals of physical optics were introduced in Chapter 7 and modal techniques were utilized in Chapters 8, 9, and 10 to analyze waveguide wave propagation. Integral equations are very popular, and they are introduced in Chapter 12. Geometrical optics and diffraction techniques are introduced and applied in Chapter 13.

In this chapter we want to examine scattering primarily by conducting objects. Each scattering problem will be analyzed using image theory, physical optics, or modal techniques. The conveniences and limitations of the applied method to each problem, as well as those of the other methods, will be stated.

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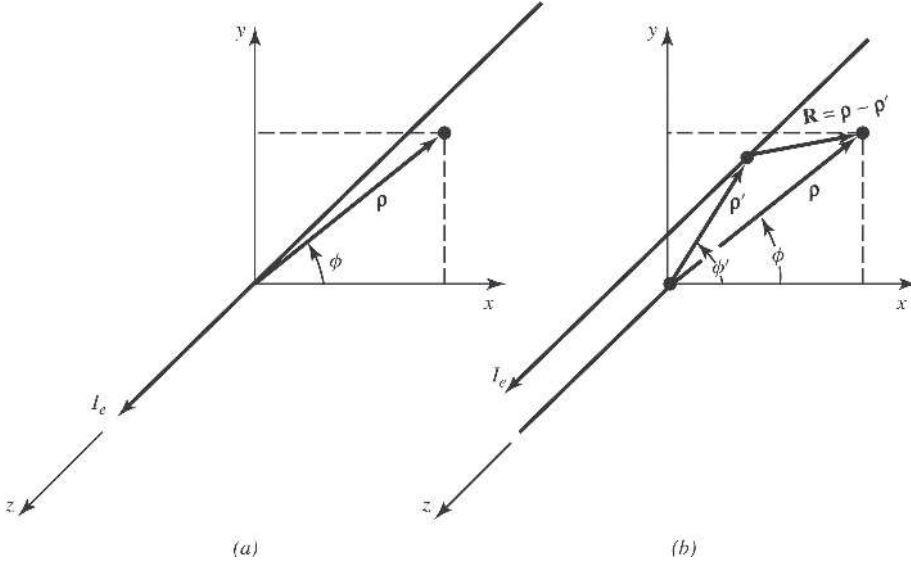


Figure 11-1 Geometry and coordinate system for an infinite electric line source. (a) At origin. (b) Offset.

11.2 INFINITE LINE-SOURCE CYLINDRICAL WAVE RADIATION

Before we examine the radiation and scattering of sources placed in the presence of scatterers, it is instructive to obtain the fields radiated by an infinite line source (both electric and magnetic) in an unbounded medium. The reason for doing this is that the infinite line source will serve as one type of source for which we will examine radiation properties in the presence of scatterers; its radiation in an unbounded medium will significantly aid in the solution of such problems.

11.2.1 Electric Line Source

The geometry of the line source is that of Figures 11-1a and 11-1b where it is assumed that its length extends to infinity and the electric current is represented by

$$\mathbf{I}_e(z') = \hat{\mathbf{a}}_z I_e \quad (11-2)$$

where I_e is a constant. Since the current is directed along the z axis, the fields radiated by the line source are TM^z and can be obtained by letting

$$\begin{aligned} \mathbf{F} &= 0 \\ \mathbf{A} &= \hat{\mathbf{a}}_z A_z(\rho, \phi, z) \\ &= \hat{\mathbf{a}}_z \left[C_1 H_m^{(1)}(\beta_\rho \rho) + D_1 H_m^{(2)}(\beta_\rho \rho) \right] \\ &\quad \times [C_2 \cos(m\phi) + D_2 \sin(m\phi)] (A_3 e^{-j\beta_z z} + B_3 e^{+j\beta_z z}) \end{aligned} \quad (11-3)$$

Since the line source is infinite in extent, the fields are two-dimensional (no z variations) so that

$$\beta_z = 0 \Rightarrow \beta_\rho^2 + \beta_z^2 = \beta^2 \Rightarrow \beta_\rho = \beta \quad (11-4)$$

In addition, since the waves radiate only in the outward direction and we choose the lowest-order mode, then

$$C_1 = 0 \quad (11-5a)$$

$$m = 0 \quad (11-5b)$$

Thus, (11-3) reduces to

$$\mathbf{A} = \hat{\mathbf{a}}_z A_z(\rho) = \hat{\mathbf{a}}_z A_0 H_0^{(2)}(\beta\rho) \quad (11-6)$$

whose corresponding electric and magnetic fields can be written using (6-70) as

$$E_\rho = -j \frac{1}{\omega\mu\varepsilon} \frac{\partial^2 A_z}{\partial\rho\partial z} = 0 \quad (11-6a)$$

$$E_\phi = -j \frac{1}{\omega\mu\varepsilon} \frac{1}{\rho} \frac{\partial^2 A_z}{\partial\phi\partial z} = 0 \quad (11-6b)$$

$$E_z = -j \frac{1}{\omega\mu\varepsilon} \left(\frac{\partial^2}{\partial z^2} + \beta^2 \right) A_z = -j\omega A_0 H_0^{(2)}(\beta\rho) \quad (11-6c)$$

$$H_\rho = \frac{1}{\mu} \frac{1}{\rho} \frac{\partial A_z}{\partial\phi} = 0 \quad (11-6d)$$

$$H_\phi = -\frac{1}{\mu} \frac{\partial A_z}{\partial\rho} = -\frac{A_0}{\mu} H_0^{(2)'}(\beta\rho) = A_0 \frac{\beta}{\mu} H_1^{(2)}(\beta\rho) \quad (11-6e)$$

$$H_z = 0 \quad (11-6f)$$

where $' \equiv \partial/\partial\rho$.

The constant A_0 can be obtained by using

$$I_e = \lim_{\rho \rightarrow 0} \oint_C \mathbf{H} \cdot d\mathbf{l} = \lim_{\rho \rightarrow 0} \int_0^{2\pi} (\hat{\mathbf{a}}_\phi H_\phi) \cdot (\hat{\mathbf{a}}_\phi \rho d\phi) = \lim_{\rho \rightarrow 0} \int_0^{2\pi} H_\phi \rho d\phi \quad (11-7)$$

Since the integration of (11-7) must be performed in the limit as $\rho \rightarrow 0$, it is convenient to represent the Hankel function of (11-6e) by its asymptotic expansion for small arguments. Using (IV-12), we can write that

$$H_1^{(2)}(\beta\rho) = J_1(\beta\rho) - jY_1(\beta\rho) \stackrel{\beta\rho \rightarrow 0}{\simeq} \frac{\beta\rho}{2} + j \frac{2}{\pi} \left(\frac{1}{\beta\rho} \right) \stackrel{\beta\rho \rightarrow 0}{\simeq} j \frac{2}{\pi} \left(\frac{1}{\beta\rho} \right) \quad (11-8)$$

Therefore, (11-7) reduces, using (11-6e) and (11-8), to

$$I_e = \lim_{\rho \rightarrow 0} \int_0^{2\pi} \left[A_0 \frac{\beta}{\mu} H_1^{(2)}(\beta\rho) \right] \rho d\phi \simeq jA_0 \frac{2}{\pi\mu} \int_0^{2\pi} \frac{1}{\rho} \rho d\phi = jA_0 \frac{4}{\mu} \quad (11-9)$$

or

$$A_0 = -j \frac{\mu}{4} I_e \quad (11-9a)$$

Thus, the nonzero electric and magnetic fields of the electric line source reduce to

$$E_z = -I_e \frac{\omega\mu}{4} H_0^{(2)}(\beta\rho) = -I_e \frac{\beta^2}{4\omega\varepsilon} H_0^{(2)}(\beta\rho) \quad (11-10a)$$

$$H_\phi = -jI_e \frac{\beta}{4} H_1^{(2)}(\beta\rho) \quad (11-10b)$$

Each of the field components is proportional to a Hankel function of the second kind whose argument is proportional to the distance from the source to the observation point. If the source is removed from the origin and it is placed as shown in Figure 11-1b, (11-10a) and (11-10b) can be written as

$$E_z = -I_e \frac{\beta^2}{4\omega\varepsilon} H_0^{(2)}(\beta R) = -I_e \frac{\beta^2}{4\omega\varepsilon} H_0^{(2)}(\beta|\boldsymbol{\rho} - \boldsymbol{\rho}'|) \quad (11-11a)$$

$$H_\psi = -jI_e \frac{\beta}{4} H_1^{(2)}(\beta R) = -jI_e \frac{\beta}{4} H_1^{(2)}(\beta|\boldsymbol{\rho} - \boldsymbol{\rho}'|) \quad (11-11b)$$

where

$$R = |\mathbf{\rho} - \mathbf{\rho}'| = \sqrt{\rho^2 + (\rho')^2 - 2\rho\rho' \cos(\phi - \phi')} \quad (11-11c)$$

ψ = circumferential angle around the source

For observations at far distances such that $\beta\rho \rightarrow \text{large}$, the Hankel functions in (11-10a) and (11-10b) can be approximated by their asymptotic expansions for large argument,

$$H_0^{(2)}(\beta\rho) \stackrel{\beta\rho \rightarrow \text{large}}{\simeq} \sqrt{\frac{2j}{\pi\beta\rho}} e^{-j\beta\rho} \quad (11-12a)$$

$$H_1^{(2)}(\beta\rho) \stackrel{\beta\rho \rightarrow \text{large}}{\simeq} j\sqrt{\frac{2j}{\pi\beta\rho}} e^{-j\beta\rho} \quad (11-12b)$$

Thus, (11-10a) and (11-10b) can be simplified for large arguments to

$$E_z = -I_e \frac{\beta^2}{4\omega\epsilon} H_0^{(2)}(\beta\rho) \stackrel{\beta\rho \rightarrow \text{large}}{\simeq} -\eta I_e \sqrt{\frac{j\beta}{8\pi}} \frac{e^{-j\beta\rho}}{\sqrt{\rho}} \quad (11-13a)$$

$$H_\phi = -jI_e \frac{\beta}{4} H_1^{(2)}(\beta\rho) \stackrel{\beta\rho \rightarrow \text{large}}{\simeq} I_e \sqrt{\frac{j\beta}{8\pi}} \frac{e^{-j\beta\rho}}{\sqrt{\rho}} \quad (11-13b)$$

The ratio (11-13a) to (11-13b) is defined as the wave impedance, which reduces to

$$Z_w^{+\rho} = \frac{E_z}{-H_\phi} = \eta \quad (11-14)$$

Since the wave impedance is equal to the intrinsic impedance, the waves radiated by the line source are TEM^{*o*}.

Example 11-1

For a displaced electric line source (at ρ', ϕ'), as shown in Figure 11-1b, of constant current I_e , derive (in terms of the cylindrical coordinates ρ, ρ', ϕ, ϕ'), the:

- Vector potential A_z .
- Electric field components (E_ρ, E_ϕ, E_z).
- Magnetic field components (H_ρ, H_ϕ, H_z).

This is an alternate solution to the expressions of (11-11a) through (11-11c).

Solution: Equations (11-6) through (11-10b) are correct when the electric field line source is located at the origin of the coordinate system, as shown in Figure 11-1a. However, when the electric line source is offset at point (ρ', ϕ', z), as shown in Figure 11-1b, the potential of (11-6) can be written as

$$\mathbf{A} = \hat{\mathbf{a}}_z A_z(\beta|\mathbf{\rho} - \mathbf{\rho}'|) = \hat{\mathbf{a}}_z A_0 H_0^{(2)}(\beta|\mathbf{\rho} - \mathbf{\rho}'|) = \hat{\mathbf{a}}_z A_0 H_0^{(2)}\left(\beta\sqrt{\rho^2 + (\rho')^2 - 2\rho\rho' \cos(\phi - \phi')}\right)$$

$$\text{where } A_0 = -j \frac{\mu}{4} I_e$$

The electric field of (6-17) or (6-70) can then be written as

$$E_z = -j \frac{1}{\omega\mu\epsilon} \beta^2 A_z = -j\omega A_z = -j\omega A_0 H_0^{(2)}\left(\beta\sqrt{\rho^2 + (\rho')^2 - 2\rho\rho' \cos(\phi - \phi')}\right)$$

$$E_z = -I_e \frac{\omega\mu}{4} H_0^{(2)}(\beta|\mathbf{p} - \mathbf{p}'|)$$

$$\text{where } |\mathbf{p} - \mathbf{p}'| = R = \sqrt{\rho^2 + (\rho')^2 - 2\rho\rho'\cos(\phi - \phi')}$$

The magnetic field can now be written based on (6-4a) and (6-70) as

$$\mathbf{H} = \frac{1}{\mu} \nabla \times \mathbf{A} = \hat{\mathbf{a}}_\rho \frac{1}{\mu} \frac{1}{\rho} \frac{\partial A_z}{\partial \phi} + \hat{\mathbf{a}}_\phi \left(-\frac{1}{\mu} \frac{\partial A_z}{\partial \rho} \right)$$

$$H_\rho = \frac{1}{\mu} \frac{1}{\rho} \frac{\partial A_z}{\partial \phi}$$

$$H_\phi = -\frac{1}{\mu} \frac{\partial A_z}{\partial \rho}$$

The individual magnetic field components of H_ρ and H_ϕ can be written using

$$\frac{d}{dx} H_0^{(2)}(\alpha x) = -\alpha H_1^{(2)}(\alpha x)$$

and the following derivatives

$$\frac{\partial A_z}{\partial \phi} = \frac{\partial R}{\partial \phi} \frac{\partial A_z}{\partial R}$$

$$\frac{\partial R}{\partial \phi} = \frac{\partial}{\partial \phi} \left[\sqrt{\rho^2 + (\rho')^2 - 2\rho\rho'\cos(\phi - \phi')} \right] = \frac{\rho\rho'\sin(\phi - \phi')}{\sqrt{\rho^2 + (\rho')^2 - 2\rho\rho'\cos(\phi - \phi')}}$$

$$\frac{\partial A_z}{\partial R} = \frac{\partial}{\partial R} \left[A_0 H_0^{(2)} \left(\beta \sqrt{\rho^2 + (\rho')^2 - 2\rho\rho'\cos(\phi - \phi')} \right) \right]$$

$$\frac{\partial A_z}{\partial R} = -\beta A_0 H_1^{(2)} \left(\beta \sqrt{\rho^2 + (\rho')^2 - 2\rho\rho'\cos(\phi - \phi')} \right)$$

$$\frac{\partial A_z}{\partial \rho} = \frac{\partial R}{\partial \rho} \frac{\partial A_z}{\partial R}$$

$$\frac{\partial R}{\partial \rho} = \frac{\partial}{\partial \rho} \left[\sqrt{\rho^2 + (\rho')^2 - 2\rho\rho'\cos(\phi - \phi')} \right] = \frac{\rho - \rho'\cos(\phi - \phi')}{\sqrt{\rho^2 + (\rho')^2 - 2\rho\rho'\cos(\phi - \phi')}}$$

$$\frac{\partial A_z}{\partial R} = \frac{\partial}{\partial R} \left[A_0 H_0^{(2)} \left(\beta \sqrt{\rho^2 + (\rho')^2 - 2\rho\rho'\cos(\phi - \phi')} \right) \right]$$

$$\frac{\partial A_z}{\partial R} = -\beta A_0 H_1^{(2)} \left(\beta \sqrt{\rho^2 + (\rho')^2 - 2\rho\rho'\cos(\phi - \phi')} \right)$$

as

$$H_\rho = \frac{1}{\mu} \frac{1}{\rho} \frac{\partial A_z}{\partial \phi} = -A_0 \frac{\beta\rho'\sin(\phi - \phi')}{\mu} \frac{H_1^{(2)} \left(\beta \sqrt{\rho^2 + (\rho')^2 - 2\rho\rho'\cos(\phi - \phi')} \right)}{\sqrt{\rho^2 + (\rho')^2 - 2\rho\rho'\cos(\phi - \phi')}}$$

$$H_\rho = jI_e \frac{\beta \rho' \sin(\phi - \phi')}{4} \frac{H_1^{(2)}(\beta |\rho - \rho'|)}{|\rho - \rho'|}$$

$$H_\phi = -\frac{1}{\mu} \frac{\partial A_z}{\partial \rho} = -A_0 \frac{\beta [\rho - \rho' \cos(\phi - \phi')]}{\mu} \frac{H_1^{(2)}(\beta \sqrt{\rho^2 + (\rho')^2 - 2\rho\rho' \cos(\phi - \phi')})}{\sqrt{\rho^2 + (\rho')^2 - 2\rho\rho' \cos(\phi - \phi')}}}$$

$$H_\phi = -\frac{1}{\mu} \frac{\partial A_z}{\partial \rho} = jI_e \frac{\beta [\rho - \rho' \cos(\phi - \phi')]}{4} \frac{H_1^{(2)}(\beta |\rho - \rho'|)}{|\rho - \rho'|}$$

11.2.2 Magnetic Line Source

Although magnetic sources as presently known are not physically realizable, they are often used to represent virtual sources in equivalent models. This was demonstrated in Chapter 7, Sections 7.7 and 7.8, where the volume and surface fields equivalence theorems were introduced. Magnetic sources can be used to represent radiating apertures.

The fields generated by magnetic sources can be obtained by solutions to Maxwell's equations or the wave equation (subject to the appropriate boundary conditions), or by using the duality theorem of Chapter 7, Section 7.2, once the solution to the same problem but with an electric source excitation is known.

Then, using the duality theorem of Section 7.2, the field generated by an infinite magnetic line source of constant current I_m can be obtained using Tables 7-1 and 7-2, (11-10a) through (11-10b), and (11-13a) through (11-13b). These can then be written as

$$E_\phi = +jI_m \frac{\beta}{4} H_1^{(2)}(\beta \rho) \stackrel{\beta \rho \rightarrow \text{large}}{\simeq} -I_m \sqrt{\frac{j\beta}{8\pi}} \frac{e^{-j\beta \rho}}{\sqrt{\rho}} \quad (11-15a)$$

$$H_z = -I_m \frac{\beta^2}{4\omega\mu} H_0^{(2)}(\beta \rho) \stackrel{\beta \rho \rightarrow \text{large}}{\simeq} -\frac{1}{\eta} I_m \sqrt{\frac{j\beta}{8\pi}} \frac{e^{-j\beta \rho}}{\sqrt{\rho}} \quad (11-15b)$$

which, when the sources are displaced from the origin, can also be expressed according to (11-11a) and (11-11b) as

$$E_\psi = +jI_m \frac{\beta}{4} H_1^{(2)}(\beta |\rho - \rho'|) \quad (11-16a)$$

$$H_z = -I_m \frac{\beta^2}{4\omega\mu} H_0^{(2)}(\beta |\rho - \rho'|) \quad (11-16b)$$

11.2.3 Electric Line Source Above an Infinite Plane Electric Conductor

When an infinite electric line source is placed at a height h above an infinite flat electric conductor, as shown in Figure 11-2a, the solution for the field components must include the presence of the conducting plane. This can be accomplished by using Maxwell's equations or the wave equation subject to the radiation conditions at infinity and boundary condition along the air-conductor interface. Instead of doing this, the same solution is obtained by introducing an equivalent model which leads to the same fields in the region of interest. Since the fields below the interface (in the electric conductor, $y < 0$) are known (they are zero), the equivalent model should be valid and

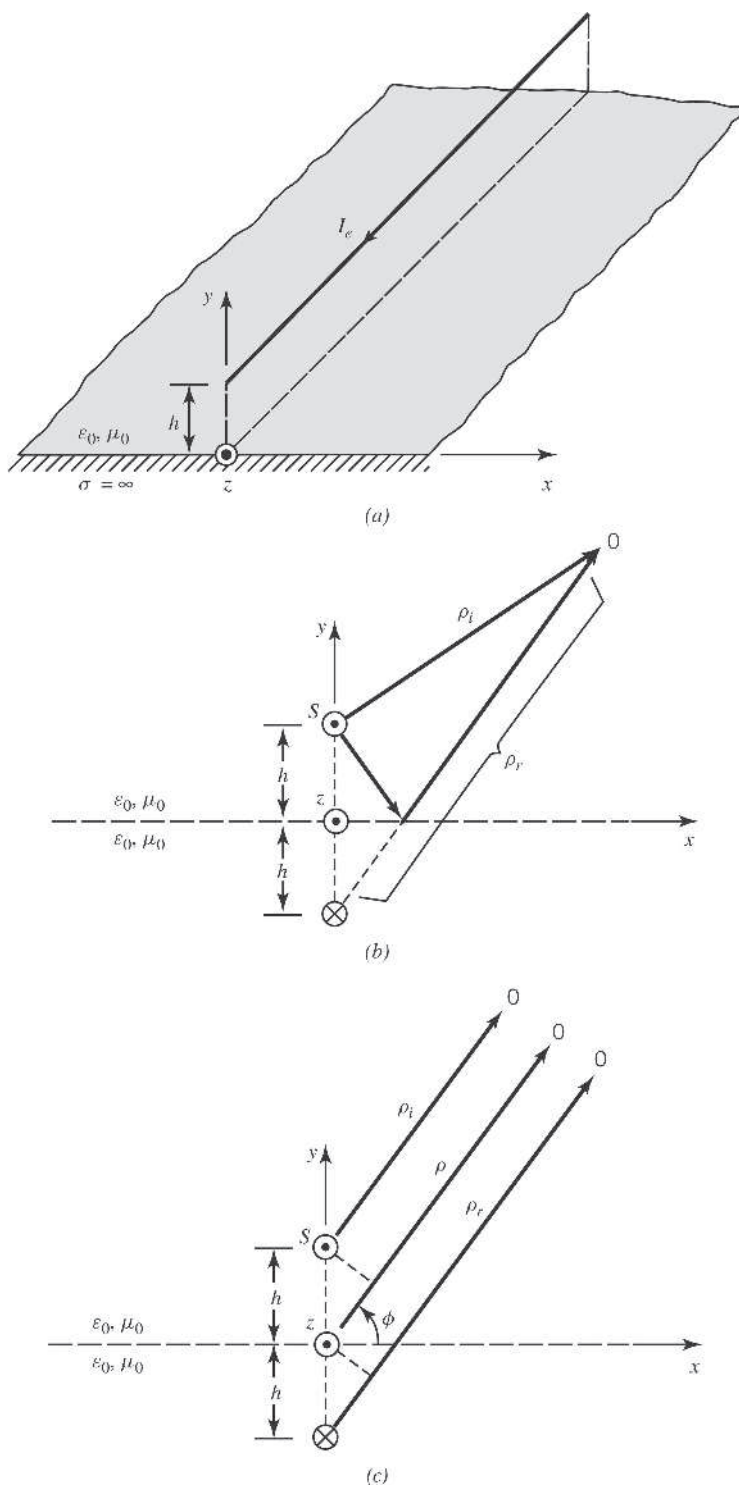


Figure 11-2 Electric line source above a flat and infinite electric ground plane. (a) Line source. (b) Equivalent (near field). (c) Equivalent (far field).

leads to the same fields as the actual physical problem on or above the interface ($y \geq 0$). In this case, as long as the equivalent model satisfies the same boundary conditions as the actual physical problem along a closed surface, according to the uniqueness theorem of Section 7.3, the solution of the equivalent model will be unique and be the same as that of the physical problem. For this problem, the closed surface that will be chosen is that of the air-conductor interface ($y = 0$), which extends on the range $-\infty \leq x \leq +\infty$.

The equivalent problem of Figure 11-2a is that of Figure 11-2b where the ground plane has been replaced by an equivalent source (usually referred to as *image* or *virtual source* or *caustic*). According to the theory of Section 7.4, the image source is introduced to account for the reflections from the surface of the ground plane. The magnitude, phase, polarization, and position of the image source must be such that the boundary conditions of the equivalent problem of Figure (11-2b) along $-\infty \leq x \leq +\infty$ are the same as those of the physical problem of Figure 11-2a. In this situation the image must have the same magnitude as the actual source, its phase must be 180° out of phase from the actual source, it must be placed below the interface at a depth h ($y = -h$) along a line perpendicular to the interface and passing through the actual source, and its length must also be parallel to the z axis. Such a system configuration, as shown in Figure 11-2b, does lead to zero tangential electric field along $-\infty \leq x \leq +\infty$, which is identical to that of Figure 11-2a along the air-conductor interface.

Therefore, according to Figures 11-2a and 11-2b and (11-13a), the total electric field is equal to

$$\mathbf{E}^t = \mathbf{E}^i + \mathbf{E}^r = \begin{cases} -\hat{\mathbf{a}}_z \frac{\beta^2 I_e}{4\omega\epsilon} [H_0^{(2)}(\beta\rho_i) - H_0^{(2)}(\beta\rho_r)], & y \geq 0 \\ 0, & y < 0 \end{cases} \quad (11-17a)$$

$$(11-17b)$$

which, for observations at large distances, as shown by Figure 11-2c, reduces using the asymptotic expansion (11-12a) to

$$\mathbf{E}^t = \mathbf{E}^i + \mathbf{E}^r = \begin{cases} -\hat{\mathbf{a}}_z \eta I_e \left(\frac{e^{-j\beta\rho_i}}{\sqrt{\rho_i}} - \frac{e^{-j\beta\rho_r}}{\sqrt{\rho_r}} \right) \sqrt{\frac{j\beta}{8\pi}}, & y \geq 0 \\ 0, & y < 0 \end{cases} \quad (11-18a)$$

$$(11-18b)$$

According to Figure 11-2c, for observations made at large distances ($\rho \gg h$)

$$\left. \begin{aligned} \rho_i &\simeq \rho - h \cos\left(\frac{\pi}{2} - \phi\right) = \rho - h \sin(\phi) \\ \rho_r &\simeq \rho + h \cos\left(\frac{\pi}{2} - \phi\right) = \rho + h \sin(\phi) \end{aligned} \right\} \text{ for phase variations} \quad (11-19a)$$

$$\rho_i \simeq \rho_r \simeq \rho \quad \text{for amplitude variations} \quad (11-19b)$$

These approximations are usually referred to in antenna and scattering theory as the *far-field approximations* [1]. Using (11-19a) and (11-19b), we can reduce (11-18a) and (11-18b) to

$$\mathbf{E}^t = \mathbf{E}^i + \mathbf{E}^r = \begin{cases} -\hat{\mathbf{a}}_z \eta I_e \sqrt{\frac{j\beta}{8\pi}} (e^{+j\beta h \sin \phi} - e^{-j\beta h \sin \phi}) \frac{e^{-j\beta\rho}}{\sqrt{\rho}} \\ -\hat{\mathbf{a}}_z j \eta I_e \sqrt{\frac{j\beta}{2\pi}} \sin(\beta h \sin \phi) \frac{e^{-j\beta\rho}}{\sqrt{\rho}}, & y \geq 0 \\ 0, & y < 0 \end{cases} \quad (11-20a)$$

$$(11-20b)$$

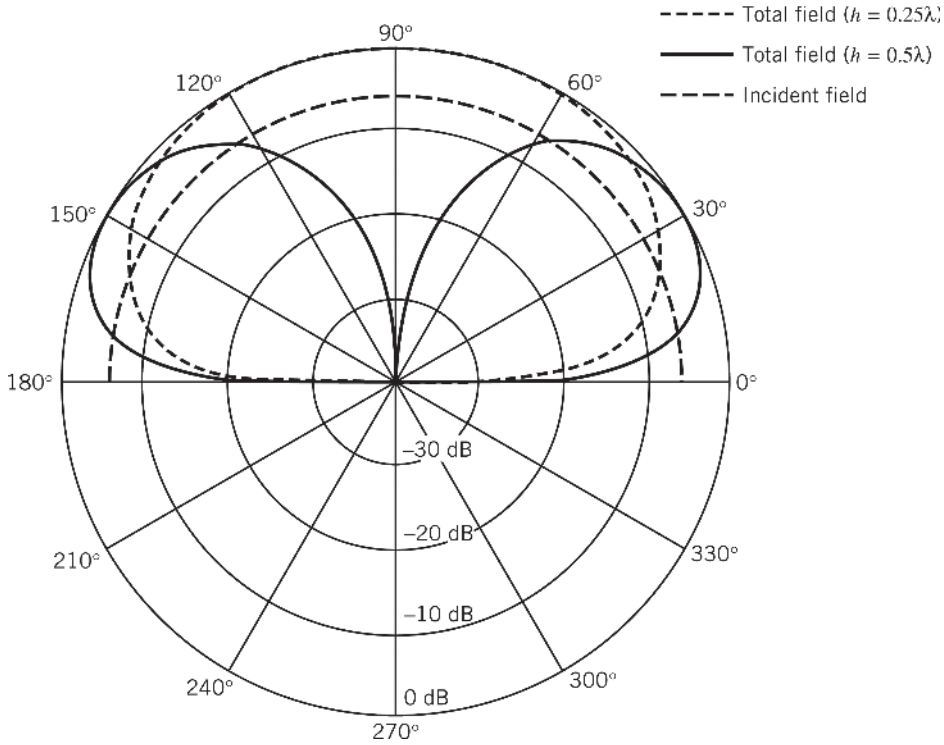


Figure 11-3 Radiation patterns of a line source above an infinite electric ground plane for $h = 0.25\lambda_0$ and $0.5\lambda_0$.

Normalized amplitude patterns (in decibels) for a source placed at a height of $h = 0.25\lambda$ and 0.5λ above the strip are shown in Figure 11-3.

11.3 PLANE WAVE SCATTERING BY PLANAR SURFACES

An important parameter in scattering is the electromagnetic scattering by a target that is usually represented by its *echo area* or *radar cross section* (RCS) (σ). The echo area or RCS is defined as “the area intercepting the amount of power that, when scattered isotropically, produces at the receiver a density that is equal to the density scattered by the actual target” [1]. For a two-dimensional target the scattering parameter is referred to as the *scattering width* (SW) or alternatively as the *radar cross section per unit length*. In equation form the scattering width and the radar cross section (σ) of a target take the form of

Scattering Width: Two-Dimensional Target

$$\sigma_{2-D} = \lim_{\rho \rightarrow \infty} \left[2\pi\rho \frac{S^s}{S^i} \right] \quad (11-21a)$$

$$\sigma_{2-D} = \lim_{\rho \rightarrow \infty} \left[2\pi\rho \frac{|\mathbf{E}^s|^2}{|\mathbf{E}^i|^2} \right] \quad (11-21b)$$

$$\sigma_{2-D} = \lim_{\rho \rightarrow \infty} \left[2\pi\rho \frac{|\mathbf{H}^s|^2}{|\mathbf{H}^i|^2} \right] \quad (11-21c)$$

Radar Cross Section: Three-Dimensional Target

$$\sigma_{3-D} = \begin{cases} \lim_{r \rightarrow \infty} \left[4\pi r^2 \frac{S^s}{S^i} \right] & (11-22a) \\ \lim_{r \rightarrow \infty} \left[4\pi r^2 \frac{|\mathbf{E}^s|^2}{|\mathbf{E}^i|^2} \right] & (11-22b) \\ \lim_{r \rightarrow \infty} \left[4\pi r^2 \frac{|\mathbf{H}^s|^2}{|\mathbf{H}^i|^2} \right] & (11-22c) \end{cases}$$

where ρ, r = distance from target to observation point

S^s, S^i = scattered, incident power densities

$\mathbf{E}^s, \mathbf{E}^i$ = scattered, incident electric fields

$\mathbf{H}^s, \mathbf{H}^i$ = scattered, incident magnetic fields

For normal incidence, the two- and three-dimensional fields, and scattering width and radar cross sections for a target of length ℓ are related by [2-4]

$$E_{3-D} \simeq \left(E_{2-D} \frac{\ell e^{j\pi/4}}{\sqrt{\lambda \rho}} \right)_{\rho=r} \quad (11-22d)$$

$$\sigma_{3-D} \simeq \sigma_{2-D} \frac{2\ell^2}{\lambda} \quad (11-22e)$$

The unit of the two-dimensional SW is length (meters in the MKS system), whereas that of the three-dimensional RCS is area (meters squared in the MKS system). A most common reference is *one meter* for the two-dimensional SW and *one meter squared* for the three-dimensional RCS. Therefore, a most common designation is dB/m (or dBm) for the two-dimensional SW and dB/(square meter) (or dBsm) for the three-dimensional RCS.

When the transmitter and receiver are at the same location, the RCS is usually referred to as *monostatic* (or *backscattered*), and it is referred to as *bistatic* when the two are at different locations. Observations made toward directions that satisfy Snell's law of reflection are usually referred to as *specular*. Therefore, the RCS of a target is a very important parameter which characterizes its scattering properties. A plot of the RCS as a function of the space coordinates is usually referred to as the *RCS pattern*. The definitions (11-21a) through (11-22c) all indicate that the SW and RCS of targets are defined under plane wave, that in practice can only be approximated when the target is placed in the far field of the source (at least $2D^2/\lambda$) where D is the largest dimension of the target [1].

In this section, physical optics (PO) techniques will be used to analyze the scattering from conducting strips and plates of finite width, neglecting edge effects. The edge effects will be considered in Chapters 12 to 14, where, respectively, moment method and geometrical theory of diffraction techniques will be utilized. Physical optics techniques are most accurate at specular directions [5].

11.3.1 TM^z Plane Wave Scattering from a Strip

Let us assume that a TM^z uniform plane wave is incident upon an electric conducting strip of width w and infinite length, as shown in Figures 11-4a and 11-4b. The incident electric and magnetic fields can be written as

$$\mathbf{E}^i = \hat{\mathbf{a}}_z E_0 e^{j\beta(x \cos \phi_i + y \sin \phi_i)} \quad (11-23a)$$

$$\mathbf{H}^i = \frac{E_0}{\eta} (-\hat{\mathbf{a}}_x \sin \phi_i + \hat{\mathbf{a}}_y \cos \phi_i) e^{j\beta(x \cos \phi_i + y \sin \phi_i)} \quad (11-23b)$$

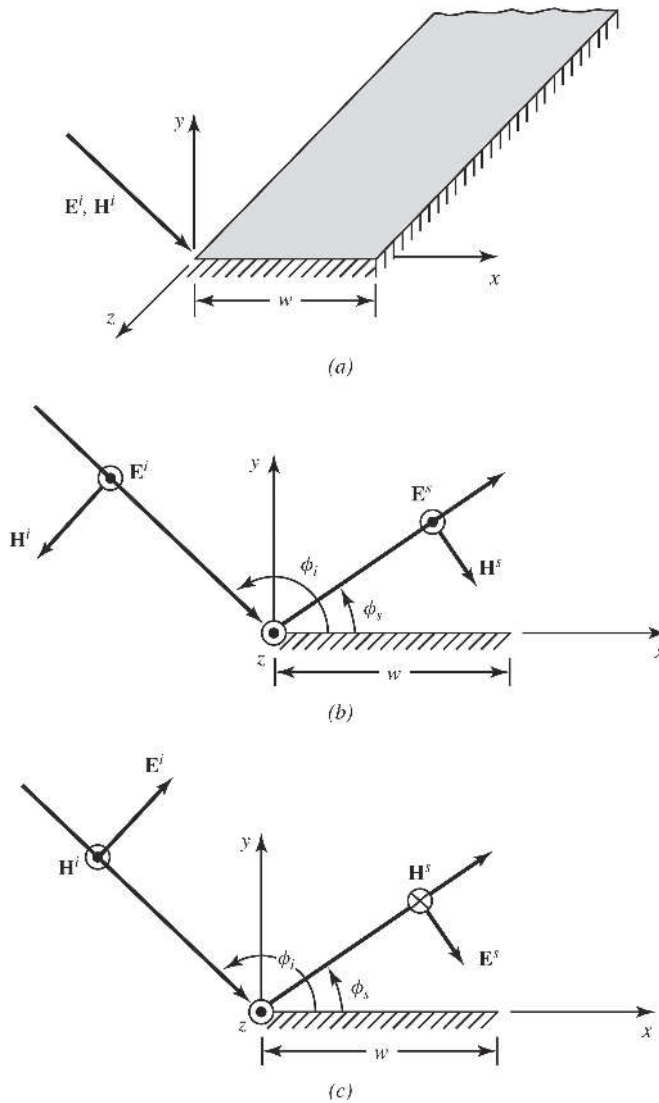


Figure 11-4 Uniform plane wave incident on a finite width strip. (a) Finite width strip. (b) TM^z polarization. (c) TE^z polarization.

where E_0 is a constant, and it represents the magnitude of the incident electric field. In Figures 11-4a and 11-4b the angle ϕ_i is shown to be greater than 90° . The reflected fields can be expressed as

$$\mathbf{E}^r = \hat{\mathbf{a}}_z \Gamma_{\perp} E_0 e^{-j\beta(x \cos \phi_r + y \sin \phi_r)} \quad (11-24a)$$

$$\mathbf{H}^r = \frac{\Gamma_{\perp} E_0}{\eta} (\hat{\mathbf{a}}_x \sin \phi_r - \hat{\mathbf{a}}_y \cos \phi_r) e^{-j\beta(x \cos \phi_r + y \sin \phi_r)} \quad (11-24b)$$

where ϕ_r is the reflection angle as determined by enforcing the boundary conditions along the interface, assuming an interface of infinite extent. For the finite-width strip the reflection angle ϕ_r is not the same as the scattering angle ϕ_s ($\phi_r \neq \phi_s$). The two coincide for an infinite-width strip

when geometrical optics and physical optics reduce to each other. Thus, we use geometrical optics to determine the reflection coefficient on the surface of the finite-width strip, and then apply physical optics to find the scattered fields. According to (5-17a) and (5-15a), the reflection coefficient for a perfectly conducting surface is equal to $\Gamma_{\perp} = -1$ and $\phi_r = \pi - \phi_i$. Thus, (11-24a) and (11-24b) reduce to

$$\mathbf{E}^r = -\hat{\mathbf{a}}_z E_0 e^{j\beta(x \cos \phi_i - y \sin \phi_i)} \quad (11-25a)$$

$$\mathbf{H}^r = \frac{E_0}{\eta} (-\hat{\mathbf{a}}_x \sin \phi_i - \hat{\mathbf{a}}_y \cos \phi_i) e^{j\beta(x \cos \phi_i - y \sin \phi_i)} \quad (11-25b)$$

Using physical optics techniques of (7-54), the current density induced on the surface of the strip can be written as

$$\begin{aligned} \mathbf{J}_s &= \hat{\mathbf{n}} \times \mathbf{H}^t \Big|_{y=0} = \hat{\mathbf{n}} \times (\mathbf{H}^i + \mathbf{H}^r) \Big|_{y=0} \\ &= 2\hat{\mathbf{n}} \times \mathbf{H}^i \Big|_{y=0} = \hat{\mathbf{a}}_y \frac{2E_0}{\eta} \times (-\hat{\mathbf{a}}_x \sin \phi_i + \hat{\mathbf{a}}_y \cos \phi_i) e^{j\beta x' \cos \phi_i} \\ \mathbf{J}_s &= \hat{\mathbf{a}}_z \frac{2E_0}{\eta} \sin \phi_i e^{j\beta x' \cos \phi_i} \end{aligned} \quad (11-26)$$

and the far-zone scattered field can be found using (6-96a), (6-101a), and (6-101b) or

$$\mathbf{A} = \frac{\mu}{4\pi} \iint_S \mathbf{J}_s(x', y', z') \frac{e^{-j\beta R}}{R} ds' \quad (11-27a)$$

$$\mathbf{E}_A \simeq -j\omega \mathbf{A} \quad (\text{for } \theta \text{ and } \phi \text{ components only}) \quad (11-27b)$$

$$\mathbf{H}_A \simeq \frac{1}{\eta} \hat{\mathbf{a}}_r \times \mathbf{E}_A = -j \frac{\omega}{\eta} \hat{\mathbf{a}}_r \times \mathbf{A} \quad (\text{for } \theta \text{ and } \phi \text{ components only}) \quad (11-27c)$$

where

$$R = \sqrt{(x-x')^2 + (y-y')^2 + (z-z')^2} = \sqrt{(|\boldsymbol{\rho} - \boldsymbol{\rho}'|)^2 + (z-z')^2} \quad (11-27d)$$

Substituting (11-26) into (11-27a) and using (11-27d), we can write that

$$\mathbf{A} = \hat{\mathbf{a}}_z \frac{\mu E_0}{2\pi\eta} \sin \phi_i \int_0^w \left\{ \int_{-\infty}^{+\infty} \frac{\exp \left[-j\beta \sqrt{(|\boldsymbol{\rho} - \boldsymbol{\rho}'|)^2 + (z-z')^2} \right]}{\sqrt{(|\boldsymbol{\rho} - \boldsymbol{\rho}'|)^2 + (z-z')^2}} dz' \right\} e^{j\beta x' \cos \phi_i} dx' \quad (11-28)$$

Since the integral with the infinite limits can be represented by a Hankel function of the second kind of zero order

$$\int_{-\infty}^{+\infty} \frac{e^{-j\alpha \sqrt{x^2 + t^2}}}{\sqrt{x^2 + t^2}} dt = -j\pi H_0^{(2)}(\alpha x) \quad (11-28a)$$

(11-28) can be reduced to

$$\mathbf{A} = -\hat{\mathbf{a}}_z j \frac{\mu E_0}{2\eta} \sin \phi_i \int_0^w H_0^{(2)}(\beta |\boldsymbol{\rho} - \boldsymbol{\rho}'|) e^{j\beta x' \cos \phi_i} dx' \quad (11-28b)$$

For far-zone observations

$$\begin{aligned}
 |\rho - \rho'| &= \sqrt{\rho^2 + (\rho')^2 - 2\rho\rho' \cos(\phi_s - \phi')} \stackrel{\rho \gg \rho'}{\simeq} \sqrt{\rho^2 - 2\rho\rho' \cos(\phi_s - \phi')} \\
 |\rho - \rho'| &\stackrel{\rho \gg \rho'}{\simeq} \rho \sqrt{1 - 2\left(\frac{\rho'}{\rho}\right) \cos(\phi_s - \phi')} \stackrel{\rho \gg \rho'}{\simeq} \rho \left[1 - \left(\frac{\rho'}{\rho}\right) \cos(\phi_s - \phi') \right] \\
 |\rho - \rho'| &\stackrel{\rho \gg \rho'}{\simeq} \rho - \rho' \cos(\phi_s - \phi')
 \end{aligned} \tag{11-29}$$

Since the source (here the current density) exists only over the width of the strip that according to Figure 11-4 lies along the x axis, then $\rho' = x'$ and $\phi' = 0$. Thus, for far-field observations (11-29) reduces to

$$|\rho - \rho'| \simeq \begin{cases} \rho - \rho' \cos(\phi_s - \phi') = \rho - x' \cos \phi_s & \text{for phase terms} \\ \rho & \text{for amplitude terms} \end{cases} \tag{11-29a}$$

$$\tag{11-29b}$$

In turn the Hankel function in the integrand of (11-28b) can be expressed, using (11-29a) and (11-29b), as

$$H_0^{(2)}(\beta |\rho - \rho'|) \stackrel{\rho \gg \rho'}{\simeq} \sqrt{\frac{2j}{\pi\beta\rho}} e^{-j\beta(\rho - x' \cos \phi_s)} = \sqrt{\frac{2j}{\pi\beta}} \frac{e^{-j\beta\rho}}{\sqrt{\rho}} e^{j\beta x' \cos \phi_s} \tag{11-30}$$

Substituting (11-30) into (11-28b) reduces it to

$$\begin{aligned}
 \mathbf{A} &\simeq -\hat{\mathbf{a}}_z j \frac{\mu E_0}{2\eta} \sqrt{\frac{2j}{\pi\beta}} \frac{e^{-j\beta\rho}}{\sqrt{\rho}} \sin \phi_i \int_0^w e^{j\beta x' (\cos \phi_s + \cos \phi_i)} dx' \\
 \mathbf{A} &\simeq -\hat{\mathbf{a}}_z j \frac{\mu w E_0}{\eta} \sqrt{\frac{j}{2\pi\beta}} e^{j(\beta w/2)(\cos \phi_s + \cos \phi_i)} \\
 &\quad \times \left\{ \sin \phi_i \left[\frac{\sin \left[\frac{\beta w}{2} (\cos \phi_s + \cos \phi_i) \right]}{\frac{\beta w}{2} (\cos \phi_s + \cos \phi_i)} \right] \frac{e^{-j\beta\rho}}{\sqrt{\rho}} \right\}
 \end{aligned} \tag{11-31}$$

Therefore, the far-zone scattered spherical components of the electric and magnetic fields of (11-27b) and (11-27c) can be written, using (11-31) and (II-12), as

$$\begin{aligned}
 E_\theta^s &\simeq j\omega A_z \sin \theta_s = -w E_0 \sqrt{\frac{j\beta}{2\pi}} e^{j(\beta w/2)(\cos \phi_s + \cos \phi_i)} \\
 &\quad \times \left\{ \sin \theta_s \sin \phi_i \left[\frac{\sin \left[\frac{\beta w}{2} (\cos \phi_s + \cos \phi_i) \right]}{\frac{\beta w}{2} (\cos \phi_s + \cos \phi_i)} \right] \frac{e^{-j\beta\rho}}{\sqrt{\rho}} \right\}
 \end{aligned} \tag{11-32a}$$

$$\begin{aligned}
 H_\phi^s &\simeq \frac{E_\theta^s}{\eta} = -\frac{w E_0}{\eta} \sqrt{\frac{j\beta}{2\pi}} e^{j(\beta w/2)(\cos \phi_s + \cos \phi_i)} \\
 &\quad \times \left\{ \sin \theta_s \sin \phi_i \left[\frac{\sin \left[\frac{\beta w}{2} (\cos \phi_s + \cos \phi_i) \right]}{\frac{\beta w}{2} (\cos \phi_s + \cos \phi_i)} \right] \frac{e^{-j\beta\rho}}{\sqrt{\rho}} \right\}
 \end{aligned} \tag{11-32b}$$

The bistatic scattering width is obtained using any of (11-21a) through (11-21c), and it is represented at $\theta_s = 90^\circ$ by

$$\sigma_{2-D}(\text{bistatic}) = \lim_{\rho \rightarrow \infty} \left[2\pi\rho \frac{|\mathbf{E}^s|^2}{|\mathbf{E}^i|^2} \right] = \frac{2\pi w^2}{\lambda} \left\{ \sin \phi_i \left[\frac{\sin \left[\frac{\beta w}{2} (\cos \phi_s + \cos \phi_i) \right]}{\frac{\beta w}{2} (\cos \phi_s + \cos \phi_i)} \right] \right\}^2 \quad (11-33)$$

which for the monostatic system configuration ($\phi_s = \phi_i$) reduces to

$$\sigma_{2-D}(\text{monostatic}) = \frac{2\pi w^2}{\lambda} \left\{ \sin \phi_i \left[\frac{\sin(\beta w \cos \phi_i)}{\beta w \cos \phi_i} \right] \right\}^2 \quad (11-33a)$$

Computed patterns of the normalized bistatic SW of (11-33) (in dB), for $0^\circ \leq \phi_s \leq 180^\circ$ when $\phi_i = 120^\circ$ and $w = 2\lambda$ and 10λ , are shown in Figure 11-5a. It is apparent that the maximum occurs when the $\sin(x)/x$ function reaches its maximum value of unity, that is, when the $x = \beta w (\cos \phi_s + \cos \phi_i) / 2 = 0$. For these examples, this occurs when $\phi_s = 180^\circ - \phi_i = 180^\circ - 120^\circ = 60^\circ$, which represents the direction of specular scattering (angle of scattering is equal to the angle of incidence). Away from the direction of maximum radiation, the pattern variations are of $\sin(x)/x$ form. The normalized monostatic SW of (11-33a) (in dB) for $w = 2\lambda$ and 10λ are shown plotted in Figure 11-5b for $0^\circ \leq \phi_i \leq 180^\circ$, where the maximum occurs when $\phi_i = 90^\circ$, which is the direction of normal incidence (the strip is viewed perpendicularly to its flat surface). Again, away from the maximum radiation, the pattern variations are approximately of $\sin(x)/x$ form.

Example 11-2

Derive the far-zone scattered fields and the associated scattering width when a TE^z uniform plane wave is incident upon a two-dimensional conducting strip of width w , as shown in Figures 11-4a and 11-4c. Use physical optics methods.

Solution: According to Figure 11-4c, the incident electric and magnetic field components for a TE^z uniform plane wave can be written as

$$\mathbf{E}^i = \eta H_0 (\hat{\mathbf{a}}_x \sin \phi_i - \hat{\mathbf{a}}_y \cos \phi_i) e^{j\beta(x \cos \phi_i + y \sin \phi_i)}$$

$$\mathbf{H}^i = \hat{\mathbf{a}}_z H_0 e^{j\beta(x \cos \phi_i + y \sin \phi_i)}$$

The current induced on the surface of the finite width strip can be approximated by the physical optics current and is equal to

$$\mathbf{J}_s \simeq 2\hat{\mathbf{n}} \times \mathbf{H}^i \Big|_{y=0} = 2\hat{\mathbf{a}}_y \times \hat{\mathbf{a}}_z H_z \Big|_{y=0} = 2\hat{\mathbf{a}}_x H_z \Big|_{y=0}$$

$$\mathbf{J}_s \simeq \hat{\mathbf{a}}_x 2H_0 e^{j\beta x' \cos \phi_i}$$

$$J_y = J_z = 0, \quad J_x = 2H_0 e^{j\beta x' \cos \phi_i}$$

Using the steps outlined by (11-27a) through (11-31), it can be shown that

$$\mathbf{A} = -\hat{\mathbf{a}}_x j \frac{\mu H_0}{2} \int_0^w H_0^{(2)}(\beta |\boldsymbol{\rho} - \boldsymbol{\rho}'|) e^{j\beta x' \cos \phi_i} dx'$$

which for far-zone observations reduces to

$$\mathbf{A} \simeq -\hat{\mathbf{a}}_x j \mu w H_0 \sqrt{\frac{j}{2\pi\beta}} e^{j(\beta w/2)(\cos \phi_s + \cos \phi_i)} \left\{ \frac{\sin \left[\frac{\beta w}{2} (\cos \phi_s + \cos \phi_i) \right]}{\frac{\beta w}{2} (\cos \phi_s + \cos \phi_i)} \right\} \frac{e^{-j\beta \rho}}{\sqrt{\rho}}$$

In spherical components this can be written, according to (II-12), as

$$\begin{aligned} A_r &= A_x \sin \theta_s \cos \phi_s \\ A_\theta &= A_x \cos \theta_s \cos \phi_s \\ A_\phi &= -A_x \sin \phi_s \end{aligned}$$

which for $\theta_s = 90^\circ$ reduce to

$$\begin{aligned} A_r &= A_x \cos \phi_s \\ A_\theta &= 0 \\ A_\phi &= -A_x \sin \phi_s \end{aligned}$$

Thus, the far-zone electric and magnetic field components in the $\theta_s = 90^\circ$ plane can be written as

$$\begin{aligned} E_r^s &\simeq E_\theta^s \simeq H_r^s \simeq H_\phi^s \simeq 0 \\ E_\phi^s &\simeq -j\omega A_\phi = j\omega A_x \sin \phi_s \\ E_\phi^s &\simeq -\eta w H_0 \sqrt{\frac{j\beta}{2\pi}} e^{j(\beta w/2)(\cos \phi_s + \cos \phi_i)} \\ &\quad \times \left\{ \sin \phi_s \left[\frac{\sin \left[\frac{\beta w}{2} (\cos \phi_s + \cos \phi_i) \right]}{\frac{\beta w}{2} (\cos \phi_s + \cos \phi_i)} \right] \frac{e^{-j\beta \rho}}{\sqrt{\rho}} \right\} \\ H_\theta^s &\simeq -\frac{E_\phi^s}{\eta} = w H_0 \sqrt{\frac{j\beta}{2\pi}} e^{j(\beta w/2)(\cos \phi_s + \cos \phi_i)} \\ &\quad \times \left\{ \sin \phi_s \left[\frac{\sin \left[\frac{\beta w}{2} (\cos \phi_s + \cos \phi_i) \right]}{\frac{\beta w}{2} (\cos \phi_s + \cos \phi_i)} \right] \frac{e^{-j\beta \rho}}{\sqrt{\rho}} \right\} \end{aligned}$$

The bistatic and monostatic (backscattering) scattering widths are given by

$$\begin{aligned} \sigma_{2-D}(\text{bistatic}) &= \lim_{\rho \rightarrow \infty} \left[2\pi \rho \frac{|H^s|^2}{|\mathbf{H}^i|^2} \right] \\ &= \frac{2\pi w^2}{\lambda} \left\{ \sin \phi_s \left[\frac{\sin \left[\frac{\beta w}{2} (\cos \phi_s + \cos \phi_i) \right]}{\frac{\beta w}{2} (\cos \phi_s + \cos \phi_i)} \right] \right\}^2 \\ \sigma_{2-D}(\text{monostatic}) &= \frac{2\pi w^2}{\lambda} \left\{ \sin \phi_i \left[\frac{\sin(\beta w \cos \phi_i)}{\beta w \cos \phi_i} \right] \right\}^2 \end{aligned}$$

The monostatic SW for the TE^z polarization is identical to that of the TM^z as given by (11-33a), and it is shown plotted in Figure 11-5b for $w = 2\lambda$ and 10λ . The bistatic SW, however, differs from that of (11-33) in that the $\sin^2 \phi_i$ term is replaced by $\sin^2 \phi_s$. Therefore, the computed normalized bistatic patterns (in dB) for $0^\circ \leq \phi_s \leq 180^\circ$ when $\phi_i = 120^\circ$ and $w = 2\lambda, 10\lambda$ are shown, respectively, in Figure 11-6. It is evident that the larger the width of the strip, the larger the maximum value of the SW and the larger the number of minor lobes in its pattern.

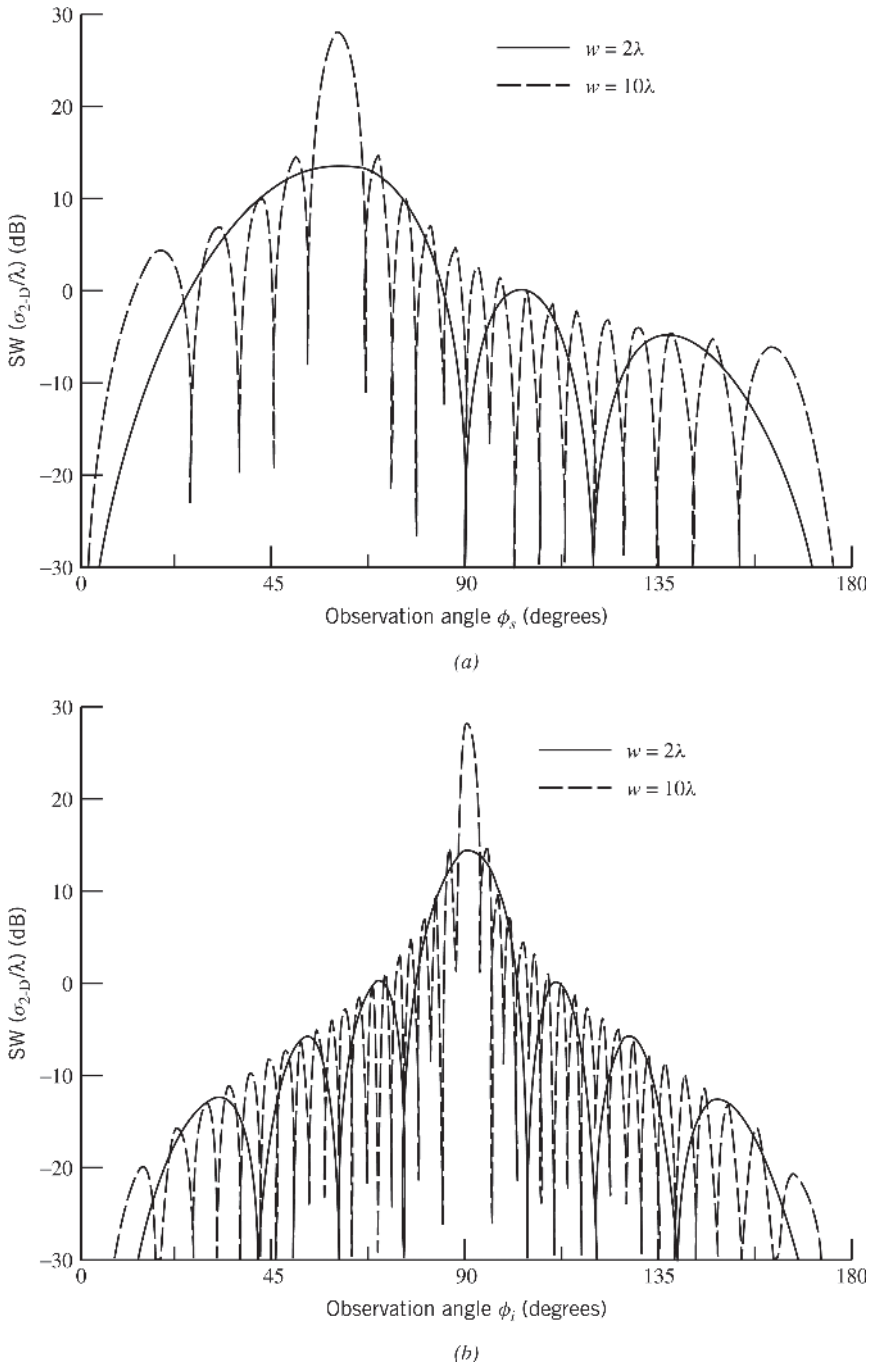


Figure 11-5 Bistatic and monostatic scattering width (SW) for a finite width strip. (a) Bistatic ($\phi_i = 120^\circ$): TM^z. (b) Monostatic: TM^z and TE^z.

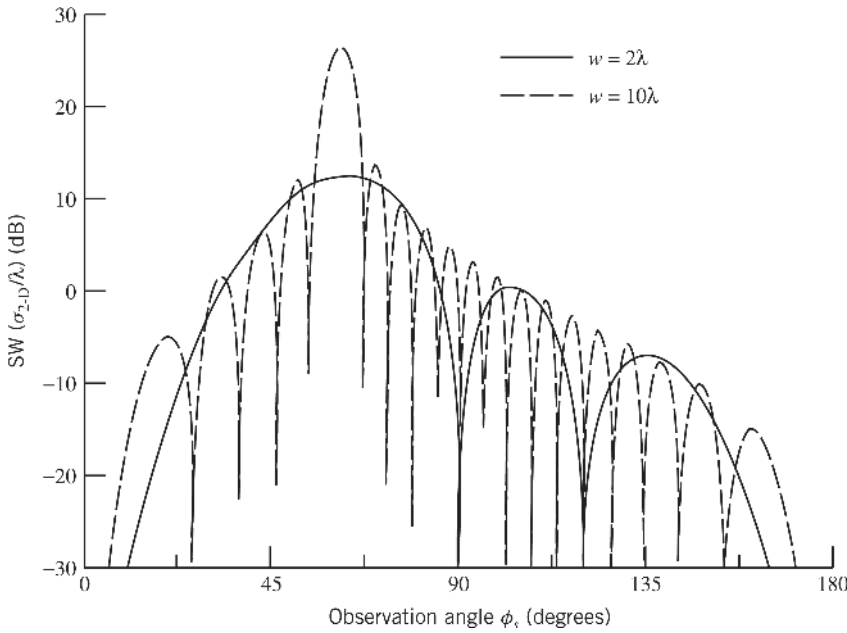


Figure 11-6 Bistatic TE^z scattering width (SW) for a finite width strip ($\phi_i = 120^\circ$).

To see that indeed the bistatic SW patterns of the TM^z and TE^z polarizations are different, we have plotted in Figure 11-7a the two for $w = 2\lambda$ when the incidence angle is $\phi_i = 120^\circ$. It is evident that the two are similar but not identical because the $\sin^2 \phi_i$ term is replaced by $\sin^2 \phi_s$, and vice versa. In addition, whereas the maximum for the TM^z occurs at the specular direction ($\phi_s = 60^\circ$ for this example), that of the TE^z occurs at an angle slightly larger than the specular direction. However, as the size of the target becomes very large, the maximum of the TE^z SW moves closer toward the specular direction and matches that of the TM^z , which always occurs at the specular direction [6]. This is illustrated in Figure 11-7b where the bistatic RCS of the two polarizations has been plotted for $w = 10\lambda$ and $\phi_i = 120^\circ$. When the size of the target is very large electrically, the $[\sin(x)/x]^2$ in the bistatic RCS expression of Example 11-1 varies very rapidly as a function of ϕ_s so that the slowly varying $\sin^2 \phi_s$ is essentially a constant near the maximum of the $[\sin(x)/x]^2$ function. This is not true when the size of the target is small electrically, as was demonstrated by the results of Figure 11-7a.

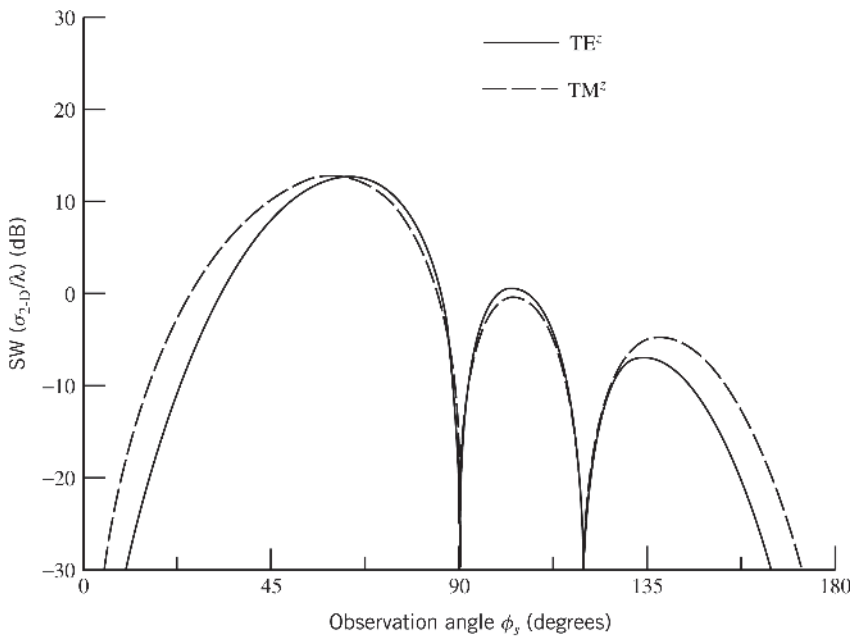
11.3.2 TE^x Plane Wave Scattering from a Flat Rectangular Plate

Let us now consider scattering from a three-dimensional scatterer, specifically uniform plane-wave scattering from a rectangular plate, as shown in Figure 11-8a. To simplify the details, let us assume that the uniform plane wave is TE^x , and that it lies on the yz plane, as shown in Figure 11-8b. The electric and magnetic fields can now be written as

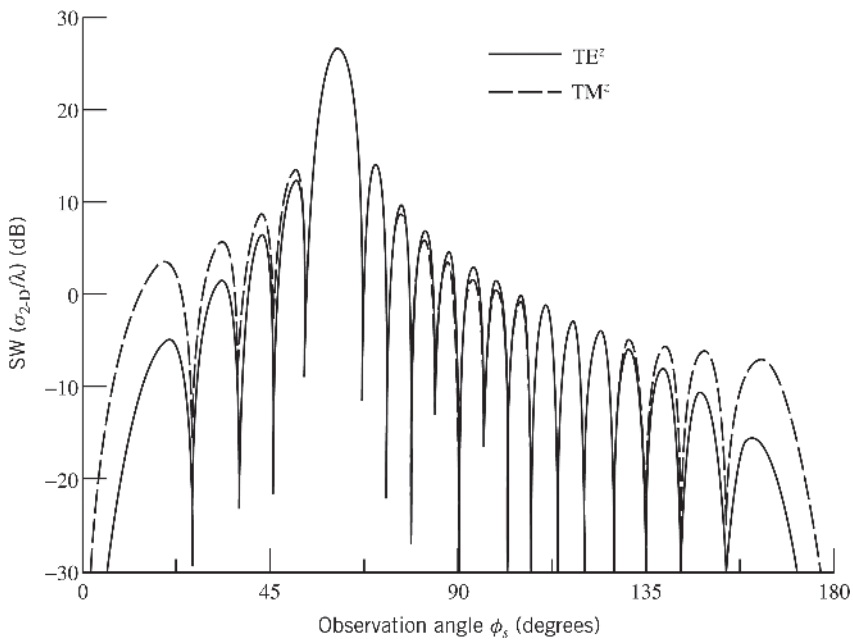
$$\mathbf{E}^i = \eta H_0 (\hat{\mathbf{a}}_y \cos \theta_i + \hat{\mathbf{a}}_z \sin \theta_i) e^{-j\beta(y \sin \theta_i - z \cos \theta_i)} \quad (11-34a)$$

$$\mathbf{H}^i = \hat{\mathbf{a}}_x H_0 e^{-j\beta(y \sin \theta_i - z \cos \theta_i)} \quad (11-34b)$$

where H_0 is a constant that represents the magnitude of the incident magnetic field.



(a)



(b)

Figure 11-7 TE^z and TM^z bistatic scattering widths (SW) for a finite-width strip ($\phi_i = 120^\circ$). (a) $w = 2\lambda$. (b) $w = 10\lambda$.

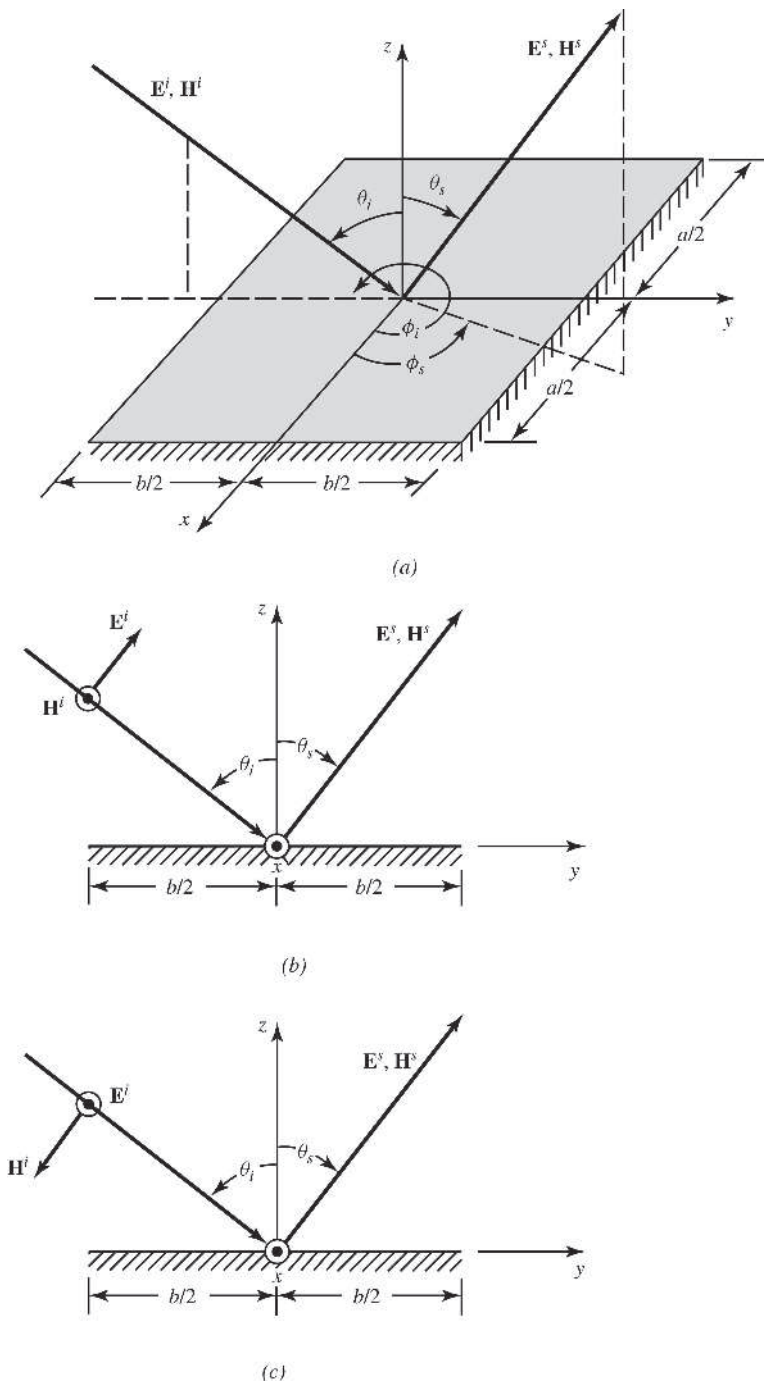


Figure 11-8 Uniform plane wave incident on a rectangular conducting plate. (a) Rectangular plate. (b) yz plane: TE^x polarization. (c) yz plane: TM^x polarization.

The scattered field can be found, by neglecting edge effects, using physical optics techniques of Section 7.10, where the current density induced at the surface of the plate is represented on the plate by

$$\mathbf{J}_s \simeq 2\hat{\mathbf{n}} \times \mathbf{H}' \Big|_{\substack{z=0 \\ y=y'}} = 2\hat{\mathbf{a}}_z \times \hat{\mathbf{a}}_x H_x \Big|_{\substack{z=0 \\ y=y'}} = \hat{\mathbf{a}}_y 2H_0 e^{-j\beta y' \sin \theta_i} \quad (11-35)$$

Thus,

$$J_x = J_z = 0 \text{ and } J_y = 2H_0 e^{-j\beta y' \sin \theta_i} \quad (11-35a)$$

For an infinite plate, the current density of (11-35) or (11-35a) yields exact field solutions. However, for finite-size plates it is approximate, and the corresponding scattered fields obtained using it are more accurate toward the specular direction, where Snell's law of reflection is satisfied. The solutions become less accurate as the observation points are removed further from the specular directions.

The scattered fields are obtained using (6-122a) through (6-122f) and (6-125a) through (6-125b), where the electric current density components are those given by (11-35a). Using (6-125a), (6-127c), (6-128c), and (11-35a), we can write that

$$\begin{aligned} N_\theta &= \iint_S [J_x \cos \theta_s \cos \phi_s + J_y \cos \theta_s \sin \phi_s - J_z \sin \theta_s]_{J_x=J_z=0} \\ &\quad \times e^{j\beta(x' \sin \theta_s \cos \phi_s + y' \sin \theta_s \sin \phi_s)} dx' dy' \\ &= 2H_0 \cos \theta_s \sin \phi_s \int_{-b/2}^{+b/2} e^{j\beta y' (\sin \theta_s \sin \phi_s - \sin \theta_i)} dy' \int_{-a/2}^{+a/2} e^{j\beta x' \sin \theta_s \cos \phi_s} dx' \end{aligned} \quad (11-36)$$

Since

$$\int_{-c/2}^{+c/2} e^{j\alpha z} dz = c \left[\frac{\sin\left(\frac{\alpha}{2}c\right)}{\frac{\alpha}{2}c} \right] \quad (11-37)$$

(11-36) reduces to

$$N_\theta = 2abH_0 \left\{ \cos \theta_s \sin \phi_s \left[\frac{\sin(X)}{X} \right] \left[\frac{\sin(Y)}{Y} \right] \right\} \quad (11-38)$$

where

$$X = \frac{\beta a}{2} \sin \theta_s \cos \phi_s \quad (11-38a)$$

$$Y = \frac{\beta b}{2} (\sin \theta_s \sin \phi_s - \sin \theta_i) \quad (11-38b)$$

In the same manner, (6-125b) can be written as

$$\begin{aligned} N_\phi &= \iint_S [-J_x \sin \phi_s + J_y \cos \phi_s]_{J_x=0} e^{j\beta(x' \sin \theta_s \cos \phi_s + y' \sin \theta_s \sin \phi_s)} dx' dy' \\ &= 2abH_0 \left\{ \cos \phi_s \left[\frac{\sin(X)}{X} \right] \left[\frac{\sin(Y)}{Y} \right] \right\} \end{aligned} \quad (11-39)$$

Therefore, the scattered fields are obtained using (6-122a) through (6-122f) and (11-38) through (11-39), and they can be expressed as

$$E_r^s \simeq 0 \quad (11-40a)$$

$$E_\theta^s \simeq -\frac{j\beta e^{-j\beta r}}{4\pi r} (L_\phi + \eta N_\theta)_{L_\phi=0} = C \frac{e^{-j\beta r}}{r} \left\{ \cos \theta_s \sin \phi_s \left[\frac{\sin(X)}{X} \right] \left[\frac{\sin(Y)}{Y} \right] \right\} \quad (11-40b)$$

$$E_\phi^s \simeq +\frac{j\beta e^{-j\beta r}}{4\pi r} (L_\theta - \eta N_\phi)_{L_\theta=0} = C \frac{e^{-j\beta r}}{r} \left\{ \cos \phi_s \left[\frac{\sin(X)}{X} \right] \left[\frac{\sin(Y)}{Y} \right] \right\} \quad (11-40c)$$

$$H_r^s \simeq 0 \quad (11-40d)$$

$$H_\theta^s \simeq -\frac{E_\phi^s}{\eta} \quad (11-40e)$$

$$H_\phi^s \simeq +\frac{E_\theta^s}{\eta} \quad (11-40f)$$

where

$$C = -j\eta \frac{ab\beta H_0}{2\pi} \quad (11-40g)$$

In the principal E plane ($\phi_s = \pi/2$) and H plane ($\theta_s = \theta_i$, $\phi_s = 0$), the electric field components reduce to

E Plane ($\phi_s = \pi/2$)

$$E_r^s \simeq E_\phi^s \simeq 0 \quad (11-41a)$$

$$E_\theta^s \simeq C \frac{e^{-j\beta r}}{r} \left\{ \cos \theta_s \left[\frac{\sin \left[\frac{\beta b}{2} (\sin \theta_s - \sin \theta_i) \right]}{\frac{\beta b}{2} (\sin \theta_s - \sin \theta_i)} \right] \right\} \quad (11-41b)$$

H Plane ($\phi_s = 0$)

$$E_r^s \simeq E_\theta^s \simeq 0 \quad (11-42a)$$

$$E_\phi^s \simeq C \frac{e^{-j\beta r}}{r} \left\{ \frac{\sin \left[\frac{\beta b}{2} (\sin \theta_i) \right]}{\frac{\beta b}{2} (\sin \theta_i)} \right\} \left\{ \frac{\sin \left[\frac{\beta a}{2} (\sin \theta_s) \right]}{\frac{\beta a}{2} (\sin \theta_s)} \right\} \quad (11-42b)$$

It can be shown that the maximum value of the total scattered field

$$E^s = \sqrt{(E_r^s)^2 + (E_\theta^s)^2 + (E_\phi^s)^2} \simeq \sqrt{(E_\theta^s)^2 + (E_\phi^s)^2} \quad (11-43)$$

for any wave incidence always lies in a scattering plane that is parallel to the incident plane [6]. For the fields of (11-40a) through (11-40f), the scattering plane that contains the maximum scattered field is that defined by $\phi_s = \pi/2$, $3\pi/2$, and $0 \leq \theta_s \leq \pi/2$, since the incident plane is that defined

by $\phi_i = 3\pi/2$, $0 \leq \theta_i \leq \pi/2$. The electric field components in the plane that contains the maximum reduce to those of (11-41a) and (11-41b) whose maximum value, when $b \gg \lambda$, occurs approximately when $\theta_s = \theta_i$ (specular reflection). For large values of b ($b \gg \lambda$) the $\sin(z)/z$ function in (11-41b) varies very rapidly compared to the $\cos\theta_s$ such that the $\cos\theta_s$ function is essentially constant near the maximum of the $\sin(z)/z$ function. For small values of b , the maximum value of (11-41b) can be found iteratively. *Thus, for the TE^x polarization, the maximum of the scattered field from a flat plate does not occur exactly at the specular direction but it approaches that value as the dimensions of the plate become large compared to the wavelength.* This is analogous to the TE^z polarization of the strip of Example 11-2. It will be shown in the example that follows that for the TM^x polarization, the maximum of the scattered field from a flat plate always occurs at the specular direction no matter what the size of the plate. This is analogous to the TM^z polarization of the strip of Section 11.3.1.

For the fields of (11-40a) through (11-40g), the radar cross section is obtained using (11-22b) or (11-22c) and can be written as

$$\begin{aligned}\sigma_{3-D} &= \lim_{r \rightarrow \infty} \left[4\pi r^2 \frac{|\mathbf{E}^s|^2}{|\mathbf{E}^i|^2} \right] = \lim_{r \rightarrow \infty} \left[4\pi r^2 \frac{|\mathbf{H}^s|^2}{|\mathbf{H}^i|^2} \right] \\ &= 4\pi \left(\frac{ab}{\lambda} \right)^2 (\cos^2 \theta_s \sin^2 \phi_s + \cos^2 \phi_s) \left[\frac{\sin(X)}{X} \right]^2 \left[\frac{\sin(Y)}{Y} \right]^2\end{aligned}\quad (11-44)$$

which in the plane that contains the maximum ($\phi_s = \pi/2$) reduces to

$$\begin{aligned}&\text{Principal Bistatic } (\phi_s = \pi/2) \\ \sigma_{3-D} &= 4\pi \left(\frac{ab}{\lambda} \right)^2 \cos^2 \theta_s \left[\frac{\sin \left[\frac{\beta b}{2} (\sin \theta_s \mp \sin \theta_i) \right]}{\frac{\beta b}{2} (\sin \theta_s \mp \sin \theta_i)} \right]^2 \begin{aligned} &- \text{ for } \phi_s = \frac{\pi}{2}, \quad 0 \leq \theta_s \leq \pi/2 \\ &+ \text{ for } \phi_s = \frac{3\pi}{2}, \quad 0 \leq \theta_s \leq \pi/2 \end{aligned}\end{aligned}\quad (11-44a)$$

while in the backscattering direction ($\phi_s = \phi_i = 3\pi/2$, $\theta_s = \theta_i$) it can be written as

$$\begin{aligned}&\text{Backscattered} \\ \sigma_{3-D} &= 4\pi \left(\frac{ab}{\lambda} \right)^2 \cos^2 \theta_i \left[\frac{\sin(\beta b \sin \theta_i)}{\beta b \sin \theta_i} \right]^2\end{aligned}\quad (11-44b)$$

Plots of (11-44a) for $a = b = 5\lambda$ and $\theta_i = 30^\circ$ ($\phi_s = 90^\circ, 270^\circ$ with $0^\circ \leq \theta_s \leq 90^\circ$) and of (11-44b) for $a = b = 5\lambda$ ($0^\circ \leq \theta_i \leq 90^\circ$) are shown, respectively, in Figures 11-9a and 11-9b. It is observed that for Figure 11-9a the maximum occurs when $\phi_s = 90^\circ$ and near $\theta_s \simeq \theta_i = 30^\circ$ while for Figure 11-9b the maximum occurs when $\theta_i = 0^\circ$ (normal incidence).

The monostatic RCS of plates using physical optics techniques is insensitive to polarization, i.e., it is the same for both polarizations [7–10], as has been demonstrated for the strip and the rectangular plate. Measurements, however, have shown that the monostatic RCS is slightly different for the two polarizations. This is one of the drawbacks of physical optics methods. In addition, the predicted RCSs using physical optics are most accurate at and near the specular directions. However, they begin to become less valid away from the specular directions, especially toward grazing incidences. This will be demonstrated in Example 12-3, and Figures 12-13 and 12-14.

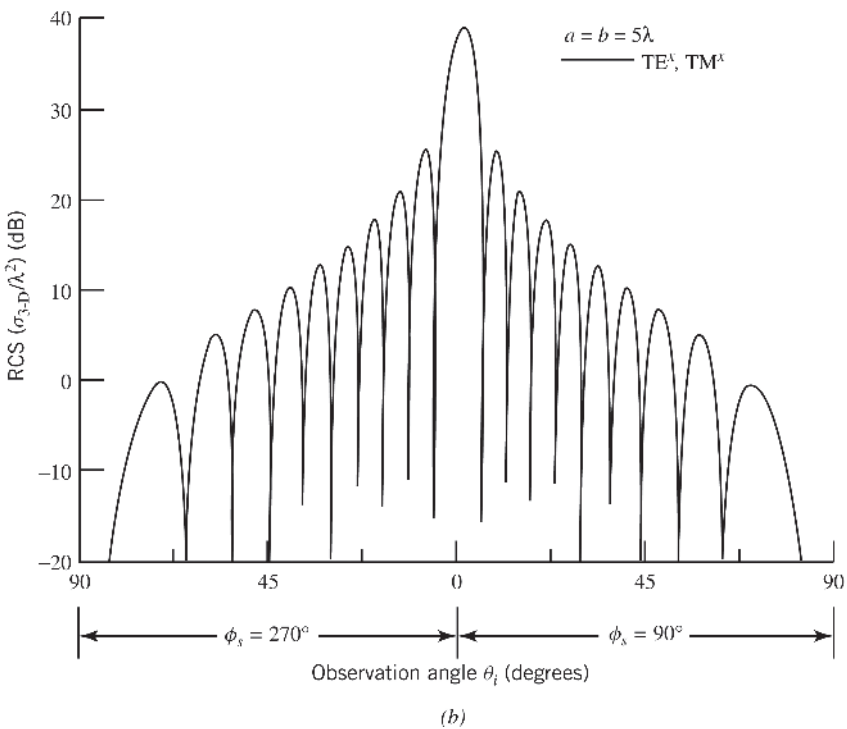
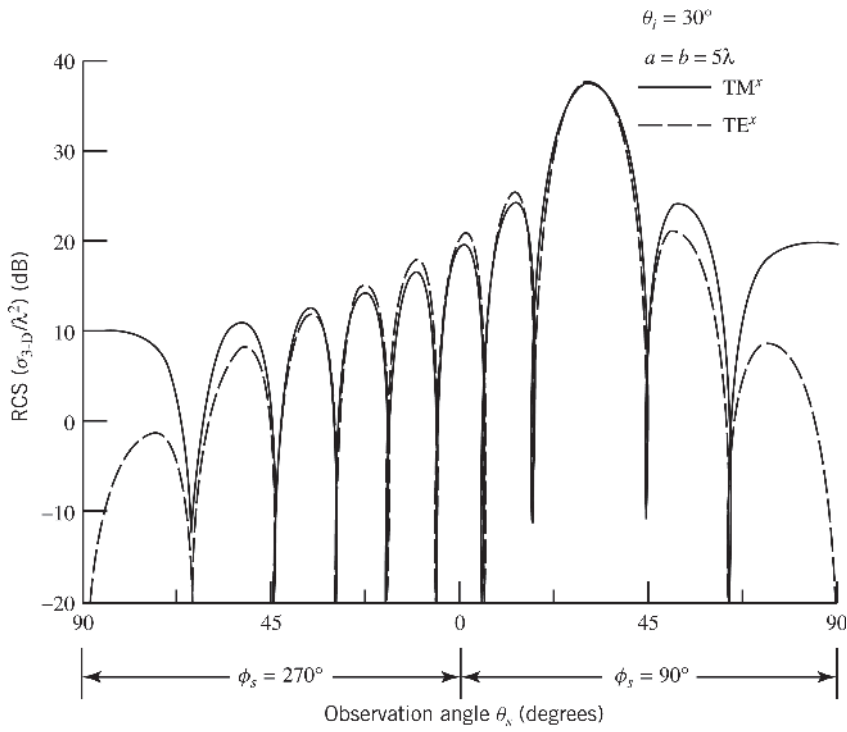


Figure 11-9 Principal plane monostatic and bistatic radar cross sections of a rectangular plate. (a) Bistatic ($\theta_i = 30^\circ$, $\phi_i = 270^\circ$, $\phi_s = 90^\circ$, 270°). (b) Monostatic ($\phi_s = 90^\circ$, 270°).

Example 11-3

Using physical optics techniques, find the scattered fields and radar cross section when a TM^x uniform plane wave is incident upon a flat rectangular plate of dimensions a and b . Assume the incident field lies on the yz plane, as shown in Figure 11-8c.

Solution: According to the geometry of Figure 11-8c, the electric and magnetic field components of the incident uniform plane wave can be written as

$$\begin{aligned}\mathbf{E}^i &= \hat{\mathbf{a}}_x E_0 e^{-j\beta(y \sin \theta_i - z \cos \theta_i)} \\ \mathbf{H}^i &= -\frac{E_0}{\eta} (\hat{\mathbf{a}}_y \cos \theta_i + \hat{\mathbf{a}}_z \sin \theta_i) e^{-j\beta(y \sin \theta_i - z \cos \theta_i)}\end{aligned}$$

Using physical optics techniques, the current density induced on the plate can be approximated by

$$\begin{aligned}\mathbf{J}_s &\simeq 2\hat{\mathbf{n}} \times \mathbf{H}^i \Big|_{z=0}^{y=y'} = 2\hat{\mathbf{a}}_z \times (\hat{\mathbf{a}}_y H_y + \hat{\mathbf{a}}_z H_z) \Big|_{z=0}^{y=y'} \\ &= -2\hat{\mathbf{a}}_x H_y \Big|_{z=0}^{y=y'} = \hat{\mathbf{a}}_x \frac{2E_0}{\eta} \cos \theta_i e^{-j\beta y' \sin \theta_i}\end{aligned}$$

or

$$J_y = J_z = 0 \quad J_x = \frac{2E_0}{\eta} \cos \theta_i e^{-j\beta y' \sin \theta_i}$$

Using (6-125a) and (6-125b), we can write that

$$\begin{aligned}N_\theta &= \iint_S J_x \cos \theta_s \cos \phi_s e^{j\beta(x' \sin \theta_s \cos \phi_s + y' \sin \theta_s \sin \phi_s)} dx' dy' \\ &= \frac{2E_0}{\eta} ab \left\{ \cos \theta_i \cos \theta_s \cos \phi_s \left[\frac{\sin(X)}{X} \right] \left[\frac{\sin(Y)}{Y} \right] \right\} \\ N_\phi &= -\iint_S J_x \sin \phi_s e^{j\beta(x' \sin \theta_s \cos \phi_s + y' \sin \theta_s \sin \phi_s)} dx' dy' \\ &= -\frac{2E_0}{\eta} ab \left\{ \cos \theta_i \sin \theta_s \left[\frac{\sin(X)}{X} \right] \left[\frac{\sin(Y)}{Y} \right] \right\}\end{aligned}$$

where

$$\begin{aligned}X &= \frac{\beta a}{2} \sin \theta_s \cos \phi_s \\ Y &= \frac{\beta b}{2} (\sin \theta_s \sin \phi_s - \sin \theta_i)\end{aligned}$$

The scattered fields are obtained using (6-122a) through (6-122f) and can be written as

$$\begin{aligned}E_r^s &\simeq H_r^s \simeq 0 \\ E_\theta^s &\simeq -C_1 \frac{e^{-j\beta r}}{r} \left\{ \cos \theta_i \cos \theta_s \cos \phi_s \left[\frac{\sin(X)}{X} \right] \left[\frac{\sin(Y)}{Y} \right] \right\} \\ E_\phi^s &\simeq C_1 \frac{e^{-j\beta r}}{r} \left\{ \cos \theta_i \sin \theta_s \left[\frac{\sin(X)}{X} \right] \left[\frac{\sin(Y)}{Y} \right] \right\} \\ H_\theta^s &\simeq -\frac{E_\phi^s}{\eta} \quad H_\phi^s \simeq +\frac{E_\theta^s}{\eta} \\ C_1 &= j \frac{ab\beta E_0}{2\pi}\end{aligned}$$

In the principal H plane ($\phi_s = \pi/2$), on which the maximum also lies, the electric field components reduce to

$$E_r^s \simeq E_\theta^s \simeq 0$$

$$E_\phi^s \simeq C_1 \frac{e^{-j\beta r}}{r} \left\{ \cos \theta_i \left[\frac{\sin \left[\frac{\beta b}{2} (\sin \theta_s - \sin \theta_i) \right]}{\frac{\beta b}{2} (\sin \theta_s - \sin \theta_i)} \right] \right\}$$

whose maximum value always occurs when $\theta_s = \theta_i$ (no matter what the size of the ground plane).

The bistatic and monostatic RCSs can be expressed, respectively, as

$$\sigma_{3-D}(\text{bistatic}) = 4\pi \left(\frac{ab}{\lambda} \right)^2 [\cos^2 \theta_i (\cos^2 \theta_s \cos^2 \phi_s + \sin^2 \phi_s)]$$

$$\times \left[\frac{\sin(X)}{X} \right]^2 \left[\frac{\sin(Y)}{Y} \right]^2$$

$$\sigma_{3-D}(\text{monostatic}) = 4\pi \left(\frac{ab}{\lambda} \right)^2 \cos^2 \theta_i \left[\frac{\sin(\beta b \sin \theta_i)}{\beta b \sin \theta_i} \right]^2$$

They are shown plotted, respectively, in Figures 11-9a and 11-9b for a square plate of $a = b = 5\lambda$.

11.4 CYLINDRICAL WAVE TRANSFORMATIONS AND THEOREMS

In scattering, it is often most convenient to express wave functions of one coordinate system in terms of wave functions of another coordinate system. An example is a uniform plane wave that can be written in a very simple form in terms of rectilinear wave functions. However, when scattering of plane waves by cylindrical structures is considered, it is most desirable to transform the rectilinear form of the uniform plane wave into terms of cylindrical wave functions. This is desirable because the surface of the cylindrical structure is most conveniently defined using cylindrical coordinates. This and other such transformations are referred to as *wave transformations* [11]. Along with these transformations, certain theorems concerning cylindrical wave functions are very desirable in describing the scattering by cylindrical structures.

11.4.1 Plane Waves in Terms of Cylindrical Wave Functions

The scattering of plane waves by cylindrical structures is considered a fundamental problem in scattering theory. To accomplish this, it is first necessary and convenient to express the plane waves by cylindrical wave functions. To demonstrate that, let us assume that a normalized uniform plane wave traveling in the $+x$ direction, as shown in Figure 11-10, can be written as

$$\mathbf{E} = \hat{\mathbf{a}}_z E_z^+ = \hat{\mathbf{a}}_z E_0 e^{-j\beta x} = \hat{\mathbf{a}}_z e^{-j\beta x} \quad (11-45)$$

The plane wave can be represented by an infinite sum of cylindrical wave functions of the form

$$E_z^+ = e^{-j\beta x} = e^{-j\beta \rho \cos \phi} = \sum_{n=-\infty}^{+\infty} a_n J_n(\beta \rho) e^{jn\phi} \quad (11-45a)$$

since it must be periodic in ϕ and finite at $\rho = 0$. The next step is to determine the amplitude coefficients, a_n .

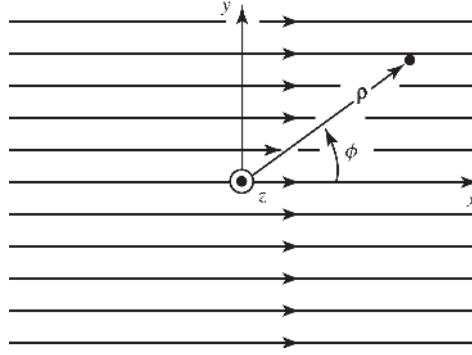


Figure 11-10 Uniform plane wave traveling in the $+x$ direction.

Multiplying both sides of (11-45a) by $e^{-jm\phi}$, where m is an integer, and integrating from 0 to 2π , we have that

$$\int_0^{2\pi} e^{-j(\beta\rho \cos\phi + m\phi)} d\phi = \int_0^{2\pi} \left[\sum_{n=-\infty}^{+\infty} a_n J_n(\beta\rho) e^{j(n-m)\phi} \right] d\phi \quad (11-46)$$

Interchanging integration and summation, we have that

$$\int_0^{2\pi} e^{-j(\beta\rho \cos\phi + m\phi)} d\phi = \sum_{n=-\infty}^{+\infty} a_n J_n(\beta\rho) \int_0^{2\pi} e^{j(n-m)\phi} d\phi \quad (11-47)$$

Using the orthogonality condition

$$\int_0^{2\pi} e^{j(n-m)\phi} d\phi = \begin{cases} 2\pi & n = m \\ 0 & n \neq m \end{cases} \quad (11-48)$$

the right side of (11-47) reduces to

$$\sum_{n=-\infty}^{+\infty} a_n J_n(\beta\rho) \int_0^{2\pi} e^{j(n-m)\phi} d\phi \stackrel{n=m}{=} 2\pi a_m J_m(\beta\rho) \quad (11-49)$$

Using the integral

$$\int_0^{2\pi} e^{+j(z \cos\phi + n\phi)} d\phi = 2\pi j^n J_n(z) \quad (11-50)$$

the left side of (11-47) can be written as

$$\int_0^{2\pi} e^{-j(\beta\rho \cos\phi + m\phi)} d\phi = 2\pi j^{-m} J_{-m}(-\beta\rho) \quad (11-51)$$

Since

$$J_{-m}(x) = (-1)^m J_m(x) \quad (11-52a)$$

and

$$J_m(-x) = (-1)^m J_m(x) \quad (11-52b)$$

(11-51) can be written as

$$\begin{aligned} \int_0^{2\pi} e^{-j(\beta\rho \cos \phi + m\phi)} d\phi &= 2\pi j^{-m} J_{-m}(-\beta\rho) = 2\pi j^{-m} (-1)^m J_m(-\beta\rho) \\ &= 2\pi j^{-m} (-1)^m (-1)^m J_m(\beta\rho) = 2\pi j^{-m} (-1)^{2m} J_m(\beta\rho) \\ \int_0^{2\pi} e^{-j(\beta\rho \cos \phi + m\phi)} d\phi &= 2\pi j^{-m} J_{-m}(-\beta\rho) = 2\pi j^{-m} J_m(\beta\rho) \end{aligned} \quad (11-53)$$

Using (11-49) and (11-53), reduce (11-47) to

$$2\pi j^{-m} J_m(\beta\rho) = 2\pi a_m J_m(\beta\rho) \quad (11-54)$$

Thus,

$$a_m = j^{-m} \quad (11-54a)$$

Therefore, (11-45a) can be written as

$$E_z^+ = e^{-j\beta x} = e^{-j\beta\rho \cos \phi} = \sum_{n=-\infty}^{+\infty} a_n J_n(\beta\rho) e^{jn\phi} = \sum_{n=-\infty}^{+\infty} j^{-n} J_n(\beta\rho) e^{jn\phi} \quad (11-55a)$$

In a similar manner it can be shown that

$$E_z^- = e^{+j\beta x} = e^{+j\beta\rho \cos \phi} = \sum_{n=-\infty}^{+\infty} j^{+n} J_n(\beta\rho) e^{jn\phi} \quad (11-55b)$$

11.4.2 Addition Theorem of Hankel Functions

A transformation that is often convenient and necessary in scattering problems is the *addition theorem of Hankel functions* [11]. Basically, it expresses the fields of a cylindrical line source located away from the origin at a radial distance ρ' , which are represented by cylindrical wave functions originating at the source, in terms of cylindrical wave functions originating at the origin ($\rho = 0$) of the coordinate system.

To derive this, let us assume that a line source of electric current I_0 is located at $\rho = \rho'$ and $\phi = \phi'$, as shown in Figure 11-11. According to (11-11a), the fields by the line source are given by

$$E_z(\rho, \phi) = -\frac{\beta^2 I_0}{4\omega\epsilon} H_0^{(2)}(\beta|\rho - \rho'|) \quad (11-56)$$

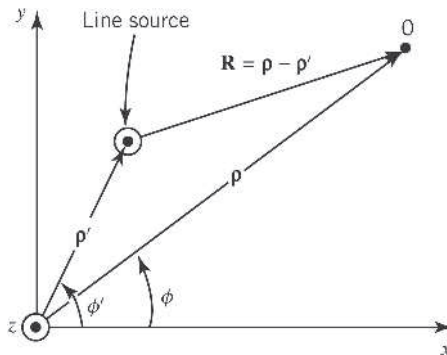


Figure 11-11 Geometry for displaced line source.

where $|\rho - \rho'|$ is the radial distance from the source to the observation point. Using Figure 11-11 and the law of cosines, (11-56) can also be written as

$$\begin{aligned} E_z(\rho, \phi) &= -\frac{\beta^2 I_0}{4\omega\epsilon} H_0^{(2)}(\beta|\rho - \rho'|) \\ &= -\frac{\beta^2 I_0}{4\omega\epsilon} H_0^{(2)}\left[\beta\sqrt{\rho^2 + (\rho')^2 - 2\rho\rho'\cos(\phi - \phi')}\right] \end{aligned} \quad (11-56a)$$

Equations 11-56 and 11-56a express the electric field in terms of a Hankel function whose radial distance originates at the source ($|\rho - \rho'| = 0$). For scattering problems, it is very convenient to express the field in terms of cylindrical wave functions, such as Bessel and Hankel functions, whose radial distance originates at the origin ($\rho = 0$). Based upon that and because of reciprocity, the field must be symmetric in terms of the primed and unprimed coordinates. Therefore, the permissible wave functions, whose radial distance originates at the origin of the coordinate system, to represent the Hankel function of (11-56a) are of the form

$$\begin{aligned} &\underline{\rho \leq \rho'} \\ &f(\beta\rho')J_n(\beta\rho)e^{jn(\phi - \phi')} \end{aligned}$$

where n is an integer, because they must be finite at $\rho = 0$ and be periodic with period 2π .

$$\begin{aligned} &\underline{\rho \geq \rho'} \\ &g(\beta\rho')H_n^{(2)}(\beta\rho)e^{jn(\phi - \phi')} \end{aligned}$$

where n is an integer, because they must represent outward traveling waves and they must be periodic with period 2π .

Thus, the Hankel function of (11-56a) can be written as

$$H_0^{(2)}(\beta|\rho - \rho'|) = \begin{cases} \sum_{n=-\infty}^{+\infty} c_n f(\beta\rho')J_n(\beta\rho)e^{jn(\phi - \phi')} & \text{for } \rho \leq \rho' \\ \sum_{n=-\infty}^{+\infty} d_n g(\beta\rho')H_n^{(2)}(\beta\rho)e^{jn(\phi - \phi')} & \text{for } \rho \geq \rho' \end{cases} \quad (11-57a)$$

$$\sum_{n=-\infty}^{+\infty} d_n g(\beta\rho')H_n^{(2)}(\beta\rho)e^{jn(\phi - \phi')} \quad \text{for } \rho \geq \rho' \quad (11-57b)$$

Since at $\rho = \rho'$ the fields of the two regions must be continuous, then from (11-57a) and (11-57b)

$$c_n f(\beta\rho')J_n(\beta\rho') = d_n g(\beta\rho')H_n^{(2)}(\beta\rho') \quad (11-58)$$

which can be satisfied provided

$$c_n = d_n = b_n \quad (11-59a)$$

$$f(\beta\rho') = H_n^{(2)}(\beta\rho') \quad (11-59b)$$

$$g(\beta\rho') = J_n(\beta\rho') \quad (11-59c)$$

Using (11-59a) through (11-59c), we can write (11-57a) and (11-57b) as

$$H_0^{(2)}(\beta|\rho - \rho'|) = \begin{cases} \sum_{n=-\infty}^{+\infty} b_n J_n(\beta\rho)H_n^{(2)}(\beta\rho')e^{jn(\phi - \phi')} & \text{for } \rho \leq \rho' \\ \sum_{n=-\infty}^{+\infty} b_n J_n(\beta\rho')H_n^{(2)}(\beta\rho)e^{jn(\phi - \phi')} & \text{for } \rho \geq \rho' \end{cases} \quad (11-60a)$$

$$\sum_{n=-\infty}^{+\infty} b_n J_n(\beta\rho')H_n^{(2)}(\beta\rho)e^{jn(\phi - \phi')} \quad \text{for } \rho \geq \rho' \quad (11-60b)$$

The only remaining part is to evaluate b_n . This can be accomplished by returning to (11-56a), according to which

$$H_0^{(2)}(\beta|\mathbf{p}-\mathbf{p}'|) = H_0^{(2)}\left[\beta\sqrt{\rho^2 + (\rho')^2 - 2\rho\rho'\cos(\phi-\phi')}\right] \quad (11-61)$$

Moving the source toward infinity ($\rho' \rightarrow \infty$) along $\phi' = 0$, the radial distance in (11-61), as represented by the square root, can be approximated using the binomial expansion

$$\begin{aligned} \sqrt{\rho^2 + (\rho')^2 - 2\rho\rho'\cos(\phi-\phi')} &\stackrel{\phi'=0}{\underset{\rho'\rightarrow\infty}{\simeq}} \sqrt{(\rho')^2 - 2\rho\rho'\cos\phi} = \rho'\sqrt{1 - 2\left(\frac{\rho}{\rho'}\right)\cos\phi} \\ &\simeq \rho'\left(1 - \frac{\rho}{\rho'}\cos\phi\right) = \rho' - \rho\cos\phi \end{aligned} \quad (11-62)$$

Using (11-62) allows us to write (11-61) as

$$\begin{aligned} H_0^{(2)}(\beta|\mathbf{p}-\mathbf{p}'|) &= H_0^{(2)}\left[\beta\sqrt{\rho^2 + (\rho')^2 - 2\rho\rho'\cos(\phi-\phi')}\right] \\ &\stackrel{\phi'=0}{\underset{\rho'\rightarrow\infty}{\simeq}} H_0^{(2)}[\beta(\rho' - \rho\cos\phi)] \end{aligned} \quad (11-63)$$

With the aid of the asymptotic form of the Hankel function for large argument

$$H_0^{(2)}(\alpha x) \stackrel{\alpha x \rightarrow \infty}{\simeq} \sqrt{\frac{2j}{\pi\alpha x}} j^n e^{-j\alpha x} \quad (11-64)$$

(11-63) reduces to

$$\begin{aligned} H_0^{(2)}(\beta|\mathbf{p}-\mathbf{p}'|) &\stackrel{\phi'=0}{\underset{\rho'\rightarrow\infty}{\simeq}} H_0^{(2)}[\beta(\rho' - \rho\cos\phi)] \simeq \sqrt{\frac{2j}{\pi\beta(\rho' - \rho\cos\phi)}} e^{-j\beta(\rho' - \rho\cos\phi)} \\ &\simeq \sqrt{\frac{2j}{\pi\beta\rho'}} e^{-j\beta\rho'} e^{+j\beta\rho\cos\phi} \end{aligned} \quad (11-65)$$

Using (11-55b) allows us to write (11-65) as

$$H_0^{(2)}(\beta|\mathbf{p}-\mathbf{p}'|) \stackrel{\phi'=0}{\underset{\rho'\rightarrow\infty}{\simeq}} \sqrt{\frac{2j}{\pi\beta\rho'}} e^{-j\beta\rho'} \sum_{n=-\infty}^{+\infty} j^n J_n(\beta\rho) e^{jn\phi} \quad (11-66)$$

Applying (11-64) to (11-60a) for $\phi' = 0$ and $\rho' \rightarrow \infty$ reduces it to

$$\begin{aligned} H_0^{(2)}(\beta|\mathbf{p}-\mathbf{p}'|) &\stackrel{\phi'=0}{\underset{\rho'\rightarrow\infty}{\simeq}} \sum_{n=-\infty}^{+\infty} b_n J_n(\beta\rho) H_n^{(2)}(\beta\rho') e^{jn\phi} \\ &\stackrel{\phi'=0}{\underset{\rho'\rightarrow\infty}{\simeq}} \sqrt{\frac{2j}{\pi\beta\rho'}} e^{-j\beta\rho'} \sum_{n=-\infty}^{+\infty} b_n j^n J_n(\beta\rho) e^{jn\phi} \end{aligned} \quad (11-67)$$

Comparing (11-66) and (11-67) leads to

$$b_n = 1 \quad (11-68)$$

Thus, the final form of (11-60a) and (11-60b) is

$$H_0^{(2)}(\beta|\mathbf{p}-\mathbf{p}'|) = \begin{cases} \sum_{n=-\infty}^{+\infty} J_n(\beta\rho)H_n^{(2)}(\beta\rho')e^{jn(\phi-\phi')} & \text{for } \rho \leq \rho' \\ \sum_{n=-\infty}^{+\infty} J_n(\beta\rho')H_n^{(2)}(\beta\rho)e^{jn(\phi-\phi')} & \text{for } \rho \geq \rho' \end{cases} \quad (11-69a)$$

$$H_0^{(2)}(\beta|\mathbf{p}-\mathbf{p}'|) = \begin{cases} \sum_{n=-\infty}^{+\infty} J_n(\beta\rho)H_n^{(2)}(\beta\rho')e^{jn(\phi-\phi')} & \text{for } \rho \leq \rho' \\ \sum_{n=-\infty}^{+\infty} J_n(\beta\rho')H_n^{(2)}(\beta\rho)e^{jn(\phi-\phi')} & \text{for } \rho \geq \rho' \end{cases} \quad (11-69b)$$

which can be used to write (11-56) as

$$E_z(\rho, \phi) = -\frac{\beta^2 I_0}{4\omega\varepsilon} H_0^{(2)}(\beta|\mathbf{p}-\mathbf{p}'|)$$

$$= -\frac{\beta^2 I_0}{4\omega\varepsilon} \begin{cases} \sum_{n=-\infty}^{+\infty} J_n(\beta\rho)H_n^{(2)}(\beta\rho')e^{jn(\phi-\phi')} & \text{for } \rho \leq \rho' \\ \sum_{n=-\infty}^{+\infty} J_n(\beta\rho')H_n^{(2)}(\beta\rho)e^{jn(\phi-\phi')} & \text{for } \rho \geq \rho' \end{cases} \quad (11-70a)$$

$$\sum_{n=-\infty}^{+\infty} J_n(\beta\rho')H_n^{(2)}(\beta\rho)e^{jn(\phi-\phi')} \quad \text{for } \rho \geq \rho' \quad (11-70b)$$

The procedure can be repeated to expand $H_0^{(1)}(\beta|\mathbf{p}-\mathbf{p}'|)$. However, it is obvious from the results of (11-69a) and (11-69b) that

$$H_0^{(1)}(\beta|\mathbf{p}-\mathbf{p}'|) = \begin{cases} \sum_{n=-\infty}^{+\infty} J_n(\beta\rho)H_n^{(1)}(\beta\rho')e^{jn(\phi-\phi')} & \text{for } \rho \leq \rho' \\ \sum_{n=-\infty}^{+\infty} J_n(\beta\rho')H_n^{(1)}(\beta\rho)e^{jn(\phi-\phi')} & \text{for } \rho \geq \rho' \end{cases} \quad (11-71a)$$

$$\sum_{n=-\infty}^{+\infty} J_n(\beta\rho')H_n^{(1)}(\beta\rho)e^{jn(\phi-\phi')} \quad \text{for } \rho \geq \rho' \quad (11-71b)$$

11.4.3 Addition Theorem for Bessel Functions

Another theorem that is often useful represents Bessel functions originating at the source, which is located away from the origin, in terms of cylindrical wave functions originating at the origin of the coordinate system. This is usually referred to as the *addition theorem for Bessel functions* [11].

We know that the Hankel functions of the first and second kinds can be written, in terms of the Bessel functions, as

$$H_0^{(1)}(\beta\rho) = J_0(\beta\rho) + jY_0(\beta\rho) \quad (11-72a)$$

$$H_0^{(2)}(\beta\rho) = J_0(\beta\rho) - jY_0(\beta\rho) \quad (11-72b)$$

Adding the two, we can write that

$$J_0(\beta\rho) = \frac{1}{2}[H_0^{(1)}(\beta\rho) + H_0^{(2)}(\beta\rho)] \quad (11-73)$$

Therefore,

$$J_0(\beta|\mathbf{p}-\mathbf{p}'|) = \frac{1}{2}[H_0^{(1)}(\beta|\mathbf{p}-\mathbf{p}'|) + H_0^{(2)}(\beta|\mathbf{p}-\mathbf{p}'|)] \quad (11-74)$$

With the aid of (11-69a), (11-69b) and (11-71a), (11-71b) we can write (11-74) as

$$J_0(\beta|\rho - \rho'|) = \begin{cases} \sum_{n=-\infty}^{+\infty} \frac{1}{2} [H_n^{(1)}(\beta\rho') + H_n^{(2)}(\beta\rho')] J_n(\beta\rho) e^{jn(\phi - \phi')} & \text{for } \rho \leq \rho' \quad (11-75a) \\ \sum_{n=-\infty}^{+\infty} \frac{1}{2} [H_n^{(1)}(\beta\rho) + H_n^{(2)}(\beta\rho)] J_n(\beta\rho') e^{jn(\phi - \phi')} & \text{for } \rho \geq \rho' \quad (11-75b) \end{cases}$$

Using the forms of (11-72a) and (11-72b) for n th order Bessel and Hankel functions, we can write (11-75a) and (11-75b) as

$$J_0(\beta|\rho - \rho'|) = \begin{cases} \sum_{n=-\infty}^{+\infty} J_n(\beta\rho') J_n(\beta\rho) e^{jn(\phi - \phi')} & \text{for } \rho \leq \rho' \quad (11-76a) \\ \sum_{n=-\infty}^{+\infty} J_n(\beta\rho) J_n(\beta\rho') e^{jn(\phi - \phi')} & \text{for } \rho \geq \rho' \quad (11-76b) \end{cases}$$

or that

$$J_0(\beta|\rho - \rho'|) = \sum_{n=-\infty}^{+\infty} J_n(\beta\rho) J_n(\beta\rho') e^{jn(\phi - \phi')} \quad \text{for } \rho \begin{matrix} \leq \\ \geq \end{matrix} \rho' \quad (11-77)$$

Subtracting (11-72b) from (11-72a), we can write that

$$\begin{aligned} H_0^{(1)}(\beta\rho) - H_0^{(2)}(\beta\rho) &= 2jY_0(\beta\rho) \\ Y_0(\beta\rho) &= \frac{1}{2j} [H_0^{(1)}(\beta\rho) - H_0^{(2)}(\beta\rho)] \end{aligned} \quad (11-78a)$$

or

$$Y_0(\beta|\rho - \rho'|) = \frac{1}{2j} [H_0^{(1)}(\beta|\rho - \rho'|) - H_0^{(2)}(\beta|\rho - \rho'|)] \quad (11-78b)$$

Using (11-69a), (11-69b), (11-71a), and (11-71b), we can write (11-78b) as

$$Y_0(\beta|\rho - \rho'|) = \begin{cases} \sum_{n=-\infty}^{+\infty} \frac{1}{2} [H_n^{(1)}(\beta\rho') - H_n^{(2)}(\beta\rho')] J_n(\beta\rho) e^{jn(\phi - \phi')} & \text{for } \rho \leq \rho' \quad (11-79a) \\ \sum_{n=-\infty}^{+\infty} \frac{1}{2} [H_n^{(1)}(\beta\rho) - H_n^{(2)}(\beta\rho)] J_n(\beta\rho') e^{jn(\phi - \phi')} & \text{for } \rho \geq \rho' \quad (11-79b) \end{cases}$$

Using the forms (11-72a) and (11-72b) for n th order Hankel functions, we can write (11-79a) and (11-79b) as

$$Y_0(\beta|\rho-\rho'|) = \begin{cases} \sum_{n=-\infty}^{+\infty} \frac{1}{2j} [J_n(\beta\rho') + jY_n(\beta\rho') - J_n(\beta\rho') + jY_n(\beta\rho')] J_n(\beta\rho') e^{jn(\phi-\phi')} \\ \quad = \sum_{n=-\infty}^{+\infty} Y_n(\beta\rho') J_n(\beta\rho) e^{jn(\phi-\phi')} & \text{for } \rho \leq \rho' \\ \sum_{n=-\infty}^{+\infty} \frac{1}{2j} [J_n(\beta\rho) + jY_n(\beta\rho) - J_n(\beta\rho) + jY_n(\beta\rho)] J_n(\beta\rho') e^{jn(\phi-\phi')} \\ \quad = \sum_{n=-\infty}^{+\infty} Y_n(\beta\rho) J_n(\beta\rho') e^{jn(\phi-\phi')} & \text{for } \rho \geq \rho' \end{cases} \quad (11-80a)$$

$$(11-80b)$$

11.4.4 Summary of Cylindrical Wave Transformations and Theorems

The following are the most prominent cylindrical wave transformations and theorems that are very convenient for scattering from cylindrical scatterers:

$$e^{-j\beta x} = e^{-j\beta\rho \cos\phi} = \sum_{n=-\infty}^{+\infty} j^{-n} J_n(\beta\rho) e^{jn\phi} \quad (11-81a)$$

$$e^{+j\beta x} = e^{+j\beta\rho \cos\phi} = \sum_{n=-\infty}^{+\infty} j^{+n} J_n(\beta\rho) e^{jn\phi} \quad (11-81b)$$

$$H_0^{(1,2)}(\beta|\rho-\rho'|) = \begin{cases} \sum_{n=-\infty}^{+\infty} J_n(\beta\rho) H_n^{(1,2)}(\beta\rho') e^{jn(\phi-\phi')} & \rho \leq \rho' \\ \sum_{n=-\infty}^{+\infty} J_n(\beta\rho') H_n^{(1,2)}(\beta\rho) e^{jn(\phi-\phi')} & \rho \geq \rho' \end{cases} \quad (11-82a)$$

$$(11-82b)$$

$$J_0(\beta|\rho-\rho'|) = \sum_{n=-\infty}^{+\infty} J_n(\beta\rho') J_n(\beta\rho) e^{jn(\phi-\phi')} \quad \rho \begin{matrix} > \\ = \\ < \end{matrix} \rho' \quad (11-83)$$

$$Y_0(\beta|\rho-\rho'|) = \begin{cases} \sum_{n=-\infty}^{+\infty} Y_n(\beta\rho') J_n(\beta\rho) e^{jn(\phi-\phi')} & \rho \leq \rho' \\ \sum_{n=-\infty}^{+\infty} Y_n(\beta\rho) J_n(\beta\rho') e^{jn(\phi-\phi')} & \rho \geq \rho' \end{cases} \quad (11-84a)$$

$$(11-84b)$$

11.5 SCATTERING BY CIRCULAR CYLINDERS

Cylinders represent one of the most important classes of geometrical surfaces. The surface of many practical scatterers, such as the fuselage of airplanes, missiles, and so on, can often be represented by cylindrical structures. The circular cylinder, because of its simplicity and the fact that its solution is represented in terms of well known and tabulated functions (such as Bessel and Hankel functions), is probably one of the geometries most widely used to represent practical scatterers [12]. Because of its importance, it will be examined here in some detail. We will consider scattering of both plane and cylindrical waves by circular conducting cylinders of infinite length at normal and oblique incidences. The solutions will be obtained using modal techniques. Scattering from finite length cylinders is obtained by transforming the scattered fields of infinite lengths using approximate relationships. Scattering by dielectric and dielectric covered cylinders are assigned to the reader as end-of-chapter exercises.

11.5.1 Normal Incidence Plane Wave Scattering by a Conducting Circular Cylinder: TM^z Polarization

Let us assume that a TM^z uniform plane wave is normally incident upon a perfectly conducting circular cylinder of radius a , as shown in Figure 11-12a, and the electric field can be written as

$$\mathbf{E}^i = \hat{\mathbf{a}}_z E_z^i = \hat{\mathbf{a}}_z E_0 e^{-j\beta x} = \hat{\mathbf{a}}_z E_0 e^{-j\beta \rho \cos \phi} \quad (11-85)$$

which, according to the transformation (11-55a) or (11-81a), can also be expressed as

$$\mathbf{E}^i = \hat{\mathbf{a}}_z E_z^i = \hat{\mathbf{a}}_z E_0 \sum_{n=-\infty}^{+\infty} j^{-n} J_n(\beta \rho) e^{jn\phi} = \hat{\mathbf{a}}_z E_0 \sum_{n=0}^{\infty} (-j)^n \varepsilon_n J_n(\beta \rho) \cos(n\phi) \quad (11-85a)$$

where

$$\varepsilon_n = \begin{cases} 1 & n = 0 \\ 2 & n \neq 0 \end{cases} \quad (11-85b)$$

The corresponding magnetic field components can be obtained by using Maxwell's Faraday equation, which for this problem reduces to

$$\mathbf{H}^i = -\frac{1}{j\omega\mu} \nabla \times \mathbf{E}^i = -\frac{1}{j\omega\mu} \left(\hat{\mathbf{a}}_\rho \frac{1}{\rho} \frac{\partial E_z^i}{\partial \phi} - \hat{\mathbf{a}}_\phi \frac{\partial E_z^i}{\partial \rho} \right) \quad (11-86)$$

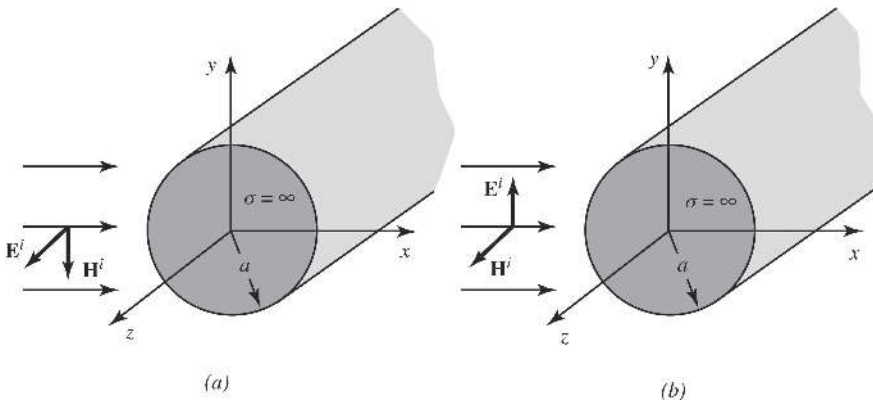


Figure 11-12 Uniform plane wave incident on a conducting circular cylinder. (a) TM^z . (b) TE^z .

or

$$H_\rho^i = -\frac{1}{j\omega\mu} \frac{1}{\rho} \frac{\partial E_z^i}{\partial \phi} = -\frac{E_0}{j\omega\mu} \frac{1}{\rho} \sum_{n=-\infty}^{+\infty} n j^{-n+1} J_n(\beta\rho) e^{jn\phi} \quad (11-86a)$$

$$H_\phi^i = \frac{1}{j\omega\mu} \frac{\partial E_z^i}{\partial \rho} = \frac{\beta E_0}{j\omega\mu} \sum_{n=-\infty}^{+\infty} j^{-n} J_n'(\beta\rho) e^{jn\phi} \quad (11-86b)$$

$$' \equiv \frac{\partial}{\partial(\beta\rho)} \quad (11-86c)$$

It should be noted here that throughout this chapter the prime indicates partial derivative with respect to the entire argument of the Bessel or Hankel function.

In the presence of the conducting cylinder, the total field E_z^t according to (11-1a) can be written as

$$\mathbf{E}^t = \mathbf{E}^i + \mathbf{E}^s \quad (11-87)$$

where $\mathbf{E}^s$ is the scattered field. Since the scattered fields travel in the outward direction, they must be represented by cylindrical traveling wave functions. Thus, we choose to represent $\mathbf{E}^s$ by

$$\mathbf{E}^s = \hat{\mathbf{a}}_z E_z^s = \hat{\mathbf{a}}_z E_0 \sum_{n=-\infty}^{+\infty} c_n H_n^{(2)}(\beta\rho) \quad (11-88)$$

where c_n represents the yet unknown amplitude coefficients. Equation 11-88 is chosen to be of similar form to (11-85a) since the two together will be used to represent the total field. This becomes convenient when we attempt to solve for the amplitude coefficients c_n .

The unknown amplitude coefficients c_n can be found by applying the boundary condition

$$\mathbf{E}^t = \hat{\mathbf{a}}_z E_z^t(\rho = a, 0 \leq \phi \leq 2\pi, z) = 0 \quad (11-89)$$

Using (11-85a), (11-88), and (11-89) we can write that

$$E_z^t(\rho = a, 0 \leq \phi \leq 2\pi, z) = E_0 \sum_{n=-\infty}^{\infty} \left[j^{-n} J_n(\beta a) e^{jn\phi} + c_n H_n^{(2)}(\beta a) \right] = 0 \quad (11-90)$$

or

$$c_n = -j^{-n} \frac{J_n(\beta a)}{H_n^{(2)}(\beta a)} e^{jn\phi} \quad (11-90a)$$

Thus, the scattered field of (11-88) reduces to

$$\begin{aligned} E_z^s &= -E_0 \sum_{n=-\infty}^{+\infty} j^{-n} \frac{J_n(\beta a)}{H_n^{(2)}(\beta a)} H_n^{(2)}(\beta\rho) e^{jn\phi} \\ &= -E_0 \sum_{n=0}^{+\infty} (-j)^n \varepsilon_n \frac{J_n(\beta a)}{H_n^{(2)}(\beta a)} H_n^{(2)}(\beta\rho) \cos(n\phi) \end{aligned} \quad (11-91)$$

where ε_n is defined by (11-85b).

The corresponding scattered magnetic field components can be obtained by using Maxwell's equation 11-86, which leads to

$$H_\rho^s = -\frac{1}{j\omega\mu} \frac{1}{\rho} \frac{\partial E_z^s}{\partial \phi} = -\frac{E_0}{j\omega\mu} \frac{1}{\rho} \sum_{n=-\infty}^{+\infty} nj^{-n+1} \frac{J_n(\beta a)}{H_n^{(2)}(\beta a)} H_n^{(2)}(\beta \rho) e^{jn\phi} \quad (11-92a)$$

$$H_\phi^s = \frac{1}{j\omega\mu} \frac{\partial E_z^s}{\partial \rho} = -\frac{\beta E_0}{j\omega\mu} \sum_{n=-\infty}^{+\infty} j^{-n} \frac{J_n(\beta a)}{H_n^{(2)}(\beta a)} H_n^{(2)'}(\beta \rho) e^{jn\phi} \quad (11-92b)$$

Thus, the total electric and magnetic field components can be written as

$$E_\rho^t = E_\phi^t = H_z^t = 0 \quad (11-93a)$$

$$E_z^t = E_0 \sum_{n=-\infty}^{+\infty} j^{-n} \left[J_n(\beta \rho) - \frac{J_n(\beta a)}{H_n^{(2)}(\beta a)} H_n^{(2)}(\beta \rho) \right] e^{jn\phi} \quad (11-93b)$$

$$H_\rho^t = -\frac{E_0}{j\omega\mu} \frac{1}{\rho} \sum_{n=-\infty}^{+\infty} nj^{-n+1} \left[J_n(\beta \rho) - \frac{J_n(\beta a)}{H_n^{(2)}(\beta a)} H_n^{(2)}(\beta \rho) \right] e^{jn\phi} \quad (11-93c)$$

$$H_\phi^t = \frac{\beta E_0}{j\omega\mu} \sum_{n=-\infty}^{+\infty} j^{-n} \left[J_n'(\beta \rho) - \frac{J_n(\beta a)}{H_n^{(2)}(\beta a)} H_n^{(2)'}(\beta \rho) \right] e^{jn\phi} \quad (11-93d)$$

On the surface of the cylinder ($\rho = a$), the total tangential magnetic field can be written as

$$\begin{aligned} H_\phi^t(\rho = a) &= \frac{\beta E_0}{j\omega\mu} \sum_{n=-\infty}^{+\infty} j^{-n} \left[J_n'(\beta a) - \frac{J_n(\beta a)}{H_n^{(2)}(\beta a)} H_n^{(2)'}(\beta a) \right] e^{jn\phi} \\ &= \frac{\beta E_0}{\omega\mu} \sum_{n=-\infty}^{+\infty} j^{-n} \left[\frac{J_n(\beta a) Y_n'(\beta a) - J_n'(\beta a) Y_n(\beta a)}{H_n^{(2)}(\beta a)} \right] e^{jn\phi} \end{aligned} \quad (11-94)$$

Using the Wronskian of Bessel functions

$$J_n(\alpha \rho) Y_n'(\alpha \rho) - Y_n(\alpha \rho) J_n'(\alpha \rho) = \frac{2}{\pi \alpha \rho} \quad (11-95)$$

reduces (11-94) to

$$H_\phi^t(\rho = a) = \frac{2E_0}{\pi a \omega \mu} \sum_{n=-\infty}^{+\infty} j^{-n} \frac{e^{jn\phi}}{H_n^{(2)}(\beta a)} \quad (11-96)$$

Thus, the current induced on the surface of the cylinder can be written as

$$\begin{aligned} \mathbf{J}_s &= \hat{\mathbf{n}} \times \mathbf{H}^t |_{\rho=a} = \hat{\mathbf{a}}_\rho \times (\hat{\mathbf{a}}_\rho H_\rho^t + \hat{\mathbf{a}}_\phi H_\phi^t) |_{\rho=a} = \hat{\mathbf{a}}_z H_\phi^t(\rho = a) \\ &= \hat{\mathbf{a}}_z \frac{2E_0}{\pi a \omega \mu} \sum_{n=-\infty}^{+\infty} j^{-n} \frac{e^{jn\phi}}{H_n^{(2)}(\beta a)} \end{aligned} \quad (11-97)$$

A. Small Radius Approximation As the radius of the cylinder increases, more terms in the infinite series of (11-97) are needed to obtain convergence. However, for very small cylinders, like a very thin wire ($a \ll \lambda$), the first term ($n=0$) in (11-97) is dominant and is often sufficient to represent the induced current. Thus, for a very thin wire (11-97) can be approximated by

$$\mathbf{J}_s \stackrel{a \ll \lambda}{\simeq} \hat{\mathbf{a}}_z \frac{2E_0}{\pi a \omega \mu} \frac{1}{H_0^{(2)}(\beta a)} \quad (11-98)$$

where

$$\begin{aligned} H_0^{(2)}(\beta a) &= J_0(\beta a) - jY_0(\beta a) \stackrel{a \ll \lambda}{\simeq} 1 - j \frac{2}{\pi} \ln \left(\frac{\gamma \beta a}{2} \right) = 1 - j \frac{2}{\pi} \ln \left(\frac{1.781 \beta a}{2} \right) \\ &\stackrel{a \ll \lambda}{\simeq} -j \frac{2}{\pi} \ln \left(\frac{1.781 \beta a}{2} \right) \end{aligned} \quad (11-98a)$$

Thus, for a very thin wire the current density (11-98) can be approximated by

$$\mathbf{J}_s \stackrel{a \ll \lambda}{\simeq} \hat{\mathbf{a}}_z j \frac{E_0}{a \omega \mu} \frac{1}{\ln \left(\frac{1.781 \beta a}{2} \right)} \quad (11-98b)$$

B. Far-Zone Scattered Field

One of the most important parameters in scattering is the scattering width, which is obtained by knowing the scattered field in the far zone. For this problem, it can be accomplished by first reducing the scattered fields for far-zone observations ($\beta \rho \rightarrow \text{large}$). Referring to (11-91), the Hankel function can be approximated for observations made in the far field by

$$H_n^{(2)}(\beta \rho) \stackrel{\beta \rho \rightarrow \text{large}}{\simeq} \sqrt{\frac{2j}{\pi \beta \rho}} j^n e^{-j\beta \rho} \quad (11-99)$$

which, when substituted in (11-91), reduces it to

$$E_z^s \stackrel{\beta \rho \rightarrow \infty}{\simeq} -E_0 \sqrt{\frac{2j}{\pi \beta}} \frac{e^{-j\beta \rho}}{\sqrt{\rho}} \sum_{n=-\infty}^{+\infty} \frac{J_n(\beta a)}{H_n^{(2)}(\beta a)} e^{jn\phi} \quad (11-100)$$

The ratio of the far-zone scattered electric field to the incident field can then be written as

$$\begin{aligned} \frac{|E_z^s|}{|E_z^i|} &\stackrel{\beta \rho \rightarrow \text{large}}{\simeq} \frac{\left| -E_0 \sqrt{\frac{2j}{\pi \beta}} \frac{e^{-j\beta \rho}}{\sqrt{\rho}} \sum_{n=-\infty}^{+\infty} \frac{J_n(\beta a)}{H_n^{(2)}(\beta a)} e^{jn\phi} \right|}{|E_0 e^{-j\beta x}|} \\ &= \sqrt{\frac{2}{\pi \beta \rho}} \left| \sum_{n=-\infty}^{+\infty} \frac{J_n(\beta a)}{H_n^{(2)}(\beta a)} e^{jn\phi} \right| \end{aligned} \quad (11-101)$$

Thus, the scattering width of (11-21b) can be expressed as

$$\begin{aligned} \sigma_{2-D} &= \lim_{\rho \rightarrow \infty} \left[2\pi \rho \frac{|E_z^s|^2}{|E_z^i|^2} \right] = \frac{4}{\beta} \left| \sum_{n=-\infty}^{+\infty} \frac{J_n(\beta a)}{H_n^{(2)}(\beta a)} e^{jn\phi} \right|^2 \\ &= \frac{2\lambda}{\pi} \left| \sum_{n=0}^{+\infty} \varepsilon_n \frac{J_n(\beta a)}{H_n^{(2)}(\beta a)} \cos(n\phi) \right|^2 \end{aligned} \quad (11-102)$$

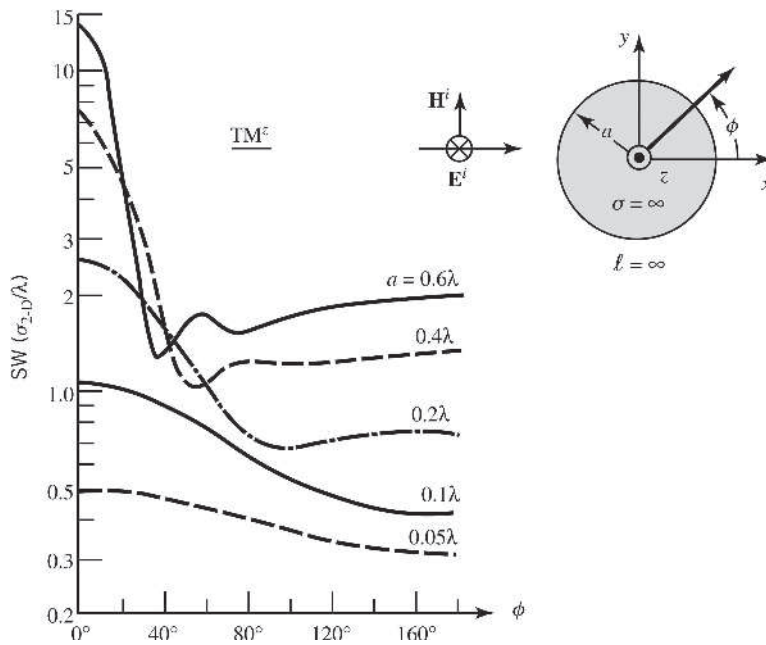


Figure 11-13 Two-dimensional TM^z bistatic scattering width (SW) of a circular conducting cylinder. (Courtesy of J. H. Richmond, Ohio State University.)

where

$$\varepsilon_n = \begin{cases} 1 & n = 0 \\ 2 & n \neq 0 \end{cases} \quad (11-102a)$$

Plots of the bistatic σ_{2-D} / λ computed using (11-102) are shown in Figure 11-13 for cylinder radii of $a = 0.05\lambda$, 0.1λ , 0.2λ , 0.4λ , and 0.6λ [13]. The backscattered ($\phi = 180^\circ$) patterns of σ_{2-D} / λ , as a function of the cylinder radius, are displayed in Figure 11-14 [13].

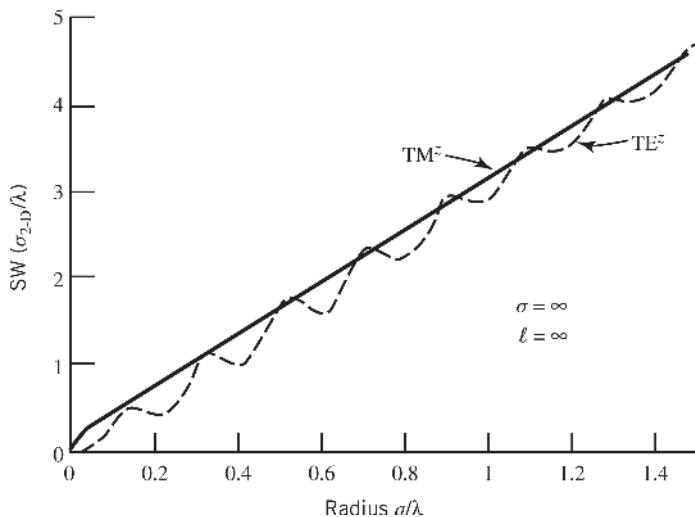


Figure 11-14 Two-dimensional monostatic (backscattered) scattering width for a circular conducting cylinder as a function of its radius. (Courtesy of J. H. Richmond, Ohio State University.)

For small radii ($a \ll \lambda$), the first term ($n=0$) in (11-102) is the dominant term, and it is sufficient to represent the scattered field. Thus, for small radii, the ratio of the Bessel to the Hankel function for $n=0$ can be approximated using (11-98a) by

$$\frac{J_0(\beta a)}{H_0^{(2)}(\beta a)} \stackrel{a \ll \lambda}{\simeq} \frac{1}{-j \frac{2}{\pi} \ln(0.89\beta a)} = j \frac{\pi}{2} \frac{1}{\ln(0.89\beta a)} \quad (11-103)$$

and (11-102) can then be reduced to

$$\sigma_{2-D} \stackrel{a \ll \lambda}{\simeq} \frac{2\lambda}{\pi} \left(\frac{\pi^2}{4} \right) \left| \frac{1}{\ln(0.89\beta a)} \right|^2 = \frac{\pi\lambda}{2} \left| \frac{1}{\ln(0.89\beta a)} \right|^2 \quad (11-103a)$$

This is independent of ϕ , which becomes evident in Figure 11-13 by the curves for the smaller values of a .

For a cylinder of finite length ℓ the three-dimensional radar cross section for normal incidence is related to the two-dimensional scattering width by (11-22e). Thus, using (11-102) and (11-103a), we can write the three-dimensional RCS (11-22e) as

$$\sigma_{3-D} \simeq \frac{4\ell^2}{\pi} \left| \sum_{n=-\infty}^{+\infty} \frac{J_n(\beta a)}{H_n^{(2)}(\beta a)} e^{jn\phi} \right|^2 \quad (11-104a)$$

$$\sigma_{3-D} \stackrel{a \ll \lambda}{\simeq} \pi \ell^2 \left| \frac{1}{\ln(0.89\beta a)} \right|^2 \quad (11-104b)$$

11.5.2 Normal Incidence Plane Wave Scattering by a Conducting Circular Cylinder: TE^z Polarization

Now let us assume that a TE^z uniform plane wave traveling in the $+x$ direction is normally incident upon a perfectly conducting circular cylinder of radius a , as shown in Figure 11-12b. The incident magnetic field can be written as

$$\begin{aligned} \mathbf{H}^i &= \hat{\mathbf{a}}_z H_0 e^{-j\beta x} = \hat{\mathbf{a}}_z H_0 e^{-j\beta \rho \cos \phi} = \hat{\mathbf{a}}_z H_0 \sum_{n=-\infty}^{+\infty} j^{-n} J_n(\beta \rho) e^{jn\phi} \\ &= \hat{\mathbf{a}}_z H_0 \sum_{n=0}^{\infty} (-j)^n \varepsilon_n J_n(\beta \rho) \cos(n\phi) \end{aligned} \quad (11-105)$$

where ε_n is defined by (11-85b). The corresponding incident electric field can be obtained by using Maxwell's Ampere equation, which for this problem reduces to

$$\mathbf{E}^i = \frac{1}{j\omega\varepsilon} \nabla \times \mathbf{H}^i = + \frac{1}{j\omega\varepsilon} \left[\hat{\mathbf{a}}_\rho \frac{1}{\rho} \frac{\partial H_z^i}{\partial \phi} - \hat{\mathbf{a}}_\phi \frac{\partial H_z^i}{\partial \rho} \right] \quad (11-106)$$

and by using (11-105) leads to

$$E_\rho^i = \frac{1}{j\omega\varepsilon} \frac{1}{\rho} \frac{\partial H_z^i}{\partial \phi} = \frac{H_0}{j\omega\varepsilon} \frac{1}{\rho} \sum_{n=-\infty}^{+\infty} n j^{-n+1} J_n(\beta \rho) e^{jn\phi} \quad (11-106a)$$

$$E_{\phi}^i = -\frac{1}{j\omega\varepsilon} \frac{\partial H_z^i}{\partial \rho} = -\frac{\beta H_0}{j\omega\varepsilon} \sum_{n=-\infty}^{+\infty} j^{-n} J_n'(\beta\rho) e^{jn\phi} \quad (11-106b)$$

The scattered magnetic field takes a form very similar to that of the scattered electric field of (11-88) for the TM^z polarization, and it can be written as

$$\mathbf{H}^s = \hat{\mathbf{a}}_z H_z^s = \hat{\mathbf{a}}_z H_0 \sum_{n=-\infty}^{+\infty} d_n H_n^{(2)}(\beta\rho) \quad (11-107)$$

where d_n represents the yet unknown amplitude coefficients that will be found by applying the appropriate boundary conditions.

Before the boundary conditions on the vanishing of the total tangential electric field on the surface of the cylinder can be applied, it is necessary to first find the corresponding electric fields. This can be accomplished by using Maxwell's equation 11-106 that, for the scattered magnetic field of (11-107), leads to

$$E_{\rho}^s = \frac{1}{j\omega\varepsilon} \frac{1}{\rho} \frac{\partial H_z^s}{\partial \phi} = \frac{H_0}{j\omega\varepsilon} \frac{1}{\rho} \sum_{n=-\infty}^{+\infty} H_n^{(2)}(\beta\rho) \frac{\partial d_n}{\partial \phi} \quad (11-108a)$$

$$E_{\phi}^s = -\frac{1}{j\omega\varepsilon} \frac{\partial H_z^s}{\partial \rho} = -\frac{\beta H_0}{j\omega\varepsilon} \sum_{n=-\infty}^{+\infty} d_n H_n^{(2)'}(\beta\rho) \quad (11-108b)$$

where ' indicates a partial derivate with respect to the entire argument of the Hankel function.

Since the cylinder is v electric field must vanish on its surface ($\rho = a$). Thus, using (11-106b) and (11-108b), we can write that

$$E_{\phi}^t(\rho = a, 0 \leq \phi \leq 2\pi, z) = -\frac{\beta H_0}{j\omega\varepsilon} \sum_{n=-\infty}^{+\infty} \left[j^{-n} J_n'(\beta a) e^{jn\phi} + d_n H_n^{(2)'}(\beta a) \right] = 0 \quad (11-109)$$

which is satisfied provided

$$d_n = -j^{-n} \frac{J_n'(\beta a)}{H_n^{(2)'}(\beta a)} e^{jn\phi} \quad (11-109a)$$

Thus, the scattered electric and magnetic fields can be written, using (11-107) and (11-109a), as

$$E_z^s = H_{\rho}^s = H_{\phi}^s = 0 \quad (11-110a)$$

$$E_{\rho}^s = -\frac{H_0}{j\omega\varepsilon} \frac{1}{\rho} \sum_{n=-\infty}^{+\infty} n j^{-n+1} \frac{J_n'(\beta a)}{H_n^{(2)'}(\beta a)} H_n^{(2)}(\beta\rho) e^{jn\phi} \quad (11-110b)$$

$$E_{\phi}^s = \frac{\beta H_0}{j\omega\varepsilon} \sum_{n=-\infty}^{+\infty} j^{-n} \frac{J_n'(\beta a)}{H_n^{(2)'}(\beta a)} H_n^{(2)'}(\beta\rho) e^{jn\phi} \quad (11-110c)$$

$$H_z^s = -H_0 \sum_{n=-\infty}^{+\infty} j^{-n} \frac{J_n'(\beta a)}{H_n^{(2)'}(\beta a)} H_n^{(2)}(\beta\rho) e^{jn\phi} \quad (11-110d)$$

The total electric and magnetic fields can now be expressed [using (11-105), (11-106a), (11-106b), and (11-110a) through (11-110d)] as

$$\begin{aligned} E_z^t &= H_\rho^t = H_\phi^t = 0 & (11-111a) \\ E_\rho^t &= \frac{H_0}{j\omega\varepsilon} \frac{1}{\rho} \sum_{n=-\infty}^{+\infty} nj^{-n+1} \left[J_n(\beta\rho) - \frac{J_n'(\beta a)}{H_n^{(2)'}(\beta a)} H_n^{(2)}(\beta\rho) \right] e^{jn\phi} & (11-111b) \\ E_\phi^t &= -\frac{\beta H_0}{j\omega\varepsilon} \sum_{n=-\infty}^{+\infty} j^{-n} \left[J_n'(\beta\rho) - \frac{J_n'(\beta a)}{H_n^{(2)'}(\beta a)} H_n^{(2)'}(\beta\rho) \right] e^{jn\phi} & (11-111c) \\ H_z^t &= H_0 \sum_{n=-\infty}^{+\infty} j^{-n} \left[J_n(\beta\rho) - \frac{J_n'(\beta a)}{H_n^{(2)'}(\beta a)} H_n^{(2)}(\beta\rho) \right] e^{jn\phi} & (11-111d) \end{aligned}$$

On the surface of the cylinder ($\rho = a$), the total tangential magnetic field can be written as

$$\begin{aligned} H_z^t(\rho = a) &= H_0 \sum_{n=-\infty}^{+\infty} j^{-n} \left[J_n(\beta a) - \frac{J_n'(\beta a)}{H_n^{(2)'}(\beta a)} H_n^{(2)}(\beta a) \right] e^{jn\phi} \\ &= -H_0 \sum_{n=-\infty}^{+\infty} j^{-n+1} \left[\frac{J_n(\beta a) Y_n'(\beta a) - J_n'(\beta a) Y_n(\beta a)}{H_n^{(2)'}(\beta\rho)} \right] e^{jn\phi} \end{aligned} \quad (11-112)$$

which reduces, using the Wronskian of (11-95), to

$$H_z^t(\rho = a) = -jH_0 \frac{2}{\pi\beta a} \sum_{n=-\infty}^{+\infty} j^{-n} \frac{e^{jn\phi}}{H_n^{(2)'}(\beta a)} \quad (11-112a)$$

Thus, the current induced on the surface of the cylinder can be written as

$$\begin{aligned} \mathbf{J}_s &= \hat{\mathbf{n}} \times \mathbf{H}^t|_{\rho=a} = \hat{\mathbf{a}}_\rho \times \hat{\mathbf{a}}_z H_z^t|_{\rho=a} = -\hat{\mathbf{a}}_\phi H_z^t(\rho = a) \\ &= \hat{\mathbf{a}}_\phi j \frac{2H_0}{\pi\beta a} \sum_{n=-\infty}^{+\infty} j^{-n} \frac{e^{jn\phi}}{H_n^{(2)'}(\beta a)} \end{aligned} \quad (11-113)$$

A. Small Radius Approximation As the radius of the cylinder increases, more terms in the infinite series of (11-113) are needed to obtain convergence. However, for very small cylinders, like very thin wires where $a \ll \lambda$, the first three terms ($n = 0$, $n = \pm 1$) in (11-113) are dominant and are sufficient to represent the induced current. Thus, for a very thin wire, (11-113) can be approximated by

$$\mathbf{J}_s \stackrel{a \ll \lambda}{\simeq} \hat{\mathbf{a}}_\phi j \frac{2H_0}{\pi\beta a} \left[\frac{1}{H_0^{(2)'}(\beta a)} + j^{-1} \frac{e^{j\phi}}{H_1^{(2)'}(\beta a)} + j^{+1} \frac{e^{-j\phi}}{H_{-1}^{(2)'}(\beta a)} \right] \quad (11-114)$$

where

$$\begin{aligned} H_0^{(2)'}(\beta a) &= -H_1^{(2)}(\beta a) = -[J_1(\beta a) - jY_1(\beta a)] \\ &\stackrel{a \ll \lambda}{\simeq} -\left[\frac{\beta a}{2} + j \frac{1}{\pi} \left(\frac{2}{\beta a} \right) \right] = -j \frac{2}{\pi\beta a} \end{aligned} \quad (11-114a)$$

$$\begin{aligned}
H_1^{(2)'}(\beta a) &= -H_2^{(2)}(\beta a) + \frac{1}{\beta a} H_1^{(2)}(\beta a) \\
&= -[J_2(\beta a) - jY_2(\beta a)] + \frac{1}{\beta a} [J_1(\beta a) - jY_1(\beta a)] \\
&\stackrel{a \ll \lambda}{\simeq} -\left[\frac{1}{2} \left(\frac{\beta a}{2} \right)^2 + j \frac{1}{\pi} \left(\frac{2}{\beta a} \right)^2 \right] + \frac{1}{\beta a} \left[\frac{\beta a}{2} + j \frac{1}{\pi} \left(\frac{2}{\beta a} \right) \right] \\
&\stackrel{a \ll \lambda}{\simeq} -\left[j \frac{1}{\pi} \left(\frac{2}{\beta a} \right)^2 \right] + \frac{1}{\beta a} \left[j \frac{1}{\pi} \left(\frac{2}{\beta a} \right) \right] = -j \frac{2}{\pi} \frac{1}{(\beta a)^2}
\end{aligned} \tag{11-114b}$$

$$H_{-1}^{(2)'}(\beta a) = -H_1^{(2)'}(\beta a) \stackrel{a \ll \lambda}{\simeq} +j \frac{2}{\pi} \frac{1}{(\beta a)^2} \tag{11-114c}$$

Therefore, (11-114) reduces to

$$\begin{aligned}
\mathbf{J}_s &\stackrel{a \ll \lambda}{\simeq} \hat{\mathbf{a}}_\phi j \frac{2H_0}{\pi \beta a} \left[-\frac{\pi \beta a}{j2} + j \frac{\pi}{j2} (\beta a)^2 e^{j\phi} + j \frac{\pi}{j2} (\beta a)^2 e^{-j\phi} \right] \\
&\stackrel{a \ll \lambda}{\simeq} \hat{\mathbf{a}}_\phi H_0 [-1 + j\beta a (e^{j\phi} + e^{-j\phi})] = \hat{\mathbf{a}}_\phi H_0 [-1 + j2(\beta a) \cos(\phi)]
\end{aligned} \tag{11-114d}$$

B. Far-Zone Scattered Field Since the scattered field, as given by (11-110a), through (11-110d), has two non-vanishing electric field components and only one magnetic field component, it is most convenient to use the magnetic field to find the far-zone scattered field pattern and the radar cross section. However, the same answer can be obtained using the electric field components.

For far-field observations, the scattered magnetic field (11-110d) can be approximated, using the Hankel function approximation (11-99), by

$$H_z^s \stackrel{\beta \rho \rightarrow \infty}{\simeq} -H_0 \sqrt{\frac{2j}{\pi \beta}} \frac{e^{-j\beta \rho}}{\sqrt{\rho}} \sum_{n=-\infty}^{+\infty} \frac{J'_n(\beta a)}{H_n^{(2)'}(\beta a)} e^{jn\phi} \tag{11-115}$$

The ratio of the far-zone scattered magnetic field to the incident field can then be written as

$$\begin{aligned}
\frac{|H_z^s|}{|H_z^i|} &\stackrel{\beta \rho \rightarrow \infty}{\simeq} \frac{\left| -H_0 \sqrt{\frac{2j}{\pi \beta}} \frac{e^{-j\beta \rho}}{\sqrt{\rho}} \sum_{n=-\infty}^{+\infty} \frac{J'_n(\beta a)}{H_n^{(2)'}(\beta a)} e^{jn\phi} \right|}{|H_0 e^{-j\beta x}|} \\
&= \sqrt{\frac{2}{\pi \beta \rho}} \left| \sum_{n=-\infty}^{+\infty} \frac{J'_n(\beta a)}{H_n^{(2)'}(\beta a)} e^{jn\phi} \right|
\end{aligned} \tag{11-116}$$

Thus, the scattering width of (11-21c) can be expressed as

$$\begin{aligned}
\sigma_{2-D} &= \lim_{\rho \rightarrow \infty} \left[2\pi \rho \frac{|H_z^s|^2}{|H_z^i|^2} \right] = \frac{4}{\beta} \left| \sum_{n=-\infty}^{+\infty} \frac{J'_n(\beta a)}{H_n^{(2)'}(\beta a)} e^{jn\phi} \right|^2 \\
&= \frac{2\lambda}{\pi} \left| \sum_{n=0}^{+\infty} \varepsilon_n \frac{J'_n(\beta a)}{H_n^{(2)'}(\beta a)} \cos(n\phi) \right|^2
\end{aligned} \tag{11-117}$$

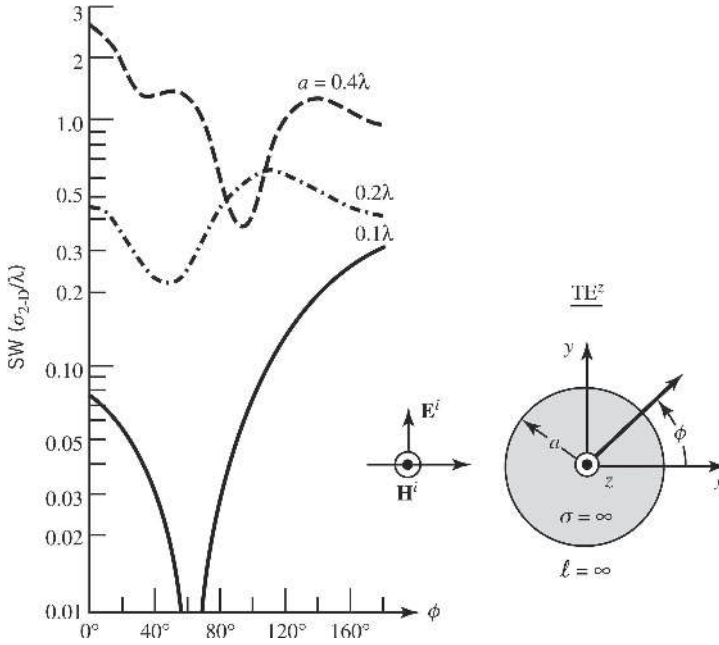


Figure 11-15 Two-dimensional TE^z bistatic scattering width (SW) of a circular conducting cylinder. (Courtesy of J. H. Richmond, Ohio State University.)

where

$$\varepsilon_n = \begin{cases} 1 & n = 0 \\ 2 & n \neq 0 \end{cases} \quad (11-117a)$$

Plots of bistatic σ_{2-D} / λ computed using (11-117) are shown in Figure 11-15 for cylinder radii of $a = 0.1\lambda$, 0.2λ , and 0.4λ while the backscattered patterns as a function of the cylinder radius are displayed in Figure 11-14 [13].

For small radii ($a \ll \lambda$), the first three terms ($n = 0$, $n = \pm 1$) in (11-117) are the dominant terms, and they are sufficient to represent the scattered field. Thus, for small radii, (11-117) can be approximated by

$$\sigma_{2-D} \stackrel{a \ll \lambda}{\simeq} \frac{2\lambda}{\pi} \left| \frac{J'_0(\beta a)}{H_0^{(2)'}(\beta a)} + \frac{J'_1(\beta a)}{H_1^{(2)'}(\beta a)} e^{j\phi} + \frac{J'_{-1}(\beta a)}{H_{-1}^{(2)'}(\beta a)} e^{-j\phi} \right|^2 \quad (11-118)$$

where

$$\begin{aligned} \frac{J'_0(\beta a)}{H_0^{(2)'}(\beta a)} &= \frac{-J_1(\beta a)}{-H_1^{(2)}(\beta a)} \simeq \frac{\frac{\beta a}{2}}{\frac{\beta a}{2} + j \frac{1}{\pi} \left(\frac{2}{\beta a} \right)} \simeq \frac{\frac{\beta a}{2}}{j \frac{1}{\pi} \left(\frac{2}{\beta a} \right)} = -j \frac{\pi}{4} (\beta a)^2 \\ \frac{J'_1(\beta a)}{H_1^{(2)'}(\beta a)} &= \frac{-J_2(\beta a) + \frac{1}{\beta a} J_1(\beta a)}{-H_2^{(2)}(\beta a) + \frac{1}{\beta a} H_1^{(2)}(\beta a)} \end{aligned} \quad (11-118a)$$

$$\begin{aligned}
\frac{J'_1(\beta a)}{H_1^{(2)'}(\beta a)} &\simeq \frac{-\frac{1}{2}\left(\frac{\beta a}{2}\right)^2 + \frac{1}{\beta a}\left(\frac{\beta a}{2}\right)}{-\left[\frac{1}{2}\left(\frac{\beta a}{2}\right)^2 + j\frac{1}{\pi}\left(\frac{2}{\beta a}\right)^2\right] + \frac{1}{\beta a}\left[\left(\frac{\beta a}{2}\right) + j\frac{1}{\pi}\left(\frac{2}{\beta a}\right)\right]} \\
&\simeq \frac{-\frac{1}{2}\left(\frac{\beta a}{2}\right)^2 + \frac{1}{2}}{-j\frac{1}{\pi}\left(\frac{2}{\beta a}\right)^2 + \frac{1}{\beta a}\left[j\frac{1}{\pi}\left(\frac{2}{\beta a}\right)\right]} \simeq \frac{\frac{1}{2}}{-j\frac{2}{\pi}\left(\frac{1}{\beta a}\right)^2} = j\frac{\pi}{4}(\beta a)^2 \quad (11-118b)
\end{aligned}$$

$$\frac{J'_{-1}(\beta a)}{H_{-1}^{(2)'}(\beta a)} = \frac{J'_1(\beta a)}{H_1^{(2)'}(\beta a)} \simeq j\frac{\pi}{4}(\beta a)^2 \quad (11-118c)$$

Thus, (11-118) reduces to

$$\begin{aligned}
\sigma_{2-D} &\stackrel{a \ll \lambda}{\simeq} \frac{2\lambda}{\pi} \left| -j\frac{\pi}{4}(\beta a)^2 + j\frac{\pi}{4}(\beta a)^2 e^{j\phi} + j\frac{\pi}{4}(\beta a)^2 e^{-j\phi} \right|^2 \\
&= \frac{\pi\lambda}{8}(\beta a)^4 [1 - 2\cos(\phi)]^2 \quad (11-118d)
\end{aligned}$$

Even for small radii ($a \ll \lambda$), σ is a function of ϕ , as is evident in Figure 11-15 by the curves for small values of a .

For a cylinder of finite length ℓ , the three-dimensional radar cross section for normal incidence is related to the two-dimensional scattering width by (11-22e). Thus, using (11-117) and (11-118d), we can write the three-dimensional RCS (11-22e) as

$$\begin{aligned}
\sigma_{3-D} &\simeq \frac{4\ell^2}{\pi} \left| \sum_{n=-\infty}^{+\infty} \frac{J'_n(\beta a)}{H_n^{(2)'}(\beta a)} e^{jn\phi} \right|^2 \\
&= \frac{4\ell^2}{\pi} \left| \sum_{n=0}^{+\infty} \varepsilon_n \frac{J'_n(\beta a)}{H_n^{(2)'}(\beta a)} \cos(n\phi) \right|^2 \quad (11-119a)
\end{aligned}$$

$$\sigma_{3-D} \stackrel{a \ll \lambda}{\simeq} \frac{\pi\ell^2}{4}(\beta a)^4 [1 - 2\cos(\phi)]^2 \quad (11-119b)$$

11.5.3 Oblique Incidence Plane Wave Scattering by a Conducting Circular Cylinder: TM^z Polarization

In the previous two sections we analyzed scattering by a conducting cylinder at normal incidence. Scattering at oblique incidence will be considered here. Let us assume that a TM^z plane wave traveling parallel to the xz plane is incident upon a circular cylinder of radius a , as shown in Figure 11-16. The incident electric field can be written as

$$\mathbf{E}^i = E_0(\hat{\mathbf{a}}_x \cos \theta_i + \hat{\mathbf{a}}_z \sin \theta_i) e^{-j\beta x \sin \theta_i} e^{+j\beta z \cos \theta_i} \quad (11-120)$$

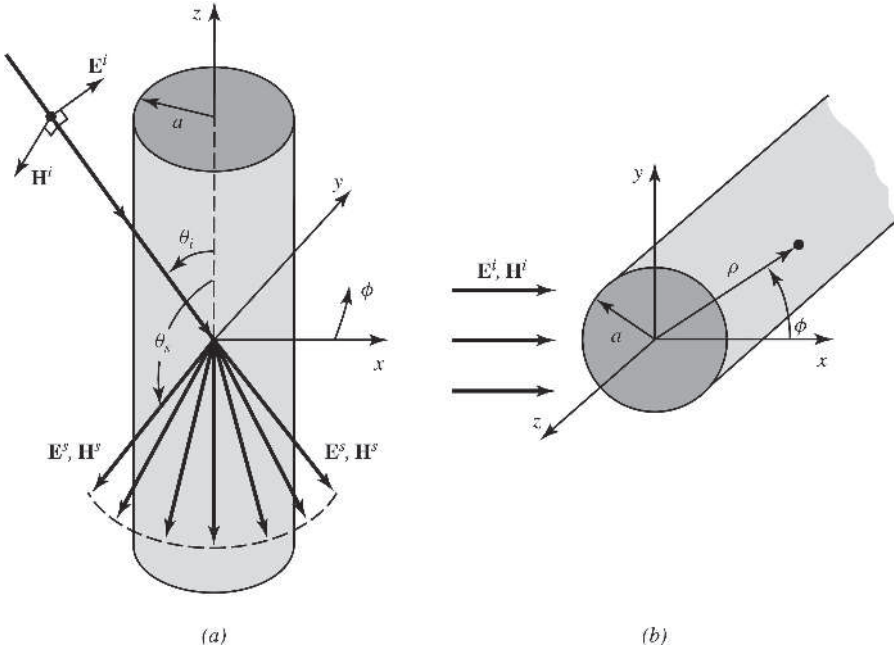


Figure 11-16 Uniform plane wave obliquely incident on a circular cylinder. (a) Side view. (b) Top view.

Using the transformation (11-81a), the z component of (11-120) can be expressed as

$$E_z^i = E_0 \sin \theta_i e^{+j\beta z \cos \theta_i} \sum_{n=-\infty}^{+\infty} j^{-n} J_n(\beta \rho \sin \theta_i) e^{jn\phi} \quad (11-120a)$$

The tangential component of the total field is composed of two parts: incident and scattered field components. The z component of the scattered field takes a form similar to (11-120a). By referring to (11-88) and Figure 11-16, where θ_s is shown to be greater than 90° , it can be written as

$$E_z^s = E_0 \sin \theta_s e^{-j\beta z \cos \theta_s} \sum_{n=-\infty}^{+\infty} c_n H_n^{(2)}(\beta \rho \sin \theta_s) \quad (11-121)$$

The Hankel function was chosen to indicate that the scattered field is a wave traveling in the outward radial direction. It should be stated at this time that smooth perfectly conducting infinite cylinders do not depolarize the oblique incident wave (i.e., do not introduce additional components in the scattered field as compared to the incident field). This, however, is not the case for homogeneous dielectric or dielectric coated cylinders that introduce cross polarization under oblique wave incidences.

When the incident electric field is decomposed into its cylindrical components, the E_x^i component of (11-120) will result in E_ρ^i and E_ϕ^i . Similarly scattered E_ρ^s and E_ϕ^s components will also exist. Therefore, the boundary conditions on the surface of the cylinder are

$$\begin{aligned} E_z^t(\rho = a, 0 \leq \theta_i, \theta_s \leq \pi, 0 \leq \phi \leq 2\pi) &= 0 \\ &= E_z^i(\rho = a, 0 \leq \theta_i \leq \pi, 0 \leq \phi \leq 2\pi) + E_z^s(\rho = a, 0 \leq \theta_s \leq \pi, 0 \leq \phi \leq 2\pi) \end{aligned} \quad (11-122a)$$

$$\begin{aligned} E_\phi^t(\rho = a, 0 \leq \theta_i, \theta_s \leq \pi, 0 \leq \phi \leq 2\pi) &= 0 \\ &= E_\phi^i(\rho = a, 0 \leq \theta_i \leq \pi, 0 \leq \phi \leq 2\pi) + E_\phi^s(\rho = a, 0 \leq \theta_s \leq \pi, 0 \leq \phi \leq 2\pi) \end{aligned} \quad (11-122b)$$

Since each is not independent of the other, either one can be used to find the unknown coefficients. Applying (11-122a) leads to

$$E_0 \left[\sin \theta_i e^{+j\beta z \cos \theta_i} \sum_{n=-\infty}^{+\infty} j^{-n} J_n(\beta a \sin \theta_i) e^{jn\phi} + \sin \theta_s e^{-j\beta z \cos \theta_s} \sum_{n=-\infty}^{+\infty} c_n H_n^{(2)}(\beta a \sin \theta_s) \right] = 0 \quad (11-123a)$$

which is satisfied provided

$$\theta_s = \pi - \theta_i \quad (11-123b)$$

$$c_n = -j^{-n} \frac{J_n(\beta a \sin \theta_i)}{H_n^{(2)}(\beta a \sin \theta_i)} e^{jn\phi} = j^{-n} a_n e^{jn\phi} \quad (11-123c)$$

$$a_n = -\frac{J_n(\beta a \sin \theta_i)}{H_n^{(2)}(\beta a \sin \theta_i)} \quad (11-123d)$$

Thus, the scattered component E_z^s of (11-121) reduces to

$$E_z^s = E_0 \sin \theta_i e^{+j\beta z \cos \theta_i} \sum_{n=-\infty}^{+\infty} j^{-n} a_n H_n^{(2)}(\beta \rho \sin \theta_i) e^{jn\phi} \quad (11-124)$$

It is apparent that the scattered field exists for all values of angle ϕ (measured from the x axis) along a cone in the forward direction whose half-angle from the z axis is equal to $\theta_s = \pi - \theta_i$.

To find the remaining $\mathbf{E}^s$ and $\mathbf{H}^s$ scattered field components, we expand Maxwell's curl equations as

$$\nabla \times \mathbf{E}^s = -j\omega\mu\mathbf{H}^s \Rightarrow \mathbf{H}^s = -\frac{1}{j\omega\mu} \nabla \times \mathbf{E}^s \quad (11-125)$$

or

$$H_\rho^s = -\frac{1}{j\omega\mu} \left(\frac{1}{\rho} \frac{\partial E_z^s}{\partial \phi} - \frac{\partial E_\phi^s}{\partial z} \right) \quad (11-125a)$$

$$H_\phi^s = -\frac{1}{j\omega\mu} \left(\frac{\partial E_\rho^s}{\partial z} - \frac{\partial E_z^s}{\partial \rho} \right) \quad (11-125b)$$

$$H_z^s = -\frac{1}{j\omega\mu\rho} \left(\frac{\partial}{\partial \rho} (\rho E_\phi^s) - \frac{\partial E_\rho^s}{\partial \phi} \right) \quad (11-125c)$$

and

$$\nabla \times \mathbf{H}^s = j\omega\varepsilon\mathbf{E}^s \Rightarrow \mathbf{E}^s = \frac{1}{j\omega\varepsilon} \nabla \times \mathbf{H}^s \quad (11-126)$$

or

$$E_\rho^s = \frac{1}{j\omega\varepsilon} \left(\frac{1}{\rho} \frac{\partial H_z^s}{\partial \phi} - \frac{\partial H_\phi^s}{\partial z} \right) \quad (11-126a)$$

$$E_\phi^s = \frac{1}{j\omega\varepsilon} \left(\frac{\partial H_\rho^s}{\partial z} - \frac{\partial H_z^s}{\partial \rho} \right) \quad (11-126b)$$

$$E_z^s = \frac{1}{j\omega\varepsilon\rho} \left(\frac{\partial}{\partial \rho} (\rho H_\phi^s) - \frac{\partial H_\rho^s}{\partial \phi} \right) \quad (11-126c)$$

Since for the TM^z solution $H_z^s = 0$, (11-126a) reduces to

$$E_\rho^s = -\frac{1}{j\omega\varepsilon} \frac{\partial H_\phi^s}{\partial z} \quad (11-127)$$

When substituted into (11-125b), we can write that

$$\begin{aligned} H_\phi^s &= -\frac{1}{j\omega\mu} \left[\frac{\partial}{\partial z} \left(-\frac{1}{j\omega\varepsilon} \frac{\partial H_\phi^s}{\partial z} \right) - \frac{\partial E_z^s}{\partial \rho} \right] \\ &= -\frac{1}{\omega^2\mu\varepsilon} \frac{\partial^2 H_\phi^s}{\partial z^2} + \frac{1}{j\omega\mu} \frac{\partial E_z^s}{\partial \rho} \end{aligned} \quad (11-128)$$

The z variations of all field components are of the same form as in (11-124) (i.e., $e^{+j\beta z \cos \theta_i}$). Thus (11-128) reduces to

$$H_\phi^s = -\frac{(j\beta \cos \theta_i)^2}{\omega^2\mu\varepsilon} H_\phi^s + \frac{1}{j\omega\mu} \frac{\partial E_z^s}{\partial \rho} = \cos^2 \theta_i H_\phi^s + \frac{1}{j\omega\mu} \frac{\partial E_z^s}{\partial \rho} \quad (11-129)$$

or

$$H_\phi^s (1 - \cos^2 \theta_i) = \sin^2 \theta_i H_\phi^s = \frac{1}{j\omega\mu} \frac{\partial E_z^s}{\partial \rho} \quad (11-129a)$$

$$\begin{aligned} H_\phi^s &= \frac{1}{j\omega\mu} \frac{1}{\sin^2 \theta_i} \frac{\partial E_z^s}{\partial \rho} = \frac{\beta}{j\omega\mu} \frac{1}{\sin \theta_i} \frac{\partial E_z^s}{\partial (\beta \rho \sin \theta_i)} \\ &= -jE_0 \sqrt{\frac{\varepsilon}{\mu}} e^{+j\beta z \cos \theta_i} \sum_{n=-\infty}^{+\infty} j^{-n} a_n H_n^{(2)'}(\beta \rho \sin \theta_i) e^{jn\phi} \end{aligned} \quad (11-129b)$$

where

$$' \equiv \frac{\partial}{\partial (\beta \rho \sin \theta_i)} \quad (11-129c)$$

In a similar manner we can solve for H_ρ^s by first reducing (11-126b) to

$$E_\phi^s = \frac{1}{j\omega\varepsilon} \left(\frac{\partial H_\rho^s}{\partial z} - \frac{\partial H_z^s}{\partial \rho} \right)_{H_z^s=0} = \frac{1}{j\omega\varepsilon} \frac{\partial H_\rho^s}{\partial z} \quad (11-130)$$

and then substituting it into (11-125a). Thus

$$\begin{aligned} H_\rho^s &= -\frac{1}{j\omega\mu} \left[\frac{1}{\rho} \frac{\partial E_z^s}{\partial \phi} - \frac{\partial}{\partial z} \left(\frac{1}{j\omega\varepsilon} \frac{\partial H_\rho^s}{\partial z} \right) \right] = -\frac{1}{j\omega\mu} \frac{1}{\rho} \frac{\partial E_z^s}{\partial \phi} - \frac{1}{\omega^2\mu\varepsilon} \frac{\partial^2 H_\rho^s}{\partial z^2} \\ &= -\frac{1}{j\omega\mu} \frac{1}{\rho} \frac{\partial E_z^s}{\partial \phi} - \frac{(j\beta \cos \theta_i)^2}{\omega^2\mu\varepsilon} H_\rho^s = -\frac{1}{j\omega\mu} \frac{1}{\rho} \frac{\partial E_z^s}{\partial \phi} + \cos^2 \theta_i H_\rho^s \end{aligned} \quad (11-131)$$

or

$$(1 - \cos^2 \theta_i) H_\rho^s = \sin^2 \theta_i H_\rho^s = -\frac{1}{j\omega\mu} \frac{1}{\rho} \frac{\partial E_z^s}{\partial \phi} \quad (11-131a)$$

$$H_\rho^s = -\frac{1}{j\omega\mu\rho\sin^2\theta_i}\frac{\partial E_z^s}{\partial\phi} = j\frac{E_0e^{+j\beta z\cos\theta_i}}{\omega\mu\rho\sin\theta_i}\sum_{n=-\infty}^{+\infty}nj^{-n+1}a_nH_n^{(2)}(\beta\rho\sin\theta_i)e^{jn\phi} \quad (11-131b)$$

Expressions for E_ρ^s and E_ϕ^s can be written using (11-126a), (11-126b), (11-129b), and (11-131b). Thus

$$E_\rho^s = -\frac{1}{j\omega\varepsilon}\frac{\partial H_\phi^s}{\partial z} = jE_0\cos\theta_i e^{+j\beta z\cos\theta_i}\sum_{n=-\infty}^{+\infty}j^{-n}a_nH_n^{(2)'}(\beta\rho\sin\theta_i)e^{jn\phi} \quad (11-132)$$

$$E_\phi^s = \frac{1}{j\omega\varepsilon}\frac{\partial H_\rho^s}{\partial z} = jE_0\frac{\cot\theta_i}{\beta\rho}e^{+j\beta z\cos\theta_i}\sum_{n=-\infty}^{+\infty}nj^{-n+1}a_nH_n^{(2)}(\beta\rho\sin\theta_i)e^{jn\phi} \quad (11-133)$$

In summary, the scattered fields can be written as

$$\begin{array}{c} \text{TM}_z \\ E_\rho^s = jE_0\cos\theta_i e^{+j\beta z\cos\theta_i}\sum_{n=-\infty}^{+\infty}j^{-n}a_nH_n^{(2)'}(\beta\rho\sin\theta_i)e^{jn\phi} \end{array} \quad (11-134a)$$

$$E_\phi^s = jE_0\frac{\cot\theta_i}{\beta\rho}e^{+j\beta z\cos\theta_i}\sum_{n=-\infty}^{+\infty}nj^{-n+1}a_nH_n^{(2)}(\beta\rho\sin\theta_i)e^{jn\phi} \quad (11-134b)$$

$$E_z^s = E_0\sin\theta_i e^{+j\beta z\cos\theta_i}\sum_{n=-\infty}^{+\infty}j^{-n}a_nH_n^{(2)}(\beta\rho\sin\theta_i)e^{jn\phi} \quad (11-134c)$$

$$H_\rho^s = jE_0\frac{e^{+j\beta z\cos\theta_i}}{\omega\mu\rho\sin\theta_i}\sum_{n=-\infty}^{+\infty}nj^{-n+1}a_nH_n^{(2)}(\beta\rho\sin\theta_i)e^{jn\phi} \quad (11-134d)$$

$$H_\phi^s = -jE_0\sqrt{\frac{\varepsilon}{\mu}}e^{+j\beta z\cos\theta_i}\sum_{n=-\infty}^{+\infty}j^{-n}a_nH_n^{(2)'}(\beta\rho\sin\theta_i)e^{jn\phi} \quad (11-134e)$$

$$H_z^s = 0 \quad (11-134f)$$

$$a_n = -\frac{J_n(\beta a\sin\theta_i)}{H_n^{(2)}(\beta a\sin\theta_i)} \quad (11-134g)$$

$$' = \frac{\partial}{\partial(\beta\rho\sin\theta_i)} \quad (11-134h)$$

A. Far-Zone Scattered Field Often it is desired to know the scattered fields at large distances. This can be accomplished by approximating in (11-134a) through (11-134f) the Hankel function and its derivative by their corresponding asymptotic expressions for large distances, as given by

$$H_n^{(2)}(\alpha x) \stackrel{\alpha x \rightarrow \infty}{\simeq} \sqrt{\frac{2j}{\pi\alpha x}}j^n e^{-j\alpha x} \quad (11-135a)$$

$$H_n^{(2)'}(\alpha x) = \frac{dH_n^{(2)}(\alpha x)}{d(\alpha x)} \stackrel{\alpha x \rightarrow \infty}{\simeq} -\sqrt{\frac{2j}{\pi\alpha x}}j^{n+1}e^{-j\alpha x} \quad (11-135b)$$

Thus, we can reduce the scattered magnetic field expressions of (11-134d) and (11-134e) by

$$\begin{aligned} H_\rho^s &\stackrel{\rho \rightarrow \infty}{\simeq} jE_0 \frac{1}{\omega\mu} \frac{1}{\rho \sin \theta_i} \sqrt{\frac{2j}{\pi\beta\rho \sin \theta_i}} e^{+j\beta(z \cos \theta_i - \rho \sin \theta_i)} \sum_{n=-\infty}^{+\infty} n a_n e^{jn\phi} \\ &\stackrel{\rho \rightarrow \infty}{\simeq} jE_0 \frac{1}{\omega\mu} \frac{1}{\rho \sin \theta_i} \sqrt{\frac{2j}{\pi\beta\rho \sin \theta_i}} e^{+j\beta(z \cos \theta_i - \rho \sin \theta_i)} \sum_{n=0}^{+\infty} n \varepsilon_n a_n \cos(n\phi) \end{aligned} \quad (11-136a)$$

$$\begin{aligned} H_\phi^s &\stackrel{\rho \rightarrow \infty}{\simeq} -E_0 \sqrt{\frac{\varepsilon}{\mu}} \sqrt{\frac{2j}{\pi\beta\rho \sin \theta_i}} e^{+j\beta(z \cos \theta_i - \rho \sin \theta_i)} \sum_{n=-\infty}^{+\infty} a_n e^{jn\phi} \\ &\stackrel{\rho \rightarrow \infty}{\simeq} -E_0 \sqrt{\frac{\varepsilon}{\mu}} \sqrt{\frac{2j}{\pi\beta\rho \sin \theta_i}} e^{+j\beta(z \cos \theta_i - \rho \sin \theta_i)} \sum_{n=0}^{+\infty} \varepsilon_n a_n \cos(n\phi) \end{aligned} \quad (11-136b)$$

where ε_n is defined in (11-102a). A comparison of (11-136a) and (11-136b) indicates that at large distances H_ρ^s is small compared to H_ϕ^s since H_ρ^s varies inversely proportional to $\rho^{3/2}$ while H_ϕ^s is inversely proportional to $\rho^{1/2}$.

The scattering width of (11-21c) can now be expressed as

$$\begin{aligned} \sigma_{2-D} &= \lim_{\rho \rightarrow \infty} \left(2\pi\rho \frac{|H_\phi^s|^2}{|\mathbf{H}^i|^2} \right) = \lim_{\rho \rightarrow \infty} \left[2\pi\rho \frac{\frac{|E_0|^2}{\eta^2} \left(\frac{2}{\pi\beta\rho \sin \theta_i} \right)}{\frac{|E_0|^2}{\eta^2}} \left| \sum_{n=-\infty}^{+\infty} a_n e^{jn\phi} \right|^2 \right] \\ &= \frac{4}{\beta \sin \theta_i} \left| \sum_{n=-\infty}^{+\infty} a_n e^{jn\phi} \right|^2 = \frac{2\lambda}{\pi} \frac{1}{\sin \theta_i} \left| \sum_{n=0}^{+\infty} \varepsilon_n a_n \cos(n\phi) \right|^2 \end{aligned} \quad (11-137)$$

where

$$a_n = -\frac{J_n(\beta a \sin \theta_i)}{H_n^{(2)}(\beta a \sin \theta_i)} \quad (11-137a)$$

$$\varepsilon_n = \begin{cases} 1 & n = 0 \\ 2 & n \neq 0 \end{cases} \quad (11-137b)$$

which is similar to (11-102) except that β in (11-102) is replaced by $\beta \sin \theta_i$.

From the results of the normal incidence case of Section 11.5.1B we can write, by referring to (11-103a), that for small radii the scattering width of (11-137) reduces to

$$\sigma_{2-D} \stackrel{a \ll \lambda}{\simeq} \frac{\pi\lambda}{2\sin \theta_i} \left| \frac{1}{\ln(0.89\beta a \sin \theta_i)} \right|^2 \quad (11-138)$$

which is independent of ϕ .

For a cylinder of finite length ℓ , the scattered fields of oblique incidence propagate in all directions, in contrast to the infinitely long cylinder where all the energy is along a conical surface formed in the forward direction whose half-angle is equal to θ_i . However, as the length of the cylinder becomes much larger than its radius ($\ell \gg a$), then the scattered fields along $\theta_s = \pi - \theta_i$ will be much greater than those in other directions. When the length of the cylinder is a multiple

of half a wavelength, resonance phenomena are exhibited in the scattered fields [12]. However, as the length increases beyond several wavelengths, the resonance phenomena disappear. For both TM^z and TE^z polarizations, the three-dimensional radar cross section for oblique wave incidence is related approximately to the two-dimensional scattering width, by referring to the geometry of Figure 11-16, by [12, 14]

$$\sigma_{3-D} = \sigma_{2-D} \left\{ \frac{2\ell^2}{\lambda} \sin^2 \theta_{s,i} \left[\frac{\sin \left[\frac{\beta\ell}{2} (\cos \theta_i + \cos \theta_s) \right]}{\frac{\beta\ell}{2} (\cos \theta_i + \cos \theta_s)} \right]^2 \right\}, \quad \ell \gg a \quad (11-139)$$

where $\sin^2 \theta_{s,i}$ is used for TM^z and $\sin^2 \theta_i$ is used for TE^z . This is analogous to the rectangular plate scattering of Section 11.3.2 and Example 11-3. This indicates that the maximum RCS occurs along the specular direction ($\theta_s = \pi - \theta_i$) and away from it follows the variations exhibited from a flat plate, as given by (11-44a). Equation 11-139 yields reasonable good results even for cylinders with lengths near one wavelength ($\ell \simeq \lambda$). Thus, using (11-139) converts (11-137) and (11-138) for three-dimensional scatterers to

$$\sigma_{3-D} \simeq \frac{4\ell^2}{\pi} \frac{\sin^2 \theta_s}{\sin \theta_i} \left| \sum_{n=0}^{\infty} \varepsilon_n a_n \cos(n\phi) \right|^2 \left[\frac{\sin \left[\frac{\beta\ell}{2} (\cos \theta_i + \cos \theta_s) \right]}{\frac{\beta\ell}{2} (\cos \theta_i + \cos \theta_s)} \right]^2 \quad (11-140a)$$

$$\sigma_{3-D} \stackrel{a \ll \lambda}{\simeq} \frac{\pi \ell^2}{\sin \theta_i} \left| \frac{\sin \theta_s}{\ln(0.89\beta a \sin \theta_i)} \right|^2 \left[\frac{\sin \left[\frac{\beta\ell}{2} (\cos \theta_i + \cos \theta_s) \right]}{\frac{\beta\ell}{2} (\cos \theta_i + \cos \theta_s)} \right]^2 \quad (11-140b)$$

11.5.4 Oblique Incidence Plane Wave Scattering by a Conducting Circular Cylinder: TE^z Polarization

TE^z scattering by a cylinder at oblique incidence can be analyzed following a procedure similar to that of TM^z scattering as discussed in the previous section. Using the geometry of Figure 11-16, we can write the incident magnetic field, for a plane wave traveling parallel to the xz plane, as

$$\mathbf{H}^i = H_0 (\hat{\mathbf{a}}_x \cos \theta_i + \hat{\mathbf{a}}_z \sin \theta_i) e^{-j\beta x \sin \theta_i} e^{+j\beta z \cos \theta_i} \quad (11-141)$$

Using the transformation (11-81a), it can also be expressed as

$$\mathbf{H}^i = H_0 (\hat{\mathbf{a}}_x \cos \theta_i + \hat{\mathbf{a}}_z \sin \theta_i) e^{+j\beta z \cos \theta_i} \sum_{n=-\infty}^{+\infty} j^{-n} J_n(\beta \rho \sin \theta_i) e^{jn\phi} \quad (11-141a)$$

Using the transformation from rectangular to cylindrical components (II-6), or

$$H_\rho = H_x \cos \phi + H_y \sin \phi = H_x \cos \phi \quad (11-142a)$$

$$H_\phi = -H_x \sin \phi + H_y \cos \phi = -H_x \sin \phi \quad (11-142b)$$

$$H_z = H_z \quad (11-142c)$$

reduces (11-141a) to

$$H_\rho^i = H_0 \cos \theta_i \cos \phi e^{+j\beta z \cos \theta_i} \sum_{n=-\infty}^{+\infty} j^{-n} J_n(\beta \rho \sin \theta_i) e^{jn\phi} \quad (11-143a)$$

$$H_\phi^i = -H_0 \cos \theta_i \sin \phi e^{+j\beta z \cos \theta_i} \sum_{n=-\infty}^{+\infty} j^{-n} J_n(\beta \rho \sin \theta_i) e^{jn\phi} \quad (11-143b)$$

$$H_z^i = H_0 \sin \theta_i e^{+j\beta z \cos \theta_i} \sum_{n=-\infty}^{+\infty} j^{-n} J_n(\beta \rho \sin \theta_i) e^{jn\phi} \quad (11-143c)$$

In the source-free region, the corresponding electric field components can be obtained using Maxwell's curl equation

$$\nabla \times \mathbf{H}^i = j\omega \varepsilon \mathbf{E}^i \Rightarrow \mathbf{E}^i = \frac{1}{j\omega \varepsilon} \nabla \times \mathbf{H}^i \quad (11-144)$$

or

$$E_\rho^i = \frac{1}{j\omega \varepsilon} \left(\frac{1}{\rho} \frac{\partial H_z^i}{\partial \phi} - \frac{\partial H_\phi^i}{\partial z} \right) \quad (11-144a)$$

$$E_\phi^i = \frac{1}{j\omega \varepsilon} \left(\frac{\partial H_\rho^i}{\partial z} - \frac{\partial H_z^i}{\partial \rho} \right) \quad (11-144b)$$

$$E_z^i = \frac{1}{j\omega \varepsilon} \frac{1}{\rho} \left[\frac{\partial(\rho H_\phi^i)}{\partial \rho} - \frac{\partial H_\rho^i}{\partial \phi} \right] \quad (11-144c)$$

To aid in doing this, we also utilize Maxwell's curl equation

$$\nabla \times \mathbf{E}^i = -j\omega \mu \mathbf{H}^i \Rightarrow \mathbf{H}^i = -\frac{1}{j\omega \mu} \nabla \times \mathbf{E}^i \quad (11-145)$$

which when expanded takes the form, for a TE^z polarization ($E_z^i = 0$), of

$$H_\rho^i = -\frac{1}{j\omega \mu} \left(\frac{1}{\rho} \frac{\partial E_z^i}{\partial \phi} - \frac{\partial E_\phi^i}{\partial z} \right)_{E_z^i=0} = \frac{1}{j\omega \mu} \frac{\partial E_\phi^i}{\partial z} \quad (11-145a)$$

$$H_\phi^i = -\frac{1}{j\omega \mu} \left(\frac{\partial E_\rho^i}{\partial z} - \frac{\partial E_z^i}{\partial \rho} \right)_{E_z^i=0} = -\frac{1}{j\omega \mu} \frac{\partial E_\rho^i}{\partial z} \quad (11-145b)$$

$$H_z^i = -\frac{1}{j\omega \mu} \frac{1}{\rho} \left[\frac{\partial(\rho E_\phi^i)}{\partial \rho} - \frac{\partial E_\rho^i}{\partial \phi} \right] \quad (11-145c)$$

Substituting (11-145a) into (11-144b), we can write that

$$E_\phi^i = \frac{1}{j\omega \varepsilon} \left[\frac{1}{j\omega \mu} \frac{\partial^2 E_\phi^i}{\partial z^2} - \frac{\partial H_z^i}{\partial \rho} \right] = -\frac{1}{\omega^2 \mu \varepsilon} \frac{\partial^2 E_\phi^i}{\partial z^2} - \frac{1}{j\omega \varepsilon} \frac{\partial H_z^i}{\partial \rho} \quad (11-146)$$

Since the z variations of all the field components are of the same form (i.e., $e^{+j\beta z \cos \theta_i}$), as given by (11-141), then (11-146) reduces to

$$E_\phi^i = \frac{\beta^2}{\omega^2 \mu \varepsilon} \cos^2 \theta_i E_\phi^i - \frac{1}{j\omega \varepsilon} \frac{\partial H_z^i}{\partial \rho} = \cos^2 \theta_i E_\phi^i - \frac{1}{j\omega \varepsilon} \frac{\partial H_z^i}{\partial \rho} \quad (11-147)$$

or

$$\begin{aligned}
 (1 - \cos^2 \theta_i) E_\phi^i &= \sin^2 \theta_i E_\phi^i = -\frac{1}{j\omega\varepsilon} \frac{\partial H_z^i}{\partial \rho} \\
 E_\phi^i &= -\frac{1}{j\omega\varepsilon} \frac{1}{\sin^2 \theta_i} \frac{\partial H_z^i}{\partial \rho} = -\frac{\beta}{j\omega\varepsilon} \frac{1}{\sin \theta_i} \frac{\partial H_z^i}{\partial (\beta \rho \sin \theta_i)} \\
 &= j\sqrt{\frac{\mu}{\varepsilon}} H_0 e^{+j\beta z \cos \theta_i} \sum_{n=-\infty}^{+\infty} j^{-n} J'_n(\beta \rho \sin \theta_i) e^{jn\phi} \quad (11-147a)
 \end{aligned}$$

In a similar manner, we can solve for E_ρ^i by substituting (11-145b) into (11-144a). Thus,

$$\begin{aligned}
 E_\rho^i &= \frac{1}{j\omega\varepsilon} \left[\frac{1}{\rho} \frac{\partial H_z^i}{\partial \phi} - \left(-\frac{1}{j\omega\mu} \frac{\partial^2 E_\rho^i}{\partial z^2} \right) \right] = \frac{1}{j\omega\varepsilon} \frac{1}{\rho} \frac{\partial H_z^i}{\partial \phi} - \frac{1}{\omega^2 \mu \varepsilon} \frac{\partial^2 E_\rho^i}{\partial z^2} \\
 &= \frac{1}{j\omega\varepsilon} \frac{1}{\rho} \frac{\partial H_z^i}{\partial \phi} - \frac{(j\beta \cos \theta_i)^2}{\omega^2 \mu \varepsilon} E_\rho^i = \frac{1}{j\omega\varepsilon} \frac{1}{\rho} \frac{\partial H_z^i}{\partial \phi} + \cos^2 \theta_i E_\rho^i \quad (11-148)
 \end{aligned}$$

or

$$\begin{aligned}
 (1 - \cos^2 \theta_i) E_\rho^i &= \sin^2 \theta_i E_\rho^i = \frac{1}{j\omega\varepsilon} \frac{1}{\rho} \frac{\partial H_z^i}{\partial \phi} \\
 E_\rho^i &= \frac{1}{j\omega\varepsilon \rho} \frac{1}{\sin^2 \theta_i} \frac{\partial H_z^i}{\partial \phi} = -j \frac{H_0 e^{+j\beta z \cos \theta_i}}{\omega \varepsilon \rho \sin \theta_i} \sum_{n=-\infty}^{+\infty} n j^{-n+1} J_n(\beta \rho \sin \theta_i) e^{jn\phi} \quad (11-148a)
 \end{aligned}$$

Since the z component of the incident $\mathbf{H}$ field is given by (11-143c), its scattered field can be written in a form similar to (11-121) or

$$H_z^s = H_0 \sin \theta_s e^{-j\beta z \cos \theta_s} \sum_{n=-\infty}^{+\infty} d_n H_n^{(2)}(\beta \rho \sin \theta_s) \quad (11-149)$$

where d_n represents unknown coefficients to be determined by boundary conditions. According to (11-147a), the ϕ component of the scattered field can be written using (11-149) as

$$\begin{aligned}
 E_\phi^s &= -\frac{1}{j\omega\varepsilon} \frac{1}{\sin^2 \theta_s} \frac{\partial H_z^s}{\partial \rho} = -\frac{\beta}{j\omega\varepsilon} \frac{1}{\sin \theta_s} \frac{\partial H_z^s}{\partial (\beta \rho \sin \theta_s)} \\
 &= jH_0 \sqrt{\frac{\mu}{\varepsilon}} e^{-j\beta z \cos \theta_s} \sum_{n=-\infty}^{+\infty} d_n H_n^{(2)'}(\beta \rho \sin \theta_s) \quad (11-150)
 \end{aligned}$$

Applying the boundary condition

$$\begin{aligned}
 E_\phi^i(\rho = a, 0 \leq \theta_i, \theta_s \leq \pi, 0 \leq \phi \leq 2\pi) &= 0 \\
 &= E_\phi^i(\rho = a, 0 \leq \theta_i \leq \pi, 0 \leq \phi \leq 2\pi) + E_\phi^s(\rho = a, 0 \leq \theta_s \leq \pi, 0 \leq \phi \leq 2\pi) \quad (11-151)
 \end{aligned}$$

leads to

$$\begin{aligned}
 jH_0 \sqrt{\frac{\mu}{\varepsilon}} \left[e^{+j\beta z \cos \theta_i} \sum_{n=-\infty}^{+\infty} j^{-n} J'_n(\beta a \sin \theta_i) e^{jn\phi} \right. \\
 \left. + e^{-j\beta z \cos \theta_s} \sum_{n=-\infty}^{+\infty} d_n H_n^{(2)'}(\beta a \sin \theta_s) \right] &= 0 \quad (11-151a)
 \end{aligned}$$

which is satisfied provided

$$\theta_s = \pi - \theta_i \quad (11-151b)$$

$$d_n = -j^{-n} \frac{J'_n(\beta a \sin \theta_i)}{H_n^{(2)'}(\beta a \sin \theta_i)} e^{jn\phi} = j^{-n} b_n e^{jn\phi} \quad (11-151c)$$

$$b_n = -\frac{J'_n(\beta a \sin \theta_i)}{H_n^{(2)'}(\beta a \sin \theta_i)} \quad (11-151d)$$

Thus, the scattered H_z^s component of (11-149) reduces to

$$H_z^s = H_0 \sin \theta_i e^{+j\beta z \cos \theta_i} \sum_{n=-\infty}^{+\infty} j^{-n} b_n H_n^{(2)}(\beta \rho \sin \theta_i) e^{jn\phi} \quad (11-152)$$

Knowing H_z^s , the remaining electric and magnetic field components can be found using (11-148a), (11-147a), and (11-145a) through (11-145c).

In summary, the scattered fields can be written as

TE^z

$$E_\rho^s = \frac{1}{j\omega\epsilon\rho} \frac{1}{\sin^2 \theta_i} \frac{\partial H_z^s}{\partial \phi} = -j \frac{H_0}{\omega\epsilon\rho} \frac{e^{+j\beta z \cos \theta_i}}{\sin \theta_i} \times \sum_{n=-\infty}^{+\infty} nj^{-n+1} b_n H_n^{(2)}(\beta \rho \sin \theta_i) e^{jn\phi} \quad (11-153a)$$

$$E_\phi^s = -\frac{1}{j\omega\epsilon} \frac{1}{\sin^2 \theta_i} \frac{\partial H_z^s}{\partial \rho} = jH_0 \sqrt{\frac{\mu}{\epsilon}} e^{+j\beta z \cos \theta_i} \times \sum_{n=-\infty}^{+\infty} j^{-n} b_n H_n^{(2)'}(\beta \rho \sin \theta_i) e^{jn\phi} \quad (11-153b)$$

$$E_z^s = 0 \quad (11-153c)$$

$$H_\rho^s = \frac{1}{j\omega\mu} \frac{\partial E_\phi^s}{\partial z} = jH_0 \cos \theta_i e^{+j\beta z \cos \theta_i} \times \sum_{n=-\infty}^{+\infty} j^{-n} b_n H_n^{(2)'}(\beta \rho \sin \theta_i) e^{jn\phi} \quad (11-153d)$$

$$H_\phi^s = -\frac{1}{j\omega\mu} \frac{\partial E_\rho^s}{\partial z} = jH_0 \frac{\cot \theta_i}{\beta \rho} e^{+j\beta z \cos \theta_i} \times \sum_{n=-\infty}^{+\infty} nj^{-n+1} b_n H_n^{(2)}(\beta \rho \sin \theta_i) e^{jn\phi} \quad (11-153h)$$

$$H_z^s = H_0 \sin \theta_i e^{+j\beta z \cos \theta_i} \sum_{n=-\infty}^{+\infty} j^{-n} b_n H_n^{(2)}(\beta \rho \sin \theta_i) e^{jn\phi} \quad (11-153e)$$

$$b_n = -\frac{J'_n(\beta a \sin \theta_i)}{H_n^{(2)'}(\beta a \sin \theta_i)} \quad (11-153f)$$

$$' \equiv \frac{\partial}{\partial(\beta \rho \sin \theta_i)} \quad (11-153g)$$

A. Far-Zone Scattered Field The scattered electric fields of (11-153a) through (11-153c) can be approximated in the far zone by replacing the Hankel function and its derivative by their asymptotic forms, as given by (11-135a) and (11-135b). Thus,

$$\begin{aligned} E_\rho^s &\simeq H_0 \frac{1}{\omega \varepsilon} \frac{1}{\rho \sin \theta_i} \sqrt{\frac{2j}{\pi \beta \rho \sin \theta_i}} e^{+j\beta(z \cos \theta_i - \rho \sin \theta_i)} \sum_{n=-\infty}^{+\infty} n b_n e^{jn\phi} \\ &\simeq H_0 \frac{1}{\omega \varepsilon} \frac{1}{\rho \sin \theta_i} \sqrt{\frac{2j}{\pi \beta \rho \sin \theta_i}} e^{+j\beta(z \cos \theta_i - \rho \sin \theta_i)} \sum_{n=0}^{+\infty} n \varepsilon_n b_n \cos(n\phi) \end{aligned} \quad (11-154a)$$

$$\begin{aligned} E_\phi^s &\simeq H_0 \sqrt{\frac{\mu}{\varepsilon}} \sqrt{\frac{2j}{\pi \beta \rho \sin \theta_i}} e^{+j\beta(z \cos \theta_i - \rho \sin \theta_i)} \sum_{n=-\infty}^{+\infty} b_n e^{jn\phi} \\ &\simeq H_0 \sqrt{\frac{\mu}{\varepsilon}} \sqrt{\frac{2j}{\pi \beta \rho \sin \theta_i}} e^{+j\beta(z \cos \theta_i - \rho \sin \theta_i)} \sum_{n=-\infty}^{+\infty} \varepsilon_n b_n \cos(n\phi) \end{aligned} \quad (11-154b)$$

where ε_n is defined in (11-102a). A comparison of (11-154a) and (11-154b) indicates that at large distances E_ρ^s is small compared to E_ϕ^s since E_ρ^s is inversely proportional to $\rho^{3/2}$, whereas E_ϕ^s is inversely proportional to $\rho^{1/2}$.

The scattering width of (11-21b) can now be expressed using (11-154b) and the incident electric field corresponding to (11-141) as

$$\begin{aligned} \sigma_{2-D} &= \lim_{\rho \rightarrow \infty} \left[2\pi\rho \frac{|E_\phi^s|^2}{|\mathbf{E}^i|^2} \right] = \lim_{\rho \rightarrow \infty} \left[2\pi\rho \frac{|H_0|^2 \frac{\mu}{\varepsilon} \left(\frac{2}{\pi \beta \rho \sin \theta_i} \right) \left| \sum_{n=-\infty}^{+\infty} b_n e^{jn\phi} \right|^2}{|H_0|^2 \frac{\mu}{\varepsilon}} \right] \\ \sigma_{2-D} &= \frac{4}{\beta \sin \theta_i} \frac{1}{\left| \sum_{n=-\infty}^{+\infty} b_n e^{jn\phi} \right|^2} = \frac{2\lambda}{\pi} \frac{1}{\sin \theta_i} \left| \sum_{n=0}^{+\infty} \varepsilon_n b_n \cos(n\phi) \right|^2 \end{aligned} \quad (11-155)$$

where

$$b_n = -\frac{J'_n(\beta a \sin \theta_i)}{H_n^{(2)'}(\beta a \sin \theta_i)} \quad (11-155a)$$

$$\varepsilon_n = \begin{cases} 1 & n = 0 \\ 2 & n \neq 0 \end{cases} \quad (11-155b)$$

which is similar to (11-117) except that β in (11-117) has been replaced by $\beta \sin \theta_i$.

From the results of the normal incidence case of Section 11.5.2B we can write by referring to (11-118d) that for small radii the scattering width of (11-155) reduces to

$$\sigma_{2-D} \stackrel{a \ll \lambda}{\simeq} \frac{\pi \lambda}{8} \frac{(\beta a \sin \theta_i)^4}{\sin \theta_i} [1 - 2 \cos(\phi)]^2 \quad (11-156)$$

which is dependent on ϕ , even for small-radii cylinders.

Using (11-139), the radar cross section at oblique incidence for a finite length ℓ cylinder can be written using (11-155) and (11-156), and referring to the geometry of Figure 11-16, as

$$\sigma_{3-D} \simeq \frac{4\ell^2}{\pi} \sin^2 \theta_i \left| \sum_{n=0}^{\infty} \varepsilon_n b_n \cos(n\phi) \right|^2 \left\{ \frac{\sin \left[\frac{\beta\ell}{2} (\cos \theta_i + \cos \theta_s) \right]}{\frac{\beta\ell}{2} (\cos \theta_i + \cos \theta_s)} \right\}^2 \quad (11-157a)$$

$$\sigma_{3-D} \stackrel{a \ll \lambda}{\simeq} \frac{\pi \ell^2}{4} \left[(\beta a \sin \theta_i)^4 \sin^2 \theta_i \right] [1 - 2 \cos(\phi)]^2 \left\{ \frac{\sin \left[\frac{\beta\ell}{2} (\cos \theta_i + \cos \theta_s) \right]}{\frac{\beta\ell}{2} (\cos \theta_i + \cos \theta_s)} \right\}^2 \quad (11-157b)$$

11.5.5 Line-Source Scattering by a Conducting Circular Cylinder

While in the previous sections we examined plane wave scattering by a conducting circular cylinder, both at normal and oblique wave incidences, a more general problem is that of line-source (both electric and magnetic) scattering. The geometry is that shown in Figure 11-17 where an infinite line of constant current (I_e for electric and I_m for magnetic) is placed in the vicinity of a circular conducting cylinder of infinite length. We will examine here the scattering by the cylinder assuming the source is either electric or magnetic.

A. Electric Line Source (TM^z Polarization) If the line source of Figure 11-17 is of constant electric current I_e , the field generated everywhere by the source in the absence of the cylinder is

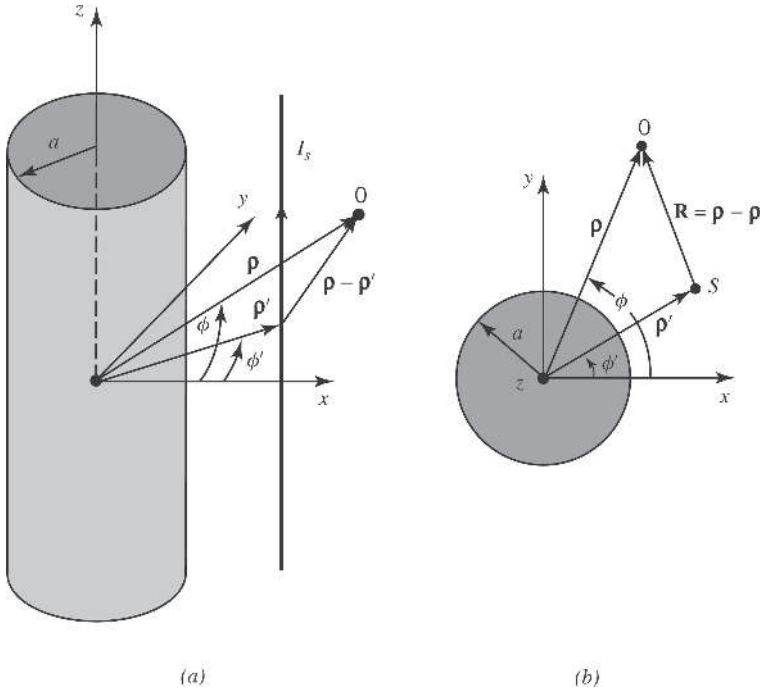


Figure 11-17 Electric line source near a circular cylinder. (a) Side view. (b) Top view.

given, according to (11-11a), by

$$E_z^i = -\frac{\beta^2 I_e}{4\omega\varepsilon} H_0^{(2)}(\beta|\rho - \rho'|) \quad (11-158)$$

which is referred to here as the incident field. By the addition theorem for Hankel functions (11-69a) and (11-69b), we can write (11-158) as

$$E_z^i = -\frac{\beta^2 I_e}{4\omega\varepsilon} \begin{cases} \sum_{n=-\infty}^{\infty} J_n(\beta\rho) H_n^{(2)}(\beta\rho') e^{jn(\phi-\phi')}, & \rho \leq \rho' \\ \sum_{n=-\infty}^{+\infty} J_n(\beta\rho') H_n^{(2)}(\beta\rho) e^{jn(\phi-\phi')}, & \rho \geq \rho' \end{cases} \quad (11-158a)$$

$$(11-158b)$$

Bessel functions $J_n(\beta\rho)$ were selected to represent the fields for $\rho < \rho'$ because the field must be finite everywhere (including $\rho = 0$) and Hankel functions were chosen for $\rho \geq \rho'$ to represent the traveling nature of the wave.

In the presence of the cylinder, the total field is composed of two parts: incident and scattered fields. The scattered field is produced by the current induced on the surface of the cylinder that acts as a secondary radiator. The scattered field also has only an E_z component (no cross polarized components are produced), and it can be expressed as

$$E_z^s = -\frac{\beta^2 I_e}{4\omega\varepsilon} \sum_{n=-\infty}^{+\infty} c_n H_n^{(2)}(\beta\rho), \quad a \leq \rho \leq \rho', \quad \rho \geq \rho' \quad (11-159)$$

The same expression is valid for $\rho \leq \rho'$ and $\rho \geq \rho'$ because the scattered field exists only when the cylinder is present, and it is nonzero only when $\rho \geq a$. Since the scattered field emanates from the surface of the cylinder, the Hankel function of the second kind in (11-159) is chosen to represent the traveling wave nature of the radiation.

The coefficients represented by c_n in (11-159) can be found by applying the boundary condition

$$\begin{aligned} E_z^i(\rho = a, 0 \leq \phi, \phi' \leq 2\pi, z) \\ = E_z^i(\rho = a, 0 \leq \phi, \phi' \leq 2\pi, z) + E_z^s(\rho = a, 0 \leq \phi, \phi' \leq 2\pi, z) = 0 \end{aligned} \quad (11-160)$$

which, by using (11-158a) and (11-159), leads to

$$-\frac{\beta^2 I_e}{4\omega\varepsilon} \sum_{n=-\infty}^{+\infty} \left[H_n^{(2)}(\beta\rho') J_n(\beta a) e^{jn(\phi-\phi')} + c_n H_n^{(2)}(\beta a) \right] = 0 \quad (11-161)$$

which is satisfied provided

$$c_n = -H_n^{(2)}(\beta\rho') \frac{J_n(\beta a)}{H_n^{(2)}(\beta a)} e^{jn(\phi-\phi')} \quad (11-161a)$$

Thus, (11-159) can be expressed as

$$E_z^s = +\frac{\beta^2 I_e}{4\omega\varepsilon} \sum_{n=-\infty}^{+\infty} H_n^{(2)}(\beta\rho') \frac{J_n(\beta a)}{H_n^{(2)}(\beta a)} H_n^{(2)}(\beta\rho) e^{jn(\phi-\phi')}, \quad a \leq \rho \leq \rho', \quad \rho \geq \rho' \quad (11-162)$$

The total electric field can then be written as

$$E_\rho^t = E_\phi^t = 0 \quad (11-163)$$

$$E_z^t = -\frac{\beta^2 I_e}{4\omega\varepsilon} \begin{cases} \sum_{n=-\infty}^{+\infty} H_n^{(2)}(\beta\rho') \left[J_n(\beta\rho) - \frac{J_n(\beta a)}{H_n^{(2)}(\beta a)} H_n^{(2)}(\beta\rho) \right] \\ \times e^{jn(\phi-\phi')} & a \leq \rho \leq \rho' \\ \sum_{n=-\infty}^{+\infty} H_n^{(2)}(\beta\rho) \left[J_n(\beta\rho') - \frac{J_n(\beta a)}{H_n^{(2)}(\beta a)} H_n^{(2)}(\beta\rho') \right] \\ \times e^{jn(\phi-\phi')} & \rho \geq \rho' \end{cases} \quad (11-164a)$$

$$\quad (11-164b)$$

where the first terms within the summations and brackets represent the incident fields and the second terms represent the scattered fields. The corresponding magnetic components can be found using Maxwell's equations 11-86 through 11-86c, which can be written as

$$H_\rho^t = -\frac{1}{j\omega\mu} \frac{1}{\rho} \frac{\partial E_z^t}{\partial \phi} \begin{cases} \sum_{n=-\infty}^{+\infty} jnH_n^{(2)}(\beta\rho') \left[J_n(\beta\rho) - \frac{J_n(\beta a)}{H_n^{(2)}(\beta a)} H_n^{(2)}(\beta\rho) \right] \\ \times e^{jn(\phi-\phi')} & a \leq \rho \leq \rho' \\ \sum_{n=-\infty}^{+\infty} jnH_n^{(2)}(\beta\rho) \left[J_n(\beta\rho') - \frac{J_n(\beta a)}{H_n^{(2)}(\beta a)} H_n^{(2)}(\beta\rho') \right] \\ \times e^{jn(\phi-\phi')} & \rho \geq \rho' \end{cases} \quad (11-165a)$$

$$= -j \frac{I_e}{4\rho} \begin{cases} \sum_{n=-\infty}^{+\infty} jnH_n^{(2)}(\beta\rho') \left[J_n(\beta\rho) - \frac{J_n(\beta a)}{H_n^{(2)}(\beta a)} H_n^{(2)}(\beta\rho) \right] \\ \times e^{jn(\phi-\phi')} & a \leq \rho \leq \rho' \\ \sum_{n=-\infty}^{+\infty} jnH_n^{(2)}(\beta\rho) \left[J_n(\beta\rho') - \frac{J_n(\beta a)}{H_n^{(2)}(\beta a)} H_n^{(2)}(\beta\rho') \right] \\ \times e^{jn(\phi-\phi')} & \rho \geq \rho' \end{cases} \quad (11-165b)$$

$$H_\phi^t = \frac{1}{j\omega\mu} \frac{\partial E_z^t}{\partial \rho} \begin{cases} \sum_{n=-\infty}^{+\infty} H_n^{(2)}(\beta\rho') \left[J_n'(\beta\rho) - \frac{J_n(\beta a)}{H_n^{(2)}(\beta a)} H_n^{(2)'}(\beta\rho) \right] \\ \times e^{jn(\phi-\phi')} & a \leq \rho \leq \rho' \\ \sum_{n=-\infty}^{+\infty} H_n^{(2)'}(\beta\rho) \left[J_n(\beta\rho') - \frac{J_n(\beta a)}{H_n^{(2)}(\beta a)} H_n^{(2)}(\beta\rho') \right] \\ \times e^{jn(\phi-\phi')} & \rho \geq \rho' \end{cases} \quad (11-166a)$$

$$= j \frac{\beta I_e}{4} \begin{cases} \sum_{n=-\infty}^{+\infty} H_n^{(2)}(\beta\rho') \left[J_n'(\beta a) - \frac{J_n(\beta a)}{H_n^{(2)}(\beta a)} H_n^{(2)'}(\beta a) \right] \\ \times e^{jn(\phi-\phi')} & a \leq \rho \leq \rho' \\ \sum_{n=-\infty}^{+\infty} H_n^{(2)'}(\beta\rho) \left[J_n(\beta\rho') - \frac{J_n(\beta a)}{H_n^{(2)}(\beta a)} H_n^{(2)}(\beta\rho') \right] \\ \times e^{jn(\phi-\phi')} & \rho \geq \rho' \end{cases} \quad (11-166b)$$

$$H_z^t = 0$$

On the surface of the cylinder, the current density can be found to be

$$\begin{aligned} \mathbf{J}_s &= \hat{\mathbf{n}} \times \mathbf{H}^t|_{\rho=a} = \hat{\mathbf{a}}_\rho \times (\hat{\mathbf{a}}_\rho H_\rho^t + \hat{\mathbf{a}}_\phi H_\phi^t)|_{\rho=a} = \hat{\mathbf{a}}_z H_\phi^t|_{\rho=a} \\ &= \hat{\mathbf{a}}_z j \frac{\beta I_e}{4} \sum_{n=-\infty}^{+\infty} H_n^{(2)}(\beta\rho') \left[J_n'(\beta a) - \frac{J_n(\beta a)}{H_n^{(2)}(\beta a)} H_n^{(2)'}(\beta a) \right] e^{jn(\phi-\phi')} \end{aligned} \quad (11-167)$$

$$\mathbf{J}_s = -\hat{\mathbf{a}}_z \frac{\beta I_e}{4} \sum_{n=-\infty}^{+\infty} H_n^{(2)}(\beta \rho') \left[\frac{J_n(\beta a) Y_n'(\beta a) - J_n'(\beta a) Y_n(\beta a)}{H_n^{(2)}(\beta a)} \right] e^{jn(\phi - \phi')} \quad (11-168)$$

which, by using the Wronskian of (11-95), reduces to

$$\mathbf{J}_s = -\hat{\mathbf{a}}_z \frac{I_e}{2\pi a} \sum_{n=-\infty}^{+\infty} \frac{H_n^{(2)}(\beta \rho')}{H_n^{(2)}(\beta a)} e^{jn(\phi - \phi')} \quad (11-168a)$$

For far-field observations ($\beta \rho \gg 1$), the total electric field of (11-164b) can be reduced by replacing the Hankel function $H_n^{(2)}(\beta \rho)$ by its asymptotic expression (11-135a). Doing this reduces (11-164b) to

$$E_z^t \stackrel{\beta \rho \gg 1}{\simeq} -\frac{\beta^2 I_e}{4\omega \varepsilon} \sqrt{\frac{2j}{\pi \beta}} \frac{e^{-j\beta \rho}}{\sqrt{\rho}} \sum_{n=-\infty}^{+\infty} j^n \left[J_n(\beta \rho') - \frac{J_n(\beta a)}{H_n^{(2)}(\beta a)} H_n^{(2)}(\beta \rho') \right] e^{jn(\phi - \phi')} \quad (11-169)$$

which can be used to compute more conveniently far-field patterns of an electric line source near a circular conducting cylinder. Plots of the normalized pattern for $a = 5\lambda$, $\phi' = 0$ with $\rho' = 5.25\lambda$ and 5.5λ are shown, respectively, in Figures 11-18a and 11-18b where they are compared with that of a planar reflector ($a = \infty$) of Figure 11-3. Because of the finite radius of the cylinder, radiation

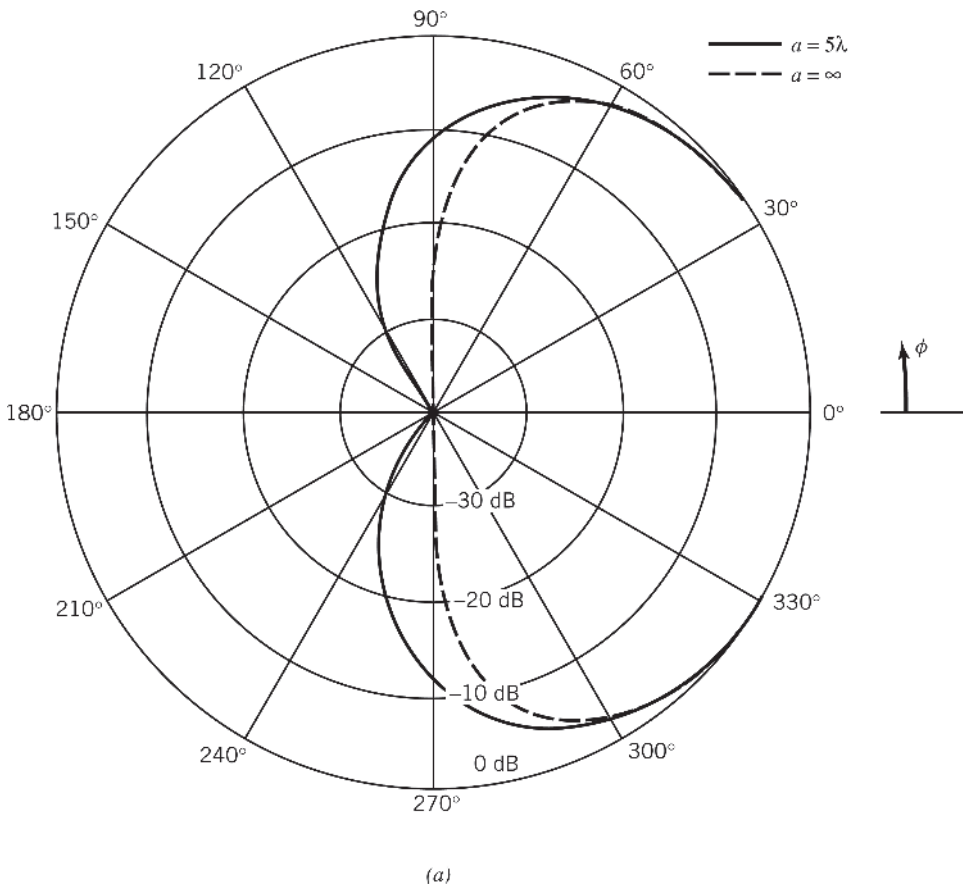


Figure 11-18 Normalized far-field pattern of an electric line source near a circular conducting cylinder. (a) $\rho' = 5.25\lambda$, $\phi' = 0^\circ$. (b) $\rho' = 5.5\lambda$, $\phi' = 0^\circ$.

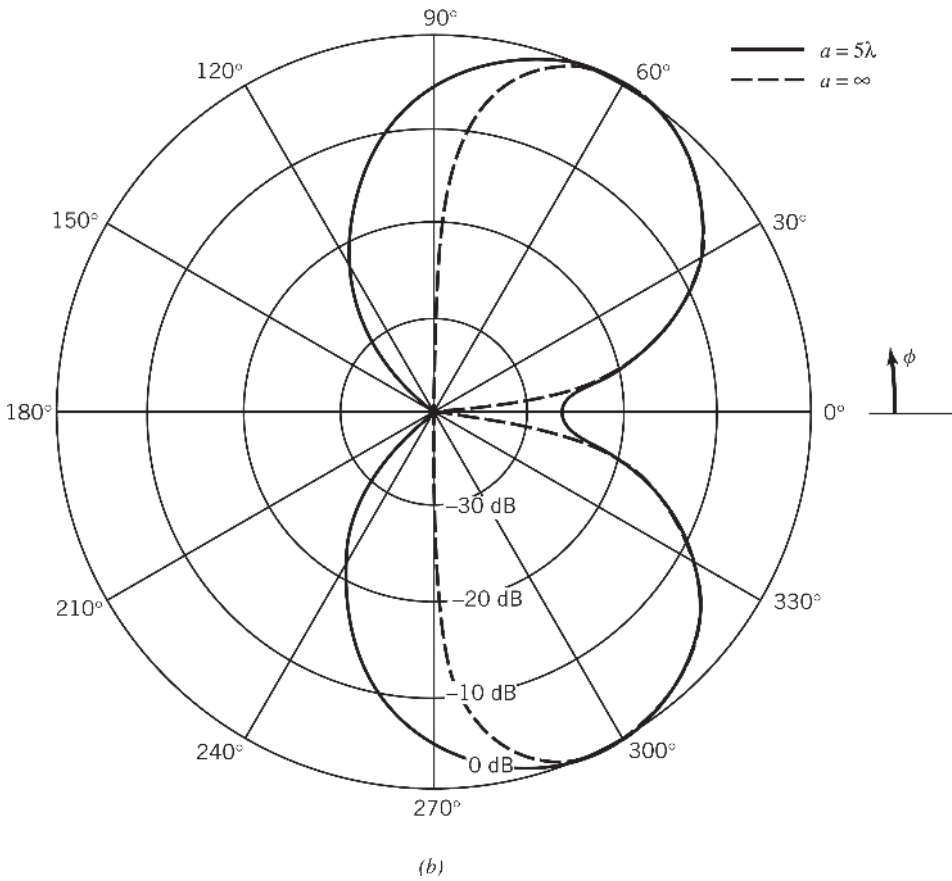


Figure 11-18 (Continued)

is allowed to “leak” around the cylinder in terms of “creeping” waves [15–17]; this is not the case for the planar reflector.

B. Magnetic Line Source (TE^z Polarization) Magnetic sources, although not physically realizable, are often used as equivalent sources to analyze aperture antennas [1, 18]. If the line source of Figure 11-17 is magnetic and it is allowed to recede to the surface of the cylinder ($\rho' = a$), the total field of the line source in the presence of the cylinder would be representative of a very thin infinite axial slot on the cylinder. Finite-width slots can be represented by a number of line sources with some amplitude and phase distribution across the width. Therefore, knowing the radiation and scattering by a magnetic line source near a cylinder allows us to solve other physical problems by using it as an equivalent.

If the line of Figure 11-17 is magnetic with a current of I_m , the fields that it radiates in the absence of the cylinder can be obtained from those of an electric line source by the use of duality (Section 7.2). Doing this, we can write the incident magnetic field, by referring to (11-158) through (11-158b), as

$$H_z^i = -\frac{\beta^2 I_m}{4\omega\mu} H_0^{(2)}(\beta|\mathbf{\rho} - \mathbf{\rho}'|) \quad (11-170)$$

which can also be expressed as

$$H_z^i = -\frac{\beta^2 I_m}{4\omega\mu} \begin{cases} \sum_{n=-\infty}^{\infty} J_n(\beta\rho) H_n^{(2)}(\beta\rho') e^{jn(\phi-\phi')}, & \rho \leq \rho' \\ \sum_{n=-\infty}^{+\infty} J_n(\beta\rho') H_n^{(2)}(\beta\rho) e^{jn(\phi-\phi')}, & \rho \geq \rho' \end{cases} \quad (11-170a)$$

$$(11-170b)$$

The scattered magnetic field takes a form similar to that of (11-159) and can be written as

$$H_z^s = -\frac{\beta^2 I_m}{4\omega\mu} \sum_{n=-\infty}^{+\infty} d_n H_n^{(2)}(\beta\rho), \quad a \leq \rho \leq \rho', \quad \rho \geq \rho' \quad (11-171)$$

where d_n is used to represent the coefficients of the scattered field. Thus, the total magnetic field can be expressed, by combining (11-170a) through (11-171), as

$$H_z^t = -\frac{\beta^2 I_m}{4\omega\mu} \begin{cases} \sum_{n=-\infty}^{+\infty} [H_n^{(2)}(\beta\rho') J_n(\beta\rho) e^{jn(\phi-\phi')} + d_n H_n^{(2)}(\beta\rho)] & a \leq \rho \leq \rho' \\ \sum_{n=-\infty}^{+\infty} [J_n(\beta\rho') e^{jn(\phi-\phi')} + d_n] H_n^{(2)}(\beta\rho) & \rho \geq \rho' \end{cases} \quad (11-172a)$$

$$(11-172b)$$

The corresponding electric field components can be found using Maxwell's equations 11-106 or 11-106a and 11-106b. Doing this, and utilizing (11-172a) and (11-172b), we can write that

$$E_\rho^t = \frac{1}{j\omega\varepsilon} \frac{1}{\rho} \frac{\partial H_z^t}{\partial \phi} = j \frac{I_m}{4\rho} \begin{cases} \sum_{n=-\infty}^{+\infty} \left[jn H_n^{(2)}(\beta\rho') J_n(\beta\rho) e^{jn(\phi-\phi')} + H_n^{(2)}(\beta\rho) \frac{\partial d_n}{\partial \phi} \right] & a \leq \rho \leq \rho' \\ \sum_{n=-\infty}^{+\infty} \left[jn J_n(\beta\rho') e^{jn(\phi-\phi')} + \frac{\partial d_n}{\partial \phi} \right] H_n^{(2)}(\beta\rho) & \rho \geq \rho' \end{cases} \quad (11-173a)$$

$$(11-173b)$$

$$E_\phi^t = -\frac{1}{j\omega\varepsilon} \frac{\partial H_z^t}{\partial \rho} = -j \frac{\beta I_m}{4} \begin{cases} \sum_{n=-\infty}^{+\infty} [H_n^{(2)}(\beta\rho') J'_n(\beta\rho) e^{jn(\phi-\phi')} + d_n H_n^{(2)'}(\beta\rho)] & a \leq \rho \leq \rho' \\ \sum_{n=-\infty}^{+\infty} [J_n(\beta\rho') e^{jn(\phi-\phi')} + d_n] H_n^{(2)'}(\beta\rho) & \rho \geq \rho' \end{cases} \quad (11-174a)$$

$$(11-174b)$$

Applying the boundary condition

$$\begin{aligned} E_\phi^t(\rho = a, 0 \leq \phi, \phi' \leq 2\pi, z) \\ = E_\phi^t(\rho = a, 0 \leq \phi, \phi' \leq 2\pi, z) + E_\phi^s(\rho = a, 0 \leq \phi, \phi' \leq 2\pi, z) = 0 \end{aligned} \quad (11-175)$$

on (11-174a) leads to

$$d_n = -H_n^{(2)}(\beta\rho') \frac{J_n'(\beta a)}{H_n^{(2)'}(\beta a)} e^{jn(\phi-\phi')} \quad (11-175a)$$

Thus, the total electric and magnetic field components can be written as

TE^z

$$E_z^t = H_\rho^t = H_\phi^t = 0 \quad (11-176a)$$

$$E_\rho^t = -\frac{I_m}{4\rho} \left\{ \begin{aligned} &\sum_{n=-\infty}^{+\infty} n H_n^{(2)}(\beta\rho') \left[J_n(\beta\rho) - \frac{J_n'(\beta a)}{H_n^{(2)'}(\beta a)} H_n^{(2)}(\beta\rho) \right] \\ &\quad \times e^{jn(\phi-\phi')} \quad a \leq \rho \leq \rho' \\ &\sum_{n=-\infty}^{+\infty} n H_n^{(2)}(\beta\rho) \left[J_n(\beta\rho') - \frac{J_n'(\beta a)}{H_n^{(2)'}(\beta a)} H_n^{(2)}(\beta\rho') \right] \\ &\quad \times e^{jn(\phi-\phi')} \quad \rho \geq \rho' \end{aligned} \right. \quad (11-176b)$$

$$\left. \begin{aligned} &\sum_{n=-\infty}^{+\infty} n H_n^{(2)}(\beta\rho) \left[J_n(\beta\rho') - \frac{J_n'(\beta a)}{H_n^{(2)'}(\beta a)} H_n^{(2)}(\beta\rho') \right] \\ &\quad \times e^{jn(\phi-\phi')} \quad \rho \geq \rho' \end{aligned} \right\} \quad (11-176c)$$

$$E_\phi^t = -j \frac{\beta I_m}{4} = \left\{ \begin{aligned} &\sum_{n=-\infty}^{+\infty} H_n^{(2)}(\beta\rho') \left[J_n'(\beta\rho) - \frac{J_n'(\beta a)}{H_n^{(2)'}(\beta a)} H_n^{(2)'}(\beta\rho) \right] \\ &\quad \times e^{jn(\phi-\phi')} \quad a \leq \rho \leq \rho' \\ &\sum_{n=-\infty}^{+\infty} H_n^{(2)'}(\beta\rho) \left[J_n(\beta\rho') - \frac{J_n'(\beta a)}{H_n^{(2)'}(\beta a)} H_n^{(2)}(\beta\rho') \right] \\ &\quad \times e^{jn(\phi-\phi')} \quad \rho \geq \rho' \end{aligned} \right. \quad (11-176d)$$

$$\left. \begin{aligned} &\sum_{n=-\infty}^{+\infty} H_n^{(2)'}(\beta\rho) \left[J_n(\beta\rho') - \frac{J_n'(\beta a)}{H_n^{(2)'}(\beta a)} H_n^{(2)}(\beta\rho') \right] \\ &\quad \times e^{jn(\phi-\phi')} \quad \rho \geq \rho' \end{aligned} \right\} \quad (11-176e)$$

$$H_z^t = -\frac{\beta^2 I_m}{4\omega\mu} = \left\{ \begin{aligned} &\sum_{n=-\infty}^{+\infty} H_n^{(2)}(\beta\rho') \left[J_n(\beta\rho) - \frac{J_n'(\beta a)}{H_n^{(2)'}(\beta a)} H_n^{(2)}(\beta\rho) \right] \\ &\quad \times e^{jn(\phi-\phi')} \quad a \leq \rho \leq \rho' \\ &\sum_{n=-\infty}^{+\infty} H_n^{(2)}(\beta\rho) \left[J_n(\beta\rho') - \frac{J_n'(\beta a)}{H_n^{(2)'}(\beta a)} H_n^{(2)}(\beta\rho') \right] \\ &\quad \times e^{jn(\phi-\phi')} \quad \rho \geq \rho' \end{aligned} \right. \quad (11-176f)$$

$$\left. \begin{aligned} &\sum_{n=-\infty}^{+\infty} H_n^{(2)}(\beta\rho) \left[J_n(\beta\rho') - \frac{J_n'(\beta a)}{H_n^{(2)'}(\beta a)} H_n^{(2)}(\beta\rho') \right] \\ &\quad \times e^{jn(\phi-\phi')} \quad \rho \geq \rho' \end{aligned} \right\} \quad (11-176g)$$

where the first terms within the summation and brackets represent the incident fields and the second terms represent the scattered fields.

On the surface of the cylinder, the current density can be found to be

$$\begin{aligned}\mathbf{J}_s &= \hat{\mathbf{n}} \times \mathbf{H}^t|_{\rho=a} = \hat{\mathbf{a}}_\rho \times \hat{\mathbf{a}}_z H_z^t|_{\rho=a} = -\hat{\mathbf{a}}_\phi H_z^t|_{\rho=a} \\ &= \hat{\mathbf{a}}_\phi \frac{\beta^2 I_m}{4\omega\mu} \sum_{n=-\infty}^{+\infty} H_n^{(2)}(\beta\rho') \left[J_n(\beta a) - \frac{J_n'(\beta a)}{H_n^{(2)'}(\beta a)} H_n^{(2)}(\beta a) \right] e^{jn(\phi-\phi')} \\ \mathbf{J}_s &= -j\hat{\mathbf{a}}_\phi \frac{\beta^2 I_m}{4\omega\mu} \sum_{n=-\infty}^{+\infty} H_n^{(2)}(\beta\rho') \left[\frac{J_n(\beta a)Y_n'(\beta a) - J_n'(\beta a)Y_n(\beta a)}{H_n^{(2)'}(\beta a)} \right] e^{jn(\phi-\phi')}\end{aligned}\quad (11-177)$$

which, by using the Wronskian (11-95), reduces to

$$\mathbf{J}_s = -j\hat{\mathbf{a}}_\phi \frac{I_m}{2\eta\pi a} \sum_{n=-\infty}^{+\infty} \frac{H_n^{(2)}(\beta\rho')}{H_n^{(2)'}(\beta a)} e^{jn(\phi-\phi')} \quad (11-177a)$$

For far-field observations ($\beta\rho \gg 1$), the total magnetic field (11-176g) can be reduced in form by replacing the Hankel function $H_n^{(2)}(\beta\rho)$ by its asymptotic expression (11-135a). Doing this reduces (11-176g) to

$$H_z^t \stackrel{\beta\rho \gg 1}{\simeq} -\frac{\beta^2 I_m}{4\omega\mu} \sqrt{\frac{2j}{\pi\beta}} \frac{e^{-j\beta\rho}}{\sqrt{\rho}} \sum_{n=-\infty}^{+\infty} j^n \left[J_n(\beta\rho') - \frac{J_n'(\beta a)}{H_n^{(2)'}(\beta a)} H_n^{(2)}(\beta\rho') \right] e^{jn(\phi-\phi')} \quad (11-178)$$

which can be used to compute, more conveniently, far-field patterns of a magnetic line source near a circular electric conducting cylinder. When the line source is moved to the surface of the cylinder ($\rho' = a$), (11-178) reduces, with the aid of the Wronskian (11-95), to

$$\begin{aligned}H_z^t \stackrel{\beta\rho \gg 1}{\simeq} j \frac{\beta^2 I_m}{4\omega\mu} \sqrt{\frac{2j}{\pi\beta}} \frac{e^{-j\beta\rho}}{\sqrt{\rho}} \sum_{n=-\infty}^{+\infty} j^n \left[\frac{J_n(\beta a)Y_n'(\beta a) - J_n'(\beta a)Y_n(\beta a)}{H_n^{(2)'}(\beta a)} \right] e^{jn(\phi-\phi')} \\ H_z^t \stackrel{\beta\rho \gg 1}{\simeq} j \frac{I_m}{\pi} \frac{1}{a} \sqrt{\frac{\varepsilon}{\mu}} \sqrt{\frac{j}{2\pi\beta}} \frac{e^{-j\beta\rho}}{\sqrt{\rho}} \sum_{n=-\infty}^{+\infty} j^n \frac{e^{jn(\phi-\phi')}}{H_n^{(2)'}(\beta a)}\end{aligned}\quad (11-178a)$$

The pattern of (11-178a) is representative of a very thin (ideally zero width) infinite-length axial slot on a circular conducting cylinder, and its normalized form is shown plotted in Figure 11-19 for $a = 2\lambda$ and 5λ . Because of the larger radius of curvature for the $a = 5\lambda$ radius, which results in larger attenuation, less energy is allowed to “creep” around the cylinder compared to that of $a = 2\lambda$.

Scattering by cylinders of other cross sections, and by dielectric and dielectric-covered cylinders, can be found in the literature [12, 19–31].

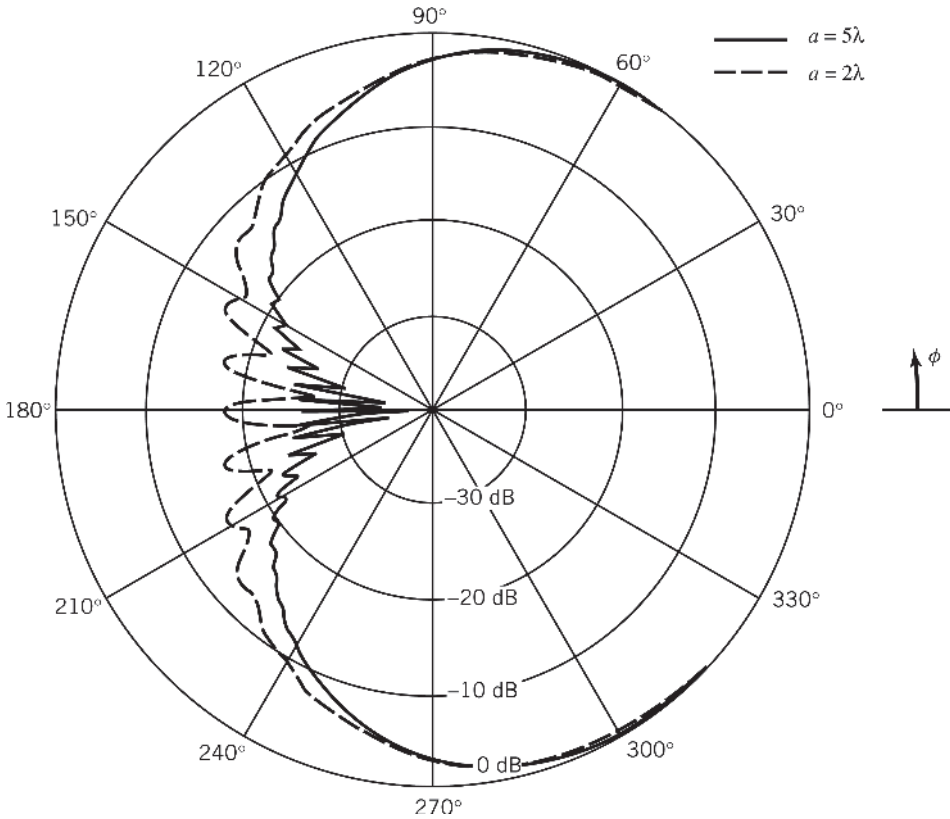


Figure 11-19 Normalized far-field amplitude pattern of a very thin axial slot on a circular conducting cylinder ($\phi' = 0^\circ$).

Example 11-4

A more practical problem is one in which an aperture of finite dimensions is mounted on a circular cylinder. This type of configuration is applicable to apertures mounted on the surface of missiles, fuselages of airplanes, and other similar structures and airframes. Practically, a rectangular aperture can be mounted on the cylinder primarily in two orientations, as shown in Figures 11-20a and 11-20b. The one in Figure 11-20a is usually referred to as a *circumferential* aperture while that in Figure 11-20b is referred to as an *axial* aperture. The choice of orientation is dictated by the desired polarization. One such decision was made in the 1970s in the development of the Microwave Landing System (MLS) [32]. The requirement for the MLS was to select antenna elements that radiate either vertical or horizontal polarization, especially near the forward direction of the aircraft. In using apertures, the choice was either of these two apertures, depending how they were mounted on the fuselage of the aircraft.

- a. Assume the electric field expressions on each of the two apertures is given, respectively, by

$$\begin{aligned} \text{Circumferential aperture:} \quad E_z &= \frac{V_0}{h} \cos\left(\frac{\pi}{2\phi_0} \phi'\right) \begin{cases} -\phi_0 \leq \phi' \leq +\phi_0 \\ -\frac{h}{2} \leq z' \leq +\frac{h}{2} \end{cases} \\ \text{Axial aperture:} \quad E_\phi &= \frac{V_0}{2a\phi_0} \cos\left(\frac{\pi}{w} z'\right) \begin{cases} -\phi_0 \leq \phi' \leq +\phi_0 \\ -\frac{w}{2} \leq z' \leq +\frac{w}{2} \end{cases} \end{aligned}$$

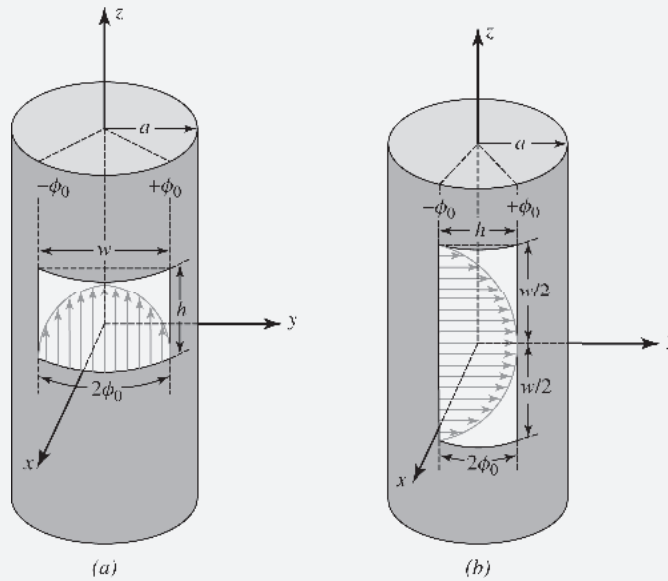


Figure 11-20 Aperture on the surface of a circular cylinder. (a) Circumferential. (b) Axial.

Write expressions for the electric field spherical components of the far-zone fields radiated by these two apertures.

- b. Assuming $f = 10$ GHz, $w = 2.286$ cm, $h = 1.016$ cm (X-band waveguide), compute and plot the:
- H-plane normalized amplitude patterns for the circumferential aperture of Figure 11-20a when it is on a circular cylinder with radii $a = 2\lambda$ and 5λ . Compare them when the same aperture is mounted on an infinite PEC flat ground plane. Compare the results and assess the effect of the cylinder curvature on the H-plane radiation patterns.
 - Repeat the previous computations and plots for the E-plane of the axial aperture of Figure 11-20b.

Solution: The fields radiated by the two apertures can be determined by replacing the apertures with equivalent currents densities $\mathbf{J}_s$ and $\mathbf{M}_s$ and then using integration over the aperture and surface of the cylinder. Since $\mathbf{J}_s$ is not known outside the aperture, an approximate equivalent will be to either assume $\mathbf{J}_s$ is small outside the aperture; however, this is not an exact equivalent. Another equivalent is to assume $\mathbf{M}_s$ only over the aperture, as is done for apertures mounted on ground planes [1]. However, for the cylinder, this also is not exact because the surface to which the aperture is mounted is not flat.

Another procedure is to use transform techniques, as it was done in [11, 33-34] where the fields external to the cylinder are expressed as the sum of TE^z and TM^z modes. This is accomplished by writing the corresponding vector potentials F_z and A_z in the transform domain and then the fields are obtained using (6-34) and (6-35). Using such a procedure, it is shown that the far-zone fields radiated by the respective apertures are:

Circumferential Aperture (circular cylinder)

$$E_\theta = -j \frac{V_0}{\pi^2} \frac{e^{-j\beta r}}{r} \left\{ \frac{\beta a}{\sin \theta} \sum_{n=0}^{+\infty} \frac{\cos\left(\frac{n\pi}{2\beta a}\right)}{[(\beta a)^2 - n^2]} \varepsilon_n j^n \cos(n\phi) \right\} H_n^{(2)}(\beta a \sin \theta)$$

$$E_\phi = -j \frac{V_0}{\pi^2} \frac{e^{-j\beta r}}{r} \left[\frac{\cot \theta}{\sin \theta} \sum_{n=1}^{+\infty} \frac{\cos\left(\frac{n\pi}{2\beta a}\right)}{[(\beta a)^2 - n^2]} \frac{2nj^n \sin(n\phi)}{H_n^{(2)'}(\beta a \sin \theta)} \right]$$

$$\varepsilon_n = \begin{cases} 1 & n=0 \\ 2 & n>0 \end{cases}; \quad H_n^{(2)'}(\beta\rho) = \frac{\partial}{\partial\rho} H_n^{(2)}(\beta\rho)$$

$$\text{When } \begin{cases} \theta = \pi/2: & E_\phi = 0 \\ \phi = 0: & E_\phi = 0 \end{cases} \quad \text{as it should be.}$$

Axial Aperture (circular cylinder)

$$E_\phi = \frac{V_0 \lambda}{2a\pi^3} \frac{e^{-j\beta r}}{r} \left[\frac{\cos\left(\frac{\beta w}{2} \cos \theta\right)}{\left[1 - \left(\frac{\beta w}{\pi} \cos \theta\right)^2\right]} \sum_{n=0}^{+\infty} \frac{\varepsilon_n j^n \cos(n\phi)}{H_n^{(2)'}(\beta a \sin \theta)} \right]$$

$$H_n^{(2)'}(\beta\rho) = \frac{\partial}{\partial\rho} H_n^{(2)}(\beta\rho)$$

$E_\theta = 0$ in all planes, as it should be.

Using the surface equivalence theorem outlined in Section 7.8 (Chapter 7), we can derive the fields when the respective apertures are mounted on infinite ground planes. The same procedure is outlined in Section 12.1 of [1]. Some of these apertures, when mounted on flat ground planes, have been assigned as end-of-the-chapter problems in Chapters 6 and 7. Following such a procedure, it is shown that the fields in the principle H- and E- planes, radiated by these two apertures when mounted on infinite flat PEC ground planes are:

Circumferential Aperture (flat PEC ground plane)

H-Plane ($\theta = 90^\circ$)

$$E_\phi = 0$$

$$E_\theta = +\frac{\pi}{2} C \left[\cos \phi \frac{\cos\left(\frac{\beta a}{2} \sin \phi\right)}{\left(\frac{\beta a}{2} \sin \phi\right)^2 - \left(\frac{\pi}{2}\right)^2} \right]$$

Axial Aperture (flat PEC ground plane)

E-Plane ($\theta = 90^\circ$)

$$E_\theta = 0$$

$$E_\phi = +\frac{2}{\pi} C \left[\frac{\sin\left(\frac{\beta b}{2} \sin \phi\right)}{\frac{\beta b}{2} \sin \phi} \right]$$

where $C = jab\beta V_0 e^{-j\beta r} / 2\pi r$. The expressions for the fields radiated in all space are found in the end-of-chapter exercises of Chapters 6 and 7.

To compute the pattern in the H-plane of the circumferential aperture and E-plane of the axial slot, for both the circular cylinder and flat infinite ground plane, a MATLAB computer program, referred to as **PEC_Cyl_Plate_Rect**, was written and it is included in the multimedia folder

associated with this book. The respective patterns are shown in Figure 11-21a for the H-plane of the circumferential aperture and in Figure 11-21b for the E-plane of the axial aperture, where they are compared with those of the flat ground plane. It is apparent, as expected, that the:

- Cylinder allows radiation on the rear region, whereas the PEC flat ground plane does not.
- Larger radius cylinder diminishes more the radiation in the rear region because the creeping waves that travel around the surface of the cylinder attenuate faster.
- Number of lobes in the rear region is greater for the larger cylinder because the two creeping waves that travel in opposite directions around the surface of the cylinder, and radiate tangentially [15–17], have a greater space separation, which allows the formation of a greater number of constructive and destructive interferences, leading to greater number of lobes.

Although the E-plane patterns of the circumferential aperture and the H-plane of the axial aperture are not computed or shown here, those of the cylinder are basically identical to those of the flat ground plane, if both are assumed to be of infinite extent.

11.6 SCATTERING BY A CONDUCTING WEDGE

Scattering of electromagnetic waves by a two-dimensional conducting wedge has received considerable attention since about the middle 1950s. Because the wedge is a canonical problem that can be used to represent locally (near the edge) the scattering of more complex structures, asymptotic forms of its solution have been utilized to solve numerous practical problems. The asymptotic forms of its solution are obtained by taking the infinite series modal solution and first transforming it into an integral by the so-called *Watson transformation* [19, 35, 36]. The integral is then evaluated by the *method of steepest descent (saddle point method)* (see Appendix VI) [37]. The resulting terms of the integral evaluation can be recognized to represent the geometrical optics fields, both incident and reflected geometrical optics fields, and the diffracted fields, both incident and reflected diffracted fields [38, 39]. These forms of the solution have received considerable attention in the geometrical theory of diffraction (GTD), which has become a generic name in the area of antennas and scattering [40–42].

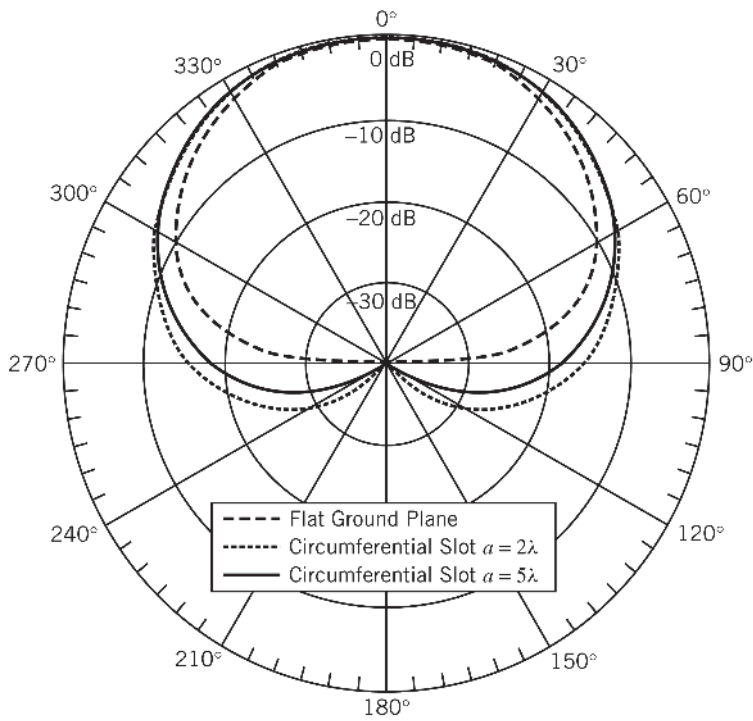
First, we will present the modal solution of the scattering by the wedge. In Chapter 13 we will briefly outline its asymptotic solution, whose form represents the geometrical optics and diffracted fields, and apply it to antenna and scattering problems.

11.6.1 Electric Line-Source Scattering by a Conducting Wedge: TM^z Polarization

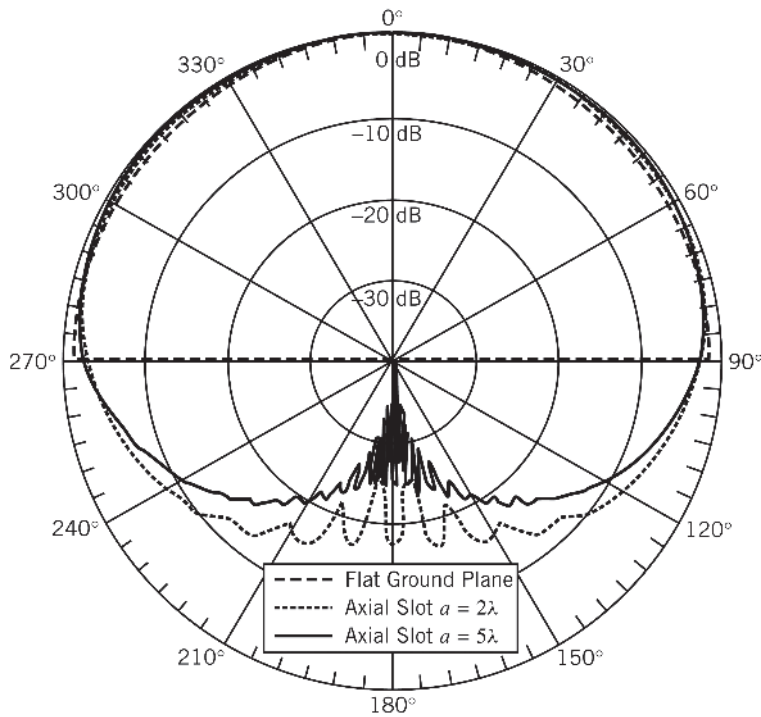
Let us assume that an infinite electric line source of electric current I_e is placed near a conducting wedge whose total inner wedge angle is $\text{WA} = 2\alpha$, as shown in Figure 11-22a. The incident field produced everywhere by the source, in the absence of the wedge, can be written according to (11-158a) and (11-158b) as

$$E_z^i = -\frac{\beta^2 I_e}{4\omega\epsilon} = \begin{cases} \sum_{m=-\infty}^{+\infty} J_m(\beta\rho) H_m^{(2)}(\beta\rho') e^{jm(\phi-\phi')}, & \rho \leq \rho' \\ \sum_{m=-\infty}^{+\infty} J_m(\beta\rho') H_m^{(2)}(\beta\rho) e^{jm(\phi-\phi')}, & \rho \geq \rho' \end{cases} \quad (11-179a)$$

$$(11-179b)$$



(a)



(b)

Figure 11-21 Normalized H-plane (for the circumferential aperture) and E-plane (for the axial aperture) amplitude pattern when mounted on a cylinder and flat ground plane. (a) H-plane (circumferential). (b) E-plane (axial).

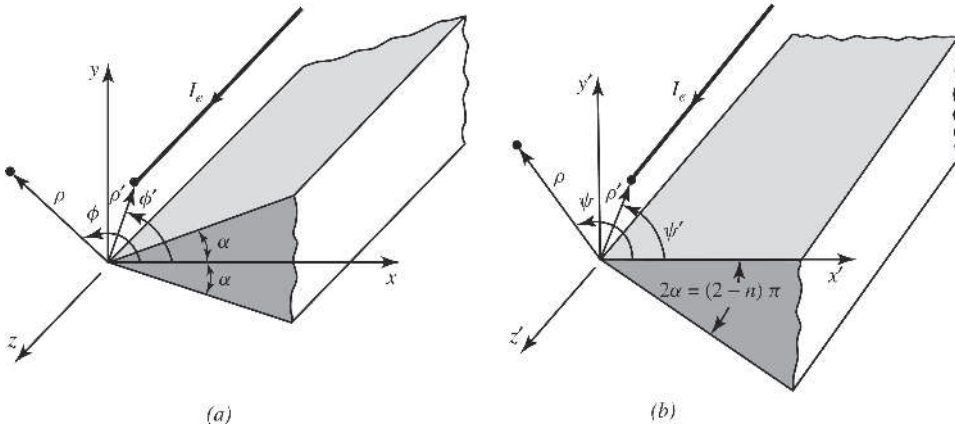


Figure 11-22 Electric line source near a two-dimensional conducting wedge. (a) Reference at bisector. (b) Reference at face.

The corresponding z component of the total scattered field must be chosen so that the sum of the two (incident plus scattered) along the faces of the wedge ($\phi = \alpha$ and $\phi = 2\pi - \alpha$) must vanish and simultaneously satisfy reciprocity (interchanging source and observation points). The ϕ variations must be represented by standing wave functions, since in the ϕ direction the waves bounce back and forth between the plates forming the wedge. It can be shown that expressions for the electric field that satisfy these conditions take the form

$$E_z^t = E_z^i + E_z^s = \begin{cases} \sum_v c_v f(\rho') J_v(\beta\rho) \sin[v(\phi' - \alpha)] \sin[v(\phi - \alpha)], & \rho \leq \rho' \\ \sum_v d_v g(\rho') H_v^{(2)}(\beta\rho) \sin[v(\phi' - \alpha)] \sin[v(\phi - \alpha)], & \rho \geq \rho' \end{cases} \quad (11-180a)$$

$$= \begin{cases} \sum_v c_v f(\rho') J_v(\beta\rho) \sin[v(\phi' - \alpha)] \sin[v(\phi - \alpha)], & \rho \leq \rho' \\ \sum_v d_v g(\rho') H_v^{(2)}(\beta\rho) \sin[v(\phi' - \alpha)] \sin[v(\phi - \alpha)], & \rho \geq \rho' \end{cases} \quad (11-180b)$$

When $\rho = \rho'$, the two must be identical. Thus,

$$\begin{aligned} \sum_v c_v f(\rho') J_v(\beta\rho') \sin[v(\phi' - \alpha)] \sin[v(\phi - \alpha)] \\ = \sum_v d_v g(\rho') H_v^{(2)}(\beta\rho') \sin[v(\phi' - \alpha)] \sin[v(\phi - \alpha)] \end{aligned} \quad (11-181)$$

which is satisfied if

$$c_v f(\rho') J_v(\beta\rho') = d_v g(\rho') H_v^{(2)}(\beta\rho') \quad (11-181a)$$

or

$$a_v = c_v = d_v \quad (11-181b)$$

$$f(\rho') = H_v^{(2)}(\beta\rho') \quad (11-181c)$$

$$g(\rho') = J_v(\beta\rho') \quad (11-181d)$$

Therefore, the total electric field (11-180a) and (11-180b) can be written as

$$E_z^t = E_z^i + E_z^s = \begin{cases} \sum_v a_v J_v(\beta\rho) H_v^{(2)}(\beta\rho') \sin[v(\phi' - \alpha)] \\ \quad \times \sin[v(\phi - \alpha)] & \rho \leq \rho' \\ \sum_v a_v J_v(\beta\rho') H_v^{(2)}(\beta\rho) \sin[v(\phi' - \alpha)] \\ \quad \times \sin[v(\phi - \alpha)] & \rho \geq \rho' \end{cases} \quad (11-182a)$$

$$(11-182b)$$

It is evident from (11-182a) and (11-182b) that, when $\phi = \alpha$, the total tangential electric field vanishes. However, when $\phi = 2\pi - \alpha$, the electric field (11-182a) and (11-182b) vanishes when

$$\sin[v(\phi - \alpha)]_{\phi=2\pi-\alpha} = \sin[v(2\pi - 2\alpha)] = \sin[2v(\pi - \alpha)] = 0 \quad (11-183)$$

or

$$2v(\pi - \alpha) = \sin^{-1}(0) = m\pi$$

$$v = \frac{m\pi}{2(\pi - \alpha)}, \quad m = 1, 2, 3 \dots \quad (11-183a)$$

Thus, in (11-182a) and (11-182b), the allowable values of v are those of (11-183a). The values of a_v depend on the type of source.

The magnetic field components can be obtained by using Maxwell's equations 11-86 through 11-86b, so that we can write that

$$H_\rho^t = -\frac{1}{j\omega\mu} \frac{1}{\rho} \frac{\partial E_z^t}{\partial \phi} = -\frac{1}{j\omega\mu} \frac{1}{\rho} \begin{cases} \sum_v v a_v J_v(\beta\rho) H_v^{(2)}(\beta\rho') \\ \quad \times \sin[v(\phi' - \alpha)] \cos[v(\phi - \alpha)] & \rho \leq \rho' \\ \sum_v v a_v J_v(\beta\rho') H_v^{(2)}(\beta\rho) \\ \quad \times \sin[v(\phi' - \alpha)] \cos[v(\phi - \alpha)] & \rho \geq \rho' \end{cases} \quad (11-184a)$$

$$(11-184b)$$

$$H_\phi^t = \frac{1}{j\omega\mu} \frac{\partial E_z^t}{\partial \rho} = \frac{\beta}{j\omega\mu} \begin{cases} \sum_v a_v J_v'(\beta\rho) H_v^{(2)}(\beta\rho') \\ \quad \times \sin[v(\phi' - \alpha)] \sin[v(\phi - \alpha)] & \rho \leq \rho' \\ \sum_v a_v J_v(\beta\rho') H_v^{(2)'}(\beta\rho) \\ \quad \times \sin[v(\phi' - \alpha)] \sin[v(\phi - \alpha)] & \rho \geq \rho' \end{cases} \quad (11-185a)$$

$$(11-185b)$$

where

$$' \equiv \frac{\partial}{\partial(\beta\rho)} \quad (11-185c)$$

At the source, the current density is obtained using

$$\begin{aligned}
 \mathbf{J}_s &= \hat{\mathbf{n}} \times \mathbf{H}^t = \hat{\mathbf{a}}_\rho \times (\hat{\mathbf{a}}_\rho H_\rho^t + \hat{\mathbf{a}}_\phi H_\phi^t)_{\rho=\rho'_+, \rho'_-} \\
 &= \hat{\mathbf{a}}_z H_\phi^t|_{\rho=\rho'_+, \rho'_-} = \hat{\mathbf{a}}_z [H_\phi^t(\rho'_+) - H_\phi^t(\rho'_-)] \\
 &= \hat{\mathbf{a}}_z \frac{\beta}{j\omega\mu} \sum_v a_v [J_v(\beta\rho') H_v^{(2)'}(\beta\rho') - H_v^{(2)}(\beta\rho') J_v'(\beta\rho')] \\
 &\quad \times \sin[v(\phi' - \alpha)] \sin[v(\phi - \alpha)] \\
 \mathbf{J}_s &= \hat{\mathbf{a}}_z \frac{\beta}{j\omega\mu} \sum_v a_v (-j) [J_v(\beta\rho') Y_v'(\beta\rho') - J_v'(\beta\rho') Y_v(\beta\rho')] \\
 &\quad \times \sin[v(\phi' - \alpha)] \sin[v(\phi - \alpha)]
 \end{aligned} \tag{11-186}$$

which, by using the Wronskian of (11-95), reduces to

$$\mathbf{J}_s = -\hat{\mathbf{a}}_z \frac{2}{\pi\omega\mu\rho'} \sum_v a_v \sin[v(\phi' - \alpha)] \sin[v(\phi - \alpha)] \tag{11-186a}$$

Since the Fourier series for a current impulse of amplitude I_e located at $\rho = \rho'$ and $\phi = \phi'$ is [11]

$$J_z = \frac{I_e}{(\pi - \alpha)\rho'} \sum_v \sin[v(\phi' - \alpha)] \sin[v(\phi - \alpha)] \tag{11-187}$$

then comparing (11-186a) and (11-187) leads to

$$-\frac{2}{\pi\omega\mu} a_v = \frac{I_e}{\pi - \alpha} \Rightarrow a_v = -\frac{\pi\omega\mu I_e}{2(\pi - \alpha)} \tag{11-188}$$

A. Far-Zone Field When the observations are made in the far zone ($\beta\rho \gg 1$, $\rho > \rho'$) the total electric field (11-182b) can be written, by replacing the Hankel function $H_v^{(2)}(\beta\rho)$ by its asymptotic expression (11-135a), as

$$\begin{aligned}
 E_z^t &\stackrel{\beta\rho \rightarrow \infty}{\simeq} \sqrt{\frac{2j}{\pi\beta\rho}} e^{-j\beta\rho} \sum_v a_v j^v J_v(\beta\rho') \sin[v(\phi' - \alpha)] \sin[v(\phi - \alpha)] \\
 &\stackrel{\beta\rho \rightarrow \infty}{\simeq} -I_e \sqrt{\frac{\pi j}{2\beta}} \frac{\omega\mu}{\pi - \alpha} \frac{e^{-j\beta\rho}}{\sqrt{\rho}} \sum_v j^v J_v(\beta\rho') \sin[v(\phi' - \alpha)] \sin[v(\phi - \alpha)] \\
 &\quad \boxed{E_z^t \stackrel{\beta\rho \rightarrow \infty}{\simeq} f_e(\rho) \sum_v j^v J_v(\beta\rho') \sin[v(\phi' - \alpha)] \sin[v(\phi - \alpha)]}
 \end{aligned} \tag{11-189}$$

where

$$f_e(\rho) = -I_e \sqrt{\frac{\pi j}{2\beta}} \frac{\omega\mu}{\pi - \alpha} \frac{e^{-j\beta\rho}}{\sqrt{\rho}} \tag{11-189a}$$

Therefore, (11-189) represents the total electric field created in the far-zone region by an electric source of strength I_e located at ρ' , ϕ' .

B. Plane Wave Scattering When the source is placed at far distances ($\beta\rho' \gg 1$ and $\rho' > \rho$) and the observations are made at any point, the total electric field of (11-182a) can be written, by replacing the Hankel function $H_v^{(2)}(\beta\rho')$ by its asymptotic form (11-135a), as

$$\begin{aligned}
 E_z^t &\stackrel{\beta\rho' \rightarrow \infty}{\simeq} -I_e \sqrt{\frac{\pi j}{2\beta}} \frac{\omega\mu}{\pi - \alpha} \frac{e^{-j\beta\rho'}}{\sqrt{\rho'}} \sum_v j^v J_v(\beta\rho) \sin[v(\phi' - \alpha)] \sin[v(\phi - \alpha)] \\
 &\simeq g_e(\rho') \sum_v j^v J_v(\beta\rho) \sin[v(\phi' - \alpha)] \sin[v(\phi - \alpha)] \\
 E_z^t &\stackrel{\beta\rho' \rightarrow \infty}{\simeq} E_0 \sum_v j^v J_v(\beta\rho) \sin[v(\phi' - \alpha)] \sin[v(\phi - \alpha)]
 \end{aligned} \tag{11-190}$$

where

$$E_0 = g_e(\rho') = -I_e \sqrt{\frac{\pi j}{2\beta}} \frac{\omega\mu}{\pi - \alpha} \frac{e^{-j\beta\rho'}}{\sqrt{\rho'}} \tag{11-190a}$$

It is evident that (11-190) can also be obtained from (11-189) by reciprocity, that is, interchanging source and observation point. This is accomplished by interchanging ρ with ρ' , and ϕ with ϕ' , or for this problem simply by interchanging ρ and ρ' only.

Equation 11-190 also represents the total electric field of a TM^z uniform plane wave of strength E_0 incident at an angle ϕ' on a conducting wedge of interior angle 2α . When the wedge is a half-plane ($\alpha = 0$) (11-190) reduces to

$$E_z^t \stackrel{\beta\rho' \rightarrow \infty}{\simeq}_{\alpha=0} E_0 \sum_v j^v J_v(\beta\rho) \sin(v\phi') \sin(v\phi) \tag{11-191}$$

which by using (11-183a) can also be expressed as

$$E_z^t \stackrel{\beta\rho' \rightarrow \infty}{\simeq}_{\alpha=0} E_0 \sum_{m=1}^{\infty} j^{m/2} J_{m/2}(\beta\rho) \sin\left(\frac{m}{2}\phi'\right) \sin\left(\frac{m}{2}\phi\right) \tag{11-191a}$$

The normalized scattering patterns at a distance λ ($\rho = \lambda$) from the edge of the wedge formed when a plane wave is incident upon a wedge of $2\alpha = 0^\circ$ ($n = 2$; half-plane) and 90° ($n = 1.5$) are shown in Figure 11-23.

11.6.2 Magnetic Line-Source Scattering by a Conducting Wedge: TE^z Polarization

When the line source of Figure 11-22a is magnetic of current I_m , the total magnetic field has only a z component, and it can be written, by referring to (11-182a) and (11-182b), as

$$\begin{aligned}
 H_z^t &= H_z^i + H_z^s \\
 &= \begin{cases} \sum_s b_s J_s(\beta\rho) H_s^{(2)}(\beta\rho') \cos[s(\phi' - \alpha)] \cos[s(\phi - \alpha)], & \rho \leq \rho' \\ \sum_s b_s J_s(\beta\rho') H_s^{(2)}(\beta\rho) \cos[s(\phi' - \alpha)] \cos[s(\phi - \alpha)], & \rho \geq \rho' \end{cases}
 \end{aligned} \tag{11-192a}$$

$$\tag{11-192b}$$

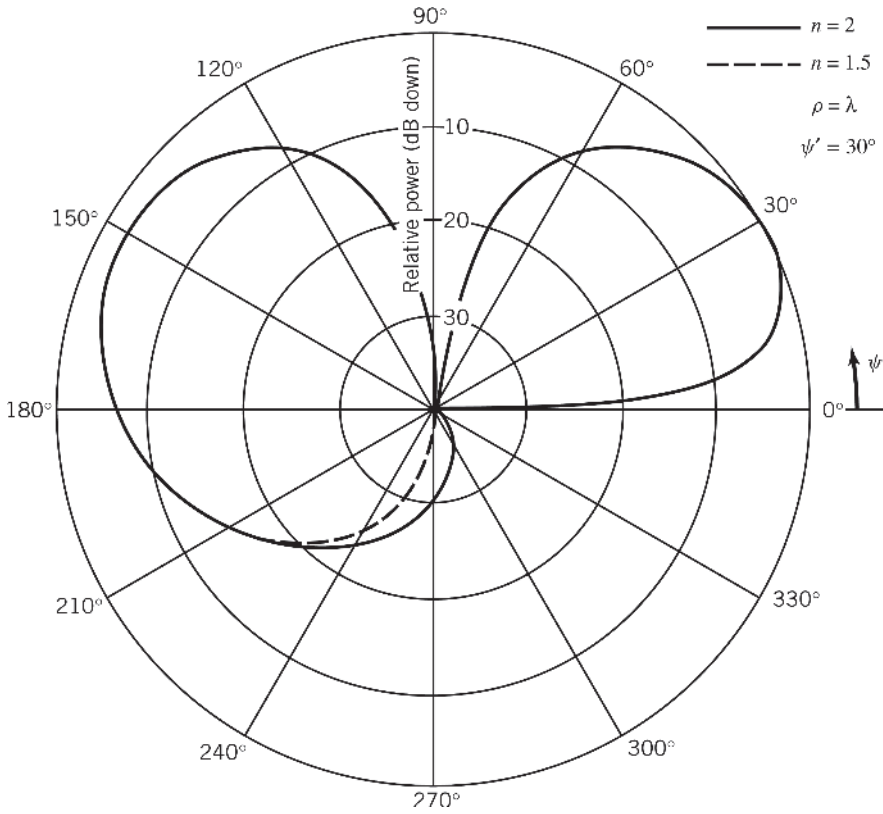


Figure 11-23 Normalized amplitude pattern of a TM^z (soft polarization) plane wave incident on a two-dimensional conducting wedge.

The allowable values of s are obtained by applying the boundary conditions, and the coefficients b_s are determined by the type of source.

To apply the boundary conditions, we first need to find the corresponding electric field components (especially the tangential components). This is accomplished by using Maxwell's equations 11-106 through 11-106b. Thus, the total radial component of the electric field can be written as

$$E_\rho^t = \frac{1}{j\omega\epsilon} \frac{1}{\rho} \frac{\partial H_z^t}{\partial \phi}$$

$$= -\frac{1}{j\omega\epsilon} \frac{1}{\rho} \begin{cases} \sum_s s b_s J_s(\beta\rho) H_s^{(2)}(\beta\rho') \\ \times \cos[s(\phi' - \alpha)] \sin[s(\phi - \alpha)], & \rho \leq \rho' \end{cases} \quad (11-193a)$$

$$\begin{cases} \sum_s s b_s J_s(\beta\rho') H_s^{(2)}(\beta\rho) \\ \times \cos[s(\phi' - \alpha)] \sin[s(\phi - \alpha)], & \rho \geq \rho' \end{cases} \quad (11-193b)$$

The boundary conditions that must be satisfied are

$$\begin{aligned} E_\rho^t(0 \leq \rho, \rho' \leq \infty, \phi = \alpha, 0 \leq \phi' \leq 2\pi) \\ = E_\rho^t(0 \leq \rho, \rho' \leq \infty, \phi = 2\pi - \alpha, 0 \leq \phi' \leq 2\pi) = 0 \end{aligned} \quad (11-194)$$

The first boundary condition of (11-194) is always satisfied regardless of the values of s . Applying the second boundary condition leads to

$$\begin{aligned}
 E_\rho^t(0 \leq \rho, \rho' \leq \infty, \phi = 2\pi - \alpha, 0 \leq \phi' \leq 2\pi) &= 0 \\
 &= -\frac{1}{j\omega\varepsilon\rho} \sum_s s b_s J_s(\beta\rho) H_s^{(2)}(\beta\rho') \cos[s(\phi' - \alpha)] \sin[2s(\pi - \alpha)] \\
 &= -\frac{1}{j\omega\varepsilon\rho} \sum_s s b_s J_s(\beta\rho') H_s^{(2)}(\beta\rho) \cos[s(\phi' - \alpha)] \sin[2s(\pi - \alpha)] \quad (11-195)
 \end{aligned}$$

which is satisfied provided

$$\sin[2s(\pi - \alpha)] = 0 \Rightarrow 2s(\pi - \alpha) = \sin^{-1}(0) = m\pi$$

$$s = \frac{m\pi}{2(\pi - \alpha)}, \quad m = 0, 1, 2, \dots$$

(11-195a)

Since the source is magnetic, the coefficients b_s take the form of

$$b_s = \varepsilon_s \left[\frac{\pi\omega\varepsilon I_m}{4(\pi - \alpha)} \right]$$

(11-196)

where

$$\varepsilon_s = \begin{cases} 1 & s = 0 \\ 2 & s \neq 0 \end{cases}$$

(11-196a)

In the far zone ($\beta\rho \gg 1$), the total field of (11-192b) reduces, by replacing the Hankel function $H_s^{(2)}(\beta\rho)$ with its asymptotic form (11-135a), to

$$\begin{aligned}
 H_z^t \stackrel{\beta\rho \rightarrow \infty}{\simeq} I_m \sqrt{\frac{\pi j}{8\beta}} \frac{\omega\varepsilon}{\pi - \alpha} \frac{e^{-j\beta\rho}}{\sqrt{\rho}} \sum_s \varepsilon_s j^s J_s(\beta\rho') \cos[s(\phi' - \alpha)] \cos[s(\phi - \alpha)] \\
 \boxed{H_z^t \stackrel{\beta\rho \rightarrow \infty}{\simeq} f_h(\rho) \sum_s \varepsilon_s j^s J_s(\beta\rho') \cos[s(\phi' - \alpha)] \cos[s(\phi - \alpha)]} \quad (11-197)
 \end{aligned}$$

where

$$f_h(\rho) = I_m \sqrt{\frac{\pi j}{8\beta}} \frac{\omega\varepsilon}{\pi - \alpha} \frac{e^{-j\beta\rho}}{\sqrt{\rho}} \quad (11-197a)$$

When the source is removed at far distances ($\beta\rho' \gg 1$ and $\rho' > \rho$), (11-192a) reduces to

$$\begin{aligned}
 H_z^t \stackrel{\beta\rho' \rightarrow \infty}{\simeq} g_h(\rho') \sum_s \varepsilon_s j^s J_s(\beta\rho) \cos[s(\phi' - \alpha)] \cos[s(\phi - \alpha)] \\
 \boxed{H_z^t \stackrel{\beta\rho' \rightarrow \infty}{\simeq} H_0 \sum_s \varepsilon_s j^s J_s(\beta\rho) \cos[s(\phi' - \alpha)] \cos[s(\phi - \alpha)]} \quad (11-198)
 \end{aligned}$$

where

$$H_0 = g_h(\rho') = I_m \sqrt{\frac{\pi j}{8\beta}} \frac{\omega \varepsilon}{\pi - \alpha} \frac{e^{-j\beta\rho'}}{\sqrt{\rho'}} \quad (11-198a)$$

Equation 11-198) also represents the total magnetic field of a TE^z uniform plane wave of strength H_0 incident at an angle ϕ' on a conducting wedge of interior angle 2α . For a wedge with zero included angle (half-plane $\alpha = 0$), (11-198) reduces to

$$H_z^t \underset{\alpha=0}{\overset{\beta\rho' \rightarrow \infty}{\simeq}} H_0 \sum_s \varepsilon_s j^s J_s(\beta\rho) \cos(s\phi') \cos(s\phi) \quad (11-199)$$

which, by using (11-195a), can also be expressed as

$$H_z^t \underset{\alpha=0}{\overset{\beta\rho' \rightarrow \infty}{\simeq}} H_0 \sum_{m=0}^{\infty} \varepsilon_{m/2} j^{m/2} J_{m/2}(\beta\rho) \cos\left(\frac{m}{2}\phi'\right) \cos\left(\frac{m}{2}\phi\right) \quad (11-199a)$$

The normalized scattering patterns at a distance λ ($\rho = \lambda$) from the edge of the wedge formed when a plane wave is incident upon a wedge of $2\alpha = 0^\circ$ ($n = 2$; half-plane) and 90° ($n = 1.5$) are shown in Figure 11-24.

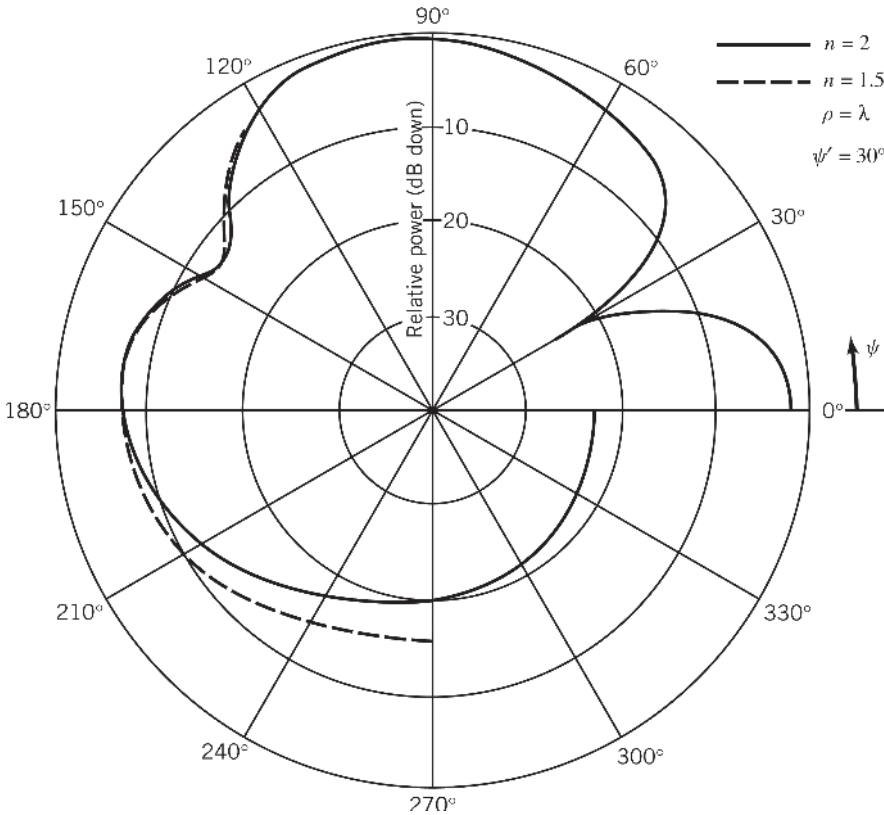


Figure 11-24 Normalized amplitude pattern of a TE^z (hard polarization) plane wave incident on a two-dimensional conducting wedge.

11.6.3 Electric and Magnetic Line-Source Scattering by a Conducting Wedge

The total electric field of an electric line source near a conducting wedge, as given by (11-182a) and (11-182b), and the total magnetic field of a magnetic line source near a conducting wedge, as given by (11-192a) and (11-192b), can both be represented by the same expression by adopting the coordinate system (x', y', z') of Figure 11-22b, instead of the (x, y, z) , which is referenced to the side of the wedge that is illuminated by the source. This is usually more convenient because similar forms of the expression can represent either polarization. Also the interior angle of the wedge will be represented by

$$2\alpha = (2 - n)\pi \Rightarrow n = 2 - \frac{2\alpha}{\pi} \quad (11-200)$$

Thus, a given value of n represents a wedge with a specific included angle: values of $n > 1$ represent wedges with included angles less than 180° (referred to as *exterior* wedges) and values of $n < 1$ represent wedges with included angles greater than 180° (referred to as *interior* wedges). Thus the allowable values of v , as given by (11-183a), and those of s , as given by (11-195a), can now be represented by

$$v = \frac{m\pi}{2(\pi - \alpha)} \Big|_{\alpha=(1-n/2)\pi} = \frac{m}{n}, \quad m = 1, 2, 3, \dots \quad (11-201a)$$

$$s = \frac{m\pi}{2(\pi - \alpha)} \Big|_{\alpha=(1-n/2)\pi} = \frac{m}{n}, \quad m = 0, 1, 2, \dots \quad (11-201b)$$

In addition, the amplitude coefficients a_v , as given by (11-188), and b_s , as given by (11-196), can now be expressed as

$$a_v = -\frac{\pi\omega\mu I_e}{2(\pi - \alpha)} \Big|_{\alpha=(1-n/2)\pi} = -\frac{\omega\mu I_e}{2} \left(\frac{2}{n} \right) \quad (11-202a)$$

$$b_s = \varepsilon_s \left[\frac{\pi\omega\varepsilon I_m}{4(\pi - \alpha)} \right]_{\alpha=(1-n/2)\pi} = \frac{\omega\varepsilon I_m}{2} \left(\frac{\varepsilon_s}{n} \right) \quad (11-202b)$$

Using the new coordinate system (x', y', z') of Figure 11-22b, we can write that

$$\phi' = \psi' + \alpha \quad (11-203a)$$

$$\phi = \psi + \alpha \quad (11-203b)$$

Therefore, the sine functions of (11-182a) and (11-182b) and the cosine functions of (11-192a) and (11-192b) can be written as

$$\begin{aligned} \sin[v(\phi' - \alpha)] \sin[v(\phi - \alpha)] &= \sin\left[\frac{m}{n}(\psi' + \alpha - \alpha)\right] \sin\left[\frac{m}{n}(\psi + \alpha - \alpha)\right] \\ &= \sin\left(\frac{m}{n}\psi'\right) \sin\left(\frac{m}{n}\psi\right) \\ &= \frac{1}{2} \left\{ \cos\left[\frac{m}{n}(\psi - \psi')\right] - \cos\left[\frac{m}{n}(\psi + \psi')\right] \right\} \end{aligned} \quad (11-204a)$$

$$\begin{aligned}
\cos[s(\phi' - \alpha)] \cos[s(\phi - \alpha)] &= \cos\left[\frac{m}{n}(\psi' + \alpha - \alpha)\right] \cos\left[\frac{m}{n}(\psi + \alpha - \alpha)\right] \\
&= \cos\left(\frac{m}{n}\psi'\right) \cos\left(\frac{m}{n}\psi\right) \\
&= \frac{1}{2} \left\{ \cos\left[\frac{m}{n}(\psi - \psi')\right] - \cos\left[\frac{m}{n}(\psi + \psi')\right] \right\} \quad (11-204b)
\end{aligned}$$

Using all these new notations, we can write for the TM^z polarization the total electric field of (11-182a) and (11-182b) and for the TE^z polarization the total magnetic field of (11-192a) and (11-192b) as

TM^z

$$E_z^t = -\frac{\omega\mu I_e}{4} \frac{1}{n} \left\{ \begin{aligned} &\sum_{m=0,1,\dots}^{\infty} 2J_{m/n}(\beta\rho) H_{m/n}^{(2)}(\beta\rho') \\ &\times \left\{ \cos\left[\frac{m}{n}(\psi - \psi')\right] - \cos\left[\frac{m}{n}(\psi + \psi')\right] \right\}, \quad \rho \leq \rho' \end{aligned} \right. \quad (11-205a)$$

$$\left\{ \begin{aligned} &\sum_{m=0,1,\dots}^{\infty} 2J_{m/n}(\beta\rho') H_{m/n}^{(2)}(\beta\rho) \\ &\times \left\{ \cos\left[\frac{m}{n}(\psi - \psi')\right] - \cos\left[\frac{m}{n}(\psi + \psi')\right] \right\}, \quad \rho \geq \rho' \end{aligned} \right. \quad (11-205b)$$

TE^z

$$H_z^t = \frac{\omega\varepsilon I_m}{4} \frac{1}{n} \left\{ \begin{aligned} &\sum_{m=0,1,\dots}^{\infty} \varepsilon_m J_{m/n}(\beta\rho) H_{m/n}^{(2)}(\beta\rho') \\ &\times \left\{ \cos\left[\frac{m}{n}(\psi - \psi')\right] + \cos\left[\frac{m}{n}(\psi + \psi')\right] \right\}, \quad \rho \leq \rho' \end{aligned} \right. \quad (11-206a)$$

$$\left\{ \begin{aligned} &\sum_{m=0,1,\dots}^{\infty} \varepsilon_m J_{m/n}(\beta\rho') H_{m/n}^{(2)}(\beta\rho) \\ &\times \left\{ \cos\left[\frac{m}{n}(\psi - \psi')\right] + \cos\left[\frac{m}{n}(\psi + \psi')\right] \right\}, \quad \rho \geq \rho' \end{aligned} \right. \quad (11-206b)$$

To make the summations in (11-205a) through (11-206b) uniform, the summations of (11-205a) and (11-205b) are noted to begin with $m=0$, even though the allowable values of m as given by (11-201a) begin with $m=1$. However, it should be noted that $m=0$ in (11-205a) and (11-205b) does not contribute, and the expressions are correct as stated.

It is apparent, by comparing (11-205a) and (11-205b) with (11-206a) and (11-206b), that they are of similar forms. Therefore, we can write both as

$$E_z^t = -\frac{\omega\mu I_e}{4} G(\rho, \rho', \psi, \psi', n) \quad \text{for } \text{TM}^z \quad (11-207a)$$

$$H_z^t = +\frac{\omega\varepsilon I_m}{4} G(\rho, \rho', \psi, \psi', n) \quad \text{for } \text{TE}^z \quad (11-207b)$$

where

$$G(\rho, \rho', \psi - \psi', n) = \frac{1}{n} \begin{cases} \sum_{m=0,1,\dots}^{\infty} \varepsilon_m J_{m/n}(\beta\rho) H_{m/n}^{(2)}(\beta\rho') \left\{ \cos\left[\frac{m}{n}(\psi - \psi')\right] \pm \cos\left[\frac{m}{n}(\psi + \psi')\right] \right\}, & \rho \leq \rho' \\ \sum_{m=0,1,\dots}^{\infty} \varepsilon_m J_{m/n}(\beta\rho') H_{m/n}^{(2)}(\beta\rho) \left\{ \cos\left[\frac{m}{n}(\psi - \psi')\right] \pm \cos\left[\frac{m}{n}(\psi + \psi')\right] \right\}, & \rho \geq \rho' \end{cases} \quad (11-208a)$$

$$(11-208b)$$

$$\varepsilon_m = \begin{cases} 1 & m = 0 \\ 2 & m \neq 0 \end{cases} \quad (11-208c)$$

The plus (+) sign between the cosine terms is used for the TE^z polarization and the minus (−) sign is used for the TM^z polarization. Again, note that the $m = 0$ terms do not contribute anything for the TM^z polarization.

The forms of (11-207a) through (11-208b) are those usually utilized in the geometrical theory of diffraction (GTD) [40–44] where $G(\rho, \rho', \psi, \psi', n)$ is usually referred to as the *Green's function*. Since the summations in (11-208a) and (11-208b) are poorly convergent when the arguments of the Bessel and/or Hankel functions are large, asymptotic forms of them will be derived in Chapter 13 that are much more computationally efficient. The various terms of the asymptotic forms will be associated with incident and reflected geometrical optics and diffracted fields. It is also convenient in diffraction theory to refer to the TM^z polarization as the *soft* polarization; the TE^z is referred to as the *hard* polarization. This is a convenient designation adopted from acoustics.

11.7 SPHERICAL WAVE ORTHOGONALITIES, TRANSFORMATIONS, AND THEOREMS

When dealing with scattering from structures whose geometry best conforms to spherical coordinates, it is often most convenient to transform wave functions (such as plane waves) from one coordinate system to another. This was done in Section 11.4 where uniform plane wave functions in rectilinear form were transformed and represented by cylindrical wave functions. This allowed convenient examination of the scattering of plane waves by cylindrical structures of circular and wedge cross sections. In addition, certain theorems concerning cylindrical wave functions were introduced, which were helpful in analyzing the scattering by cylindrical structures of circular cross sections of waves emanating from line sources.

In this section we want to introduce some orthogonality relationships, wave transformations, and theorems that are very convenient for examining scattering of plane waves from spherical structures and waves emanating from finite sources placed in the vicinity of spherical scatterers. First of all, let us examine radiation from a finite source radiating in an unbounded medium.

11.7.1 Vertical Dipole Spherical Wave Radiation

There are many sources of spherical wave radiation. In fact, almost all sources used in practice are considered to excite spherical waves. One of the most prominent is that of a finite length wire whose total radiation can be obtained as a superposition of radiation from a very small linear

current element of length $\Delta\ell$ and constant electric current $\mathbf{I}_e = \hat{\mathbf{a}}_z I_e$. This is usually referred to as an infinitesimal dipole [1]. The radiation of other sources can be obtained by knowing the radiation from an *infinitesimal dipole*. Therefore, it is important that we briefly examine the radiation from such a source.

It is usually most convenient to place the linear element at the origin of the coordinate system and have its length and current flow along the z axis, as shown in Figure 11-25a. To find the fields radiated by this source, we resort to the techniques of Chapter 6, Sections 6.4 and 6.6, where we first specify the currents I_e and I_m of the source. Then we find the potentials $\mathbf{A}$ and $\mathbf{F}$ [using (6-97a) and (6-97b)], and determine the radiated $\mathbf{E}$ and $\mathbf{H}$ [using (6-34) and (6-35)].

Following such a procedure, the electric and magnetic fields radiated by the infinitesimal dipole of Figure 11-25 were derived in Example 6-3. In terms of spherical wave functions, the vector potential $\mathbf{A}$ can also be written as

$$\mathbf{A} = \hat{\mathbf{a}}_z \frac{\mu I_e \Delta\ell}{4\pi} \frac{e^{-j\beta r}}{r} = -\hat{\mathbf{a}}_z j \frac{\mu \beta I_e \Delta\ell}{4\pi} h_0^{(2)}(\beta r) \quad (11-209)$$

where $h_0^{(2)}(\beta r)$ is the spherical Hankel function of order zero, given by

$$h_0^{(2)}(\beta r) = \frac{e^{-j\beta r}}{-j\beta r} \quad (11-209a)$$

For the dual problem of the linear magnetic current element of current I_m , the vector potential function $\mathbf{F}$ takes the form of

$$\mathbf{F} = \hat{\mathbf{a}}_z \frac{\varepsilon I_m \Delta\ell}{4\pi} \frac{e^{-j\beta r}}{r} = -\hat{\mathbf{a}}_z j \frac{\varepsilon \beta I_m \Delta\ell}{4\pi} h_0^{(2)}(\beta r) \quad (11-210)$$

If the source is removed from the origin, as shown in Figure 11-25b, then the potentials $\mathbf{A}$ and $\mathbf{F}$ of (11-209) and (11-210) take the form of

$$\mathbf{A} = -\hat{\mathbf{a}}_z j \frac{\mu \beta I_e \Delta\ell}{4\pi} h_0^{(2)}(\beta |\mathbf{r} - \mathbf{r}'|) \quad (11-211a)$$

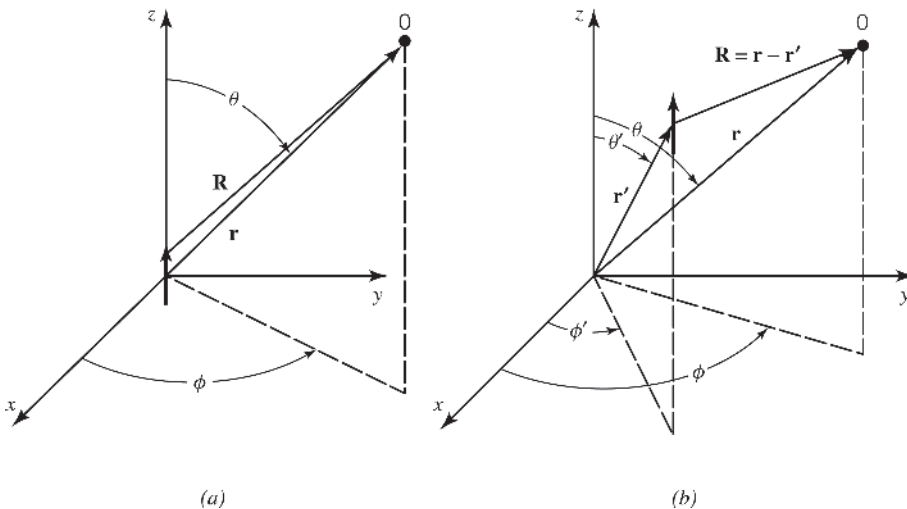


Figure 11-25 Geometry and coordinate system for vertical dipole radiation. (a) At origin. (b) Offset.

$$\mathbf{F} = -\hat{\mathbf{a}}_z j \frac{\varepsilon \beta I_m \Delta \ell}{4\pi} h_0^{(2)}(\beta |\mathbf{r} - \mathbf{r}'|) \quad (11-211b)$$

where

$$h_0^{(2)}(\beta |\mathbf{r} - \mathbf{r}'|) = \frac{e^{-j\beta |\mathbf{r} - \mathbf{r}'|}}{-j\beta |\mathbf{r} - \mathbf{r}'|} \quad (11-211c)$$

11.7.2 Orthogonality Relationships

When solving electromagnetic wave problems dealing with spherical structures (either waveguides, cavities, or scatterers), the θ variations are represented, as illustrated in Chapters 3 (Section 3.4.3) and 10, by Legendre polynomials $P_n(\cos\theta)$ and associated Legendre functions $P_n^m(\cos\theta)$ (see Appendix V).

The Legendre polynomials $P_n(\cos\theta)$ are often called *zonal harmonics* [45, 46], and they form a complete orthogonal set in the interval $0 \leq \theta \leq \pi$. Therefore, in this interval any arbitrary wave function can be represented by a series of Legendre polynomials. This is similar to the representation of any periodic function by a series of sines and cosines (Fourier series), since Legendre polynomials are very similar in form to cosinusoidal functions. In addition, the products of associated Legendre functions $P_n^m(\cos\theta)$ with sines and cosines [$P_n^m(\cos\theta)\cos(m\phi)$ and $P_n^m(\cos\theta)\sin(m\phi)$] are often referred to as *tesseral harmonics* [45, 46], and they form a complete orthogonal set on the surface of a sphere. Therefore, any wave function that is defined over a sphere can be expressed by a series of tesseral harmonics.

Some of the most important and necessary orthogonality relationships that are necessary to solve wave scattering by spheres will be stated here. The interested reader is referred to [11, 45–48] for more details and derivations.

In the interval $0 \leq \theta \leq \pi$, the integral of the product of Legendre polynomials is equal to

$$\int_0^\pi P_n(\cos\theta)P_m(\cos\theta)\sin\theta\,d\theta = \begin{cases} 0, & n \neq m \\ \frac{2}{2n+1}, & n = m \end{cases} \quad (11-212a)$$

$$(11-212b)$$

Any function $f(\theta)$ defined in the interval of $0 \leq \theta \leq \pi$ can be represented by a series of Legendre polynomials

$$f(\theta) = \sum_{n=0}^{\infty} a_n P_n(\cos\theta), \quad 0 \leq \theta \leq \pi \quad (11-213)$$

where

$$a_n = \frac{2n+1}{2} \int_0^\pi f(\theta)P_n(\cos\theta)\sin\theta\,d\theta \quad (11-213a)$$

which is known as the *Fourier–Legendre* series.

Defining the tesseral harmonics by

$$T_{mn}^e(\theta, \phi) = P_n^m(\cos\theta)\cos(m\phi) \quad (11-214a)$$

$$T_{mn}^o(\theta, \phi) = P_n^m(\cos\theta)\sin(m\phi) \quad (11-214b)$$

and because

$$\begin{aligned} \int_0^{2\pi} \sin(p\phi)\sin(q\phi)\,d\phi &= \int_0^{2\pi} \cos(p\phi)\cos(q\phi)\,d\phi \\ &= \begin{cases} 0, & p \neq q \\ \pi, & p = q \neq 0 \end{cases} \end{aligned} \quad (11-215a)$$

$$(11-215b)$$

it can be shown that

$$\int_0^{2\pi} \left[\int_0^\pi T_{mn}^e(\theta, \phi) T_{pq}^0(\theta, \phi) \sin \theta d\theta \right] d\phi = 0 \quad (11-216a)$$

$$\int_0^{2\pi} \left[\int_0^\pi T_{mn}^i(\theta, \phi) T_{pq}^i(\theta, \phi) \sin \theta d\theta \right] d\phi = 0 \quad \begin{matrix} mn \neq pq \\ i = e \text{ or } 0 \end{matrix} \quad (11-216b)$$

$$\int_0^{2\pi} \left[\int_0^\pi \left[T_{mn}^i(\theta, \phi) \right]^2 \sin \theta d\theta \right] d\phi = \begin{cases} \frac{4\pi}{2n+1}, & m=0, \quad i=e \\ \frac{2\pi}{2n+1} \frac{(n+m)!}{(n-m)!}, & m \neq 0, \quad i=e \text{ or } 0 \end{cases} \quad (11-216c)$$

$$(11-216d)$$

11.7.3 Wave Transformations and Theorems

As was done for cylindrical wave functions in Section 11.4, in scattering it is often most convenient to express wave functions in one coordinate system in terms of wave functions of another coordinate system. The same applies to wave scattering by spherical structures. Therefore, it is convenient for plane wave scattering by spherical geometries to express the plane waves, which are most conveniently written in rectilinear form, in terms of spherical wave functions.

To demonstrate that, let us assume that a uniform plane wave is traveling along the $+z$ direction, as shown in Figure 11-26, and it can be written as

$$\mathbf{E}^+ = \hat{\mathbf{a}}_x E_x^+ = \hat{\mathbf{a}}_x E_0 e^{-j\beta z} = \hat{\mathbf{a}}_x e^{-j\beta z} \quad (11-217)$$

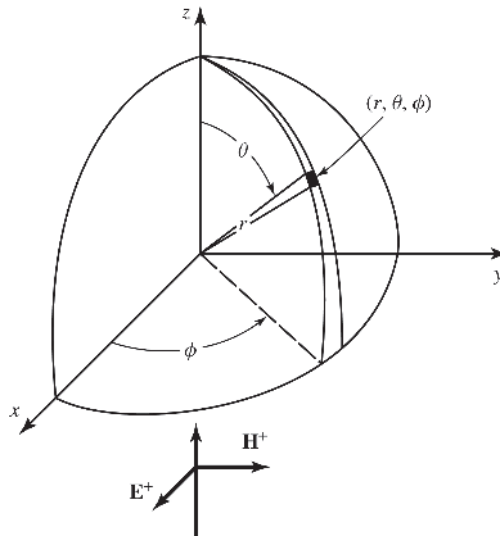


Figure 11-26 Uniform plane wave traveling in the $+z$ direction.

The plane wave can be represented by an infinite sum of spherical wave functions of the form

$$E_x^+ = e^{-j\beta z} = e^{-j\beta r \cos \theta} = \sum_{n=0}^{\infty} a_n j_n(\beta r) P_n(\cos \theta) \quad (11-217a)$$

since it must be independent of ϕ and be finite at the origin ($r = 0$). The next step is to determine the amplitude coefficients a_n . This can be accomplished as follows.

Multiplying both sides of (11-217a) by $P_m(\cos \theta) \sin \theta$ and integrating in θ from 0 to π , we have that

$$\int_0^{\pi} e^{-j\beta r \cos \theta} P_m(\cos \theta) \sin \theta d\theta = \int_0^{\pi} \left[\sum_{n=0}^{\infty} a_n j_n(\beta r) P_n(\cos \theta) P_m(\cos \theta) \sin \theta \right] d\theta \quad (11-218)$$

Interchanging integration and summation, we have that

$$\int_0^{\pi} e^{-j\beta r \cos \theta} P_m(\cos \theta) \sin \theta d\theta = \sum_{n=0}^{\infty} a_n j_n(\beta r) \int_0^{\pi} P_n(\cos \theta) P_m(\cos \theta) \sin \theta d\theta \quad (11-218a)$$

Using the orthogonality (11-212b) reduces (11-218a) to

$$\int_0^{\pi} e^{-j\beta r \cos \theta} P_m(\cos \theta) \sin \theta d\theta = \frac{2a_m}{2m+1} j_m(\beta r) \quad (11-219)$$

Since the integral of the left side of (11-219) is equal to

$$\int_0^{\pi} e^{-j\beta r \cos \theta} P_m(\cos \theta) \sin \theta d\theta = 2j^{-m} j_m(\beta r) \quad (11-219a)$$

equating (11-219) and (11-219a) leads to

$$\frac{2a_m}{2m+1} j_m(\beta r) = 2j^{-m} j_m(\beta r) \Rightarrow a_m = j^{-m} (2m+1) \quad (11-220)$$

Thus, (11-217a) reduces to

$$E_x^+ = e^{-j\beta z} = e^{-j\beta r \cos \theta} = \sum_{n=0}^{\infty} a_n j_n(\beta r) P_n(\cos \theta) \quad (11-221)$$

where

$$a_n = j^{-n} (2n+1) \quad (11-221a)$$

In a similar manner, it can be shown that

$$E_x^- = e^{+j\beta z} = e^{+j\beta r \cos \theta} = \sum_{n=0}^{\infty} b_n j_n(\beta r) P_n(\cos \theta) \quad (11-222)$$

where

$$b_n = j^n (2n+1) \quad (11-222a)$$

When dealing with spherical wave scattering of waves generated by linear dipole radiators, it is convenient to express their radiation, determined using (11-211a) through (11-211c), in terms of spherical wave functions. This can be accomplished using the *addition theorem* [11] of spherical wave functions, which states that (11-211c) can be expressed by referring to the geometry of Figure 11-25b as

$$h_0^{(2)}(\beta |\mathbf{r} - \mathbf{r}'|) = \begin{cases} \sum_{n=0}^{\infty} (2n+1) h_n^{(2)}(\beta r') j_n(\beta r) P_n(\cos \xi), & r < r' \\ \sum_{n=0}^{\infty} (2n+1) h_n^{(2)}(\beta r) j_n(\beta r') P_n(\cos \xi), & r > r' \end{cases} \quad (11-223a)$$

$$\text{where} \quad \cos \xi = \cos \theta \cos \theta' + \sin \theta \sin \theta' \cos(\phi - \phi') \quad (11-223c)$$

Similarly,

$$h_0^{(1)}(\beta |\mathbf{r} - \mathbf{r}'|) = \begin{cases} \sum_{n=0}^{\infty} (2n+1) h_n^{(1)}(\beta r') j_n(\beta r) P_n(\cos \xi), & r < r' \\ \sum_{n=0}^{\infty} (2n+1) h_n^{(1)}(\beta r) j_n(\beta r') P_n(\cos \xi), & r > r' \end{cases} \quad (11-224a)$$

$$(11-224b)$$

11.8 SCATTERING BY A SPHERE

Plane wave scattering by a sphere is a classic problem in scattering and has been addressed by many authors [11, 14, 20, 45, 46, 49–54]. Here we will outline one that parallels that of [11]. Scattering by other sources of excitation such as dipoles, both radial and tangential to the surface of the sphere, has also been addressed. Because of its symmetry, a PEC sphere is often used as a reference scatterer to calibrate and measure the scattering properties (such as RCS) of other radar targets (missiles, airplanes, helicopters, etc.). A set of five aluminum RCS calibration spheres is shown in Figure 11-27. Apart from its symmetry, the RCS of a sphere can be calculated, which allows the user to set on the display, during measurements, a baseline to compare the RCS of other targets to that of the sphere. It is therefore very important that we know and understand the scattering characteristics of a sphere. We will consider both PEC and lossy dielectric spheres.

11.8.1 Perfect Electric Conducting (PEC) Sphere

Let us assume that the electric field of a uniform plane wave is polarized in the x direction, and it is traveling along the z axis, as shown in Figure 11-28. The electric field of the incident wave upon a PEC sphere can then be expressed as

$$\mathbf{E}^i = \hat{\mathbf{a}}_x E_x^i = \hat{\mathbf{a}}_x E_0 e^{-j\beta z} = \hat{\mathbf{a}}_x E_0 e^{-j\beta r \cos \theta} \quad (11-225)$$

Using the transformation (II-12), the x component of (11-225) can be transformed in spherical components to

$$\mathbf{E}^i = \hat{\mathbf{a}}_r E_r^i + \hat{\mathbf{a}}_\theta E_\theta^i + \hat{\mathbf{a}}_\phi E_\phi^i \quad (11-226)$$

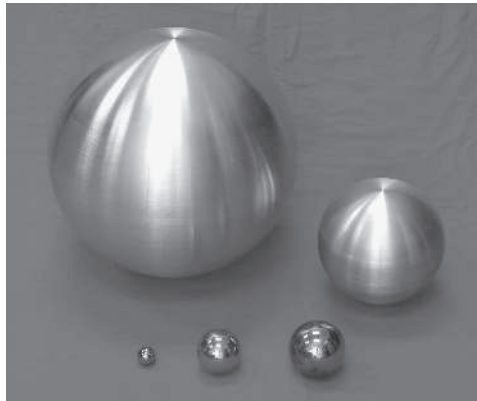


Figure 11-27 A set of five aluminum RCS calibration spheres.

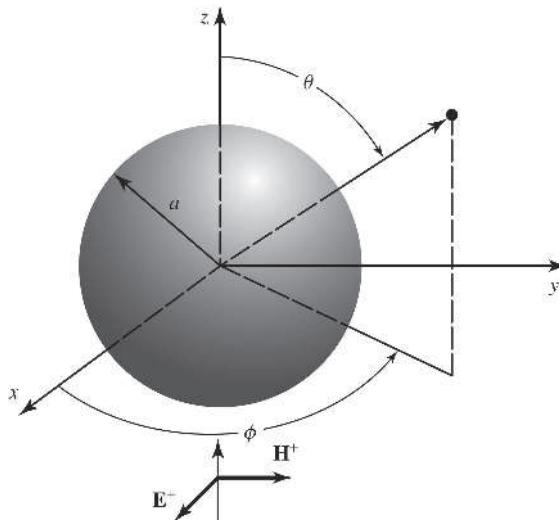


Figure 11-28 Uniform plane wave incident on a conducting sphere.

where

$$E_r^i = E_x^i \sin \theta \cos \phi = E_0 \sin \theta \cos \phi e^{-j\beta r \cos \theta} = E_0 \frac{\cos \phi}{j\beta r} \frac{\partial}{\partial \theta} (e^{-j\beta r \cos \theta}) \quad (11-226a)$$

$$E_\theta^i = E_x^i \cos \theta \cos \phi = E_0 \cos \theta \cos \phi e^{-j\beta r \cos \theta} \quad (11-226b)$$

$$E_\phi^i = -E_x^i \sin \phi = -E_0 \sin \phi e^{-j\beta r \cos \theta} \quad (11-226c)$$

Each of the spherical components of the preceding incident electric field can be expressed using the transformation (11-221) and (11-221a) as

$$E_r^i = E_0 \frac{\cos \phi}{j\beta r} \sum_{n=0}^{\infty} j^{-n} (2n+1) j_n(\beta r) \frac{\partial}{\partial \theta} [P_n(\cos \theta)] \quad (11-227a)$$

$$E_{\theta}^i = E_0 \cos \theta \cos \phi \sum_{n=0}^{\infty} j^{-n} (2n+1) j_n(\beta r) P_n(\cos \theta) \quad (11-227b)$$

$$E_{\phi}^i = -E_0 \sin \phi \sum_{n=0}^{\infty} j^{-n} (2n+1) j_n(\beta r) P_n(\cos \theta) \quad (11-227c)$$

Since according to (10-31),

$$j_n(\beta r) = \frac{1}{\beta r} \hat{J}_n(\beta r) \quad (11-228a)$$

and

$$\frac{\partial P_n}{\partial \theta} = P_n^1(\cos \theta) \quad (11-228b)$$

$$P_0^1 = 0 \quad (11-228c)$$

we can rewrite (11-227a) through (11-227c) as

$$E_r^i = -jE_0 \frac{\cos \phi}{(\beta r)^2} \sum_{n=1}^{\infty} j^{-n} (2n+1) \hat{J}_n(\beta r) P_n^1(\cos \theta) \quad (11-229a)$$

$$E_{\theta}^i = E_0 \frac{\cos \theta \cos \phi}{\beta r} \sum_{n=0}^{\infty} j^{-n} (2n+1) \hat{J}_n(\beta r) P_n^0(\cos \theta) \quad (11-229b)$$

$$E_{\phi}^i = -E_0 \frac{\sin \phi}{\beta r} \sum_{n=0}^{\infty} j^{-n} (2n+1) \hat{J}_n(\beta r) P_n^0(\cos \theta) \quad (11-229c)$$

The incident and scattered fields by the sphere can be expressed as a superposition of TE^r and TM^r as outlined, respectively, in Sections 10.2.4 and 10.2.5. The TE^r fields are constructed by letting the vector potentials $\mathbf{A}$ and $\mathbf{F}$ be equal to $\mathbf{A} = 0$ and $\mathbf{F} = \hat{\mathbf{a}}_r F_r(r, \theta, \phi)$. The TM^r fields are constructed when $\mathbf{A} = \hat{\mathbf{a}}_r A_r(r, \theta, \phi)$ and $\mathbf{F} = 0$. For example, the incident radial electric field component E_r^i can be obtained by expressing it in terms of TM^r modes or A_r^i . Thus, using A_r^i , we can write, according to (10-27a), the incident electric field as

$$E_r^i = \frac{1}{j\omega\mu\epsilon} \left(\frac{\partial^2}{\partial r^2} + \beta^2 \right) A_r^i \quad (11-230)$$

Equating (11-230) to (11-229a), it can be shown that A_r^i takes the form

$$A_r^i = E_0 \frac{\cos \phi}{\omega} \sum_{n=1}^{\infty} a_n \hat{J}_n(\beta r) P_n^1(\cos \theta) \quad (11-231)$$

where

$$a_n = j^{-n} \frac{(2n+1)}{n(n+1)} \quad (11-231a)$$

This potential component A_r^i will give the correct value of E_r^i , and it will lead to $H_r^i = 0$.

The correct expression for the radial component of the incident magnetic field can be obtained by following a similar procedure but using TE^r modes or F_r^i of Section 10.2.4. This allows us to show that

$$F_r^i = E_0 \frac{\sin \phi}{\omega \eta} \sum_{n=1}^{\infty} a_n \hat{J}_n(\beta r) P_n^1(\cos \theta) \quad (11-232)$$

where a_n is given by (11-231a). This expression leads to the correct H_r^i and to $E_r^i = 0$. Therefore, the sum of (11-231) and (11-232) will give the correct E_r^i, H_r^i and the remaining electric and magnetic components.

Since the incident electric and magnetic field components of a uniform plane wave can be represented by TM^r and TE^r modes, that can be constructed using the potentials A_r^i and F_r^i of (11-231) and (11-232), the scattered fields can also be represented by TM^r and TE^r modes and be constructed using potentials A_r^s and F_r^s . The forms of A_r^s and F_r^s are similar to those of A_r^i and F_r^i of (11-231) and (11-232), and we can represent them by

$$A_r^s = E_0 \frac{\cos \phi}{\omega} \sum_{n=1}^{\infty} b_n \hat{H}_n^{(2)}(\beta r) P_n^1(\cos \theta) \quad (11-233a)$$

$$F_r^s = E_0 \frac{\sin \phi}{\omega \eta} \sum_{n=1}^{\infty} c_n \hat{H}_n^{(2)}(\beta r) P_n^1(\cos \theta) \quad (11-233b)$$

where the coefficients b_n and c_n will be found using the appropriate boundary conditions. In (11-233a) and (11-233b), the spherical Hankel function of the second kind $\hat{H}_n^{(2)}(\beta r)$ has replaced the spherical Bessel function $\hat{J}_n(\beta r)$ in (11-231) and (11-232) in order to represent outward traveling waves. Thus, all the components of the total field, incident plus scattered, can be found using the sum of (10-23) through (10-24c) and (10-27) through (10-28c), or

$$E_r^t = \frac{1}{j\omega\mu\epsilon} \left(\frac{\partial^2}{\partial r^2} + \beta^2 \right) A_r^t \quad (11-234a)$$

$$E_\theta^t = \frac{1}{j\omega\mu\epsilon} \frac{1}{r} \frac{\partial^2 A_r^t}{\partial r \partial \theta} - \frac{1}{\epsilon} \frac{1}{r \sin \theta} \frac{\partial F_r^t}{\partial \phi} \quad (11-234b)$$

$$E_\phi^t = \frac{1}{j\omega\mu\epsilon} \frac{1}{r \sin \theta} \frac{\partial^2 A_r^t}{\partial r \partial \phi} + \frac{1}{\epsilon} \frac{1}{r} \frac{\partial F_r^t}{\partial \theta} \quad (11-234c)$$

$$H_r^t = \frac{1}{j\omega\mu\epsilon} \left(\frac{\partial^2}{\partial r^2} + \beta^2 \right) F_r^t \quad (11-234d)$$

$$H_\theta^t = \frac{1}{\mu} \frac{1}{r \sin \theta} \frac{\partial A_r^t}{\partial \phi} + \frac{1}{j\omega\mu\epsilon} \frac{1}{r} \frac{\partial^2 F_r^t}{\partial r \partial \theta} \quad (11-234e)$$

$$H_\phi^t = -\frac{1}{\mu} \frac{1}{r} \frac{\partial A_r^t}{\partial \theta} + \frac{1}{j\omega\mu\epsilon} \frac{1}{r \sin \theta} \frac{\partial^2 F_r^t}{\partial r \partial \phi} \quad (11-234f)$$

where A_r^t and F_r^t are each equal to the sum of (11-231), (11-232), (11-233a), and (11-233b), or

$$A_r^t = A_r^i + A_r^s = E_0 \frac{\cos \phi}{\omega} \sum_{n=1}^{\infty} \left[a_n \hat{J}_n(\beta r) + b_n \hat{J}_n^{(2)}(\beta r) \right] P_n^1(\cos \theta) \quad (11-235a)$$

$$F_r^t = F_r^i + F_r^s = E_0 \frac{\sin \phi}{\omega \eta} \sum_{n=1}^{\infty} \left[a_n \hat{J}_n(\beta r) + c_n \hat{J}_n^{(2)}(\beta r) \right] P_n^1(\cos \theta) \quad (11-235b)$$

$$a_n = j^{-n} \frac{2n+1}{n(n+1)} \quad (11-235c)$$

To determine the coefficients b_n and c_n , the boundary conditions

$$E_\theta^t(r=a, 0 \leq \theta \leq \pi, 0 \leq \phi \leq 2\pi) = 0 \quad (11-236a)$$

$$E_\phi^t(r=a, 0 \leq \theta \leq \pi, 0 \leq \phi \leq 2\pi) = 0 \quad (11-236b)$$

must be applied. Using (11-235a) and (11-235b), we can write (11-234b) as

$$\begin{aligned} E_\theta^t = & j \frac{E_0}{\omega \mu \epsilon r \sin \theta} \left\{ \frac{\beta}{\omega} \cos \phi \sum_{n=1}^{\infty} \left[a_n \hat{J}_n'(\beta r) + b_n \hat{H}_n^{(2)'}(\beta r) \right] P_n^{1'}(\cos \theta) \right\} \\ & - \frac{E_0}{\epsilon r \sin \theta} \left\{ \frac{1}{\omega \eta} \cos \phi \sum_{n=1}^{\infty} \left[a_n \hat{J}_n(\beta r) + c_n \hat{H}_n^{(2)}(\beta r) \right] P_n^1(\cos \theta) \right\} \end{aligned} \quad (11-237)$$

where in (11-237)

$$' \equiv \frac{\partial}{\partial(\beta r)} \quad \text{for the spherical Bessel or Hankel function} \quad (11-237a)$$

$$' \equiv \frac{\partial}{\partial(\cos \theta)} = -\frac{1}{\sin \theta} \frac{\partial}{\partial \theta} \quad \text{for the associated Legendre functions} \quad (11-237b)$$

Using (11-237), the boundary condition of (11-236a) is satisfied provided that

$$a_n \hat{J}_n'(\beta a) + b_n \hat{H}_n^{(2)'}(\beta a) = 0 \Rightarrow b_n = -a_n \frac{\hat{J}_n'(\beta a)}{\hat{H}_n^{(2)'}(\beta a)} \quad (11-238a)$$

$$a_n \hat{J}_n(\beta a) + c_n \hat{H}_n^{(2)}(\beta a) = 0 \Rightarrow c_n = -a_n \frac{\hat{J}_n(\beta a)}{\hat{H}_n^{(2)}(\beta a)} \quad (11-238b)$$

The scattered electric field components can be written, using (11-233a) and (11-233b), as

$$E_r^s = -jE_0 \cos \phi \sum_{n=1}^{\infty} b_n \left[\hat{H}_n^{(2)''}(\beta r) + \hat{H}_n^{(2)}(\beta r) \right] P_n^1(\cos \theta) \quad (11-239a)$$

$$E_\theta^s = \frac{E_0}{\beta r} \cos \phi \sum_{n=1}^{\infty} \left[j b_n \hat{H}_n^{(2)'}(\beta r) \sin \theta P_n^{1'}(\cos \theta) - c_n \hat{H}_n^{(2)}(\beta r) \frac{P_n^1(\cos \theta)}{\sin \theta} \right] \quad (11-239b)$$

$$E_\phi^s = \frac{E_0}{\beta r} \sin \phi \sum_{n=1}^{\infty} \left[j b_n \hat{H}_n^{(2)'}(\beta r) \frac{P_n^1(\cos \theta)}{\sin \theta} - c_n \hat{H}_n^{(2)}(\beta r) \sin \theta P_n^{1'}(\cos \theta) \right] \quad (11-239c)$$

where in (11-239a) through (11-239c)

$$' \equiv \frac{\partial}{\partial(\beta r)} \quad \text{for the spherical Hankel functions} \quad (11-239d)$$

$$'' \equiv \frac{\partial^2}{\partial(\beta r)^2} \quad \text{for the spherical Hankel functions} \quad (11-239e)$$

$$' \equiv \frac{\partial}{\partial(\cos \theta)} = -\frac{1}{\sin \theta} \frac{\partial}{\partial \theta} \quad \text{for the associated Legendre functions} \quad (11-239f)$$

The spherical Hankel function is related to the regular Hankel function by (10-31) or

$$\hat{H}_n^{(2)}(\beta r) = \sqrt{\frac{\pi \beta r}{2}} H_{n+1/2}^{(2)}(\beta r) \quad (11-240)$$

Since for large values of βr the regular Hankel function can be represented by

$$H_{n+1/2}^{(2)}(\beta r) \stackrel{\beta r \rightarrow \infty}{\simeq} \sqrt{\frac{2j}{\pi \beta r}} j^{n+1/2} e^{-j\beta r} = j \sqrt{\frac{2}{\pi \beta r}} j^n e^{-j\beta r} \quad (11-241)$$

then the spherical Hankel function of (11-240) and its partial derivatives can be approximated by

$$\hat{H}_n^{(2)}(\beta r) \stackrel{\beta r \rightarrow \infty}{\simeq} j^{n+1} e^{-j\beta r} \quad (11-241a)$$

$$\hat{H}_n^{(2)'}(\beta r) = \frac{\partial \hat{H}_n^{(2)}(\beta r)}{\partial(\beta r)} \stackrel{\beta r \rightarrow \infty}{\simeq} -j^2 j^n e^{-j\beta r} = j^n e^{-j\beta r} \quad (11-241b)$$

$$\hat{H}_n^{(2)''}(\beta r) = \frac{\partial^2 \hat{H}_n^{(2)}(\beta r)}{\partial(\beta r)^2} \stackrel{\beta r \rightarrow \infty}{\simeq} -j^{n+1} e^{-j\beta r} \quad (11-241c)$$

For far-field observations ($\beta r \rightarrow \text{large}$), the electric field components of (11-239a) through (11-239c) can be simplified using the approximations (11-241a) through (11-241c). Since the radial component E_r^s of (11-239a) reduces with the approximations of (11-241a) through (11-241c) to zero, then in the far zone (11-239a) through (11-239c) can be approximated by

Far-Field Observations ($\beta r \rightarrow \text{large}$)

$$E_r^s \simeq 0 \quad (11-242a)$$

$$E_\theta^s \simeq jE_0 \frac{e^{-j\beta r}}{\beta r} \cos \phi \sum_{n=1}^{\infty} j^n \left[b_n \sin \theta P_n^{1'}(\cos \theta) - c_n \frac{P_n^1(\cos \theta)}{\sin \theta} \right] \quad (11-242b)$$

$$E_\phi^s \simeq jE_0 \frac{e^{-j\beta r}}{\beta r} \sin \phi \sum_{n=1}^{\infty} j^n \left[b_n \frac{P_n^1(\cos \theta)}{\sin \theta} - c_n \sin \theta P_n^{1'}(\cos \theta) \right] \quad (11-242c)$$

where b_n and c_n are given by (11-238a) and (11-238b), respectively.

The bistatic radar cross section is obtained using (11-22b), and it can be written, using (11-225) and (11-242a) through (11-242c), as

$$\sigma(\text{bistatic}) = \lim_{r \rightarrow \infty} \left[4\pi r^2 \frac{|\mathbf{E}^s|^2}{|\mathbf{E}^i|^2} \right] = \frac{\lambda^2}{\pi} \left[\cos^2 \phi |A_\theta|^2 + \sin^2 \phi |A_\phi|^2 \right] \quad (11-243)$$

where

$$|A_\theta|^2 = \left| \sum_{n=1}^{\infty} j^n \left[b_n \sin \theta P_n^{1'}(\cos \theta) - c_n \frac{P_n^1(\cos \theta)}{\sin \theta} \right] \right|^2 \quad (11-243a)$$

$$|A_\phi|^2 = \left| \sum_{n=1}^{\infty} j^n \left[b_n \frac{P_n^1(\cos \theta)}{\sin \theta} - c_n \sin \theta P_n^{1'}(\cos \theta) \right] \right|^2 \quad (11-243b)$$

The monostatic radar cross section can be found by first reducing the field expressions for observations toward $\theta = \pi$. In that direction the scattered electric field of interest is the copolar component, E_x^s , and it can be found using (11-242a) through (11-242c) and the transformation (11-13b), by evaluating either

$$E_x^s = E_\theta^s \cos \theta \cos \phi \Big|_{\substack{\theta=\pi \\ \phi=\pi}} = E_\theta^s \Big|_{\substack{\theta=\pi \\ \phi=\pi}} \quad (11-244a)$$

or

$$E_x^s = -E_\phi^s \sin \phi \Big|_{\substack{\theta=\pi \\ \phi=3\pi/2}} = E_\phi^s \Big|_{\substack{\theta=\pi \\ \phi=3\pi/2}} \quad (11-244b)$$

To accomplish either (11-244a) or (11-244b), we need to first evaluate the associated Legendre function and its derivative when $\theta = \pi$. It can be shown that [11, 45]

$$\frac{P_n^1(\cos \theta)}{\sin \theta} \Big|_{\theta=\pi} = (-1)^n \frac{n(n+1)}{2} \quad (11-245a)$$

$$\sin \theta P_n^{1'}(\cos \theta) \Big|_{\theta=\pi} = \sin \theta \frac{dP_n^1}{d(\cos \theta)} = -\frac{dP_n^1(\cos \theta)}{d\theta} \Big|_{\theta=\pi} = (-1)^n \frac{n(n+1)}{2} \quad (11-245b)$$

Thus, (11-242b) can be expressed using (11-235c), (11-238a), (11-238b), (11-245a), and (11-245b), as

$$\begin{aligned} E_\theta^s \Big|_{\substack{\theta=\pi \\ \phi=\pi}} &= jE_0 \frac{e^{-j\beta r}}{\beta r} \sum_{n=1}^{\infty} j^n (-1)^n \frac{n(n+1)}{2} [b_n - c_n] \\ &= -jE_0 \frac{e^{-j\beta r}}{\beta r} \sum_{n=1}^{\infty} j^n (-1)^n \frac{n(n+1)}{2} a_n \left[\frac{\hat{J}_n'(\beta a)}{\hat{H}_n^{(2)'}(\beta a)} - \frac{\hat{J}_n(\beta a)}{\hat{H}_n^{(2)}(\beta a)} \right] \\ E_\theta^s \Big|_{\substack{\theta=\pi \\ \phi=\pi}} &= -jE_0 \frac{e^{-j\beta r}}{\beta r} \sum_{n=1}^{\infty} (-1)^n \frac{2(n+1)}{2} \left[\frac{\hat{J}_n'(\beta a) \hat{H}_n^{(2)}(\beta a) - \hat{J}_n(\beta a) \hat{H}_n^{(2)'}(\beta a)}{\hat{H}_n^{(2)'}(\beta a) \hat{H}_n^{(2)}(\beta a)} \right] \end{aligned} \quad (11-246)$$

which reduces, using the Wronskian for spherical Bessel functions

$$\hat{J}_n'(\beta a) \hat{H}_n^{(2)}(\beta a) - \hat{J}_n(\beta a) \hat{H}_n^{(2)'}(\beta a) = j[\hat{J}_n(\beta a) \hat{Y}_n'(\beta a) - \hat{J}_n'(\beta a) \hat{Y}_n(\beta a)] = j \quad (11-246a)$$

to

$$E_\theta^s \Big|_{\substack{\theta=\pi \\ \phi=\pi}} = E_0 \frac{e^{-j\beta r}}{2\beta r} \sum_{n=1}^{\infty} \frac{(-1)^n (2n+1)}{\hat{H}_n^{(2)'}(\beta a) \hat{H}_n^{(2)}(\beta a)} \quad (11-246b)$$

Thus, the monostatic radar cross section of (11-22b) can be expressed, using (11-246b), by

$$\sigma_{3-D}(\text{monostatic}) = \lim_{r \rightarrow \infty} \left[4\pi r^2 \frac{|\mathbf{E}^s|^2}{|\mathbf{E}^i|^2} \right] = \frac{\lambda^2}{4\pi} \left| \sum_{n=1}^{\infty} \frac{(-1)^n (2n+1)}{\hat{H}_n^{(2)'}(\beta a) \hat{H}_n^{(2)}(\beta a)} \right|^2 \quad (11-247)$$

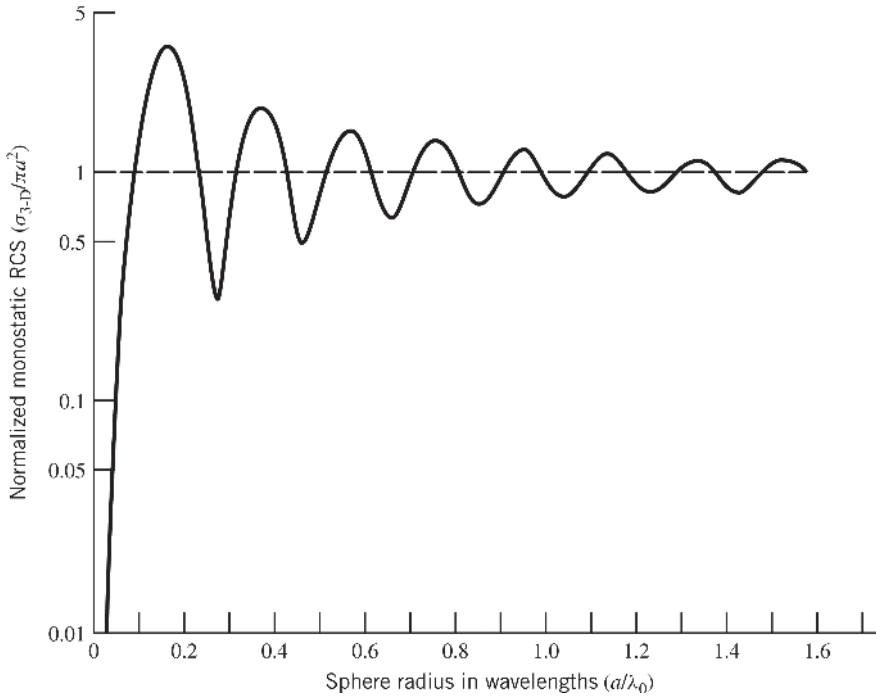


Figure 11-29 Normalized monostatic radar cross section for a conducting sphere as a function of its radius. (Source: G. T. Ruck, D. E. Barrick, W. D. Stuart, and C. K. Krichbaum, *Radar Cross Section Handbook*, Vol. 1, 1970, Plenum Publishing Co.)

A plot of (11-247) as a function of the sphere radius is shown in Figure 11-29 [49]. This is a classic signature that can be found in any literature dealing with electromagnetic scattering. The total curve can be subdivided into three regions; the *Rayleigh*, the *Mie* (or *resonance*), and the *optical* regions. The Rayleigh region represents the part of the curve for small values of the radius ($a < 0.1\lambda$) and the optical region represents the RCS of the sphere for large values of the radius (typically $a > 2\lambda$). The region between those two extremes is the Mie or resonance region. It is apparent that for small values of the radius the RCS is linear, for intermediate values it is oscillatory about πa^2 , and for large values it approaches πa^2 , which is the physical area of the cross section of the sphere.

For very small values of the radius a , the first term of (11-247) is sufficient to accurately represent the RCS. Doing this, we can approximate (11-247) by

$$\sigma_{3-D} \text{ (monostatic)} \stackrel{a \rightarrow 0}{\simeq} \frac{\lambda^2}{4\pi} \left| \frac{3}{\hat{H}_1^{(2)'}(\beta a) \hat{H}_1^{(2)}(\beta a)} \right|^2 \quad (11-248)$$

Since

$$\hat{H}_1^{(2)}(\beta a) \stackrel{a \rightarrow 0}{\simeq} -j \hat{Y}_1(\beta a) = -j \sqrt{\frac{\pi \beta a}{2}} Y_{3/2}(\beta a) \simeq -j \sqrt{\frac{\pi \beta a}{2}} \left[-\frac{1}{\pi} \left(\frac{2}{\beta a} \right)^{3/2} \right] = j \frac{1}{\beta a} \quad (11-248a)$$

$$\hat{H}_1^{(2)}(\beta a) \stackrel{a \rightarrow 0}{\simeq} -j \hat{Y}_1'(\beta a) \simeq -j \frac{1}{(\beta a)^2} \quad (11-248b)$$

$$\frac{1}{2}! = \frac{1}{2} \sqrt{\pi} \quad (11-248c)$$

(11-248) reduces to

$$\sigma_{3-D}(\text{monostatic}) \stackrel{a \rightarrow 0}{\simeq} \frac{9\lambda^2}{4\pi} (\beta a)^6 \quad (11-248d)$$

which is representative of the Rayleigh region scattering.

For very large values of the radius a , we can approximate the spherical Hankel function and its derivative in (11-247) by their asymptotic forms (11-241a) and (11-241b), or

$$\hat{H}_n^{(2)}(\beta a) \stackrel{\beta a \rightarrow \infty}{\simeq} \frac{e^{-j[\beta a(\sin \alpha - \alpha \cos \alpha) - \pi/4]}}{\sqrt{\sin \alpha}} \quad (11-249a)$$

$$\hat{H}_n^{(2)'}(\beta a) \stackrel{\beta a \rightarrow \infty}{\simeq} \sqrt{\sin \alpha} e^{-j[\beta a(\sin \alpha - \alpha \cos \alpha) + \pi/4]} \quad (11-249b)$$

$$\cos \alpha = (n + 1/2)/\beta a \quad (11-249c)$$

Thus, (11-247) reduces for very large values of the radius a to

$$\sigma_{3-D}(\text{monostatic}) = \frac{\lambda^2}{4\pi} \left| \sum_{n=1}^{\infty} \frac{(-1)^n (2n+1)}{\hat{H}_n^{(2)'}(\beta a) \hat{H}_n^{(2)}(\beta a)} \right|^2 \stackrel{a \rightarrow \infty}{\simeq} \pi a^2 \quad (11-250)$$

which is representative of the optical region scattering, and is also equal to the physical area of the cross section of the sphere.

11.8.2 Lossy Dielectric Sphere

The development of the scattering by a lossy dielectric sphere follows that of a PEC sphere, which was outlined in the previous section. The major difference is that now electric and magnetic fields penetrate the sphere, and we need to write expressions to properly represent them. The expressions for the fields outside the sphere will be of similar forms as those of the PEC sphere. To relate the fields outside and inside the sphere, the appropriate boundary conditions must be applied on the surface of the sphere; continuity of the tangential electric and magnetic fields, in contrast to the vanishing of the tangential electric fields on the surface of the sphere for the PEC case.

To start the development, we will use the geometry of Figure 11-28 and assume that the medium outside the sphere is free space (wave number β_o) and inside is a lossy dielectric (wave number β_d) represented by a relative complex permittivity $\hat{\epsilon}_r$ ($\hat{\epsilon}_r = \epsilon_r' - j\epsilon_r''$) and relative complex permeability $\hat{\mu}_r$ ($\hat{\mu}_r = \mu_r' - j\mu_r''$). The total, incident, and scattered fields outside the sphere can be represented by the vector potentials, (11-235a) through (11-235c) and the corresponding electric and magnetic fields by (11-234a) through (11-234f). Inside the sphere the vector potentials should be similar to those outside the sphere, but chosen to represent standing waves in the radial direction, instead of traveling waves. This is accomplished by choosing the vector potentials for the total fields inside the sphere to be similar to the first terms of (11-235a) and (11-235b), and written as

$$A_r^{t-} = E_0 \frac{\cos \phi}{\omega} \sum_{n=1}^{\infty} d_n \hat{J}_n(\beta_d r) P_n^{(1)}(\cos \theta) \quad (11-251a)$$

$$F_r^{t-} = E_0 \frac{\sin \phi}{\omega \eta} \sum_{n=1}^{\infty} e_n \hat{J}_n(\beta_d r) P_n^{(1)}(\cos \theta) \quad (11-251b)$$

The modal coefficients, b_n and c_n (for the fields outside the sphere) and d_n and e_n (for the fields inside the sphere), can be determined by application of the boundary conditions. The superscript minus (−) is used to identify the vector potentials and associated fields on and within the sphere ($r \leq a$), while the plus (+) is used to identify those on and outside the sphere ($r \geq a$).

Based on the fields of (11-234a) through (11-234f), there exist two tangential electric (E_θ, E_ϕ) and two tangential magnetic (H_θ, H_ϕ) field components. For the PEC sphere of the previous section, the boundary conditions were (11-236a) and (11-236b). However, the appropriate boundary conditions for the lossy dielectric sphere require the continuity of the tangential electric and magnetic fields on the surface of the sphere, which can be written as

$$E_\theta^{t-}(r=a, 0 \leq \theta \leq \pi, 0 \leq \phi \leq 2\pi) = E_\theta^{t+}(r=a, 0 \leq \theta \leq \pi, 0 \leq \phi \leq 2\pi) \quad (11-252a)$$

$$E_\phi^{t-}(r=a, 0 \leq \theta \leq \pi, 0 \leq \phi \leq 2\pi) = E_\phi^{t+}(r=a, 0 \leq \theta \leq \pi, 0 \leq \phi \leq 2\pi) \quad (11-252b)$$

$$H_\theta^{t-}(r=a, 0 \leq \theta \leq \pi, 0 \leq \phi \leq 2\pi) = H_\theta^{t+}(r=a, 0 \leq \theta \leq \pi, 0 \leq \phi \leq 2\pi) \quad (11-252c)$$

$$H_\phi^{t-}(r=a, 0 \leq \theta \leq \pi, 0 \leq \phi \leq 2\pi) = H_\phi^{t+}(r=a, 0 \leq \theta \leq \pi, 0 \leq \phi \leq 2\pi) \quad (11-252d)$$

The modal coefficients, b_n and c_n (for the fields outside the sphere), and d_n and e_n (for the fields inside the sphere), can be determined by enforcing this set of boundary conditions. To accomplish this, the two tangential electric (E_θ, E_ϕ) and two tangential magnetic (H_θ, H_ϕ) field components, both inside and outside the sphere, must first be written using (11-234b)–(11-234c) and (11-234e)–(11-234f) with the vector potentials of (11-235a)–(11-235b) for outside the sphere and (11-251a)–(11-251b) for inside the sphere.

The enforcement of the boundary conditions (11-252a) through (11-252d) is straightforward but cumbersome. Because of space limitations, and as a practice to the reader, the procedure will not be detailed here but left as an end-of-chapter exercise. Following the procedure outlined here, it can be shown that b_n , c_n , d_n , and e_n can be written and related to a_n by

$$b_n = \frac{-\sqrt{\hat{\epsilon}_r} \hat{J}'_n(\beta_0 a) \hat{J}_n(\dot{\beta}_d a) + \sqrt{\dot{\mu}_r} \hat{J}_n(\beta_0 a) \hat{J}'_n(\dot{\beta}_d a)}{\sqrt{\hat{\epsilon}_r} \hat{H}_n^{(2)'}(\beta_0 a) \hat{J}_n(\dot{\beta}_d a) - \sqrt{\dot{\mu}_r} \hat{H}_n^{(2)}(\beta_0 a) \hat{J}'_n(\dot{\beta}_d a)} a_n \quad (11-253a)$$

$$c_n = \frac{-\sqrt{\hat{\epsilon}_r} \hat{J}_n(\beta_0 a) \hat{J}'_n(\dot{\beta}_d a) + \sqrt{\dot{\mu}_r} \hat{J}'_n(\beta_0 a) \hat{J}_n(\dot{\beta}_d a)}{\sqrt{\hat{\epsilon}_r} \hat{H}_n^{(2)}(\beta_0 a) \hat{J}'_n(\dot{\beta}_d a) - \sqrt{\dot{\mu}_r} \hat{H}_n^{(2)'}(\beta_0 a) \hat{J}_n(\dot{\beta}_d a)} a_n \quad (11-253b)$$

$$d_n = -j \frac{\dot{\mu}_r \sqrt{\hat{\epsilon}_r}}{\sqrt{\hat{\epsilon}_r} \hat{H}_n^{(2)'}(\beta_0 a) \hat{J}_n(\dot{\beta}_d a) - \sqrt{\dot{\mu}_r} \hat{H}_n^{(2)}(\beta_0 a) \hat{J}'_n(\dot{\beta}_d a)} a_n \quad (11-253c)$$

$$e_n = +j \frac{\dot{\mu}_r \sqrt{\hat{\epsilon}_r}}{\sqrt{\hat{\epsilon}_r} \hat{H}_n^{(2)}(\beta_0 a) \hat{J}'_n(\dot{\beta}_d a) - \sqrt{\dot{\mu}_r} \hat{H}_n^{(2)'}(\beta_0 a) \hat{J}_n(\dot{\beta}_d a)} a_n \quad (11-253d)$$

where a_n is given by (11-231a). For a dielectric sphere of small radius, the first term ($n=1$) may be sufficient to represent the fields. It can be shown that for $n=1$ (11-253a) through (11-253d) reduce to

$$b_1 \approx \frac{\beta_0 a \rightarrow 0}{-(\beta_0 a)^3} \frac{\dot{\epsilon}_r - 1}{\dot{\epsilon}_r + 2} \quad (11-254a)$$

$$c_1 \approx \frac{\beta_0 a \rightarrow 0}{-(\beta_0 a)^3} \frac{\dot{\mu}_r - 1}{\dot{\mu}_r + 2} \quad (11-254b)$$

$$d_1 \approx \frac{\beta_0 a \rightarrow 0}{j2(\dot{\epsilon}_r + 2)} \quad (11-254c)$$

$$e_1 \underset{\beta_0 a \rightarrow 0}{\approx} \frac{9\sqrt{\dot{\mu}_r}}{j2\sqrt{\dot{\epsilon}_r}(\dot{\mu}_r + 2)} \quad (11-254d)$$

Two MATLAB computer programs, **PEC_DIEL_Sphere_Fields** and **Sphere_RCS**, have been written. The first one, **PEC_DIEL_Sphere_Fields**, allows the visualization of the total fields, within and outside the sphere, for both PEC and lossy dielectric spheres. For static fields, a PEC sphere can be represented solely by letting $\dot{\epsilon}_r \rightarrow$ very large (ideally infinity). However, at rf , the PEC sphere must be represented as a special case of the lossy dielectric sphere by allowing both $\dot{\epsilon}_r \rightarrow$ very large (ideally infinity) and $\dot{\mu}_r \rightarrow$ very small (ideally zero) so that β_d remains finite. The second computer program, **Sphere_RCS**, is based on the formulation of Section 11.8.2. It computes and plots the normalized amplitude scattering pattern, and bistatic and monostatic RCSs of a plane wave scattered by PEC and lossy dielectric spheres based on the geometry of Figure 11-28.

11.9 MULTIMEDIA

On the website that accompanies this book, the following multimedia resources are included for the review, understanding, visualization, and presentation of the material of this chapter.

- **MATLAB** computer programs:
 - a. **PEC_Strip_SW**: Computes, using PO, the TM^z and TE^z 2-D scattering width (SW), monostatic and bistatic, of a PEC strip of Figure 11-4.
 - b. **PEC_Rect_Plate_RCS**: Computes, using PO, the TE^x and TM^x 3-D radar cross section (RCS), monostatic and bistatic, of a PEC rectangular plate of Figure 11-8.
 - c. **PEC_Circ_Plate_RCS**: Computes, using PO, the TE^x and TM^x 3-D radar cross section (RCS), monostatic and bistatic, of a PEC circular plate of Figure P11-7.
 - d. **PEC_Cyl_Normal_Fields**: Visualizes the TM^z and TE^z scattered fields of a uniform plane wave incident, at normal incidence angles, upon the PEC cylinder of circular cross section of Figure 11-12.
 - e. **PEC_Cyl_Normal_SW**: Computes the 2-D scattering width (SW) of a uniform plane TM^z and TE^z wave incident, at normal incidence angles, upon the PEC cylinder of circular cross section of Figure 11-12.
 - f. **PEC_Cyl_Oblique_Fields**: Visualizes the TM^z and TE^z scattered fields of a uniform plane wave incident, at oblique incidence angles, upon the PEC cylinder of circular cross section of Figure 11-16.
 - g. **PEC_Cyl_Oblique_SW**: Computes the 2-D scattering width (SW) of a uniform plane TM^z and TE^z wave incident, at oblique incidence angles, upon the PEC cylinder of circular cross section of Figure 11-16.
 - h. **PEC_Cyl_Oblique_RCS**: Computes and plots the 2-D and 3-D RCS of a uniform plane TM^z and TE^z wave incident, at oblique incidence angles, upon the PEC cylinder of circular cross section and finite length of Figure 11-16.
 - i. **Cylinder_RCS**: Computes and plots the normalized amplitude scattering pattern, and bistatic and monostatic RCSs of a TM^z and TE^z plane wave scattered by a:
 - PEC 2-D cylinder, based on geometry of Figures 11-22a and 11-22b and formulations of Sections 11.5.1 and 11.5.2, respectively.
 - Lossy 2-D dielectric cylinder based on the end-of-chapter Problems 11.58 and 11.60 and associated Figures P11-58 and P11-62, respectively.
 - j. **Circum_Axial_Slot**: Computes and plots the radiation patterns of a circumferential and an axial slot/aperture on a PEC cylinder of circular cross section in the H- and E-planes,

respectively. The program also computes the radiation patterns with the ones of a rectangular aperture on a flat PEC ground plane.

- k. **PEC_DIEL_Sphere_Fields:** Visualizes the scattered fields of a uniform plane wave by a lossy dielectric sphere of Figure 11-25.
- l. **Sphere_RCS:** Computes and plots the normalized amplitude scattering pattern, and bistatic and monostatic RCSs of a uniform plane wave scattered by a:
 - PEC sphere, based on the geometry of Figure 11-28 and formulations of Section 11.8.1.
 - Lossy dielectric sphere based on the geometry of Figure 11.28 and formulations of Section 11.8.2
- **PowerPoint (PPT)** viewgraphs, in multicolor.

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PROBLEMS

- 11.1. Repeat Example 11-1 for a displaced *magnetic* line source of constant current I_m at ρ', ϕ' .
- 11.2. A magnetic line source of infinite length and constant magnetic current I_m is placed parallel to the z axis at a height h above a PEC ground plane of infinite extent, as shown in Figure 11-2 except that we now have a magnetic line source.
 - (a) Determine the total magnetic field at ρ, ϕ for $0 \leq \phi \leq 180^\circ$.
 - (b) Simplify the expressions when the observations are made at very large distances (far field).
 - (c) Determine the smallest height h (in λ) that will introduce a null in the far field amplitude pattern at:
 - $\phi = 30^\circ$
 - $\phi = 90^\circ$
- 11.3. Repeat Problem 11.2 for the electric field of an electric line source of infinite length and constant electric current I_e placed parallel to the z axis at a height h above a perfect magnetic conducting (PMC) ground plane of infinite extent, as shown in Figure 11-2, except that we now have a PMC ground plane.
- 11.4. Repeat Problem 11.3 for the magnetic field of a magnetic line source of constant current I_m above a PMC.
- 11.5. For the problem of Figure 11-2, where the electric line source is placed at a height $h \ll \lambda$, and observations are made at any point (including near field):
 - (a) Show that the corresponding magnetic vector potential of the line source at $y = h$ in the absence of the ground plane is given by

$$A_z^{(1)} = A_z^{(0)}(x, y - h)$$
 where $A_z^{(0)}$ is the magnetic vector potential of an isolated line source at $z = 0$.
 - (b) Write a simplified closed-form expression for the total magnetic vector potential of the line source above the ground plane valid for $0 \leq \phi \leq 180^\circ$.
 - (c) Determine the angles ϕ ($0 \leq \phi \leq 180^\circ$) where the total far-zone electric field, at a constant observation distance ρ , vanishes.
- 11.6. Repeat Problem 1.5 for a magnetic line source, and corresponding electric vector potential F_z and magnetic field, above a PMC ground plane.

- 11.7. Two constant current, infinite length electric line sources, are displaced along the x axis a distance s apart, as shown in Figure P11-7. Use superposition and neglect mutual coupling between the lines.

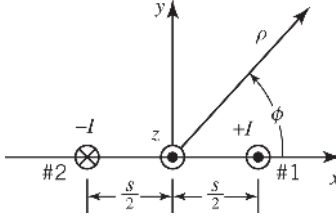


Figure P11-7

- (a) Show that the magnetic vector potential for the two lines can be written as

$$A_z^t = A_z^{(1)}\left(x - \frac{s}{2}, y\right) - A_z^{(2)}\left(x + \frac{s}{2}, y\right)$$

- (b) Show that for small spacings (in the limit as $s \rightarrow 0$), the vector potential of part (a) reduces to

$$A_z^t \stackrel{s \rightarrow 0}{\simeq} \frac{\mu\beta Is}{4j} H_1^{(2)}(\beta\rho) \cos\phi$$

- (c) Determine the electric and magnetic field components associated with the two line sources when $s \rightarrow 0$.

- 11.8. Repeat Problem 11.7 for the electric vector potential, and the electric and magnetic fields, when the two sources are magnetic line sources displaced symmetrically along the x axis a distance s apart, as shown in Figure P11-7.

- 11.9. Two electric line sources, of infinite length and constant current I_e , are displaced along the y axis a distance s apart, as shown in Figure P11-9.

- (a) Show that the total magnetic vector potential A_z^t for the two line sources can be written as

$$A_z^t = A_z^{(1)}\left(y - \frac{s}{2}, x\right) - A_z^{(2)}\left(y + \frac{s}{2}, x\right)$$

- (b) Show that, for $s \rightarrow 0$, the total magnetic vector potential can be written in simplified form as the product of one Hankel function and cosine/sine functions.

- (c) Determine the total electric and magnetic field components for the two sources for $s \rightarrow 0$.

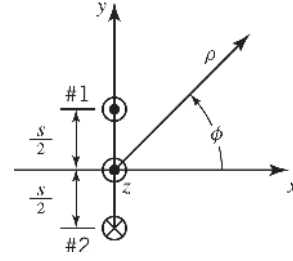


Figure P11-9

- 11.10. Repeat Problem 11.9 for the electric vector potential F_z^t , and electric and magnetic fields for two magnetic line sources displaced along the y axis a distance s apart.

- 11.11. Four constant current, infinite length electric line sources, of phase as indicated, are displaced along the x axis, as shown in Figure P11-11. Assume that the spacings s_1 and s_2 are very small.

- (a) Find an approximate closed-form expression for the magnetic vector potential for the entire array by using the procedure of Problem 11.7. First consider the pairs on each side as individual arrays and then combine the results to form a new array.

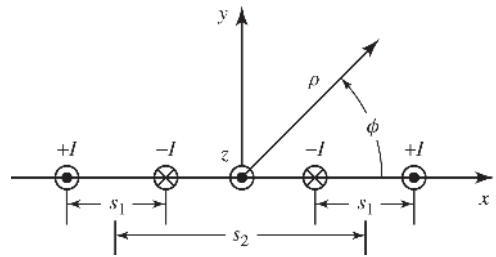


Figure P11-11

- (b) Determine in terms of ρ and ϕ the electric and magnetic field components associated with the four line sources when $\rho \gg s_1$ and $\rho \gg s_2$.

- 11.12. Four constant current, infinite length electric line sources, of phase as indicated, are displaced along the x and y axes, as shown in Figure P11-12. Assume that the spacings s

and h are very small and neglect any mutual coupling between the lines.

- (a) Show, by using the procedure of Problem 11.7, that the magnetic vector potential for the entire array can be written as

$$A_z \underset{h \rightarrow 0}{\overset{s \rightarrow 0}{\simeq}} -\frac{\mu\beta shI}{4j} \frac{\partial}{\partial y} [H_1^{(2)}(\beta\rho) \cos\phi] \\ = \frac{\mu\beta^2 shI}{8j} H_2^{(2)}(\beta\rho) \sin(2\phi)$$

- (b) Determine in terms of ρ and ϕ the electric and magnetic field components associated with the four line sources when $\rho \gg s$ and $\rho \gg h$.

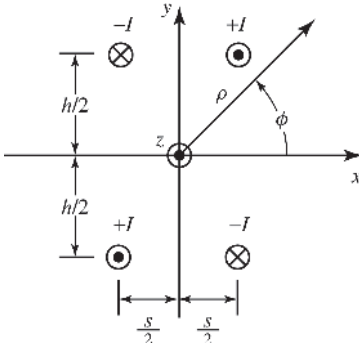


Figure P11-12

- 11.13.** Two constant current, infinite length electric line sources are placed above an infinite electric ground plane, as shown in Figure P11-13. Assume that the spacings s_1 and s_2 are very small and neglect any mutual coupling between the lines.

- (a) Show, by using the procedure of Problem 11.7, that the magnetic vector potential for the entire array can be written as

$$A_z \underset{h \rightarrow 0}{\overset{s \rightarrow 0}{\simeq}} -\frac{\mu\beta shI}{4j} \frac{\partial}{\partial y} [H_1^{(2)}(\beta\rho) \cos\phi] \\ = \frac{\mu\beta^2 shI}{8j} H_2^{(2)}(\beta\rho) \sin(2\phi)$$

- (b) Determine in terms of ρ and ϕ the electric and magnetic field components

associated with the two line sources and ground plane when $\rho \gg s$ and $\rho \gg h$.

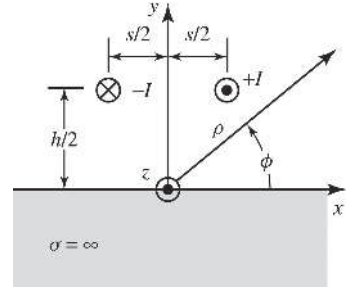


Figure P11-13

- 11.14.** An infinite length and constant current electric line source is placed parallel to the plates of a 90° conducting corner reflector, as shown in Figure P11-14. Assume that the plates of the wedge are infinite in extent and the distance s from the apex to the source is very small ($s \ll \lambda$).

- (a) Show that the magnetic vector potential for the line source and corner reflector can be written, using the procedure of Problem 11.7, as

$$A_z = \frac{\mu\beta^2 s^2 I}{4j} H_2^{(2)}(\beta\rho) \sin(2\phi)$$

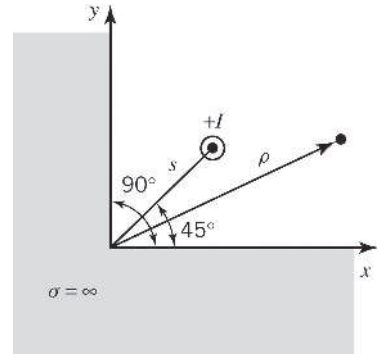


Figure P11-14

- (b) Determine the electric and magnetic field components.
(c) Find the angles ϕ of observation (for a constant value of ρ) where the electric field vanishes.

- 11.15.** Two parallel slots, identical, very thin ($w \rightarrow 0$), of infinite length and uniform electric field, but

directed in opposite directions, are positioned on an infinite PEC ground plane and symmetrically displaced a distance s apart, as shown in Figure P11-15.

- Show that the total vector potential for the two line sources can be written as the sum/difference of two magnetic vector potentials. Indicate which vector potential, electric or magnetic, should be used.
- Show that for $s \rightarrow 0$, the appropriate total vector potential can be written in simplified form as the product of one Hankel function and cosine/sine functions.
- Determine the total electric and magnetic field components for the two sources for $s \rightarrow 0$.

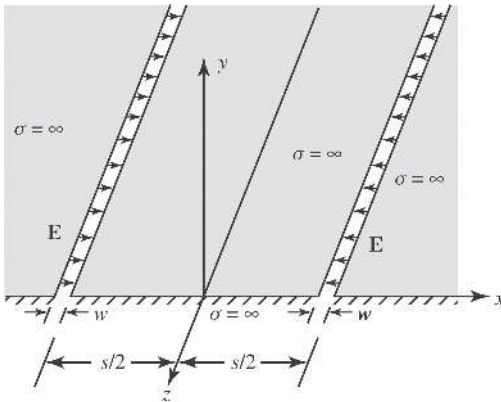


Figure P11-15

- An infinite-length electric line source of constant current I_e is placed a height h above a flat and infinite in extent PEC ground plane, as shown in Figure 11-2. For far-field observations and constant radius $\rho = \rho_0$, it is desired to place a null in the amplitude pattern at an angle $\phi = 60^\circ$. Determine the smallest height $h > 0$ (in λ) that will accomplish this. Assume geometrical optics.
- For Problem 11.16 determine the first two smallest heights (in λ) so that the far-field amplitude pattern at $\phi = 60^\circ$ is -3 dB from the maximum.
- Using the geometry shown in Figure P11-18 of a TE^z of a uniform plane wave incident upon a 2-D PMC strip of infinite width,

- Write a complete vector expression for the incident magnetic field in terms of the angle ϕ_i . The magnitude of the incident magnetic field is H_0 .
- Write a complete vector expression for the reflected magnetic field in terms of the magnitude H_0 of the incident magnetic field, a reflection coefficient Γ , and the angle ϕ_r .
- Based on Snell's law, determine the value of the reflection coefficient (magnitude and phase).

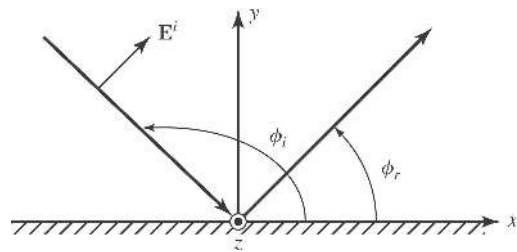


Figure P11-18

- A uniform plane wave is incident upon a flat PEC rectangular plate whose dimensions are very large compared to the wavelength, as shown in Figure 11-8. It is desired to reduce the maximum value of the bistatic RCS of the plate by 10 dB, compared to normal incidence, by illuminating it at an oblique angle. Determine the angle of incidence (in degrees), measured from the normal to the plate, to accomplish this. You can assume that the plate is sufficiently large such that physical optics (PO) is a good approximation. Do this for a polarization:

(a) TE^x (b) TM^x

- For a strip of width $w = 2\lambda$, plot the RCS / λ_0^2 (in dB) when the length of the strip is $l = 5\lambda$, 10λ , and 20λ (plot all three graphs on the same figure). Use the approximate relation between the 2-D SW and the 3-D RCS. Assume normal incidence.
- Repeat Problem 11.20 by treating the strip of finite length as a rectangular plate. Compare the results of the previous two problems. Are they different? Please comment.

11.22. Design a flat PEC flat plate radar target so that the maximum normalized monostatic RCS (σ/λ^2) at an angle of 30° from the normal of the plate is $+20$ dB. Assuming physical optics (PO):

- Determine the area (in λ^2) of the plate.
- At 10 GHz, what is the:
 - Area of the plate (in cm^2)?
 - RCS in dB/sm (dB/square meter)?

11.23. Show that for normal incidence, the two-dimensional scattering width and three-dimensional RCS are related by (11-22e).

11.24. A uniform plane wave on the yz plane is obliquely incident at an angle θ_i from the vertical z axis upon a perfectly electric conducting circular ground plane of radius a , as shown in Figure P11-24. Assume TE^x polarization for the incident field.

- Determine the physical optics current density induced on the plate.
- Determine the far-zone bistatic scattered electric and magnetic fields based on the physical optics current density of part (a).
- Determine the bistatic and monostatic RCSs of the plate. Plot the normalized monostatic RCS (σ_{3-D}/λ^2) in dB for plates with radii of $a = \lambda$ and 5λ .

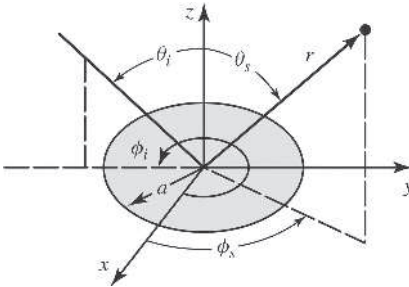


Figure P11-24

11.25. Repeat Problem 11.24 for TM^x plane wave incidence.

11.26. Show that for normal incidence the monostatic RCS of a flat plate of area A and any cross section, based on physical optics, is equal to a $\sigma_{3-D} = 4\pi(A/\lambda)^2$.

11.27. A uniform plane wave traveling in the $-z$ direction is incident upon a perfectly electric conducting curved surface, as shown in Figure P11-27, with radii of curvature

sufficiently large, usually greater than about one wavelength, so that at each point the surface can be considered locally flat. For such a surface, the induced currents and the fields radiated from each infinitesimal area can be represented if the same area were part of an infinite plane that was tangent to the surface at the same location.

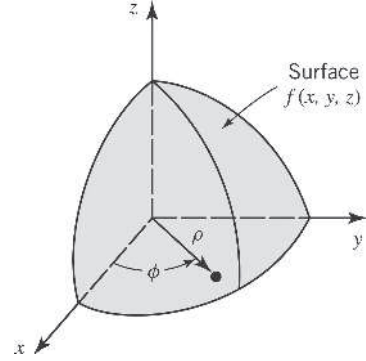


Figure P11-27

- Show that the monostatic RCS can be written as

$$\sigma_{3-D} = \frac{4\pi}{\lambda^2} \left| \iint_A e^{+j2\beta z} dA \right|^2$$

where z is any point on the surface of the scatterer and the integration on dA is performed in the xy plane.

- If in part (a) the differential area in the xy plane is expressed in terms of the polar coordinates ρ , ϕ , then show that the monostatic RCS can be expressed as

$$\sigma_{3-D} = \frac{\pi}{\lambda^2} \left| \int_0^{2\pi} \int_0^{z(\phi)} \frac{d\rho^2}{dz} e^{+j2\beta z} dz d\phi \right|^2$$

11.28. If the scattering conducting curved surface of Problem 11.27, Figure P11-27, is a quadric surface, the integration to find the RCS can be performed in closed form. Assume that the scattering surface is an elliptic paraboloid opening downward along the positive z axis, and that it is represented by

$$\left(\frac{x}{a}\right)^2 + \left(\frac{y}{b}\right)^2 = -\frac{z}{c}$$

where a , b , and c are constants. Show the following.

- (a) The radii of curvature a_1 and a_2 in the xz and yz planes are given by

$$a_1 = \frac{a^2}{2c}, \quad a_2 = \frac{b^2}{2c}$$

- (b) The equation of the elliptic paraboloid surface transformed to polar coordinates can be expressed as

$$z = -\frac{\rho^2}{2a_1} \left[1 - \left(1 - \frac{a_1}{a_2} \right) \sin^2 \phi \right]$$

- (c) The RCS of Problem 11.27 reduces to

$$\begin{aligned} a_{3-D} &= \pi a_1 a_2 |e^{-j2\beta h} - 1| \\ &= 4\pi a_1 a_2 \sin^2(\beta h) \end{aligned}$$

- (d) The RCS of part (c) reduces to $\sigma_{3-D} = \pi a_1 a_2$ if the height of the paraboloid is cut very irregular so that the contributions from the last zone(s) would tend to cancel. To account for this, the exponential term in part (c) disappears.

- 11.29.** If the scattering conducting curved surface of Problem 11.27, Figure P11-27, is a closed surface, such as an ellipsoid represented by

$$\left(\frac{x}{a} \right)^2 + \left(\frac{y}{b} \right)^2 + \left(\frac{z}{c} \right)^2 = 1$$

demonstrate the following.

- (a) The distance z at any point on the surface can be represented by

$$z^2 = c^2 - \rho^2 \frac{c}{a_1} \left[1 - \left(1 - \frac{a_1}{a_2} \right) \sin^2 \phi \right]$$

where $a_1 = a^2/c$ and $a_2 = b^2/c$.

- (b) The RCS of Problem 11.27 from the upper part of the ellipsoid from $z = h$ to $z = c$ reduces, neglecting $(2\beta c)^{-1}$ terms, to

$$\begin{aligned} \sigma_{3-D} &= \pi a_1 a_2 \left[1 + \left(\frac{h}{c} \right)^2 \right. \\ &\quad \left. - 2 \left(\frac{h}{c} \right) \cos[2\beta(c-h)] \right] \end{aligned}$$

where $h(h < c)$ is the distance along the z axis from the origin to the point of integration. If $h = 0$ or if h is very irregular around the periphery of the ellipsoid, the preceding equation reduces to $\sigma_{3-D} = \pi a_1 a_2$. If $a_1 = a_2 = a$, like for a sphere, then $\sigma_{3-D} = \pi a^2$.

- 11.30.** Show that $H_{-n}^{(2)}(x) = (-1)^n H_n^{(2)}(x)$. This identity is often used for the computation of fields scattered by circular cylinders based on modal solutions.

- 11.31.** Verify (11-55a).

- 11.32.** Verify (11-71a) and (11-71b).

- 11.33.** Verify that the infinite summation from minus to plus infinity for the incident electric field of (11-85a) can also be written as an infinite summation from $n = 0$ to $n = \infty$.

- 11.34.** Write the current density expression of (11-97) as an infinite summation from $n = 0$ to $n = \infty$.

- 11.35.** Refer to Figure 11-12a for the TM^z uniform plane wave scattering by a circular conducting cylinder.

- (a) Determine the normalized induced current density based on the physical optics approximation of Section 7.10.
(b) Plot and compare for $0 \leq \phi \leq 180^\circ$ the normalized induced current density based on the physical optics approximation and on the modal solution of (11-97). Do this for cylinders with radii of $a = \lambda$ and 5λ .

- 11.36.** For the TM^z uniform plane wave scattering by a circular conducting cylinder of Figure 11-12a, plot and compare the normalized induced current density based on the exact modal solution of (11-97) and its small argument approximation of (11-98b) for radii of $a = 0.01\lambda$, 0.01λ , and λ .

- 11.37.** Verify that the infinite summation from minus to plus infinity for the scattering width of (11-102) can also be written as an infinite summation from zero to infinity.

- 11.38.** Using the definition (11-21c), instead of (11-21b), show that the TM^z polarization radar cross section reduces to (11-102).

- 11.39.** Write the current density expression of (11-113) as an infinite summation from $n = 0$ to $n = \infty$.

- 11.40.** Refer to Figure 11-12b for the TE^z uniform plane wave scattering by a circular conducting cylinder.

- (a) Determine the normalized induced current density based on the physical optics approximation of Section 7.10.
(b) Plot and compare for $0^\circ \leq \phi \leq 180^\circ$ the normalized induced current density

based on the physical optics approximation and on the modal solution of (11-113). Do this for cylinders with radii of $a = \lambda$ and 5λ .

11.41. For the TE^z uniform plane wave scattering by a circular conducting cylinder of Figure 11-12b, plot and compare the normalized induced current density based on the exact modal solution of (11-113) and its small argument approximation of (11-114d) for radii of $a = 0.01\lambda$, 0.1λ , and λ .

11.42. Verify that the infinite summation from minus to plus infinity for the scattering width of (11-117) can also be written as an infinite summation from zero to infinity.

11.43. Using the definition (11-21b), instead of (11-21c), show that the TE^z polarization radar cross section reduces to (11-117).

11.44. A right-hand circularly polarized uniform plane wave with an electric field equal to

$$\mathbf{E}^i = (\hat{\mathbf{a}}_y + j\hat{\mathbf{a}}_z)e^{-j\beta x}$$

is incident, at normal incidence, upon a PEC cylinder of circular cross section and radius a , as shown in Figure 11-12. Assuming far-field observations, determine the:

- Cylindrical components of the total scattered electric field.
- Polarization of the total scattered electric field toward:
 - $\phi = 0^\circ$.
 - $\phi = 180^\circ$.

11.45. Repeat Problem 11.44 when the circular cylinder is PMC.

11.46. Repeat the plots of Figure 11-13 (both dimensionless and in dB).

11.47. For the same cases of Figure 11-13, plot the magnitude of the induced electric current density (in A/m). Assume $f = 10$ GHz and an incident electric field of 1×10^{-3} V/m.

11.48. Repeat the plots of Fig. 11-15 (both dimensionless and in dB).

11.49. For the same cases of Figure 11-15, plot the magnitude of the induced electric current density (in A/m). Assume $f = 10$ GHz and an incident electric field of 1×10^{-3} V/m.

11.50. A TM^z uniform plane wave is normally incident, upon a very thin PEC wire of radius a ($a \ll \lambda$), as shown in Figure 11-12a. Determine

the values of ϕ (in degrees) where the bistatic scattering width of the wire is:

- Maximum.
- Zero.

11.51. Repeat Problem 11.50 when the incident uniform plane wave has TE^z polarization, as shown in Figure 11-12b.

11.52. A PEC wire of circular cross section, and of radius $a \ll \lambda$, is used as a radar target. It is desired to maintain the normalized maximum bistatic SW (σ/λ) at any angle ϕ at a level not greater than -20 dB. When the incident uniform plane wave is TM^z polarized:

- What should the maximum radius (in λ) of the wire be to meet the desired specifications?
- At what observation angle ϕ would this maximum occur? Identify the angle ϕ (in degrees) graphically.

11.53. Repeat Problem 11.52 for a TE^z polarized incident uniform plane wave.

11.54. A very long (ideally infinite in length) thin (radius $a \ll \lambda$) PEC wire is attached to an airplane and is used as a trailing antenna. In order for the wire not to be very visible to radar, it is desired for the wire to have a normalized scattering width (σ/λ) not to exceed -10 dB. Determine the largest radius of the wire (in λ) when the incident uniform plane wave is TM^z polarized, as shown in Figure 11-12a.

11.55. Repeat Problem 11.54 when the incident uniform plane wave is TE^z polarized, as shown in Figure 11-12b.

11.56. A TM^z uniform plane wave traveling along the $+x$ direction is normally incident upon a PMC cylinder, as shown in Figure 11-12a.

- Derive the two-dimensional RCS (SW) expressed as SW/λ .
- Derive an expression for electric current density $\mathbf{J}_s$ and numerically evaluate $\mathbf{J}_s$ on the surface of the cylinder ($\rho = a$). Justify the numerical value; i.e., is it what you were expecting? Should it be that value?

11.57. Repeat Problem 11.56 for TE^z polarization, as shown in Figure 11-12b.

11.58. A TM^z uniform plane wave traveling in the $+x$ direction in free space is incident normally on a lossless dielectric circular cylinder of radius a , as shown in Figure P11-58.

Assume that the incident, scattered, and transmitted (into the cylinder) electric fields can be written as

$$\begin{aligned}\mathbf{E}^i &= \hat{\mathbf{a}}_z E_0 \sum_{n=-\infty}^{+\infty} j^{-n} J_n(\beta_0 \rho) e^{jn\phi} \\ \mathbf{E}^s &= \hat{\mathbf{a}}_z E_0 \sum_{n=-\infty}^{+\infty} a_n H_n^{(2)}(\beta_0 \rho) e^{jn\phi} \\ \mathbf{E}^d &= \hat{\mathbf{a}}_z E_0 \sum_{n=-\infty}^{+\infty} [b_n J_n(\beta_1 \rho) + c_n Y_n(\beta_1 \rho)] e^{jn\phi}\end{aligned}$$

- Derive expressions for the incident, scattered, and transmitted magnetic field components.
- Show that the wave amplitude coefficients are equal to

$$c_n = 0$$

$$\begin{aligned}a_n &= j^{-n} \frac{J'_n(\beta_0 a) J_n(\beta_1 a) - \sqrt{\varepsilon_r / \mu_r} J_n(\beta_0 a) J'_n(\beta_1 a)}{\sqrt{\varepsilon_r / \mu_r} J'_n(\beta_1 a) H_n^{(2)}(\beta_0 a) - J_n(\beta_1 a) H_n^{(2)'}(\beta_0 a)} \\ b_n &= j^{-n} \frac{J_n(\beta_0 a) H_n^{(2)}(\beta_0 a) - J'_n(\beta_0 a) H_n^{(2)'}(\beta_0 a)}{J_n(\beta_1 a) H_n^{(2)'}(\beta_0 a) - \sqrt{\varepsilon_r / \mu_r} J'_n(\beta_1 a) H_n^{(2)}(\beta_0 a)}\end{aligned}$$

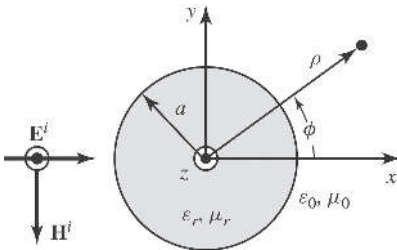


Figure P11-58

- For Problem 11.58,
 - Derive an expression for the scattering width (SW).
 - Plot SW / λ_0 (in dB) for $a = 2\lambda$, dielectric constants of $\varepsilon_r = 4$ and 9 ($0 \leq \phi \leq 180^\circ$).
- A TM^z uniform plane wave traveling along the $+x$ direction, is normally incident upon a dielectric cylinder with $\varepsilon_r \gg 1$ and $\mu_r = 1$, as

shown in Figure P11-58. Write an expression for the:

- Two-dimensional RCS (SW) expressed in SW / λ_0 .
- Electric current density $\mathbf{J}_s$ and evaluate $\mathbf{J}_s$ on the surface of the cylinder ($\rho = a$). Is it what you were expecting?
- Repeat part (b) for magnetic current density $\mathbf{M}_s$.

For parts (a)–(c), you do not have to derive the equations, but must justify the answers.

- Repeat Problem 11-60 when the plane wave is normally incident upon a dielectric cylinder with $\mu_r \gg 1$ and $\varepsilon_r = 1$, as shown in Figure P11-58.

- A TE^z uniform plane wave traveling in the $+x$ direction in free space is incident normally upon a lossless dielectric circular cylinder of radius a , as shown in Figure P11-62. Assume that the incident, scattered, and transmitted (into the cylinder) magnetic fields can be written as

$$\begin{aligned}\mathbf{H}^i &= \hat{\mathbf{a}}_z H_0 \sum_{n=-\infty}^{+\infty} j^{-n} J_n(\beta_0 \rho) e^{jn\phi} \\ \mathbf{H}^s &= \hat{\mathbf{a}}_z H_0 \sum_{n=-\infty}^{+\infty} a_n H_n^{(2)}(\beta_0 \rho) e^{jn\phi} \\ \mathbf{H}^d &= \hat{\mathbf{a}}_z H_0 \sum_{n=-\infty}^{+\infty} [b_n J_n(\beta_1 \rho) + c_n Y_n(\beta_1 \rho)] e^{jn\phi}\end{aligned}$$

- Derive expressions for the incident, scattered, and transmitted electric field components.
- Show that the wave amplitude coefficients are equal to

$$c_n = 0$$

$$\begin{aligned}a_n &= j^{-n} \frac{J'_n(\beta_0 a) J_n(\beta_1 a) - \sqrt{\mu_r / \varepsilon_r} J_n(\beta_0 a) J'_n(\beta_1 a)}{\sqrt{\mu_r / \varepsilon_r} J'_n(\beta_1 a) H_n^{(2)}(\beta_0 a) - J_n(\beta_1 a) H_n^{(2)'}(\beta_0 a)} \\ b_n &= j^{-n} \frac{J_n(\beta_0 a) H_n^{(2)'}(\beta_0 a) - J'_n(\beta_0 a) H_n^{(2)}(\beta_0 a)}{J_n(\beta_1 a) H_n^{(2)'}(\beta_0 a) - \sqrt{\mu_r / \varepsilon_r} J'_n(\beta_1 a) H_n^{(2)}(\beta_0 a)}\end{aligned}$$

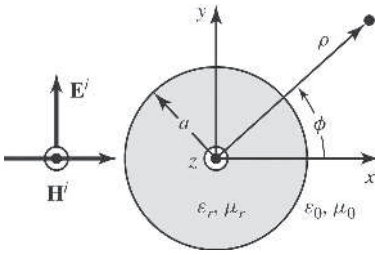


Figure P11-62

11.63. A TE^z uniform plane wave traveling along the $+x$ direction is normally incident upon a dielectric cylinder with $\epsilon_r \gg 1$ and $\mu_r = 1$, as shown in Figure P11-62. Write an expression for the:

- Two-dimensional RCS (SW) expressed in SW / λ_0 .
- Electric current density $\mathbf{J}_s$ and evaluate $\mathbf{J}_s$ on the surface of the cylinder ($\rho = a$). Is it what you were expecting?
- Repeat part (b) for magnetic current density $\mathbf{M}_s$.

For parts (a)–(c), you do not have to derive the equations, but must justify the answers.

11.64. Repeat Problem 11.63 when the plane wave is normally incident upon a dielectric cylinder with $\mu \gg 1$ and $\epsilon_r = 1$, as shown in Figure P11-62.

11.65. A TM^z uniform plane wave traveling in the $+x$ direction in free space is incident normally upon a dielectric-coated conducting circular cylinder of radius a as shown in Figure P11-65. The thickness of the lossless dielectric coating is $b - a$. Assume that the incident, reflected, and transmitted (into the coating) electric fields can be written as shown in Problem 11.58.

- Write expressions for the incident, scattered, and transmitted magnetic field components.

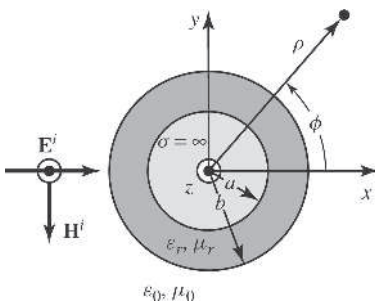


Figure P11-65

- Determine the wave amplitude coefficients a_n , b_n , and c_n . Write them in their simplest forms.

11.66. Repeat Problem 11.65 for a TE^z uniform plane wave incidence. Assume that the incident, scattered, and transmitted (into the coating) magnetic fields can be written as shown in Problem 11.62.

11.67. Using the definition of (11-21b), instead of (11-21c), show that the TM^z polarization scattering width reduces to (11-137) through (11-137b).

11.68. Using the definition of (11-21c), instead of (11-21b), show that the TE^z polarization scattering width reduces to (11-155) through (11-155b).

11.69. An electric line source of constant current is placed above a circular PEC cylinder of infinite length, as shown in Figure 11-17 where $\phi' = 90^\circ$. The radius of the cylinder is $a = 50\lambda$. Determine the approximate smallest height (in number of λ) of the line source above the cylinder that will allow the normalized total amplitude pattern to be at $\phi = 90^\circ$:

- Maximum.
- Minimum.
- −3 dB.

Indicate how you arrive at your answers.

11.70. Two infinite length line sources of constant current I and of the same phase are placed near a conducting cylinder along the x axis (one on each side) a distance s from the center of the cylinder, as shown in Figure P11-70.

- Neglecting coupling between the sources, write an expression for the total electric field for both sources (assume $\rho > s$).

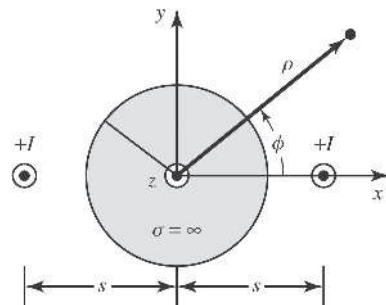


Figure P11-70

- (b) Assuming the observations are made at large distances from the cylinder ($\rho \gg s$) and the radius of the cylinder as well as the distance s are very small ($a \ll \lambda$ and $s \ll \lambda$), find the distance s that the sources must be placed so that the electric field at any observation point will vanish. Explain.

11.71. Three infinite length line sources carrying constant magnetic currents of I_m , $2I_m$, and I_m , respectively, are positioned a distance b near a perfect electric conducting cylinder, as shown in Figure P11-71. Neglecting mutual coupling between the sources, find the following.

- (a) The total scattered magnetic field when $\rho > b$.
 (b) The magnitude of the ratio of the scattered to the incident magnetic field for $\rho > b$.
 (c) The normalized total magnetic field pattern when $\beta\rho \rightarrow \text{large}$.

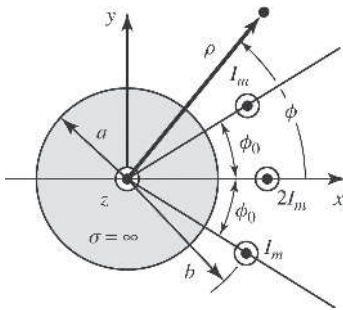


Figure P11-71

11.72. A TM^z uniform plane wave is incident at an angle ϕ' upon a half plane, as shown in Figure P11-72. Show that the current density on the upper side of the half plane is

$$J_z = \frac{E_0}{j2\omega\mu\rho} \sum_{m=1}^{\infty} m j^{m/2} J_{m/2}(\beta\rho) \times \sin\left(\frac{m\phi'}{2}\right) \quad \text{for any } \rho$$

$$J_z \simeq \frac{E_0}{2\eta} \sqrt{\frac{2}{j\pi\beta\rho}} \sin\left(\frac{\phi'}{2}\right) \quad \text{for } \beta\rho \rightarrow 0$$

where E_0 is the amplitude of the incident electric field.

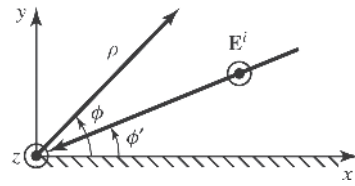


Figure P11-72

11.73. Derive (11-198) from (11-192a).

11.74. Repeat Problem 11.72 for TE^z uniform plane wave incidence. Show that the current density on the upper side of the half plane is

$$J_\rho = H_0 \sum_{m=0}^{\infty} \varepsilon_m j^{m/2} J_{m/2}(\beta\rho) \times \cos\left(\frac{m\phi'}{2}\right) \quad \text{for any } \rho$$

$$J_\rho = H_0 \quad \text{for } \beta\rho \rightarrow 0$$

where H_0 is the amplitude of the incident field and ε_m is defined by (11-196a).

11.75. Verify (11-222) and (11-222a).

11.76. A uniform plane wave is incident upon a conducting sphere of radius a . Assume that the diameter of the sphere is 1.128 m and the frequency is 8.5 GHz.

- (a) Determine the monostatic radar cross section of the sphere in decibels per square meter.
 (b) Find the area (in square meters) of a flat plate whose normal-incidence monostatic RCS is the same as that of the sphere in part (a).

11.77. A uniform plane wave is normally incident, on a circular PEC ground plane of radius a_p . The same wave is also incident upon a PEC sphere of radius a_s .

- (a) Determine the radius of the sphere a_s , in terms of the radius a_p of the circular plate, so that the RCS of the plate and sphere are identical.
 (b) If the radius of the plate is 3λ , determine the radius of the sphere so that the plate and sphere will have equal RCS. Assume that the radii of the plate and sphere are sufficiently large, compared to the wavelength, so that both geometrical and physical optics are good approximations.

11.78. A uniform plane wave is incident, at normal incidence, upon a flat PEC plate of area $25\lambda^2$. Determine:

- The 3-D monostatic RCS (in dBsm) of the plate at $f = 10$ GHz based on physical optics.
- The radius (in λ) of a sphere so that it has equal backscattered RCS as the plate. Assume the radius of the sphere is large compared to the wavelength.

11.79. Repeat the calculations of Figure 11-29 by plotting the normalized RCS $[\sigma / (\pi a^2)]$:

- Dimensionless.
- In dB.

11.80. For the scattering of a plane wave by a PEC sphere of radius a , with the incident electric field with only one component (E_x^i), as outlined in Section 11.8:

- Derive an expression, in simplified form, for the cross-polarized component (E_y) of the far-zone scattered electric field in the monostatic direction only.
- Derive an expression, in simplified form, of the 3-D monostatic RCS for the cross-polarized field.
- Plot the normalized RCS $[\sigma / (\pi a^2)]$ of the cross-polarized component E_y for $0 \leq a \leq 2\lambda_0$ (λ_0 is the free-space wavelength), similar to Figure 11-29:
 - Dimensionless.
 - In dB.

If you have any comments to make concerning the monostatic cross-polarized field and associated RCS, please do so.

11.81. Applying the boundary conditions (11-252a) through (11-252d) on the internal and external fields of a dielectric sphere, show that the modal coefficients of the vector potential and fields are those of (11-253a) through (11-253d).

11.82. Show that for $n=1$, (11-253a) through (11-253d) reduce to (11-254a) through (11-254d).

11.83. A dielectric sphere of radius a , as shown in Figure 11-28, is illuminated by a plane wave with an electric field given by

$$\mathbf{E}^i = \hat{\mathbf{a}}_x E_x^i = \hat{\mathbf{a}}_x E_o e^{-j\beta z} = \hat{\mathbf{a}}_x E_o e^{-j\beta r \cos \theta}$$

The dielectric constant and relative permeability of the material of the sphere are $\epsilon_r \gg 1$, $\mu_r = 1$. Write an expression for the three-dimensional monostatic RCS of the sphere valid for:

- Any radius.
- Small radius ($a \ll \lambda$; valid only in the so-called Rayleigh region). This expression must be simple.
- Large radius ($a \gg \lambda$; valid only in the so-called optical region). This expression must be simple.

You do not have to derive the expressions, if you think you know what they are. However, justify their use.

CHAPTER 12



Integral Equations and the Moment Method

12.1 INTRODUCTION

In Chapter 11 we discussed scattering from conducting objects, such as plates, circular cylinders, and spheres, using geometrical optics, physical optics, and modal solutions. For the plates and cylinders, we assumed that their dimensions were of infinite extent. In practice, however, the dimensions of the objects are always finite, although some of them may be very large. Expressions for the radar cross section of finite-size scatterers were introduced in the previous chapter. These, however, represent approximate forms, and more accurate expressions are sometimes desired.

The physical optics method of Chapter 7, Section 7.10, was used in the previous chapter to approximate the current induced on the surface of a finite-size target, such as the strip and rectangular plate. Radiation integrals were then used to find the field scattered by the target. To derive a more accurate representation of the current induced on the surface of the finite-size target, and thus, of the scattered fields, two methods will be examined in this book.

One method, referred to here as the *integral equation* (IE) technique, casts the solution for the induced current in the form of an integral equation (hence its name) where the unknown induced current density is part of the integrand. Numerical techniques, such as the *moment method* (MM) [1–6], can then be used to solve for the current density. Once this is accomplished, the fields scattered by the target can be found using the traditional radiation integrals. The total induced current density will be the sum of the physical optics current density and a *fringe wave* current density [7–13], which can be thought of as a perturbation current density introduced by the edge diffractions of the finite-size structure. This method will be introduced and applied in this chapter.

The other method, referred to here as the *geometrical theory of diffraction* (GTD) [14–17], is an extension of geometrical optics and accounts for the contributions from the edges of the finite structure using diffraction theory. This method will be introduced and applied in Chapters 13 and 14. More extensive discussions of each can be found in the open literature.

12.2 INTEGRAL EQUATION METHOD

The objective of the integral equation (IE) method for scattering is to cast the solution for the unknown current density, which is induced on the surface of the scatterer, in the form of an integral equation where the unknown induced current density is part of the integrand. The integral equation is then solved for the unknown induced current density using numerical techniques such as the *moment method* (MM). To demonstrate the technique, we will initially consider some specific problems. We will start with an electrostatics problem and follow it with time-harmonic problems.

12.2.1 Electrostatic Charge Distribution

In electrostatics, the problem of finding the potential, that is due to a given charge distribution, is often considered. In physical situations, however, it is seldom possible to specify a charge distribution. Whereas we may connect a conducting body to a voltage source, and thus, specify the potential throughout the body, the distribution of charge is obvious only for a few rotationally symmetric canonical geometries. In this section we will consider an integral equation approach to solve for the electric charge distribution, once the electric potential is specified. Some of the material here and in other sections is drawn from [18, 19].

From statics, we know that a linear electric charge distribution $\rho(\mathbf{r}')$ will create an electric potential, $V(\mathbf{r})$, according to [20]

$$V(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \int_{\text{source (charge)}} \frac{\rho(\mathbf{r}')}{R} d\ell' \quad (12-1)$$

where $\mathbf{r}'(x', y', z')$ denotes the source coordinates, $\mathbf{r}(x, y, z)$ denotes the observation coordinates, $d\ell'$ is the path of integration, and R is the distance from any point on the source to the observation point, which is generally represented by

$$R(\mathbf{r}, \mathbf{r}') = |\mathbf{r} - \mathbf{r}'| = \sqrt{(x - x')^2 + (y - y')^2 + (z - z')^2} \quad (12-1a)$$

We see that (12-1) may be used to calculate the potentials that are due to any known line charge density. However, the charge distribution on most configurations of practical interest, i.e., complex geometries, is not usually known, even when the potential on the source is given. It is the nontrivial problem of determining the charge distribution, for a specified potential, that is to be solved here using an integral equation approach.

A. Finite Straight Wire Consider a straight wire of length ℓ and radius a , placed along the y axis, as shown in Figure 12-1a. The wire is maintained at a normalized constant electric potential of 1 V.

Note that (12-1) is valid everywhere, including on the wire itself ($V_{\text{wire}} = 1$ V). Thus, choosing the observation along the wire axis ($x = z = 0$) and representing the charge density on the surface of the wire by $\rho(y')$, (12-1) can be expressed as

$$1 = \frac{1}{4\pi\epsilon_0} \int_0^\ell \frac{\rho(y')}{R(y, y')} dy' \quad 0 \leq y \leq \ell \quad (12-2)$$

where

$$R(y, y') = R(\mathbf{r}, \mathbf{r}')|_{x=z=0} = \sqrt{(y - y')^2 + [(x')^2 + (z')^2]} = \sqrt{(y - y')^2 + a^2} \quad (12-2a)$$

The observation point is chosen along the wire axis and the charge density is represented along the surface of the wire to avoid $R(y, y') = 0$, which would introduce a singularity in the integrand of (12-2). If the radius of the wire is zero ($a = 0$), then the source (current density) and observation

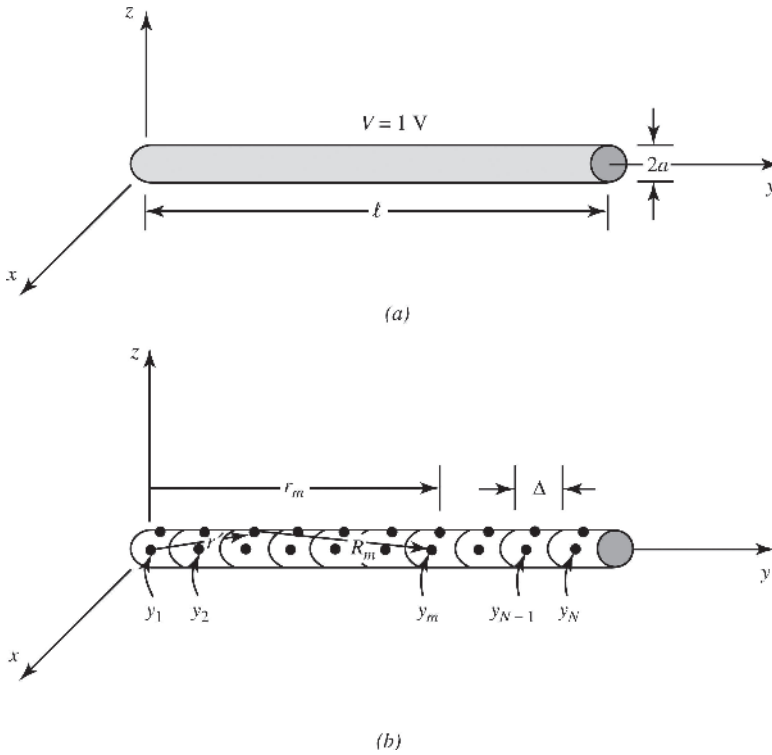


Figure 12-1 (a) Straight wire of constant potential. (b) Its segmentation.

points will be along the same line (y axis). In that case, to avoid singularities on the distance $R(y, y')$ [$R(y, y') = 0$], the source (y') and observation point (y) should be chosen not to coincide.

It is necessary to solve (12-2) for the unknown $\rho(y')$ (an inverse problem). Equation 12-2 is an integral equation that can be used to find the charge density $\rho(y')$ based on the 1-V potential. The solution may be reached numerically by reducing (12-2) to a series of linear algebraic equations that may be solved by conventional matrix-equation techniques. To facilitate this, let us approximate the unknown charge distribution $\rho(y')$ by an expansion of N known terms with constant, but unknown, coefficients; that is

$$\rho(y') = \sum_{n=1}^N a_n g_n(y') \quad (12-3)$$

Thus, (12-2) may be written, using (12-3), as

$$4\pi\epsilon_0 = \int_0^\ell \frac{1}{R(y, y')} \left[\sum_{n=1}^N a_n g_n(y') \right] dy' \quad (12-4)$$

Because (12-4) is a nonsingular integral, its integration and summation can be interchanged and it can be written as

$$4\pi\epsilon_0 = \sum_{n=1}^N a_n \int_0^\ell \frac{g_n(y')}{\sqrt{(y - y')^2 + a^2}} dy' \quad (12-4a)$$

The wire is now divided into N uniform segments, each of length $\Delta = \ell/N$, as illustrated in Figure 12-1b. The $g_n(y')$ functions in the expansion (12-3) are chosen for their ability to accurately model the unknown quantity, while minimizing computation. They are often referred to as *basis* (or expansion) functions, and they will be discussed further in Section 12.2.5. To avoid

complexity in this solution, subdomain piecewise constant (or “pulse”) functions will be used. These functions, shown in Figure 12-6, are defined to be of a constant value over one segment and zero elsewhere, or

$$g_n(y') = \begin{cases} 0 & y' < (n-1)\Delta \\ 1 & (n-1)\Delta \leq y' \leq n\Delta \\ 0 & n\Delta < y' \end{cases} \quad (12-5)$$

Many other basis functions are possible, some of which will be introduced later in Section 12.2.5.

Replacing y in (12-4) by a fixed point on the surface of the wire, such as y_m , results in an integrand that is solely a function of y' , so the integral may be evaluated. Obviously, (12-4) leads to one equation with N unknowns a_n written as

$$\begin{aligned} 4\pi\epsilon_0 = & a_1 \int_0^\Delta \frac{g_1(y')}{R(y_m, y')} dy' + a_2 \int_\Delta^{2\Delta} \frac{g_2(y')}{R(y_m, y')} dy' + \cdots \\ & + a_n \int_{(n-1)\Delta}^{n\Delta} \frac{g_n(y')}{R(y_m, y')} dy' + \cdots + a_N \int_{(N-1)\Delta}^\ell \frac{g_N(y')}{R(y_m, y')} dy' \end{aligned} \quad (12-6)$$

In order to obtain a solution for these N amplitude coefficients, N linearly independent equations are necessary. These equations may be produced by choosing an observation point y_m on the surface of the wire and at the center of each Δ length element as shown in Figure 12-1b. This will result in one equation of the form of (12-6) corresponding to each observation point. For N such observation points, we can reduce (12-6) to

$$\begin{aligned} 4\pi\epsilon_0 = & a_1 \int_0^\Delta \frac{g_1(y')}{R(y_1, y')} dy' + \cdots + a_N \int_{(N-1)\Delta}^\ell \frac{g_N(y')}{R(y_1, y')} dy' \\ & \vdots \\ 4\pi\epsilon_0 = & a_1 \int_0^\Delta \frac{g_1(y')}{R(y_N, y')} dy' + \cdots + a_N \int_{(N-1)\Delta}^\ell \frac{g_N(y')}{R(y_N, y')} dy' \end{aligned} \quad (12-6a)$$

We may write (12-6a) more concisely using matrix notation as

$$[V_m] = [Z_{mn}][I_n] \quad (12-7)$$

where each Z_{mn} term is equal to

$$Z_{mn} = \int_0^\ell \frac{g_n(y')}{\sqrt{(y_m - y')^2 + a^2}} dy' = \int_{(n-1)\Delta}^{n\Delta} \frac{1}{\sqrt{(y_m - y')^2 + a^2}} dy' \quad (12-7a)$$

and

$$[I_n] = [a_n] \quad (12-7b)$$

$$[V_m] = [4\pi\epsilon_0] \quad (12-7c)$$

The V_m column matrix has all terms equal to $4\pi\epsilon_0$, and the $I_n = a_n$ values are the unknown charge distribution coefficients. Solving (12-7) for $[I_n]$ gives

$$[I_n] = [a_n] = [Z_{mn}]^{-1}[V_m] \quad (12-8)$$

Either (12-7) or (12-8) may readily be solved on a digital computer by using any of a number of matrix-inversion or equation-solving routines. Whereas the integrals involved here may be evaluated in closed form by making appropriate approximations, this is not usually possible with more

complicated problems. Efficient numerical integration computer subroutines are commonly available in easy-to-use forms.

One closed form evaluation of (12-7a) is to reduce the integral and represent it by

$$Z_{mn} = \begin{cases} 2 \ln \left[\frac{\frac{\Delta}{2} + \sqrt{a^2 + \left(\frac{\Delta}{2}\right)^2}}{a} \right] & m = n \end{cases} \quad (12-9a)$$

$$Z_{mn} = \begin{cases} \ln \left[\frac{d_{mn}^+ + [(d_{mn}^+)^2 + a^2]^{1/2}}{d_{mn}^- + [(d_{mn}^-)^2 + a^2]^{1/2}} \right] & m \neq n \text{ but } |m - n| \leq 2 \end{cases} \quad (12-9b)$$

$$Z_{mn} = \begin{cases} \ln \left(\frac{d_{mn}^+}{d_{mn}^-} \right) & |m - n| > 2 \end{cases} \quad (12-9c)$$

where

$$d_{mn}^+ = \ell_m + \frac{\Delta}{2} \quad (12-9d)$$

$$d_{mn}^- = \ell_m - \frac{\Delta}{2} \quad (12-9e)$$

ℓ_m is the distance between the m th matching point and the center of the n th source point.

In summary, the solution of (12-2) for the charge distribution on a wire has been accomplished by approximating the unknown with some basis functions, dividing the wire into segments, and then sequentially enforcing (12-2) at the center of each segment to form a set of linear equations.

Even for the relatively simple straight wire geometry we have discussed, the exact form of the charge distribution is not intuitively apparent. To illustrate the principles of the numerical solution, an example is now presented.

Example 12-1

A 1-m-long straight wire of radius $a = 0.001$ m is maintained at a constant potential of 1 V. Determine the linear charge distribution on the wire by dividing the length into 5 and 20 uniform segments. Assume subdomain pulse basis functions.

Solution:

1. $N = 5$. When the 1-m-long wire is divided into five uniform segments each of length $\Delta = 0.2$ m, (12-7) reduces to

$$\begin{bmatrix} 10.60 & 1.10 & 0.51 & 0.34 & 0.25 \\ 1.10 & 10.60 & 1.10 & 0.51 & 0.34 \\ 0.51 & 1.10 & 10.60 & 1.10 & 0.51 \\ 0.34 & 0.51 & 1.10 & 10.60 & 1.10 \\ 0.25 & 0.34 & 0.51 & 1.10 & 10.60 \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \\ a_3 \\ a_4 \\ a_5 \end{bmatrix} = \begin{bmatrix} 1.11 \times 10^{-10} \\ 1.11 \times 10^{-10} \\ \vdots \\ 1.11 \times 10^{-10} \end{bmatrix}$$

Inverting this matrix leads to the amplitude coefficients and subsequent charge distribution of

$$a_1 = 8.81 \text{ pC/m}$$

$$a_2 = 8.09 \text{ pC/m}$$

$$a_3 = 7.97 \text{ pC/m}$$

$$a_4 = 8.09 \text{ pC/m}$$

$$a_5 = 8.81 \text{ pC/m}$$

The charge distribution is shown plotted in Figure 12-2a.

2. $N = 20$. Increasing the number of segments to 20 results in a much smoother distribution, as shown plotted in Figure 12-2b. As more segments are used, a better approximation of the actual charge distribution is attained, which has smaller discontinuities over the length of the wire.

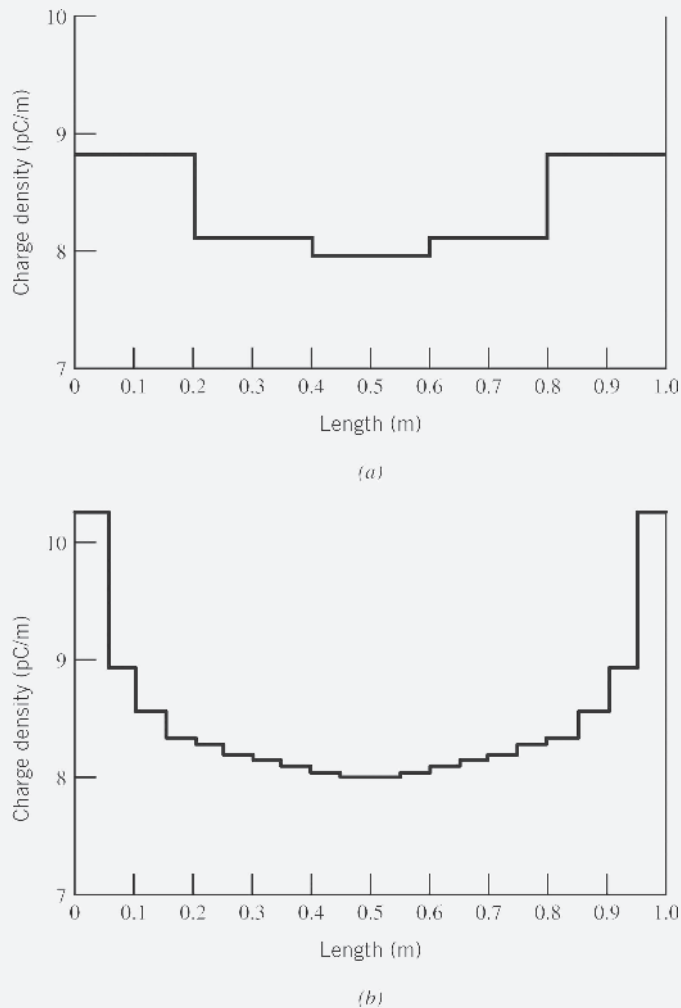


Figure 12-2 Charge distribution on a 1-m straight wire at 1 V. (a) $N = 5$. (b) $N = 20$.

B. Bent Wire In order to illustrate the solution for a more complex structure, let us analyze a body composed of two noncollinear straight wires, that is, a bent wire. If a straight wire is bent, the charge distribution will be altered, although the solution to find it will differ only slightly from the straight wire case. We will assume a bend of angle α , which remains in the yz plane, as shown in Figure 12-3.

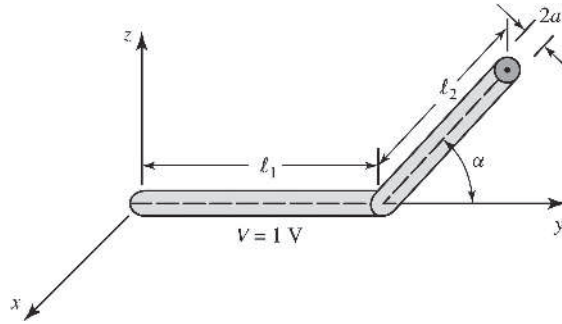


Figure 12-3 Geometry for bent wire.

For the first segment ℓ_1 of the wire, the distance R can be represented by (12-2a). However, for the second segment ℓ_2 , we can express the distance as

$$R = \sqrt{(y - y')^2 + (z - z')^2} \quad (12-10)$$

Also because of the bend, the integral in (12-7a) must be separated into two parts of

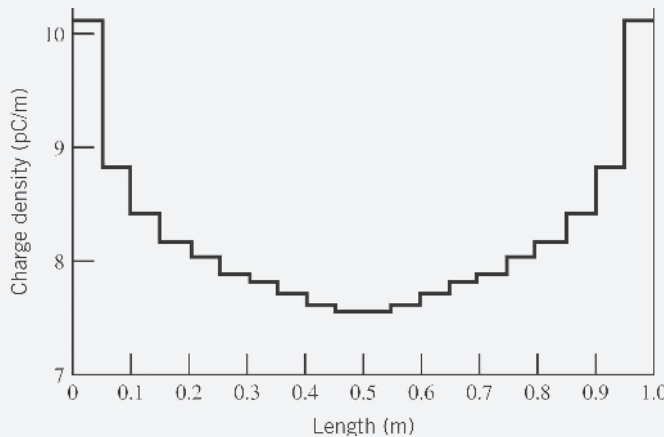
$$Z_{mm} = \int_0^{\ell_1} \frac{\rho_n(\ell'_1)}{R} d\ell'_1 + \int_0^{\ell_2} \frac{\rho_n(\ell'_2)}{R} d\ell'_2 \quad (12-11)$$

where ℓ_1 and ℓ_2 are measured along the corresponding straight sections from their left ends.

Example 12-2

Repeat Example 12-1 assuming that the wire has been bent 90° at its midpoint. Subdivide the entire wire into 20 uniform segments.

Solution: The charge distribution for this case, calculated using (12-10) and (12-11), is plotted in Figure 12-4 for $N = 20$ segments. Note that the charge is relatively more concentrated near the ends of this structure than was the case for a straight wire of Figure 12-2b. Further, the overall charge density, and thus capacitance, on the structure has decreased.


 Figure 12-4 Charge distribution on a 1-m bent wire ($\alpha = 90^\circ$, $N = 20$).

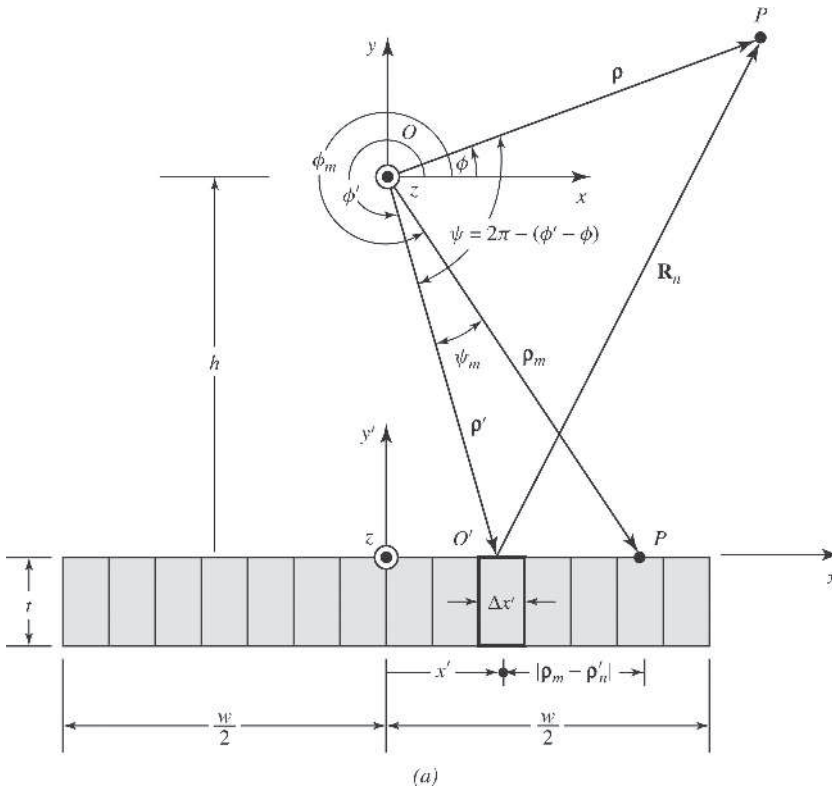


Figure 12-5 Geometry of a line source above a two-dimensional finite width strip. (a) Boundary conditions and integration on the same surface. (b) Boundary conditions and integration not on the same surface.

Arbitrary wire configurations, including numerous bends and even curved sections, may be analyzed by the methods already outlined here. As with the simple bent wire, the only alterations generally necessary are those required to describe the geometry analytically.

12.2.2 Integral Equation

Now that we have demonstrated the numerical solution of a well-known electrostatics integral equation, we will derive and solve a time-harmonic integral equation for an infinite line source above a two-dimensional conducting strip, as shown in Figure 12-5a. Once this is accomplished, we will generalize the integral equation formulation for three-dimensional problems in Section 12.3.

Referring to Figure 12-5a, the field radiated by a line source of constant current I_z in the absence of the strip (referred to as E_z^d) is given by (11-10a) or

$$E_z^d(\rho) = -\frac{\beta^2 I_z}{4\omega\epsilon} H_0^{(2)}(\beta\rho) \quad (12-12)$$

where $H_0^{(2)}(\beta\rho)$ is the Hankel function of the second kind of order zero. Part of the field given by (12-12) is directed toward the strip, and it induces on it a linear current density J_z (in amperes per meter) such that

$$J_z(x')\Delta x' = \Delta I_z(x') \quad (12-13)$$

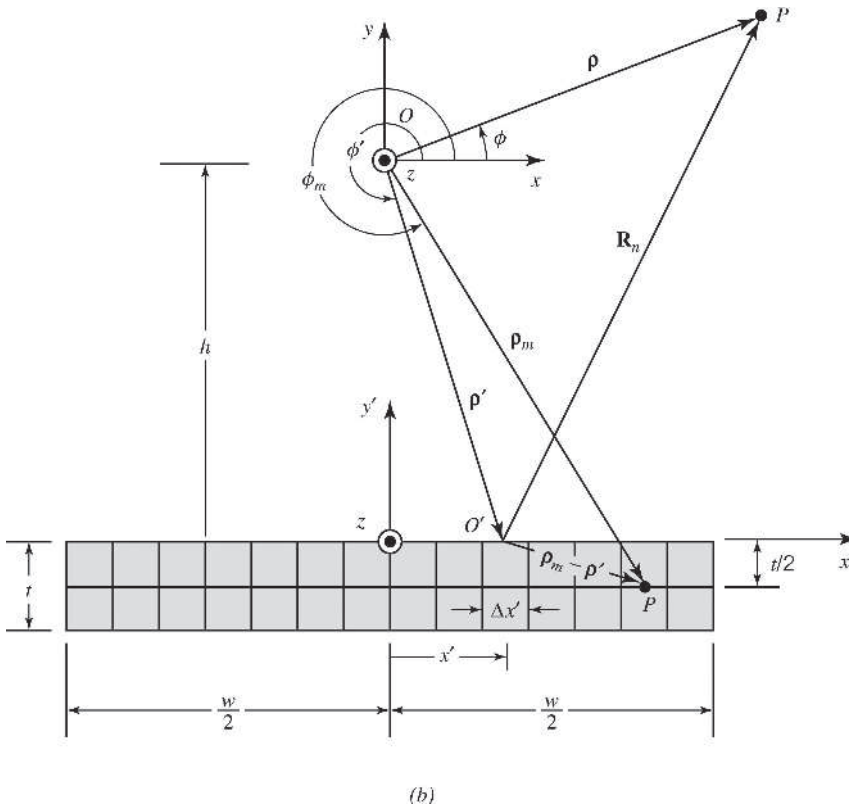


Figure 12-5 (Continued)

which, as $\Delta x' \rightarrow 0$, can be written as

$$J_z(x') dx' = dI_z(x') \quad (12-13a)$$

The induced current of (12-13a) reradiates and produces an electric field component that will be referred to as *reflected* (or *scattered*) and designated as $E_z^r(\rho)$ [or $E_z^s(\rho)$]. If the strip is subdivided into N segments, each of width $\Delta x'_n$ as shown in Figure 12-5b, the scattered field can be written, according to (12-12), as

$$E_z^s(\rho) = -\frac{\beta^2}{4\omega\epsilon} \sum_{n=1}^N H_0^{(2)}(\beta R_n) \Delta I_z(x'_n) = -\frac{\beta^2}{4\omega\epsilon} \sum_{n=1}^N H_0^{(2)}(\beta R_n) J_z(x'_n) \Delta x'_n \quad (12-14)$$

where x'_n is the position of the n th segment. In the limit, as each segment becomes very small ($\Delta x'_n \rightarrow 0$), (12-14) can be written as

$$E_z^s(\rho) = -\frac{\beta^2}{4\omega\epsilon} \int_{\text{strip}} H_0^{(2)}(\beta R) dI_z = -\frac{\beta^2}{4\omega\epsilon} \int_{-w/2}^{w/2} J_z(x') H_0^{(2)}(\beta |\rho - \rho'|) dx' \quad (12-15)$$

since

$$R = |\rho - \rho'| = \sqrt{\rho^2 + (\rho')^2 - 2\rho\rho' \cos(\phi - \phi')} \quad (12-15a)$$

The total field at any observation point, including the strip itself, will be the sum of the direct component E_z^d of (12-12) and the scattered component E_z^s of (12-15). However, to determine the scattered component, we need to know the induced current density $J_z(x')$. The objective here then will be to find an equation, which in this case will be in terms of an integral and will be referred to as an *integral equation*, that can be used to determine $J_z(x')$. This can be accomplished by choosing the observation point on the strip itself. Doing this, we have that for any observation point $\rho = \rho_m$ on the strip, the total tangential electric field vanishes and it is given by

$$E_z^t(\rho = \rho_m)|_{\text{strip}} = [E_z^d(\rho = \rho_m) + E_z^s(\rho = \rho_m)]_{\text{strip}} = 0 \quad (12-16)$$

or

$$E_z^d(\rho = \rho_m)|_{\text{strip}} = -E_z^s(\rho = \rho_m)|_{\text{strip}} \quad (12-16a)$$

Using (12-12) and (12-15), we can write (12-16a) as

$$-\frac{\beta^2 I_z}{4\omega\epsilon} H_0^{(2)}(\beta|\rho_m|) = +\frac{\beta^2}{4\omega\epsilon} \int_{-w/2}^{w/2} J_z(x') H_0^{(2)}(\beta|\rho_m - \rho'|) dx' \quad (12-17)$$

which for a unit current I_z (i.e., $I_z = 1$) reduces to

$$\boxed{H_0^{(2)}(\beta|\rho_m|) = -\int_{-w/2}^{w/2} J_z(x') H_0^{(2)}(\beta|\rho_m - \rho'|) dx'} \quad (12-17a)$$

Equation 12-17a is the *electric field integral equation* (EFIE) for the line source above the strip, and it can be used to find the current density $J_z(x')$ based upon a unit current I_z . If I_z is of any other constant value, then all the values of $J_z(x')$ must be multiplied by that same constant value. Electric field integral equations (EFIE) and magnetic field integral equations (MFIE) are discussed in more general forms in Section 12.3.

12.2.3 Radiation Pattern

Once J_z is found, we can then determine the total field radiated of the entire system for any observation point. The total field is composed of two parts: the field radiated from the line source itself (E_z^d) and that which is scattered (reradiated) from the strip (E_z^s). Thus, using (12-12) and (12-15), we can write the total field as

$$\begin{aligned} E_z^t(\rho) &= E_z^d(\rho) + E_z^s(\rho) \\ &= -\frac{\beta^2 I_z}{4\omega\epsilon} H_0^{(2)}(\beta\rho) - \frac{\beta^2}{4\omega\epsilon} \int_{-w/2}^{w/2} J_z(x') H_0^{(2)}(\beta|\rho - \rho'|) dx' \end{aligned} \quad (12-18)$$

which for a unit amplitude current I_z ($I_z = 1$) reduces to

$$\boxed{E_z^t(\rho) = -\frac{\beta^2}{4\omega\epsilon} \left[H_0^{(2)}(\beta\rho) + \int_{-w/2}^{w/2} J_z(x') H_0^{(2)}(\beta|\rho - \rho'|) dx' \right]} \quad (12-18a)$$

Equation 12-18a can be used to find the total field at any observation point, near or far field. The current density $J_z(x')$ can be found using (12-17a). However, for far-field observations, (12-18a) can be approximated and written in a more simplified form. In general, the distance

R is given by (12-15a). However, for far-field observations ($\rho \gg \rho'$), (12-15a) reduces, using the binomial expansion, to

$$R \simeq \begin{cases} \rho - \rho' \cos(\phi - \phi') & \text{for phase terms} \\ \rho & \text{for amplitude terms} \end{cases} \quad (12-19a)$$

$$(12-19b)$$

For large arguments, the Hankel functions in (12-18a) can be replaced by their asymptotic form

$$H_n^{(2)}(\beta z) \stackrel{\beta z \rightarrow \infty}{\simeq} \sqrt{\frac{2j}{\pi\beta z}} j^n e^{-j\beta z} \quad (12-20)$$

For $n = 0$, (12-20) reduces to

$$H_0^{(2)}(\beta z) \simeq \sqrt{\frac{2j}{\pi\beta z}} e^{-j\beta z} \quad (12-20a)$$

Using (12-19a) through (12-20a), we can write the Hankel functions in (12-18a) as

$$H_0^{(2)}(\beta \rho) \simeq \sqrt{\frac{2j}{\pi\beta \rho}} e^{-j\beta \rho} \quad (12-21a)$$

$$\begin{aligned} H_0^{(2)}(\beta |\rho - \rho'|) &\simeq \sqrt{\frac{2j}{\pi\beta \rho}} e^{-j\beta [\rho - \rho' \cos(\phi - \phi')]} \\ &\simeq \sqrt{\frac{2j}{\pi\beta \rho}} e^{-j\beta \rho + j\beta \rho' \cos(\phi - \phi')} \end{aligned} \quad (12-21b)$$

When (12-21a) and (12-21b) are substituted into (12-18a), they reduce it to

$$E_z^t(\rho) \simeq -\frac{\beta^2}{4\omega\epsilon} \sqrt{\frac{2j}{\pi\beta \rho}} e^{-j\beta \rho} \left[1 + \int_{-w/2}^{+w/2} J_z(x') e^{j\beta \rho' \cos(\phi - \phi')} dx' \right] \quad (12-22)$$

which in normalized form can be written as

$$E_z^t(\text{normalized}) \simeq 1 + \int_{-w/2}^{+w/2} J_z(x') e^{j\beta \rho' \cos(\phi - \phi')} dx' \quad (12-22a)$$

Equation 12-22a represents the normalized pattern of the line above the strip. It is based on the linear current density $J_z(x')$ that is induced by the source on the strip. The current density can be found using approximate methods or, more accurately, using the electric field integral equation 12-17a.

12.2.4 Point-Matching (Collocation) Method

The next step will be to use a numerical technique to solve the electric field integral equation 12-17a for the unknown current density $J_z(x')$. We first expand $J_z(x')$ into a finite series of the form

$$J_z(x') \simeq \sum_{n=1}^N a_n g_n(x') \quad (12-23)$$

where $g_n(x')$ represents *basis (expansion)* functions [1, 2]. When (12-23) is substituted into (12-17a), we can write it as

$$\begin{aligned} H_0^{(2)}(\beta|\rho_m|) &= -\int_{-w/2}^{w/2} \sum_{n=1}^N a_n g_n(x') H_0^{(2)}(\beta|\rho_m - \rho'_n|) dx' \\ H_0^{(2)}(\beta|\rho_m|) &= -\sum_{n=1}^N a_n \int_{-w/2}^{w/2} g_n(x') H_0^{(2)}(\beta|\rho_m - \rho'_n|) dx' \end{aligned} \quad (12-24)$$

which takes the general form

$$h = \sum_{n=1}^N a_n F(g_n) \quad (12-25)$$

where

$$h = H_0^{(2)}(\beta|\rho_m|) \quad (12-25a)$$

$$F(g_n) = -\int_{-w/2}^{w/2} g_n(x') H_0^{(2)}(\beta|\rho_m - \rho'_n|) dx' \quad (12-25b)$$

In (12-25), F is referred to as a *linear integral operator*, g_n represents the response function, and h is the known excitation function.

Equation (12-17a) is an electric field integral equation derived by enforcing the boundary conditions of vanishing total tangential electric field on the surface of the conducting strip. A numerical solution of (12-17a) is (12-24) or (12-25) through (12-25b), which, for a given observation point $\rho = \rho_m$, leads to one equation with N unknowns. This can be repeated N times by choosing N observation points. Such a procedure leads to a system of N linear equations each with N unknowns of the form

$$H_0^{(2)}(\beta|\rho_m|) = \sum_{n=1}^N a_n \left[-\int_{-w/2}^{w/2} g_n(x') H_0^{(2)}(\beta|\rho_m - \rho'_n|) dx' \right] \quad m = 1, 2, \dots, N \quad (12-26)$$

which can also be written as

$$V_m = \sum_{n=1}^N I_n Z_{mn} \quad (12-27)$$

where

$$V_m = H_0^{(2)}(\beta|\rho_m|) \quad (12-27a)$$

$$I_n = a_n \quad (12-27b)$$

$$Z_{mn} = -\int_{-w/2}^{w/2} g_n(x') H_0^{(2)}(\beta|\rho_m - \rho'_n|) dx' \quad (12-27c)$$

In matrix form, (12-27) can be expressed as

$$[V_m] = [Z_{mm}][I_n] \quad (12-28)$$

where the unknown is $[I_n]$ and can be found by solving (12-28), or

$$[I_n] = [Z_{mm}]^{-1}[V_m] \quad (12-28a)$$

Since the system of N linear equations with N unknowns—as given by (12-26), (12-27), or (12-28)—was derived by applying the boundary conditions at N discrete points, the technique is referred to as the *point-matching* (or *collocation*) *method* [1, 2].

Thus, by finding the elements of the $[V]$ and $[Z]$, and then the inverse $[Z]^{-1}$, we can determine the elements a_n of the $[I]$ matrix. This in turn allows us to approximate $J_z(x')$ using (12-23), which can then be used in (12-18a) to find the total field everywhere. However, for far-field observations, the total field can be found more easily using (12-22) or, in normalized form, (12-22a).

12.2.5 Basis Functions

One very important step in any numerical solution is the choice of basis functions. In general, one chooses as basis functions the set that has the ability to accurately represent and resemble the anticipated unknown function, while minimizing the computational effort required to employ it [21–23]. Do not choose basis functions with smoother properties than the unknown being represented.

Theoretically, there are many possible basis sets. However, only a limited number are discussed here. These sets may be divided into two general classes. The first class consists of subdomain functions, which are nonzero only over a part of the domain of the function $g(x')$; its domain is the surface of the structure. The second class contains entire domain functions that exist over the entire domain of the unknown function. The entire domain basis function expansion is analogous to the well-known Fourier series expansion method.

A. Subdomain Functions Of the two types of basis functions, subdomain functions are the most common. Unlike entire domain bases, they may be used without prior knowledge of the nature of the function that they must represent.

The subdomain approach involves subdivision of the structure into N nonoverlapping segments, as illustrated on the axis in Figure 12-6a. For clarity, the segments are shown here to be collinear and of equal length, although neither condition is necessary. The basis functions are defined in conjunction with the limits of one or more of the segments.

Perhaps the most common of these basis functions is the conceptually basic piecewise constant, or “pulse” function, shown in Figure 12-6a. It is defined by

$$\begin{array}{c} \text{Piecewise Constant} \\ g_n(x') = \begin{cases} 1 & x'_{n-1} \leq x' \leq x'_n \\ 0 & \text{elsewhere} \end{cases} \end{array} \quad (12-29)$$

Once the associated coefficients are determined, this function will produce a staircase representation of the unknown function, similar to that in Figures 12-6b and 12-6c.

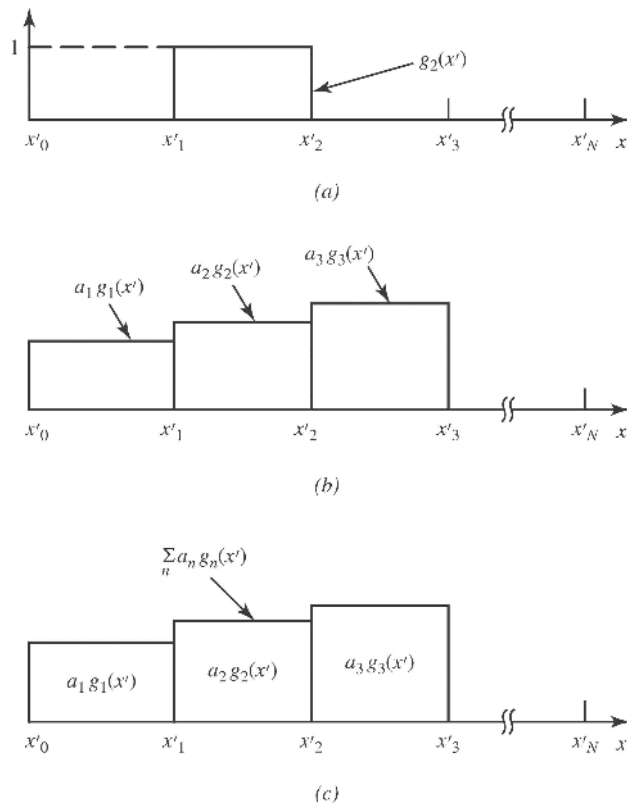


Figure 12-6 Piecewise constant subdomain functions. (a) Single. (b) Multiple. (c) Function representation.

Another common basis set is the piecewise linear, or “triangle,” functions seen in Figure 12-7a. These are defined by

$$\text{Piecewise Linear}$$

$$g_n(x') = \begin{cases} \frac{x' - x'_{n-1}}{x'_n - x'_{n-1}} & x'_{n-1} \leq x' \leq x'_n \\ \frac{x'_{n+1} - x'}{x'_{n+1} - x'_n} & x'_n \leq x' \leq x'_{n+1} \\ 0 & \text{elsewhere} \end{cases} \quad (12-30)$$

and are seen to cover two segments, and overlap adjacent functions (Figure 12-7b). The resulting representation (Figures 12-7b and 12-7c) is smoother than that for “pulses,” but at the cost of somewhat increased computational complexity.

Increasing the sophistication of subdomain basis functions beyond the level of the “triangle” may not be warranted by the possible improvement in accuracy. However, there are cases where more specialized functions are useful for other reasons. For example, some integral operators may be evaluated without numerical integration when their integrands are multiplied by a $\sin(kx')$ or $\cos(kx')$ function, where x' is the variable of integration. In such examples, considerable advantages in computation time and resistance to errors can be gained by using basis functions

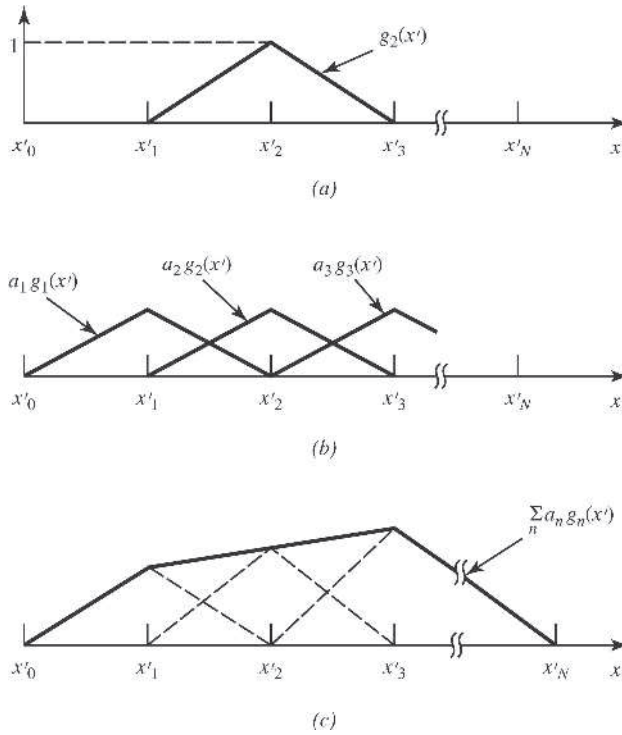


Figure 12-7 Piecewise linear subdomain functions. (a) Single. (b) Multiple. (c) Function representation.

like the piecewise sinusoid of Figure 12-8 or truncated cosine of Figure 12-9. These functions are defined by

Piecewise Sinusoid

$$g_n(x') = \begin{cases} \frac{\sin[\beta(x' - x'_{n-1})]}{\sin[\beta(x'_n - x'_{n-1})]} & x'_{n-1} \leq x' \leq x'_n \\ \frac{\sin[\beta(x'_{n+1} - x')]}{\sin[\beta(x'_{n+1} - x'_n)]} & x'_n \leq x' \leq x'_{n+1} \\ 0 & \text{elsewhere} \end{cases} \quad (12-31)$$

Truncated Cosine

$$g_n(x') = \begin{cases} \cos\left[\beta\left(x' - \frac{x'_n + x'_{n-1}}{2}\right)\right] & x'_{n-1} \leq x' \leq x'_n \\ 0 & \text{elsewhere} \end{cases} \quad (12-32)$$

B. Entire-Domain Functions Entire domain basis functions, as their name implies, are defined and are nonzero over the entire length of the structure being considered. Thus, no segmentation is involved in their use.

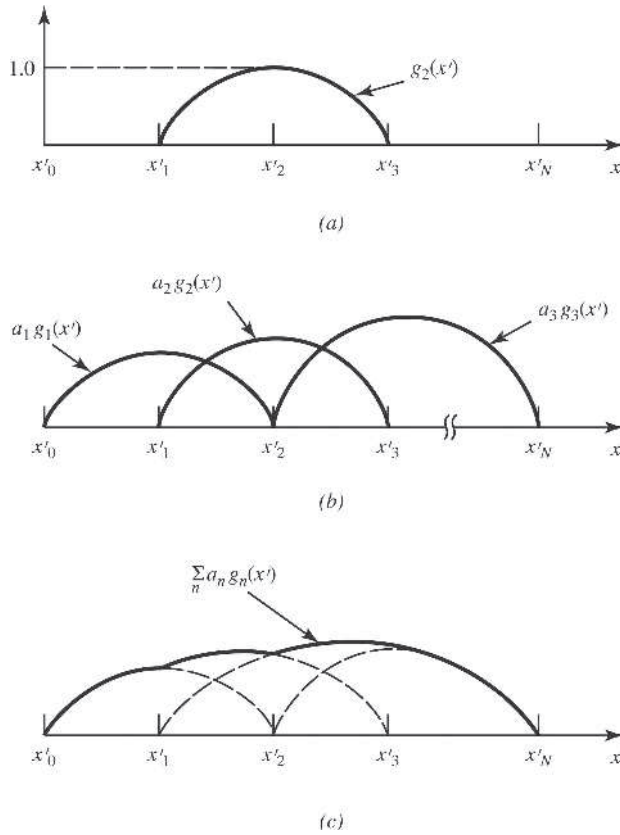


Figure 12-8 Piecewise sinusoids subdomain functions. (a) Single. (b) Multiple. (c) Function representation.

A common entire domain basis set is that of sinusoidal functions, where

$$\begin{aligned} & \text{Entire Domain} \\ g_n(x') &= \cos \left[\frac{(2n-1)\pi x'}{\ell} \right] \quad -\frac{\ell}{2} \leq x' \leq \frac{\ell}{2} \end{aligned} \quad (12-33)$$

Note that this basis set would be particularly useful for modeling the current distribution on a wire dipole, which is known to have primarily sinusoidal distribution. The main advantage of entire domain basis functions lies in problems where the unknown function is known *a priori* to follow a certain pattern. Such entire-domain functions may render an acceptable representation of the unknown while using far fewer terms in the expansion of (12-23) than would be necessary for subdomain bases. Representation of a function by entire domain cosine and/or sine functions is similar to the Fourier series expansion of arbitrary functions.

Because we are constrained to use a finite number of functions (or *modes*, as they are sometimes called), entire domain basis functions usually have difficulty in modeling arbitrary or complicated unknown functions.

Entire domain basis functions, sets like (12-33), can be generated using Tschebyscheff, Maclaurin, Legendre, and Hermite polynomials, or other convenient functions.

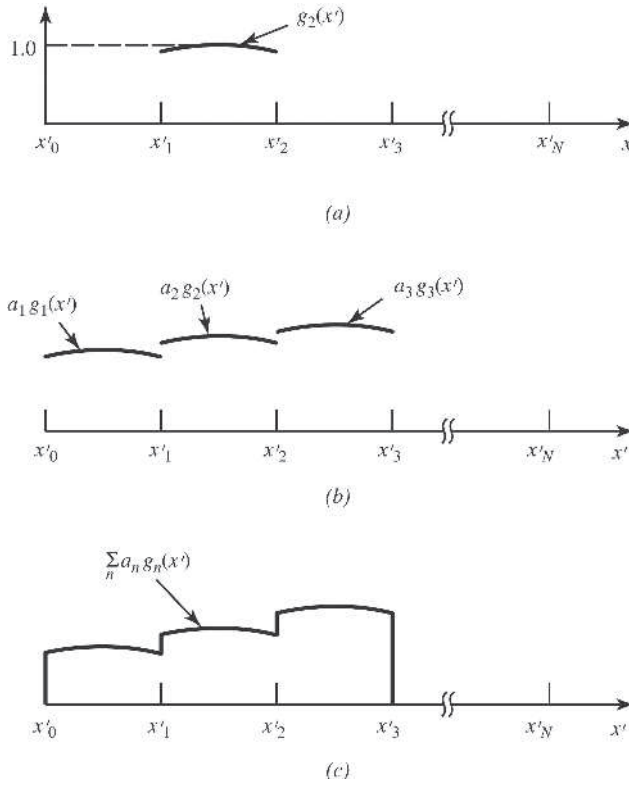


Figure 12-9 Truncated cosines subdomain functions. (a) Single. (b) Multiple. (c) Function representation.

12.2.6 Application of Point Matching

If each of the expansion functions $g_n(x')$ in (12-23) is of the subdomain type, where each exists only over one segment of the structure, then Z_{mn} of (12-27c) reduces to

$$Z_{mn} = - \int_{x_n}^{x_{n+1}} g_n(x') H_0^{(2)}(\beta |\rho_m - \rho'_n|) dx' \quad (12-34)$$

where x_n and x_{n+1} represent, respectively, the lower and upper limits of the segment over which each of the subdomain expansion functions $g_n(x')$ exists. If, in addition, the g_n are subdomain pulse expansion functions of the form

$$g_n(x') = \begin{cases} 1 & x_n \leq x' \leq x_{n+1} \\ 0 & \text{elsewhere} \end{cases} \quad (12-35)$$

then (12-34) reduces to

$$Z_{mn} = - \int_{x_n}^{x_{n+1}} H_0^{(2)}(\beta |\rho_m - \rho'_n|) dx' \quad (12-36)$$

The preceding integral cannot be evaluated exactly in closed form. However, there exist various approximations for its evaluation.

In solving (12-17a) using (12-24) or (12-26), there are few problems that must be addressed. Before we do that, let us first state in words what (12-24) and (12-26) represent. Each equation is

a solution to (12-17a), which was derived by enforcing the boundary conditions. These conditions required the total tangential electric field to vanish on the surface of the conductor. For each observation point, the total field consists of the sum of the direct (E_z^d) and scattered (E_z^s) components. Thus, to find the total scattered field at *each observation point*, we must add the contributions of the scattered field components *from all* the segments of the strip, which also includes those coming from the segment where the observations are made (referred to as *self-terms*). When the contributions from the segment over which the observation point lies are considered, the distance $R_{mn} = R_{mm} = |\rho_m - \rho'_m|$ used for evaluating the self-term Z_{mm} in (12-27c) will become zero. This introduces a singularity in the integrand of (12-36) because the Hankel function defined as

$$H_0^{(2)}(\beta\rho) = J_0(\beta\rho) - jY_0(\beta\rho) \quad (12-37)$$

is infinite since $Y_0(0) = \infty$.

For finite thickness strips, the easiest way to get around the problem of evaluating the Hankel function for the self-terms will be to choose observation points away from the surface of the strip over which the integration in (12-36) is performed. For example, the observation points can be selected at the center of each segment along a line that divides the thickness of the strip, while the integration is performed along the upper surface of the strip. These points are designated in Figure 12-5b by the distance $|\rho_m - \rho'|$.

Even if the aforementioned procedure is implemented for the evaluation of all the terms of Z_{mn} , including the self-terms, the distance $R_{mm} = |\rho_m - \rho'_m|$ for the self-terms (and some from the neighboring elements) will sometimes be sufficiently small that standard algorithms for computing Bessel functions, and thus, Hankel functions, may not be very accurate. For these cases the Hankel functions can be evaluated using asymptotic expressions for small arguments. That is, for cases where the argument of the Hankel functions in (12-36) is small, which may include the self-terms and some of the neighboring elements, the Hankel function can be computed using [24]

$$H_0^{(2)}(\beta\rho) = J_0(\beta\rho) - jY_0(\beta\rho) \stackrel{\beta\rho \rightarrow 0}{\simeq} 1 - j\frac{2}{\pi} \ln\left(\frac{1.781\beta\rho}{2}\right) \quad (12-38)$$

The integral of (12-36) can be evaluated approximately in closed form, even if the observation and source points are chosen to be along the same line. This can be done not only for diagonal (self, i.e., $m = n$) terms but also for the nondiagonal ($m \neq n$) terms. For the diagonal terms ($m = n$), the Hankel function of (12-36) has an integrable singularity, and the integral can be evaluated analytically in closed form using the small argument approximation of (12-38) for the Hankel function. When (12-38) is used, it can be shown that (12-36) reduces to [2]

Diagonal Terms Approximation

$$Z_{mn} \simeq -\Delta x_n \left[1 - j\frac{2}{\pi} \ln\left(\frac{1.781\beta\Delta x_n}{4e}\right) \right] \quad (12-39)$$

where

$$\Delta x_n = x_{n+1} - x_n \quad (12-39a)$$

$$e \approx 2.718 \quad (12-39b)$$

For evaluation of the nondiagonal terms of (12-36), the crudest approximation would be to consider the Hankel function over each segment to be essentially constant [2]. To minimize the error using such an approximation, it is recommended that the argument of the Hankel function in (12-36) be represented by its average value over each segment. For straight line segments that

average value will be representative of the distance from the center of the segment to the observation point. Thus, for the nondiagonal terms, (12-36) can be approximated by

Nondiagonal Terms Approximation

$$Z_{mn} \simeq -\Delta x_n H_0^{(2)}(\beta |R_{mn}|_{av}) = -\Delta x_n H_0^{(2)}(\beta |\rho_m - \rho'_n|_{av}) \quad m \neq n \quad (12-40)$$

The average-value approximation for the distance R_{mn} in the Hankel function evaluation of (12-36) can also be used for curved surface scattering by approximating each curved segment by a straight line segment. Crude as it may seem, the average-value approximation for the distance yields good results.

12.2.7 Weighting (Testing) Functions

Application of (12-24) for one observation point leads to one equation with N unknowns. It alone is not sufficient to determine the N unknown a_n ($n=1, 2, \dots, N$) constants. To resolve the N constants, it is necessary to have N linearly independent equations. This can be accomplished by evaluating (12-24) (i.e., applying boundary conditions) at N different points, as represented by (12-26). To improve the point-matching solution, an inner product $\langle w, g \rangle$ can be defined, which is a scalar operation satisfying the laws of

$$\langle w, g \rangle = \langle g, w \rangle \quad (12-41a)$$

$$\langle bf + cg, w \rangle = b\langle f, w \rangle + c\langle g, w \rangle \quad (12-41b)$$

$$\langle g^*, g \rangle > 0 \quad \text{if } g \neq 0 \quad (12-41c)$$

$$\langle g^*, g \rangle = 0 \quad \text{if } g = 0 \quad (12-41d)$$

where b and c are scalars and the asterisk (*) indicates complex conjugation. Note that the functions w and g can be vectors. A typical, but not unique, inner product is

$$\langle \mathbf{w}, \mathbf{g} \rangle = \iint_S \mathbf{w}^* \cdot \mathbf{g} \, ds \quad (12-42)$$

where the w 's are the *weighting (testing)* functions and S is the surface of the structure being analyzed. This technique is known better as the *moment method* or *method of moments* (MM, MoM) [1, 2].

12.2.8 Moment Method

The collocation (point-matching) method is a numerical technique whose solutions satisfy the electromagnetic boundary conditions (e.g., vanishing tangential electric fields on the surface of an electric conductor) only at discrete points. Between these points the boundary conditions may not be satisfied, and we define the deviation as a *residual* [e.g., residual $= \Delta E|_{\text{tan}} = E(\text{scattered})|_{\text{tan}} + E(\text{incident})|_{\text{tan}} \neq 0$ on the surface of an electric conductor]. For a half-wavelength dipole, a typical residual is shown in Figure 12-10a for pulse-basis functions and point matching and Figure 12-10b exhibits the residual for piecewise sinusoidal Galerkin's method [25]. As expected, the pulse-basis point matching exhibits the most ill-behaved residual and the piecewise sinusoidal Galerkin's method indicates an improved residual. To minimize the residual in such a way that its overall average over the entire structure approaches zero, the method of *weighted residuals* is utilized in conjunction with the inner product of (12-42). This technique, referred to as the

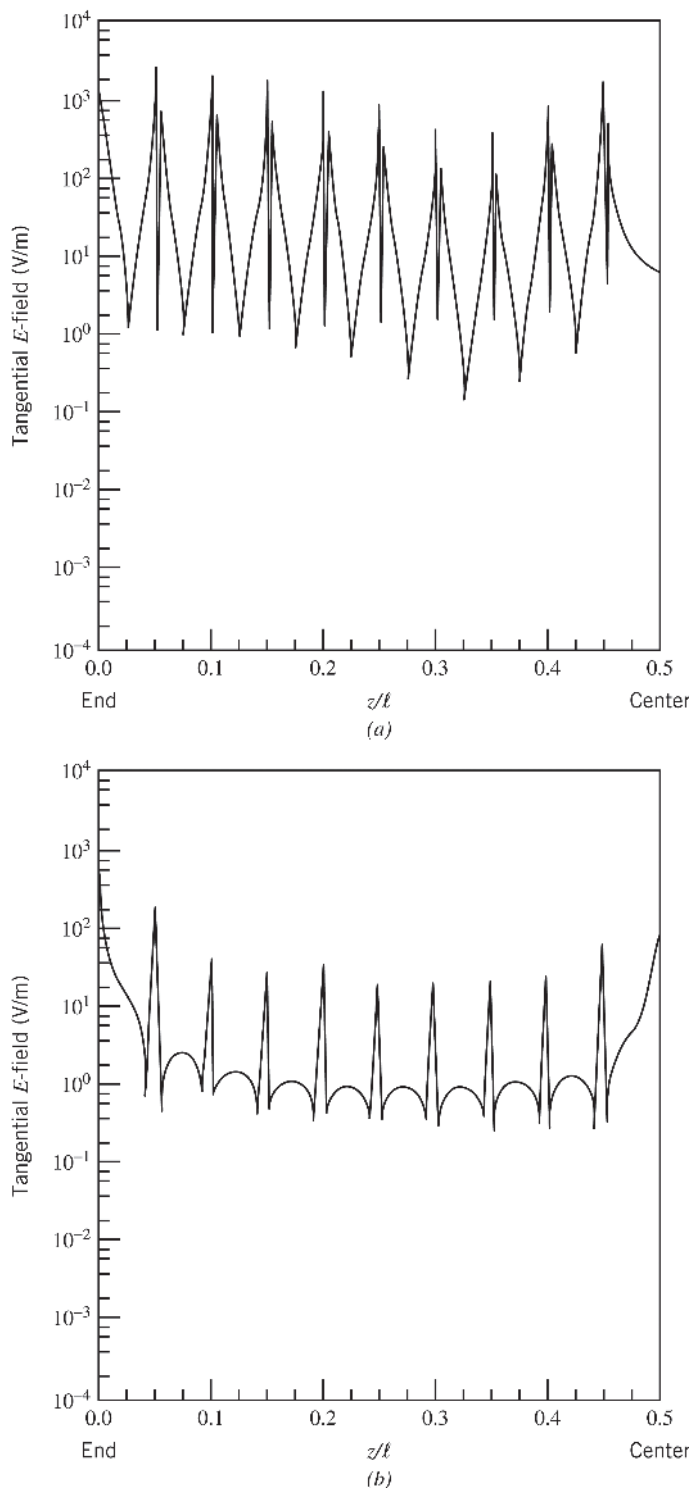


Figure 12-10 Tangential electric field on the conducting surface of the $\lambda/2$ dipole. (a) Pulse basis-point matching. (b) Piecewise sinusoidal Galerkin's method. (Source: E. K. Miller and F. J. Deadrack, "Some computational aspects of thin-wire modeling" in *Numerical and Asymptotic Techniques in Electromagnetics*, 1975, Springer-Verlag.)

Moment Method (MM), does not lead to a vanishing residual at every point on the surface of a conductor, but it forces the boundary conditions to be satisfied in an average sense over the entire surface.

To accomplish this, we define a set of N *weighting* (or testing) functions $\{w_m\}$ ($m = 1, 2, \dots, N$) in the domain of the operator F . Forming the inner product between each of these functions and the excitation function h , (12-25) reduces to

$$\langle w_m, h \rangle = \sum_{n=1}^N a_n \langle w_m, F(g_n) \rangle \quad m = 1, 2, \dots, N \quad (12-43)$$

This set of N equations may be written in matrix form as

$$[h_m] = [F_{mn}] [a_n] \quad (12-44)$$

where

$$[F_{mn}] = \begin{bmatrix} \langle w_1, F(g_1) \rangle & \langle w_1, F(g_2) \rangle & \dots \\ \langle w_2, F(g_1) \rangle & \langle w_2, F(g_2) \rangle & \\ \vdots & & \ddots \end{bmatrix} \quad (12-44a)$$

$$[a_n] = \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_N \end{bmatrix} \quad h_m = \begin{bmatrix} \langle w_1, h \rangle \\ \langle w_2, h \rangle \\ \vdots \\ \langle w_N, h \rangle \end{bmatrix} \quad (12-44b)$$

The matrix of (12-44) may be solved for the a_n by inversion, and it can be written as

$$[a_n] = [F_{mn}]^{-1} [h_m] \quad (12-45)$$

The choice of weighting functions is important in that the elements of $\{w_n\}$ must be linearly independent, so that the N equations in (12-43) will be linearly independent [1–3, 22, 23]. Further, it will generally be advantageous to choose weighting functions that minimize the computations required to evaluate the inner products.

The condition of linear independence between elements and the advantage of computational simplicity are also important characteristics of basis functions. Because of this, similar types of functions are often used for both weighting and expansion. A particular choice of functions may be to let the weighting and basis function be the same, that is, $w_n = g_n$. This technique is known as *Galerkin's method* [26].

It should be noted that there are N^2 terms to be evaluated in (12-44a). Each term usually requires two or more integrations; at least one to evaluate each $F(g_n)$, and one to perform the inner product of (12-42). When these integrations are to be done numerically, as is often the case, vast amounts of computation time may be necessary.

There is, however, a unique set of weighting functions that reduce the number of required integrations. This is the set of Dirac delta weighting functions

$$\{w_m\} = \{\delta(p - p_m)\} = \{\delta(p - p_1), \delta(p - p_2), \dots, \delta(p - p_N)\} \quad (12-46)$$

where p specifies a position with respect to some reference (origin), and p_m represents a point at which the boundary condition is enforced. Using (12-42) and (12-46) reduces (12-43) to

$$\begin{aligned}\langle \delta(p - p_m), h \rangle &= \sum_n a_n \langle \delta(p - p_m), F(g_n) \rangle, \quad m = 1, 2, \dots, N \\ \iint_S \delta(p - p_m) h \, ds &= \sum_n a_n \iint_S \delta(p - p_m) F(g_n) \, ds, \quad m = 1, 2, \dots, N \\ \boxed{h|_{p=p_m} = \sum_n a_n F(g_n)|_{p=p_m}, \quad m = 1, 2, \dots, N} \end{aligned} \quad (12-47)$$

Hence, the only remaining integrations are those specified by $F(g_n)$. This simplification may make it possible to obtain some solutions that would be unattainable if other weighting functions were used. Physically, the use of Dirac delta weighting functions is seen as a relaxation of boundary conditions so that they are enforced only at discrete points on the surface of the structure, hence the name *point matching*.

An important consideration when using point matching is the positioning of the N points (p_m). While equally spaced points often yield good results, much depends on the basis functions used. When using subsectional basis functions in conjunction with point matching, one match point should be placed on each segment to maintain linear independence. Placing the points at the center of the segments usually produces the best results. It is important that a match point does not coincide with the “peak” of a triangle or any other point, where the basis function is not differentiable. Ignoring this would cause errors in some situations.

Because it provides acceptable accuracy along with obvious computational advantages, point matching is the most popular testing technique for moment-method solutions to electromagnetics problems. The analysis presented here, along with most problems considered in the literature, proceed via point matching.

For the strip problem, a convenient inner product of the form (12-42) is

$$\langle w_m, g_n \rangle = \int_{-w/2}^{w/2} w_m^*(x) g_n(x) \, dx \quad (12-48)$$

Taking the inner product (12-48) with $w_m^*(x)$ on both sides of (12-26), we can write it as

$$V'_m = \sum_{n=1}^N I_n Z'_{mn}, \quad m = 1, 2, \dots, N \quad (12-49)$$

where

$$V'_m = \int_{-w/2}^{w/2} w_m^*(x) H_0^{(2)}(\beta \rho_m) \, dx \quad (12-49a)$$

$$Z'_{mn} = - \int_{-w/2}^{w/2} w_m^*(x) \left[\int_{-w/2}^{w/2} g_n(x') H_0^{(2)}(\beta |\rho_m - \rho'_n|) \, dx' \right] \, dx \quad (12-49b)$$

or in matrix form as

$$[V'_m] = [Z'_{mn}] [I_n] \quad (12-50)$$

If the w_m weighting functions are Dirac delta functions [i.e., $w_m(y) = \delta(y - y_m)$], then (12-49) reduces to (12-27) or

$$V'_m = V_m \quad (12-51a)$$

and

$$Z'_{mn} = Z_{mn} \quad (12-51b)$$

The *method of weighted residuals* (moment method) was introduced to minimize the average deviation from the actual values of the boundary conditions over the entire structure. However, it is evident that it has complicated the formulation by requiring an integration in the evaluation of the elements of the V' matrix [as given by (12-49a)] and an additional integration in the evaluation of the elements of the Z'_{mn} matrix [as given by (12-49b)]. Therein lies the penalty that is paid to improve the solution.

If both the expansion g_n and the weighting w_m functions are of the subdomain type, each of which exists only over one of the strip segments, then (12-49b) can be written as

$$Z'_{mn} = - \int_{x_m}^{x_{m+1}} w_m^*(x) \left[\int_{x'_n}^{x'_{n+1}} g_n(x') H_0^{(2)}(\beta |\rho_m - \rho'|) dx' \right] dx \quad (12-52)$$

where (x_m, x_{m+1}) and (x'_n, x'_{n+1}) represent, respectively, the lower and upper limits of the strip segments over which the weighting w_m and expansion g_n functions exist. To evaluate the m th element of Z'_{mn} from (12-49b) or (12-52), we first choose the weighting function w_m , and the region of the segment over which it exists, and weigh the contributions from the g_n expansion function over the region in which it exists. To find the next element $Z'_{m(n+1)}$, we maintain the same weighting function w_m , and the region over which it exists, and weigh the contributions from the g_{n+1} expansion function. We repeat this until the individual contributions from all the N expansion functions (g_n) are weighted by the w_m weighting function. Then we choose the w_{m+1} weighting function, and the region over which it exists, and we weigh individually the contributions from each of the N expansion functions (g_n). We repeat this until all the N weighting functions (w_m), and the regions of the strip over which they exist, are individually weighted by the N expansion functions (g_n). This procedure allows us to form N linear equations, each with N unknowns, that can be solved using matrix inversion methods.

Example 12-3

For the electric line source of Figure 12-5 with $w = 2\lambda$, $t = 0.001\lambda$, and $h = 0.5\lambda$ perform the following:

1. Compute the equivalent current density induced on the open surface of the strip. This equivalent current density is representative of the vector sum of the current densities that flow on the opposite sides of the strip. Use subdomain pulse expansion functions and point matching. Subdivide the strip into 150 segments.
2. Compare the current density of part 1 with the physical optics current density.
3. Compute the normalized far-field amplitude pattern of (12-22a) using the current densities of parts 1 and 2. Compare these patterns with those obtained using a combination of geometrical optics (GO) and geometrical theory of diffraction (GTD) techniques of Chapter 13 and physical optics (PO) and physical theory of diffraction (PTD) techniques of [13].

Solution:

1. Utilizing (12-27) through (12-27b) and (12-36), the current density of (12-23) is computed using (12-28a). It is plotted in Figure 12-11. It is observed that the current density exhibits singularities toward the edges of the strip.

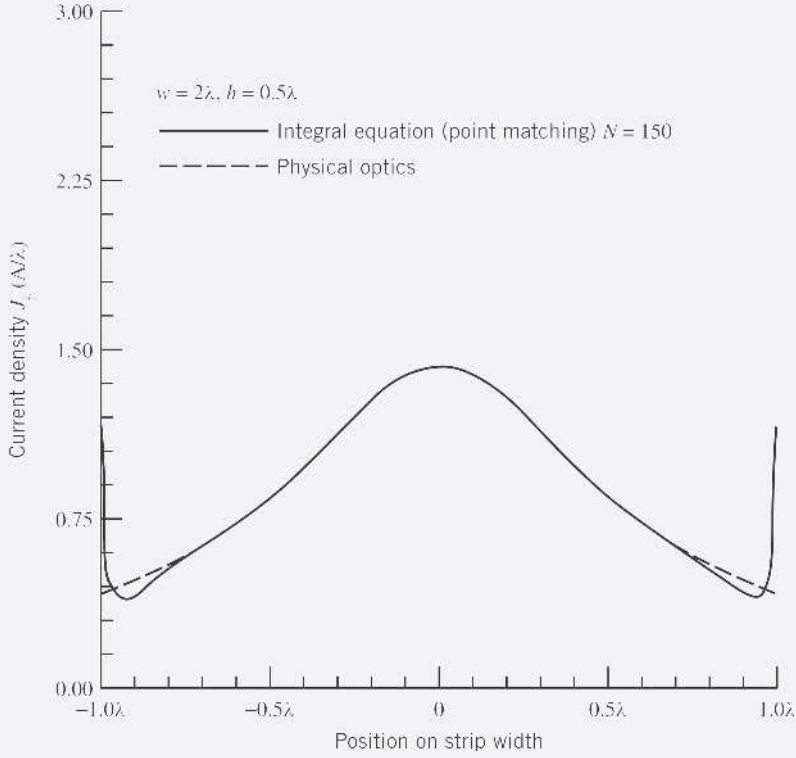


Figure 12-11 Current density on a finite-width strip that is due to the electric line source above the strip.

2. The physical optics current density is found using

$$\mathbf{J}_s^{\text{PO}} \simeq 2\hat{\mathbf{n}} \times \mathbf{H}^i$$

which reduces using (11-10b) to

$$\begin{aligned} \mathbf{J}_s^{\text{PO}} &\simeq 2\hat{\mathbf{a}}_y \times \hat{\mathbf{a}}_\phi H_\phi^i|_{\text{strip}} = 2\hat{\mathbf{a}}_y \times (-\hat{\mathbf{a}}_x \sin\phi + \hat{\mathbf{a}}_y \cos\phi) H_\phi^i|_{\text{strip}} \\ &= \hat{\mathbf{a}}_z 2\sin\phi H_\phi^i|_{\text{strip}} \simeq -j\hat{\mathbf{a}}_z I_z \frac{\beta}{z} \sin\phi_m H_1^{(2)}(\beta\rho_m) \\ \mathbf{J}_s^{\text{PO}} &= -j\hat{\mathbf{a}}_z I_z \frac{\beta}{2} \left(\frac{y_m}{\rho_m} \right) H_1^{(2)}(\beta\rho_m) \end{aligned}$$

The normalized value of this has also been plotted in Figure 12-11 so that it can be compared with the more accurate one obtained in part 1 using the integral equation.

3. The far-field amplitude patterns, based on the current densities of parts 1 and 2, are plotted in Figure 12-12. In addition to the normalized radiation patterns obtained using the current densities of parts 1 and 2, the pattern obtained using geometrical optics (GO) plus first-order diffractions by the geometrical theory of diffraction (GTD), to be discussed in Chapter 13, is also displayed in Figure 12-12. There is an excellent agreement between the IE and the GO plus GTD patterns. The pattern obtained using physical optics, supplemented by first-order diffractions of the physical theory of diffraction (PTD) [13], is also displayed in Figure 12-12 for comparison purposes. It also

compares extremely well with the others. As expected, the only one that does not compare well with the others is that of PO. Its largest differences are in the back lobes.

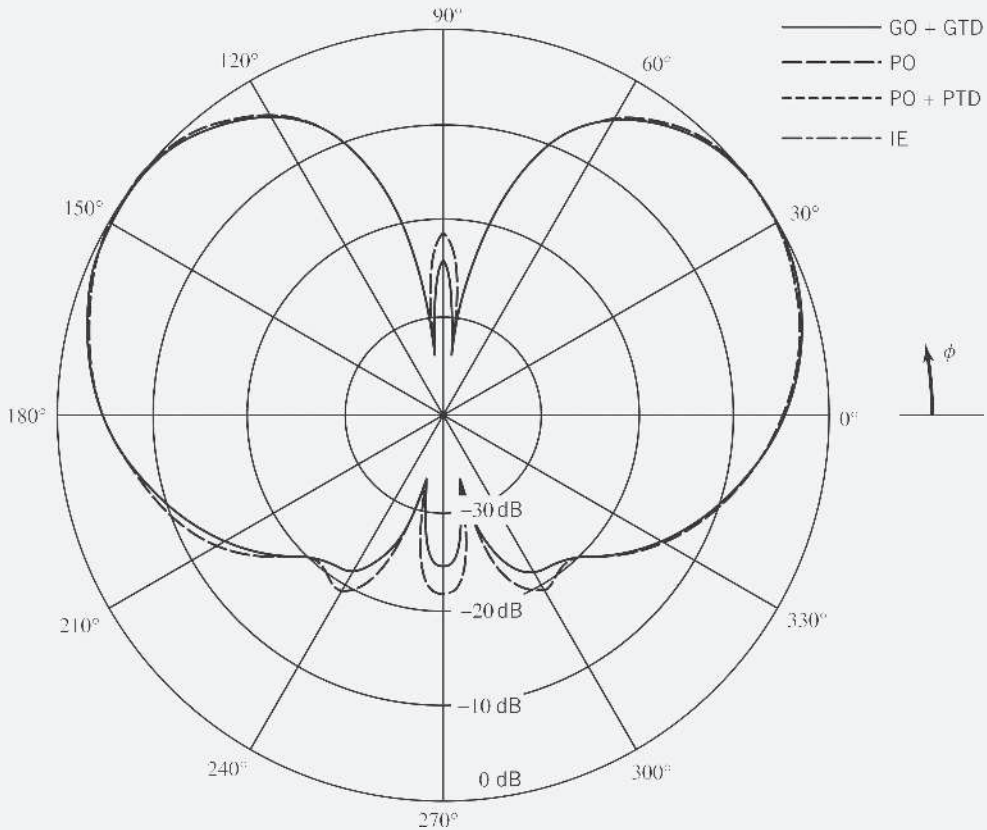


Figure 12-12 Normalized amplitude pattern of the line source above the finite-width strip ($w = 2\lambda$, $h = 0.5\lambda$).

12.3 ELECTRIC AND MAGNETIC FIELD INTEGRAL EQUATIONS

The key to the solution of any antenna or scattering problem is a knowledge of the physical or equivalent current density distributions on the volume or surface of the antenna or scatterer. Once these are known, the radiated or scattered fields can be found using the standard radiation integrals of Chapter 6. A main objective then of any solution method is to be able to predict accurately the current densities over the antenna or scatterer. This can be accomplished by the integral-equation (IE) method. One form of IE, for a two-dimensional structure, was discussed in Section 12.2.2 and represented by the integral equation 12-17a.

In general, there are many forms of integral equations. Two of the most popular forms for time-harmonic electromagnetics are the *electric field integral equation* (EFIE) and the *magnetic field integral equation* (MFIE). The EFIE enforces the boundary condition on the tangential electric field while the MFIE enforces the boundary condition on the tangential components of the magnetic field. Both of these will be discussed here as they apply to perfectly conducting structures.

12.3.1 Electric Field Integral Equation

The electric field integral equation (EFIE) is based on the boundary condition that the total tangential electric field on a perfectly electric conducting (PEC) surface of an antenna or scatterer is zero. This can be expressed as

$$\mathbf{E}_t^i(r=r_s) = \mathbf{E}_t^i(r=r_s) + \mathbf{E}_t^s(r=r_s) = 0 \quad \text{on } S \quad (12-53)$$

or

$$\mathbf{E}_t^s(r=r_s) = -\mathbf{E}_t^i(r=r_s) \quad \text{on } S \quad (12-53a)$$

where S is the conducting surface of the antenna or scatterer and $r=r_s$ is the distance from the origin to any point on the surface of the antenna or scatterer. The subscript t indicates tangential components.

The incident field that impinges on the surface S of the antenna or scatterer induces on it an electric current density $\mathbf{J}_s$ which in turn radiates the scattered field. If $\mathbf{J}_s$ is known, the scattered field everywhere, that is due to $\mathbf{J}_s$, can be found using (6-32b), or

$$\mathbf{E}^s(r) = -j\omega\mathbf{A} - j\frac{1}{\omega\mu\epsilon}\nabla(\nabla\cdot\mathbf{A}) = -j\frac{1}{\omega\mu\epsilon}\left[\omega^2\mu\epsilon\mathbf{A} + \nabla(\nabla\cdot\mathbf{A})\right] \quad (12-54)$$

where, according to (6-96a),

$$\mathbf{A}(r) = \frac{\mu}{4\pi} \iint_S \mathbf{J}_s(r') \frac{e^{-j\beta R}}{R} ds' = \mu \iint_S \mathbf{J}_s(r') \frac{e^{-j\beta R}}{4\pi R} ds' \quad (12-54a)$$

Equations 12-54 and 12-54a can also be expressed, by referring to Figure 6-2b, as

$$\mathbf{E}^s(r) = -j\frac{\eta}{\beta} \left[\beta^2 \iint_S \mathbf{J}_s(r') G(\mathbf{r}, \mathbf{r}') ds' + \nabla \iint_S \nabla' \cdot \mathbf{J}_s(r') G(\mathbf{r}, \mathbf{r}') ds' \right] \quad (12-55)$$

where

$$G(\mathbf{r}, \mathbf{r}') = \frac{e^{-j\beta R}}{4\pi R} = \frac{e^{-j\beta|\mathbf{r}-\mathbf{r}'|}}{4\pi|\mathbf{r}-\mathbf{r}'|} \quad (12-55a)$$

$$R = |\mathbf{r} - \mathbf{r}'| \quad (12-55b)$$

In (12-55) ∇ and ∇' are, respectively, the gradients with respect to the observation (unprimed) and source (primed) coordinates and $G(\mathbf{r}, \mathbf{r}')$ is referred to as the *Green's function* for a three-dimensional radiator or scatterer.

If the observations are restricted on the surface of the antenna or scatterer ($r=r_s$), then (12-55) and (12-55b) can be expressed, using (12-53a), as

$$\boxed{j\frac{\eta}{\beta} \left[\beta^2 \iint_S \mathbf{J}_s(r') G(\mathbf{r}_s, \mathbf{r}') ds' + \nabla \iint_S \nabla' \cdot \mathbf{J}_s(r') G(\mathbf{r}_s, \mathbf{r}') ds' \right]_t = \mathbf{E}_t^i(r=r_s)} \quad (12-56)$$

Because the right side of (12-56) is expressed in terms of the known incident electric field, it is referred to as the *electric field integral equation* (EFIE). It can be used to find the current density $\mathbf{J}_s(r')$ at any point $r=r'$ on the antenna or scatterer. It should be noted that (12-56) is actually an integro-differential equation, but usually it is referred to as an integral equation.

Equation 12-56 can be used for closed or open surfaces. Once $\mathbf{J}_s$ is determined, the scattered field is found using (6-32b) and (6-96a) or (12-54) and (12-54a), which assume that $\mathbf{J}_s$ radiates in one medium. Because of this, $\mathbf{J}_s$ in (12-56) represents the physical equivalent electric current

density of (7-53a) in Section 7.10. For open surfaces, $\mathbf{J}_s$ is also the physical equivalent current density that represents the vector sum of the equivalent current densities on the opposite sides of the surface. Whenever this equivalent current density represents open surfaces, a boundary condition supplemental to (12-56) must be enforced to yield a unique solution for the normal component of the current density to vanish on S .

Equation 12-56 is a general surface EFIE for three-dimensional problems, and its form can be simplified for two-dimensional geometries. To demonstrate this, let us derive the two-dimensional EFIEs for both TM^z and TE^z polarizations.

A. Two-Dimensional EFIE: TM^z Polarization The best way to demonstrate the derivation of the two-dimensional EFIE for TM^z polarization is to consider a specific example. Its form can then be generalized to more complex geometries. The example to be examined here is that of a TM^z uniform plane wave incident on a finite-width strip, as shown in Figure 12-13a.

By referring to Figure 12-13a, the incident electric field can be expressed as

$$\mathbf{E}^i = \hat{\mathbf{a}}_z E_0 e^{-j\beta^i \cdot \mathbf{r}} = \hat{\mathbf{a}}_z E_0 e^{j\beta(x \cos \phi_i + y \sin \phi_i)} \quad (12-57)$$

which at the surface of the strip ($y = 0, 0 \leq x \leq w$) reduces to

$$\mathbf{E}^i(y = 0, 0 \leq x \leq w) = \hat{\mathbf{a}}_z E_0 e^{j\beta x \cos \phi_i} \quad (12-57a)$$

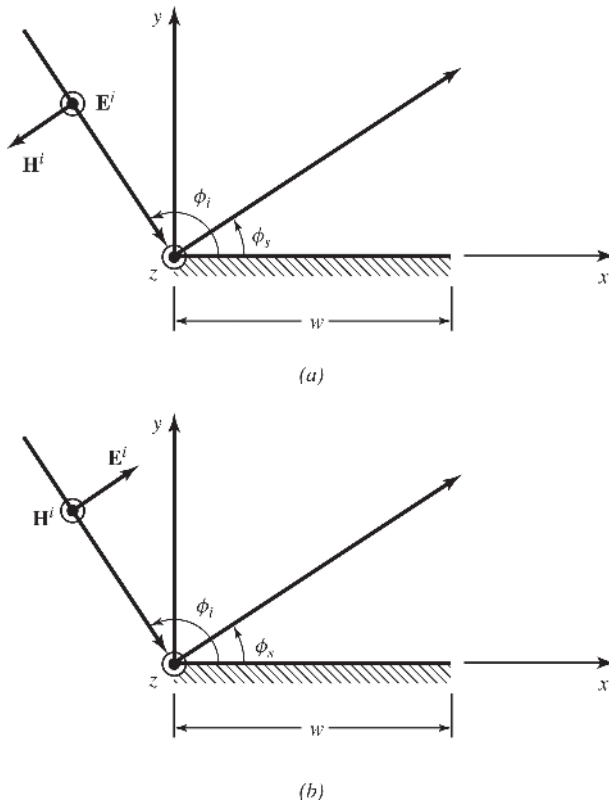


Figure 12-13 Uniform plane wave incident on a conducting strip of finite width. (a) TM^z polarization. (b) TE^z polarization.

Since the incident electric field has only a z component, the scattered and total fields each also has only a z component that is independent of z variations (two dimensional). Therefore the scattered field can be found by expanding (12-54), assuming $\mathbf{A}$ has only a z component that is independent of z variations. Doing this reduces (12-54) to

$$\mathbf{E}^s = -\hat{\mathbf{a}}_z j\omega A_z \quad (12-58)$$

The vector potential component A_z is obtained using (12-54a), which in conjunction with (11-27d), (11-28a) and (12-55b) reduces to

$$\begin{aligned} A_z &= \frac{\mu}{4\pi} \iint_S J_z(x') \frac{e^{-j\beta R}}{R} ds' = \frac{\mu}{4\pi} \int_0^w J_z(x') \left[\int_{-\infty}^{+\infty} \frac{e^{-j\beta \sqrt{|\boldsymbol{\rho}-\boldsymbol{\rho}'|^2 + (z-z')^2}}}{\sqrt{|\boldsymbol{\rho}-\boldsymbol{\rho}'|^2 + (z-z')^2}} dz' \right] dx' \\ &= -j \frac{\mu}{4} \int_0^w J_z(x') H_0^{(2)}(\beta |\boldsymbol{\rho}-\boldsymbol{\rho}'|) dx' \end{aligned} \quad (12-59)$$

where J_z is a linear current density (measured in amperes per meter). Thus, we can write the scattered electric field at any observation point, using the geometry of Figure 12-14, as

$$\begin{aligned} \mathbf{E}^s &= -\hat{\mathbf{a}}_z j\omega A_z = -\hat{\mathbf{a}}_z \frac{\omega\mu}{4} \int_0^w J_z(x') H_0^{(2)}(\beta |\boldsymbol{\rho}-\boldsymbol{\rho}'|) dx' \\ &= -\hat{\mathbf{a}}_z \frac{\beta\eta}{4} \int_0^w J_z(x') H_0^{(2)}(\beta |\boldsymbol{\rho}-\mathbf{x}'|) dx' \end{aligned} \quad (12-60)$$

For far-field observations we can reduce (12-60), using the Hankel function approximation (12-21b) for $\phi' = 0$, to

$$\mathbf{E}^s \simeq -\hat{\mathbf{a}}_z \eta \sqrt{\frac{j\beta}{8\pi}} \frac{e^{-j\beta\rho}}{\sqrt{\rho}} \int_0^w J_z(x') e^{j\beta x' \cos\phi} dx' \quad (12-60a)$$

To evaluate the integral in (12-60a) in order to find the scattered field, we must know the induced current density $J_z(x')$ over the extent of the strip ($0 \leq x' \leq w$). This can be accomplished by observing the field on the surface of the strip ($\rho = x_m$). Under those conditions, the total field over the strip must vanish. Thus,

$$E_z^t(0 \leq x_m \leq w, y=0) = E_z^i(0 \leq x_m \leq w, y=0) + E_z^s(0 \leq x_m \leq w, y=0) = 0 \quad (12-61)$$

or

$$E_z^s(0 \leq x_m \leq w, y=0) = -E_z^i(0 \leq x_m \leq w, y=0)$$

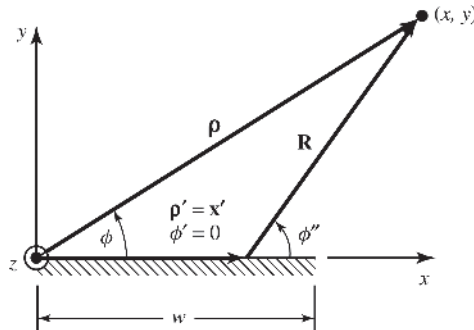


Figure 12-14 Geometry of the finite-width strip for scattering.

Since over the strip $\rho = x_m$, we can write the scattered field over the strip as

$$\begin{aligned} E_z^s(0 \leq x_m \leq w, y = 0) &= -E_z^i(0 \leq x_m \leq w, y = 0) \\ &= -\frac{\beta\eta}{4} \int_0^w J_z(x') H_0^{(2)}(\beta |\mathbf{x}_m - \mathbf{x}'|) dx' \end{aligned} \quad (12-62)$$

or

$$\begin{aligned} \frac{\beta\eta}{4} \int_0^w J_z(x') H_0^{(2)}(\beta |\mathbf{x}_m - \mathbf{x}'|) dx' \\ = E_z^i(0 \leq x_m \leq w, y = 0) = E_0 e^{j\beta x_m \cos \phi_i} \end{aligned} \quad (12-62a)$$

For a normalized field of unity amplitude ($E_0 = 1$), (12-62a) reduces to

$$\boxed{\frac{\beta\eta}{4} \int_0^w J_z(x') H_0^{(2)}(\beta |\mathbf{x}_m - \mathbf{x}'|) dx' = e^{j\beta x_m \cos \phi_i}} \quad (12-63)$$

This is the desired two-dimensional electric field integral equation (EFIE) for the TM^z polarization of the conducting strip, and it is equivalent to (12-56) for the general three-dimensional case. This EFIE can be solved for $J_z(x')$ using techniques similar to those used to solve the EFIE of (12-17a). It must be used to solve for the induced current density $J_z(x')$ over the surface of the strip. Since the surface of the strip is open, the aforementioned $J_z(x')$ represents the equivalent vector current density that flows on the opposite sides of the surface. For a more general geometry, the EFIE of (12-63) can be written as

$$\boxed{\frac{\beta\eta}{4} \int_C J_z(\rho') H_0^{(2)}(\beta |\boldsymbol{\rho}_m - \boldsymbol{\rho}'|) d\mathbf{c}' = E_z^i(\boldsymbol{\rho}_m)} \quad (12-64)$$

where $\boldsymbol{\rho}_m$ = any observation point on the scatterer

$\boldsymbol{\rho}'$ = any source point on the scatterer

C = perimeter of the scatterer

The solution of the preceding integral equations for the equivalent linear current density can be accomplished by using either the point-matching (collocation) method of Section 12.2.4 or the weighted residual of the moment method of Section 12.2.8. However, using either method for the solution of the integral equation 12-63 for the strip of Figure 12-13a, we encounter the same problems as for the evaluation of the integral equation 12-17a for the finite strip of Figure 12-5, which are outlined in Section 12.2.6. However, these problems are overcome here using the same techniques that were outlined in Section 12.2.6, namely, choosing the observation points along the bisector of the width of the strip, or using the approximations (12-39) and (12-40).

Example 12-4

For the TM^z plane wave incidence on the conducting strip of Figure 12-13a perform the following:

1. Plot the induced equivalent current density for normal incidence ($\phi_i = 90^\circ$) obtained using the EFIE of (12-63). Assume the strip has a width of $w = 2\lambda$ and 0.001λ thickness. Use subdomain pulse expansion functions and point matching. Subdivide the strip into 250 segments.
2. Compare the current density of part 1 with the physical optics current density.

3. Compute the monostatic scattering width pattern for $0 \leq \phi_i \leq 180^\circ$ using current density obtained using the EFIE of (12-63). Compare this pattern with those obtained with physical optics (PO), geometrical theory of diffraction (GTD) of Chapter 13, and physical optics (PO) plus physical theory of diffraction (PTD) techniques [13].

Solution:

1. Using the EFIE of (12-63) and applying point-matching methods with subdomain pulse expansion functions, the current density of Figure 12-15 for $\phi_i = 90^\circ$ is obtained for a strip of $w = 2\lambda$. It is observed that the current density exhibits singularities toward the edges of the strip.
2. The physical optics current density is represented by

$$\begin{aligned} \mathbf{J}_s^{\text{PO}} &\simeq 2\hat{\mathbf{n}} \times \mathbf{H}^i|_{\text{strip}} = 2\hat{\mathbf{a}}_y \times (\hat{\mathbf{a}}_x H_x^i + \hat{\mathbf{a}}_y H_y^i)|_{\text{strip}} \\ &= -\hat{\mathbf{a}}_z 2H_x^i|_{\text{strip}} = \hat{\mathbf{a}}_z 2 \frac{E_0}{\eta} \sin \phi_i e^{j\beta x \cos \phi_i} \end{aligned}$$

which is shown plotted in Figure 12-15 for $\phi_i = 90^\circ$. It is apparent that the PO current density does not compare well with that obtained using the IE, especially toward the edges of the strip. Therefore, it does not provide a good representation of the equivalent current density induced on the strip. In Figure 12-15 we also display the equivalent current density for TE^z polarization which will be discussed in the next section and in Example 12-5.

3. The monostatic scattering width patterns for $0^\circ \leq \phi_i \leq 180^\circ$ obtained using the methods of IE, PO, GTD, and PO plus PTD are all shown in Figure 12-16. As expected, the only one that differs from the others is that due to the PO; the other three are indistinguishable from each other and are represented by the solid curve. The pattern for the TE^z polarization for the IE method is also displayed in Figure 12-16. This will be discussed in the next section and in Example 12-5.

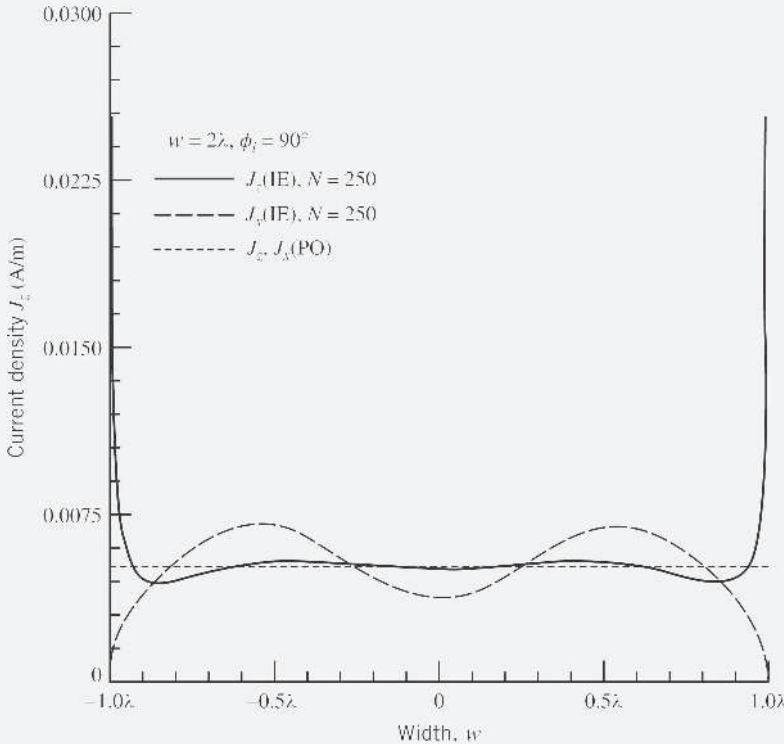


Figure 12-15 Current density induced on a finite-width strip by a plane wave at normal incidence.

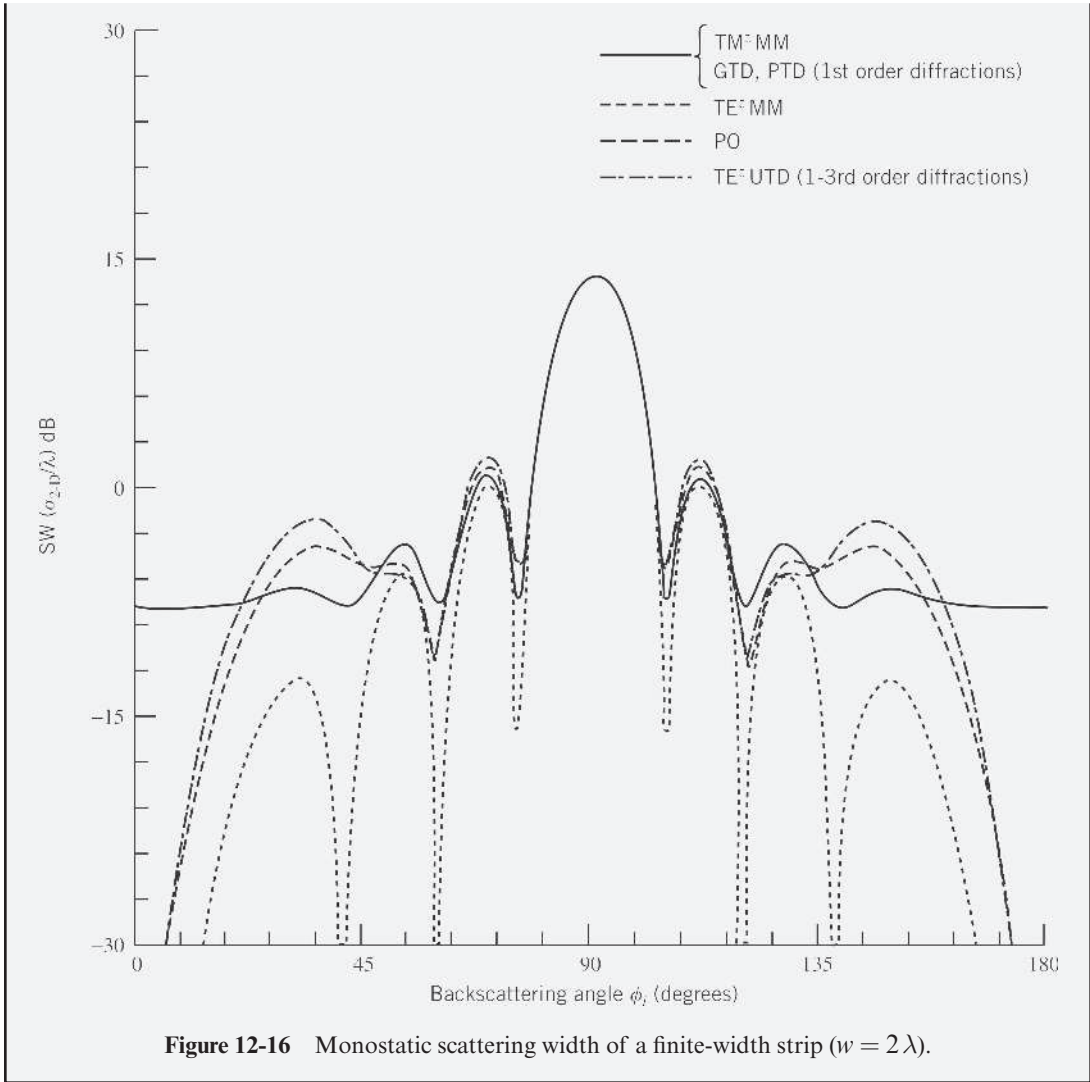


Figure 12-16 Monostatic scattering width of a finite-width strip ($w = 2\lambda$).

B. Two-Dimensional EFIE: TE^z Polarization As in the previous section, the derivation of the EFIE for TE^z polarization is best demonstrated by considering a uniform plane wave incident on the strip, as shown in Figure 12-13b. Its form can then be generalized to more complex geometries.

By referring to Figure 12-13b, the incident electric field can be expressed as

$$\mathbf{E}^i = E_0(\hat{\mathbf{a}}_x \sin \phi_i - \hat{\mathbf{a}}_y \cos \phi_i) e^{-j\beta_i \cdot \mathbf{r}} = E_0(\hat{\mathbf{a}}_x \sin \phi_i - \hat{\mathbf{a}}_y \cos \phi_i) e^{j\beta(x \cos \phi_i + y \sin \phi_i)} \quad (12-65)$$

which on the surface of the strip ($y = 0, 0 \leq x \leq w$) reduces to

$$\mathbf{E}^i = E_0(\hat{\mathbf{a}}_x \sin \phi_i - \hat{\mathbf{a}}_y \cos \phi_i) e^{j\beta x \cos \phi_i} \quad (12-65a)$$

On the surface of the strip ($0 \leq x \leq w, y = 0$) the tangential components of the total field, incident plus scattered, must vanish. This can be written as

$$\begin{aligned} \hat{\mathbf{n}} \times \mathbf{E}'|_{\text{strip}} &= \hat{\mathbf{n}} \times (\mathbf{E}^i + \mathbf{E}^s)|_{\text{strip}} \\ &= \hat{\mathbf{a}}_y \times \left[\left(\hat{\mathbf{a}}_x E_x^i + \hat{\mathbf{a}}_y E_y^i \right) + \left(\hat{\mathbf{a}}_x E_x^s + \hat{\mathbf{a}}_y E_y^s \right) \right]_{\text{strip}} = 0 \end{aligned} \quad (12-66)$$

which leads to

$$-\hat{\mathbf{a}}_z(E_x^i + E_x^s)|_{\text{strip}} = 0 \Rightarrow E_x^s(0 \leq x \leq w, y = 0) = -E_x^i(0 \leq x \leq w, y = 0) \quad (12-66a)$$

or

$$E_x^s(0 \leq x \leq w, y = 0) = -E_x^i(0 \leq x \leq w, y = 0) = -E_0 \sin \phi_i e^{j\beta x \cos \phi_i} \quad (12-66b)$$

The x and y components of the scattered electric field, which are independent of z variations, are obtained using (12-54), which when expanded reduce to

$$\begin{aligned} E_x^s &= -j\omega A_x - j \frac{1}{\omega\mu\epsilon} \frac{\partial^2 A_x}{\partial x^2} = -j \frac{1}{\omega\mu\epsilon} \left[\beta^2 A_x + \frac{\partial^2 A_x}{\partial x^2} \right] \\ &= -j \frac{1}{\omega\mu\epsilon} \left(\beta^2 + \frac{\partial^2}{\partial x^2} \right) A_x \end{aligned} \quad (12-67a)$$

$$E_y^s = -j \frac{1}{\omega\mu\epsilon} \frac{\partial^2 A_x}{\partial x \partial y} \quad (12-67b)$$

The vector potential A_x is obtained using (12-54a), which, in conjunction with (11-27d) and (11-28a), reduces to

$$A_x = -j \frac{\mu}{4} \int_0^w J_x(x') H_0^{(2)}(\beta |\mathbf{p} - \mathbf{p}'|) dx' \quad (12-68)$$

Thus, we can write that the x and y components of the scattered field can be expressed as

$$\begin{aligned} E_x^s &= -j \frac{1}{\omega\mu\epsilon} \left(-j \frac{\mu}{4} \right) \left(\beta^2 + \frac{\partial^2}{\partial x^2} \right) \int_0^w J_x(x') H_0^{(2)}(\beta |\mathbf{p} - \mathbf{p}'|) dx' \\ &= -\frac{\eta}{4\beta} \left(\beta^2 + \frac{\partial^2}{\partial x^2} \right) \int_0^w J_x(x') H_0^{(2)}(\beta |\mathbf{p} - \mathbf{p}'|) dx' \end{aligned} \quad (12-69a)$$

$$E_y^s = -\frac{\eta}{4\beta} \frac{\partial^2}{\partial x \partial y} \int_0^w J_x(x') H_0^{(2)}(\beta |\mathbf{p} - \mathbf{p}'|) dx' \quad (12-69b)$$

Interchanging integration and differentiation and letting $\rho' = x'$, we can rewrite the x and y components as

$$E_x^s = -\frac{\eta}{4\beta} \int_0^w J_x(x') \left[\left(\frac{\partial^2}{\partial x^2} + \beta^2 \right) H_0^{(2)}(\beta R) \right] dx' \quad (12-70a)$$

$$E_y^s = -\frac{\eta}{4\beta} \int_0^w J_x(x') \left[\frac{\partial^2}{\partial x \partial y} H_0^{(2)}(\beta R) \right] dx' \quad (12-70b)$$

where

$$R = |\mathbf{p} - \mathbf{x}'| \quad (12-70c)$$

It can be shown, using the geometry of Figure 12-14, that

$$\left[\frac{\partial^2}{\partial x^2} + \beta^2 \right] H_0^{(2)}(\beta R) = \frac{\beta^2}{2} \left[H_0^{(2)}(\beta R) + H_2^{(2)}(\beta R) \cos(2\phi'') \right] \quad (12-71a)$$

$$\frac{\partial^2}{\partial x \partial y} H_0^{(2)}(\beta R) = \frac{\beta^2}{2} H_2^{(2)}(\beta R) \sin(2\phi'') \quad (12-71b)$$

Thus, the x and y components of the electric field reduce to

$$E_x^s = -\frac{\beta\eta}{8} \int_0^w J_x(x') [H_0^{(2)}(\beta R) + H_2^{(2)}(\beta R) \cos(2\phi'')] dx' \quad (12-72a)$$

$$E_y^s = -\frac{\beta\eta}{8} \int_0^w J_x(x') H_2^{(2)}(\beta R) \sin(2\phi'') dx' \quad (12-72b)$$

The next objective is to solve for the induced current density, that can then be used to find the scattered field. This can be accomplished by applying the boundary conditions on the x component of the electric field. When the observations are restricted to the surface of the strip ($\rho = x_m$), the x component of the scattered field over the strip can be written as

$$\begin{aligned} E_x^s(0 \leq x_m \leq w, y=0) \\ = -E_x^i(0 \leq x_m \leq w, y=0) = -E_0 \sin \phi_i e^{j\beta x_m \cos \phi_i} \\ = -\frac{\beta\eta}{8} \int_0^w J_x(x') [H_0^{(2)}(\beta R_m) + H_2^{(2)}(\beta R_m) \cos(2\phi_m'')] dx' \end{aligned} \quad (12-73)$$

or

$$\frac{\beta\eta}{8} \int_0^w J_x(x') [H_0^{(2)}(\beta R_m) + H_2^{(2)}(\beta R_m) \cos(2\phi_m'')] dx' = E_0 \sin \phi_i e^{j\beta x_m \cos \phi_i} \quad (12-73a)$$

For a normalized field of unity amplitude ($E_0 = 1$), (12-73a) reduces to

$$\boxed{\frac{\beta\eta}{8} \int_0^w J_x(x') [H_0^{(2)}(\beta R_m) + H_2^{(2)}(\beta R_m) \cos(2\phi_m'')] dx' = \sin \phi_i e^{j\beta x_m \cos \phi_i}} \quad (12-74)$$

where

$$R_m = |\rho_m - \mathbf{x}'| \quad (12-74a)$$

This is the desired two-dimensional electric field integral equation (EFIE) for the TE^z polarization of the conducting strip, and it is equivalent to (12-56) for the general three-dimensional case. This EFIE must be used to solve for the induced current density $J_x(x')$ over the surface of the strip using techniques similar to those used to solve the EFIE of (12-17a). For a more general geometry, the EFIE can be written as

$$\boxed{\begin{aligned} & \frac{\eta}{4\beta} \left\{ \beta^2 \int_C J_c(\rho') [\hat{\mathbf{c}}_m \cdot \hat{\mathbf{c}}' H_0^{(2)}(\beta |\rho_m - \rho'|)] dc' \right. \\ & \left. + \frac{d}{dc} \left[\nabla \cdot \int_C J_c(\rho') [\hat{\mathbf{c}}' H_0^{(2)}(\beta |\rho_m - \rho'|)] dc' \right] \right\} = -E_c^i(\rho_m) \end{aligned}} \quad (12-75)$$

where ρ_m = any observation point on the scatterer

ρ' = any source point on the scatterer

C = perimeter of the scatterer

$\hat{\mathbf{c}}_m, \hat{\mathbf{c}}'$ = unit vector tangent to scatterer perimeter at observation, source points

The linear current density J_x is obtained by solving the integral equation 12-74 using either the point-matching (collocation) method of Section 12.2.4 or the weighted residual moment method of Section 12.2.8. Using either method, the solution of the preceding integral equation for J_x is more difficult than that of the TM^z polarization of the previous example. There exist various approaches (either exact or approximate) that can be used to accomplish this.

To demonstrate this, we will discuss one method that can be used to solve the integral equation 12-74. Let us assume that the current density $J_x(x')$ is expanded into a finite series similar to (12-23). Then the integral equation can be written as

$$\sin \phi_i e^{j\beta x_m \cos \phi_i} = \frac{\beta \eta}{8} \sum_{n=1}^N a_n \int_0^w g_n(x') \left[H_0^{(2)}(\beta R_m) + H_2^{(2)}(\beta R_m) \cos(2\phi_m'') \right] dx' \quad (12-76)$$

If, in addition, the basis functions are subdomain pulse functions, as defined by (12-35), then (12-76) using point matching reduces for each observation point to

$$\sin \phi_i e^{j\beta x_m \cos \phi_i} = \frac{\beta \eta}{8} \sum_{n=1}^N a_n \int_{x_n}^{x_{n+1}} \left[H_0^{(2)}(\beta R_{mn}) + H_2^{(2)}(\beta R_{mn}) \cos(2\phi_{mn}'') \right] dx' \quad (12-77)$$

If N observations are selected, then we can write (12-77) as

$$[\sin \phi_i e^{j\beta x_m \cos \phi_i}] = \sum_{n=1}^N a_n \left\{ \frac{\beta \eta}{8} \int_{x_n}^{x_{n+1}} \left[H_0^{(2)}(\beta R_{mn}) + H_2^{(2)}(\beta R_{mn}) \cos(2\phi_{mn}'') \right] dx' \right\} \quad (12-78)$$

$m = 1, 2, \dots, N$

or, in matrix form,

$$[V_m] = [Z_{mn}] [I_n] \quad (12-78a)$$

where

$$V_m = \sin \phi_i e^{j\beta x_m \cos \phi_i} \quad (12-78b)$$

$$I_n = a_n \quad (12-78c)$$

$$Z_{mn} = \frac{\beta \eta}{8} \int_{x_n}^{x_{n+1}} \left[H_0^{(2)}(\beta R_{mn}) + H_2^{(2)}(\beta R_{mn}) \cos(2\phi_{mn}'') \right] dx' \quad (12-78d)$$

One of the tasks here will be the evaluation of the integral for Z_{mn} . We will examine one technique that requires Z_{mn} to be evaluated using three different expressions depending upon the position of the segment relative to the observation point. We propose here that Z_{mn} is evaluated using

$$\left[\frac{\beta \eta \Delta x_n}{8} \left\{ 1 - j \frac{1}{\pi} \left[3 + 2 \ln \left(\frac{1.781 \beta \Delta x_n}{4e} \right) + \frac{16}{(\beta \Delta x_n)^2} \right] \right\} \right] \quad (12-79a)$$

$e \approx 2.718 \quad m = n$

$$Z_{mn} = \frac{\beta \eta \Delta x_n}{8} \left\{ 1 + j \frac{4}{\pi \beta^2} \frac{1}{|x_m - x_n|^2 - \frac{(\Delta x_n)^2}{4}} \right\} \quad (12-79b)$$

$|m - n| \leq 2, \quad m \neq n$

$$\frac{\beta \eta}{4} \int_{-\Delta x_n/2}^{\Delta x_n/2} \frac{H_1^{(2)}[\beta(|x_m - x_n| + x')] }{\beta(|x_m - x_n| + x')} dx' \quad (12-79c)$$

$|m - n| > 2$

where x_m and x_n are measured from the center of their respective segments.

The current density J_x obtained from the preceding integral equation also represents the total current density $\mathbf{J}_s$ induced on the strip. This is evident from the induced current density equation

$$\mathbf{J}_s = \hat{\mathbf{n}} \times \mathbf{H}^t = \hat{\mathbf{a}}_y \times \hat{\mathbf{a}}_z H_z^t = \hat{\mathbf{a}}_x H_z^t = \hat{\mathbf{a}}_x (H_z^i + H_z^s) \quad (12-80)$$

Example 12-5

For the TE^z plane wave incident on the conducting strip of Figure 12-13b, perform the following tasks.

1. Plot the induced equivalent current density for normal incidence ($\phi_i = 90^\circ$) obtained using the EFIE of (12-74) or (12-78) through (12-78d). Assume a width of $w = 2\lambda$ and thickness equal to 0.001λ . Use subdomain pulse expansion functions and point matching. Subdivide the strip into 250 segments.
2. Compare the current density of part 1 with the physical optics current density.
3. Compute the monostatic scattering width pattern for $0^\circ \leq \phi_i \leq 180^\circ$ using the current density obtained using the EFIE of (12-78) through (12-78d). Compare this pattern with those obtained with physical optics (PO), geometrical theory of diffraction (GTD) of Chapter 13, and physical optics (PO) plus physical theory of diffraction (PTD) techniques [13].

Solution:

1. Using the EFIE of (12-78) through (12-78d), the current density of Figure 12-15 for $\phi_i = 90^\circ$ is obtained for a strip of $w = 2\lambda$. It is observed that the current density vanishes toward the edges of the strip.
2. The physical optics current density is represented by

$$\mathbf{J}_s^{\text{PO}} \simeq 2\hat{\mathbf{n}} \times \mathbf{H}^i|_{\text{strip}} = 2\hat{\mathbf{a}}_y \times \hat{\mathbf{a}}_z H_z^i|_{\text{strip}} = \hat{\mathbf{a}}_x 2H_z^i|_{\text{strip}} = \hat{\mathbf{a}}_x 2 \frac{E_0}{\eta} e^{j\beta x \cos \phi_i}$$

which for normal incidence ($\phi_i = 90^\circ$) is identical to that for the TM^z polarization, and it is shown plotted in Figure 12-15. As for the TM^z polarization, the PO TE^z polarization current density does not compare well with that obtained using the IE method. Therefore, it does not provide a good representation of the equivalent current density induced on the strip. In Figure 12-15 the TE^z polarization current density is compared with that of the TM^z polarization using the different methods.

3. The monostatic scattering width pattern for $0^\circ \leq \phi_i \leq 180^\circ$ obtained using the IE method is shown plotted in Figure 12-16 where it is compared to those obtained by PO, PO plus PTD (first-order diffractions), and GTD (first-order diffractions) techniques. It is observed that the patterns of PO, PO plus PTD, and GTD (using first-order diffractions only) are insensitive to polarization whereas those of the integral equation with moment method solution vary with polarization. The SW patterns should vary with polarization. Therefore, those obtained using the integral-equation method are more accurate. It can be shown that if higher-order diffractions are included, the patterns of the PO plus PTD, and GTD will also vary with polarization. Higher-order diffractions are greater contributors to the overall scattering pattern for the TE^z polarization than for the TM^z . This is demonstrated by including in Figure 12-16 the monostatic SW for TE^z polarization obtained using higher-order GTD (UTD) diffractions [27]. It is apparent that this pattern agrees quite well with that of the IE method.

12.3.2 Magnetic Field Integral Equation

The magnetic field integral equation (MFIE) is expressed in terms of the known incident magnetic field. It is based on the boundary condition that expresses the total electric current density induced at any point $r = r'$ on a conducting surface S

$$\mathbf{J}_s(r') = \mathbf{J}_s(r = r') = \hat{\mathbf{n}} \times \mathbf{H}^i(r = r') = \hat{\mathbf{n}} \times [\mathbf{H}^i(r = r') + \mathbf{H}^s(r = r')] \quad (12-81)$$

Once the current density is known or determined, the scattered magnetic field can be obtained using (6-32a) and (6-96a), or

$$\mathbf{H}^s(r) = \frac{1}{\mu} \nabla \times \mathbf{A} = \nabla \times \iint_S \mathbf{J}_s(r') \frac{e^{-j\beta R}}{4\pi R} ds' = \nabla \times \iint_S \mathbf{J}_s(r') G(\mathbf{r}, \mathbf{r}') ds' \quad (12-82)$$

where $G(\mathbf{r}, \mathbf{r}')$ is the Green's function of (12-55a). Interchanging differentiation with integration and using the vector identity

$$\nabla \times (\mathbf{J}_s G) = G \nabla \times \mathbf{J}_s - \mathbf{J}_s \times \nabla G \quad (12-83)$$

where

$$\nabla \times \mathbf{J}_s(r') = 0 \quad (12-83a)$$

$$\nabla G = -\nabla' G \quad (12-83b)$$

(12-82) reduces to

$$\mathbf{H}^s(r) = \iint_S \mathbf{J}_s(r') \times [\nabla' G(\mathbf{r}, \mathbf{r}')] ds' \quad (12-84)$$

On the surface S of the conductor, the tangential magnetic field is discontinuous by the amount of the current density induced on the surface of the conductor. Therefore, the current density is determined by (12-81) but with $\mathbf{H}^s$ found using (12-84). Thus, we can write that

$$\begin{aligned} \mathbf{J}_s(r') &= \hat{\mathbf{n}} \times \mathbf{H}^i(r = r') + \lim_{r \rightarrow S} [\hat{\mathbf{n}} \times \mathbf{H}^s(r = r')] \\ &= \hat{\mathbf{n}} \times \mathbf{H}^i(r = r') + \lim_{r \rightarrow S} \left\{ \hat{\mathbf{n}} \times \iint_S \mathbf{J}_s(r') \times [\nabla' G(\mathbf{r}, \mathbf{r}')] ds' \right\} \end{aligned} \quad (12-85)$$

or

$$\boxed{\mathbf{J}_s(r') - \lim_{r \rightarrow S} \left\{ \hat{\mathbf{n}} \times \iint_S \mathbf{J}_s(r') \times [\nabla' G(\mathbf{r}, \mathbf{r}')] ds' \right\} = \hat{\mathbf{n}} \times \mathbf{H}^i(r = r')} \quad (12-85a)$$

Since in (12-85a)

$$\lim_{r \rightarrow S} \left\{ \hat{\mathbf{n}} \times \iint_S \mathbf{J}_s(r') \times [\nabla' G(\mathbf{r}, \mathbf{r}')] ds' \right\} = \frac{\mathbf{J}_s(r')}{2} + \hat{\mathbf{n}} \times \iint_S \mathbf{J}_s(r') \times [\nabla' G(\mathbf{r}, \mathbf{r}')] ds' \quad (12-85b)$$

then (12-85a) can be written, in a more useful form, as

$$\boxed{\mathbf{J}_s(r') = 2\hat{\mathbf{n}} \times \mathbf{H}^i(r = r') + 2\hat{\mathbf{n}} \times \iint_S \mathbf{J}_s(r') \times [\nabla' G(\mathbf{r}, \mathbf{r}')] ds'} \quad (12-85c)$$

where $r \rightarrow S$ indicates that S is approached by r from the outside.

Equation 12-85a is referred to as the *magnetic field integral equation* (MFIE) because its right side is in terms of the incident magnetic field, and it is valid only for *closed* surfaces. Once the current density distribution can be found using (12-85a)–(12-85c), then the scattered fields can be found using standard radiation integrals. It should be noted that the integral of (12-85a) or (12-85b) must be carefully evaluated. The MFIE is the most popular for TE^z polarizations, although it can be used for both TE^z and TM^z cases. Since (12-85a)–(12-85c) are only valid for closed surfaces, the current density obtained using (12-85a)–(12-85b) is the actual current density induced on the surface of the conducting obstacle. Usually, the MFIE is well posed while the EFIE is ill posed.

At this point, before proceeding any further, it may be appropriate to comment as to why the EFIE is valid for *open and closed* surfaces while the MFIE is valid only for *closed* surfaces [28]. This can be accomplished by referring to Figure 12-17, which represents a thin closed surface of

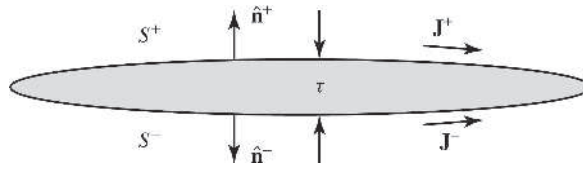


Figure 12-17 Thin closed structure with corresponding current densities, surfaces, and unit vectors [28].

thickness τ with the two sides of the surface and unit vectors represented, respectively, by S^+ , $\hat{n}^+$ and S^- , $\hat{n}^-$. In the limit, as $\tau \rightarrow 0$, $S^+ \rightarrow S^-$, and $\hat{n}^+ \rightarrow -\hat{n}^-$, the surface reduces to an *open* structure, and then the MFIE cannot be used because of what follows.

In applying the EFIE of (12-56) and MFIE of (12-85a) to open thin surfaces, one finds that each equation reduces to the same form on opposite sides of the conducting surface. Hence, neither equation can be used to solve for the two opposite-side surface current densities ($\mathbf{J}^+$, $\mathbf{J}^-$). In the case of the EFIE, however, the unknown surface conduction current density appears only in the integrand as a sum of the opposite-side surface currents (it does not appear outside the integral), and it is this sum, or total, surface current density that can be solved for. The total equivalent current density is then sufficient to find all fields radiated by the conducting structure. This is different from the MFIE case, in which the sum of the opposite-side surface current densities also appears inside the integral, but their difference appears outside the integral. Thus, the MFIE reduces to an identity relating the surface current densities on opposite sides of the surface, but does not contain sufficient information to completely determine them and, hence, cannot be used to find the surface current densities on open conducting surfaces. The missing information could be provided, for example, by combining the EFIE with the MFIE and solving the two equations as coupled equations for the opposite-side current densities—something that cannot be done using the EFIE alone. There is no problem in applying the MFIE for closed surfaces as long as care is exercised in evaluating the integral in (12-85a) or (12-85b).

An alternate MFIE can be derived using the *null field approach* [28], which utilizes the surface equivalence theorem of Chapter 7, Section 7.8, which requires that the tangential magnetic field vanishes *just inside* the conductor. Therefore, the alternate MFIE, based on the *null field approach*, is derived using $\hat{n} \times (\mathbf{H}^{inc} + \mathbf{H}^{scat}) = 0$, where the surface S is approached from the *interior* instead of the exterior; when using $\hat{n} \times (\mathbf{H}^{inc} + \mathbf{H}^{scat}) = \mathbf{J}$, the surface is approached from the *exterior*; hence, the problem for open surfaces.

Whereas (12-85a) is a general MFIE for three-dimensional problems, its form can be simplified for two-dimensional MFIEs for both TM^z and TE^z polarizations.

A. Two-Dimensional MFIE: TM^z Polarization The best way to demonstrate the derivation of the two-dimensional MFIE for TM^z polarization is to consider a TM^z uniform plane wave incident upon a two-dimensional smooth curved surface, as shown in Figure 12-18.

Since the incident field has only a z component of the electric field, and x and y components of the magnetic field, the electric current density induced on the surface of the scatterer will only have a z component. That is

$$\mathbf{J}_s(\rho) = \hat{\mathbf{a}}_z J_z(\rho)|_C \quad (12-86)$$

On the surface of the scatterer, the current density is related to the incident and scattered magnetic fields by (12-81), which, for the geometry of Figure 12-18, can be written as

$$\mathbf{J}_s(\rho)|_C = \hat{\mathbf{a}}_z J_z(\rho)|_C = \hat{\mathbf{n}} \times (\mathbf{H}^i + \mathbf{H}^s)|_C = \hat{\mathbf{n}} \times \mathbf{H}^i + \lim_{\rho \rightarrow C} (\hat{\mathbf{n}} \times \mathbf{H}^s) \quad (12-87)$$

we can write that

$$\begin{aligned}\hat{\mathbf{n}} \times [\hat{\mathbf{a}}_z \times \nabla H_0^{(2)}(\beta R)] &= \hat{\mathbf{a}}_z [\hat{\mathbf{n}} \cdot \nabla H_0^{(2)}(\beta R)] - (\hat{\mathbf{n}} \cdot \hat{\mathbf{a}}_z) \nabla H_0^{(2)}(\beta R) \\ &= \hat{\mathbf{a}}_z [\hat{\mathbf{n}} \cdot \nabla H_0^{(2)}(\beta R)]\end{aligned}\quad (12-93)$$

since $\hat{\mathbf{n}} \cdot \hat{\mathbf{a}}_z = 0$. Substituting (12-93) into (12-91) reduces it to

$$\begin{aligned}\hat{\mathbf{a}}_z \cdot (\hat{\mathbf{n}} \times \mathbf{H}^s) &= j \frac{1}{4} \int_C J_z(\rho') [\hat{\mathbf{n}} \cdot \nabla H_0^{(2)}(\beta R)] dc' \\ &= j \frac{1}{4} \int_C J_z(\rho') [-\beta \cos \psi H_1^{(2)}(\beta R)] dc' \\ \hat{\mathbf{a}}_z \cdot (\hat{\mathbf{n}} \times \mathbf{H}^s) &= -j \frac{\beta}{4} \int_C J_z(\rho') \cos \psi H_1^{(2)}(\beta R) dc'\end{aligned}\quad (12-94)$$

where the angle ψ is defined in Figure 12-18. Thus, we can write (12-88), using (12-94), as

$$J_z(\rho)|_C = H_c^i(\rho)|_C + \lim_{\rho \rightarrow C} \left[-j \frac{\beta}{4} \int_C J_z(\rho') \cos \psi H_1^{(2)}(\beta R) dc' \right] \quad (12-95)$$

or

$$\boxed{J_z(\rho)|_C + j \frac{\beta}{4} \lim_{\rho \rightarrow C} \left[\int_C J_z(\rho') \cos \psi H_1^{(2)}(\beta R) dc' \right] = H_c^i(\rho)|_C} \quad (12-95a)$$

B. Two-Dimensional MFIE: TE^z Polarization To derive the MFIE for the TE^z polarization, let us consider a TE^z uniform plane wave incident upon a two-dimensional curved surface, as shown in Figure 12-19. Since the incident field has only a z component of the magnetic field, the current induced on the surface of the scatterer will have only a component that is tangent to C and it will coincide with the surface of the scatterer. That is

$$\mathbf{J}_s = \hat{\mathbf{c}} J_c(\rho) \quad (12-96)$$

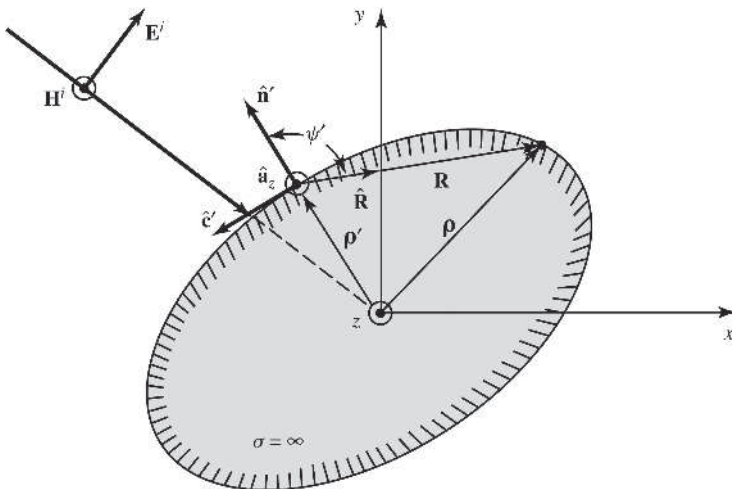


Figure 12-19 Geometry for two-dimensional MFIE TE^z polarization scattering.

On the surface of the scatterer the current density is related to the incident and scattered magnetic fields by (12-81), which for the geometry of Figure 12-19 can be written as

$$\begin{aligned}\mathbf{J}_s|_C &= \hat{\mathbf{c}} J_c(\rho)|_C = \hat{\mathbf{n}} \times (\mathbf{H}^i + \mathbf{H}^s)|_C = \hat{\mathbf{n}} \times \mathbf{H}^i + \lim_{\rho \rightarrow C} (\hat{\mathbf{n}} \times \mathbf{H}^s) \\ &= \hat{\mathbf{n}} \times \hat{\mathbf{a}}_z H^i|_C + \lim_{\rho \rightarrow C} (\hat{\mathbf{n}} \times \mathbf{H}^s) \\ \mathbf{J}_s|_C &= \hat{\mathbf{c}} J_c(\rho)|_C = -\hat{\mathbf{c}} H_z^i|_C + \lim_{\rho \rightarrow C} (\hat{\mathbf{n}} \times \mathbf{H}^s)\end{aligned}\quad (12-97)$$

where $\rho \rightarrow C$ indicates that the boundary C is approached by ρ from the outside. Since the left side and the first term of the right side of (12-97) have only C components, then the second term of the right side of (12-97) must also have only a C component. Thus, we can write (12-97) as

$$J_c(\rho)|_C = -H_z^i(\rho)|_C + \lim_{\rho \rightarrow C} [\hat{\mathbf{c}} \cdot (\hat{\mathbf{n}} \times \mathbf{H}^s)] \quad (12-98)$$

Using the scattered magnetic field of (12-90), we can write the second term within the brackets of (12-98) as

$$\begin{aligned}\hat{\mathbf{c}} \cdot (\hat{\mathbf{n}} \times \mathbf{H}^s) &= \hat{\mathbf{c}} \cdot \left\{ j \frac{1}{4} \hat{\mathbf{n}} \times \int_C [\hat{\mathbf{c}}' J_c(\rho') \times \nabla H_0^{(2)}(\beta R)] dc' \right\} \\ &= j \frac{1}{4} \int_C J_c(\rho') \left\{ \hat{\mathbf{c}} \cdot \hat{\mathbf{n}} \times [\hat{\mathbf{c}}' \times \nabla H_0^{(2)}(\beta R)] \right\} dc'\end{aligned}\quad (12-99)$$

Since from Figure 12-19

$$\hat{\mathbf{c}}' = -\hat{\mathbf{n}}' \times \hat{\mathbf{a}}'_z = -\hat{\mathbf{n}}' \times \hat{\mathbf{a}}_z \quad (12-100)$$

with the aid of (12-92)

$$\begin{aligned}\hat{\mathbf{c}}' \times \nabla H_0^{(2)}(\beta R) &= (-\hat{\mathbf{n}}' \times \hat{\mathbf{a}}_z) \times \nabla H_0^{(2)}(\beta R) = \nabla H_0^{(2)}(\beta R) \times (\hat{\mathbf{n}}' \times \hat{\mathbf{a}}_z) \\ &= -\hat{\mathbf{a}}_z [\hat{\mathbf{n}}' \cdot \nabla H_0^{(2)}(\beta R)] + \hat{\mathbf{n}}' [\hat{\mathbf{a}}_z \cdot \nabla H_0^{(2)}(\beta R)] \\ &= -\hat{\mathbf{a}}_z [\hat{\mathbf{n}} \cdot \nabla H_0^{(2)}(\beta R)]\end{aligned}\quad (12-100a)$$

since $\hat{\mathbf{a}}_z \cdot \nabla H_0^{(2)}(\beta R) = 0$. Thus, the terms within the brackets in (12-99) can be written as

$$\begin{aligned}\hat{\mathbf{c}} \cdot \hat{\mathbf{n}} \times [\hat{\mathbf{c}}' \times \nabla H_0^{(2)}(\beta R)] &= -\hat{\mathbf{c}} \cdot (\hat{\mathbf{n}} \times \hat{\mathbf{a}}_z) [\hat{\mathbf{n}}' \cdot \nabla H_0^{(2)}(\beta R)] = (\hat{\mathbf{c}} \cdot \hat{\mathbf{c}}) [\hat{\mathbf{n}}' \cdot \nabla H_0^{(2)}(\beta R)] \\ &= \hat{\mathbf{n}}' \cdot \nabla H_0^{(2)}(\beta R)\end{aligned}\quad (12-101)$$

since $-\hat{\mathbf{c}} = \hat{\mathbf{n}} \times \hat{\mathbf{a}}_z$. Substituting (12-101) into (12-99) reduces it to

$$\begin{aligned}\hat{\mathbf{c}} \cdot \hat{\mathbf{n}} \times \mathbf{H}^s &= j \frac{1}{4} \int_C J_c(\rho') [\hat{\mathbf{n}}' \cdot \nabla H_0^{(2)}(\beta R)] dc' = j \frac{1}{4} \int_C J_c(\rho') [-\beta \cos \psi' H_1^{(2)}(\beta R)] dc' \\ \hat{\mathbf{c}} \cdot \hat{\mathbf{n}} \times \mathbf{H}^s &= -j \frac{\beta}{4} \int_C J_c(\rho') \cos \psi' H_1^{(2)}(\beta R) dc'\end{aligned}\quad (12-102)$$

where the angle ψ' is defined in Figure 12-19. Thus, we can write (12-98), using (12-102), as

$$J_c(\rho)|_C = -H_z^i(\rho)|_C + \lim_{\rho \rightarrow C} \left[-j \frac{\beta}{4} \int_C J_c(\rho') \cos \psi' H_1^{(2)}(\beta R) dc' \right] \quad (12-103)$$

or

$$J_c(\rho)|_C + j\frac{\beta}{4}\lim_{\rho \rightarrow C}\left[\int_C J_c(\rho')\cos\psi' H_1^{(2)}(\beta R)dc'\right] = -H_z^i|_C \quad (12-103a)$$

C. Solution of the Two-Dimensional MFIE TE^z Polarization The two-dimensional MFIEs of (12-95a) for TM^z polarization and (12-103a) for TE^z polarization are of identical form and their solutions are then similar. Since TM^z polarizations are very conveniently solved using the EFIE, usually the MFIEs are mostly applied to TE^z polarization problems where the magnetic field has only a z component. Therefore, we will demonstrate here the solution of the TE^z MFIE of (12-103a).

In the evaluation of the scattered magnetic field at $\rho = \rho_m$ from all points on C (including the point $\rho = \rho_m$ where the observation is made), the integral of (12-103a) can be split into two parts; one part coming from ΔC and the other part outside ΔC ($C - \Delta C$), as shown in Figure 12-20. Thus, we can write the integral of (12-103a) as

$$\begin{aligned} & j\frac{\beta}{4}\lim_{\rho \rightarrow C}\int_C J_c(\rho')\cos\psi' H_1^{(2)}(\beta R)dc' \\ &= j\frac{\beta}{4}\lim_{\rho \rightarrow C}\left\{\int_{\Delta C} J_c(\rho')\cos\psi' H_1^{(2)}(\beta R)dc' + \int_{C-\Delta C} J_c(\rho')\cos\psi' H_1^{(2)}(\beta R)dc'\right\} \end{aligned} \quad (12-104)$$

In a solution of (12-103a), where C is subdivided into segments, ΔC would typically represent one segment (the self-term) and $C - \Delta C$ would represent the other segments (the nonself-terms). Let us now examine the evaluation of each of the integrals in (12-104).

At any point, the total magnetic field is equal to the sum of the incident and scattered parts. Within the scattered conducting obstacle, the total magnetic field is zero, whereas above the conducting surface the field is nonzero. The discontinuity of the two along C is used to represent the current density along C . Within the thin rectangular box with dimensions of h and ΔC (as $h \rightarrow 0$), the total magnetic field, ($\mathbf{H}_1^i$) above and ($\mathbf{H}_2^i$) below the interface, can be written as

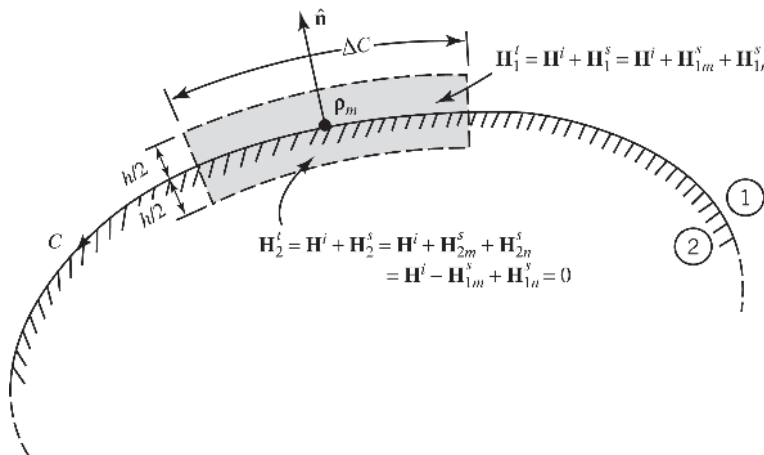


Figure 12-20 Geometry and fields along the scattering surface for a two-dimensional MFIE.

$$\mathbf{H}_1^t = \mathbf{H}^i + \mathbf{H}_1^s = \mathbf{H}^i + (\mathbf{H}_{1m}^s + \mathbf{H}_{1n}^s) \quad (12-105a)$$

$$\mathbf{H}_2^t = \mathbf{H}^i + \mathbf{H}_2^s = \mathbf{H}^i + (\mathbf{H}_{2m}^s + \mathbf{H}_{2n}^s) = \mathbf{H}^i + (-\mathbf{H}_{1m}^s + \mathbf{H}_{1n}^s) = 0 \quad (12-105b)$$

$$\text{or } \mathbf{H}_{1n}^s = \mathbf{H}_{1m}^s - \mathbf{H}^i \quad (12-105c)$$

where $\mathbf{H}_{1m}^s$ ($\mathbf{H}_{2m}^s$) = scattered field in region 1 (2) within the box that is due to ΔC (self-term), which is discontinuous across the boundary along ΔC

$\mathbf{H}_{1n}^s$ ($\mathbf{H}_{2n}^s$) = scattered field in region 1 (2) within the box that is due to $C - \Delta C$ (non-self-terms), which is continuous across the boundary along ΔC

It is assumed here that ΔC along C becomes a straight line as the segment becomes small. The current density along ΔC can then be represented using (12-105a) and (12-105b) by

$$\mathbf{J}_c(\rho) |_{\Delta C} = \hat{\mathbf{n}} \times (\mathbf{H}_1^t - \mathbf{H}_2^t) |_{\Delta C} = \hat{\mathbf{n}} \times (\mathbf{H}_{1m}^s + \mathbf{H}_{1n}^s) = 2\hat{\mathbf{n}} \times \mathbf{H}_{1m}^s = -2\hat{\mathbf{c}} H_{1m}^s \quad (12-106)$$

or

$$J_c(\rho_m) = -2H_{1m}^s(\rho_m) \Rightarrow H_{1m}^s(\rho_m) = -\frac{J_c(\rho_m)}{2} \quad (12-106a)$$

Therefore, the integral along ΔC in (12-104), which can be used to represent the scattered magnetic field at $\rho = \rho_m$ that is due to the ΔC , can be replaced by (12-106a). The nonself-terms can be found using the integral along $C - \Delta C$ in (12-104). Thus, using (12-104) and (12-106a), we can reduce (12-103a) for $\rho = \rho_m$ to

$$J_c(\rho_m) - \frac{J_c(\rho_m)}{2} + j \frac{\beta}{4} \int_{C-\Delta C} J_c(\rho') \cos \psi'_m H_1^{(2)}(\beta R_m) dc' = -H_z^i(\rho_m) \quad (12-107)$$

or

$$\boxed{\frac{J_c(\rho_m)}{2} + j \frac{\beta}{4} \int_{C-\Delta C} J_c(\rho') \cos \psi'_m H_1^{(2)}(\beta R_m) dc' = -H_z^i(\rho_m)} \quad (12-107a)$$

An analogous procedure can be used to reduce (12-95a) to a form similar to that of (12-107a).

Let us now represent the current density $J_c(\rho)$ of (12-107a) by the finite series of (12-23)

$$J_c(\rho) \simeq \sum_{n=1}^N a_n g_n(\rho) \quad (12-108)$$

where $g_n(\rho)$ represents the basis (expansion) functions. Substituting (12-108) into (12-107a) and interchanging integration and summation, we can write that, at any point $\rho = \rho_m$ on C (12-107a) can be written as

$$-H_z^i(\rho_m) = \frac{1}{2} \sum_{n=1}^N a_n g_n(\rho_m) + j \frac{\beta}{4} \sum_{n=1}^N a_n \int_{C-\Delta C} g_n(\rho') \cos \psi'_m H_1^{(2)}(\beta R_m) dc' \quad (12-109)$$

If the g_n 's are subdomain piecewise constant pulse functions with each basis function existing only over its own segment, then (12-109) reduces to

$$-H_z^i(\rho_m) = \frac{\delta_{mm}}{2} a_m + j \frac{\beta}{4} \sum_{\substack{n=1 \\ n \neq m}}^N a_n \int_{\rho_n}^{\rho_{n+1}} \cos \psi'_{mn} H_1^{(2)}(\beta R_{mn}) dc' \quad (12-110)$$

or

$$-H_z^i(\rho_m) = \sum_{n=1}^N a_n \left[\frac{\delta_{mn}}{2} + j \frac{\beta}{4} \int_{\rho_n}^{\rho_{n+1}} \cos \psi'_{mn} H_1^{(2)}(\beta R_{mn}) dc' \right] \quad (12-110a)$$

where δ_{mn} is the Kronecker delta function, defined by

$$\delta_{mn} = \begin{cases} 1 & m = n \\ 0 & m \neq n \end{cases} \quad (12-110b)$$

The Kronecker delta function is used to indicate that for a given observation point m only the segment itself ($n = m$) contributes to the first term on the right side of (12-110a).

If (12-110a) is applied to m points on C , it can be written as

$$\boxed{\begin{aligned} [-H_z^i(\rho_m)] &= \sum_{n=1}^N a_n \left[\frac{\delta_{mn}}{2} + j \frac{\beta}{4} \int_{\rho_n}^{\rho_{n+1}} \cos \psi'_{mn} H_1^{(2)}(\beta R_{mn}) dc' \right] \\ &\quad m = 1, 2, \dots, N \end{aligned}} \quad (12-111)$$

In general matrix notation, (12-111) can be expressed as

$$[V_m] = [Z_{mn}][I_n] \quad (12-112)$$

where

$$V_m = -H_z^i(\rho_m) \quad (12-112a)$$

$$Z_{mn} = \left[\frac{\delta_{mn}}{2} + j \frac{\beta}{4} \int_{\rho_n}^{\rho_{n+1}} \cos \psi'_{mn} H_1^{(2)}(\beta R_{mn}) dc' \right] \quad (12-112b)$$

$$I_n = a_n \quad (12-112c)$$

To demonstrate the applicability of (12-111), let us consider an example.

Example 12-6

A TE^z uniform plane wave is normally incident upon a circular conducting cylinder of radius a , as shown in Figure 12-21.

1. Using the MFIE of (12-107a), determine and plot the current density induced on the surface of the cylinder when $a = 2\lambda$. Assume the incident magnetic field is of unity amplitude. Use subdomain piecewise constant pulse functions. Subdivide the circumference into 540 segments. Compare the current density obtained using the IE with the exact modal solution of (11-113).
2. Based on the electric current density, derive and then plot the normalized (σ_{2-D} / λ) bistatic scattering width (in decibels) for $0^\circ \leq \phi \leq 360^\circ$ when $a = 2\lambda$. Compare these values with those obtained using the exact modal solution of (11-117).

Solution:

1. Since for subdomain piecewise constant pulse functions (12-103a) or (12-107a) reduces to (12-111) or (12-112) through (12-112c), then a solution of (12-112) for I_n leads to the current density shown in Figure 12-22. In the same figure we have plotted the current density of (11-113) based on the modal solution, and we can see an excellent agreement between the two. We also have plotted the

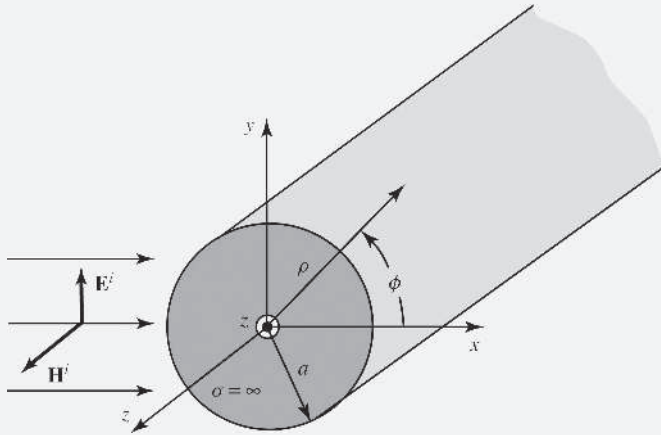


Figure 12-21 TE^z uniform plane wave incident on a circular conducting cylinder.

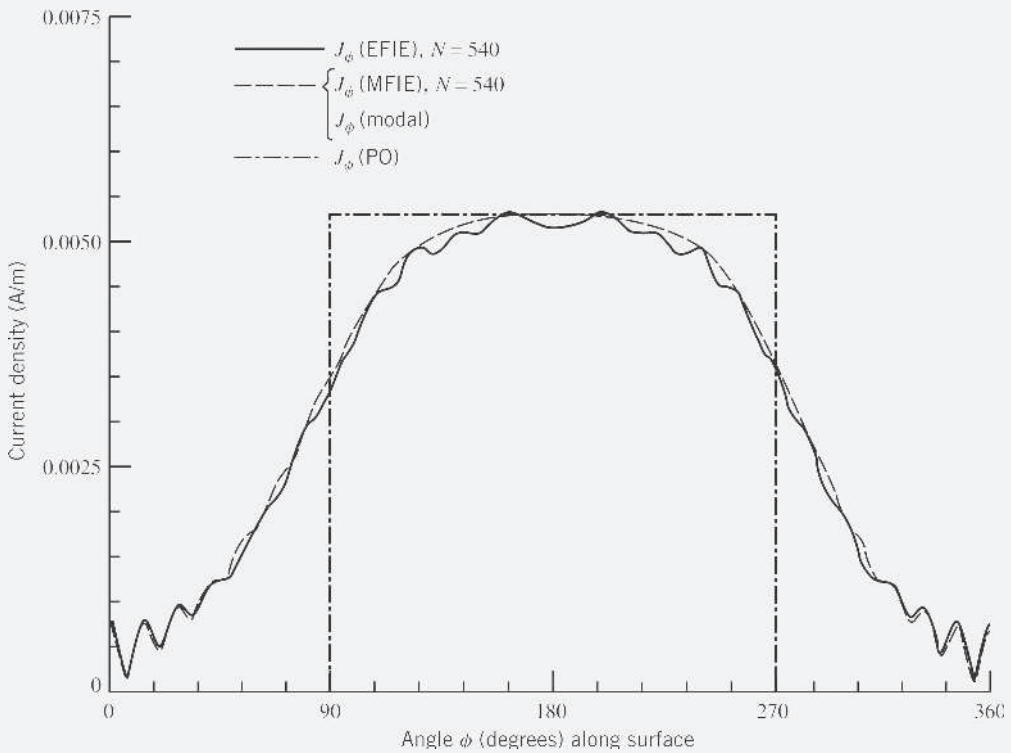


Figure 12-22 Current density induced on the surface of a circular conducting cylinder by TE^z plane wave incidence ($a = 2\lambda$).

current densities based on the EFIE for TE^z polarization of Section 12.3.1B and on the physical optics of (7-54) over the illuminated portion of the cylinder surface. The results of the EFIE do not agree with the modal solution as accurately as those of the MFIE. However, they still are very good. As expected, the physical optics current density is not representative of the true current density.

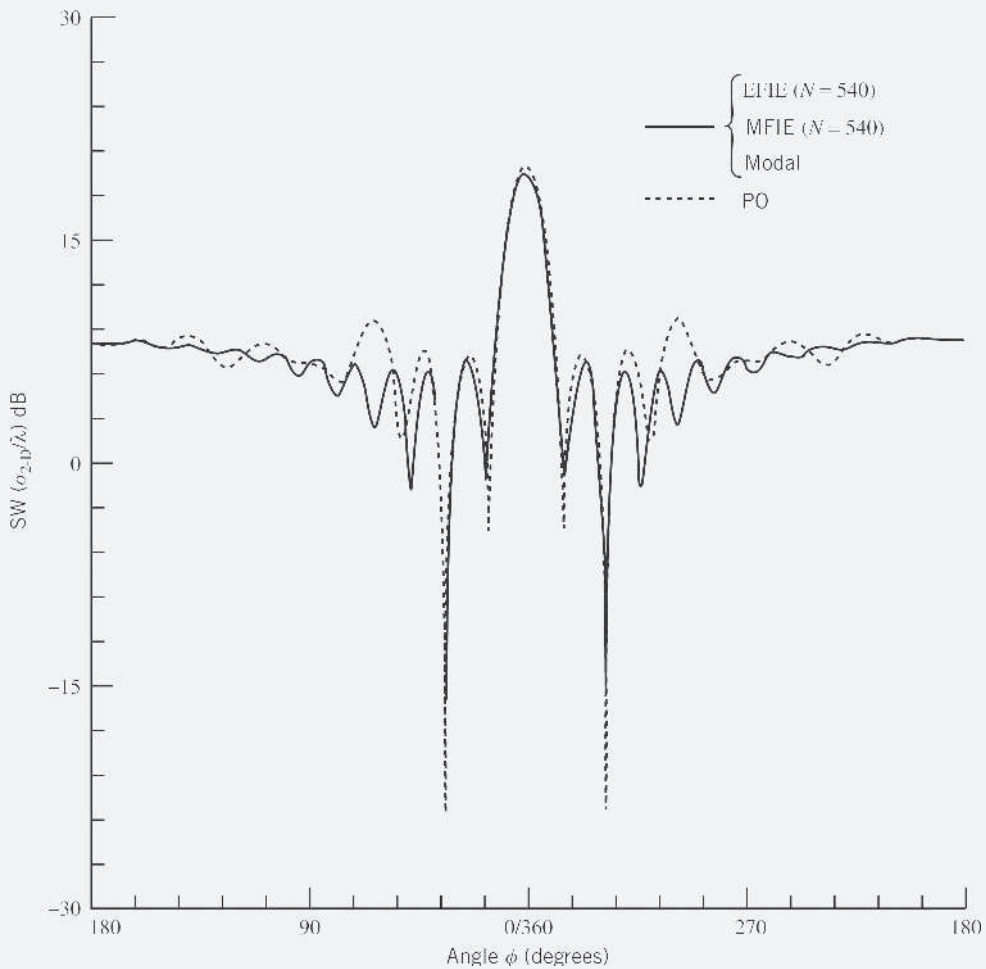


Figure 12-23 TE^z bistatic scattering width of a circular conducting cylinder ($a = 2\lambda$).

2. Based on the current densities obtained in part 1, the far-zone scattered field was derived and the corresponding bistatic scattering width was formulated. The computed SW results are shown in Figure 12-23. Besides the results based on the physical optics approximation, the other three (MFIE, EFIE, and modal solution) give almost indistinguishable data and are indicated in Figure 12-23 by basically one curve.

12.4 FINITE-DIAMETER WIRES

In this section we want to derive and apply two classic three-dimensional integral equations, referred to as *Pocklington's integro-differential equation* and *Hallén's integral equation* [29–37], that can be used most conveniently to find the current distribution on conducting wires. Hallén's equation is usually restricted to the use of a *delta-gap* voltage source model at the feed of a wire antenna. Pocklington's equation, however, is more general and it is adaptable to many types of feed sources (through alteration of its excitation function or excitation matrix), including a

magnetic frill [38]. In addition, Hallén's equation requires the inversion of an $N + 1$ order matrix (where N is the number of divisions of the wire) while Pocklington's equation requires the inversion of an N order matrix.

For very thin wires, the current distribution is usually assumed to be of sinusoidal form [24]. For finite diameter wires (usually diameters $d > 0.05\lambda$), the sinusoidal current distribution is representative but not accurate. To find a more accurate current distribution on a cylindrical wire, an integral equation is usually derived and solved. Previously, solutions to the integral equation were obtained using iterative methods [31]; presently, it is most convenient to use moment method techniques [1–3].

If we know the voltage at the feed terminals of a wire antenna and find the current distribution, we can obtain the input impedance and radiation pattern. Similarly if a wave impinges upon the surface of a wire scatterer, it induces a current density that in turn is used to find the scattered field. Whereas the linear wire is simple, most of the information presented here can be readily extended to more complicated structures.

12.4.1 Pocklington's Integral Equation

In deriving Pocklington's integral equation, we will use the integral equation approach of Section 12.3.1. However, each step, as applied to the wire scatterer, will be repeated here to show the simplicity of the method.

Refer to Figure 12-24a. Let us assume that an incident wave impinges on the surface of a conducting wire. The total tangential electric field (E_z) at the surface of the wire is given by (12-53) or (12-53a), that is

$$E_z^t(r = r_s) = E_z^i(r = r_s) + E_z^s(r = r_s) = 0 \quad (12-113)$$

or

$$E_z^s(r = r_s) = -E_z^i(r = r_s) \quad (12-113a)$$

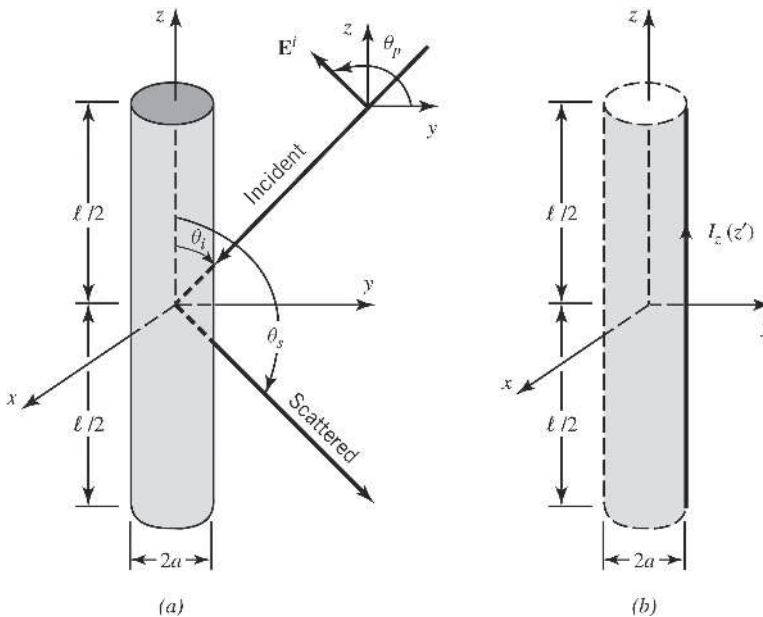


Figure 12-24 (a) Uniform plane wave obliquely incident on a conducting wire. (b) Equivalent current.

At any observation point, the field scattered by the induced current density on the surface of the wire is given by (12-54). However, for observations at the wire surface, only the z component of (12-54) is needed, and we can write it as

$$E_z^s(r) = -j \frac{1}{\omega \mu \varepsilon} \left(\beta^2 A_z + \frac{\partial^2 A_z}{\partial z^2} \right) \quad (12-114)$$

According to (12-54a) and neglecting edge effects

$$A_z = \frac{\mu}{4\pi} \iint_S J_z \frac{e^{-j\beta R}}{R} ds' = \frac{\mu}{4\pi} \int_{-\ell/2}^{+\ell/2} \int_0^{2\pi} J_z \frac{e^{-j\beta R}}{R} a d\phi' dz' \quad (12-115)$$

If the wire is very thin, the current density J_z is not a function of the azimuthal angle ϕ , and we can write it as

$$2\pi a J_z = I_z(z') \Rightarrow J_z = \frac{1}{2\pi a} I_z(z') \quad (12-116)$$

where $I_z(z')$ is assumed to be an equivalent filament line-source current located a radial distance $\rho = a$ from the z axis, as shown in Figure 12-24b. Thus, (12-115) reduces to

$$A_z = \frac{\mu}{4\pi} \int_{-\ell/2}^{+\ell/2} \left[\frac{1}{2\pi a} \int_0^{2\pi} I_z(z') \frac{e^{-j\beta R}}{R} a d\phi' \right] dz' \quad (12-117)$$

$$\begin{aligned} R &= \sqrt{(x-x')^2 + (y-y')^2 + (z-z')^2} \\ &= \sqrt{\rho^2 + a^2 - 2\rho a \cos(\phi - \phi') + (z-z')^2} \end{aligned} \quad (12-117a)$$

where ρ is the radial distance to the observation point and a is the radius.

Because of the symmetry of the scatterer, the observations are not a function of ϕ . For simplicity, let us then choose $\phi = 0$. For observations at the surface ($\rho = a$) of the scatterer (12-117) and (12-117a) reduce to

$$A_z(\rho = a) = \mu \int_{-\ell/2}^{+\ell/2} I_z(z') \left(\frac{1}{2\pi} \int_0^{2\pi} \frac{e^{-j\beta R}}{4\pi R} d\phi' \right) dz' = \mu \int_{-\ell/2}^{+\ell/2} I_z(z') G(z, z') dz' \quad (12-118)$$

$$\boxed{G(z, z') = \frac{1}{2\pi} \int_0^{2\pi} \frac{e^{-j\beta R}}{4\pi R} d\phi'} \quad (12-118a)$$

$$\boxed{R(\rho = a) = \sqrt{4a^2 \sin^2\left(\frac{\phi'}{2}\right) + (z - z')^2}} \quad (12-118b)$$

Thus, for observations at the surface ($\rho = a$) of the scatterer, the z component of the scattered electric field can be expressed as

$$E_z^s(\rho = a) = -j \frac{1}{\omega \varepsilon} \left(\beta^2 + \frac{d^2}{dz^2} \right) \int_{-\ell/2}^{+\ell/2} I_z(z') G(z, z') dz' \quad (12-119)$$

which by using (12-113a) reduces to

$$-j \frac{1}{\omega \varepsilon} \left(\frac{d^2}{dz^2} + \beta^2 \right) \int_{-\ell/2}^{+\ell/2} I_z(z') G(z, z') dz' = -E_z^i(\rho = a) \quad (12-120)$$

or

$$\left(\frac{d^2}{dz^2} + \beta^2 \right) \int_{-\ell/2}^{+\ell/2} I_z(z') G(z, z') dz' = -j\omega\epsilon E_z^i(\rho = a) \quad (12-120a)$$

Interchanging integration with differentiation, we can rewrite (12-120a) as

$$\int_{-\ell/2}^{+\ell/2} I_z(z') \left[\left(\frac{\partial^2}{\partial z^2} + \beta^2 \right) G(z, z') \right] dz' = -j\omega\epsilon E_z^i(\rho = a) \quad (12-121)$$

where $G(z, z')$ is given by (12-118a).

Equation 12-121 is referred to as *Pocklington's integro-differential equation* [29], and it can be used to determine the equivalent filamentary line-source current of the wire, and thus current density on the wire, by knowing the incident field on the surface of the wire. It is a simplified form of (12-56) as applied to a wire scatterer, and it could have been derived directly from (12-56).

If we assume that the wire is very thin ($a \ll \lambda$), such that (12-118a) reduces to

$$G(z, z') = G(R) = \frac{e^{-j\beta R}}{4\pi R} \quad (12-122)$$

(12-121) can also be expressed in a more convenient form as [33]

$$\int_{-\ell/2}^{+\ell/2} I_z(z') \frac{e^{-j\beta R}}{4\pi R^5} \left[(1 + j\beta R)(2R^2 - 3a^2) + (\beta a R)^2 \right] dz' = -j\omega\epsilon E_z^i(\rho = a) \quad (12-123)$$

where, for observations along the center of the wire ($\rho = 0$),

$$R = \sqrt{a^2 + (z - z')^2} \quad (12-123a)$$

In (12-121) or (12-123), $I_z(z')$ represents the equivalent filamentary line-source current located on the surface of the wire, as shown in Figure 12-24b, and it is obtained by knowing the incident electric field at the surface of the wire. By point-matching techniques, this is solved by matching the boundary conditions at discrete points on the surface of the wire. Often it is easier to choose the matching points to be at the interior of the wire, especially along the axis, as shown in Figure 12-25a, where $I_z(z')$ is located on the surface of the wire. By reciprocity, the configuration of Figure 12-25a is analogous to that of Figure 12-25b, where the equivalent filamentary line-source current is assumed to be located along the center axis of the wire and the matching points are selected on the surface of the wire. Either of the two configurations can be used to determine the equivalent filamentary line-source current $I_z(z')$; the choice is left to the individual.

Pocklington's integral equation of (12-121) is derived methodically based on (12-114). Eventually (12-121), using (12-122), reduces to (12-123). While the derivation of (12-121) and (12-123) is straightforward, their numerical evaluation may be more difficult because it involves double differentiation of the kernel, which leads to a non-integrable singularity, especially of the $1/R^5$ order. However, choosing the observation point along the axis of the wire while the current is on its surface or vice versa, as illustrated in Figure 12-25, mitigates the problem to some extent. Another way to derive such an integral equation, although it may not be as straightforward, will be to represent the electric field as a combination of both the vector $\mathbf{A}$ and scalar ϕ potentials, as given by (6-9a) ($\mathbf{E} = -j\omega\mathbf{A} - \nabla\phi$), instead of (12-114), and use the continuity equation $\nabla \cdot \mathbf{J} = -j\omega q$ to relate the charge q within ϕ as the divergence of $\mathbf{J}$ [28]. Using such a procedure, the resulting singularity in the kernel for both the vector $\mathbf{A}$ and scalar ϕ potentials is no more singular than $1/R$, which can be evaluated numerically more accurately.

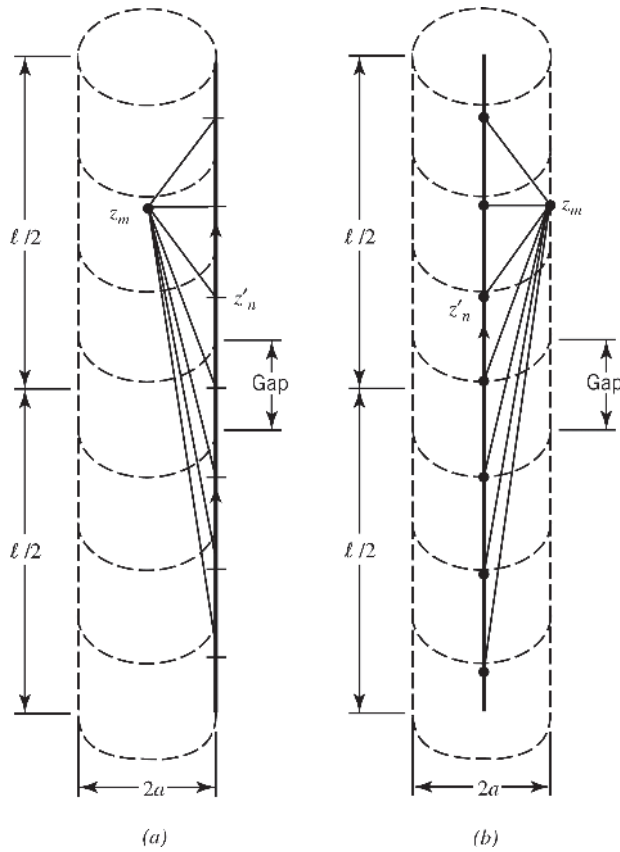


Figure 12-25 Dipole segmentation and its equivalent current. (a) On the surface. (b) Along its center.

12.4.2 Hallén's Integral Equation

Referring again to Figure 12-24a, let us assume that the length of the cylinder is much larger than its radius ($\ell \gg a$) and its radius is much smaller than the wavelength ($a \ll \lambda$), so that the effects of the end faces of the cylinder can be neglected. Therefore, the boundary conditions for a wire with infinite conductivity are those of vanishing total tangential E fields on the surface of the cylinder and vanishing current at the ends of the cylinder [$I_z(z' = \pm \ell/2) = 0$].

Since only an electric current density flows on the cylinder and it is directed along the z axis ($\mathbf{J} = \hat{\mathbf{a}}_z J_z$), according to (6-30) and (6-96a), $\mathbf{A} = \hat{\mathbf{a}}_z A_z(z')$, which for small radii is assumed to be only a function of z' . Thus, (6-34) reduces for $\mathbf{F} = 0$ to

$$E_z^t = -j\omega A_z - j \frac{1}{\omega\mu\epsilon} \frac{\partial^2 A_z}{\partial z'^2} = -j \frac{1}{\omega\mu\epsilon} \left[\frac{d^2 A_z}{dz'^2} + \omega^2 \mu\epsilon A_z \right] \quad (12-124)$$

Since the total tangential electric field E_z^t vanishes on the surface of the cylinder, (12-124) reduces to

$$\frac{d^2 A_z}{dz'^2} + \beta^2 A_z = 0 \quad (12-124a)$$

Because the current density on the cylinder is symmetrical [$J_z(z') = J_z(-z')$], the potential A_z is also symmetrical [i.e., $A_z(z') = A_z(-z')$]. Thus, the solution of (12-124a) is given by

$$A_z(z) = -j\sqrt{\mu\epsilon} [B_1 \cos(\beta z) + C_1 \sin(\beta |z|)] \quad (12-125)$$

where B_1 and C_1 are constants. For a current-carrying wire, its potential is also given by (6-97a). Equating (12-125) to (6-97a) leads to

$$\int_{-\ell/2}^{+\ell/2} I_z(z') \frac{e^{-j\beta R}}{4\pi R} dz' = -j \sqrt{\frac{\epsilon}{\mu}} [B_1 \cos(\beta z) + C_1 \sin(\beta |z|)] \quad (12-126)$$

If a voltage V_i is applied at the input terminals of the wire, it can be shown that the constant $C_1 = V_i/2$. The constant B_1 is determined from the boundary condition that requires the current to vanish at the end points of the wire.

Equation 12-126 is referred to as *Hallén's integral equation* for a perfectly conducting wire. It was derived by solving the differential equation 6-34 or 12-124a with the enforcement of the appropriate boundary conditions.

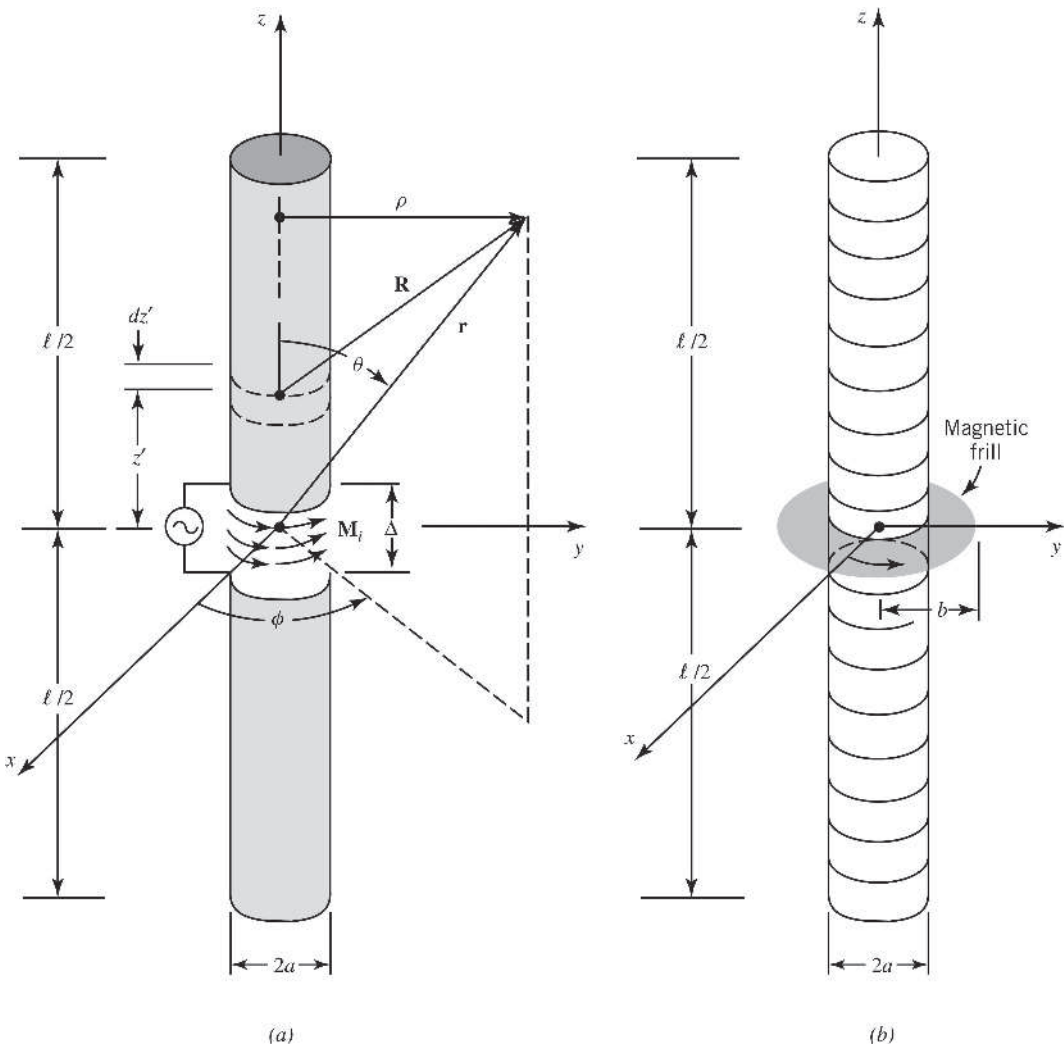


Figure 12-26 Cylindrical dipole, its segmentation, and gap modeling. (a) Cylindrical dipole. (b) Segmented dipole. (Source: C. A. Balanis, *Antenna Theory: Analysis and Design*, Third Edition, copyright © 2005, John Wiley & Sons, Inc. Reprinted by permission of John Wiley & Sons, Inc.)

12.4.3 Source Modeling

Let us assume that the wire of Figure 12-24 is symmetrically fed by a voltage source, as shown in Figure 12-26a, and the element is acting as a dipole antenna. To use, for example, Pocklington's integro-differential equation 12-121 or 12-123, we need to know how to express $E_z^i(\rho = a)$. Traditionally, there have been two methods used to model the excitation to represent $E_z^i(\rho = a, 0 \leq \phi \leq 2\pi, -\ell/2 \leq z \leq +\ell/2)$ at all points on the surface of the dipole: one is referred to as the *delta-gap* excitation and the other as the *equivalent magnetic ring current* (better known as *magnetic frill generator*) [38].

A. Delta Gap The delta-gap source modeling is the simplest and most widely used of the two, but it is also the least accurate, especially for impedances. Usually it is most accurate for smaller width gaps. Using the delta gap, it is assumed that the excitation voltage at the feed terminals is of a constant V_i value, and zero elsewhere. Therefore, the incident electric field $E_z^i(\rho = a, 0 \leq \phi \leq 2\pi, -\ell/2 \leq z \leq +\ell/2)$ is also a constant (V_i/Δ where Δ is the gap width) over the feed gap and zero elsewhere, hence the name delta gap. For the delta-gap model, the feed gap Δ of Figure 12-26a is replaced by a narrow band of strips of equivalent magnetic current density of

$$\mathbf{M}_i = -\hat{\mathbf{n}} \times \mathbf{E}^i = -\hat{\mathbf{a}}_\rho \times \hat{\mathbf{a}}_z \frac{V_i}{\Delta} = \hat{\mathbf{a}}_\phi \frac{V_i}{\Delta}, \quad -\frac{\Delta}{2} \leq z' \leq \frac{\Delta}{2} \quad (12-127)$$

The magnetic current density $\mathbf{M}_i$ is sketched in Figure 12-26a.

B. Magnetic Frill Generator The magnetic frill generator was introduced to calculate the near- as well as the far-zone fields from coaxial apertures [38]. To use this model, the feed gap is replaced with a circumferentially directed magnetic current density that exists over an annular aperture with inner radius a , which is usually chosen to be the radius of the inner wire, and an outer radius b , as shown in Figure 12-26b. Since the dipole is usually fed by transmission lines, the outer radius b of the equivalent annular aperture of the magnetic frill generator is found using the expression for the characteristic impedance of the transmission line.

Over the annular aperture of the magnetic frill generator, the electric field is represented by the TEM mode field distribution of a coaxial transmission line given by

$$\mathbf{E}_f = \hat{\mathbf{a}}_\rho \frac{V_i}{2\rho' \ln(b/a)} \quad (12-128)$$

Therefore, the corresponding equivalent magnetic current density $\mathbf{M}_f$ for the magnetic frill generator, used to represent the aperture, is equal to

$$\mathbf{M}_f = -2\hat{\mathbf{n}} \times \mathbf{E}_f = -2\hat{\mathbf{a}}_z \times \hat{\mathbf{a}}_\rho E_\rho = -\hat{\mathbf{a}}_\phi \frac{V_i}{\rho' \ln(b/a)} \quad (12-129)$$

The fields generated by the magnetic frill generator of (12-129) on the surface of the wire are found using [38]

$$\begin{aligned} E_z^i \left(\rho = a, 0 \leq \phi \leq 2\pi, -\frac{\ell}{2} \leq z \leq \frac{\ell}{2} \right) \\ \simeq -V_i \left[\frac{\beta(b^2 - a^2)e^{-j\beta R_0}}{8\ln(b/a)r_0^2} \left\{ 2 \left[\frac{1}{\beta R_0} + j \left(1 - \frac{b^2 - a^2}{2R_0^2} \right) \right] \right. \right. \\ \left. \left. + \frac{a^2}{R_0} \left[\left(\frac{1}{\beta R_0} + j \left(1 - \frac{(b^2 + a^2)}{2R_0^2} \right) \right) \left(-j\beta - \frac{2}{R_0} \right) + \left(-\frac{1}{\beta R_0^2} + j \frac{b^2 + a^2}{R_0^3} \right) \right] \right\} \right] \end{aligned} \quad (12-130)$$

where

$$R_0 = \sqrt{z^2 + a^2} \quad (12-130a)$$

The fields generated on the surface of the wire computed using (12-130) can be approximated by those found along the axis ($\rho = 0$). Doing this leads to a simpler expression of the form [38]

$$E_z^i \left(\rho = 0, -\frac{\ell}{2} \leq z \leq \frac{\ell}{2} \right) = -\frac{V_i}{2 \ln(b/a)} \left[\frac{e^{-j\beta R_1}}{R_1} - \frac{e^{-j\beta R_2}}{R_2} \right] \quad (12-131)$$

where

$$R_1 = \sqrt{z^2 + a^2} \quad (12-131a)$$

$$R_2 = \sqrt{z^2 + b^2} \quad (12-131b)$$

The following example compares the results obtained using the two source modelings (delta gap and magnetic frill generator).

Example 12-7

Assume a center-fed linear dipole of $\ell = 0.47\lambda$ and $a = 0.005\lambda$.

1. Determine the voltage and normalized current distribution over the length of the dipole using $N = 21$ segments to subdivide the length. Plot the current distribution.
2. Determine the input impedance using segments of $N = 7, 11, 21, 29, 41, 51, 61, 71$, and 79.

Use Pocklington's integro-differential equation 12-123 with piecewise constant subdomain basis functions and point matching to solve the problem, model the gap with one segment, and use both the delta gap and magnetic frill generator to model the excitation. Use (12-131) for the magnetic frill generator. Because the current at the ends of the wire vanishes, the piecewise constant subdomain basis functions are not the most judicious choices. However, because of their simplicity, they are chosen here to illustrate the principles, even though the results are not the most accurate. Assume that the characteristic impedance of the annular aperture is 50 ohms and the excitation voltage V_i is 1 V.

Solution:

1. Since the characteristic impedance of the annular aperture (coaxial line) is 50 ohms, then

$$Z_c = \sqrt{\frac{\mu_0}{\epsilon_0}} \frac{\ln(b/a)}{2\pi} = 50 \Rightarrow \frac{b}{a} = 2.3$$

Subdividing the total length ($\ell = 0.47\lambda$) of the dipole into 21 segments makes the gap and each segment equal to

$$\Delta = \frac{0.47\lambda}{21} = 0.0224\lambda$$

Using (12-131) to compute E_z^i , the corresponding induced voltages, obtained by multiplying the value of $-E_z^i$ at each segment by the length of the segment, are found listed in Table 12-1, where they are compared with those of the delta gap. $N = 1$ represents the outermost segment and $N = 11$ represents the center segment. Because of the symmetry, only values for the center segment and half of the other segments are shown. Although the two distributions are not identical, the magnetic frill distribution voltages decay quite rapidly away from the center segment, and they very quickly reach almost vanishing values.

TABLE 12-1 Unnormalized and normalized dipole induced voltage^a differences for delta gap and magnetic frill generator ($\ell = 0.47\lambda$, $a = 0.005\lambda$, $N = 21$)

Segment number n	Delta gap voltage		Magnetic frill generator voltage	
	Unnormalized	Normalized	Unnormalized	Normalized
1	0	0	$1.11 \times 10^{-4} \angle -26.03^\circ$	$7.30 \times 10^{-5} \angle -26.03^\circ$
2	0	0	$1.42 \times 10^{-4} \angle -20.87^\circ$	$9.34 \times 10^{-5} \angle -20.87^\circ$
3	0	0	$1.89 \times 10^{-4} \angle -16.13^\circ$	$1.24 \times 10^{-4} \angle -16.13^\circ$
4	0	0	$2.62 \times 10^{-4} \angle -11.90^\circ$	$1.72 \times 10^{-4} \angle -11.90^\circ$
5	0	0	$3.88 \times 10^{-4} \angle -8.23^\circ$	$2.55 \times 10^{-4} \angle -8.23^\circ$
6	0	0	$6.23 \times 10^{-4} \angle -5.22^\circ$	$4.10 \times 10^{-4} \angle -5.22^\circ$
7	0	0	$1.14 \times 10^{-3} \angle -2.91^\circ$	$7.5 \times 10^{-4} \angle -2.91^\circ$
8	0	0	$2.52 \times 10^{-3} \angle -1.33^\circ$	$1.66 \times 10^{-3} \angle -1.33^\circ$
9	0	0	$7.89 \times 10^{-3} \angle -0.43^\circ$	$5.19 \times 10^{-3} \angle -0.43^\circ$
10	0	0	$5.25 \times 10^{-2} \angle -0.06^\circ$	$3.46 \times 10^{-2} \angle -0.06^\circ$
11	1	1	1.52 $\angle 0^\circ$	1.0 $\angle 0^\circ$

^aVoltage differences as defined here represent the product of the incident electric field at the center of each segment and the corresponding segment length.

The corresponding unnormalized and normalized currents, obtained using (12-123) with piecewise constant pulse functions and the point-matching technique for both the delta gap and magnetic frill generator, are listed in Table 12-1. The normalized magnitudes of these currents are shown plotted in Figure 12-27. It is apparent that the two distributions are almost identical in shape, and they resemble that of the ideal sinusoidal current distribution which is more valid for very thin wires and very small gaps. The distributions obtained using Pocklington's integral equation do not vanish at the ends because of the use of piecewise constant subdomain basis functions, which make the dipole look longer by either a half or full subdomain length depending whether the reference of each subdomain function is taken at its middle or end point. A more accurate modeling will be to use higher-order subdomain basic functions, such as the piecewise linear function of Figure 12-7 or piecewise sinusoids of Figure 12-8.

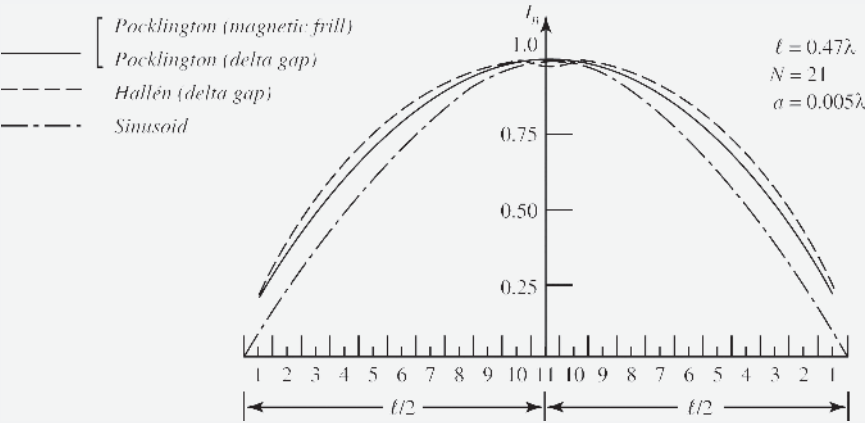


Figure 12-27 Current distribution along a dipole antenna.

2. The input impedances, computed using both the delta gap and the magnetic frill generator, are shown listed in Table 12-2. It is evident that the values begin to stabilize and compare favorably to each other once 61 or more segments are used.

TABLE 12-2 Dipole input impedance for delta gap and magnetic frill generator using Pocklington's integral equation ($\ell = 0.47\lambda$, $a = 0.005\lambda$)

N	Delta gap	Magnetic frill
7	$122.8 + j113.9$	$26.8 + j24.9$
11	$94.2 + j49.0$	$32.0 + j16.7$
21	$77.7 - j0.8$	$47.1 - j0.2$
29	$75.4 - j6.6$	$57.4 - j4.5$
41	$75.9 - j2.4$	$68.0 - j1.0$
51	$77.2 + j2.4$	$73.1 + j4.0$
61	$78.6 + j6.1$	$76.2 + j8.5$
71	$79.9 + j7.9$	$77.9 + j11.2$
79	$80.4 + j8.8$	$78.8 + j12.9$

12.5 COMPUTER CODES

With the advent of the computer there has been a proliferation of computer program development. Many of these programs are based on algorithms that are suitable for efficient computer programming for the analysis and synthesis of electromagnetic boundary-value problems. Some of these computer programs are very sophisticated and can be used to solve complex radiation and scattering problems. Others are much simpler and have limited applications. Many programs are public domain; others are restricted.

Five computer programs based on integral equation formulations and moment method solutions will be described here. The first computes the radiation or scattering by a two-dimensional perfectly electric conducting (PEC) body. It is referred to here as TDRS (two-dimensional radiation and scattering), and it is based on the two-dimensional formulations of the electric field integral equation (EFIE) of Section 12.3.1. It can be used for both electric and magnetic line-source excitation or TM^z and TE^z plane wave incidence. The second program, referred to here as PWRS (Pocklington's wire radiation and scattering) is based on Pocklington's integral equation of Section 12.4.1, and it is used for both radiation and scattering by a perfect electric conducting (PEC) wire.

The remaining three programs are more general, public domain moment method programs. A very brief description of these programs is given here. Information as to where these programs can be obtained is also included. It should be stated, however, that there are numerous other codes, public domain and restricted, that utilize moment method and other techniques, such as geometrical optics, geometrical theory of diffraction, physical optics, and physical theory of diffraction, which are too numerous to mention here.

Both the TDRS and PWRS codes are part of the Multimedia for this chapter. These two codes were initially developed in Fortran and have been translated to MATLAB for this edition. Both versions of each are included in this edition.

12.5.1 Two-Dimensional Radiation and Scattering

The two-dimensional radiation and scattering (TDRS) program is used to analyze four different two-dimensional perfectly electric conducting problems: the strip, and the circular, elliptical, and

rectangular cylinders. The algorithm is based on the electric field integral equation of Section 12.3.1, and it is used for both electric and magnetic line-source excitation, or plane wave incidence of arbitrary polarization. For simplicity, piecewise constant pulse expansion functions and point-matching techniques have been adopted.

A. Strip For the strip problem, the program can analyze either of the following:

1. A line source (electric or magnetic). It computes the electric current density over the width of the strip and the normalized radiation amplitude pattern (in decibels) for $0^\circ \leq \phi \leq 360^\circ$. The user must specify the width of the strip (in wavelengths), the type of line source (either electric or magnetic), and the location x_s, y_s of the source (in wavelengths).
2. Plane wave incidence of arbitrary polarization. The program can analyze either monostatic or bistatic scattering.

For monostatic scattering, the program computes the two-dimensional normalized (with respect to λ) monostatic SW σ_{2-D}/λ (in decibels) for all angles of incidence ($0^\circ \leq \phi \leq 360^\circ$). The program starts at $\phi = 0^\circ$ and then completes the entire 360° monostatic scattering pattern. The user must specify the width w of the strip (in wavelengths) and the polarization angle θ_p (in degrees) of the incident plane wave. The polarization of the incident wave is specified by the direction θ_p of the incident electric field relative to the z axis ($\theta_p = 0^\circ$ implies TM^z ; $\theta_p = 90^\circ$ implies TE^z ; other values of θ_p represent an arbitrary polarization). The polarization angle θ_p needs to be specified only when the polarization is neither TM^z nor TE^z .

For bistatic scattering, the program computes for the specified incidence angle the current density over the width of the strip and the two-dimensional normalized (with respect to λ) bistatic SW σ_{2-D}/λ (in decibels) for all angles of observation ($0^\circ \leq \phi_s \leq 360^\circ$). The user must specify the width w of the strip (in wavelengths), the angle of incidence ϕ_i (in degrees), and the polarization angle θ_p (in degrees) of the incident plane wave. The polarization angle of the incident wave is specified in the same manner as for the monostatic case.

B. Circular, Elliptical, or Rectangular Cylinder For the cylinder program, the program can analyze either a line source (electric or magnetic) or plane wave scattering of arbitrary polarization by a two-dimensional circular, elliptical, or rectangular cylinder.

1. For the line-source excitation, the program computes the current distribution over the entire surface of the cylinder and the normalized radiation amplitude pattern (in decibels). The user must specify, for each cylinder, the type of line source (electric or magnetic), the location x_s, y_s of the line source, and the size of the cylinder. For the circular cylinder, the size is specified by its radius a (in wavelengths) and for the elliptical and rectangular cylinders by the principal semiaxes lengths a and b (in wavelengths), with a measured along the x axis and b along the y axis.
2. For the plane wave incidence, the program computes monostatic or bistatic scattering of arbitrary polarization by a circular, elliptical, or rectangular cylinder.

For monostatic scattering, the program computes the two-dimensional normalized (with respect to λ) monostatic SW σ_{2-D}/λ (in decibels) for all angles of incidence ($0^\circ \leq \phi \leq 360^\circ$). The program starts at $\phi = 0^\circ$ and then computes the entire 360° monostatic scattering pattern. The user must specify the size of the cylinder, as was done for the line-source excitation, and the polarization angle θ_p (in degrees) of the incident plane wave. The polarization of the incident wave is specified by the direction θ_p of the incident electric field relative to the z axis ($\theta_p = 0^\circ$ implies TM^z ; $\theta_p = 90^\circ$ implies TE^z ; other values of θ_p represent an arbitrary polarization). The polarization angle θ_p needs to be specified only when the polarization is neither TM^z nor TE^z .

For bistatic scattering, the program computes for the specified incidence angle the current density over the entire surface of the cylinder and the two-dimensional normalized (with respect to λ) bistatic SW σ_{2-D} / λ (in decibels) for all angles of observation ($0^\circ \leq \phi_s \leq 360^\circ$). The user must specify the size of the cylinder, as was done for the line-source excitation, the incidence angle ϕ_i (in degrees), and the polarization angle θ_p (in degrees) of the incident plane wave. The polarization angle of the incident wave is specified in the same manner as for the monostatic case.

12.5.2 Pocklington's Wire Radiation and Scattering

Pocklington's wire radiation and scattering (PWRS) program computes the radiation characteristics of a center-fed wire antenna and the scattering characteristics of a perfectly electric conducting (PEC) wire, each of radius a and length ℓ . Both are based on Pocklington's integral equation 12-123.

A. Radiation For the wire antenna of Figure 12-26, the excitation is modeled by either a delta gap or a magnetic frill feed modeling, and it computes the current distribution, normalized amplitude radiation pattern, and the input impedance. The user must specify the length of the wire, its radius (both in wavelengths), and the type of feed modeling (delta gap or magnetic frill). A computer program based on Hallén's integral equation can be found in [24].

B. Scattering The geometry for the plane wave scattering by the wire is shown in Figure 12-24(a). The program computes the monostatic or bistatic scattering of arbitrary polarization.

For monostatic scattering the program computes the normalized (with respect to m^2) RCS σ_{3-D} / m^2 (in dBsm) for all angles of incidence ($0^\circ \leq \theta_i \leq 180^\circ$). The program starts at $\theta_i = 0^\circ$ and then computes the entire 180° monostatic scattering pattern. The user must specify the length and radius of the wire (both in wavelengths) and the polarization angle θ_p (in degrees) of the incident plane wave. The polarization of the incident wave is specified by the direction θ_p of the incident electric field relative to the plane of incidence, where the plane of incidence is defined as the plane that contains the vector of the incident wave and the wire scatterer ($\theta = 0^\circ$ implies that the electric field is on the plane of incidence; $\theta = 90^\circ$ implies that the electric field is perpendicular to the plane of incidence and to the wire; thus no scattering occurs for this case).

For bistatic scattering, the program computes for the specified incidence angle the current distribution over the length of the wire and the normalized (with respect to m^2) bistatic RCS σ_{3-D} / m^2 (in dBsm) for all angle of observation ($0^\circ \leq \theta_s \leq 180^\circ$). The user must specify the length and radius of the wire (both in wavelengths), the angle of incidence θ_i (in degrees), and the polarization angle θ_p of the incident plane wave. The polarization angle is specified in the same manner as for the monostatic case.

12.5.3 Numerical Electromagnetics Code

Over the years there have been a number of computational electromagnetic codes developed, both personal and commercial, based on the Integral Equation/Method of Moments method. An attempt is made here to indicate the genesis of EM code development. The process started with the development of the Numerical Electromagnetic Code (NEC) [39] and the Mini-Numerical Electromagnetic Code (MININEC) [40, 41]. The NEC code analyzes the interaction of electromagnetic waves with arbitrary structures consisting of conducting wires and surfaces. It uses the EFIE for thin wires and the MFIE for surfaces. The initial MININEC was a user-oriented compact version of the NEC, and it was coded in BASIC. Since the initial introduction of these two codes, there have been various versions of them.

After the NEC code, the Electromagnetic Surface Patch (ESP) code [42] was introduced. The ESP is a method of moments surface patch code based on the piecewise sinusoidal reaction formulation, which is basically equivalent to the EFIE. It can be used for the analysis of the radiation and scattering from 3-D geometries consisting of an interconnection of thin wires, perfectly conducting (or thin dielectric) polygonal plates, thin wires, wire/plate and plate/plate junctions, and polygonal dielectric material volumes. Numerous other codes, including [43], based on IE/MoM have been developed since the NEC and ESP. Even as of the writing of this edition of the book, there are efforts underway for the development of other personal and commercial codes. They are too numerous to mention all of them here. The reader is directed to the internet for the search of such, and other, electromagnetic codes. Some basic student-oriented MATLAB codes have been developed to complement the material of this chapter. They are listed under the 12.6 Multimedia section, and they are available to the reader on the website that accompanies this book.

12.6 MULTIMEDIA

On the website that accompanies this book, the following multimedia resources are included for the review, understanding, and presentation of the material of this chapter.

- **MATLAB** computer programs:
 - a. **Wire_Charge:** Computes the charge distribution on a straight or bent PEC wire, of Figures 12-1 and 12-3, based on the Integral Equation (IE) of Section 12.2.
 - b. **PEC_Strip_Line_MoM:** Computes the far-zone amplitude radiation pattern and current density of a line source above a PEC strip, of finite width, based on the Integral Equation (IE) of Sections 12.2.2 through 12.2.8 and Physical Optics (PO) of Section 11.2.3, and Figures 12-5 and 11-2.
 - c. **PEC_Strip_SW_MoM:** Computes the TM^z and TE^z 2-D scattering width (SW), monostatic and bistatic, and current density of a uniform plane wave incident upon a PEC strip, of finite width, based on the Integral Equations (IE) of Sections 12.3.1 and 12.3.2, and Physical Optics of Section 11.3.1, and Figures 12-13 and 11-4.
 - d. **TDRS:** Computes the radiation and scattering of a plane wave incident of a 2-D scatterer (strip, cylinder) based on the Integral Equation (IE) as outlined in Section 12.4.1.
 - e. **PWRS:** Computes the radiation characteristics of a symmetrical dipole of Figure 12-25 or scatterer of Figure 12-23, based on Pocklington's Integral Equation (IE) of (12-13) in Section 12.4.1.
- **PowerPoint (PPT)** viewgraphs, in multicolor.

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PROBLEMS

- 12.1. A circular loop of radius $a = 0.2$ m is constructed out of a wire of radius $b = 10^{-3}$ m, as shown in Figure P12-1. The entire loop is maintained at a constant potential of 1 V. Using integral equation techniques, determine and plot for $0^\circ \leq \phi \leq 360^\circ$ the surface charge density on the wire. Assume that at any given angle the charge is uniformly distributed along the circumference of the wire.
- 12.2. Repeat Problem 12.1 when the loop is split into two parts; one part (from 0 to 180°) is maintained at a constant potential of 1 V and the other part (from 180 to 360°) is maintained at a constant potential of 2 V.
- 12.3. A linear charge ρ_c is distributed along a very thin wire circular loop of radius a which is discontinuous (open) for $355^\circ \leq \phi \leq 360^\circ$ and $0^\circ \leq \phi \leq 5^\circ$. The static potential V_z produced by this charge distribution on a very thin line along the z axis, passing through the origin of the circular loop is given by

$$V_z = \frac{10a}{\sqrt{a^2 + z^2}}, \quad 0 \leq z \leq 7 \text{ meters}$$

- (a) Subdivide the line for $0 \leq z \leq 7$ meters in 70 segments and compute the potential V_z at the center of each of the 70 segments when the radius of the loop is $a = 1$ meter.

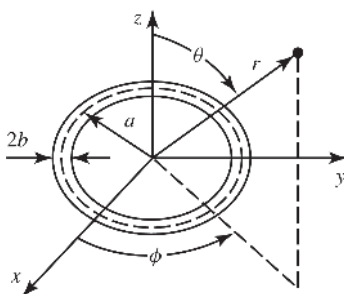


Figure P12-1

- (b) Plot the potential V_z of part (a) on a linear plot for $0 \leq z \leq 7$ meters.
- (c) Write an integral equation based on Poisson's differential equation to solve this problem.
- (d) For a loop of radius $a = 1$ meter, determine the linear charge ρ_c that produces the stated potential along $0 \leq z \leq 7$ meters that is computed in part (a) and plotted in part (b).
- (e) Plot the linear charge distribution of part (d) on a linear plot of $5^\circ \leq \phi \leq 355^\circ$.

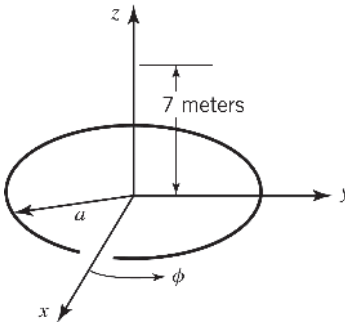


Figure P12-3

Use pulse expansion functions and $N = 70$ segments along the circumference of the loop.

- 12.4. Repeat Example 12-3 for a strip with $w = 2\lambda$, $h = 0.25\lambda$, and $t = 0.01\lambda$.
- 12.5. A magnetic line source of constant current I_m is placed a height h above a Perfect Magnetic Conducting (PMC) strip of width w . The geometry is the same as that of Figure 12-5. Write a normalized Integral Equation (IE) that can be used to solve for the linear magnetic current density M_z on the strip. You do not have to derive it as long as you justify it.
- 12.6. A TE^z uniform plane wave of the form shown in Figure 12-13b, with a normalized z component of the incident magnetic field of magnitude H_0 is incident upon a Perfect Magnetic Conducting (PMC) strip of finite width w . Write a normalized Integral Equation (IE) that can be used to solve for the magnetic current M_z which is induced on the PMC strip. You do not have to derive it as long as you justify it.
- 12.7. Derive (12-71a) and (12-71b).

12.8. Instead of using the electric field components of (12-72a) and (12-72b) to formulate the two-dimensional SW of a PEC strip for TE^z polarization, derive an integral expression for H_z^s in terms of J_x and then use the definition for SW of (11-21c). This requires only one component of the scattered H-field while using the definition of (11-21b) requires two scattered electric field components as given by (12-71a) and (12-71b).

12.9. An infinite electric line source of constant current I_e is placed next to a circular conducting cylinder of radius a , as shown in Figure P12-9. The line source is positioned a distance b ($b > a$) from the center of the cylinder. Use the EFIE, piecewise constant subdomain basis functions, and point-matching techniques.

- (a) Formulate the current density induced on the surface of the cylinder.
- (b) Compute the induced current density when $a = 5\lambda$ and $b = 5.25\lambda$. Assume a unity line-source current. Compare with the modal solution current density of (11-168a).
- (c) For part (b), compute the normalized far-zone amplitude pattern (in decibels). Normalize so that the maximum is 0 dB.

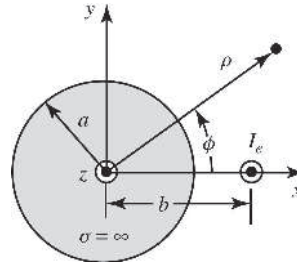


Figure P12-9

- 12.10. Repeat Problem 12.9 for $a = 5\lambda$ and $b = 5.5\lambda$.
- 12.11. An infinite electric line source of constant current I_e is placed next to a rectangular cylinder of dimensions a and b , as shown in Figure P12-11. The line source is positioned a distance c ($c > a$) from the center of the cylinder along the x axis. Use the EFIE and piecewise subdomain basis functions and point-matching techniques, and do the following.

- (a) Compute the induced current on the surface of the cylinder when $a = 5\lambda$, $b = 2.5$, and $c = 5.25\lambda$. Assume a unity line-source current.
- (b) Compute for part (a) the normalized far-zone amplitude pattern (in decibels). Normalize so that the maximum is 0 dB.

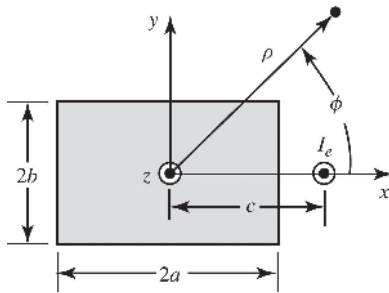


Figure P12-11

- 12.12.** Repeat Problem 12.11 for an electric line source near an elliptic cylinder with $a = 5\lambda$, $b = 2.5\lambda$, and $c = 5.25\lambda$.

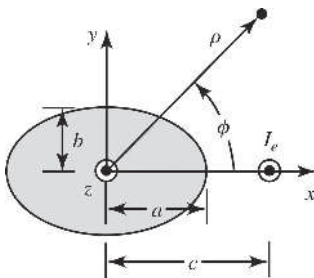


Figure P12-12

- 12.13.** A TM^z uniform plane wave traveling in the $+x$ direction is normally incident upon a conducting circular cylinder of radius a , as shown in Figure P12-13. Using the EFIE, piecewise constant subdomain basis functions, and point-matching techniques, write your own program, and do the following.
- (a) Plot the current density induced on the surface of the cylinder when $a = 2\lambda$. Assume the incident field is of unity amplitude. Compare with the modal solution current density of (11-97).
- (b) Plot the normalized σ_{2-D}/λ bistatic scattering width (in decibels) for $0^\circ \leq \phi \leq 360^\circ$ when $a = 2\lambda$. Compare with the modal solution of (11-102).

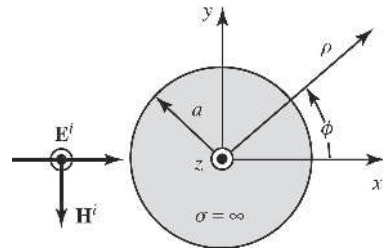


Figure P12-13

- 12.14.** A TM^z uniform plane wave traveling in the $+x$ direction is normally incident upon a conducting rectangular cylinder of dimensions a and b , as shown in Figure P12-14. Using the EFIE, piecewise constant subdomain basis functions, and point-matching techniques, write your own program, and do the following.
- (a) Compute the induced current density on the surface of the cylinder when $a = 5\lambda$ and $b = 2.5\lambda$. Assume a unity line-source current.
- (b) For part (a), compute and plot the two-dimensional normalized σ_{2-D}/λ bistatic scattering width (in dB) for $0^\circ \leq \phi \leq 360^\circ$.

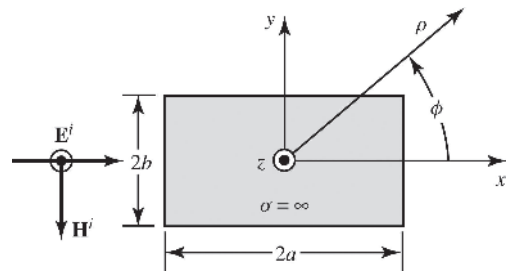


Figure P12-14

- 12.15.** Repeat Problem 12.14 for a TM^z uniform plane wave impinging upon an elliptic conducting cylinder with $a = 5\lambda$ and $b = 2.5\lambda$.

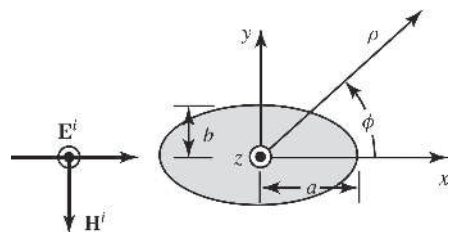


Figure P12-15

- 12.16.** Using the geometry of Figure 12-14, verify (12-71a) and (12-71b), and that (12-70a) reduces to (12-72a) and (12-70b) to (12-72b).
- 12.17.** Show that the integral of (12-78d) can be evaluated using (12-79a) through (12-79c).
- 12.18.** Repeat Problem 12.9 for a magnetic line source of constant current $I_m = 1$ when $a = b = 5\lambda$. This problem is representative of a very thin axial slot on the surface of the cylinder. Compare the current density on the surface of the cylinder from part (b) with that of the modal solution of (11-117a).
- 12.19.** Repeat Problem 12.11 for a magnetic line source of constant current $I_m = 1$ when $a = 5\lambda$, $b = 2.5\lambda$, and $c = 5\lambda$. This problem is representative of a very thin axial slot on the surface of the cylinder.
- 12.20.** Repeat Problem 12.12 for a magnetic line source of constant current $I_m = 1$ when $a = 5\lambda$, $b = 2.5\lambda$, and $c = 5\lambda$. This problem is representative of a very thin axial slot on the surface of the cylinder.
- 12.21.** Repeat Problem 12.13 for a TE^z uniform plane wave of unity amplitude. Compare the current density with the modal solution of (11-113) and the normalized σ_{2-D} / λ bistatic scattering width with the modal solution of (11-117).
- 12.22.** Repeat Problem 2.14 for a TE^z uniform plane wave at unity amplitude.
- 12.23.** Repeat Problem 12.15 for a TE^z uniform plane wave of unity amplitude.
- 12.24.** Using the geometry of Figure 12-17, show that
$$\hat{\mathbf{n}} \cdot \nabla H_0^{(2)}(\beta R) = -\beta \cos \psi H_1^{(2)}(\beta R)$$
- 12.25.** Repeat Problem 12.9 using the MFIE.
- 12.26.** Repeat Problem 12.13 using the MFIE. You must write your own computer program to solve this problem.
- 12.27.** Using the geometry of Figure 12-18, show that
$$\hat{\mathbf{n}}' \cdot \nabla H_0^{(2)}(\beta R) = -\beta \cos \psi' H_1^{(2)}(\beta R)$$
- 12.28.** Repeat Problem 12.18 using the MFIE.
- 12.29.** Derive Pocklington's integral equation 12-123 using (12-121) and (12-122).
- 12.30.** Derive the solution of (12-125) to the differential equation (12-124a). Show that Hallén's integral equation can be written as (12-126).
- 12.31.** Show that the incident tangential electric field (E_z^i) generated on the surface of a wire of radius a by a magnetic field generator of (12-129) is given by (12-130).
- 12.32.** Reduce (12-130) to (12-131) valid only along the z axis ($\rho = 0$).
- 12.33.** For the center-fed dipole of Example 12-7, write the $[Z]$ matrix for $N = 21$ using for the gap the delta-gap generator and the magnetic frill generator.
- 12.34.** For an infinitesimal center fed dipole of $\ell = \lambda / 50$ and radius $a = 0.005\lambda$, derive the input impedance using Pocklington's integral equation with piecewise constant subdomain basis functions and point matching. Use $N = 21$ and model the gap as a delta-gap generator and as a magnetic-frill generator. Use the MATLAB **PWRS** computer program listed in Multimedia at the end of the chapter.
- 12.35.** A conducting wire of length $\ell = 0.47\lambda$ and radius $a = 0.005\lambda$ is placed symmetrically along the z axis. Assuming a TM^z uniform plane wave is incident on the wire at an angle $\theta_i = 30^\circ$ from the z axis, do the following.
 (a) Compute and plot the current induced on the surface of the wire.
 (b) Compute and plot the bistatic RCS for $0^\circ \leq \theta_s \leq 180^\circ$.
 (c) Compute and plot the monostatic RCS for $0^\circ \leq \theta_i = \theta_s \leq 180^\circ$.
 The amplitude of the incident electric field is 10^{-3} V/m. Use Pocklington's integral equation and the **PWRS** computer program. Determine the number of segments that leads to a stable solution.
- 12.36.** Repeat Problem 12.35 for a TE^z uniform plane wave incidence.

CHAPTER 13



Geometrical Theory of Diffraction

13.1 INTRODUCTION

The treatment of the radiation and scattering characteristics from radiating and scattering systems using modal solutions is limited to objects whose surfaces can be described by orthogonal curvilinear coordinates. Moreover, most of the solutions are in the form of infinite series, which are poorly convergent when the dimensions of the object are greater than about a wavelength. These limitations, therefore, exclude rigorous analyses of many practical radiating and scattering systems.

A method that describes the solution in the form of an integral equation has received considerable attention. Whereas arbitrary shapes can be handled by this method, it mostly requires the use of a digital computer for numerical computations and therefore, is most convenient for objects that are not too many wavelengths in size because of the capacity limitations of computers. This method is usually referred to as the *integral equation* (IE) method, and its solution is generally accomplished by the *moment method* (MM) [1–4]. These were discussed in Chapter 12.

When the dimensions of the radiating or scattering object are many wavelengths, high-frequency asymptotic techniques can be used to analyze many problems that are otherwise mathematically intractable. Two such techniques, which have received considerable attention in the past few years, are the *geometrical theory of diffraction* (GTD) and the *physical theory of diffraction* (PTD). The GTD, originated by Keller [5, 6] and extended by Kouyoumjian and Pathak [7–10], is an extension of the classical *geometrical optics* (GO) (direct, reflected, and refracted rays), and it overcomes some of the limitations of geometrical optics by introducing a diffraction mechanism [11]. The PTD, introduced by Ufimtsev [12–14], supplements *physical optics* (PO) to provide corrections that are due to diffractions at edges of conducting surfaces. Ufimtsev suggested the existence of nonuniform (“fringe”) edge currents in addition to the uniform physical optics surface currents [15–18]. The PTD bears some resemblance to GTD in its method of application.

At high frequencies, diffraction—like reflection and refraction—is a local phenomenon and it depends on two things:

1. The geometry of the object at the point of diffraction (edge, vertex, curved surface).
2. The amplitude, phase, and polarization of the incident field at the point of diffraction.

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A field is associated with each diffracted ray, and the total field at a point is the sum of all the rays at that point. Some of the diffracted rays enter the shadow regions and account for the field intensity there. The diffracted field, which is determined by a generalization of *Fermat's principle* [6, 7], is initiated at points on the surface of the object that create a discontinuity in the incident GO field (incident and reflected shadow boundaries).

The phase of the field on a ray is assumed to be equal to the product of the optical length of the ray from some reference point and the wave number of the medium. Appropriate phase jumps must be added as rays pass through caustics (defined in Section 13.2.1). The amplitude is assumed to vary in accordance with the principle of conservation of energy in a narrow tube of rays.

The initial value of the field on a diffracted ray is determined from the incident field with the aid of an appropriate diffraction coefficient that is a dyadic for electromagnetic fields. This is analogous to the manner reflected fields are determined using the reflection coefficient. The rays also follow paths that make the optical distance from the source to the observation point an extremum (usually a minimum). This leads to straight-line propagation within homogeneous media and along geodesics (surface extrema) on smooth surfaces. The field intensity also attenuates exponentially as it travels along surface geodesics.

The diffraction and attenuation coefficients are usually determined from the asymptotic solutions of the simplest boundary-value problems, which have the same local geometry at the points of diffraction as the object at the points of interest. Geometries of this type are referred to as *canonical* problems. One of the simplest geometries that will be discussed in this chapter is a conducting wedge. The primary objective in using the GTD is to resolve each problem to smaller components [19–25], each representing a canonical geometry with a known solution. The ultimate solution is a superposition of the contributions from each canonical problem.

Some of the advantages of GTD are given in the following list.

1. It is simple to apply.
2. It can be used to solve complicated problems that do not have exact solutions.
3. It provides physical insight into the radiation and scattering mechanisms from the various parts of the structure.
4. It yields accurate results that compare quite well with experiments and other methods.
5. It can be combined with other techniques such as the moment method [26–28].

One of the main interests of diffraction by wedges is that engineers and scientists have investigated how the shape and material properties of complex structures affect their backscattering characteristics. The attraction in this area is primarily aimed toward designs of low-profile (stealth) technology by using appropriate shaping along with lossy or coated materials to reduce the radar visibility, as represented by radar cross section (RCS), of complex radar targets, such as aircraft, spacecraft, and missiles. A good example is the F-117 shown in Figure 13-1, whose surface is primarily structured by a number of faceted flat plates and wedges because, as will become evident from the developments, formulations, examples, and problems of this chapter (see also Problem 13.50), the backscatter from exterior wedges is lower than that of convex curved surfaces. In addition, the plates are oriented judiciously so that the maximum scattered field is toward the specular direction and away from the source of detection. While in this chapter we will focus on the diffraction by PEC wedges, the diffraction by wedges with impedance surfaces, to represent lossy and composite wedge surfaces, is the subject of Chapter 14.

13.2 GEOMETRICAL OPTICS

Geometrical optics (GO) is an approximate high-frequency method for determining wave propagation for incident, reflected, and refracted fields. Because it uses ray concepts, it is often referred



Figure 13-1 F-117 Nighthawk. (Printed with permission of Lockheed Martin Corporation © 2010).

to as *ray optics*. Originally, geometrical optics was developed to analyze the propagation of light at sufficiently high frequencies where it was not necessary to consider the wave nature of light. Instead, the transport of energy from one point to another in an isotropic lossless medium is accomplished using the conservation of energy flux in a tube of rays. For reflection problems, geometrical optics approximates the scattered fields only toward specular directions as determined by Snell's law of reflection: the angle of reflection is equal to the angle of incidence. For sufficiently high frequencies, geometrical optics fields may dominate the scattering phenomena and may not require any corrections. This is more evident for backscattering from smooth curved surfaces whose curvature is large compared to the wavelength.

According to classical geometrical optics, the rays between any two points P_1 and P_2 follow a path that makes the optical distance between them an extremum (usually a minimum). In equation form, this is expressed as

$$\delta \int_{P_1}^{P_2} n(s) ds = 0 \quad (13-1)$$

where δ represents what is referred to in the calculus of variations as the *variational differential* and $n(s)$ is the index of refraction of the medium, $\beta(s)/\beta_0 = n(s)$. If the medium is homogeneous, $n(s) = n = \text{constant}$, the paths are straight lines. Equation 13-1 is a mathematical representation of *Fermat's principle*. In addition, the light intensity, power per unit solid angle, between any two points is also governed by the conservation of energy flux in a tube of rays.

To demonstrate the principles of geometrical optics, let us consider a *primary wave front* surface ψ_0 , as shown in Figure 13-2, formed at $t = t_0$ by the motion of light propagating in an isotropic lossless medium. The objectives here are:

1. To determine the *secondary wave front* surfaces ψ_n formed at $t = t_{n+1} > t_n$, $n = 0, 1, 2, 3, \dots$
2. To relate the power density and field intensity on the secondary wave fronts to those of the primary or previous wave fronts.

The secondary wave fronts can be determined by first selecting a number of discrete points on the primary wave front. If the medium of wave propagation is also assumed to be homogeneous, ray paths from the primary to the secondary wave front are drawn as straight lines that at each point are normal to the surface of the primary wave front.

Since the wave travels in the medium with the speed of light given by $v = c/n$, where c is the speed of light in free space and n is the index of refraction, then at $\Delta t = t_1 - t_0$ ($t_1 > t_0$) the wave would have traveled a distance $\Delta \ell = v \Delta t$. Along each of the normal rays a distance $\Delta \ell$ is marked,

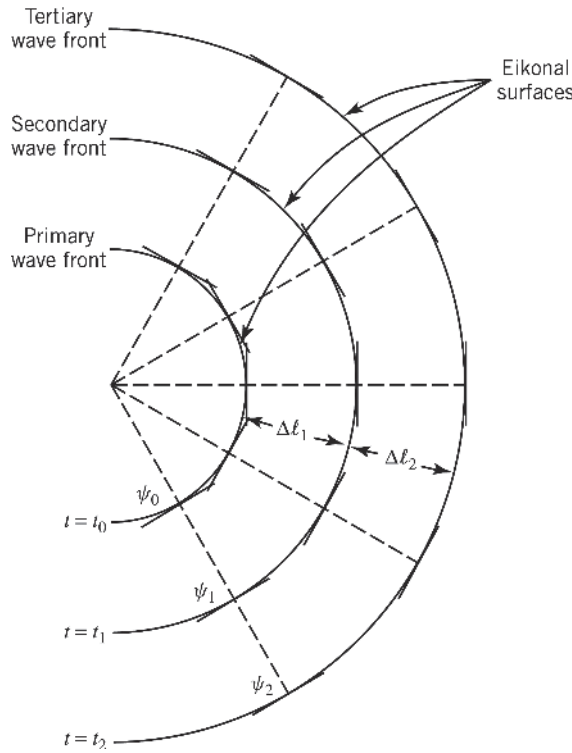


Figure 13-2 Primary and secondary wave front (eikonal surfaces) of a radiated wave.

and a surface perpendicular to each ray is drawn. The surfaces normal to each of the rays are then connected to form the secondary wave front ψ_1 , as shown in Figure 13-2. The same procedure can be repeated to determine the subsequent wave front surfaces $\psi_2, \psi_3, \dots$

The family of wave front surfaces $\psi_n(x, y, z)$, $n = 0, 1, 2, 3, \dots$, that are normal to each of the radial rays is referred to as the *eikonal* surfaces, and they can be determined using the *eikonal equation* [7]

$$\|\nabla \psi_n(x, y, z)\|^2 = \left\{ \frac{\partial \psi_n}{\partial x} \right\}^2 + \left\{ \frac{\partial \psi_n}{\partial y} \right\}^2 + \left\{ \frac{\partial \psi_n}{\partial z} \right\}^2 = n^2(s) \quad (13-2)$$

Since the rays normal to the wave fronts and the eikonal surfaces are uniquely related, it is only necessary to know one or the other when dealing with geometrical optics.

Extending this procedure to approximate the wave motion of electromagnetic waves of lower frequencies, it is evident that:

1. The eikonal surfaces for plane waves are planar surfaces perpendicular to the direction of wave travel.
2. The eikonal surfaces for cylindrical waves are cylindrical surfaces perpendicular to the cylindrical radial vectors.
3. The eikonal surfaces for spherical waves are spherical surfaces perpendicular to the spherical radial vectors.

Each of these is demonstrated, respectively, in Figures 13-3a, 13-3b, and 13-3c.

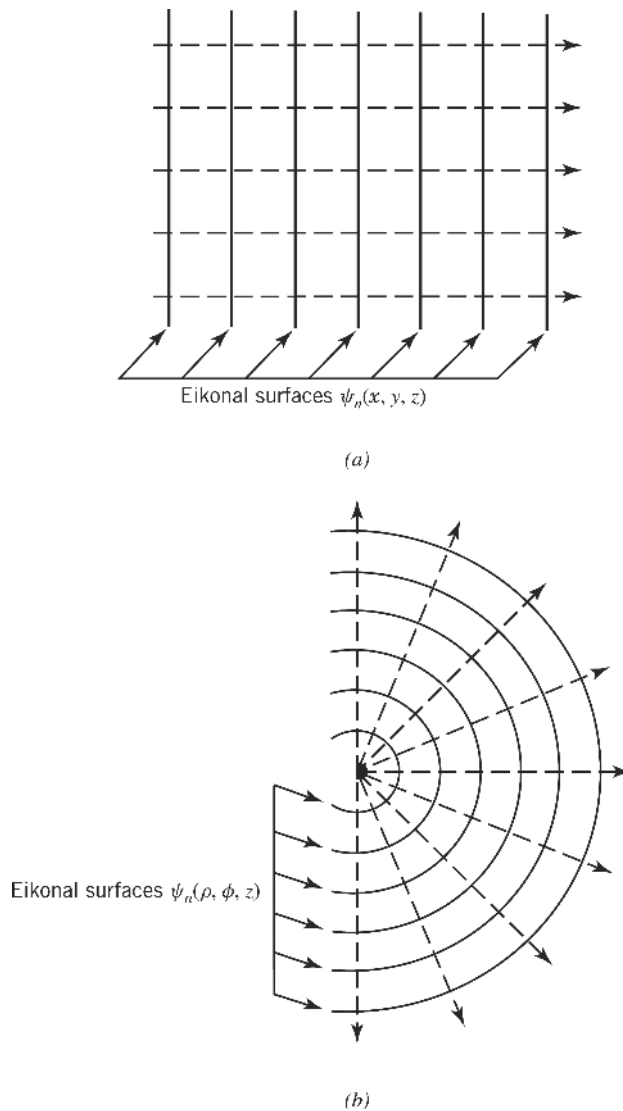


Figure 13-3 Eikonal surfaces. (a) Plane waves. (b) Cylindrical waves. (c) Spherical waves.

13.2.1 Amplitude Relation

In geometrical optics, the light intensity (power per unit solid angle) between two points is also governed by the conservation of energy flux in a tube of rays. To demonstrate that, let us assume that a point source, as shown in Figure 13-4, emanates isotropically spherical waves. Within a tube of rays, the cross-sectional areas at some reference point $s = 0$ and at s are given, respectively, by dA_0 and dA . The radiation density S_0 at $s = 0$ is related to the radiation density S at s by

$$S_0 dA_0 = S dA \quad (13-3)$$

or

$$\frac{S(s)}{S_0(0)} = \frac{dA_0}{dA} \quad (13-3a)$$

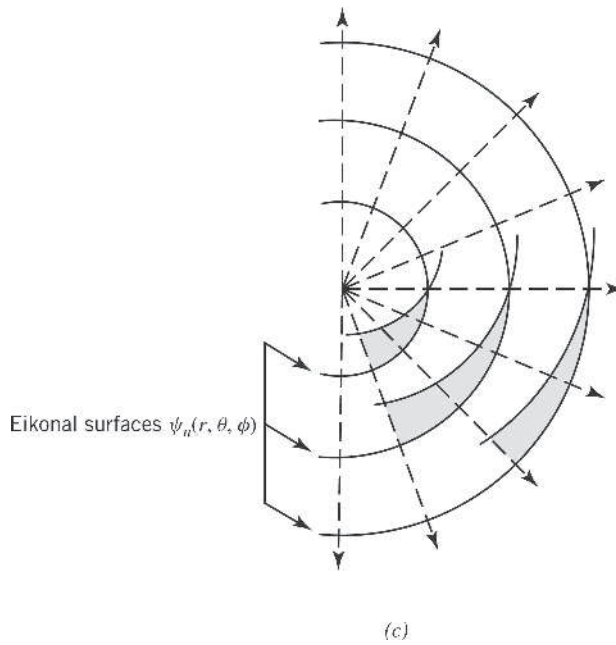


Figure 13-3 (Continued)

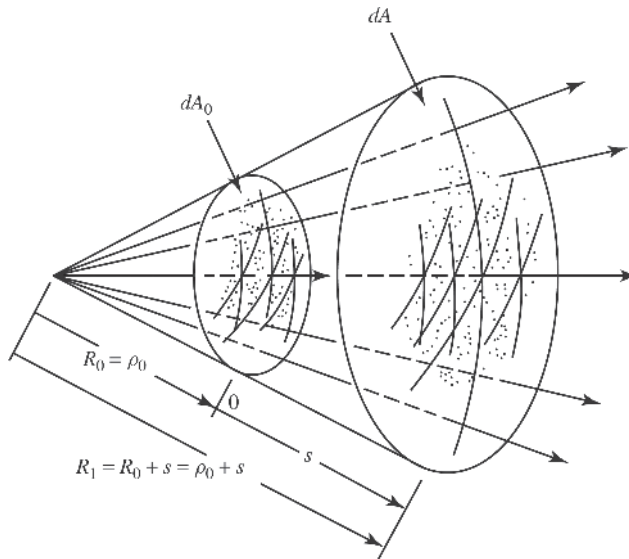


Figure 13-4 Tube of rays for a spherical radiated wave.

It has been assumed that S_0 and S are constant, respectively, throughout the cross-sectional areas dA_0 and dA , and that no power flows across the sides of the conical tube.

For electromagnetic waves, the far-zone electric field $\mathbf{E}(r, \theta, \phi)$ is related to the radiation density $S(r, \theta, \phi)$ by [7]

$$S(r, \theta, \phi) = \frac{1}{2\eta} |\mathbf{E}(r, \theta, \phi)|^2 = \frac{1}{2} \sqrt{\frac{\epsilon}{\mu}} |\mathbf{E}(r, \theta, \phi)|^2 \quad (13-4)$$

Therefore, (13-3a) can also be written, using (13-4), as

$$\frac{|\mathbf{E}|^2}{|\mathbf{E}_0|^2} = \frac{dA_0}{dA} \quad (13-5)$$

or

$$\frac{|\mathbf{E}|}{|\mathbf{E}_0|} = \sqrt{\frac{dA_0}{dA}} \quad (13-5a)$$

Since in the tube of rays in Figure 13-4, the differential surface areas dA_0 and dA are patches of spherical surfaces with radii of $R_0 = \rho_0$ and $R_1 = R_0 + s = \rho_0 + s$, respectively, then (13-5a) can be written in terms of the radii of curvature of the wave fronts at $s=0$ and s . Thus, (13-5a) reduces to

$$\frac{|\mathbf{E}|}{|\mathbf{E}_0|} = \sqrt{\frac{dA_0}{dA}} = \sqrt{\frac{4\pi R_0^2/C_0}{4\pi R_1^2/C_0}} = \frac{R_0}{R_1} = \frac{\rho_0}{\rho_0 + s} \quad (13-6)$$

and it indicates that the electric field varies, as expected, inversely proportional to the distance of travel; C_0 is a proportionality constant.

If the eikonal surfaces of the radiated fields are cylindrical surfaces, representing the wave fronts of cylindrical waves, then the field relation (13-5a) takes the form

$$\frac{|\mathbf{E}|}{|\mathbf{E}_0|} = \sqrt{\frac{dA_0}{dA}} = \sqrt{\frac{2\pi R_0/C_1}{2\pi R_1/C_1}} = \sqrt{\frac{R_0}{R_1}} = \sqrt{\frac{\rho_0}{\rho_0 + s}} \quad (13-7)$$

where C_1 is a proportionality constant. Relation (13-7) indicates that the electric field for cylindrical waves varies, as expected, inversely to the square root of the distance of travel. For planar eikonal surfaces, representing plane waves, (13-5a) simplifies to

$$\frac{|\mathbf{E}|}{|\mathbf{E}_0|} = 1 \quad (13-8)$$

For the previous three cases, the eikonal surfaces were, respectively, spherical, cylindrical, and planar. Let us now consider a more general configuration in which the eikonal surfaces (wave fronts) are not necessarily spherical. This is illustrated in Figure 13-5a where the wave front is represented by a radius of curvature R_1 in the xz and R_2 in the yz planes, which are not equal ($R_1 \neq R_2$). To determine the focusing characteristics of such a surface, let us trace the focusing diagram of rays 1, 2, 3, and 4 from the four corners of the wave front. It is apparent that the rays focus (cross) at different points. For this example, rays 1 and 2 focus at P , rays 3 and 4 focus at P' , rays 2 and 3 focus at Q' , and rays 1 and 4 focus at Q . This system of a tube of rays is referred to as *astigmatic* (not meeting at a single point) and the lines PP' and QQ' are called *caustics*.¹

Referring to the geometry of Figure 13-5b, it can be shown that for a wave whose eikonal surface (wave front) forms an astigmatic tube of rays the electric field intensity from one surface relative to that of another, as related by (13-5a), takes the form

$$\frac{|\mathbf{E}|}{|\mathbf{E}_0|} = \sqrt{\frac{dA_0}{dA}} = \sqrt{\frac{\rho_1 \rho_2}{(\rho_1 + s)(\rho_2 + s)}} \quad (13-9)$$

¹ A *caustic* is a point, a line, or a surface through which all the rays of a wave pass. Examples of it are the focal point of a paraboloid (parabola of revolution) and the focal line of a parabolic cylinder. The field at a caustic is infinite, in principle, because an infinite number of rays pass through it.

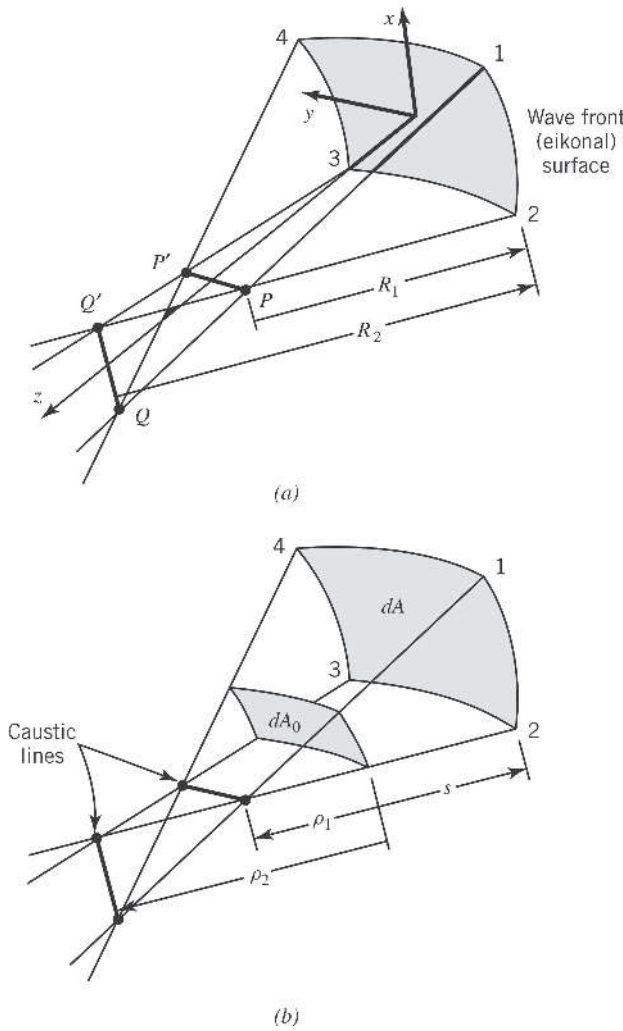


Figure 13-5 Astigmatic tube of rays. (a) Eikonal surface. (b) Caustic lines.

It is apparent that (13-9) reduces to the following equations:

1. (13-6) if the wave front is spherical ($\rho_1 = \rho_2 = \rho_0$).
2. (13-7) if the wave front is cylindrical ($\rho_1 = \infty, \rho_2 = \rho_0$ or $\rho_2 = \infty, \rho_1 = \rho_0$).
3. (13-8) if the wave front is planar ($\rho_1 = \rho_2 = \infty$).

Expressions (13-6) through (13-9) correctly relate the *magnitudes* of the high-frequency electric field at one wave front surface to that of another. These were derived using geometrical optics based on the principle of conservation of energy flux through a tube of rays. Although these may be valid high-frequency approximations for light waves, they are not accurate relations for electromagnetic waves of lower frequencies. Two apparent missing properties in these relations are those of *phase* and *polarization*.

13.2.2 Phase and Polarization Relations

Phase and polarization information can be introduced to the relations (13-6) through (13-9) by examining the approach introduced by Luneberg [29] and Kline [30, 31] to develop high-frequency solutions of electromagnetic problems. The works of Luneberg and Kline, referred to as the *Luneberg–Kline high-frequency expansion*, best bridge the gap between geometrical (ray) optics and wave propagation phenomena.

The Luneberg–Kline series expansion solution begins by assuming that the electric field for large ω can be written as a series

$$\mathbf{E}(\mathbf{R}, \omega) = e^{-j\beta_0\psi(\mathbf{R})} \sum_{m=0}^{\infty} \frac{\mathbf{E}_m(\mathbf{R})}{(j\omega)^m} \quad (13-10)$$

where $\mathbf{R}$ = position vector

β_0 = phase constant for free-space

Substituting (13-10) into the wave equation

$$\nabla^2 \mathbf{E} + \beta^2 \mathbf{E} = 0 \quad (13-11)$$

subject to Maxwell's equation

$$\nabla \cdot \mathbf{E} = 0 \quad (13-12)$$

it can be shown, by equating like powers of ω , that one obtains the following.

1. The eikonal equation 13-2 or

$$\|\nabla\psi\|^2 = n^2 \quad (13-13a)$$

where ψ = eikonal (wave front) surface

n = index of refraction

2. The transport equations

$$\frac{\partial \mathbf{E}_0}{\partial s} + \frac{1}{2} \left\{ \frac{\nabla^2 \psi}{n} \right\} \mathbf{E}_0 = 0 \quad \text{for first-order terms} \quad (13-13b)$$

$$\frac{\partial \mathbf{E}_m}{\partial s} + \frac{1}{2} \left\{ \frac{\nabla^2 \psi}{n} \right\} \mathbf{E}_m = \frac{v_p}{2} \nabla^2 \mathbf{E}_{m-1} \quad \text{for higher-order terms} \quad (13-13c)$$

where $m = 1, 2, 3 \dots$

v_p = speed of light in medium

3. The conditional equations

$$\hat{\mathbf{s}} \cdot \mathbf{E}_0 = 0 \quad \text{for first-order terms} \quad (13-13d)$$

$$\hat{\mathbf{s}} \cdot \mathbf{E}_m = v_p \nabla \cdot \mathbf{E}_{m-1} \quad \text{for higher-order terms} \quad (13-13e)$$

$$m = 1, 2, 3 \dots$$

where

$$\hat{\mathbf{s}} = \frac{\nabla \psi}{n} = \text{unit vector in the direction path (normal to the wave front } \psi) \quad (13-13f)$$

s = distance along the ray path

At the present time we are interested mainly in first-order solutions for the electric field of (13-10) that can be approximated and take the form of

$$\mathbf{E}(s) \simeq e^{-j\beta_0\psi(s)} \mathbf{E}_0(s=0) \quad (13-14)$$

Integrating the first-order transport equation 13-13b along s and referring to the geometry of Figure 13-5a, it can be shown that (13-14) can be written as [30, 31]

$$\mathbf{E}(s) \simeq \mathbf{E}_0(0) e^{-j\beta_0\psi(0)} \sqrt{\frac{\rho_1\rho_2}{(\rho_1+s)(\rho_2+s)}} e^{-j\beta s} \quad (13-15)$$

where $s=0$ is taken as a reference point. Since $\mathbf{E}_0(0)$ is complex, the phase term $e^{-j\beta_0\psi(0)}$ can be combined with $\mathbf{E}_0(0)$ and (13-15) rewritten as

$\mathbf{E}(s) = \underbrace{\mathbf{E}'_0(0)e^{j\phi_0(0)}}_{\text{Field at reference point (s=0)}} \underbrace{\sqrt{\frac{\rho_1\rho_2}{(\rho_1+s)(\rho_2+s)}}}_{\text{Spatial attenuation factor (divergence, spreading) factor}} \underbrace{e^{-j\beta s}}_{\text{Phase factor}}$	(13-15a)
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where $\mathbf{E}'_0(0)$ = field amplitude at reference point ($s=0$)
 $\phi_0(0)$ = field phase at reference point ($s=0$)

Comparing (13-15a) to (13-9), it is evident that the leading term of the Luneberg–Kline series expansion solution for large ω predicts the spatial attenuation relation between the electric fields of two points as obtained by classical geometrical optics, as given by (13-9), which ignores both the polarization and the wave motion (phase) of electromagnetic fields. It also predicts their phase and polarization relations, as given by (13-15a). Obviously, (13-15a) could have been obtained from (13-9) by artificially converting the magnitudes of the fields to vectors (to account for polarization) and by introducing a complex exponential to account for the phase delay of the field from $s=0$ to s . This was not necessary since (13-15a) was derived here rigorously using the leading term of the Luneberg–Kline expansion series for large ω subject to the wave and Maxwell's equations. It should be pointed out, however, that (13-15a) is only a high-frequency approximation and it becomes more accurate as the frequency approaches infinity. However, for many practical engineering problems, it does predict quite accurate results that compare well with measurements.

In principle, more accurate expressions to the geometrical optics approximation can be obtained by retaining higher-order terms $\mathbf{E}_1(\mathbf{R}_1), \mathbf{E}_2(\mathbf{R}_2), \dots$ in the Luneberg–Kline series expansion (13-10), and in the transport (13-13c) and conditional (13-13e) equations. However, such a procedure is very difficult. In addition, the resulting terms do not remove the discontinuities introduced by geometrical optics fields along the incident and reflection boundaries, and the method does not lend itself to other improvements in the geometrical optics, such as those of diffraction. Therefore, no such procedure will be pursued here.

It should be noted that when the observation point is chosen so that $s = -\rho_1$ or $s = -\rho_2$, (13-15a) possesses singularities representing the congruence of the rays at the caustic lines PP' and QQ' . Therefore, (13-15a) is not valid along the caustics and not very accurate near them, and it should not be used in those regions. Other methods should be utilized to find the fields at and near caustics [32–36]. In addition, it is observed that when $-\rho_2 < s < -\rho_1$ the sign in the $(\rho_1 + s)$ term of the

denominator of (13-15a) changes. Similar changes of sign occur in the $(\rho_1 + s)$ and $(\rho_2 + s)$ terms when $s < -\rho_2 < -\rho_1$. Therefore, (13-15a) correctly predicts $+90^\circ$ phase jumps each time a caustic is crossed in the direction of propagation.

13.2.3 Reflection from Surfaces

Geometrical optics can be used to compute high-frequency approximations to the fields reflected from surfaces, the directions of which are determined by Snell's law of reflection. To demonstrate the procedure, let us assume that a field impinges on a smooth conducting surface S , where it undergoes a reflection at point Q_R . This is illustrated in Figure 13-6a where $\hat{s}^i$ is the unit vector in the direction of incidence, $\hat{s}^r$ is the unit vector in the direction of reflection, $\hat{e}_\parallel^i, \hat{e}_\parallel^r$ are unit vectors, for incident and reflected electric fields, parallel to the planes of incidence and reflection, and $\hat{e}_\perp^i, \hat{e}_\perp^r$ are unit vectors, for incident and reflected electric fields, perpendicular to the planes of incidence and reflection. The plane of incidence is formed by the unit vector $\hat{n}$ normal to the surface at the point of reflection Q_R and the unit vector $\hat{s}^i$, and the plane of reflection is formed by the unit vectors $\hat{n}$ and $\hat{s}^r$. The angle of incidence θ_i is measured between $\hat{n}$ and $\hat{s}^i$ whereas θ_r is measured between $\hat{n}$ and $\hat{s}^r$, and they are equal ($\theta_i = \theta_r$).

The polarization unit vectors are chosen so that

$$\hat{e}_\perp^i \times \hat{s}^i = \hat{e}_\parallel^i \quad (13-16a)$$

$$\hat{e}_\perp^r \times \hat{s}^r = \hat{e}_\parallel^r \quad (13-16b)$$

and the incident and reflected electric fields can be expressed as

$$\mathbf{E}_0^i = \hat{e}_\parallel^i E_{0\parallel}^i + \hat{e}_\perp^i E_{0\perp}^i \quad (13-17a)$$

$$\mathbf{E}_0^r = \hat{e}_\parallel^r E_{0\parallel}^r + \hat{e}_\perp^r E_{0\perp}^r \quad (13-17b)$$

The incident and reflected fields at the point of reflection can be related by applying the boundary conditions of vanishing tangential components of the electric field at the point of reflection (Q_R). Doing this, we can write that

$$\mathbf{E}_0^r(s=0) = \mathbf{E}_0^i(Q_R) \cdot \bar{\mathbf{R}} = \mathbf{E}_0^i(Q_R) \cdot [\hat{e}_\parallel^i \hat{e}_\parallel^r - \hat{e}_\perp^i \hat{e}_\perp^r] \quad (13-18)$$

where $\mathbf{E}_0^r(s=0)$ = reflected field at the point of reflection (the reference point for the reflected ray is taken on the reflecting surface so that $s=0$)

$\mathbf{E}_0^i(Q_R)$ = incident field at the point of reflection Q_R

$\bar{\mathbf{R}}$ = dyadic reflection coefficient

In matrix notation, the reflection coefficient can be written as

$$\mathbf{R} = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \quad (13-19)$$

which is identical to the Fresnel reflection coefficients of electromagnetic plane waves reflected from plane, perfectly conducting surfaces. This is quite acceptable in practice since at high frequencies, reflection, as well as diffraction, is a local phenomenon, and it depends largely on the geometry of the surface in the immediate neighborhood of the reflection point. Therefore, near the reflection point Q_R , the following approximations can be made:

1. The reflecting surface can be approximated by a plane tangent at Q_R .
2. The wave front of the incident field can be assumed to be planar.

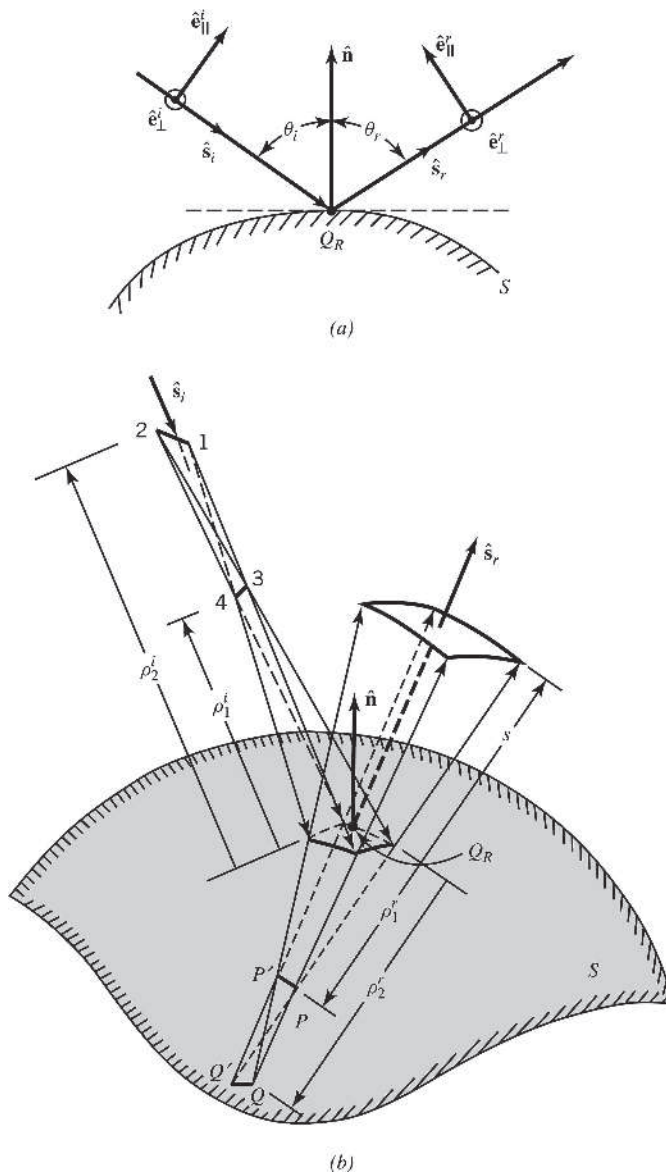


Figure 13-6 Reflection from a curved surface. (a) Reflection point. (b) Astigmatic tube of rays.

With the aid of (13-15a) and (13-18), it follows that the reflected field $\mathbf{E}^r(s)$ at a distance s from the point of reflection Q_R can be written as

$$\mathbf{E}^r(s) = \underbrace{\mathbf{E}^i(Q_R)}_{\text{Field at reference point } (Q_r)} \cdot \underbrace{\bar{\mathbf{R}}}_{\text{Reflection coefficient}} \underbrace{\sqrt{\frac{\rho_1^r \rho_2^r}{(\rho_1^r + s)(\rho_2^r + s)}}}_{\text{Spatial attenuation (divergence, spreading) factor}} \underbrace{e^{-j\beta s}}_{\text{Phase factor}} \quad (13-20)$$

where $\rho_1^r, \rho_2^r =$ principal radii of curvature of the reflected wave front at the point of reflection

The astigmatic tube of rays for the reflected fields is shown in Figure 13-6b where the reference surface is taken at the reflecting surface.

The principal radii of curvature of the reflected wave front, ρ_1^r and ρ_2^r , are related to the principal radii of curvature of the incident wave front, ρ_1^i and ρ_2^i , the aspect of wave incidence, and the curvature of the reflecting surface at Q_R . It can be shown that ρ_1^r and ρ_2^r can be expressed as [10]

$$\frac{1}{\rho_1^r} = \frac{1}{2} \left\{ \frac{1}{\rho_1^i} + \frac{1}{\rho_2^i} \right\} + \frac{1}{f_1} \quad (13-21a)$$

$$\frac{1}{\rho_2^r} = \frac{1}{2} \left\{ \frac{1}{\rho_1^i} + \frac{1}{\rho_2^i} \right\} + \frac{1}{f_2} \quad (13-21b)$$

where ρ_1^i, ρ_2^i = principal radii of curvature of incident wave front ($\rho_1^i = \rho_2^i = s'$ for spherical incident wave front; $\rho_1^i = \rho', \rho_2^i = \infty$ or $\rho_1^i = \infty, \rho_2^i = \rho'$ for cylindrical incident wave front, and $\rho_1^i = \rho_2^i = \infty$ for planar incident wave front)

Equations 13-21a and 13-21b are similar in form to the simple lens and mirror formulas of elementary physics. In fact, when the incident ray is spherical ($\rho_1^i = \rho_2^i = s'$), f_1 and f_2 represent focal distances that are independent of the source range that is creating the spherical wave.

When the incident field has a spherical wave front, $\rho_1^i = \rho_2^i = s'$, then f_1 and f_2 simplify to

$$\frac{1}{f_1} = \frac{1}{\cos \theta_i} \left\{ \frac{\sin^2 \theta_2}{R_1} + \frac{\sin^2 \theta_1}{R_2} \right\} + \sqrt{\frac{1}{\cos^2 \theta_i} \left\{ \frac{\sin^2 \theta_2}{R_1} + \frac{\sin^2 \theta_1}{R_2} \right\}^2 - \frac{4}{R_1 R_2}} \quad (13-22a)$$

$$\frac{1}{f_2} = \frac{1}{\cos \theta_i} \left\{ \frac{\sin^2 \theta_2}{R_1} + \frac{\sin^2 \theta_1}{R_2} \right\} - \sqrt{\frac{1}{\cos^2 \theta_i} \left\{ \frac{\sin^2 \theta_2}{R_1} + \frac{\sin^2 \theta_1}{R_2} \right\}^2 - \frac{4}{R_1 R_2}} \quad (13-22b)$$

where R_1, R_2 = radii of curvature of the reflecting surface

θ_1 = angle between the direction of the incident ray $\hat{s}^i$ and $\hat{u}_1$

θ_2 = angle between the direction of the incident ray $\hat{s}^i$ and $\hat{u}_2$

$\hat{u}_1$ = unit vector in the principal direction of S at Q_R with principal radius of curvature R_1

$\hat{u}_2$ = unit vector in the principal direction of S at Q_R with principal radius of curvature R_2

The geometrical arrangement of these is exhibited in Figure 13-7.

If the incident wave form is a plane wave, then $\rho_1^i = \rho_2^i = \infty$ and according to (13-21a) and (13-21b)

$$\frac{1}{\rho_1^r \rho_2^r} = \frac{1}{f_1 f_2} = \frac{4}{R_1 R_2} \quad (13-23)$$

or

$$\rho_1^r \rho_2^r = \frac{R_1 R_2}{4} \quad (13-23a)$$

The relation of (13-23a) is very useful to calculate the far-zone reflected fields. Now if either R_1 and/or R_2 becomes infinite, as is the case for flat plates or cylindrical scatterers, the geometrical optics field fails to predict the scattered field when the incident field is a plane wave.

To give some physical insight into the principal radii of curvature R_1 and R_2 of a reflecting surface, let us assume that the reflecting surface is well behaved (continuous and smooth). At each point on the surface there exists a unit normal vector. If through that point a plane intersects the reflecting surface, it generates on the reflecting surface a curve as shown in Figure 13-7b. If in addition the intersecting plane contains the unit normal to the surface at that point, the curve generated on the reflecting surface by the intersecting plane is known as the *normal section curve*, as shown in Figure 13-7b. If the intersecting plane is rotated about the surface normal at that point, a number of unique normal sections are generated, one on each orientation of the intersecting plane. Associated with each normal section curve, there is a radius of curvature. It can be shown that for each point on an arbitrary well-behaved curved reflecting surface, there is one intersecting plane that maximizes the radius of curvature of its corresponding normal section curve while there is another intersecting plane at the same point that minimizes the radius of curvature of its corresponding normal section curve. For each point on the reflecting surface there are two normal section radii of curvature, denoted here as R_1 and R_2 and referred to as *principal radii of curvature*, and the two corresponding planes are known as the *principal planes*. For an arbitrary surface, the two principal planes are perpendicular to each other. Also for each principal plane we can define unit vectors $\hat{\mathbf{u}}_1$ and $\hat{\mathbf{u}}_2$ that are tangent to each normal section curve generated

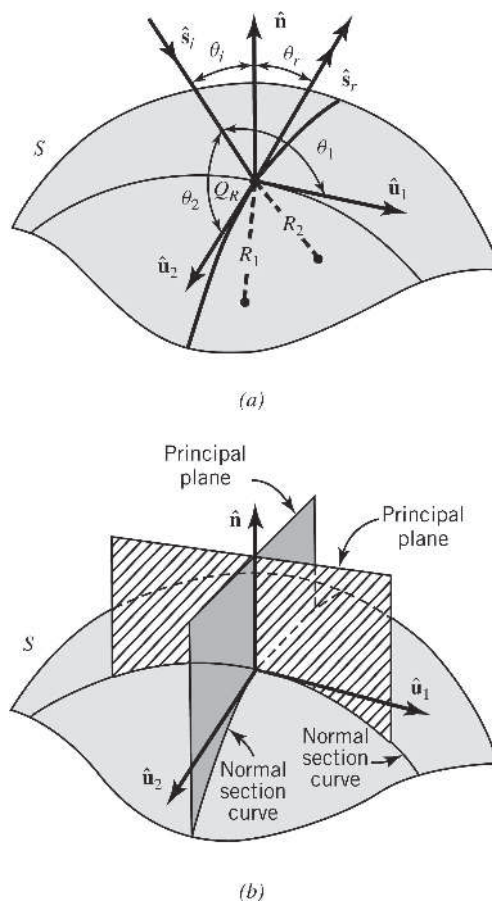


Figure 13-7 Geometry for reflection from a three-dimensional curved surface. (a) Principal radii of curvature. (b) Normal section curves and principal planes.

by each intersecting principal plane. Also the unit vectors $\hat{\mathbf{u}}_1$ and $\hat{\mathbf{u}}_2$, which lie in the principal plane, point along the principal directions whose normal section curves have radii of curvature R_1 and R_2 , respectively. Expressions for determining the principal unit vectors $\hat{\mathbf{u}}_1$, $\hat{\mathbf{u}}_2$ and the principal radii of curvature R_1 , R_2 of arbitrary surfaces of revolution are given in Problem 13.2 at the end of the chapter.

In general, however, f_1 and f_2 can be obtained using [10]

$$\begin{aligned} \frac{1}{f_{1(2)}} = & \frac{\cos \theta_i}{|\boldsymbol{\theta}|^2} \left(\frac{\theta_{22}^2 + \theta_{12}^2}{R_1} + \frac{\theta_{21}^2 + \theta_{11}^2}{R_2} \right) \\ & \pm \frac{1}{2} \left\{ \left(\frac{1}{\rho_1^i} - \frac{1}{\rho_2^i} \right)^2 + \left(\frac{1}{\rho_1^i} - \frac{1}{\rho_2^i} \right) \frac{4 \cos \theta_i}{|\boldsymbol{\theta}|^2} \left(\frac{\theta_{22}^2 - \theta_{12}^2}{R_1} + \frac{\theta_{21}^2 - \theta_{11}^2}{R_2} \right) \right. \\ & \left. + \frac{4 \cos^2 \theta_i}{|\boldsymbol{\theta}|^4} \left[\left(\frac{\theta_{22}^2 + \theta_{12}^2}{R_1} + \frac{\theta_{21}^2 + \theta_{11}^2}{R_2} \right)^2 - \frac{4 |\boldsymbol{\theta}|^2}{R_1 R_2} \right] \right\}^{1/2} \end{aligned} \quad (13-24)$$

where the plus sign is used for f_1 and the minus for f_2 . In (13-24), $|\boldsymbol{\theta}|$ is the determinant of

$$[\boldsymbol{\theta}] = \begin{vmatrix} \hat{\mathbf{X}}_1^i \cdot \hat{\mathbf{u}}_1 & \hat{\mathbf{X}}_1^i \cdot \hat{\mathbf{u}}_2 \\ \hat{\mathbf{X}}_2^i \cdot \hat{\mathbf{u}}_1 & \hat{\mathbf{X}}_2^i \cdot \hat{\mathbf{u}}_2 \end{vmatrix} \quad (13-24a)$$

or

$$|\boldsymbol{\theta}| = (\hat{\mathbf{X}}_1^i \cdot \hat{\mathbf{u}}_1)(\hat{\mathbf{X}}_2^i \cdot \hat{\mathbf{u}}_2) - (\hat{\mathbf{X}}_2^i \cdot \hat{\mathbf{u}}_1)(\hat{\mathbf{X}}_1^i \cdot \hat{\mathbf{u}}_2) \quad (13-24b)$$

and

$$\theta_{jk} = \hat{\mathbf{X}}_j^i \cdot \hat{\mathbf{u}}_k \quad (13-24c)$$

The vectors $\hat{\mathbf{X}}_1^i$ and $\hat{\mathbf{X}}_2^i$ represent the principal directions of the incident wave front at the reflection point Q_R with principal radii of curvature ρ_1^i and ρ_2^i .

Equations 13-24 through 13-24c can be used to find single, first-order reflections by a reflecting surface. The process, using basically the same set of equations, must be repeated if second- and higher-order reflections are required. However, to accomplish this the principal plane directions $\hat{\mathbf{X}}_1^r$ and $\hat{\mathbf{X}}_2^r$ of the reflected fields from the previous reflection must be known. For example, second-order reflections can be found provided the principal plane directions $\hat{\mathbf{X}}_1^r$ and $\hat{\mathbf{X}}_2^r$ of the first-order reflected field are found. To do this, we first introduce

$$\mathbf{Q}^r = \begin{bmatrix} Q_{11}^r & Q_{12}^r \\ Q_{12}^r & Q_{22}^r \end{bmatrix} \quad (13-25)$$

where $\mathbf{Q}^r$ is defined as the curvature matrix for the reflected wave front whose entries are

$$Q_{11}^r = \frac{1}{\rho_1^i} + \frac{2 \cos \theta_i}{|\boldsymbol{\theta}|^2} \left(\frac{\theta_{22}^2}{R_1} + \frac{\theta_{21}^2}{R_2} \right) \quad (13-26a)$$

$$Q_{12}^r = -\frac{2 \cos \theta_i}{|\boldsymbol{\theta}|^2} \left(\frac{\theta_{22} \theta_{12}}{R_1} + \frac{\theta_{11} \theta_{21}}{R_2} \right) \quad (13-26b)$$

$$Q_{22}^r = \frac{1}{\rho_2^i} + \frac{2 \cos \theta_i}{|\boldsymbol{\theta}|^2} \left(\frac{\theta_{12}^2}{R_1} + \frac{\theta_{11}^2}{R_2} \right) \quad (13-26c)$$

Then the principal directions $\hat{\mathbf{X}}_1^r$ and $\hat{\mathbf{X}}_2^r$ of the reflected wave front, with respect to the x_1^r and x_2^r coordinates, can be written as

$$\hat{\mathbf{X}}_1^r = \frac{\left(Q_{22}^r - \frac{1}{\rho_1^r}\right)\hat{\mathbf{x}}_1^r - Q_{12}^r\hat{\mathbf{x}}_2^r}{\sqrt{\left(Q_{22}^r - \frac{1}{\rho_1^r}\right)^2 + (Q_{12}^r)^2}} \quad (13-27a)$$

$$\hat{\mathbf{X}}_2^r = -\hat{\mathbf{s}}^r \times \hat{\mathbf{X}}_1^r \quad (13-27b)$$

where $\hat{\mathbf{x}}_1^r$ and $\hat{\mathbf{x}}_2^r$ are unit vectors perpendicular to the reflected ray, and they are determined using

$$\hat{\mathbf{x}}_1^r = \hat{\mathbf{X}}_1^i - 2(\hat{\mathbf{n}} \cdot \hat{\mathbf{X}}_1^i)\hat{\mathbf{n}} \quad (13-28a)$$

$$\hat{\mathbf{x}}_2^r = \hat{\mathbf{X}}_2^i - 2(\hat{\mathbf{n}} \cdot \hat{\mathbf{X}}_2^i)\hat{\mathbf{n}} \quad (13-28b)$$

with $\hat{\mathbf{n}}$ being a unit vector normal to the surface at the reflection point.

To demonstrate the application of these formulations, let us consider a problem that is classified as a classic example in scattering.

Example 13-1

A linearly polarized uniform plane wave of amplitude E_0 is incident on a conducting sphere of radius a , as shown in Figures 11-26 and 11-28. Using geometrical optics methods, determine the:

1. Far-zone ($s \gg \rho_1^r$ and ρ_2^r) fields that are reflected from the surface of the sphere.
2. Backscatter radar cross section.

Solution: For a linearly polarized uniform plane wave incident upon a conducting sphere (13-20) reduces in the far zone to

$$E^r(s) = E_0(-1)\sqrt{\frac{\rho_1^r \rho_2^r}{(\rho_1^r + s)(\rho_2^r + s)}} e^{-j\beta s} \stackrel{s \gg \rho_1^r, \rho_2^r}{\simeq} -E_0 \frac{\sqrt{\rho_1^r \rho_2^r}}{s} e^{-j\beta s}$$

According to (13-23a)

$$\rho_1^r \rho_2^r = \frac{a^2}{4}$$

Thus,

$$E^r(s) = -E_0 \frac{a}{2s} e^{-j\beta s} = -\frac{E_0}{2} \left(\frac{a}{s}\right) e^{-j\beta s}$$

In turn, the backscatter radar cross section, according to (11-22b), can be written as

$$\sigma = \lim_{s \rightarrow \infty} \left[4\pi s^2 \frac{|E^r(s)|^2}{|E^i(Q_R)|^2} \right] \simeq 4\pi s^2 \frac{\left| -\frac{E_0}{2} \left(\frac{a}{s}\right) e^{-j\beta s} \right|^2}{|E_0|^2} = \pi a^2$$

It is recognized that the geometrical optics radar cross section of a sphere is equal to its physical cross-sectional area [see Figure 11.29 and (11-250)]. This is a well-known relation, and it is valid when the radius of the sphere is large compared to the wavelength.

The variation of the normalized radar cross section of a sphere as a function of its radius is displayed in Figure 11-29, and it is obtained by solving the wave equation in exact form. The Rayleigh, Mie (resonance), and geometrical optics regions represent three regimes in the figure.

For a cylindrical reflected field with radii of curvature $\rho_1^r = \rho^r$ and $\rho_2^r = \infty$, the reflected field of (13-20) reduces to

$$\mathbf{E}^r(s) = \mathbf{E}^i(Q_R) \cdot \bar{\mathbf{R}} \sqrt{\frac{\rho^r}{\rho^r + s}} e^{-j\beta s} \quad (13-29)$$

where ρ^r is the radius of curvature of the reflected field wave front.

An expression for ρ^r can be derived [37] by assuming a cylindrical wave, radiated by a line source at ρ_0 , incident upon a two-dimensional curved surface S with positive radius of curvature ρ_a , as shown in Figure 13-8a. The rays that are reflected from the surface S diverge for positive values of ρ_a , as shown in Figure 13-8b, and appear to be emanating from a caustic a distance ρ^r from the reflecting surface S . It can be shown that the wave front curvature of the reflected field can be determined using

$$\frac{1}{\rho^r} = \frac{1}{\rho_0} + \frac{2}{\rho_a \cos \theta_i} \quad (13-30)$$

This is left as an end-of-chapter exercise for the reader.

The expression for ρ^r has been developed for a cylindrical wave incident on a two-dimensional convex scattering surface corresponding to positive values of ρ_a . For this arrangement, the caustic resides within the reflecting surface, and the rays seem to emanate from a virtual source (image) located at the caustic. This is equivalent to the Cassegrain reflector arrangement where the virtual feed focal point for the main reflector is behind the convex subreflector as shown in Figures 15-30 and 15-31 of [38]. If the curved scattering surface is concave, corresponding to negative values of ρ_a , the value of ρ^r is obtained by making ρ_a negative. In this case the caustic resides outside the reflecting surface, and the rays seem to emanate from that location. This is analogous to the Gregorian reflector arrangement where the effective focal point of the main reflector is between the concave subreflector and main reflector, as shown in Figure 15-31 of [38]. Although ρ^r of (13-30) was derived

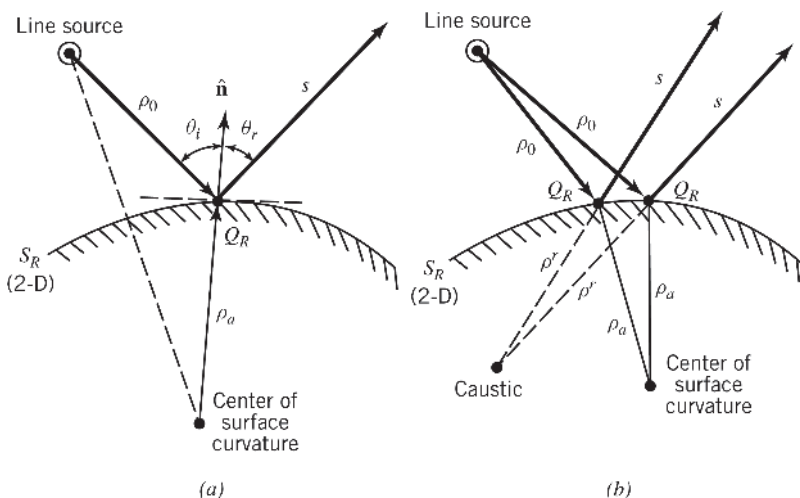


Figure 13-8 Line source near a two-dimensional curved surface. (a) Reflection point. (b) Caustic.

for cylindrical wave incidence on a two-dimensional convex curved surface, it can also be used when the plane of incidence coincides with any of the principal planes of curvature of the reflecting surface.

The application of (13-29) and (13-30) can best be demonstrated by an example.

Example 13-2

An electric line source of infinite length and constant current I_0 is placed symmetrically a distance h above an electric conducting strip of width w and infinite length, as shown in Figure 13-9a. The length of the line is placed parallel to the z axis. Assuming a free-space medium and far-field observations ($\rho \gg w$, $\rho \gg h$), derive expressions for the incident and reflected electric field components. Then compute and plot the normalized amplitude distribution (in decibels) of the incident, reflected, and incident plus reflected geometrical optics fields for $h = 0.5\lambda$ when $w = \text{infinite}$ and $w = 2\lambda$. Normalize the fields with respect to the maximum of the total geometrical optics field.

Solution: The analysis begins by first determining the incident (direct) field radiated by the source in the absence of the strip, which is given by (11-10a), or

$$E_z^i = -\frac{\beta^2 I_0}{4\omega\epsilon} H_0^{(2)}(\beta\rho_i)$$

where $H_0^{(2)}(\beta\rho_i)$ is the Hankel function of the second kind and of order zero, and ρ_i is the distance from the source to the observation point.

For far-zone observations ($\beta\rho_i \rightarrow \text{large}$) the Hankel function can be replaced by its asymptotic expansion

$$H_0^{(2)}(\beta\rho_i) \stackrel{\beta\rho_i \rightarrow \text{large}}{\simeq} \sqrt{\frac{2j}{\pi\beta\rho_i}} e^{-j\beta\rho_i}$$

This allows us to write the incident field as

$$E_z^i = E_0 \frac{e^{-j\beta\rho_i}}{\sqrt{\rho_i}}$$

where

$$E_0 = \left\{ -\frac{\beta^2 I_0}{4\omega\epsilon} \sqrt{\frac{2j}{\pi\beta}} \right\}$$

Using (13-29) and referring to Figure 13-9b, the reflected field can be written as

$$E_z^r = E_z^i(\rho_i = s')(-1) \sqrt{\frac{\rho^r}{\rho^r + s}} e^{-j\beta s} = -E_0 \frac{e^{-j\beta s'}}{\sqrt{s'}} \sqrt{\frac{\rho^r}{\rho^r + s}} e^{-j\beta s}$$

The wave front radius of curvature of the reflected field can be found using (13-30), that is,

$$\frac{1}{\rho^r} = \frac{1}{s'} + \frac{2}{\infty \cos \theta_i} = \frac{1}{s'} \Rightarrow \rho^r = s'$$

Therefore, the reflected field can now be written as

$$E_z^r = -E_0 \frac{e^{-j\beta s'}}{\sqrt{s'}} \sqrt{\frac{s'}{s' + s}} e^{-j\beta s} = -E_0 \frac{e^{-j\beta(s+s')}}{\sqrt{s+s'}} = -E_0 \frac{e^{-j\beta\rho_r}}{\sqrt{\rho_r}}$$

It is apparent from Figure 13-9b that the reflected rays seem to emanate from a virtual (image) source which is also a caustic for the reflected fields. The wave front radius of curvature $\rho^r = s'$ of the reflected field also represents the distance of the caustic (image) from the point of reflection. The reflected field

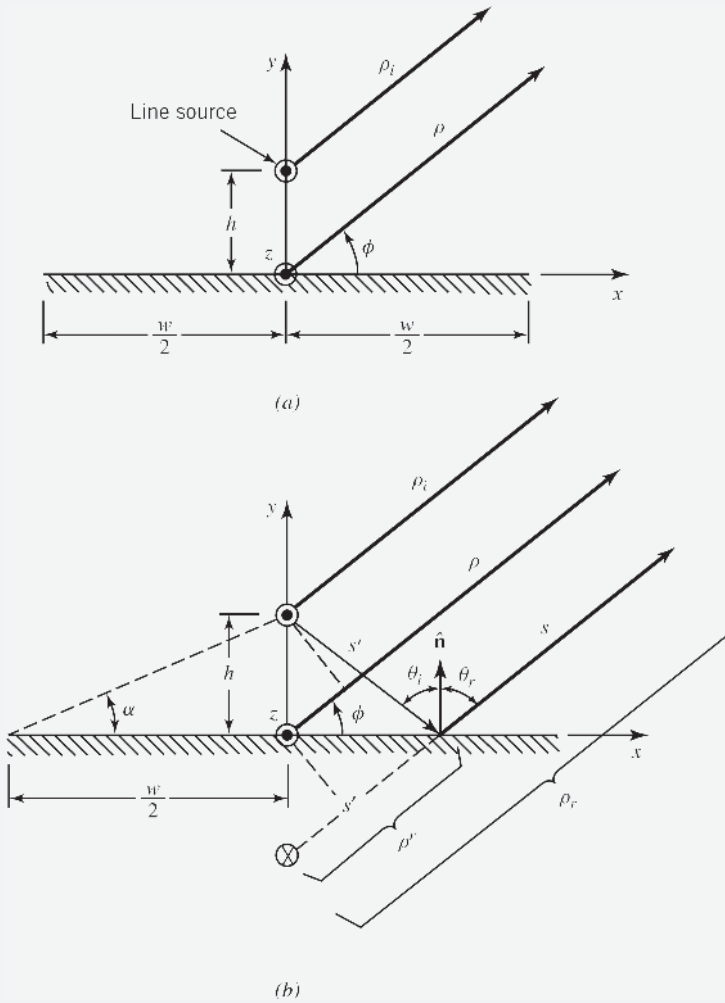


Figure 13-9 Line source above a finite-width strip. (a) Coordinate system. (b) Reflection geometry.

could also have been obtained very simply by using image theory. However, we chose to use the equations of geometrical optics to demonstrate the principles and applications of geometrical optics.

For far-field observations ($\rho_i \gg w$, $\rho_i \gg h$),

$$\left. \begin{aligned} \rho_i &= \rho - h \cos\left(\frac{\pi}{2} - \phi\right) = \rho - h \sin \phi \\ \rho_r &= \rho + h \cos\left(\frac{\pi}{2} - \phi\right) = \rho + h \sin \phi \end{aligned} \right\} \text{for phase variations}$$

$$\rho_i \simeq \rho_r \simeq \rho \quad \text{for amplitude variations}$$

Therefore, the incident (direct) and reflected fields can be reduced to

$$E_z^i = E_0 e^{+j\beta h \sin \phi} \frac{e^{-j\beta \rho}}{\sqrt{\rho}}, \quad 0 \leq \phi \leq \pi + \alpha, \quad 2\pi - \alpha \leq \phi \leq 2\pi$$

$$E_z^r = -E_0 e^{-j\beta h \sin \phi} \frac{e^{-j\beta \rho}}{\sqrt{\rho}}, \quad \alpha \leq \phi \leq \pi - \alpha$$

and the total field can be written as

$$E_z^t = \begin{cases} E_0 e^{+j\beta h \sin \phi} \frac{e^{-j\beta \rho}}{\sqrt{\rho}} & 0 \leq \phi \leq \alpha, \\ E_0 e^{+j\beta h \sin \phi} \frac{e^{-j\beta \rho}}{\sqrt{\rho}} & \pi - \alpha \leq \phi \leq \pi + \alpha, \\ E_0 e^{+j\beta h \sin \phi} \frac{e^{-j\beta \rho}}{\sqrt{\rho}} & 2\pi - \alpha \leq \phi \leq 2\pi \\ E_z^i + E_z^r = 2jE_0 \sin(\beta h \sin \phi) \frac{e^{-j\beta \rho}}{\sqrt{\rho}} & \alpha \leq \phi \leq \pi - \alpha \\ 0 & \pi + \alpha \leq \phi \leq 2\pi - \alpha \end{cases}$$

Normalized amplitude patterns for $w = 2\lambda$ and $h = 0.5\lambda$, computed using the preceding geometrical optics fields in their respective regions, are plotted (in decibels) in Figure 13-10 where discontinuities created along the incident and reflected shadow boundaries by the geometrical optics fields are apparent. The total amplitude pattern assuming an infinite ground plane is also displayed in Figure 13-10.

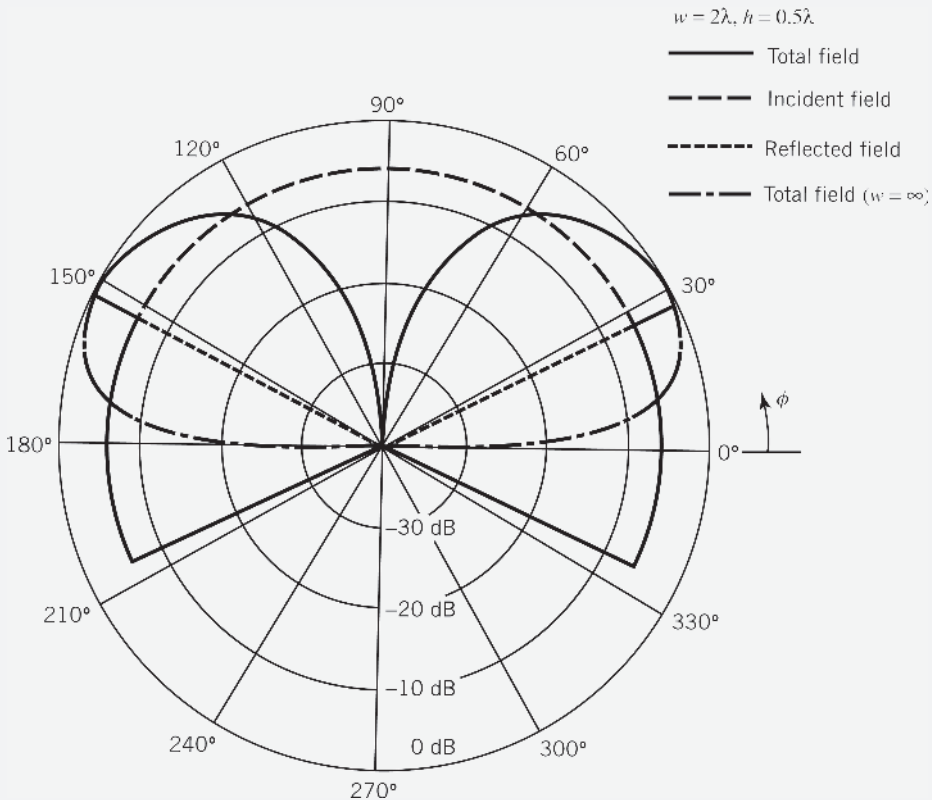


Figure 13-10 Amplitude radiation pattern of an electric line source above a finite-width strip.

In summary, geometrical optics methods approximate the fields by the leading term of the Luneberg–Kline expansion for large ω , but they fail along caustics. Improvements can be incorporated into the solutions by finding higher-order terms $E_1(\mathbf{R}_1)$, $E(\mathbf{R}_2)$,... in the Luneberg–Kline series expansion. Higher-order Luneberg–Kline expansions have been derived for fields scattered

from spheres, cylinders, and other curved surfaces with simple geometries [30, 31]. These solutions exhibit the following tendencies.

1. They improve the high-frequency field approximations if the observation specular point is not near edges, shadow boundaries, or other surface discontinuities.
2. They become singular as the observation specular point approaches a shadow boundary on the surface.
3. They do not correct for geometrical optics discontinuities along incident and reflection shadow boundaries.
4. They do not describe the diffracted fields in the shadow region.

Because of some of these deficiencies, in addition to being quite complex, higher-order Luneberg–Kline expansion methods cannot be used to treat diffraction. Therefore other approaches, usually somewhat heuristic in nature, must be used to introduce diffraction in order to improve geometrical optics approximations. It should be noted, however, that for sufficiently large ω , geometrical optics fields may dominate the scattering phenomena and may alone provide results that often agree quite well with measurements. This is more evident for backscattering from smooth curved surfaces with large radii of curvature. In those instances, corrections to the fields predicted by geometrical optics methods may not be necessary. However, for other situations where such solutions are inaccurate, corrections are usually provided by including diffraction. Therefore, a combination of geometrical optics and diffraction techniques often leads to solutions of many practical engineering problems whose results agree extremely well with measurements. This has been demonstrated in many applications [19–25, 39]. Because of their extreme importance and ease of application, diffraction techniques will be next introduced, discussed, and applied.

13.3 GEOMETRICAL THEORY OF DIFFRACTION: EDGE DIFFRACTION

Examining high-frequency diffraction problems has revealed that their solutions contain terms of fractional power that are not always included in the geometrical optics expression (13-15a) or in the Luneberg–Kline series solution. For example, geometrical optics fails to account for the energy diffracted into the shadow region when the incident rays are tangent to the surface of a curved object and for the diffracted energy when the surface contains an edge, vertex, or corner. In addition, caustics of the diffracted fields are located at the boundary surface. Therefore, some semi-heuristic approaches must be used to provide correction factors that improve the geometrical optics approximation.

13.3.1 Amplitude, Phase, and Polarization Relations

To introduce diffraction, let us assume that the smooth surface S of Figure 13-6a has a curved edge as shown in Figure 13-11a. When an electromagnetic wave impinges on this curved edge, diffracted rays emanate from the edge whose leading term of the high-frequency solution for the electric field takes the form

$$\mathbf{E}^d(\mathbf{R}) \simeq \frac{e^{-j\beta\psi_d(\mathbf{R})}}{\sqrt{\beta}} \mathbf{A}(\mathbf{R}) \quad (13-31)$$

where ψ_d = eikonal surface for the diffracted rays

$\mathbf{A}(\mathbf{R})$ = field factor for the diffracted rays

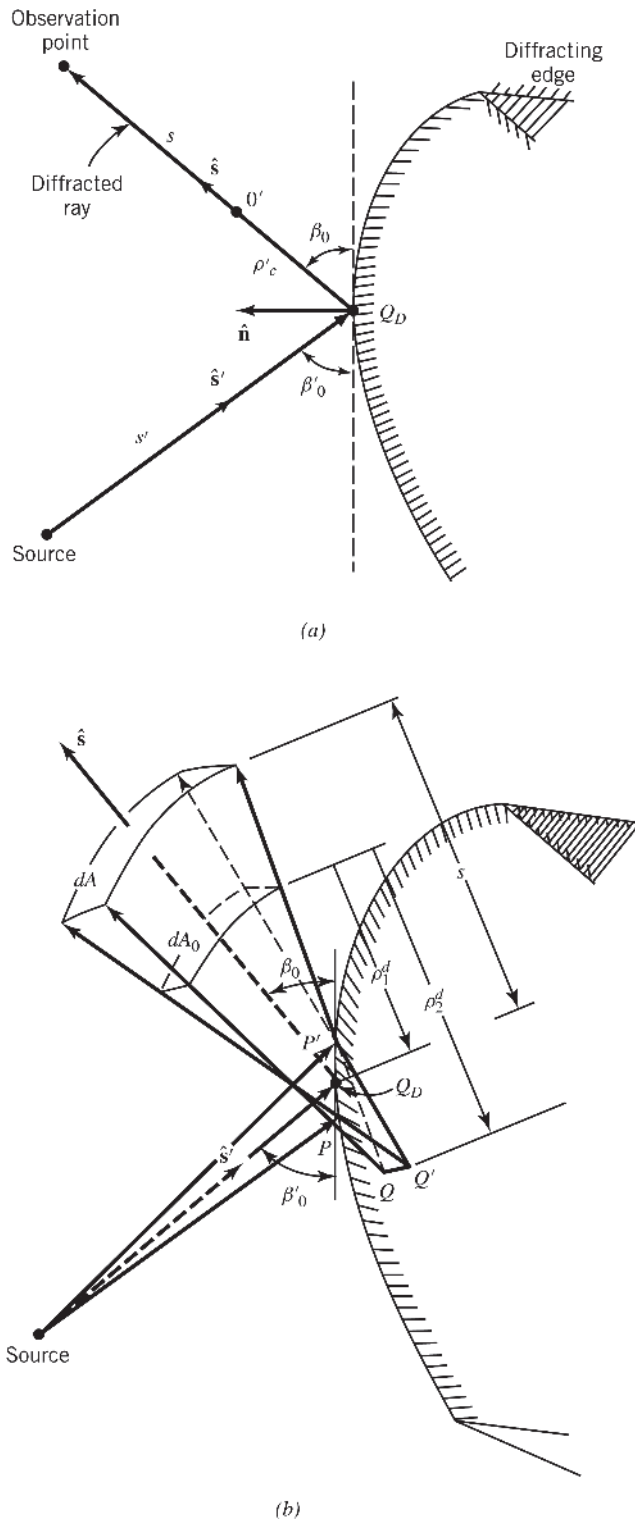


Figure 13-11 Geometry for diffraction by a curved edge. (a) Diffraction point. (b) Astigmatic tube of rays.

Substituting (13-31) into (13-11) and (13-12), and referring to the geometry of Figure 13-11a, it can be shown that the diffracted field can be written as

$$\begin{aligned}\mathbf{E}^d(s) &= \left[\frac{\mathbf{A}(0')}{\sqrt{\beta}} e^{-j\beta\psi_d(0')} \right] \sqrt{\frac{\rho'_c \rho_c}{(\rho'_c + s)(\rho_c + s)}} e^{-j\beta s} \\ \mathbf{E}^d(s) &= [\mathbf{E}^d(0')] \sqrt{\frac{\rho'_c \rho_c}{(\rho'_c + s)(\rho_c + s)}} e^{-j\beta s}\end{aligned}\quad (13-32)$$

where $\mathbf{E}^d(0')$ = diffracted field at the reference point $0'$

s = distance along the diffracted ray from the reference point $0'$

ρ'_c = distance from diffraction point Q_D (first caustic of diffracted field) to reference point $0'$

ρ_c = distance between the second caustic of diffracted field and reference point $0'$

It would have been more convenient to choose the reference point $0'$ to coincide with the diffraction point Q_D , located on the diffracting edge. However, like the geometrical optics rays, the diffracted rays form an astigmatic tube of the form shown in Figure 13-11b where the caustic line PP' coincides with the diffracting edge. Because the diffraction point is a caustic of the diffracted field, it is initially more straightforward to choose the reference point away from the edge diffraction caustic Q_D . However, the diffracted field of (13-32) should be independent of the location of the reference point $0'$, including $\rho'_c = 0$. Therefore, the diffracted field of (13-32) must be such that

$$\lim_{\rho'_c \rightarrow 0} \mathbf{E}^d(0') \sqrt{\rho'_c} = \text{finite} \quad (13-33)$$

and must be equal to

$$\lim_{\rho'_c \rightarrow 0} \mathbf{E}^d(0') \sqrt{\rho'_c} = \mathbf{E}^i(Q_D) \cdot \bar{\mathbf{D}} \quad (13-33a)$$

where $\mathbf{E}^i(Q_D)$ = incident field at the point of diffraction

$\bar{\mathbf{D}}$ = dyadic diffraction coefficient (analogous to dyadic reflection coefficient)

Using (13-33a), the diffracted field of (13-32) reduces to

$$\begin{aligned}\mathbf{E}^d(s) &= \lim_{\rho'_c \rightarrow 0} \left\{ [\mathbf{E}^d(0') \sqrt{\rho'_c}] \sqrt{\frac{\rho_c}{(\rho'_c + s)(\rho_c + s)}} e^{-j\beta s} \right\} \\ \mathbf{E}^d(s) &= \mathbf{E}^i(Q_D) \cdot \bar{\mathbf{D}} \sqrt{\frac{\rho_c}{s(\rho_c + s)}} e^{-j\beta s}\end{aligned}\quad (13-34)$$

which has the form

$\mathbf{E}^d(s)$	$=$	$\underbrace{\mathbf{E}^i(Q_D)}$	$\cdot$	$\underbrace{\bar{\mathbf{D}}}$	$\underbrace{A(\rho_c, s)}$	$\underbrace{e^{-j\beta s}}$	(13-34a)
		Field at reference point		Diffraction coefficient (usually a dyadic)	Spatial attenuation (spreading, divergence) factor	Phase factor	

and compares with that of (13-20) for reflection. In (13-34) and (13-34a)

$$A(\rho_c, s) = \sqrt{\frac{\rho_c}{s(\rho_c + s)}} \quad (13-34b)$$

= spatial attenuation (spreading, divergence) factor for a curved surface

ρ_c = distance between the reference point $Q_D(s=0)$ at the edge (also first caustic of the diffracted rays) and the second caustic of the diffracted rays.

In general, ρ_c is a function of the following:

1. Wave front curvature of the incident field.
2. Angles of incidence and diffraction, relative to unit vector normal to edge at the point Q_D of diffraction.
3. Radius of curvature of diffracting edge at point Q_D of diffraction.

An expression for ρ_c is given by (13-100a) in Section 13.3.4.

The diffracted field, which is determined by a generalization of Fermat's principle [7, 9], is initiated at a point on the surface of the object where discontinuities are formed along the incident and reflected shadow boundaries. As represented in (13-34) and (13-34a), the initial value of the field of a diffracted ray is determined from the incident field with the aid of an appropriate diffraction coefficient $\bar{\mathbf{D}}$, which in general is a dyadic for electromagnetic fields. The amplitude is assumed to vary in accordance with the principle of conservation of energy flux along a tube of rays. Appropriate phase jumps of $+90^\circ$ are added each time a ray passes through a caustic at $s=0$ and $s=-\rho_c$, as properly accounted for in (13-34) and (13-34a). The phase of the field on a diffracted ray is assumed to be equal to the product of the optical lengths of the ray, from some reference point Q_D , and the phase constant β of the medium.

When the edge is straight, the source is located a distance s' from the point of diffraction, and the observations are made at a distance s from it—as shown in Figure 13-12—the diffracted field can be written as

$$\mathbf{E}^d(s) = \mathbf{E}^i(Q_D) \cdot \bar{\mathbf{D}}\mathbf{A}(s', s) e^{-j\beta s} \quad (13-35)$$

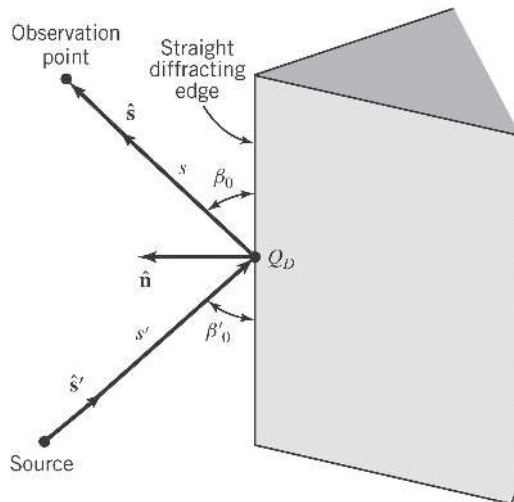


Figure 13-12 Diffraction by a wedge with a straight edge.

where

$$A(s', s) = \begin{cases} \frac{1}{\sqrt{s}} & \text{for plane and conical wave incidences} \\ \frac{1}{\sqrt{\rho}}, \rho = s \sin \beta_0 & \text{for cylindrical wave incidence} \\ \sqrt{\frac{s'}{s(s+s')}} \stackrel{s \gg s'}{\simeq} \frac{\sqrt{s'}}{s} & \text{for spherical wave incidence} \end{cases} \quad \begin{matrix} (13-35a) \\ (13-35b) \\ (13-35c) \end{matrix}$$

The $A(s', s)$ formulas of (13-35a) through (13-35c) are obtained by letting ρ_c in (13-34) tend to infinity ($\rho_c = \infty$) for plane, cylindrical, and conical wave incidence and $\rho_c = s'$ for spherical wave incidence.

The diffraction coefficients are usually determined from the asymptotic solutions of canonical problems that have the same local geometry at the points of diffraction as the object(s) of investigation. One of the simplest geometries, which will be discussed in this chapter, is a conducting wedge [40–42]. Another is that of a conducting, smooth, and convex surface [43–46]. The main objectives in the remaining part of this chapter are to introduce and apply the diffraction coefficients for the canonical problem of the conducting wedge. Curved surface diffraction is derived in [43–46].

13.3.2 Straight Edge Diffraction: Normal Incidence

In order to examine the manner in which fields are diffracted by edges, it is necessary to have a diffraction coefficient available. To derive a diffraction coefficient for an edge, we need to consider a canonical problem that has the same local geometry near the edge, like that shown in Figure 13-13a.

Let us begin by assuming that a source is placed near the two-dimensional electric conducting wedge of included angle $WA = (2 - n)\pi$ radians. If observations are made on a circle of constant radius ρ from the edge of the wedge, it is quite clear that, in addition to the direct ray (OP), there are rays that are reflected from the side of the wedge ($OQ_R P$), which contribute to the intensity at point P . These rays obey Fermat's principle, that is, they minimize the path between points O and P by including points on the side of the wedge, and deduce Snell's law of reflection. It would then seem appropriate to extend the class of such points to include in the trajectory rays that pass through the edge of the wedge ($OQ_D P$), leading to the generalized Fermat's principle [7]. This class of rays is designated as diffracted rays and they lead to the *law of diffraction*.

By considering rays that obey only geometrical optics radiation mechanisms (direct and reflected), we can separate the space surrounding the wedge into three different field regions. Using the geometrical coordinates of Figure 13-13b, the following geometrical optics fields will contribute to the corresponding regions:

Region I	Region II	Region III
$0 < \phi < \pi - \phi'$	$\pi - \phi' < \phi < \pi < \phi'$	$\pi + \phi' < \phi < n\pi$
Direct Reflected	Direct	...

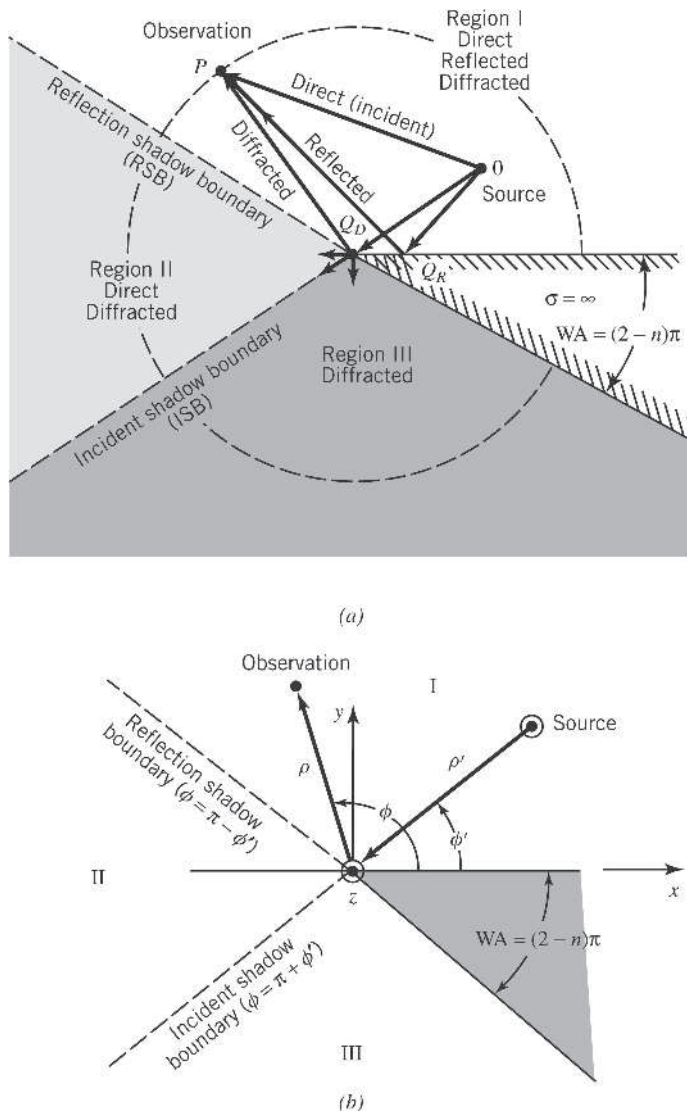


Figure 13-13 Line source near a two-dimensional conducting wedge. (a) Region separation. (b) Coordinate system. Region I: $0 \leq \phi < \pi - \phi'$. Region II: $\pi - \phi' < \phi < \pi + \phi'$. Region III: $\pi + \phi' < \phi < n\pi$.

With these fields, it is evident that the following will occur:

1. Discontinuities in the field will be formed along the RSB (Reflection Shadow Boundary) separating regions I and II ($\phi = \pi - \phi'$), and along the ISB (Incident Shadow Boundary) separating regions II and III ($\phi = \pi + \phi'$).
2. No field will be present in region III (shadow region).

Since neither of the preceding results should be present in a physically realizable field, modifications and/or additions need to be made.

To *remove* the discontinuities along the boundaries and to *modify* the fields in *all* three regions, diffracted fields must be included. To obtain expressions for the diffracted field distribution, we assume that the source in Figure 13-13 is an infinite line source (either electric or magnetic).

The fields of an electric line source satisfy the homogeneous Dirichlet boundary conditions $E_z = 0$ on both faces of the wedge and the fields of the magnetic line source satisfy the homogeneous Neumann boundary condition $\partial E_z / \partial \phi = 0$ or $H_z = 0$ on both faces of the wedge. The faces of the wedge are formed by two semi-infinite intersecting planes. The infinitely long line source is parallel to the edge of the wedge, and its position is described by the coordinate (ρ', ϕ') . The typical field point is denoted by (ρ, ϕ) , as shown in Figure 13-13b. The line source is assumed to have constant current.

Initially, we will consider only normal incidence diffraction by a straight edge, as shown in Figure 13-14. For this situation, the plane of diffraction is perpendicular to the edge of the wedge. Oblique incidence diffraction (Figure 13-31) and curved-edge diffraction (Figure 13-35) will be discussed, respectively, in Sections 13.3.3 and 13.3.4.

The diffraction coefficient for the geometries of Figures 13-13 and 13-14 is obtained by [42]:

1. Finding the Green's function solution in the form of an infinite series using modal techniques and then approximating it for large values of $\beta\rho$ (far-field observations).
2. Converting the infinite series Green's function solution into an integral.
3. Performing, on the integral Green's function, a high-frequency asymptotic expansion (in inverse powers of $\beta\rho$) using standard techniques, such as the *method of steepest descent*.

An abbreviated derivation of this procedure will now be presented.

A. Modal Solution Using modal techniques, the total radiation electric field for an electric line source of current I_e was found in Chapter 11, Section 11.6.3, to be that of (11-207a), or

$$E_z^e = -\frac{\omega\mu I_e}{4}G \Rightarrow \mathbf{H}^e = -\frac{1}{j\omega\mu}\nabla \times \mathbf{E} \quad (13-36)$$

and the total magnetic field for a magnetic line source of current I_m to be that of (11-207b), or

$$H_z^m = \frac{\omega\varepsilon I_m}{4}G \Rightarrow \mathbf{E}^m = +\frac{1}{j\omega\varepsilon}\nabla \times \mathbf{H} \quad (13-37)$$

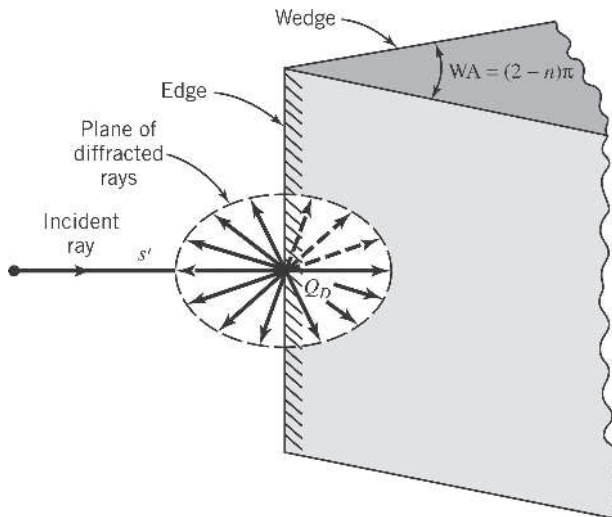


Figure 13-14 Wedge plane of diffraction for normal incidence.

where G is referred to as the Green's function, and it is given by (11-208a) as

$$G = \frac{1}{n} \sum_{m=0}^{\infty} \varepsilon_m J_{m/n}(\beta\rho) H_{m/n}^{(2)}(\beta\rho') \left[\cos \frac{m}{n}(\phi - \phi') \pm \cos \frac{m}{n}(\phi + \phi') \right] \quad \text{for } \rho \leq \rho' \quad (13-38)$$

$$\varepsilon_m = \begin{cases} 1 & m = 0 \\ 2 & m \neq 0 \end{cases} \quad (13-38a)$$

For the case $\rho \geq \rho'$, ρ and ρ' are interchanged. The plus sign between the two cosine terms is used if the boundary condition is of the homogeneous Neumann type, $\partial G / \partial \phi = 0$, on both faces of the wedge. For the homogeneous Dirichlet boundary condition, $G = 0$, on both faces of the wedge, the minus sign is used. In acoustic terminology, the Neumann boundary condition is referred to as *hard* polarization and the Dirichlet boundary condition is referred to as *soft* polarization. This series is an exact solution to the time-harmonic, inhomogeneous wave equation of a radiating line source and wedge embedded in a linear, isotropic, homogeneous, lossless medium.

B. High-Frequency Asymptotic Solution Many times it is necessary to determine the total radiation field when the line source is far removed from the vertex of the wedge. In such cases, (13-38) can be simplified by replacing the Hankel function by the first term of its asymptotic expansion, that is, by

$$H_{m/n}^{(2)}(\beta\rho') \stackrel{\beta\rho' \rightarrow \infty}{\simeq} \sqrt{\frac{2}{\pi\beta\rho'}} e^{-j[\beta\rho' - \pi/4 - (m/n)(\pi/2)]} \quad (13-39)$$

This substitution reduces G to

$$\begin{aligned} G &= \sqrt{\frac{2}{\pi\beta\rho'}} e^{-j(\beta\rho' - \pi/4)} \frac{1}{n} \sum_{m=0}^{\infty} \varepsilon_m J_{m/n}(\beta\rho) e^{+j(m/n)(\pi/2)} \\ &\quad \times \left[\cos \frac{m}{n}(\phi - \phi') \pm \cos \frac{m}{n}(\phi + \phi') \right] \\ G &= \sqrt{\frac{2}{\pi\beta\rho'}} e^{-j(\beta\rho' - \pi/4)} F(\beta\rho) \end{aligned} \quad (13-40)$$

where

$$F(\beta\rho) = \frac{1}{n} \sum_{m=0}^{\infty} \varepsilon_m J_{m/n}(\beta\rho) e^{+j(m/n)(\pi/2)} \left[\cos \frac{m}{n}(\phi - \phi') \pm \cos \frac{m}{n}(\phi + \phi') \right] \quad (13-40a)$$

$F(\beta\rho)$ is used to represent either the normalized total E_z^e , when the source is an electric line (soft polarization), or H_z^m , when the source is a magnetic line (hard polarization).

The infinite series of (13-40a) converges rapidly for small values of $\beta\rho$. For example, if $\beta\rho$ is less than 1, less than 15 terms are required to achieve a five-significant-figure accuracy. However, at least 40 terms should be included when $\beta\rho$ is 10 to achieve the five-significant-figure accuracy. To demonstrate the variations of (13-40a), the patterns of a unit amplitude plane wave of hard polarization incident upon a half-plane ($n = 2$) and 90° wedge ($n = 3/2$), computed using (13-40a), are shown in Figure 13-15. Computed patterns for the soft polarization are shown in Figure 13-16.

Whereas (13-40a) represents the normalized total field, in the space around a wedge of included angle $WA = (2 - n)\pi$, of a unity amplitude plane wave incident upon the wedge, the field of a unit amplitude cylindrical wave incident upon the same wedge and observations made

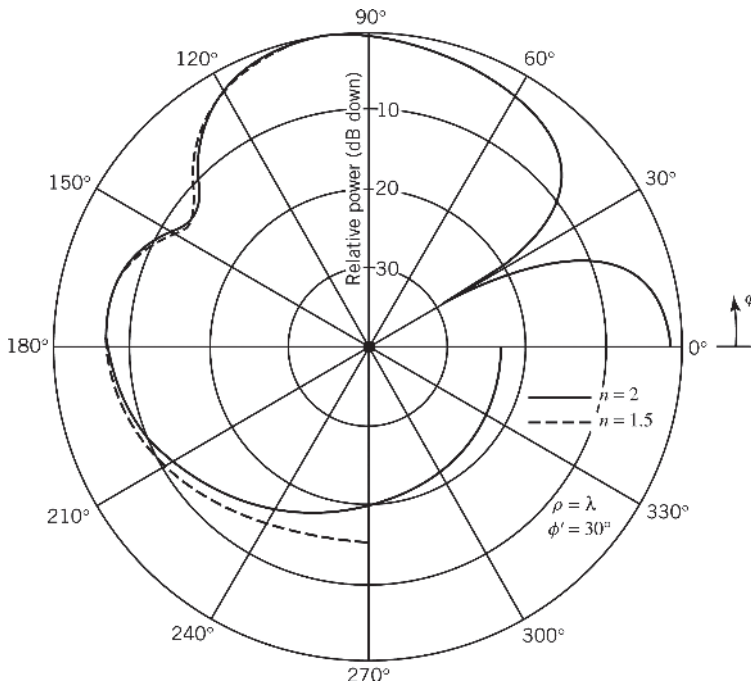


Figure 13-15 Normalized amplitude pattern of a hard polarization plane wave incident normally on a two-dimensional conducting wedge.

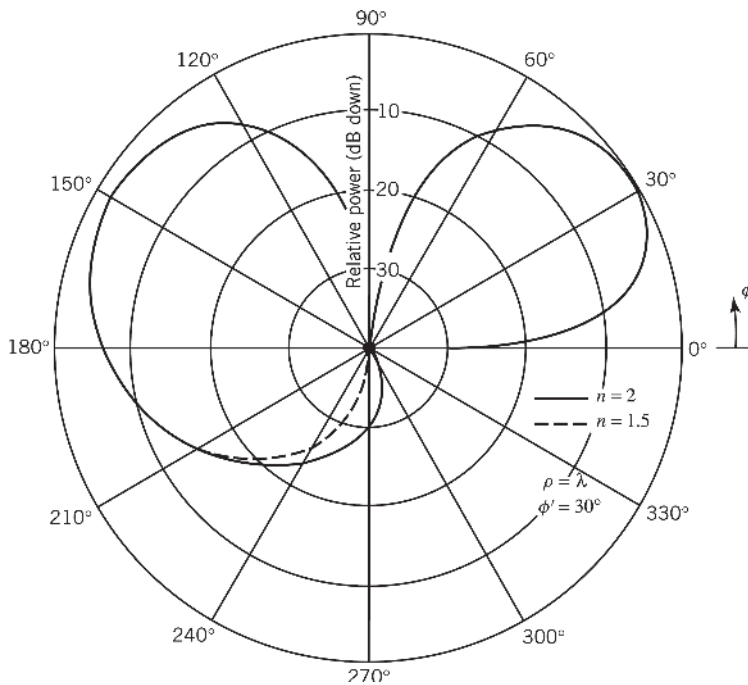


Figure 13-16 Normalized amplitude pattern of a soft polarization plane wave incident normally on a two-dimensional conducting wedge.

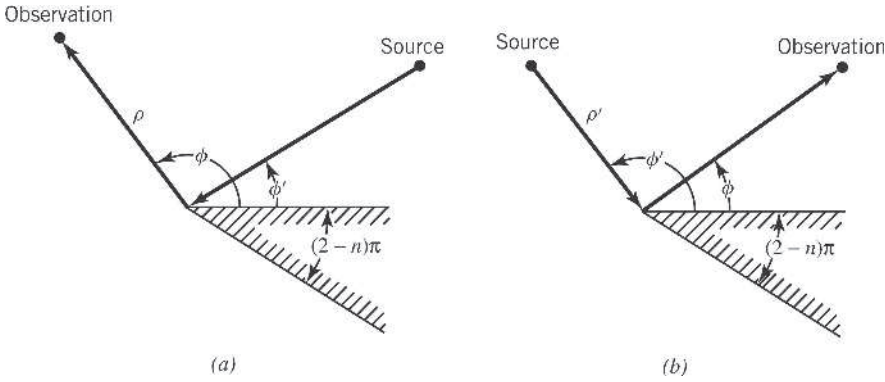


Figure 13-17 Principle of reciprocity in diffraction. (a) Plane wave incidence. (b) Cylindrical wave incidence.

at very large distances can be obtained by the reciprocity principle. Graphically, this is illustrated in Figures 13-17a and 13-17b. Analytically, the cylindrical wave incidence fields of Figure 13-17b can be obtained from (13-40a) and Figure 13-17a by substituting in (13-40a) $\rho = \rho'$, $\phi = \phi'$, and $\phi' = \phi$.

A high-frequency asymptotic expansion for $F(\beta\rho)$ in inverse powers of $\beta\rho$ is very useful for computational purposes, because of the slow convergence of (13-40a) for large values of $\beta\rho$. In order to derive an asymptotic expression for $F(\beta\rho)$ by the *conventional method of steepest descent* for isolated poles and saddle points (see Appendix VI), it must first be transformed into an integral or integrals of the form

$$P(\beta\rho) = \int_C H(z) e^{\beta\rho h(z)} dz \quad (13-41)$$

and then evaluated for large $\beta\rho$ by means of the method of steepest descent [8, 47].

To accomplish this, first the cosine terms are expressed in complex form by

$$\cos\left(\frac{m}{n}\xi^\mp\right) = \frac{1}{2} \left[e^{j(m/n)\xi^\mp} + e^{-j(m/n)\xi^\mp} \right] \quad (13-42)$$

where

$$\xi^\mp = \phi \mp \phi' \quad (13-42a)$$

and the Bessel functions are replaced by contour integrals in the complex z plane, of the form

$$J_{m/n}(\beta\rho) = \frac{1}{2\pi} \int_C e^{j[\beta\rho \cos z + m/n(z - \pi/2)]} dz \quad (13-43a)$$

or

$$J_{m/n}(\beta\rho) = \frac{1}{2\pi} \int_{C'} e^{j[\beta\rho \cos z - m/n(z + \pi/2)]} dz \quad (13-43b)$$

where the paths C and C' are shown in Figure 13-18a. Doing this and interchanging the order of integration and summation, it can be shown that (13-40a) can be written as

$$F(\beta\rho) = I(\beta\rho, \phi - \phi', n) \pm I(\beta\rho, \phi + \phi', n) \quad (13-44)$$

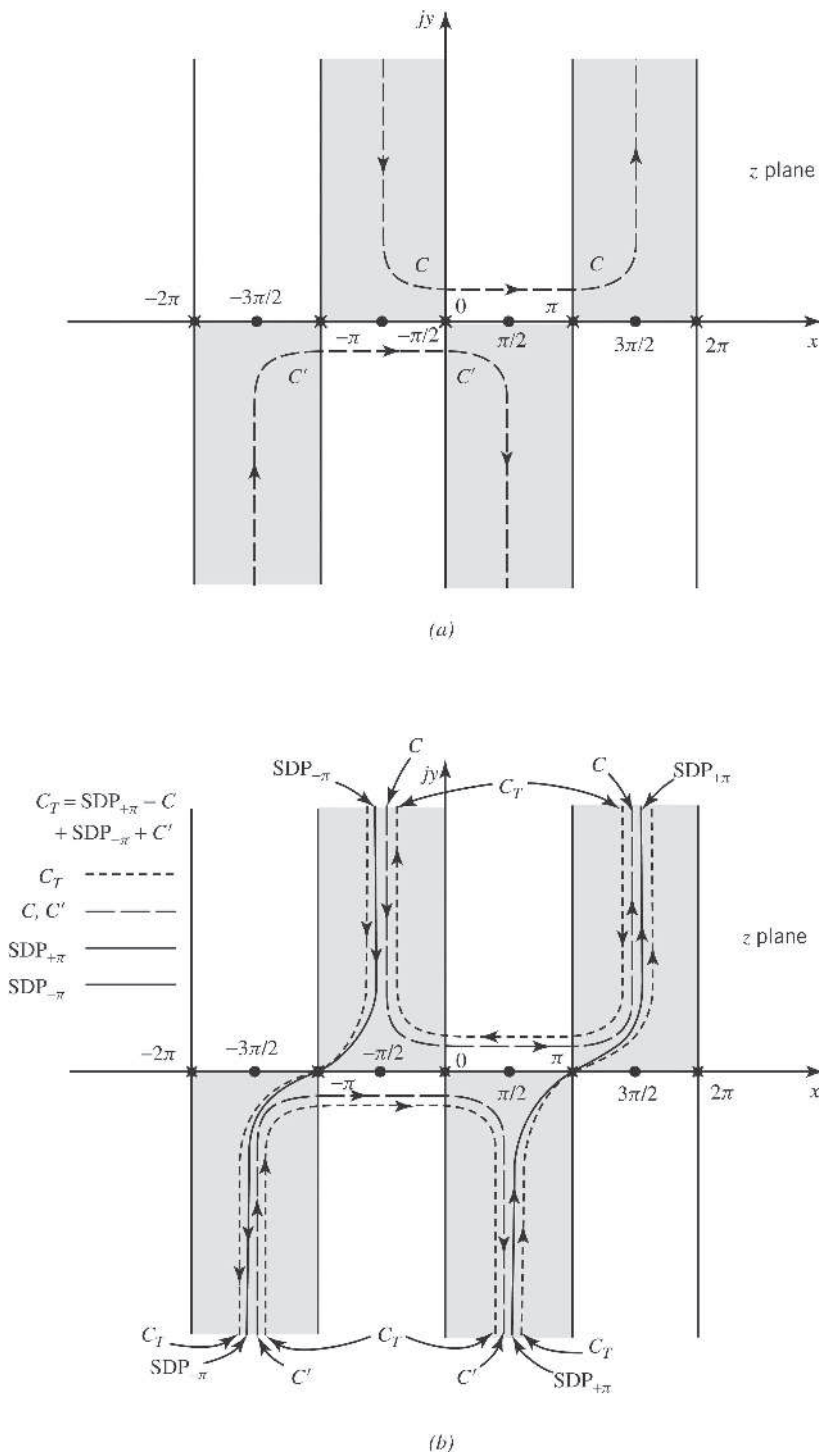


Figure 13-18 Line contours, steepest descent paths, saddle points, and poles in complex z plane for asymptotic evaluation of wedge diffraction formulas. (a) Contours for Bessel function. (b) Steepest descent paths, saddle points, and poles.

where

$$I(\beta\rho, \xi^\mp, n) = \frac{1}{2\pi n} \int_C e^{j\beta\rho \cos z} \sum_{m=1}^{\infty} e^{+j(m/n)(\xi^\mp + z)} dz \\ + \frac{1}{2\pi n} \int_{C'} e^{j\beta\rho \cos z} \sum_{m=0}^{\infty} e^{-j(m/n)(\xi^\mp + z)} dz \quad (13-44a)$$

with $\xi^\mp = \phi \mp \phi'$.

Using the series expansions of

$$-\frac{1}{1-x^{-1}} = x(1+x+x^2+x^3+\cdots) = \sum_{m=1}^{\infty} x^m \quad (13-45a)$$

$$\frac{1}{1-x^{-1}} = 1+x^{-1}+x^{-2}+x^{-3}+\cdots = \sum_{m=0}^{\infty} x^{-m} \quad (13-45b)$$

and that

$$\frac{1}{1-e^{-j(\xi^\mp + z)/n}} = \frac{e^{j(\xi^\mp + z)/2n}}{e^{j(\xi^\mp + z)/2n} - e^{-j(\xi^\mp + z)/2n}} = \frac{1}{2} + \frac{1}{2j} \cot\left(\frac{\xi^\mp + z}{2n}\right) \quad (13-46)$$

we can ultimately write (13-44) or (13-40a) as

$$F(\beta\rho) = \frac{1}{4\pi jn} \int_{(C'-C)} \cot\left(\frac{\phi - \phi' + z}{2n}\right) e^{j\beta\rho \cos z} dz \\ \pm \frac{1}{4\pi jn} \int_{(C'-C)} \cot\left(\frac{\phi + \phi' + z}{2n}\right) e^{j\beta\rho \cos z} dz \quad (13-47)$$

where the negative sign before C indicates that this integration path is to be traversed in the direction opposite to that shown in Figure 13-18a. It is now clear that $F(\beta\rho)$ has been written in an integral of the form of (13-41), which can be evaluated asymptotically by contour integration and by the method of steepest descent.

C. Method of Steepest Descent Equation 13-47 can also be written as

$$F(\beta\rho) = F_1(\beta\rho) \pm F_2(\beta\rho) \quad (13-48)$$

$$F_1(\beta\rho) = \frac{1}{4\pi jn} \int_{(C'-C)} H_1(z) e^{\beta\rho h_1(z)} dz \quad (13-48a)$$

$$F_2(\beta\rho) = \frac{1}{4\pi jn} \int_{(C'-C)} H_2(z) e^{\beta\rho h_2(z)} dz \quad (13-48b)$$

$$H_1(z) = \cot\left[\frac{(\phi - \phi') + z}{2n}\right] = \cot\left(\frac{\xi^- + z}{2n}\right) \quad (13-48c)$$

$$H_2(z) = \cot\left[\frac{(\phi + \phi') + z}{2n}\right] = \cot\left(\frac{\xi^+ + z}{2n}\right) \quad (13-48d)$$

$$h_1(z) = h_2(z) = j \cos(z) \quad (13-48e)$$

The evaluation of (13-48) can be accomplished by evaluating separately (13-48a) and (13-48b) and summing the results. Let us examine first the evaluation of (13-48a) in detail. A similar procedure can be used for (13-48b).

Using the complex z -plane closed contour C_T , we can write by referring to Figure 13-18b that

$$\begin{aligned} \frac{1}{4\pi jn} \oint_{C_T} H_1(z) e^{\beta \rho h_1(z)} dz &= \frac{1}{4\pi jn} \int_{(C'-C)} H_1(z) e^{\beta \rho h_1(z)} dz \\ &+ \frac{1}{4\pi jn} \int_{SDP_{+\pi}} H_1(z) e^{\beta \rho h_1(z)} dz \\ &+ \frac{1}{4\pi jn} \int_{SDP_{-\pi}} H_1(z) e^{\beta \rho h_1(z)} dz \end{aligned} \quad (13-49)$$

The closed contour C_T is equal to the sum of

$$C_T = C' + SDP_{+\pi} - C + SDP_{-\pi} \quad (13-49a)$$

where $SDP_{\pm\pi}$ is used to represent the steepest descent paths passing through the saddle points $\pm\pi$. We can rewrite (13-49) as

$$\begin{aligned} F_1(\beta\rho) &= \frac{1}{4\pi jn} \int_{(C'-C)} H_1(z) e^{\beta \rho h_1(z)} dz = \frac{1}{4\pi jn} \oint_{C_T} H_1(z) e^{\beta \rho h_1(z)} dz \\ &- \frac{1}{4\pi jn} \int_{SDP_{+\pi}} H_1(z) e^{\beta \rho h_1(z)} dz - \frac{1}{4\pi jn} \int_{SDP_{-\pi}} H_1(z) e^{\beta \rho h_1(z)} dz \end{aligned} \quad (13-50)$$

which is the same as (13-48a). Therefore, (13-48a) can be integrated by evaluating the three terms on the right side of (13-50). The closed contour of the first term on the right side of (13-50) is evaluated using residue calculus and the other two terms are evaluated using the method of steepest descent. When evaluated, it will be shown that the first term on the right side of (13-50) will represent the incident geometrical optics and the other two terms will represent what will be referred to as the incident diffracted field. A similar interpretation will be given when (13-48b) is evaluated; it represents the reflected geometrical optics and reflected diffracted fields.

Using residue calculus, we can write the first term on the right side of (13-50) as [48]

$$\begin{aligned} \frac{1}{4\pi jn} \oint_{C_T} H_1(z) e^{\beta \rho h_1(z)} dz &= 2\pi j \sum_p \text{Res}(z = z_p) \\ &= 2\pi j \sum_p (\text{residues of the poles enclosed by } C_T) \end{aligned} \quad (13-51)$$

To evaluate (13-51), we first rewrite it as

$$\begin{aligned} \frac{1}{4\pi jn} \oint_{C_T} H_1(z) e^{\beta \rho h_1(z)} dz &= \frac{1}{4\pi jn} \oint_{C_T} \cot \left[\frac{(\phi - \phi') + z}{2n} \right] e^{j\beta\rho \cos(z)} dz \\ &= \oint_{C_T} \frac{1}{4\pi jn} \cot \left[\frac{(\phi - \phi') + z}{2n} \right] e^{j\beta\rho \cos(z)} dz = \oint_{C_T} \frac{N(z)}{D(z)} dz \end{aligned} \quad (13-52)$$

where

$$\frac{N(z)}{D(z)} = \frac{\cos\left[\frac{(\phi - \phi') + z}{2n}\right] e^{j\beta\rho \cos(z)}}{4\pi jn \sin\left[\frac{(\phi - \phi') + z}{2n}\right]} \quad (13-52a)$$

$$N(z) = \cos\left[\frac{(\phi - \phi') + z}{2n}\right] e^{j\beta\rho \cos(z)} \quad (13-52b)$$

$$D(z) = 4\pi jn \sin\left[\frac{(\phi - \phi') + z}{2n}\right] \quad (13-52c)$$

Equation 13-52 has simple poles that occur when

$$\left[\frac{(\phi - \phi') + z}{2n}\right]_{z=z_p} = \pi N, \quad N = 0, \pm 1, \pm 2, \dots \quad (13-53)$$

or

$$z_p = -(\phi - \phi') + 2\pi nN \quad (13-53a)$$

provided that

$$-\pi \leq z_p = -(\phi - \phi') + 2\pi nN \leq +\pi \quad (13-53b)$$

Using residue calculus, the residues of (13-51) or (13-52) for simple poles (no branch points, etc.) can be found using [48]

$$\text{Res}(z = z_p) = \frac{N(z)}{dD(z)} \bigg|_{z=z_p} = \frac{N(z)}{D'(z)} \bigg|_{z=z_p} \quad (13-54)$$

where

$$\begin{aligned} N(z) \big|_{z=z_p} &= \cos\left[\frac{(\phi - \phi') + z}{2n}\right] e^{j\beta\rho \cos(z)} \bigg|_{z=-(\phi - \phi') + 2\pi nN} \\ &= \cos(\pi N) e^{j\beta\rho \cos[-(\phi - \phi') + 2\pi nN]} \end{aligned} \quad (13-54a)$$

$$D'(z) \big|_{z=z_p} = 2\pi j \cos\left[\frac{(\phi - \phi') + z}{2n}\right] \bigg|_{z=-(\phi - \phi') + 2\pi nN} = 2\pi j \cos(\pi N) \quad (13-54b)$$

Thus, (13-54) and (13-51) can be written, respectively, as

$$\text{Res}(z = z_p) = \frac{1}{2\pi j} e^{j\beta\rho \cos[-(\phi - \phi') + 2\pi nN]} \quad (13-55a)$$

$$\frac{1}{4\pi jn} \oint_{C_T} H_1(z) e^{\beta\rho h(z)} dz = e^{j\beta\rho \cos[-(\phi - \phi') + 2\pi nN]} U[\pi - |-(\phi - \phi') + 2\pi nN|] \quad (13-55b)$$

The $U(t - t_0)$ function in (13-55b) is a unit step function defined as

$$U(t - t_0) = \begin{cases} 1 & t > t_0 \\ \frac{1}{2} & t = t_0 \\ 0 & t < t_0 \end{cases} \quad (13-56)$$

The unit step function is introduced in (13-55b) so that (13-53b) is satisfied. When $z = \pm\pi$, (13-53b) is expressed as

$$2\pi nN^+ - (\phi - \phi') = +\pi \quad (13-57a)$$

for $z_p = +\pi$ and

$$2\pi n N^- - (\phi - \phi') = -\pi \quad (13-57b)$$

for $z_p = -\pi$. For the principal value of $N(N^\pm = 0)$, (13-55b) reduces to

$$F_1(\beta\rho)|_{C_T} = \frac{1}{4\pi j n} \oint_{C_T} H_1(z) e^{\beta\rho h_1(z)} dz = e^{j\beta\rho \cos(\phi - \phi')} U[\pi - |-(\phi - \phi')|] \quad (13-58)$$

This is referred to as the *incident geometrical optics* field, and it exists provided $|\phi - \phi'| < \pi$. This completes the evaluation of (13-51). Let us now evaluate the other two terms on the right side of (13-50).

In evaluating the last two terms on the right side of (13-50), the contributions from all the saddle points along the steepest descent paths must be accounted for. In this situation, however, only saddle points at $z = z_s = \pm\pi$ occur. These are found by taking the derivative of (13-48e) and setting it equal to zero. Doing this leads to

$$h_1'(z)|_{z=z_s} = -j \sin(z)|_{z=z_s} = 0 \Rightarrow z_s = \pm\pi \quad (13-59)$$

The form used to evaluate the last two terms on the right side of (13-50) depends on whether the poles of (13-53a), which contribute to the geometrical optics field, are near or far removed from the saddle points at $z = z_s = \pm\pi$. Let us first evaluate each of the last two terms on the right side of (13-50) when the poles of (13-53a) are far removed from the saddle point of (13-59).

When the poles of (13-53a) are far removed from the saddle points of (13-59), then the last two terms on the right side of (13-50) are evaluated using the *conventional steepest descent method* for isolated poles and saddle points. Doing this, we can write the last two terms on the right side of (13-50), using the saddle points of (13-59), as [8, 47]:

$$\begin{aligned} \frac{1}{4\pi j n} \int_{\text{SDP}_{+\pi}} H_1(z) e^{\beta\rho h_1(z)} dz &\stackrel{\beta\rho \rightarrow \text{large}}{\simeq} \frac{1}{4\pi j n} \left| \sqrt{\frac{2\pi}{-\beta\rho h_1''(z_s = +\pi)}} \right| e^{j\pi/4} \\ &\times H_1(z_s = +\pi) e^{\beta\rho h_1(z_s = \pi)} = \frac{e^{-j\pi/4}}{2n\sqrt{2\pi\beta}} \cot\left[\frac{\pi + (\phi - \phi')}{2n}\right] \frac{e^{-j\beta\rho}}{\sqrt{\rho}} \end{aligned} \quad (13-60a)$$

$$\begin{aligned} \frac{1}{4\pi j n} \int_{\text{SDP}_{-\pi}} H_1(z) e^{\beta\rho h_1(z)} dz &\stackrel{\beta\rho \rightarrow \text{large}}{\simeq} \frac{1}{4\pi j n} \left| \sqrt{\frac{2\pi}{-\beta\rho h_1''(z_s = -\pi)}} \right| e^{-j3\pi/4} \\ &\times H_1(z_s = -\pi) e^{\beta\rho h_1(z_s = -\pi)} = \frac{e^{-j\pi/4}}{2n\sqrt{2\pi\beta}} \cot\left[\frac{\pi - (\phi - \phi')}{2n}\right] \frac{e^{-j\beta\rho}}{\sqrt{\rho}} \end{aligned} \quad (13-60b)$$

Combining (13-60a) and (13-60b), it can be shown that the sum of the two can be written as

$$\begin{aligned} F_1(\beta\rho)|_{\text{SDP}_{\pm\pi}} &= -\frac{1}{4\pi j n} \int_{\text{SDP}_{+\pi}} H_1(z) e^{\beta\rho h_1(z)} dz - \frac{1}{4\pi j n} \int_{\text{SDP}_{-\pi}} H_1(z) e^{\beta\rho h_1(z)} dz \\ &\simeq \frac{e^{-j\pi/4}}{\sqrt{2\pi\beta}} \frac{\frac{1}{n} \sin\left(\frac{\pi}{n}\right)}{\cos\left(\frac{\pi}{n}\right) - \cos\left(\frac{\phi - \phi'}{n}\right)} \frac{e^{-j\beta\rho}}{\sqrt{\rho}} \end{aligned} \quad (13-61)$$

This is referred to as the *incident diffracted field*, and its form, as given by (13-61), is valid provided that the poles of (13-53a) are not near the saddle points of (13-59). Another way to say this is that (13-61) is valid provided that the observations are not made at or near the incident shadow boundary of Figure 13-13. When the observations are made at the incident shadow boundary $\phi - \phi' = \pi$, (13-61) becomes infinite. Therefore, another form must be used for such situations.

Following a similar procedure for the evaluation of (13-48b), it can be shown that its contributions along C_T and the saddle points can be written in forms corresponding to (13-58) and (13-61) for the evaluation of (13-48a). Thus, we can write that

$$\begin{aligned} F_2(\beta\rho)|_{C_T} &= \frac{1}{4\pi jn} \oint_{C_T} H_2(z) e^{\beta\rho h_2(z)} dz \\ &= e^{j\beta\rho \cos(\phi+\phi')} U[\pi - (\phi + \phi')] \end{aligned} \quad (13-62a)$$

$$\begin{aligned} F_2(\beta\rho)|_{\text{SDP}_{\pm\pi}} &= -\frac{1}{4\pi jn} \int_{\text{SDP}_{+\pi}} H_2(z) e^{\beta\rho h_2(z)} dz - \frac{1}{4\pi jn} \int_{\text{SDP}_{-\pi}} H_2(z) e^{\beta\rho h_2(z)} dz \\ &\simeq \frac{e^{-j\pi/4}}{\sqrt{2\pi\beta}} \frac{\frac{1}{n} \sin\left(\frac{\pi}{n}\right)}{\cos\left(\frac{\pi}{n}\right) - \cos\left(\frac{\phi+\phi'}{n}\right)} \frac{e^{-j\beta\rho}}{\sqrt{\rho}} \end{aligned} \quad (13-62b)$$

Equation 13-62b is valid provided the observations are not made at or near the reflection shadow boundary of Figure 13-13. When the observations are made at the reflection shadow boundary $\phi + \phi' = \pi$, (13-62b) becomes infinite. Another form must be used for such cases.

If the poles of (13-53a) are near the saddle points of (13-59), then the conventional steepest descent method of (13-60a) and (13-60b) cannot be used for the evaluation of the last two terms of (13-50) for $F_1(\beta\rho)$, and similarly for $F_2(\beta\rho)$ of (13-48b). One method that can be used for such cases is the so-called *Pauli-Clemmow modified method of steepest descent* [8, 47]. The main difference between the two methods is that the solution provided by the Pauli-Clemmow modified method of steepest descent has an additional discontinuous function that compensates for the singularity along the corresponding shadow boundaries introduced by the conventional steepest-descent method for isolated poles and saddle points. This factor is usually referred to as the *transition function*, and it is proportional to a Fresnel integral. Away from the corresponding shadow boundaries these transition functions are nearly unity, and the Pauli-Clemmow modified method of steepest descent reduces to the conventional method of steepest descent.

It can be shown that by using the Pauli-Clemmow modified method of steepest descent, (13-60a) and (13-60b) are evaluated using

$$\begin{aligned} \frac{1}{4\pi jn} \int_{\text{SDP}_{+\pi}} H_1(z) e^{\beta\rho h_1(z)} dz &\stackrel{\beta\rho \rightarrow \text{large}}{\simeq} \frac{1}{4\pi jn} \left| \sqrt{\frac{2\pi}{-\beta\rho h_1''(z_s = +\pi)}} \right| \\ &\quad \times e^{j\pi/4} H_1(z_s = +\pi) e^{\beta\rho h_1(z_s = \pi)} F[\beta\rho g^+(\xi^-)] \\ &= \frac{e^{-j\pi/4}}{2n\sqrt{2\pi\beta}} \cot\left[\frac{\pi + (\phi - \phi')}{2n}\right] \\ &\quad \times F[\beta\rho g^+(\phi - \phi')] \frac{e^{-j\beta\rho}}{\sqrt{\rho}} \end{aligned} \quad (13-63a)$$

$$\begin{aligned}
 \frac{1}{4\pi jn} \int_{\text{SDP}_{-\pi}} H_1(z) e^{\beta \rho h_1(z)} dz &\stackrel{\beta \rho \rightarrow \text{large}}{\simeq} \frac{1}{4\pi jn} \left| \sqrt{\frac{2\pi}{-\beta \rho h_1''(z_s = -\pi)}} \right| \\
 &\times e^{-j3\pi/4} H_1(z_s = -\pi) e^{\beta \rho h_1(z_s = -\pi)} F[\beta \rho g^-(\xi^-)] \\
 &= \frac{e^{-j\pi/4}}{2n\sqrt{2\pi\beta}} \cot \left[\frac{\pi - (\phi - \phi')}{2n} \right] \\
 &\times F[\beta \rho g^-(\phi - \phi')] \frac{e^{-j\beta \rho}}{\sqrt{\rho}}
 \end{aligned} \tag{13-63b}$$

where

$$\begin{aligned}
 F[\beta \rho g^\pm(\phi - \phi')] &\equiv j[h_1(z_s) - h_1(z_p)] \\
 &\equiv \text{measure of separation between saddle points and poles} \\
 &= 2j \left| \sqrt{\beta \rho g^\pm(\phi - \phi')} \right| e^{+j\beta \rho g^\pm} \int_{\sqrt{\beta \rho g^\pm(\phi - \phi')}}^{\infty} e^{-j\tau^2} d\tau
 \end{aligned} \tag{13-63c}$$

$$g^\pm(\phi - \phi') = 1 + \cos[(\phi - \phi') - 2\pi n N^\pm] \tag{13-63d}$$

$$2\pi n N^+ - (\phi - \phi') = +\pi \tag{13-63e}$$

$$2\pi n N^- - (\phi - \phi') = -\pi \tag{13-63f}$$

Similar forms are used for the evaluation of $F_2(\beta \rho)$ along the steepest descent path using the Pauli–Clemmow modified method of steepest descent. In (13-63e) and (13-63f), $N^\pm$ represents integer values that most closely satisfy the equalities. Such a procedure accounts for the poles that are nearest to the saddle point at $x = \pm\pi$, either from outside or within $-\pi \leq x \leq +\pi$, of Figure 13-18*b*. In general, there are two such poles associated with $F_1(\beta \rho)$ and two with $F_2(\beta \rho)$. More details about the transition function will follow.

D. Geometrical Optics and Diffracted Fields After the contour integration and the method of steepest descent have been applied in the evaluation of (13-47), as discussed in the previous section, it can be shown that for large values of $\beta \rho$, (13-47) is separated into

$$F(\beta \rho) = F_G(\beta \rho) + F_D(\beta \rho) \tag{13-64}$$

where $F_G(\beta \rho)$ and $F_D(\beta \rho)$ represent, respectively, the total geometrical optics and total diffracted fields created by the incidence of a unit amplitude plane wave upon a two-dimensional wedge, as shown in Figure 13-13.

In summary, then, the geometrical optics fields (F_G) and the diffracted fields (F_D) are represented, respectively, by

Geometrical Optics Fields		
<i>Incident GO</i>	<i>Reflected GO</i>	<i>Region</i>
$e^{j\beta \rho \cos(\phi - \phi')}$	$\pm e^{j\beta \rho \cos(\phi + \phi')}$	$0 < \phi < \pi - \phi'$
$e^{j\beta \rho \cos(\phi - \phi')}$		$\pi - \phi' < \phi < \pi + \phi'$
0		$\pi + \phi' < \phi < n\pi$

$$F_G(\beta \rho) = \begin{cases} e^{j\beta \rho \cos(\phi - \phi')} & \pm e^{j\beta \rho \cos(\phi + \phi')} \\ e^{j\beta \rho \cos(\phi - \phi')} & \\ 0 & \end{cases} \tag{13-65}$$

<u>Diffracted Fields</u>		
<i>Total diffracted field</i>	=	<i>Incident diffracted</i> $\pm$ <i>Reflected diffracted</i>
$F_D(\beta\rho) = F_D(\rho, \phi, \phi', n) = V_B(\rho, \phi, \phi', n)$	=	$V_B^i(\rho, \phi - \phi', n) \pm V_B^r(\rho, \phi + \phi', n)$

(13-66)

where

$$V_B^{i,r}(\rho, \phi \mp \phi', n) = I_{-\pi}(\rho, \phi \mp \phi', n) + I_{+\pi}(\rho, \phi \mp \phi', n) \quad (13-66a)$$

$$I_{\pm\pi}(\rho, \phi \mp \phi', n) \simeq \frac{e^{-j(\beta\rho + \pi/4)}}{jn\sqrt{2\pi}} \sqrt{g^\pm} \cot\left[\frac{\pi \pm (\phi \pm \phi')}{2n}\right] \\ \times e^{+j\beta\rho g^\pm} \int_{\sqrt{\beta\rho g^\pm}}^{\infty} e^{-j\tau^2} d\tau + (\text{higher-order terms}) \quad (13-66b)$$

$$g^+ = 1 + \cos[(\phi \mp \phi') - 2n\pi N^+] \quad (13-66c)$$

$$g^- = 1 + \cos[(\phi \mp \phi') - 2n\pi N^-] \quad (13-66d)$$

with N^+ or N^- being a positive or negative integer or zero that most closely satisfies the equation

$$2n\pi N^+ - (\phi \mp \phi') = +\pi \quad \text{for } g^+ \quad (13-66e)$$

$$2n\pi N^- - (\phi \mp \phi') = -\pi \quad \text{for } g^- \quad (13-66f)$$

Each of the diffracted fields (incident and reflected) exists in all space surrounding the wedge. Equation 13-66b contains the leading term of the diffracted field plus higher-order terms that are negligible for large values of $\beta\rho$. The integral in (13-66b) is a Fresnel integral (see Appendix III). In (13-65) and (13-66), the plus (+) sign is used for the hard polarization and the minus sign is used for the soft polarization.

If observations are made away from each of the shadow boundaries so that $\beta\rho g^\pm \gg 1$, (13-66a) and (13-66b) reduce, according to (13-61) and (13-62b), to

$V_B^{h,s}(\rho, \phi \mp \phi', n) = V_B^i(\rho, \phi - \phi', n) \pm V_B^r(\rho, \phi + \phi', n) \\ = \frac{e^{-j\pi/4}}{\sqrt{2\pi\beta}} \frac{1}{n} \sin\left(\frac{\pi}{n}\right) \left[\frac{1}{\cos\left(\frac{\pi}{n}\right) - \cos\left(\frac{\phi - \phi'}{n}\right)} \pm \frac{1}{\cos\left(\frac{\pi}{n}\right) - \cos\left(\frac{\phi + \phi'}{n}\right)} \right] \frac{e^{-j\beta\rho}}{\sqrt{\rho}} \quad (13-67)$
--

which is an expression of much simpler form, even for computational purposes. It is quite evident that when the observations are made at the incident shadow boundary (ISB, where $\phi = \pi + \phi'$), $V_B^i(\rho, \phi - \phi', n)$ of (13-67) becomes infinite because $\phi - \phi'$ is equal to π and the two cosine terms in the denominator of (13-67) are identical. Similarly, $V_B^r(\rho, \phi + \phi', n)$ of (13-67) becomes infinite when observations are made at the reflected shadow boundary (RSB, where $\phi = \pi - \phi'$). The incident and reflected diffraction functions of the form in equation 13-67 are referred to as *Keller's diffraction functions* and possess singularities along the incident and reflection shadow boundaries. The diffraction functions of (13-66a) through (13-66f) are representatives of the *uniform theory of diffraction* (UTD). The regions in the neighborhood of the incident and reflection shadow boundaries are referred to as the *transition regions*, and in these regions the fields undergo their most rapid changes.

The functions g^+ and g^- of (13-66c) and (13-66d) are representative of the angular separation between the observation point and the incident or reflection shadow boundary. In fact, when observations are made along the shadow boundaries, the $g^\pm$ functions are equal to zero. For *exterior wedges* ($1 \leq n \leq 2$), the values of N^+ and N^- in (13-66e) and (13-66f) are equal to $N^+ = 0$ or 1 and $N^- = -1, 0$, or 1. The values of n are plotted, as a function of ξ , where $-2\pi \leq \xi^\pm = \phi \pm \phi' \leq 4\pi$; in Figure 13-19a for $N^- = -1, 0, 1$ and in Figure 13-19b for $N^+ = 0, 1$. These integral values of $N^\pm$ are particularly important along the shadow boundaries, which are represented by the dotted lines. The variations of $N^\pm$ as a function of ϕ near the shadow boundaries are not abrupt, and this is a desirable property. The permissible values of $\xi^\pm = \phi \pm \phi'$ for $0 \leq \phi, \phi' \leq n\pi$ when $1 \leq n \leq 2$ are those bounded by the trapezoids formed by the solid straight lines in Figures 13-19a and 13-19b.

In order for (13-67) to be valid, $\beta\rho g^\pm \gg 1$. This can be achieved by having one of the following conditions:

1. $\beta\rho$ and $g^\pm$ large. This is satisfied if the distance ρ to the observation point is large and the observation angle ϕ is far away from either of the two shadow boundaries.
2. $\beta\rho$ large and $g^\pm$ small. This is satisfied if the distance ρ to the observation point is large and the observation angle ϕ is near either one or both of the shadow boundaries.
3. $\beta\rho$ small and $g^\pm$ large. This is satisfied if the distance ρ to the observation point is small and the observation angle ϕ is far away from either of the two shadow boundaries.

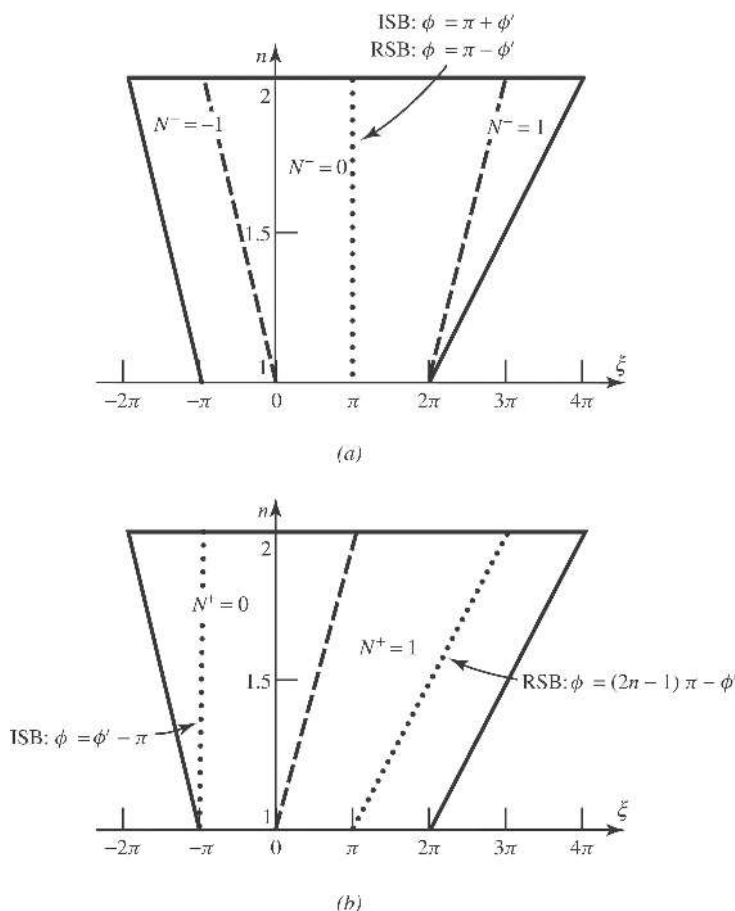


Figure 13-19 Graphical representation of (a) N^- and (b) N^+ as a function of ξ and n . (Source: R. G. Kouyoumjian and P. H. Pathak, "A uniform geometrical theory of diffraction for an edge in a perfectly conducting surface," *Proc. IEEE*, © 1974, IEEE.)

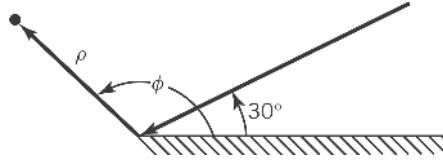


Figure 13-20 Plane wave incident on a half-plane.

To demonstrate, we consider the problem of a plane wave, of incidence angle $\phi' = 30^\circ$, impinging on a half-plane ($n = 2$), shown in Figure 13-20. In Figures 13-21a and 13-21b we have plotted as a function of ϕ , respectively, the magnitude of the incident V_B^i and the reflected V_B^r diffracted fields, as given by (13-66a) through (13-66f) for $\rho = \lambda$, and 100λ when $\phi' = 30^\circ$ and $n = 2$. In the same figures, these results are compared with those obtained using (13-67). It is apparent that as the observation distance ρ increases, the angular sector near the incident and reflected shadow boundaries over which (13-67) becomes invalid decreases; in the limit as $\rho \rightarrow \infty$, both give the same results.

The expression of (13-66) represents the diffraction of the unity strength incident plane wave with observations made at $P(\rho, \phi)$, as shown in Figure 13-13. Diffraction solutions of cylindrical waves, with observations at large distances, can be obtained by the use of the reciprocity principle along with the solution of the diffraction of an incident plane wave by a wedge as given by (13-66) through (13-66f). Using the geometry of Figure 13-17b, it is evident that cylindrical wave incidence diffraction can be obtained by substituting in (13-66) through (13-66f) $\rho = \rho'$, $\phi = \phi'$, and $\phi' = \phi$.

E. Diffraction Coefficients The incident diffraction function V_B^i of (13-66) can also be written, using (13-66a) through (13-66f), as

$$V_B^i(\rho, \phi - \phi', n) = V_B^i(\rho, \xi^-, n) = \frac{e^{-j\beta\rho}}{\sqrt{\rho}} D^i(\rho, \xi^-, n) \quad (13-68)$$

where

$$\xi^- = \phi - \phi' \quad (13-68a)$$

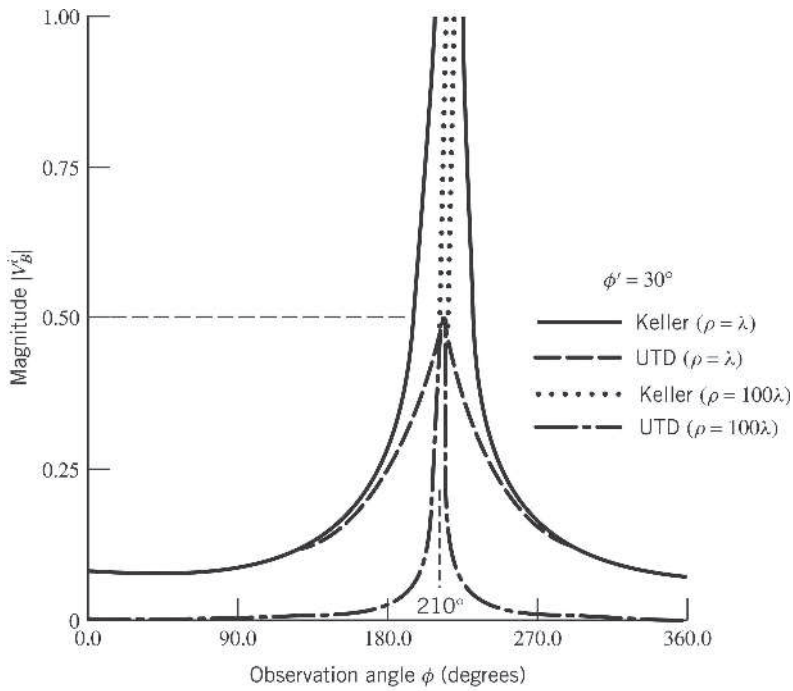
$$D^i(\rho, \xi^-, n) = -\frac{e^{-j\pi/4}}{2n\sqrt{2\pi\beta}} \{C^+(\xi^-, n)F[\beta\rho g^+(\xi^-)] + C^-(\xi^-, n)F[\beta\rho g^-(\xi^-)]\} \quad (13-68b)$$

$$C^+(\xi^-, n) = \cot\left(\frac{\pi + \xi^-}{2n}\right) \quad (13-68c)$$

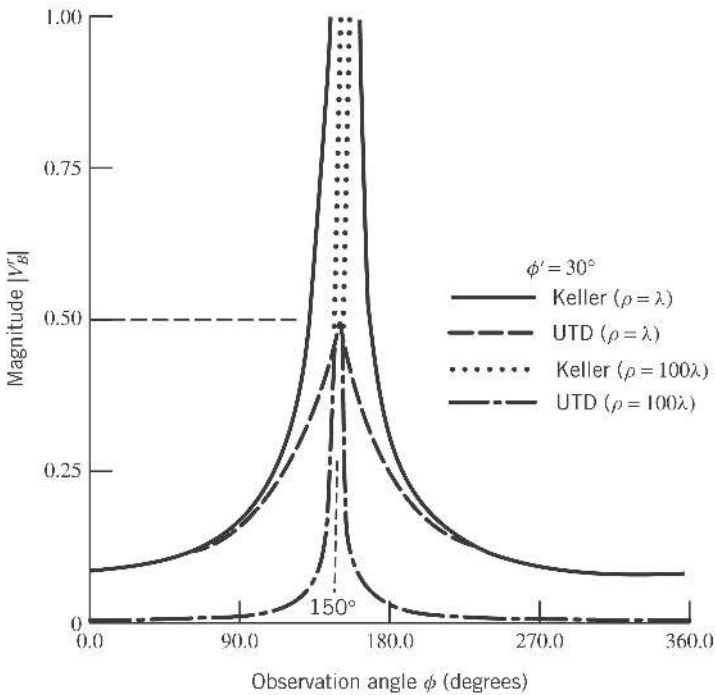
$$C^-(\xi^-, n) = \cot\left(\frac{\pi - \xi^-}{2n}\right) \quad (13-68d)$$

$$F[\beta\rho g^+(\xi^-)] = 2j\sqrt{\beta\rho g^+(\xi^-)} e^{+j\beta\rho g^+(\xi^-)} \int_{\sqrt{\beta\rho g^+(\xi^-)}}^{\infty} e^{-j\tau^2} d\tau \quad (13-68e)$$

$$F[\beta\rho g^-(\xi^-)] = 2j\sqrt{\beta\rho g^-(\xi^-)} e^{+j\beta\rho g^-(\xi^-)} \int_{\sqrt{\beta\rho g^-(\xi^-)}}^{\infty} e^{-j\tau^2} d\tau \quad (13-68f)$$



(a)



(b)

Figure 13-21 Plane wave diffraction by a half-plane. (a) Incident diffracted field. (b) Reflected diffracted field.

The functions $g^+(\xi^-)$ and $g^-(\xi^-)$ are given by (13-66c) through (13-66f). In (13-68), $D^i(\rho, \xi^-, n)$ is defined as the diffraction coefficient for the incident diffracted field, and it will be referred to as the *incident diffraction coefficient* for a unit amplitude incident plane wave.

In a similar manner, the reflected diffraction function V_B^r of (13-66) can also be written, using (13-66a) through (13-66f), as

$$V_B^r(\rho, \phi + \phi', n) = V_B^r(\rho, \xi^+, n) = \frac{e^{-j\beta\rho}}{\sqrt{\rho}} D^r(\rho, \xi^+, n) \quad (13-69)$$

where

$$\xi^+ = \phi + \phi' \quad (13-69a)$$

$$D^r(\rho, \xi^+, n) = -\frac{e^{-j\pi/4}}{2n\sqrt{2\pi\beta}} \left\{ C^+(\xi^+, n) F[\beta\rho g^+(\xi^+)] + C^-(\xi^+, n) F[\beta\rho g^-(\xi^+)] \right\} \quad (13-69b)$$

$$C^+(\xi^+, n) = \cot \left(\frac{\pi + \xi^+}{2n} \right) \quad (13-69c)$$

$$C^-(\xi^+, n) = \cot \left(\frac{\pi - \xi^+}{2n} \right) \quad (13-69d)$$

$$F[\beta\rho g^+(\xi^+)] = 2j\sqrt{\beta\rho g^+(\xi^+)} e^{+j\beta\rho g^+(\xi^+)} \int_{\sqrt{\beta\rho g^+(\xi^+)}}^{\infty} e^{-j\tau^2} d\tau \quad (13-69e)$$

$$F[\beta\rho g^-(\xi^+)] = 2j\sqrt{\beta\rho g^-(\xi^+)} e^{+j\beta\rho g^-(\xi^+)} \int_{\sqrt{\beta\rho g^-(\xi^+)}}^{\infty} e^{-j\tau^2} d\tau \quad (13-69f)$$

$D^r(\rho, \xi^+, n)$ will be referred to as the *reflection diffraction coefficient* for a unit amplitude incident plane wave.

With the aid of (13-68) and (13-69), the total diffraction function $V_B(\rho, \phi \mp \phi', n) = V_B^i(\rho, \phi - \phi', n) \mp V_B^r(\rho, \phi + \phi', n)$ of (13-66) can now be written as

$$\begin{aligned} V_B(\rho, \phi, \phi', n) &= V_B^i(\rho, \phi - \phi', n) \mp V_B^r(\rho, \phi + \phi', n) \\ &= \frac{e^{-j\beta\rho}}{\sqrt{\rho}} \left[D^i(\rho, \phi - \phi', n) \mp D^r(\rho, \phi + \phi', n) \right] \end{aligned} \quad (13-70)$$

or

$$\begin{aligned} V_{Bs}(\rho, \phi, \phi', n) &= V_B^i(\rho, \phi - \phi', n) - V_B^r(\rho, \phi + \phi', n) \\ &= \frac{e^{-j\beta\rho}}{\sqrt{\rho}} \left[D^i(\rho, \phi - \phi', n) - D^r(\rho, \phi + \phi', n) \right] \\ &= \frac{e^{-j\beta\rho}}{\sqrt{\rho}} D_s(\rho, \phi, \phi', n) \end{aligned} \quad (13-70a)$$

$$\begin{aligned}
 V_{Bh}(\rho, \phi, \phi', n) &= V_B^i(\rho, \phi - \phi', n) + V_B^r(\rho, \phi + \phi', n) \\
 &= \frac{e^{-j\beta\rho}}{\sqrt{\rho}} \left[D^i(\rho, \phi - \phi', n) + D^r(\rho, \phi + \phi', n) \right] \\
 &= \frac{e^{-j\beta\rho}}{\sqrt{\rho}} D_h(\rho, \phi, \phi', n)
 \end{aligned} \tag{13-70b}$$

$$D_s(\rho, \phi, \phi', n) = D^i(\rho, \phi - \phi', n) - D^r(\rho, \phi + \phi', n) \tag{13-70c}$$

$$D_h(\rho, \phi, \phi', n) = D^i(\rho, \phi - \phi', n) + D^r(\rho, \phi + \phi', n) \tag{13-70d}$$

where $V_{Bs} = V_B^i - V_B^r$ = soft (polarization) diffraction function
 $V_{Bh} = V_B^i + V_B^r$ = hard (polarization) diffraction function
 V_B^i = incident diffraction function
 V_B^r = reflection diffraction function
 $D_s = D^i - D^r$ = soft (polarization) diffraction coefficient
 $D_h = D^i + D^r$ = hard (polarization) diffraction coefficient
 D^i = incident diffraction coefficient
 D^r = reflection diffraction coefficient

Using (13-68b) and (13-69b) in expanded form, the soft (D_s) and hard (D_h) diffraction coefficients can ultimately be written, respectively, as

$$\begin{aligned}
 D_s(\rho, \phi, \phi', n) &= -\frac{e^{-j\pi/4}}{2n\sqrt{2\pi\beta}} \\
 &\times \left\{ \cot \left[\frac{\pi + (\phi - \phi')}{2n} \right] F[\beta\rho g^+(\phi - \phi')] + \cot \left[\frac{\pi - (\phi - \phi')}{2n} \right] F[\beta\rho g^-(\phi - \phi')] \right\} \\
 &- \left\{ \cot \left[\frac{\pi + (\phi + \phi')}{2n} \right] F[\beta\rho g^+(\phi + \phi')] + \cot \left[\frac{\pi - (\phi + \phi')}{2n} \right] F[\beta\rho g^-(\phi + \phi')] \right\}
 \end{aligned} \tag{13-71a}$$

$$\begin{aligned}
 D_h(\rho, \phi, \phi', n) &= -\frac{e^{-j\pi/4}}{2n\sqrt{2\pi\beta}} \\
 &\times \left\{ \cot \left[\frac{\pi + (\phi - \phi')}{2n} \right] F[\beta\rho g^+(\phi - \phi')] + \cot \left[\frac{\pi - (\phi - \phi')}{2n} \right] F[\beta\rho g^-(\phi - \phi')] \right\} \\
 &+ \left\{ \cot \left[\frac{\pi + (\phi + \phi')}{2n} \right] F[\beta\rho g^+(\phi + \phi')] + \cot \left[\frac{\pi - (\phi + \phi')}{2n} \right] F[\beta\rho g^-(\phi + \phi')] \right\}
 \end{aligned} \tag{13-71b}$$

where

$$\phi - \phi' = \xi^- \tag{13-71c}$$

$$\phi + \phi' = \xi^+ \tag{13-71d}$$

The formulations of (13-68) through (13-71d) are part of the often referred to *uniform theory of diffraction* (UTD) [10], which are extended to include oblique incidence and curved edge diffraction. These will be discussed in Sections 13.3.3 and 13.3.4, respectively.

A Fortran and MATLAB computer program designated WDC, Wedge Diffraction Coefficients, computes the soft and hard polarization diffraction coefficients of (13-71a) and (13-71b) (*actually the diffraction coefficients normalized by $\sqrt{\lambda}$*). The Fortran program was initially developed and reported in [53]. The program also accounts for oblique wave incidence, and it is based on the more general formulation of (13-89a) through (13-90b). The main difference in the two sets of equations is the $\sin\beta'_0$ function found in the denominator of (13-90a) and (13-90b); it has been introduced to account for the oblique wave incidence. In (13-90a) and (13-90b), L is used as the distance parameter and ρ is used in (13-71a) and (13-71b).

Therefore, to use the subroutine WDC to compute (13-71a) and (13-71b) let the oblique incidence angle β'_0 , referred to as BTD, be 90° . The distance parameter R should represent ρ (in wavelengths). This program uses the complex function FTF (Fresnel transition function) to complete its computations. The FTF program computes (13-68e), (13-68f), (13-69e), and (13-69f) based on the asymptotic expressions of (13-74a), (13-74b), and a linear interpolation for intermediate arguments. Computations of the Fresnel integral can also be made on an algorithm reported in [49] as well as on approximate expressions of [50].

To use the WDC subroutine, the user must specify $R = \rho$ (in wavelengths), $\text{PHID} = \phi$ (in degrees), $\text{PHIPD} = \phi'$ (in degrees), $\text{BTD} = \beta'_0$ (in degrees), and $\text{FN} = n$ (dimensionless). The program subroutine computes the normalized (with respect to $\sqrt{\lambda}$) diffraction coefficients $\text{CDCS} = D_s$ and $\text{CDCH} = D_h$. The angles represented by ϕ and ϕ' should be referenced from the face of the wedge, as shown in Figure 13-13b. For normal incidence, $\beta'_0 = 90^\circ$. This computer subroutine has been used successfully in a multitude of problems; it is very efficient, and the user is encouraged to utilize it effectively.

Following a similar procedure, the incident and reflected diffraction functions and the incident, reflected, soft, and hard diffraction coefficients using Keller's diffraction functions of (13-67) can be written, respectively, as

$$V_B^i(\rho, \phi - \phi', n) = V_B^i(\rho, \xi^-, n) = \frac{e^{-j\beta\rho}}{\sqrt{\rho}} D^i(\rho, \phi - \phi', n) = \frac{e^{-j\beta\rho}}{\sqrt{\rho}} D^i(\rho, \xi^-, n) \quad (13-72a)$$

$$V_B^r(\rho, \phi + \phi', n) = V_B^r(\rho, \xi^+, n) = \frac{e^{-j\beta\rho}}{\sqrt{\rho}} D^r(\rho, \phi - \phi', n) = \frac{e^{-j\beta\rho}}{\sqrt{\rho}} D^r(\rho, \xi^+, n) \quad (13-72b)$$

$$D^i(\rho, \phi - \phi', n) = \frac{e^{-j\pi/4}}{\sqrt{2\pi\beta}} \frac{\frac{1}{n} \sin\left(\frac{\pi}{n}\right)}{\cos\left(\frac{\pi}{n}\right) - \cos\left(\frac{\phi - \phi'}{n}\right)} \quad (13-72c)$$

$$D^r(\rho, \phi + \phi', n) = \frac{e^{-j\pi/4}}{\sqrt{2\pi\beta}} \frac{\frac{1}{n} \sin\left(\frac{\pi}{n}\right)}{\cos\left(\frac{\pi}{n}\right) - \cos\left(\frac{\phi + \phi'}{n}\right)} \quad (13-72d)$$

$$D_s(\rho, \phi, \phi', n) = D^i(\rho, \phi - \phi', n) - D^r(\rho, \phi + \phi', n)$$

$$= \frac{e^{-j\pi/4} \frac{1}{n} \sin\left(\frac{\pi}{n}\right)}{\sqrt{2\pi\beta}} \left[\frac{1}{\cos\left(\frac{\pi}{n}\right) - \cos\left(\frac{\phi - \phi'}{n}\right)} - \frac{1}{\cos\left(\frac{\pi}{n}\right) - \cos\left(\frac{\phi + \phi'}{n}\right)} \right] \quad (13-72e)$$

$$D_h(\rho, \phi, \phi', n) = D^i(\rho, \phi - \phi', n) + D^r(\rho, \phi + \phi', n)$$

$$= \frac{e^{-j\pi/4} \frac{1}{n} \sin\left(\frac{\pi}{n}\right)}{\sqrt{2\pi\beta}} \left[\frac{1}{\cos\left(\frac{\pi}{n}\right) - \cos\left(\frac{\phi - \phi'}{n}\right)} + \frac{1}{\cos\left(\frac{\pi}{n}\right) - \cos\left(\frac{\phi + \phi'}{n}\right)} \right] \quad (13-72f)$$

The diffraction coefficients of (13-72c) through (13-72f) are referred to as *Keller's diffraction coefficients*, and they possess singularities along the incident and reflection shadow boundaries.

The wedge diffraction coefficients described previously assume that the orientational direction for the incident and diffracted electric and magnetic field components of the soft and hard polarized fields are those shown, respectively, in Figures 13-22a and 13-22b. A negative value in the diffraction coefficients will reverse the directions of the appropriate fields.

The function $F(X)$ of (13-63c), (13-68e), (13-68f), (13-69e), and (13-69f) is known as a *Fresnel transition function* and it involves a Fresnel integral. Its magnitude and phase for $0.001 \leq X \leq 10$ are shown plotted in Figure 13-23. It is evident that

$$\left. \begin{aligned} |F(X)| &\leq 1 \\ 0 \leq \text{Phase of } F(X) &\leq \pi/4 \end{aligned} \right\}, \quad \text{for } 0.001 \leq X \leq 10 \quad (13-73a)$$

and

$$F(X) \simeq 1, \quad \text{for } X > 10 \quad (13-73b)$$

Thus, if the argument X of the transition function exceeds 10, it can be replaced by unity. Then the expressions for the diffraction coefficients of (13-68b), (13-69b), (13-71a), and (13-71b) reduce, respectively, to those of (13-72c) through (13-72f). Asymptotic expressions for the transition function $F(X)$ are [53]:

$$\begin{aligned} &\text{For small } X(X < 0.3) \\ F(X) &\simeq \left[\sqrt{\pi X} - 2X e^{j\pi/4} - \frac{2}{3} X^2 e^{-j\pi/4} \right] e^{j(\pi/4 + X)} \end{aligned} \quad (13-74a)$$

$$\begin{aligned} &\text{For large } X(X > 5.5) \\ F(X) &\simeq \left[1 + j \frac{1}{2X} - \frac{3}{4} \frac{1}{X^2} - j \frac{15}{8} \frac{1}{X^3} + \frac{75}{16} \frac{1}{X^4} \right] \end{aligned} \quad (13-74b)$$

To facilitate the reader in the computations, a Fortran and MATLAB computer function program designated as FTF, for Fresnel Transition Function, computes the wedge transition function $F(X)$ of (13-68e), (13-68f), (13-69e), or (13-69f). The algorithm is based on the approximations of

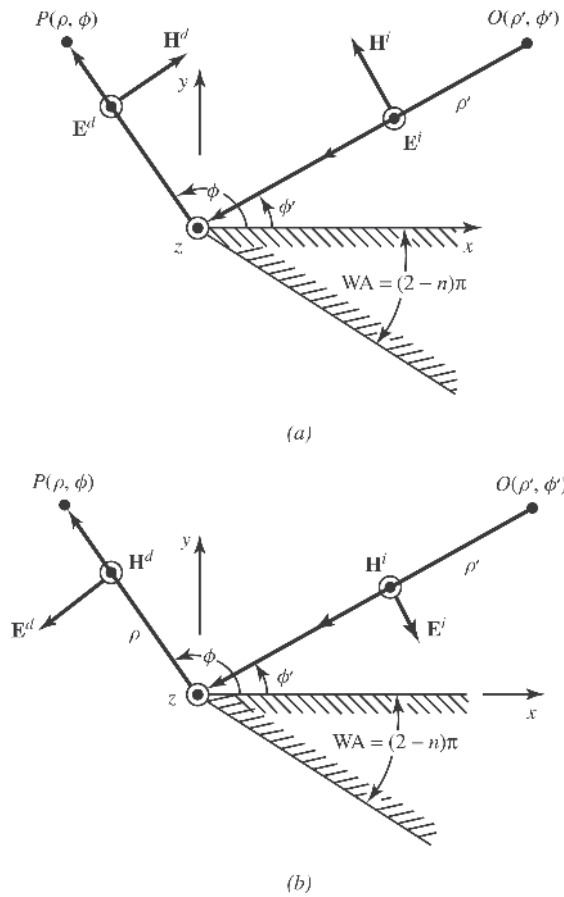


Figure 13-22 Polarization of incident and diffracted fields. (a) Soft polarization. (b) Hard polarization.

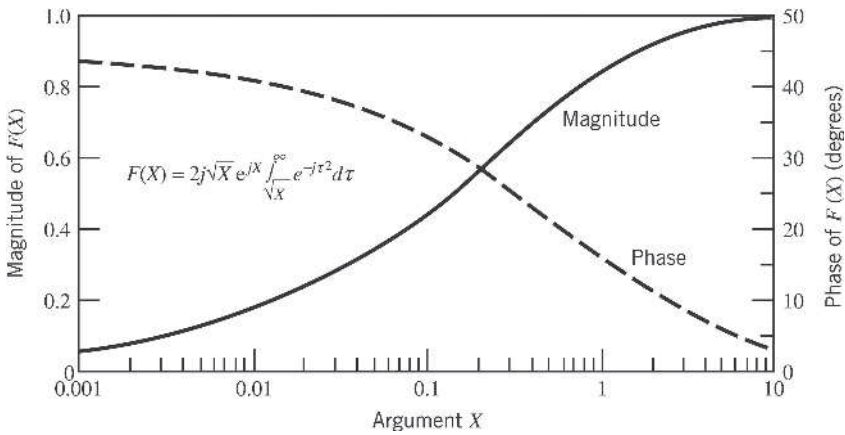


Figure 13-23 Variations of magnitude and phase of transition function $F(X)$ as a function of X . (Source: R. G. Kouyoumjian and P. H. Pathak, "A uniform geometrical theory of diffraction for an edge in a perfectly conducting surface," *Proc. IEEE*, © 1974, IEEE.)

(13-74a) for small arguments ($X < 0.3$) and on (13-74b) for large arguments ($X > 5.5$). For intermediate values ($0.3 \leq X \leq 5.5$), a linear interpolation scheme is used. The program was developed and reported in [53].

To have a better understanding of the UTD diffraction coefficients $D^i(\rho, \phi - \phi', n)$ and $D^r(\rho, \phi + \phi', n)$ of (13-68b) and (13-69b), let us examine their behavior around the incident and reflection shadow boundaries. This will be accomplished by considering only exterior wedges ($n \geq 1$) and examining separately the geometry where the incident wave illuminates the $\phi = 0$ side of the wedge [$\phi' \leq (n-1)\pi$], as shown in Figure 13-24a, and the geometry where the incident wave illuminates the $\phi = n\pi$ side of the wedge [$\phi' \geq (n-1)\pi$], as shown in Figure 13-24b.

Case A [$\phi' \leq (n-1)\pi$], Figure 13-24a

For this case the incident shadow boundary (ISB) occurs when $\phi = \pi + \phi'$ (or $\phi - \phi' = \pi$) and the reflection shadow boundary (RSB) occurs when $\phi = \pi - \phi'$ (or $\phi + \phi' = \pi$).

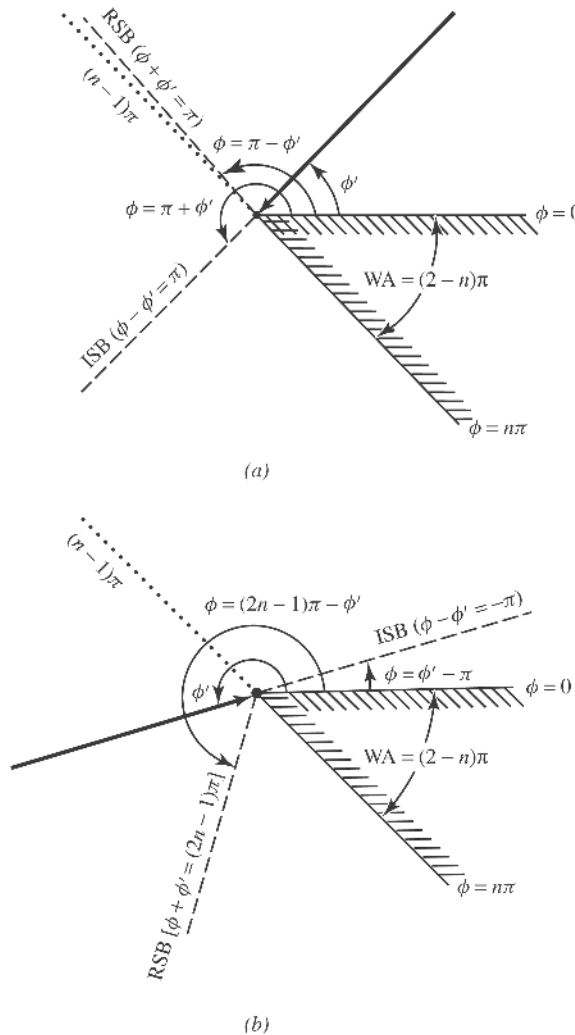


Figure 13-24 Incident and reflection shadow boundaries for wedge diffraction. (a) Case A: $\phi' \leq (n-1)\pi$. (b) Case B: $\phi' \geq (n-1)\pi$.

1. Along the ISB ($\phi = \pi + \phi'$ or $\phi - \phi' = \pi$), the second cotangent function of (13-68b) becomes singular and the first cotangent function in (13-68b), and both of them in (13-69b), remain bounded. That is, from (13-68b),

$$C^-(\xi^-, n)|_{\phi-\phi'=\pi} = \cot\left[\frac{\pi-\xi^-}{2n}\right]\bigg|_{\phi-\phi'=\pi} = \cot\left[\frac{\pi-(\phi-\phi')}{2n}\right]\bigg|_{\phi-\phi'=\pi} = \infty \quad (13-75a)$$

In addition, the value of N^- from (13-66f) is equal to 0. That is,

$$2n\pi N^- - (\phi - \phi')|_{\phi-\phi'=\pi} = 2n\pi N^- - \pi = -\pi \Rightarrow N^- = 0 \quad (13-75b)$$

2. Along the RSB ($\phi = \pi - \phi'$ or $\phi + \phi' = \pi$), the second cotangent function of (13-69b) becomes singular and the first cotangent function in (13-69b) and both of them in (13-68b) remain bounded. That is, from (13-69b),

$$C^-(\xi^+, n)|_{\phi+\phi'=\pi} = \cot\left[\frac{\pi-\xi^+}{2n}\right]\bigg|_{\phi+\phi'=\pi} = \cot\left[\frac{\pi-(\phi+\phi')}{2n}\right]\bigg|_{\phi+\phi'=\pi} = \infty \quad (13-76a)$$

In addition, the value of N^- from (13-66f) is equal to 0. That is,

$$2n\pi N^- - (\phi + \phi')|_{\phi+\phi'=\pi} = 2n\pi N^- - \pi = -\pi \Rightarrow N^- = 0 \quad (13-76b)$$

Case B [$\phi' \geq (n-1)\pi$], Figure 13-24b

For this case, the incident shadow boundary (ISB) occurs when $\phi = \phi' - \pi$ (or $\phi - \phi' = -\pi$) and the reflection shadow boundary (RSB) occurs when $\phi = (2n-1)\pi - \phi'$ [or $\phi + \phi' = (2n-1)\pi$].

1. Along the ISB ($\phi = \phi' - \pi$ or $\phi - \phi' = -\pi$), the first cotangent function in (13-68b) becomes singular and the second cotangent function of (13-68b), and both of them in (13-69b), remain bounded. That is, from (13-68b),

$$\begin{aligned} C^+(\xi^-, n)|_{\phi-\phi'=-\pi} &= \cot\left[\frac{\pi+\xi^-}{2n}\right]\bigg|_{\phi-\phi'=-\pi} \\ &= \cot\left[\frac{\pi+(\phi-\phi')}{2n}\right]\bigg|_{\phi-\phi'=-\pi} = \infty \end{aligned} \quad (13-77a)$$

In addition, the value of N^+ from (13-66e) is equal to 0. That is,

$$2n\pi N^+ - (\phi - \phi')|_{\phi-\phi'=-\pi} = 2n\pi N^+ + \pi = +\pi \Rightarrow N^+ = 0 \quad (13-77b)$$

2. Along the RSB [$\phi = (2n-1)\pi - \phi'$ or $\phi + \phi' = (2n-1)\pi$], the first cotangent function of (13-69b) becomes singular and the second cotangent function of (13-69b), and both of them in (13-68b), remain bounded. That is, from (13-69b),

$$\begin{aligned} C^+(\xi^+, n)|_{\phi+\phi'=(2n-1)\pi} &= \cot\left[\frac{\pi+\xi^+}{2n}\right]\bigg|_{\phi+\phi'=(2n-1)\pi} \\ &= \cot\left[\frac{\pi+(\phi+\phi')}{2n}\right]\bigg|_{\phi+\phi'=(2n-1)\pi} = \infty \end{aligned} \quad (13-78a)$$

In addition, the value of N^+ from (13-66e) is equal to 1. That is,

$$2n\pi N^+ - (\phi + \phi')|_{\phi+\phi'=(2n-1)\pi} = 2n\pi N^+ - (2n-1)\pi = +\pi \Rightarrow N^+ = 1 \quad (13-78b)$$

The results for Cases A and B are summarized in Table 13-1.

TABLE 13-1 Cotangent function behavior and values of $N^\pm$ along the shadow boundaries

	The cotangent function becomes singular when	Value of $N^\pm$ at the shadow boundary
$\cot \left[\frac{\pi - (\phi - \phi')}{2n} \right]$	$\phi = \pi + \phi'$ or $\phi - \phi' = \pi$ ISB of Case A Figure 13-24a	$N^- = 0$
$\cot \left[\frac{\pi - (\phi + \phi')}{2n} \right]$	$\phi = \pi - \phi'$ or $\phi + \phi' = \pi$ RSB of Case A Figure 13-24a	$N^- = 0$
$\cot \left[\frac{\pi + (\phi - \phi')}{2n} \right]$	$\phi = \phi' - \pi$ or $\phi - \phi' = -\pi$ ISB of Case B Figure 13-24b	$N^+ = 0$
$\cot \left[\frac{\pi + (\phi + \phi')}{2n} \right]$	$\phi = (2n - 1)\pi - \phi'$ or $\phi + \phi' = (2n - 1)\pi$ RSB of Case B Figure 13-24b	$N^+ = 1$

Whereas in the diffraction coefficients of UTD one of the cotangent functions becomes singular along the incident or reflection shadow boundary, while the other three cotangent functions are bounded, the product of the cotangent function along with its corresponding Fresnel transition function along that shadow boundary is discontinuous but bounded. It is this finite discontinuity created by the singular cotangent term and its corresponding Fresnel transition function that removes the bounded geometrical optics discontinuity along that boundary.

To demonstrate, let us consider one of the four shadow boundaries created in Figure 13-24. We choose the ISB ($\phi = \pi + \phi'$ or $\phi - \phi' = \pi$) of Figure 13-24a where $\phi' \leq (n - 1)\pi$. Similar results are found for the other three choices. At the ISB of Figure 13-24a

$$\xi^- = \phi - \phi' = \pi \quad (13-79)$$

and in the neighborhood of it

$$\xi^- = \phi - \phi' = \pi - \varepsilon \quad (13-79a)$$

where ε is positive on the illuminated side of the incident shadow boundary. Using (13-79a), we can write (13-66f) as

$$2n\pi N^- - (\phi - \phi') = 2n\pi N^- - (\pi - \varepsilon) = (2nN^- - 1)\pi + \varepsilon = -\pi \quad (13-80)$$

For this situation, the cotangent function that becomes singular is that shown in the first row of Table 13-1 whose N^- value is 0. Therefore, that cotangent function near the ISB of Figure 13-24a can be written using (13-80) with $N^- = 0$ (or $\phi - \phi' = \pi - \varepsilon$) as

$$\begin{aligned} C^-(\phi - \phi', n) &= \cot \left[\frac{\pi - (\phi - \phi')}{2n} \right] \\ &= \cot \left[\frac{\pi - \pi + \varepsilon}{2n} \right] = \cot \left(\frac{\varepsilon}{2n} \right) \simeq \frac{2n}{\varepsilon} = \frac{2n}{|\varepsilon| \operatorname{sgn}(\varepsilon)} \end{aligned} \quad (13-80a)$$

where sgn is the sign function. According to (13-66d),

$$\begin{aligned} g^-(\xi^-) &= g^-(\phi - \phi') = 1 + \cos[(\phi - \phi') - 2\pi n N^-] = 1 + \cos(\phi - \phi') \\ g^-(\phi - \phi') &= 1 + \cos(\pi - \varepsilon) = 1 - \cos(\varepsilon) \xrightarrow{\varepsilon \rightarrow 0} 1 - \left(1 - \frac{\varepsilon^2}{2} \right) = \frac{\varepsilon^2}{2} \end{aligned} \quad (13-80b)$$

The transition function of (13-68f) can also be written using (13-80b) as

$$F[\beta\rho g^-(\xi^-)] = F[\beta\rho g^-(\phi - \phi')] = F\left[\beta\rho\left(\frac{\varepsilon^2}{2}\right)\right] \quad (13-80c)$$

which for small values of its argument can be approximated by the first term of its small-argument asymptotic form (13-74a). That is,

$$F\left(\frac{\beta\rho\varepsilon^2}{2}\right) \simeq \sqrt{\frac{\pi\beta\rho\varepsilon^2}{2}} e^{j\pi/4} = |\varepsilon| \sqrt{\frac{\pi\beta\rho}{2}} e^{j\pi/4} \quad (13-81)$$

Thus, the product of $C^-(\phi - \phi', n)F[\beta\rho g^-(\phi - \phi')]$, as each is given by (13-80a) and (13-81), can be approximated by

$$\cot\left[\frac{\pi - (\phi - \phi')}{2n}\right] F[\beta\rho g^-(\phi - \phi')] = n\sqrt{2\pi\beta\rho} \operatorname{sgn}(\varepsilon) e^{j\pi/4} \quad (13-82)$$

It is apparent that (13-82) exhibits a finite discontinuity that is positive along the illuminated side of the incident shadow boundary and negative on the other side.

The corresponding incident diffracted field (13-68) along the incident shadow boundary can be approximated, using (13-82) and only the second term within the brackets in (13-68b), as

$$V_B^i(\rho, \phi - \phi' = \pi - \varepsilon, n) \simeq \frac{e^{-j\beta\rho}}{\sqrt{\rho}} \left[-\frac{e^{-j\pi/4}}{2n\sqrt{2\pi\beta}} n\sqrt{2\pi\beta\rho} \operatorname{sgn}(\varepsilon) e^{j\pi/4} \right] = -\frac{e^{-j\beta\rho}}{2} \operatorname{sgn}(\varepsilon) \quad (13-83)$$

Apart from the phase factor, this function is equal to -0.5 , on the illuminated side of the incident shadow boundary and $+0.5$ on the other side. Clearly, such a bounded discontinuity possesses the proper magnitude and polarity to compensate for the discontinuity created by the geometrical optics field. A similar procedure can be used to demonstrate the discontinuous nature of the diffracted field along the other shadow boundaries of Figure 13-24.

To illustrate the principles of geometrical optics (GO) and geometrical theory of diffraction (GTD), an example will be considered next.

Example 13-3

A plane wave of unity amplitude is incident upon a half-plane ($n = 2$) at an incidence angle of $\phi' = 30^\circ$, as shown in Figure 13-20. At a distance of one wavelength ($\rho = \lambda$) from the edge of the wedge, compute and plot the amplitude and phase of the following:

1. Total (incident plus reflected) geometrical optics field.
2. Incident diffracted field.
3. Reflected diffracted field.
4. Total field (geometrical optics plus diffracted).

Do these for both *soft* and *hard* polarizations.

Solution: The geometrical optics field components are computed using (13-65); incident and reflected diffracted fields are computed using (13-68) through (13-69f). These are plotted in Figure 13-25a and 13-25b for *soft* and *hard* polarizations, respectively.

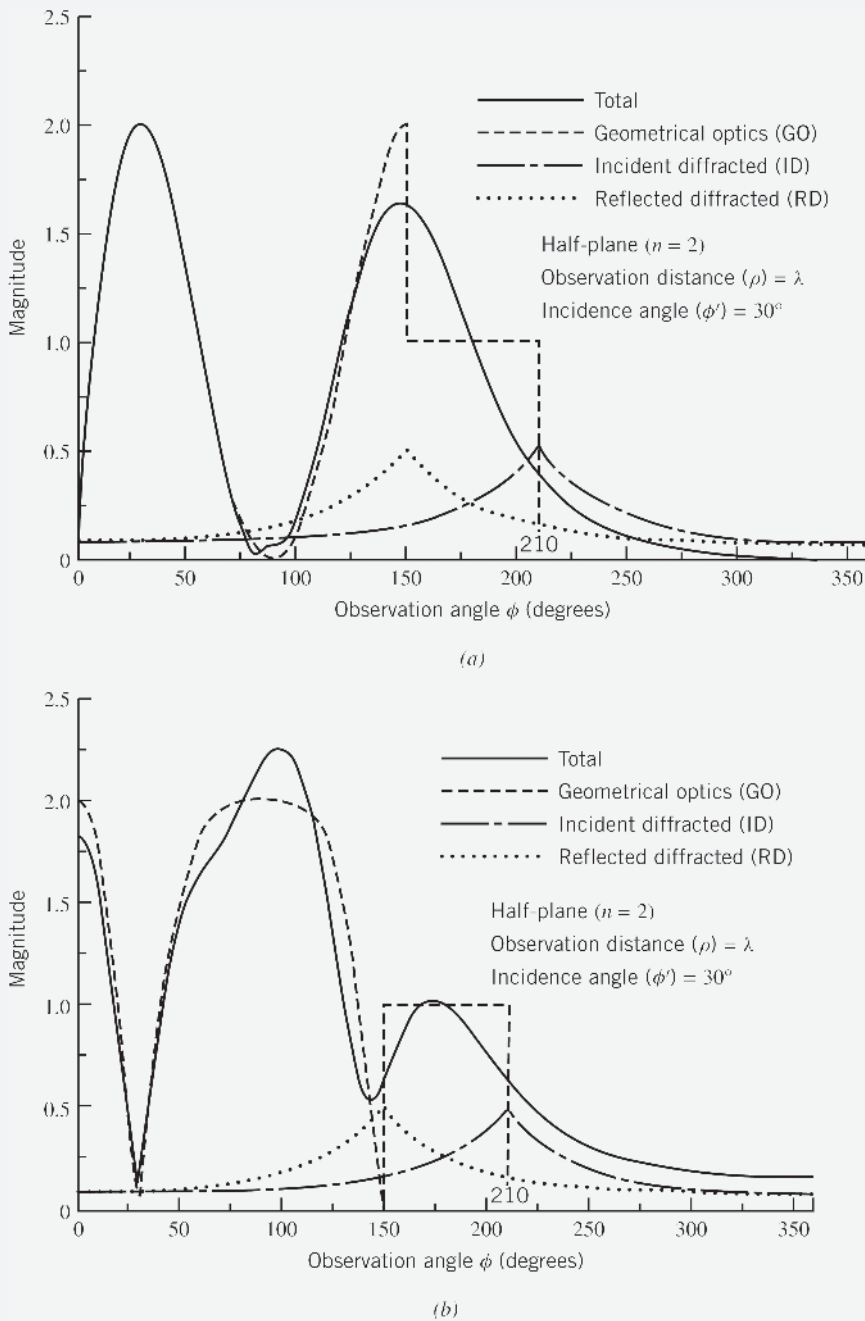


Figure 13-25 Field distribution of various components of a plane wave incident normally on a conducting half-plane. (a) Soft polarization. (b) Hard polarization. (Source: C. A. Balanis, *Antenna Theory: Analysis and Design*, copyright © 1982, John Wiley & Sons, Inc. Reprinted by permission of John Wiley & Sons, Inc.)

In Figure 13-25a the amplitude patterns of the geometrical optics and diffracted fields for the soft polarization are displayed as follows. The

1. Dashed curve (----) represents the total geometrical optics field (incident and reflected) computed using (13-65).
2. Dash-dot curve (- · - · -) represents the amplitude of the incident diffracted (ID) field V_B^i computed using (13-68) through (13-68f).
3. Dotted curve (····) represents the amplitude of the reflected diffracted (RD) field V_B^r computed using (13-69) through (13-69f).
4. Solid curve (—) represents the total amplitude pattern for soft polarization computed using results from parts 1 through 3.

Observing the data of Figure 13-25a it is evident that the

1. GO field is discontinuous at the reflection shadow boundary (RSB) ($\phi = 180^\circ - \phi' = 180^\circ - 30^\circ = 150^\circ$) and at the incident shadow boundary (ISB) ($\phi = 180^\circ + \phi' = 180^\circ + 30^\circ = 210^\circ$).
2. GO field in the shadow region ($210^\circ < \phi < 360^\circ$) is zero.
3. Reflected diffracted (RD) field, although it exists everywhere, predominates around the reflection shadow boundary ($\phi = 150^\circ$) with values of -0.5 for $\phi = (150^\circ)^-$ and $+0.5$ for $\phi = (150^\circ)^+$. The total discontinuity at $\phi = 150^\circ$ occurs because the phase undergoes a phase jump of 180° .
4. Incident diffracted (ID) field also exists everywhere but it predominates around the incident shadow boundary ($\phi = 210^\circ$) with values of -0.5 for $\phi = (210^\circ)^-$ and $+0.5$ for $\phi = (210^\circ)^+$. The total discontinuity at $\phi = 210^\circ$ occurs because the phase undergoes a phase jump of 180° .
5. Total amplitude field pattern is continuous everywhere with the discontinuities of the GO field compensated with the inclusion of the diffracted fields. It should be emphasized that the GO discontinuity at the RSB was removed by the inclusion of the reflected diffracted (RD) field and that at the ISB was compensated by the incident diffracted (ID) field. The GO field was also modified in all space with the addition of the diffracted fields, and radiation intensity is present in the shadow region ($210^\circ < \phi < 360^\circ$).

Computations for the same geometry were also carried out for the hard polarization and the amplitude is shown in Figure 13-25b. The same phenomena observed for soft polarization are also evident for the hard polarization.

The geometrical optics fields of Example 13-2 and Figure 13-9, displayed in Figure 13-10, exhibit discontinuities. To remove the discontinuities, diffracted fields must be included. This can be accomplished using the formulations for diffracted fields that have been developed up to this point.

Example 13-4

For the geometry of Figure 13-9, repeat the formulations of Example 13-2, including the fields diffracted from the edges of the strip.

Solution: According to the solution of Example 13-2, the normalized incident (direct) and reflected fields of the line source above an infinite-width strip are given, respectively, by

$$E_z^i = E_0 \frac{e^{-j\beta\rho_i}}{\sqrt{\rho_i}}$$

and

$$E_z^r = -E_0 \frac{e^{-j\beta\rho_r}}{\sqrt{\rho_r}}$$

where ρ_i and ρ_r are, respectively, the distances from the source and image (caustic) to the observation point, as shown in Figure 13-26a.

To take into account the finite width of the strip, we assume that the far-zone geometrical optics field components (direct and reflected) are the same as for the infinite-width strip and the field intensity at the edges of the strip is the same as for the infinite-width strip.

These assumptions, which become more valid for larger width strips, allow us to determine the diffraction contributions from each of the edges. Because of the geometrical symmetry, we can separate the space surrounding the strip only into four regions, as shown in Figure 13-26a. The angular bounds and the components that contribute to each are as follows:

Region	Angular space	Components
I	$\alpha \leq \phi \leq \pi - \alpha$	Direct, reflected, diffracted (1 and 2)
II	$2\pi - \alpha \leq \phi \leq 2\pi, 0 \leq \phi \leq \alpha$	Direct, diffracted (1 and 2)
III	$\pi + \alpha \leq \phi \leq 2\pi - \alpha$	Diffracted (1 and 2)
IV	$\pi - \alpha \leq \phi \leq \pi + \alpha$	Direct, diffracted (1 and 2)

Because of symmetry, we need only consider half of the total space for computations.

To determine the first-order diffractions from each of the edges, we also assume that each forms a wedge (in this case a half space) that initially is isolated from the other. This allows us to use for each the diffraction properties of the canonical problem (wedge) discussed in the previous section. Thus, the field diffracted from wedge 1 is equal to the product of:

1. The direct (incident) field E_z evaluated at the point of diffraction.
2. The diffraction coefficient as given by (13-70c).
3. The spatial attenuation factor as given by (13-35b).
4. The phase factor as given by (13-34a).

In equation form, it is similar to (13-34a), and it is written as

$$E_{z1}^d(\rho_1, \phi) = E_z^i(\rho_d = s') D_s(s', \psi_1, \alpha, 2) A_1(\rho_1) e^{-j\beta\rho_1}$$

where

$$\begin{aligned} E_z^i(\rho_d, s') &= E_0 \frac{e^{-j\beta s'}}{\sqrt{s'}} \\ D_s(s', \psi_1, \alpha, 2) &= D^i(s', \psi_1 - \alpha, 2) - D^r(s', \psi_1 + \alpha, 2) \\ A_1(\rho_1) &= \frac{1}{\sqrt{\rho_1}} \end{aligned}$$

Using the preceding equations, we can write that

$$\begin{aligned} E_{z1}^d(\rho_1, \phi) &= E_0 \left\{ \frac{e^{-j\beta s'}}{\sqrt{s'}} [D^i(s', \psi_1 - \alpha, 2) - D^r(s', \psi_1 + \alpha, 2)] \right\} \frac{e^{-j\beta\rho_1}}{\sqrt{\rho_1}} \\ E_{z1}^d(\rho_1, \phi) &= E_0 [V_B^i(s', \psi_1 - \alpha, 2) - V_B^r(s', \psi_1 + \alpha, 2)] \frac{e^{-j\beta\rho_1}}{\sqrt{\rho_1}} \end{aligned}$$

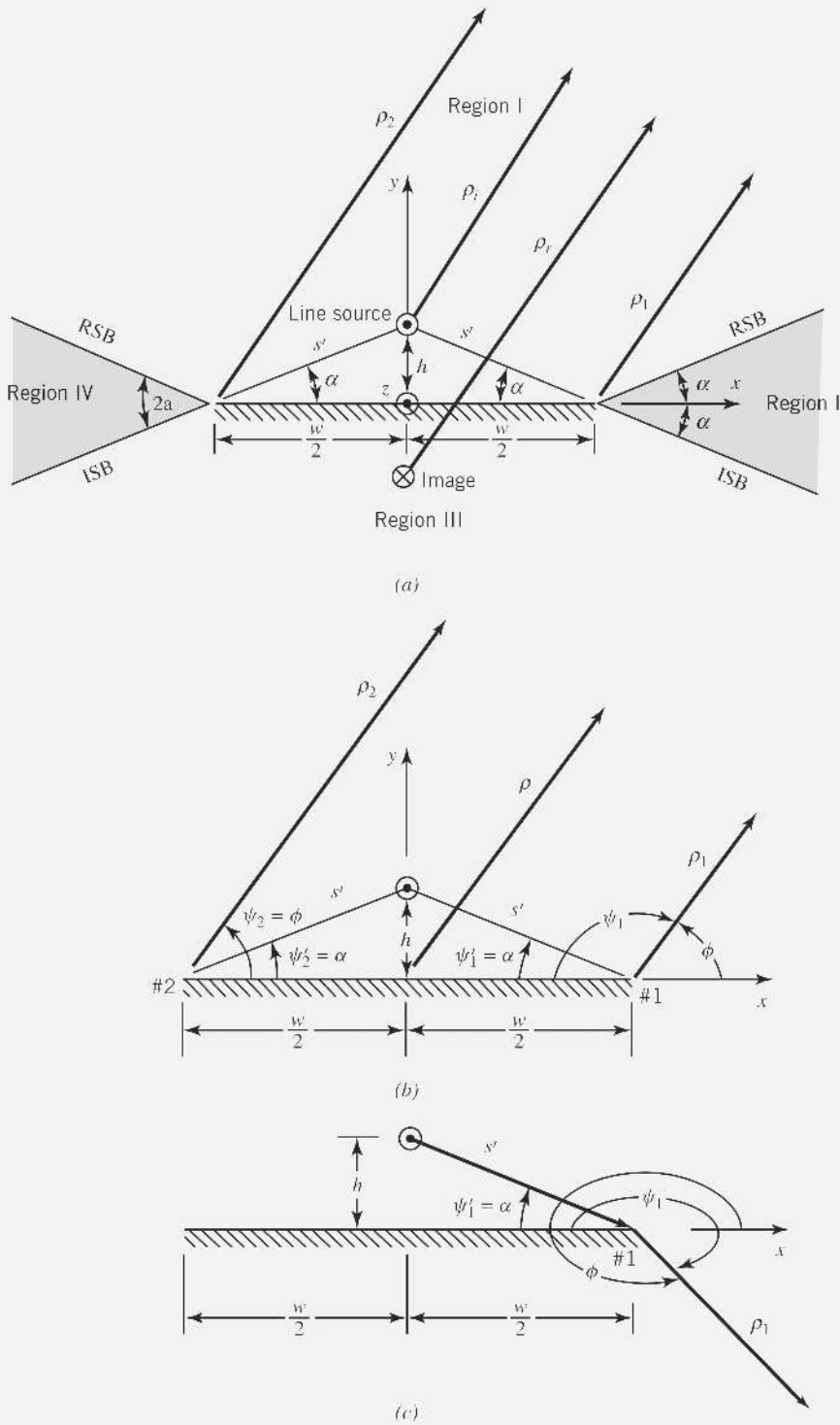


Figure 13-26 Line source above a finite-width strip. (a) Region separation. (b) Diffraction by edges 1 and 2. (c) Diffraction by edge 1 in region III.

where V_B^i and V_B^r are the diffraction functions of (13-68) and (13-69). According to the geometry of Figures 13-26*b* and 13-26*c*

$$\psi_1 = \begin{cases} \pi - \phi & 0 \leq \phi \leq \pi \text{ (Figure 13-26*b*)} \\ 3\pi - \phi & \pi \leq \phi \leq 2\pi \text{ (Figure 13-26*c*)} \end{cases}$$

In a similar manner, it can be shown that the field diffracted by wedge 2 is given by

$$E_{z2}^d = E_0 [V_B^i(s', \psi_2 - \alpha, 2) - V_B^r(s', \psi_2 + \alpha, 2)] \frac{e^{-j\beta\rho_2}}{\sqrt{\rho_2}}$$

$$\psi_2 = \phi \quad \text{for } 0 < \phi < 2\pi \text{ (Figure 13-26*b*)}$$

For far-field observations

$$\left. \begin{aligned} \rho_t &\simeq \rho - h \sin \phi \\ \rho_r &\simeq \rho + h \sin \phi \\ \rho_l &\simeq \rho - \frac{w}{2} \cos \phi \\ \rho_2 &\simeq \rho + \frac{w}{2} \cos \phi \end{aligned} \right\} \text{ for phase variations}$$

$$\rho_l \simeq \rho_r \simeq \rho_1 \simeq \rho_2 \simeq \rho \quad \text{for amplitude variations}$$

which allow the fields to be written as

$$E_z^i = E_0 e^{+j\beta h \sin \phi} \quad 0 < \phi < \pi + \alpha, \quad 2\pi - \alpha < \phi < 2\pi \quad (\text{direct})$$

$$E_z^r = -E_0 e^{-j\beta h \sin \phi} \quad \alpha < \phi < \pi - \alpha \quad (\text{reflected})$$

$$E_{z1}^d = E_0 [V_B^i(s', \psi_1 - \alpha, 2) - V_B^r(s', \psi_1 + \alpha, 2)] e^{+j(\beta w/2) \cos \phi} \quad (\text{diffracted from wedge \# 1})$$

$$\psi_1 = \begin{cases} \pi - \phi & 0 < \phi < \pi \\ 3\pi - \phi & \pi < \phi < 2\pi \end{cases}$$

$$E_{z2}^d = E_0 [V_B^i(s', \psi_2 - \alpha, 2) - V_B^r(s', \psi_2 + \alpha, 2)] e^{-j(\beta w/2) \cos \phi} \quad (\text{diffracted from wedge \# 2})$$

$$\psi_2 = \phi \quad 0 < \phi < 2\pi$$

where the $e^{-j\beta\rho}/\sqrt{\rho}$ factor has been suppressed.

It should be stated that the preceding equations represent only first-order diffractions, which usually provide sufficient accuracy for many high-frequency applications. Multiple diffractions between the edges occur and should be included when the strip is electrically small and when more accurate results are required.

For a strip of width $w = 2\lambda$ and with the source at a height of $h = 0.5\lambda$, the normalized pattern computed using these equations is shown in Figure 13-27, where it is compared with GO patterns for infinite- and finite-width strips. It is evident that there is a significant difference between the three, especially in the lower hemisphere, where for the most part the GO pattern exhibits no radiation, and in the regions where there are discontinuities in the GO pattern.

In addition to antenna pattern prediction, diffraction techniques are extremely well suited for scattering problems. To demonstrate the applicability and versatility of diffraction techniques to scattering, let us consider such an example.

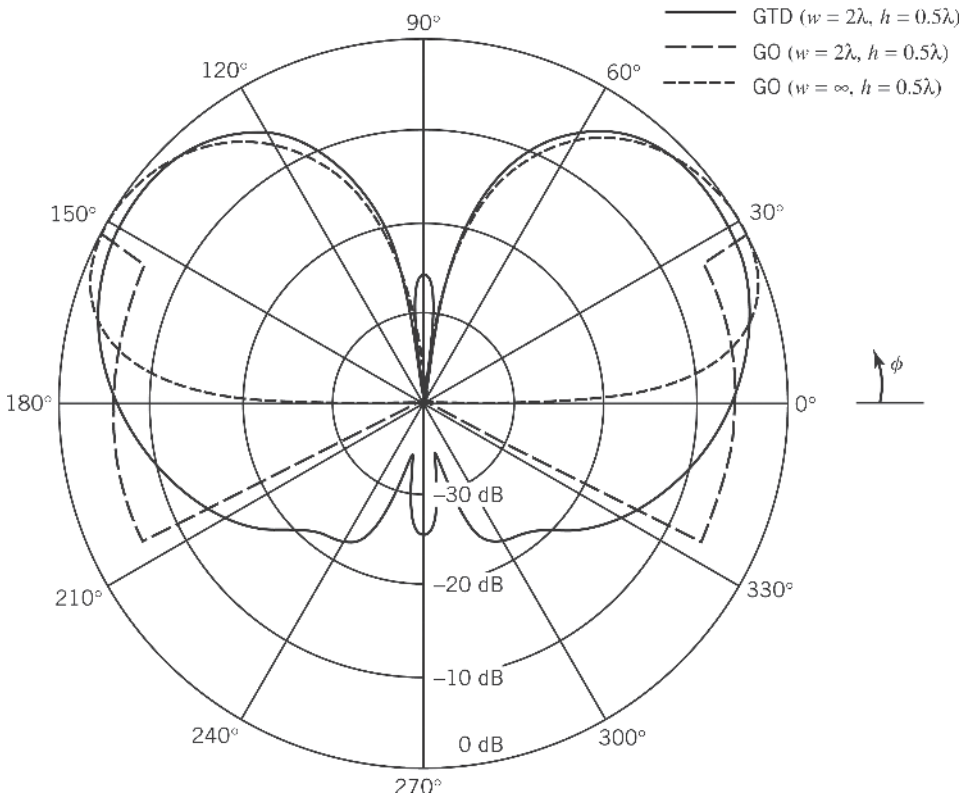


Figure 13-27 Radiation amplitude pattern of electric line source above a finite-width strip.

Example 13-5

A soft polarized uniform plane wave, whose electric field amplitude is E_0 , is incident upon a two-dimensional electrically conducting strip of width w , as shown in Figure 13-28a.

1. Determine the backscattered ($\phi = \phi'$) electric field and its backscattered scattering width (SW).
2. Compute and plot the normalized SW (σ_{2-D} / λ) in dB when $w = 2\lambda$ and the SW (σ_{2-D}) in dB/m (dBm) when $w = 2\lambda$ and $f = 10$ GHz.

Solution: For a soft polarized field, the incident electric field can be written, according to the geometry of Figure 13-28a, as

$$\mathbf{E}^i = \hat{\mathbf{a}}_z E_0 e^{-j\beta' \cdot \mathbf{r}} = \hat{\mathbf{a}}_z E_0 e^{j\beta(x \cos \phi' + y \sin \phi')}$$

The backscattered field diffracted from wedge 1 can be written, by referring to the geometry of Figure 13-28b, as

$$\mathbf{E}_1^d = \mathbf{E}^i(Q_1) \cdot \bar{D}_1^s A_1(\rho_1) e^{-j\beta \rho_1}$$

where

$$E^i(Q_1) = E^i \bigg|_{\substack{x = w/2 \\ y = 0 \\ \phi = \phi'}} = \hat{\mathbf{a}}_z E_0 e^{j(\beta w/2) \cos \phi'}$$

In a similar manner, the fields diffracted from wedge 2 can be written, by referring to the geometry of Figure 13-28b, as

$$\mathbf{E}_2^d = -\hat{\mathbf{a}}_z E_0 \frac{e^{-j\pi/4} e^{-j(\beta w/2) \cos \phi}}{2\sqrt{2\pi\beta}} \left(1 - \frac{1}{\cos \phi}\right) \frac{e^{-j\beta \rho_2}}{\sqrt{\rho_2}}$$

When both of the diffracted fields are referred to the center of the coordinate system, they can be written using

$$\left. \begin{aligned} \rho_1 &\simeq \rho + \frac{w}{2} \cos(\pi - \phi) = \rho - \frac{w}{2} \cos(\phi) \\ \rho_2 &\simeq \rho - \frac{w}{2} \cos(\pi - \phi) = \rho + \frac{w}{2} \cos(\phi) \end{aligned} \right\} \text{for phase terms}$$

$$\rho_1 \simeq \rho_2 \simeq \rho \quad \text{for amplitude terms}$$

as

$$\begin{aligned} \mathbf{E}_1^d &= -\hat{\mathbf{a}}_z E_0 \frac{e^{-j\pi/4}}{2\sqrt{2\pi\beta}} \left(1 + \frac{1}{\cos \phi}\right) e^{j\beta w \cos \phi} \frac{e^{-j\beta \rho}}{\sqrt{\rho}} \\ \mathbf{E}_2^d &= -\hat{\mathbf{a}}_z E_0 \frac{e^{-j\pi/4}}{2\sqrt{2\pi\beta}} \left(1 - \frac{1}{\cos \phi}\right) e^{-j\beta w \cos \phi} \frac{e^{-j\beta \rho}}{\sqrt{\rho}} \end{aligned}$$

When the two diffracted fields are combined, the sum can be expressed as

$$\begin{aligned} \mathbf{E}^d &= \mathbf{E}_1^d + \mathbf{E}_2^d = -\hat{\mathbf{a}}_z E_0 \frac{e^{-j\pi/4}}{2\sqrt{2\pi\beta}} [(e^{j\beta w \cos \phi} + e^{-j\beta w \cos \phi}) \\ &\quad + \frac{1}{\cos \phi} (e^{j\beta w \cos \phi} - e^{-j\beta w \cos \phi})] \frac{e^{-j\beta \rho}}{\sqrt{\rho}} \\ \mathbf{E}^d &= -\hat{\mathbf{a}}_z E_0 \frac{e^{-j\pi/4}}{\sqrt{2\pi\beta}} \left[\cos(\beta w \cos \phi) + j\beta w \frac{\sin(\beta w \cos \phi)}{(\beta w \cos \phi)} \right] \frac{e^{-j\beta \rho}}{\sqrt{\rho}} \end{aligned}$$

Since there are no geometrical optics fields in the backscattered direction (Snell's law is not satisfied) when $\phi = \phi' \neq \pi/2$, the total diffracted field also represents the total field. In the limit as $\phi = \phi' = \pi/2$, each diffracted field exhibits a singularity; however, the total diffracted field is finite because the singularity of one diffracted field compensates for the singularity of the other. This is always evident at normal incidence as long as the edges of the two diffracted wedges are parallel to each other, even though the included angles of the two wedges are not necessarily the same [39]. In addition, the limiting value of the total diffracted field at normal incidence reduces and represents also the geometrical optics scattered (reflected) field.

The two-dimensional backscattered scattering width σ_{2-D} of (11-21b) can now be written as

$$\sigma_{2-D} = \lim_{\rho \rightarrow \infty} \left[2\pi \rho \frac{|\mathbf{E}^s|^2}{|\mathbf{E}^i|^2} \right] = \frac{\lambda}{2\pi} \left| \cos(\beta w \cos \phi) + j\beta w \frac{\sin(\beta w \cos \phi)}{\beta w \cos \phi} \right|^2$$

The limiting value, as $\phi \rightarrow \pi/2$, reduces to

$$\sigma_{2-D}|_{\phi=\pi/2} = \frac{\lambda}{2\pi} |1 + j\beta w|^2 = \frac{\lambda}{2\pi} [1 + (\beta w)^2] \stackrel{\beta w \gg 1}{\simeq} \beta w^2$$

which agrees with the physical optics expression. Computed results for σ_{2-D}/λ (in decibels) and σ_{2-D} (in decibels per meter or dBm) at $f = 10$ GHz when $w = 2\lambda$ are shown in Figure 13-29.

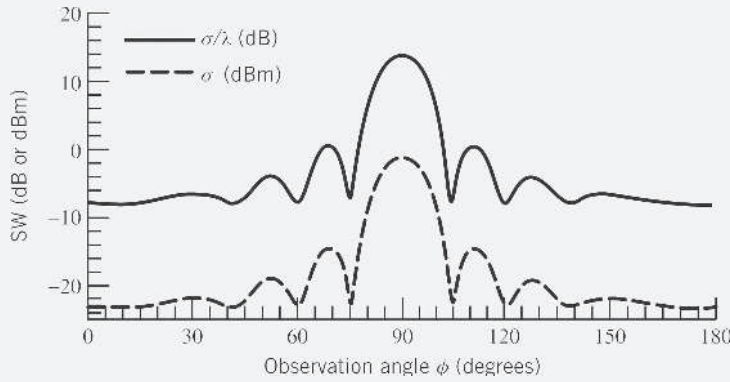


Figure 13-29 Two-dimensional monostatic scattering width for soft polarization of a finite-width strip of $w = 2\lambda$ at $f = 10$ GHz.

Before proceeding to discuss other topics in diffraction, such as oblique incidence, curved edge diffraction, equivalent currents, slope diffraction, and multiple diffraction, let us complete our two-dimensional diffraction by addressing some modifications and extensions to the concepts covered in this section.

For a plane wave incidence, ρ in (13-68) through (13-68f) and in (13-69) through (13-69f) represents the distance from the edge of the wedge to the observation point. According to the principle of reciprocity illustrated in Figure 13-17, ρ in (13-68) through (13-69f) must be replaced by ρ' to represent the diffraction of a cylindrical wave whose source is located a distance ρ' from the edge of the wedge and the observations made in the far zone (ideally at infinity). If both the source and observation point are located at finite distances from the edge of the wedge, represented, respectively, by ρ' and ρ , then a better estimate of the distance would be to introduce a so-called *distance parameter* L , which in this case takes the form of

$$L = \frac{\rho\rho'}{\rho + \rho'} \begin{cases} \rho' \rightarrow \infty \\ \rho \rightarrow \infty \end{cases} \begin{cases} \simeq \rho \\ \simeq \rho' \end{cases} \quad (13-84)$$

Thus, the incident and reflected diffracted fields and coefficients of (13-68) and (13-69) can be written as

$$V_B^i(L, \phi - \phi', n) = V_B^i(L, \xi^-, n) = \frac{e^{-j\beta\rho}}{\sqrt{\rho}} D^i(L, \xi^-, n) \quad (13-84a)$$

$$V_B^r(L, \phi + \phi', n) = V_B^r(L, \xi^+, n) = \frac{e^{-j\beta\rho}}{\sqrt{\rho}} D^r(L, \xi^+, n) \quad (13-84b)$$

It is observed in (13-68) and (13-69) that for grazing angle incidence, $\phi' = 0$ or $\phi' = n\pi$ of Figure 13-24 (where n represents the wedge angle), then $\xi^- = \xi^+ = \phi - \phi' = \phi - \phi'$ and $V_B^i = V_B^r$, $D^i = D^r$. Therefore, here D_s of (13-70c) or (13-72e) is equal to zero ($D_s = 0$) and D_h of (13-70d) or (13-72f) is equal to twice D^i or twice D^r ($D_h = 2D^i = 2D^r$). Since grazing is a limiting situation, the incident and reflected fields combine to make the total geometrical optics field effectively incident at the observation point. Therefore, one-half of the total field propagating along the face of

the wedge toward the edge is the incident field and the other one-half represents the reflected field. The diffracted fields for this case can properly be accounted for by doing *either* of the following:

1. Let the total GO field represent the incident GO field but multiply the diffraction coefficients by a factor of $\frac{1}{2}$.
2. Multiply the total GO field by a factor of $\frac{1}{2}$ and let the product represent the incident field. The diffraction coefficients should not be modified.

Either procedure produces the same results, and the choice is left to the reader.

For grazing incidence, the diffraction coefficient of (13-70c) or (13-72e) are equal to zero. These diffraction coefficients, as well as those of (13-70d) and (13-72f), account for the diffracted fields based on the value of the field at the point of diffraction. This is formulated using (13-34a). There are other higher-order diffraction coefficients that account for the diffracted fields based on the *rate of change (slope)* of the field at the point of diffraction. These diffraction coefficients are referred to as the *slope diffraction coefficients* [52], and they yield nonzero (even though small) fields for soft polarization at grazing incidence. The slope diffraction coefficients exist also for hard polarization, but they are not as dominant as they are for soft polarization.

Up to now, we have restricted our attention to exterior wedge ($1 < n \leq 2$) diffraction. However, the theory of diffraction can be applied also to interior wedge diffraction ($0 \leq n \leq 1$). When $n = 1$, the wedge reduces to an infinite flat plate and the diffraction coefficients reduce to zero [as seen better by examining (13-72e) and (13-72f)] since $\sin(\pi/n) = 0$. The incident and reflected fields for $n = 1$ (half-plane), $n = \frac{1}{2}$ (90° interior wedge), and $n = 1/M$, $M = 3, 4, \dots$ (acute interior wedges) can be found exactly by image theory. In each of these, the number of finite images is determined by the included angle of the interior wedge [38]. As $n \rightarrow 0$ the geometrical optics (incident and reflected) fields become more dominant compared to the nonvanishing diffracted fields.

13.3.3 Straight Edge Diffraction: Oblique Incidence

The normal incidence and diffraction formulations of the previous section are convenient to analyze radiation characteristics of antennas and structures primarily in principal planes. However, a complete analysis of an antenna or scatterer requires examination not only in principal planes but also in nonprincipal planes, as shown in Figure 13-30 for an aperture and a horn antenna each mounted on a finite-size ground plane.

Whereas the diffraction of a normally incident wave discussed in the previous section led to scalar diffraction coefficients, the diffraction of an obliquely incident wave by a two-dimensional wedge can be derived using the geometry of Figure 13-31. To accomplish this, it is most convenient to define ray-fixed coordinate systems (s', β'_0, ϕ') for the source and (s, β_0, ϕ) for the observation point [8, 10], in contrast to the edge-fixed coordinate system $(\rho', \phi', z'; \rho, \phi, z)$. By doing this, it can be shown that the diffracted field, in a general form, can be written as

$$\mathbf{E}^d(s) = \mathbf{E}^i(Q_D) \cdot \bar{\mathbf{D}}(L; \phi, \phi'; n; \beta'_0) \sqrt{\frac{s'}{s(s' + s)}} e^{-j\beta s} \quad (13-85)$$

where $\bar{\mathbf{D}}(L; \phi, \phi'; n; \beta'_0)$ is the dyadic edge diffraction coefficient for illumination of the wedge by plane, cylindrical, conical, or spherical waves.

Introducing an edge-fixed plane of incidence with the unit vectors $\hat{\beta}'_0$ and $\hat{\phi}'$ parallel and perpendicular to it, and a plane of diffraction with the unit vectors $\hat{\beta}_0$ and $\hat{\phi}$ parallel

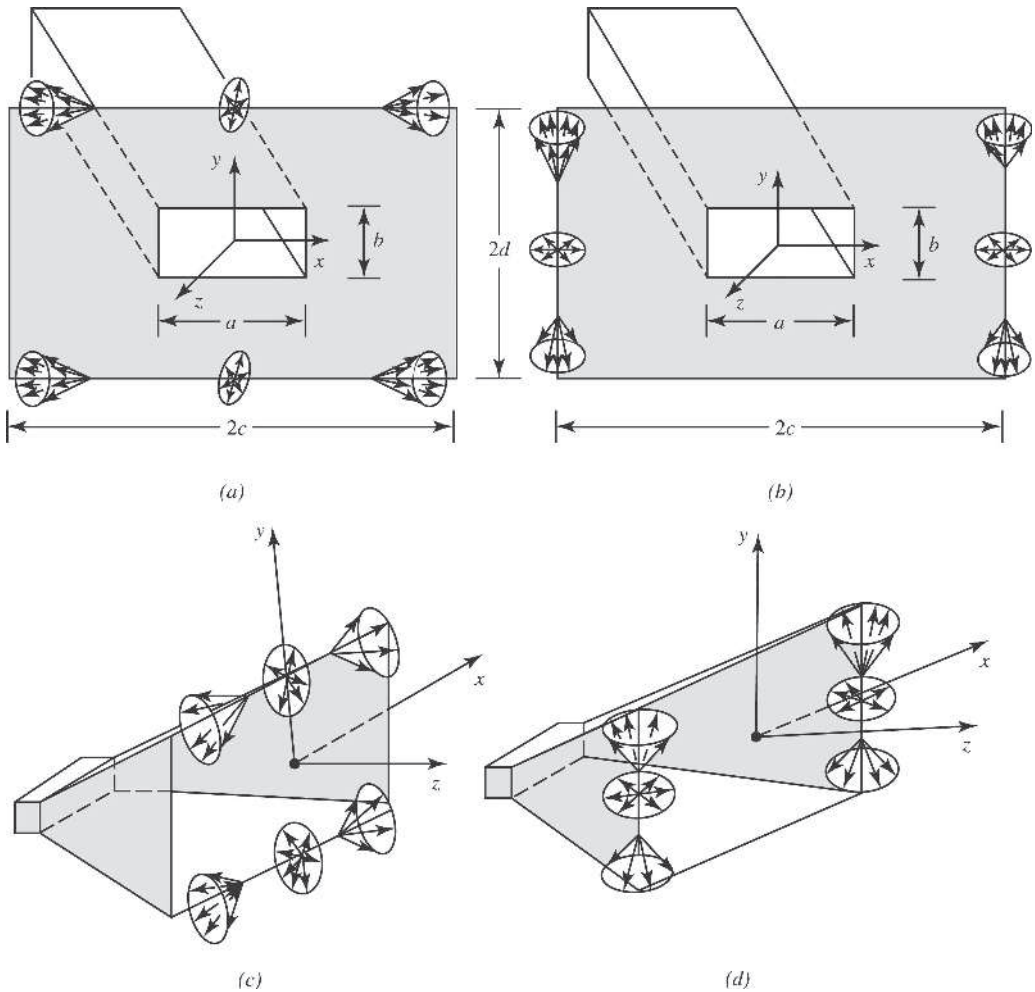


Figure 13-30 *E*- and *H*-plane diffraction by rectangular waveguide and pyramidal horn. Waveguide: (a) *E*-plane diffraction. (b) *H*-plane diffraction. Horn: (c) *E*-plane diffraction. (d) *H*-plane diffraction.

and perpendicular to it, we can write the radial unit vectors of incidence and diffraction, respectively, as

$$\hat{s}' = \hat{\phi}' \times \hat{\beta}'_0 \quad (13-86a)$$

$$\hat{s} = \hat{\phi} \times \hat{\beta}_0 \quad (13-86b)$$

where $\hat{s}'$ points toward the point of diffraction. With the adoption of the ray-fixed coordinate systems, the dyadic diffraction coefficient can be represented by

$$\bar{\mathbf{D}}(L; \phi, \phi'; n; \beta'_0) = -\hat{\beta}'_0 \hat{\beta}_0 D_s(L; \phi, \phi'; n; \beta'_0) - \hat{\phi}' \hat{\phi} D_h(L; \phi, \phi'; n; \beta'_0) \quad (13-87)$$

where D_s and D_h are, respectively, the scalar diffraction coefficients for soft and hard polarizations. If an edge-fixed coordinate system were adopted, the dyadic coefficient would be the sum of seven dyads that in matrix notation would be represented by a 3×3 matrix with seven nonvanishing elements instead of the 2×2 matrix with two nonvanishing elements.

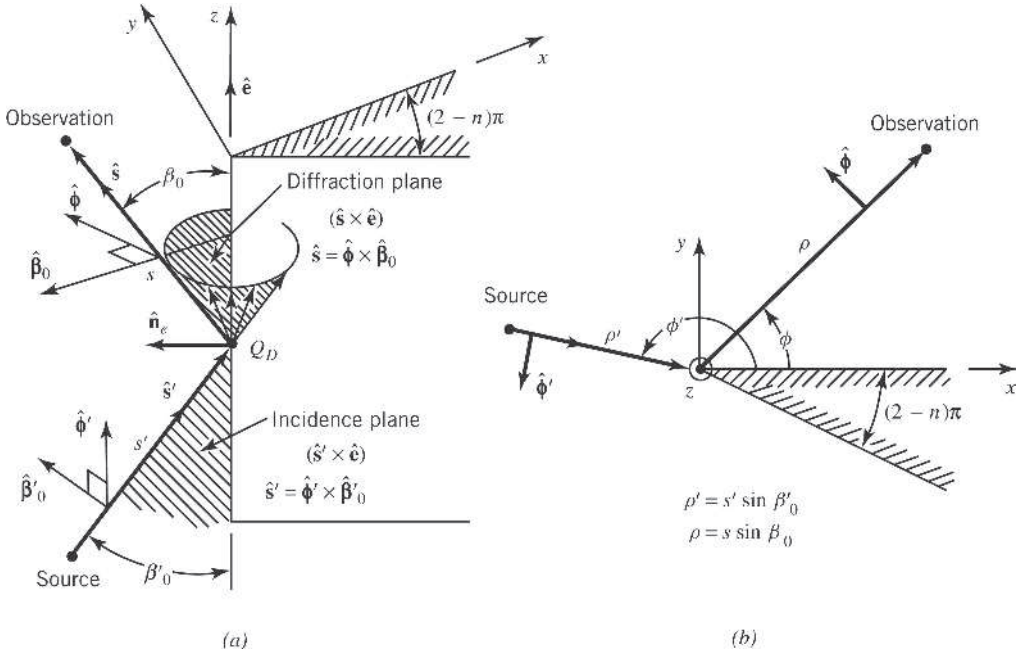


Figure 13-31 Oblique incidence wedge diffraction. (a) Oblique incidence. (b) Top view.

For the diffraction shown in Figure 13-31, we can write in matrix form the diffracted E -field components that are parallel ($E_{\beta_0}^d$) and perpendicular (E_{ϕ}^d) to the plane of diffraction as

$$\begin{bmatrix} E_{\beta_0}^d(s) \\ E_{\phi}^d(s) \end{bmatrix} = - \begin{bmatrix} D_s & 0 \\ 0 & D_h \end{bmatrix} \begin{bmatrix} E_{\beta_0'}^i(Q_D) \\ E_{\phi'}^i(Q_D) \end{bmatrix} A(s', s) e^{-j\beta s} \quad (13-88)$$

where

$$E_{\beta_0'}^i(Q_D) = \hat{\beta}_0' \cdot \mathbf{E}^i$$

= component of the incident $\mathbf{E}$ field parallel to the plane of incidence at the point of diffraction Q_D (13-88a)

$$E_{\phi'}^i(Q_D) = \hat{\phi}' \cdot \mathbf{E}^i$$

= component of the incident $\mathbf{E}$ field perpendicular to the plane of incidence at the point of diffraction Q_D (13-88b)

D_s and D_h are the scalar diffraction coefficients that take the form

$$D_s(L; \phi, \phi'; n; \beta_0') = D^i(L, \phi - \phi', n, \beta_0') - D^r(L, \phi + \phi', n, \beta_0') \quad (13-89a)$$

$$D_h(L; \phi, \phi'; n; \beta_0') = D^i(L, \phi - \phi', n, \beta_0') + D^r(L, \phi + \phi', n, \beta_0') \quad (13-89b)$$

where

$$D^i(L, \phi - \phi', n, \beta'_0) = -\frac{e^{-j\pi/4}}{2n\sqrt{2\pi\beta} \sin \beta'_0} \left\{ \cot \left[\frac{\pi + (\phi - \phi')}{2n} \right] F[\beta L g^+(\phi - \phi')] + \cot \left[\frac{\pi + (\phi - \phi')}{2n} \right] F[\beta L g^-(\phi - \phi')] \right\} \quad (13-90a)$$

$$D^r(L, \phi + \phi', n, \beta'_0) = -\frac{e^{-j\pi/4}}{2n\sqrt{2\pi\beta} \sin \beta'_0} \left\{ \cot \left[\frac{\pi + (\phi + \phi')}{2n} \right] F[\beta L g^+(\phi + \phi')] + \cot \left[\frac{\pi - (\phi + \phi')}{2n} \right] F[\beta L g^-(\phi + \phi')] \right\} \quad (13-90b)$$

To facilitate the reader in the computations a Fortran and MATLAB computer subroutine designated **WDC**, **W**edge **D**iffraction **C**oefficient, computes the normalized (with respect to $\sqrt{\lambda}$) wedge diffraction coefficients based on (13-89a) through (13-90b). The subroutine utilizes the Fresnel transition function program FTF. Both programs were developed and reported in [53].

In general, L is a distance parameter that can be found by satisfying the condition that the total field (the sum of the geometrical optics and the diffracted fields) must be continuous along the incident and reflection shadow boundaries. Doing this, it can be shown that a general form of L is

$$L = \frac{s(\rho_e^i + s)\rho_1^i \rho_2^i \sin^2 \beta'_0}{\rho_e^i(\rho_1^i + s)(\rho_2^i + s)} \quad (13-91)$$

where ρ_1^i, ρ_2^i = radii of curvature of the incident wave front at Q_D

ρ_e^i = radius of curvature of the incident wave front in the edge-fixed plane of incidence

For oblique incidence upon a wedge, as shown in Figure 13-31, the distance parameter can be expressed in the ray-fixed coordinate system as

$$L = \begin{cases} \frac{s \sin^2 \beta'_0}{\rho \rho'} & \text{plane wave incidence} \\ \frac{\rho \rho'}{\rho + \rho'} & \text{cylindrical wave incidence} \\ & (\rho = s \sin \beta_0, \rho' = s' \sin \beta'_0) \\ \frac{ss' \sin^2 \beta'_0}{s + s'} & \text{conical and spherical wave incidences} \end{cases} \quad (13-92)$$

The spatial attenuation factor $A(s', s)$, which describes how the field intensity varies along the diffracted ray, is given by

$$A(s', s) = \begin{cases} \frac{1}{\sqrt{s}} & \text{plane and conical wave incidences} \\ \frac{1}{\sqrt{\rho}} & \rho = s \sin \beta_0; \text{ cylindrical wave incidence} \\ \sqrt{\frac{s'}{s(s' + s)}} & \text{spherical wave incidence} \end{cases} \quad (13-93)$$

If the observations are made in the far field ($s \gg s'$ or $\rho \gg \rho'$), the distance parameter L and spatial attenuation factor $A(s', s)$ reduce, respectively, to

$$L = \begin{cases} s \sin^2 \beta'_0 & \text{plane wave incidence} \\ \rho' & \text{cylindrical wave incidence} \\ s' \sin^2 \beta'_0 & \text{conical and spherical wave incidences} \end{cases} \quad (13-94)$$

$$A(s', s) = \begin{cases} \frac{1}{\sqrt{s}} & \text{plane and conical wave incidences} \\ \frac{1}{\sqrt{\rho}} & \rho = s \sin \beta'_0; \text{ cylindrical wave incidence} \\ \frac{\sqrt{s'}}{s} & \text{spherical wave incidence} \end{cases} \quad (13-95)$$

For normal incidence, $\beta_0 = \beta'_0 = \pi/2$.

To demonstrate the principles of this section, an example will be considered.

Example 13-6

To determine the far-zone elevation plane pattern, in the principal planes, of a $\lambda/4$ monopole mounted on a finite-size square ground plane of width w on each of its sides, refer to Figure 13-32a. Examine the contributions from all four edges.

Solution: In addition to the direct and reflected field contributions (referred to as geometrical optics, GO), there are diffracted fields from the edges of the ground plane. The radiation mechanisms from the two edges that are perpendicular to the principal plane of observation are illustrated graphically in Figure 13-32b. It is apparent that from these two edges only two points contribute to the radiation in the principal plane. These two points occur at the intersection of the principal plane with the edges.

The incident and reflected fields are obtained by assuming the ground plane is infinite in extent. Using the coordinate system of Figure 13-32a, and the image theory of Section 7.4, the total geometrical optics field of the $\lambda/4$ monopole above the ground plane can be written as [38]

$$E_{\theta G}(r, \theta) = E_0 \left[\frac{\cos\left(\frac{\pi}{2} \cos \theta\right)}{\sin \theta} \right] \frac{e^{-j\beta r}}{r}, \quad 0 \leq \theta \leq \pi/2$$

The field diffracted from wedge 1 can be obtained using the formulation of (13-88) through (13-95). Referring to the geometry of Figure 13-32b, the direct field is incident normally ($\beta'_0 = \pi/2$) on the edge of the ground plane along the principal planes, and the diffracted field from wedge 1 can be written as

$$E_{\theta 1}^d(\theta) = +E^i(Q_1)D_h(L, \xi_1^\pm, \beta'_0 = \pi/2, n=2)A_1(w, r_1)e^{-j\beta r_1}$$

The total field can be assumed to all emanate from the base of the monopole. This is a good approximation whose modeling has agreed well with measurements. Thus,

$$E^i(Q_1) = \frac{1}{2}E_{\theta G}\left(r = \frac{w}{2}, \theta = \frac{\pi}{2}\right) = \frac{E_0}{2} \frac{e^{-j\beta w/2}}{w/2}$$

$$D_h(L, \xi_1^\pm, \beta'_0 = \frac{\pi}{2}, n=2) = D^i(L, \xi_1^-, n=2) + D^r(L, \xi_1^+, n=2)$$

Since the incident wave is of spherical waveform and observations are made in the far field, the distance parameter L and spatial attenuation factor $A_1(w, r_1)$ can be expressed, according to (13-94)

and (13-95) for $\beta'_0 = \pi/2$, as

$$L = s' \sin^2 \beta'_0 \bigg|_{\substack{s' = w/2 \\ \beta'_0 = \pi/2}} = \frac{w}{2}$$

$$A_1(w, r_1) = \frac{\sqrt{s'}}{s} \bigg|_{\substack{s' = w/2 \\ s = r_1}} = \frac{\sqrt{w/2}}{r_1}$$

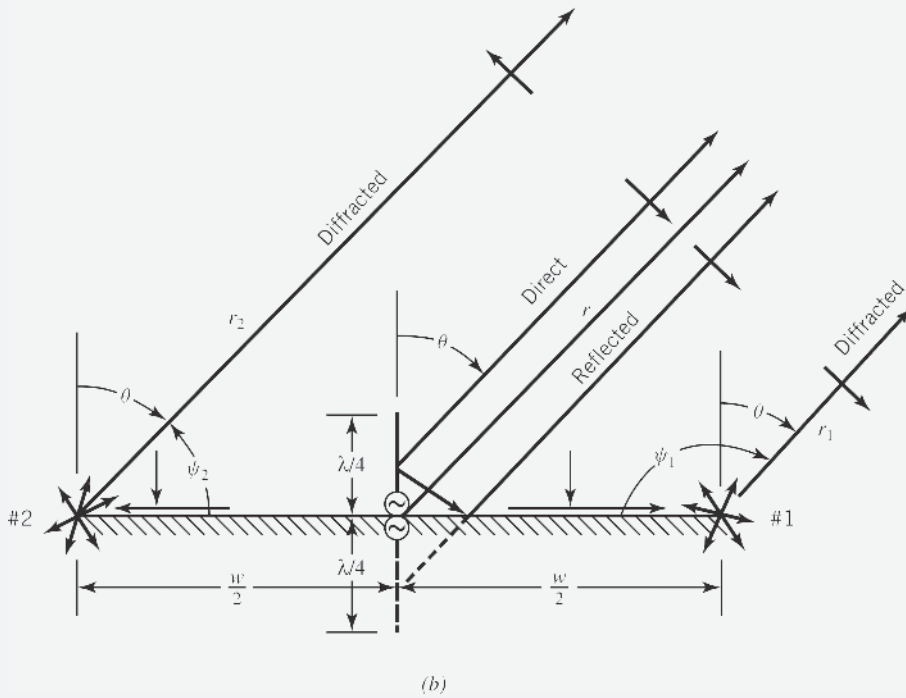
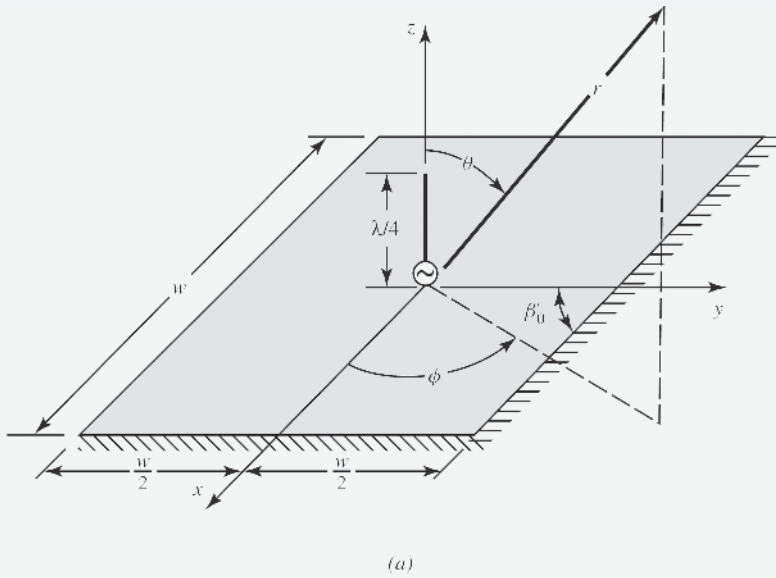


Figure 13-32 Vertical monopole on a square ground plane, and reflection and diffraction mechanisms. (a) Monopole on ground plane. (b) Reflection and diffraction mechanisms.

Since the angle of incidence ψ_0 from the main source toward the point of diffraction Q_1 is zero degrees ($\psi_0 = 0$), then

$$\xi_1^- = \psi_1 - \psi_0 = \psi_1 = \theta + \frac{\pi}{2} = \xi_1$$

$$\xi_1^+ = \psi_1 + \psi_0 = \psi_1 = \theta + \frac{\pi}{2} = \xi_1$$

Therefore,

$$\begin{aligned} D_h(L, \xi_1^\pm, \beta'_0 = \frac{\pi}{2}, n=2) &= 2D^i\left(\frac{w}{2}, \theta + \frac{\pi}{2}, n=2\right) \\ &= 2D^r\left(\frac{w}{2}, \theta + \frac{\pi}{2}, n=2\right) \end{aligned}$$

The total diffracted field can now be written as

$$\begin{aligned} E_{\theta 1}^d(\theta) &= \frac{E_0}{2} \frac{e^{-j\beta w/2}}{w/2} 2D^{i,r}\left(\frac{w}{2}, \theta + \frac{\pi}{2}, n=2\right) \frac{\sqrt{w/2}}{r_1} e^{-j\beta r_1} \\ &= E_0 \left[\frac{e^{-j\beta w/2}}{\sqrt{w/2}} D^{i,r}\left(\frac{w}{2}, \theta + \frac{\pi}{2}, n=2\right) \right] \frac{e^{-j\beta r_1}}{r_1} \\ E_{\theta 1}^d(\theta) &= E_0 V_B^{i,r}\left(\frac{w}{2}, \theta + \frac{\pi}{2}, n=2\right) \frac{e^{-j\beta r_1}}{r_1} \end{aligned}$$

Using a similar procedure, the field diffracted from wedge 2 can be written, by referring to the geometry of Figure 13-32b, as

$$\begin{aligned} E_{\theta 2}^d(\theta) &= -E_0 \left[\frac{e^{-j\beta w/2}}{\sqrt{w/2}} D^{i,r}\left(\frac{w}{2}, \xi_2, n=2\right) \right] \frac{e^{-j\beta r_2}}{r_2} \\ E_{\theta 2}^d(\theta) &= -E_0 V_B^{i,r}\left(\frac{w}{2}, \xi_2, n=2\right) \frac{e^{-j\beta r_2}}{r_2} \end{aligned}$$

where

$$\xi_2 = \psi_2 = \begin{cases} \frac{\pi}{2} - \theta, & 0 \leq \theta \leq \frac{\pi}{2} \\ \frac{5\pi}{2} - \theta, & \frac{\pi}{2} < \theta < \pi \end{cases}$$

For far-field observations

$$\begin{aligned} r_1 &\simeq r - \frac{w}{2} \cos\left(\frac{\pi}{2} - \theta\right) = r - \frac{w}{2} \sin\theta \\ r_2 &\simeq r + \frac{w}{2} \cos\left(\frac{\pi}{2} - \theta\right) = r + \frac{w}{2} \sin\theta \end{aligned} \quad \left. \begin{array}{l} \text{for phase terms} \\ \text{for amplitude terms} \end{array} \right\}$$

Therefore, the diffracted fields from wedges 1 and 2 reduce to

$$\begin{aligned} E_{\theta 1}^d(\theta) &= +E_0 V_B^{i,r}\left(\frac{w}{2}, \theta + \frac{\pi}{2}, n=2\right) e^{j(\beta w/2) \sin\theta} \frac{e^{-j\beta r}}{r} \\ E_{\theta 2}^d(\theta) &= -E_0 V_B^{i,r}\left(\frac{w}{2}, \xi_2, n=2\right) e^{-j(\beta w/2) \sin\theta} \frac{e^{-j\beta r}}{r} \end{aligned}$$

It should be noted that there are oblique incidence diffractions from the other two edges of the ground plane that are parallel to the principal plane of observation. However, the diffracted field from these edges is primarily cross-polarized (E_ϕ component) to the incident E_θ field and to the E_θ field produced in the

principal plane. The cross-polarized E_ϕ components produced by diffractions from these two sides cancel each other out so that in the principal plane there is primarily an E_θ component.

Using the total geometrical optics field and the field diffracted from wedges 1 and 2, a normalized amplitude pattern was computed for a $\lambda/4$ monopole mounted on a square ground plane of width $w = 4 \text{ ft} = 1.22 \text{ m}$ at a frequency of $f = 1 \text{ GHz}$. This pattern is shown in Figure 13-33 where it is compared with the computed GO (assuming an infinite ground plane) and measured patterns. A very good agreement is seen between the GO + GTD and measured patterns, which are quite different from that of the GO pattern.

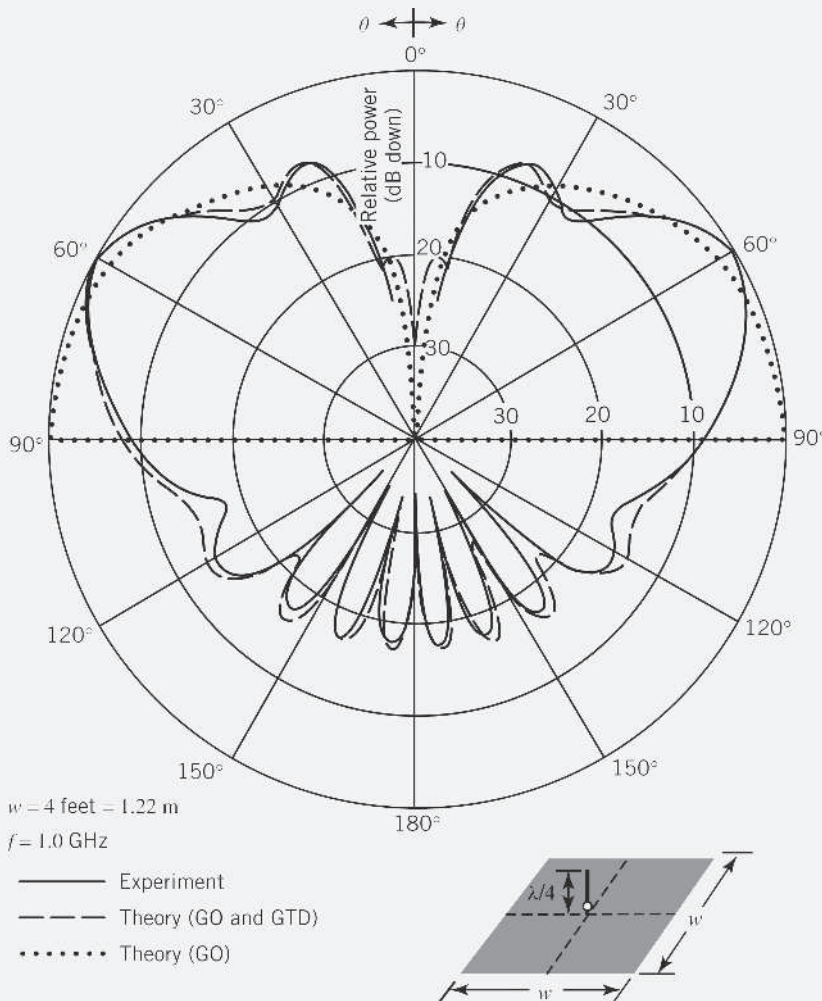


Figure 13-33 Measured and computed principal elevation plane amplitude patterns of a $\lambda/4$ monopole above infinite and finite square ground planes. (Source: C. A. Balanis, *Antenna Theory: Analysis and Design*, Third Edition, copyright © 2005, John Wiley & Sons, Inc. Reprinted by permission of John Wiley & Sons, Inc.)

13.3.4 Curved Edge Diffraction: Oblique Incidence

The edges of many practical antenna or scattering structures are not straight, as demonstrated in Figure 13-34 by the edges of a circular ground plane, a paraboloidal reflector, and a conical horn. In order to account for the diffraction phenomenon from the edges of these structures, even in their principal planes, curved edge diffraction must be utilized.

Curved edge diffraction can be derived by assuming an oblique wave incidence (at an angle β'_0) on a curved edge, as shown in Figure 13-35, where the surfaces (sides) forming the curved edge in general may be convex, concave, or plane. Since diffraction is a local phenomenon, the curved edge geometry can be approximated at the point of diffraction Q_D by a wedge whose straight edge is tangent to the curved edge at that point and whose plane surfaces are tangent to the curved surfaces forming the curved edge. This allows wedge diffraction theory to be applied directly to curved edge diffraction by simply representing the curved edge by an equivalent wedge. Analytically, this is accomplished simply by generalizing the expressions for the distance parameter L that appear in the arguments of the transition functions.

The general form of oblique incidence curved edge diffraction can be expressed in matrix form as in (13-88). However, the diffraction coefficients, distance parameters, and spatial spreading factor must be modified to account for the curvature of the edge and its curved surfaces (sides).

The diffraction coefficients D_s and D_h are those of (13-89a) and (13-89b), where D^i and D^r can be found by imposing the continuity conditions on the total field across the incident and reflection shadow boundaries. Doing this, we can show that D^i and D^r of (13-90a) and (13-90b) take the form of [10]

$$D^i(L^i, \phi - \phi', n) = -\frac{e^{-j\pi/4}}{2n\sqrt{2\pi\beta} \sin \beta'_0} \left\{ \cot \left[\frac{\pi + (\phi - \phi')}{2n} \right] F[\beta L^i g^+(\phi - \phi')] + \cot \left[\frac{\pi - (\phi - \phi')}{2n} \right] F[\beta L^i g^-(\phi - \phi')] \right\} \quad (13-96a)$$

$$D^r(L^r, \phi + \phi', n) = -\frac{e^{-j\pi/4}}{2n\sqrt{2\pi\beta} \sin \beta'_0} \left\{ \cot \left[\frac{\pi + (\phi + \phi')}{2n} \right] F[\beta L^{rn} g^+(\phi + \phi')] + \cot \left[\frac{\pi - (\phi + \phi')}{2n} \right] F[\beta L^{r0} g^-(\phi + \phi')] \right\} \quad (13-96b)$$

where

$$L^i = \frac{s(\rho_e^i + s)\rho_1^i \rho_2^i \sin^2 \beta'_0}{\rho_e^i(\rho_1^i + s)(\rho_2^i + s)} \quad (13-97a)$$

$$L^{r0, rn} = \frac{s(\rho_e^r + s)\rho_1^r \rho_2^r \sin^2 \beta'_0}{\rho_e^r(\rho_1^r + s)(\rho_2^r + s)} \quad (13-97b)$$

ρ_1^i, ρ_2^i = radii of curvature of the incident wave front at Q_D

ρ_e^i = radius of curvature of the incident wave front in the edge fixed plane of incidence

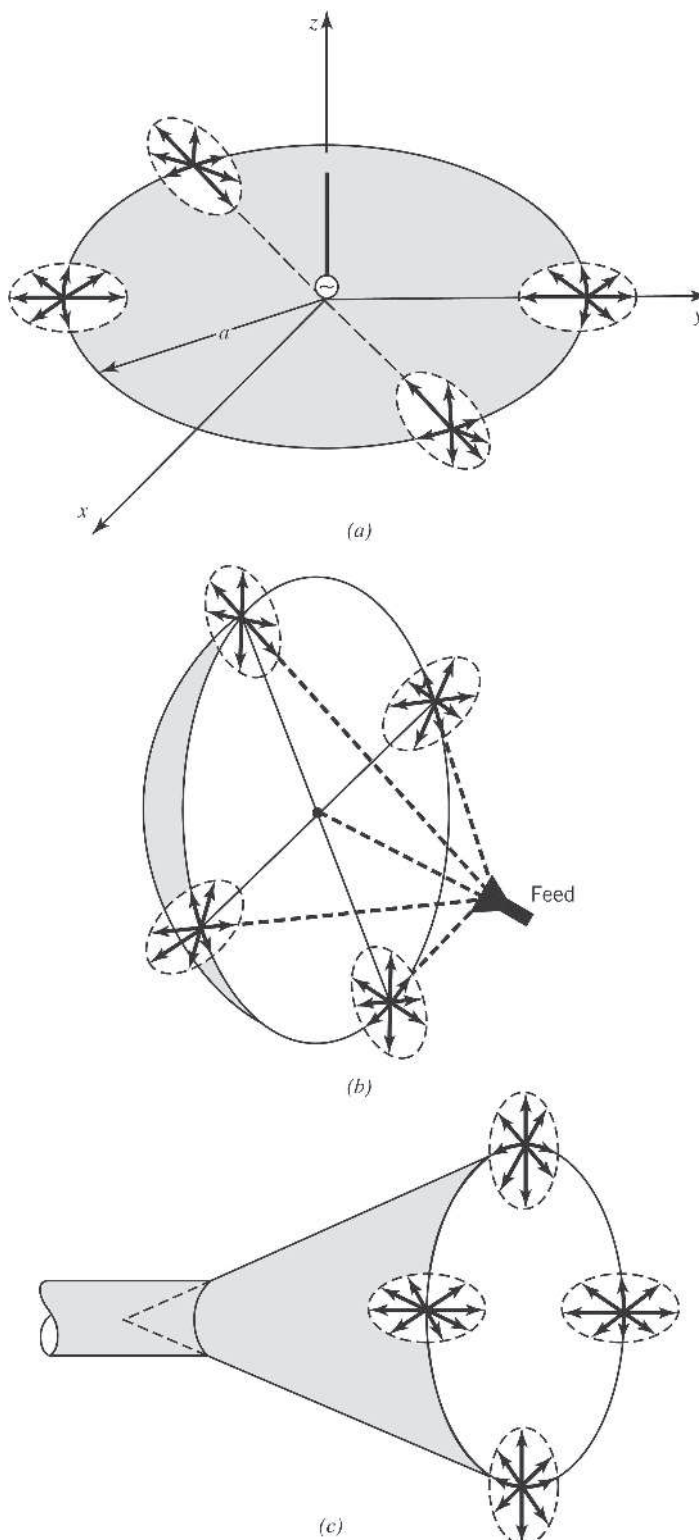


Figure 13-34 Diffraction by curved-edge structures. (a) Circular ground plane. (b) Paraboloidal reflector. (c) Conical horn.

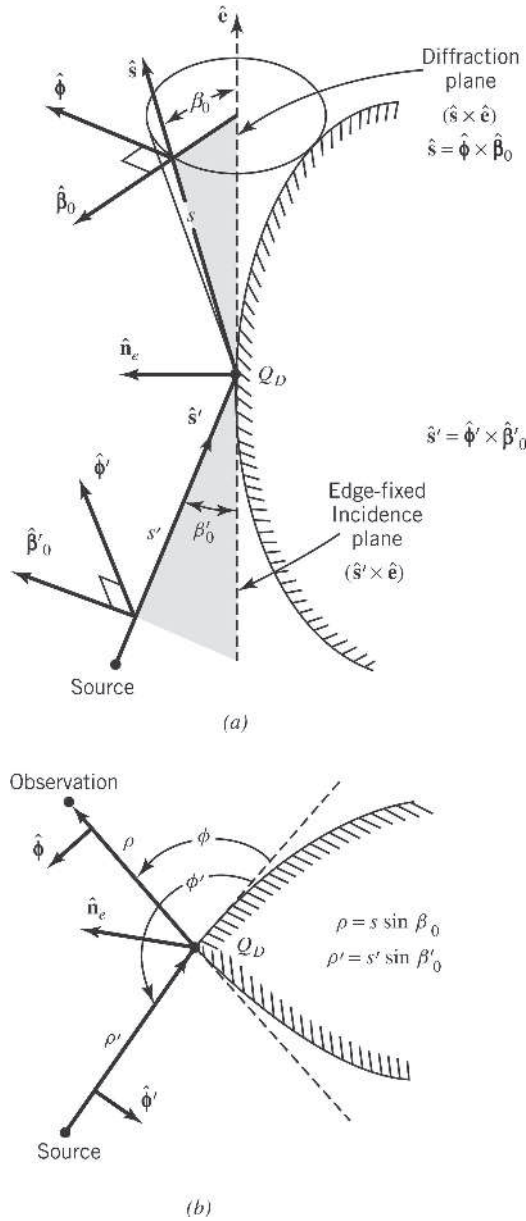


Figure 13-35 Oblique incidence diffraction by a curved edge. (a) Oblique incidence. (b) Top view.

ρ_1^r, ρ_2^r = principal radii of curvature of the reflected wave front at Q_D [found using (13-21a) and (13-21b)]
 ρ_e^r = radius of curvature of the reflected wave front in the plane containing the diffracted ray and edge

The superscripts *ro* and *rn* of L in (13-96b) and (13-97a) denote that the radii of curvature ρ_1^r, ρ_2^r , and ρ_e^r must be calculated for *ro* at the reflection boundary $\pi - \phi'$ of Figure 13-24a

and for rn at the reflection boundary $(2n-1)\pi - \phi'$ of Figure 13-24b. For far-field observation, where $s \gg \rho_e^i, \rho_1^i, \rho_2^i, \rho_e^r, \rho_1^r, \rho_2^r$ (13-97a), (13-97b) simplify to

$$L^i = \frac{\rho_1^i \rho_2^i}{\rho_e^i} \sin^2 \beta'_0 \quad (13-98a)$$

$$L^{r0,m} = \frac{\rho_1^r \rho_2^r}{\rho_e^r} \sin^2 \beta'_0 \quad (13-98b)$$

If the intersecting curved surfaces forming the curved edge in Figure 13-35 are plane surfaces that form an ordinary wedge, then the distance parameters in (13-97a) and (13-97b) or (13-98a) and (13-98b) are equal, that is,

$$L^{r0} = L^m = L^i \quad (13-99)$$

Using the geometries of Figure 13-36, it can be shown that the spatial spreading factor $A(\rho_c, s)$ of (13-35) for the curved edge diffraction takes the form

$$A(\rho_c, s) = \sqrt{\frac{\rho_c}{s(\rho_c + s)}} \stackrel{s \gg \rho_c}{\simeq} \frac{1}{s} \sqrt{\rho_c} \quad (13-100)$$

$$\frac{1}{\rho_c} = \frac{1}{\rho_e} - \frac{\hat{\mathbf{n}}_e \cdot (\hat{\mathbf{s}}' - \hat{\mathbf{s}})}{\rho_g \sin^2 \beta'_0} \quad (13-100a)$$

where ρ_c = distance between caustic at edge and second caustic of diffracted ray

ρ_e = radius of curvature of incidence wave front in the edge-fixed plane of incidence which contains unit vectors $\hat{\mathbf{s}}'$ and $\hat{\mathbf{e}}$ (infinity for plane, cylindrical, and conical waves; $\rho_e = s'$ for spherical waves)

ρ_g = radius of curvature of the edge at the diffraction point

$\hat{\mathbf{n}}_e$ = unit vector normal to the edge at Q_D and directed away from the center of curvature

$\hat{\mathbf{s}}'$ = unit vector in the direction of incidence

$\hat{\mathbf{s}}$ = unit vector in the direction of diffraction

β'_0 = angle between $\hat{\mathbf{s}}'$ and tangent to the edge at the point of diffraction

$\hat{\mathbf{e}}$ = unit vector tangent to the edge at the point of diffraction

For normal incidence, $\beta'_0 = \pi/2$.

The spatial attenuation factor (13-100) creates additional caustics, other than the ones that occur at the points of diffraction. Each caustic occurs at a distance ρ_c from the one at the diffraction point. Diffracted fields in the regions of the caustics must be corrected to remove the discontinuities and inaccuracies from them.

To demonstrate the principles of curved edge diffraction, let us consider Example 13-7.

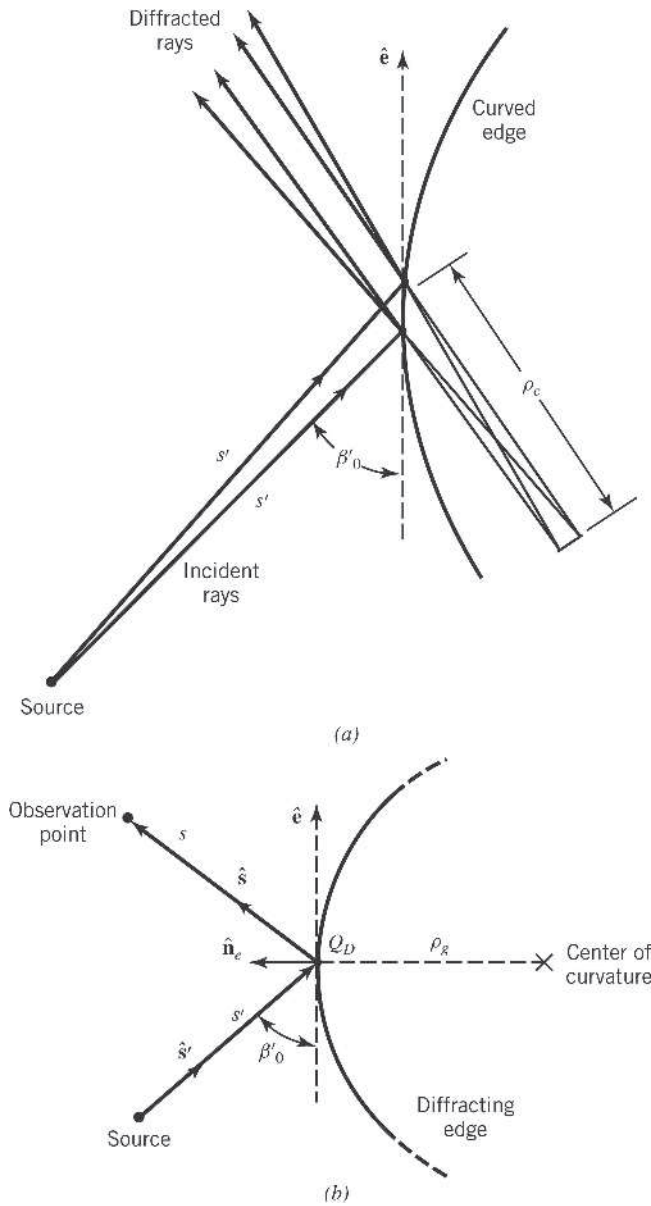


Figure 13-36 (a) Caustic distance. (b) Center of curvature. Caustic distance and center of curvature for curved-edge diffraction. (Source: C. A. Balanis, *Antenna Theory: Analysis and Design*, copyright © 1982, John Wiley & Sons, Inc. Reprinted by permission of John Wiley & Sons, Inc.)

Example 13-7

Determine the far-zone elevation plane pattern of a $\lambda/4$ monopole mounted on a circular electrically conducting ground plane of radius a , as shown in Figure 13-37a.

Solution: Because of the symmetry of the structure, the diffraction mechanism in any of the elevation planes is the same. Therefore, the principal yz plane is chosen here. For observations made away from the symmetry axis of the ground plane ($\theta \neq 0^\circ$ and 180°), it can be shown [33] that most of the diffraction radiation from the rim of the ground plane comes from the two diametrically opposite points of the rim

that coincide with the observation plane. Therefore, for points removed from the symmetry axis ($\theta \neq 0^\circ$ and 180°) the overall formulation of this problem, and that of Example 13-6, is identical other than the amplitude spreading factor, which now must be computed using (13-100) and (13-100a) instead of (13-95).

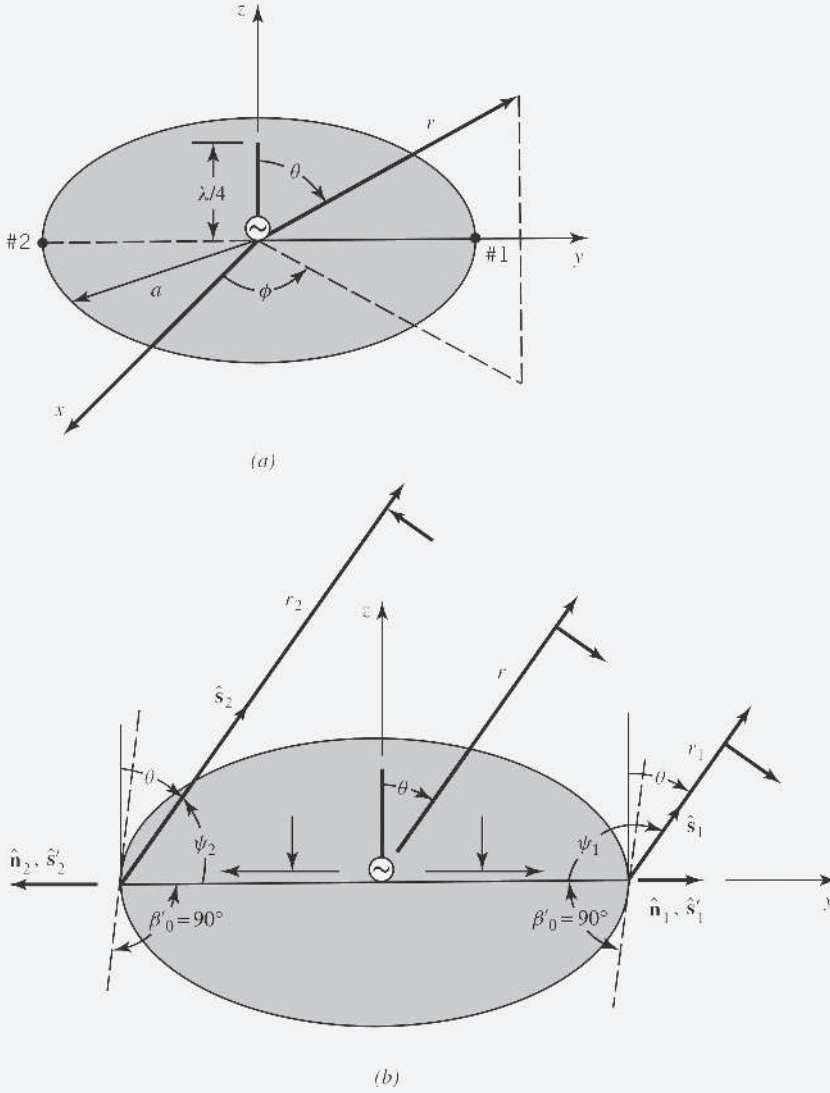


Figure 13-37 Quarter-wavelength monopole on a circular ground plane and diffraction mechanism. (a) $\lambda/4$ monopole. (b) Diffraction mechanism.

Referring to the geometry of Figure 13-37b, and using (13-100) and (13-100a), the amplitude spreading factor for wedge 1 can be written as

$$A_1(r_1, a) = \frac{1}{r_1} \sqrt{\rho_{c1}}$$

where

$$\frac{1}{\rho_{c1}} = \frac{1}{a} - \frac{\hat{\mathbf{n}}_1 \cdot (\hat{\mathbf{s}}_1' - \hat{\mathbf{s}}_1)}{a} = \frac{1 - [1 - \cos(\frac{\pi}{2} - \theta)]}{a} = \frac{\sin \theta}{a} \Rightarrow \rho_{c1} = \frac{a}{\sin \theta}$$

Therefore,

$$A_1(r_1, a) = \frac{1}{r_1} \sqrt{\frac{a}{\sin \theta}} \simeq \frac{1}{r} \sqrt{\frac{a}{\sin \theta}}$$

In a similar manner, the amplitude spreading factor for wedge 2 can be expressed as

$$A_2(r_2, a) = \frac{1}{r_2} \sqrt{\rho_{c2}}$$

where

$$\frac{1}{\rho_{c2}} = \frac{1}{a} - \frac{\hat{\mathbf{n}}_2 \cdot (\hat{\mathbf{s}}'_2 - \hat{\mathbf{s}}_2)}{a} = \frac{1 - [1 - \cos(\frac{\pi}{2} + \theta)]}{a} = -\frac{\sin \theta}{a} \Rightarrow \rho_{c2} = -\frac{a}{\sin \theta}$$

This reduces the amplitude spreading factor to

$$A_2(r_2, a) = \frac{1}{r_2} \sqrt{-\frac{a}{\sin \theta}} \simeq \frac{1}{r} \sqrt{-\frac{a}{\sin \theta}}$$

Using the results from Example 13-6, the fields for this problem can be written as

$$E_{\theta G}(r, \theta) = E_0 \left[\frac{\cos\left(\frac{\pi}{2} \cos \theta\right)}{\sin \theta} \right] \frac{e^{-j\beta r}}{r}, \quad 0 \leq \theta \leq \pi/2$$

$$E_{\theta 1}^d(r, \theta) = E_0 V_B^{i,r} \left(a, \theta + \frac{\pi}{2}, n=2 \right) \frac{e^{j\beta a \sin \theta}}{\sqrt{\sin \theta}} \frac{e^{-j\beta r}}{r}, \quad \theta_0 \leq \theta \leq \pi - \theta_0$$

$$E_{\theta 2}^d(r, \theta) = -E_0 V_B^{i,r} (a, \xi_2, n=2) \frac{e^{-j\beta a \sin \theta}}{\sqrt{-\sin \theta}} \frac{e^{-j\beta r}}{r}, \quad \theta_0 \leq \theta \leq \pi - \theta_0$$

where

$$\xi_2 = \psi_2 = \begin{cases} \frac{\pi}{2} - \theta, & \theta_0 \leq \theta \leq \frac{\pi}{2} \\ \frac{5\pi}{2} - \theta, & \frac{\pi}{2} < \theta \leq \pi - \theta_0 \end{cases}$$

It is noted that at $\theta = 0^\circ$ or 180° , the diffracted fields become singular because along these directions there are caustics for the diffracted fields. The rim of the ground plane acts as a *ring radiator*, which is illustrated graphically in Figure 13-38. The ring radiator can be formulated analytically

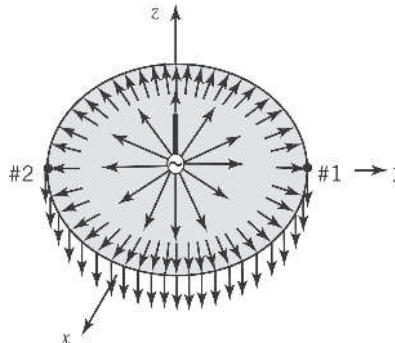


Figure 13-38 Uniform ring radiator representing diffractions around the rim of the circular ground plane.

(see Problem (13.49) as a continuous symmetrical and constant source of diffraction around the rim of the circular ground plane. This can be cast as an integral with uniform excitation around the rim of the circular ground plane, and it can be treated similarly as a circular loop of uniform current [38]. Toward and near $\theta = 0^\circ$ and 180° , the integral reduces to a Bessel function of the first kind of order one. However, it can be shown, using the method of steepest descent (saddle point method), that the integral representing the continuous ring radiator reduces to a two-point diffraction for angles away from $\theta = 0^\circ$ and 180° [33]. Due to the ring radiator characteristics, the radiation of the monopole toward and near $\theta = 0^\circ$ and 180° is much more intense for this geometry compared to when the monopole is mounted on a rectangular/square ground plane, as shown in Figure 13-33. Similarly, the *scattering* from circular ground planes and apertures is much more intense toward and near $\theta = 0^\circ$ and 180° than that from rectangular/square ground planes and apertures. Therefore, to reduce the radar signature/visibility of the engine inlets (apertures) of the F-117, they may have been chosen to be rectangular/square, as shown in Figure 13-1. Toward $\theta = 0^\circ$ and 180° the infinite number of diffracted rays from the rim are identical in amplitude and phase and lead to the caustics. Therefore, the diffracted fields from the aforementioned two points of the rim are invalid within a cone of half included angle θ_0 , which is primarily a function of the radius of curvature of the rim. For most moderate size ground planes, θ_0 is in the range of $10^\circ < \theta_0 < 30^\circ$.

To make corrections for the diffracted field singularity and inaccuracy at and near the symmetry axis ($\theta = 0^\circ$ and 180°), due to axial caustics, the rim of the ground plane must be modeled as a ring radiator [32, 33]. This can be accomplished by using “equivalent” current concepts in diffraction, which will be discussed in the next section.

A pattern based on the formulations of the preceding two-point diffraction was computed for a ground plane of 4.064λ diameter. This pattern is shown in Figure 13-39 where it is compared with measurements. It should be noted that this pattern was computed using the two-point diffraction for $10^\circ \lesssim \theta \lesssim 170^\circ$ ($\theta_0 \simeq 10^\circ$); the remaining parts were computed using equivalent current concepts that will be discussed next.

13.3.5 Equivalent Currents in Diffraction

In contrast to diffraction by straight edges, diffraction by curved edges creates caustics. If observations are not made at or near caustics, ordinary diffraction techniques can be applied; otherwise, corrections must be made.

One technique that can be used to correct for caustic discontinuities and inaccuracies is the concept of the *equivalent currents* [32–36, 54–61]. To apply this principle, the two-dimensional wedge of Figure 13-13 is replaced by one of the following two forms:

1. An equivalent two-dimensional electric line source of equivalent electric current I^e , for soft polarization diffraction.
2. An equivalent two-dimensional magnetic line source of equivalent magnetic current I^m , for hard polarization diffraction.

This is illustrated in Figure 13-40. The equivalent currents I^e and I^m are adjusted so that the field radiated by each of the line sources is equal to the diffracted field of the corresponding polarization.

The electric field radiated by a two-dimensional electric line source placed along the z axis with a constant current I_z^e is given by (11-10a), or

$$E_z = -\frac{\beta^2 I_z^e}{4\omega\epsilon} H_0^{(2)}(\beta\rho) \stackrel{\beta\rho \rightarrow \infty}{\simeq} -I_z^e \frac{\eta\beta}{2} \sqrt{\frac{j}{2\pi\beta}} \frac{e^{-j\beta\rho}}{\sqrt{\rho}} \quad (13-101a)$$

where $H_0^{(2)}(\beta\rho)$ is the Hankel function of the second kind of order zero. The approximate form of (13-101a) is valid for large distances of observation (far field), and it is obtained by replacing

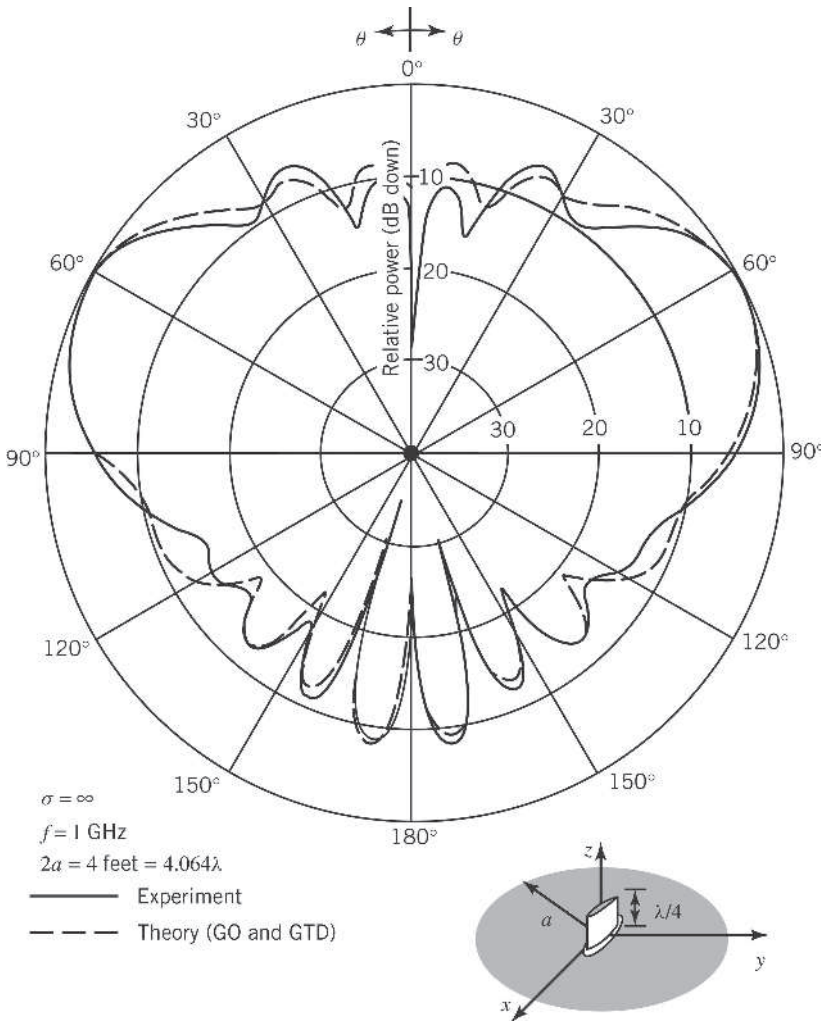


Figure 13-39 Measured and computed principal elevation plane amplitude patterns of a $\lambda/4$ monopole (blade) above a circular ground plane. (Source: C. A. Balanis, *Antenna Theory: Analysis and Design*, Third Edition, copyright © 2005, John Wiley & Sons, Inc. Reprinted by permission of John Wiley & Sons, Inc.)

the Hankel function by its asymptotic formula for large argument (see Appendix IV, Equation IV-17).

The magnetic field radiated by a two-dimensional magnetic line source placed along the z axis with a constant current I_z^m can be obtained using the duality theorem (Section 7.2, Table 7-2) and (13-101a). Thus,

$$H_z = -\frac{\beta^2 I_z^m}{4\omega\mu} H_0^{(2)}(\beta\rho) \stackrel{\beta\rho \rightarrow \infty}{\simeq} -I_z^m \frac{\beta}{2\eta} \sqrt{\frac{j}{2\pi\beta}} \frac{e^{-j\beta\rho}}{\sqrt{\rho}} \quad (13-101b)$$

To determine the equivalent electric current I_z^e , (13-101a) is equated to the field diffracted by a wedge when the incident field is of soft polarization. A similar procedure is used for the equivalent magnetic I_z^m of (13-101b). Using (13-34), (13-34a), (13-95), (13-101a), and (13-101b), and assuming normal incidence, we can write that

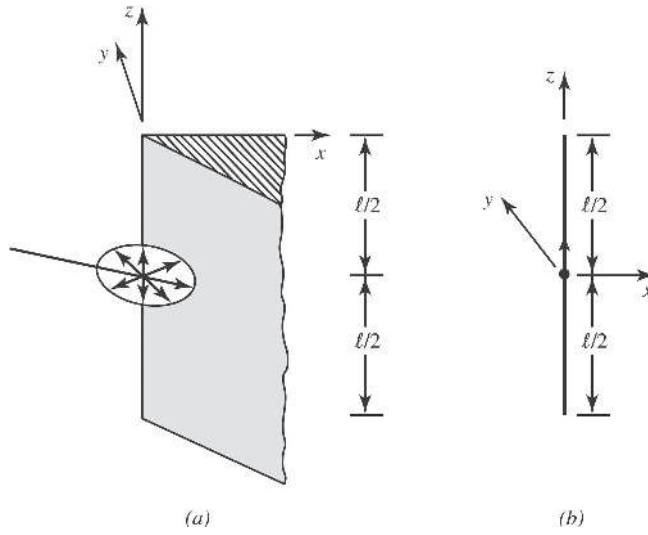


Figure 13-40 Wedge diffraction at normal incidence and its equivalent. (a) Actual wedge. (b) Equivalent.

$$E_z^i(Q_d)D_s(\xi^-, \xi^+, n)\frac{e^{-j\beta\rho}}{\sqrt{\rho}} = -I_z^e \frac{\eta\beta}{2} \sqrt{\frac{j}{2\pi\beta}} \frac{e^{-j\beta\rho}}{\sqrt{\rho}} \quad (13-102a)$$

$$H_z^i(Q_d)D_h(\xi^-, \xi^+, n)\frac{e^{-j\beta\rho}}{\sqrt{\rho}} = -I_z^m \frac{\beta}{2\eta} \sqrt{\frac{j}{2\pi\beta}} \frac{e^{-j\beta\rho}}{\sqrt{\rho}} \quad (13-102b)$$

where $E_z^i(Q)$ = incident electric field at the diffraction point Q_d

$H_z^i(Q)$ = incident magnetic field at the diffraction point Q_d

D_s = diffraction coefficient for soft polarization [(13-71a) or (13-71e)]

D_h = diffraction coefficient for hard polarization [(13-71b) or (13-72f)]

Solving (13-102a) and (13-102b) for I_z^e and I_z^m respectively, leads to

$$I_z^e = -\frac{\sqrt{8\pi\beta}}{\eta\beta} e^{-j\pi/4} E_z^i(Q)D_s(\xi^-, \xi^+, n) \quad (13-103a)$$

$$I_z^m = -\frac{\eta\sqrt{8\pi\beta}}{\beta} e^{-j\pi/4} H_z^i(Q)D_h(\xi^-, \xi^+, n) \quad (13-103b)$$

If the wedge of Figure 13-40 is of finite length ℓ , its equivalent current will also be of finite length. The far-zone field radiated by each can be obtained by using techniques similar to those of Chapter 4 of [38]. Assuming the edge is along the z axis, the far-zone electric field radiated by an electric line source of length ℓ can be written using (4-58a) of [38] as

$$E_\theta^e = j\eta \frac{\beta e^{-j\beta r}}{4\pi r} \sin\theta \int_{-\ell/2}^{\ell/2} I_z^e(z') e^{j\beta z' \cos\theta} dz' \quad (13-104a)$$

Using duality, the magnetic field of a magnetic line source can be written as

$$H_{\theta}^m = j \frac{\beta e^{-j\beta r}}{4\pi\eta r} \sin\theta \int_{-\ell/2}^{\ell/2} I_z^m(z') e^{j\beta z' \cos\theta} dz' \quad (13-104b)$$

For a constant equivalent current, the integrals in (13-104a) and (13-104b) reduce to a $\sin(\zeta)/\zeta$ form.

If the equivalent current is distributed along a circular loop of radius a and it is parallel to the xy plane, the field radiated by each of the equivalent currents can be obtained using the techniques of Chapter 5, Section 5.3, of [38]. Thus,

$$E_{\phi}^e = \frac{-j\omega\mu a e^{-j\beta r}}{4\pi r} \int_0^{2\pi} I_{\phi}^e(\phi') \cos(\phi - \phi') e^{j\beta a \sin\theta \cos(\phi - \phi')} d\phi' \quad (13-105a)$$

$$H_{\phi}^m = \frac{-j\omega\varepsilon a e^{-j\beta r}}{4\pi r} \int_0^{2\pi} I_{\phi}^m(\phi') \cos(\phi - \phi') e^{j\beta a \sin\theta \cos(\phi - \phi')} d\phi' \quad (13-105b)$$

If the equivalent currents are constant, the field is not a function of the azimuthal observation angle ϕ , and (13-105a) and (13-105b) reduce to

$$E_{\phi}^e = \frac{a\omega\mu e^{-j\beta r}}{2r} I_{\phi}^e J_1(\beta a \sin\theta) \quad (13-106a)$$

$$H_{\phi}^m = \frac{a\omega\varepsilon e^{-j\beta r}}{2r} I_{\phi}^m J_1(\beta a \sin\theta) \quad (13-106b)$$

where $J_1(x)$ is the Bessel function of the first kind of order 1.

For diffraction by an edge of finite length, the equivalent current concept for diffraction assumes that each incremental segment of the edge radiates as would a corresponding segment of a two-dimensional edge of infinite length. Similar assumptions are used for diffraction from finite-length curved edges. The concepts, although approximate, have been shown to yield very good results.

For oblique plane wave incidence diffraction by a wedge of finite length ℓ , as shown in Figure 13-41, the equivalent currents of (13-103a) and (13-103b) take the form

$$I_z^e = -\frac{\sqrt{8\pi\beta}}{\eta\beta} e^{-j\pi/4} E_z^i(Q_D) D_s(\xi^-, \xi^+, n; \beta_0') \quad (13-107a)$$

$$\stackrel{s' \gg z'}{\simeq} -\frac{\sqrt{8\pi\beta}}{\eta\beta} e^{-j\pi/4} E_z^i(0) D_s(\xi^-, \xi^+, n; \beta_0') e^{-j\beta z' \cos\beta_0'}$$

$$I_z^m = -\frac{\eta\sqrt{8\pi\beta}}{\beta} e^{-j\pi/4} H_z^i(Q_D) D_h(\xi^-, \xi^+, n; \beta_0') \quad (13-107b)$$

$$\stackrel{s' \gg z'}{\simeq} -\frac{\eta\sqrt{8\pi\beta}}{\beta} e^{-j\pi/4} H_z^i(0) D_h(\xi^-, \xi^+, n; \beta_0') e^{-j\beta z' \cos\beta_0'}$$

where $-\ell/2 \leq z' \leq \ell/2$ (ℓ = length of wedge) and D_s and D_h are formed by (13-96a) and (13-96b). The far-zone fields associated with the equivalent currents of (13-107a) and (13-107b) can be found using, respectively, (13-104a) and (13-104b).

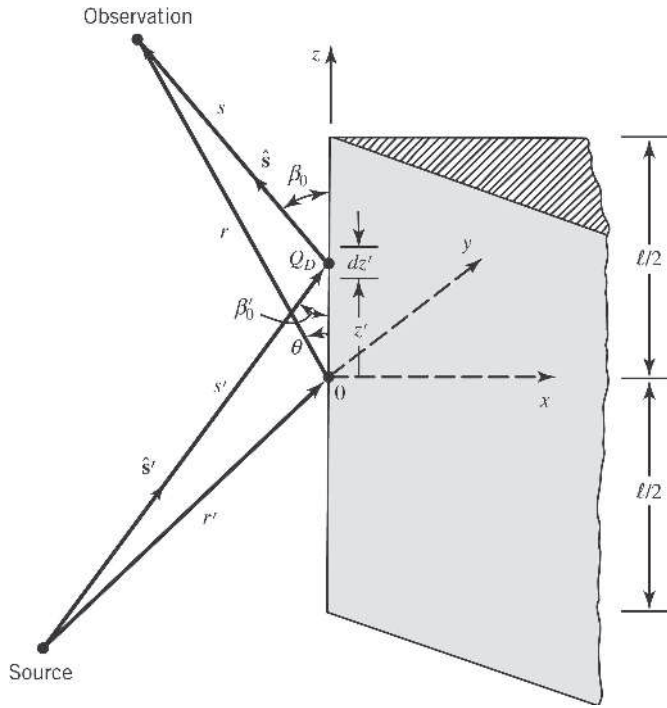


Figure 13-41 Oblique incidence diffraction by a finite-length wedge.

To demonstrate the technique of curved edge diffraction and the equivalent current concept, the radiation of a $\lambda/4$ monopole (blade) mounted on a circular ground plane was modeled. The analytical formulation is assigned as a problem at the end of the chapter. The computed pattern is shown in Figure 13-39 where it is compared with measurements.

To make corrections for the diffracted field discontinuity and inaccuracy at and near the symmetry axis ($\theta = 0^\circ$ and 180°), due to axial caustics, the rim of the ground plane was modeled as a ring radiator [32, 33]. Equivalent currents were used to compute the pattern in the region given by $0^\circ \leq \theta \leq \theta_0$ and $180^\circ - \theta_0 \leq \theta \leq 180^\circ$. In the other space, a two-point diffraction was used. The two points were taken diametrically opposite to each other, and they were contained in the plane of observation. The value of θ_0 depends upon the curvature of the ground plane. For most ground planes of moderate size, θ_0 is in the range $10^\circ < \theta_0 < 30^\circ$.

A very good agreement between theory and experiment is exhibited in Figure 13-39. For a ground plane of this size, the blending of the two-point diffraction pattern and the pattern from the ring source radiator was performed at $\theta_0 \simeq 10^\circ$. It should be noted that the minor lobes near the symmetry axis ($\theta \simeq 0^\circ$ and $\theta \simeq 180^\circ$) for the circular ground plane are more intense than the corresponding ones for the square plane of Figure 13-33. In addition, the back lobe nearest $\theta = 180^\circ$ is of greater magnitude than the one next to it. These effects are due to the ring source radiation by the rim [33] of the circular ground plane toward the symmetry axis.

13.3.6 Slope Diffraction

Until now the field diffracted by an edge has been found based on (13-34a), (13-85), or (13-88) where $\mathbf{E}^i(Q_D)$ represents the incident field at the point of diffraction. This type of formulation indicates that if the incident field $\mathbf{E}^i(Q_D)$ at the point of diffraction Q_D is zero, then the diffracted field will be zero. In addition to this type of diffraction, there is an additional diffraction term that

is based not on the magnitude of the incident field at the point of diffraction but rather on the slope (rate of change, or directional derivative) of the incident field at the point of diffraction. This is a higher-order diffraction, and it becomes more significant when the incident field at the point of diffraction vanishes. It is referred to as *slope diffraction*, and it creates currents on the wedge surface that result in a diffracted field [52].

By referring to the geometry of Figure 13-42, the slope diffracted field can be computed using

Soft Polarization

$$E^d = \frac{1}{j\beta} \left[\frac{\partial E^i(Q_D)}{\partial n} \right] \left(\frac{\partial D_s}{\partial \phi'} \right) \sqrt{\frac{\rho_c}{s(\rho_c + s)}} e^{-j\beta s} \quad (13-108)$$

$$\frac{\partial E^i(Q_D)}{\partial n} = \frac{1}{s'} \frac{\partial E^i}{\partial \phi'} \bigg|_{Q_D} = \text{slope of the incident field} \quad (13-108a)$$

$$\frac{\partial D_s}{\partial \phi'} = \text{slope diffraction coefficient} \quad (13-108b)$$

Hard Polarization

$$H^d = \frac{1}{j\beta} \left[\frac{\partial H^i(Q_D)}{\partial n} \right] \left(\frac{\partial D_h}{\partial \phi'} \right) \sqrt{\frac{\rho_c}{s(\rho_c + s)}} e^{-j\beta s} \quad (13-109)$$

$$\frac{\partial H^i(Q_D)}{\partial n} = \frac{1}{s'} \frac{\partial H^i}{\partial \phi'} \bigg|_{Q_D} = \text{slope of the incident field} \quad (13-109a)$$

$$\frac{\partial D_h}{\partial \phi'} = \text{slope diffraction coefficient} \quad (13-109b)$$

Therefore, in general, the total diffracted field can be found using

$$U^d = \left[U^i(Q_D) D_{s,h} + \frac{1}{j\beta} \frac{\partial U^i(Q_D)}{\partial n} \frac{\partial D_{s,h}}{\partial \phi'} \right] \sqrt{\frac{\rho_c}{s(\rho_c + s)}} e^{-j\beta s} \quad (13-110)$$

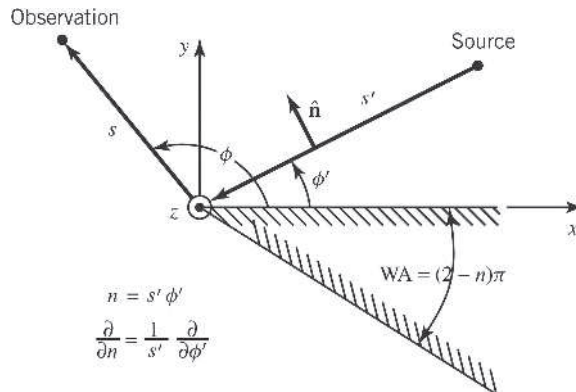


Figure 13-42 Wedge geometry for slope diffraction.

where the first term represents the contribution to the total diffracted field due to the magnitude of the incident field and the second accounts for the contribution due to the slope (rate of change) of the incident field. In (13-110), U represents the electric field for soft polarization and the magnetic field for hard polarization. Similarly $D_{s,h}$ is used to represent D_s for soft polarization and D_h for hard polarization.

The slope diffraction coefficients for soft and hard polarizations can be written, respectively, as [52]

$$\begin{aligned} & \frac{\partial D_s(\phi, \phi', n; \beta'_0)}{\partial \phi'} \\ &= -\frac{e^{-j\pi/4}}{4n^2 \sqrt{2\pi\beta} \sin \beta'_0} \left\{ \left[\csc^2 \left[\frac{\pi + (\phi - \phi')}{2n} \right] F_s[\beta L g^+(\phi - \phi')] \right. \right. \\ & \quad \left. \left. - \csc^2 \left[\frac{\pi - (\phi - \phi')}{2n} \right] F_s[\beta L g^-(\phi - \phi')] \right] \right. \\ & \quad \left. - \left[\csc^2 \left[\frac{\pi + (\phi + \phi')}{2n} \right] F_s[\beta L g^+(\phi + \phi')] \right. \right. \\ & \quad \left. \left. - \csc^2 \left[\frac{\pi - (\phi + \phi')}{2n} \right] F_s[\beta L g^-(\phi + \phi')] \right] \right\} \end{aligned} \quad (13-111a)$$

$$\begin{aligned} & \frac{\partial D_h(\phi, \phi', n; \beta'_0)}{\partial \phi'} \\ &= -\frac{e^{-j\pi/4}}{4n^2 \sqrt{2\pi\beta} \sin \beta'_0} \left\{ \left[\csc^2 \left[\frac{\pi + (\phi - \phi')}{2n} \right] F_s[\beta L g^+(\phi - \phi')] \right. \right. \\ & \quad \left. \left. - \csc^2 \left[\frac{\pi - (\phi - \phi')}{2n} \right] F_s[\beta L g^-(\phi - \phi')] \right] \right. \\ & \quad \left. - \left[\csc^2 \left[\frac{\pi + (\phi + \phi')}{2n} \right] F_s[\beta L g^+(\phi + \phi')] \right. \right. \\ & \quad \left. \left. - \csc^2 \left[\frac{\pi - (\phi + \phi')}{2n} \right] F_s[\beta L g^-(\phi + \phi')] \right] \right\} \end{aligned} \quad (13-111b)$$

where

$$F_s(X) = 2jX \left[1 - j2\sqrt{X} e^{jX} \int_{\sqrt{X}}^{\infty} e^{-j\tau^2} d\tau \right] = 2jX[1 - F(X)] \quad (13-111c)$$

A Fortran and MATLAB computer subroutine designated as **SWDC**, for Slope Wedge Diffraction Coefficients, computes the normalized (with respect to $\sqrt{\lambda}$) slope diffraction coefficients based on (13-111a) through (13-111c). It was developed and reported in [53]. This program uses the complex function FTF (Fresnel transition function) to complete its computations.

To use the subroutine, the user must specify $R = L$ (in wavelengths), $\text{PHID} = \phi$ (in degrees), $\text{PHIPD} = \phi'$ (in degrees), $\text{BTD} = \beta'_0$ (in degrees), and $\text{FN} = n$ (dimensionless) and the subroutine computes the normalized (with respect to $\sqrt{\lambda}$) slope diffraction coefficients $\text{CSDCS} = \partial D_s / \partial \phi'$ and $\text{CSDCH} = \partial D_h / \partial \phi'$.

13.3.7 Multiple Diffractions

Until now we have considered single-order diffractions from each of the edges of a structure. If the structure is composed of multiple edges (as is the case for infinitely thin strips, rectangular and

circular ground planes, etc.), then coupling between the edges will take place. For finite thickness ground planes coupling is evident not only between diametrically opposite edges but also between edges on the same side of the ground plane. Coupling plays a bigger role when the separation between the edges is small, and it should then be taken into account.

A. Higher-Order Diffractions For structures with multiple edges, coupling is introduced in the form of higher-order diffractions. To illustrate this point, let us refer to Figure 13-43a, where a plane wave of hard polarization, represented by a magnetic field parallel to the edge of the wedges, is incident upon a two-dimensional PEC structure composed of three wedges.

The diffraction mechanism of this system can be outlined as follows: The plane wave incident on wedge 1, represented by wedge angle WA_1 , will be diffracted as shown in Figure 13-43a. This is referred to as *first-order diffraction*. The field diffracted by wedge 1 in the direction of wedge 2 (WA_2) will be diffracted again, as shown in Figures 13-43a and 13-43b. This is referred to as *second-order diffraction*, because it is the result of diffraction from diffraction. In turn, the field diffracted from wedge 2 toward wedges 1 and 3 will be diffracted again. The same procedure can be followed for second-order diffractions from wedge 3 due to first-order diffractions from wedge 1. Second- and higher-order diffractions are all referred to as *higher-order diffractions*, and they account for coupling between the edges and are more important for bistatic than monostatic scattering.

Following the procedures that have been outlined for diffractions from two-dimensional PEC wedges, the first-order diffractions from wedge 1, first-order diffractions from wedge 1 toward

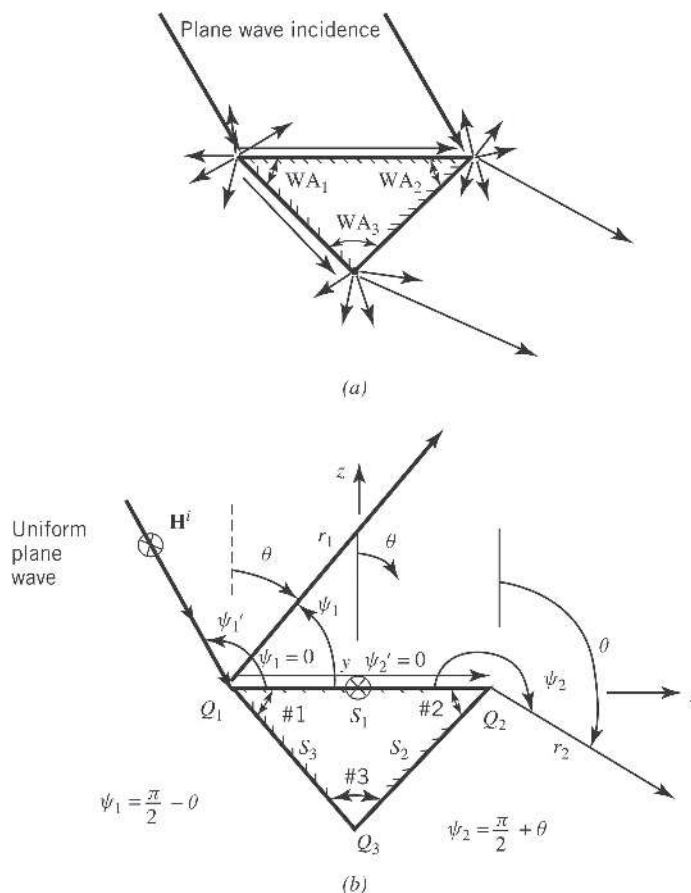


Figure 13-43 Higher-order diffractions from a two-dimensional wedged geometry. (a) Plane wave incidence and diffraction by wedges. (b) Second-order diffraction by wedge 2 due to diffractions from wedge 1.

wedge 2, and second-order diffractions from wedge 2 due to first-order diffractions from wedge 1, can be written, using the geometries of Figures 13-43a and 13-43b, as:

First-order diffractions from wedge 1

$$\begin{aligned}\mathbf{H}_{y1}^{d1} &= \mathbf{H}_1^i(Q_1) \cdot \left[\hat{\mathbf{a}}_y \hat{\mathbf{a}}_y D_1^h(r_1, \psi_1, \psi'_1, n_1) \right] \frac{1}{\sqrt{r_1}} e^{-j\beta r_1} \\ \mathbf{H}_{y1}^{d1} &= \hat{\mathbf{a}}_y H_1^i(Q_1) \cdot \left[\hat{\mathbf{a}}_y \hat{\mathbf{a}}_y \left\{ \begin{aligned} &D_1^i(r_1, \psi_1 - \psi'_1, n_1) \\ &+ D_1^r(r_1, \psi_1 + \psi'_1, n_1) \end{aligned} \right\} \right] \frac{1}{\sqrt{r_1}} e^{-j\beta r_1} \\ \mathbf{H}_{y1}^{d1} &= +\hat{\mathbf{a}}_y H_1^i(Q_1) \left\{ \begin{aligned} &D_1^i(r_1, \psi_1 - \psi'_1, n_1) \\ &+ D_1^r(r_1, \psi_1 + \psi'_1, n_1) \end{aligned} \right\} \frac{e^{-j\beta r_1}}{\sqrt{r_1}} \end{aligned} \quad (13-112a)$$

First-order diffractions from wedge 1 toward wedge 2

$$\begin{aligned}\mathbf{H}_{y1}^{d1}(r_1 = s_1, \psi_1 = 0, n_1) \Big|_{r_1=s_1, \psi_1=0} &= +\hat{\mathbf{a}}_y H_1^i(Q_1) \left\{ \begin{aligned} &D_1^i(s_1, -\psi'_1, n_1) \\ &+ D_1^r(s_1, \psi'_1, n_1) \end{aligned} \right\} \frac{e^{-j\beta s_1}}{\sqrt{s_1}} \\ \mathbf{H}_{y1}^{d1}(r_1 = s_1, \psi_1 = 0, n_1) \Big|_{r_1=s_1, \psi_1=0} &= +\hat{\mathbf{a}}_y H_1^i(Q_1) \left\{ \begin{aligned} &V_1^i(s_1, -\psi'_1, n_1) \\ &+ V_1^r(s_1, \psi'_1, n_1) \end{aligned} \right\} \end{aligned} \quad (13-112b)$$

Equation 13-112b represents the total diffracted field; half of it is the incident diffracted field and the other half is the reflected diffracted field.

Second-order diffractions from wedge 2 due to first-order diffractions from wedge 1

$$\begin{aligned}\mathbf{H}_{y2}^{d2} &= \mathbf{H}_{21}^i(Q_2) \cdot \hat{\mathbf{a}}_y \hat{\mathbf{a}}_y D_2^h(s_2, \psi_2, \psi'_2, n_2) \frac{1}{\sqrt{r_2}} e^{-j\beta r_2} \Big|_{\psi'_2=0, \psi_2=\frac{\pi}{2}+\theta} \\ \mathbf{H}_{y2}^{d2} &= \frac{\mathbf{H}_{y1}^{d1}(r_1 = s_1, \psi_1 = 0, n_1)}{2} \cdot \left[\hat{\mathbf{a}}_y \hat{\mathbf{a}}_y D_2^h(s_2, \psi_2, \psi'_2, n_2) \right] \frac{e^{-j\beta r_2}}{\sqrt{r_2}} \\ \mathbf{H}_{y2}^{d2} &= +\hat{\mathbf{a}}_y \frac{H_1^i(Q_1)}{2} \left\{ V_{B1}^i(s_1, -\psi'_1, n_1) + V_B^r(s_1, \psi'_1, n_1) \right\} \\ &\quad \cdot \hat{\mathbf{a}}_y \hat{\mathbf{a}}_y \left[D_2^i(s_2, \psi_2, n_2) + D_2^r(s_2, \psi_2, n_2) \right] \frac{e^{-j\beta r_2}}{\sqrt{r_2}} \\ \mathbf{H}_{y2}^{d2} &= +\hat{\mathbf{a}}_y \frac{H_1^i(Q_1)}{2} [V_{B1}^i(s_1, -\psi'_1, n_1) + V_B^r(s_1, \psi'_1, n_1)] \\ &\quad \cdot [D_2^i(s_2, \psi_2, n_2) + D_2^r(s_2, \psi_2, n_2)] \frac{e^{-j\beta r_2}}{\sqrt{r_2}} \\ \mathbf{H}_{y2}^{d2} &= +\hat{\mathbf{a}}_y \frac{1}{2} H_1^i(Q_1) [V_{B1}^i(s_1, -\psi'_1, n_1) + V_B^r(s_1, \psi'_1, n_1)] \\ &\quad \cdot [2D_2^i(s_2, \psi_2, n_2)] \frac{e^{-j\beta r_2}}{\sqrt{r_2}} \\ \mathbf{H}_{y2}^{d2} &= +\hat{\mathbf{a}}_y H_1^i(Q_1) [V_B^i(s_1, -\psi'_1, n_1) + V_B^r(s_1, \psi'_1, n_1)] \\ &\quad \cdot D_2^i(s_2, \psi_2, n_2) \frac{e^{-j\beta r_2}}{\sqrt{r_2}} \end{aligned} \quad (13-112c)$$

The $\frac{1}{2}$ factor in the development of (13-112c) is used to represent the incident diffracted field of (13-112b) from wedge 1 toward wedge 2.

The procedure needs to be repeated for first- and second-order diffractions due to direct wave incidence to wedge 2. The method was developed for hard polarization as there are no higher-order diffractions for soft polarization, based on *regular* diffraction, since the diffracted field from any of the wedges toward the others will be zero due to the vanishing of the tangential electric field along the PEC surface of the structure.

B. Self-Consistent Method It becomes apparent that the procedure for accounting for higher-order diffractions, especially for third and higher orders, can be very tedious, although straightforward. It is recommended that when third- and even higher-order diffractions are of interest, a procedure be adopted that accounts for all (infinite) orders of diffraction. This procedure is known as the *self-consistent method* [62], which is used in scattering theory [63]. It can be shown that the interactions between the edges can also be expressed in terms of a geometrical progression, which in scattering theory is known as the *successive scattering procedure* [63].

Let us now illustrate the self-consistent method as applied to the diffractions of Figure 13-43a. According to Figures 13-43a and 13-43b diffractions by wedge 1 that are due to radiation from the source and that are due to all orders of diffraction from wedge 2 can be written as

$$\begin{aligned} U_1^{s,h}(r_1, \phi) = & U_0^{s,h}(Q_1) D_{10}^{s,h} \left(L_{10}, \psi_{10} = \frac{\pi}{2} + \phi, \psi'_{10} = \delta, n_1 \right) A_{10}(r_1) e^{-j\beta r_1} \\ & + \frac{1}{2} [U_2^{s,h}(r_2 = d, \phi = 0)] D_{12}^{s,h}(L_{12}, \psi_{12} = \pi - \phi, \psi'_{12} = 0, n_1) \\ & \times A_{12}(r_1) e^{-j\beta r_1} \end{aligned} \quad (13-113a)$$

where $U^{s,h}$ is used to represent here the electric field for soft polarization and the magnetic field for hard polarization. In (13-113a),

$U_1^{s,h}(r_1, \phi)$ = total diffracted field by wedge 1

$U_0^{s,h}(Q_1)$ = field from source at wedge 1

$U_2^{s,h}(r_2 = d, \phi = 0)$ = total diffracted field (including all orders of diffraction) by wedge 2 toward wedge 1

$D_{10}^{s,h}$ = diffraction coefficient (for soft or hard polarization) of wedge 1 that is due to radiation from the source

$D_{12}^{s,h}$ = diffraction coefficient (for soft or hard polarization) of wedge 1 that is due to radiation from wedge 2

A_{10} = amplitude spreading factor of wedge 1 that is due to radiation from the source

A_{12} = amplitude spreading factor of wedge 1 that is due to radiation from wedge 2

The unknown part in (13-113a) is $U_2^{s,h}(r_2 = d, \phi = 0)$, and the self-consistent method will be used to determine it.

Using a similar procedure and referring to Figure 13-44c, the total diffracted field by wedge 2 that is due to all orders of diffraction from wedge 1 can be written as

$$\begin{aligned} U_2^{s,h}(r_2, \phi) = & \frac{1}{2} [U_1^{s,h}(r_1 = d, \phi = \pi)] \\ & \times D_{21}^{s,h}(L_{21}, \psi_{21} = \phi, \psi'_{21} = 0, n_2) A_{21}(r_2) e^{-j\beta r_2} \end{aligned} \quad (13-113b)$$

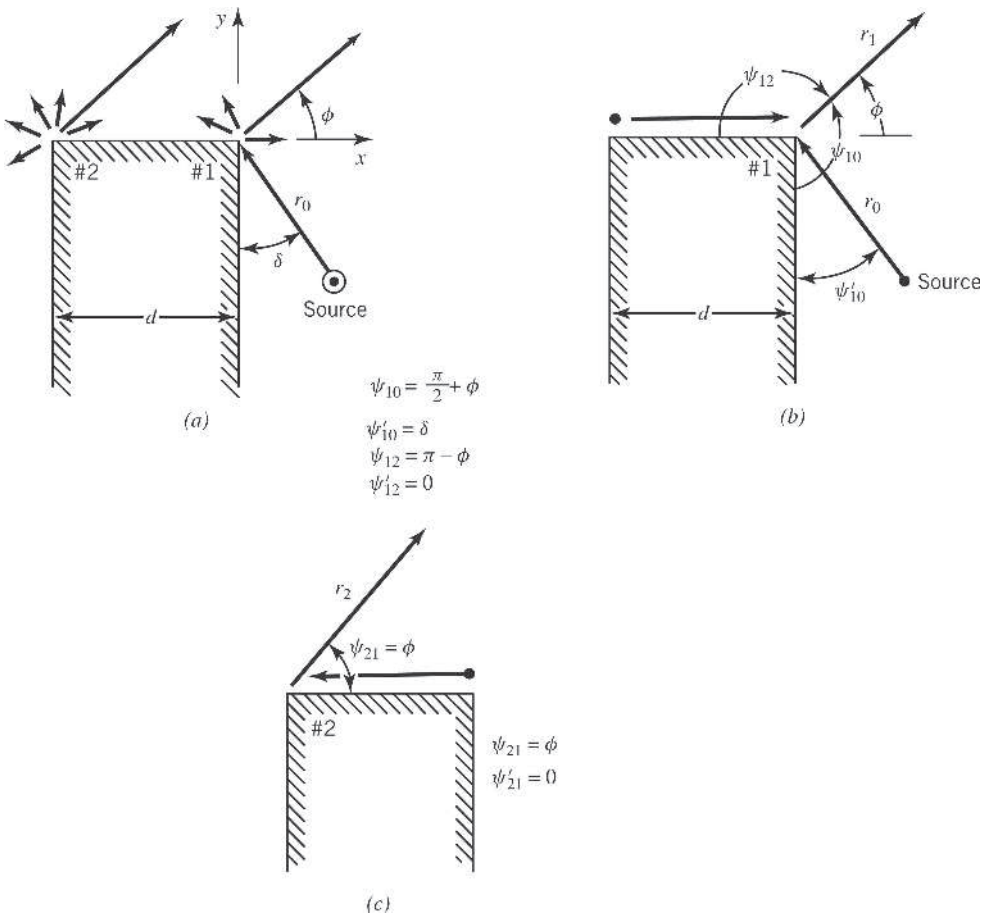


Figure 13-44 Finite thickness edge for multiple diffractions. (a) Source incidence. (b) Diffraction by edge 1. (c) Diffraction by edge 2.

where $U_2^{s,h}(r_2, \phi) =$ total diffracted field by wedge 2

$U_1^{s,h}(r_1 = d, \phi = \pi) =$ total diffracted field (including all orders of diffraction) by wedge 1 toward wedge 2

$D_{21}^{s,h} =$ diffraction coefficient (for soft or hard polarization) of wedge 2 due to radiation from wedge 1

$A_{21} =$ amplitude spreading factor of wedge 2 due to radiation from wedge 1

In (13-113b), the unknown part is $U_1^{s,h}(r_1 = d, \phi = \pi)$, and it will be determined using the self-consistent method.

Equations 13-113a and 13-113b form a consistent pair where there are two unknowns, that is, $U_2^{s,h}(r_2 = d, \phi = 0)$ in (13-113a) and $U_1^{s,h}(r_1 = d, \phi = \pi)$ in (13-113b). If these two unknowns can be found, then (13-113a) and (13-113b) can be used to predict the diffracted fields from each of the wedges taking into account all (infinite) orders of diffraction. These two unknowns can be

found by doing the following. At the position of wedge 2 ($r_1 = d, \phi = \pi$) the total diffracted field by wedge 1, as given by (13-113a), can be reduced to

$$\begin{aligned} [U_1^{s,h}(r_1 = d, \phi = \pi)] \\ = U_0^{s,h}(Q_1) \left\{ D_{10}^{s,h} \left(L_{10}, \psi_{10} = \frac{3\pi}{2}, \psi'_{10} = \delta, n_1 \right) A_{10}(r_1 = d) e^{-j\beta d} \right\} \\ + [U_2^{s,h}(r_2 = d, \phi = 0)] \left\{ \frac{1}{2} D_{12}^{s,h} (L_{12}, \psi_{12} = 0, \psi'_{12} = 0, n_1) A_{12}(r_1 = d) e^{-j\beta d} \right\} \end{aligned} \quad (13-114a)$$

In a similar manner, at the position of wedge 1 ($r_2 = d, \phi = 0$) the total diffracted field by wedge 2, as given by (13-113b), can be reduced to

$$\begin{aligned} [U_2^{s,h}(r_2 = d, \phi = 0)] = U_1^{s,h}(r_1 = d, \phi = \pi) \\ \times \left\{ \frac{1}{2} D_{21}^{s,h} (L_{21}, \psi_{21} = 0, \psi'_{21} = 0, n_2) A_{21}(r_2 = d) e^{-j\beta d} \right\} \end{aligned} \quad (13-114b)$$

Equations 13-114a and (13-114b) can be rewritten, respectively, in simplified form as

$$[U_1^{s,h}(r_1 = d, \phi = \pi)] = U_0^{s,h}(Q_1) T_{10}^{s,h} + [U_2^{s,h}(r_2 = d, \phi = 0)] R_{12}^{s,h} \quad (13-115a)$$

$$[U_2^{s,h}(r_2 = d, \phi = 0)] = [U_1^{s,h}(r_1 = d, \phi = \pi)] R_{12}^{s,h} \quad (13-115b)$$

where

$$\begin{aligned} T_{10}^{s,h} &= D_{10}^{s,h} \left(L_{10}, \psi_{10} = \frac{3\pi}{2}, \psi'_{10} = \delta, n_1 \right) A_{10}(r_1 = d) e^{-j\beta d} \\ &= \text{transmission coefficient from wedge 1 toward} \\ &\quad \text{wedge 2 due to radiation from main source} \end{aligned} \quad (13-115c)$$

$$\begin{aligned} R_{12}^{s,h} &= \frac{1}{2} D_{12}^{s,h} (L_{12}, \psi_{12} = 0, \psi'_{12} = 0, n_1) A_{12}(r_1 = d) e^{-j\beta d} \\ &= \text{reflection coefficient from wedge 1 toward} \\ &\quad \text{wedge 2 due to diffractions from wedge 2} \end{aligned} \quad (13-115d)$$

$$\begin{aligned} R_{21}^{s,h} &= \frac{1}{2} D_{21}^{s,h} (L_{21}, \psi_{21} = 0, \psi'_{21} = 0, n_2) A_{21}(r_2 = d) e^{-j\beta d} \\ &= \text{reflection coefficient from wedge 2 toward} \\ &\quad \text{wedge 1 due to diffractions from wedge 1} \end{aligned} \quad (13-115e)$$

The self-consistent pair of (13-115a) and (13-115b) contains the two unknowns that are needed to predict the total diffracted field as given by (13-113a) and (13-113b). Solving (13-115a) and (13-115b) for $U_1^{s,h}(r_1 = d, \phi = \pi)$ and $U_2^{s,h}(r_2 = d, \phi = 0)$, we can show that

$$U_1^{s,h}(r_1 = d, \phi = \pi) = U_0^{s,h}(Q_1) \frac{T_{10}^{s,h}}{1 - R_{21}^{s,h} R_{12}^{s,h}} \quad (13-116a)$$

$$U_2^{s,h}(r_2 = d, \phi = 0) = U_0^{s,h}(Q_1) \frac{T_{10}^{s,h} R_{21}^{s,h}}{1 - R_{21}^{s,h} R_{12}^{s,h}} \quad (13-116b)$$

When expanded, it can be shown that (13-116a) and (13-116b) can be written as a geometric series of the form

$$U_1^{s,h}(r_1 = d, \phi = \pi) = U_0^{s,h}(Q_1)T_{10}^{s,h}[1 + x_0 + x_0^2 + \cdots] \quad (13-117a)$$

$$U_2^{s,h}(r_2 = d, \phi = 0) = U_0^{s,h}(Q_1)T_{10}^{s,h}R_{21}^{s,h}[1 + x_0 + x_0^2 + \cdots] \quad (13-117b)$$

where

$$x_0 = R_{21}^{s,h}R_{12}^{s,h} \quad (13-117c)$$

Each term of the geometric series can be related to an order of diffraction by the corresponding wedge.

In matrix form, the self-consistent set of equations as given by (13-115a) and (13-115b) can be written as

$$\begin{bmatrix} 1 & -R_{12}^{s,h} \\ -R_{21}^{s,h} & 1 \end{bmatrix} \begin{bmatrix} U_1^{s,h} \\ U_2^{s,h} \end{bmatrix} = \begin{bmatrix} U_0^{s,h}T_{10}^{s,h} \\ 0 \end{bmatrix} \quad (13-118)$$

which can be solved using standard matrix inversion methods.

The outlined self-consistent method can be extended and applied to the interactions between a larger number of edges. However, the order of the system of equations to be solved will also increase and will be equal to the number of interactions between the various edge combinations.

C. Overlap Transition Diffraction Region The UTD diffraction coefficients fail to predict accurately the field diffracted near grazing angles. This is best illustrated in Figure 13-45, where a uniform plane wave is incident on a two-dimensional PEC strip.

The field diffracted by wedge 1 toward wedge 2 creates a Transition Region (TR), shown cross-hatched, over which the diffracted field is non-ray optical and the second-order and successive diffractions are not accurately predicted using the traditional GTD/UTD procedure outlined previously. The same is true for diffractions from other wedges with similar angles of incidence. However, as the angle of incidence moves away from grazing, the GTD/UTD diffractions become more valid. This is illustrated with some examples that follow.

To overcome the issue of the non-ray optical nature of the first-order diffractions and the inaccurate predictions by standard GTD/UTD of the higher-order successive diffractions by wedges, the following two methods can be used.

1. Extended Spectral Theory of Diffraction (ESTD) [64]
2. Extended Physical Theory of Diffraction (EPTD) [65]

While the GTD and UTD are considered to be somewhat heuristic, they are more general and less cumbersome in their application to multiple diffractions. The ESTD and EPTD are more

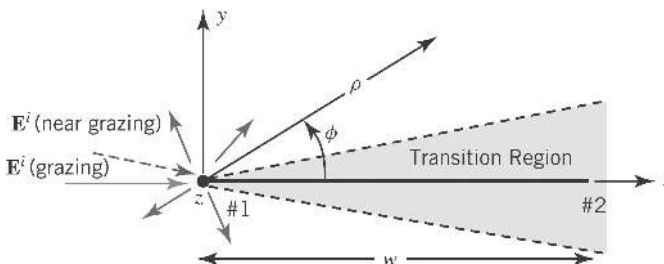


Figure 13-45 Diffraction by strip at and near grazing.

rigorous and more accurate but less general and more complex. The ESTD is an extension of the STD method [66]. Using the ESTD, the current density induced on the scatter of interest is transformed in the spectral domain. The radiation integral is then asymptotically evaluated in the spectral domain, after the induced current density is multiplied by a spectral diffraction coefficient. The original STD was limited to a half-plane and aperture scattering for plane wave incidence. The ESTD extends the STD to general double-wedge configuration, and it can be used for plane, cylindrical, and spherical wave incidence for both normal and oblique incidences.

The EPTD is an alternative transition-region method based upon a different evaluation of the surface radiation integral. The induced current density is approximated using the PTD fringe currents [67], whereas the ESTD uses the UTD diffraction coefficients. The resulting radiation integral is evaluated asymptotically to obtain the second-order field diffracted by the double wedge structure. The EPTD formulation is limited to plane-wave incidence, far-field observation in the plane of the structure. As with the ESTD, the EPTD doubly-diffracted field expression can be greatly simplified for certain geometries, such as the strip.

To demonstrate the concepts of near grazing-angle-incidence diffraction, a numbers of examples are considered for monostatic and bistatic scattering for both soft (TM^z) and hard (TE^z) polarizations [68, 69]. For monostatic scattering, the patterns are illustrated in Figures 13-46a and 13-46b for a strip of width $w = 2\lambda$. A width of 2λ is chosen for all cases so that the GTD/UTD diffraction coefficients are valid. For the soft polarization, only first-order UTD diffractions are

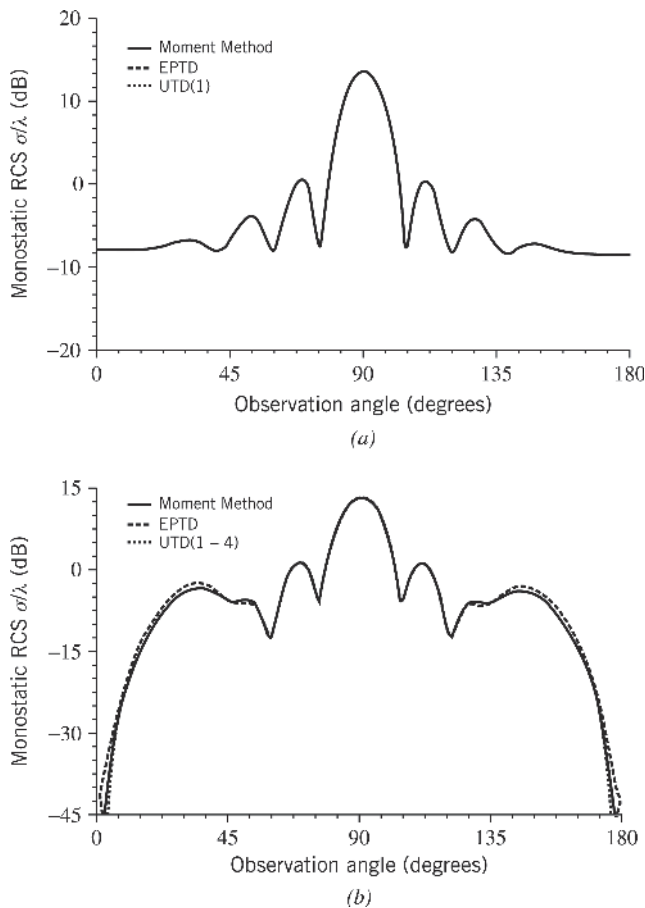


Figure 13-46 Monostatic RCS by a two-dimensional strip of width $w = 2\lambda$ [68, 69]. (a) Soft (TM^z) polarization. (b) Hard (TE^z) polarization.

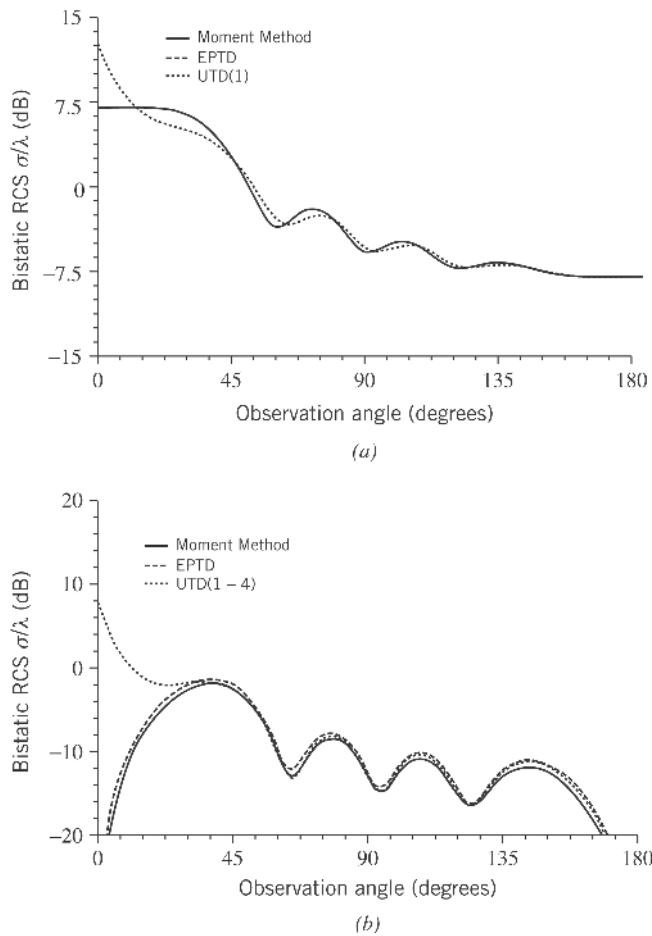


Figure 13-47 Bistatic RCS by a two-dimensional strip of width $w = 2\lambda$ at $\phi_i = 170^\circ$ [68, 69]. (a) Soft (TM^z) polarization. (b) Hard (TE^z) polarization.

considered since higher-orders are not applicable; however, for the hard polarization, up to fourth-order UTD diffractions are included. It is clear that the results of all three methods (MM, EPTD, UTD) for both monostatic cases are in very good agreement.

The bistatic scattering results for incidence angles of near grazing ($\phi_i = 170^\circ$) and away from grazing ($\phi_i = 135^\circ$) are shown in Figures 13-47a, 13-47b, 13-48a, and 13-48b, respectively. As expected, because of the UTD diffracted fields near grazing angle ($\phi_i = 170^\circ$) are non-ray optical, the patterns of the UTD results are not in very good agreement with those of the MM and EPTD as indicated in Figures 13-47a and 13-47b for both polarizations. However, the comparison of not near-grazing incidence ($\phi_i = 135^\circ$) of all three methods is very good for both polarizations, as indicated in Figures 13-48a and 13-48b.

13.4 COMPUTER CODES

Using geometrical optics and wedge diffraction techniques, a number of computer codes have been developed over the years to compute the radiation and scattering characteristics of simple and complex antenna and scattering systems. Some are in the form of subroutines that are used primarily to compute wedge diffraction coefficients and associated functions. Others are very

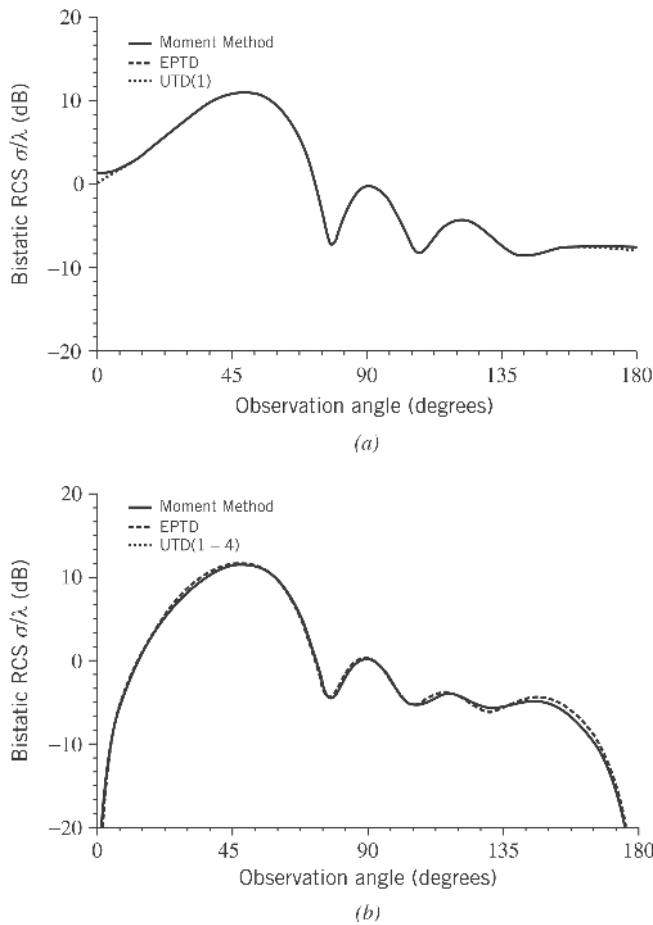


Figure 13-48 Bistatic RCS by a two-dimensional strip of width $w = 2\lambda$ at $\phi_i = 135^\circ$ [68, 69]. (a) Soft (TM^z) polarization. (b) Hard (TE^z) polarization.

sophisticated codes that can be used to analyze very complex radiation and scattering problems. We will describe here two wedge diffraction subroutines and a subfunction, which the readers can utilize to either develop their own codes or to solve problems of interest. These subroutines were initially written in Fortran but were translated to MATLAB for this edition.

13.4.1 Wedge Diffraction Coefficients

The **Wedge Diffraction Coefficients (WDC)** subroutine computes the soft and hard polarization wedge diffraction coefficients based on (13-89a) through (13-90b). To complete the computations, the program uses the complex function FTF (Fresnel transition function). This program was developed at the ElectroScience Laboratory at the Ohio State University, and it was reported in [53].

To use this subroutine, the user must specify

- $R = L$ (in wavelengths) = distance parameter
- $\text{PHID} = \phi$ (in degrees) = observation angle
- $\text{PHIPD} = \phi'$ (in degrees) = incident angle
- $\text{BTD} = \beta'_0$ (in degrees) = oblique angle
- $\text{FN} = n$ (dimensionless) = wedge angle factor

and the program computes the complex diffraction coefficients

$CDCS = D_s =$ complex diffraction coefficient (soft polarization)

$CDCH = D_h =$ complex diffraction coefficient (hard polarization)

13.4.2 Fresnel Transition Function

The **Fresnel Transition Function (FTF)** computes the Fresnel transition functions $F(X)$ of (13-68e), (13-68f), and (13-69e), (13-69f). The program is based on the asymptotic expression of (13-74a) for small arguments ($X < 0.3$) and on (13-74b) for large arguments ($X > 5.5$). For intermediate values ($0.3 \leq X \leq 5.5$), linear interpolation is used. The program was developed at the ElectroScience Laboratory of the Ohio State University, and it was reported in [53].

13.4.3 Slope Wedge Diffraction Coefficients

The **Slope Wedge Diffraction Coefficients (SWDC)** subroutine computes the soft and hard polarization wedge diffraction coefficients based on (13-111a) through (13-111c). To complete the computations, the program uses the complex function FTF (Fresnel transition function) of Section 13.4.2. This program was developed at the ElectroScience Laboratory at the Ohio State University, and it was reported in [53].

To use this subroutine, the user must specify

$R = L$ (in wavelengths) = distance parameter

$PHID = \phi$ (in degrees) = observation angle

$PHIPD = \phi'$ (in degrees) = incident angle

$BTD = \beta'_0$ (in degrees) = oblique angle

$FN = n$ (dimensionless) = wedge angle factor

and the program computes the complex slope diffraction coefficients

$CSDCS = \frac{\partial D_s}{\partial \phi'} =$ complex slope diffraction coefficient (soft polarization)

$CSDCH = \frac{\partial D_h}{\partial \phi'} =$ complex slope diffraction coefficient (hard polarization)

13.5 MULTIMEDIA

On the website that accompanies this book, the following multimedia resources are included for the review, understanding, and presentation of the material of this chapter.

- **MATLAB** computer programs (the first two also in Fortran):
 - a. **WDC:** (Both MATLAB and Fortran). Computes the first-order wedge diffraction coefficient based on (13-89a) through (13-90b). The initial Fortran algorithm was developed and reported in [53].
 - b. **SWDC:** (Both MATLAB and Fortran). Computes the first slope wedge diffraction coefficient based on (13-111a) through (13-111c). The initial Fortran algorithm was developed and reported in [53].

- c. **PEC_Wedge:** Computes, based on the exact solution of (13-40a), the normalized amplitude pattern of a uniform plane wave incident upon a two-dimensional PEC wedge, as shown in Figure 13-13.
- d. **PEC_Strip_Line_UTD:** Computes, using UTD, the normalized amplitude radiation pattern of a line source based on the UTD of Example 13-4. It is compared with that based of the Integral Equation (IE) of Sections 12.2.2 through 12.2.8 and Physical Optics (PO) of Section 11.2.3.
- e. **PEC_Strip_SW_UTD:** Computes, using UTD, the TM^z and TE^z 2-D scattering width (SW), monostatic and bistatic, of a PEC strip of finite width, based on the UTD of Example 13-5. It is compared with that of the Integral Equation (IE) of Section 12.3.1 and Physical Optics of Section 11.3.1, and Figures 12-13 and 11-4.
- f. **Monopole_GP_UTD:** Computes, using UTD, the normalized amplitude radiation pattern of a $\lambda/4$ monopole on a rectangular or circular ground plane based on UTD and Figures 13-32 and 13-37.
- g. **Aperture_GP_UTD:** Computes, using UTD, the normalized amplitude radiation pattern of a rectangular or circular aperture, with either a uniform or dominant mode aperture field distribution, on a rectangular ground plane based on UTD and Figure 13-32, where the monopole is replaced by an aperture, as shown in Figure P13-41.
- h. **PEC_Rect_RCS_UTD:** Computes, using UTD, the TE^x and TM^x bistatic and monostatic RCS of a PEC rectangular plate using UTD. It is compared with the Physical Optics (PO) of Section 11.2.3.
- i. **PEC_Circ_RCS_UTD:** Computes the TE^x and TM^x monostatic RCS of a PEC circular plate using UTD. It is compared with the Physical Optics (PO) of Chapter 11 and Problem 11.24.
- j. **PEC_Square_Circ_RCS_UTD.** Computes, using UTD, the TE^x and TM^x monostatic RCS of PEC square and circular plates, which have the same area and equal maximum monostatic RCS at normal incidence. The UTD patterns of the two plates, square and circular, are compared with the Physical Optics (PO) of Chapter 11.
- **PowerPoint (PPT)** viewgraphs, in multicolor.

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PROBLEMS

- 13.1. Using the geometry of Figure 13-5, derive (13-9).
- 13.2. An arbitrary surface of revolution can be represented by

$$z = g(u) \quad \text{where} \quad u = \frac{x^2 + y^2}{2}$$

Define

$$K^2 = 1 + 2u \left[\frac{dg(u)}{du} \right]^2$$

Then the unit vectors $\hat{\mathbf{u}}_1$ and $\hat{\mathbf{u}}_2$ of Figure 13-7 in the directions of the principal radii of curvature R_1 and R_2 , respectively, can be determined using

$$\hat{\mathbf{u}}_1 = \frac{\hat{\mathbf{a}}_x y - \hat{\mathbf{a}}_y x}{\sqrt{x^2 + y^2}}$$

$$\hat{\mathbf{u}}_2 = \frac{\hat{\mathbf{a}}_x x + \hat{\mathbf{a}}_y y + \hat{\mathbf{a}}_z \left[(x^2 + y^2) \frac{dg(u)}{du} \right]}{K \sqrt{x^2 + y^2}}$$

and the R_1 and R_2 can be found using

$$\frac{1}{R_1} = \frac{1}{K} \frac{dg(u)}{du}$$

$$\frac{1}{R_2} = \frac{1}{K^3} \left[\frac{dg(u)}{du} + 2u \frac{d^2g(u)}{du^2} \right]$$

For a paraboloidal reflector (parabola of revolution), widely used as a microwave reflector antenna, whose surface can be represented by

$$z = f - \frac{x^2 + y^2}{4f}$$

where f is the focal distance, show that the principal radii of curvature are given by

$$\frac{1}{R_1} = -\frac{1}{2f} \frac{1}{\left[1 + \frac{x^2 + y^2}{4f^2} \right]^{1/2}}$$

$$\frac{1}{R_2} = -\frac{1}{2f} \frac{1}{\left[1 + \frac{x^2 + y^2}{4f^2}\right]^{3/2}}$$

- 13.3. The intensity radiated by an electromagnetic source is contained within a cone of total included angle $\alpha = 60^\circ$. Assuming that the radiation density at a radial distance of 5 meters from the vertex of the cone is uniformly distributed over the spherical cap of the cone and it is 10 milliwatts/cm², determine, using classical geometrical optics and conservation of energy within a tube of rays, the:
- Total power radiated by the source.
 - Radiation density (in milliwatts/cm²) over the spherical cap of the cone at a radial distance of 50 meters from the vertex of the cone.

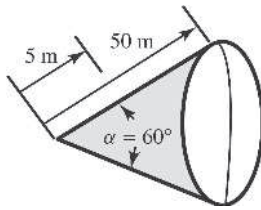


Figure P13-3

- 13.4. Derive (13-30) using the geometry of Figure 13-8.
- 13.5. An electric line source is placed in front of a 30° convex segment of an infinite length conducting circular arc, as shown in Figure P13-5. Assume the line source is placed symmetrically about the arc, its position coincides with the origin of the coordinate system, and it is parallel to the length of the arc. Using geometrical optics determine the following:

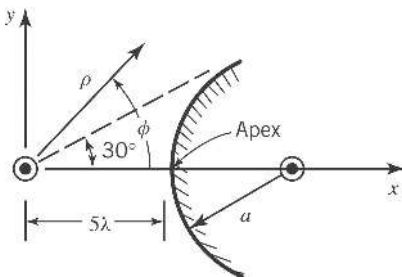


Figure P13-5

- The location of the caustic for fields reflected by the surface of the cap.
- The far-zone backscattered electric field at a distance of 50λ from the center of the apex of the arc.

Assume that the radius of the arc is 5λ and the incident electric field at the apex of the arc is

$$\mathbf{E}^i(\rho = 5\lambda, \phi = 0^\circ) = \hat{\mathbf{a}}_z 10^{-3} \text{ V/m}$$

- 13.6. Repeat Problem 13.5 for a 30° concave segment of an infinite length conducting circular arc, as shown in Figure P13-6.

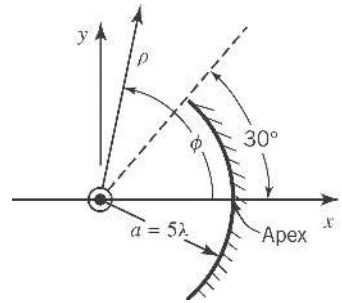


Figure P13-6

- 13.7. A uniform plane wave traveling in free space with an incident electric field given by

$$\mathbf{E}^i = \hat{\mathbf{a}}_x e^{-j\beta z}$$

is incident upon a smooth curved conducting surface as shown in Figure P13-7.

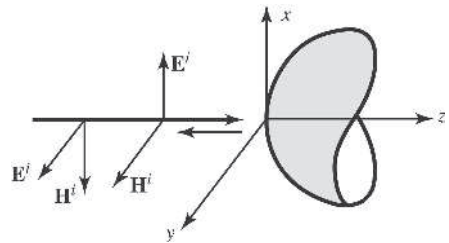


Figure P13-7

Assuming the radii of curvature in the two principal planes are: xz plane $= 10\lambda$ and yz plane $= 19.6\lambda$, utilize the simplest and sufficiently accurate method to:

- Write an expression, in vector form, of the backscattered electric field.

- (b) Compute the normalized monostatic RCS (σ/λ^2) in dB.
 (c) Repeat parts (a) and (b) when the incident electric field is given by

$$\mathbf{E}^i = \hat{\mathbf{a}}_y e^{-j\beta z}$$

- 13.8.** An infinite-length electric line source is placed a distance s' from a two-dimensional flat PEC strip of finite width in the x direction and infinite length in the z direction, as shown in Figure P13-8. To reduce the backscattered field, the flat strip is replaced with a two-dimensional semi-circular PEC arc of radius a , as shown dashed in the figure. Determine, using GO, the radius (in wavelengths) of the PEC arc so that the backscattered electric field from the arc is 20 dB down (or -20 dB) relative to the electric field backscattered from the flat PEC strip. Assume that $s' = 10\lambda$ and that $s \gg s' \gg a$.

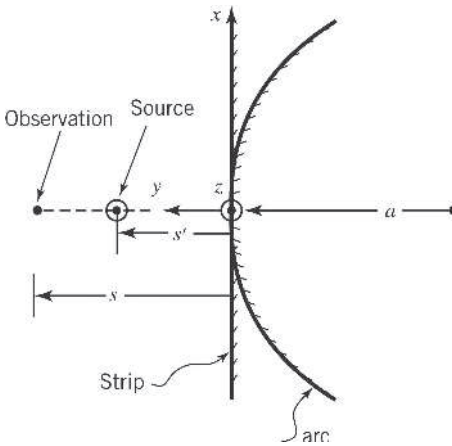


Figure P13-8

- 13.9.** Repeat the calculations of Example 13-2 for $h = 0.25$ and $w = 2\lambda$.
13.10. Show that by combining (13-60a) and (13-60b) you get (13-61).
13.11. Show that evaluation of (13-48b) along C_T leads to (13-62a) and along $\text{SDP}_{\pm\pi}$ leads to (13-62b).
13.12. When the line source is in the vicinity of the edge of the wedge and the observations are made at large distances ($\rho \gg \rho'$) in Figure 13-38, show the following:

- (a) Green's function of (13-38) can be approximated by

$$G \simeq \sqrt{\frac{2}{\pi\beta\rho}} e^{-j(\beta\rho - \pi/4)} F(\beta\rho')$$

where

$$F(\beta\rho') = \frac{1}{n} \sum_{m=0}^{\infty} \varepsilon_m J_{m/n}(\beta\rho') e^{+j(m/n)(\pi/2)} \times \left[\cos \frac{m}{n}(\phi - \phi') \pm \cos \frac{m}{n}(\phi + \phi') \right]$$

- (b) Geometrical optics fields of (13-65) can be written as

$$F_G(\beta\rho') = \begin{cases} e^{j\beta\rho' \cos(\phi - \phi')} \pm e^{j\beta\rho' \cos(\phi + \phi')} & \text{for } 0 < \phi < \pi - \phi' \\ e^{j\beta\rho' \cos(\phi - \phi')} & \text{for } \pi - \phi' < \phi < \pi + \phi' \\ 0 & \text{for } \pi + \phi' < \phi < 2\pi \end{cases}$$

- (c) Diffracted fields of (13-67) can be written as

$$V_D^{i,r}(\rho', \phi \mp \phi', n) = V_D^i(\rho', \phi - \phi', n) \pm V_D^r(\rho', \phi + \phi', n)$$

$$V_D^{i,r}(\rho', \phi \mp \phi', n) = \frac{e^{-j\pi/4}}{\sqrt{2\pi\beta}} \frac{1}{n} \sin\left(\frac{\pi}{n}\right)$$

$$\times \left[\frac{1}{\cos\left(\frac{\pi}{n}\right) - \cos\left(\frac{\phi - \phi'}{n}\right)} \pm \frac{1}{\cos\left(\frac{\pi}{n}\right) - \cos\left(\frac{\phi + \phi'}{n}\right)} \right] \times \frac{e^{-j\beta\rho'}}{\sqrt{\rho'}}$$

- 13.13.** A unity amplitude uniform plane wave of soft polarization is incident normally on a half-plane at an angle of 45° , as shown in Figure P13-13. At an observation point P of coordinates $\rho = 5\lambda$, $\phi = 180^\circ$ from the edge of the half-plane, determine the following:
 (a) Incident geometrical optics field.
 (b) Reflected geometrical optics field.
 (c) Total geometrical optics field.
 (d) Incident diffracted field.

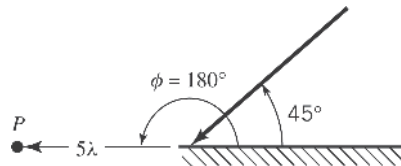


Figure P13-13

- (e) Reflected diffracted field.
- (f) Total diffracted field.
- (g) Total field (geometrical optics plus diffracted).

13.14. Repeat Problem 13.13 for a hard polarization uniform plane wave.

13.15. An electric line source, whose normalized electric field at the origin is unity, is placed at a distance of $\rho' = 5\lambda$, $\phi' = 180^\circ$ from the edge of a half-plane, as shown in Figure P13-15. Using the reciprocity principle of Figure 13-17 and the results of Problem 13.12, determine at large distances the following:

- (a) Incident geometrical optics electric field.
- (b) Reflected geometrical optics electric field.
- (c) Total geometrical optics electric field.
- (d) Incident diffracted electric field.
- (e) Reflected diffracted electric field.
- (f) Total diffracted electric field.
- (g) Total electric field (geometrical optics plus diffracted).

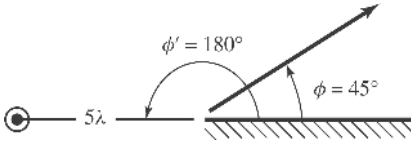


Figure P13-15

13.16. Repeat Problem 13.15 for a magnetic line source whose normalized magnetic field at the origin is unity. At each observation determine the magnetic field.

13.17. A uniform plane wave of soft polarization ($\mathbf{E} = \hat{\mathbf{a}}_z E^i$), whose electric field normalized amplitude at the diffraction edge is 2, is incident upon a half-plane at an incidence angle of 60° . When $\rho = 5.5\lambda$, determine (in vector form) approximate values for the:

- (a) Incident GO electric field at $\phi = 120^\circ^-$ and $\phi = 120^\circ^+$.
- (b) Reflected GO electric field at $\phi = 120^\circ^-$ and $\phi = 120^\circ^+$.
- (c) Incident diffracted electric field at $\phi = 120^\circ^-$ and $\phi = 120^\circ^+$.
- (d) Reflected diffracted electric field at $\phi = 120^\circ^-$ and $\phi = 120^\circ^+$.
- (e) Total electric field at $\phi = 120^\circ^-$ and $\phi = 120^\circ^+$.

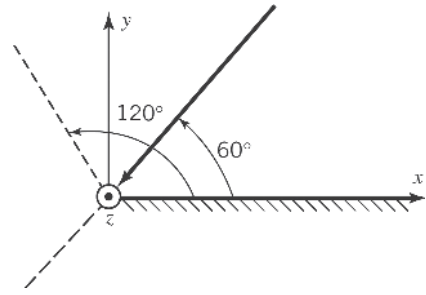


Figure P13-17

13.18. A uniform plane wave, of unity amplitude, is incident upon a PEC 90° wedge at an angle of $\phi' = 60^\circ$. At an observation distance of $\rho = 81\lambda$ from the edge of the wedge and observation angle of 180° , compute the:

- (a) Incident diffracted field using GTD.
- (b) Approximate incident diffracted field using UTD.
- (c) Reflected diffracted field using GTD.
- (d) Approximate reflected diffracted field using UTD.

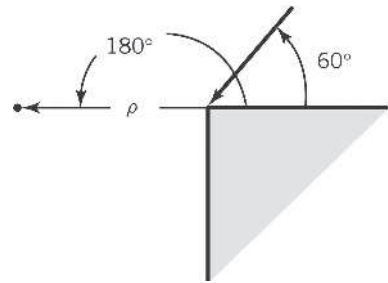


Figure P13-18

If you make any approximations, state as to why you think they are valid. Compare the corresponding answers of parts (a) through (d). Should they be approximately the same or different, and why?

13.19. A hard-polarized uniform plane wave, with the magnetic field directed in the $+z$ direction and traveling in the $+x$ direction, is incident upon a half-plane (knife edge), as shown in Figure P13-19. Assume an observation distance s from the edge of the wedge. Also assume that the magnitude of the incident magnetic field is unity.

Using exclusively the coordinate system shown in the figure, write vector expressions for the following fields:

- Incident magnetic field.
- Incident electric field.
- Backscattered diffracted magnetic field.
- Backscattered diffracted electric field.
- Magnitude (in dB) of the ratio of the backscattered magnetic field to the incident magnetic field at a distance of $s = 100\lambda$.

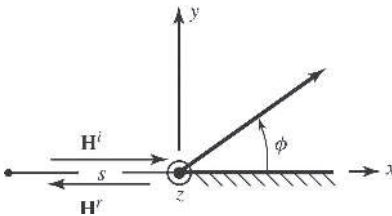


Figure P13-19

- 13.20.** An electric line source, whose normalized electric field at the origin (leading edge of half-plane) is unity, is placed a distance 5λ from the edge of the half-plane, as shown in Figure P13-20. Determine, in the far field, the following:

- Incident GO electric field.
- Reflected GO electric field.
- Total GO electric field.
- Incident diffracted electric field.
- Reflected diffracted electric field.
- Total diffracted electric field.
- Total electric field (GO plus diffracted).

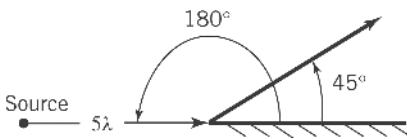


Figure P13-20

- 13.21.** A uniform plane wave, with the electric/magnetic field directed in the $+z$ direction and traveling in the $+x$ direction, is incident upon a half-plane (knife edge), as shown in Figure 13-21. Assume an observation distance s from the edge of the wedge, and that the magnitude of the incident electric/magnetic field is unity at the leading edge.

Using exclusively the coordinate system shown in the figure, write vector expressions for the following fields. Assume $\phi = 90^\circ$, $s = 9\lambda$.

- Soft Polarization:* GTD diffracted electric field.
- Hard Polarization:* GTD diffracted magnetic field.

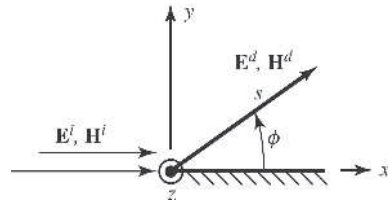


Figure P13-21

- 13.22.** A unity amplitude uniform plane wave is incident normally on a 30° conducting wedge. Assume that the incident electric field is polarized in the z direction and the incident angle is 45° . Then determine at $\rho = 5.5\lambda$ the following:

- Incident GO electric field at $\phi = 225^\circ^-$.
- Incident GO electric field at $\phi = 225^\circ^+$.
- Approximate incident diffracted field at $\phi = 225^\circ^-$.
- Approximate incident diffracted field at $\phi = 225^\circ^+$.
- Approximate total electric field at $\phi = 225^\circ^-$.
- Approximate total electric field at $\phi = 225^\circ^+$.

Plus and minus refer to angles slightly greater or smaller than the designated values.

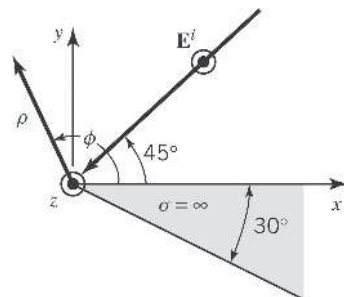


Figure P13-22

13.23. Repeat Problem 13.22 when the incident magnetic field of the unity amplitude uniform plane wave is polarized in the z direction. At each point determine the corresponding GO, diffracted, or total magnetic field.

13.24. A unity amplitude uniform plane wave of soft polarization is incident upon a half-plane, as shown in Figure P13-24. At a plane parallel and behind the half-plane perform the following tasks.

- Formulate expressions for the incident and reflected geometrical optics fields, incident and reflected diffracted fields, and total field.
- Plot along the observation plane the total field (geometrical optics plus diffracted fields) when $y_0 = 5\lambda$, $-5\lambda \leq x_0 \leq 5\lambda$.
- Determine the total field at $y_0 = 5\lambda$ and $x_0 = 0$. Explain the result.

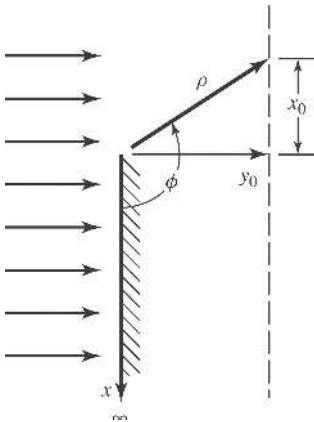


Figure P13-24

13.25. Repeat Problem 13.24 for a hard polarized uniform plane wave.

13.26. A unity amplitude uniform plane wave is incident normally on a two-dimensional conducting wedge, as shown in Figure P13-26.

- Formulate expressions that can be used to determine the incident, reflected, and total diffracted fields away from the incident and reflected shadow boundaries.
- Plot the soft and hard polarization normalized diffraction coefficients ($D_{s,h}/\sqrt{\lambda}$) as a function of ϕ for $n = 1.5$ and $n = 2$.
- Simplify the expressions of part (a) when $n = 2$ (half-plane).

(d) Formulate expressions for the two-dimensional scattering width of the half-plane ($n = 2$) for soft and hard polarizations.

(e) Simplify the expressions of part (d) for backscattering observations ($\phi = \phi'$). How can the results of this part be used to design low-observable radar targets?

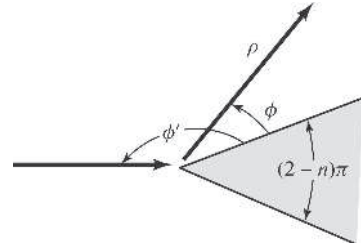


Figure P13-26

13.27. For the geometry of Problem 13.26 plot at $\phi = \phi'$ the normalized diffraction coefficient ($D_{s,h}/\sqrt{\lambda}$) as a function of the wedge angle ($1 \leq n \leq 2$) when the polarization of the wave is (a) soft and (b) hard.

13.28. Repeat Problem 13.27 when the observations are made along the surface of the wedge ($\phi = 0^\circ$).

13.29. By approximating the integrand of the Fresnel integral of (13-63c) with a truncated Taylor series

$$e^{-j\tau^2} \simeq \sum_{n=0}^M \frac{(-j\tau^2)^n}{n!}$$

derive the small argument approximation of (13-74a) for the transition function.

13.30. By repeatedly integrating by parts the Fresnel integral of (13-63c), derive the large argument approximation of (13-74b) for the transition function.

13.31. Show that there is a finite discontinuity of unity amplitude with the proper polarity, similar to (13-83), along the following boundaries.

- RSB of Figure 13-24a.
- ISB of Figure 13-24b.
- RSB of Figure 13-24b.

13.32. Using the Uniform Theory of Diffraction (UTD), the diffraction coefficient, assuming the incident wave illuminates the upper face

of the PEC wedge, as shown in Figure P13-32, is given by (13-71a) and (13-71b). Identify which of the above cotangent functions and corresponding Fresnel transition functions F are used to eliminate the discontinuity introduced by the GO field along the:

- Incident Shadow Boundary (ISB).
- Reflection Shadow Boundary (RSB).

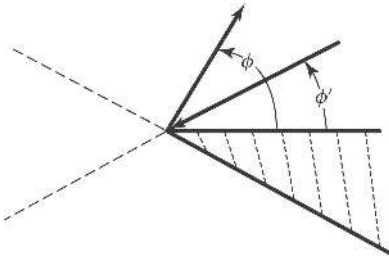


Figure P13-32

Identify the ISB and RSB in terms of angles, and indicate what happens individually, and as a product, to those two functions along the corresponding boundaries.

- Repeat Problem 13.32 when the incident wave illuminates the lower face of the PEC wedge, as shown in Figure P13-33. The equations for the diffraction coefficients are the same, (13-71a) and (13-71b), as those given in Problem 13.32.

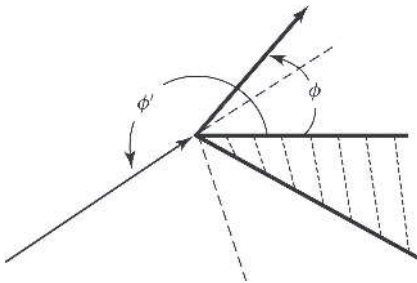


Figure P13-33

- Compute the corresponding phases for the fields, GO and diffracted, for Figures 13-25a and 13-25b.
- Repeat the calculations of Example 13-4 for a strip of width $w = 2\lambda$ and with the line source at a height of $h = 0.25\lambda$.
- Repeat Example 13-5 for a hard-polarized uniform plane wave.

- A uniform plane wave is incident upon a two-dimensional PEC strip, as shown in Figure 13-28. Formulate the problem, using GTD, for:
 - Hard polarization (assume incident magnetic field of H_0 amplitude):
 - Backscattered/monostatic ($\phi = \phi'$) magnetic field and its backscattering/monostatic scattering width SW.
 - Bistatic scattered magnetic field and its bistatic scattering width SW.
 - Soft polarization (assume incident electric field of E_0 amplitude).
 - Bistatic scattered electric field and its bistatic scattering width SW.
 - Plot the bistatic and backscattering/monostatic patterns for $w = 2\lambda$, $f = 10$ GHz, and $\phi' = 120^\circ$ for both soft- and hard-polarized uniform plane waves. For each polarization (soft and hard), the backscattering/monostatic patterns should be in two separate figures; one figure for the soft and the other for the hard. Similarly, the bistatic cases should be in two separate figures; one figure for the soft and the other for the hard. In each of the four figures, plot and compare, for the respective polarizations, the patterns based on EFIE/MoM SW/ λ (dB) and SW(dBm).

- A z -polarized electric-field uniform plane wave of unity amplitude is traveling in the negative y direction, as shown in Figure P13-38. In order to introduce a blockage to the wave, a half-plane (knife edge) is placed, as shown in the figure.

Assuming an observation point of $\phi = 330^\circ$ and $s = 9\lambda$, determine, at that point, the:

- Vector electric field in the presence of the knife edge.

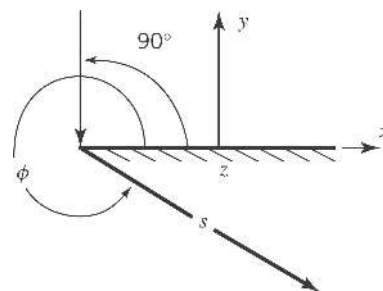


Figure P13-38

- (b) Blockage (in dB) introduced to the wave by the knife-edge.

13.39. A unity amplitude uniform plane wave is incident normally on a two-dimensional strip of width w as shown in Figure P13-39.

- (a) Formulate expressions for the fields when the observations are made below the strip along its axis of symmetry.
 (b) Compute the field when $w = 3\lambda$ and $d = 5\lambda$ for soft and hard polarizations.

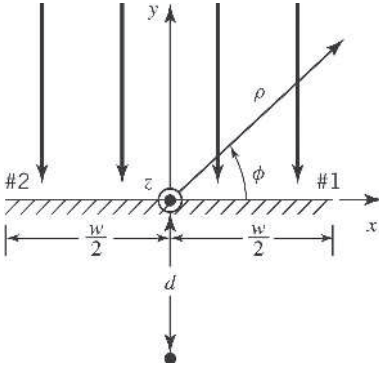


Figure P13-39

13.40. A transmitter and a receiver are placed on either side of a mountain that can be modeled as a perfectly conducting half-plane, as shown in Figure P13-40. Assume that the transmitting source is isotropic.

- (a) Derive an expression for the field at the receiver that is diffracted from the top of the mountain. Assume the field is soft or hard polarized.
 (b) Compute the power loss (in decibels) at the receiver that is due to the presence of the mountain when $h = 5\lambda$ and $d = 5\lambda$. Do this when the transmitter and receiver are both, for each case, either soft or hard polarized.

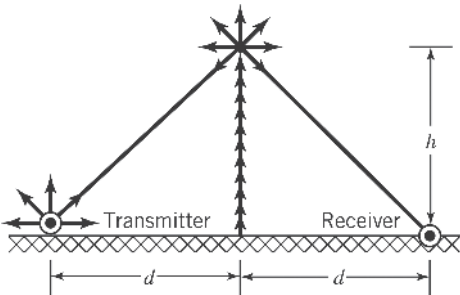


Figure P13-40

13.41. A rectangular waveguide of dimensions a and b , operating in the dominant TE_{10} mode, is mounted on a square ground plane with dimensions w on each of its sides, as shown in Figure P13-41. Using the geometry of Example 13-6 and assuming that the total geometrical optics field above the ground plane in the principal yz plane is given by

$$E_{\theta G}(\theta) = E_0 \left[\frac{\sin\left(\frac{\beta b}{2} \sin \theta\right)}{\frac{\beta b}{2} \sin \theta} \right] \frac{e^{-j\beta r}}{r}$$

$$0 \leq \theta \leq \frac{\pi}{2}$$

- (a) Show that the fields diffracted from edges 1 and 2 in the principal yz plane are given by

$$E_{\theta 1}^d(\theta) = E_0 \frac{\sin\left(\frac{\beta b}{2}\right)}{\frac{\beta b}{2}} \times V_B^i\left(\frac{w}{2}, \psi_1, n_1 = 2\right) e^{+j(\beta w/2) \sin \theta} \frac{e^{-j\beta r}}{r}$$

$$\psi_1 = \frac{\pi}{2} + \theta, \quad 0 \leq \theta \leq \pi$$

$$E_{\theta 2}^d(\theta) = E_0 \frac{\sin\left(\frac{\beta b}{2}\right)}{\frac{\beta b}{2}} \times V_B^i\left(\frac{w}{2}, \psi_2, n_2 = 2\right) e^{-j(\beta w/2) \sin \theta} \frac{e^{-j\beta r}}{r}$$

$$\psi_2 = \begin{cases} \frac{\pi}{2} - \theta, & 0 \leq \theta \leq \frac{\pi}{2} \\ \frac{5\pi}{2} - \theta, & \frac{\pi}{2} \leq \theta \leq \pi \end{cases}$$

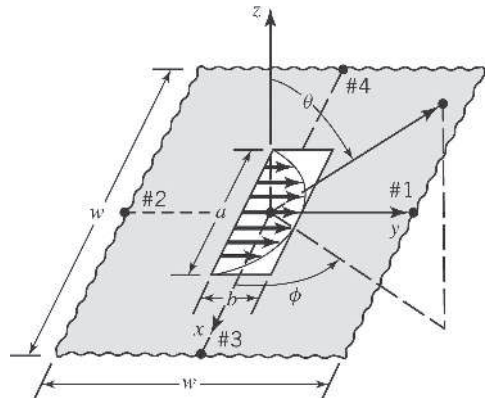


Figure P13-41

- (b) Plot the normalized amplitude pattern (in decibels) for $0^\circ \leq \theta \leq 180^\circ$ when $w/2 = 14.825\lambda$ and $b = 0.42\lambda$.

13.42. A uniform plane wave is incident upon a square ground plane, of dimensions a and b in the x and y directions, respectively. The plane of incidence is the yz plane and the plane of observation is also the yz plane. For a hard-polarized (TE^x) shown in Figure 11-8b and Figure P13-42:

- Model/formulate the first-order diffractions from edges #1 and #2 for general bistatic scattering, and eventually 3-D RCS.
- Model/formulate the second-order bistatic diffractions from edges #1 and #2 due to first-order diffractions from edges #2 and #1, respectively, and eventually 3-D RCS.
- Plot, in one figure and three patterns, the 3-D RCS scattering bistatic patterns (σ_{3-D}/λ^2) (in dB) one due to first-order diffractions, and one due to first- plus second-order diffractions (σ_{3-D}/λ^2) (in dB) for an incidence angle of $\theta_i = 30^\circ$ and $a = b = 5\lambda$. On the same figure plot the 3-D RCS pattern based on Physical Optics (σ_{3-D}/λ^2) (in dB) shown in Figure 11-9a.
- Plot, in one figure and three patterns, the 3-D RCS scattering monostatic patterns (σ_{3-D}/λ^2) (in dB), in the upper 180° region, one due to first-order diffractions, and one due to first- plus second-order diffractions (σ_{3-D}/λ^2) (in dB) for $a = b = 5\lambda$. On the

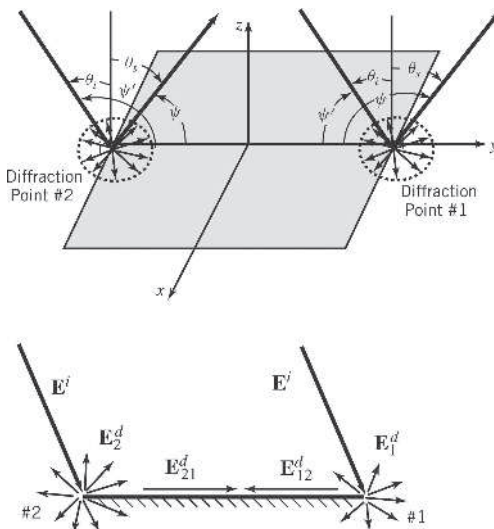


Figure P13-42

same figure, plot the 3-D RCS pattern based on Physical Optics (σ_{3-D}/λ^2) (in dB) shown in Figure 11-9b.

Use (11-22e) to convert the 2-D RCS to 3-D RCS.

13.43. For Problem 13.42 and a soft-polarized (TM^x) shown in Figure 11-8(c), including the figures in Problem 13.42:

- Model/formulate the first-order diffractions from edges #1 and #2 for general bistatic scattering, and eventually 3-D RCS. There are no second- or higher-order regular diffractions for this polarization.
- Plot, in one figure and two patterns, the 3-D RCS scattering bistatic patterns (σ_{3-D}/λ^2) (in dB) due to first-order diffractions for an incidence angle of $\theta_i = 30^\circ$ and $a = b = 5\lambda$. On the same figure plot the 3-D RCS based on PO (σ_{3-D}/λ^2) (in dB) shown in Fig. 11-9a.
- Plot, in one figure and two patterns, the 3-D RCS scattering monostatic patterns (σ_{3-D}/λ^2) (in dB), in the upper 180° region, due to first-order diffractions for $a = b = 5\lambda$. On the same figure, plot the 3-D RCS pattern based on Physical Optics (σ_{3-D}/λ^2) (in dB) shown in Fig. 11-9b.

Use (11-22e) to convert the 2-D RCS to 3-D RCS.

13.44. A soft-polarized spherical wave is incident, at a normal incidence angle, upon a half-plane with a straight edge. The wave emanates from a source a distance s' from the edge of the half-plane. In order to reduce the diffracted field, the straight edge is replaced with a curved edge of radius a .

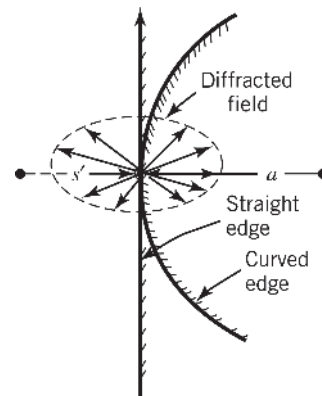


Figure P13-44

- (a) Derive a simplified expression for the ratio of amplitude spreading factor reduction for the backscattered field, by a curved edge relative to that by the straight edge, at a large distance s from the edge of the half-plane.
- (b) Determine, in dB, the relative amplitude reduction of the backscattered diffracted field by a curved edge of radius $a = 25\lambda$ and a source distance of $s' = 100\lambda$.

13.45. A uniform plane wave with an electric field given by

$$\mathbf{E}^i = \hat{\mathbf{a}}_z e^{-j\beta x}$$

is incident upon a flat conducting square plate, as shown in Figure P13-45. To reduce the backscattered diffractions from the leading edge, the square plate is replaced by a circular plate. Determine the radius of curvature a of the square plate so that the backscattered diffractions at a distance of 50λ from the leading edge diffraction point of the curved edge are -20 dB than those of the leading straight edge.

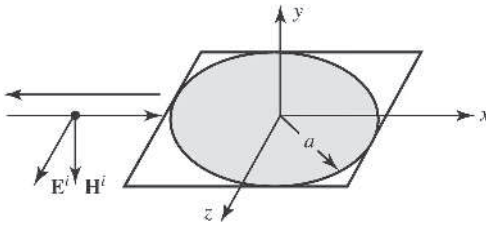


Figure P13-45

13.46. A unity amplitude uniform plane wave is incident at a grazing angle on a circular ground plane of radius a , as shown in Figure P13-46.

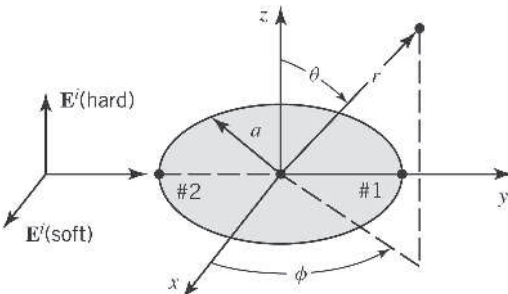


Figure P13-46

- (a) Formulate expressions for the field diffracted from the leading edge (#2) of the ground plane along the principal yz plane for both soft and hard polarizations.
- (b) Locate the position of the caustic for the leading edge.
- (c) Assuming the incident electric field amplitude at the leading edge of the plate is 10^{-3} V/m and its phase is zero, compute for each polarization at $r = 50\lambda$ when $a = 2\lambda$ the backscattered electric field that is due to the leading edge of the plate.

13.47. Show that the fields radiated by a circular loop with nonuniform equivalent electric and magnetic current I_ϕ^e and I_ϕ^m are given, respectively, by (13-105a) and (13-105b). If the equivalent currents I_ϕ^e and I_ϕ^m are uniform, show that (13-105a) and (13-105b) reduce, respectively, to (13-106a) and (13-106b).

13.48. An infinitesimal dipole is placed on the tip of a finite cone, as shown in Figure P13-48. The total-geometrical optics magnetic field radiated by the source in the presence of the cone, referred to the center of the base of the cone, is given by

$$H_{\phi G} = R(\theta) e^{j\beta s \cos \theta \cos(\alpha/2)}$$

where $R(\theta)$ is the field distribution when the cone is infinite in length ($s = \infty$). For the finite-length cone there is also a diffracted field forming a ring source at the base of the cone.

- (a) Using (13-105b), show that the diffracted magnetic field is given by

$$H_\phi^d = b R \left(\theta = \pi - \frac{\alpha}{2} \right) V_B^i(s, \psi, n) \times \int_0^{2\pi} \cos \phi e^{j(\beta b) \sin \theta \cos \phi} d\phi, \quad \psi = \frac{\alpha}{2} + \theta$$

where $V_B^i(s, \psi, n)$ is the incident diffraction function and ϕ is the azimuthal observation angle.

- (b) Assuming that the base of the cone is large ($b \gg \lambda$) and the observations are made away from the symmetry axis so that $\sin \theta > 0$, show that by using the method of steepest descent of Appendix VI the integral formed by the ring source diffracted field from the rim of the cone

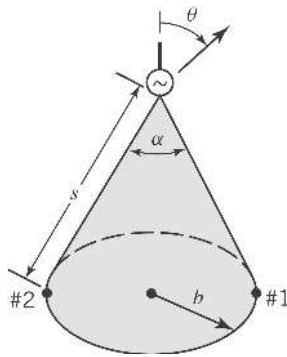


Figure P13-48

reduces to the field diffracted from two diametrically opposite points on the rim (i.e., points #1 and #2 of Figure P13-48).

13.49. For the $\lambda/4$ monopole on the circular ground plane of Example 13-7, model the rim of the ground plane as a ring source radiator using equivalent current concepts of Section 13.3.5 to correct for the caustic formed when the observations are made near the axis ($\theta \simeq 0^\circ$ and 180°) of the ground plane.

- Derive expressions for the diffracted field from the rim using the ring radiator of Figure 13-38.
- Show, using the method of steepest descent (saddle point method), that the ring radiator radiation of part a reduces to a two-point diffraction away from the symmetry axis.
- Compute and plot the pattern near the axis ($\theta \simeq 0^\circ$ and 180°) of the ground plane when the diameter of the ground plane is $d = 4.064\lambda$. Use this part of the pattern to complement that computed using the two-point diffraction of Example 13-7.

13.50. We all have seen the general geometrical shapes of the stealth bomber and fighter (see Figure 13-1); most of the fuselage consists of wedge-type shapes. There must be a reason for that. To answer some of the questions, formulate the problem, and show why a wedge-type of geometry is desired, assume that a uniform plane wave, of either soft or hard polarization, impinges either upon a 2-D PEC wedge of included angle WA or a 2-D PEC cylinder of circular cross section and radius a , as shown in Figure P13-50:

- In one figure with two curves, plot the normalized 2-D monostatic scattering width $SW[\sigma_{2-D}/\lambda(\text{dB})]$ of the wedge vs. wedge angle WA (in degrees) $[0 \leq WA \leq 60^\circ]$ for both hard and soft polarization.
- In one figure with two curves, plot the normalized 2-D monostatic scattering width $SW[\sigma_{2-D}/\lambda(\text{dB})]$ of the cylinder vs. cylinder radius a wavelengths $[0 \leq a \leq 10\lambda]$ for both hard and soft polarization.
- In one figure with two curves, plot the normalized 2-D monostatic scattering width $SW[\sigma_{2-D}/\lambda(\text{dB})]$ of the wedge WA (in degrees) $[0 \leq WA \leq 60^\circ]$ and that of cylinder with radius a $[0 \leq a \leq 10\lambda]$ for hard polarization.
- In one figure with two curves, plot the normalized 2-D monostatic scattering width $SW[\sigma_{2-D}/\lambda(\text{dB})]$ of the wedge WA (in degrees) $[0 \leq WA \leq 60^\circ]$ and that of the cylinder with radius a $[0 \leq a \leq 10\lambda]$ for soft polarization.

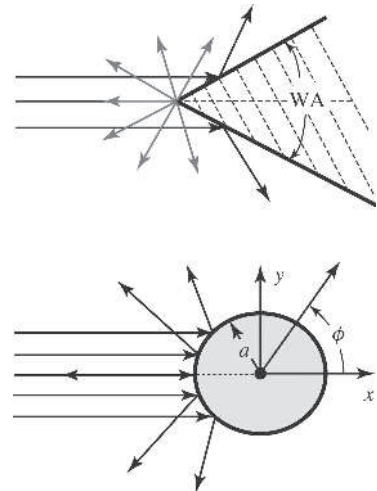


Figure P13-50

13.51. A uniform plane wave is incident upon a circular ground plane of radius a , as shown in the figure that follows. The plane of incidence is the yz plane and the plane of observation is also the yz plane. Referring to [36], Figure P13-51, and for hard-polarized (TE^x) wave:

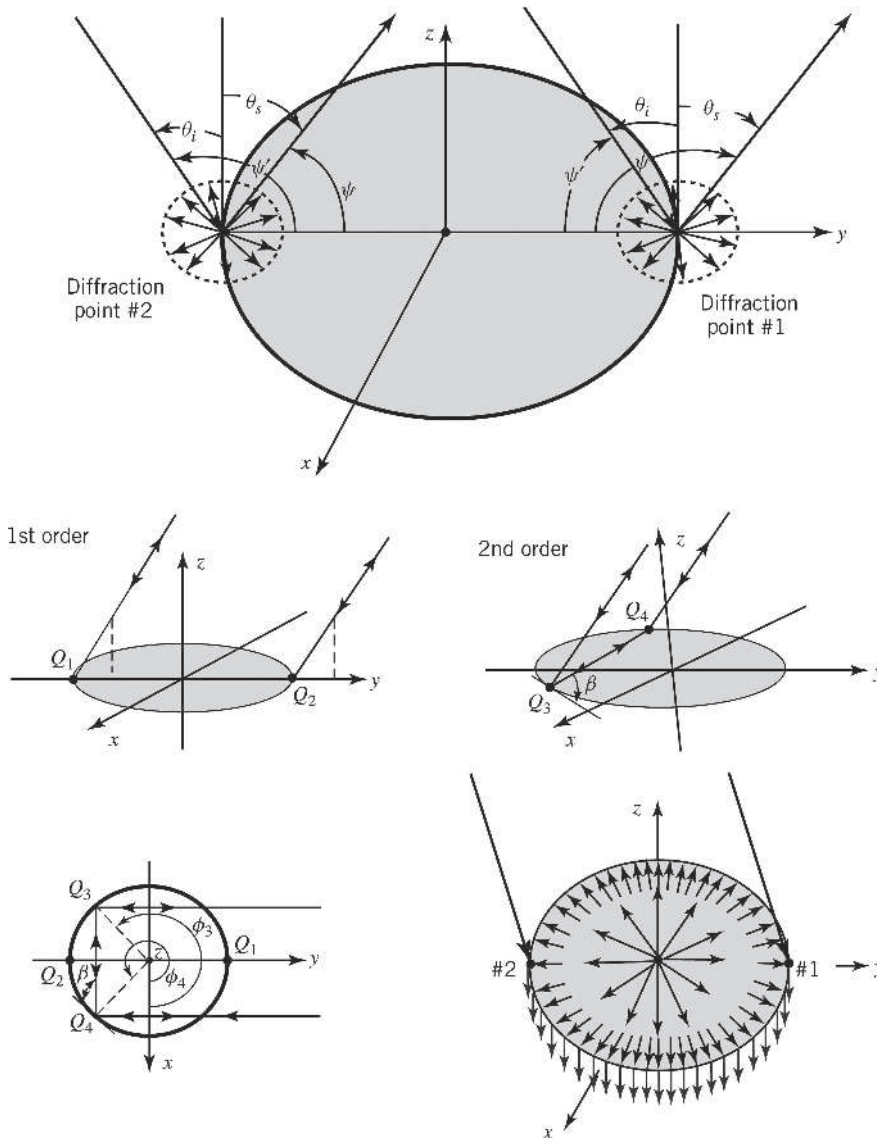


Figure P13-51

- Model the first-order 3-D diffractions from edges #1 and #2 for monostatic scattering. Use the middle of the plate as the reference point.
- Model the second-order monostatic 3-D diffractions from edges #1 and #2 due to first-order diffractions from edges #2 and #1, respectively.
- Model the second-order 3-D diffractions from the migrating points #3 and #4.
- Model the 'ring-radiator' contributions. In [36] they are referred to as

Axial Caustic Correction formulated using *Equivalent Currents*.

- Plot, in one figure and two patterns, the 3-D RCS scattering monostatic patterns (σ_{3-D}/λ^2) (in dB), in the upper 180° region, due to first-order (#1 and #2) diffractions plus ring radiator for $a = 3.516\lambda$. On the same figure, plot the 3-D RCS pattern based on Physical Optics (σ_{3-D}/λ^2) (in dB).
- Plot, in one figure and three patterns, the 3-D RCS scattering monostatic patterns (σ_{3-D}/λ^2) (in dB), in the

upper 180° region, one due to first-order (#1 and #2) diffractions, and one due to first- (#1 and #2) plus all second-order (#1, #2, #3, and #4 points) diffractions plus ring radiator (σ_{3-D}/λ^2) (in dB) for $a = 3.516\lambda$. On the same figure, plot the 3-D RCS pattern based on Physical Optics (σ_{3-D}/λ^2) (in dB).

13.52. For Problem 13.51 and a soft-polarized (TM^x) wave, along with the figures of Problem 13.51:

- Model/formulate the first-order 3-D diffractions from edges #1 and #2 for monostatic scattering. For this polarization, there are no second-order regular diffractions from edges #1 and #2. Use the middle of the plate as the reference point.
- Model/formulate the second-order 3-D diffractions from the migrating points #3 and #4.
- Model/formulate the ‘ring-radiator’ contributions. In [36] they are referred to as *Axial Caustic Correction* formulated using *Equivalent Currents*.
- Plot, in one figure and two patterns, the 3-D RCS scattering monostatic patterns (σ_{3-D}/λ^2) (in dB), in the upper 180° region, due to first-order diffractions (#1 and #2) plus the ‘ring-radiator’ for $a = 3.516\lambda$. On the same figure, plot the 3-D RCS pattern based on Physical Optics (σ_{3-D}/λ^2) (in dB).
- Plot, in one figure and three patterns, the 3-D RCS scattering monostatic patterns (σ_{3-D}/λ^2) (in dB), in the upper 180° region, one due to first-order diffractions (#1 and #2), and one due to first- (#1 and #2) plus second-order (#3 and #4) diffractions plus the ring radiator for $a = 3.516\lambda$. On the same figure, plot the 3-D RCS pattern based on Physical Optics (σ_{3-D}/λ^2) (in dB).

13.53. According to PO, the maximum monostatic scattering RCS of flat plates, irrespective of polarization and plate configuration (rectangular, square, circular, elliptical, or any other shape), occurs at normal incidence and it is proportional to the square of the plate. Assuming a square ground plane and specifying the total length of one of its sides, w (in λ), you can determine the radius a (in λ) of

the circular ground plane so that both have the same area, as shown in Figure P13-53. For the circular ground plane the blending angle between the two-point diffraction and the ‘ring radiator’ is referred to as *thetao* (typically $10\text{--}30$ degrees). The MATLAB program **Rect_Circ_Scat**, for both TE^x and TM^x polarizations, performs this task. Using the MATLAB **Rect_Circ_Scat**.

- Compute and plot the monostatic 3-D RCS (σ_{3-D}/λ^2) (in dB), in the upper 180° region, for hard polarization and $w = 5\lambda$ that will include four curves on the same figure.
 - One curve of a square ground plane using diffraction modeling that includes both first- and second-order diffractions from points #1 and #2.
 - One curve of a square ground plane using PO.
 - One curve of a circular ground plane using diffraction modeling that includes both first- and second-order diffractions from points #1 and #2, second-order diffractions from points #3 and #4, and ring radiator.
 - One curve of a circular ground plane using PO.
- Compute and plot the monostatic 3-D RCS (σ_{3-D}/λ^2) (in dB), in the upper 180° region, for soft polarization and $w = 5\lambda$ that will include four curves on the same figure.
 - One curve of a square ground plane using diffraction modeling that includes first-order diffractions from points #1 and #2.

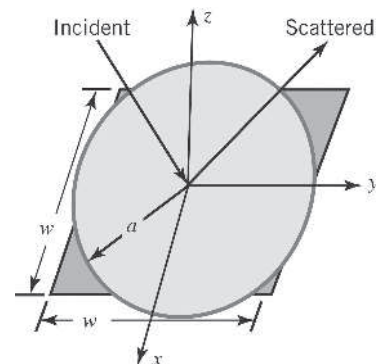


Figure P13-53

- One curve of a square ground plane using PO.
- One curve of a circular ground plane using diffraction modeling that includes first-order diffractions from points #1 and #2, second-order diffractions from points #3 and #4, and ring radiator.
- One curve of a circular ground plane using PO.

13.54. The normalized total geometrical optics field radiated in the principal xz plane (H plane; $\phi = 0^\circ, 180^\circ$) by the rectangular waveguide of Problem 13.41 (Figure P13-41) is given by

$$E_{\phi G} = E_0 \cos \theta \frac{\cos\left(\frac{\beta a}{2} \sin \theta\right)}{\left(\frac{\beta a}{2} \sin \theta\right)^2 - \left(\frac{\pi}{2}\right)^2} \frac{e^{-j\beta r}}{r}$$

$$0^\circ \leq \theta \leq 90^\circ$$

Use the slope diffraction concepts of Section 13.3.6.

- Formulate the field diffracted along the xz plane using two-point diffraction (points #3 and #4 of Figure P13-41).
- Plot (in decibels) the normalized amplitude pattern when $a = \lambda/2$, $w = 4\lambda$.

13.55. An infinite magnetic line source is placed on a two-dimensional square conducting cylinder at the center of its top side. Use successive single-order diffractions on each of the edges of the cylinder.

- Formulate expressions for the magnetic field that would be observed at the center of the bottom side of the cylinder.
- Compute the power loss (in decibels) at the observation point that is due to the presence of the cylinder when $w = 5\lambda$. Assume far-field observation approximations.

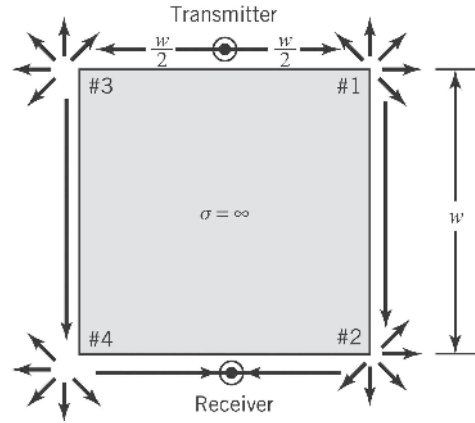


Figure P13-55

CHAPTER 14



Diffraction by a Wedge with Impedance Surfaces

14.1 INTRODUCTION

In Chapter 11 we introduced the exact solution of wave scattering by a perfectly conducting wedge while in Chapter 13 we examined its asymptotic high-frequency solution and reduced it to geometrical optics (GO) fields, incident and reflected, and diffracted fields (GTD and UTD) (incident and reflected diffracted fields). The solution of electromagnetic scattering from a penetrable or lossy wedge has not been solved using exact boundary conditions based on Maxwell's equations. It has always been necessary to use some approximate boundary conditions from which the exact solution can be obtained. Hence, it is important to recognize that when an exact solution is discussed, it is for a boundary condition that in reality can only be approximated by physical materials or structures.

In recent years great interest has been generated, because of the design of low-profile (stealth) radar targets and the modeling of wave propagation for wireless communication in the presence of penetrable structures (such as buildings, ridges, hills, and various other structures), to extend the theories of perfectly conducting materials to include materials with penetrable characteristics. Many aircraft are now built with composite materials which, with appropriate shaping and integration of materials, can reduce the radar echo, RCS, and thus minimize the probability of detection. See Figure 13-1 of the F-117 Nighthawk, whose surface is composed primarily of faceted flat plates and wedges judiciously oriented so that the maximum scattered field is toward specular direction and away from source of detection. In wireless communication, the wave propagation, direct and indirect (multipath), plays a pivotal role in the performance of the wireless mobile communication system using, for example, the Bit Error Ratio (BER) as a metric.

The most common of these boundary conditions is the *impedance boundary condition*, which relates the tangential components of the electric and magnetic fields by a surface impedance. The impedance surface boundary condition was first used in electromagnetic scattering by a number of Russian authors in the early 1940s [1], and it is generally attributed to Leontovich [2, 3], and commonly referred to as the *Leontovich boundary condition*. It was initially used to describe wave propagation over the earth's surface by specifying an impedance boundary for the ground surface. It was later used for other electromagnetics problems and was an effort to simplify very complex

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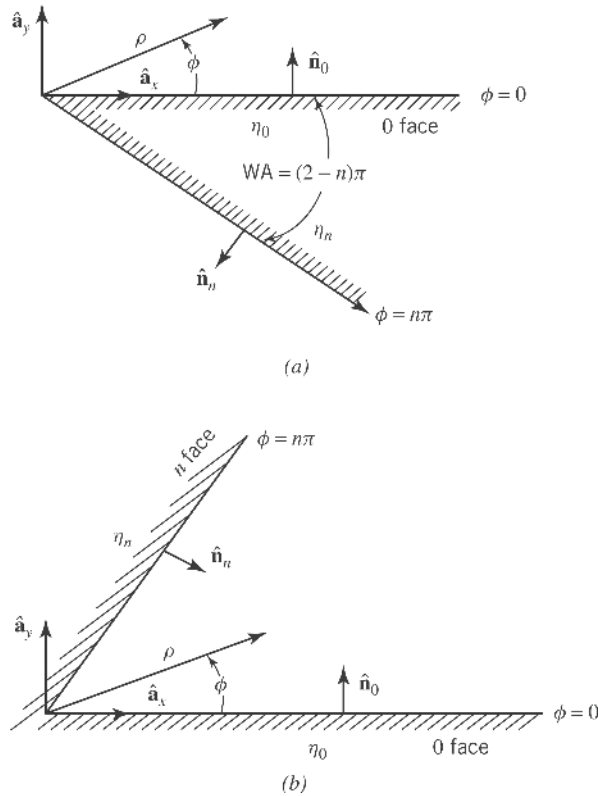


Figure 14-1 Exterior and interior wedges with impedance surfaces [28]. (a) Exterior wedge. (b) Interior wedge.

electromagnetics boundary-value problems, which were difficult to solve exactly. It is especially useful for surfaces that support surface waves.

The solution of a half-plane with impedance boundary conditions has been examined and reported in [4–8]. The more general problem of plane wave incidence on a wedge of arbitrary interior angle was solved by Maliuzhinets [9, 10]. The wedge was permitted to have different impedances on each face but only normal incidence was considered. No solution for oblique incidence on a general wedge has been obtained, except for a very few special cases. Maliuzhinets' exact solution of the wedge was accompanied by a steepest-descent analysis, very similar to Keller's theory for the PEC wedge, which decomposed the exact solution into *incident*, *reflected*, *diffracted*, and *surface wave* components. These Keller-type diffraction terms are not valid near the shadow boundaries and do not eliminate the discontinuities in the GO and surface wave terms; they had to be extended to include UTD type of formulations. There have been numerous publications [11–61] addressing various special cases of the diffraction by an impedance wedge, including that of the half plane. This chapter follows the presentation of [28, 30].

The impedance wedge, shown in Figure 14-1, is a canonical geometry whose faces can be represented by surface impedances. The exact solution for this geometry for the fields both interior and exterior to the wedge, similar to that of the PEC wedge of Chapter 11, does not exist because appropriate eigenfunctions and eigenvalues that satisfy Maxwell's equations and the boundary conditions cannot be found. However, an exact solution for the fields exterior to the wedge, for normal plane wave incidence and assuming uniform, but not identical surface impedances to the faces of the wedge, does exist and it was first presented by Maliuzhinets [9, 10]. The wedge, with wedge angle $WA = (2 - n)\pi$ and $0 < n \leq 2$, has two faces located at $\phi = 0$ and $\phi = n\pi$ with normalized

uniform surface impedances of η_0 and η_m , respectively. An exterior wedge (Figure 14-1a) has values of n in the range of $1 \leq n \leq 2$ while for an interior wedge (Figure 14-1b) $0 < n < 1$. Maliuzhinets' solution can be used to extract geometrical optics, diffracted fields, and surface waves, as was done for the PEC wedge in Chapter 13.

14.2 IMPEDANCE SURFACE BOUNDARY CONDITIONS

The impedance surface boundary condition is simply a statement that the tangential electric and magnetic field vectors are related by a constant impedance that is related only to the properties and configurations of the material and is independent of the source illumination. Mathematically, it is essentially a boundary condition of the third kind, as it relates a function and its derivative at the surface through some constant. As such, it is only an approximation to the actual boundary conditions, developed through Maxwell's equations, which exist on complex structures. The accuracy of the approximation depends on the composition, geometry, and illumination of the actual surface of interest. The impedance boundary condition has been applied in the past to homogeneous and inhomogeneous materials, layered structures, lossy materials, and randomly rough surfaces.

Physically, it is easy to understand where a concept, like the impedance boundary condition, originates. For plane, cylindrical, and spherical waves in inhomogeneous media and sufficiently far from the source, the electric and magnetic field vectors are mutually orthogonal and are related to each other by the intrinsic impedance of the homogeneous media through which they propagate. For free space, this impedance is $Z_0 = 377$ ohms. In vector form, this statement is expressed as

$$\mathbf{E} = -Z_0 \hat{\mathbf{a}}_r \times \mathbf{H} \quad (14-1)$$

where $\mathbf{E}$ and $\mathbf{H}$ are the electric and magnetic field vectors, Z_0 is the intrinsic impedance of free space (377 ohms), and $\hat{\mathbf{a}}_r$ is a unit vector in the direction of wave propagation.

The impedance boundary condition simply extends this wave property to surface interfaces with which the wave interacts. The boundary condition requires that the tangential electric and magnetic field vectors at the interface be mutually orthogonal and can be related by the impedance Z of the surface. The normalized impedance η , relative to the free-space value Z_0 , is commonly specified, and is given by $\eta = Z/Z_0$. This notation can cause confusion because many electromagnetics texts, as this one in previous chapters, use both η and Z as the total intrinsic impedance of the medium. Nonetheless, this notation has been utilized for surface impedances as early as 1959 [6] and has been used consistently since. Hence, in this chapter, η refers to a normalized impedance and Z refers to the total impedance, $Z = \eta Z_0$. For passive materials, the real part of η must be nonnegative.

In a vector equation form, the impedance surface boundary condition, referred to also as the *Leontovich Boundary Condition*, can be written as [1, 6]

$$\mathbf{E} - (\hat{\mathbf{n}} \cdot \mathbf{E}) \hat{\mathbf{n}} = Z \hat{\mathbf{n}} \times \mathbf{H} = \eta Z_0 (\hat{\mathbf{n}} \times \mathbf{H}) \quad (14-2)$$

where

- $\mathbf{E}$ = the electric field vector
- $\mathbf{H}$ = the magnetic field vector
- $\hat{\mathbf{n}}$ = the outward unit normal to the surface

It is noted that $(\hat{\mathbf{n}} \cdot \mathbf{E}) \hat{\mathbf{n}}$ is simply the normal component of $\mathbf{E}$; hence, the left side of (14-2) represents the tangential electric field. The term $\hat{\mathbf{n}} \times \mathbf{H}$ is a vector whose magnitude equals the tangential magnetic field; however, its direction is perpendicular to both the tangential magnetic field and the unit normal. Hence, $\hat{\mathbf{n}} \times \mathbf{H}$ is also tangential to the surface. The tangential electric and magnetic fields are then related by the constant of proportionality Z , the surface impedance $Z = \eta Z_0$. In words, the tangential electric and magnetic fields are related by the surface impedance of the material.

14.3 IMPEDANCE SURFACE REFLECTION COEFFICIENTS

To determine the reflection coefficients appropriate for an impedance surface, the reflection of a plane wave at a planar interface is examined. At high frequencies, these reflection coefficients are also appropriate for curved boundaries for which the radius of curvature is relatively large compared to the wavelength. In Figure 14-2, a plane wave in free space and incident on an impedance boundary is illustrated. For soft (perpendicular) polarization, Figure 14-2a, $\mathbf{E}$ is perpendicular to the plane of incidence, and for hard (parallel) polarization $\mathbf{E}$ is parallel to it.

For *soft polarization*, Figure 14-2a, the impedance boundary condition of (14-2) (using the incident electric and magnetic fields as reference and $\phi_r = \phi_i$), reduces to

$$E^i + E^r = \eta Z_0 (H^i - H^r) \sin \phi_i \quad (14-3a)$$

while for the *hard polarization*, Figure 14-2b, the impedance boundary condition of (14-2) (using $\phi_r = \phi_i$), simplifies to

$$(E^i + E^r) \sin \phi_i = \eta Z_0 (H^i - H^r) \quad (14-3b)$$

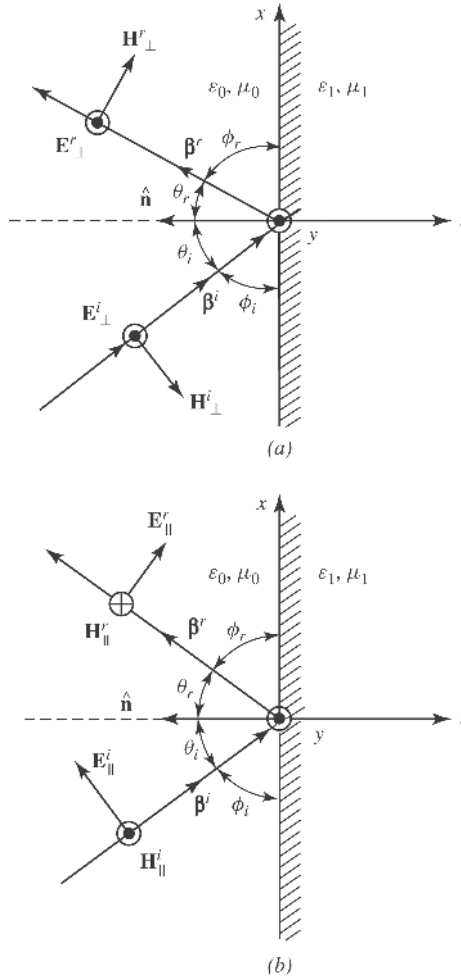


Figure 14-2 Oblique incident and reflected fields from a planar interface with impedance boundary for soft and hard polarizations and direction of their components. (a) Soft (perpendicular) polarization. (b) Hard (parallel) polarization.

where $\eta = Z_1/Z_0 = \eta_1/Z_0$ ($Z_1 = \eta_1$) is the normalized surface impedance of the medium. The reader should pay attention to the directions of the incident and reflected fields, as indicated in Figures 14-2a and 14-2b. Using $E^i = Z_0 H^i$ and $E^r = Z_0 H^r$ in (14-3a) and (14-3b), the reflection coefficients for soft and hard polarization become, respectively

$$\Gamma_s = \frac{E^r}{E^i} = \frac{\eta \sin \phi_i - 1}{\eta \sin \phi_i + 1} = \frac{\sin \phi_i - \frac{1}{\eta}}{\sin \phi_i + \frac{1}{\eta}} \quad (14-4a)$$

$$\Gamma_h = \frac{E^r}{E^i} = \frac{\eta - \sin \phi_i}{\eta + \sin \phi_i} = \frac{1 - \frac{\sin \phi_i}{\eta}}{1 + \frac{\sin \phi_i}{\eta}} \quad (14-4b)$$

For a perfect electric conductor (PEC), $\eta = 0$, the reflection coefficients of (14-4a) and (14-4b) reduce, respectively, to $\Gamma_s = -1$ and $\Gamma_h = -1$. This implies, as it should, that the tangential electric field components vanish on the PEC surface. Similarly, for a nonphysical perfect magnetic conductor (PMC), $\eta = \infty$, the reflection coefficients of (14-4a) and (14-4b) reduce, respectively, to $\Gamma_s = +1$ and $\Gamma_h = +1$. This implies, as it should, that the tangential magnetic field components vanish on the PMC surface. At grazing incidence, $\sin \phi_i = 0$, $\Gamma_s = -1$ and $\Gamma_h = +1$ for the lossy impedance surface. These are based on the directions of the incident and reflected fields of Figures 14-2a and 14-2b.

The reflection coefficients of (14-4a) and (14-4b) are, respectively, the same as those of (5-17a) and (5-24c), which are derived based on Snell's laws, provided that $\sin \phi_i = \sin(\pi/2 - \theta_i) = 1$. This occurs when the constitutive parameters of the two media forming the interface are such that $\sin \phi_i = \sqrt{1 - \frac{\mu_0 \varepsilon_0}{\mu_1 \varepsilon_1} \cos^2 \phi_i} \approx 1$, which is satisfied provided $\mu_1 \varepsilon_1 \gg \mu_0 \varepsilon_0$. Therefore, (14-4a) and (14-4b) are valid provided this relationship is satisfied.

For both polarizations, (14-4a) and (14-4b) show that there may be a particular Brewster angle ϕ_B [see definition in (5-33a), (5-33b)] for which the reflected field is zero. For the soft polarization, the Brewster angle $\phi_B = \sin^{-1}(1/\eta)$, while for the hard polarization $\phi_B = \sin^{-1}(\eta)$. If ϕ_B is complex, no physical Brewster angle exists. If η is complex, ϕ_B will be complex (and nonphysical) for both polarizations. If η is real and $0 \leq \eta \leq 1$, ϕ_B is real for the hard polarization and complex for the soft polarization. If η is real and $\eta \geq 1$, then ϕ_B is real for the soft polarization and complex for the hard polarization. Hence, a real Brewster angle cannot exist for both polarizations except when $\eta = 1$, and for this normalized impedance ($\eta = 1$), $\phi_B = \pi/2$ for both polarizations. Obviously, this is just the case of normal incidence on a matched surface for which there is no reflection for either polarization. Since the real part of η must be nonnegative, and since the inverse sine function maps the right half-plane into the strip $0 \leq \text{Re}[\sin^{-1} \eta] < \pi/2$, then it must always be true that $0 \leq \text{Re}[\phi_B] < \pi/2$.

In practice, these reflection coefficients are quite accurate for many scattering geometries. However, for one special case, some intrinsic difficulties may arise; this special case is the hard polarization near grazing incidence for an imperfect electric conductor. Similar problems arise for the imperfect magnetic conductor. It is noted that for perfect conductors, $\eta = 0$ and $\Gamma_h = -1$. However, at grazing incidence $\sin \phi_i = 0$ and $\Gamma_h = +1$. For imperfect conductors (η complex) near grazing incidence ($\sin \phi_i \simeq 0$), the hard reflection coefficient may change very rapidly from $+1$ to -1 for very small changes in the grazing angle or the conductivity. In general, it is difficult to develop methods for the imperfectly conducting case that revert uniformly to the perfectly conducting case as the surface impedance approaches zero for the hard polarization at grazing incidence.

In passing, it is worth mentioning that the grazing incidence case is especially important to the radar community and hence deserves attention. It is well known that the lossy earth or sea surface

near grazing incidence ($\sin \phi_i \simeq 0$) has reflection coefficients of $\Gamma_s = -1$ [i.e., the incident and reflected electric fields are in opposite directions using the direction designation of Figure 14-2a and definition of (14-4a)], and $\Gamma_h = +1$ [i.e., the incident and reflected electric fields are also in opposite directions using the direction designation of Figure 14-2b and definition of (14-4b)]. This implies that radar targets flying very near the ground or sea surface ($\sin \phi_i \simeq 0$) are especially difficult to detect because, at grazing incidence, the incident reflected electric fields are nearly equal in magnitude and opposite in phase. Hence, these two components effectively cancel each other, giving no radar echo/return, which makes low-flying radar targets difficult to track [62]. This physical phenomenon is exploited in many sea skimming missiles that fly within a few meters of the ocean surface.

14.4 THE MALIUZHINETS IMPEDANCE WEDGE SOLUTION

An exact solution for the fields exterior to the wedge, for normal plane wave incidence and assuming uniform, but not identical, surface impedances on the faces of the wedge, does exist and it was first presented by Maliuzhinets [9, 10]. The wedge has two faces located at $\phi = 0$ and $\phi = n\pi$, with normalized uniform surface impedances of η_0 and η_n , respectively. An interior wedge has values of n in the range of $0 < n < 1$ while for an exterior wedge $1 \leq n \leq 2$. Maliuzhinets' solution can be used to extract geometrical optics, diffracted fields, as was done for the PEC wedge, and surface waves.

Let us assume that a plane wave is incident from ϕ' and the observation point P is at a distance ρ from the edge of the wedge at an angle ϕ , where ϕ' and ϕ are both measured from the 0 face, as shown in Figure 14-3. Assuming a field of amplitude U_0 , the exact solution for the total field U_t , including the incident and reflected fields is [9]

$$U_t(\rho, \phi) = jU_0 \frac{1}{2n\pi} \int_{\gamma} \frac{\Psi\left(z + \frac{n\pi}{2} - \phi\right)}{\Psi\left(\frac{n\pi}{2} - \phi'\right)} \frac{\sin\left(\frac{\phi'}{n}\right)}{\cos\left(\frac{z - \phi}{n}\right) - \cos\left(\frac{\phi'}{n}\right)} e^{j\beta\rho \cos z} dz \quad (14-5)$$

where γ is the contour shown in Figure 14-4. *It should be remembered that this is an exact solution for an approximate boundary condition (i.e., the impedance boundary condition), and it is not an exact solution for the exact boundary conditions derived from Maxwell's equations.* Also in order to match the notation of Chapter 13 for the PEC wedge, the complex plane is represented by z , instead of α used in the corresponding published literature.

In (14-5), $\Psi(z)$ is the *auxiliary* Maliuzhinets function [9], and it depends explicitly on the integration variable z and implicitly on the parameters n , θ_0 and θ_n . Perhaps it would be more evident by writing $\Psi(z)$ as $\Psi(z; n, \theta_0, \theta_n)$; however, in the existing literature, it is exclusively represented by $\Psi(z)$. The function $\Psi(z)$ and its properties are discussed in [28]. Both the E_y (soft) and

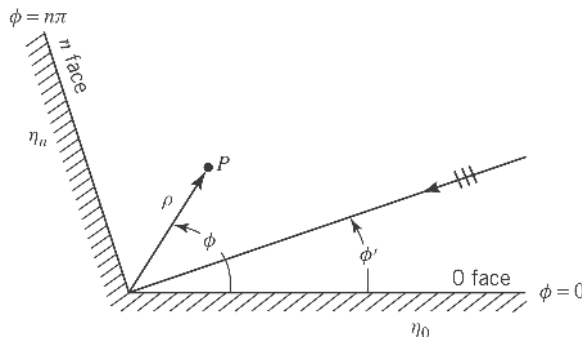


Figure 14-3 Interior wedge with impedance surfaces.

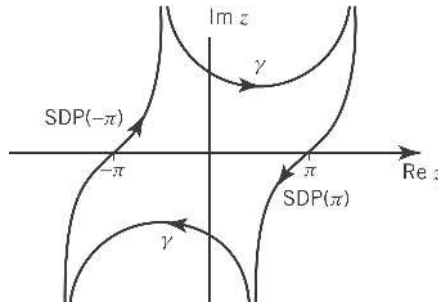


Figure 14-4 Integration contours for exact Maliuzhinets solution of wedge with impedance surfaces [28, 30]. (Source: T. Griesser and C. A. Balanis, “Reflections, diffractions, and surface waves for an interior impedance wedge of arbitrary angle,” *IEEE Trans. Antennas Propagat.*, © 1989, IEEE.)

H_y (hard) are included in this representation [58] by simply using the normalized surface impedance, or its inverse, in determining the Brewster angles θ_0 and θ_n . The Brewster angles for each polarization are determined as follows.

For the *soft* (TM^z , E_z) polarization, set (14-4a) equal to zero and solve for ϕ_i ; it leads to:

$$U_t(\rho, \phi) = E_z(\rho, \phi) \quad (14-6)$$

$$\theta_0 = \phi_{i0} = \sin^{-1} \left(\frac{1}{\eta_0} \right) \quad (14-6a)$$

$$\theta_n = \phi_{in} = \sin^{-1} \left(\frac{1}{\eta_n} \right) \quad (14-6b)$$

while for the *hard* (TE^z , H_z) polarization, set (14-4b) equal to zero and solve for ϕ_i ; it leads to:

$$U_t(\rho, \phi) = H_z(\rho, \phi) \quad (14-7)$$

$$\theta_0 = \phi_{i0} = \sin^{-1}(\eta_0) \quad (14-7a)$$

$$\theta_n = \phi_{in} = \sin^{-1}(\eta_n) \quad (14-7b)$$

Since η_0 and η_n must be nonnegative real for passive materials, θ_0 and θ_n will always lie in the strip ($0 < \text{Re}[\theta_0] < \pi/2$) and ($0 < \text{Re}[\theta_n] < \pi/2$) due to the principal branch cuts of the inverse sine. For the soft polarization, θ will be real if η is real and $\eta \geq 1$. Similarly, for hard polarization, θ will be real if η is real and $0 \leq \eta \leq 1$. Otherwise the inverse sine will be complex and the corresponding Brewster angle θ_B will also be complex.

To calculate the inverse sine of a complex number or a real number with magnitude greater than unity, the following identities can be used since many computer software programs do not provide a complex inverse sine function.

$$\sin^{-1}(z) = \frac{\pi}{2} - \cos^{-1}(z) \quad (14-8)$$

$$\cos^{-1}(z) = \text{Arg}[G(z)] - j \ln |G(z)| \quad (14-8a)$$

$$G(z) = \begin{cases} z - \sqrt{z^2 - 1} & (\text{Re}[z])(\text{Im}[z]) < 0 \\ z + \sqrt{z^2 - 1} & (\text{Re}[z])(\text{Im}[z]) \geq 0 \end{cases} \quad (14-8b)$$

$$G(z) = \begin{cases} z - \sqrt{z^2 - 1} & (\text{Re}[z])(\text{Im}[z]) < 0 \\ z + \sqrt{z^2 - 1} & (\text{Re}[z])(\text{Im}[z]) \geq 0 \end{cases} \quad (14-8c)$$

$$-\pi < \text{Arg } G(z) \leq \pi \quad (14-8d)$$

$$-\frac{\pi}{2} < \text{Arg}[\sqrt{z^2 - 1}] \leq \frac{\pi}{2} \quad (14-8e)$$

The exact wedge solution in its integral form of (14-5) cannot be easily evaluated. The integrand contains a ratio of *auxiliary Maliuzhinets functions*, each of which can be written as a product of four *Maliuzhinets functions*. Each Maliuzhinets function can only be written in a complex integral form or as an infinite product, and hence the Maliuzhinets function is not easily evaluated for arbitrary argument and wedge angle. Finally, the very complicated integrand must be integrated along the complex contour γ of Figure 14-4. Overall, the integral of (14-5) is difficult to evaluate efficiently.

To construct a more useful high-frequency asymptotic expansion of the integral, the *Method of Steepest Descent* (also known as the *Saddle Point Method*) of Chapter 13 will be used in this chapter. The steepest descent paths must first be located, and then the exact solution must be transformed to an integral along the steepest descent paths, as was done in Chapter 13 for the PEC wedge. Once the steepest descent paths are identified, the contour can then be closed to evaluate the exact solution. The sum of the integrals along the exact solution contour γ , plus along the steepest descent paths, must equal the sum of the residues enclosed. That is,

$$\int_{\gamma} (\text{integrand}) dz + \int_{SDPs} (\text{integrand}) dz = \int_{\text{closed path}} (\text{integrand}) dz = \sum_{p=1}^N \text{Residues} \quad (14-9a)$$

where $SDPs = SDP(+\pi) + SDP(-\pi)$. Alternatively, the integral along the exact solution contour γ of (14-9a) can be written as the sum of the residues *minus* the contributions along the steepest descent paths, or

$$\int_{\gamma} (\text{integrand}) dz = \sum_{p=1}^N \text{Residues} - \int_{SDPs} (\text{integrand}) dz \quad (14-9b)$$

Therefore, the exact solution of (14-5) can now be written as

$$\begin{aligned} U_t(\rho, \phi) &= jU_0 \frac{1}{2n\pi} \int_{\gamma} \frac{\Psi\left(z + \frac{n\pi}{2} - \phi\right)}{\Psi\left(\frac{n\pi}{2} - \phi'\right)} \frac{\sin\left(\frac{\phi'}{n}\right)}{\cos\left(\frac{z-\phi}{n}\right) - \cos\left(\frac{\phi'}{n}\right)} e^{j\beta\rho \cos z} dz \\ &= \frac{U_0}{n} \sum_{p=1}^N \text{Res} \left[\frac{\Psi\left(z + \frac{n\pi}{2} - \phi\right)}{\Psi\left(\frac{n\pi}{2} - \phi'\right)} \frac{\sin\left(\frac{\phi'}{n}\right)}{\cos\left(\frac{z-\phi}{n}\right) - \cos\left(\frac{\phi'}{n}\right)} e^{j\beta\rho \cos z}, z_p \right] \\ &\quad - jU_0 \frac{1}{2n\pi} \int_{SDPs} \frac{\Psi\left(z + \frac{n\pi}{2} - \phi\right)}{\Psi\left(\frac{n\pi}{2} - \phi'\right)} \frac{\sin\left(\frac{\phi'}{n}\right)}{\cos\left(\frac{z-\phi}{n}\right) - \cos\left(\frac{\phi'}{n}\right)} e^{j\beta\rho \cos z} dz \quad (14-10) \end{aligned}$$

where z_p are the poles of the integrand enclosed by the steepest descents paths while the notation $\text{Res}[f(z), z_p]$ represents the residue of $f(z)$ at the pole z_p .

14.5 GEOMETRICAL OPTICS

Using the canonical geometry of Figure 14-5, the solution of (14-10) can be decomposed into geometrical optics (incident and reflected), diffracted (incident and reflected), as was done in Chapter 13 for the PEC wedge, and surface wave fields. Because the wedge has impedance surfaces, surface waves must also be included. The region of interest is outside the wedge ($0 \leq \phi \leq n\pi$), which has been subdivided into three regions (I, II, III), as was done in Figure 13-13a of Chapter 13.

Maliuzhinets gives the geometrical optics terms as ratios of auxiliary Maliuzhinets functions. However, a more efficient and accurate method is to use a reflection coefficient at each reflection. In this section, based on the work of Griesser, et al. [28, 30], it is shown that the pole residue of the exact solution gives identically the same geometrical optics field as the simple ray tracing model for any number of interior reflections. Both methods give identical results in magnitude, phase, and also in angular range over which a particular reflection mechanism exists. Previously this was only performed for the two singly-reflected fields of the half plane [23, 24] and the general wedge [26].

For a geometrical analysis of the multiple reflected fields, the incident field is multiplied by the appropriate reflection coefficients. The surface impedance reflection coefficients for infinite planar boundaries and plane wave incidence are

$$\Gamma_{0,n}(\phi) = \frac{\sin \phi - \sin \theta_{0,n}}{\sin \phi + \sin \theta_{0,n}} \quad \text{soft polarization} \quad (14-11a)$$

$$\Gamma_{0,n}(\phi) = \frac{\sin \theta_{0,n} - \sin \phi}{\sin \theta_{0,n} + \sin \phi} \quad \text{hard polarization} \quad (14-11b)$$

where ϕ is the angle measured from the planar surface to the incident ray, as indicated in Figure 14-3.

Let us now consider a typical reflection mechanism. In particular, consider the third-order reflection, which is initially incident on face 0, as shown in Figure 14-6 for an interior wedge. The plane wave is incident and reflects at an angle ϕ' from the 0 face. Next it reflects at an angle $\pi - n\pi - \phi'$ from the n face, and lastly at an angle $\pi - 2n\pi - \phi'$ from the 0 face. Hence, the reflected field includes the product Γ_{Π} of these three reflection coefficients

$$\Gamma_{\Pi} = \Gamma_0(\phi')\Gamma_n[\pi - (n\pi + \phi')]\Gamma_0[\pi - (2n\pi + \phi')] \quad (14-12a)$$

or equivalently

$$\Gamma_{\Pi} = \Gamma_0(\phi')\Gamma_n(n\pi + \phi')\Gamma_0(2n\pi + \phi') \quad (14-12b)$$

The distance the third-order reflected ray has traveled can be determined by tracing the image ray through the 0 face. Taking the phase reference at the vertex of the wedge, the extra distance

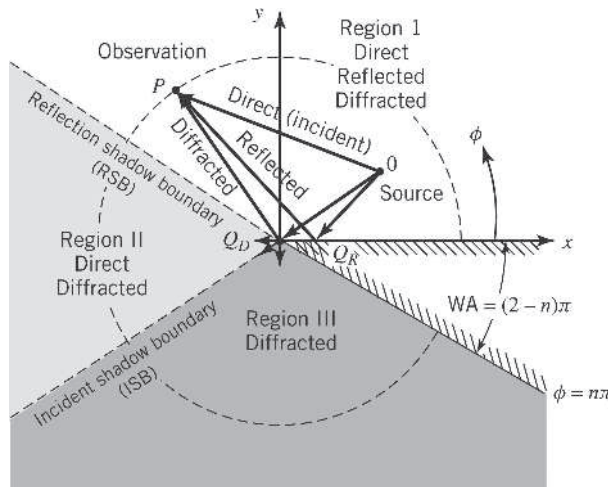


Figure 14-5 Wedge geometry and region separation for geometrical optics and diffracted fields.

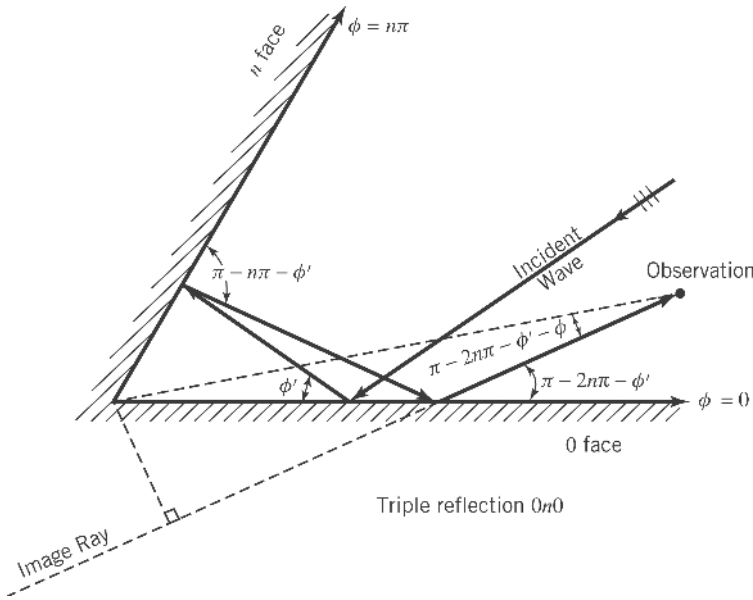


Figure 14-6 Typical third-order reflection from an interior wedge with impedance surfaces [28].

traveled is found by generating a perpendicular line to the image ray from the vertex, as shown in Figure 14-6, and it is equal to $\rho \cos(\pi - 2n\pi - \phi' - \phi)$.

Therefore, an additional phase factor of

$$e^{-j\beta\rho \cos(\pi - 2n\pi - \phi' - \phi)} = e^{-j\beta\rho \cos[\pi - (2n\pi + \phi' + \phi)]} = e^{j\beta\rho \cos(\phi + \phi' + 2n\pi)} \quad (14-13)$$

must be included for the GO field. Finally, the total field for this third-order reflected component can be written as

$$U_{GO} = U_0 \Gamma_0(\phi') \Gamma_n(n\pi + \phi') \Gamma_0(2n\pi + \phi') e^{j\beta\rho \cos(\phi + \phi' + 2n\pi)} \quad (14-14)$$

This third-order component has reflection shadow boundaries and does not exist for all ϕ . In fact, from the geometry of Figure 14-6, it exists only if $\phi < \pi - 2n\pi - \phi'$, or equivalently, if $\phi + \phi' < \pi - 2n\pi$.

In general, all the reflected components can be analyzed using the same procedure: ray tracing. The geometrical-optics reflected field for any multiple-reflected component C can be written as

$$U_{GO}^C = U_0 \Gamma_{\Pi}(\phi', \theta_0, \theta_n) e^{+j\beta\rho \cos(z_p)} \quad (14-15)$$

where $\Gamma_{\Pi}(\phi', \theta_0, \theta_n)$ is a product of reflection coefficients and $\rho \cos(z_p)$ is a distance factor that yields the appropriate phase delay. Later, the component C will be shown to correspond to an angle $\xi^{\pm}$ and an integer m , and will be written as $U_{GO}^{m,\xi}$. The multiple-reflected field is identified by a sequence of 0's and n 's indicating the order of reflection. As an example, component $0n0n$ is a quadruple-reflected field incident on face 0 which, in sequence, reflects from face 0 to n to 0 to n .

In Table 14-1, the term $\Gamma_{\Pi}(\phi', \theta_0, \theta_n)$ and z_p are listed for all reflection mechanisms up to fourth-order. In addition, the range over which these terms exist is listed, with the implied conditions that $0 \leq \phi \leq n\pi$ and $0 \leq \phi' \leq n\pi$. The number of terms presented is sufficient to identify the pattern by which the table can be expanded. The ordering of this table has been selected to correspond to the positions of the poles of the exact solution.

TABLE 14-1 Geometrical optics reflection coefficients and associated pole residues [28, 30]

Comp	Pole	$\Gamma_\pi(\phi', \theta_0, \theta_n)$	z_p	Existence
0n0n	$\xi^-, m = -2$	$\Gamma_0(\phi')\Gamma_n(n\pi + \phi')\Gamma_0(2n\pi + \phi')\Gamma_n(3n\pi + \phi')$	$\phi - \phi' - 4n\pi$	$\phi - \phi' > 4n\pi - \pi$
n0n	$\xi^+, m = -2$	$\Gamma_n(n\pi - \phi')\Gamma_0(2n\pi - \phi')\Gamma_n(3n\pi - \phi')$	$\phi + \phi' - 4n\pi$	$\phi + \phi' > 4n\pi - \pi$
0n	$\xi^-, m = -1$	$\Gamma_0(\phi')\Gamma_n(n\pi + \phi')$	$\phi - \phi' - 2n\pi$	$\phi - \phi' > 2n\pi - \pi$
n	$\xi^+, m = -1$	$\Gamma_n(n\pi - \phi')$	$\phi + \phi' - 2n\pi$	$\phi + \phi' > 2n\pi - \pi$
Inc	$\xi^-, m = 0$	1	$\phi - \phi'$	$-\pi < \phi - \phi' < \pi$
0	$\xi^+, m = 0$	$\Gamma_0(\phi')$	$\phi + \phi'$	$\phi + \phi' < \pi$
n0	$\xi^-, m = 1$	$\Gamma_n(n\pi - \phi')\Gamma_0(2n\pi - \phi')$	$\phi - \phi' + 2n\pi$	$\phi - \phi' < \pi - 2n\pi$
0n0	$\xi^+, m = 1$	$\Gamma_0(\phi')\Gamma_n(n\pi + \phi')\Gamma_0(2n\pi + \phi')$	$\phi + \phi' + 2n\pi$	$\phi + \phi' < \pi - 2n\pi$
n0n0	$\xi^-, m = 2$	$\Gamma_n(n\pi - \phi')\Gamma_0(2n\pi - \phi')\Gamma_n(3n\pi - \phi')\Gamma_0(4n\pi - \phi')$	$\phi - \phi' + 4n\pi$	$\phi - \phi' < \pi - 4n\pi$

(Source: T. Griesser and C. A. Balanis, "Reflections, diffractions, and surface waves for an interior impedance wedge of arbitrary angle," *IEEE Trans. Antennas Propagat.*, © 1989, IEEE.)

When considering the exact solution of (14-5) for the impedance wedge, the geometrical optics poles must be identified. The GO poles of (14-5) are those for which

$$\cos\left(\frac{z_p - \phi}{n}\right) = \cos\left(\frac{\phi'}{n}\right) \quad (14-16)$$

This equation can be inverted to solve for z_p by considering every value of the multi-valued inverse cosine. The GO poles are represented by

$$z_p = \phi \pm \phi' + 2mn\pi = \xi^\pm + 2mn\pi, \quad \xi^\pm = \phi \pm \phi' \quad (14-17)$$

where m is an integer, and n is real and depends on the wedge angle. The notation $\xi^\pm = \phi \pm \phi'$ is also conveniently used in the PEC wedge of Chapter 13. The choice of the notation z_p in this equation and in the GO phase factor will become evident when the pole residues are evaluated.

The geometrical optics poles of (14-17) appear in two sets of equally spaced poles corresponding to the upper and lower sign. For each set, the spacing between poles is $2n\pi$, or twice the exterior wedge angle $n\pi$. If ϕ' is considered to be fixed and ϕ varies from 0 to $n\pi$, then the GO poles move from $\pm\phi' + 2mn\pi$ to $\pm\phi' + 2mn\pi + n\pi$. These pole loci are located along the real axis of the complex z plane and are plotted in Figure 14-7 for $\phi' = 30^\circ$ and an 85° interior wedge. The movement of the poles with increasing ϕ is indicated by the arrows, and it is noted that a given pole can only move half the distance to the next pole of the same set. In addition, the surface wave poles for $\eta_0 = \eta_n = 0.2 + j0.8$, although they have not yet been discussed, are indicated in Figure 14-7.

A particular pole contributes to the exact integral solution of (14-10) only if it lies between the steepest descent paths. Hence, it is possible to identify each pole with a specific reflection mechanism by the angular ranges of existence listed in Table 14-1. The example of the 0n0 term considered previously will be examined here again. This third-order term exists for $\phi + \phi' < \pi - 2n\pi$. Consider the pole for which $\xi = \xi^+ = \phi + \phi'$ and $m = 1$. This pole contributes to the exact solution if it lies between the steepest descent paths that cross the real axis at $z = \pm\pi$. Since the steepest descent paths cross the real axis at $z_s = \pm\pi$, the pole contributes if

$$-\pi < z = \phi + \phi' + 2n\pi < \pi \quad (14-18)$$

or equivalently if

$$-\pi - 2n\pi < \phi + \phi' < \pi - 2n\pi \quad (14-19)$$

Since $0 \leq \phi \leq n\pi$ and $0 \leq \phi' \leq n\pi$, then it is always true that $0 \leq \phi + \phi' \leq 2n\pi$. Therefore, the lower limit of (14-19) adds no new information. Hence, the pole contributes for $\phi + \phi' < \pi - 2n\pi$, which

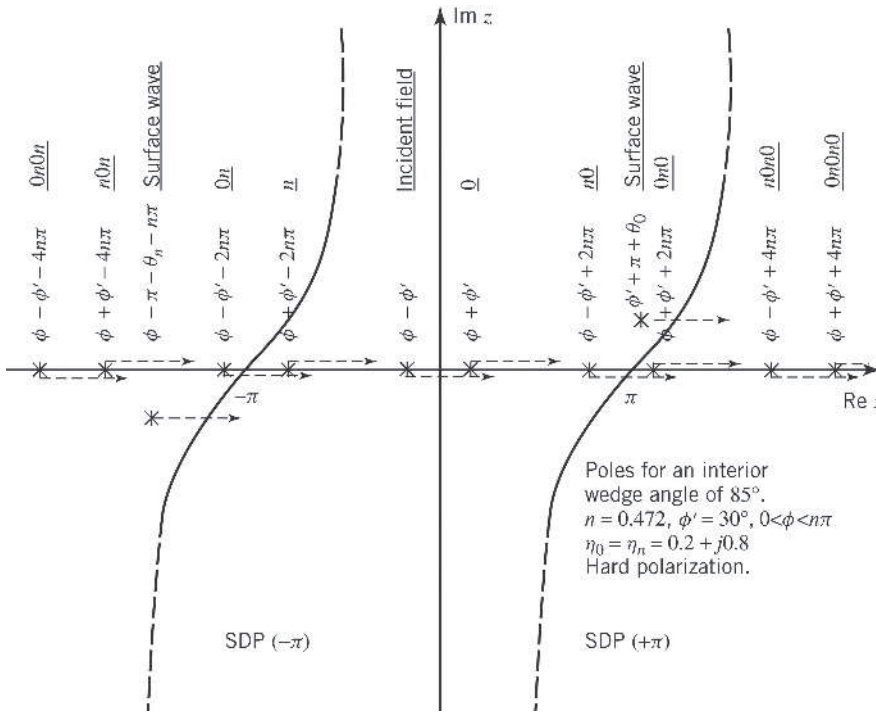


Figure 14-7 Geometrical optics and surface wave poles in complex plane [28, 30]. (Source: T. Griesser and C. A. Balanis, “Reflections, diffractions, and surface waves for an interior impedance wedge of arbitrary angle,” *IEEE Trans. Antennas Propagat.*, © 1989, IEEE.)

is exactly the GO result determined after (14-14). The reflection component corresponding to each pole is labeled in Figure 14-7. For each GO reflection, the appropriate values of ξ and m are listed in Table 14-1.

From the exact solution, the contribution of the GO pole is $-j2\pi$ times the residue of the pole, where the minus (–) sign is attributed to the clockwise contour encircling the pole in Figure 14-4. For the pole $z_p = \phi \pm \phi' + 2mn\pi$, the residue is

$$U_{GO}^C = \frac{U_0}{n} \text{Res} \left[\frac{\Psi \left(z + \frac{n\pi}{2} - \phi \right)}{\Psi \left(\frac{n\pi}{2} - \phi' \right)} \frac{\sin \left(\frac{\phi'}{n} \right)}{\cos \left(\frac{z - \phi}{n} \right) - \cos \left(\frac{\phi'}{n} \right)} e^{j\beta\rho \cos z}, z_p \right] \quad (14-20)$$

Assuming that no poles of the Maliuzhinets function coincide with the GO poles, and since the exponential has no finite poles, the residue can be determined by

$$\begin{aligned} U_{GO}^C &= \frac{U_0}{n} \lim_{z \rightarrow z_p} \left\{ (z - z_p) \left[\frac{\Psi \left(z + \frac{n\pi}{2} - \phi \right)}{\Psi \left(\frac{n\pi}{2} - \phi' \right)} \frac{\sin \left(\frac{\phi'}{n} \right)}{\cos \left(\frac{z - \phi}{n} \right) - \cos \left(\frac{\phi'}{n} \right)} e^{j\beta\rho \cos z} \right] \right\} \\ &= \frac{U_0}{n} \frac{\Psi \left(z + \frac{n\pi}{2} - \phi \right)}{\Psi \left(\frac{n\pi}{2} - \phi' \right)} \sin \left(\frac{\phi'}{n} \right) e^{j\beta\rho \cos z} \lim_{z \rightarrow z_p} \text{Res} \left[\frac{(z - z_p)}{\cos \left(\frac{z - \phi}{n} \right) - \cos \left(\frac{\phi'}{n} \right)} \right] \end{aligned} \quad (14-21)$$

From (14-16) and (14-17)

$$\lim_{z \rightarrow z_p} \operatorname{Res} \left[\frac{(z - z_p)}{\cos\left(\frac{z - \phi}{n}\right) - \cos\left(\frac{\phi'}{n}\right)} \right] = \lim_{z \rightarrow \xi^{\pm} + 2mn\pi} \operatorname{Res} \left[\frac{[z - (\xi^{\pm} + 2mn\pi)]}{\cos\left(\frac{z - \phi}{n}\right) - \cos\left(\frac{\phi'}{n}\right)} \right] \quad (14-22)$$

This is an indeterminate form that can be evaluated by L' Hopital's Rule. Then

$$\begin{aligned} \lim_{z \rightarrow z_p} \left[\frac{(z - z_p)}{\cos\left(\frac{z - \phi}{n}\right) - \cos\left(\frac{\phi'}{n}\right)} \right] &= \lim_{z \rightarrow \xi^{\pm} + 2mn\pi} \left[\frac{1}{-\frac{1}{n} \sin\left(\frac{z - \phi}{n}\right)} \right] \\ &= \frac{-n}{\sin\left(\frac{\pm\phi'}{n}\right)} = \frac{\mp n}{\sin\left(\frac{\phi'}{n}\right)} \end{aligned} \quad (14-23)$$

where the upper sign represents ξ^+ poles and the lower sign represents ξ^- poles.

Finally, (14-21) becomes

$$\begin{aligned} U_{\text{GO}}^C &= \mp U_0 \frac{\Psi\left(z_p + \frac{n\pi}{2} - \phi\right)}{\Psi\left(\frac{n\pi}{2} - \phi'\right)} e^{j\beta\rho \cos z} \\ &= \mp U_0 \frac{\Psi\left(\pm\phi' + 2mn\pi + \frac{n\pi}{2}\right)}{\Psi\left(\frac{n\pi}{2} - \phi'\right)} e^{j\beta\rho \cos(\phi \pm \phi' + 2mn\pi)} \end{aligned} \quad (14-24)$$

corresponding to the poles $z_p = (\phi \pm \phi') + 2mn\pi$. By comparison with Table 14-1, it is clear that the phase factor matches the geometrical-optics phase term. However, it remains to be shown that the ratio of auxiliary Maliuzhinets functions in (14-24) is a product of reflection coefficients, as in the geometrical analysis of (14-15). For the incident GO field, which corresponds to $\xi = \xi^- = \phi - \phi'$ and $m = 0$, the ratio is

$$\frac{\Psi\left(\pm\phi' + 2mn\pi + \frac{n\pi}{2}\right)}{\Psi\left(\frac{n\pi}{2} - \phi'\right)} = \frac{\Psi\left(-\phi' + \frac{n\pi}{2}\right)}{\Psi\left(\frac{n\pi}{2} - \phi'\right)} = 1 \quad (14-25)$$

So the incident field from the exact solution matches the GO incident field as expected, and no further consideration is necessary for $m = 0$.

To consider the many possible multiple reflections, the ratio of auxiliary Maliuzhinets functions (and leading sign) of (14-24) is denoted by $\Gamma_{\Pi}(\phi', \theta_0, \theta_n)$, since it is shown next that the ratio indeed reduces to the products of reflection coefficients of Table 14-1:

$$\Gamma_{\Pi}(\phi', \theta_0, \theta_n) = \mp \frac{\Psi\left(z_p + \frac{n\pi}{2} - \phi\right)}{\Psi\left(\frac{n\pi}{2} - \phi'\right)} \quad (14-26)$$

where $z_p = (\phi \pm \phi') + 2mn\pi$. The numerator and denominator are expanded using [9, 26]

$$\begin{aligned} \Psi(z) = & \Psi_n \left(z + \frac{n\pi}{2} + \frac{\pi}{2} - \theta_0 \right) \Psi_n \left(z + \frac{n\pi}{2} - \frac{\pi}{2} + \theta_0 \right) \\ & \cdot \Psi_n \left(z - \frac{n\pi}{2} + \frac{\pi}{2} - \theta_n \right) \Psi_n \left(z - \frac{n\pi}{2} - \frac{\pi}{2} + \theta_n \right) \end{aligned} \quad (14-27)$$

and (14-26) takes two forms depending on whether $\phi + \phi'$ or $\phi - \phi'$ is selected.

For $\xi = \xi^- = \phi - \phi'$:

$$\begin{aligned} \Gamma_{\Pi}(\phi', \theta_0, \theta_n) = & \frac{\Psi_n \left(-\phi' + n\pi + \frac{\pi}{2} - \theta_0 + 2mn\pi \right) \Psi_n \left(-\phi' + n\pi - \frac{\pi}{2} + \theta_0 + 2mn\pi \right)}{\Psi_n \left(-\phi' + n\pi + \frac{\pi}{2} - \theta_0 \right) \Psi_n \left(-\phi' + n\pi - \frac{\pi}{2} + \theta_0 \right)} \\ & \cdot \frac{\Psi_n \left(-\phi' + \frac{\pi}{2} - \theta_n + 2mn\pi \right) \Psi_n \left(-\phi' - \frac{\pi}{2} + \theta_n + 2mn\pi \right)}{\Psi_n \left(-\phi' + \frac{\pi}{2} - \theta_n \right) \Psi_n \left(-\phi' - \frac{\pi}{2} + \theta_n \right)} \end{aligned} \quad (14-28a)$$

For $\xi = \xi^+ = \phi + \phi'$:

$$\begin{aligned} \Gamma_{\Pi}(\phi', \theta_0, \theta_n) = & - \frac{\Psi_n \left(\phi' + n\pi + \frac{\pi}{2} - \theta_0 + 2mn\pi \right) \Psi_n \left(\phi' + n\pi - \frac{\pi}{2} + \theta_0 + 2mn\pi \right)}{\Psi_n \left(\phi' + n\pi + \frac{\pi}{2} - \theta_0 \right) \Psi_n \left(\phi' + n\pi - \frac{\pi}{2} + \theta_0 \right)} \\ & \cdot \frac{\Psi_n \left(\phi' + \frac{\pi}{2} - \theta_n + 2mn\pi \right) \Psi_n \left(\phi' - \frac{\pi}{2} + \theta_n + 2mn\pi \right)}{\Psi_n \left(\phi' + \frac{\pi}{2} - \theta_n \right) \Psi_n \left(\phi' - \frac{\pi}{2} + \theta_n \right)} \end{aligned} \quad (14-28b)$$

The fact that $\Psi_n(z)$ is an even function of z has also been utilized here. Next, (14-28a) and (14-28b) are shown to reduce to products of reflection coefficients for multiple reflected rays.

The two single reflections are considered first, since both are exceptional cases. Each can be reduced to a single reflection coefficient using the trigonometric identity

$$\tan \left[\frac{1}{2}(x - y) \right] \cot \left[\frac{1}{2}(x + y) \right] = \frac{\sin x - \sin y}{\sin x + \sin y} \quad (14-29)$$

and the Maliuzhinets identity [15]

$$\frac{\Psi_n(z + n\pi)}{\Psi_n(z - n\pi)} = \cot \left[\frac{1}{2} \left(z + \frac{\pi}{2} \right) \right] \quad (14-30)$$

For the reflection from face 0 ($\xi = \xi^+$, $m = 0$), the last two terms in (14-28b) reduce to unity since $m = 0$. Then, using (14-30) in (14-28b), the ratio of Maliuzhinets functions can be reduced to the single reflection coefficient $\Gamma_{\Pi}(\phi', \theta_0, \theta_n) = \Gamma_0(\phi')$ from the 0 face by applying (14-29) and (14-11a), in that order. Similarly, for the single-reflected field from face n ($\xi = \xi^+$, $m = -1$), the first two terms in (14-28b) reduce to unity since $m = -1$. Again, using (14-30) in (14-28b), the ratio of Maliuzhinets

functions can be reduced to the single reflection coefficient $\Gamma_{\Pi}(\phi', \theta_0, \theta_n) = \Gamma_n(n\pi - \phi')$ from the n face by applying (14-29) and (14-11b), in that order.

By comparing with Table 14-1, it is evident that the ratio of auxiliary Maliuzhinets functions reduces to the two single reflection coefficients for the $(\xi = \xi^+, m = 0)$ and $(\xi = \xi^+, m = -1)$ poles. This equality of the ratios of Maliuzhinets functions with the single reflection coefficients has been demonstrated in [23, 24, 26], but it is more difficult and less accurate for calculation purposes. The higher-order reflection coefficients were not considered elsewhere, and it was demonstrated in [28, 30] that they can be reduced to products of reflection coefficients for any order of reflections. The products appear in four forms, depending on whether the number of reflections is even or odd and whether the 0 and n face is initially illuminated for the multiple-reflected fields.

To obtain the appropriate products of reflection coefficients in (14-28a) and (14-28b) for the higher-order reflections, the following identities are utilized:

$$\frac{\Psi_n(x + 2mn\pi)}{\Psi_n(x)} = \prod_{p=1}^m \frac{\Psi_n[x + (2p-1)n\pi + n\pi]}{\Psi_n[x + (2p-1)n\pi - n\pi]}, \quad m > 0 \quad (14-31a)$$

$$\frac{\Psi_n(x + 2mn\pi)}{\Psi_n(x)} = \prod_{p=1}^{|m|} \frac{\Psi_n[x + (1-2p)n\pi - n\pi]}{\Psi_n[x + (1-2p)n\pi + n\pi]}, \quad m < 0 \quad (14-31b)$$

These are algebraic identities, valid for any function, which can be verified by expanding the product. Their value lies in the fact that they correctly separate the Maliuzhinets function into appropriate forms so that the recursion relations can be applied to reduce the Maliuzhinets ratios to products of reflection coefficients.

Using (14-29) through (14-31b) in (14-28a) and (14-28b), the GO residues are reduced to products of reflection coefficients. Four different forms are possible, depending on the choices of ξ and m . Consider first $\xi = \xi^- = \phi - \phi'$ and $m > 0$. Then (14-28a) can be written, using (14-31a), as

$$\begin{aligned} \Gamma_{\Pi}(\phi', \theta_0, \theta_n) = & \prod_{p=1}^m \frac{\Psi_n \left[-\phi' + n\pi + \frac{\pi}{2} - \theta_0 + (2p-1)n\pi + n\pi \right]}{\Psi_n \left[-\phi' + n\pi + \frac{\pi}{2} - \theta_0 + (2p-1)n\pi - n\pi \right]} \\ & \cdot \prod_{p=1}^m \frac{\Psi_n \left[-\phi' + n\pi - \frac{\pi}{2} + \theta_0 + (2p-1)n\pi + n\pi \right]}{\Psi_n \left[-\phi' + n\pi - \frac{\pi}{2} + \theta_0 + (2p-1)n\pi - n\pi \right]} \\ & \cdot \prod_{p=1}^m \frac{\Psi_n \left[-\phi' + \frac{\pi}{2} - \theta_n + (2p-1)n\pi + n\pi \right]}{\Psi_n \left[-\phi' + \frac{\pi}{2} - \theta_n + (2p-1)n\pi - n\pi \right]} \\ & \cdot \prod_{p=1}^m \frac{\Psi_n \left[-\phi' - \frac{\pi}{2} + \theta_n + (2p-1)n\pi + n\pi \right]}{\Psi_n \left[-\phi' - \frac{\pi}{2} + \theta_n + (2p-1)n\pi - n\pi \right]} \end{aligned} \quad (14-32)$$

Now using (14-30), (14-32) can be expressed as

$$\begin{aligned}\Gamma_{\Pi}(\phi', \theta_0, \theta_n) = & \prod_{p=1}^m \cot \left\{ \frac{1}{2} [-\phi' + n\pi + \pi - \theta_0 + (2p-1)n\pi] \right\} \\ & \cdot \prod_{p=1}^m \cot \left\{ \frac{1}{2} [-\phi' + n\pi + \theta_0 + (2p-1)n\pi] \right\} \\ & \cdot \prod_{p=1}^m \cot \left\{ \frac{1}{2} [-\phi' + \pi - \theta_n + (2p-1)n\pi] \right\} \\ & \cdot \prod_{p=1}^m \cot \left\{ \frac{1}{2} [-\phi' + \theta_n + (2p-1)n\pi] \right\}\end{aligned}\quad (14-33)$$

Next, some of the cotangents are changed to tangents by trigonometric identities, and (14-33) can be written as

$$\begin{aligned}\Gamma_{\Pi}(\phi', \theta_0, \theta_n) = & \prod_{p=1}^m -\tan \left\{ \frac{1}{2} [-\phi' + n\pi - \theta_0 + (2p-1)n\pi] \right\} \\ & \cdot \prod_{p=1}^m \cot \left\{ \frac{1}{2} [-\phi' + n\pi + \theta_0 + (2p-1)n\pi] \right\} \\ & \cdot \prod_{p=1}^m -\tan \left\{ \frac{1}{2} [-\phi' - \theta_n + (2p-1)n\pi] \right\} \\ & \cdot \prod_{p=1}^m \cot \left\{ \frac{1}{2} [-\phi' + \theta_n + (2p-1)n\pi] \right\}\end{aligned}\quad (14-34)$$

Using (14-29), all the terms in (14-34) can be combined, and (14-34) can then be expressed as

$$\Gamma_{\Pi}(\phi', \theta_0, \theta_n) = (-1)^{2m} \prod_{p=1}^m \frac{\sin(2pn\pi - \phi') - \sin \theta_0}{\sin(2pn\pi - \phi') + \sin \theta_0} \prod_{p=1}^m \frac{\sin[(2p-1)n\pi - \phi'] - \sin \theta_n}{\sin[(2p-1)n\pi - \phi'] + \sin \theta_n} \quad (14-35)$$

Finally, using (14-11a) and (14-11b), the two ratios of (14-35) reduce to products of reflection coefficients for the 0 and n faces, or

$$\Gamma_{\Pi}(\phi', \theta_0, \theta_n) = \prod_{p=1}^m \Gamma_0(2pn\pi - \phi') \prod_{p=1}^m \Gamma_n[(2p-1)n\pi - \phi'] \quad (14-36)$$

Again, (14-36) is only valid for $\xi = \xi^- = \phi - \phi'$ and $m > 0$. There are still three more cases to consider. However, all three cases reduce to similar products of reflection coefficients, and the final forms obtained from the GO pole residues can be written as:

For $\xi = \xi^- = \phi - \phi'$ and $m > 0$:

$$\Gamma_{\Pi}(\phi', \theta_0, \theta_n) = \prod_{p=1}^m \Gamma_0(2pn\pi - \phi') \prod_{p=1}^m \Gamma_n[(2p-1)n\pi - \phi'] \quad (14-37)$$

For $\xi = \xi^- = \phi - \phi'$ and $m < 0$:

$$\Gamma_{\Pi}(\phi', \theta_0, \theta_n) = \prod_{p=1}^{|m|} \Gamma_0[\phi' + 2(p-1)n\pi] \prod_{p=1}^{|m|} \Gamma_n[\phi' + (2p-1)n\pi] \quad (14-38)$$

For $\xi = \xi^+ = \phi + \phi'$ and $m > 0$:

$$\Gamma_{\Pi}(\phi', \theta_0, \theta_n) = \prod_{p=1}^{m+1} \Gamma_0[\phi' + 2(p-1)n\pi] \prod_{p=1}^m \Gamma_n[\phi' + (2p-1)n\pi] \quad (14-39)$$

For $\xi = \xi^+ = \phi + \phi'$ and $m < -1$:

$$\Gamma_{\Pi}(\phi', \theta_0, \theta_n) = \prod_{p=1}^{|m+1|} \Gamma_0(2pn\pi - \phi') \prod_{p=1}^{|m|} \Gamma_n[2(p-1)n\pi - \phi'] \quad (14-40)$$

The cases $(\xi^-, m = 0)$, $(\xi^+, m = 0)$, and $(\xi^+, m = -1)$ are the incident and single reflected terms, which have already been considered. By comparison with Table 14-1, it is evident that (14-37) through (14-40) are identical to the geometrical ray-tracing analysis.

14.6 SURFACE WAVE TERMS

The surface wave is a wave that propagates along one face of the wedge and typically decays exponentially in a vertical direction away from the face. It is confined to a particular angular range from the wedge face, whenever it exists. Since the wave may decay slowly along the face, its contribution may be more dominant than other scattering mechanisms near the wedge surface. Hence, it is often important to include this contribution for reactive surfaces.

The surface wave is determined by considering the contributions of the residues of enclosed complex poles of the Maliuzhinets function between the steepest descent paths. The surface wave poles were identified by Maliuzhinets, and they are located at [9, 27]

$$z_0 = \phi + \pi + \theta_0 \quad (14-41a)$$

$$z_n = \phi - n\pi - \pi - \theta_n \quad (14-41b)$$

for the 0 and n faces, respectively. If ϕ is allowed to vary from 0 to $n\pi$, the z_0 pole moves from $\pi + \theta_0$ to $n\pi + \pi + \theta_0$ and the z_n pole moves from $-n\pi - \pi - \theta_n$ to $-\pi - \theta_n$. The loci are shown in Figure 14-7 for $\phi' = 30^\circ$ for a wedge with an 85° interior angle and normalized surface impedances of $\eta_0 = \eta_n = 0.2 + j0.8$. The z_0 pole can only lie within the steepest descent paths for ϕ less than some maximum value. Similarly, the z_n pole can only lie within the steepest descent paths for ϕ greater than some minimum value. Hence, the surface wave term is bounded to a finite angular range near the corresponding face.

In general, the surface wave component corresponding to the pole z exists if it lies within the steepest descent paths of (14-18). Then

$$-\pi < z_r - \cos^{-1}\left(\frac{1}{\cosh z_i}\right) \operatorname{sgn}(z_i) < \pi \quad (14-42)$$

where $z = z_r + jz_i$. To determine the pole residue, the following Maliuzhinets function identity is utilized [15, 18]:

$$\Psi_n\left[z \pm \left(n\pi + \frac{3\pi}{2}\right)\right] = \pm \sin\left(\frac{\pi \pm z}{2n}\right) \csc\left(\frac{z}{2n}\right) \Psi_n\left(n\pi - \frac{\pi}{2} \pm z\right) \quad (14-43)$$

This expression isolates the singular part of the pole in the cosecant function, and hence the residue is readily calculated. The cosecant is singular at $z = 0$ and therefore $\Psi\left[\pm\left(n\pi + \frac{3\pi}{2}\right)\right]$ is also singular. The example of the z_0 pole is presented next, and the calculations for the z_n pole are similar.

Example 14-1

Evaluate the surface wave residue of U_{SW}^0 .

Solution: The surface wave residue for the z_0 pole is given by

$$U_{\text{SW}}^0 = \frac{U_0}{n} \text{Res}_{z \rightarrow z_0} \left[\frac{\Psi\left(z + \frac{n\pi}{2} - \phi\right)}{\Psi\left(\frac{n\pi}{2} - \phi'\right)} \frac{\sin\left(\frac{\phi'}{n}\right)}{\cos\left(\frac{z-\phi}{n}\right) - \cos\left(\frac{\phi'}{n}\right)} e^{j\beta\rho \cos z}, z_0 \right]$$

Assuming that no GO poles coincide with the surface wave pole, and since the exponential has no finite poles, the residue can be determined as

$$U_{\text{SW}}^0 = \frac{U_0}{n} \lim_{z \rightarrow z_0} \left[(z - z_0) \frac{\Psi\left(z + \frac{n\pi}{2} - \phi\right)}{\Psi\left(\frac{n\pi}{2} - \phi'\right)} \frac{\sin\left(\frac{\phi'}{n}\right)}{\cos\left(\frac{z-\phi}{n}\right) - \cos\left(\frac{\phi'}{n}\right)} e^{j\beta\rho \cos z} \right]$$

In the limit, $(z - z_0)$ has a zero and $\Psi\left(z + \frac{n\pi}{2} - \phi\right)$ contains the complex pole. All the other terms can be brought outside the limit. Thus,

$$U_{\text{SW}}^0 = \frac{U_0}{n} \left[\frac{e^{j\beta\rho \cos(\phi + \pi + \theta_0)}}{\Psi\left(\frac{n\pi}{2} - \phi'\right)} \frac{\sin\left(\frac{\phi'}{n}\right)}{\cos\left(\frac{\pi + \theta_0}{n}\right) - \cos\left(\frac{\phi'}{n}\right)} \right] \lim_{z \rightarrow z_0} \left[(z - z_0) \Psi\left(z + \frac{n\pi}{2} - \phi\right) \right]$$

Next, the auxiliary Maliuzhinets function within the limit is decomposed into four Maliuzhinets functions using (14-27). That is,

$$\begin{aligned} \Psi\left(z + \frac{n\pi}{2} - \phi\right) &= \Psi_n\left(z + n\pi - \phi + \frac{\pi}{2} - \theta_0\right) \Psi_n\left(z + n\pi - \phi - \frac{\pi}{2} + \theta_0\right) \\ &\quad \cdot \Psi_n\left(z - \phi + \frac{\pi}{2} - \theta_n\right) \Psi_n\left(z - \phi - \frac{\pi}{2} + \theta_n\right) \end{aligned}$$

As $z \rightarrow z_0 = \phi + \pi + \theta_0$, it is evident that the first Maliuzhinets function becomes $\Psi_n\left(n\pi + \frac{3\pi}{2}\right)$, which was shown to be singular in (14-43). Since only the first Maliuzhinets function contains the complex pole, the other three can be moved outside the limit. Thus

$$\begin{aligned} U_{\text{SW}}^0 &= \frac{U_0}{n} \left[\frac{e^{-j\beta\rho \cos(\phi + \theta_0)}}{\Psi\left(\frac{n\pi}{2} - \phi'\right)} \frac{\sin\left(\frac{\phi'}{n}\right)}{\cos\left(\frac{\pi + \theta_0}{n}\right) - \cos\left(\frac{\phi'}{n}\right)} \right] \Psi_n\left(n\pi + \frac{\pi}{2} + 2\theta_0\right) \cdot \Psi_n\left(\frac{3\pi}{2} + \theta_0 - \theta_n\right) \\ &\quad \cdot \Psi_n\left(\frac{\pi}{2} + \theta_0 + \theta_n\right) \lim_{z \rightarrow z_0} \left[(z - z_0) \Psi_n\left(z + n\pi - \phi + \frac{\pi}{2} - \theta_0\right) \right] \end{aligned}$$

The limit of the above equation can be determined using (14-43). Thus, replacing $Z - Z_0 = \alpha$

$$\begin{aligned} \lim_{z \rightarrow z_0} \left[(z - z_0) \Psi_n\left(z + n\pi - \phi + \frac{\pi}{2} - \theta_0\right) \right] &= \lim_{z \rightarrow z_0} \left[(z - z_0) \Psi_n\left(z - z_0 + n\pi + \frac{3\pi}{2}\right) \right] \\ &= \lim_{\alpha \rightarrow 0} \left[\alpha \Psi_n\left(\alpha + n\pi + \frac{3\pi}{2}\right) \right] = \sin\left(\frac{\pi}{2n}\right) \Psi_n\left(n\pi - \frac{\pi}{2}\right) \lim_{\alpha \rightarrow 0} \left[\alpha \csc\left(\frac{\alpha}{2n}\right) \right] \\ &= 2n \sin\left(\frac{\pi}{2n}\right) \Psi_n\left(n\pi - \frac{\pi}{2}\right) \end{aligned}$$

which, when substituted into the previous equation, leads to the surface wave contribution.

The final form of the surface waves from face 0 is

$$U_{SW}^0 = U_0 \left[\frac{2 \sin\left(\frac{\pi}{2n}\right) \sin\left(\frac{\phi'}{n}\right) e^{-j\beta\rho \cos(\phi+\theta_0)}}{\Psi\left(\frac{n\pi}{2} - \phi'\right) \cos\left(\frac{\pi+\theta_0}{n}\right) - \cos\left(\frac{\phi'}{n}\right)} \right] \Psi_n\left(n\pi - \frac{\pi}{2}\right) \Psi_n\left(n\pi + \frac{\pi}{2} + 2\theta_0\right) \\ \cdot \Psi_n\left(\frac{3\pi}{2} + \theta_0 - \theta_n\right) \Psi_n\left(\frac{\pi}{2} + \theta_0 + \theta_n\right)$$

In a similar manner, based on the procedure used in Example 14-1, the surface wave contribution U_{SW}^n from face n can be written as

$$U_{SW}^n = U_0 \left[\frac{-2 \sin\left(\frac{\pi}{2n}\right) \sin\left(\frac{\phi'}{n}\right) e^{-j\beta\rho \cos(\phi-n\pi-\theta_n)}}{\Psi\left(\frac{n\pi}{2} - \phi'\right) \cos\left(\frac{n\pi + \pi + \theta_n}{n}\right) - \cos\left(\frac{\phi'}{n}\right)} \right] \Psi_n\left(n\pi - \frac{\pi}{2}\right) \Psi_n\left(\frac{\pi}{2} + n\pi + 2\theta_n\right) \\ \cdot \Psi_n\left(\frac{3\pi}{2} + \theta_n - \theta_0\right) \Psi_n\left(\frac{\pi}{2} + \theta_n + \theta_0\right) \quad (14-44)$$

By the symmetry of the wedge geometry, it is noted that (14-44) can be obtained from the U_{SW}^0 of Example 14-1 by replacing ϕ by $n\pi - \phi$, ϕ' by $n\pi - \phi'$, θ_0 by θ_n , and θ_n by θ_0 . The angular ranges over which U_{SW}^0 and U_{SW}^n exist are given by (14-41a) through (14-42).

The GO component of (14-21) and the surface wave U_{SW}^0 of Example 14-1 were obtained assuming the GO poles and surface wave poles did not coincide. This is a valid assumption because the GO poles are always real, and if a surface wave pole coincides with a GO pole, it must necessarily be real. However, from (14-41a) and (14-41b), it is evident that real surface wave poles always lie outside the steepest descent paths since $0 < \phi < n\pi$. Therefore, if a surface wave pole and a GO pole coincide, they must lie outside the steepest descent paths, and their residues do not contribute to the exact solution.

14.7 DIFFRACTED FIELDS

In the previous sections, Maliuzhinets' exact solution was examined for both interior and exterior impedance wedges. It was shown that the exact solution can be written as a sum of residues from the GO and surface wave poles plus contribution along the steepest descent paths. In this section, the steepest descent paths contribution is examined thoroughly using high-frequency asymptotic expansions. The high-frequency asymptotic expansion gives the diffracted field and the surface wave transition field. The diffracted field is similar in form to the diffraction coefficients for the PEC theory, with a suitable multiplying factor which includes ratios of auxiliary Maliuzhinets functions.

14.7.1 Diffraction Terms

Maliuzhinets' exact integral solution of (14-5) has been manipulated in (14-10) to follow the steepest descent paths through the saddle points at $z = \pm\pi$. The exact solution from (14-10) can be written as

$$\begin{aligned}
U_t(\rho, \phi) &= -jU_0 \frac{1}{2n\pi} \int_{\gamma} \frac{\Psi\left(z + \frac{n\pi}{2} - \phi\right)}{\Psi\left(\frac{n\pi}{2} - \phi'\right)} \frac{\sin\left(\frac{\phi'}{n}\right)}{\cos\left(\frac{z-\phi}{n}\right) - \cos\left(\frac{\phi'}{n}\right)} e^{j\beta\rho \cos z} dz \\
&= \frac{U_0}{n} \sum_{p=1}^N \text{Res} \left[\frac{\Psi\left(z + \frac{n\pi}{2} - \phi\right)}{\Psi\left(\frac{n\pi}{2} - \phi'\right)} \frac{\sin\left(\frac{\phi'}{n}\right)}{\cos\left(\frac{z-\phi}{n}\right) - \cos\left(\frac{\phi'}{n}\right)} e^{j\beta\rho \cos z}, z_p \right] \\
&\quad - jU_0 \frac{1}{2n\pi} \int_{SDPs} \frac{\Psi\left(z + \frac{n\pi}{2} - \phi\right)}{\Psi\left(\frac{n\pi}{2} - \phi'\right)} \frac{\sin\left(\frac{\phi'}{n}\right)}{\cos\left(\frac{z-\phi}{n}\right) - \cos\left(\frac{\phi'}{n}\right)} e^{j\beta\rho \cos z} dz \quad (14-45)
\end{aligned}$$

$$\begin{aligned}
U_t(\rho, \phi) &= \sum \left[U_{GO}^{m,\xi} + U_{SW}^0 + U_{SW}^n \right] \\
&\quad - jU_0 \frac{1}{2n\pi} \int_{SDPs} \frac{\Psi\left(z + \frac{n\pi}{2} - \phi\right)}{\Psi\left(\frac{n\pi}{2} - \phi'\right)} \frac{\sin\left(\frac{\phi'}{n}\right)}{\cos\left(\frac{z-\phi}{n}\right) - \cos\left(\frac{\phi'}{n}\right)} e^{j\beta\rho \cos z} dz \quad (14-45a)
\end{aligned}$$

$$\begin{aligned}
U_t(\rho, \phi) &= \sum \left[U_{GO}^{m,\xi} + U_{SW}^0 + U_{SW}^n + U_{SDP} \right] \\
&= \sum \left[U_{GO}^{m,\xi} + U_{SW}^0 + U_{SW}^n + U_D + U_{SWTR}^0 + U_{SWTR}^n \right] \quad (14-45b)
\end{aligned}$$

where

$U_{GO}^{m,\xi}$ = GO incident or reflected field corresponding to integer m and $\xi = \xi^+$ or $\xi = \xi^-$ as given by (14-15) and in Table 14-1

U_{SW}^0, U_{SW}^n = surface wave contributions, as given by Example 14-1 and (14-44)

U_D = diffracted field, examined asymptotically in Section 14.7.2

U_{SWTR}^0, U_{SWTR}^n = surface wave transition field terms

The canonical wedge geometry of Figures 14-1a and 14-5 illustrate the shadow boundaries of the GO field for the exterior wedge. Similar shadow boundaries exist for multiple reflected fields for the interior wedge of Figures 14-1b and 14-6. All the GO shadow boundaries are compensated by the diffracted field. Similarly, the surface wave transition field compensates for discontinuities at the surface wave boundary.

14.7.2 Asymptotic Expansions

To determine an asymptotic expansion of a steepest descent integral, such as that of (14-45a), or

$$\begin{aligned}
U_{SDP}(\rho, \phi) &= -jU_0 \frac{1}{2n\pi} \int_{SDPs} \frac{\Psi\left(z + \frac{n\pi}{2} - \phi\right)}{\Psi\left(\frac{n\pi}{2} - \phi'\right)} \frac{\sin\left(\frac{\phi'}{n}\right)}{\cos\left(\frac{z-\phi}{n}\right) - \cos\left(\frac{\phi'}{n}\right)} e^{j\beta\rho \cos z} dz \\
&= -jU_0 \frac{1}{2n\pi} \frac{\sin\left(\frac{\phi'}{n}\right)}{\Psi\left(\frac{n\pi}{2} - \phi'\right)} \int_{SDPs} \frac{\Psi\left(z + \frac{n\pi}{2} - \phi\right)}{\cos\left(\frac{z-\phi}{n}\right) - \cos\left(\frac{\phi'}{n}\right)} e^{j\beta\rho \cos z} dz \quad (14-46)
\end{aligned}$$

requires careful consideration of poles that may lie close to the steepest descent paths. The contribution of the steepest descent integral should be discontinuous as a pole crosses the path to compensate for the addition or loss of the pole residue that corresponds to a GO or surface wave component. In this way, the total solution, which is the sum of the residues plus the integral contribution, is always continuous.

Two asymptotic expansions are used: the *Modified Pauli-Clemmow* method [63–67] for the GO poles and the *Felsen-Marcuvitz* method [68, 69] for the surface wave poles. The Modified Pauli-Clemmow method yields a diffracted field which is analogous to the PEC case for the real GO poles. A diffraction coefficient is formulated, which is similar to the UTD diffraction coefficient with multiplying factors that include suitable ratios of auxiliary Maliuzhinets functions. The method of Felsen-Marcuvitz is used for the complex surface wave poles to construct a surface wave transition field that yields the proper continuity. The method of Felsen-Marcuvitz [68] or that of [70] could be used for all the poles. However, there would be two disadvantages. First, the method would not be analogous to the PEC UTD, which is a powerful tool in modern diffraction. Second, the formulation would require that the contributions of all the poles for all the multiple reflections be added together. This is the approach used in [27], and the equations become very complicated after adding only the poles of the incident field, the two single-reflected fields, and the two surface waves. For an interior wedge with many multiple reflections, the Felsen-Marcuvitz method is more cumbersome than the Modified Pauli-Clemmow method for the GO poles.

The two methods of asymptotic approximation have been shown to be the same [70, 71], provided that the asymptotic expansions are complete; that is, they contain an infinite number of higher-order terms. In practice, this is usually not the case because only the first and second terms are retained and all others are omitted. Consequently, one or the other may be more appropriate for a given problem. However, either will give the correct discontinuity as the pole crosses the saddle point, and both reduce to the same expression when no pole is near the steepest descent path.

The Modified Pauli-Clemmow method of steepest descent was introduced in Chapter 13 to evaluate an integral of the (13-41) form, or

$$P(\beta\rho) = \int_C H(z) e^{\beta\rho h(z)} dz \stackrel{\beta\rho \rightarrow \text{large}}{\approx} e^{\beta\rho h(z_s)} H(z_s) e^{j\phi_s} \left| \sqrt{\frac{-2\pi}{\beta\rho h''(z_s)}} \right| F[\beta\rho g(\xi)] \quad (14-47)$$

$$F[\beta\rho g(\xi)] \equiv j[h(z_s) - h(z_p)] = 2j\sqrt{\beta\rho g(\xi)} \int_{\sqrt{\beta\rho g(\xi)}}^{\infty} e^{-j\tau^2} d\tau$$

≡ a measure of separation between saddle points and poles (14-47a)

and to reduce (13-60a)–(13-60b) to (13-63a)–(13-63b), and eventually to the UTD diffraction coefficients.

The Felsen-Marcuvitz method is used to evaluate an integral of the (13-41) or (14-47) form, and it is expressed as

$$P(\beta\rho) = \int_C H(z) e^{\beta\rho h(z)} dz \stackrel{\beta\rho \rightarrow \text{large}}{\approx} e^{\beta\rho h(z_s)} H(z_s) e^{j\phi_s} \left| \sqrt{\frac{-2\pi}{\beta\rho h''(z_s)}} \right|$$

$$+ \frac{h_1 \sqrt{\pi} e^{\beta\rho h(z_s)}}{\sqrt{\beta\rho [h(z_s) - h(z_p)]}} \{F[\beta\rho g(\xi)] - 1\} \quad (14-48)$$

where h_1 is the residue of $H(z)$ at $z = z_p$; all other quantities are the same as for the Modified Pauli-Clemmow method.

14.7.3 Diffracted Field

Using a procedure similar to that of Section 13.3.2, it can be shown, through some trigonometric identities, that the integral of (14-46) can be written in a more convenient form as

$$\begin{aligned}
 U_{\text{SDP}}(\rho, \phi) &= -jU_0 \frac{1}{2n\pi} \int_{SDPs} \frac{\Psi\left(z + \frac{n\pi}{2} - \phi\right)}{\Psi\left(\frac{n\pi}{2} - \phi'\right)} \frac{\sin\left(\frac{\phi'}{n}\right)}{\cos\left(\frac{z-\phi}{n}\right) - \cos\left(\frac{\phi'}{n}\right)} e^{j\beta\rho \cos z} dz \\
 &= -jU_0 \frac{1}{4n\pi} \int_{SDPs} \frac{\Psi\left(z + \frac{n\pi}{2} - \phi\right)}{\Psi\left(\frac{n\pi}{2} - \phi'\right)} \left\{ \cot\left[\frac{(\phi + \phi') - z}{2n}\right] - \cot\left[\frac{(\phi - \phi') - z}{2n}\right] \right\} e^{j\beta\rho \cos z} dz
 \end{aligned} \quad (14-49)$$

where the following substitution, through trigonometric identities, has been used.

$$\frac{\sin\left(\frac{\phi'}{n}\right)}{\cos\left(\frac{z-\phi}{n}\right) - \cos\left(\frac{\phi'}{n}\right)} = \frac{1}{2} \left\{ \cot\left[\frac{(\phi + \phi') - z}{2n}\right] - \cot\left[\frac{(\phi - \phi') - z}{2n}\right] \right\} \quad (14-50)$$

By the convention of the PEC wedge formulation, $\xi^+ = (\phi + \phi')$ and $\xi^- = (\phi - \phi')$. The poles of the first cotangent occur at

$$z = (\phi + \phi') - 2\pi Nn = \xi^+ - 2\pi Nn \quad (14-51a)$$

and correspond to the GO components that include an odd number of reflections (i.e., single reflections, triple reflections, etc.). The poles of the second cotangent occur at

$$z = (\phi - \phi') - 2\pi Nn = \xi^- - 2\pi Nn \quad (14-51b)$$

and correspond to the GO terms that include an even number of reflections (i.e., incident, double reflected, etc.). In (14-51a) and (14-51b), N can be any integer, and hence there are an infinite number of poles of the integrand for both ξ^+ and ξ^- .

The PEC UTD diffraction considers only the four dominant poles of the integrand of (14-49). These four are the poles corresponding to ξ^+ that are the closest to the saddle points at $\pm\pi$, and the poles corresponding to ξ^- that are closest to the same saddle points. Hence, four values of N are chosen, where the N 's are the integers that most closely satisfy

$$2\pi nN_{-}^{+} - \xi^{-} = +\pi \quad (14-52a)$$

$$2\pi nN_{-}^{-} - \xi^{-} = -\pi \quad (14-52b)$$

$$2\pi nN_{+}^{+} - \xi^{-} = +\pi \quad (14-52c)$$

$$2\pi nN_{+}^{-} - \xi^{-} = -\pi \quad (14-52d)$$

where the subscript in N indicates the choice of ξ . The superscript of N corresponds to the sign of $\pm\pi$ on the right side of (14-52). Therefore, based on (14-52a)–(14-52d), (14-49) can be decomposed into four integrals corresponding to the four selected poles; that is,

$$\begin{aligned}
U_{\text{SDP}}(\rho, \phi) = & +jU_0 \frac{1}{4n\pi} \int_{\text{SDP}(-\pi)} \frac{\Psi\left(z + \frac{n\pi}{2} - \phi\right)}{\Psi\left(\frac{n\pi}{2} - \phi'\right)} \left\{ \cot\left[\frac{\xi^- - z}{2n}\right] \right\} e^{j\beta\rho \cos z} dz \\
& + jU_0 \frac{1}{4n\pi} \int_{\text{SDP}(+\pi)} \frac{\Psi\left(z + \frac{n\pi}{2} - \phi\right)}{\Psi\left(\frac{n\pi}{2} - \phi'\right)} \left\{ \cot\left[\frac{\xi^- - z}{2n}\right] \right\} e^{j\beta\rho \cos z} dz \\
& - jU_0 \frac{1}{4n\pi} \int_{\text{SDP}(-\pi)} \frac{\Psi\left(z + \frac{n\pi}{2} - \phi\right)}{\Psi\left(\frac{n\pi}{2} - \phi'\right)} \left\{ \cot\left[\frac{\xi^+ - z}{2n}\right] \right\} e^{j\beta\rho \cos z} dz \\
& - jU_0 \frac{1}{4n\pi} \int_{\text{SDP}(+\pi)} \frac{\Psi\left(z + \frac{n\pi}{2} - \phi\right)}{\Psi\left(\frac{n\pi}{2} - \phi'\right)} \left\{ \cot\left[\frac{\xi^+ - z}{2n}\right] \right\} e^{j\beta\rho \cos z} dz \quad (14-53)
\end{aligned}$$

where the order of the terms of (14-53) corresponds to the order of the N 's in (14-52). Each term is individually evaluated asymptotically using (14-47).

Example 14-2

To demonstrate the asymptotic procedure, evaluate the first term of (14-53).

Solution:

$$\begin{aligned}
H_1(z) &= \frac{\Psi\left(z + \frac{n\pi}{2} - \phi\right)}{\Psi\left(\frac{n\pi}{2} - \phi'\right)} \cot\left(\frac{\xi^- - z}{2n}\right) \\
h_1(z) &= j \cos z \\
z_s &= -\pi \\
z_p &= \xi^- - 2\pi N_-^+ n \\
g &= j[-j - j \cos(\xi^- - 2\pi N_-^+ n)] = [1 + \cos(\xi^- - 2\pi N_-^+ n)] \\
h''(z) &= -j \cos z \\
\left| \sqrt{\frac{-2\pi}{\beta\rho h''(z_s)}} \right| &= \left| \sqrt{\frac{-2\pi}{j\beta\rho}} \right| = \sqrt{\frac{2\pi}{\beta\rho}}
\end{aligned}$$

where the correct branch of the radical is determined by the angle of the integration path in the direction of integration. The asymptotic expansion for the first term of (14-53) can now be constructed using (14-47)–(14-47a) with $\phi_s = \pi/4$ and can be written, using the above, as

$$\begin{aligned}
U_{\text{SDP}}^1(\rho, \phi) = & +jU_0 \frac{1}{4n\pi} e^{-j\beta\rho} \frac{\Psi\left(-\pi + \frac{n\pi}{2} - \phi\right)}{\Psi\left(\frac{n\pi}{2} - \phi'\right)} \cot\left(\frac{\xi^- + \pi}{2n}\right) \sqrt{\frac{2\pi}{\beta\rho}} e^{j\frac{\pi}{4}} \\
& \cdot F\{\beta\rho[1 + \rho \cos(\xi^- - 2\pi N_-^+ n)]\}
\end{aligned}$$

The other three terms of (14-53) are evaluated in the same manner, remembering that for the steepest descent path through $+\pi$, the exponential $e^{j\phi_s}$ is written as $e^{j\phi_s} = e^{-j3\pi/4}$ because of the direction of the integration contour of Figure 14-4 (use $\phi_s = \pi/4$ for SDP through $-\pi$). The final total diffracted field can now be written as

$$U_{\text{SDP}}^{s,h}(\rho, \phi) = U_0 \frac{e^{-j\beta\rho}}{\sqrt{\rho}} \left[-\frac{e^{-j\frac{\pi}{4}}}{2n\sqrt{2\pi\beta}} \right] \left\{ \left[\frac{\Psi\left(-\pi + \frac{n\pi}{2} - \phi\right)}{\Psi\left(\frac{n\pi}{2} - \phi'\right)} \cot\left(\frac{\pi + \xi^-}{2n}\right) F\{\beta\rho[1 + \cos(\xi^- - 2\pi N_-^+ n)]\} \right. \right. \\ \left. \left. + \frac{\Psi\left(\pi + \frac{n\pi}{2} - \phi\right)}{\Psi\left(\frac{n\pi}{2} - \phi'\right)} \cot\left(\frac{\pi - \xi^-}{2n}\right) F\{\beta\rho[1 + \cos(\xi^- - 2\pi N_-^- n)]\} \right] \right. \\ \left. - \left[\frac{\Psi\left(-\pi + \frac{n\pi}{2} - \phi\right)}{\Psi\left(\frac{n\pi}{2} - \phi'\right)} \cot\left(\frac{\pi + \xi^+}{2n}\right) F\{\beta\rho[1 + \cos(\xi^+ - 2\pi N_+^+ n)]\} \right. \right. \\ \left. \left. + \frac{\Psi\left(\pi + \frac{n\pi}{2} - \phi\right)}{\Psi\left(\frac{n\pi}{2} - \phi'\right)} \cot\left(\frac{\pi - \xi^+}{2n}\right) F\{\beta\rho[1 + \cos(\xi^+ - 2\pi N_+^- n)]\} \right] \right\} \quad (14-54)$$

where $F(z)$ is the Fresnel transition function. By comparing (14-54) with the PEC case of Chapter 13, it is evident that (14-54) is the same but with the introduction of suitable ratios of auxiliary Maliuzhinets functions as multiplying factors.

At the reflection and incident shadow boundaries, $\phi = +\pi \pm \phi' + 2n\pi N$ or $\phi = -\pi \pm \phi' + 2n\pi N$, the ratios of the auxiliary Maliuzhinets become

$$\left. \frac{\Psi\left(\pm\pi + \frac{n\pi}{2} - \phi\right)}{\Psi\left(\frac{n\pi}{2} - \phi'\right)} \right|_{\phi = \pm\pi \pm \phi' + 2n\pi N} = \frac{\Psi\left(\mp\phi' - 2\pi Nn + \frac{n\pi}{2}\right)}{\Psi\left(\frac{n\pi}{2} - \phi'\right)} \quad (14-55)$$

In Section 14.5 it was verified that the ratio (14-55) is simply a product of reflection coefficients. The interesting conclusion drawn is that the diffraction coefficient is modified by the appropriate ratios that reduce the diffraction discontinuities to account for reflection discontinuities for the lossy wedge.

The diffracted field of (14-54) is a valid asymptotic expansion, but is not the best possible asymptotic expansion because it does not reduce to the PEC diffraction for the hard polarization. It has been verified, by numerical integration [28], that these expressions work well for the soft polarization or the imperfectly conducting hard polarization. However, for the hard polarization, the expressions fail as the conductivity increases since they do not revert to the PEC case.

To evaluate the auxiliary Maliuzhinets function ratio in (14-54), the following identity is utilized for the hard polarized PEC case $\theta_0 = \theta_n = 0$. That is,

$$\begin{aligned} \frac{\Psi\left(\pm\pi + \frac{n\pi}{2} - \phi\right)}{\Psi\left(\frac{n\pi}{2} - \phi'\right)} &= \frac{\cos\left(\frac{\pm\pi + n\pi - \phi}{2n}\right)\cos\left(\frac{\pm\pi - \phi}{2n}\right)}{\cos\left(\frac{n\pi - \phi'}{2n}\right)\cos\left(\frac{\phi'}{2n}\right)} = \frac{\cos\left(\frac{\pm\pi - \phi}{n} + \frac{\pi}{2}\right)}{\cos\left(\frac{\phi'}{n} - \frac{\pi}{2}\right)} \\ &= \frac{-\sin\left(\frac{\pm\pi - \phi}{n}\right)}{\sin\left(\frac{\phi'}{n}\right)} \end{aligned} \quad (14-56)$$

Hence, as $\theta_0 = \theta_n \rightarrow 0$, this formulation differs from the PEC UTD by

$$\frac{-\sin\left(\frac{\pm\pi - \phi}{n}\right)}{\sin\left(\frac{\phi'}{n}\right)} \quad (14-57)$$

For the soft polarization, there is no problem because the ratio of the left side of (14-56) reduces, for $\theta_0 = \theta_n \rightarrow \infty$, to

$$\lim_{\theta_0 = \theta_n \rightarrow \infty} \frac{\Psi\left(\pm\pi + \frac{n\pi}{2} - \phi\right)}{\Psi\left(\frac{n\pi}{2} - \phi'\right)} = 1 \quad (14-58)$$

Example 14-3

Since (14-54) does not reduce to the PEC diffraction for the hard polarization, reformulate the asymptotic expansion in a different manner so that when (14-54) is written in a slightly different form, it reduces to the PEC case for both soft and hard polarizations.

Solution: To achieve diffraction coefficients that reduce to the PEC case for both polarizations, soft and hard, all that is needed is a different subdivision of the singular portions of the integrand of (14-49). Instead of substituting (14-50)

$$\frac{\sin\left(\frac{\phi'}{n}\right)}{\cos\left(\frac{z - \phi}{n}\right) - \cos\left(\frac{\phi'}{n}\right)} = \frac{1}{2} \left\{ \cot\left[\frac{(\phi + \phi') - z}{2n}\right] - \cot\left[\frac{(\phi - \phi') - z}{2n}\right] \right\}$$

into (14-46), it was recommended in [26] that the following trigonometric identity be used. That is,

$$\frac{\sin\left(\frac{\phi'}{n}\right)}{\cos\left(\frac{z - \phi}{n}\right) - \cos\left(\frac{\phi'}{n}\right)} = \frac{1}{2} \left\{ \frac{\sin\left(\frac{\phi'}{n}\right) + \sin\left(\frac{\theta_0}{n}\right)}{\sin\left(\frac{\phi - z}{n}\right) + \sin\left(\frac{\theta_0}{n}\right)} \cot\left[\frac{z - (\phi - \phi')}{2n}\right] + \frac{\sin\left(\frac{\phi'}{n}\right) - \sin\left(\frac{\theta_0}{n}\right)}{\sin\left(\frac{\phi - z}{n}\right) + \sin\left(\frac{\theta_0}{n}\right)} \cot\left[\frac{z - (\phi + \phi')}{2n}\right] \right\}$$

where the saddle points are at $z_s = \pm\pi$. Also for $\theta_0 = \theta_n = 0$, the sine ratios become

$$\frac{\sin\left(\frac{\phi'}{n}\right) \pm \sin\left(\frac{\theta_0}{n}\right)}{\sin\left(\frac{\phi-z}{n}\right) + \sin\left(\frac{\theta_0}{n}\right)} \bigg|_{\substack{z=\pm\pi \\ \theta_0=0}} = \frac{\sin\left(\frac{\phi'}{n}\right)}{-\sin\left(\frac{\pm\pi-\phi}{n}\right)}$$

which is exactly the proper term to remove the problematic multiplying factor (14-57). Hence, the resultant expression for the diffracted field will reduce to the PEC forms for the hard polarization as $\theta_0 = \theta_n \rightarrow 0$. For the soft polarization, where $\theta_0 = \theta_n \rightarrow \infty$ with $0 < \text{Re}[\theta_0] < \pi/2$ and $0 < \text{Re}[\theta_n] < \pi/2$, the sine ratios on the left side of the above equation reduce to ± 1 and hence, are identical to the second equation in this example. Therefore, the correct expressions, which have already been established for the soft polarization, are retained.

It is interesting that the asymptotic expansion is not unique but depends upon the subdivision of the poles into individual terms. While the diffracted field from the subdivision of (14-50) does not reduce to the PEC case for one polarization (the hard one), the diffracted field based on the second equation in Example 14-3 reduces to the PEC forms for both principal polarizations, and it is the proper identity to subdivide the singular parts of the Maliuzhinets function.

Substituting the correct expression (the second equation of Example 14-3) into (14-49), it is evident that the final expression for the diffracted field to replace (14-54) is

$$U_{\text{SDP}}^{s,h}(\rho, \phi) = U_0 \frac{e^{-j\beta\rho}}{\sqrt{\rho}} \left[-\frac{e^{-j\frac{\pi}{4}}}{2n\sqrt{2\pi\beta}} \right. \\ \left. \left\{ \begin{aligned} & \frac{\Psi\left(-\pi + \frac{n\pi}{2} - \phi\right)}{\Psi\left(\frac{n\pi}{2} - \phi'\right)} \frac{\sin\left(\frac{\phi'}{n}\right) + \sin\left(\frac{\theta_0}{n}\right)}{\sin\left(\frac{\phi+\pi}{n}\right) + \sin\left(\frac{\theta_0}{n}\right)} \cot\left(\frac{\pi + \xi^-}{2n}\right) \\ & \quad \cdot F\{\beta\rho[1 + \cos(\xi^- - 2\pi N_-^+ n)]\} \\ & + \frac{\Psi\left(\pi + \frac{n\pi}{2} - \phi\right)}{\Psi\left(\frac{n\pi}{2} - \phi'\right)} \frac{\sin\left(\frac{\phi'}{n}\right) + \sin\left(\frac{\theta_0}{n}\right)}{\sin\left(\frac{\phi+\pi}{n}\right) + \sin\left(\frac{\theta_0}{n}\right)} \cot\left(\frac{\pi - \xi^-}{2n}\right) \\ & \quad \cdot F\{\beta\rho[1 + \cos(\xi^- - 2\pi N_-^- n)]\} \end{aligned} \right\} \\ + \left\{ \begin{aligned} & \frac{\Psi\left(-\pi + \frac{n\pi}{2} - \phi\right)}{\Psi\left(\frac{n\pi}{2} - \phi'\right)} \frac{\sin\left(\frac{\phi'}{n}\right) - \sin\left(\frac{\theta_0}{n}\right)}{\sin\left(\frac{\phi+\pi}{n}\right) + \sin\left(\frac{\theta_0}{n}\right)} \cot\left(\frac{\pi + \xi^+}{2n}\right) \\ & \quad \cdot F\{\beta\rho[1 + \cos(\xi^+ - 2\pi N_+^+ n)]\} \\ & + \frac{\Psi\left(\pi + \frac{n\pi}{2} - \phi\right)}{\Psi\left(\frac{n\pi}{2} - \phi'\right)} \frac{\sin\left(\frac{\phi'}{n}\right) - \sin\left(\frac{\theta_0}{n}\right)}{\sin\left(\frac{\phi+\pi}{n}\right) + \sin\left(\frac{\theta_0}{n}\right)} \cot\left(\frac{\pi - \xi^+}{2n}\right) \\ & \quad \cdot F\{\beta\rho[1 + \cos(\xi^+ - 2\pi N_+^- n)]\} \end{aligned} \right\} \right] \quad (14-59)$$

Equation (14-59), although cast in a different form, is identical to those in [26]. However, in [26], a great deal of effort was focused on manipulating these expressions to a form that was symmetric with respect to θ_0 and θ_n , and to reduce the number of times the Maliuzhinets function needed to be calculated. The formulation of (14-59) does not appear explicitly symmetric with respect to θ_0 and θ_n , but it is symmetric because the $\Psi(z)$ auxiliary Maliuzhinets function includes θ_0 and θ_n implicitly. There are twelve Maliuzhinets functions that need to be calculated. Four are necessary for each auxiliary Maliuzhinets function, and there are three auxiliary functions:

$$\Psi\left(\frac{n\pi}{2} - \phi'\right), \Psi\left(\pi + \frac{n\pi}{2} - \phi'\right), \text{ and } \Psi\left(-\pi + \frac{n\pi}{2} - \phi'\right)$$

In [26], it was shown that the Maliuzhinets functions reduce to a double-nested integration; however, this double-nested integral is exceedingly difficult to evaluate, even numerically. Hence, [26] only demonstrated results for three special wedge angles, corresponding to the half-plane ($WA = 0^\circ$), the exterior right angle ($WA = 90^\circ$), and the planar interface discontinuity ($WA = 180^\circ$). It is shown in [28] that this double-nested integral can be written as eight Maliuzhinets functions, all of which are readily available for an arbitrary wedge angle by methods in Appendix B of [28].

For the integral of [26], which is denoted $M_n(\phi, \phi'; \theta_0, \theta_n)$, the simpler expression is

$$M_n(\phi, \phi'; \theta_0, \theta_n) = \frac{\Psi_n\left(n\pi - \phi' + \frac{\pi}{2} + \theta_n\right) \Psi_n\left(\phi' + \frac{\pi}{2} + \theta_n\right)}{\Psi_n\left(n\pi - \phi' + \frac{\pi}{2} - \theta_0\right) \Psi_n\left(\phi' + \frac{\pi}{2} - \theta_n\right)} \cdot \frac{\Psi_n\left(n\pi - \phi - \frac{\pi}{2} - \theta_0\right) \Psi_n\left(\phi - \frac{\pi}{2} - \theta_n\right)}{\Psi_n\left(n\pi - \phi - \frac{\pi}{2} + \theta_0\right) \Psi_n\left(\phi' - \frac{\pi}{2} + \theta_n\right)} \quad (14-60)$$

In this manner, the necessary evaluations of Maliuzhinets functions are reduced from twelve to eight, and the results are obtainable for arbitrary interior and exterior wedge angles rather than for only the three specific wedge angles [half-plane ($WA = 0^\circ$), exterior right angle ($WA = 90^\circ$), and planar interface discontinuity ($WA = 180^\circ$)]. This allows for the modeling of more general structures.

The Keller-type diffraction coefficients, which are the nonuniform versions, are obtained from (14-59) by setting every Fresnel transition function to unity.

14.8 SURFACE WAVE TRANSITION FIELD

The surface waves of the exact impedance wedge solution exist only for a limited angular range near the associated wedge face. It was shown in Section 14.6 that this angular range was determined by the position of the surface wave pole relative to the steepest descent path. As the pole moves outside the region between the steepest descent paths, the surface wave term vanishes. This is completely analogous to the GO shadow boundaries that arise when a pole crosses a steepest descent path. The major difference is that the surface wave poles are generally complex, whereas the GO poles are always real.

The surface wave transition field provides continuity across the surface wave boundary by uniformly accounting for the complex pole in the vicinity of the steepest descent path. It performs the same function as the diffracted field in providing continuity for the GO field. This component of the exact solution has also been referred to as the *surface ray field* [27].

To determine the contribution of the complex surface wave pole in the steepest descent integral, the method of Felsen-Marcuvitz [68] is utilized. In (14-48), the integral was shown to be

asymptotically approximated by the sum of two terms; the first corresponding to a first-order saddle point evaluation with no pole, and the second corresponding to the pole contribution. In (14-47), the modified Pauli-Clemmow method was shown to reduce precisely to the first term of (14-48) when the GO pole was not near the steepest descent path. Hence, a method to include the surface wave transition field is to use the second term of (14-48) added to the diffraction contribution of Section 14.7. This is valid provided the poles approach the steepest descent path individually.

The contribution of the surface wave transition field is then given by [28]

$$U_{\text{SWTR}}^0(\rho, \phi) = U_0 \frac{e^{-j\beta\rho}}{\sqrt{\rho}} \left[\frac{-\sqrt{\frac{j}{\pi}} \sin\left(\frac{\pi}{2n}\right)}{\Psi\left(\frac{n\pi}{2} - \phi'\right)} \frac{\sin\left(\frac{\phi'}{n}\right)}{\cos\left(\frac{\pi + \theta_0}{n}\right) - \cos\left(\frac{\phi'}{n}\right)} \right] \\ \cdot \Psi_n\left(n\pi - \frac{\pi}{2}\right) \Psi_n\left(n\pi + \frac{\pi}{2} + 2\theta_0\right) \Psi_n\left(\frac{3\pi}{2} + \theta_0 - \theta_n\right) \Psi_n\left(\frac{\pi}{2} + \theta_0 + \theta_n\right) \\ \cdot \frac{F\{\beta\rho[1 - \cos(\phi + \theta_0)]\} - 1}{\sqrt{\beta[\cos(\phi + \theta_0) - 1]}} \quad (14-61a)$$

$$U_{\text{SWTR}}^n(\rho, \phi) = U_0 \frac{e^{-j\beta\rho}}{\sqrt{\rho}} \left[\frac{\sqrt{\frac{j}{\pi}} \sin\left(\frac{\pi}{2n}\right)}{\Psi\left(\frac{n\pi}{2} - \phi'\right)} \frac{\sin\left(\frac{\phi'}{n}\right)}{\cos\left(\frac{n\pi + \pi + \theta_n}{n}\right) - \cos\left(\frac{\phi'}{n}\right)} \right] \\ \cdot \Psi_n\left(n\pi - \frac{\pi}{2}\right) \Psi_n\left(n\pi + \frac{\pi}{2} + 2\theta_n\right) \Psi_n\left(\frac{3\pi}{2} + \theta_n - \theta_0\right) \Psi_n\left(\frac{\pi}{2} + \theta_0 + \theta_n\right) \\ \cdot \frac{F\{\beta\rho[1 - \cos(\phi - n\pi - \theta_n)]\} - 1}{\sqrt{\beta[\cos(\phi - n\pi - \theta_n) - 1]}} \quad (14-61b)$$

The pole residue has been determined as in (14-43) and Example 14-1. By the symmetry of the wedge geometry, it is noted that (14-61b) can be determined from (14-61a) by replacing ϕ by $n\pi - \phi$, ϕ' by $n\pi - \phi'$, θ_0 by θ_n , and θ_n by θ_0 . This Fresnel transition function is as defined in [66] but extended for complex arguments. The $\sqrt{z}$ in the definition of $F(z)$ should have a branch cut along the positive imaginary axis so that $-3\pi/4 < [\text{Arg}\sqrt{z}] < \pi/4$. It is the discontinuity in $F(z)$ at the branch cut that provides the discontinuity in the integral contribution. When the pole is far from the steepest descent path, the Fresnel transition function is approximately unity, and hence, the surface wave transition field contribution is zero.

A troublesome case occurs when the surface wave pole coincides with a GO pole. Since the GO poles are always real, this can only occur for real surface impedances. Real surface impedances, however, cannot support surface waves, as the surface wave pole will never lie within the steepest descent paths. However, if the surface wave transition field is calculated by blindly applying (14-61a) and (14-61b), erroneous results will occur whenever the surface wave pole is far from the steepest descent path, yet near a GO pole. Indeed, the surface wave transition field should be zero when the surface wave pole is far from the steepest descent paths. *Hence, (14-61a) and (14-61b) should only be used when the surface wave pole is closer to a steepest descent path than to a GO pole. When the surface wave pole is near both a GO pole and the steepest descent path, then (16) of [26] should be used. When the surface wave pole is far from the steepest descent path, the surface wave transition field is taken as zero.*

A complete asymptotic expansion, which uniformly accounts for coalescing GO and surface wave poles, can be found in [70]. However, this type of formulation would not retain the form of

the PEC UTD diffraction coefficient [26, 66], in which the four dominant poles are accounted for in four cotangent-Fresnel products. Retaining only the four dominant poles is sufficient for all shadow boundaries of all the multiple reflected fields.

14.9 COMPUTATIONS

Based on the analytical formulations developed in the previous sections, a number of computations were performed and a sample of them are presented here. Some of the others will be assigned as end-of-chapter problems.

The amplitude radiation patterns of an infinitesimal dipole placed at a height of 6λ above a square ground plane (10.6λ on each side) with impedance surface are shown in Figure 14-8. We see that the pattern variations are not severely affected by the changes in the normalized surface impedance, ranging from 0.001 to 0.5 (unnormalized impedances of 0.377–188.5 ohms). Most of the variations occur above the ground plane within $\theta = \pm 60^\circ$, where the field intensity is weaker. The amplitude radiation patterns, predicted [28] and measured [49], of a $\lambda/4$ monopole above a square graphite ground plane [normalized surface impedance of $0.001668(1 + j)$] are exhibited in Figure 14-9. It is evident that there is a very good agreement between simulations and measurements.

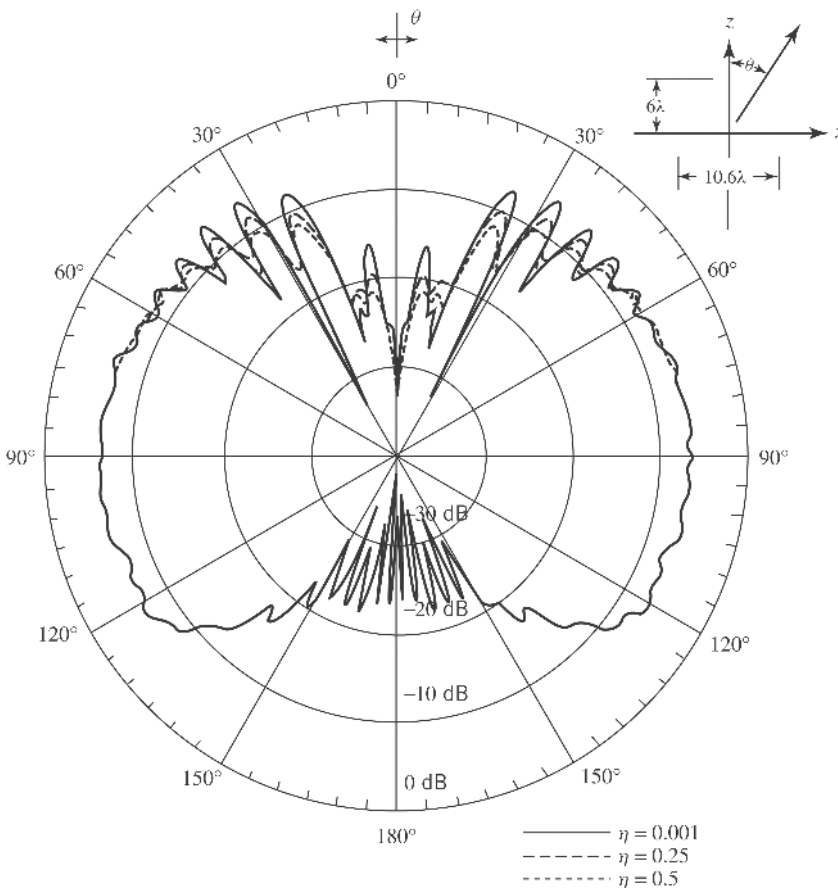


Figure 14-8 Amplitude radiation patterns of infinitesimal vertical dipole above square ground plane with normalized impedance surface [28].

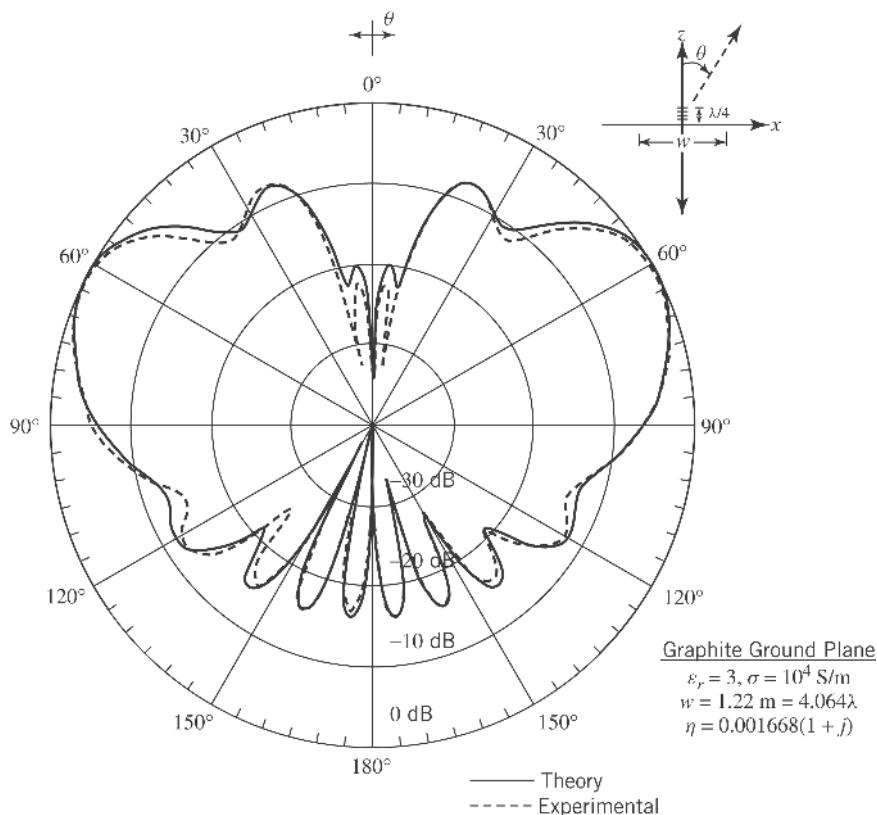


Figure 14-9 Amplitude radiation patterns of $\lambda/4$ vertical monopole above square graphite ground plane [28, 49]. (Source: C. A. Balanis and D. DeCarlo, "Monopole antenna patterns on finite size composite ground planes," *IEEE Trans. Antennas Propagat.*, © 1982, IEEE.)

A dihedral corner reflector is often used as a reference for RCS measurements. The geometry of a dihedral corner reflector, and its dimensions, are shown in Figure 14-10. A very thorough examination of a PEC dihedral corner reflector can be found in [72–74]. To reduce the RCS, the interior faces of the dihedral corner reflector can be coated with Radio Absorbing Material (RAM) with

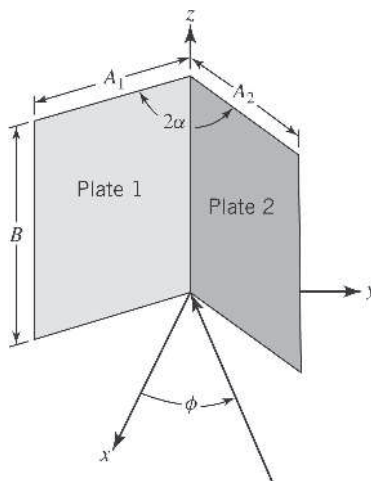


Figure 14-10 Geometry of dihedral corner reflector.

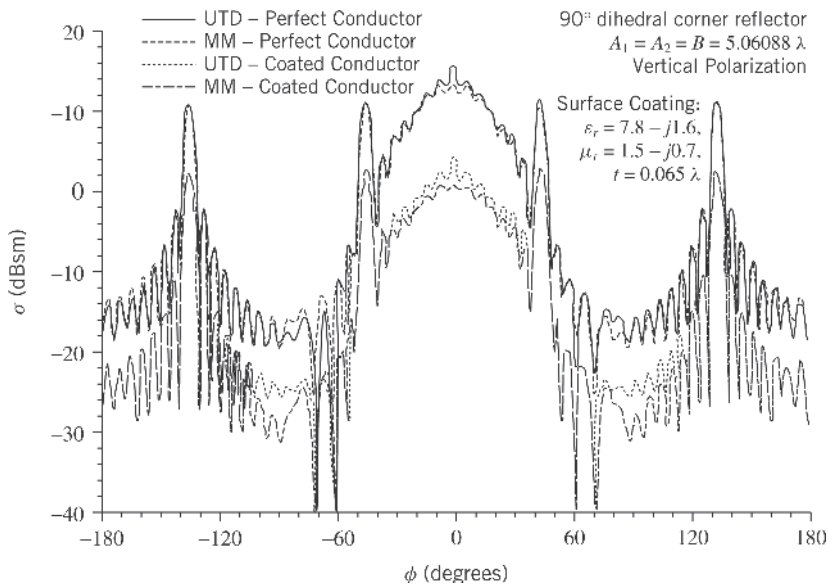


Figure 14-11 RCS of PEC and coated 90° corner reflector [28, 29]. (Source: T. Griesser, C. A. Balanis, and K. Liu, “RCS analysis and reduction for lossy dihedral corner reflectors,” *Proceedings of the IEEE*, © 1989, IEEE.)

complex permittivity and permeability. This is a procedure used in practice to reduce the RCS of radar targets and make them more stealthy. The RCS patterns of a 90° corner reflector, PEC and coated, are shown in Figure 14-11 and are based on computations using the Uniform Theory of Diffraction (UTD) and Method of Moments (MoM). A 90° dihedral corner reflector possesses a very high RCS near its axis primarily because of the second reflection, which is directed toward the incident direction when the interior angle of the dihedral corner reflector is 90° . This is not the case for interior angles other than 90° . This can be verified by simply using Snell’s law of reflection. It is also apparent from the patterns in Figure 14-11 that there is a very good agreement between the two methods, and both predict reductions of nearly 10 dB at the flare spots (directions where the RCS is the most intense; along the axis of the 90° corner reflector and in directions perpendicular to its faces; both interior and exterior). The RCS of 77° and 98° dihedral corner reflectors were also performed, and they are left as end-of-chapter exercises. We should expect that for these two dihedral corner reflectors (77° and 98°) the RCS at and near the axis is not very intense because the second reflection is not directed toward the direction of incidence.

14.10 MULTIMEDIA

On the website that accompanies this book, the following multimedia resources are included for the review, understanding, and presentation of the material of this chapter.

- **MATLAB** computer programs:
 - a. **Monopole:** Computes the normalized amplitude pattern of a monopole on a rectangular ground plane with an impedance surface. The geometry is the same as that of Figure 13-32, where the PEC ground plane is replaced by one with an equivalent surface impedance. The pattern is computed using the UTD formulation of Chapter 14.

- b. **Dipole_Vertical:** Computes the normalized amplitude pattern of a vertical dipole placed at a height h above a rectangular ground plane with an equivalent surface impedance. The pattern is computed using the UTD formulation of Chapter 14.
- c. **Dipole_Horizontal_H_Plane:** Computes the normalized H-plane amplitude pattern of a horizontal dipole placed at a height h above a rectangular ground plane with an equivalent surface impedance. The pattern is computed using the UTD formulation of Chapter 14.
- **PowerPoint (PPT)** viewgraphs, in multicolor.

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PROBLEMS

- 14.1. Using duality, derive the dual of (14-2) for the other polarization. Show that the impedance boundary condition has the same duality as Maxwell's equations.

- 14.2. The impedance boundary condition can be written in a different form than the vector form of (14-2) to conveniently solve the wedge scattering problem. For a wedge, in which the wedge vertex is along the z axis in a cylindrical coordinate system, the wedge faces are located at $\phi = 0$ and $\phi = n\pi$. This wedge geometry is illustrated in Figure 14-1, for both the exterior and interior wedges. For the soft polarization (TM^{*z*}), the impedance boundary conditions are

$$E_z = -\eta_0 Z_0 H_\phi \quad \text{for } \phi = 0$$

$$E_z = +\eta_n Z_0 H_\phi \quad \text{for } \phi = n\pi$$

on the faces 0 and n , respectively. Rewrite/reduce each of these equations in terms of the E_z and its partial derivative on the respective faces.

- 14.3. Repeat Problem 14.2 for the hard polarization Polarization (TE^{*z*}) for which, using Figure 14-1, the impedance boundary conditions are

$$E_\rho = +\eta_0 Z_0 H_z \quad \text{for } \phi = 0$$

$$E_\rho = -\eta_n Z_0 H_z \quad \text{for } \phi = n\pi$$

on the faces 0 and n , respectively. Rewrite/reduce each of these equations in terms of the H_z and its partial derivative on the respective faces, usually referred to as a boundary condition of the third kind.

- 14.4. For the soft (perpendicular) polarization of Figure 14-2a, derive an expression for the reflection coefficient for oblique incidence assuming the reflecting planar surface is a PEC ground plane covered with a lossless dielectric slab of thickness t and constitutive parameters ϵ_1 and μ_1 . Use transmission line

theory to represent the normalized equivalent surface impedance at the leading interface formed by free space and the dielectric slab.

- 14.5. Repeat Problem 14.4 for the hard (parallel) polarization of Figure 14-2b.

- 14.6. For the method of steepest descent (saddle point method), the saddle point z_s (assuming only one saddle point) is the point in the complex plane at which $|e^{\beta \phi(z_s)}|$ is a maximum of $|e^{\beta \phi(z)}|$ along one direction and a minimum of $|e^{\beta \phi(z)}|$ along a perpendicular direction in the complex plane. In three dimensions, a plot of $|e^{\beta \phi(z_s)}|$ would look like a saddle with z_s at the center of the saddle. Assuming $H(z)$ in (14-71) is smoothly varying near the saddle point, the integrand achieves its maximum at the saddle point, and it is basically negligible elsewhere. The steepest descent and ascent paths can be found if the relationship $\text{Im}[h(z)] = \text{Im}[h(z_s)]$ is a constant. Derive equations for the steepest descent and ascent paths in the complex z plane.

- 14.7. In contrast to the steepest descent and ascent paths of Problem 14.6, the stationary phase paths are those for which $\text{Re}[h(z)] = \text{Re}[h(z_s)]$ is a constant. Derive equations for the stationary paths in the complex z plane.

- 14.8. A uniform plane wave of unity amplitude is incident upon a 90° dielectric wedge at an angle of $\phi' = 60^\circ$ (Figure P14-8). At an observation distance of $\rho = 81\lambda$ from the edge of the wedge and an observation angle of 180° , compute for *hard* polarization the approximate:

(a) Incident diffracted field.

(b) Reflected diffracted field.

The electrical properties of the dielectric are: $\epsilon_r = 4$, $\mu_r = 1$. Assume that the surface impedance of the faces of the wedge are each

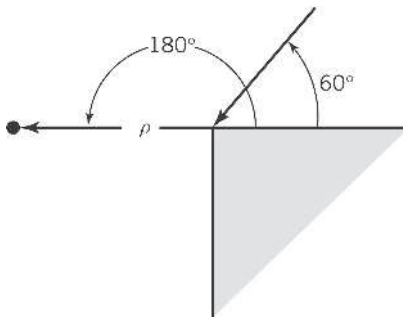


Figure P14-8

equal to the intrinsic impedance of the dielectric material. Clearly indicate what equations you are using. If you make any

approximations, state as to why you think they are valid.

- 14.9. Repeat Problem 14.8 for *soft* polarization.
- 14.10. Derive (14-44) for U_{SW}^n using a similar procedure as in Example 14-1 for U_{SW}^0 .
- 14.11. Using the procedure of Example 14-2, derive the second term of (14-54) from the second term of (14-53).
- 14.12. Using the procedure of Example 14-2, derive the third term of (14-54) from the third term of (14-53).
- 14.13. Using the procedure of Example 14-2, derive the fourth term of (14-54) from the fourth term of (14-53).

CHAPTER 15



Green's Functions

15.1 INTRODUCTION

In the area of electromagnetics, solutions to many problems are obtained using a second-order, uncoupled partial differential equation, derived from Maxwell's equations, and the appropriate boundary conditions. The form of most solutions of this type is an infinite series, provided the partial differential equation and the boundary conditions representing the problems are separable in the chosen coordinate system. The difficulty in using these types of solutions to obtain an insight into the behavior of the function is that they are usually slowly convergent, especially at regions where rapid changes occur. It would then seem appropriate, at least for some problems and associated regions, that closed-form solutions would be desirable. Even solutions in the form of integrals would be acceptable. The technique known as the *Green's function* accomplishes this goal.

Before proceeding with the presentation of the Green's function solution, let us briefly describe what the Green's functions represent and how they are used to obtain the overall solution to the problem.

With the Green's function technique, a solution to the partial differential equation is obtained using a unit source (impulse, Dirac delta) as the driving function. *This is known as the Green's function.* The solution to the actual driving function is written as a superposition of the impulse response solutions (Green's function) with the Dirac delta source at different locations, which in the limit reduces to an integral. The contributions to the overall solution from the general source may be greater or smaller than that of the impulse response depending on the strength of the source at that given location. In engineering terminology, then, the Green's function is nothing more but the *impulse response* of a system; in system theory, this is better known as the *transfer function*.

For a given problem, the Green's function can take various forms. One form of its solution can be expressed in terms of finite explicit functions, and it is obtained based on a procedure that will be outlined later. This procedure for developing the Green's function can be used only if the solution to the homogeneous differential equation is known. Another form of the Green's function is to construct its solution by an infinite series of suitably chosen orthonormal functions. The boundary conditions determine the eigenvalues of the eigenfunctions, and the strength of the sources influences the coefficients of these eigenfunctions. Integral forms can also be used to represent the

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Green's function, especially when the eigenvalue spectrum is continuous. All solutions, although different in form, give the same results. The form of the Green's function that is most appropriate will depend on the problem in question. *The representation of the actual source plays a significant role as to which form of the Green's function may be most convenient for a given problem.*

Usually, there is as much work involved in finding the Green's function as there is in obtaining the infinite-series solution. However, the major advantages of the Green's function technique become evident when the same problem is to be solved for a variety of driving sources and when the sources are in the presence of boundaries [1–8].

In this chapter we shall initially study the one-dimensional differential equation

$$[L + \lambda r(x)]y(x) = f(x) \quad (15-1)$$

where L is the Sturm-Liouville operator and λ is a constant. It is hoped that an understanding of the Green's function method for this equation will lead to a better understanding of the equations that occur in electromagnetic field theory applied to homogeneous media such as

$$\nabla^2 \phi(\mathbf{R}) + \beta^2 \phi(\mathbf{R}) = p(\mathbf{R}) \quad (\text{scalar wave equation}) \quad (15-2a)$$

$$\nabla \times \nabla \times \psi(\mathbf{R}) + \beta^2 \psi(\mathbf{R}) = \mathbf{F}(\mathbf{R}) \quad (\text{vector wave equation}) \quad (15-2b)$$

particularly since (15-2a) and (15-2b) can often be reduced to several equations similar to (15-1) by the separation of variables technique. Before proceeding to the actual solution of (15-1) by the Green's function method, we shall first consider several examples of Green's functions in other areas of electrical and general engineering. These will be followed by some topics associated with the Sturm-Liouville operator L before we embark on the solution of (15-2a).

15.2 GREEN'S FUNCTIONS IN ENGINEERING

The Green's function approach to solution of differential equations has been used in many areas of engineering, physics, and elsewhere [9–18]. Before we embark on constructing Green's function solutions to electromagnetic boundary-value problems, let us consider two other problems, one dealing with electric circuit theory and the other with mechanics. This will give the reader a better appreciation of the Green's function concept.

15.2.1 Circuit Theory

Analysis of lumped electric element circuits is a fundamental method of electrical engineering. Therefore, we will relate the Green's function to the solution of a very simple lumped-element circuit problem.

Let us assume that a voltage source $v(t)$ is connected to a resistor R and inductor L , as shown in Figure 15-1a. The equation that governs the solution to that circuit can be written as

$$L \frac{di}{dt} + Ri = v(t) \quad (15-3)$$

where $v(t)$ is the excitation voltage source that is turned on at $t = t'$. Initially, ($t < t'$) the circuit is at rest and at $t = t'$ the voltage is suddenly turned on by an impulse V_0 of a very short duration $\Delta t'$. For $t > t' + \Delta t'$, when $v(t) = 0$, the circuit performance is governed by the homogeneous equation

$$L \frac{di}{dt} + Ri(t) = 0 \quad \text{for } t > t' + \Delta t' \quad (15-4)$$

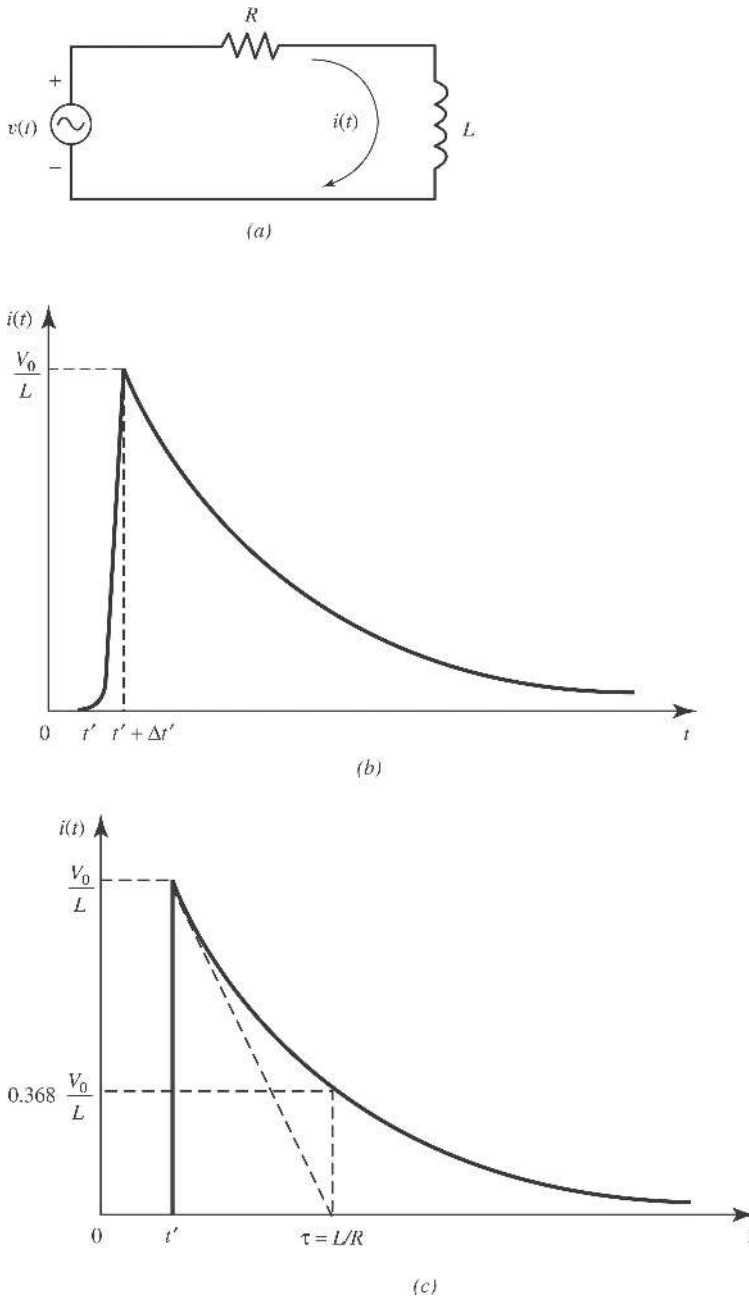


Figure 15-1 (a) RL series circuit. (b) Current response. (c) Time constant.

whose solution for $i(t)$ can be written as

$$i(t) = I_0 e^{-(R/L)t} \quad \text{for } t > t' + \Delta t' \quad (15-4a)$$

where I_0 is a constant and L/R is referred to as the *time constant* τ of the circuit.

Since the voltage excitation $v(t)$ during $\Delta t'$ was an impulse V_0 , then

$$\int_{t'}^{t'+\Delta t'} v(t) dt = V_0 \quad (15-5)$$

where V_0 is the voltage for duration $\Delta t'$ (volts $\cdot$ Δt = volts-sec). Therefore, between $t' \leq t \leq t' + \Delta t'$, (15-3) can be written using (15-5) as

$$\begin{aligned} L \int_{t'}^{t'+\Delta t'} di + R \int_{t'}^{t'+\Delta t'} i(t) dt &= \int_{t'}^{t'+\Delta t'} v(t) dt \\ L[i(t' + \Delta t') - i(t')] + R \int_{t'}^{t'+\Delta t'} i(t) dt &= V_0 \end{aligned} \quad (15-6)$$

Because $\Delta t'$ is very small, we assume that $i(t)$ during the excitation $\Delta t'$ of the voltage source is not exceedingly large, and it behaves as shown in Figure 15-1b. Therefore, during $\Delta t'$

$$\lim_{\Delta t' \rightarrow 0} R \int_{t'}^{t'+\Delta t'} i(t) dt \simeq 0 \quad (15-7)$$

so that the terms on the left side of (15-6) reduce using (15-4a) to

$$i(t') = 0 \quad (15-8a)$$

$$i(t' + \Delta t') = I_0 e^{-(R/L)(t' + \Delta t')} \stackrel{\Delta t' \rightarrow 0}{\simeq} I_0 e^{-(R/L)t'} \quad (15-8b)$$

Using (15-7) through (15-8b), we can express (15-6) as

$$L I_0 e^{-(R/L)t'} = V_0 \quad (15-9)$$

or

$$I_0 = \frac{V_0}{L} e^{+(R/L)t'} \quad (15-9a)$$

Therefore, (15-4a) can be written, using (15-9a), as

$$i(t) = \begin{cases} 0 & t < t' \\ \frac{V_0}{L} e^{-(R/L)(t-t')} & t \geq t' \end{cases} \quad (15-10a)$$

$$i(t) = \begin{cases} 0 & t < t' \\ \frac{V_0}{L} e^{-(R/L)(t-t')} & t \geq t' \end{cases} \quad (15-10b)$$

which is shown plotted in Figure 15-1c.

If the circuit is subjected to N voltage impulses each of duration Δt and amplitude V_i occurring at $t = t'_i$, ($i = 0, \dots, N$), then the current response can be written as

$$i(t) = \begin{cases} 0 & t < t'_0 \\ \frac{V_0}{L} e^{-(R/L)(t-t'_0)} & t'_0 < t < t'_1 \\ \frac{V_0}{L} e^{-(R/L)(t-t'_0)} + \frac{V_1}{L} e^{-(R/L)(t-t'_1)} & t'_1 < t < t'_2 \\ \vdots & \vdots \\ \sum_{i=0}^N \frac{V_i}{L} e^{-(R/L)(t-t'_i)} & t'_N < t < t'_{N+1} \end{cases} \quad (15-11)$$

If the circuit is subjected to a continuous voltage source $v(t)$ starting at t'_0 such that at an instant of time $t = t'$ and short interval $\Delta t'$ would produce an impulse of

$$dV = v(t') dt' \quad (15-12)$$

then the response of the system for $t \geq t'$ can be expressed, provided that $i(t) = v(t) = 0$ for $t < t'$, as

$$i(t) = \int_{t'}^t \left[\frac{v(t') dt'}{L} e^{-(R/L)(t-t')} \right] = \int_{t'}^t v(t') \frac{e^{-(R/L)(t-t')}}{L} dt'$$

$$\boxed{i(t) = \int_{t'}^t v(t') G(t, t') dt'} \quad (15-13)$$

where

$$\boxed{G(t, t') = \frac{e^{-(R/L)(t-t')}}{L} \quad \text{for } t > t'}$$
(15-13a)

In (15-13), $G(t, t')$ of (15-13a), is referred to as the *Green's function*, and it represents the response of the system for $t > t'$ when an excitation voltage $v(t)$ at $t = t'$ is an impulse (Dirac delta) function. Knowing the response of the system to an impulse function, represented by the Green's function of (15-13a), the response $i(t)$ to any voltage source $v(t)$ can then be obtained by convolving the voltage source excitation with the Green's function according to (15-13).

15.2.2 Mechanics

Another problem that the reader may be familiar with is that of a string of length ℓ that is connected at the two ends and is subjected to external force per unit length (load) of $F(x)$. The objective is to find the displacement $u(x)$ of the string. If the load $F(x)$ is assumed to be acting down (negative direction), the displacement $u(x)$ of the string is governed by the differential equation

$$T \frac{d^2 u}{dx^2} = F(x) \quad (15-14)$$

or

$$\frac{d^2 u}{dx^2} = \frac{1}{T} F(x) = f(x) \quad (15-14a)$$

where T is the uniform tensile force of the string. If the string is stationary at the two ends, then the displacement function $u(x)$ satisfies the boundary conditions

$$u(x=0) = u(x=\ell) = 0 \quad (15-15)$$

Initially, instead of solving the displacement $u(x)$ of the string subject to the load $F(x)$, let us assume that the load to which the string is subjected is a concentrated load (impulse) of $F(x=x') = \delta(x-x')$ at a point $x = x'$, as shown in Figure 15-2. For the impulse load, the differential equation 15-14a can be written as

$$\frac{d^2 G(x, x')}{dx^2} = \frac{1}{T} \delta(x - x') \quad (15-16)$$

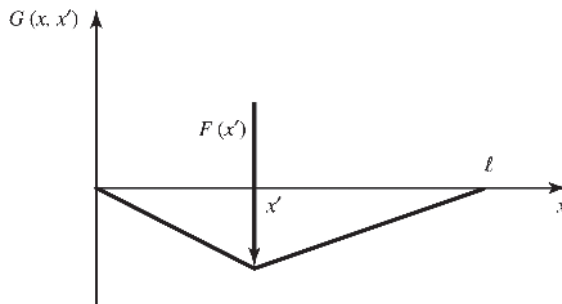


Figure 15-2 Attached string subjected to a load force.

subject to the boundary conditions

$$G(x=0, x') = G(0, x') = 0 \quad (15-16a)$$

$$G(x=\ell, x') = G(\ell, x') = 0 \quad (15-16b)$$

In (15-16), $G(x, x')$ represents the displacement of the string when it is subject to an impulse load of $1/T$ at $x = x'$, and it is referred to as the *Green's function* for the string. Once this is found, the displacement $u(x)$ of the string subjected to the load $F(x)$ can be determined by convolving the load $F(x)$ with the Green's function $G(x, x')$, as was done for the circuit problem by (15-13).

The solution to (15-16) is accomplished by following the procedure outlined here. Away from the load at $x = x'$, the differential equation 15-16 reduces to the homogeneous form

$$\frac{d^2 G(x, x')}{dx^2} = 0 \quad (15-17)$$

which has solutions of the form

$$G(x, x') = \begin{cases} A_1 x + B_1 & 0 \leq x \leq x' \\ A_2 x + B_2 & x' \leq x \leq \ell \end{cases} \quad (15-17a)$$

$$(15-17b)$$

Applying the boundary conditions (15-16a) and (15-16b) leads to

$$G(x=0, x') = A_1(0) + B_1 = 0 \Rightarrow B_1 = 0 \quad (15-18a)$$

$$G(x=\ell, x') = A_2 \ell + B_2 = 0 \Rightarrow B_2 = -A_2 \ell \quad (15-18b)$$

Thus, (15-17a) and (15-17b) reduce to

$$G(x, x') = \begin{cases} A_1 x & 0 \leq x \leq x' \\ A_2 (x - \ell) & x' \leq x \leq \ell \end{cases} \quad (15-19a)$$

$$(15-19b)$$

where A_1 and A_2 have not been determined yet.

At $x = x'$ the displacement $u(x)$ of the string must be continuous. Therefore, the Green's function of (15-19a) and (15-19b) must also be continuous at $x = x'$. Thus,

$$A_1 x' = A_2 (x' - \ell) \Rightarrow A_2 = A_1 \frac{x'}{x' - \ell} \quad (15-20)$$

According to (15-16), the second derivative of $G(x, x')$ is equal to an impulse function. Therefore, the first derivative of $G(x, x')$, obtained by integrating (15-16), must be discontinuous by an amount equal to $1/T$. Thus,

$$\lim_{\varepsilon \rightarrow 0} \left[\frac{dG(x' + \varepsilon, x')}{dx} - \frac{dG(x' - \varepsilon, x')}{dx} \right] = \frac{1}{T} \quad (15-21)$$

or

$$\frac{dG(x'_+, x')}{dx} - \frac{dG(x'_-, x')}{dx} = \frac{1}{T} \quad (15-21a)$$

Using (15-19a), (15-19b), and (15-20), we can write that

$$\frac{dG(x'_-, x')}{dx} = A_1 \quad (15-22a)$$

$$\frac{dG(x'_+, x')}{dx} = A_2 = A_1 \frac{x'}{x' - \ell} \quad (15-22b)$$

Thus, using (15-22a) and (15-22b), (15-21a) leads to

$$A_1 \frac{x'}{x' - \ell} - A_1 = \frac{1}{T} \Rightarrow A_1 \frac{\ell}{x' - \ell} = \frac{1}{T} \Rightarrow A_1 = \frac{1}{T} \frac{x' - \ell}{\ell} \quad (15-23)$$

Therefore, the Green's function of (15-19a) and (15-19b) can be written, using (15-20) and (15-23), as

$$G(x, x') = \begin{cases} \frac{1}{T} \left(\frac{x' - \ell}{\ell} \right) x & 0 \leq x \leq x' \\ \frac{1}{T} \left(\frac{x - \ell}{\ell} \right) x' & x' \leq x \leq \ell \end{cases} \quad (15-24a)$$

$$(15-24b)$$

The displacement $u(x)$ subject to the load $F(x)$, governed by (15-14a), can now be written as

$$\begin{aligned} u(x) &= \int_0^\ell F(x') G(x, x') dx' \\ &= \frac{1}{T} \int_0^x F(x') \left(\frac{x' - \ell}{\ell} \right) x dx' \\ &\quad + \frac{1}{T} \int_x^\ell F(x') \left(\frac{x - \ell}{\ell} \right) x' dx' \end{aligned} \quad (15-25)$$

15.3 STURM-LIOUVILLE PROBLEMS

Now that we illustrated the Green's function development for two specific problems, one for an *RL* electrical circuit and the other for a stretched string, let us consider the construction of Green's functions for more general differential equations subject to appropriate boundary conditions. Specifically in this section we want to consider Green's functions for the one-dimensional differential equation of the *Sturm-Liouville* form [1, 7, 13].

A one-dimensional differential equation of the form

$$\frac{d}{dx} \left[p(x) \frac{dy}{dx} \right] - q(x)y = f(x) \quad (15-26)$$

subject to homogeneous boundary conditions, is a *Sturm-Liouville* problem. This equation can also be written as

$$Ly = f(x) \quad (15-27)$$

where L is the Sturm-Liouville operator

$$L \equiv \left\{ \frac{d}{dx} \left[p(x) \frac{d}{dx} \right] - q(x) \right\} \quad (15-27a)$$

Every general one-dimensional, source-excited, second-order differential equation of the form

$$A(x) \frac{d^2 y}{dx^2} + B(x) \frac{dy}{dx} + C(x)y = S(x) \quad (15-28)$$

or

$$Dy = S(x) \quad (15-28a)$$

where

$$D \equiv \left[A(x) \frac{d^2}{dx^2} + B(x) \frac{d}{dx} + C(x) \right] \quad (15-28b)$$

can be converted to a Sturm-Liouville form. This can be accomplished by following the procedure outlined here.

First, expand (15-26) and write it as

$$p(x) \frac{d^2 y}{dx^2} + \frac{dp}{dx} \frac{dy}{dx} - q(x)y = f(x) \quad (15-29)$$

Dividing (15-28) by $A(x)$ and (15-29) by $p(x)$, we have that

$$\frac{d^2 y}{dx^2} + \frac{B(x)}{A(x)} \frac{dy}{dx} + \frac{C(x)}{A(x)} y = \frac{S(x)}{A(x)} \quad (15-30a)$$

$$\frac{d^2 y}{dx^2} + \frac{1}{p(x)} \frac{dp}{dx} \frac{dy}{dx} - \frac{q(x)}{p(x)} y = \frac{f(x)}{p(x)} \quad (15-30b)$$

Comparing (15-30a) and (15-30b), we see that

$$\frac{B(x)}{A(x)} = \frac{1}{p(x)} \frac{dp(x)}{dx} \quad (15-31a)$$

$$\frac{C(x)}{A(x)} = -\frac{q(x)}{p(x)} \quad (15-31b)$$

$$\frac{S(x)}{A(x)} = \frac{f(x)}{p(x)} \quad (15-31c)$$

From (15-31a) we have that

$$\frac{dp(x)}{dx} = p(x) \frac{B(x)}{A(x)} \quad (15-32)$$

which is a linear first-order differential equation, a particular solution of which is

$$p(x) = \exp \left[\int^x \frac{B(t)}{A(t)} dt \right] \quad (15-32a)$$

From (15-31b)

$$q(x) = -p(x) \frac{C(x)}{A(x)} = -\frac{C(x)}{A(x)} \exp \left[\int^x \frac{B(t)}{A(t)} dt \right] \quad (15-32b)$$

and from (15-31c)

$$f(x) = p(x) \frac{S(x)}{A(x)} = \frac{S(x)}{A(x)} \exp \left[\int^x \frac{B(t)}{A(t)} dt \right] \quad (15-32c)$$

In summary then, a one-dimensional, source-excited, second-order differential equation of the form (15-28) is converted to a Sturm-Liouville form (15-26) by letting $p(x)$ be that of (15-32a), $q(x)$ that of (15-32b), and $f(x)$ that of (15-32c).

To demonstrate, let us consider an example.

Example 15-1

Convert the Bessel differential equation

$$x^2 \frac{d^2 y}{dx^2} + x \frac{dy}{dx} + (x^2 - \lambda^2)y = 0$$

to a Sturm-Liouville form.

Solution: Since

$$A(x) = x^2$$

$$B(x) = x$$

$$C(x) = x^2 - \lambda^2$$

$$S(x) = 0$$

and according to (15-32a), (15-32b), and (15-32c)

$$p(x) = \exp \left[\int^x \frac{B(t)}{A(t)} dt \right] = \exp \left[\int^x \frac{t}{t^2} dt \right] = \exp \left[\int^x \frac{dt}{t} \right] = e^{\ln(x)} = x$$

$$q(x) = -p(x) \frac{C(x)}{A(x)} = -x \frac{(x^2 - \lambda^2)}{x^2} = - \left(\frac{x^2 - \lambda^2}{x} \right)$$

$$f(x) = p(x) \frac{S(x)}{A(x)} = 0$$

Thus, using (15-26), Bessel's differential equation takes the Sturm-Liouville form

$$\frac{d}{dx} \left(x \frac{dy}{dx} \right) + \left(\frac{x^2 - \lambda^2}{x} \right) y = 0$$

As a check, when the preceding equation is expanded and is multiplied by x , it reduces to the usual form of Bessel's differential equation.

15.3.1 Green's Function in Closed Form

Now that we have shown that each general second-order, source-excited differential equation can be converted to a Sturm-Liouville form, let us develop a procedure to construct the Green's function of a Sturm-Liouville differential equation represented by (15-26) or more generally by

$$\left\{ \frac{d}{dx} \left[p(x) \frac{dy}{dx} \right] - q(x)y \right\} + \lambda r(x)y = f(x) \quad (15-33)$$

which can be written as

$$\left[\left\{ \frac{d}{dx} \left[p(x) \frac{d}{dx} \right] - q(x) \right\} + \lambda r(x) \right] y = f(x) \quad (15-33a)$$

or simply

$$[L + \lambda r(x)]y = f(x) \quad (15-33b)$$

where L is the Sturm-Liouville operator of (15-27a). In (15-33), $r(x)$ and $f(x)$ are assumed to be piecewise continuous in the region of interest ($a \leq x \leq b$) and λ is a parameter to be determined

by the nature and boundary of the region of interest. *It should be noted that throughout this chapter λ is used to represent eigenvalues, and it should not be confused with wavelength. The use of λ to represent eigenvalues is a very common practice in Green's function theory.* The differential equations 15-33 through 15-33b possess a Green's function for all values of λ except those that are eigenvalues of the homogeneous equation

$$[L + \lambda r(x)]y = 0 \quad (15-34)$$

For values of λ for which (15-34) has nontrivial solutions, a Green's function will not exist. This is analogous to a system of linear equations represented by

$$Dy = f \quad (15-35)$$

which has a solution of

$$y = D^{-1}f \quad (15-35a)$$

provided D^{-1} exists (i.e., D nonsingular). If D is singular (so D^{-1} does not exist), then (15-35) does not possess a solution. This occurs when $Dy = 0$, which has a nontrivial solution when the determinant of D is zero [so D is written $\det(D) = 0$].

According to (15-25), the solution of (15-33b) can be written as

$$y(x) = \int_a^b f(x')G(x, x')dx' \quad (15-36)$$

where $G(x, x')$ is the Green's function of (15-33) or (15-33b). Since (15-35a) is a solution to (15-35), and it exists only if D is nonsingular, then (15-36) is a solution to (15-33b) if the inverse of the operator $[L + \lambda r(x)]$ exists. Then (15-33b) can be written as

$$y(x) = [L + \lambda r(x)]^{-1}f \quad (15-36a)$$

By comparing (15-36) to (15-36a), then $G(x, x')$ is analogous to the inverse of the operator $[L + \lambda r(x)]$.

Whenever λ is equal to an eigenvalue of the operator $[L + \lambda r(x)]$, obtained by setting the determinant of (15-34) equal to zero, i.e.,

$$\det[L + \lambda r(x)] = 0 \quad (15-37)$$

then the inverse of $[L + \lambda r(x)]$ does not exist, and (15-36) and (15-36a) are not valid. Thus, for values of λ equal to the eigenvalues of the operator $[L + \lambda r(x)]$, the Green's function does not exist.

For a unit impulse driving function, the Sturm-Liouville equation 15-33 can be written as

$$\frac{d}{dx} \left[p(x) \frac{dG}{dx} \right] - q(x)G + \lambda r(x)G = \delta(x - x') \quad (15-38)$$

where G is the Green's function. At points removed from the impulse driving function, (15-38) reduces to

$$\left\{ \frac{d}{dx} \left[p(x) \frac{dG}{dx} \right] - q(x)G \right\} + \lambda r(x)G = 0 \quad (15-38a)$$

As can be verified by the Green's function (15-24a) and (15-24b) of the mechanics problems in Section 15.2.2, the Green's functions of (15-38a), in general, exhibit the following properties:

Properties of Green's Functions

1. $G(x, x')$ satisfies the *homogeneous* differential equation *except* at $x = x'$.
2. $G(x, x')$ is symmetrical with respect to x and x' .
3. $G(x, x')$ satisfies certain *homogeneous* boundary conditions.
4. $G(x, x')$ is continuous at $x = x'$.
5. $[dG(x, x')]/dx$ has a discontinuity of $1/[p(x')]$ at $x = x'$.

The discontinuity of the derivative of $G(x, x')$ at $x = x'$ ($[dG(x, x')]/dx = 1/[p(x')]$) can be derived by first integrating the differential equation 15-38 between $x = x' - \varepsilon$ and $x = x' + \varepsilon$. Doing this leads to

$$\begin{aligned} \lim_{\varepsilon \rightarrow 0} \left\{ \int_{x' - \varepsilon}^{x' + \varepsilon} \frac{d}{dx} \left[p(x) \frac{dG(x, x')}{dx} \right] dx + \int_{x' - \varepsilon}^{x' + \varepsilon} [-q(x) + \lambda r(x)] G(x, x') dx \right\} \\ = \int_{x' - \varepsilon}^{x' + \varepsilon} \delta(x - x') dx \\ \lim_{\varepsilon \rightarrow 0} \left\{ p(x) \frac{dG(x, x')}{dx} \Big|_{x' - \varepsilon}^{x' + \varepsilon} + \int_{x' - \varepsilon}^{x' + \varepsilon} [-q(x) + \lambda r(x)] G(x, x') dx \right\} = 1 \end{aligned} \quad (15-39)$$

Since $q(x)$, $r(x)$, and $G(x, x')$ are continuous at $x = x'$, then

$$\lim_{\varepsilon \rightarrow 0} \int_{x' - \varepsilon}^{x' + \varepsilon} [-q(x) + \lambda r(x)] G(x, x') dx = 0 \quad (15-40)$$

Using (15-40) reduces (15-39) to

$$\begin{aligned} \lim_{\varepsilon \rightarrow 0} \left\{ p(x) \left[\frac{dG(x' + \varepsilon, x')}{dx} - \frac{dG(x' - \varepsilon, x')}{dx} \right] \right\} = 1 \\ p(x) \left[\frac{dG(x'_+, x')}{dx} - \frac{dG(x'_-, x')}{dx} \right] = 1 \end{aligned} \quad (15-41)$$

or

$$\frac{dG(x'_+, x')}{dx} - \frac{dG(x'_-, x')}{dx} = \frac{1}{p(x)} \quad (15-41a)$$

which proves the discontinuity of the derivative of $G(x, x')$ at $x = x'$.

The Green's function must satisfy the differential equation 15-38a, the five general properties listed previously, and the appropriate boundary conditions. We propose to construct the Green's function solution in two parts: one that is valid for $a \leq x \leq x'$ and the other for $x' \leq x \leq b$ where a and b are the limits of the region of interest. For the homogeneous equation 15-33, valid at all points except $x = x'$:

1. Let $y_1(x)$ represent a nontrivial solution of the homogeneous differential equation 15-33 in the interval $a \leq x < x'$ satisfying the boundary conditions at $x = a$. Since both $y_1(x)$ and $G(x, x')$ satisfy the same differential equation in the interval $a \leq x < x'$, they are related to each other by a constant, that is,

$$G(x, x') = A_1 y_1(x), \quad a \leq x < x' \quad (15-42a)$$

2. Let $y_2(x)$ represent a nontrivial solution of the homogeneous differential equation of 15-33 in the interval $x' < x \leq b$ satisfying the boundary conditions at $x = b$. Since both $y_2(x)$ and $G(x, x')$ satisfy the same differential equation in the interval $x' < x \leq b$, they are related to each other by a constant, that is

$$G(x, x') = A_2 y_2(x), \quad x' < x \leq b \quad (15-42b)$$

Since one of the general properties of the Green's function is that it must be continuous at $x = x'$, then using (15-42a) and (15-42b)

$$A_1 y_1(x') = A_2 y_2(x') \Rightarrow -A_1 y_1(x') + A_2 y_2(x') = 0 \quad (15-43a)$$

Also one of the properties of the derivative of the Green's function is that it must be discontinuous at $x = x'$ by an amount of $1/p(x')$. Thus applying (15-42a) and (15-42b) into (15-41a) leads to

$$-A_1 y_1'(x') + A_2 y_2'(x') = \frac{1}{p(x')} \quad (15-43b)$$

Solving (15-43a) and (15-43b) simultaneously leads to

$$A_1 = \frac{y_2(x')}{p(x')W(x')} \quad (15-44a)$$

$$A_2 = \frac{y_1(x')}{p(x')W(x')} \quad (15-44b)$$

where $W(x')$ is the Wronskian of y_1 and y_2 at $x = x'$, defined as

$$W(x') \equiv y_1(x') y_2'(x') - y_2(x') y_1'(x') \quad (15-44c)$$

Using (15-44a) through (15-44c), the closed form Green's function of (15-42a) and (15-42b) for the differential equation 15-33 or (15-38) can be written as

$$G(x, x') = \begin{cases} \frac{y_2(x')}{p(x')W(x')} y_1(x), & a \leq x \leq x' \\ \frac{y_1(x')}{p(x')W(x')} y_2(x), & x' \leq x \leq b \end{cases} \quad (15-45a)$$

$$(15-45b)$$

where $y_1(x)$ and $y_2(x)$ are two independent solutions of the homogeneous form of the differential equation 15-33, each satisfying, respectively, the boundary conditions at $x = a$ and $x = b$.

The preceding recipe can be used to construct in closed form the Green's function of a differential equation of the form (15-33). It is convenient to use this procedure when the following conditions are satisfied.

1. The solution to the homogeneous differential equation is known.
2. The Green's function is desired in closed form, instead of an infinite series of orthonormal functions, which will be shown in the next section.

If this procedure is used for the mechanics problem of (15-14a), Section 15.2.2, the same answer [as given by (15-24a) and (15-24b)] will be obtained. In the next section we want to present an alternate procedure for constructing the Green's function. By this other method, the Green's function will be represented by an infinite series of orthonormal functions. Whether one form of

the Green's function is more suitable than the other will depend on the problem in question. Remember, however, that the closed-form procedure just derived can only be used provided the solution to the homogeneous differential equation is known.

Before we proceed, let us illustrate that the Sturm-Liouville operator L exhibits *Hermitian* (or *symmetrical*) properties [1]. These are very important, and they establish the relationships that are used in the construction of the Green's function.

Example 15-2

Show that the Sturm-Liouville operator L of (15-27a) or (15-33) through (15-33b) exhibits Hermitian (symmetrical) properties. Assume that in the interval $a \leq x \leq b$ any solution $y_i(x)$ to (15-33b) satisfies the boundary conditions (where $i = 1, 2, \dots, n$)

$$\begin{aligned}\alpha_1 y_i(x=a) + \alpha_2 \frac{dy_i(x=a)}{dx} &= \alpha_1 y_i(a) + \alpha_2 y_i'(a) = 0 \\ \beta_1 y_i(x=b) + \beta_2 \frac{dy_i(x=b)}{dx} &= \beta_1 y_i(b) + \beta_2 y_i'(b) = 0\end{aligned}$$

Solution: Let us assume that $y_1(x)$ and $y_2(x)$ are two solutions to (15-33) through (15-33b) each satisfying the boundary conditions. Then, according to (15-27a)

$$\begin{aligned}Ly_1(x) &= \frac{d}{dx} \left[p(x) \frac{dy_1(x)}{dx} \right] - q(x)y_1(x) \\ Ly_2(x) &= \frac{d}{dx} \left[p(x) \frac{dy_2(x)}{dx} \right] - q(x)y_2(x)\end{aligned}$$

Multiplying the first by $y_2(x)$ and the second by $y_1(x)$, we can write each using a shorthand notation as

$$\begin{aligned}y_2 Ly_1 &= y_2 (py_1')' - y_2 q y_1 \\ y_1 Ly_2 &= y_1 (py_2')' - y_1 q y_2\end{aligned}$$

where ' indicates d/dx . Subtracting the two and integrating between a and b leads to

$$\int_a^b (y_2 Ly_1 - y_1 Ly_2) dx = \int_a^b [y_2 (py_1')' - y_1 (py_2')'] dx$$

Since

$$(y_2 py_1')' = y_2 (py_1')' + y_2' py_1'$$

and

$$(y_1 py_2')' = y_1 (py_2')' + y_1' py_2'$$

then by subtracting the two,

$$(y_2 py_1') - (y_1 py_2')' = y_2 (py_1')' - y_1 (py_2')'$$

Thus, the integral reduces to

$$\int_a^b (y_2 Ly_1 - y_1 Ly_2) dx = \int_a^b [(y_2 py_1')' - (y_1 py_2')'] dx = [p(y_2 y_1' - y_1 y_2')]_a^b$$

Each of the solutions, $y_1(x)$ and $y_2(x)$, satisfies the same boundary conditions that can be written as

$$\begin{aligned}\alpha_1 y_1(a) + \alpha_2 y_1'(a) &= 0 \Rightarrow \alpha_1 y_1(a) = -\alpha_2 y_1'(a) \\ \beta_1 y_1(b) + \beta_2 y_1'(b) &= 0 \Rightarrow \beta_1 y_1(b) = -\beta_2 y_1'(b) \\ \alpha_1 y_2(a) + \alpha_2 y_2'(a) &= 0 \Rightarrow \alpha_1 y_2(a) = -\alpha_2 y_2'(a) \\ \beta_1 y_2(b) + \beta_2 y_2'(b) &= 0 \Rightarrow \beta_1 y_2(b) = -\beta_2 y_2'(b)\end{aligned}$$

Dividing the first by the third and the second by the fourth, we can write that

$$\frac{y_1(a)}{y_2(a)} = \frac{y_1'(a)}{y_2'(a)} \Rightarrow y_1(a)y_2'(a) = y_2(a)y_1'(a)$$

$$\frac{y_1(b)}{y_2(b)} = \frac{y_1'(b)}{y_2'(b)} \Rightarrow y_1(b)y_2'(b) = y_2(b)y_1'(b)$$

Using these relations, it is apparent that the right side of the previous integral equation vanishes, so we can write it as

$$\int_a^b (y_2 L y_1 - y_1 L y_2) dx = [p(y_2 y_1' - y_1 y_2')]_a^b = 0$$

or

$$\boxed{\int_a^b (y_2 L y_1) dx = \int_a^b (y_1 L y_2) dx}$$

This illustrates that the operator L exhibits Hermitian (symmetrical) properties with respect to the solutions $y_1(x)$ and $y_2(x)$.

15.3.2 Green's Function in Series

The procedure outlined in the previous section can only be used to derive in closed form the Green's function for differential equations whose homogeneous form solution is known. Otherwise, other techniques must be used. Even for equations whose homogeneous form solution is known, the closed-form representation of the Green's function may not be the most convenient one. Therefore, an alternate representation may be attractive even for those cases.

An alternate form of the Green's function is to represent it as a series of orthonormal functions. The most appropriate orthonormal functions would be those that satisfy the boundary conditions. To demonstrate the procedure, let us initially rederive the Green's function of the mechanics problem of Section 15.2.2 but this time represented as a series of orthonormal functions. We will then generalize the method to (15-33) through (15-33b).

A. Vibrating String For the differential equation 15-14a, subject to the boundary conditions (15-15), its Green's function must satisfy (15-16) subject to the boundary conditions (15-16a) and (15-16b). Since the Green's function $G(x, x')$ must vanish at $x = 0$ and ℓ , it is most convenient to represent $G(x, x')$ as an infinite series of $\sin(n\pi x / \ell)$ orthonormal functions, that is

$$G(x, x') = \sum_{n=1}^{\infty} a_n(x') \sin\left(\frac{n\pi}{\ell} x\right) \quad (15-46)$$

where $a_n(x')$ represents the amplitude expansion coefficients that will be a function of the position x' of the excitation source.

Substituting (15-46) into (15-16), multiplying both sides by $\sin(m\pi x / \ell)$, and then integrating in x from 0 to ℓ leads to

$$-\sum \left(\frac{n\pi}{\ell}\right)^2 a_n(x') \int_0^{\ell} \sin\left(\frac{n\pi}{\ell} x\right) \sin\left(\frac{m\pi}{\ell} x\right) dx = \frac{1}{T} \int_0^{\ell} \delta(x - x') \sin\left(\frac{m\pi}{\ell} x\right) dx \quad (15-47)$$

Because the orthogonality conditions of sine functions state that

$$\int_0^\ell \sin\left(\frac{n\pi}{\ell}x\right) \sin\left(\frac{m\pi}{\ell}x\right) dx = \begin{cases} \frac{\ell}{2} & m = n \\ 0 & m \neq n \end{cases} \quad (15-48a)$$

then (15-47) reduces to

$$-\left(\frac{n\pi}{\ell}\right)^2 \frac{\ell}{2} a_n(x') = \frac{1}{T} \sin\left(\frac{n\pi}{\ell}x'\right) \quad (15-49)$$

that is,

$$a_n(x') = -\frac{2\ell}{\pi^2 T} \frac{1}{n^2} \sin\left(\frac{n\pi}{\ell}x'\right) \quad (15-49a)$$

Thus, the Green's function (15-46) can be expressed, using (15-49a), as

$$G(x, x') = -\frac{2\ell}{\pi^2 T} \sum_{n=1}^{\infty} \frac{1}{n^2} \sin\left(\frac{n\pi}{\ell}x'\right) \sin\left(\frac{n\pi}{\ell}x\right) \quad (15-50)$$

This is an alternate form to (15-24a) and (15-24b), but one that leads to the same results, even though its form looks quite different. The displacement $u(x)$ subject to the load $F(x)$, governed by (15-14a), can now be written as

$$u(x) = \int_0^\ell F(x') G(x, x') dx' = -\frac{2\ell}{\pi^2 T} \sum_{n=1}^{\infty} \frac{1}{n^2} \sin\left(\frac{n\pi}{\ell}x\right) \int_0^\ell F(x') \sin\left(\frac{n\pi}{\ell}x'\right) dx' \quad (15-51)$$

B. Sturm-Liouville Operator Let us now generalize the Green's function series expansion method of the vibrating string, as given by (15-50) and (15-51), to (15-33b) where L is a Sturm-Liouville operator and λ is an arbitrary parameter to be determined by the nature and boundary of the region of interest. We seek a solution to solve the differential equation 15-33b

$$[L + \lambda r(x)]y(x) = f(x) \quad (15-52)$$

in the interval $a \leq x \leq b$ subject to the general boundary conditions

$$\alpha_1 y(x)|_{x=a} + \alpha_2 \frac{dy(x)}{dx} \Big|_{x=a} = \alpha_1 y(a) + \alpha_2 \frac{dy(a)}{dx} = 0 \quad (15-52a)$$

$$\beta_1 y(x)|_{x=b} + \beta_2 \frac{dy(x)}{dx} \Big|_{x=b} = \beta_1 y(b) + \beta_2 \frac{dy(b)}{dx} = 0 \quad (15-52b)$$

which are usually referred to as the *mixed* boundary conditions. In (15-52a) at least one of the constants α_1 or α_2 , if not both of them, is nonzero. The same is true for (15-52b). The Green's function $G(x, x')$, if it exists, will satisfy the differential equation

$$[L + \lambda r(x)] G(x, x') = \delta(x - x') \quad (15-53)$$

subject to the boundary conditions

$$\alpha_1 G(a, x') + \alpha_2 \frac{dG(a, x')}{dx} = 0 \quad (15-53a)$$

$$\beta_1 G(b, x') + \beta_2 \frac{dG(b, x')}{dx} = 0 \quad (15-53b)$$

If $\{\psi_n(x)\}$ represents a complete set of orthonormal eigenfunctions for the Sturm-Liouville operator L , then it must satisfy the differential equation

$$\boxed{[L + \lambda_n r(x)]\psi_n(x) = 0} \quad (15-54)$$

subject to the same boundary conditions of (15-52a) or (15-52b) for the Green's function (15-53), that is,

$$\alpha_1 \psi_n(a) + \alpha_2 \frac{d\psi_n(a)}{dx} = 0 \quad (15-54a)$$

$$\beta_1 \psi_n(b) + \beta_2 \frac{d\psi_n(b)}{dx} = 0 \quad (15-54b)$$

The boundary conditions (15-54a) and (15-54b) of $\psi_n(x)$ are also used to determine the eigenvalues λ_n . In the finite interval $a \leq x \leq b$ the complete set of orthonormal eigenfunctions $\{\psi_n(x)\}$, and their amplitude coefficients, must satisfy the orthogonality condition of

$$\boxed{\int_a^b \psi_m(x) \psi_n(x) r(x) dx = \delta_{mn} = \begin{cases} 1 & m = n \\ 0 & m \neq n \end{cases}} \quad (15-55)$$

where δ_{mn} is the Kronecker delta function.

If the Green's function exists, it can be represented in series form in terms of the orthonormal eigenfunctions $\{\psi_n(x)\}$ as

$$G(x, x') = \sum_n a_n(x') \psi_n(x) \quad (15-56)$$

where $a_n(x')$ are the amplitude coefficients. These can be obtained by multiplying both sides of (15-56) by $\psi_m(x)r(x)$, integrating from a to b , and then using (15-55). It can be shown that

$$a_n(x') = \int_a^b G(x, x') \psi_n(x) r(x) dx \quad (15-56a)$$

Since $G(x, x')$ satisfies (15-53) and $\psi_n(x)$ satisfies (15-54), then we can rewrite each as

$$LG(x, x') = -\lambda r(x)G(x, x') + \delta(x - x') \quad (15-57a)$$

$$L\psi_n(x) = -\lambda_n r(x)\psi_n(x, x') \quad (15-57b)$$

Multiplying (15-57a) by $\psi_n(x)$, (15-57b) by $G(x, x')$, and then subtracting the two equations leads to

$$\psi_n(x) LG(x, x') - G(x, x') L\psi_n(x) = -(\lambda - \lambda_n) G(x, x') \psi_n(x) r(x) + \delta(x - x') \psi_n(x) \quad (15-58)$$

Integrating (15-58) between a and b , we can write that

$$\begin{aligned} \int_a^b [\psi_n(x) L G(x, x') - G(x, x') L \psi_n(x)] dx \\ = -(\lambda - \lambda_n) \int_a^b G(x, x') \psi_n(x) r(x) dx + \int_a^b \delta(x - x') \psi_n(x) dx \end{aligned} \quad (15-59)$$

which by using (15-56a) reduces to

$$\int_a^b [\psi_n(x) L G(x, x') - G(x, x') L \psi_n(x)] dx = -(\lambda - \lambda_n) a_n(x') + \psi_n(x') \quad (15-59a)$$

By the symmetrical (Hermitian) property of the operator L , as derived in Example 15-2, with respect to the functions $y_1 = G(x, x')$ and $y_2 = \psi_n(x)$, the left side of (15-59) or (15-59a) vanishes. Therefore, (15-59) and (15-59a) reduce to

$$-(\lambda - \lambda_n) a_n(x') + \psi_n(x') = 0 \quad (15-60)$$

and therefore,

$$a_n(x') = \frac{\psi_n(x')}{\lambda - \lambda_n}, \quad \text{where } \lambda \neq \lambda_n \quad (15-60a)$$

Thus, the series form of the Green's function (15-56) can ultimately be expressed as

$$G(x, x') = \sum_n \frac{\psi_n(x') \psi_n(x)}{(\lambda - \lambda_n)} \quad (15-61)$$

where $\{\psi_n(z)\}$ represents a complete set of orthonormal eigenfunctions for the Sturm-Liouville operator L , which satisfies the differential equation 15-54 subject to the boundary conditions (15-54a) and (15-54b). This is also referred to as the *bilinear* formula, and it represents, aside from (15-45a) and (15-45b), the second form that can be used to derive the Green's function for the differential equation 15-33 or 15-33b as a series solution in the finite interval $a \leq x \leq b$. It should be noted that at $\lambda = \lambda_n$ the Green's function of (15-61) possesses singularities. Usually, these singularities are simple poles although in some cases the λ_n 's are branch points whose branch cuts represent a continuous spectrum of eigenvalues. In those cases the Green's function may involve a summation, for the discrete spectrum of eigenvalues, and an integral, for the continuous spectrum of eigenvalues.

Example 15-3

A very common differential equation in solutions of transmission-line and antenna problems (such as metallic waveguides, microstrip antennas, etc.) that exhibit rectangular configurations is

$$\frac{d^2 \varphi(x)}{dx^2} + \beta^2 \varphi(x) = f(x)$$

subject to the boundary conditions

$$\varphi(0) = \varphi(\ell) = 0$$

where $\beta^2 = \omega^2 \mu \varepsilon$. Derive in closed and series forms the Green's functions for the given equation.

Solution: For the given equation, the Green's function must satisfy the differential equation

$$\frac{d^2 G(x, x')}{dx^2} + \beta^2 G(x, x') = \delta(x - x')$$

subject to the boundary conditions

$$G(0) = G(\ell) = 0$$

The differential equation is of the Sturm-Liouville form (15-33) with

$$\left. \begin{aligned} p(x) &= 1 \\ q(x) &= 0 \\ r(x) &= 1 \\ \lambda &= \beta^2 \\ y(x) &= \varphi(x) \\ L &= \frac{d^2}{dx^2} \end{aligned} \right\} \Rightarrow \frac{d^2 \varphi}{dx^2} + \beta^2 \varphi = f(x)$$

A. Closed-Form Solution: This form of the solution will be obtained using the recipe of (15-45a) and (15-45b) along with (15-44c). The homogeneous differential equation for $\varphi(x)$ reduces to

$$\frac{d^2 \varphi}{dx^2} + \beta^2 \varphi = 0$$

Two independent solutions, one $\phi_1(x)$ valid in the interval $0 \leq x \leq x'$ and that vanishes at $x = 0$, and the other $\phi_2(x)$ valid in the interval $x' \leq x \leq \ell$ and that vanishes at $x = \ell$, take the form

$$\begin{aligned} \phi_1(x) &= \sin(\beta x) \\ \phi_2(x) &= \sin[\beta(\ell - x)] \end{aligned}$$

According to (15-44c), the Wronskian can be written as

$$\begin{aligned} W(x') &= -\beta \{ \sin(\beta x') \cos[\beta(\ell - x')] + \sin[\beta(\ell - x')] \cos(\beta x') \} \\ W(x') &= -\beta \sin(\beta x' + \beta\ell - \beta x') = -\beta \sin(\beta\ell) \end{aligned}$$

Thus, the Green's function in closed form can be expressed, using (15-45a) and (15-45b), as

$$G(x, x') = \begin{cases} -\frac{\sin[\beta(\ell - x')]}{\beta \sin(\beta\ell)} \sin(\beta x), & 0 \leq x \leq x' \\ -\frac{\sin(\beta x')}{\beta \sin(\beta\ell)} \sin[\beta(\ell - x)], & x' \leq x \leq \ell \end{cases}$$

This form of the Green's function indicates that it possesses singularities (poles) when

$$\beta\ell = \beta_r \ell = n\pi \Rightarrow \beta_r = \omega_r \sqrt{\mu\varepsilon} = 2\pi f_r \sqrt{\mu\varepsilon} = \frac{n\pi}{\ell}$$

that is,

$$f_r = \frac{n}{2\ell\sqrt{\mu\varepsilon}} \quad n = 1, 2, 3, \dots$$

B. Series-Form Solution: This form of the solution will be obtained using (15-61). The complete set of eigenfunctions $\{\psi_n(x)\}$ must satisfy the differential equation 15-54, or

$$\frac{d^2\psi_n(x)}{dx^2} + \beta_n^2\psi_n(x) = 0$$

subject to the boundary conditions

$$\psi_n(0) = \psi_n(\ell) = 0$$

The most appropriate solution of $\psi_n(x)$ is to represent it in terms of standing wave eigenfunctions which in a rectangular coordinate system are sine and cosine functions, as discussed in Chapters 3 and 8, Sections 3.4.1 and 8.2.1.

Thus, according to (3-28b) and (8-4a), we can write that

$$\psi_n(x) = A \cos(\beta_n x) + B \sin(\beta_n x)$$

The allowable eigenvalues of β_n are found by applying the boundary conditions. Since $\psi_n(0) = 0$, then

$$\psi_n(0) = A + B(0) = 0 \Rightarrow A = 0$$

Also since $\psi_n(\ell) = 0$, then

$$\psi_n(\ell) = B \sin(\beta_n \ell) = 0 \Rightarrow \beta_n \ell = \sin^{-1}(0) = n\pi$$

that is,

$$\beta_n = \frac{n\pi}{\ell}, \quad n = 1, 2, 3, \dots \quad (\text{for nontrivial solutions})$$

Thus,

$$\psi_n(x) = B \sin(\beta_n x) = B \sin\left(\frac{n\pi}{\ell} x\right)$$

The amplitude constant B is such that (15-55) is satisfied. Therefore,

$$B^2 \int_0^\ell \sin^2\left(\frac{n\pi}{\ell} x\right) dx = 1$$

$$\frac{B^2}{2} \int_0^\ell \left[1 - \cos\left(\frac{2n\pi}{\ell} x\right)\right] dx = B^2 \left(\frac{\ell}{2}\right) = 1 \Rightarrow B = \sqrt{\frac{2}{\ell}}$$

Thus, the complete set of the orthonormal eigenfunctions of $\{\psi_n(x)\}$ is represented by

$$\psi_n(x) = \sqrt{\frac{2}{\ell}} \sin\left(\frac{n\pi}{\ell} x\right), \quad n = 1, 2, 3, \dots$$

with

$$\lambda = \beta^2 = \omega^2 \mu \varepsilon$$

$$\lambda_n = \beta_n^2 = \left(\frac{n\pi}{\ell}\right)^2, \quad n = 1, 2, 3, \dots$$

In series-form, the Green's function (15-61) can then be written as

$$G(x, x') = \frac{2}{\ell} \sum_{n=1}^{\infty} \frac{\sin\left(\frac{n\pi}{\ell} x'\right) \sin\left(\frac{n\pi}{\ell} x\right)}{\beta^2 - \left(\frac{n\pi}{\ell}\right)^2}$$

which yields the same results as the closed-form solution of part A, even though it looks quite different analytically. It is apparent that the Green's function is symmetrical. Also it possesses a singularity, and it fails to exist when

$$\beta = \beta_r = \omega_r \sqrt{\mu\varepsilon} = 2\pi f_r \sqrt{\mu\varepsilon} = \frac{n\pi}{\ell}$$

that is,

$$f_r = \frac{n\pi/\ell}{2\pi\sqrt{\mu\varepsilon}} = \frac{n}{2\ell\sqrt{\mu\varepsilon}}$$

which is identical to the condition obtained by the closed-form solution in part A.

This is in accordance with (15-34), which states that the Green's function (15-33b) exists for all values of λ , in this case $\lambda = \beta^2 = \omega^2 \mu\varepsilon$, except those that are eigenvalues of the homogeneous equation 15-34. For our case (15-34) reduces to

$$\frac{d^2\phi(x)}{dx^2} + \beta^2\phi(x) = 0$$

whose nontrivial solution takes the form

$$\phi(x) = C \sin\left(\frac{n\pi}{\ell}x\right)$$

with eigenvalues $\beta = n\pi/\ell$, $n = 1, 2, 3, \dots$

It should be noted that when

$$\beta = \beta_r = \omega_r \sqrt{\mu\varepsilon} = 2\pi f_r \sqrt{\mu\varepsilon} = \frac{n\pi}{\ell} \Rightarrow f_r = \frac{n}{2\ell\sqrt{\mu\varepsilon}}$$

the Green's function singularity consists of simple poles. At those frequencies the external frequencies of the source match the natural (characteristic) frequencies of the system, in this instance the transmission line. This is referred to as *resonance*. When this occurs, the field of the mode whose natural frequency matches the source excitation frequency (resonance condition) will continuously increase without any bounds, in the limit reaching values of infinity. For those situations, no steady-state solutions can exist. One way to contain the field amplitude is to introduce damping.

15.3.3 Green's Function in Integral Form

In the previous two sections we outlined procedures that can be used to derive the Green's function in closed and series forms. The bilinear formula (15-61) of Section 15.3.2 is used to derive the Green's function (15-33) through (15-33b) when the eigenvalue spectrum, represented by the λ_n 's in (15-61), is discrete. However, often the eigenvalue spectrum is continuous, and it can be represented in (15-61) by an integral. In the limit, the infinite summation of the bilinear formula reduces to an integral. This form is usually desirable when at least one of the boundary conditions is at infinity. This would be true when a source placed at the origin is radiating in an unbounded medium.

To demonstrate the derivation, let us construct the Green's function of the one-dimensional scalar Helmholtz equation

$$\frac{d^2\varphi}{dx^2} + \beta_0^2\varphi = f(x) \quad (15-62)$$

subject to the boundary (radiation) conditions of

$$\varphi(+\infty) = \varphi(-\infty) = 0 \quad (15-62a)$$

The Sommerfield radiation condition [19] could also be used instead of (15-62a).

The Green's function $G(x, x')$ will satisfy the differential equation

$$\frac{d^2 G(x, x')}{dx^2} + \beta_0^2 G(x, x') = \delta(x - x') \quad (15-63)$$

subject to the boundary conditions

$$G(+\infty) = G(-\infty) = 0 \quad (15-63a)$$

The complete set of orthonormal eigenfunctions, represented here by $\{\psi(x)\}$, must satisfy the differential equation

$$\frac{d^2 \psi}{dx^2} = -\lambda \psi = -\beta^2 \psi \quad (15-64)$$

where $\lambda = +\beta^2$, subject to the boundary conditions of

$$\psi(+\infty) = \psi(-\infty) = 0 \quad (15-64a)$$

Since the source is radiating in an unbounded medium, represented here by the boundary (radiation) conditions, the most appropriate eigenfunctions are those representing traveling waves, instead of standing waves. Thus, a solution for (15-64), subject to (15-64a), is

$$\psi(x, x') = C(x') e^{\mp j\beta x} \quad \begin{array}{l} - \text{ for } x > x' \\ + \text{ for } x < x' \end{array} \quad (15-65)$$

where for an $e^{j\omega t}$ time convention the upper sign (minus) represents waves traveling in the $+x$ direction, satisfying the boundary condition at $x = +\infty$, and the lower sign (plus) represents waves traveling in the $-x$ direction, satisfying the boundary condition at $x = -\infty$. Let us assume that the waves of interest here are those traveling in the $+x$ direction, represented in (15-65) by the upper sign, that is,

$$\psi(x, x') = C(x') e^{-j\beta x} \quad (15-65a)$$

which represents a plane wave of amplitude $C(x')$.

The Green's function can be represented by a continuous spectrum of plane waves or by a Fourier integral

$$G(x, x') = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} g(\beta, x') e^{-j\beta x} d\beta \quad (15-66)$$

whose Fourier transform pair is

$$g(\beta, x') = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} G(x, x') e^{+j\beta x} dx \quad (15-66a)$$

In (15-66) $G(x, x')$ is represented by a continuous spectrum of plane waves each of the form of (15-65a) and each with an amplitude coefficient of $g(\beta, x')$. Using (15-66a), we can write the transform $\tilde{\delta}(\beta, x')$ of the Dirac delta function $\delta(x - x')$ as

$$\tilde{\delta}(\beta, x') = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} \delta(x, x') e^{+j\beta x} dx = \frac{1}{\sqrt{2\pi}} e^{+j\beta x'} \quad (15-67)$$

Thus, according to (15-66), $\delta(x, x')$ can then be written using (15-67) as

$$\delta(x, x') = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} \tilde{\delta}(\beta, x') e^{-j\beta x} d\beta = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} \left(\frac{1}{\sqrt{2\pi}} e^{+j\beta x'} \right) e^{-j\beta x} d\beta \quad (15-67a)$$

The amplitude coefficients $g(\beta, x')$ in (15-66) can be determined by substituting (15-66) and (15-67a) into (15-63). Then it can be shown that

$$\begin{aligned} \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} (-\beta^2 + \beta_0^2) g(\beta, x') e^{-j\beta x} d\beta &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} \left[\frac{1}{\sqrt{2\pi}} e^{+j\beta x'} \right] e^{-j\beta x} d\beta \\ \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} \left\{ (\beta_0^2 - \beta^2) g(\beta, x') - \frac{1}{\sqrt{2\pi}} e^{+j\beta x'} \right\} e^{-j\beta x} d\beta &= 0 \end{aligned} \quad (15-68)$$

which is satisfied provided

$$g(\beta, x') = \frac{1}{\sqrt{2\pi}} \frac{e^{j\beta x'}}{\beta_0^2 - \beta^2} \quad (15-68a)$$

Thus, the Green's function of (15-66) reduces to

$$G(x, x') = \frac{1}{2\pi} \int_{-\infty}^{+\infty} \frac{e^{-j\beta(x-x')}}{\beta_0^2 - \beta^2} d\beta \quad (15-69)$$

which is a generalization of the bilinear formula (15-61).

The integrand in (15-69) has poles at $\beta = \pm\beta_0$ and can be evaluated using residue calculus [5]. In the evaluation of (15-69), the contour along a circular arc C_R of radius $R \rightarrow \infty$ with center at the origin should close in the lower half plane for $x > x'$, as shown in Figure 15-3a, and should close in the upper half plane for $x < x'$, as shown in Figure 15-3b. This is necessary so that the contribution of the integral along the circular arc C_R of radius $R \rightarrow \infty$ is equal to zero. In general then, by residue calculus, the integral of (15-69) can be evaluated using the geometry of Figure 15-3, and it can be written as

$$\begin{aligned} G(x, x') &= \frac{1}{2\pi} \int_{-\infty}^{+\infty} \frac{e^{-j\beta(x-x')}}{(\beta_0^2 - \beta^2)} d\beta \\ &= \mp 2\pi j \left[\text{residue}(\beta = \pm\beta_0) \right] - \frac{1}{2\pi} \int_{C_R} \frac{e^{-j\beta(x-x')}}{\beta_0^2 - \beta^2} d\beta \\ G(x, x') &= \mp 2\pi j \left[\text{residue}(\beta = \pm\beta_0) \right] \begin{array}{ll} \text{upper signs for} & x > x' \\ \text{lower signs for} & x < x' \end{array} \end{aligned} \quad (15-70)$$

since the contribution along C_R is zero.

It is apparent that the Green's function of (15-69) possesses pole singularities at $\beta = +\beta_0$ and $\beta = -\beta_0$. If these were allowed to contribute, then the exponentials in the integral of (15-69) for an $e^{j\omega t}$ time convention would be represented by either

$$e^{-j\beta_0(x-x')} e^{j\omega t} = e^{+j\beta_0 x'} e^{j(-\beta_0 x + \omega t)} \quad \text{for } \beta = +\beta_0 \quad (15-70a)$$

or

$$e^{+j\beta_0(x-x')} e^{j\omega t} = e^{-j\beta_0 x'} e^{j(\beta_0 x + \omega t)} \quad \text{for } \beta = -\beta_0 \quad (15-70b)$$

Thus, (15-70a) represents waves traveling in the $+x$ direction and (15-70b) represents waves traveling in the $-x$ direction. In the contour evaluation of (15-69), the contour should do the following:

1. For $x > x'$, pass the pole at $\beta = -\beta_0$ from below, the one at $\beta = +\beta_0$ from above, and then close down so that only the latter contributes, as shown in Figure 15-3a.
2. For $x < x'$, pass the pole at $\beta = -\beta_0$ from below, the one at $\beta = +\beta_0$ from above, and then close up so that only the former contributes, as shown in Figure 15-3b.

Sometimes an integral Green's function can be used to represent both discrete and continuous spectra whereby part of the integral would represent the discrete spectrum and the remainder would represent the continuous spectrum. Typically, the discrete spectrum would represent a finite number of propagating modes and an infinite number of evanescent modes in the closed regions, and the continuous spectrum would represent radiation in open regions.

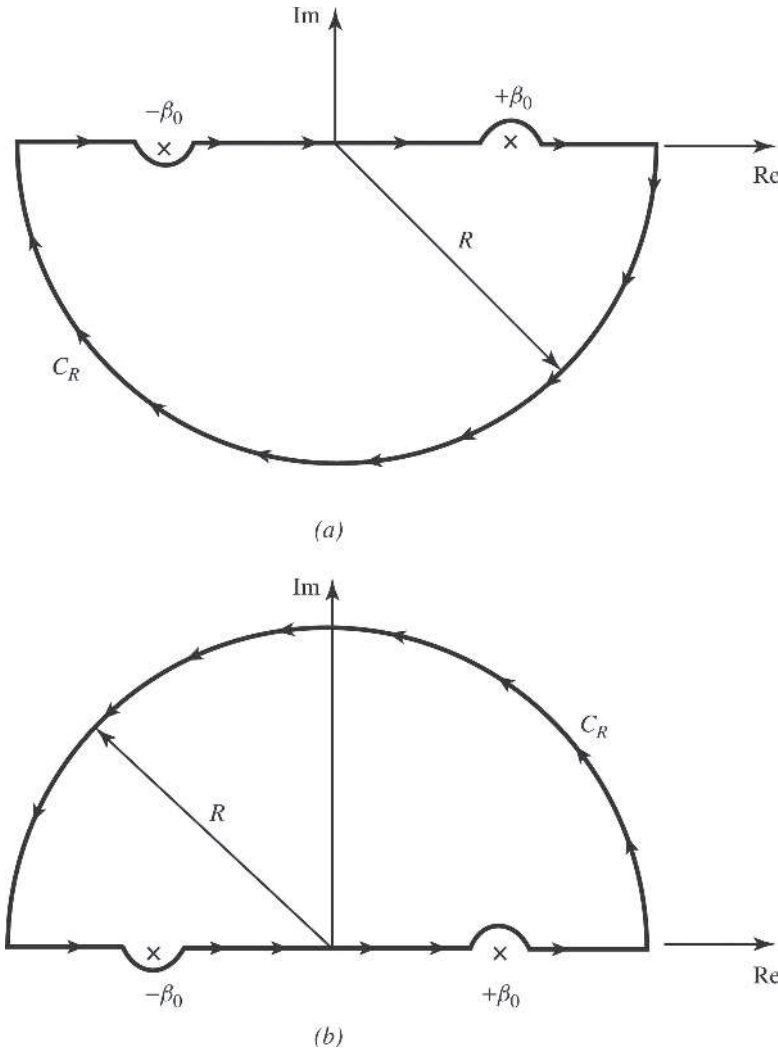


Figure 15-3 Residue calculus for contour integration (a) $x > x'$. (b) $x < x'$.

15.4 TWO-DIMENSIONAL GREEN'S FUNCTION IN RECTANGULAR COORDINATES

Until now we have considered the construction of Green's functions for problems involving a single space variable. Let us now consider problems involving two space variables, both for static and for time-varying fields.

15.4.1 Static Fields

A two-dimensional partial differential equation often encountered in static electromagnetics is Poisson's equation

$$\frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} = f(x, y) = -\frac{1}{\varepsilon_0} q(x, y) \quad (15-71)$$

subject to the boundary conditions

$$V(x=0, 0 \leq y \leq b) = V(x=a, 0 \leq y \leq b) = 0 \quad (15-71a)$$

$$V(0 \leq x \leq a, y=0) = V(0 \leq x \leq a, y=b) = 0 \quad (15-71b)$$

In (15-71), V can represent the electric potential distribution on a rectangular structure of dimensions a along the x direction and b along the y direction and $f(x, y) = q(x, y)$ can represent the electric charge distribution along the structure. The objective here is to obtain the Green's function and ultimately the potential distribution. The Green's function $G(x, y; x', y')$ will satisfy the partial differential equation

$$\frac{\partial^2 G}{\partial x^2} + \frac{\partial^2 G}{\partial y^2} = \delta(x - x') \delta(y - y') \quad (15-72)$$

subject to the boundary conditions

$$G(x=0, 0 \leq y \leq b) = G(x=a, 0 \leq y \leq b) = 0 \quad (15-72a)$$

$$G(0 \leq x \leq a, y=0) = G(0 \leq x \leq a, y=b) = 0 \quad (15-72b)$$

and the potential distribution $V(x, y)$ will be represented by

$$V(x, y) = -\frac{1}{\varepsilon_0} \int_0^b \int_0^a q(x', y') G(x, y; x', y') dx' dy' \quad (15-73)$$

We will derive the Green's function here in two forms.

1. *Closed form*, similar to that of Section 15.3.1 but utilizing (15-44c) and (15-45a) through (15-45b) for a two space variable problem.
2. *Series form*, similar to that of Section 15.3.2 but using basically two-dimensional expressions for (15-54), (15-55), and (15-61).

A. Closed Form The Green's function of (15-72) for the closed-form solution can be formulated by choosing functions that initially satisfy the boundary conditions either along the x direction, at $x=0$ and $x=a$, or along the y direction, at $y=0$ and $y=b$. Let us begin here the development of the Green's function of (15-72) by choosing functions that initially satisfy the boundary

conditions along the x direction. This is accomplished by initially representing the Green's function by a normalized single function Fourier series of sine functions that satisfy the boundary conditions at $x = 0$ and $x = a$, that is,

$$G(x, y; x', y') = \sum_{m=1}^{\infty} g_m(y; x', y') \sin\left(\frac{m\pi}{a} x\right) \quad (15-74)$$

The coefficients $g_m(y; x', y')$ of the Fourier series will be determined by first substituting (15-74) into (15-72). This leads to

$$\sum_{m=1,2,\dots}^{\infty} \left[-\left(\frac{m\pi}{a}\right)^2 g_m(y; x', y') \sin\left(\frac{m\pi}{a} x\right) + \sin\left(\frac{m\pi}{a} x\right) \frac{d^2 g_m(y; x', y')}{dy^2} \right] = \delta(x - x') \delta(y - y') \quad (15-75)$$

Multiplying both sides of (15-75) by $\sin(n\pi x/a)$, integrating with respect to x from 0 to a , and using (15-48a) and (15-48b), we can write that

$$\frac{d^2 g_m(y; x', y')}{dy^2} - \left(\frac{m\pi}{a}\right)^2 g_m(y; x', y') = \frac{2}{a} \sin\left(\frac{m\pi}{a} x'\right) \delta(y - y') \quad (15-76)$$

Equation 15-76 is recognized as a one-dimensional differential equation for $g_m(y; x', y')$, which can be solved using the recipe of Section 15.3.1 as provided by (15-44c) and (15-45a) through (15-45b).

For the homogeneous form of (15-76), or

$$\frac{d^2 g_m(y; x', y')}{dy^2} - \left(\frac{m\pi}{a}\right)^2 g_m(y; x', y') = 0 \quad (15-77)$$

two solutions that satisfy, respectively, the boundary conditions at $y = 0$ and $y = b$ are

$$g_m^{(1)}(y; x', y') = A_m(x', y') \sinh\left(\frac{m\pi}{a} y\right) \quad \text{for } y \leq y' \quad (15-78a)$$

$$g_m^{(2)}(y; x', y') = B_m(x', y') \sinh\left[\frac{m\pi}{a}(b - y)\right] \quad \text{for } y \geq y' \quad (15-78b)$$

The hyperbolic functions were chosen as solutions to (15-77), instead of real exponentials, so that (15-78a) satisfies the boundary condition of (15-71b) at $y = 0$ and (15-78b) satisfies the boundary condition of (15-71b) at $y = b$.

Using (15-44c) where $y_1 = g_m^{(1)}$ and $y_2 = g_m^{(2)}$, we can write the Wronskian as

$$\begin{aligned} W(y; x', y') &= -\left(\frac{m\pi}{a}\right) A_m B_m \left\{ \sinh\left(\frac{m\pi}{a} y'\right) \cosh\left[\frac{m\pi}{a}(b - y')\right] \right. \\ &\quad \left. + \cosh\left(\frac{m\pi}{a} y'\right) \sinh\left[\frac{m\pi}{a}(b - y')\right] \right\} \\ W(y; x', y') &= -\left(\frac{m\pi}{a}\right) A_m B_m \sinh\left(\frac{m\pi b}{a}\right) \end{aligned} \quad (15-79)$$

By comparing (15-76) with the form of (15-33), it is apparent that

$$\begin{aligned} p(y) &= 1 \\ q(y) &= 0 \\ r(y) &= 1 \\ \lambda &= -\left(\frac{m\pi}{a}\right)^2 \end{aligned} \quad (15-80)$$

Using (15-78a) through (15-80), the solution for $g_m(y; x', y')$ of (15-76) can be written, by referring to (15-45a) and (15-45b), as

$$g_m(y; x', y') = \begin{cases} -\frac{2}{m\pi} \sin\left(\frac{m\pi}{a} x'\right) \frac{\sinh\left[\frac{m\pi}{a}(b-y')\right]}{\sinh\left(\frac{m\pi b}{a}\right)} \sinh\left(\frac{m\pi}{a} y\right), & (15-81a) \\ 0 \leq y \leq y' \end{cases}$$

$$g_m(y; x', y') = \begin{cases} -\frac{2}{m\pi} \sin\left(\frac{m\pi}{a} x'\right) \frac{\sinh\left(\frac{m\pi}{a} y'\right)}{\sinh\left(\frac{m\pi b}{a}\right)} \sinh\left[\frac{m\pi}{a}(b-y)\right], & (15-81b) \\ y' \leq y \leq b \end{cases}$$

Thus, the Green's function of (15-74) can be written as

$$G(x, y; x', y') = \begin{cases} -\frac{2}{\pi} \sum_{m=1}^{\infty} \frac{\sin\left(\frac{m\pi}{a} x'\right) \sinh\left[\frac{m\pi}{a}(b-y')\right]}{m \sinh\left(\frac{m\pi b}{a}\right)} \\ \quad \times \sin\left(\frac{m\pi}{a} x\right) \sinh\left(\frac{m\pi}{a} y\right) & (15-82a) \\ \text{for } 0 \leq x \leq a, \quad 0 \leq y \leq y' \end{cases}$$

$$G(x, y; x', y') = \begin{cases} -\frac{2}{\pi} \sum_{m=1}^{\infty} \frac{\sin\left(\frac{m\pi}{a} x'\right) \sinh\left(\frac{m\pi}{a} y'\right)}{m \sinh\left(\frac{m\pi b}{a}\right)} \\ \quad \times \sin\left(\frac{m\pi}{a} x\right) \sinh\left[\frac{m\pi}{a}(b-y)\right] & (15-82b) \\ \text{for } 0 \leq x \leq a, \quad y' \leq y \leq b \end{cases}$$

which is a series of sine functions in x' and x , and hyperbolic sine functions in y' and y .

If the Green's function solution were developed by selecting and writing initially (15-74) by functions that satisfy the boundary conditions at $y = 0$ and $y = b$, then it can be shown that the Green's function can be written as a series of hyperbolic sine functions in x' and x , and ordinary

sine functions in y' and y , or

$$G(x, y; x', y') = \begin{cases} -\frac{2}{\pi} \sum_{n=1,2,\dots}^{\infty} \frac{\sinh\left[\frac{n\pi}{b}(a-x')\right] \sin\left(\frac{n\pi}{b}y'\right)}{n \sinh\left(\frac{n\pi a}{b}\right)} \\ \quad \times \sinh\left(\frac{n\pi}{b}x\right) \sin\left(\frac{n\pi}{b}y\right) \\ \quad \text{for } 0 \leq x \leq x', \quad 0 \leq y \leq b \\ -\frac{2}{\pi} \sum_{n=1,2,\dots}^{\infty} \frac{\sinh\left(\frac{n\pi}{b}x'\right) \sin\left(\frac{n\pi}{b}y'\right)}{n \sinh\left(\frac{n\pi a}{b}\right)} \\ \quad \times \sinh\left[\frac{n\pi}{b}(a-x)\right] \sin\left(\frac{n\pi}{b}y\right) \\ \quad \text{for } x' \leq x \leq a, \quad 0 \leq y \leq b \end{cases} \quad (15-83a)$$

(15-83b)

The derivation of this is left to the reader as an end-of-chapter exercise.

Example 15-4

Electric charge is uniformly distributed along an infinitely long conducting wire positioned at $\rho = \rho'$, $\phi = \phi'$ and circumscribed by a grounded ($V = 0$) electric conducting circular cylinder of radius a and infinite length, as shown in Figure 15-4. Find series-form expressions for the Green's function and potential distribution. Assume free space within the cylinder.

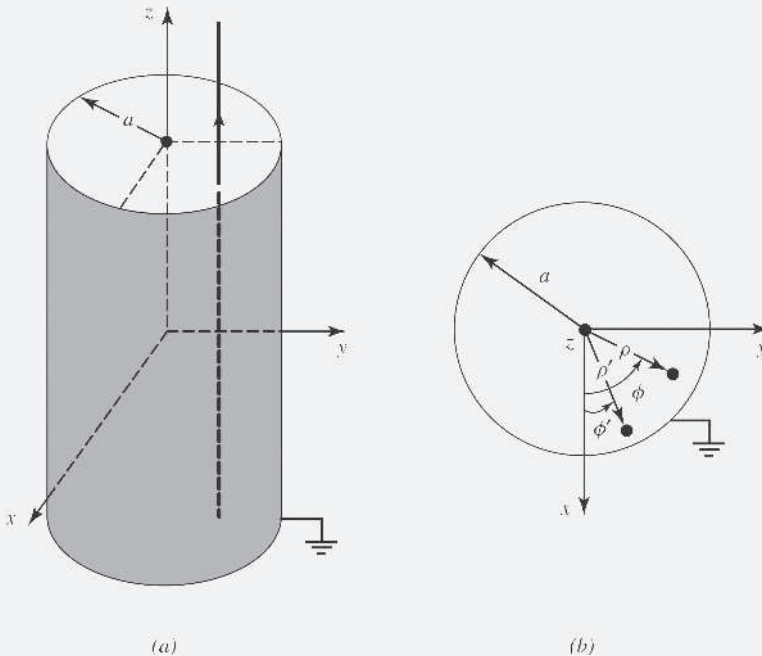


Figure 15-4 Long wire within a grounded circular conducting cylinder. (a) Wire and grounded cylinder. (b) Top view.

Solution: The potential distribution $V(\rho, \phi, z)$ must satisfy Poisson's equation

$$\nabla^2 V(\rho, \phi, z) = -\frac{1}{\varepsilon_0} q(\rho, \phi, z)$$

subject to the boundary condition

$$V(\rho = a, 0 \leq \phi \leq 2\pi, z) = 0$$

Since the wire is infinitely long, the solutions for the potential will not be functions of z . Thus, an expanded form of Poisson's equation reduces to

$$\frac{1}{\rho} \frac{\partial}{\partial \rho} \left(\rho \frac{\partial V}{\partial \rho} \right) + \frac{1}{\rho^2} \frac{\partial^2 V}{\partial \phi^2} = -\frac{1}{\varepsilon_0} q(\rho, \phi)$$

The Green's function $G(\rho, \phi; \rho', \phi')$ must satisfy the partial differential equation

$$\nabla^2 G(\rho, \phi; \rho', \phi') = \delta(\rho - \rho')$$

which in expanded form reduces for this problem to

$$\frac{1}{\rho} \frac{\partial}{\partial \rho} \left(\rho \frac{\partial G}{\partial \rho} \right) + \frac{1}{\rho^2} \frac{\partial^2 G}{\partial \phi^2} = \delta(\rho - \rho')$$

For the series solution of the Green's function, the complete set of orthonormal eigenfunctions $\{\psi_{mn}(\rho, \phi)\}$ can be obtained by considering the homogeneous form of Poisson's equation, or

$$\frac{1}{\rho} \frac{\partial}{\partial \rho} \left(\rho \frac{\partial \psi_{mn}}{\partial \rho} \right) + \frac{1}{\rho^2} \frac{\partial^2 \psi_{mn}}{\partial \phi^2} = -\lambda_{mn} \psi_{mn}$$

subject to the boundary condition

$$\psi_{mn}(\rho = a, 0 \leq \phi \leq 2\pi, z) = 0$$

Using the separation-of-variables method of Section 3.4.2, we can express $\psi_{mn}(\rho, \phi)$ by

$$\psi_{mn}(\rho, \phi) = f(\rho)g(\phi)$$

Following the method outlined in Section 3.4.2, the functions $f(\rho)$ and $g(\phi)$ satisfy, respectively, the differential equations (3-66a) and (3-66b), that is,

$$\rho^2 \frac{d^2 f}{d\rho^2} + \rho \frac{df}{d\rho} + (\lambda_{mn} \rho^2 - m^2) f = 0$$

and

$$\frac{d^2 g}{d\phi^2} = -m^2 g$$

whose appropriate solutions for this problem, according to (3-67a) and (3-68b), are, respectively,

$$\begin{aligned} f &= A J_m(\sqrt{\lambda_{mn}} \rho) + B Y_m(\sqrt{\lambda_{mn}} \rho) \\ g &= C \cos(m\phi) + D \sin(m\phi) \end{aligned}$$

Since ψ_{mn} must be periodic in ϕ , then m must take integer values, $m = 0, 1, 2, \dots$, and both the $\cos(m\phi)$ and $\sin(m\phi)$ variations (modes) exist simultaneously; see Chapter 9. Also since ψ_{mn} must be finite everywhere, including $\rho = 0$, then $B = 0$. Thus, the eigenfunctions are reduced to either of two forms, that is,

$$\psi_{mn}^{(1)} = A_{mn} J_m(\sqrt{\lambda_{mn}} \rho) \cos(m\phi)$$

or

$$\psi_{mn}^{(2)} = A_{mn} J_m(\sqrt{\lambda_{mn}} \rho) \sin(m\phi)$$

The eigenvalues λ_{mn} are found by applying the boundary condition at $\rho = a$, that is,

$$\psi_{mn}(\rho = a, 0 \leq \phi \leq 2\pi) = A_{mn} J_m(\sqrt{\lambda_{mn}} a) = 0$$

or

$$\sqrt{\lambda_{mn}} a = \chi_{mn} \Rightarrow \lambda_{mn} = \left(\frac{\chi_{mn}}{a} \right)^2$$

where χ_{nm} represents the n zeroes of the Bessel function J_m of the first kind of order m . These are listed in Table 9-2.

The complete set of orthonormal eigenfunctions must be normalized so that

$$\int_0^{2\pi} \int_0^a \psi_{mn}^{(1)}(\rho, \phi) \psi_{mp}^{(1)}(\rho, \phi) \rho \, d\rho \, d\phi = \int_0^{2\pi} \int_0^a \psi_{mn}^{(2)}(\rho, \phi) \psi_{mp}^{(2)}(\rho, \phi) \rho \, d\rho \, d\phi = 1$$

Thus,

$$A_{mn}^2 \int_0^{2\pi} \int_0^a \rho J_m(\sqrt{\lambda_{mn}} \rho) J_m(\sqrt{\lambda_{mp}} \rho) \cos^2(m\phi) \, d\rho \, d\phi = 1$$

or

$$A_{mn}^2 \int_0^{2\pi} \int_0^a \rho J_m(\sqrt{\lambda_{mn}} \rho) J_m(\sqrt{\lambda_{mp}} \rho) \sin^2(m\phi) \, d\rho \, d\phi = 1$$

Since

$$\int_0^{2\pi} \cos^2(m\phi) \, d\phi = \begin{cases} 2\pi & m=0 \\ \pi & m \neq 0 \end{cases}$$

$$\int_0^{2\pi} \sin^2(m\phi) \, d\phi = \begin{cases} 0 & m=0 \\ \pi & m \neq 0 \end{cases}$$

and

$$\int_0^a \rho J_m(\sqrt{\lambda_{mn}} \rho) J_m(\sqrt{\lambda_{mp}} \rho) \, d\rho = \begin{cases} \frac{a^2}{2} [J'_m(\sqrt{\lambda_{mn}} a)]^2 & p=n \\ 0 & p \neq n \end{cases}$$

then

$$A_{mn}^2 \varepsilon_m \frac{\pi a^2}{2} [J'_m(\sqrt{\lambda_{mp}} a)]^2 = 1$$

hence,

$$A_{mn} = \sqrt{\frac{2}{\varepsilon_m \pi}} \frac{1}{a J'_m(\sqrt{\lambda_{mn}} a)}$$

where

$$\varepsilon_m = \begin{cases} 2 & m=0 \\ 1 & m \neq 0 \end{cases}$$

Thus, the complete set of orthonormal eigenfunctions can be written as

$$\psi_{mn}^{(1)} = \sqrt{\frac{2\varepsilon_m}{\pi}} \frac{1}{a J'_m(\sqrt{\lambda_{mn}} a)} J_m(\sqrt{\lambda_{mp}} \rho) \cos(m\phi)$$

or

$$\psi_{mn}^{(2)} = \sqrt{\frac{2\varepsilon_m}{\pi}} \frac{1}{aJ'_m(\sqrt{\lambda_{mn}}a)} J_m(\sqrt{\lambda_{mn}}\rho) \sin(m\phi)$$

The Green's function can now be written using the bilinear formula (15-94) with $\lambda = 0$ as

$$\begin{aligned} G(\rho, \phi; \rho', \phi') &= -\frac{2}{\pi} \varepsilon_m \frac{1}{a^2 \left[J'_m(\sqrt{\lambda_{mn}}a) \right]^2} \sum_{m=0}^{\infty} \sum_{n=1}^{\infty} \frac{J_m(\sqrt{\lambda_{mn}}\rho') J_m(\sqrt{\lambda_{mn}}\rho)}{\lambda_{mn}} \\ &\quad \times [\cos(m\phi) \cos(m\phi') + \sin(m\phi) \sin(m\phi')] \\ G(\rho, \phi; \rho', \phi') &= -\frac{2}{\pi} \varepsilon_m \frac{1}{a^2 \left[J'_m(\sqrt{\lambda_{mn}}a) \right]^2} \sum_{m=0}^{\infty} \sum_{n=1}^{\infty} \frac{J_m(\sqrt{\lambda_{mn}}\rho') J_m(\sqrt{\lambda_{mn}}\rho)}{\lambda_{mn}} \\ &\quad \times \cos[m(\phi - \phi')] \end{aligned}$$

where

$$\lambda_{mn} = \left(\frac{\chi_{mn}}{a} \right)^2$$

Both the $\cos(m\phi)$ and $\sin(m\phi)$ field variations were included in the final expression for the Green's function.

Finally, the potential distribution $V(\rho, z)$ can be written as

$$V(\rho, \phi) = -\frac{1}{\varepsilon_0} \int_0^{2\pi} \int_0^a q(\rho', \phi') G(\rho, \phi; \rho', \phi') \rho' d\rho' d\phi'$$

where $G(\rho, \phi; \rho', \phi')$ is the Green's function and $q(\rho', \phi')$ is the linear charge distribution.

The Green's function of Example 15-4 can also be developed in closed form. This is done in Section 15.6.2 for a time-harmonic electric line source inside a circular cylinder. The statics solution is obtained by letting $\beta_0 = 0$.

B. Series Form For the series solution of the Green's function of (15-72), the complete set of orthonormal eigenfunctions $\{\psi_{mn}(x, y)\}$ can be obtained by considering the homogeneous form of (15-71), i.e.,

$$\frac{\partial^2 \psi_{mn}}{\partial x^2} + \frac{\partial^2 \psi_{mn}}{\partial y^2} = -\lambda_{mn} \psi_{mn} \quad (15-84)$$

subject to the boundary conditions

$$\psi_{mn}(x=0, 0 \leq y \leq b) = \psi_{mn}(x=a, 0 \leq y \leq b) = 0 \quad (15-84a)$$

$$\psi_{mn}(0 \leq x \leq a, y=0) = \psi_{mn}(0 \leq x \leq a, y=b) = 0 \quad (15-84b)$$

Using the method of separation of variables of Section 3.4.1, we can represent $\psi_{mn}(x, y)$ by

$$\psi_{mn}(x, y) = f(x)g(y) \quad (15-85)$$

Substituting (15-85) into (15-84) reduces to

$$\frac{1}{f} \frac{d^2 f}{dx^2} = -p^2 \Rightarrow \frac{d^2 f}{dx^2} = -p^2 f \Rightarrow f(x) = A \cos(px) + B \sin(px) \quad (15-85a)$$

$$\frac{1}{g} \frac{d^2 g}{dy^2} = -q^2 \Rightarrow \frac{d^2 g}{dy^2} = -q^2 g \Rightarrow g(y) = C \cos(qy) + D \sin(qy) \quad (15-85b)$$

where the system eigenvalues are those of

$$\lambda_{mn} = p^2 + q^2 \quad (15-85c)$$

Thus, (15-85) can be represented by

$$\psi_{mn}(x, y) = [A \cos(px) + B \sin(px)][C \cos(qy) + D \sin(qy)] \quad (15-86)$$

Applying the boundary conditions of (15-84a) on (15-86) leads to

$$\psi_{mn}(x=0, 0 \leq y \leq b) = [A(1) + B(0)][C \cos(qy) + D \sin(qy)] = 0 \Rightarrow A = 0 \quad (15-87a)$$

$$\psi_{mn}(x=a, 0 \leq y \leq b) = B \sin(pa)[C \cos(qy) + D \sin(qy)] = 0 \Rightarrow \sin(pa) = 0$$

$$\begin{aligned} pa &= \sin^{-1}(0) = m\pi \\ p &= \frac{m\pi}{a}, \quad m = 1, 2, 3, \dots \end{aligned} \quad (15-87b)$$

Similarly, applying the boundary conditions of (15-84b) into (15-86), using (15-87a) and (15-87b), leads to

$$\psi_{mn}(0 \leq x \leq a, y=0) = B \sin\left(\frac{m\pi}{a}x\right)[C(1) + D(0)] = 0 \Rightarrow C = 0 \quad (15-88a)$$

$$\psi_{mn}(0 \leq x \leq a, y=b) = BD \sin\left(\frac{m\pi}{a}x\right) \sin(qb) = 0 \Rightarrow \sin(qb) = 0$$

$$\begin{aligned} qb &= \sin^{-1}(0) = n\pi \\ q &= \frac{n\pi}{b}, \quad n = 1, 2, 3, \dots \end{aligned} \quad (15-88b)$$

Thus, the eigenfunctions of (15-85) reduce to

$$\psi_{mn}(x, y) = BD \sin\left(\frac{m\pi}{a}x\right) \sin\left(\frac{n\pi}{b}y\right) = B_{mn} \sin\left(\frac{m\pi}{a}x\right) \sin\left(\frac{n\pi}{b}y\right) \quad (15-89)$$

where the eigenvalues of (15-85c) are equal to

$$\lambda_{mn} = (p^2 + q^2) = \left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2, \quad \begin{aligned} m &= 1, 2, 3, \dots \\ n &= 1, 2, 3, \dots \end{aligned} \quad (15-89a)$$

To form the Green's function, the eigenfunctions of (15-89) must be normalized so that

$$\boxed{\int_0^b \int_0^a \psi_{mn}(x, y) \psi_{rs}(x, y) \, dx \, dy = \begin{cases} 1 & m=r, \quad n=s \\ 0 & m \neq r, \quad n \neq s \end{cases}} \quad (15-90)$$

Equation 15-90 is similar, and is an expanded form in two space variables, of (15-55). Substituting (15-89) into (15-90), we can write that

$$B_{mn}^2 \int_0^b \int_0^a \sin\left(\frac{m\pi}{a}x\right) \sin\left(\frac{n\pi}{b}y\right) \sin\left(\frac{r\pi}{a}x\right) \sin\left(\frac{s\pi}{b}y\right) dx dy = 1 \quad (15-91)$$

Using (15-48a) and (15-48b) reduces (15-91) to

$$B_{mn}^2 \left(\frac{ab}{4}\right) = 1 \quad (15-92)$$

hence,

$$B_{mn} = \frac{2}{\sqrt{ab}} \quad (15-92a)$$

Thus, (15-89) can be written as

$$\psi_{mn}(x, y) = \frac{2}{\sqrt{ab}} \sin\left(\frac{m\pi}{a}x\right) \sin\left(\frac{n\pi}{b}y\right) \quad (15-93)$$

The Green's function can be expressed as a double summation of two-dimensional eigenfunctions,

$$G(x, y; x', y') = \sum_m \sum_n \frac{\psi_{mn}(x', y') \psi_{mn}(x, y)}{\lambda - \lambda_{mn}} \quad (15-94)$$

Equation 15-94 is an expanded version in two space variables of the bilinear equation of (15-61). Thus, we can write the Green's function of (15-94), using (15-89a), (15-93), and $\lambda = 0$, as

$$G(x, y; x', y') = -\frac{4}{ab} \sum_{m=1}^{\infty} \sum_{n=1}^{\infty} \frac{\sin\left(\frac{m\pi}{a}x'\right) \sin\left(\frac{n\pi}{b}y'\right)}{\left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2} \sin\left(\frac{m\pi}{a}x\right) \sin\left(\frac{n\pi}{b}y\right) \quad (15-95)$$

The electric potential of (15-73), due to the static electric charge density of $q(x', y')$, can be expressed as

$$V(x, y) = -\frac{1}{\epsilon_0} \frac{4}{ab} \sum_{m=1}^{\infty} \sum_{n=1}^{\infty} \frac{\sin\left(\frac{m\pi}{a}x\right) \sin\left(\frac{n\pi}{b}y\right)}{\left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2} \times \int_0^b \int_0^a q(x', y') \sin\left(\frac{m\pi}{a}x'\right) \sin\left(\frac{n\pi}{b}y'\right) dx' dy' \quad (15-96)$$

15.4.2 Time-Harmonic Fields

For time-harmonic fields, a popular partial differential equation is

$$\frac{\partial^2 E_z}{\partial x^2} + \frac{\partial^2 E_z}{\partial y^2} + \beta^2 E_z = f(x, y) = j\omega\mu J_z(x, y) \quad (15-97)$$

subject to the boundary conditions

$$E_z(x=0, 0 \leq y \leq b) = E_z(x=a, 0 \leq y \leq b) = 0 \quad (15-97a)$$

$$E_z(0 \leq x \leq a, y=0) = E_z(0 \leq x \leq a, y=b) = 0 \quad (15-97b)$$

In (15-97), E_z can represent the electric field component of a TM^z field configuration (mode) with no z variations inside a rectangular metallic cavity of dimensions a, b, c in the x, y, z directions, respectively, as shown in Figure 15-5. The function $f(x, y) = j\omega\mu J_z$ can represent the normalized electric current density component of the feed probe that is used to excite the fields within the metallic cavity. The objective here is to obtain the Green's function of the problem and ultimately the electric field component represented in (15-97) by $E_z(x, y)$.

The Green's function $G(x, y; x', y')$ will satisfy the partial differential equation

$$\frac{\partial^2 G}{\partial x^2} + \frac{\partial^2 G}{\partial y^2} + \beta^2 G = \delta(x - x')\delta(y - y') \quad (15-98)$$

subject to the boundary conditions

$$G(x=0, 0 \leq y \leq b) = G(x=a, 0 \leq y \leq b) = 0 \quad (15-98a)$$

$$G(0 \leq x \leq a, y=0) = G(0 \leq x \leq a, y=b) = 0 \quad (15-98b)$$

and the electric field distribution $E_z(x, y)$ will be represented by

$$E_z(x, y) = j\omega\mu \int_0^b \int_0^a J_z(x', y') G(x, y; x', y') dx' dy' \quad (15-99)$$

The Green's function of (15-98) can be derived either in closed form, as was done in Section 15.4.1A for the statics problem, or in series form, as was done in Section 15.4.1B for the statics

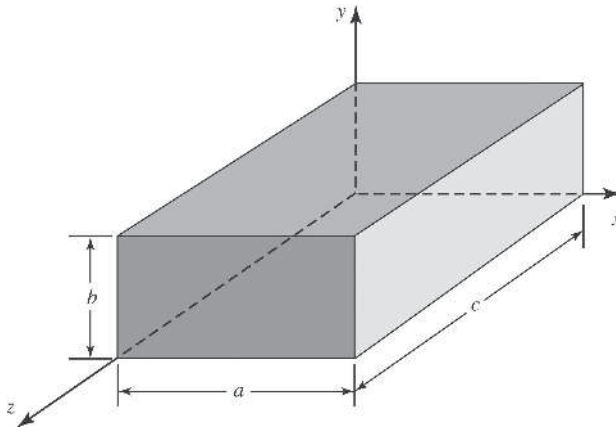


Figure 15-5 Rectangular waveguide cavity geometry.

problem. We will derive the Green's function here by a series form. The closed form is left as an end-of-chapter exercise for the reader.

For the solution of the Green's function, the complete set of eigenfunctions $\{\psi_{mn}(x, y)\}$ can be obtained by considering the homogeneous form of (15-97),

$$\frac{\partial^2 \psi_{mn}}{\partial x^2} + \frac{\partial^2 \psi_{mn}}{\partial y^2} + \beta_{mn}^2 \psi_{mn} = 0 \quad (15-100)$$

subject to the boundary conditions

$$\psi_{mn}(x=0, 0 \leq y \leq b) = \psi_{mn}(x=a, 0 \leq y \leq b) = 0 \quad (15-100a)$$

$$\psi_{mn}(0 \leq x \leq a, y=0) = \psi_{mn}(0 \leq x \leq a, y=b) = 0 \quad (15-100b)$$

Using the separation of variables of Section 3.4.1, we can represent $\psi_{mn}(x, y)$ by

$$\psi_{mn}(x, y) = f(x)g(y) \quad (15-101)$$

Substituting (15-101) into (15-100) and applying the boundary conditions of (15-100a) and (15-100b), it can be shown that $\psi_{mn}(x, y)$ reduces to

$$\psi_{mn}(x, y) = B_{mn} \sin(\beta_x x) \sin(\beta_y y) = B_{mn} \sin\left(\frac{m\pi}{a} x\right) \sin\left(\frac{n\pi}{b} y\right) \quad (15-102)$$

where

$$\beta_x = \frac{m\pi}{a}, \quad m = 1, 2, 3, \dots \quad (15-102a)$$

$$\beta_y = \frac{n\pi}{b}, \quad n = 1, 2, 3, \dots \quad (15-102b)$$

The eigenvalues of the system are equal to

$$\lambda_{mn} = \beta_{mn}^2 = (\beta_x)^2 + (\beta_y)^2 = \left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2 \quad (15-103a)$$

and

$$\lambda = \beta^2 \quad (15-103b)$$

The eigenfunctions of (15-102) must satisfy an equation similar to (15-90) but over a volume integral. That is,

$$\int_0^c \int_0^b \int_0^a \psi_{mn}(x, y, z) \psi_{rs}(x, y, z) dx dy dz = \begin{cases} 1 & m=r, \quad n=s \\ 0 & m \neq r, \quad n \neq s \end{cases} \quad (15-104)$$

which leads to

$$B_{mn} = \frac{2}{\sqrt{abc}} \quad (15-104a)$$

Thus,

$$\psi_{mn}(x, y) = \frac{2}{\sqrt{abc}} \sin\left(\frac{m\pi}{a} x\right) \sin\left(\frac{n\pi}{b} y\right) \quad (15-105)$$

The Green's function is obtained using (15-94), (15-103a), (15-103b), and (15-105). Doing this, we can write that

$$G(x, y; x', y') = \frac{4}{abc} \sum_{m=1}^{\infty} \sum_{n=1}^{\infty} \frac{\sin\left(\frac{m\pi}{a} x'\right) \sin\left(\frac{n\pi}{b} y'\right)}{\beta^2 - \left[\left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2\right]} \times \sin\left(\frac{m\pi}{a} x\right) \sin\left(\frac{n\pi}{b} y\right) \quad (15-106)$$

and the electric field component of (15-99), due to the normalized electric current density represented by $J_n(x', y')$, can be written as

$$E_z(x, y) = \frac{4}{abc} \sum_{m=1}^{\infty} \sum_{n=1}^{\infty} \frac{\sin\left(\frac{m\pi}{a} x\right) \sin\left(\frac{n\pi}{b} y\right)}{\beta^2 - \left[\left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2\right]} \times \int_0^b \int_0^a J_z(x', y') \sin\left(\frac{m\pi}{a} x'\right) \sin\left(\frac{n\pi}{b} y'\right) dx' dy' \quad (15-107)$$

It is apparent that the Green's function possesses a singularity, and it fails when

$$\beta = \beta_r = \omega_r \sqrt{\mu\varepsilon} = 2\pi f_r \sqrt{\mu\varepsilon} = \sqrt{\left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2} \quad (15-108)$$

that is, when

$$f_r = \frac{1}{2\pi\sqrt{\mu\varepsilon}} \sqrt{\left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2} \quad (15-108a)$$

This is in accordance with (15-34), which states that the Green's function of (15-33b) exists for all values of λ (here $\lambda = \beta^2 = \omega^2 \mu\varepsilon$) except those that are eigenvalues λ_{mn} of the homogeneous equation 15-34 [here λ_{mn} given by (15-103a)]. At those eigenvalues for which (15-106) and (15-107) possess singularities (simple poles here), the frequencies of the excitation source match the natural (characteristic) frequencies of the system. As explained in Section 1.3.2, this is referred to as *resonance*, and the field will continuously increase without any bounds (in the limit reaching infinity). For these cases, no steady-state solutions exist. One way to contain the field is to introduce damping. In practice, for metallic cavities, damping is introduced by the losses due to nonperfectly conducting walls.

15.5 GREEN'S IDENTITIES AND METHODS

Now that we have derived Green's functions, both for single and two space variables in rectangular coordinates for the general Sturm-Liouville self-adjoint operator L , let us generalize the procedure for the development of the Green's function for the three-dimensional scalar Helmholtz partial differential equation

$$\nabla^2 \phi(\mathbf{r}) + \beta^2 \phi(\mathbf{r}) = f(\mathbf{r}) \quad (15-109)$$

subject to the generalized homogeneous boundary conditions

$$\alpha_1 \phi(\mathbf{r}_s) + \alpha_2 \frac{\partial \phi(\mathbf{r}_s)}{\partial n_s} = 0, \quad s = 1, 2, \dots, N \quad (15-109a)$$

where $\mathbf{r}_s$ is on S and $\hat{\mathbf{n}}$ is an outward directed unit vector. In electromagnetics these are referred to not only as the *mixed* boundary conditions [7] but also as the *impedance* boundary conditions.

The Green's function $G(\mathbf{r}, \mathbf{r}')$ of (15-109) must satisfy the partial differential equation

$$\nabla^2 G(\mathbf{r}, \mathbf{r}') + \beta^2 G(\mathbf{r}, \mathbf{r}') = \delta(\mathbf{r} - \mathbf{r}') \quad (15-110)$$

subject to the generalized homogeneous boundary conditions

$$\alpha_1 G(\mathbf{r}_s, \mathbf{r}') + \alpha_2 \frac{\partial G(\mathbf{r}_s, \mathbf{r}')}{\partial n_s} = 0, \quad s = 1, 2, \dots, N \quad (15-110a)$$

To accomplish this, we will need two identities from vector calculus that are usually referred to as *Green's first* and *second identities*. We will state them first before proceeding with the development of the generalized Green's function.

15.5.1 Green's First and Second Identities

Within a volume V , conducting bodies with surfaces $S_1, S_2, S_3, \dots, S_n$ are contained, as shown in Figure 15-6. By introducing appropriate cuts, the volume V is bounded by a regular surface S that consists of surfaces $S_1 - S_n$, the surfaces along the cuts, and the surface S_a of an infinite radius sphere that encloses all the conducting bodies. A unit vector $\hat{\mathbf{n}}$ normal to S is directed inward to the volume V , as shown in Figure 15-6.

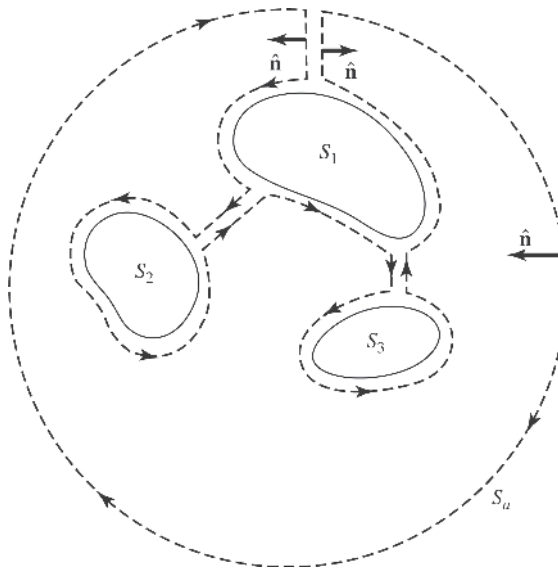


Figure 15-6 Conducting surfaces and appropriate cuts for application of Green's theorem.

Let us introduce within V two scalar functions ϕ and ψ , which, along with their first and second derivatives, are continuous within V and on the surface S . To the vector $\phi \nabla \psi$, we apply the divergence theorem (1-8),

$$\oint_S (\phi \nabla \psi) \cdot d\mathbf{s} = \oint_S (\phi \nabla \psi) \cdot \hat{\mathbf{n}} \, da = \iiint_V \nabla \cdot (\phi \nabla \psi) \, dv \quad (15-111)$$

When expanded, the integrand of the volume integral can be written as

$$\nabla \cdot (\phi \nabla \psi) = \phi \nabla \cdot (\nabla \psi) + \nabla \phi \cdot \nabla \psi = \phi \nabla^2 \psi + \nabla \phi \cdot \nabla \psi \quad (15-112)$$

Thus (15-111) can be expressed as

$$\boxed{\oint_S (\phi \nabla \psi \cdot d\mathbf{s}) = \iiint_V (\phi \nabla^2 \psi) \, dv + \iiint_V (\nabla \phi \cdot \nabla \psi) \, dv} \quad (15-113)$$

which is referred to as *Green's first identity*. Since

$$(\nabla \psi) \cdot \hat{\mathbf{n}} = \frac{\partial \psi}{\partial n} \quad (15-114)$$

where the derivative $\partial \psi / \partial n$ is taken in the direction of positive normal, (15-113) can also be written as

$$\boxed{\oint_S \left(\phi \frac{\partial \psi}{\partial n} \right) ds = \iiint_V (\phi \nabla^2 \psi) \, dv + \iiint_V (\nabla \phi \cdot \nabla \psi) \, dv} \quad (15-115)$$

which is an *alternate form of Green's first identity*.

If we repeat the procedure but apply the divergence theorem (15-111) to the vector $\psi \nabla \phi$, then we can write, respectively, Green's first identity (15-113) and its alternate form (15-115) as

$$\oint_S (\psi \nabla \phi \cdot d\mathbf{s}) = \iiint_V (\psi \nabla^2 \phi) \, dv + \iiint_V (\nabla \psi \cdot \nabla \phi) \, dv \quad (15-116)$$

and

$$\oint_S \left(\psi \frac{\partial \phi}{\partial n} \right) ds = \iiint_V (\psi \nabla^2 \phi) \, dv + \iiint_V (\nabla \psi \cdot \nabla \phi) \, dv \quad (15-117)$$

Subtracting (15-116) from (15-113), we can write that

$$\boxed{\oint_S (\phi \nabla \psi - \psi \nabla \phi) \cdot d\mathbf{s} = \iiint_V (\phi \nabla^2 \psi - \psi \nabla^2 \phi) \, dv} \quad (15-118)$$

which is referred to as *Green's second identity*. Its alternate form

$$\boxed{\oint_S \left(\phi \frac{\partial \psi}{\partial n} - \psi \frac{\partial \phi}{\partial n} \right) ds = \iiint_V (\phi \nabla^2 \psi - \psi \nabla^2 \phi) \, dv} \quad (15-119)$$

is obtained by subtracting (15-117) from (15-115).

Green's first and second identities expressed, respectively, either as (15-113) and (15-118) or (15-115) and (15-119), will be used to develop the formulation for the more general Green's function.

15.5.2 Generalized Green's Function Method

With the introduction of Green's first and second identities in the previous section, we are now ready to develop the formulation of the generalized Green's function method of φ for the partial differential equation 15-109 whose Green's function $G(\mathbf{r}, \mathbf{r}')$ satisfies (15-110).

Let us multiply (15-109) by $G(\mathbf{r}, \mathbf{r}')$ and (15-110) by $\varphi(\mathbf{r})$. Doing this leads to

$$G\nabla^2\varphi + \beta^2\varphi G = fG \quad (15-120a)$$

$$\varphi\nabla^2G + \beta^2\varphi G = \varphi\delta(\mathbf{r} - \mathbf{r}') \quad (15-120b)$$

Subtracting (15-120a) from (15-120b) and integrating over the volume V , we can write that

$$\iiint_V \varphi\delta(\mathbf{r} - \mathbf{r}') dv - \iiint_V fG dv = \iiint_V (\varphi\nabla^2G - G\nabla^2\varphi) dv \quad (15-121)$$

or

$$\begin{aligned} \varphi(\mathbf{r} = \mathbf{r}') &= \varphi(\mathbf{r}') = \iiint_V f(\mathbf{r})G(\mathbf{r}, \mathbf{r}') dv \\ &+ \iiint_V [\varphi(\mathbf{r})\nabla^2G(\mathbf{r}, \mathbf{r}') - G(\mathbf{r}, \mathbf{r}')\nabla^2\varphi(\mathbf{r})] dv \end{aligned} \quad (15-121a)$$

Applying Green's second identity (15-118) reduces (15-121a) to

$$\varphi(\mathbf{r}') = \iiint_V f(\mathbf{r})G(\mathbf{r}, \mathbf{r}') dv + \iint_S [\varphi(\mathbf{r})\nabla G(\mathbf{r}, \mathbf{r}') - G(\mathbf{r}, \mathbf{r}')\nabla\varphi(\mathbf{r})] \cdot d\mathbf{s} \quad (15-122)$$

Since $\mathbf{r}'$ is an arbitrary point within V and $\mathbf{r}$ is a dummy variable, we can also write (15-122) as

$$\boxed{\varphi(\mathbf{r}) = \iiint_V f(\mathbf{r}')G(\mathbf{r}, \mathbf{r}') dv' + \iint_S [\varphi(\mathbf{r}')\nabla'G(\mathbf{r}, \mathbf{r}') - G(\mathbf{r}, \mathbf{r}')\nabla'\varphi(\mathbf{r}')] \cdot d\mathbf{s}'} \quad (15-123)$$

where ∇' indicates differentiation with respect to the prime coordinates.

Equation 15-123 is a generalized formula for the development of the Green's function for a three-dimensional scalar Helmholtz equation. It can be simplified depending on the boundary conditions of φ and G , and their derivatives on S . The objective then will be to judiciously choose the boundary conditions on the development of G , once the boundary conditions on φ are stated, so as to simplify, if not completely eliminate, the surface integral contribution in (15-123). We will demonstrate here some combinations of boundary conditions on φ and G , and the simplifications of (15-123), based on those boundary conditions.

A. Nonhomogeneous Partial Differential Equation with Homogeneous Dirichlet Boundary Conditions If the nonhomogeneous form of the partial differential equation of 15-109 satisfies the homogeneous Dirichlet boundary condition

$$\varphi(\mathbf{r}_s) = 0, \quad \text{where } \mathbf{r}_s \text{ is on } S \quad (15-124a)$$

then it is reasonable to construct a Green's function with the same boundary condition

$$G(\mathbf{r}_s, \mathbf{r}') = 0, \quad \text{where } \mathbf{r}_s \text{ is on } S \quad (15-124b)$$

so as to simplify the surface integral contributions in (15-123).

For these boundary conditions on φ and G , both terms in the surface integral of (15-123) vanish, so that (15-123) reduces to

$$\boxed{\varphi(\mathbf{r}) = \iiint_V f(\mathbf{r}') G(\mathbf{r}, \mathbf{r}') dv'} \quad (15-125)$$

The Green's function $G(\mathbf{r}, \mathbf{r}')$ needed in (15-125) can be obtained using any of the previous methods developed in Sections 15.3.1 through 15.3.2. In many cases, the bilinear form (15-61) or (15-94) or its equivalent, in the desired coordinate system and number of space variables, is appropriate for forming the Green's function. Its existence will depend upon the eigenvalues of the homogeneous partial differential equation, as discussed in Section 15.3.1.

B. Nonhomogeneous Partial Differential Equation with Nonhomogeneous Dirichlet Boundary Conditions If the nonhomogeneous partial differential equation 15-109 satisfies the nonhomogeneous Dirichlet boundary condition

$$\varphi(\mathbf{r}_s) = g(\mathbf{r}_s), \quad \text{where } \mathbf{r}_s \text{ is on } S \quad (15-126a)$$

then we can still construct a Green's function that satisfies the boundary condition

$$G(\mathbf{r}_s, \mathbf{r}') = 0, \quad \text{where } \mathbf{r}_s \text{ is on } S \quad (15-126b)$$

For these boundary conditions on φ and G , the second term in the surface integral of (15-123) vanishes, so that (15-123) reduces to

$$\boxed{\varphi(\mathbf{r}) = \iiint_V f(\mathbf{r}') G(\mathbf{r}, \mathbf{r}') dv' + \iint_S \varphi(\mathbf{r}') \nabla' G(\mathbf{r}_s, \mathbf{r}') \cdot d\mathbf{s}'} \quad (15-127)$$

The Green's function $G(\mathbf{r}, \mathbf{r}')$ needed in (15-127) can be determined using any of the previous methods developed in Sections 15.3.1 through 15.3.2.

C. Nonhomogeneous Partial Differential Equation with Homogeneous Neumann Boundary Conditions When Neumann boundary conditions are involved, the solutions become more complicated primarily because the normal gradients of $\varphi(\mathbf{r})$ are not independent of the partial differential equation. If the nonhomogeneous form of the partial differential equation 15-109 satisfies the homogeneous Neumann boundary condition

$$[\nabla' \varphi(\mathbf{r}_s)] \cdot \hat{\mathbf{n}} = \frac{\partial \varphi(\mathbf{r}_s)}{\partial n} = 0 \quad \text{where } \mathbf{r}_s \text{ is on } S \quad (15-128)$$

then we *cannot*, in general, construct a Green's function with a boundary condition $[\nabla' G(\mathbf{r}_s, \mathbf{r}')] \cdot \hat{\mathbf{n}} = [\partial G(\mathbf{r}_s, \mathbf{r}') / \partial n] = 0$. This is evident from what follows.

If we apply the divergence theorem (1-8) to the vector $\nabla G(\mathbf{r}, \mathbf{r}')$, we can write that

$$\iint_S \nabla G(\mathbf{r}, \mathbf{r}') \cdot d\mathbf{s} = \iiint_V \nabla \cdot \nabla G(\mathbf{r}, \mathbf{r}') dv = \iiint_V \nabla^2 G(\mathbf{r}, \mathbf{r}') dv \quad (15-129)$$

Taking the volume integral of (15-110), we can express it as

$$\iiint_V \nabla^2 G(\mathbf{r}, \mathbf{r}') dv + \beta^2 \iiint_V G(\mathbf{r}, \mathbf{r}') dv = \iiint_V \delta(\mathbf{r} - \mathbf{r}') dv \quad (15-130)$$

Using (15-129) reduces (15-130) to

$$\oiint_S \nabla G(\mathbf{r}, \mathbf{r}') \cdot d\mathbf{s} + \beta^2 \iiint_V G(\mathbf{r}, \mathbf{r}') dv = 1 \quad (15-131)$$

If we choose

$$[\nabla G(\mathbf{r}, \mathbf{r}')|_{\mathbf{r}=\mathbf{r}_s}] \cdot \hat{\mathbf{n}} = [\nabla G(\mathbf{r}_s, \mathbf{r}')] \cdot \hat{\mathbf{n}} = \frac{\partial G(\mathbf{r}_s, \mathbf{r}')}{\partial n} = 0 \quad (15-132)$$

as a boundary condition for $G(\mathbf{r}, \mathbf{r}')$, then (15-131) reduces to

$$\beta^2 \iiint_V G(\mathbf{r}, \mathbf{r}') dv = 1 \quad (15-133)$$

which cannot be satisfied if $\beta = 0$. When $\beta = 0$, (15-131) reduces to

$$\oiint_S \nabla G(\mathbf{r}, \mathbf{r}') \cdot d\mathbf{s} = 1 \quad (15-134)$$

or

$$|\nabla G(\mathbf{r}_s, \mathbf{r}')| S_0 = 1 \quad (15-134a)$$

where S_0 is the area of the surface. This implies that a consistent boundary condition for the normal gradient of $G(\mathbf{r}, \mathbf{r}')$ on S to satisfy (15-134) or (15-134a) would be

$$\nabla' G(\mathbf{r}, \mathbf{r}')|_{\mathbf{r}=\mathbf{r}_s} = \frac{1}{S_0} = \nabla' G(\mathbf{r}_s, \mathbf{r}') \quad (15-135)$$

Substituting (15-128) and (15-135) into (15-123) leads to

$$\varphi(\mathbf{r}) = \iiint_V f(\mathbf{r}') G(\mathbf{r}, \mathbf{r}') dv' + \frac{1}{S_0} \oiint_S \varphi(\mathbf{r}') ds' \quad (15-136)$$

The second term on the right side of (15-136) is a constant, and it can be dropped since $\varphi(\mathbf{r})$ is undetermined by the boundary conditions up to an additive constant.

D. Nonhomogeneous Partial Differential Equation with Mixed Boundary Conditions

When the boundary conditions on $\varphi(\mathbf{r})$ are such that $\varphi(\mathbf{r}_s)$ is specified in part of the surface S and $[\nabla' \varphi(\mathbf{r}_s)] \cdot \hat{\mathbf{n}} = \partial \varphi(\mathbf{r}_s) / \partial n$ is specified over the remaining part of S , it is referred to as having *mixed* boundary conditions. Then it is desirable to construct a Green's function so that it vanishes on that part of S over which $\varphi(\mathbf{r}_s)$ is specified, and its normal derivative $[\partial G(\mathbf{r}_s, \mathbf{r}') / \partial n]$ vanishes over the remaining part of S over which $\partial \varphi(\mathbf{r}_s) / \partial n$ is specified. Although this is a more complex procedure, it does provide a method to derive the Green's function even under those mixed boundary conditions.

15.6 GREEN'S FUNCTIONS OF THE SCALAR HELMHOLTZ EQUATION

Now that we have derived the development of the generalized Green's function, let us apply the formulation to the scalar Helmholtz equation in three-dimensional problems of rectangular, cylindrical, and spherical coordinates.

15.6.1 Rectangular Coordinates

The development of Green's functions in rectangular coordinates has already been applied for one and two space variables in almost all of the previous sections. In this section we want to derive it for a three-dimensional problem. Specifically, let us derive the Green's function for the electric field component E_y that satisfies the partial differential equation

$$\nabla^2 E_y + \beta_0^2 E_y = \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} \right) E_y + \beta_0^2 E_y = j\omega\mu J_y(x, y, z) \quad (15-137)$$

subject to the boundary conditions

$$E_y(x=0, 0 \leq y \leq b, -\infty \leq z \leq +\infty) = E_y(x=a, 0 \leq y \leq b, -\infty \leq z \leq +\infty) = 0 \quad (15-137a)$$

In (15-137), E_y represents the electric field component of a TM^z field configuration (subject to $\nabla \cdot \mathbf{J} = 0$) inside a metallic waveguide of dimensions a, b in the x, y directions, respectively, as shown in Figure 15-7. Also $J_y(x, y, z)$ represents the electric current density of the feed probe that is used to excite the fields within the metallic waveguide. It is assumed that the wave is traveling in the z direction. The time-harmonic variations are of $e^{+j\omega t}$, and they are suppressed.

The Green's function must satisfy the partial differential equation

$$\nabla^2 G(x, y, z; x', y', z') + \beta_0^2 G(x, y, z; x', y', z') = \delta(x - x')\delta(y - y')\delta(z - z') \quad (15-138)$$

subject to the boundary conditions

$$\begin{aligned} G(x=0, 0 \leq y \leq b, -\infty \leq z \leq +\infty) \\ = G(x=a, 0 \leq y \leq b, -\infty \leq z \leq +\infty) = 0 \end{aligned} \quad (15-138a)$$

Since the electric field and Green's function satisfy, respectively, the Dirichlet boundary conditions (15-124a) and (15-124b), then, according to (15-123) or (15-125), the electric field is obtained using

$$E_y(x, y, z) = j\omega\mu \iiint_V J_y(x', y', z') G(x, y, z; x', y', z') dx' dy' dz' \quad (15-139)$$

The Green's function can be derived either in closed, series, or integral form. We will choose here the series form. We begin the development of the Green's function by assuming its solution can be represented by a two-function Fourier series of sine function in x and cosine function in y which satisfy, respectively, the boundary conditions at $x=0, a$. Thus we can express $G(x, y, z; x', y', z')$ as

$$G(x, y, z; x', y', z') = \sum_{m=1}^{\infty} \sum_{n=1}^{\infty} g_{mn}(z; x', y', z') \sin\left(\frac{m\pi}{a} x\right) \cos\left(\frac{n\pi}{b} y\right) \quad (15-140)$$

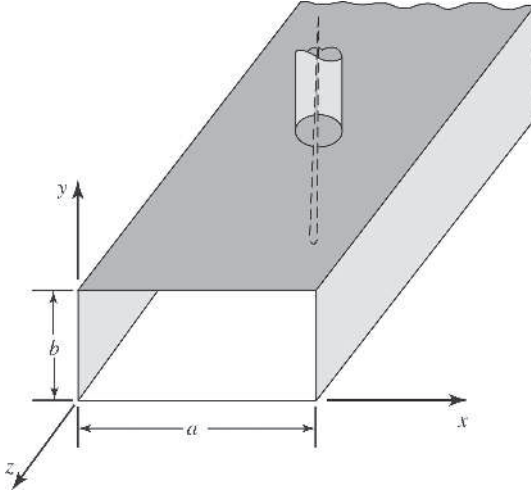


Figure 15-7 Rectangular waveguide excited by linear electric probe.

Substituting (15-140) into (15-138) leads to

$$\begin{aligned} & \sum_{m=1}^{\infty} \sum_{n=1}^{\infty} \left[-\left(\frac{m\pi}{a}\right)^2 - \left(\frac{n\pi}{b}\right)^2 + \beta_0^2 + \frac{\partial^2}{\partial z^2} \right] \\ & \quad \times g_{mn}(z; x', y', z') \sin\left(\frac{m\pi}{a} x'\right) \cos\left(\frac{n\pi}{b} y'\right) \\ & = \delta(x - x') \delta(y - y') \delta(z - z') \end{aligned} \quad (15-141)$$

Multiplying both sides of (15-141) by $\sin(p\pi x/a) \cos(q\pi y/b)$, integrating from 0 to a in x and 0 to b in y , and using (8-56a), (8-56b), (15-48a), and (15-48b), we can reduce (15-141) to

$$\frac{ab}{4} \left(\frac{\partial^2}{\partial z^2} + \beta_z^2 \right) g_{mn}(z; x', y', z') = \sin\left(\frac{m\pi}{a} x'\right) \cos\left(\frac{n\pi}{b} y'\right) \delta(z - z') \quad (15-142)$$

or

$$\left(\frac{d^2}{dz^2} + \beta_z^2 \right) g_{mn}(z; x', y', z') = \frac{4}{ab} \sin\left(\frac{m\pi}{a} x'\right) \cos\left(\frac{n\pi}{b} y'\right) \delta(z - z') \quad (15-142a)$$

where

$$\beta_z^2 = \beta_0^2 - \left[\left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2 \right] = \beta_0^2 - (\beta_x^2 + \beta_y^2) \quad (15-142b)$$

$$\beta_x = \frac{m\pi}{a}, \quad m = 1, 2, 3, \dots \quad (15-142c)$$

$$\beta_y = \frac{n\pi}{b}, \quad n = 1, 2, 3, \dots \quad (15-142d)$$

The function $g_{mn}(z; x', y', z')$ satisfies the single variable differential equation 15-142a, and it can be found by using the recipe of Section 15.3.1 as represented by (15-44c), (15-45a), and (15-45b). Two solutions of the homogeneous differential equation

$$\left(\frac{d^2}{dz^2} + \beta_z^2 \right) g_{mn}(z; x', y', z') = 0 \quad (15-143)$$

of (15-142a) are

$$g_{mn}^{(1)} = A_{mn} e^{+j\beta_z z} \quad \text{for } z < z' \quad (15-143a)$$

$$g_{mn}^{(2)} = B_{mn} e^{-j\beta_z z} \quad \text{for } z > z' \quad (15-143b)$$

Using (15-44c) where $y_1 = g_{mn}^{(1)}$ and $y_2 = g_{mn}^{(2)}$, we can write the Wronskian as

$$\begin{aligned} W(z') &= A_{mn} B_{mn} (-j\beta_z) e^{j\beta_z z'} e^{-j\beta_z z'} - A_{mn} B_{mn} (j\beta_z) e^{-j\beta_z z'} e^{+j\beta_z z'} \\ W(z') &= -j2\beta_z A_{mn} B_{mn} \end{aligned} \quad (15-144)$$

By comparing (15-142a) to (15-33), it is apparent that

$$\begin{aligned} p(z) &= 1 \\ q(z) &= 0 \\ r(z) &= 1 \\ \lambda &= \beta_z^2 \end{aligned} \quad (15-145)$$

Using (15-143a) through (15-145), the solution for $g_{mn}(z; x', y', z')$ of (15-142a) can be written, by referring to (15-45a) and (15-45b), as

$$g_{mn}(z; x', y', z') = \begin{cases} j \frac{2}{ab} \frac{\sin\left(\frac{m\pi}{a} x'\right) \cos\left(\frac{n\pi}{b} y'\right)}{\beta_z} e^{-j\beta_z(z-z')} & \text{for } z > z' \end{cases} \quad (15-146a)$$

$$g_{mn}(z; x', y', z') = \begin{cases} j \frac{2}{ab} \frac{\sin\left(\frac{m\pi}{a} x'\right) \cos\left(\frac{n\pi}{b} y'\right)}{\beta_z} e^{-j\beta_z(z-z')} & \text{for } z > z' \end{cases} \quad (15-146b)$$

or

$$g_{mn}(z; x', y', z') = j \frac{2}{ab} \frac{\sin\left(\frac{m\pi}{a} x'\right) \cos\left(\frac{n\pi}{b} y'\right)}{\beta_z} e^{-j\beta_z|z-z'|} \quad \text{for } z < z', \quad z > z' \quad (15-146c)$$

Thus, the Green's function of (15-140) can now be expressed as

$$\begin{aligned} G(x, y, z; x', y', z') &= j \frac{2}{ab} \sum_{m=1,2,\dots}^{\infty} \sum_{n=1,2,\dots}^{\infty} \frac{\sin\left(\frac{m\pi}{a} x'\right) \cos\left(\frac{n\pi}{b} y'\right)}{\beta_z} \\ &\quad \times \sin\left(\frac{m\pi}{a} x\right) \cos\left(\frac{n\pi}{b} y\right) e^{-j\beta_z|z-z'|} \quad \text{for } z < z', \quad z > z' \end{aligned} \quad (15-147)$$

where

$$\beta_z = \begin{cases} \sqrt{\beta_0^2 - (\beta_x^2 + \beta_y^2)} & \text{for } \beta_0^2 > (\beta_x^2 + \beta_y^2) \\ -j\sqrt{(\beta_x^2 + \beta_y^2) - \beta_0^2} & \text{for } \beta_0^2 < (\beta_x^2 + \beta_y^2) \end{cases} \quad (15-147a)$$

$$(15-147b)$$

It is evident from (15-147) through (15-147b) that when $\beta_0^2 > (\beta_x^2 + \beta_y^2)$ the modes are propagating and when $\beta_0^2 < (\beta_x^2 + \beta_y^2)$ the modes are not propagating (evanescent). The nonpropagating modes converge very rapidly when $|z - z'|$ is very large.

Once the Green's function is formulated as in (15-147), the electric field can be found using (15-139).

15.6.2 Cylindrical Coordinates

Until now we have concentrated on developing primarily Green's functions of problems dealing with rectangular coordinates. This was done to maintain simplicity in the mathematics so that the analytical formulations would not obscure the fundamental concepts. Now we are ready to deal with problems expressed by other coordinate systems, such as cylindrical and spherical.

Let us assume that an infinite electric line source of constant current I_z is placed at $\rho = \rho'$, $\phi = \phi'$ inside a circular waveguide of radius a , as shown in Figure 15-8. The electric field component E_z satisfies the partial differential equation

$$\nabla^2 E_z + \beta_0^2 E_z = f(\rho, \phi) = j\omega\mu I_z \quad (15-148)$$

subject to the boundary condition

$$E_z(\rho = a, 0 \leq \phi \leq 2\pi, z) = 0 \quad (15-148a)$$

The Green's function of this problem will satisfy the partial differential equation

$$\nabla^2 G + \beta_0^2 G = \delta(\mathbf{p} - \mathbf{p}') \quad (15-149)$$

The boundary condition for the Green's function can be chosen so that

$$G(\rho = a, 0 \leq \phi \leq 2\pi, z) = 0 \quad (15-149a)$$

Since the boundary conditions (15-148a) and (15-149a) on E_z and G , respectively, are of the Dirichlet type, then according to (15-123) or (15-125)

$$E_z(\rho, \phi) = \iint_S f(\rho', \phi') G(\rho, \phi; \rho', \phi') ds' = j\omega\mu \iint_S I_z(\rho', \phi') G(\rho, \phi; \rho', \phi') ds' \quad (15-150)$$

Since both the current source and the circular waveguide are of infinite length, the problem reduces to a two-dimensional one. Thus, we can express initially the Green's function by an infinite Fourier series whose eigenvalues in ϕ satisfy the periodicity requirements. That is,

$$G(\rho, \phi; \rho', \phi') = \sum_{m=-\infty}^{+\infty} g_m(\rho; \rho', \phi') e^{jm\phi} \quad (15-151)$$

In cylindrical coordinates, the delta function $\delta(\mathbf{p} - \mathbf{p}')$ in (15-149) can be expressed, in general, as [10, 13]

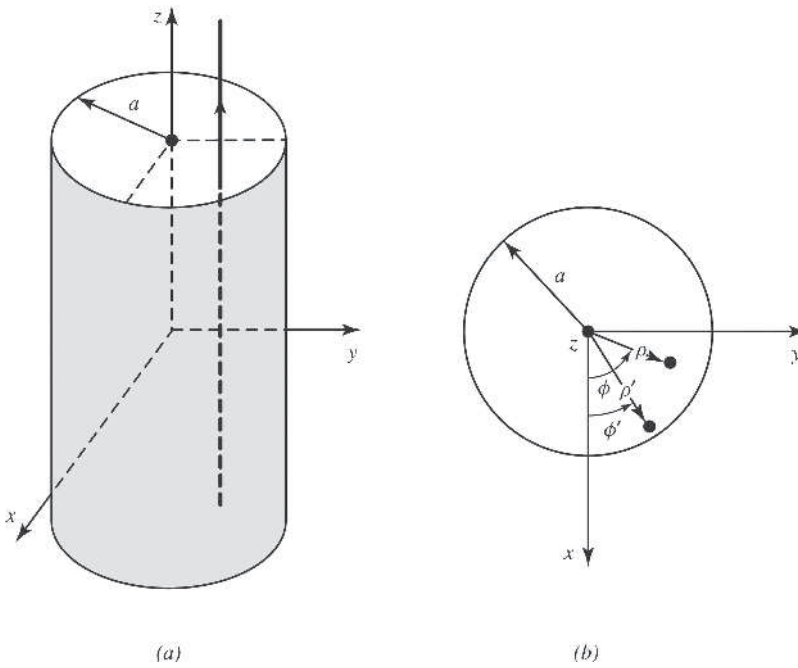


Figure 15-8 Electric line source within a circular conducting cylinder. (a) Line source and conducting cylinder. (b) Top view.

$$\delta(\mathbf{r}-\mathbf{r}') = \begin{cases} \frac{1}{\rho} \delta(\rho-\rho') \delta(\phi-\phi') \delta(z-z') & (15-152a) \\ \frac{1}{2\pi\rho} \delta(\rho-\rho') \delta(z-z') & \text{for no } \phi \text{ dependence} & (15-152b) \\ \frac{1}{2\pi\rho} \delta(\rho-\rho') & \text{for neither } \phi \text{ nor } z \text{ dependence} & (15-152c) \\ \frac{1}{\rho} \delta(\rho-\rho') \delta(\phi-\phi') & \text{for no } z \text{ dependence} & (15-152d) \end{cases}$$

In expanded form, the Green's function of (15-149) can now be written, using (15-152d) and assuming no z variations, as

$$\frac{\partial^2 G}{\partial \rho^2} + \frac{1}{\rho} \frac{\partial G}{\partial \rho} + \frac{1}{\rho^2} \frac{\partial^2 G}{\partial \phi^2} + \beta_0^2 G = \frac{1}{\rho} \delta(\rho-\rho') \delta(\phi-\phi') \quad (15-153)$$

Substituting (15-151) into (15-153) leads to

$$\sum_{m=-\infty}^{+\infty} \left[\frac{\partial^2}{\partial \rho^2} + \frac{1}{\rho} \frac{\partial}{\partial \rho} - \frac{m^2}{\rho^2} + \beta_0^2 \right] g_m(\rho; \rho', \phi') e^{jm\phi} = \frac{1}{\rho} \delta(\rho-\rho') \delta(\phi-\phi') \quad (15-154)$$

Multiplying both sides of (15-154) by $e^{-jn\phi}$, integrating both sides from 0 to 2π in ϕ , and using the orthogonality condition

$$\int_0^{2\pi} e^{j(m-n)\phi} d\phi = \begin{cases} 2\pi & m = n \\ 0 & m \neq n \end{cases} \quad (15-155a)$$

$$(15-155b)$$

reduces (15-154) to

$$2\pi \left[\frac{\partial^2 g_m}{\partial \rho^2} + \frac{1}{\rho} \frac{\partial g_m}{\partial \rho} + \left(\beta_0^2 - \frac{m^2}{\rho^2} \right) g_m \right] = \frac{1}{\rho} e^{-jm\phi'} \delta(\rho - \rho')$$

or

$$\boxed{\rho \frac{d^2 g_m}{d\rho^2} + \frac{dg_m}{d\rho} + \left(\rho \beta_0^2 - \frac{m^2}{\rho} \right) g_m = \frac{e^{-jm\phi'}}{2\pi} \delta(\rho - \rho')} \quad (15-156)$$

where the partial derivatives have been replaced by ordinary derivatives.

The function $g_m(\rho; \rho', \phi')$ satisfies the differential equation 15-156, and its solution can be obtained using the closed-form recipe of Section 15.3.1 represented by (15-44c) and (15-45a) through (15-45b). The homogeneous equation 15-156 can be written, by multiplying through by ρ , as

$$\rho^2 \frac{d^2 g_m}{d\rho^2} + \rho \frac{dg_m}{d\rho} + (\beta_0^2 \rho^2 - m^2) g_m = 0 \quad (15-157)$$

or

$$\rho \frac{d^2 g_m}{d\rho^2} + \frac{dg_m}{d\rho} + \left(\beta_0^2 \rho - \frac{m^2}{\rho} \right) g_m = 0 \quad (15-157a)$$

which is of the one-dimensional Sturm-Liouville form of (15-26) or (15-33) (see Example 15-1) where

$$\begin{aligned} p(\rho) &= \rho \\ q(\rho) &= \frac{m^2}{\rho} \\ r(\rho) &= \rho \\ \lambda &= \beta_0^2 \end{aligned} \quad (15-158)$$

Equation 15-157 is recognized as being Bessel's differential equation 3-64 whose two solutions can be written according to (3-67a) as

$$g_m^{(1)} = A_m J_m(\beta_0 \rho) + B_m Y_m(\beta_0 \rho) \quad \text{for } \rho < \rho' \quad (15-159a)$$

$$g_m^{(2)} = C_m J_m(\beta_0 \rho) + D_m Y_m(\beta_0 \rho) \quad \text{for } \rho > \rho' \quad (15-159b)$$

These two solutions were chosen because the fields within the waveguide form standing waves instead of traveling waves.

Since the Green's function of (15-151) must represent, according to (15-150), the field everywhere, including the origin, then $B_m = 0$ in (15-159a) since $Y_m(\beta_0 \rho)$ possesses a singularity at $\rho = 0$. Also since the Green's function must satisfy the boundary condition (15-149a), then the solution of $g_m^{(2)}$ of (15-159b) must also satisfy (15-149a). Thus,

$$g_m^{(2)}(\rho = a) = C_m J_m(\beta_0 a) + D_m Y_m(\beta_0 a) = 0$$

hence,

$$D_m = -C_m \frac{J_m(\beta_0 a)}{Y_m(\beta_0 a)} \quad (15-160)$$

Thus, (15-159a) and (15-159b) can be reduced to

$$g_m^{(1)} = A_m J_m(\beta_0 \rho) \quad \text{for } \rho < \rho' \quad (15-161a)$$

$$g_m^{(2)} = C_m \left[J_m(\beta_0 \rho) - \frac{J_m(\beta_0 a)}{Y_m(\beta_0 a)} Y_m(\beta_0 \rho) \right] \quad \text{for } \rho > \rho' \quad (15-161b)$$

Using (15-44c) where $y_1 = g_m^{(1)}$ and $y_2 = g_m^{(2)}$, we can write the Wronskian as

$$W(\rho') = \beta_0 A_m C_m \frac{J_m(\beta_0 a)}{Y_m(\beta_0 a)} [J'_m(\beta_0 \rho') Y_m(\beta_0 \rho') - J_m(\beta_0 \rho') Y'_m(\beta_0 \rho')] \quad (15-162)$$

where the prime indicates partial with respect to the entire argument [$' \equiv \partial / \partial(\beta_0 \rho')$]. Using the Wronskian for the Bessel functions of (11-95), we can reduce (15-162) to

$$W(\rho') = -\frac{2}{\pi} A_m C_m \frac{J_m(\beta_0 a)}{Y_m(\beta_0 a)} \frac{1}{\rho'} \quad (15-162a)$$

Finally, $g_m(\rho; \rho', \phi')$ of (15-156) can be written using (15-158), (15-161a) through (15-161b), and (15-162a), by referring to (15-45a) and (15-45b), as

$$g_m(\rho; \rho', \phi') = \begin{cases} -\frac{1}{4} [J_m(\beta_0 \rho') Y_m(\beta_0 a) - J_m(\beta_0 a) Y_m(\beta_0 \rho')] \frac{J_m(\beta_0 \rho)}{J_m(\beta_0 a)} e^{-jm\phi'} & \text{for } \rho < \rho' \\ -\frac{1}{4} [J_m(\beta_0 \rho) Y_m(\beta_0 a) - J_m(\beta_0 a) Y_m(\beta_0 \rho)] \frac{J_m(\beta_0 \rho')}{J_m(\beta_0 a)} e^{-jm\phi'} & \text{for } \rho > \rho' \end{cases} \quad (15-163a)$$

$$(15-163b)$$

Thus, the Green's function (15-151) can then be written as

$$G(\rho, \phi; \rho', \phi') = -\frac{1}{4} \begin{cases} \sum_{m=-\infty}^{+\infty} [J_m(\beta_0 \rho') Y_m(\beta_0 a) - J_m(\beta_0 a) Y_m(\beta_0 \rho')] \times \frac{J_m(\beta_0 \rho)}{J_m(\beta_0 a)} e^{jm(\phi - \phi')} & \text{for } \rho < \rho' \\ \sum_{m=-\infty}^{+\infty} [J_m(\beta_0 \rho) Y_m(\beta_0 a) - J_m(\beta_0 a) Y_m(\beta_0 \rho)] \times \frac{J_m(\beta_0 \rho')}{J_m(\beta_0 a)} e^{jm(\phi - \phi')} & \text{for } \rho > \rho' \end{cases} \quad (15-164a)$$

$$(15-164b)$$

or in cosine terms as

$$G(\rho, \phi; \rho', \phi') = -\frac{1}{2} \begin{cases} \sum_{m=-\infty}^{+\infty} \frac{[J_m(\beta_0 \rho') Y_m(\beta_0 a) - J_m(\beta_0 a) Y_m(\beta_0 \rho')]}{\varepsilon_m} \times \frac{J_m(\beta_0 \rho)}{J_m(\beta_0 a)} \cos[m(\phi - \phi')] & \text{for } \rho < \rho' \\ \sum_{m=-\infty}^{+\infty} \frac{[J_m(\beta_0 \rho) Y_m(\beta_0 a) - J_m(\beta_0 a) Y_m(\beta_0 \rho)]}{\varepsilon_m} \times \frac{J_m(\beta_0 \rho')}{J_m(\beta_0 a)} \cos[m(\phi - \phi')] & \text{for } \rho > \rho' \end{cases} \quad (15-165a)$$

$$(15-165b)$$

where

$$\varepsilon_m = \begin{cases} 2 & m = 0 \\ 0 & m \neq 0 \end{cases} \quad (15-165c)$$

$$(15-165d)$$

The Green's function for this problem can also be derived using the two space variable series expansion method whereby it is represented by orthonormal expansion functions. This is left to the reader as an end-of-chapter exercise.

Example 15-5

An infinite electric line source of constant current I_z is located at $\rho = \rho'$, $\phi = \phi'$, as shown in Figure 15-9, and it is radiating in an unbounded free-space medium. Derive its Green's function in closed form.

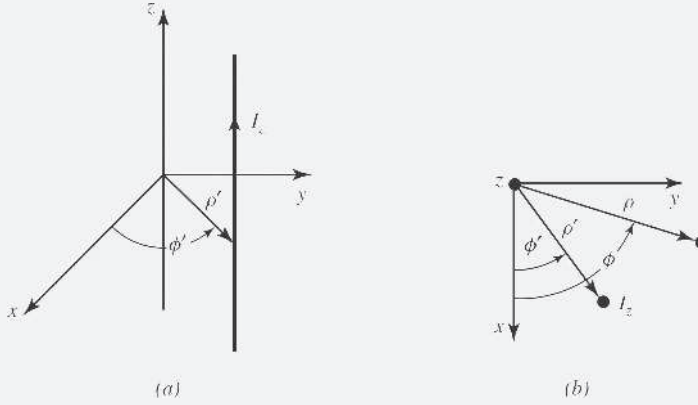


Figure 15-9 Electric line source displaced from the origin. (a) Perspective view. (b) Top view.

Solution: Since the line source is removed from the origin, its Green's function will be a function of ϕ and ϕ' . Thus, it takes the form of (15-151), and it satisfies the differential equations 15-153 through 15-157. However, since the Green's function must satisfy the radiation conditions at infinity ($G \rightarrow 0$ as $\rho \rightarrow \infty$), the two solutions to the homogeneous differential equation 15-157 can be written as

$$\begin{aligned} g_m^{(1)} &= A_m J_m(\beta_0 \rho) + B_m Y_m(\beta_0 \rho) & \text{for } \rho < \rho' \\ g_m^{(2)} &= C_m H_m^{(1)}(\beta_0 \rho) + D_m H_m^{(2)}(\beta_0 \rho) & \text{for } \rho > \rho' \end{aligned}$$

Because the fields must be finite everywhere, including $\rho = 0$, $g_m^{(1)}$ reduces to

$$g_m^{(1)} = A_m J_m(\beta_0 \rho) \quad \text{for } \rho < \rho'$$

In addition, for $\rho > \rho'$ the wave functions must represent outwardly traveling waves. Thus, for $e^{j\omega t}$ time variations, $g_m^{(2)}$ reduces to

$$g_m^{(2)} = D_m H_m^{(2)}(\beta_0 \rho) \quad \text{for } \rho > \rho'$$

Using (15-44c) where $y_1 = g_m^{(1)}$ and $y_2 = g_m^{(2)}$, we can write the Wronskian as

$$\begin{aligned} W(\rho') &= \beta_0 A_m D_m \left[J_m(\beta_0 \rho') H_m^{(2)'}(\beta_0 \rho') - H_m^{(2)}(\beta_0 \rho') J_m'(\beta_0 \rho') \right] \\ &= -j\beta_0 A_m D_m \left[J_m(\beta_0 \rho') Y_m'(\beta_0 \rho') - J_m'(\beta_0 \rho') Y_m(\beta_0 \rho') \right] \end{aligned}$$

which by using the Wronskian of (11-95) for Bessel functions can be expressed as

$$W(\rho') = -j \frac{2}{\pi \rho'} A_m D_m$$

Thus, $g_m(\rho; \rho', \phi')$ of (15-156) can be written, using (15-45a) through (15-45b) and (15-158) along with the preceding expressions for $g_m^{(1)}$, $g_m^{(2)}$, and $W(\rho')$, as

$$g_m(\rho; \rho', \phi') = -\frac{1}{4j} \begin{cases} J_m(\beta_0 \rho) H_m^{(2)}(\beta_0 \rho') e^{-jm\phi'} & \text{for } \rho < \rho' \\ J_m(\beta_0 \rho') H_m^{(2)}(\beta_0 \rho) e^{-jm\phi'} & \text{for } \rho > \rho' \end{cases}$$

Thus, the Green's function of (15-151) can be written as

$$G(\rho, \phi; \rho', \phi') = -\frac{1}{4j} \begin{cases} \sum_{m=-\infty}^{+\infty} J_m(\beta_0 \rho) H_m^{(2)}(\beta_0 \rho') e^{jm(\phi - \phi')} & \text{for } \rho < \rho' \\ \sum_{m=-\infty}^{+\infty} J_m(\beta_0 \rho') H_m^{(2)}(\beta_0 \rho) e^{jm(\phi - \phi')} & \text{for } \rho > \rho' \end{cases}$$

which, by the addition theorem for Hankel functions of (11-69a) through (11-69b) or (11-82a) through (11-82b), can be expressed in succinct form as

$$G(\rho, \phi; \rho', \phi') = -\frac{1}{4j} H_0^{(2)}(\beta_0 |\rho - \rho'|)$$

This is the well-known two-dimensional Green's function for cylindrical waves.

15.6.3 Spherical Coordinates

The development of the Green's function for problems represented by spherical coordinates is more complex, and it must be expressed, in general, in terms of spherical Bessel and Hankel functions, Legendre functions, and complex exponentials or cosinusoids (see Chapter 10). In order to minimize the mathematical complexities here, we will develop the Green's function of a source positioned at r', θ', ϕ' , inside a sphere of radius a and with free-space, as shown in Figure 15-10.

The Green's function must satisfy the partial differential equation

$$\boxed{\nabla^2 G + \beta_0^2 G = \delta(\mathbf{r} - \mathbf{r}')} \quad (15-166)$$

subject to the boundary condition

$$G(r = a, 0 \leq \theta \leq \pi, 0 \leq \phi \leq 2\pi) = 0 \quad (15-166a)$$

In spherical coordinates the delta function $\delta(\mathbf{r} - \mathbf{r}')$ in (15-166) can be expressed, in general, as [10, 13]

$$\delta(\mathbf{r} - \mathbf{r}') = \begin{cases} \frac{1}{r^2 \sin \theta} \delta(r - r') \delta(\theta - \theta') \delta(\phi - \phi') & (15-167a) \\ \frac{1}{2r^2} \delta(r - r') \delta(\phi - \phi') & \text{for no } \theta \text{ dependence} & (15-167b) \\ \frac{1}{2\pi r^2 \sin \theta} \delta(r - r') \delta(\theta - \theta') & \text{for no } \phi \text{ dependence} & (15-167c) \\ \frac{1}{4\pi r^2} \delta(r - r') & \text{for neither } \theta \text{ nor } \phi \text{ dependence} & (15-167d) \end{cases}$$

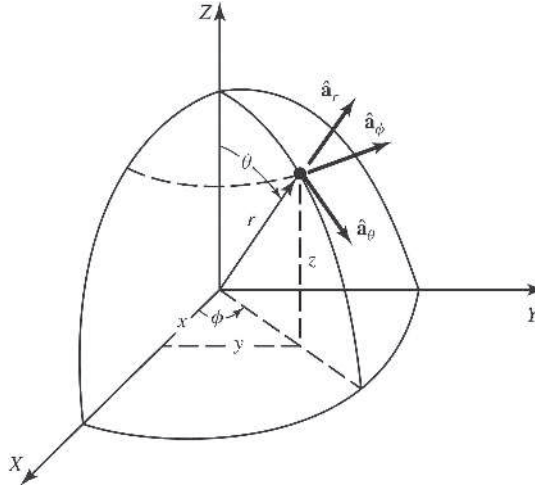


Figure 15-10 Spherical coordinate system.

In expanded form, the Green's function of (15-166) can now be written using (15-167a) as

$$\begin{aligned} \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial G}{\partial r} \right) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial G}{\partial \theta} \right) + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2 G}{\partial \phi^2} + \beta_0^2 G \\ = \frac{1}{r^2 \sin \theta} \delta(r - r') \delta(\theta - \theta') \delta(\phi - \phi') \end{aligned} \quad (15-168)$$

Since the spherical harmonics form a complete set for functions of the angles θ and ϕ , the Green's function can be represented by a double summation of an infinite series

$$\begin{aligned} G(r, \theta, \phi; r', \theta', \phi') &= \sum_{n=0}^{\infty} \sum_{m=-n}^n g_{mn}(r; r', \theta', \phi') P_n^m(\cos \theta) e^{jm\phi} \\ &= \sum_{n=0}^{\infty} \sum_{m=-n}^n g_{mn}(r; r', \theta', \phi') T_{mn}(\theta, \phi) \end{aligned} \quad (15-169)$$

where $T_{mn}(\theta, \phi)$ represents the tesseral harmonics of (11-214a) and (11-214b), or

$$T_{mn}(\theta, \phi) = C_{mn} P_n^m(\cos \theta) e^{jm\phi} \quad (15-169a)$$

where

$$C_{mn} = \sqrt{\frac{(2n+1)(n-m)!}{4\pi(n+m)!}} \quad (15-169b)$$

Multiplying (15-168) by r^2 and then substituting (15-169) into (15-168), we can write that

$$\begin{aligned} \sum_{n=0}^{\infty} \sum_{m=-n}^n \left[T_{mn} \frac{\partial}{\partial r} \left(r^2 \frac{\partial g_{mn}}{\partial r} \right) + \frac{g_{mn}}{\sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial T_{mn}}{\partial \theta} \right) - \frac{m^2}{\sin^2 \theta} g_{mn} T_{mn} + (\beta_0 r)^2 g_{mn} T_{mn} \right] \\ = \frac{1}{\sin \theta} \delta(r - r') \delta(\theta - \theta') \delta(\phi - \phi') \end{aligned} \quad (15-170)$$

Dividing both sides of (15-170) by $g_{mn}T_{mn}$, we can write that

$$\sum_{n=0}^{\infty} \sum_{m=-n}^{\infty} \left[\frac{1}{g_{mn}} \frac{\partial}{\partial r} \left(r^2 \frac{\partial g_{mn}}{\partial r} \right) + \frac{1}{T_{mn} \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial T_{mn}}{\partial \theta} \right) - \frac{m^2}{\sin^2 \theta} + (\beta_0 r)^2 \right] \\ = \frac{1}{g_{mn} T_{mn} \sin \theta} \delta(r-r') \delta(\theta-\theta') \delta(\phi-\phi') \quad (15-170a)$$

Using (3-86b), we can write that

$$\frac{1}{T_{mn} \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial T_{mn}}{\partial \theta} \right) - \frac{m^2}{\sin^2 \theta} = -n(n+1) \quad (15-171)$$

hence,

$$\frac{1}{\sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial T_{mn}}{\partial \theta} \right) + \left[n(n+1) - \left(\frac{m}{\sin \theta} \right)^2 \right] T_{mn} = 0 \quad (15-171a)$$

Thus, (15-170a) reduces, by substituting (15-171) into it and then multiplying through by $g_{mn}T_{mn}$, to

$$\sum_{n=0}^{\infty} \sum_{m=-n}^n \left\{ \frac{\partial}{\partial r} \left(r^2 \frac{\partial g_{mn}}{\partial r} \right) + [(\beta_0 r)^2 + n(n+1)] g_{mn} \right\} T_{mn} \\ = \frac{1}{\sin \theta} \delta(r-r') \delta(\theta-\theta') \delta(\phi-\phi') \quad (15-172)$$

From (11-214a) through (11-216d) and the definitions of the tesseral harmonics and Legendre functions, it can be shown that the orthogonality conditions of the tesseral harmonics are [20]

$$\int_0^{2\pi} \int_0^\pi T_{mn}(\theta, \phi) T_{pq}^*(\theta, \phi) \sin \theta d\theta d\phi = \delta_{mp} \delta_{nq} \quad (15-173)$$

where

$$T_{pq}^*(\theta, \phi) = (-1)^p T_{(-p)q}(\theta, \phi) \quad (15-173a)$$

$$\delta_{rs} = \begin{cases} 1 & r = s \\ 0 & r \neq s \end{cases} \quad (15-173b)$$

Let

$$g_{mn}(r; r', \theta', \phi') = h_{mn}(r, r') T^*(\theta', \phi') \quad (15-173c)$$

Multiplying both sides of (15-172) by $T_{pq}^*(\theta, \phi) \sin \theta$, integrating from 0 to π in θ and 0 to 2π in ϕ , and using the orthogonality condition of (15-173), it can be shown that (15-172) reduces to

$$\boxed{\frac{d}{dr} \left(r^2 \frac{dh_{mn}}{dr} \right) + [(\beta_0 r)^2 - n(n+1)] h_{mn} = \delta(r-r')} \quad (15-174)$$

where the partial derivative $\partial/\partial r$ has been replaced by ordinary derivatives.

The function $h_{mn}(r, r')$ satisfies the differential equation 15-174, and its solution can be obtained using the closed-form recipe of Section 15.3.1 represented by (15-44c) and (15-45a) through (15-45b). The homogeneous equation 15-174 can be written as

$$\frac{d}{dr} \left(r^2 \frac{dh_{mn}}{dr} \right) + [(\beta_0 r)^2 - n(n+1)] h_{mn} = 0 \quad (15-175)$$

which is of the one-dimensional Sturm-Liouville form of (15-26) or (15-33) where

$$\begin{aligned} p(r) &= r^2 \\ q(r) &= -n(n+1) \\ r(x) &= r^2 \\ \lambda &= \beta^2 \end{aligned} \quad (15-176)$$

Equation 15-175 is recognized as being (3-83) or (3-86a) whose solution can be represented by either (3-87a) or (3-87b). We choose here the form of (3-87a) since we need to represent the Green's function within the sphere by standing wave functions. Thus, the two solutions of (15-175) can be written as

$$h_{mn}^{(1)} = A_m j_n(\beta_0 r) + B_m y_n(\beta_0 r) \quad \text{for } r < r' \quad (15-177a)$$

$$h_{mn}^{(2)} = C_m j_n(\beta_0 r) + D_m y_n(\beta_0 r) \quad \text{for } r > r' \quad (15-177b)$$

where $j_n(\beta_0 r)$ and $y_n(\beta_0 r)$ are, respectively, spherical Bessel functions of the first and second kind.

Since the Green's function of (15-169) must be finite everywhere, including the origin, then $B_m = 0$ since $y_n(\beta_0 r)$ possesses a singularity at $r = 0$. Also the Green's function must satisfy the boundary condition (15-166a). Therefore, $h_{mn}^{(2)}$ of (15-177b) at $r = a$ reduces to

$$h_{mn}^{(2)}(r = a) = C_m j_n(\beta_0 a) + D_m y_n(\beta_0 a) = 0 \quad (15-178)$$

hence,

$$D_m = -C_m \frac{j_n(\beta_0 a)}{y_n(\beta_0 a)} \quad (15-178a)$$

Thus, (15-177a) and (15-177b) are reduced to

$$h_{mn}^{(1)} = A_m j_n(\beta_0 r) \quad \text{for } r < r' \quad (15-179a)$$

$$h_{mn}^{(2)} = C_m \left[j_n(\beta_0 r) - \frac{j_n(\beta_0 a)}{y_n(\beta_0 a)} y_n(\beta_0 r) \right] \quad \text{for } r > r' \quad (15-179b)$$

Using (15-44c) where $y_1 = h_{mn}^{(1)}$ and $y_2 = h_{mn}^{(2)}$, we can write the Wronskian as

$$W(r') = \beta_0 A_m C_m \frac{j_n(\beta_0 a)}{y_n(\beta_0 a)} [j'_n(\beta_0 r') y_n(\beta_0 r') - j_n(\beta_0 r') y'_n(\beta_0 r')] \quad (15-180)$$

Using the Wronskian for spherical Bessel functions of

$$j_n(\beta_0 r') y'_n(\beta_0 r') - j'_n(\beta_0 r') y_n(\beta_0 r') = \frac{1}{(\beta_0 r')^2} \quad (15-180a)$$

reduces (15-180) to

$$W(r') = -\frac{1}{\beta_0} A_m C_m \frac{j_n(\beta_0 a)}{y_n(\beta_0 a)} \frac{1}{(r')^2} \quad (15-180b)$$

Finally, $h_{mn}(r, r')$ of (15-174) can be written using (15-176), (15-179a) through (15-179b), and (15-180b), by referring to (15-45a) and (15-45b), as

$$h_{mn}(r, r') = \begin{cases} -\beta_0 C_{mn}^2 [j_n(\beta_0 r') y_n(\beta_0 a) - j_n(\beta_0 a) y_n(\beta_0 r')] \frac{j_n(\beta_0 r)}{j_n(\beta_0 a)} & \text{for } r < r' \\ -\beta_0 C_{mn}^2 [j_n(\beta_0 r) y_n(\beta_0 a) - j_n(\beta_0 a) y_n(\beta_0 r')] \frac{j_n(\beta_0 r')}{j_n(\beta_0 a)} & \text{for } r > r' \end{cases} \quad (15-181a)$$

$$(15-181b)$$

Thus, the Green's function of (15-169) can be written as

$$G(r, \theta, \phi; r', \theta', \phi') = -\beta_0 \begin{cases} \sum_{n=0}^{\infty} \sum_{m=-n}^n (-1)^m C_{mn}^2 [j_n(\beta_0 r') y_n(\beta_0 a) - j_n(\beta_0 a) y_n(\beta_0 r')] \\ \times \frac{j_n(\beta_0 r)}{j_n(\beta_0 a)} P_n^m(\cos \theta) P_n^{-m}(\cos \theta') e^{jm(\phi - \phi')} & \text{for } r < r' \\ \sum_{n=0}^{\infty} \sum_{m=-n}^n (-1)^m C_{mn}^2 [j_n(\beta_0 r) y_n(\beta_0 a) - j_n(\beta_0 a) y_n(\beta_0 r')] \\ \times \frac{j_n(\beta_0 r')}{j_n(\beta_0 a)} P_n^m(\cos \theta) P_n^{-m}(\cos \theta') e^{jm(\phi - \phi')} & \text{for } r > r' \end{cases} \quad (15-182a)$$

$$(15-182b)$$

15.7 DYADIC GREEN'S FUNCTIONS

The Green's function development of the previous sections can be used for the solution of electromagnetic problems that satisfy the scalar wave equation. The most general Green's function development and electromagnetic field solution, for problems that satisfy the vector wave equation, will be to use *vectors and dyadics* [14–18]. Before we briefly discuss such a procedure, let us first introduce and define dyadics.

15.7.1 Dyadics

Vectors and dyadics are used, in general, to describe linear transformations *within a given orthogonal coordinate system*, and they simplify the manipulations of mathematical relations, compared to using tensors. For electromagnetic problems, where linear transformations between sources and fields *within a given orthogonal coordinate system* are often necessary, vectors and dyadics are very convenient to use.

A *dyad* is defined by the juxtaposition $\mathbf{AB}$ of the vectors $\mathbf{A}$ and $\mathbf{B}$, with no dot or cross product between them. In general, a dyad has nine terms and in matrix form can be represented by

$$(\mathbf{AB}) = \begin{pmatrix} A_1 B_1 & A_1 B_2 & A_1 B_3 \\ A_2 B_1 & A_2 B_2 & A_2 B_3 \\ A_3 B_1 & A_3 B_2 & A_3 B_3 \end{pmatrix} \quad (15-183)$$

A *dyadic* $\bar{\mathbf{D}}$ can be defined by the sum of N dyads. That is,

$$\bar{\mathbf{D}} = \sum_{n=1}^N \mathbf{A}^n \mathbf{B}^n \quad (15-184)$$

In general, no more than three dyads are required to represent a dyadic, that is, $N_{\max} = 3$.

Let us now define the vectors $\mathbf{A}$, $\mathbf{C}$, $\mathbf{D}_1$, $\mathbf{D}_2$, and $\mathbf{D}_3$ in a general coordinate system with unit vectors $\hat{\mathbf{a}}_1, \hat{\mathbf{a}}_2$, and $\hat{\mathbf{a}}_3$. That is,

$$\mathbf{A} = \hat{\mathbf{a}}_1 A_1 + \hat{\mathbf{a}}_2 A_2 + \hat{\mathbf{a}}_3 A_3 \quad (15-185a)$$

$$\mathbf{C} = \hat{\mathbf{a}}_1 C_1 + \hat{\mathbf{a}}_2 C_2 + \hat{\mathbf{a}}_3 C_3 \quad (15-185b)$$

$$\mathbf{D}_1 = \hat{\mathbf{a}}_1 D_{11} + \hat{\mathbf{a}}_2 D_{12} + \hat{\mathbf{a}}_3 D_{13} \quad (15-185c)$$

$$\mathbf{D}_2 = \hat{\mathbf{a}}_1 D_{21} + \hat{\mathbf{a}}_2 D_{22} + \hat{\mathbf{a}}_3 D_{23} \quad (15-185d)$$

$$\mathbf{D}_3 = \hat{\mathbf{a}}_1 D_{31} + \hat{\mathbf{a}}_2 D_{32} + \hat{\mathbf{a}}_3 D_{33} \quad (15-185e)$$

Let us now write that

$$\begin{aligned} \mathbf{C} &= (\mathbf{A} \cdot \hat{\mathbf{a}}_1) \mathbf{D}_1 + (\mathbf{A} \cdot \hat{\mathbf{a}}_2) \mathbf{D}_2 + (\mathbf{A} \cdot \hat{\mathbf{a}}_3) \mathbf{D}_3 \\ &= \mathbf{A} \cdot (\hat{\mathbf{a}}_1 \mathbf{D}_1) + \mathbf{A} \cdot (\hat{\mathbf{a}}_2 \mathbf{D}_2) + \mathbf{A} \cdot (\hat{\mathbf{a}}_3 \mathbf{D}_3) \\ \mathbf{C} &= \mathbf{A} \cdot (\hat{\mathbf{a}}_1 \mathbf{D}_1 + \hat{\mathbf{a}}_2 \mathbf{D}_2 + \hat{\mathbf{a}}_3 \mathbf{D}_3) \end{aligned} \quad (15-186)$$

or

$$\mathbf{C} = \mathbf{A} \cdot \bar{\mathbf{D}} \quad (15-186a)$$

where

$$\bar{\mathbf{D}} = \hat{\mathbf{a}}_1 \mathbf{D}_1 + \hat{\mathbf{a}}_2 \mathbf{D}_2 + \hat{\mathbf{a}}_3 \mathbf{D}_3 \quad (15-186b)$$

In (15-186) through (15-186b) $\bar{\mathbf{D}}$ is a dyadic, and it is defined by the sum of the three dyads $\hat{\mathbf{a}}_n \mathbf{D}_n$, $n = 1, 2, 3$. In matrix form (15-186) or (15-186a) can be written as

$$(C_1 \ C_2 \ C_3) = (A_1 \ A_2 \ A_3) \begin{pmatrix} D_{11} & D_{12} & D_{13} \\ D_{21} & D_{22} & D_{23} \\ D_{31} & D_{32} & D_{33} \end{pmatrix} \quad (15-187)$$

where the dyadic $\bar{\mathbf{D}}$ has nine elements.

Just like vectors, dyadics satisfy a number of identities involving dot and cross products, differentiations, and integrations. The uninformed reader should refer to the literature [10–13] for such relations.

15.7.2 Green's Functions

In electromagnetics it is often desirable to solve, using the Green's functions approach, the linear vector problem of

$$\boxed{\mathcal{L} \mathbf{h} = \mathbf{f}} \quad (15-188)$$

where $\mathcal{L}$ is a differential operator. Equation 15-188 is a more general and vector representation of (15-27). It should be noted here that the solution of (15-188) *cannot*, in general, be represented by

$$\mathbf{h}(\mathbf{r}) \neq \iiint_V \mathbf{f}(\mathbf{r}') G(\mathbf{r}, \mathbf{r}') dv' \quad (15-189)$$

where $G(\mathbf{r}, \mathbf{r}')$ is a single scalar Green's function. The relation (15-189) would imply that a component of the source $\mathbf{f}$ parallel to a given axis produces a response (field) $\mathbf{h}$ parallel to the same axis. This, in general, is not true.

A more appropriate representation of the solution of (15-188), in a rectangular coordinate system, will be

$$h_x(\mathbf{r}) = \iiint_V [f_x(\mathbf{r}') G_{xx}(\mathbf{r}, \mathbf{r}') + f_y(\mathbf{r}') G_{xy}(\mathbf{r}, \mathbf{r}') + f_z(\mathbf{r}') G_{xz}(\mathbf{r}, \mathbf{r}')] dv' \quad (15-190a)$$

$$h_y(\mathbf{r}) = \iiint_V [f_x(\mathbf{r}') G_{yx}(\mathbf{r}, \mathbf{r}') + f_y(\mathbf{r}') G_{yy}(\mathbf{r}, \mathbf{r}') + f_z(\mathbf{r}') G_{yz}(\mathbf{r}, \mathbf{r}')] dv' \quad (15-190b)$$

$$h_z(\mathbf{r}) = \iiint_V [f_x(\mathbf{r}') G_{zx}(\mathbf{r}, \mathbf{r}') + f_y(\mathbf{r}') G_{zy}(\mathbf{r}, \mathbf{r}') + f_z(\mathbf{r}') G_{zz}(\mathbf{r}, \mathbf{r}')] dv' \quad (15-190c)$$

which, in a more compact form, can be written as

$$\mathbf{h}(\mathbf{r}) = \iiint_V [f_x(\mathbf{r}') \mathbf{G}_x(\mathbf{r}, \mathbf{r}') + f_y(\mathbf{r}') \mathbf{G}_y(\mathbf{r}, \mathbf{r}') + f_z(\mathbf{r}') \mathbf{G}_z(\mathbf{r}, \mathbf{r}')] dv' \quad (15-191)$$

where, as in (15-185c) through (15-185e)

$$\mathbf{G}_x(\mathbf{r}, \mathbf{r}') = \hat{\mathbf{a}}_x G_{xx}(\mathbf{r}, \mathbf{r}') + \hat{\mathbf{a}}_y G_{yx}(\mathbf{r}, \mathbf{r}') + \hat{\mathbf{a}}_z G_{zx}(\mathbf{r}, \mathbf{r}') \quad (15-191a)$$

$$\mathbf{G}_y(\mathbf{r}, \mathbf{r}') = \hat{\mathbf{a}}_x G_{xy}(\mathbf{r}, \mathbf{r}') + \hat{\mathbf{a}}_y G_{yy}(\mathbf{r}, \mathbf{r}') + \hat{\mathbf{a}}_z G_{zy}(\mathbf{r}, \mathbf{r}') \quad (15-191b)$$

$$\mathbf{G}_z(\mathbf{r}, \mathbf{r}') = \hat{\mathbf{a}}_x G_{xz}(\mathbf{r}, \mathbf{r}') + \hat{\mathbf{a}}_y G_{yz}(\mathbf{r}, \mathbf{r}') + \hat{\mathbf{a}}_z G_{zz}(\mathbf{r}, \mathbf{r}') \quad (15-191c)$$

In (15-190a) through (15-191c), the $G_{ij}(\mathbf{r}, \mathbf{r}')$'s are the elements of the dyadic $\bar{\mathbf{G}}(\mathbf{r}, \mathbf{r}')$, which is referred to here as the *dyadic Green's function*. In (15-191) through (15-191c), the $\mathbf{G}_i(\mathbf{r}, \mathbf{r}')$'s are the column vectors of the dyadic Green's function $\bar{\mathbf{G}}(\mathbf{r}, \mathbf{r}')$.

Using the notation of (15-186) through (15-186b), the solution of (15-191) can also be written as

$$\begin{aligned} \mathbf{h}(\mathbf{r}) &= \iiint_V \{ [\mathbf{f}(\mathbf{r}') \cdot \hat{\mathbf{a}}_x] \mathbf{G}_x(\mathbf{r}, \mathbf{r}') + [\mathbf{f}(\mathbf{r}') \cdot \hat{\mathbf{a}}_y] \mathbf{G}_y(\mathbf{r}, \mathbf{r}') + [\mathbf{f}(\mathbf{r}') \cdot \hat{\mathbf{a}}_z] \mathbf{G}_z(\mathbf{r}, \mathbf{r}') \} dv' \\ &= \iiint_V \mathbf{f}(\mathbf{r}') \cdot [\hat{\mathbf{a}}_x \mathbf{G}_x(\mathbf{r}, \mathbf{r}') + \hat{\mathbf{a}}_y \mathbf{G}_y(\mathbf{r}, \mathbf{r}') + \hat{\mathbf{a}}_z \mathbf{G}_z(\mathbf{r}, \mathbf{r}')] dv' \end{aligned}$$

$$\mathbf{h}(\mathbf{r}) = \iiint_V \mathbf{f}(\mathbf{r}') \cdot \bar{\mathbf{G}}(\mathbf{r}, \mathbf{r}') dv' \quad (15-192)$$

where $\bar{\mathbf{G}}(\mathbf{r}, \mathbf{r}')$ is the dyadic Green's function

$$\bar{\mathbf{G}}(\mathbf{r}, \mathbf{r}') = \hat{\mathbf{a}}_x \mathbf{G}_x(\mathbf{r}, \mathbf{r}') + \hat{\mathbf{a}}_y \mathbf{G}_y(\mathbf{r}, \mathbf{r}') + \hat{\mathbf{a}}_z \mathbf{G}_z(\mathbf{r}, \mathbf{r}') \quad (15-192a)$$

The dyadic Green's function $\bar{\mathbf{G}}(\mathbf{r}, \mathbf{r}')$ can be found by first finding the vectors $\mathbf{G}_x(\mathbf{r}, \mathbf{r}')$, $\mathbf{G}_y(\mathbf{r}, \mathbf{r}')$, and $\mathbf{G}_z(\mathbf{r}, \mathbf{r}')$ each satisfying the homogeneous form of the partial differential equation 15-188, that is,

$$\mathcal{L}\mathbf{G}_x(\mathbf{r}, \mathbf{r}') = \hat{\mathbf{a}}_x \delta(\mathbf{r} - \mathbf{r}') \quad (15-193a)$$

$$\mathcal{L}\mathbf{G}_y(\mathbf{r}, \mathbf{r}') = \hat{\mathbf{a}}_y \delta(\mathbf{r} - \mathbf{r}') \quad (15-193b)$$

$$\mathcal{L}\mathbf{G}_z(\mathbf{r}, \mathbf{r}') = \hat{\mathbf{a}}_z \delta(\mathbf{r} - \mathbf{r}') \quad (15-193c)$$

and the appropriate boundary conditions, and then using (15-192a) to form the dyadic Green's function. Any of the methods of the previous sections can be used to find the Green's function of (15-193a) through (15-193c).

An example for the potential use of the dyadic Green's function is the solution for the electric and magnetic fields due to a source represented by the electric current density $\mathbf{J}$. According to (6-32a) and (6-32b), the electric and magnetic fields can be written as

$$\mathbf{H}(\mathbf{r}) = \frac{1}{\mu} \nabla \times \mathbf{A} \quad (15-194a)$$

$$\mathbf{E}(\mathbf{r}) = -j\omega\mathbf{A} - j \frac{1}{\omega\mu\epsilon} \nabla(\nabla \cdot \mathbf{A}) \quad (15-194b)$$

where the vector potential $\mathbf{A}$ satisfies the partial differential equation 6-30 or

$$\nabla^2 \mathbf{A} + \beta^2 \mathbf{A} = -\mu \mathbf{J} \quad (15-195)$$

Using the dyadic Green's function approach, the vector potential $\mathbf{A}$ can be found using

$$\mathbf{A} = -\mu \iiint_V \mathbf{J}(\mathbf{r}') \cdot \bar{\mathbf{G}}(\mathbf{r}, \mathbf{r}') dv' \quad (15-196)$$

where the dyadic Green's function must satisfy the partial differential equation

$$\nabla^2 \bar{\mathbf{G}} + \beta^2 \bar{\mathbf{G}} = \bar{\delta}(\mathbf{r} - \mathbf{r}') \quad (15-197)$$

and the appropriate boundary conditions.

Because of the complexity for the development of the dyadic Green's function, it will not be pursued any further here. The interested reader is referred to the literature [10–18] for more details.

15.8 MULTIMEDIA

On the website that accompanies this book, the following multimedia resources are included for the review, understanding, and presentation of the material of this chapter.

- **PowerPoint (PPT)** viewgraphs, in multicolor.

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PROBLEMS

- 15.1.** Using the procedure of Section 15.3.1, as represented by (15-44c) and (15-45a) through (15-45b), derive the Green's function of the mechanics problem of Section 15.2.2 as given by (15-24a) and (15-24b).
- 15.2.** The displacement $u(x)$ of a string of length ℓ subjected to a cosinusoidal force

$$f(x, t) = f(x) e^{+j\omega t}$$

is determined by

$$u(x, t) = u(x) e^{+j\omega t}$$

where $u(x)$ satisfies the differential equation

$$\frac{d^2 u(x)}{dx^2} + \beta^2 u(x) = f(x), \quad \beta^2 = \omega / c$$

Assuming the ends of the string are fixed

$$u(x=0) = u(x=\ell) = 0$$

determine, in closed, form the Green's function of the system.

- 15.3.** Two PEC semi-infinite plates, which both are grounded ($V = 0$), are separated by a distance w in the x direction, as shown in Figure P15-3. A d. c. infinite line source of constant charge density $q(x')$ is positioned at x' between the two plates and extends to infinity in the y direction. Assume the plates are infinite in the y direction and free space exists between the two plates. For this problem:
- (a) Specify the appropriate boundary conditions for the potential and the associated Green's function.
 - (b) Derive the Green's function in the x - y plane in series form, including an expression for the eigenvalues with the appropriate indices.
 - (c) Based on this Green's function and stated charge distribution, write an expression for the potential $V(x)$ between the plates. Show all the steps.

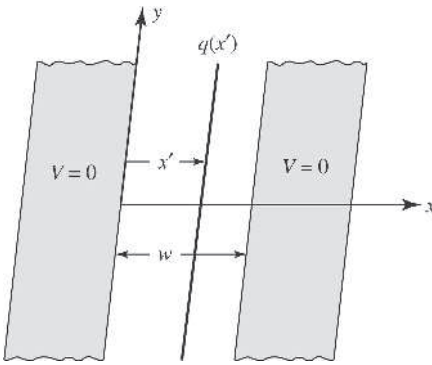


Figure P15-3

- 15.4. Derive in series form the Green's function of Problem 15.3.
- 15.5. Two infinite radial plates with an interior angle of α , as shown in Figure P15-5, are both maintained at a potential of $V=0$. Determine in closed form the Green's function for the electric potential distribution between the plates.

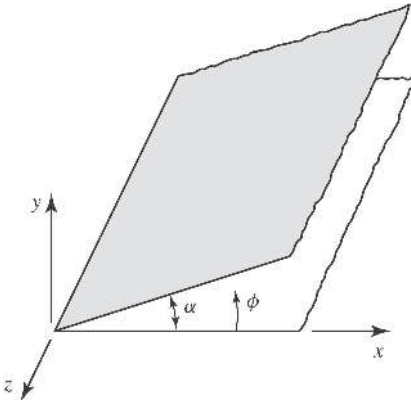


Figure P15-5

- 15.6. Repeat Problem 15.5 when the two plates are isolated from each other with the plate of $\phi=0$ grounded while that at $\phi=\alpha$ is maintained at a constant potential V_0 .
- 15.7. The top side of a rectangular cross-section, infinite length pipe is insulated from the other three, and it is maintained at a constant potential V_0 . The other three are held at a grounded potential of zero as shown in Figure P15-7. Determine in closed form the Green's function for the electric potential distribution within the pipe.

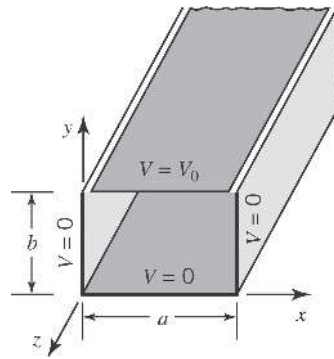


Figure P15-7

- 15.8. The three sides of an infinite length and infinite height trough are maintained at a grounded potential of zero as shown in Figure P15-8. The width of the trough is a . Determine in closed form the Green's function for the electric field distribution within the trough.

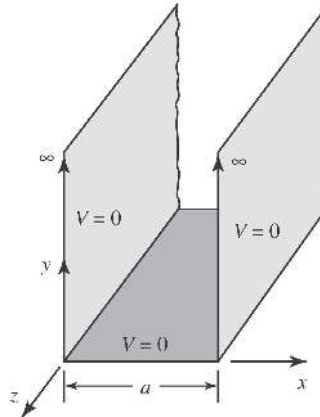


Figure P15-8

- 15.9. Derive the Green's function of (15-83a) through (15-83b) by initially choosing a solution for (15-74) that satisfies the boundary condition at $y=0$ and $y=b$.
- 15.10. An infinitely long conducting wire positioned at $\rho=\rho'$, $\phi=\phi'$ is circumscribed by a grounded ($V=0$) electric conducting circular cylinder of radius a and infinite length, as shown in Figure 15-4. Derive in closed form the Green's function for the potential distribution within the cylinder. Assume free space within the cylinder.

15.11. Derive in closed form the Green's function of the time-harmonic problem represented by (15-97) subject to the boundary conditions of (15-97a) and (15-97b). This would be an alternate representation of (15-106).

15.12. For Figure 15-7, derive the Green's function in closed form subject to the appropriate boundary conditions.

15.13. Repeat Example 15-5 for an infinite magnetic line source of constant current I_m located at $\rho = \rho'$, $\phi = \phi'$, as shown in Figure 15-9.

15.14. An annular microstrip antenna fed by a coaxial line is composed of an annular conducting circular strip, with inner and outer radii of a and b , placed on the top surface of a lossless substrate of height h and electrical parameters ϵ_s, μ_s as shown in Figure P15-14. The substrate is supported by a ground plane. Assuming the microstrip antenna can be modeled as a cavity with ideal open circuits of vanishing tangential magnetic fields at the inner ($\rho = a$) and outer ($\rho = b$) edges,

$$H_\phi(\rho = a, 0 \leq \phi \leq 2\pi, 0 \leq z \leq h) \\ = H_\phi(\rho = b, 0 \leq \phi \leq 2\pi, 0 \leq z \leq h) = 0$$

and vanishing tangential electric fields on its top and bottom sides, determine the Green's function for the TM^z modes (subject to $\nabla \cdot \mathbf{J} = 0$) with independent z variations within the cavity. For such modes the electric field must have only a z component of

$$\mathbf{E} = \hat{\mathbf{a}}_z E_z(\rho, \phi)$$

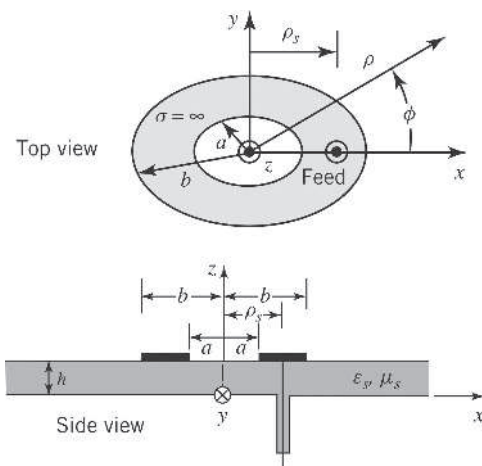


Figure P15-14

which must satisfy the partial differential equation

$$\nabla^2 E_z + \beta^2 E_z = j\omega\mu J_z(\rho_f, \phi_f), \quad \beta^2 = \omega^2 \mu \epsilon$$

15.15. Repeat Problem 15.14 for an annular sector microstrip antenna whose geometry is shown in Figure P15-15.

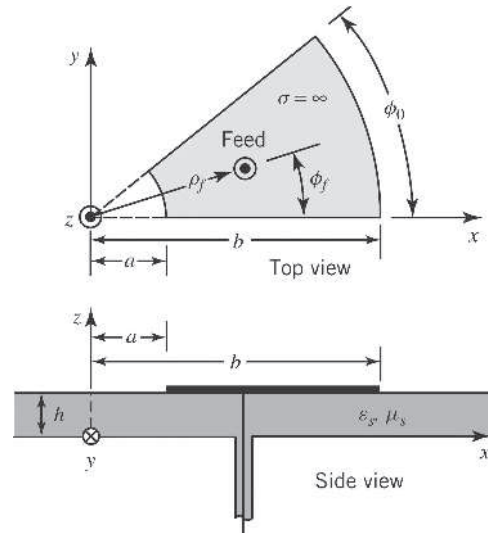


Figure P15-15

15.16. Repeat Problem 15.14 for a circular sector microstrip antenna whose geometry is shown in Figure P15-16.

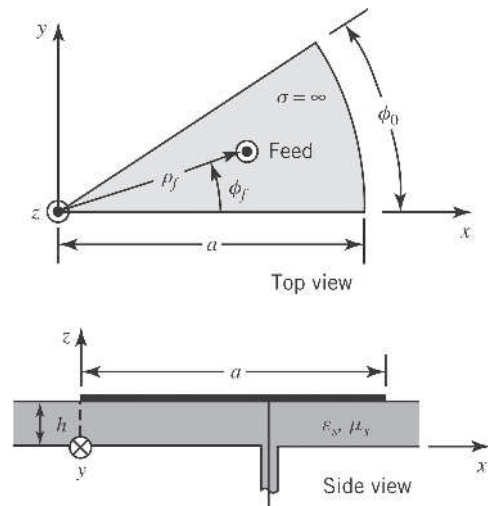


Figure P15-16

- 15.17.** Derive the Green's function represented by (15-165a) and (15-165b) in terms of the two space variable series expansion method using orthonormal expansion functions.
- 15.18.** An infinite length electric line source of constant current I_e is placed near a conducting circular cylinder of infinite length, as shown in Figure P15-18. Derive, in closed form, the Green's function for the fields in the space surrounding the cylinder.

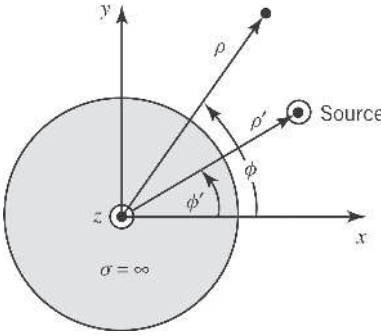


Figure P15-18

- 15.19.** Repeat Problem 15.18 for a magnetic line source of constant current I_m .
- 15.20.** An infinite length electric line source of constant electric current I_e is placed near a two-dimensional conducting wedge of interior angle 2α as shown in Figure P15-20. Derive, in closed form, the Green's function for the fields in the space surrounding the wedge. Compare with the expressions of (11-182a) and (11-182b).

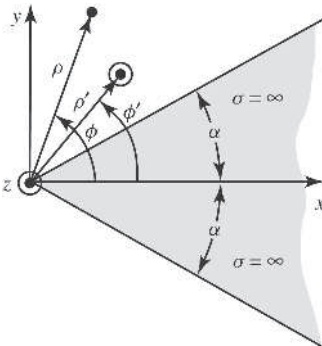


Figure P15-20

- 15.21.** Repeat Problem 15.20 for an infinite length magnetic line source of constant magnetic current I_m . Compare the answers with the expressions of (11-192a) through (11-192b) or (11-193a) through (11-193b).
- 15.22.** A point source placed at x', y', z' is radiating in free space as shown in Figure P15-22.

(a) Derive its Green's function of

$$G = -\frac{1}{4\pi} \frac{e^{-j\beta R}}{R}$$

where R is the radial distance from the point source to the observation point.

(b) By using the integral of (11-28a) or

$$\int_{-\infty}^{+\infty} \frac{e^{-j\beta R}}{R} dz = -j\pi H_0^{(2)}(\beta R)$$

show that the three-dimensional Green's function of the point source reduces to the two-dimensional Green's function of the line source derived in Example 15-5.

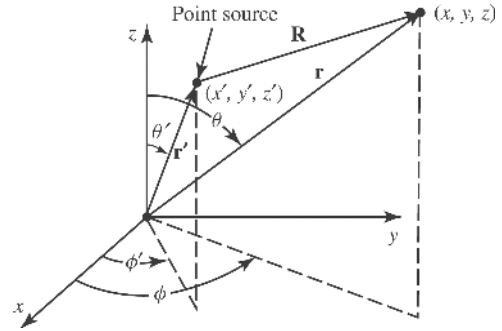


Figure P15-22

CHAPTER 16



Artificial Impedance Surfaces

16.1 INTRODUCTION

Metasurfaces, also referred to as artificial impedance surfaces and/or engineered electromagnetic surfaces, have been developed over the past few decades to alter the impedance boundary conditions of the surface of a structure and, thus, control the radiation characteristics, such as radiation efficiency and pattern, of antenna elements placed at or near them, or the scattering of impinging electromagnetic waves [1–6]. When electromagnetic waves interact with surfaces that exhibit geometrical periodicity, they result in some interesting and exciting characteristics, which typically have numerous applications that have captured the attention and imagination of engineers and scientists. Using a “broad brush” designation, these surfaces can also be referred to as metamaterials, which were discussed in Section 5.7. Since metamaterials may be used to designate double negative (DNG) type materials, there have been other designations of artificial impedance or engineered electromagnetic surfaces. These designations began initially as photonic bandgap (PBG) structures [7, 8], which targeted primarily optics type of structures and frequencies. PBG structures are 1-D, 2-D, and 3-D periodic configurations, both dielectric and conducting, which have the ability to manipulate the electromagnetic radiation so as to not allow it to propagate within certain frequency ranges or bandgaps. Most of the focus of PBG has been devoted to dielectric structures, although the applications have expanded to metallic structures, such as waveguides, resonators, filters, and antennas, which usually exhibit metallic losses that are more dominant at higher frequencies, compared to dielectric losses. The PBG structures are analogous to semiconductor materials that manipulate the electrons to exhibit electronic bandgaps. The PBG designation was expanded to include other type of structures and frequencies, such as electromagnetic bandgap (EBG) structures, frequency-selective surfaces (FSSs), high-impedance surfaces (HISs), artificial magnetic conductors (AMCs), perfect magnetic conductors (PMCs), etc. A comprehensive list of various EBG designations and references, organized by topics, can be found in the appendix of [6]. In [3], the EBG designation was introduced as a broader classification to encompass the others. Artificial impedance surfaces can be used for, but are not limited to:

- Changing the surface impedance
- Controlling the phase of the reflection coefficient

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- Manipulating the propagation of surface waves
- Controlling the frequency band (stopband, passband, bandgaps)
- Controlling the edge diffractions, especially of horns and reflectors
- Designing new boundary conditions to control the radiation pattern of small antennas
- Providing detailed control over the scattering properties
- Designing tunable impedance surfaces to be used as:
 - a. steerable reflectors
 - b. steerable leaky-wave antennas (LWAs)

This is accomplished by altering the surface of a structure, by modifying its geometry and/or adding other layers, so that the surface waves and/or the phase of the reflection coefficient of the modified surface can be controlled. Although the magnitude of the reflection coefficient will also be affected, it is the phase that primarily has the most significant impact. While an ideal PMC surface introduces, through its image, a zero-phase shift in the reflected field, in contrast to a perfect electric conductor (PEC), which presents a 180° phase shift, the reflection phase of an EBG surface can, in general, vary from -180° to $+180^\circ$, which makes the EBG more versatile and unique [9, 10]. This will be demonstrated in Section 16.3.

While, in general, PEC, PMC, and EBG surfaces possess individually attractive characteristics, they also exhibit shortcomings when electromagnetic radiating elements are mounted on such structures, especially when the designs are judged using aerodynamic, stealth, and conformal criteria. For example, when an electric element is mounted vertically on a PEC surface, its image reinforces its radiation and system efficiency; however, its geometry is not low-profile, an undesirable characteristic for aerodynamic, stealth, and conformal designs. However, when the same electric radiating element is placed horizontally on a PEC surface, its radiation efficiency suffers because its image possesses a 180° phase shift and its radiation cancels that of the actual element; however, the design exhibits low-profile characteristics usually desirable for aerodynamic, stealth, and conformal applications. In contrast, when the same electric radiating element is placed horizontally on a PMC surface, its image possesses a 0° phase and reinforces the direct radiation of the actual element, in addition to having low-profile characteristics. The characteristics of vertical and horizontal electric elements placed vertically and horizontally on PEC and PMC surfaces are based on the image theory of Figure 7-2, and they are visually contrasted in Figure 16-1 [11].

While EBG surfaces exhibit similar characteristics as PMCs when radiating elements are mounted on them, they also have the ability to suppress surface waves of low-profile antenna designs, such as microstrip arrays. Surface waves are introduced in microstrip arrays, which primarily travel within the substrate and are instrumental in developing coupling between the array elements. This can limit the beam scanning capabilities of the microstrip arrays; ultimately, surface waves and coupling may even lead to *scan blindness*, discussed in Section 16.10.5.

An EBG surface emulates a nearly PMC surface and suppresses surface waves only over a frequency range; thus, it is usually referred to as a bandgap structure. In general, the frequency range (bandgap) over which an EBG structure operates more efficiently depends upon the application. For example, the frequency range over which a radiating element in the presence of an EBG possesses a good impedance match, which in [5] is referred to as the input-match frequency band, may not be the same frequency range over which a microstrip array suppresses the surface waves, which in [5] is referred to as the surface-wave frequency bandgap. The operational band was introduced to define “the frequency region within which a low-profile wire antenna radiates efficiently, namely, having a good return loss and radiation patterns. The operational band is the overlap of the input-match frequency band and the surface-wave frequency bandgap” [5]. Generally, for EBG structures, the input-match frequency band and the surface-wave frequency bandgap are not necessarily the same. However, for a mushroom type of EBG surface, which will be introduced later, the input-match and surface-wave bands are nearly the same, which results in an overall

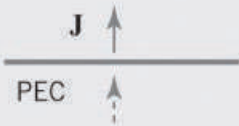
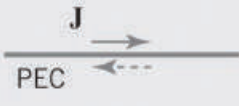
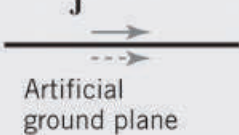
Options	Efficiency	Low Profile
		
		
		

Figure 16-1 Efficiency and conformal characteristics of a vertical and horizontal electric J current source at and near PEC and PMC surfaces. (Source: C. A. Balanis, *Antenna Theory: Analysis and Design*, Fourth Edition, John Wiley & Sons, New York, 2016, Reproduced with permission from John Wiley & Sons, Inc.)

operational band, which is a near overlap of the two other bands. It should be stated that for a mushroom type of EBG surface, the frequency band of its surface-wave suppression capability is determined by simulating and/or measuring its insertion loss amplitude, as shown later in Figure 16-7a, or by simulating the dispersion diagram of Figure 16-12. It should also be pointed out that the surface-wave suppression bandwidth of an EBG surface is not necessarily the same as the bandwidth over which the EBG surface behaves as a PMC type of surface. When a plane wave is normally incident upon a surface with a surface impedance Z_s , the $+90^\circ$ to -90° phase variation is also evident when the magnitude of the surface impedance exceeds the free-space intrinsic impedance, η [2]. An EBG surface that does not include the vias does not suppress the surface waves, even though its reflection phase changes between $+180^\circ$ and -180° . Better representatives of the surface-wave suppression ability of an EBG surface is the dispersion diagram, which for Example 16-1 is displayed in Figure 16-12.

Now that we have introduced some of the basic definitions of artificial impedance surfaces, we examine some basic structures – corrugations in Section 16.2 and mushroom EBG surfaces in Section 16.3 – that exhibit such characteristics. Application of mushroom EBG surfaces to antenna technology is discussed in Section 16.10. A semiempirical procedure for the design of mushroom EBG surfaces is outlined in Section 16.4. The design leads, in some cases, to rather excellent results when compared to simulations based on a full-wave electromagnetic solver. The limitations of the design are summarized in Section 16.6.

16.2 CORRUGATIONS

There are a number of surfaces that have been developed over the years whose surface impedance has been altered by introducing changes on the surface of the structure. One such surface alteration, which has been in use for many decades, is the introduction of grooves, usually referred to as *corrugations*, with a depth at or near quarter of a wavelength, as shown in Figure 16-2. Since the width of the corrugations, w , is usually equal to or less than about $\lambda/10$ ($w \leq \lambda/10$ and the thickness t is also about $1/10$ of the width, or $t \leq w/10 \leq \lambda/100$), each corrugation can

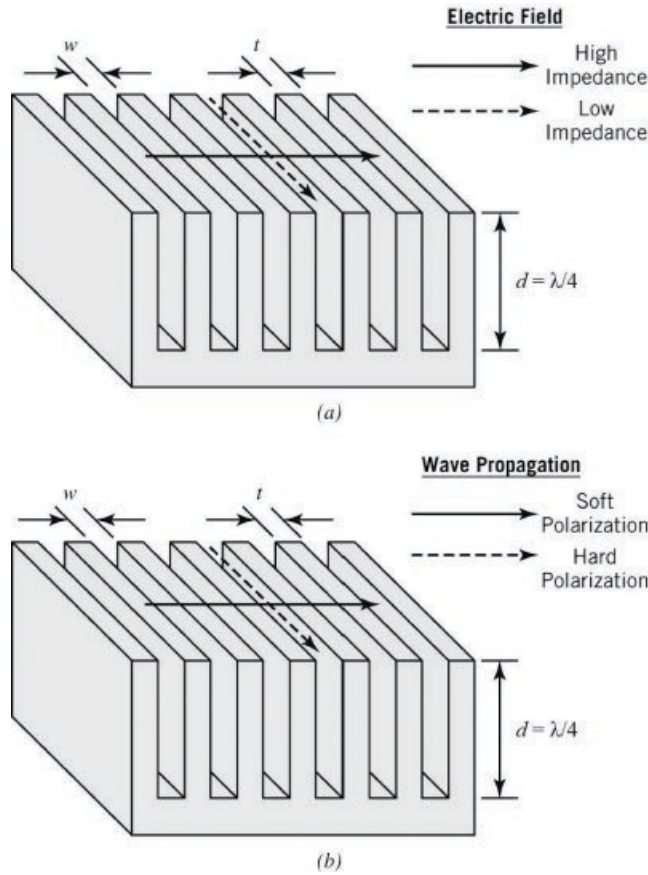


Figure 16-2 Corrugated surface and electric field direction for high- and low-impedance surface, and direction of wave propagation for soft and hard polarization impedance surfaces. (a) Electric field directions for high- and low-impedance surfaces. (b) Wave velocity for soft and hard polarization impedance surfaces. (Source: D. Sievenpiper, “Artificial impedance surfaces,” Chapter 15 in *Modern Antenna Handbook*, C. A. Balanis (Ed.), John Wiley & Sons, pp. 737–777, 2008. Reproduced with permission from John Wiley & Sons, Inc.)

be treated as a shorted transmission line. Since the input impedance of a shorted transmission line of length l is

$$Z_{\text{in}} = jZ_c \tan(\beta l) \quad (16-1)$$

then, for a corrugation of depth $d = l = \lambda/4$, the input impedance at the surface is ideally infinity ($Z_{\text{in}} = \infty$). Such a structure is classified as an anisotropic impedance surface since its impedance is high when the polarization of the electric field is perpendicular to the grooves, and it is a low-impedance surface when the electric field is parallel to the grooves, as shown in Figure 16-2a.

No matter what the polarization of the wave is, the surface is referred to as “hard” when the wave propagates parallel to the grooves and “soft” when the wave propagates perpendicular to the grooves, as shown in Figure 16-2b [2, 12, 13]. This type of designation of the surface is used to match the corresponding boundary conditions from acoustics, and it has been used extensively to design corrugated horns (see Figure 16-3) whose radiation characteristics are controlled by the design of the corrugations [9].

For a pyramidal horn, for example, the corrugations on the upper and lower walls of the horn are introduced to create a surface that nearly nulls the vertical electric field components of the



Figure 16-3 Conical corrugated horn antenna. (Source: C. A. Balanis, *Antenna Theory: Analysis and Design*, Fourth Edition, John Wiley & Sons, New York, 2016. Reproduced with permission from John Wiley & Sons, Inc.)

wave [9]. Therefore, the impedance of the upper and lower walls of the pyramidal horn nearly match those of the side walls, and they can be used to create nearly identical E- and H-plane patterns and nearly rotational by symmetric patterns in all the planes, especially in conical horns such as the one in Figure 16-3. Such an antenna element is widely used in many practical applications, especially as a feed for reflector antennas (dishes). A number of designs of corrugations can be found in [9, 10, 12–20], and they have been used to control the radiation characteristics of the horns, especially to lower the minor lobes, provide better impedance match, minimize the diffractions from the edges at the aperture of the horn, and attempt to synthesize a nearly symmetrical amplitude pattern by equalizing those of the E- and H-planes.

Just like corrugations/grooves have been introduced on PEC surfaces, conducting strips have also been placed circumferentially (T-strips) and axially (L-strips) on the surface of dielectric circular cylinders to create anisotropic boundary conditions and polarization-selective surfaces to control the scattering from cylinders [21]. Electric fields that are perpendicular to the length of the strips are ideally transmitted, while those parallel to the length of the strips are ideally reflected. This procedure creates an impedance surface that can perform as dichroic polarization-sensitive filter for electromagnetic waves.

16.3 ARTIFICIAL MAGNETIC CONDUCTORS, ELECTROMAGNETIC BANDGAP, AND PHOTONIC BANDGAP SURFACES

While perfect electric conductors (PECs) exist in nature, perfect magnetic conductors (PMCs) do not. However, it will be of benefit to fabricate PMCs, even artificially. From the electromagnetic boundary conditions, PEC surfaces are those over which the tangential components of electric fields vanish. Therefore, this precludes placement at, or even near, their surfaces radiating elements such as horizontal electric dipoles, spirals, etc., because their radiated fields over them will be shorted out, or nearly so. This of course is obvious from image theory where the actual source and its image are next to each other but are oriented in opposite directions; this is illustrated in Figure 7-2. Such arrangements even exhibit low radiation efficiency for low heights because of the 180° phase reversal of its image. In fact, when locating a horizontal electric element, such as a horizontal dipole, next to a PEC, it must be placed at a height $h = \lambda/4$ above it in order for the

radiation, in a direction normal to the surface, to be maximum. Such an arrangement is usually not desired, especially when the elements are placed on spaceborne platforms, because of aerodynamic considerations. Also, for stealth type of targets, such configurations are quite 'visible' to radar and create a large radar cross-section (RCS) signature. Therefore, it is very beneficial if PMC surfaces can be created, even artificially.

Within recent years, PMC surfaces have been synthesized and fabricated artificially and exhibit PMC-type properties within a frequency range; therefore, these surfaces often are referred to as bandgap or band-limited surfaces. There have been many such surfaces – too numerous to mention here. The reader is referred to the literature, especially [1–8, 11–13, 22–24]. One of the first and most widely utilized PMC surfaces is that shown in Figure 16-4. This surface consists of an array of periodic patches of different shapes, in this case hexagons, placed above a very thin dielectric (which could be air) and connected to the ground plane by posts through vias. The height of the substrate is usually less than a tenth of a wavelength ($h < \lambda/10$). The vias are necessary to suppress surface waves within the substrate.

This structure is also referred to as EBG and PBG. It is a practical form of engineered textured surfaces or metamaterials, discussed in Section 5.7. Because of the directional characteristics of EBG/PBG structures, the integration of antenna elements with such structures can have some unique characteristics [7, 8, 25]. A semiempirical model of the mushroom EBG surface of Figure 16-4 was developed in [1, 2]. The presentation here follows that of [2].

Of the mushroom AMC/EBG/PBG structure shown in Figure 16-4, a unit cell of its structure is displayed in Figure 16-5a. When a wave impinges upon an array of such unit cells, electric fields are created across the gap of the unit cells that can be represented by an effective capacitance C .

Also, such impinging fields create currents that circulate between adjacent unit cells. The effects of these current paths through the neighboring walls or vias can be represented by an equivalent inductance L . Therefore, the equivalent circuit of the unit cell of Figure 16-5a is shown in Figure 16-5b, which consists of a capacitance C in parallel with an inductance L [2].

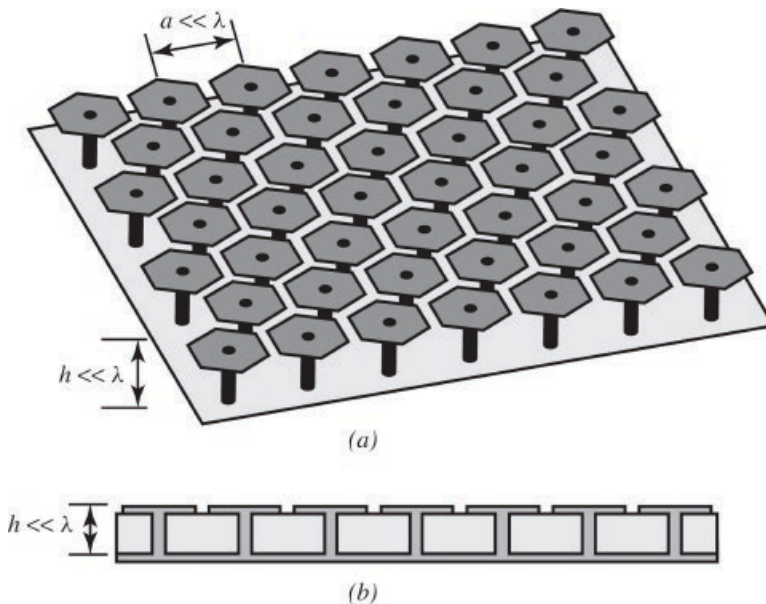


Figure 16-4 Mushroom engineered texture surface. (a) Perspective view. (b) Side view. (Source: D. Sievenpiper, "Artificial impedance surfaces," Chapter 15 in *Modern Antenna Handbook*, C. A. Balanis (Ed.), John Wiley & Sons, pp. 737–777, 2008. Reproduced with permission from John Wiley & Sons.)

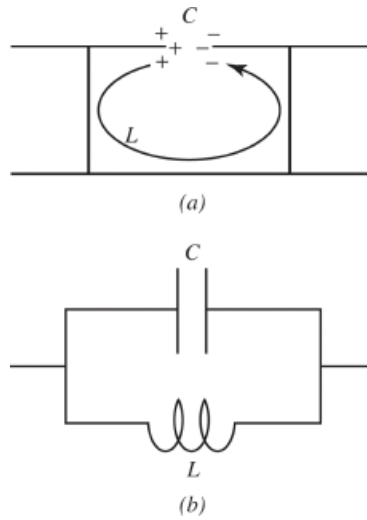


Figure 16-5 Unit cell and equivalent circuit of mushroom textured surface. (Source: D. Sievenpiper, “Artificial impedance surfaces,” Chapter 15 in *Modern Antenna Handbook*, C. A. Balanis (Ed.), John Wiley & Sons, pp. 737–777, 2008. Reproduced with permission from John Wiley & Sons.)

The surface impedance of the individual unit cell of Figure 16-5 is given by

$$Z_s = j \frac{\omega L}{1 - \omega^2 LC} \quad (16-2)$$

while its resonant frequency is represented by

$$\omega_0 = \frac{1}{\sqrt{LC}} \quad (16-3)$$

However, for design purposes, as it will be shown in Section 16.4, sheet inductance L_s and sheet capacitance C_s must be used to define the resonant frequency. The sheet inductance and capacitance take into account not only the geometry of the individual unit cells but also the geometrical arrangement of the unit cells [2]. It is apparent from (16-2) that the surface of the unit cell is inductive below the resonant frequency, capacitive above the resonant frequency, infinity at resonance, and very high near resonance. Based on (8-158c) of Section 8.7.2, inductive surfaces support transverse magnetic (TM) types of surface waves, while based on (8-169i) of Section 8.7.3, capacitive surfaces support transverse electric (TE) type of surface waves.

The support of either TE or TM surface-wave modes, or both, was verified by measuring the transmission between a pair of coaxial probes placed near the surface of a fabricated 12-cm high-impedance surface reported in [2]. The artificially fabricated PMC surface consisted of a triangular lattice of metallic hexagons placed on the surface of a grounded substrate with a dielectric constant of 2.2, as shown in Figure 16-6. The excitation of the surface waves, TE or TM modes, is controlled by the orientation of the probes. The amplitude of the transmission between the probes is shown in Figure 16-7a while the phase, based on a plane wave normal incidence, is displayed in Figure 16-7b [2]. It is apparent from Figure 16-7a that the fabricated and tested surface exhibits high impedance between approximately 11 and 16 GHz (bandgap), while it supports TM surface waves (inductive surface) below the bandgap and TE surface waves (capacitive surface) above the bandgap. By examining the plane wave incidence phase response of the mushroom surface in Figure 16-7b, it is apparent that the edges of the bandgap occur where the phase varies nearly from $+90^\circ$ to -90° , and it is basically zero at resonance. While the transmission amplitude

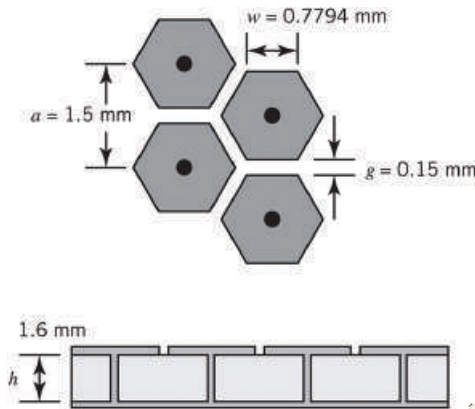


Figure 16-6 Triangular lattice of hexagons built on a grounded substrate with a relative dielectric constant of 2.2. (Source: D. Sievenpiper, “Artificial impedance surfaces,” Chapter 15 in *Modern Antenna Handbook*, C. A. Balanis (Ed.), John Wiley & Sons, pp. 737–777, 2008. Reproduced with permission from John Wiley & Sons.)

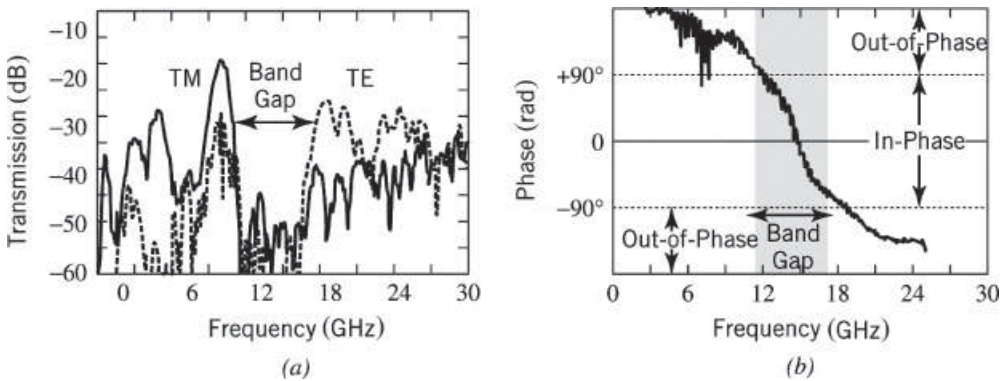


Figure 16-7 Transmission characteristics, amplitude and phase, of mushroom textured surface. (a) Amplitude: TM modes (solid), TE (dotted). (b) Phase. (Source: D. Sievenpiper, “Artificial impedance surfaces,” Chapter 15 in *Modern Antenna Handbook*, C. A. Balanis (Ed.), John Wiley & Sons, pp. 737–777, 2008. Reproduced with permission from John Wiley & Sons.)

and phase characteristics shown in Figures 16-7a and 16-7b are those of a mushroom type of high-impedance surface, they may be different for other type of surfaces, particularly those that may not include vertical vias [2]. A semiempirical design procedure for high-impedance surfaces can be found in [2], and it will be presented here in Section 16.4 that follows.

16.4 DESIGN OF MUSHROOM AMC

Now that the metasurface and its general properties and applications have been introduced, an objective is how to design such a surface based on desired specifications. Given the procedure reported in [1, 2], the design is outlined here. Other design methods can be found in [6].

The design center-radian frequency, ω_0 of (16-3), where the phase of the reflection coefficient is zero and where the AMC surface exhibits PMC characteristics, can be written as

$$\omega_0 = \frac{1}{\sqrt{L_s C_s}} \quad (16-4)$$

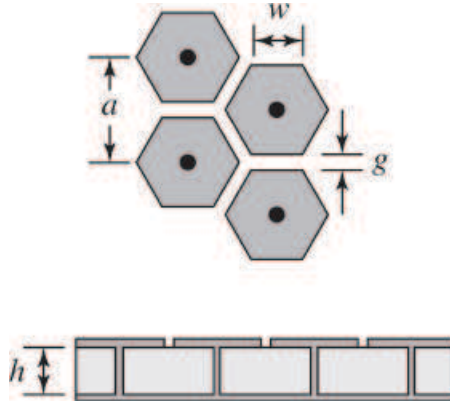


Figure 16-8 Geometry of hexagons lattice built on a substrate of height h . (Source: D. Sievenpiper, “Artificial impedance surfaces,” Chapter 15 in *Modern Antenna Handbook*, C. A. Balanis (Ed.), John Wiley & Sons, pp. 737–777, 2008. Reproduced with permission from John Wiley & Sons.)

where L_s and C_s are, respectively, the sheet inductance and capacitance, which take into account not only the geometry of the individual unit cells of Figure 16-5 but also the geometrical arrangement of the unit cells, as shown in Figure 16-8.

The fractional bandwidth BW , over which the phase of the reflection coefficient is between $+90^\circ$ and -90° , can be expressed as

$$BW = \frac{\Delta\omega}{\omega_0} = \frac{\sqrt{L_s / C_s}}{\sqrt{\mu_2 / \varepsilon_2}} \quad (16-5)$$

where ε_2 and μ_2 are, respectively, the permittivity and permeability of the superstrate (upper layer). Specifying the design radian frequency of (16-4) and the fractional bandwidth of (16-5), the sheet inductance L_s and sheet capacitance C_s can be determined.

Equations 16-4 and 16-5 can be solved for L_s and C_s in terms of ω_0 and $\Delta\omega / \omega_0$, that is,

$$L_s = f(\omega_0, \Delta\omega / \omega_0) = \frac{BW}{\omega_0} \sqrt{\mu_2 / \varepsilon_2} \quad (16-6a)$$

$$C_s = g(\omega_0, \Delta\omega / \omega_0) = \frac{1}{\omega_0 BW \sqrt{\mu_2 / \varepsilon_2}} \quad (16-6b)$$

The sheet inductance L_s can also be written as

$$L_s = \mu_1 h \quad (16-7)$$

Equating (16-6a) to (16-7), it can be shown that the fractional bandwidth can be written in terms of the substrate height h , or

$$BW = \frac{\Delta\omega}{\omega_0} = \left(\frac{\mu_1}{\mu_2} \right) \beta_2 h = \left(\frac{\mu_1}{\mu_2} \right) \omega_0 \sqrt{\mu_2 \varepsilon_2} \Rightarrow h = \left(\frac{\mu_2}{\mu_1} \right) \frac{BW}{\omega_0 \sqrt{\mu_2 \varepsilon_2}} \stackrel{\mu_1 = \mu_2}{=} \frac{BW}{\omega_0 \sqrt{\mu_2 \varepsilon_2}} \quad (16-8)$$

Substituting (16-8) for the height in (16-7), the sheet inductance can be written as

$$L_s = \mu_1 h = \mu_1 \left(\frac{\mu_2}{\mu_1} \right) \frac{BW}{\omega_0 \sqrt{\mu_2 \varepsilon_2}} = \frac{\mu_2}{\sqrt{\mu_2 \varepsilon_2}} \left(\frac{BW}{\omega_0} \right) \stackrel{\mu_1 = \mu_2}{=} \eta_2 \left(\frac{BW}{\omega_0} \right) \quad (16-9)$$

In turn, substituting (16-9) into (16-6b), the sheet capacitance can be expressed as

$$C_s = \frac{1}{(\omega_0)^2 L_s} = \frac{1}{\omega_0 \eta_2 \cdot BW} \quad (16-10)$$

To design each unit cell of Figure 16-8 (i.e., find its dimensions), the capacitance C of each unit cell is related to the sheet capacitance C_s by

$$C_s = C \times F \quad (16-11)$$

where F is a geometrical correction factor given by [1, 2].

Geometry	Geometrical correction factor (F)
Square	1
Triangle	$\sqrt{3}$
Hexagon	$1/\sqrt{3}$

Once the individual capacitance C of each unit cell is determined using (16-10) and the geometrical correction factor of the desired unit geometry, then the dimensions of the unit cell of Figure 16-8, for a two-layer design (superstrate with permittivity ε_2 and substrate with permittivity ε_1), can be determined using [1, 2]

$$C = \frac{w(\varepsilon_1 + \varepsilon_2)}{\pi} \cosh^{-1} \left(\frac{a}{g} \right) \quad (16-12)$$

If the upper layer (superstrate) of the two-layer structure is free space ($\varepsilon_2 = \varepsilon_0$), then (12-12) reduces to

$$C = \frac{w\varepsilon_0(\varepsilon_r + 1)}{\pi} \cosh^{-1} \left(\frac{a}{g} \right) \quad (16-13)$$

where ε_r is the dielectric constant of the substrate. Using Figure 16-8, the designer chooses one of the three dimensions (a , g , and w) and determines the other two using either (16-12) or (16-13) and one of the two following relationships:

$$\text{For square patch : } a = w + g \quad (16-14a)$$

$$\text{For hexagonal patch : } a = \sqrt{3}w + g \quad (16-14b)$$

According to [1, 2], the thickness of the substrate is usually chosen to be much smaller than the operating wavelength, which reduces the bandwidth since the inductance is related by (16-9) to the thickness and inversely proportional to the capacitance for a constant center frequency. Because the thickness of a two-layer structure is very small, large inductance values cannot be achieved. Thus, low frequencies are achieved by loading the structure with large capacitances. On the other hand, large capacitances usually cannot be achieved by two-layer structures. Therefore, the designer is encouraged to use three-layer structures, with overlapping plates/patches, for low-frequency applications [1, 2]. The trade-off in designs is between thickness and bandwidth. By selecting different geometries and materials with dielectric constants in the range of 2–10, it is possible to obtain capacitances on the order of 0.01–1 pF. With conventional printed circuit fabrication facilities and techniques, minimum gap separations between metallic regions should be around 100–200 μm (microns). The design of a two-layer mushroom textured surface based on a desired resonant frequency is demonstrated with a MATLAB program titled **AMC_Designs** that is included with this chapter.

Example 16-1

Design a two-layer mushroom textured surface with square patches and air as the upper layer ($\varepsilon_2 = \varepsilon_0$, $\mu_2 = \mu_0$), as shown in Figure 16-9, to exhibit PMC characteristics between 10 and 14 GHz with a center frequency of $f_0 = 12$ GHz. Use a Rogers RT/Duroid 5880 with a dielectric constant of $\varepsilon_r = 2.2$. The square patches are supported by metallic circular posts that connect the patches to the bottom ground plane through vias in the substrate.

Solution: Based on the specifications with air as the upper layer and a fractional bandwidth of

$$BW = \frac{\Delta f}{f_0} = \frac{(14 - 10)10^9}{12 \times 10^9} = \frac{1}{3}, \quad \varepsilon_r = 2.2, \text{ and using (16-8), the height of the substrate is}$$

$$h = \frac{BW}{\omega_0 \sqrt{\mu_2 \varepsilon_2}} = \frac{BW}{\omega_0 \sqrt{\mu_0 \varepsilon_0}} = \frac{3 \times 10^8}{3(2\pi \times 12 \times 10^9)} = 1.3263 \times 10^{-3} = 1.3263 \text{ mm}$$

The sheet inductance and sheet capacitance based, respectively, on (16-9) and (16-10) are

$$L_s = \mu_1 h = \mu_0 h = 4\pi \times 10^{-7} (1.3263 \times 10^{-3}) = 16.6668 \times 10^{-10} = 1.66668 \text{ nH}$$

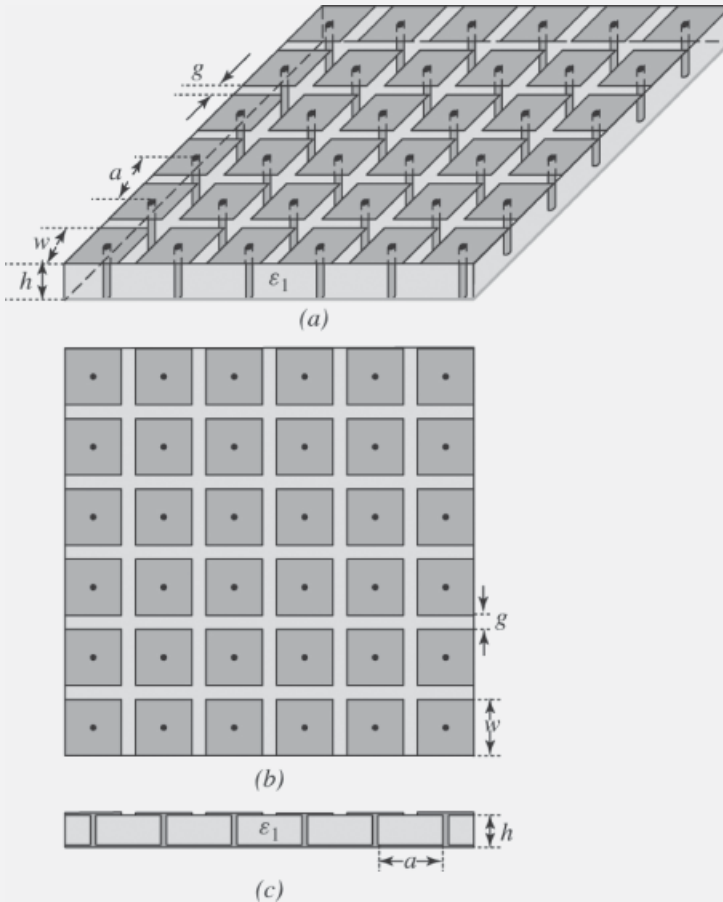


Figure 16-9 Geometry of PMC textured surface of square patches. (a) Perspective view. (b) Top view. (c) Side view.

$$C_s = \frac{1}{\omega_0 \eta_2 \cdot BW} = \frac{1}{\omega_0 \eta_0 \cdot BW} = \frac{3}{2\pi \times 12 \times 10^9 (377)} \\ = 0.10554 \times 10^{-2} = 0.10554 \text{ pF}$$

Since we are using square patches, the geometrical factor F is unity and, according to (16-11), the capacitance of each unit cell C is equal to the sheet capacitance C_s , that is, $C = C_s$.

A suitable value for the gap spacing g for integrated circuit technology between patches is $100 \mu\text{m}$ or $g = 100 \mu\text{m}$. Using the previously obtained values and the geometrical relation of (16-14a), $a = w + g$ for a square patch, the nonlinear design of (16-13)

$$C = \frac{w\epsilon_0(\epsilon_r + 1)}{\pi} \cosh^{-1}\left(\frac{a}{g}\right) = \frac{w\epsilon_0(2.2 + 1)}{\pi} \cosh^{-1}\left(\frac{w + 0.1 \times 10^{-3}}{0.1 \times 10^{-3}}\right)$$

can be solved, using a nonlinear solver, for w of the PMC surface. Doing this leads to

$$w = 2.85 \text{ mm}, \text{ and then } a = w + g = 2.95 \text{ mm}$$

To verify the performance of the mushroom PMC surface based on the specified and obtained geometrical dimensions, the commercial software HFSS [26] was used to simulate it. The plane wave normal incidence reflection phase variations of S_{11} of the mushroom textured surface of square patches of Figure 16-9 between $+90^\circ$ and -90° , similar to those of Figure 16-7b, are shown in Figure 16-10 where they are compared with the results based on the design equations of Section 16.4; a very good agreement is indicated between the two. The simulated data indicate a bandwidth of 3.9 GHz ($f_0 = 10.35$ GHz and $f_h = 14.25$ GHz), compared to the specified one of 4 GHz ($f_l = 10$ GHz and $f_h = 14$ GHz), a center frequency of 12.15 GHz (compared to 12 GHz), and a fractional bandwidth of 0.321 (compared to 0.333). Overall, the performance indicates a very favorable design.

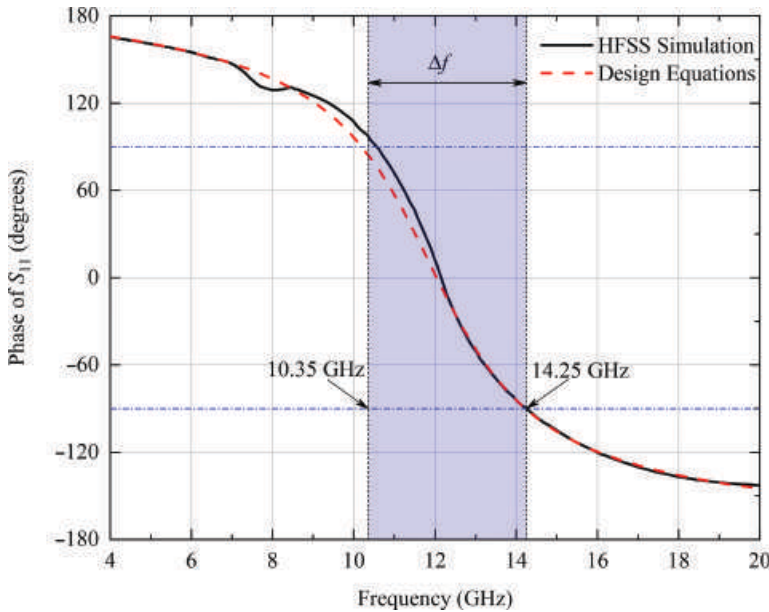


Figure 16-10 Phase of reflection coefficient S_{11} of PMC textured surface with square patches simulated using HFSS and design equations.

16.5 SURFACE-WAVE DISPERSION CHARACTERISTICS

The plane wave normal incidence reflection phase characteristics of the mushroom surface of Figure 16-9, as displayed in Figure 16-10, can be simulated by using the geometrical arrangement illustrated in Figure 16-11a. Since the structure is periodic, the problem can be solved by considering only a unit cell and by assigning the proper boundary conditions. To emulate periodicity based on the polarization of the plane wave incidence indicated in Figure 16-11a, with the electric field $\mathbf{E}$ parallel to the yz plane and the magnetic field $\mathbf{H}$ parallel to the xy plane, PEC boundary conditions are assigned on the front and rear walls of the unit cell, and PMC boundary conditions are assigned on the left and right walls of the unit cell. The assignment of the boundary conditions should be based on the polarization of the incident electric ($\mathbf{E}$) and magnetic ($\mathbf{H}$) fields. The images of $\mathbf{E}$ and $\mathbf{H}$ fields should be in the same direction as the actual field.

In addition to their unique plane wave normal incidence reflection phase characteristics, mushroom surfaces generate forbidden bandgaps within which the propagation of surface waves is suppressed. These forbidden bands can be observed via dispersion diagrams or directly by the amplitude variation of surface waves. A dispersion diagram is a plot that displays the relation between the frequency of the propagating modes and the wave vector. The dispersion diagram of a mushroom surface can be obtained by computing the allowable frequencies of different modes for certain values of the amplitude of the wave vector. This can be achieved by computing the eigenmodes of the periodic structure.

To understand the numerical approach, let us consider the vector wave equation for the electric field $\nabla^2 \mathbf{E} = -\omega^2 \mu \varepsilon \mathbf{E}$ as a starting point, where $-\omega^2 \mu \varepsilon$ are the eigenvalues and the solutions to the wave equation are the corresponding eigenvectors. For the case of an infinite medium, the eigenvalues, as well as the ω 's, have a continuous spectrum on the complex plane. On the other hand, in the case of a periodic structure, particularly a mushroom surface and because of the translational symmetry of the geometry, the solutions to the wave equation can be written in the form $\mathbf{E} = \mathbf{E}_n e^{j\phi}$, where $\mathbf{E}_n$ is a periodic function, which is referred to as *Bloch mode* [27]. The periodic

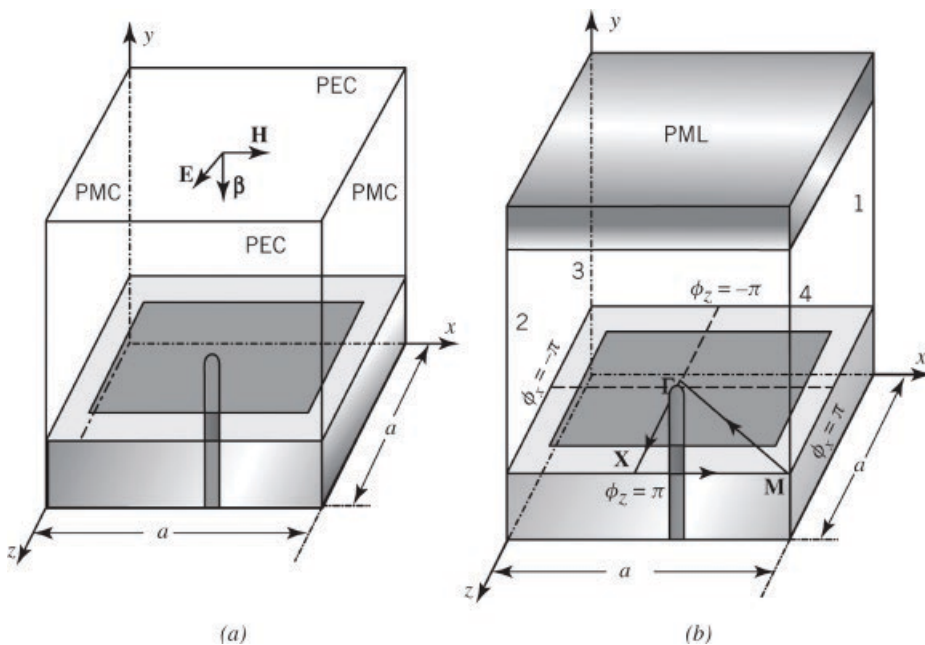


Figure 16-11 Unit cells for reflection phase and dispersion diagram simulations. (a) Reflection phase. (b) Dispersion diagram.

function $\mathbf{E}_n$ is a solution to another problem referred to as the *reduced Hermitian eigenproblem* [28]. Again, due to the periodicity of the structure, the wave vectors in the x and z directions can be written, respectively, as $\beta_x^n = \beta_x + m\left(\frac{2\pi}{a}\right)$ and $\beta_z^m = \beta_z + n\left(\frac{2\pi}{a}\right)$. The wave vectors can also be

obtained by the spatial harmonic expansion of $\mathbf{E}$ [29]. Since $\mathbf{E}_n$ is periodic, we can solve the eigenproblem only over the finite domain referred to as the *Brillouin zone*, in which β_x and β_z take values between $-\pi/a$ and $+\pi/a$. As a consequence of the finite nature of this problem, the eigenvalues would generate discrete frequency spectrum bands. Furthermore, due to the other types of symmetries including rotation, mirror reflection, and inversion, the domain of the problem can be reduced into a smaller one that is called the *irreducible Brillouin zone* [27]. Figure 16-11b shows the Brillouin and irreducible Brillouin zones of a unit cell of a mushroom EBG structure. Surface 1 represents the rear wall, surface 2 the front wall, surface 3 the left wall, and surface 4 the right wall of the unit cell. The unit cell is truncated in the y direction by a PML design. To be able to plot the band structure in a regular two-dimensional (2-D) diagram, it is sufficient to consider only the extrema of the frequency bands. This can be obtained by solving the problem on the boundary of the irreducible Brillouin zone.

For numerical computation of the eigenvalues, surfaces 1 (rear) and 2 (front) of Figure 16-11b should be connected to each other through the relation $\mathbf{E}_2 = \mathbf{E}_1 e^{j\phi_z}$. Similarly, surfaces 3 (left) and 4 (right) can be related by $\mathbf{E}_4 = \mathbf{E}_3 e^{j\phi_x}$. This is accomplished by assigning Bloch boundary conditions [sometimes referred to as linked boundary conditions (LBCs)] to each pair. In contrast to the PEC or PMC boundary conditions, LBC supports both tangential and perpendicular components of the electric ($\mathbf{E}$) and magnetic ($\mathbf{H}$) fields, over the surface where they are defined. For Figure 16-11b, ϕ_x and ϕ_z should take values between $-\pi$ and π , which is a direct consequence of the periodicity of the structure, within the Brillouin zone. Since the wave vector is also related to the phase of the fields, the inputs of the numerical simulation can be these phase terms. In the path from Γ to $\mathbf{X}$, ϕ_z should be varied from 0 to π , while ϕ_x is maintained constant at 0. Similarly, in the path from $\mathbf{X}$ to $\mathbf{M}$, ϕ_x should be varied from 0 to π , while ϕ_z is kept constant at π . Finally, from $\mathbf{M}$ back to Γ both ϕ_x and ϕ_z should be varied from 0 to π , simultaneously. Using the HFSS software for simulations, surface 1 is referred to as the *master* and surface 2 as the *slave* for the z variations. Similarly, for the x variations, surface 3 is referred to as the *master* and surface 4 as the *slave*.

The dispersion diagram, obtained by HFSS [26] simulation for the geometry of Example 16-1, is illustrated in Figure 16-12. It is evident that the bandgap for the surface-wave suppression is approximately between 9.5 and 13.5 GHz, which nearly matches that based on the phase diagram of Figure 16-10. That is, the bandwidth over which the mushroom EBG structure of Figure 16-9 and Example 16-1 behaves nearly as a PMC, which is based on the phase diagram of a plane wave normal incidence (see Figure 16-10), is nearly the same as the surface-wave suppression bandwidth indicated in the dispersion diagram of Figure 16-12. The lower end of the bandgap of the surface-wave suppression is determined from the dispersion diagram of Figure 16-12, when the group velocity of the TM surface waves becomes zero. The upper bound of the bandgap is the frequency at which the TE mode crosses the light line because the surface waves can propagate when their group velocity is smaller than the velocity of light (slow waves). However, particularly for this example, the upper bound is selected as the point where the group velocity of the TE mode significantly deviates from that of light. Since the attenuation constant of (8-179) will be very small up to this point, practically, there will not be any surface-wave propagation.

To create dispersion diagrams, such as the one of Figure 16-12 and similar ones, which shed insight into the dispersion characteristics of a metasurface, two programs have been developed and are included in the **Multimedia** of Section 16.13. The first one, entitled **Dispersion_Diagram_Matlab**, is a MATLAB program that generates a dispersion diagram based on the analytical expressions of (16-2), (16-9), and (16-13), and on the formulation reported in [1]. The second dispersion diagram, entitled **Dispersion_Diagram_HFSS**, is based on Figure 16-11 using the Floquet

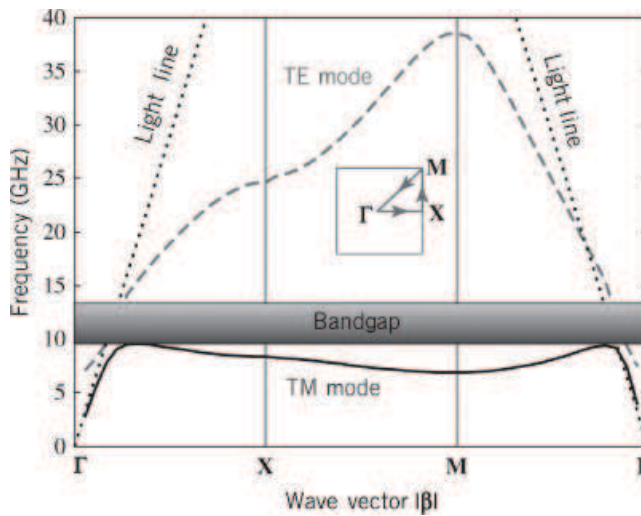


Figure 16-12 Dispersion diagram.

port simulation in Ansys HFSS. As part of the package of this program, two files, *Unit Cell Design* and *Graphical User Interface (GUI)*, are included to aid in the generation of the dispersion diagram.

Another evidence of the EBG structure bandgap characteristics is to model and simulate the amplitude transmission of the TM and TE modes, as was accomplished in Figure 16-7 through measurements.

It is worth pointing out that the dispersion diagram is not limited to metasurface structures, but it can be utilized to graphically describe the propagation constant for any bounded or unbounded medium. Figures 8-16 and 8-17 illustrate the dispersion behavior in a partially filled rectangular waveguide. Furthermore, the dispersion diagram is plotted in Figure 8-23 for a dielectric slab bounded above and below by air. Similarly, the dispersion diagram for a dielectric-covered ground plane can be calculated utilizing the characteristic equations 8-178a and 8-178b for TM and TE modes, respectively.

The characteristic/transcendental equation can be formulated based on the solution of the wave equation or by applying the transverse resonance method (TRM). Since the objective here is to find the propagation constants, the formulation based on the TRM is much simpler to apply. Section 8.6 outlined the procedure to calculate the transcendental equation for a partially filled waveguide utilizing the TRM. Furthermore, an approximate model using the TRM was developed in [30] to calculate the dispersion diagram for a metasurface in absence of vias. To illustrate the process, Example 16-2 follows for a dielectric-covered PEC [31].

Example 16-2

Using the TRM, determine for a PEC ground plane covered with a dielectric slab:

1. Derive the characteristic equation, for the TM mode, to calculate the propagation constants when the dielectric slab thickness is h . To simplify the analysis, assume an infinite flat PEC on the xy plane, no field variations in the x direction ($\frac{\partial}{\partial x} = 0$), and the waves traveling in the y direction.
2. Plot the dispersion diagram for a dielectric slab of thickness $h = 5.08$ mm and dielectric constant $\epsilon_r = 2.2$.

Solution:

1. The dielectric-covered PEC can be represented as a shorted transmission line displayed in Figure 16-13a. The input impedance looking downward (Z_m^-) can be approximated as a TM wave impedance (Z_{wd}^{TM}) of a semi-infinite shorted dielectric, while the input impedance looking upward (Z_m^+) is the TM free-space wave impedance (Z_{w0}^{TM}) as shown in Figure 16-13b. Hence, the characteristic equation for the grounded dielectric can be written as

$$Z_m^+ = -Z_m^- \quad (16-15)$$

$$\frac{\beta_{z0}}{\omega\epsilon_0} = -j \left(\frac{\beta_{zd}}{\omega\epsilon_d} \right) \tan(h\beta_{zd}) \quad (16-16)$$

$$\frac{\sqrt{\beta_0^2 - \beta_y^2}}{\omega\epsilon_0} = -j \frac{\sqrt{\beta_d^2 - \beta_y^2}}{\omega\epsilon_d} \tan\left(h\sqrt{\beta_d^2 - \beta_y^2}\right). \quad (16-17)$$

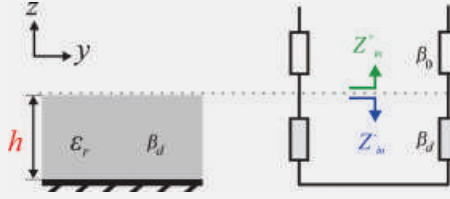


Figure 16-13 Geometry of a grounded dielectric, with its equivalent transmission line model. (Source: M. Alharbi, “High-directive printed antennas for low-profile applications,” Ph.D. dissertation, School of Electrical, Computer, and Energy Engineering, Arizona State University, 2020.)

2. The solution of (16-17) is numerically evaluated and the data are compared to those obtained by HFSS, as presented in Figure 16-14. An excellent agreement is indicated between the two sets of data.

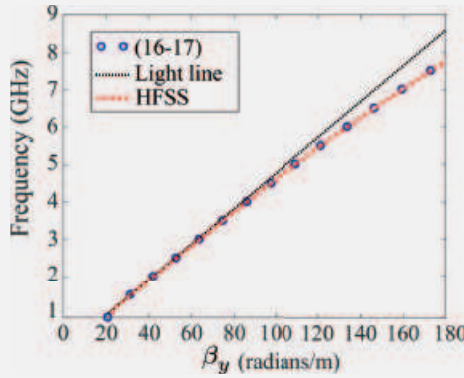


Figure 16-14 Dispersion diagram of a dielectric covered PEC. (Source: M. Alharbi, “High-directive printed antennas for low-profile applications,” Ph.D. dissertation, School of Electrical, Computer, and Energy Engineering, Arizona State University, 2020.)

Two MATLAB programs, for the calculation and development of a dispersion diagram, are provided in this chapter. The first one, entitled **TRM_Metasurface_Dispersion**, is based on the model developed in [30] to calculate the dispersion of a metasurface, and the second one, entitled **TRM_Grounded_Dielectric_Dispersion**, is based on (16-17) and plots the dispersion diagram of a dielectric-covered PEC. For details, see their description in the **Multimedia** of Section 16.13.

16.6 LIMITATIONS OF THE DESIGN

The design procedure, which has been outlined and demonstrated by Example 16-1, is limited by the bandwidth that is independent of all of the physical dimensions of the structure except the thickness of the substrate. Therefore, special care must be exercised to design the physical geometry of the mushroom type surface for a desired center frequency and bandwidth. Indeed, for a fixed substrate thickness, we do not have simultaneous control of both the center frequency and bandwidth. The design is based primarily on three parameters: *substrate thickness*, *center frequency*, and *bandwidth*. Once two of these three parameters are specified, we have no control of the third one. In addition to this, if the thickness of the substrate is fixed, the entire frequency range cannot be covered by using reasonable patch dimensions. Furthermore, the design equations, outlined in the design procedure, are valid only if the wavelength within the substrate is much larger than the dimensions of the unit cell. Hence, if very large patches are used for the design to cover a large frequency range, the design equations will lead to less accurate designs.

To overcome some of the limitations mentioned above, there are other methods that can be used but are more complex. One method is the so-called *dynamic model* [32]. In this method, a different expression is used for the surface capacitance while the expression for the inductance is the same as in [1, 2]. In this technique, the capacitance is expressed in terms of an infinite summation without assuming that the dimensions of the unit cell are much smaller than the wavelength. This model can also take into account the influence of the higher-order modes generated within the high-impedance surface. If the dimensions of the unit cell are much smaller than the wavelength, the dynamic model reduces to a simpler expression for the surface capacitance, which leads to similar results as [1, 2]. However, because of its complexity and space limitations, it will not be presented here. The reader is directed to [32] for the details.

16.7 APPLICATIONS OF AMCs

Given that perfect magnetic conductors do not exist in nature, the significance of these AMCs becomes even more pivotal in some applications. Image theory indicates that a perfect magnetic conductor (PMC) surface would be an efficient candidate in the applications where the radiating element is very close to the ground plane, unlike a perfect electric conductor (PEC) surface whose radiation efficiency is very poor [5, 33]. Thus, in recent years, AMCs have been considered a major breakthrough in antenna and electromagnetics engineering. This chapter reviews some of the applications of AMCs, particularly:

- *RCS reduction using checkerboard artificial impedance surfaces*: Conventional methods of radar cross section (RCS) reduction are primarily based on two mechanisms:
 - Absorption of the incoming waves using bulky radar-absorbing material (RAM).
 - Reduction of the scattered waves by altering the physical geometry of the original structure; such geometries and analytical techniques are discussed in Chapter 14.

The functionality of the former mechanism is limited to a narrow bandwidth, and it requires bulky lossy material; the latter mechanism is not aerodynamically efficient. However, the introduction of AMCs not only eliminates some of the limitations of the conventional RCS reduction techniques but also improves their RCS reduction performance. For example, in [34], AMC-based thin RAMs are presented, which eliminate their bulky nature. Later, in [35], the scattered waves are redirected using AMC-based tailored metasurfaces without altering the physical geometry of the original structure. However, such metasurfaces reduce the RCS only in a narrow frequency band. Recent developments [36–41] lead to ultrabroadband RCS reduction.

- *Curvilinear AMCs for low-profile superdirective loop and spiral antennas:* AMCs are judicious ground planes to be used in low-profile applications. These structures are also utilized for miniaturization and bandwidth enhancement [42–46]. Although most of the literature focuses on rectangular AMCs, curved and circularly symmetric ground planes are addressed in [47–53]; however, they mostly concentrate on the electromagnetic bandgap properties rather than on reflection characteristics, which take place at different frequency bands. Loop and spiral elements are placed above curvilinear-designed AMCs to illustrate the superior bandwidth and gain of such ground planes, compared to their rectangular counterparts [54–56]. It should also be noted that when AMCs are used as antenna ground planes, they are usually referred to as high-impedance surfaces (HISs).
- *High-gain leaky-wave antennas using holography:* Holographic artificial impedance surfaces (HAISs) are leaky-wave antennas that are designed by applying the holographic principle to AMC-based surfaces [57–61]. Along with achieving high-gain pencil beam formation, the unidirectional beam scanning ability of the leaky waves grants HAISs the advantage over conventional arrays. The surface-wave propagation control using electrically small patches can be further exploited and extended to form two-dimensional surface waveguides [62].

16.8 RCS REDUCTION USING CHECKERBOARD METASURFACES

16.8.1 Introduction

Since the radar cross section (RCS) is a critical parameter in characterizing a target's scattering properties, there has been an increasing interest in the development of RCS reduction techniques utilizing metasurfaces. Conventionally, the RCS of radar targets can be reduced primarily by two main techniques: coating the target with *radar-absorbing materials* (RAMs) and shaping the target geometry. RAMs reduce the RCS by absorbing the incident power by converting the electromagnetic energy into thermal energy (heat) that is dissipated by the lossy material. A classic example of a RAM is the Salisbury screen [63], which is realized by covering the target with a lossy resistive sheet placed a quarter-wavelength above the dielectric ground plane; ideally, the total absorption of the incident wave is achieved by impedance matching. However, Salisbury screens have a narrow bandwidth, and they add an undesired thickness to the target. The thickness of a Salisbury screen can be reduced by placing it above a magnetic surface [34, 64]. However, such designs are sensitive to the angle of incidence and operate over narrow frequency bandwidth. The second technique, namely target geometrical shaping, reduces the RCS by redirecting the scattered fields away from the radar receiver. However, this often conflicts with the aerodynamic requirements of the structure/scatterer.

An alternative technique that reduces the RCS of a target by redirecting the waves away from the radar receiver, without geometrical reshaping of the target, is by placing an array of patches of perfect magnetic conductors (PMCs) and perfect electric conductors (PECs) on the same plane to achieve scattering cancellation between the fields scattered from each of the two surfaces.

16.8.2 Plane Wave Scattering by PEC-PMC Hybrid Surfaces

To demonstrate how an array of PEC and PMC patches on the same ground plane can reduce the RCS of a radar target, let us initially consider scattering by two different rectangular flat plates, which are pure PEC (see Figure 16-15a) and pure PMC (see Figure 16-15b) [65]. To illustrate the process and reduce the details, we assume that the incident wave is a TE^x polarized uniform plane wave on the yz plane, as shown in Figure 16-15a. The associated incident electric and magnetic fields can be written as

$$\mathbf{E}^i = \eta H_0 (\hat{\mathbf{a}}_y \cos \theta_i + \hat{\mathbf{a}}_z \sin \theta_i) e^{-j\beta(y \sin \theta_i - z \sin \theta_i)} \quad (16-18a)$$

$$\mathbf{H}^i = \hat{\mathbf{a}}_x H_0 e^{-j\beta(y \sin \theta_i - z \sin \theta_i)} \quad (16-18b)$$

where H_0 is a constant which represents the amplitude of the incident magnetic field. The overall finite dimensions of the plates are the same for both cases. The scattered fields, and thus the associated RCS, are obtained using Physical Optics (PO), as detailed in Section 11.3.2 of Chapter 11.

The first scattering structure is a PEC rectangular flat surface of dimensions a and b , as shown in Figure 16-15a. The closed-form bistatic RCS, based on PO, is given by (11-44a) and repeated here as

$$\sigma_{3-D} = 4\pi \left(\frac{ab}{\lambda} \right)^2 \cos^2 \theta_s \left[\frac{\sin \left[\frac{\beta b}{2} (\sin \theta_s \mp \sin \theta_i) \right]}{\frac{\beta b}{2} (\sin \theta_s \mp \sin \theta_i)} \right]^2 \quad \begin{array}{l} - \text{ for } \phi_s = \frac{\pi}{2}, \quad 0 \leq \theta_s \leq \pi/2 \\ + \text{ for } \phi_s = \frac{3\pi}{2}, \quad 0 \leq \theta_s \leq \pi/2 \end{array} \quad (16-19)$$

The details are outlined in Section 11.3.2.

The associated plots for $a = b = 5\lambda$ and $\phi_i = 270^\circ$ ($\phi_s = 90^\circ, 270^\circ$ with $0 \leq \theta_s \leq 90^\circ$) are shown by the solid line (—) curves in Figure 16-15c for $\theta_i = 0^\circ$ and Figure 16-15d for $\theta_i = 15^\circ$. It is clear that the bistatic RCS pattern of the PEC plate follows *Snell's law of reflection* (see Section 5.3) where its maximum, for electrically large large plates, is near the specular direction (i.e., $\theta_i = \theta_s = 0^\circ$ for Figure 16-15c and $\theta_i = \theta_s = 15^\circ$ for Figure 16-15d).

Now, let us repeat the procedure but for a rectangular and flat pure PMC ground plane ($\Gamma = 1$), as shown in Figure 16-15b. The closed-form principal plane bistatic RCS can be obtained by following the same steps outlined in Section 11.3.2, and the radar cross section can then be written as

$$\sigma_{3-D} = 4\pi \left(\frac{ab}{\lambda} \right)^2 \cos^2 \theta_i \left[\frac{\sin \left[\frac{\beta b}{2} (\sin \theta_s \mp \sin \theta_i) \right]}{\frac{\beta b}{2} (\sin \theta_s \mp \sin \theta_i)} \right]^2 \quad \begin{array}{l} - \text{ for } \phi_s = \frac{\pi}{2}, \quad 0 \leq \theta_s \leq \pi/2 \\ + \text{ for } \phi_s = \frac{3\pi}{2}, \quad 0 \leq \theta_s \leq \pi/2 \end{array} \quad (16-20)$$

The associated plots for $a = b = 5\lambda$ and $\phi_i = 270^\circ$ ($\phi_s = 90^\circ, 270^\circ$ with $0 \leq \theta_s \leq 90^\circ$) are shown by the dashed line (---) curves in Figure 16-15c for $\theta_i = 0^\circ$ and Figure 16-15d for $\theta_i = 15^\circ$. The maximum RCS of the PMC ground plane is exactly along the specular direction following Snell's law of reflection (i.e., $\theta_i = \theta_s = 0^\circ$ for Figure 16-15c and $\theta_i = \theta_s = 15^\circ$ for Figure 16-15d).

From (16-19), it is apparent that the maximum RCS occurs in the principal plane (yz plane) approximately when ($\theta_s = \theta_i$) for large values of b ($b \gg \lambda$). When b is large, the values of the $\sin(z)/z$ function of (16-19) vary more rapidly compared to the $\cos \theta_s$ function, such that $\cos \theta_s$

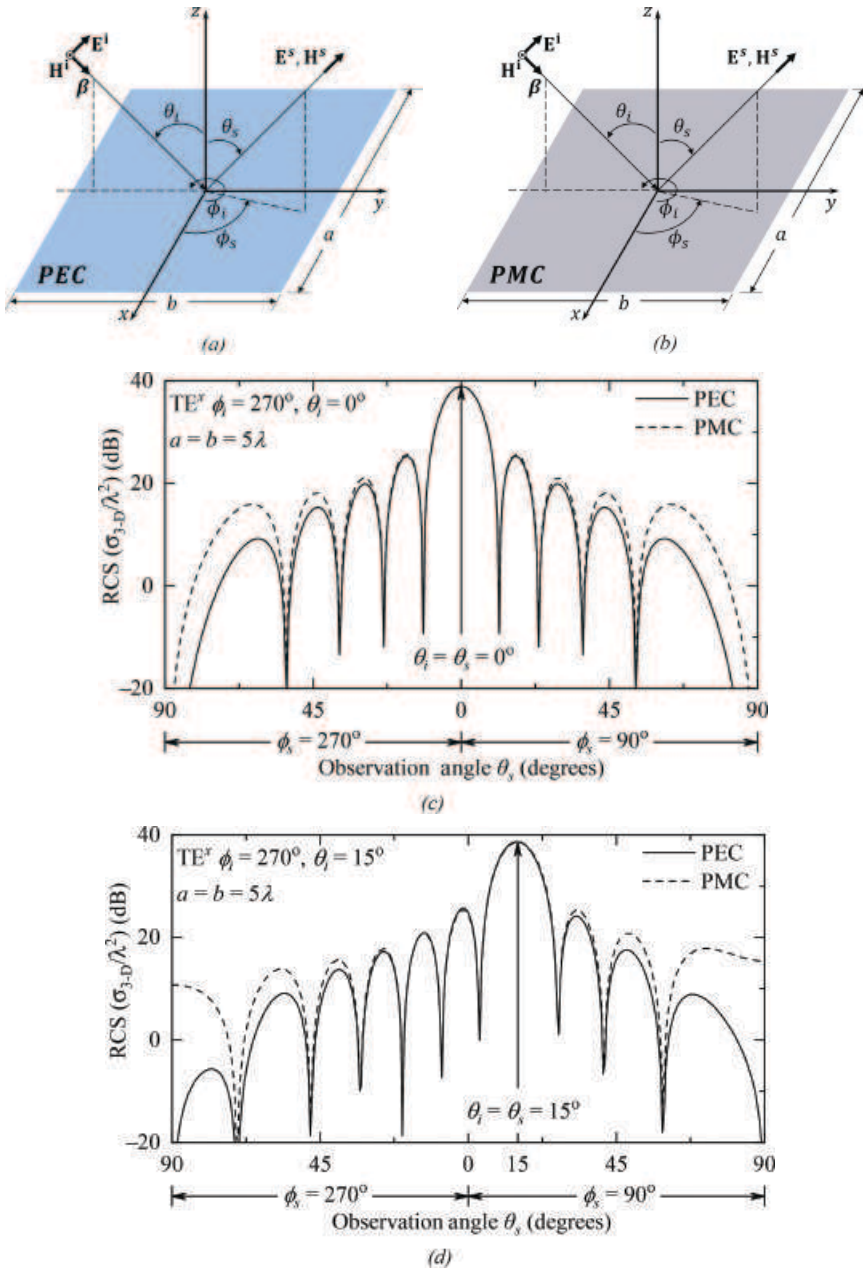


Figure 16-15 Uniform plane wave (TE^x) incident on a rectangular conducting plate (a) PEC and (b) PMC. Principal plane (yz plane) bistatic RCS ($\phi_i = 270^\circ$; $\phi_s = 90^\circ, 270^\circ$) for (c) $\theta_i = 0^\circ$ and (d) $\theta_i = 15^\circ$.

is essentially constant at and near the maximum of $\sin(z)/z$. However, for the smaller values of b , the exact maximum of (16-19) can be found iteratively. Accordingly, the maximum bistatic RCS of a flat PEC of TE^x polarization does not occur exactly at the specular direction, but it approaches that value as the dimensions of the plate become large compared to the wavelength. However, from (16-20), since $\cos^2 \theta_i$ is a function of the incident angle θ_i and not the scattering angle θ_s , the maximum value of (16-20) occurs when $\theta_s = \theta_i$ no matter what the size of the scattering plate.

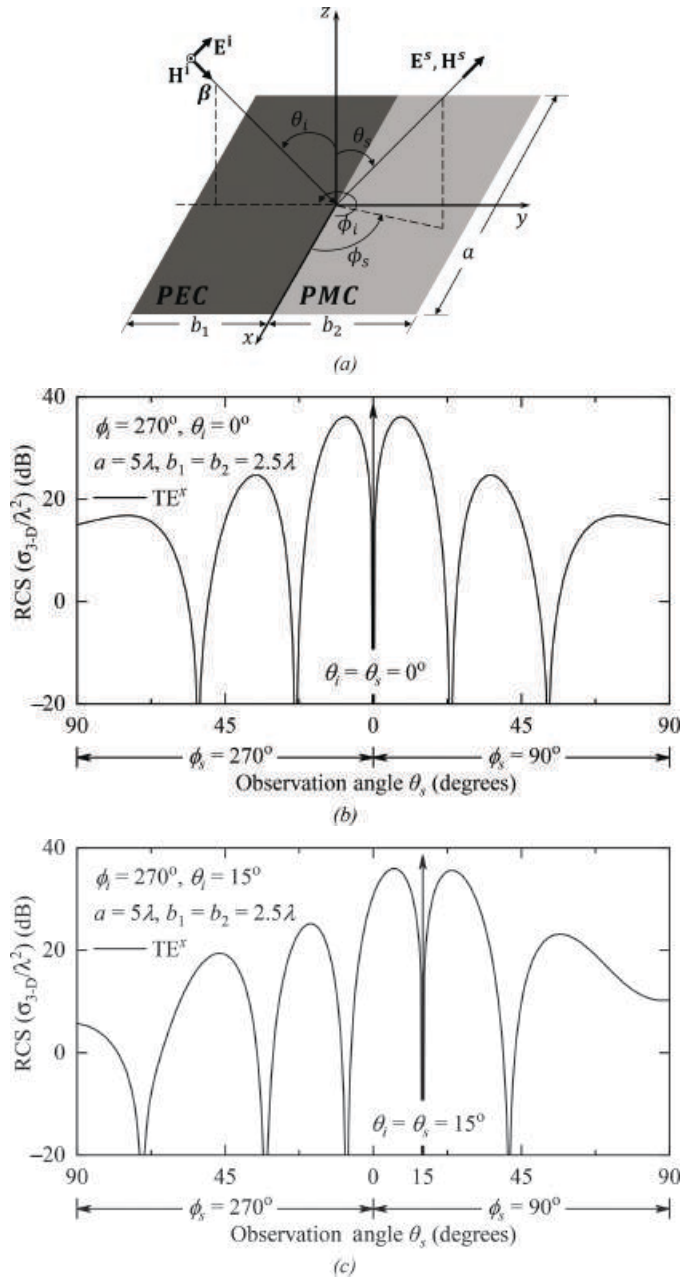


Figure 16-16 (a) Uniform plane wave incident on a rectangular hybrid conducting plate (PEC/PMC). And its principal plane (yz plane) bistatic RCS ($\phi_i = 270^\circ, \phi_s = 90^\circ, 270^\circ$) for (b) $\theta_i = 0^\circ$ and (c) $\theta_i = 15^\circ$.

Now let us consider a hybrid scattering plate, combining both surfaces (i.e., PEC and PMC) on the same scattering plate, as shown in Figure 16-16a, and by following the steps detailed in Section 11.3, the closed-form bistatic RCS of the hybrid ground plane combination in the principal plane (yz plane) can be written as [65]

$$\sigma_{3-D} = 4\pi \left(\frac{a}{\lambda} \right)^2 \left| b_2 \cos \theta_i [M_2 - jN_2] - b_1 \cos \theta_s [M_1 + jN_1] \right|^2 \quad (16-21a)$$

where

$$\begin{aligned} M_1 &= \frac{\sin(2Y_1)}{2Y_1}, \quad N_1 = \sin(Y_1) \frac{\sin(Y_1)}{Y_1} \quad \text{and} \quad Y_1 = \frac{\beta b_1}{2} (\sin \theta_s \mp \sin \theta_i) \\ M_2 &= \frac{\sin(2Y_2)}{2Y_2}, \quad N_2 = \sin(Y_2) \frac{\sin(Y_2)}{Y_2} \quad \text{and} \quad Y_2 = \frac{\beta b_2}{2} (\sin \theta_s \mp \sin \theta_i) \end{aligned} \quad (16-21b)$$

with $-$ for $\phi_s = 90^\circ, 0 \leq \theta_s \leq 90^\circ$, and $+$ for $\phi_s = 270^\circ, 0 \leq \theta_s \leq 90^\circ$.

Plots of (16-21a) for a hybrid scattering plate with of $a = 5\lambda$ and $b_1 = b_2 = 2.5\lambda$, having overall dimensions equal to the previous two cases, are depicted in Figure 16-16b for $\theta_i = 0^\circ, \phi_i = 270^\circ$ and Figure 16-16c for $\theta_i = 15^\circ, \phi_i = 270^\circ$ ($\phi_s = 90^\circ, 270^\circ$ with $0 \leq \theta_s \leq 90^\circ$). Due to the 180° phase difference between the reflection coefficients of the PEC ($\Gamma = -1$) and the PMC ($\Gamma = 1$) parts of the hybrid scattering plate, a destructive interference occurs along the specular direction; $\theta_s = 0^\circ$ for Figure 16-16b and $\theta_s = 15^\circ$ for Figure 16-16c. This destructive interference has led to the concept of RCS reduction using hybrid scattering surfaces where two different AMCs are arranged in a checkerboard architecture to create destructive interference between the fields scattered by the hybrid scattering plate.

16.8.3 Fundamentals of Conventional Checkerboard Metasurfaces

The concept of hybrid PEC/PMC scattering plates can be extended in two orthogonal directions and introduces the architecture of conventional checkerboard metasurfaces illustrated in Figure 16-17a; PMC scattering was synthesized by AMC-type metasurface unit cells. This leads to metasurfaces that consist of PEC and AMC patches configured in checkerboard pattern, as shown in Figure 16-17b. The primary appealing feature of this technique is its ability to provide RCS reduction over a broadband frequency range [36–38].

When a plane wave is incident, at normal incidence, on a PEC surface, as demonstrated in Figure 16-15a, the major lobe of the scattered fields is along the normal direction, as illustrated by the three-dimensional RCS pattern of Figure 16-18a. However, when the checkerboard surface of Figure 16-17b is illuminated by a broadside incident plane wave, the fields scattered from each of the AMC supercells are 180° out of phase and result in destructive interferences along two orthogonal planes. This forces the scattered waves to be redirected, for $d_x = d_y = 2.5\lambda$, along four quadrants with four major lobes along $\theta_0 = 12^\circ; \phi_0 = 45^\circ, 135^\circ, 225^\circ$, and 315° , as exhibited in Figure 16-18b. Figure 16-19b shows the two-dimensional RCS patterns along the principal plane (yz plane; $\phi_s = 90^\circ, 270^\circ$) and also along the two planes where the major lobes are directed (i.e., $\phi_s = 45^\circ, 225^\circ$

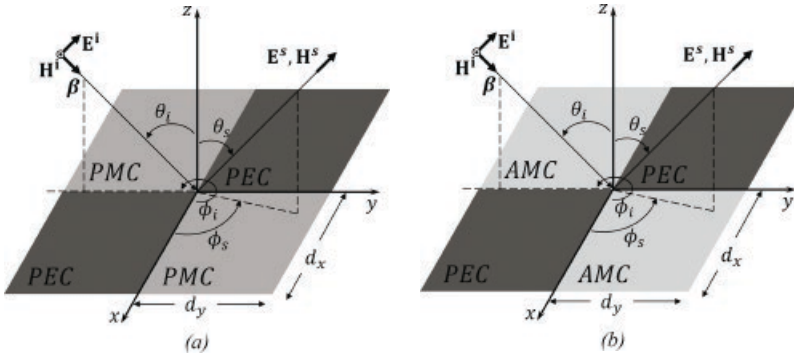


Figure 16-17 Comparison between (a) generic design of checkerboard surface of PEC/PMC and (b) conventional checkerboard surface of PEC/AMC.

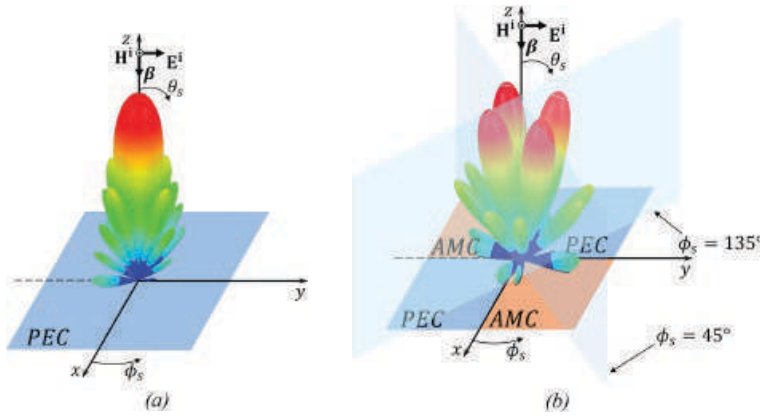


Figure 16-18 Simulated 3-D bistatic RCS patterns for (a) PEC and (b) conventional checkerboard surface with the same physical dimensions [66].

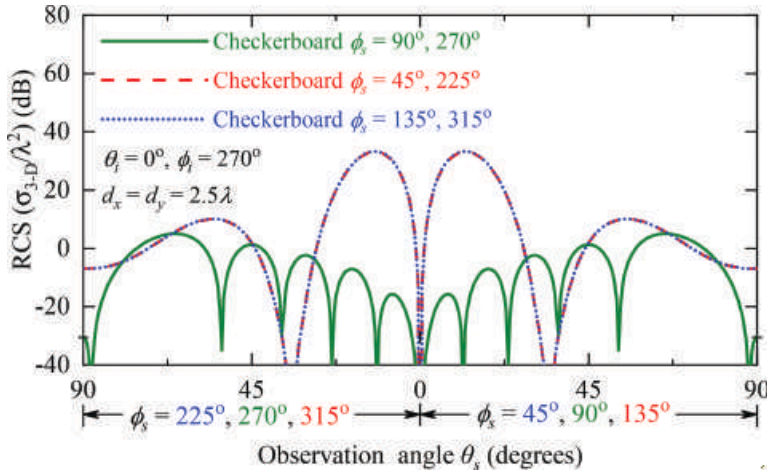


Figure 16-19 Bistatic RCS patterns of a plane wave incident on conventional checkerboard surface (see Figure 16-18b) along the principal plane (yz plane) $\phi_s = 90^\circ, 270^\circ$, $\phi_s = 45^\circ, 225^\circ$ and $\phi_s = 135^\circ, 315^\circ$ [66].

and $\phi_s = 135^\circ, 315^\circ$). The direction of all the maxima of the major lobes (θ_0, ϕ_0) can be determined approximately using array theory, particularly using [9]

$$\tan \phi_0 = \frac{\sin \theta_i \sin \phi_i \pm \frac{\delta_y}{\beta d_y}}{\sin \theta_i \cos \phi_i \pm \frac{\delta_x}{\beta d_x}} \quad (16-22)$$

$$\sin^2 \theta_0 = \left[\sin \theta_i \sin \phi_i \pm \frac{\delta_y}{\beta d_y} \right]^2 + \left[\sin \theta_i \cos \phi_i \pm \frac{\delta_x}{\beta d_x} \right]^2 \quad (16-23)$$

where (θ_0, ϕ_0) defines the angular direction of the major lobes, while (θ_i, ϕ_i) represents the angle of incidence; δ_x and δ_y represent the phase difference of the reflection coefficients of the adjacent supercells in the x and y directions, respectively, while d_x and d_y are the center-to-center separation distances between the adjacent supercells in the x and y directions, respectively. A MATLAB

program titled **Angular_Direction_Checkerboard**, utilizing (16-22) and (16-23), determines the angular directions of the scattered fields and is included at the end of this chapter.

The four major lobes formed by the scattering from the checkerboard surface of Figure 16-18b are attributed to the fields scattered by each patch having 180° phase difference between them; $\Gamma = -1$ (PEC) and $\Gamma = +1$ (PMC). This can also be verified by examining the electric current density induced on the surface of the checkerboard structure. Based on boundary conditions (see Section 1.7.2) and the incident fields of (16-18), the tangential components of the magnetic field vanish on the PMC surface and double on the PEC, such that the electric current density induced on the PEC patches of the plate is represented by

$$\mathbf{J}_s|_{\text{PMC}(z=0)} \approx 0 \quad (16-24a)$$

$$\mathbf{J}_s|_{\text{PEC}(z=0)} \approx 2\hat{\mathbf{n}} \times \mathbf{H}^{\text{total}} = 2\hat{\mathbf{a}}_z \times \hat{\mathbf{a}}_x H_x = \hat{\mathbf{a}}_y 2H_0 e^{-j\beta y' \sin \theta_i} \quad (16-24b)$$

To verify the performance of the conventional checkerboard surfaces (see Figure 16-17b) with $d_x = d_y = 2.5\lambda$ and $\delta_x = \delta_y = 180^\circ$, the induced electric current density $\mathbf{J}_s$ was simulated by the commercial software HFSS [26], and it is exhibited in Figure 16-20. It can be seen that the electric current density basically exists only on the PEC patches while that on the PMC patches is of very low intensity. This induced current density creates four major lobes in the RCS pattern where the angular directions of their maxima can be determined approximately using array theory [i.e., (16-22) and (16-23)]. Thus, the angular directions of the major lobes are $\theta_0 = 16.43^\circ$ and $\phi_0 = 45^\circ, 135^\circ, 225^\circ$, and 315° . However, by using the finite-element method solution of HFSS [26], the angular locations of the four maxima are directed along $\theta_0 = 12^\circ$ and $\phi_0 = 45^\circ, 135^\circ, 225^\circ$, and 315° .

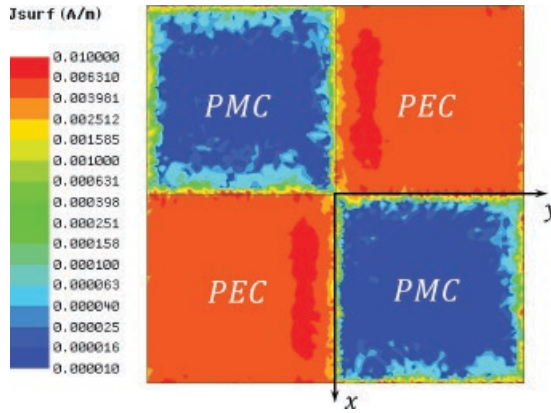


Figure 16-20 Induced surface current density on a conventional checkerboard surface (see Figure 16-18b) with $d_x = d_y = 2.5\lambda$.

Example 16-3

Determine the angular elevation direction (θ_0) of the four major lobes when a plane wave incident normally ($\theta_i = \phi_i = 0^\circ$) on a checkerboard surface (Figure 16-17b, $\delta_x = \delta_y = 180^\circ$) when the spacing/size of both PEC and PMC surfaces is uniform ($d_x = d_y$) and equal to

- $d_x = d_y = 2\lambda$
- $d_x = d_y = 3\lambda$
- $d_x = d_y = 5\lambda$

Explain the dependency of the angular directions of the major lobes on the size of the scattering surfaces.

Solution: The approximate elevation angle (θ_0) of the maxima of the four formed scattering lobes can be found based on the array theory of (16-23) and is tabulated below for each case:

$d_x = d_y$	θ_0 based on (16-23)
2λ	20.70°
3λ	13.63°
5λ	8.13°

It is observed that as the size of the PEC and PMC cells of the hybrid scattering ground plane increases, the major lobes' angular direction approaches broadside ($\theta_0 = 0^\circ$). This is expected because as $d_x = d_y \Rightarrow$ large and $\theta_i = \phi_i = 0^\circ$, (16-23) reduces to $\sin^2 \theta_0 \approx 0 \Rightarrow \sin \theta_0 \approx 0 \Rightarrow \theta_0 \approx 0^\circ$.

16.8.4 Broadband RCS Reduction Metasurfaces

A. Conventional Broadband Checkerboard Metasurfaces As illustrated in the previous section, when PEC and PMC patches are placed in a checkerboard pattern on a rectangular flat plate, the scattered fields are redirected and form four main beams, and the direction of each of the main beams can be predicted by antenna array theory [9]. One can also employ PO for more accurate predictions as recently reported in [67, 68]. The other key parameter is the RCS reduction bandwidth of these metasurfaces. Since most of the radar systems operate over a large frequency range, it will be desirable to design and realize RCS reduction metasurfaces with large frequency bandwidth. As reported in [36, 37], for checkerboard metasurfaces, this can be achieved by configuring two AMC supercells with each other in checkerboard-type architectures, as illustrated in Figure 16-17b. The primary appealing feature of this technique is its ability to provide RCS reduction over a broadband frequency range [36–38].

For such RCS reduction checkerboard metasurfaces, an analytical expression that approximates the 10-dB RCS reduction in its simplest form is given by [36, 37]

$$\text{RCS}_{\text{red}} = \left| \frac{A_1 e^{j\Phi_1} + A_2 e^{j\Phi_2}}{2} \right|^2 \quad (16-25)$$

where Φ_1 and Φ_2 are the phases of the fields reflected by AMC-1 and AMC-2, respectively, while A_1 and A_2 are their respective reflection coefficient amplitudes [36]. Therefore, to reduce the RCS by at least 10 dB, a phase difference of about $(180 \pm 37)^\circ$ must be obeyed between the AMC supercells [36]. The RCS reduction versus the phase difference between the two AMCs patches is illustrated in Figure 16-21. Designing a conventional checkerboard surface for RCS reduction utilizing AMC surfaces of Section 16.8.4 and (16-25) is demonstrated with a MATLAB program entitled **Conventional_Checkerboard_RCS_Reduction** included at the end of this chapter.

As a result, the selection of the AMCs is a critical step in realizing a broadband RCS reduction metasurface. In [36], two single-band AMCs were considered for designing a wideband checkerboard surface (see Figure 16-22), for which a nearly 60% bandwidth of 10-dB RCS reduction was reported (see Figure 16-23). The induced electric surface current density is illustrated in Figure 16-24 at 4.65 GHz when the phase difference between the selected AMCs is 180° . Such a checkerboard architecture is classified as *conventional* design [37]. Similarly, in [38, 39], two dual-band AMCs were utilized to reduce the RCS in two separate wide frequency bands, for which about 60% and 26% 10-dB RCS reduction bandwidths were achieved.

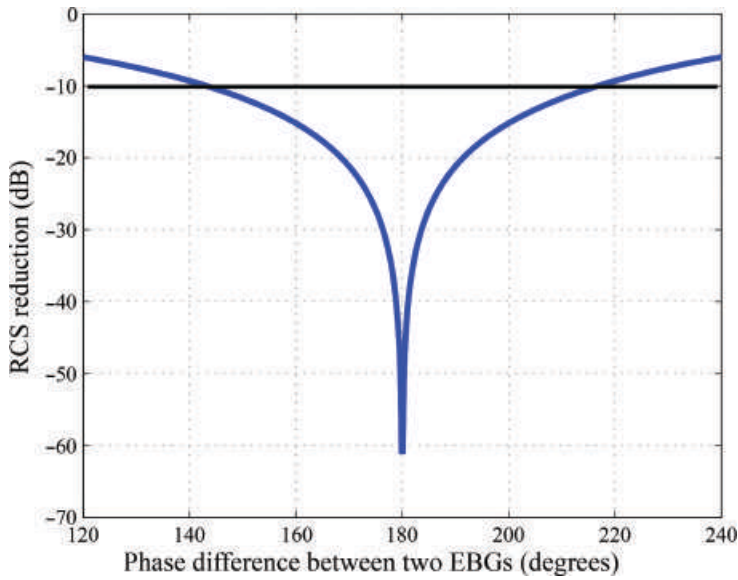


Figure 16-21 RCS reduction using (16-25) versus the phase difference between two AMC supercells. (Source: W. Chen, C. A. Balanis, and C. R. Birtcher, “Checkerboard EBG surfaces for wideband radar cross section reduction,” *IEEE Trans. Antennas Propagat.*, vol. 63, no. 3, pp. 2636–2645, 2015. Reproduced with permission from IEEE.)

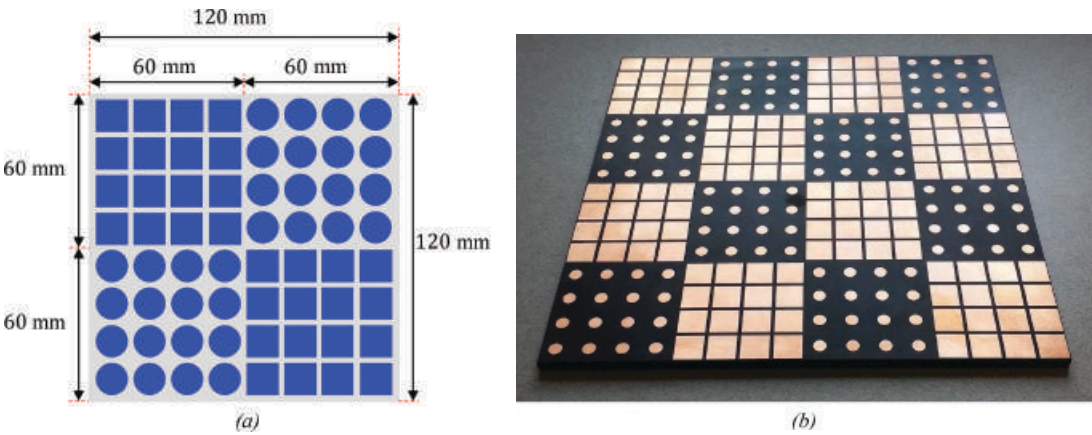


Figure 16-22 (a) Design and (b) fabricated prototypes of conventional checkerboard surface for wideband RCS reduction. (Source: W. Chen, C. A. Balanis, and C. R. Birtcher, “Checkerboard EBG surfaces for wideband radar cross section reduction,” *IEEE Trans. Antennas Propagat.*, vol. 63, no. 3, pp. 2636–2645, 2015. Reproduced with permission from IEEE.)

Once the size of the AMC unit cells is selected, the number of unit cells in each individual AMC supercell defines the distances (d_x, d_y) between adjacent supercells. According to (16-22) and (16-23), since the distances between adjacent supercells are not critical for the RCS reduction under normal incidence, the number of unit cells in an individual AMC supercell is not as pivotal. However, according to (16-25), these distances still determine the angular direction of the lobe maxima (θ_0, ϕ_0) of the scattered fields [37].

B. Judicious Selection of AMC Unit Cells A judicious selection of AMCs was investigated [37, 40], and a guideline for the selection of AMCs was presented to synthesize ultrabroadband

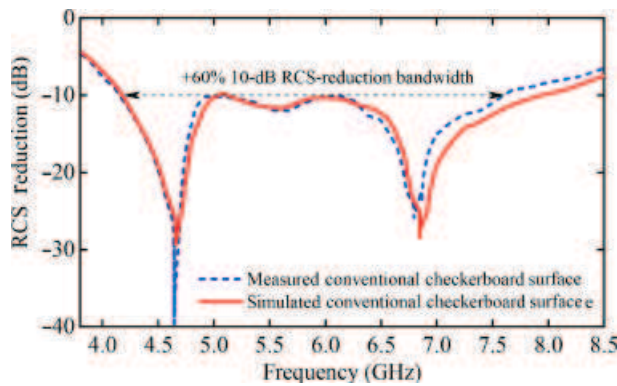


Figure 16-23 Simulated and measured RCS reduction using conventional checkerboard surface. (Source: W. Chen, C. A. Balanis, and C. R. Birtcher, “Checkerboard EBG surfaces for wideband radar cross section reduction,” *IEEE Trans. Antennas Propagat.*, vol. 63, no. 3, pp. 2636–2645, 2015. Reproduced with permission from IEEE.)

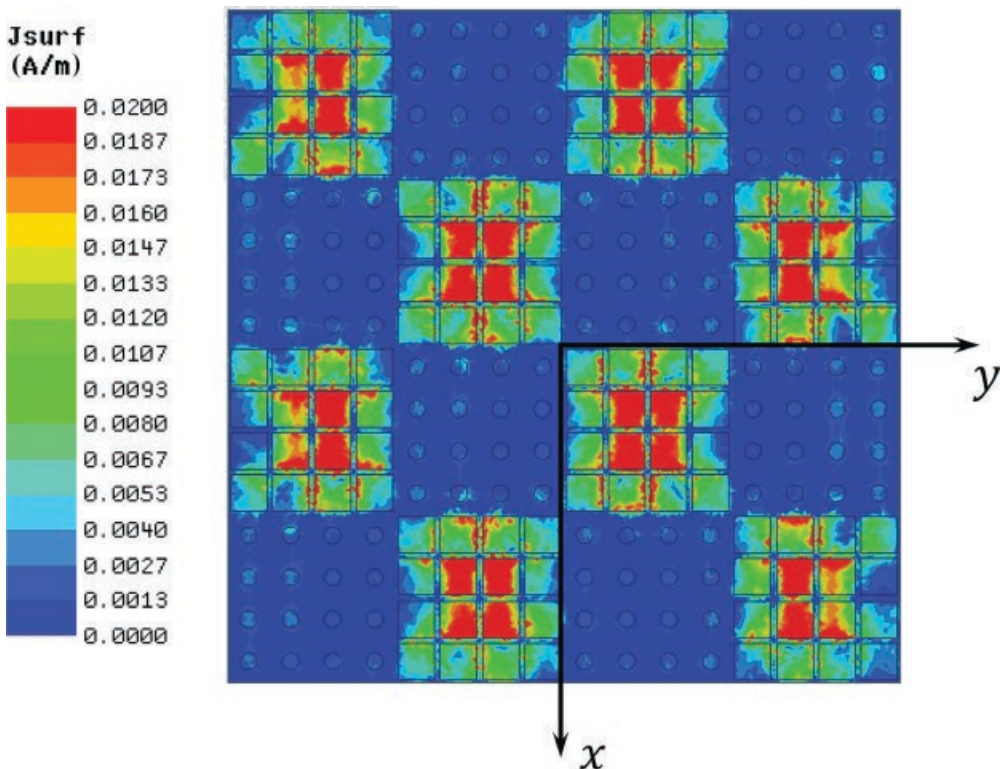


Figure 16-24 Induced surface current density on conventional checkerboard surface for wideband RCS reduction of Figure 16-22b at $f = 4.65$ GHz.

RCS reduction checkerboard surfaces. It was concluded that a combination of single- and dual-band AMCs, instead of two single-band AMCs, provides broader RCS reduction bandwidth. Such checkerboard surfaces with a combination of single- and dual-band AMCs are called *blended designs* [37]. For validation, the conventional checkerboard surface reported in [36] was transformed to a blended design, with a combination of single- and dual-band AMCs [37, 40].

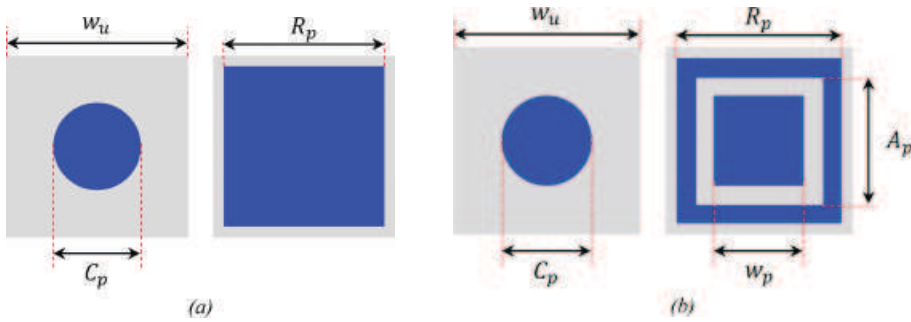


Figure 16-25 Selection of AMC unit cell designs to convert a conventional checkerboard surface into a blended checkerboard surface: (a) AMC-1 and (b) AMC-2. (Source: Adapted from A. Y. Modi, C. A. Balanis, C. R. Birtcher, and H. Shaman, “Novel design of ultrabroadband radar cross section reduction surfaces using artificial magnetic conductors,” *IEEE Trans. Antennas Propagat.*, vol. 65, no. 10, pp. 5406–5417, 2017.)

The single-band AMC (AMC-1) design consisting of single circular and square patches [36], and shown in Figure 16-25a, was converted to a dual-band AMC (AMC-2), as shown in Figure 16-25b, by introducing an outer ring to the single-band square AMC patch; the circular patch is not altered. As illustrated in Figure 16-26, the $(180 \pm 37)^\circ$ phase difference of the single band is satisfied only from 3.6 to 6.5 GHz, which results in about 60% fractional 10-dB RCS reduction bandwidth. However, after the transformation of AMC-1 from single-band AMC (AMC-1) to dual-band AMC (AMC-2), as illustrated in Figure 16-25, the required phase difference of $(180 \pm 37)^\circ$ was maintained from 3.6 to 9.4 GHz, an increase of +23% in 10-dB RCS reduction bandwidth (from 60% to 83%). This suggests that the conventional checkerboard surface should first be designed and optimized with two single-band AMCs. After its optimization, to extend the RCS reduction bandwidth, one of the single-band AMCs should be transformed to a dual-band AMC.

The RCS reduction of the blended checkerboard surface (see Figure 16-27a) was verified using a prototype displayed in Figure 16-27b, which was fabricated on a standard Rogers RT/duroid-5880 dielectric substrate backed by a PEC ground plane (with thickness of 6.35 mm and dielectric constant of 2.2) [37]. The simulated and measured RCS data, with a 10-dB reduction bandwidth of 83%, are displayed in Figure 16-27c; an excellent agreement between simulations and measurements is indicated [37].

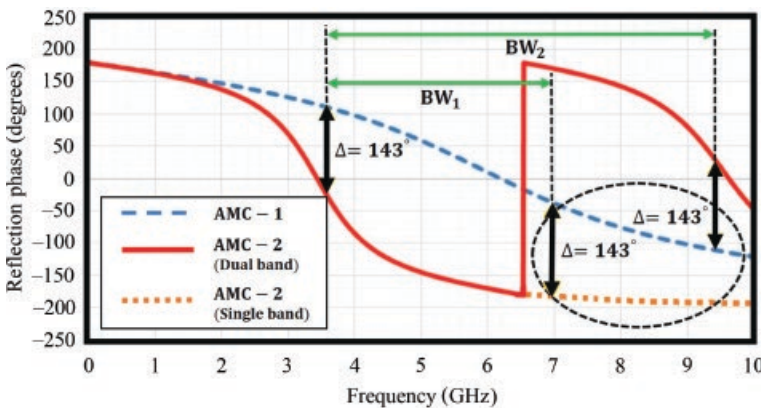


Figure 16-26 Typical reflection phase for AMC-1 and AMC-2 as a function of frequency. (Source: A. Y. Modi, C. A. Balanis, C. R. Birtcher, and H. Shaman, “Novel design of ultrabroadband radar cross section reduction surfaces using artificial magnetic conductors,” *IEEE Trans. Antennas Propagat.*, vol. 65, no. 10, pp. 5406–5417, 2017. Reproduced with permission from IEEE.)

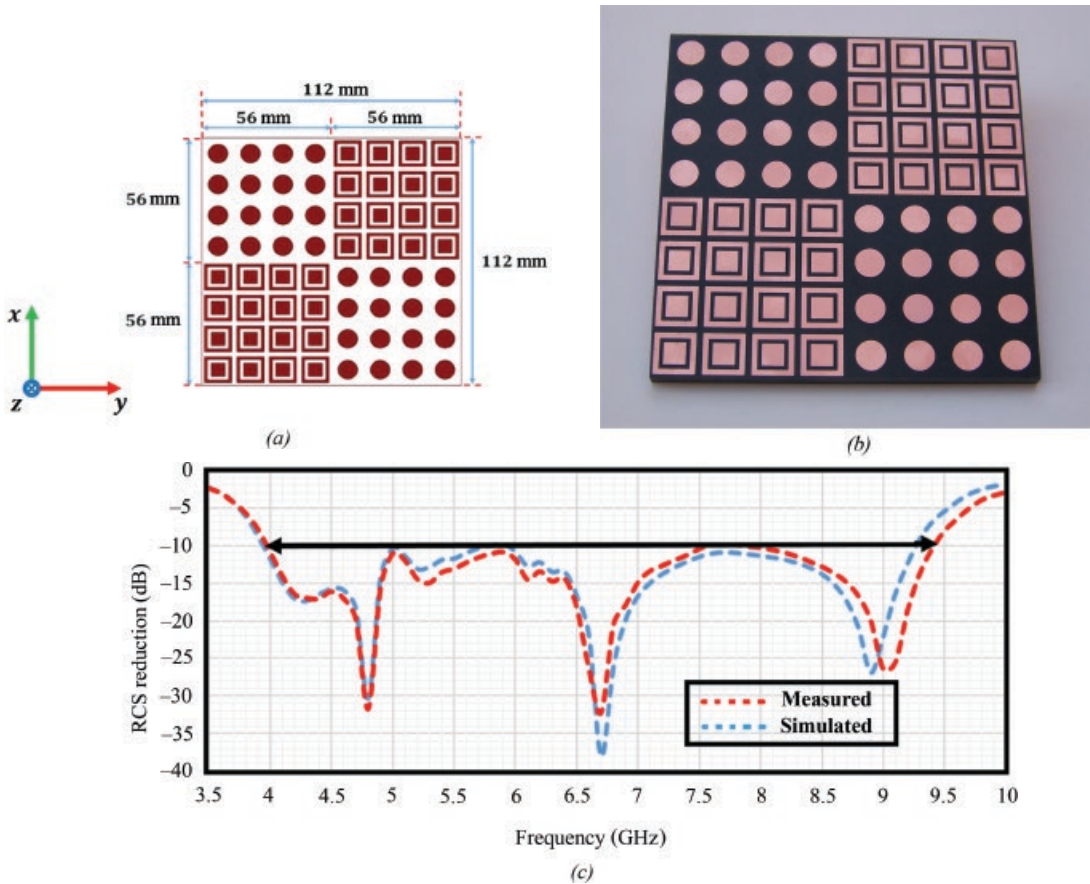


Figure 16-27 Blended checkerboard surface: (a) Design and (b) fabricated prototypes. (c) Measured and simulated RCS reduction results. (Source: A. Y. Modi, C. A. Balanis, C. R. Birtcher, and H. Shaman, “Novel design of ultrabroadband radar cross section reduction surfaces using artificial magnetic conductors,” *IEEE Trans. Antennas Propagat.*, vol. 65, no. 10, pp. 5406–5417, 2017. Reproduced with permission from IEEE.)

However, in both the conventional and blended designs, the checkerboard surfaces have the same design architecture. In each of them, only the selection of the building supercells is different to achieve the required destructive interference (i.e., two different single-band AMCs in conventional; a combination of single- and dual-band AMCs in the blended). Therefore, in both of these checkerboard metasurfaces, to obtain greater than 10-dB RCS reduction, the $(180 \pm 37)^\circ$ phase difference still must be obeyed between the building supercells. This is a primary limitation of such checkerboard designs.

Example 16-4

Design a checkerboard metasurface by synthesizing an AMC unit cell such that it exhibits zero reflection phase at 8 GHz. Use a Rogers RT/duroid 5880 with a dielectric constant of $\epsilon_r = 2.2$ and substrate thickness of $h = 2$ mm. Plot the 10-dB RCS reduction bandwidth of the synthesized checkerboard metasurface comprised of PEC (metallic surface) and AMC surface. Repeat the same exercise for substrates with thicknesses of $h = 2$ mm and $h = 4$ mm. Explain the relationship between the thickness of the substrate and the 10-dB RCS reduction bandwidth. Use the MATLAB computer program *Conventional_Checkerboard_RCS_Reduction* to verify the analytical design.

Solution: Specifying a resonant frequency of $f_0 = 8$ GHz and a substrate thickness $h = 2$ mm, the sheet inductance and sheet capacitance based, respectively, on (16-9) and (16-10) are

$$\begin{aligned} L_s &= \mu_1 h = \mu_0 h = 4\pi \times 10^{-7} (2.0 \times 10^{-3}) \\ &= 2.5133 \times 10^{-9} = 2.5133 \text{ nH} \\ C_s &= \frac{1}{(\omega_0)^2 L_s} = \frac{1}{(2\pi \times 8 \times 10^9)^2 \times 2.5133 \times 10^{-9}} \\ &= 1.5748 \times 10^{-13} = 0.15748 \text{ pF}. \end{aligned}$$

Then, the dimensions of the of AMC surface (i.e., gap spacing g , patch width w , and unit cell size a) can be determined by choosing an initial gap spacing g of 1 mm and determining the patch width w from (16-13), or

$$C = \frac{w\epsilon_0(\epsilon_r + 1)}{\pi} \cosh^{-1}\left(\frac{a}{g}\right) = \frac{w\epsilon_0(2.2 + 1)}{\pi} \cosh^{-1}\left(\frac{w + 1 \times 10^{-3}}{1 \times 10^{-3}}\right)$$

Using a nonlinear solver to the above equation leads to a patch width of $w = 6.5$ mm and $a = w + g = 7.5$ mm.

For the same design requirements and with a substrate thickness of $h = 4$ mm and $h = 6$ mm, the dimensions of the of AMC surface can be determined following the same procedure, and they are:

Substrate thickness h	Gap spacing g	Patch width w	Unit cell size a
(mm)	(mm)	(mm)	(mm)
2.0	1.0	6.5	7.5
4.0	1.5	4.3	5.8
6.0	2.0	3.47	5.47

Figure 16-28 illustrates the phase of the reflection coefficient S_{11} of the designed metasurfaces for $h = 2, 4$, and 6 mm.

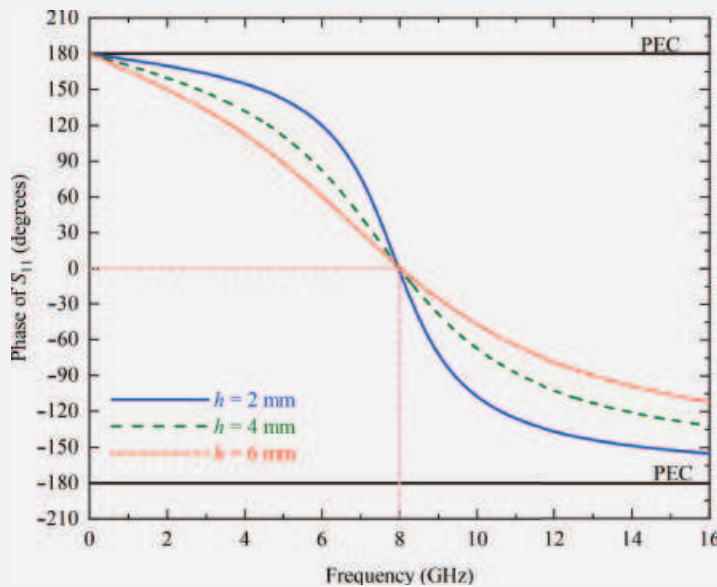


Figure 16-28 Phase versus frequency of reflection coefficient S_{11} of AMC surface with square patches of the three different substrate thicknesses.

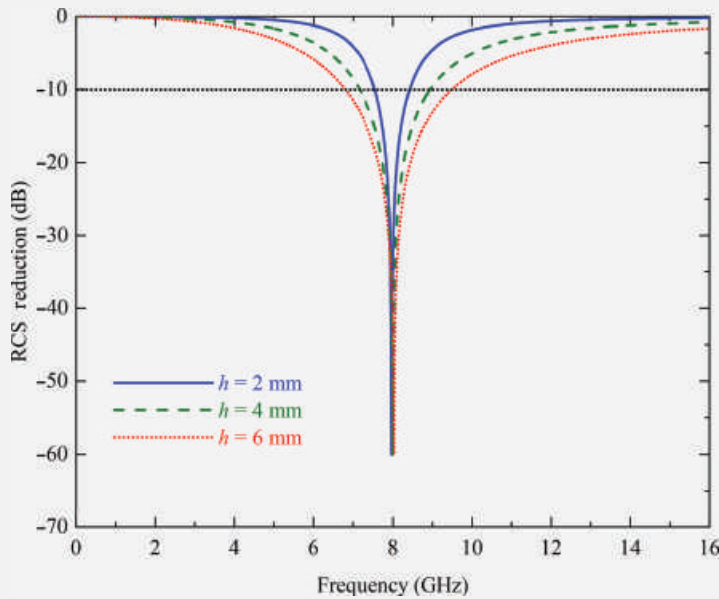


Figure 16-29 RCS reduction versus frequency of the designed AMCs using (16-25).

The corresponding 10-dB RCS reductions of the designed AMCs, when implemented with PEC ground plane in a checkerboard patterned design, are illustrated in Figure 16-29. It is clear that as the substrate thickness increases, the 10-dB RCS reduction bandwidth increases. This can be explained by examining the bandwidth on the designed AMCs. As the substrate thickness increases, the sheet inductance L_s of (16-9) increases and, thus, expands the bandwidth of the AMC surface as represented by $S_{11} = \pm 90^\circ$. Therefore, slower variance in the reflection phase over the frequency range translates into broader 10-dB RCS reduction bandwidth.

C. Modified Checkerboard Surfaces To overcome the limitation of maintaining the required phase difference of $(180 \pm 37)^\circ$, the design architecture of these surfaces can be modified [41]. As exhibited in Figure 16-30a, applicable for both *conventional* and *blended* checkerboard surfaces, the destructive interference is not necessary in both the orthogonal planes. To provide the same RCS reduction under normal incidence, only one of these two destructive interferences, along the x and y directions, is required [37]. Consequently, the surface of Figure 16-30b, where destructive interference is developed only along the x direction, is equivalent to the original conventional/blended checkerboard surface of Figure 16-30a for reducing the RCS under normal incidence [37]. As depicted in Figure 16-31a, this then allows the introduction of another type of destructive interference. To further increase the RCS reduction bandwidth, the second combination of AMCs (AMC-3 and AMC-4) should be selected such that the destructive interference produced by them reduces the RCS by more than 10 dB outside the 10-dB RCS reduction bandwidth of the first combination (AMC-1 and AMC-2) [41]. In addition, the frequency ranges of the two combinations should overlap each other over the majority of their respective frequency ranges. The RCS reduction bandwidth can be enhanced even further by altering the respective portions of the total area covered by the individual combination of AMCs, as suggested in Figure 16-31b.

In [37], the same technique was implemented on a blended checkerboard surface with a combination of single- and dual-band AMCs, as illustrated in Figure 16-32a, to convert it from a *blended* design of equal size supercells into a *modified* checkerboard surface of unequal size

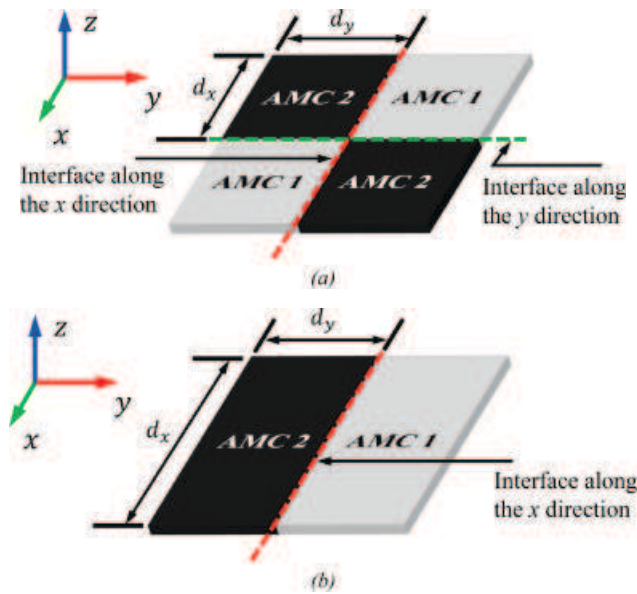


Figure 16-30 Designs of RCS reduction. (a) Conventional/blended checkerboard surface. (b) Surface with only one destructive interference. (Source: A. Y. Modi, “Metasurface-based techniques for broadband radar cross-section reduction of complex structures,” Ph.D. dissertation, School of Electrical, Computer, and Energy Engineering, Arizona State University, 2020.)

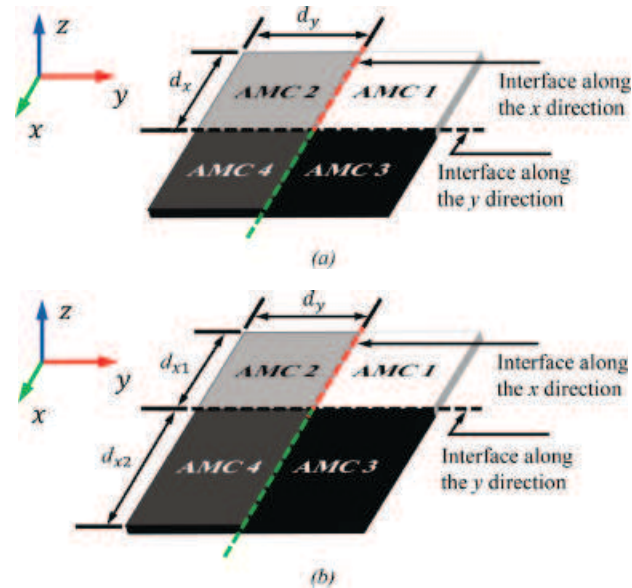


Figure 16-31 Designs of RCS reduction. (a) Modified surface with equal sized supercells. (b) Modified surface with unequal sized supercells. (Source: A. Y. Modi, “Metasurface-based techniques for broadband radar cross-section reduction of complex structures,” Ph.D. dissertation, School of Electrical, Computer, and Energy Engineering, Arizona State University, 2020.)

supercells shown in Figure 16-32d. To understand the rationale and evolution process, the design is broken down into four (4) steps leading from Figure 16-32a to Figure 16-32d, and the design steps are illustrated in Figure 16-32. To increase the bandwidth, the second combination of AMCs (AMC-3 and AMC-4) with orthogonal destructive interference, shown in Figure 16-32c, was added to the first combination of AMCs (AMC-1 and AMC-2) of Figure 16-32b. When implemented as an independent blended checkerboard surface, the second combination of AMCs

(AMC-3 and AMC-4) attains an RCS reduction of more than 27 dB near 9.7 GHz. Therefore, in the final design of Figure 16-32d, the area covered by the second combination was increased. As a result of the two AMC combinations, the RCS reduction bandwidth was extended beyond 9.3 GHz, which was the limit of the first AMC combination (AMC-1 and AMC-2). Due the $(180 \pm 37)^\circ$ phase difference limitation, the 10-dB RCS reduction bandwidth of the first combination of AMS (AMC-1 and AMC-2), of Figure 16-32a, is 83%. However, after employing the proposed design technique using both AMC combinations, (AMC-1 + AMC-2) and (AMC-3 + AMC-4) with two different orthogonal destructive interferences and unequal areas between them, of Figure 16-32d, the -10-dB RCS reduction fractional bandwidth was increased from 83% to 91%, an 8% enhancement from the design of Figure 16-32a to that of Figure 16-32d. A fabricated prototype is shown in Figure 16-33. To validate the design process, a prototype was fabricated, simulated, and measured; the data are illustrated and compared in Figure 16-34 [69] for both TM and TE polarizations; a very good agreement is indicated.

D. Generalized Approach to Synthesize Ultrabroadband RCS Reduction Checkerboard Surfaces In summary, the steps of a generalized approach to synthesize an ultrabroadband RCS reduction checkerboard surface are outlined in what follows:

1. Initially design and optimize two single-band AMC surfaces to achieve maximum 10-dB RCS reduction bandwidth, with a combination of two single-band AMC structures, as shown in Figure 16-25a.
2. Alter one of these single-band AMCs to a dual-band AMC (see Figure 16-25b) to convert a *conventional* checkerboard surface into *blended* design (see Figure 16-32a) to improve the RCS reduction bandwidth. Then, convert the design of Figure 16-32a to Figure 16-32b, since both give identical 10-dB RCS reduction.

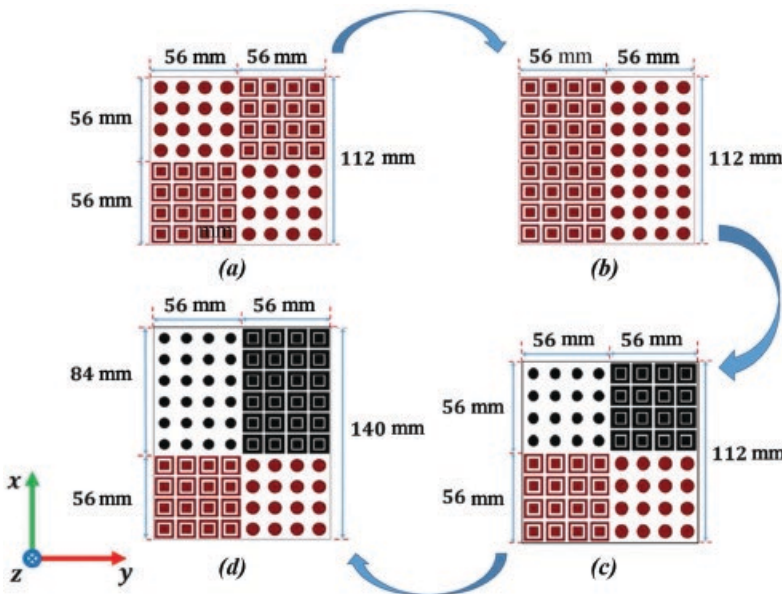


Figure 16-32 Evolution of modified checkerboard surfaces. Designs of (a) blended checkerboard surface, (b) surface with only one destructive interference, (c) modified surface with equal sized supercells, and (d) modified surface with unequal-sized supercells. (Source: A. Y. Modi, C. A. Balanis, and C. R. Birtcher, "Novel technique for enhancing RCS reduction bandwidth of checkerboard surfaces," in *Proc. IEEE Int. Symp. Antennas Propagat.*, San Diego, CA, pp. 1911–1912, 2017. Reproduced with permission from IEEE.)

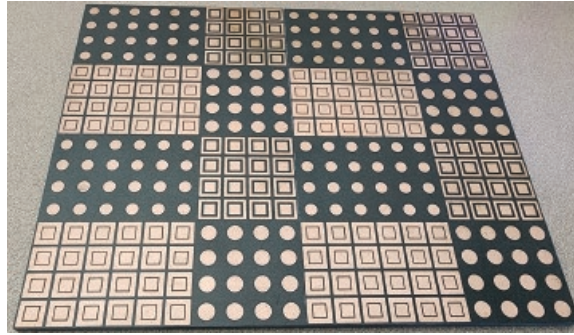


Figure 16-33 Fabricated prototype of modified checkerboard surface. (Source: A. Y. Modi, C. A. Balanis, C. R. Birtcher, and H. Shaman, “Novel design of ultrabroadband radar cross section reduction surfaces using artificial magnetic conductors,” *IEEE Trans. Antennas Propagat.*, vol. 65, no. 10, pp. 5406–5417, 2017. Reproduced with permission from IEEE.)

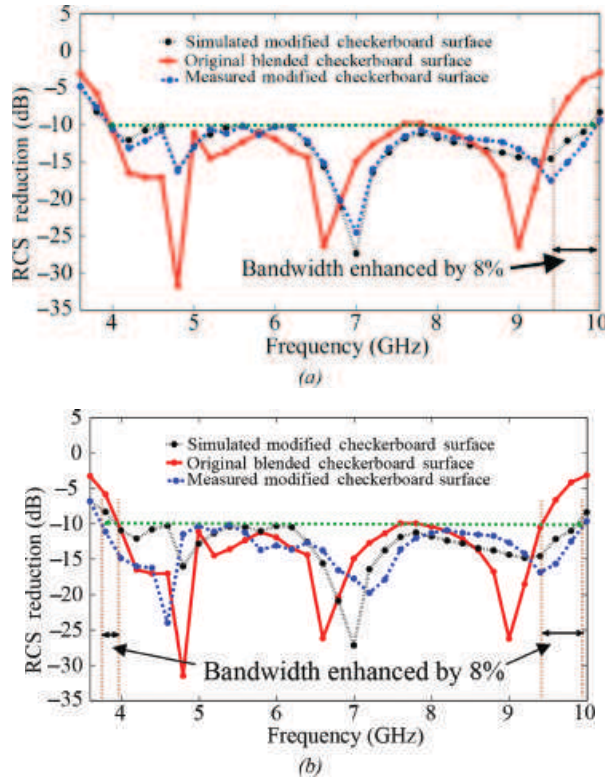


Figure 16-34 RCS reduction of modified checkerboard surface under normal wave incidence with (a) TM and (b) TE polarizations. (Source: C. A. Balanis, M. A. Amiri, A. Y. Modi, S. Pandi, and C. R. Birtcher, “Applications of AMC-based impedance surfaces,” *EPJ Applied Metamaterials*, vol. 5, no. 3, pp. 1–15, 2018.)

3. Design and introduce another *blended* checkerboard surface with the second combination of AMCs (i.e., AMC-3 and AMC-4) having orthogonal destructive interference, as shown in Figure 16-32c, where the selection criterion for the second combination was established in the previous subsection.

4. Finally, to further extend the RCS reduction bandwidth, alter the respective areas covered by the two individual AMC combinations of these *blended* checkerboard surfaces to develop a *modified* checkerboard surface (see Figure 16-32d).

16.8.5 Broadband RCS Reduction of Complex Targets

As we have discussed in Section 16.8.4, the simplicity of the checkerboard concept makes it easy to modify the designs to achieve ultrabroadband RCS reduction bandwidth. For the same reason, it also makes it easier to implement it on complex structures to reduce their RCS. For example, the concept of checkerboard metasurfaces can be generalized and utilized to reduce the radar signature of antennas [70]. Further, such metasurfaces have also been implemented and investigated on curved structures, like cylindrical geometries as well as corner reflectors [68, 71]. In the subsections that follow, each of these complex structures is reviewed.

A. Antenna and Antenna Array Antennas are primary contributors to high RCS signatures of radar targets. Consequently, the synthesis of low-RCS antennas has been of paramount interest. The checkerboard metasurface concept was generalized to synthesize low-RCS antennas [69, 72, 73]. However, it is important that modifications made to reduce the RCS of an antenna should not impact its radiation performance. In Figure 16-35, the method is implemented to reduce the RCS of a high-gain microstrip-patch antenna array of 10 elements in a (5×2) arrangement. As illustrated in Figure 16-35, the appealing feature of the proposed method is that it enables both in-band and out-of-band RCS reductions, without disturbing the radiation performance of the original antenna/antenna array [70, 73].

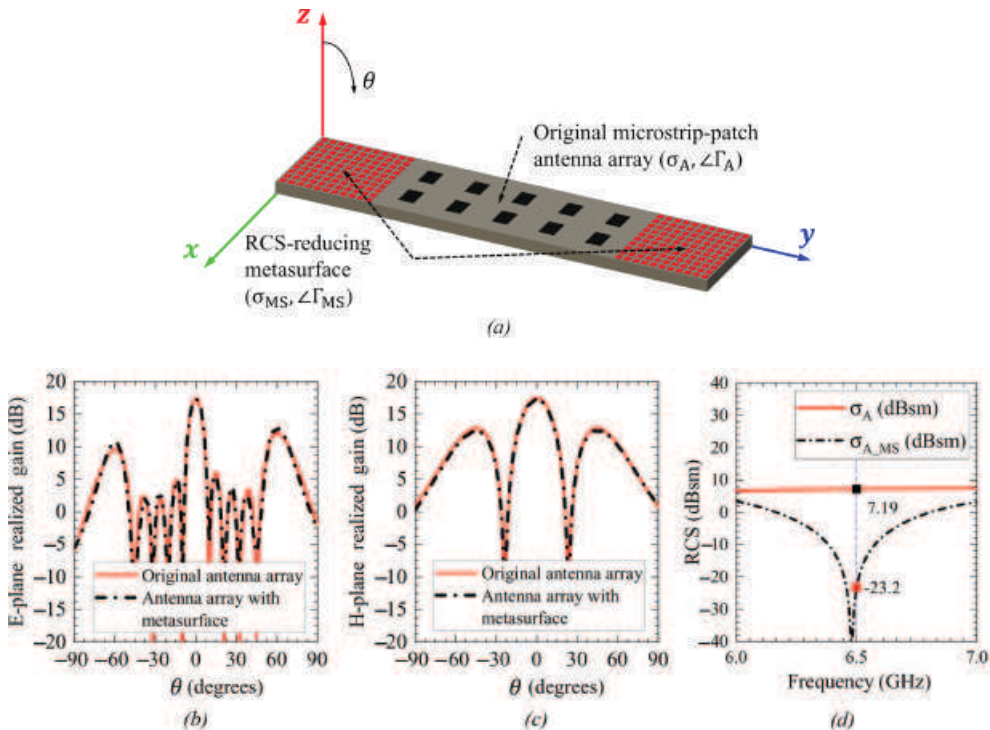


Figure 16-35 (a) Geometry of the final antenna array with metasurface, (b) its radiation performance comparison in E-plane, (c) its radiation performance comparison in H-plane, and (d) its RCS reduction performance. (Source: A. Y. Modi, C. A. Balanis, C. R. Birtcher, and H. N. Shaman, “Robust method for synthesizing low-RCS high-gain antennas using metasurfaces,” in *Proc. of IEEE Int. Symp. Antennas Propagat.*, Atlanta, GA, pp. 627–628, 2019. Reproduced with permission from IEEE.)

B. Cylindrical Structures Most of our previous discussions focused on planar metasurfaces. However, for practical applications, these metasurfaces have to perform on conformal structures. Therefore, checkerboard metasurfaces were investigated on flexible curvilinear structures [71]. To examine such structures, a prototype checkerboard metasurface was built on a flexible substrate. The prototype was then bent and mounted on the structure with two different radii of curvature. The metasurface was evaluated under normal incidence for both polarizations (i.e., HH and VV).

In [71], a flat checkerboard surface (see Figure 16-36a) was designed to obtain an RCS reduction of more than 5 dB over a bandwidth of 25% (as illustrated in Figure 16-36b). As summarized in Table 16-1, the same 5-dB RCS reduction bandwidth of 25% was approximately maintained (for both HH and VV polarizations) for the cylindrically curved checkerboard metasurface of Figure 16-36b with two orthogonal radii of curvature. It was observed, based on the basic laws of GO for curved surfaces, that the separation angle between the two adjacent innermost lobes on either side of the xy plane is inversely proportional to the radius of curvature.

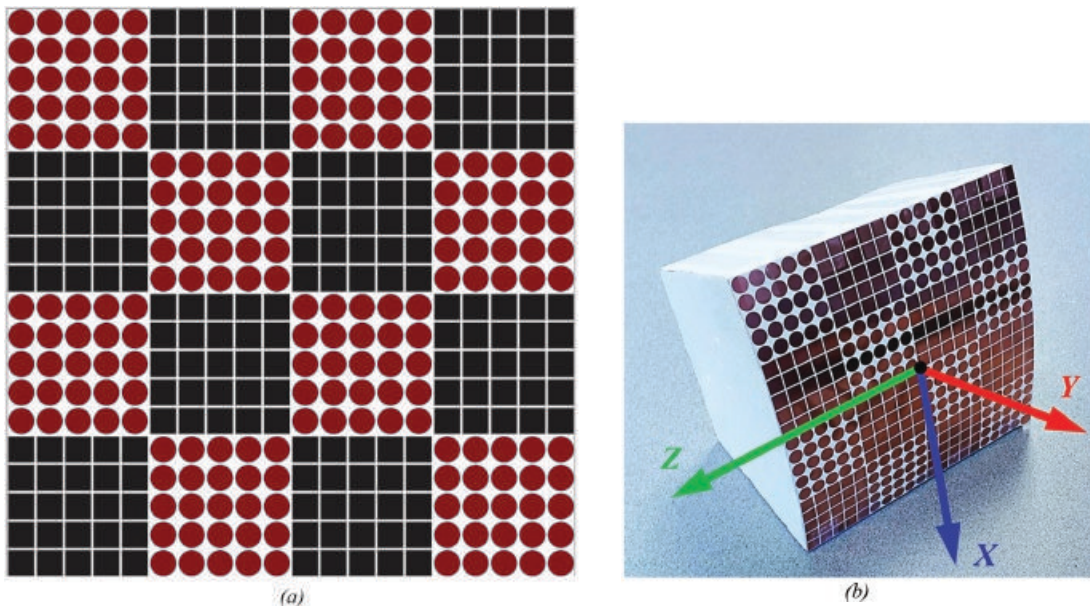


Figure 16-36 (a) Design of flat checkerboard surface. (b) Fabricated prototype of curved checkerboard architecture mounted on cylindrical structure. (Source: W. Chen, C. A. Balanis, C. R. Birtcher, and A. Y. Modi, “Cylindrically curved checkerboard surfaces for radar cross-section reduction,” *IEEE Antennas Wireless Propagat. Lett.*, vol. 17, no. 2, pp. 343–346, Feb. 2018. Reproduced with permission from IEEE.)

TABLE 16-1 Summary of RCS reduction bandwidth and separation angle between the center two lobes

Radius of curvature		10λ	5λ
HH polarization	5-dB RCS reduction BW	22.2%	25%
	Separation angle	40°	52°
VV polarization	5-dB RCS reduction BW	23.6%	22.2%
	Separation angle	40°	52°

(Source: W. Chen, C. A. Balanis, C. R. Birtcher, and A. Y. Modi, “Cylindrically curved checkerboard surfaces for radar cross-section reduction,” *IEEE Antennas Wireless Propagat. Lett.*, vol. 17, no. 2, pp. 343–346, Feb. 2018.)

C. Corner Reflectors Since corners are building blocks of complex targets, their scattering behavior was investigated [68, 74]. Further, it is extremely important to reduce the RCS of corner reflectors since they possess large RCS [68]. While RCS reduction methods, like checkerboard metasurfaces, can successfully reduce the RCS over a broad frequency range when applied to planar or convex-curved cylindrical surfaces, they fail to reduce the RCS of complex targets like corner reflectors. This is demonstrated for the 90° corner reflector of Figure 16-37a with simulated results illustrated in Figure 16-38a using the shooting and bouncing rays (SBR) method [26]. For complex structures like corner reflectors, the incident fields undergo multiple bounces [68], and these methods require additional treatment to address unique challenges presented by multiple bounces [68]. It is demonstrated that these additional challenges introduced by the multiple-bounce mechanism can successfully be addressed using orthogonality principles and array theory, as shown in Figure 16-37b [68] and illustrated in Figure 16-38b using SBR-based simulated RCS data.

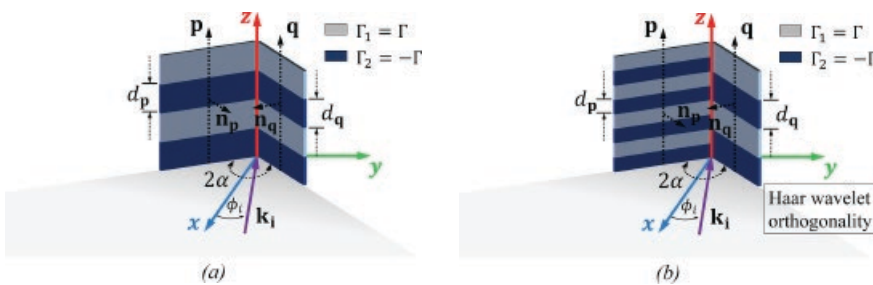


Figure 16-37 (a) Implementation of the one-dimensional checkerboard arrangement with the metasurfaces oriented perpendicularly to the plane of incidence (Configuration-2). (b) Checkerboard covered dihedral corner reflector with one-dimensional Haar wavelet orthogonality, in addition to the metasurfaces oriented perpendicularly to the plane of incidence (Configuration-4). (Source: A. Y. Modi, M. A. Alyahya, C. A. Balanis, and C. R. Birtcher, “Metasurface-based method for broadband RCS reduction of dihedral corner reflectors with multiple bounces,” *IEEE Trans. Antennas Propagat.*, vol. 68, no. 3, pp. 1436–1447, Mar. 2020. Reproduced with permission from IEEE.)

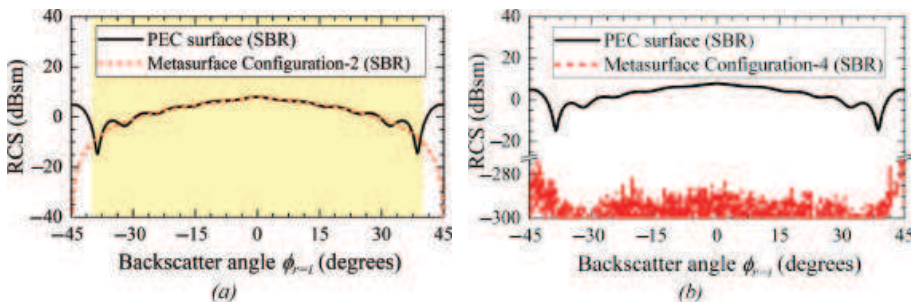


Figure 16-38 Comparison between the SBR-predicted RCSs of the target covered with PEC (solid line), and with the synthesized metasurfaces (dashed line). (a) Configuration-2. (b) Configuration-4. (Source: A. Y. Modi, M. A. Alyahya, C. A. Balanis, and C. R. Birtcher, “Metasurface-based method for broadband RCS reduction of dihedral corner reflectors with multiple bounces,” *IEEE Trans. Antennas Propagat.*, vol. 68, no. 3, pp. 1436–1447, Mar. 2020. Reproduced with permission from IEEE.)

16.9 ANTENNA FUNDAMENTAL PARAMETERS AND FIGURES-OF-MERIT

Antenna performance is described in terms of fundamental parameters and figures-of-merit. These are many and are described in detail in Chapter 2 of [9]. Four of these fundamental parameters (figures-of-merit), which will be used in this section to judge the performance of metasurface-based radiating elements, are: *radiation resistance*, *maximum directivity*, *maximum gain*, and *maximum realized gain*. To illustrate some of them, we display in Figure 16-39 an equivalent circuit of a radiating element (antenna) connected as a load to a generator through a transmission line. The antenna (load) is represented by impedance Z_A

$$Z_A = (R_r + R_L) + jX_A \quad (16-26)$$

where R_r = radiation resistance through which the antenna radiates
 R_L = antenna loss resistance, representing the losses of the antenna
 X_A = antenna reactance

Once the far-zone electric field components of a radiating element (antenna) are determined, which in the spherical coordinate system are represented by E_θ and E_ϕ , we can compute the normalized radiation intensity $U_n(\theta, \phi)$ using [9]

$$U_n(\theta, \phi) = [E_\theta(\theta, \phi)]^2 + [E_\phi(\theta, \phi)]^2 \quad \text{watts/steradian (unit solid angle)} \quad (16-27)$$

In turn, (16-27) can be used to compute the power radiated by the radiating element (antenna) using [9]

$$P_{\text{rad}} = \int_0^{2\pi} \int_0^\pi U_n(\theta, \phi) \sin \theta d\theta d\phi \quad \text{watts} \quad (16-28)$$

where $(0 \leq \theta \leq \pi)$ and $(0 \leq \phi \leq 2\pi)$ are, respectively, the elevation and azimuthal angles of a spherical coordinate system.

With $U_n(\theta, \phi)$ and P_{rad} defined by (16-27) and (16-28), respectively, we can represent the radiation resistance R_r and the maximum directivity $D_0 = D_{\text{max}}$ by [9]

$$R_r = \frac{2P_{\text{rad}}}{|I_o|^2} \quad (16-29)$$

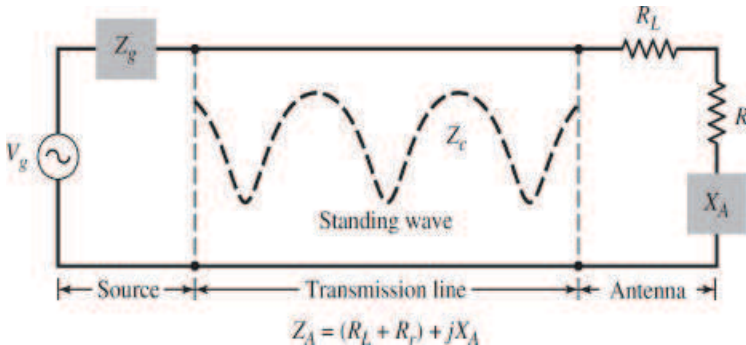


Figure 16-39 Equivalent circuit of an antenna connected to a generator. (Source: C. A. Balanis, *Antenna Theory: Analysis and Design*, Fourth Edition, John Wiley & Sons, New York, 2016. Reproduced with permission from John Wiley & Sons.)

$$D_{\max} = D_0 = \frac{U_{\max}(\theta, \phi)}{U_0} = \frac{4\pi U_{\max}(\theta, \phi)}{P_{\text{rad}}} \quad \text{dimensionless} \quad (16-30a)$$

$$U_0 = \frac{P_{\text{rad}}}{4\pi} \quad \text{watts/steradian} \quad (16-30b)$$

where I_0 is the maximum amplitude of the current, and U_0 is the radiation intensity of an isotropic source which radiates equally well in all directions. The maximum directivity D_0 is the parameter that describes the *directional* characteristics (collimating capabilities) of the radiating element (antenna).

Once the directivity has been established, another important parameter is the maximum gain G_0 , which is related to the maximum directivity D_0 by [9]

$$G_0 = e_{cd} D_0 = \frac{R_r}{R_r + R_L} D_0 \quad \text{dimensionless} \quad (16-31)$$

$$e_{cd} = \frac{R_r}{R_r + R_L} \quad (16-32)$$

where e_{cd} = *radiation efficiency* of the radiating element (antenna). For a lossless (no losses) radiating element, $R_L = 0$, the maximum gain G_0 is identical to the maximum directivity D_0 ($G_0 = D_0$).

The maximum directivity D_0 is referenced at the output terminals of the antenna, while the maximum gain G_0 is referenced at the input terminals of the antenna (power that has crossed over the intersection from the transmission line to the input side of the antenna), as illustrated in Figure 16-40a. The losses of the antenna, represented in Figure 16-39 by R_L , are highlighted in Figure 16-40b and can be either conducting losses (represented by i_c) or dielectric losses (represented by i_d).

The maximum gain G_0 , as defined by (16-31), does not account for the matching losses at the intersection between the transmission line and the radiating element (antenna). To account for the mismatch losses between the transmission line (represented by the characteristic impedance Z_c)

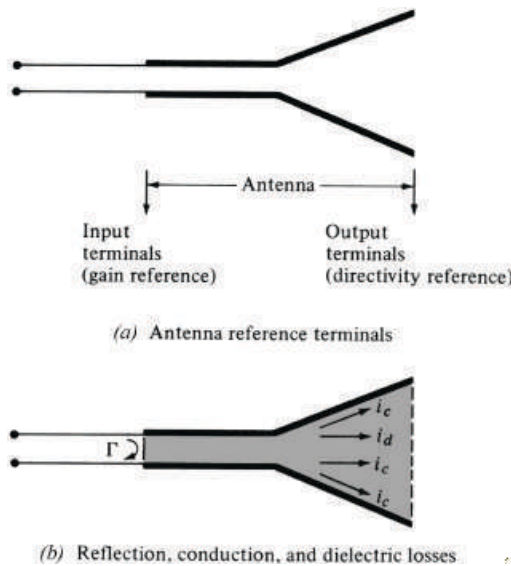


Figure 16-40 Reference terminals and losses of an antenna. (Source: C. A. Balanis, *Antenna Theory: Analysis and Design*, Fourth Edition, John Wiley & Sons, New York, 2016. Reproduced with permission from John Wiley & Sons.)

and the input impedance of the antenna represented by $Z_{in} = (R_r + R_L) + jX_A$, as illustrated in Figure 16-39, we introduce the reflection (mismatch) efficiency e_r defined as [9]

$$e_r = 1 - |\Gamma|^2 = \text{reflection efficiency (dimensionless)} \quad (16-33a)$$

$$\Gamma = \frac{Z_{in} - Z_c}{Z_{in} + Z_c} = \text{reflection coefficient (dimensionless)} \quad (16-33b)$$

$$Z_{in} = (R_r + R_L) + jX_A = \text{antenna input impedance} \quad (16-33c)$$

Once the reflection (mismatch) efficiency is introduced by (16-33a), the maximum realized gain G_{re0} is defined as [9]

$$G_{re0} = [1 - |\Gamma|^2] G_0 = [1 - |\Gamma|^2] e_{cd} D_0 = e_r e_{cd} D_0 \quad (16-34)$$

The reflection (mismatch) efficiency e_r between the antenna and the transmission line is represented in Figure 16-40b by the reflection coefficient Γ , defined by (16-33b).

It is apparent from (16-34) that for a lossless ($e_{cd} = 0$) and a matched to the transmission line ($e_r = 1$, $\Gamma = 0$) radiating element (antenna), the maximum directivity D_0 , maximum gain G_0 , and maximum realized gain G_{re0} are all equal ($D_0 = G_0 = G_{re0}$).

16.10 ANTENNA APPLICATIONS

As indicated previously, artificially fabricated surfaces have many applications, especially related to suppression or enhancement of surface waves and/or controlling the phase characteristics of the reflection coefficient. In this section, we will consider some basic radiating elements and their integration and interaction with metasurfaces.

16.10.1 Monopole

A basic illustrative example is to examine the radiation characteristics of a monopole mounted on a finite-size ground plane, PEC, and high-impedance surface (HIS). The monopole and its measured radiation patterns are shown in Figures 16-41a–16-41d. The basic geometry, shown in Figure 16-41a, consists of a 3-mm monopole mounted on a 5-cm ground plane; its amplitude patterns were measured at 35 GHz. When mounted on a PEC ground plane, the amplitude pattern displayed in Figure 16-41b is basically that which is expected and shown in Figure 13-33. It exhibits radiation not only in the upper hemisphere but also in the lower, due to diffractions from the edges of the finite ground plane. However, when the monopole is mounted on a high-impedance surface designed at the bandgap frequency of 35 GHz, the radiation in the lower hemisphere, as displayed in Figure 16-41c, is diminished as the diffractions from the edges of the ground plane have been reduced because the high-impedance surface suppresses the surface waves as they travel from the center of the ground plane toward its edges. Even in the upper hemisphere, the pattern is very smooth compared with that of Figure 16-41b, because the diffractions from the edges of the ground plane are basically insignificant compared to the direct radiation (geometrical optics) from the radiating element and its image. However, when the monopole is operated below the bandgap ($f = 26$ GHz), as shown in Figure 16-41d, the pattern is scalloped both in the upper and lower hemispheres, with considerable radiation in the lower hemisphere. The measured patterns in Figure 16-41 exhibit slight asymmetries, probably due to system errors introduced by cables and mounting structures.

Based on orientation and placement on ground planes, vertical monopole antennas of Figure 16-41a are not suitable for low-profile applications; however, horizontally oriented dipoles, or other radiators, may provide an alternative. To assess alternate geometries, let us first examine the

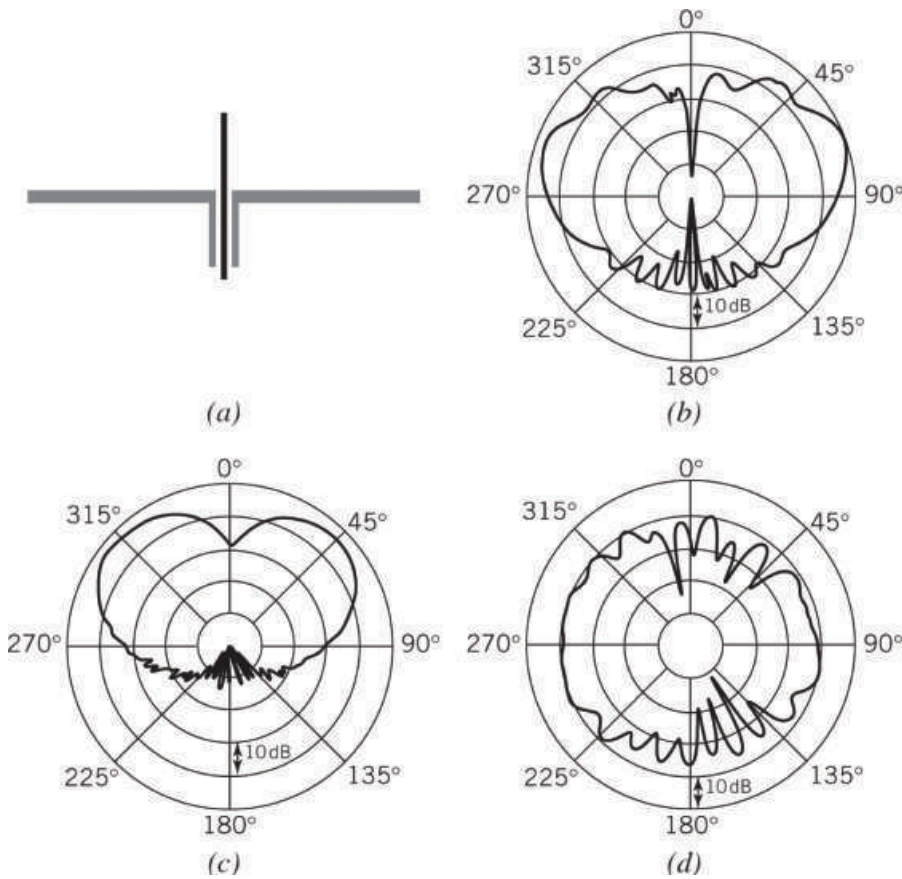


Figure 16-41 Monopole and its patterns when mounted on different ground planes; PEC and PMC. (a) Monopole geometry (3-mm monopole, 5-cm ground plane). (b) PEC ground plane ($f = 35$ GHz). (c) EBG ground plane ($f = 35$ GHz). (d) EBG ground plane ($f = 26$ GHz). (Source: D. Sievenpiper, “Artificial impedance surfaces,” Chapter 15 in *Modern Antenna Handbook*, C. A. Balanis (Ed.), John Wiley & Sons, pp. 737–777, 2008. Reproduced with permission from John Wiley & Sons.)

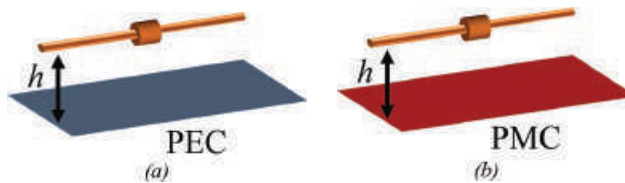


Figure 16-42 Horizontal electric dipole above PEC and PMC ground planes.

performance of a horizontal electric dipole placed above PEC and PMC ground planes, as shown in Figure 16-42; their performance will be examined based on the radiation resistance R_r and maximum directivity D_0 .

16.10.2 Horizontal Dipole

A dipole antenna is a classical radiating element, and it has been thoroughly investigated when it is isolated in free space [9]. The radiation performance and the 50- Ω matching are positively/negatively disturbed when another object/surface is placed in proximity to the radiating element. The

influence of the surface on the dipole performance depends on different parameters such as size of the surface, the height of the dipole above the surface, and the composition of the surface material.

To demonstrate how the spacing and the ground plane influence the dipole performance, based on radiation resistance R_r and maximum directivity D_0 , a horizontal electric dipole in the vicinity of infinite PEC and PMC is depicted in Figure 16-42.

For the PEC ground plane, the radiation resistance of a horizontal dipole is represented by [9]

$$R_r = \eta\pi \left(\frac{l}{\lambda} \right)^2 \left[\frac{2}{3} - \frac{\cos(2\beta h)}{(2\beta h)^2} - \frac{\sin(2\beta h)}{2\beta h} + \frac{\sin(2\beta h)}{(2\beta h)^3} \right] \quad (16-35)$$

and the maximum directivity, for $h \leq \lambda/4$, by [9]

$$D_0 = \frac{4 \sin^2(\beta h)}{\left[\frac{2}{3} - \frac{\cos(2\beta h)}{(2\beta h)^2} - \frac{\sin(2\beta h)}{2\beta h} + \frac{\sin(2\beta h)}{(2\beta h)^3} \right]}, \quad h \leq \frac{\lambda}{4} \quad (16-36)$$

However, when the horizontal dipole is positioned in close proximity to an infinite PMC ground, the radiation resistance and the maximum directivity, for $h \leq \lambda/4$, are [9]

$$R_r = \eta\pi \left(\frac{l}{\lambda} \right)^2 \left[\frac{2}{3} + \frac{\cos(2\beta h)}{(2\beta h)^2} + \frac{\sin(2\beta h)}{2\beta h} - \frac{\sin(2\beta h)}{(2\beta h)^3} \right] \quad (16-37)$$

$$D_0 = \frac{4}{\left[\frac{2}{3} + \frac{\cos(2\beta h)}{(2\beta h)^2} + \frac{\sin(2\beta h)}{2\beta h} - \frac{\sin(2\beta h)}{(2\beta h)^3} \right]}, \quad h \leq \frac{\lambda}{4} \quad (16-38)$$

The radiation resistance and maximum directivity [see (16-35)–(16-38)] for a $\lambda/10$ horizontal dipole above PEC and PMC ground planes are plotted in Figure 16-43. As illustrated, the PEC ground plane severely impacts the radiation resistance when the dipole is positioned very close to it ($h < 0.1\lambda$); however, the peak directivity is at maximum for smaller spacings. Unlike the PEC surface, the dipole in the presence of a PMC ground plane possesses a greater radiation resistance with small maximum directivity when h is less than 0.1λ . For low-profile applications, a preferable ground plane is one that leads to a radiation resistance comparable to the PMC surface, and at the

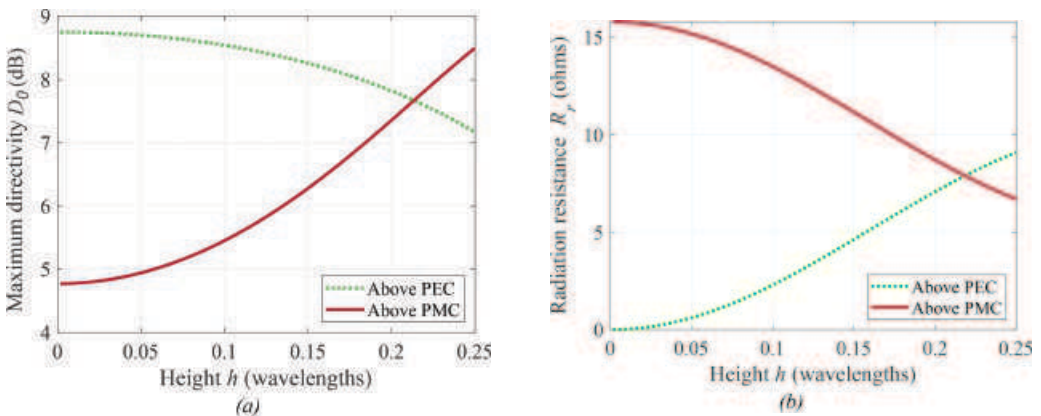


Figure 16-43 Maximum directivity and radiation resistance of horizontal electric dipole above PEC and PMC ground planes.

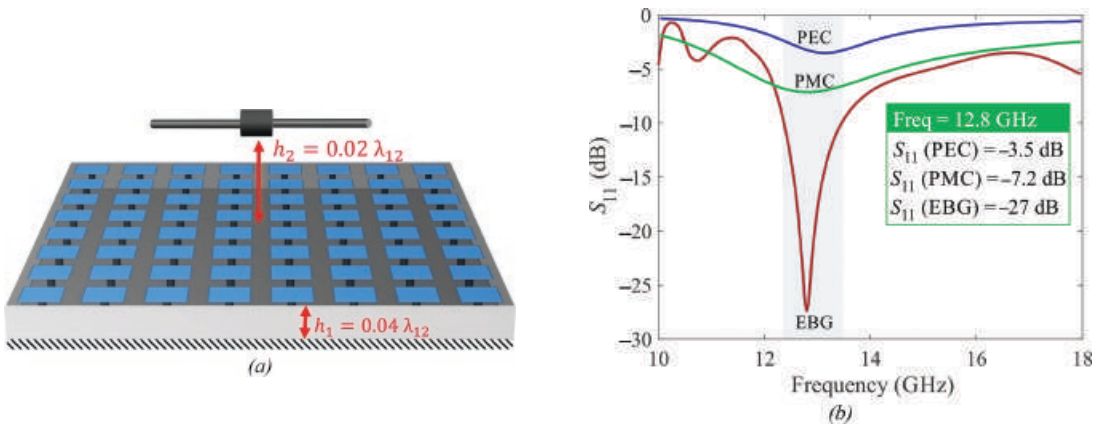


Figure 16-44 Geometry and S_{11} of horizontal electric dipole above PEC, PMC, and EBG surfaces. (a) Geometry. (b) S_{11} . (Source: Y. Rahmat-Samii and F. Yang, “Development of complex artificial ground planes in antenna engineering,” Chapter 12 in *Metamaterials: Physics and Engineering Explorations*, N. Engheta and R. W. Ziolkowski (Eds.), IEEE Press, Wiley Inter-Science, New York, 2006. Reproduced with permission from John Wiley & Sons, Inc.)

same time, it results in high directivity, as the one attained with a PEC surface. This can be realized by the introduction of artificial magnetic conductors (AMCs) as ground planes.

Whether a PEC, PMC, or EBG surface outperforms the others as a ground plane depends upon the application. This is best illustrated by a basic example; a $l = 0.4\lambda_{12}$ dipole (λ_{12} is the free-space wavelength at $f = 12$ GHz) placed horizontally above PEC, PMC, and EBG surfaces, as shown in Figure 16-44; the EBG surface has a height of $0.04\lambda_{12}$. The dipole is placed at a height h of $0.06\lambda_{12}$ ($h_2 = 0.06\lambda_{12}$) above $\lambda_{12} \times \lambda_{12}$ PEC and PMC square surfaces.

The S_{11} of this system was simulated, using the FDTD method, over a frequency range of 10–18 GHz [5, 11]. Based on a 50- Ω line impedance, the results are shown in Figure 16-44b, where it is clear that the EBG surface (which has a reflection coefficient phase variation from $+180^\circ$ to -180°) exhibits a best return loss of -27 dB, while the PMC (which has a reflection phase of 0°) has a best return loss of -7.2 dB and the PEC (which has a reflection phase of 180°) has a best return loss of only -3.5 dB. For the PMC surface, the return loss is influenced by the mutual coupling, due to the close proximity between the main element and its in-phase image, whereas for the PEC the return loss is influenced by the 180° phase reversal, which severely impacts the radiation efficiency. In this example, the EBG surface, because of its $+180^\circ$ to -180° phase variation over the frequency bandgap of the EBG design, outperforms the PEC and PMC and serves as a good ground plane. The other two, the PEC and PMC, possess constant out-of-phase and in-phase phase characteristics, respectively, over the entire frequency range.

Now that we have examined a horizontal oriented infinitesimal electric dipole above PEC and PMC ground planes based on their radiation resistance R_r and maximum directivity D_o , as well as an $l = 0.4\lambda$ dipole above PEC, PMC, and EBG surfaces, the results are summarized on a graphical illustration in Figure 16-45 based on radiation efficiency, maximum directivity, overall height, and low profile. It is clear that a horizontal electric radiating element above a PMC/EBG surface is the most attractive low-profile configuration. This configuration will now be pursued in detail with other radiating elements, such as an electric circular loop, placed above square and circular AMC surfaces.

16.10.3 Circular Loop

Another popular and classical radiating element is the circular loop. Depending on the electrical size of their circumference, loops can be classified as electrically small or large. Electrically small loops ($C < \lambda/10$) suffer from low radiation resistance; thus, they are classified as poor radiators [9].

Structure	Source and image current	For low-profile configuration ($h \leq 0.1 \lambda$)			
		Efficiency	Directivity	Overall height	Low-profile
Vertical dipole above infinite PEC		✓	Moderate	Large	✗
Horizontal dipole above infinite PEC		✗	High	Low	✓
Horizontal dipole above infinite PMC		✓	Moderate	Low	✓

Figure 16-45 Performance summary of vertical and horizontal electric dipoles above PEC and PMC ground plane.

As the circumference approaches λ , the radiation resistance increases, and the loop becomes a better match to a 50-ohm feed line.

An alternate approach to alter the radiation resistance, besides increasing the circumference, is to introduce an AMC metasurface ground plane [54, 75]. The metasurface ground plane can favorably be utilized to enhance the radiating element performance over a certain operational bandwidth. One way to examine the aforementioned bandwidth is to place the loop near the metasurface and observe the input reflection coefficient, as the circumference of the loop is varied. Following this approach, the loop reflection coefficient has been examined when it is positioned at 1 mm above two different metasurface ground planes: circular and square [54], as depicted in Figures 16-46a and 16-47a.

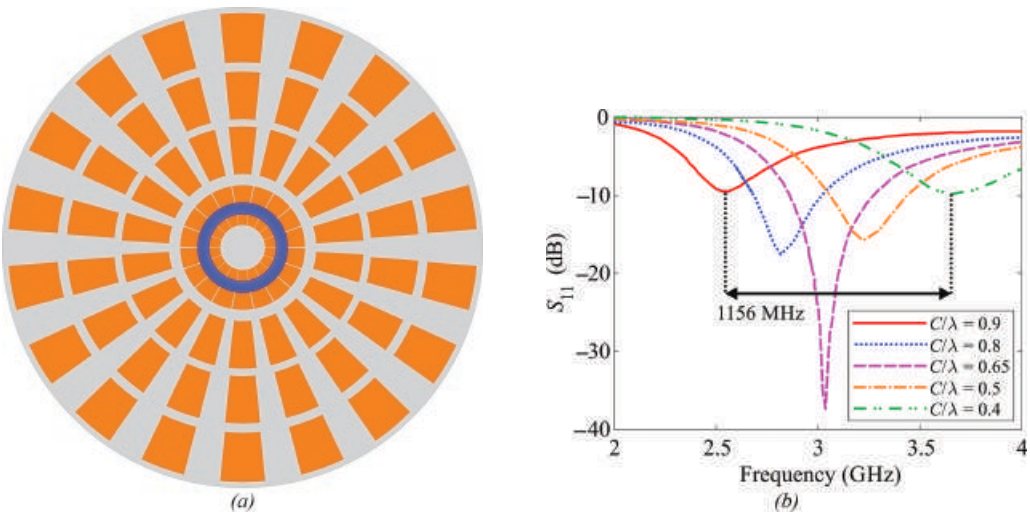


Figure 16-46 (a) Wire circular loop above a circular HIS. (b) Associated S_{11} . (Source: Adapted from M. A. Amiri, C. A. Balanis, and C. R. Birtcher, “Analysis, design and measurements of circularly symmetric high impedance surfaces for loop antenna applications,” *IEEE Trans. Antennas Propagat.*, vol. 64, pp. 618–629, 2016.)

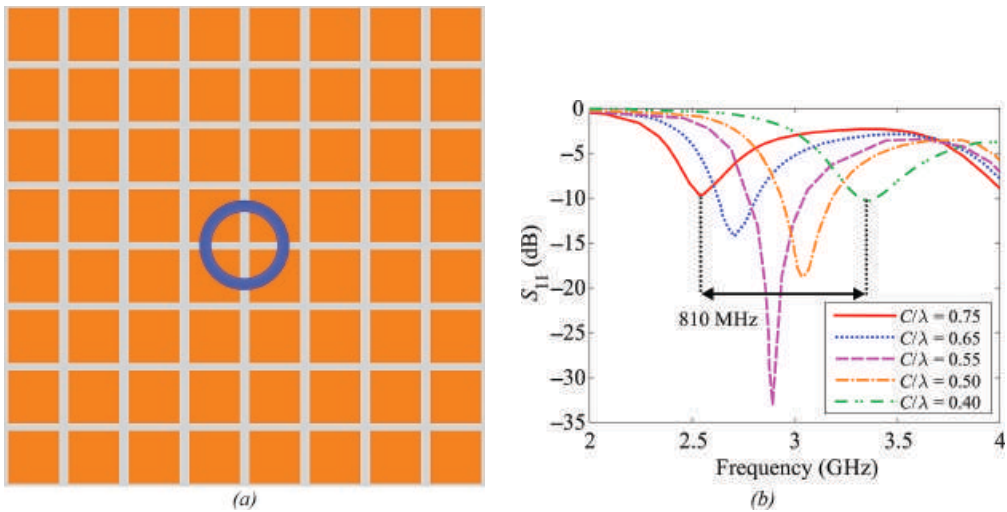


Figure 16-47 (a) Wire circular loop above a square HIS with square patches. (b) Associated S_{11} . (Source: M. A. Amiri, C. A. Balanis, and C. R. Birtcher, “Analysis, design and measurements of circularly symmetric high impedance surfaces for loop antenna applications,” *IEEE Trans. Antennas Propagat.*, vol. 64, pp. 618–629, 2016. Reproduced with permission from IEEE.)

As illustrated in Figures 16-46b and 16-47b, the operational bandwidth of the circular loop above circular high-impedance surfaces (CHIS) is greater than their square counterparts. For further comparison, the realized gain pattern at 3 GHz for a circular loop with a radius of 1 cm, placed above circular and square HISs, was simulated, and it is shown in Figure 16-48. As observed from Figure 16-48, the gain patterns of the circular loop are more symmetric on different elevation planes when the loop is placed above the circular HIS. This is expected as the fields generated by the circular loop interact more efficiently with the circular HIS [54] and result in more symmetric patterns.

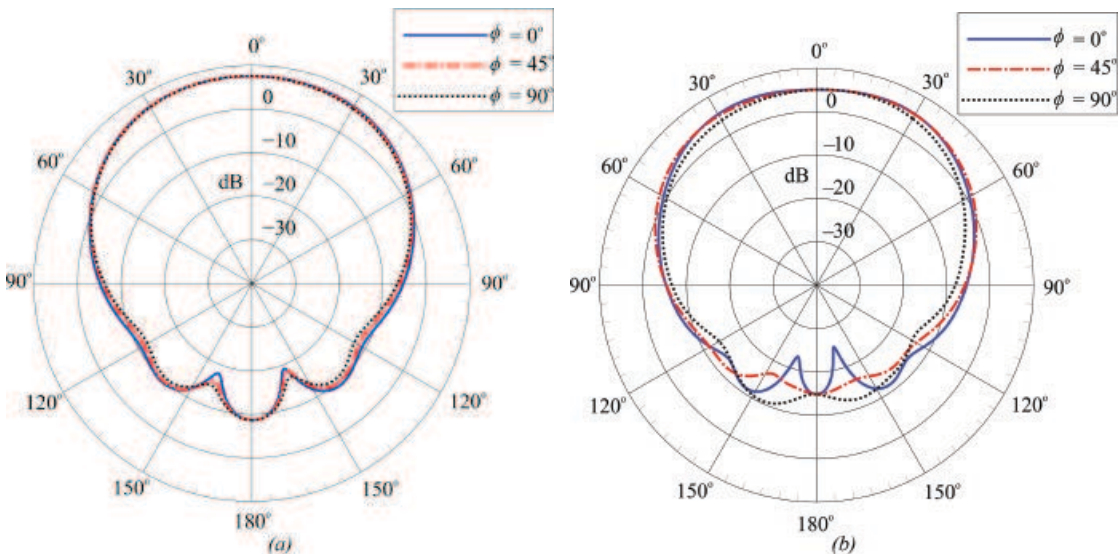


Figure 16-48 Realized gain of a circular loop along different ϕ planes, when it is located above circular and square HISs. (Source: M. A. Amiri, C. A. Balanis, and C. R. Birtcher, “Analysis, design and measurements of circularly symmetric high impedance surfaces for loop antenna applications,” *IEEE Trans. Antennas Propagat.*, vol. 64, pp. 618–629, 2016. Reproduced with permission from IEEE.)

The broadside realized gain for the loop element is investigated for different configurations: in free space, and above PMC, *square* high impedance (SHIS), *circular* high-impedance (CHIS) ground planes; the simulated data is plotted in Figure 16-49. As expected, at 3 GHz, the realized gain is around 2 dB when the loop is isolated in free space [9]. However, as the different ground planes are introduced adjacent to the loop, the magnitude and the phase of the reflected fields change accordingly and lead to different realized gains. When the PMC ground plane is placed below the loop, the broadside realized gain increases by 1.3 dB. However, when the square HIS and circular HIS are utilized, the gain increases by 3 and 5.5 dB, respectively. From these results, it is clear that the circular HIS is a better ground plane for circular loops.

To validate the simulated results, a prototype of the circular loop above a circular metasurface was fabricated and measured, as depicted in Figure 16-50. The S_{11} of two different loops, with radii of 3 and 2.7 cm, was simulated and measured (see Figure 16-51a). Furthermore, the input impedance and the angular gain pattern for the 2.7-cm loop radius were also plotted in Figures 16-51b and 16-52, respectively. As observed, an excellent agreement is indicated between simulations and measurements.

To further enhance the radiation performance, a hybrid circular ground plane was proposed as an alternate ground plane for loop radiators [75]. The loop element was positioned on the top

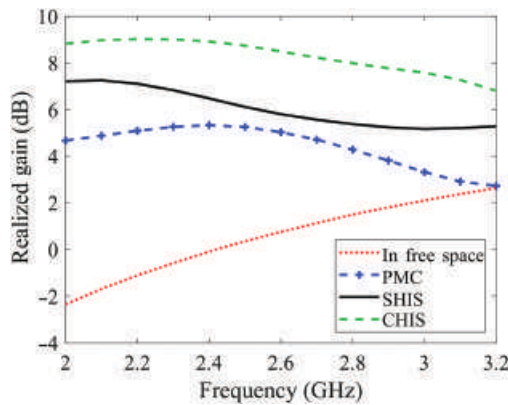


Figure 16-49 Broadside realized gain of a circular loop in free space and above different ground planes.



Figure 16-50 Fabricated circular HIS with the loop antenna at a height of 0.01λ . (Source: M. A. Amiri, C. A. Balanis, and C. R. Birtcher, "Analysis, design and measurements of circularly symmetric high impedance surfaces for loop antenna applications," *IEEE Trans. Antennas Propagat.*, vol. 64, pp. 618–629, 2016. Reproduced with permission from IEEE.)

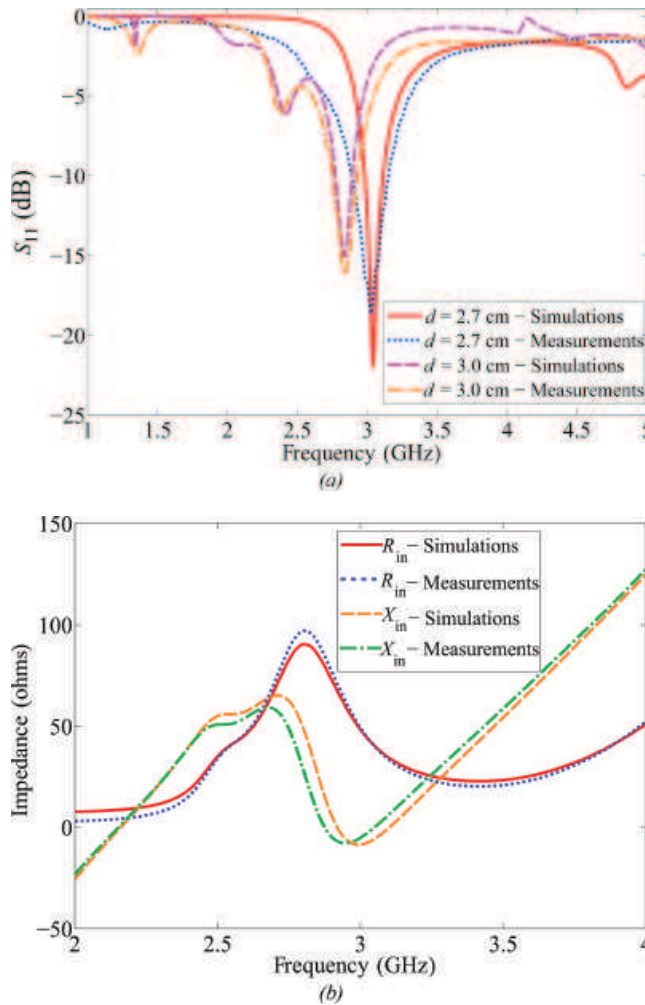


Figure 16-51 Simulations and measurements of the loop element above a CHIS. (Source: M. A. Amiri, C. A. Balanis, and C. R. Birtcher, “Analysis, design and measurements of circularly symmetric high impedance surfaces for loop antenna applications,” *IEEE Trans. Antennas Propagat.*, vol. 64, pp. 618–629, 2016. Reproduced with permission from IEEE.)

surface of a hybrid ground with no spacing between them, as demonstrated in Figure 16-53. The loop was designed with a circumference of λ_0 , since broadside radiation is desired at 2.9 GHz. At one-wavelength circumference, the loop exhibits capacitive reactance [9]; thus, an inductive load is required to resonate the loop. The hybrid structure incorporates two different metallic rings, as shown in Figure 16-53, printed on a grounded dielectric: circular HIS and annular circular ring. The radial length of the HIS ring (r_{HIS}) is 22 mm, while it is 24 mm for the annular ring (r_{annular}). The circular HIS ring was designed to provide an inductive impedance at 2.9 GHz; therefore, by placing the HIS ring close to the loop radiating element, resonance can be achieved. Once the mismatch losses are minimized, the next step was to enhance the broadside directivity. This can be realized by increasing the effective area of the structure. Therefore, the gaps were omitted in the annular ring, and it was designed with a slightly larger radial length. The combination of the two rings surrounding the loop element enhances both the matching and the broadside realized gain.

To further demonstrate the contribution of each ring of the hybrid ground plane, the antenna performance was investigated for the three designs shown in Figure 16-54. The S_{11} and broadside directivity for the three designs are presented in Figure 16-55. As observed, the presence of the HIS

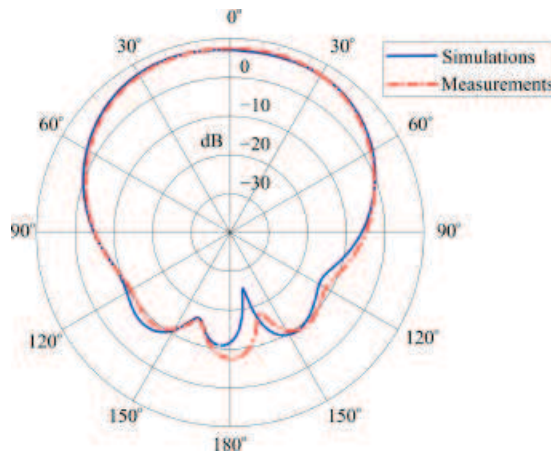


Figure 16-52 Simulations and measurements of the loop element above CHIS: Gain pattern at 3 GHz. (Source: M. A. Amiri, C. A. Balanis, and C. R. Birtcher, “Analysis, design and measurements of circularly symmetric high impedance surfaces for loop antenna applications,” *IEEE Trans. Antennas Propagat.*, vol. 64, pp. 618–629, 2016. Reproduced with permission from IEEE.)

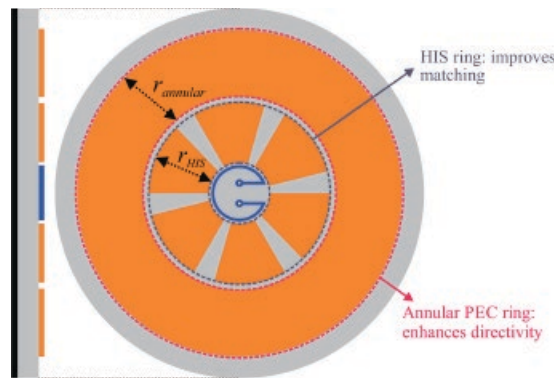


Figure 16-53 Geometry of a hybrid ground plane with printed loop element.

ring alleviates the mismatch losses and improves the reflection coefficient (S_{11}). However, at resonance, it only contributes to slight improvement in the broadside directivity. When the two rings are utilized to form the hybrid ground plane, in addition to possess better matching bandwidth, a 3-dB directivity increase is observed at resonance, primarily due to the presence of the outside PEC ring.

The hybrid ground plane was fabricated along with a printed loop element, as illustrated in Figure 16-56. The comparison between simulated and measured results for the reflection coefficient and broadside realized gain are plotted in Figure 16-57; an excellent agreement is indicated.

The superior performance of the circular metasurface ground plane inspired interest to investigate circular HISs when utilized with different curvilinear elements, such as a spiral and circular patch [55, 76].

16.10.4 Aperture Antenna

Another example of diffraction control that can be provided by a textured high-impedance surface is to examine the radiation of an aperture mounted on a high-impedance surface ground

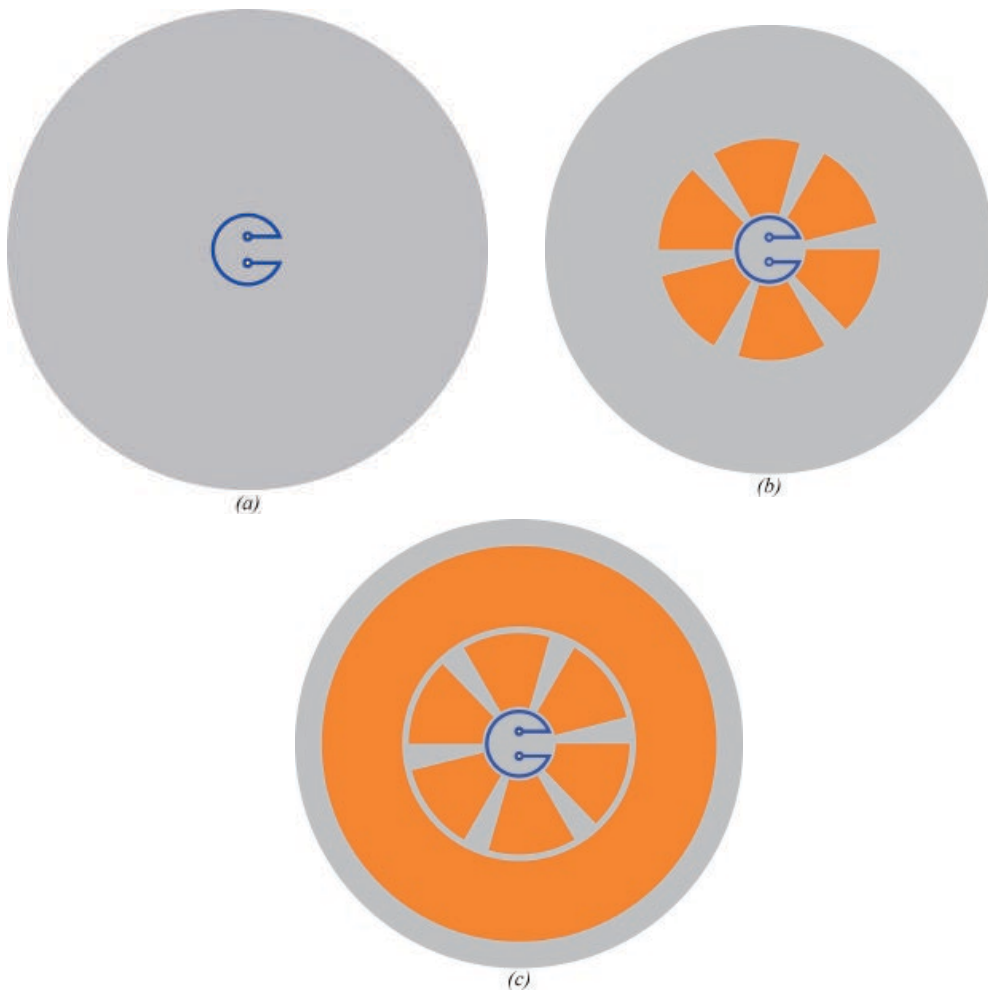


Figure 16-54 Printed loop antenna: (a) above grounded dielectric, (b) surrounded by a HIS section on grounded dielectric. (Source: M. Alharbi, C. A. Balanis, C. R. Birtcher, and H. N. Shaman, “Hybrid circular ground planes for high-realized-gain low-profile loop antennas,” *IEEE Antennas Wireless Propagat. Lett.*, vol. 17, no. 8, pp. 1426–1429, Aug. 2018. Reproduced with permission from IEEE.)

plane. The geometry of the radiator is shown in Figure 16-58, and it consists of a rectangular aperture mounted on a 12.7-cm ground plane, both PEC and high impedance [2]. Each unit cell measures 3.7 mm and the textured high-impedance surface has been designed for a bandgap in the range of 12–18 GHz. The measured patterns, both for the PEC and high-impedance surface, are shown in Figure 16-59, and they were measured at 13 GHz, which is within the designed bandgap. In general, the shape of the radiation pattern is influenced not only by the shape and size of the aperture but also by the ground plane and its texture.

The patterns in Figure 16-59a, E- and H-planes, are those of the aperture on a PEC ground plane, and they are representative, as shown in Figures 16-59a and 16-59b [9]. The E-plane is usually broader than the H-plane, because for the E-plane its vertical polarized fields are not shorted out by the PEC ground plane, while for the H-plane its horizontally polarized fields are, ideally, nulled. When the aperture is mounted on a textured high-impedance surface and operated within its bandgap (specifically 13 GHz), its patterns in both the E- and H-planes, shown in Figure 16-16b,

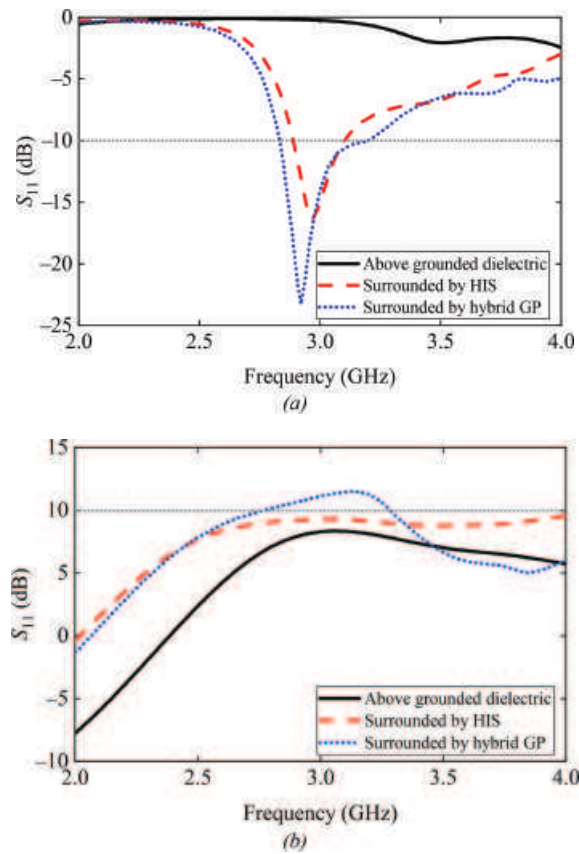


Figure 16-55 Simulated S_{11} and broadside directivity of loop element with three ground plane configurations. (Source: M. Alharbi, C. A. Balanis, C. R. Birtcher, and H. N. Shaman, “Hybrid circular ground planes for high-realized-gain low-profile loop antennas,” *IEEE Antennas Wireless Propagat. Lett.*, vol. 17, no. 8, pp. 1426–1429, Aug. 2018. Reproduced with permission from IEEE.)

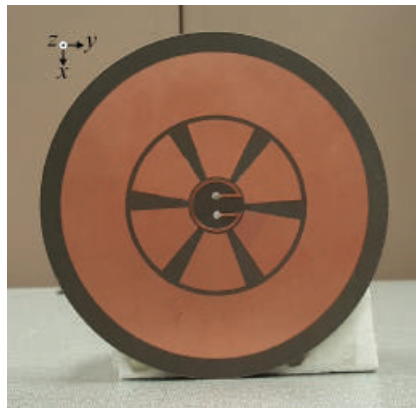


Figure 16-56 A photograph of the fabricated antenna. (Source: M. Alharbi, C. A. Balanis, C. R. Birtcher, and H. N. Shaman, “Hybrid circular ground planes for high-realized-gain low-profile loop antennas,” *IEEE Antennas Wireless Propagat. Lett.*, vol. 17, no. 8, pp. 1426–1429, Aug. 2018. Reproduced with permission from IEEE.)

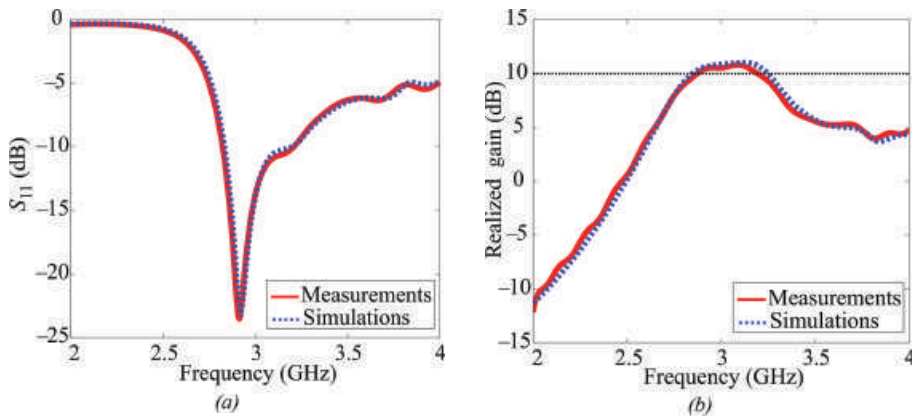


Figure 16-57 Comparisons between the simulated and measured results for the loop element above the hybrid ground plane. (Source: M. Alharbi, C. A. Balanis, C. R. Birtcher, and H. N. Shaman, “Hybrid circular ground planes for high-realized-gain low-profile loop antennas,” *IEEE Antennas Wireless Propagat. Lett.*, vol. 17, no. 8, pp. 1426–1429, Aug. 2018. Reproduced with permission from IEEE.)

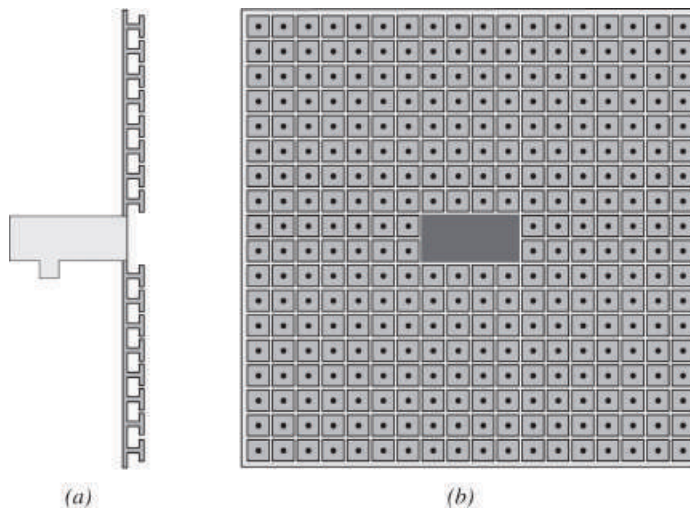


Figure 16-58 Aperture antenna in a high-impedance surface. The unit cells of the high-impedance surface measure 3.7 mm, and the size of the ground plane (not shown to scale) is 12.7 cm. The aperture is fed by a coax to Ku-band rectangular waveguide transition. (a) Side view. (b) Front view. (Source: D. Sievenpiper, “Artificial impedance surfaces,” Chapter 15 in *Modern Antenna Handbook*, C. A. Balanis (Ed.), John Wiley & Sons, pp. 737–777, 2008. Reproduced with permission from John Wiley & Sons.)

are nearly the same and symmetrical because the textured surface suppresses both the TM and TE surface waves near the resonant frequency. However, when the aperture is mounted on the same textured surface but operated at the leading edge of the TE band where TM waves are suppressed, the H-plane pattern is broader than the E-plane (see Figure 16-59c), which is the opposite of that observed for the patterns in Figure 16-59a for the PEC ground plane. Since the behavior of the E- and H-plane patterns in Figure 16-59c is opposite of that of the patterns in Figure 16-59a, the textured ground plane acts as a PMC (ideally shorts out the tangential magnetic fields), which is the opposite of that for the PEC.

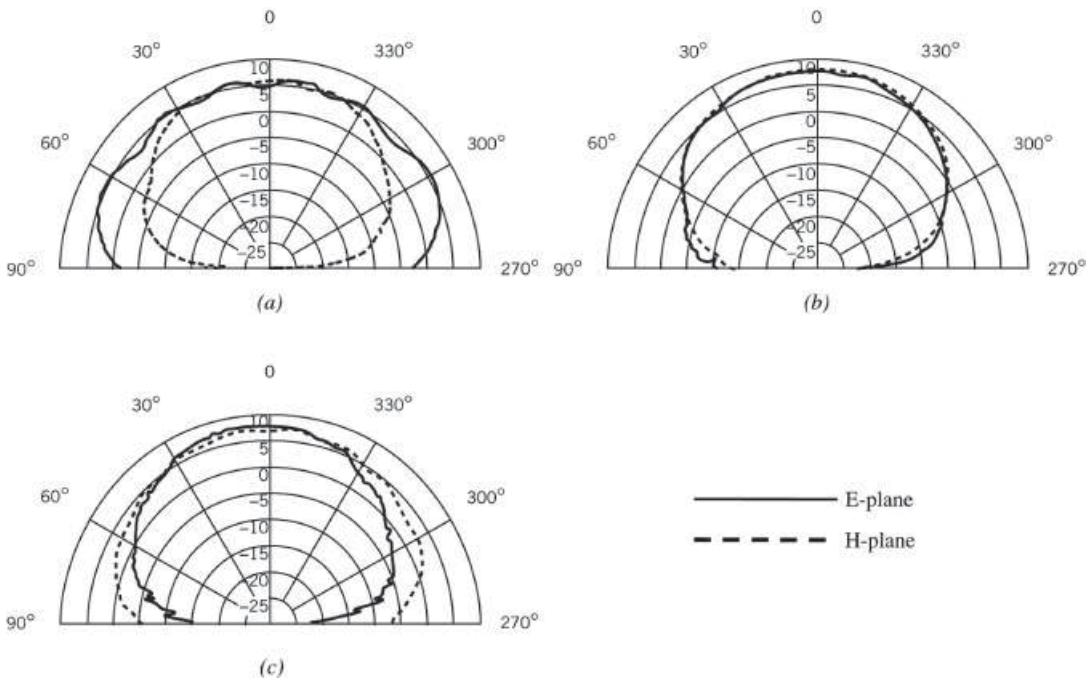


Figure 16-59 Amplitude far-field pattern of an aperture on a ground plane; PEC and AMC. (a) PEC. (b) AMC ($f = 13$ GHz) (high-impedance). (c) AMC (high-impedance near edge of TE band). (Source: D. Sievenpiper, “Artificial impedance surfaces,” Chapter 15 in *Modern Antenna Handbook*, C. A. Balanis (Ed.), John Wiley & Sons, pp. 737–777, 2008. Reproduced with permission from John Wiley & Sons.)

16.10.5 Microstrip Array

In microstrip arrays, a detrimental phenomenon that leads to an increase in the input reflection coefficient of the array, as a function of the scan angle, is the surface waves created, sustained, and traveling within the substrate. Space waves also contribute but they are not as dominant as the surface waves, which can also lead to *scan blindness* [77]. One way to eliminate the surface waves and reduce the input reflection coefficient, and even eliminate scan blindness, is to use cavities to surround each of the patches [78]. This is a rather expensive design but it works.

Another way to minimize the surface waves, without the use of cavities, is to mount the patches on EBG-textured surfaces, as shown in Figure 16-60a for a 3×3 array. A 2×2 unit cell of the EBG surface, with a dipole in its middle, is displayed in Figure 16-60b [79]. Such surfaces have the ability to control and minimize surface waves in substrates. A microstrip dipole element, placed within a textured high-impedance surface, was designed, simulated, fabricated, and measured [79]. It consists of a dipole patch of length 9.766 mm placed in the middle of 4×4 EBG unit cells; each cell had dimensions of $w = l = 1.22$ mm and a separation gap between them of $g = 1.66$ mm. The substrate had a dielectric constant of $\epsilon_r = 2.2$ and a height of $h = 4.771$ mm. The scanned magnitude of the simulated reflection coefficient at 13 GHz of such a design is shown in Figure 16-61; the bandgap frequency range was 9.7–15.1 GHz. The E-plane curves are presented by the E curves, the H-plane by the H curves, and the diagonal (45°) plane by the D curves. It is clear that using conventional substrates, the reflection coefficient varies as a function of the scan angle and, in fact, creates scan blindness around 50° . However, when the same elements were placed on a textured high-impedance surface, the reflection coefficient was reduced, especially in the E-plane, and in fact, the scan blindness was eliminated.

There are many other applications where textured high-impedance surfaces can be used to control the radiation characteristics of electromagnetic problems. Such applications include, but are

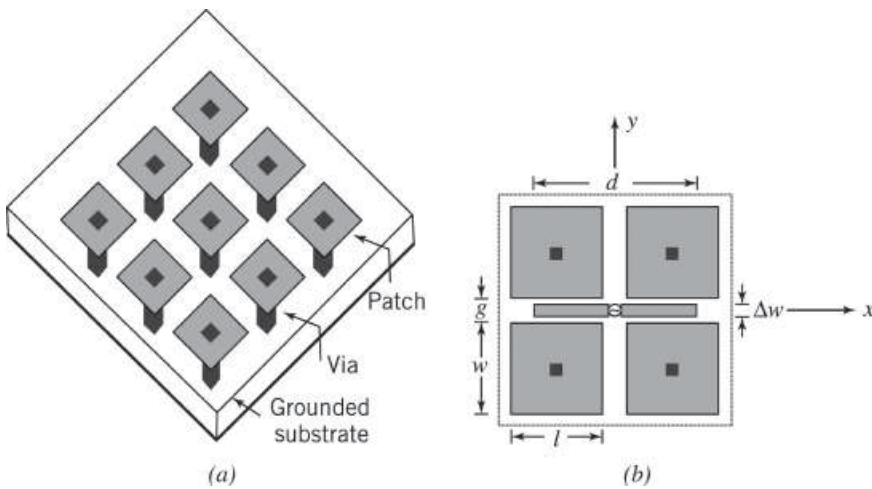


Figure 16-60 Dipole phased array on a PBG surface. (a) 3×3 array. (b) Dipole between 2×2 unit cell. (Source: L. Zhang, J. A. Castaneda, and N. G. Alexopoulos, "Scan blindness free phased array design using PBG materials," *IEEE Trans. Antennas Propagat.*, vol. 52, no. 8, pp. 2000–2007, Aug. 2004. Reproduced with permission from IEEE.)

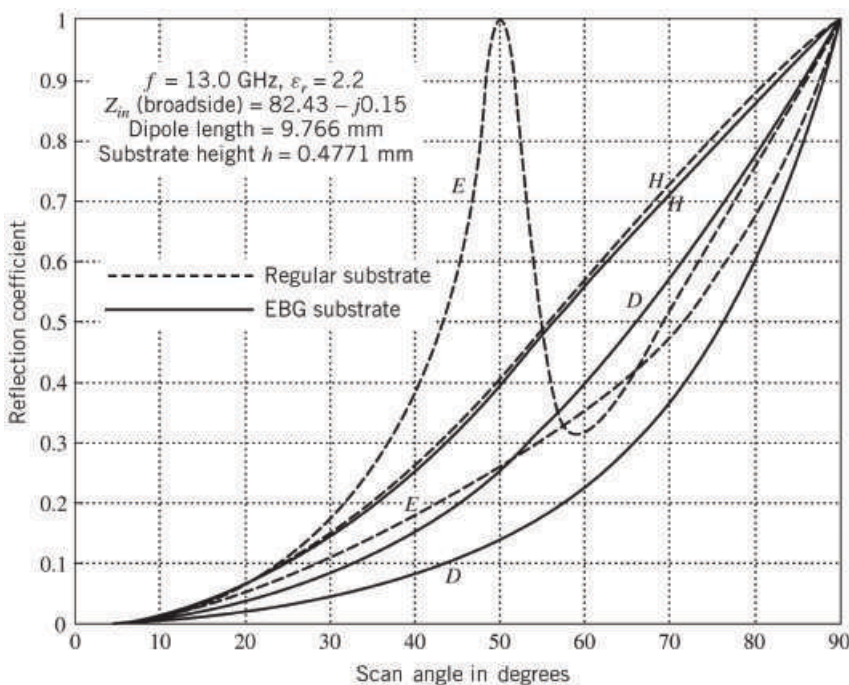


Figure 16-61 Reflection coefficient of phase array of dipoles on a regular and 4×4 unit cell PBG substrate. (Source: L. Zhang, J. A. Castaneda, and N. G. Alexopoulos, "Scan blindness free phased array design using PBG materials," *IEEE Trans. Antennas Propagat.*, vol. 52, no. 8, pp. 2000–2007, Aug. 2004. Reproduced with permission from IEEE.)

not limited to, reflective beam steering, microwave holography, low-profile antennas (see Section 16.10.3), and leaky-wave beam steering (see Section 16.12 that follows). These are accomplished by using textured high-impedance surfaces to control, and even suppress, the surface waves within the bandgap and/or the phase of the reflection coefficient, which can even be made zero (PMC

surface) at resonance. Other examples of interest utilizing EBG surfaces to suppress surface waves can be found in [6].

16.10.6 Surface-Wave Antennas

When a radiating element is placed adjacent to a grounded dielectric slab, besides radiating into free space, some of the power excites surface waves into the dielectric. For some antenna applications, the excitation of the surface waves is not desired as it degrades the overall radiation behavior, as the case for monopole discussed earlier. However, in some other applications, these surface waves are intentionally excited to advance the overall antenna performance [3, 80–83]. As metasurfaces can be utilized to suppress surface waves, they can also be designed to support and guide surface waves.

With proper arrangement of the periodic patches, the guided waves radiate as they propagate along the ground plane and dictate the radiation pattern. To achieve an omnidirectional, or monopole-like radiation pattern, a 0.26λ horizontal dipole antenna is placed 0.02λ above a metasurface ground plane to launch surface waves [81]. Since most of the radiation is contributed by the radiation of surface waves, the dipole length can be smaller than a half wavelength as it is only used as a surface-wave exciter. Nevertheless, the small radiation contributed by the dipole introduces a higher cross-polarization level compared to the monopole. To overcome this limitation, the dipole is replaced with a center-fed circular patch, and the square geometry of the grounded slab is replaced with a circular-shaped ground plane [82]. The overall radiation possesses a symmetrical omnidirectional pattern in the principal and diagonal planes. The attained patterns are very similar to those of a monopole with only a 0.05λ overall height profile.

Surface-wave antennas can also be designed to achieve a broadside radiation. This was accomplished by exciting surface waves utilizing different radiators such as dipole [84], truncated patch [85], diamond-slotted patch [85], and square ring elements [80]. As the field generated by different radiators varies, the launch and the excitation of surface waves will differ accordingly leading to unique radiation patterns. Moreover, the surface-wave resonances change according to the size of the metasurfaces and the dimensions of the periodic patches. As the size of the metasurfaces increases, more surface-wave resonances begin to contribute. For example, a dipole was positioned above square periodic patches (see Figure 16-62a) of different sizes (5×5 , 7×7 , and 9×9), and it was observed that the surface-wave resonances shift toward lower frequencies and become more closely spaced as the size of the metasurface increases. In addition to the metasurface size, the dimensions of the periodic patches control the surface-wave resonances, as was demonstrated in [79]. A square ring element was placed within the metasurface, as depicted in Figure 16-62b. The metasurface is formed with rectangular patches to possess different surface-wave resonances along two orthogonal planes. The resonance that is attributed to the square ring element, along with the surface-wave resonances, is designed to be closely spaced to improve the matching bandwidth by 26%. Furthermore, the proper excitation of surface waves in the design resulted in stable broadside radiation within the designed frequency range [79].

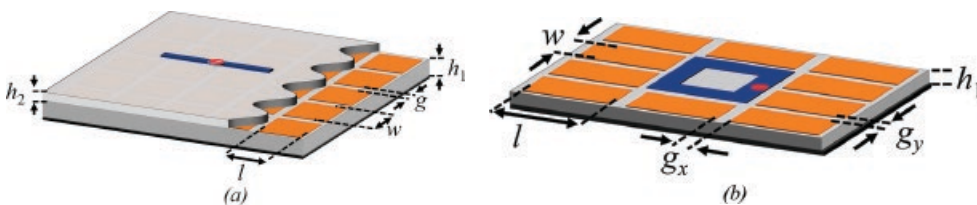


Figure 16-62 Geometry of surface-wave antennas for broadside radiation. (a) Dipole above metasurface; (b) Square ring within metasurface. (Source: M. Alharbi, C. A. Balanis, and C. R. Birtcher, “Performance enhancement of square-ring antennas exploiting surface-wave metasurfaces,” *IEEE Antennas Wireless Propagat. Lett.*, vol. 18, no. 10, pp. 1991–1995, Oct. 2019, Reproduced with permission from IEEE.)

16.11 HIGH-GAIN PRINTED LEAKY-WAVE ANTENNAS USING METASURFACES

Leaky-wave antennas (LWAs) are traveling-wave antennas [9] that can be considered as waveguide structures. Radiation is achieved in the form of leakage from the propagating wave as it travels through the guiding structure. The propagation constant of leaky waves is complex ($\gamma = \alpha + j\beta$), where α is the attenuation constant that corresponds to the leakage (or radiation), and β is the phase constant.

LWAs can be classified as *one-dimensional* (1-D) and *two-dimensional* (2-D) [86]. In 1-D LWAs, the leaky waves are guided along a single direction, while in 2-D LWAs, the waves are guided in two-dimensional guiding structures. One-dimensional LWAs can be either *uniform* or *periodic* [86, 87]. Uniform leaky-wave antennas are uniform throughout their length, with no periodic modulation on their geometries. An example of a uniform leaky-wave antenna is the meandering long slot leaky-wave antenna shown in Figure 16-63.

The antenna is realized by the continuous long slot along the broad wall of a rectangular waveguide. The slot should be offset from the center of the rectangular waveguide to achieve radiation [88]. While straight-line slots have higher sidelobe levels, the offset of the meandering long slot, from the waveguide centerline toward the side wall, controls the coupling and, thus, the aperture distribution over the slot that is responsible for the shape of the amplitude radiation pattern, especially the minor lobe structure.

The fundamental mode of a uniform leaky-wave antenna is a *fast wave*, where the longitudinal phase constant (β) is smaller than the free-space phase constant (β_0), and the antenna radiates from the discontinuities. Contrary to uniform LWAs, periodic leaky-wave antennas have some form of a periodic modulation throughout their structure. Examples of such periodic LWAs are undulating conductors [89], periodic holes or slits on waveguides [90], periodically asymmetric trough waveguides [91], and strip gratings [58]. A classical waveguide LWA with discrete slots is shown in Figure 16-64 [9].



Figure 16-63 A continuous slot leaky-wave rectangular waveguide antenna.

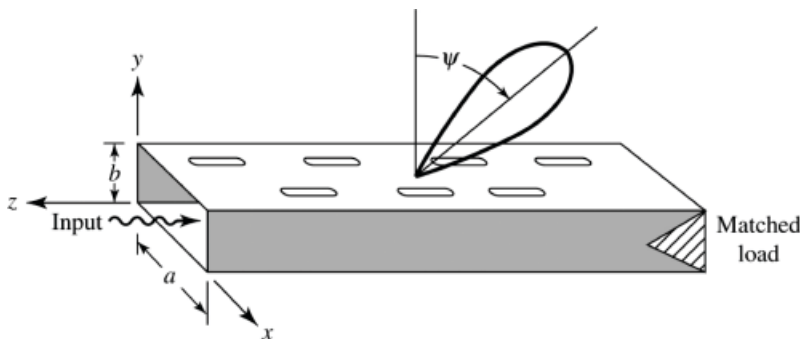


Figure 16-64 A discrete-slot leaky-wave rectangular waveguide antenna. (Source: C. A. Balanis, *Antenna Theory: Analysis and Design*, Fourth Edition, John Wiley & Sons, New York, 2016. Reproduced with permission from John Wiley & Sons.)

The antenna is based on a rectangular waveguide of lateral dimensions a and b , similar to the waveguide shown in Figure 8-3. The LWA forms a beam along an angle ψ from broadside, which can be scanned as the frequency is varied.

16.11.1 Floquet-Bloch Modes

In periodic structures, the periodic modulation at a given frequency causes the modal electromagnetic fields of the guided structure at any two points separated by the modulation period to be related by the same constant. In other words, the periodicity in the structure enables the electromagnetic fields to be represented as the product of an exponential factor corresponding to the traveling-wave nature of the fields and a periodic function with the same periodicity as that of the structure. For instance, a one-dimensionally modulated structure along the y axis with a periodicity of p is exhibited in Figure 16-65.

The electric field at any point on the surface can be represented by

$$\mathbf{E}(x, y, z) = \mathbf{E}_p(x, y, z) e^{-\gamma y} \quad (16-39)$$

where $\mathbf{E}_p(x, y, z)$ is periodic with a period p , which can be expressed as

$$\mathbf{E}_p(x, y + p, z) = \mathbf{E}_p(x, y, z) \quad (16-40)$$

$\mathbf{E}_p(x, y, z)$ can be expanded using Fourier series and represented as

$$\mathbf{E}_p(x, y, z) = \sum_{n=-\infty}^{+\infty} \mathbf{E}_{pn}(x, z) e^{-j\frac{2\pi}{p}ny} \quad (16-41)$$

Substituting (16-41) into (16-39) yields

$$\mathbf{E}(x, y, z) = \sum_{n=-\infty}^{+\infty} \mathbf{E}_{pn}(x, z) e^{-\left[\alpha + j\left(\beta + \frac{2\pi}{p}n\right)\right]y} \quad (16-42)$$

From (16-42), the electromagnetic fields can be characterized as an infinite number of space harmonics ($n = -\infty, \dots, -1, 0, 1, \dots, \infty$) referred to as *Floquet-Bloch modes*; this representation is referred to as the *Floquet-Bloch theorem*. The phase constant of the n th Floquet-Bloch mode can be expressed, using (16-42), as

$$\beta_n = \beta + \frac{2\pi}{p}n \quad (16-43)$$

Note the following notation for leaky-wave propagation and radiation: β_n for Floquet modes, β for the ($n = 0$) Floquet mode, and β_0 for the free-space constant.

In a periodic leaky-wave antenna, the fundamental Floquet-Bloch mode ($n = 0$) is usually a slow wave, where the longitudinal phase constant (β) is greater than the free-space phase constant (β_0), and it does not radiate continuously along the structure of the antenna. However, one or many of the space harmonics (for instance, $n = -1$) introduced by the periodic modulation are designed to be

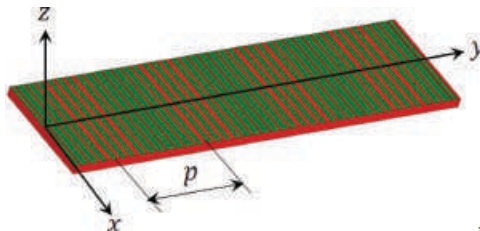


Figure 16-65 One-dimensional periodic structure of periodicity p .

fast waves, which radiate. The attenuation constant of the radiating harmonic controls the beam-width, while the phase constant determines the angular direction of the formed beam [92]. Because the phase constant β is a function of frequency, the angular direction of the formed beam varies with the frequency; hence, leaky-wave antennas are capable of *frequency scanning*.

Uniform leaky-wave antennas can beam-form only in the forward quadrant (near broadside to near forward endfire), while periodic LWAs can frequency scan in both the forward and backward quadrants. However, conventional 1-D leaky-wave antennas fed from one end cannot radiate along the broadside direction. This is due to the *open stopband phenomenon*, where, as the beam approaches the broadside direction, the attenuation constant varies rapidly and eventually reduces to zero, due to mode coupling. The impedance of the Floquet-Bloch mode is reactive and causes a stopband; this phenomenon is similar to scan blindness due to mutual coupling in arrays [77]. This issue has been addressed by adding impedance transformers between radiating elements to keep the impedance real, and by employing metamaterials [93, 94]. Broadside radiation can be achieved by other configurations such as feeding the antenna at the center or both ends, thereby forming two beams close to the broadside, which combine to form a single broadside beam [95]. Endfire radiation by conventional leaky-wave antennas is usually limited by the element pattern of the radiating elements.

16.12 METASURFACE LEAKY-WAVE ANTENNAS

Printed leaky-wave antennas can be implemented using high-impedance surfaces (HISs) with the surface impedance modulated with a periodic function; these are referred to as *metasurface leaky-wave antennas*.

16.12.1 Holographic Principle on Antennas

The surface-impedance modulation function required to form the desired beam in the desired direction can also be obtained from the *holographic principle*. This technique was inspired by the concept of optical holography [96] in which the interference pattern of the reflection from the object when illuminated by a laser source (object wave) and the illumination from the source (reference wave) is captured on a photographic plate, called a *hologram*, as illustrated in Figure 16-66.

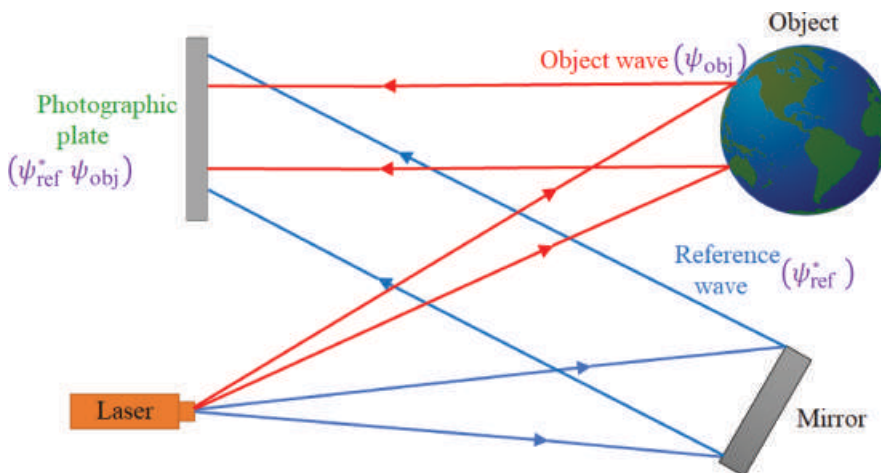


Figure 16-66 Optical holography: making a hologram. (Source: S. Ramalingam, “Impedance modulated metasurface antennas,” Ph.D. dissertation, School of Electrical, Computer, and Energy Engineering, Arizona State University, 2020.)

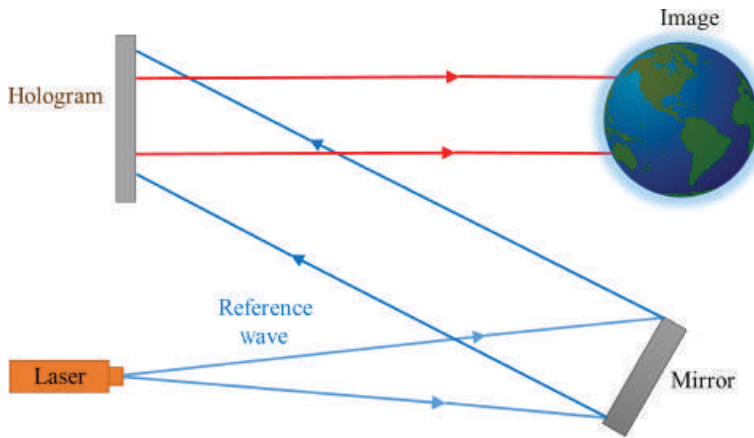


Figure 16-67 Optical holography: illuminating the hologram. (Source: S. Ramalingam, “Impedance modulated metasurface antennas,” Ph.D. dissertation, School of Electrical, Computer, and Energy Engineering, Arizona State University, 2020.)

When the hologram is illuminated by the laser source an image of the object is produced, as illustrated in Figure 16-67.

Employing this technique on antennas, the surface impedance of the metasurface can be modulated as an interference pattern of the phase of the reference wave (Ψ_{ref}) and that of the object wave (Ψ_{obj}). The reference wave corresponds to the surface waves excited on the surface of the LWA, and the object wave corresponds to the aperture fields needed to form a fan beam or a pencil beam in the desired direction. The surface impedance is modulated as the interference pattern, which contains a term proportional to $\Psi_{\text{obj}}\Psi_{\text{ref}}^*$. When this interference pattern (impedance modulated surface) is illuminated by the reference wave (surface wave) Ψ_{ref} , the interference pattern produces a copy of the object wave as [57]

$$(\Psi_{\text{obj}}\Psi_{\text{ref}}^*)\Psi_{\text{ref}} = |\Psi_{\text{ref}}|^2 \Psi_{\text{obj}} \quad (16-44)$$

16.12.2 One-Dimensional Periodic Metasurface LWAs

One-dimensionally periodic metasurface LWAs can be realized as arrays of metallic patches or strips on a dielectric-covered ground plane with varying gap sizes between them. The surface-impedance profile of the metasurface is modulated with a periodic function like a sinusoidal wave or a square wave, and it is realized using HISs. A periodic modulation function can be represented as a sum of sinusoids using Fourier series. Each fast-wave mode of every sinusoid contributes to a fan beam.

The application of the holographic principle for a 1-D planar LWA fed at one end to form a fan beam at the desired angular direction results in a sinusoidal impedance modulation function, as illustrated in Figure 16-68.

The surface-impedance modulation function $Z_s(x, y)$ is expressed as

$$Z_s(x, y) = \pm jX_a \left[1 + M \cos\left(\frac{2\pi}{p}y\right) \right] \quad (16-45)$$

where X_a is the average surface reactance, M is the modulation index, and p is the modulation period. The surface reactance needs to be inductive to obtain radiation from TM-mode leaky waves and capacitive for TE-mode leaky waves. The average surface reactance and the modulation index control the gain of the fan beam, while the period of sinusoidal modulation, along with the

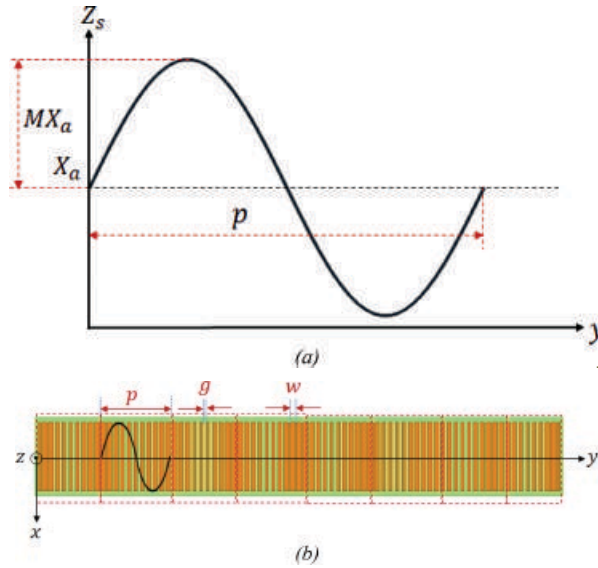


Figure 16-68 (a) The parameters in a sinusoidal impedance modulation. (b) The top view of a sinusoidally modulated planar 1-D LWA.

average surface reactance, controls the phase constant of the Floquet-Bloch mode and, therefore, controls the angular direction of the formed fan beam. The phase constant β_n of the n th Floquet-Bloch mode is related approximately to the modulation parameters by

$$\beta_n = \beta_0 \sqrt{1 + \left(\frac{X_a}{\eta_0}\right)^2} + \frac{2\pi n}{p} \quad (\text{TM mode}) \quad (16-46)$$

$$\beta_n = \beta_0 \sqrt{1 + \left(\frac{\eta_0}{X_a}\right)^2} + \frac{2\pi n}{p} \quad (\text{TE mode}) \quad (16-47)$$

A fan beam formed by a sinusoidally modulated 1-D metasurface LWA along the desired angular direction θ_d from broadside is demonstrated in Figure 16-69; θ_d is related to the phase constant of the Floquet mode by

$$\theta_d = \sin^{-1} \left(\frac{\beta_n}{\beta_0} \right) \quad (16-48)$$

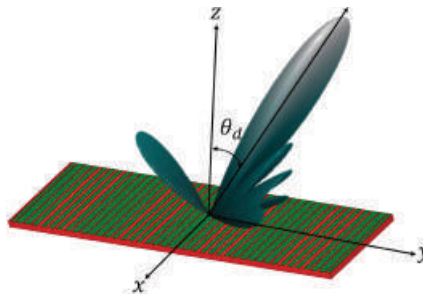


Figure 16-69 A fan beam formed by a 1-D metasurface LWA. (Source: S. Ramalingam, “Impedance modulated metasurface antennas,” Ph.D. dissertation, School of Electrical, Computer, and Energy Engineering, Arizona State University, 2020.)

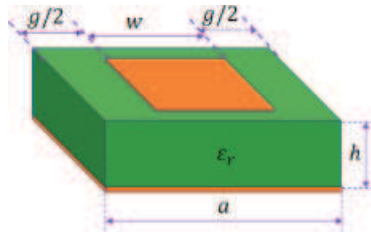


Figure 16-70 A HIS unit cell. The width of the patch is $w = a - g$. (Source: S. Ramalingam, “Impedance modulated metasurface antennas,” Ph.D. dissertation, School of Electrical, Computer, and Energy Engineering, Arizona State University, 2020.)

The design of one-dimensional periodic metasurface LWAs to form a fan beam in the desired angular direction is also demonstrated using the MATLAB program entitled **One_Dim_LWA** included at the end of this chapter.

Each period of the modulation function is sampled at multiple points. The required surface impedance at each point is realized using unit cells, which consist of metallic strips of width w , separated by a gap g on a dielectric-covered ground plane of thickness h , as illustrated in Figure 16-70.

The gap size g can be varied to achieve the desired surface impedance at each unit cell. For a unit cell of length a and gap size g , the surface reactance X_s , at a given frequency of operation f and a phase constant along the surface β , can be computed using the transverse resonance method (TRM) [59] of Section 8.6 and illustrated in Figure 16-71.

According to the transverse resonance method the impedance looking upward (Z_{up}) and the impedance looking downward (Z_{down}) are equal in magnitude but of opposite signs. The impedance looking downward $Z_{text{down}}$ is the surface impedance of the PEC terminated high-impedance surface given by [30, 59]

$$Z_{\text{down}} = jX_s(f, \beta) = \frac{j\eta_0\beta_{z1}\tan(\beta_{z1}h)}{\beta_0\epsilon_r - \beta_0(\epsilon_r + 1)\delta\beta_{z1}\tan(\beta_{z1}h)} \quad (16-49)$$

where η_0 is the free-space intrinsic impedance and h is the thickness of the substrate, as shown in Figures 16-70 and 16-71, and

$$\beta_{z1} = \sqrt{\beta_0^2\epsilon_r - \beta^2} \quad (16-50)$$

$$\delta = \left(\frac{a}{\pi}\right) \ln \left\{ \frac{1}{\sin[\pi(a-g)/2a]} \right\} \quad (16-51)$$

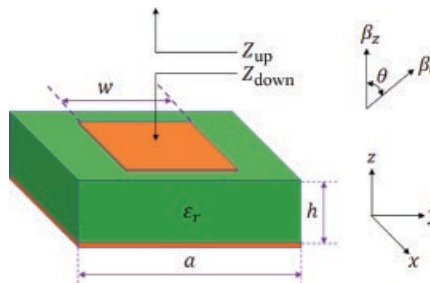


Figure 16-71 Transverse resonance method to obtain the surface impedance of a unit cell of 1-D LWA. (Source: S. Ramalingam, “Impedance modulated metasurface antennas,” Ph.D. dissertation, School of Electrical, Computer, and Energy Engineering, Arizona State University, 2020.)

Equation (16-49) is applicable to both TE and TM modes. Z_{up} is the wave-impedance for both TE and TM modes, as discussed in Section 4.2.2B, and it is expressed as

$$Z_{\text{up}}^{\text{TE}} = \frac{\eta_0}{\cos \theta} = \frac{\eta_0 \beta_0}{\beta_z} \quad (16-52)$$

$$Z_{\text{up}}^{\text{TM}} = \eta_0 \cos \theta = \frac{\eta_0 \beta_z}{\beta_0} \quad (16-53)$$

where

$$\beta_z = \sqrt{\beta_0^2 - \beta^2} \quad (16-54)$$

A relation between the surface reactance X_s and the gap sizes g can be derived by plotting the magnitude of the impedances looking upward Z_{up} and downward Z_{down} against the longitudinal phase constant β . The intersecting point of the two curves is the surface reactance of the unit cell with a gap size g . Using this technique on a unit cell of size $a = 3$ mm, with a substrate of dielectric constant $\epsilon_r = 3$ and thickness $h = 1.524$ mm, the variation of the TM-mode surface reactance X_s as a function of the gap size g is displayed in Figure 16-72 [97]. A similar plot can be obtained for TE-mode surface waves, where the surface reactance is capacitive.

By using the relationship between X_s and g , the surface reactance of the LWA can be realized as an array of unit cells to build the 1-D metasurface LWA. The application of the transverse resonance method to determine the surface reactance of a high-impedance surface unit cell is also demonstrated with a MATLAB program entitled **TRM_HIS** that is included at the end of this chapter.

Conventionally, one-dimensional periodic metasurface LWAs have been impedance-modulated with a sinusoidal function. However, a square-wave modulated metasurface, shown in Figure 16-73, is a proposed design for low-observable applications [98].

These metasurface leaky-wave antennas can form high-gain fan beams while inherently having a low monostatic radar cross section (RCS) for parallel polarization. The modulation parameters such as the average surface reactance, the modulation index, and the modulation period control both the radiation and RCS characteristics. In this way, these low-observable LWAs are codesigned for both leaky-wave behavior and RCS reduction. High-resolution artwork used to fabricate prototypes of such antennas, with square-wave modulation, is displayed

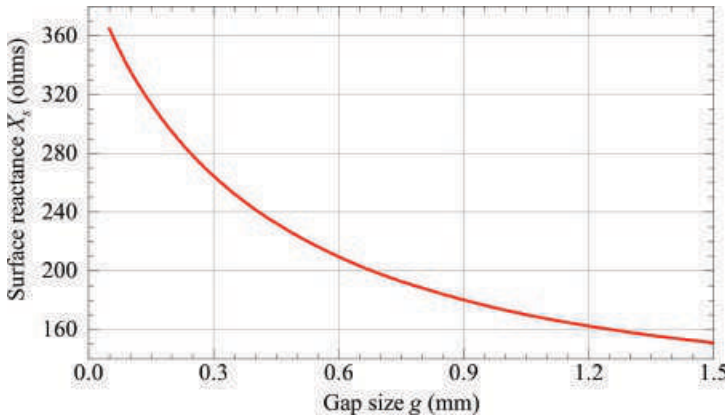


Figure 16-72 Plot of TM-mode surface reactance X_s versus gap size g for a unit cell of size $a = 3$ mm (the width of the patch $w = a - g$) at 15 GHz, with a substrate of dielectric constant $\epsilon_r = 3$ and thickness $h = 1.524$ mm. (Source: S. Ramalingam, C. A. Balanis, C. R. Birtcher, and S. Pandi, “Analysis and design of checkerboard leaky-wave antennas with low radar cross section,” *IEEE Open J. Antennas Propagat.*, vol. 1, pp. 26–40, 2020.)

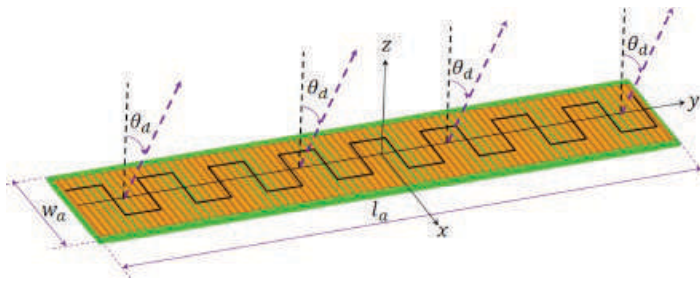


Figure 16-73 A square-wave modulated one-dimensionally periodic metasurface LWA. (Source: S. Ramalingam, “Impedance modulated metasurface antennas,” Ph.D. dissertation, School of Electrical, Computer, and Energy Engineering, Arizona State University, 2020.)



Figure 16-74 Artwork used to fabricate a prototype of the square-wave modulated LWA. (Source: S. Ramalingam, C. A. Balanis, C. R. Birtcher, and S. Pandi, “Analysis and design of checkerboard leaky-wave antennas with low radar cross section,” *IEEE Open J. Antennas Propagat.*, vol. 1, pp. 26–40, 2020.)

in Figure 16-74 [97]. The measured and simulated E-plane radiation patterns and the monostatic RCS patterns are shown in Figure 16-75 [97].

While sinusoidally modulated metasurfaces usually form fan beams with linear polarization, circularly polarized fan beams can be formed by multilayered *cascaded metasurfaces*, which consist of patterned anisotropic metallic layers with tensor impedances separated by dielectrics on a dielectric-covered ground plane [99].

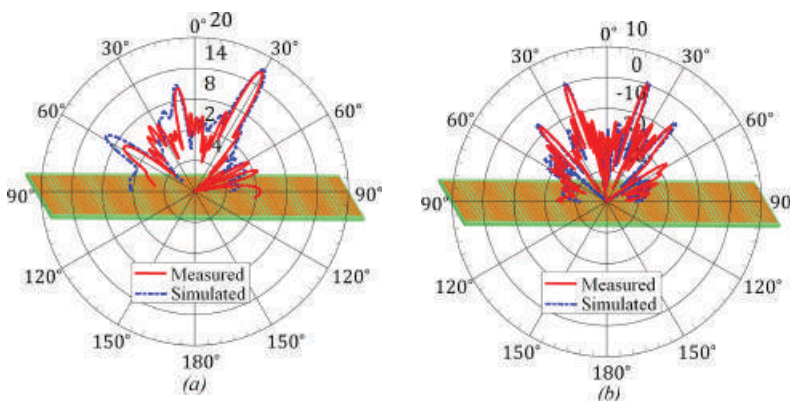


Figure 16-75 Measured (a) E-plane radiation and (b) monostatic RCS patterns compared to simulations. (Source: S. Ramalingam, C. A. Balanis, C. R. Birtcher, and S. Pandi, “Analysis and design of checkerboard leaky-wave antennas with low radar cross section,” *IEEE Open J. Antennas Propagat.*, vol. 1, pp. 26–40, 2020.)

Example 16-5

Design a TM-mode planar metasurface leaky-wave antenna to form a fan beam along 45° from the broad-side direction at 15 GHz. The variation of the surface impedance with the gap size is shown in Figure 16-72. Determine the modulation period required to form a fan beam using the $n = -1$ Floquet-Bloch mode.

Solution: From Figure 16-72, a value of the surface reactance in the middle of the plot is 230 ohms. Assume a modulation index $M = 0.25$.

Based on these and a frequency of 15 GHz,

$$\beta_0 = \omega \sqrt{\mu_0 \epsilon_0} = 2\pi f \sqrt{\mu_0 \epsilon_0} = 100\pi = 314.16 \text{ rad/m}$$

Based on (16-48)

$$\beta_{-1} = \beta_0 \sin 45^\circ = 222.14 \text{ rad/m}$$

Using (16-46)

$$222.11 = 314.16 \sqrt{1 + \left(\frac{230}{377}\right)^2} - \frac{2\pi}{p}$$

leads to a modulation period of $p = 43.1 \text{ mm}$.

Metasurface LWAs can be conformed on curved surfaces, which makes them suitable for vehicular applications. One of the features of such an application, referred to as *automotive radar technology*, is illustrated in Figure 16-76.

Automotive radar technology is employed for driver assistance and safety applications. Some important features of this technology include collision avoidance, lane change assistance, parking aid, adaptive cruise control, and blind spot detection.

When the metasurface LWA is conformed on a curved aperture, where the direction of surface-wave propagation is nonplanar, the curvature causes defocusing. The phase constant can be varied locally to compensate for the defocusing introduced by the curvature [100, 101]. Alternatively, the impact of curvature can be compensated by using the holographic principle to design the conformal LWAs [102, 103]. On a cylindrically curved surface, the surface impedance can be modulated along the axis or the circumference of the cylinder, as shown in Figures 16-77a and 16-77b.

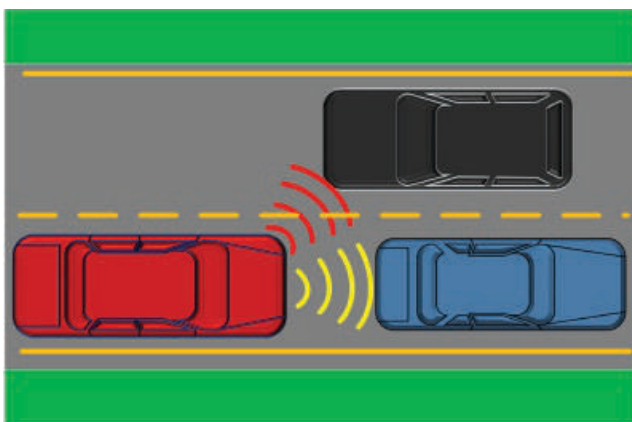


Figure 16-76 Automotive radars employed for driver safety and assistance. (Source: S. Ramalingam, C. A. Balanis, C. R. Birtcher, and S. Pandi, “Analysis and design of checkerboard leaky-wave antennas with low radar cross section,” *IEEE Open J. Antennas Propagat.*, vol. 1, pp. 26–40, 2020.)

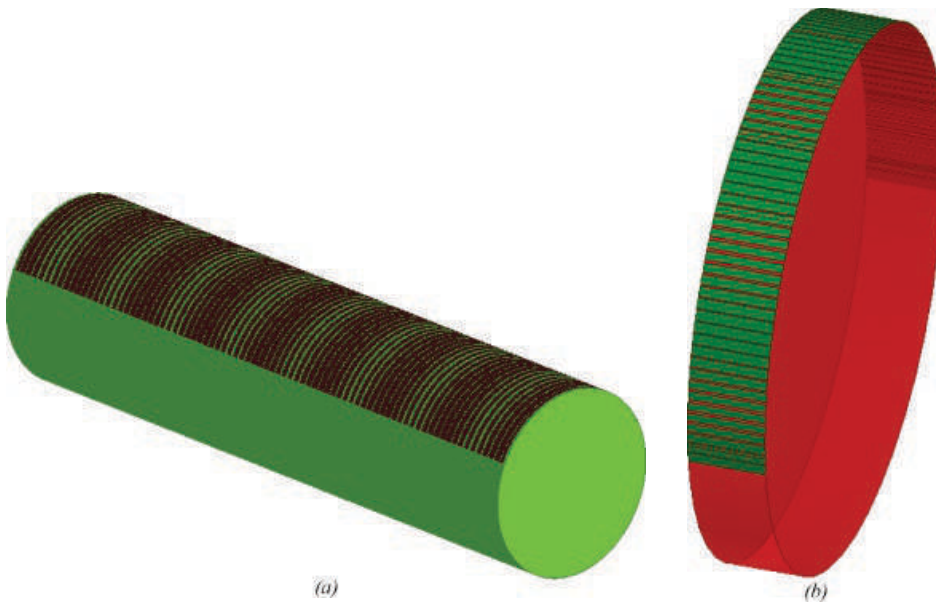


Figure 16-77 (a) Axially and (b) circumferentially modulated LWA. (Source: S. Ramalingam, “Impedance modulated metasurface antennas,” Ph.D. dissertation, School of Electrical, Computer, and Energy Engineering, Arizona State University, 2020.)

In a circumferentially modulated LWA, the surface impedance is modulated along the *circumference* of the cylindrical surface. The electrically long flat metallic strips are placed transverse to the direction of surface-wave propagation. In an axially modulated LWA, the metasurface is impedance modulated along the *axis* of the cylindrical surface. In this case, the electrically long metallic strips are curved, and the antenna is flat along the direction of propagation of the surface waves. A fabricated axially modulated metasurface and a comparison between its measured and simulated E-plane radiation patterns are displayed in Figure 16-78 [103].

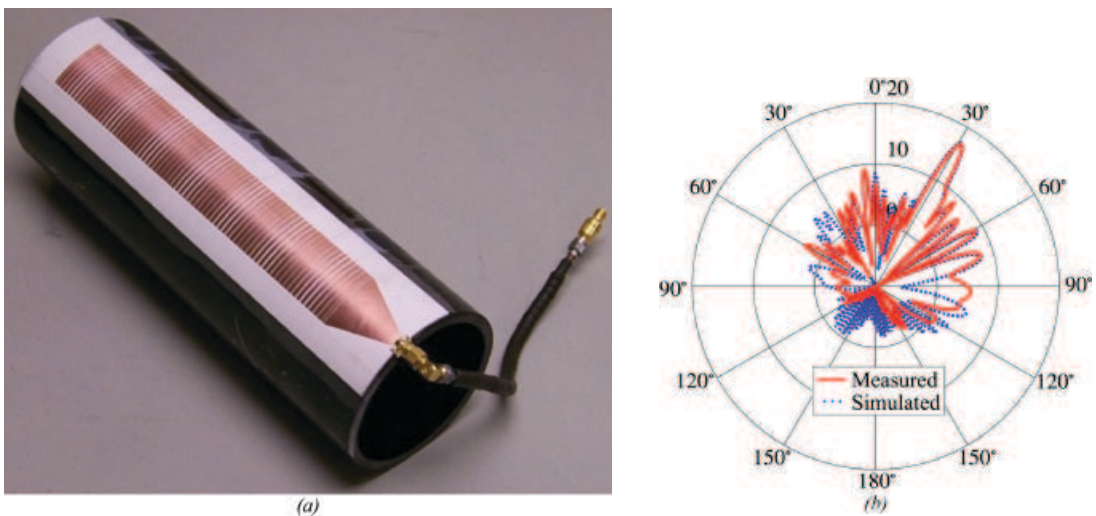


Figure 16-78 (a) A fabricated prototype of an axially modulated LWA and (b) its measured E-plane radiation pattern. (Source: S. Ramalingam, C. A. Balanis, C. R. Birtcher, S. Pandi, and H. N. Shaman, “Axially modulated cylindrical metasurface leaky-wave antennas,” *IEEE Antennas Wireless Propagat. Lett.*, vol. 17, no. 1, pp. 130–133, Jan. 2018. Reproduced with permission from IEEE.)

16.12.3 Two-Dimensional Holographic Metasurface LWAs

In two-dimensional (2-D) LWAs, the leaky waves are guided in a 2-D guiding structure. The LWAs are usually excited at the center using a monopole or a patch, and the leaky waves propagate radially, as shown in Figure 16-79.

For 2-D metasurface LWAs, the period of radial surface-impedance modulation p needed to form a pencil beam along an angle θ_d , in the $\phi = \phi_d$ plane, is represented by [104, 105]

$$p(\phi) = \frac{2\pi}{\beta_n - \beta_0 \sin \theta_d \cos(\phi - \phi_d)} \quad (16-55)$$

The surface-impedance modulation function Z_s needed to form a pencil beam along the desired angular direction (θ_d, ϕ_d) can be derived using the holographic principle [57]. The reference wave Ψ_{ref} is the radially propagating leaky-wave mode expressed as

$$\Psi_{\text{ref}} = e^{-j\beta\sqrt{(x-x_c)^2 + (y-y_c)^2}} \quad (16-56)$$

while the objective wave is a plane wave propagating along the desired direction represented by

$$\Psi_{\text{obj}} = e^{-j\beta_0(x \sin \theta_d \cos \phi_d + y \sin \theta_d \sin \phi_d)} \quad (16-57)$$

The surface-impedance modulation function $Z_s(x, y)$ can be obtained as an interference pattern of the reference and objective waves using [57]

$$Z_s(x, y) = \pm jX_s(x, y) = \pm jX_a \left[1 + M \text{Re}(\Psi_{\text{ref}}^* \Psi_{\text{obj}}) \right] \quad (16-58)$$

where X_a and M are the average surface reactance and modulation index, as defined in (16-45) and Figure 16-68, and (x_c, y_c) represents the location of the source. The holographic pattern can be viewed as concentric ellipses, whose major axis and the normal to the surface form the elevation plane of the formed pencil beam; this can also be observed in Figure 16-79. The surface-reactance profile derived using the holographic principle can be realized using high-impedance surface unit cells, of the form shown in Figure 16-70 as detailed in Section 16.12.2.

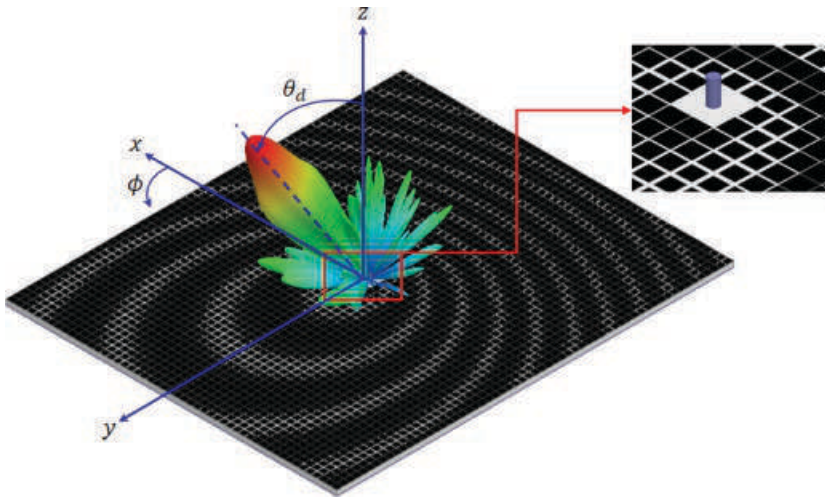


Figure 16-79 A 2-D holographic metasurface LWA fed at the center using a monopole. (Source: Adapted from S. Ramalingam, “Impedance modulated metasurface antennas,” Ph.D. dissertation, School of Electrical, Computer, and Energy Engineering, Arizona State University, 2020.)

Example 16-6

Derive the surface-reactance modulation function for a TM-mode two-dimensional holographic metasurface to form a pencil beam along ($\theta_d = 45^\circ$, $\phi_d = 0^\circ$). The substrate used is the same as that in Example 16-5, $\varepsilon_r = 3$ and $h = 1.524$ mm, whose surface-reactance plot is shown in Figure 16-72. The metasurface aperture is centered at the origin, and it is excited using a monopole, as shown in Figure 16-79. Assume a frequency of 15 GHz.

Solution: From Example 16-5, $X_a = 230$ ohms and $M = 0.25$. At a frequency of 15 GHz,

$$\beta_0 = 100\pi = 314.16 \text{ rad/m}$$

The phase constant β of the fundamental Floquet mode can be determined from (16-46) as

$$\beta = 314.16 \sqrt{1 + \left(\frac{230}{377}\right)^2} = 368 \text{ rad/m}$$

From (16-56)

$$\Psi_{\text{ref}} = e^{-j368r}, r = \sqrt{x^2 + y^2}$$

From (16-57)

$$\Psi_{\text{obj}} = e^{-j314.16 \sin 45^\circ x} = e^{-j222.14x}$$

For the TM mode, the surface reactance is inductive. Therefore, the surface-reactance modulation function can be expressed as

$$\begin{aligned} X_s(x, y) &= 230 + 57.5 \cos(368r - 222.14x) \\ &= 230 + 57.5 \cos(368\sqrt{x^2 + y^2} - 222.14x) \text{ ohms} \end{aligned}$$

While mapping the impedance profile $Z_s(x, y)$ of the metasurface, the position of any point (x, y) must be expressed in meters measured from the center/origin of the metasurface shown in Figure 16-79.

16.12.4 Radiation Mechanism of 2-D Holographic Metasurfaces

When a holographic metasurface is excited at the center to form a pencil beam in the desired direction (θ_d, ϕ_d), as illustrated in Figure 16-79, the pencil beam is formed from the constructive interference of the radiation from the radially propagating leaky waves. This necessitates the leaky waves in one half of the metasurface to radiate along the direction of propagation, and those in the other half to radiate in the opposite direction; these types of leaky waves are referred to as *forward* and *backward leaky waves*, respectively. For example, if the holographic metasurface is designed to form a pencil beam in the $\phi = 90^\circ$ plane as shown in Figure 16-79, radiation is facilitated by forward leaky waves in the $y > 0$ plane and backward leaky waves in the $y < 0$ plane, as illustrated in Figure 16-80.

It can be shown from (16-48) that, with an increase in frequency, the beam maximum formed by the forward leaky-waves shifts toward the endfire direction, while that formed by the backward leaky waves shifts toward the broadside direction. This phenomenon forces the holographic metasurface to operate at the frequency, referred to as *phase-crossover frequency*, at which the respective maxima of the radiation from the forward and backward leaky waves are along the same angular direction [105]. Frequencies at which the respective maxima of the forward and backward leaky-wave radiation are not along the same angular direction are referred to as *nonphase-crossover frequencies*. When a holographic metasurface is operated at a nonphase-crossover frequency, the pencil beams formed by forward and backward leaky waves shift in opposite directions, causing a null, instead of a maximum, along the desired angular direction θ_d [60, 105].

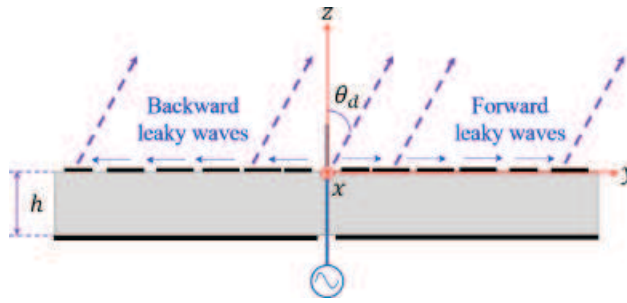


Figure 16-80 Forward and backward leaky waves in a holographic metasurface forming a pencil beam along θ_d from broadside.

16.12.5 Polarization-Diverse Holographic Metasurfaces

The polarization of the pencil beam formed by the antenna depends on the propagating leaky-wave mode, which in turn is a function of the frequency of operation and the high-impedance surface geometry. The first two modes of high-impedance surfaces are TM and TE, as detailed in this section. TM-polarized leaky waves form pencil beams with vertical polarization, while TE-polarized leaky waves form pencil beams with horizontal polarization [57]. Scalar impedance surfaces cannot independently control both TM- and TE-mode leaky waves. At the dominant TM mode, the horizontal polarized (E_ϕ) fields are canceled due to the symmetry of the metasurface along the major axis of the ellipses. However, as detailed in the previous section, the operation of conventional holographic metasurfaces is limited to the phase-crossover frequency. The concepts to control the polarization of the pencil beam formed by TM-polarized leaky waves in scalar impedance surfaces will be discussed.

A. Vertical Polarization Holographic metasurfaces form a pencil beam with vertical (E_θ) polarization while operating at a non-phase crossover frequency when the surface-reactance profile on one part of the surface (split by the line passing through the position of the feed and along the minor axis of the ellipses) is shifted by 180° with respect to the other. As discussed in the previous section, one part of the metasurface radiates using forward leaky waves, while the other uses backward leaky waves; these are 180° out of phase. Hence, the additional 180° phase difference reinforces radiation from forward and backward leaky waves.

The objective wave in (16-57) to form a pencil beam with vertical polarization can be rewritten as either (16-59) or (16-60) [60, 104]

$$\Psi_{\text{obj}} = \begin{cases} e^{-j\beta_0(x\sin\theta_d\cos\phi_d + y\sin\theta_d\sin\phi_d)} & \forall y \geq y_c \\ -e^{-j\beta_0(x\sin\theta_d\cos\phi_d + y\sin\theta_d\sin\phi_d)} & \forall y < y_c \end{cases} \quad (16-59)$$

or

$$\Psi_{\text{obj}} = \begin{cases} -e^{-j\beta_0(x\sin\theta_d\cos\phi_d + y\sin\theta_d\sin\phi_d)} & \forall y \geq y_c \\ e^{-j\beta_0(x\sin\theta_d\cos\phi_d + y\sin\theta_d\sin\phi_d)} & \forall y < y_c \end{cases} \quad (16-60)$$

The holographic surface-reactance pattern that includes the 180° phase difference to form a pencil beam along the desired direction θ_d in the $\phi = 90^\circ$ plane is shown in Figure 16-81. It can be seen from Figure 16-81 that the symmetric halves (split along the major axis of the ellipses) are modulated with the same surface-reactance modulation. Hence, the E_ϕ components cancel in the far-field region resulting in a vertically (E_θ) polarized pencil beam along the desired angular direction.

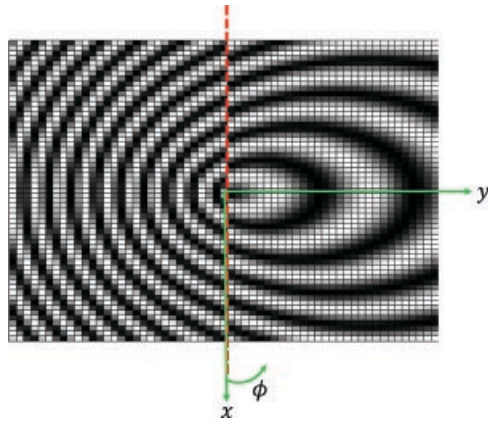


Figure 16-81 The holographic surface-reactance pattern to form a vertically polarized pencil beam along the $\phi = 90^\circ$ plane. (Source: S. Pandi, C. A. Balanis, and C. R. Birtcher, “Design of scalar impedance holographic meta-surfaces for antenna beam formation with desired polarization,” *IEEE Trans. Antennas Propagat.*, vol. 63, pp. 3016–3024, 2015. Reproduced with permission from IEEE.)

B. Horizontal Polarization Horizontally (E_ϕ) polarized pencil beams are obtained by disturbing the symmetry, along the major axis of the ellipses, to avoid the cancellation of the E_ϕ components in the far-field region. The surface-reactance profile in one half (split along the major axis of the ellipses) is shifted by 180° , with respect to the other to reinforce the E_ϕ components.

The corresponding objective wave can be represented by either (16-61) or (16-62) [104].

$$\Psi_{\text{obj}} = \begin{cases} e^{-j\beta_0(x\sin\theta_d\cos\phi_d + y\sin\theta_d\sin\phi_d)} & \forall x \geq x_m \\ -e^{-j\beta_0(x\sin\theta_d\cos\phi_d + y\sin\theta_d\sin\phi_d)} & \forall x < x_m \end{cases} \quad (16-61)$$

or

$$\Psi_{\text{obj}} = \begin{cases} -e^{-j\beta_0(x\sin\theta_d\cos\phi_d + y\sin\theta_d\sin\phi_d)} & \forall x \geq x_m \\ e^{-j\beta_0(x\sin\theta_d\cos\phi_d + y\sin\theta_d\sin\phi_d)} & \forall x < x_m \end{cases} \quad (16-62)$$

where $x_m = (y - y_c)\cot\phi_d + x_c$ is the major axis of the ellipses. The holographic surface-reactance pattern to obtain a horizontally polarized pencil beam, at a desired angular direction θ_d in the $\phi = 90^\circ$ plane, is shown in Figure 16-82.

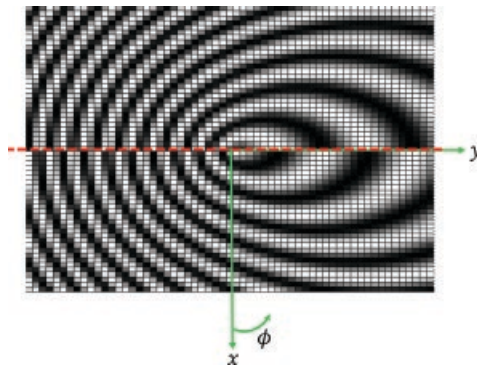


Figure 16-82 The holographic surface-reactance pattern to form a horizontally polarized pencil beam along the $\phi = 90^\circ$ plane. (Source: Adapted from S. Pandi, C. A. Balanis, and C. R. Birtcher, “Design of scalar impedance holographic meta-surfaces for antenna beam formation with desired polarization,” *IEEE Trans. Antennas Propagat.*, vol. 63, pp. 3016–3024, 2015. Reproduced with permission from IEEE.)

As the metasurface is operated at a nonphase-crossover frequency, the radiation from the forward and backward leaky waves are out of phase. Hence, E_θ fields cancel in the far-field region, resulting in a horizontally (E_ϕ) polarized pencil beam.

C. Circular Polarization As discussed in Section 4.4.2, circular polarization can be achieved if there are two orthogonal linearly polarized field components, equal in magnitude and with a 90° phase difference. Hence, the conventional metasurface is geometrically modified to reinforce both E_θ and E_ϕ components in the far-field region, and a 90° phase difference included in their objective functions. Hence, the resulting objective function can be expressed by [60]

$$\Psi_{\text{obj}} = 0.5(\Psi_{\text{obj1}} \pm j\Psi_{\text{obj2}}) \quad (16-63)$$

$$\Psi_{\text{obj1}} = \begin{cases} e^{-j\beta_0(x\sin\theta_d\cos\phi_d+y\sin\theta_d\sin\phi_d)} & \forall y \geq y_c \\ -e^{-j\beta_0(x\sin\theta_d\cos\phi_d+y\sin\theta_d\sin\phi_d)} & \forall y < y_c \end{cases} \quad (16-64)$$

$$\Psi_{\text{obj2}} = \begin{cases} e^{-j\beta_0(x\sin\theta_d\cos\phi_d+y\sin\theta_d\sin\phi_d)} & \forall x \geq x_m \\ -e^{-j\beta_0(x\sin\theta_d\cos\phi_d+y\sin\theta_d\sin\phi_d)} & \forall x < x_m \end{cases} \quad (16-65)$$

A $\pm j$ is included in (16-63) to introduce a phase difference of 90° between the E_θ and E_ϕ components; the $\pm$ sign also determines the sense of rotation. The holographic pattern needed to form a pencil beam along $(\theta = \theta_d, \phi = 90^\circ)$ with circular polarization is shown in Figure 16-83.

D. Multiple Polarizations The concepts discussed so far can be used to design a single holographic metasurface that can form, one at a time, a pencil beam with a vertical, horizontal, or circular polarization. The antenna is fed by two or more sources instead of one, to achieve polarization diversity. Each source or a combination of sources is designed to enable the metasurface to form a pencil beam with the desired polarization. When a holographic metasurface is fed by multiple sources the total surface-reactance modulation function $X_s(x, y)$ can be obtained as [106]

$$X_s(x, y) = X_a + M_1 \text{Re}(\Psi_{\text{ref}_1}^* \Psi_{\text{obj}_1}) + M_2 \text{Re}(\Psi_{\text{ref}_2}^* \Psi_{\text{obj}_2}) \quad (16-66)$$

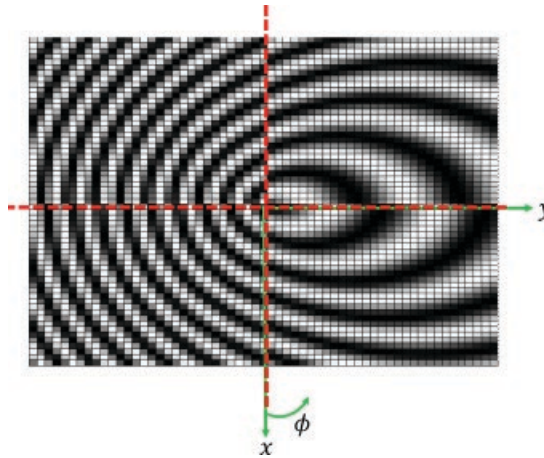


Figure 16-83 The holographic surface-reactance pattern to form a circularly polarized pencil beam along the $\phi = 0$ plane. (Source: S. Pandi, C. A. Balanis, and C. R. Birtcher, “Design of scalar impedance holographic metasurfaces for antenna beam formation with desired polarization,” *IEEE Trans. Antennas Propagat.*, vol. 63, pp. 3016–3024, 2015. Reproduced with permission from IEEE.)

where subscripts 1 and 2 denote the parameters pertaining to the first and second sources, respectively. Hence, a polarization-diverse metasurface can be designed by feeding the metasurface with two sources and modulating the surface reactance with the function [104]

$$X_s(x, y) = X_a + M_1 \text{Re}(\Psi_{\text{ref}_1}^* \Psi_{\text{obj}_1}) \pm j M_2 \text{Re}(\Psi_{\text{ref}_2}^* \Psi_{\text{obj}_2}) \quad (16-67)$$

where Ψ_{ref_1} and Ψ_{ref_2} are defined similar to (16-56), and Ψ_{obj_1} and Ψ_{obj_2} are expressed similar to (16-59)–(16-65). The metasurface can be designed to form a pencil beam with vertical polarization when excited by source 1 and horizontal polarization when excited by source 2, as illustrated in Figure 16-84.

The $\pm j$ term is included to enable the metasurface to form a circularly polarized pencil beam when excited by both sources and the $\pm$ sign also determines the sense of rotation; clockwise or counterclockwise. Holographic metasurfaces can be designed to include circular polarization with both senses of rotation, besides vertical and horizontal, by feeding them with three sources. A fabricated prototype of a polarization-diverse holographic metasurface and the measured circular polarized patterns, compared to the simulations, are shown in Figure 16-85. The design of two-dimensional periodic metasurface LWAs to form a pencil beam in the desired angular direction with the desired polarization is demonstrated using the MATLAB program entitled **Two_Dim_LWA** that is included at the end of this chapter.

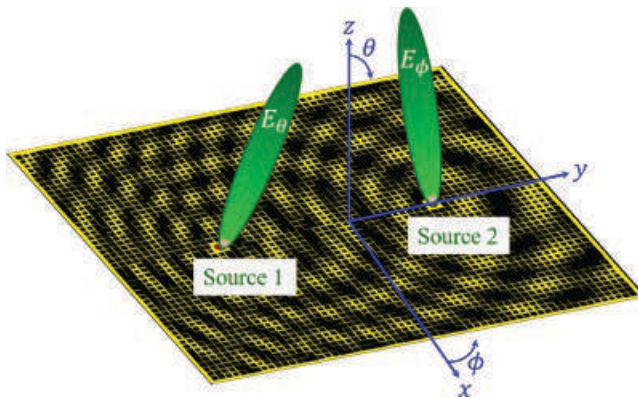


Figure 16-84 A polarization-diverse holographic metasurface. (Source: S. Ramalingam, “Impedance modulated metasurface antennas,” Ph.D. dissertation, School of Electrical, Computer, and Energy Engineering, ASU, 2020.)

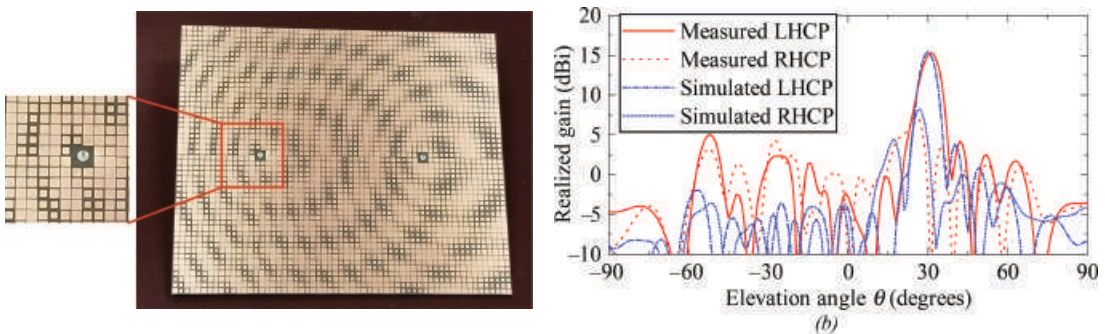


Figure 16-85 (a) A fabricated prototype of a polarization-diverse metasurface LWA. (b) Comparison of measured and simulated circularly polarized radiation patterns. (Source: S. Ramalingam, C. A. Balanis, C. R. Birtcher, and H. N. Shaman, “Polarization-diverse holographic metasurfaces,” *IEEE Antennas Wireless Propagat. Lett.*, vol. 18, no. 2, pp. 264–268, Feb. 2019. Reproduced with permission from IEEE.)

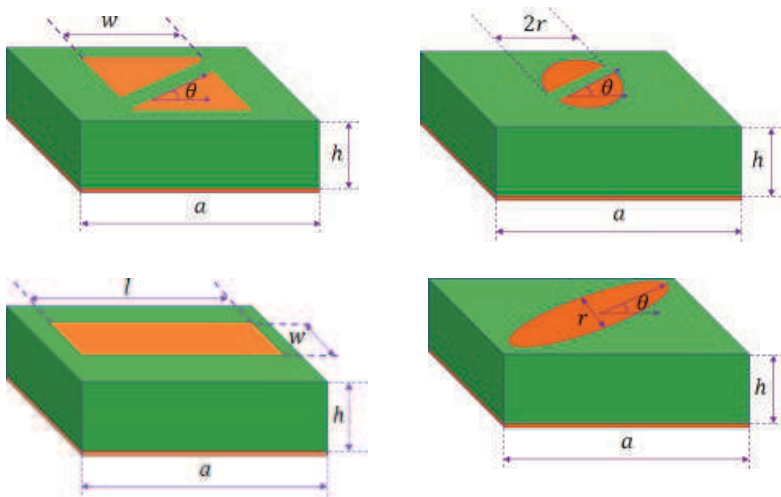


Figure 16-86 Anisotropic unit cells, proposed in the literature, that are used to realize a tensor surface reactance. (Source: S. Ramalingam, “Impedance modulated metasurface antennas,” Ph.D. dissertation, School of Electrical, Computer, and Energy Engineering, Arizona State University, 2020.)

16.12.6 Tensor Impedance Surfaces

In tensor impedance surfaces, the electric and magnetic fields are related by an impedance tensor expressed as

$$\bar{\mathbf{Z}} = \begin{pmatrix} Z_{xx} & Z_{xy} \\ Z_{yx} & Z_{yy} \end{pmatrix} \quad (16-68)$$

The impedance tensor can be determined using the holographic principle and realized using anisotropic impedance surfaces. Some anisotropic unit cells that have been employed in metasurface antennas [57, 107] are exhibited in Figure 16-86.

Tensor impedance surfaces support both TE and TM modes as degenerate modes and can control both modes independently [108]. Hence, the aperture efficiency of anisotropic metasurfaces is much higher than that of scalar impedance metasurfaces.

16.13 MULTIMEDIA

On the website that accompanies this book, the following multimedia resources are included for the review, understanding, and presentation of the material of this chapter.

- **MATLAB** computer programs:
 - a. **HIS_Mush** designs and computes the characteristics of a high-impedance surface (HIS) of Figures (16-4) and (16-9) based on the procedure and designs closed-form equations of Section 16.3.
 - b. **TRM_Metasurface_Dispersion** plots the dispersion diagram for metasurface using the approximate TRM model developed in [30].
 - c. **TRM_Grounded_Dielectric_Dispersion** develops and plots the dispersion diagram for dielectric-covered PEC based on the TRM as formulated and represented by (16-17).
 - d. **AMC_Designs** designs a two-layer mushroom textured surface based on a desired resonant frequency and following Section 16.4. The program also plots the reflection phase versus frequency of the designed AMC surface.

- e. **Dispersion_Diagram_Matlab** generates a dispersion diagram based on (16-2), (16-9), and (16-13), and on the formulation reported in [1].
- f. **Dispersion_Diagram_HFSS**: As part of the package of this program, two files, *Unit Cell Design* and *Graphical User Interface (GUI)*, are included to aid in the generation of the dispersion diagram in HFSS. It is based on a simulation setup shown in Figure 16-11 using the Floquet ports and periodic boundary conditions in Ansys HFSS.
- g. **Angular_Direction_Checkerboard** determines the angular directions of the scattered fields based on array theory of (16-22) and (16-23), and similar to Example 16-3.
- h. **Conventional_Checkerboard_RCS_Reduction** determines RCS reduction of a conventional checkerboard surface based on an AMC design that is also entered by the user. This program is similar to Example 16-4.
- i. **TRM_HIS** computes the surface reactance of a high-impedance surface unit cell based on the transverse-resonance method detailed in Section 16.12.2.
- j. **One_Dim_LWA** determines the modulation period of a 1-D LWA to form a fan beam in the desired angular direction and plots the surface-reactance profile for both TM and TE modes.
- k. **Two_Dim_LWA** computes and plots the surface-reactance profile to form a pencil beam in the desired angular direction with the desired polarization, with polarization diversity, for both TM and TE modes.
- **PowerPoint (PPT)** viewgraphs, in multicolor.

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PROBLEMS

- 16.1.** Design a two-layer mushroom textured surface with hexagonal patches and air as the upper layer ($\varepsilon_2 = \varepsilon_0$, $\mu_2 = \mu_0$), as shown in Figures 16-4 and 16-8 and outlined in Section 16.4 to exhibit PMC characteristics between:
- (a) 14 and 16 GHz
 - (b) 12 and 18 GHz

with a center frequency of $f_0 = 15$ GHz. Use a Rogers RT/duroid 3006 with a dielectric constant of $\varepsilon_r = 6.15$. The hexagonal patches are supported by metallic circular posts that connect the patches to the bottom ground plane through vias in the substrate. Use the MATLAB computer program **AMC_Designs** to verify the solution design.

- 16.2.** Derive, showing all the details, the RCS expression in closed form for the PMC ground plane of Figure 16-15b as given by (16-18) for TE^x polarization.
- 16.3.** Derive, showing all the details, the RCS expression in closed form for the PMC ground plane of Figure 16.15b for the perpendicular polarization (TM^x), as shown in Figure 11-8, where incident electric and magnetic fields can now be written as

$$\begin{aligned}\mathbf{E}^i &= \hat{\mathbf{a}}_x E_0 e^{-j\beta(y \sin \theta_i - z \sin \theta_i)} \\ \mathbf{H}^i &= -\frac{E_0}{\eta} (\hat{\mathbf{a}}_y \cos \theta_i + \hat{\mathbf{a}}_z \sin \theta_i) \\ &\quad \times e^{-j\beta(y \sin \theta_i - z \sin \theta_i)}\end{aligned}$$

- 16.4.** Derive, showing all the details, the RCS expression in closed form for the PEC/PMC ground plane of Figure 16-16a as given by (16-21a) for TE^x polarization.
- 16.5.** The electric field radiated by an electrical horizontal dipole in free space is given by:

$$E_{\psi}^d = j\eta \frac{kI_0 l e^{-jk_1 r}}{4\pi r} \sqrt{1 - (\sin \theta \sin \phi)^2}$$

Determine the:

- (a) Reflected electric field when the dipole is placed a height h above an infinite PEC plate.
- (b) Total electric field.
- (c) Power radiated (P_{rad}) by the total radiated field.

$$\text{Hint: } P_{\text{rad}} = \frac{1}{2\eta} \int_0^{2\pi} \int_0^{\pi/2} |E_\psi|^2 \times r^2 \sin \theta d\theta d\phi$$

- 16.6.** Repeat Problem 16.5 when the horizontal electric dipole is placed a height h above an infinite PMC.
- 16.7.** The radiation resistance of a radiating element is defined as

$$R_r = \frac{2P_{\text{rad}}}{(I_0)^2}$$

Find the radiation resistance of a horizontal electric dipole when it is placed a height h above:

- (a) Infinite PEC ground plane.
(b) Infinite PMC ground plane.

- 16.8.** For Rogers RT/duroid 5880 slab ($\epsilon_r = 2.2$) covered PEC, perform the following:
(a) Plot the dispersion diagram when the slab thickness h is 1 mm.
(b) Highlight the tightly and loosely bound regions.
- 16.9.** For the design in Example 16-6, examine if the second Floquet mode ($n = -2$) radiates. If so, determine (in degrees) the angular direction (from the normal to the strip) of the formed fan beam. (Note: If the second Floquet mode radiates, the formed fan beam is a minor lobe while the radiation from the first Floquet mode is the major lobe.)
- 16.10.** Derive (16-45) using the holographic principle for a planar one-dimensional (1-D) periodic metasurface leaky-wave antenna.
- 16.11.** Prove that for an axially modulated metasurface LWA, whose geometry is shown in Figure P16-11, the surface-impedance modulation function derived using the holographic principle is same as that of a planar LWA given by

$$Z_s(y, z) = jX_a \left[1 + M \cos \left(\frac{2\pi}{p} z \right) \right]$$

(Note: The effect of curvature can be modeled using the projected-aperture method. For further details, refer to [103].)

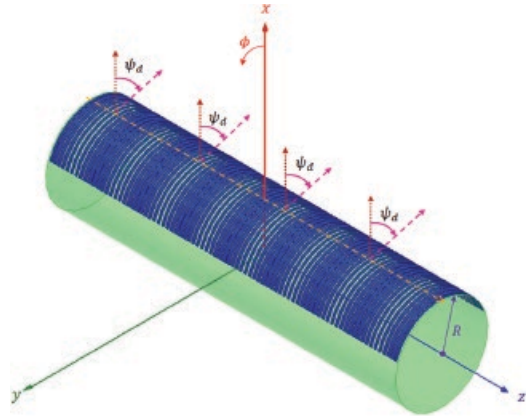


Figure P16-11 (Source: S. Ramalingam, C. A. Balanis, C. R. Birtcher, S. Pandi, and H. N. Shaman, “Axially modulated cylindrical metasurface leaky-wave antennas,” *IEEE Antennas Wireless Propagat. Lett.*, vol. 18, no. 2, pp. 264–268, Feb. 2019. Reproduced with permission from IEEE.)

- 16.12** Show that the surface-impedance modulation function for a circumferentially modulated LWA, whose geometry is shown in Figure P16-12, to form a fan beam in the desired angular direction $\phi = \phi_d$ (ϕ is

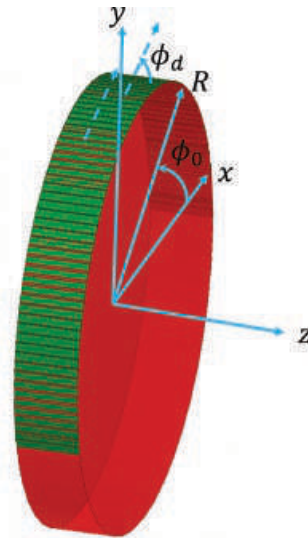


Figure P16-12 (Source: S. Pandi, C. A. Balanis, and C. R. Birtcher, “Curvature modeling in design of circumferentially modulated cylindrical metasurface LWA,” *IEEE Antennas Wireless Propagat. Lett.*, vol. 16, pp. 1024–1027, 2017. Reproduced with permission from IEEE.)

measured from the x axis along the xy plane) is represented by

$$X_s(\phi) = X_a \left\{ 1 + M \cos[\beta R(\phi_0 - \phi) - \beta_0 R \cos(\phi_d - \phi)] \right\}$$

where β is the phase constant along the direction of leaky-wave propagation in the LWA.

- 16.13.** Repeat Example 16-6, if the source exciting the metasurface is off-centered by y_c along the y axis [location of the source is $(0, y_c)$].
- 16.14.** Repeat Example 16-6 for a holographic metasurface to form a pencil beam with:
- Vertical polarization.
 - Horizontal polarization.
 - Circular polarization.
- 16.15.** For the polarization-diverse metasurface shown in Figure 16-84, derive the surface-impedance modulation function to form a pencil beam on the yz plane at 45° from broadside ($\phi = 90^\circ$). Assume that the position of sources 1 and 2 to be at $(0, y_{c1})$ and $(0, y_{c2})$, respectively.
- 16.16.** The metasurface of Example 16-5 is impedance modulated with a square wave of

$$Z_s(y) = jX_a \left\{ 1 + M \operatorname{sgn} \left[\cos \left(\frac{2\pi y}{p} \right) \right] \right\}$$

Assuming that X_a , M , and p are the same values as in Example 16-5, determine the angular directions (in degrees) of the fan beams formed by the first and the third Fourier harmonics. Assume that only the first Floquet-Bloch mode of each harmonic radiates. *Hint:* Using Fourier series, the square-wave surface impedance modulation can be rewritten as

$$Z_s(y) = jX_a \left\{ 1 + M \left(\frac{4}{\pi} \right) \times \sum_{q=1,3,5,\dots}^{\infty} \frac{1}{q} \sin \left(\frac{2q\pi y}{p} \right) \right\}$$

(For further details, refer to [97]).

- 16.17.** Derive the surface-impedance modulation function of a polarization-diverse holographic metasurface that forms a pencil beam with a left-hand circular polarization when excited by source 1, and a right-hand circular polarization for source 2, as illustrated in Figure 16-84. Assume that the position of sources 1 and 2 are at $(0, -y_c)$ and $(0, y_c)$, respectively. What is the polarization of the pencil beam when the metasurface is excited by both the sources? Assume that the two sources are orthogonal to each other ($\Psi_{\text{ref}2}^* \Psi_{\text{ref}1} = \Psi_{\text{ref}1}^* \Psi_{\text{ref}2} = 0$).
- 16.18.** Derive the surface-impedance modulation function of a polarization-diverse holographic metasurface excited by three sources as shown in Figure P16-18. The metasurface should form a pencil beam with vertical polarization when excited by source 1, and horizontal polarization for the other two sources. It should also form a left-hand circularly polarized pencil beam when excited by both sources 1 and 2, and a right-hand circular polarization for excitation by sources 1 and 3. Assume that the position of sources 1, 2, and 3 to be at $(0, 0)$, $(-x_c, 0)$, and $(-x_c, 0)$, respectively.

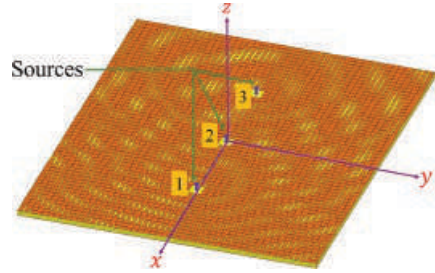


Figure P16-18 (Source: S. Ramalingam, C. A. Balanis, C. R. Birtcher, and H. N. Shaman, "Polarization-diverse holographic metasurfaces," *IEEE Antennas Wireless Propagat. Lett.*, vol. 18, no. 2, pp. 264–268, Feb. 2019. Reproduced with permission from IEEE.)

APPENDIX I



Identities

I.1 TRIGONOMETRIC

1. Sum or difference:

a. $\sin(x + y) = \sin x \cos y + \cos x \sin y$

b. $\sin(x - y) = \sin x \cos y - \cos x \sin y$

c. $\cos(x + y) = \cos x \cos y - \sin x \sin y$

d. $\cos(x - y) = \cos x \cos y + \sin x \sin y$

e. $\tan(x + y) = \frac{\tan x + \tan y}{1 - \tan x \tan y}$

f. $\tan(x - y) = \frac{\tan x - \tan y}{1 + \tan x \tan y}$

g. $\sin^2 x + \cos^2 x = 1$

h. $\tan^2 x - \sec^2 x = -1$

i. $\cot^2 x - \csc^2 x = -1$

2. Sum or difference into products:

a. $\sin x + \sin y = 2 \sin \frac{1}{2}(x + y) \cos \frac{1}{2}(x - y)$

b. $\sin x - \sin y = 2 \cos \frac{1}{2}(x + y) \sin \frac{1}{2}(x - y)$

c. $\cos x + \cos y = 2 \cos \frac{1}{2}(x + y) \cos \frac{1}{2}(x - y)$

d. $\cos x - \cos y = -2 \sin \frac{1}{2}(x + y) \sin \frac{1}{2}(x - y)$

3. Products into sum or difference:

a. $2 \sin x \cos y = \sin(x + y) + \sin(x - y)$

b. $2 \cos x \sin y = \sin(x + y) - \sin(x - y)$

c. $2 \cos x \cos y = \cos(x + y) + \cos(x - y)$

d. $2 \sin x \sin y = -\cos(x + y) + \cos(x - y)$

4. Double and half-angles:

- a. $\sin 2x = 2 \sin x \cos x$
- b. $\cos 2x = \cos^2 x - \sin^2 x = 2 \cos^2 x - 1 = 1 - 2 \sin^2 x$
- c. $\tan 2x = \frac{2 \tan x}{1 - \tan^2 x}$
- d. $\sin \frac{1}{2}x = \pm \sqrt{\frac{1 - \cos x}{2}}$ or $2 \sin^2 \theta = 1 - \cos 2\theta$
- e. $\cos \frac{1}{2}x = \pm \sqrt{\frac{1 + \cos x}{2}}$ or $2 \cos^2 \theta = 1 + \cos 2\theta$
- f. $\tan \frac{1}{2}x = \pm \sqrt{\frac{1 - \cos x}{1 + \cos x}} = \frac{\sin x}{1 + \cos x} = \frac{1 - \cos x}{\sin x}$

5. Series:

- a. $\sin x = \frac{e^{jx} - e^{-jx}}{2j} = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \dots$
- b. $\cos x = \frac{e^{jx} + e^{-jx}}{2} = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \dots$
- c. $\tan x = \frac{e^{jx} - e^{-jx}}{j(e^{jx} + e^{-jx})} = x + \frac{x^3}{3} + \frac{2x^5}{15} + \frac{17x^7}{315} + \dots$

I.2 HYPERBOLIC

1. Definitions:

- a. Hyperbolic sine: $\sinh x = \frac{1}{2}(e^x - e^{-x})$
- b. Hyperbolic cosine: $\cosh x = \frac{1}{2}(e^x + e^{-x})$
- c. Hyperbolic tangent: $\tanh x = \frac{\sinh x}{\cosh x}$
- d. Hyperbolic cotangent: $\coth x = \frac{1}{\tanh x} = \frac{\cosh x}{\sinh x}$
- e. Hyperbolic secant: $\operatorname{sech} x = \frac{1}{\cosh x}$
- f. Hyperbolic cosecant: $\operatorname{csch} x = \frac{1}{\sinh x}$

2. Sum or difference:

- a. $\cosh(x + y) = \cosh x \cosh y + \sinh x \sinh y$
- b. $\sinh(x - y) = \sinh x \cosh y - \cosh x \sinh y$
- c. $\cosh(x - y) = \cosh x \cosh y - \sinh x \sinh y$
- d. $\tanh(x + y) = \frac{\tanh x + \tanh y}{1 + \tanh x \tanh y}$
- e. $\tanh(x - y) = \frac{\tanh x - \tanh y}{1 - \tanh x \tanh y}$
- f. $\cosh^2 x - \sinh^2 x = 1$
- g. $\tanh^2 x + \operatorname{sech}^2 x = 1$
- h. $\coth^2 x - \operatorname{csch}^2 x = 1$

- i. $\cosh(x \pm jy) = \cosh x \cos y \pm j \sinh x \sin y$
- j. $\sinh(x \pm jy) = \sinh x \cos y \pm j \cosh x \sin y$

3. Series:

$$\text{a. } \sinh x = \frac{e^x - e^{-x}}{2} = x + \frac{x^3}{3!} + \frac{x^5}{5!} + \frac{x^7}{7!} + \dots$$

$$\text{b. } \cosh x = \frac{e^x + e^{-x}}{2} = 1 + \frac{x^2}{2!} + \frac{x^4}{4!} + \frac{x^6}{6!} + \dots$$

$$\text{c. } e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \dots$$

I.3 LOGARITHMIC

1. $\log_b(MN) = \log_b M + \log_b N$
2. $\log_b(M/N) = \log_b M - \log_b N$
3. $\log_b(1/N) = -\log_b N$
4. $\log_b(M^n) = n \log_b M$
5. $\log_b(M^{1/n}) = \frac{1}{n} \log_b M$
6. $\log_a N = \log_b N \cdot \log_a b = \log_b N / \log_b a$
7. $\log_e N = \log_{10} N \cdot \log_e 10 = 2.302585 \log_{10} N$
8. $\log_{10} N = \log_e N \cdot \log_{10} e = 0.434294 \log_e N$

APPENDIX II



Vector Analysis

II.1 VECTOR TRANSFORMATIONS

In this appendix we will indicate the vector transformations from rectangular to cylindrical (and vice versa), from cylindrical to spherical (and vice versa), and from rectangular to spherical (and vice versa). The three coordinate systems are shown in Figure II-1.

II.1.1 Rectangular to Cylindrical (and Vice Versa)

The coordinate transformation from rectangular (x, y, z) to cylindrical (ρ, ϕ, z) is given, referring to Figure II-1(b), by:

$$\begin{aligned} x &= \rho \cos \phi \\ y &= \rho \sin \phi \\ z &= z \end{aligned} \quad (\text{II-1})$$

In the rectangular coordinate system, we express a vector $\mathbf{A}$ as

$$\mathbf{A} = \hat{\mathbf{a}}_x A_x + \hat{\mathbf{a}}_y A_y + \hat{\mathbf{a}}_z A_z \quad (\text{II-2})$$

where $\hat{\mathbf{a}}_x$, $\hat{\mathbf{a}}_y$, $\hat{\mathbf{a}}_z$ are the unit vectors and A_x , A_y , A_z are the components of the vector $\mathbf{A}$ in the rectangular coordinate system. We wish to write $\mathbf{A}$ as

$$\mathbf{A} = \hat{\mathbf{a}}_\rho A_\rho + \hat{\mathbf{a}}_\phi A_\phi + \hat{\mathbf{a}}_z A_z \quad (\text{II-3})$$

where $\hat{\mathbf{a}}_\rho$, $\hat{\mathbf{a}}_\phi$, $\hat{\mathbf{a}}_z$ are the unit vectors and A_ρ , A_ϕ , A_z are the vector components in the cylindrical coordinate system. The z axis is common to both of them.

Referring to Figure II-2, we can write

$$\begin{aligned} \hat{\mathbf{a}}_x &= \hat{\mathbf{a}}_\rho \cos \phi - \hat{\mathbf{a}}_\phi \sin \phi \\ \hat{\mathbf{a}}_y &= \hat{\mathbf{a}}_\rho \sin \phi + \hat{\mathbf{a}}_\phi \cos \phi \\ \hat{\mathbf{a}}_z &= \hat{\mathbf{a}}_z \end{aligned} \quad (\text{II-4})$$

Balanis' Advanced Engineering Electromagnetics, Third Edition. Constantine A. Balanis.

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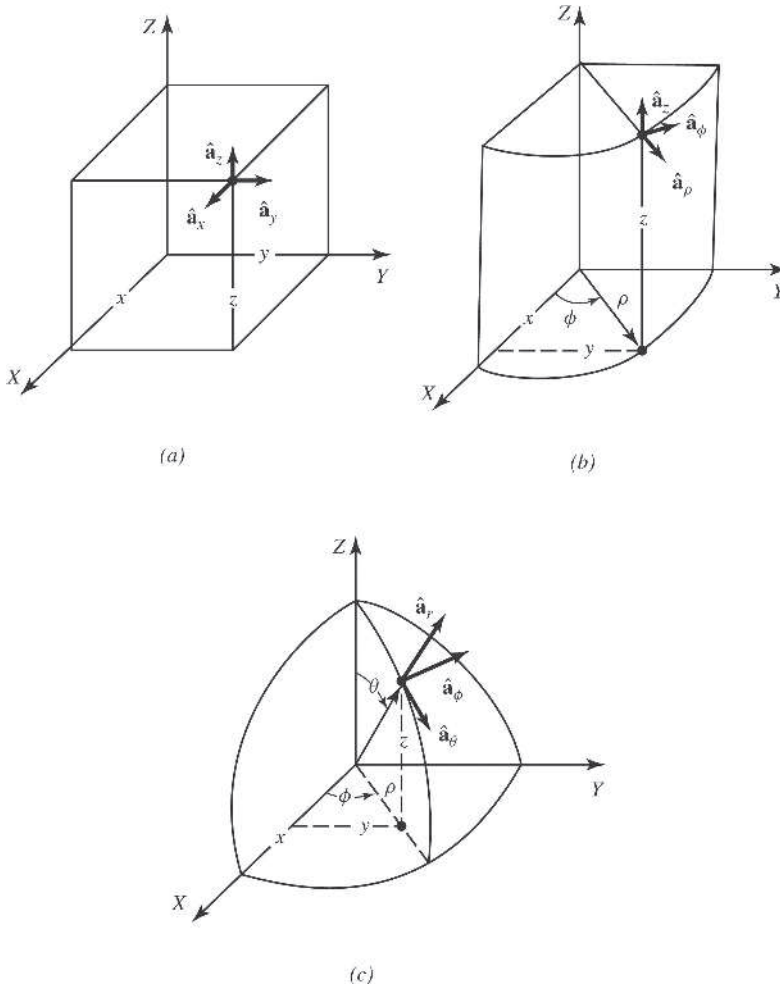


Figure II-1 Coordinate systems. (a) Rectangular. (b) Cylindrical. (c) Spherical. (Source: C. A. Balanis, *Antenna Theory: Analysis and Design*; Third Edition, copyright © 2005, John Wiley & Sons, Inc.; reprinted by permission of John Wiley & Sons, Inc.)

Using (II-4) reduces (II-2) to

$$\begin{aligned} \mathbf{A} &= (\hat{a}_\rho \cos \phi - \hat{a}_\phi \sin \phi)A_x + (\hat{a}_\rho \sin \phi + \hat{a}_\phi \cos \phi)A_y + \hat{a}_z A_z \\ \mathbf{A} &= \hat{a}_\rho (A_x \cos \phi + A_y \sin \phi) + \hat{a}_\phi (-A_x \sin \phi + A_y \cos \phi) + \hat{a}_z A_z \end{aligned} \quad (\text{II-5})$$

which when compared with (II-3) leads to

$$\begin{aligned} A_\rho &= A_x \cos \phi + A_y \sin \phi \\ A_\phi &= -A_x \sin \phi + A_y \cos \phi \\ A_z &= A_z \end{aligned} \quad (\text{II-6})$$

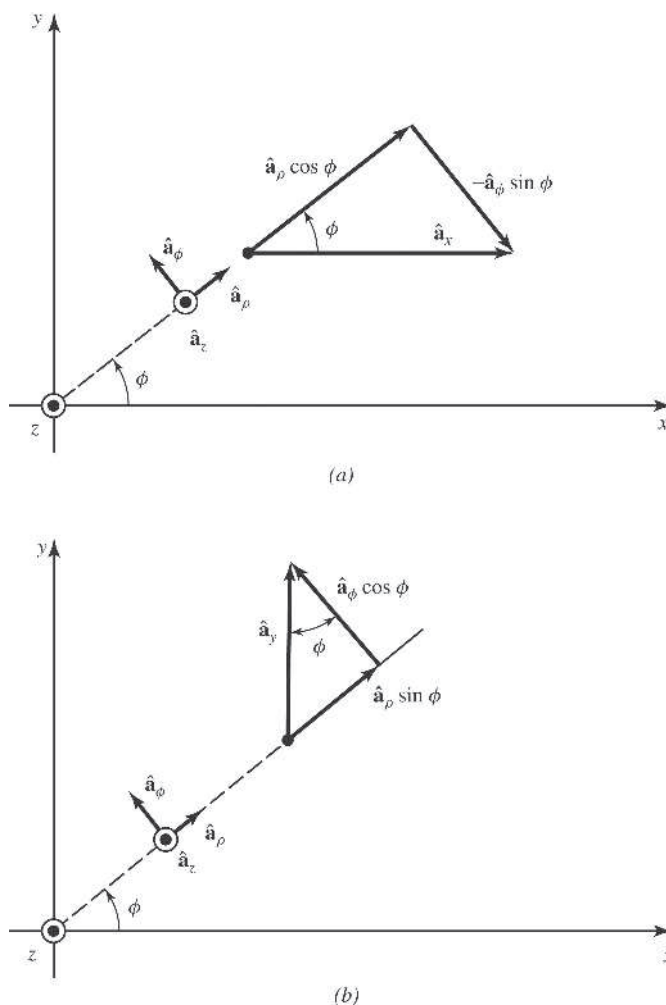


Figure II-2 Geometrical representation of transformation between unit vectors of rectangular and cylindrical coordinate systems. (a) Geometry for unit vector $\hat{a}_x$. (b) Geometry for unit vector $\hat{a}_y$. (Source: C. A. Balanis, *Antenna Theory: Analysis and Design*; Third Edition, copyright © 2005, John Wiley & Sons, Inc.; reprinted by permission of John Wiley & Sons, Inc.)

In matrix form, (II-6) can be written as

$$\begin{pmatrix} A_\rho \\ A_\phi \\ A_z \end{pmatrix} = \begin{pmatrix} \cos \phi & \sin \phi & 0 \\ -\sin \phi & \cos \phi & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} A_x \\ A_y \\ A_z \end{pmatrix} \quad (\text{II-6a})$$

where

$$[A]_{rc} = \begin{pmatrix} \cos \phi & \sin \phi & 0 \\ -\sin \phi & \cos \phi & 0 \\ 0 & 0 & 1 \end{pmatrix} \quad (\text{II-6b})$$

is the transformation matrix for rectangular-to-cylindrical components.

Since $[A]_{rc}$ is an orthonormal matrix (its inverse is equal to its transpose), we can write the transformation matrix for cylindrical-to-rectangular components as

$$[A]_{cr} = [A]_{rc}^{-1} = [A]_{rc}^t = \begin{bmatrix} \cos \phi & -\sin \phi & 0 \\ \sin \phi & \cos \phi & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad (\text{II-7})$$

or

$$\begin{pmatrix} A_x \\ A_y \\ A_z \end{pmatrix} = \begin{pmatrix} \cos \phi & -\sin \phi & 0 \\ \sin \phi & \cos \phi & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} A_\rho \\ A_\phi \\ A_z \end{pmatrix} \quad (\text{II-7a})$$

or

$$\begin{aligned} A_x &= A_\rho \cos \phi - A_\phi \sin \phi \\ A_y &= A_\rho \sin \phi + A_\phi \cos \phi \\ A_z &= A_z \end{aligned} \quad (\text{II-7b})$$

II.1.2 Cylindrical to Spherical (and Vice Versa)

Referring to Figure II-1c, we can write that the cylindrical and spherical coordinates are related by

$$\begin{aligned} \rho &= r \sin \theta \\ z &= r \cos \theta \end{aligned} \quad (\text{II-8})$$

In a geometrical approach, similar to the one employed in the previous section, we can show that the cylindrical-to-spherical transformation of vector components is given by

$$\begin{aligned} A_r &= A_\rho \sin \theta + A_z \cos \theta \\ A_\theta &= A_\rho \cos \theta - A_z \sin \theta \\ A_\phi &= A_\phi \end{aligned} \quad (\text{II-9})$$

or in matrix form by

$$\begin{pmatrix} A_r \\ A_\theta \\ A_\phi \end{pmatrix} = \begin{pmatrix} \sin \theta & 0 & \cos \theta \\ \cos \theta & 0 & -\sin \theta \\ 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} A_\rho \\ A_\phi \\ A_z \end{pmatrix} \quad (\text{II-9a})$$

Thus, the cylindrical-to-spherical transformation matrix can be written as

$$[A]_{cs} = \begin{bmatrix} \sin \theta & 0 & \cos \theta \\ \cos \theta & 0 & -\sin \theta \\ 0 & 1 & 0 \end{bmatrix} \quad (\text{II-9b})$$

The $[A]_{cs}$ matrix is also orthonormal so that its inverse is given by

$$[A]_{sc} = [A]_{cs}^{-1} = [A]_{cs}^t = \begin{bmatrix} \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \\ \cos \theta & -\sin \theta & 0 \end{bmatrix} \quad (\text{II-10})$$

and the spherical-to-cylindrical transformation is accomplished by

$$\begin{pmatrix} A_\rho \\ A_\phi \\ A_z \end{pmatrix} = \begin{pmatrix} \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \\ \cos \theta & -\sin \theta & 0 \end{pmatrix} \begin{pmatrix} A_r \\ A_\theta \\ A_\phi \end{pmatrix} \quad (\text{II-10a})$$

or

$$\begin{aligned} A_\rho &= A_r \sin \theta + A_\theta \cos \theta \\ A_\phi &= A_\phi \\ A_z &= A_r \cos \theta - A_\theta \sin \theta \end{aligned} \quad (\text{II-10b})$$

This time the component A_ϕ and coordinate ϕ are the same in both systems.

II.1.3 Rectangular to Spherical (and Vice Versa)

Many times it may be required that a transformation be performed directly from rectangular to spherical components. By referring to Figure II-1c, we can write that the rectangular and spherical coordinates are related by

$$\begin{aligned} x &= r \sin \theta \cos \phi \\ y &= r \sin \theta \sin \phi \\ z &= r \cos \theta \end{aligned} \quad (\text{II-11})$$

and the rectangular and spherical components by

$$\begin{aligned} A_r &= A_x \sin \theta \cos \phi + A_y \sin \theta \sin \phi + A_z \cos \theta \\ A_\theta &= A_x \cos \theta \cos \phi + A_y \cos \theta \sin \phi - A_z \sin \theta \\ A_\phi &= -A_x \sin \phi + A_y \cos \phi \end{aligned} \quad (\text{II-12})$$

which can also be obtained by substituting (II-6) into (II-9). In matrix form (II-12) can be written as

$$\begin{pmatrix} A_r \\ A_\theta \\ A_\phi \end{pmatrix} = \begin{pmatrix} \sin \theta \cos \phi & \sin \theta \sin \phi & \cos \theta \\ \cos \theta \cos \phi & \cos \theta \sin \phi & -\sin \theta \\ -\sin \phi & \cos \phi & 0 \end{pmatrix} \begin{pmatrix} A_x \\ A_y \\ A_z \end{pmatrix} \quad (\text{II-12a})$$

with the rectangular-to-spherical transformation matrix being

$$[A]_{rs} = \begin{bmatrix} \sin \theta \cos \phi & \sin \theta \sin \phi & \cos \theta \\ \cos \theta \cos \phi & \cos \theta \sin \phi & -\sin \theta \\ -\sin \phi & \cos \phi & 0 \end{bmatrix} \quad (\text{II-12b})$$

The transformation matrix of (II-12b) is also orthonormal so that its inverse can be written as

$$[A]_{sr} = [A]_{rs}^{-1} = [A]_{rs}^t = \begin{bmatrix} \sin \theta \cos \phi & \cos \theta \cos \phi & -\sin \phi \\ \sin \theta \sin \phi & \cos \theta \sin \phi & \cos \phi \\ \cos \theta & -\sin \theta & 0 \end{bmatrix} \quad (\text{II-13})$$

and the spherical to rectangular transformation is accomplished by

$$\begin{pmatrix} A_x \\ A_y \\ A_z \end{pmatrix} = \begin{pmatrix} \sin \theta \cos \phi & \cos \theta \cos \phi & -\sin \phi \\ \sin \theta \sin \phi & \cos \theta \sin \phi & \cos \phi \\ \cos \theta & -\sin \theta & 0 \end{pmatrix} \begin{pmatrix} A_r \\ A_\theta \\ A_\phi \end{pmatrix} \quad (\text{II-13a})$$

or

$$\begin{aligned}
A_x &= A_r \sin \theta \cos \phi + A_\theta \cos \theta \cos \phi - A_\phi \sin \phi \\
A_y &= A_r \sin \theta \sin \phi + A_\theta \cos \theta \sin \phi + A_\phi \cos \phi \\
A_z &= A_r \cos \theta - A_\theta \sin \theta
\end{aligned} \tag{II-13b}$$

II.2 VECTOR DIFFERENTIAL OPERATORS

The differential operators of gradient of a scalar ($\nabla\psi$), divergence of a vector ($\nabla \cdot \mathbf{A}$), curl of a vector ($\nabla \times \mathbf{A}$), Laplacian of a scalar ($\nabla^2\psi$), and Laplacian of a vector ($\nabla^2\mathbf{A}$), frequently encountered in electromagnetic field analysis, will be listed in the rectangular, cylindrical, and spherical coordinate systems.

II.2.1 Rectangular Coordinates

$$\nabla\psi = \hat{\mathbf{a}}_x \frac{\partial\psi}{\partial x} + \hat{\mathbf{a}}_y \frac{\partial\psi}{\partial y} + \hat{\mathbf{a}}_z \frac{\partial\psi}{\partial z} \tag{II-14}$$

$$\nabla \cdot \mathbf{A} = \frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z} \tag{II-15}$$

$$\nabla \times \mathbf{A} = \hat{\mathbf{a}}_x \left(\frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z} \right) + \hat{\mathbf{a}}_y \left(\frac{\partial A_x}{\partial z} - \frac{\partial A_z}{\partial x} \right) + \hat{\mathbf{a}}_z \left(\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y} \right) \tag{II-16}$$

$$\nabla \cdot \nabla\psi = \nabla^2\psi = \frac{\partial^2\psi}{\partial x^2} + \frac{\partial^2\psi}{\partial y^2} + \frac{\partial^2\psi}{\partial z^2} \tag{II-17}$$

$$\nabla^2\mathbf{A} = \hat{\mathbf{a}}_x \nabla^2 A_x + \hat{\mathbf{a}}_y \nabla^2 A_y + \hat{\mathbf{a}}_z \nabla^2 A_z \tag{II-18}$$

II.2.2 Cylindrical Coordinates

$$\nabla\psi = \hat{\mathbf{a}}_\rho \frac{\partial\psi}{\partial \rho} + \hat{\mathbf{a}}_\phi \frac{1}{\rho} \frac{\partial\psi}{\partial \phi} + \hat{\mathbf{a}}_z \frac{\partial\psi}{\partial z} \tag{II-19}$$

$$\nabla \cdot \mathbf{A} = \frac{1}{\rho} \frac{\partial}{\partial \rho} (\rho A_\rho) + \frac{1}{\rho} \frac{\partial A_\phi}{\partial \phi} + \frac{\partial A_z}{\partial z} \tag{II-20}$$

$$\begin{aligned}
\nabla \times \mathbf{A} &= \hat{\mathbf{a}}_\rho \left(\frac{1}{\rho} \frac{\partial A_z}{\partial \phi} - \frac{\partial A_\phi}{\partial z} \right) + \hat{\mathbf{a}}_\phi \left(\frac{\partial A_\rho}{\partial z} - \frac{\partial A_z}{\partial \rho} \right) \\
&\quad + \hat{\mathbf{a}}_z \left(\frac{1}{\rho} \frac{\partial (\rho A_\phi)}{\partial \rho} - \frac{1}{\rho} \frac{\partial A_\rho}{\partial \phi} \right)
\end{aligned} \tag{II-21}$$

$$\nabla^2\psi = \frac{1}{\rho} \frac{\partial}{\partial \rho} \left(\rho \frac{\partial\psi}{\partial \rho} \right) + \frac{1}{\rho^2} \frac{\partial^2\psi}{\partial \phi^2} + \frac{\partial^2\psi}{\partial z^2} \tag{II-22}$$

$$\nabla^2\mathbf{A} = \nabla(\nabla \cdot \mathbf{A}) - \nabla \times \nabla \times \mathbf{A} \tag{II-23}$$

or in an expanded form

$$\begin{aligned}\nabla^2 \mathbf{A} = & \hat{\mathbf{a}}_\rho \left(\frac{\partial^2 A_\rho}{\partial \rho^2} + \frac{1}{\rho} \frac{\partial A_\rho}{\partial \rho} - \frac{A_\rho}{\rho^2} + \frac{1}{\rho^2} \frac{\partial^2 A_\rho}{\partial \phi^2} - \frac{2}{\rho^2} \frac{\partial A_\phi}{\partial \phi} + \frac{\partial^2 A_\rho}{\partial z^2} \right) \\ & + \hat{\mathbf{a}}_\phi \left(\frac{\partial^2 A_\phi}{\partial \rho^2} + \frac{1}{\rho} \frac{\partial A_\phi}{\partial \rho} - \frac{A_\phi}{\rho^2} + \frac{1}{\rho^2} \frac{\partial^2 A_\phi}{\partial \phi^2} + \frac{2}{\rho^2} \frac{\partial A_\rho}{\partial \phi} + \frac{\partial^2 A_\phi}{\partial z^2} \right) \\ & + \hat{\mathbf{a}}_z \left(\frac{\partial^2 A_z}{\partial \rho^2} + \frac{1}{\rho} \frac{\partial A_z}{\partial \rho} + \frac{1}{\rho^2} \frac{\partial^2 A_z}{\partial \phi^2} + \frac{\partial^2 A_z}{\partial z^2} \right)\end{aligned}\quad (\text{II-23a})$$

In the cylindrical coordinate system $\nabla^2 \mathbf{A} \neq \hat{\mathbf{a}}_\rho \nabla^2 A_\rho + \hat{\mathbf{a}}_\phi \nabla^2 A_\phi + \hat{\mathbf{a}}_z \nabla^2 A_z$ because the orientation of the unit vectors $\hat{\mathbf{a}}_\rho$ and $\hat{\mathbf{a}}_\phi$ varies with the ρ and ϕ coordinates.

II.2.3 Spherical Coordinates

$$\nabla \psi = \hat{\mathbf{a}}_r \frac{\partial \psi}{\partial r} + \hat{\mathbf{a}}_\theta \frac{1}{r} \frac{\partial \psi}{\partial \theta} + \hat{\mathbf{a}}_\phi \frac{1}{r \sin \theta} \frac{\partial \psi}{\partial \phi} \quad (\text{II-24})$$

$$\nabla \cdot \mathbf{A} = \frac{1}{r^2} \frac{\partial}{\partial r} (r^2 A_r) + \frac{1}{r \sin \theta} \frac{\partial}{\partial \theta} (\sin \theta A_\theta) + \frac{1}{r \sin \theta} \frac{\partial A_\phi}{\partial \phi} \quad (\text{II-25})$$

$$\begin{aligned}\nabla \times \mathbf{A} = & \frac{\hat{\mathbf{a}}_r}{r \sin \theta} \left[\frac{\partial}{\partial \theta} (A_\phi \sin \theta) - \frac{\partial A_\theta}{\partial \phi} \right] + \frac{\hat{\mathbf{a}}_\theta}{r} \left[\frac{1}{\sin \theta} \frac{\partial A_r}{\partial \phi} - \frac{\partial}{\partial r} (r A_\phi) \right] \\ & + \frac{\hat{\mathbf{a}}_\phi}{r} \left[\frac{\partial}{\partial r} (r A_\theta) - \frac{\partial A_r}{\partial \theta} \right]\end{aligned}\quad (\text{II-26})$$

$$\nabla^2 \psi = \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial \psi}{\partial r} \right) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial \psi}{\partial \theta} \right) + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2 \psi}{\partial \phi^2} \quad (\text{II-27})$$

$$\nabla^2 \mathbf{A} = \nabla (\nabla \cdot \mathbf{A}) - \nabla \times \nabla \times \mathbf{A} \quad (\text{II-28})$$

or in an expanded form

$$\begin{aligned}\nabla^2 \mathbf{A} = & \hat{\mathbf{a}}_r \left(\frac{\partial^2 A_r}{\partial r^2} + \frac{2}{r} \frac{\partial A_r}{\partial r} - \frac{2}{r^2} A_r + \frac{1}{r^2} \frac{\partial^2 A_r}{\partial \theta^2} + \frac{\cot \theta}{r^2} \frac{\partial A_r}{\partial \theta} + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2 A_r}{\partial \phi^2} \right. \\ & \left. - \frac{2}{r^2} \frac{\partial A_\theta}{\partial \theta} - \frac{2 \cot \theta}{r^2} A_\theta - \frac{2}{r^2 \sin \theta} \frac{\partial A_\phi}{\partial \phi} \right) \\ & + \hat{\mathbf{a}}_\theta \left(\frac{\partial^2 A_\theta}{\partial r^2} + \frac{2}{r} \frac{\partial A_\theta}{\partial r} - \frac{A_\theta}{r^2 \sin^2 \theta} + \frac{1}{r^2} \frac{\partial^2 A_\theta}{\partial \theta^2} + \frac{\cot \theta}{r^2} \frac{\partial A_\theta}{\partial \theta} \right. \\ & \left. + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2 A_\theta}{\partial \phi^2} + \frac{2}{r^2} \frac{\partial A_r}{\partial \theta} - \frac{2 \cot \theta}{r^2 \sin \theta} \frac{\partial A_\phi}{\partial \phi} \right) \\ & + \hat{\mathbf{a}}_\phi \left(\frac{\partial^2 A_\phi}{\partial r^2} + \frac{2}{r} \frac{\partial A_\phi}{\partial r} - \frac{1}{r^2 \sin^2 \theta} A_\phi + \frac{1}{r^2} \frac{\partial^2 A_\phi}{\partial \theta^2} \right. \\ & \left. + \frac{\cot \theta}{r^2} \frac{\partial A_\phi}{\partial \theta} + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2 A_\phi}{\partial \phi^2} + \frac{2}{r^2 \sin \theta} \frac{\partial A_r}{\partial \phi} + \frac{2 \cot \theta}{r^2 \sin \theta} \frac{\partial A_\theta}{\partial \phi} \right)\end{aligned}\quad (\text{II-28a})$$

Again note that $\nabla^2 \mathbf{A} \neq \hat{\mathbf{a}}_r \nabla^2 A_r + \hat{\mathbf{a}}_\theta \nabla^2 A_\theta + \hat{\mathbf{a}}_\phi \nabla^2 A_\phi$ since the orientation of the unit vectors $\hat{\mathbf{a}}_r, \hat{\mathbf{a}}_\theta$, and $\hat{\mathbf{a}}_\phi$ varies with the r, θ , and ϕ coordinates.

II.3 VECTOR IDENTITIES

II.3.1 Addition and Multiplication

$$\mathbf{A} \cdot \mathbf{A} = |\mathbf{A}|^2 \quad (\text{II-29})$$

$$\mathbf{A} \cdot \mathbf{A}^* = |\mathbf{A}|^2 \quad (\text{II-30})$$

$$\mathbf{A} + \mathbf{B} = \mathbf{B} + \mathbf{A} \quad (\text{II-31})$$

$$\mathbf{A} \cdot \mathbf{B} = \mathbf{B} \cdot \mathbf{A} \quad (\text{II-32})$$

$$\mathbf{A} \times \mathbf{B} = -\mathbf{B} \times \mathbf{A} \quad (\text{II-33})$$

$$(\mathbf{A} + \mathbf{B}) \cdot \mathbf{C} = \mathbf{A} \cdot \mathbf{C} + \mathbf{B} \cdot \mathbf{C} \quad (\text{II-34})$$

$$(\mathbf{A} + \mathbf{B}) \times \mathbf{C} = \mathbf{A} \times \mathbf{C} + \mathbf{B} \times \mathbf{C} \quad (\text{II-35})$$

$$\mathbf{A} \cdot \mathbf{B} \times \mathbf{C} = \mathbf{B} \cdot \mathbf{C} \times \mathbf{A} = \mathbf{C} \cdot \mathbf{A} \times \mathbf{B} \quad (\text{II-36})$$

$$\mathbf{A} \times (\mathbf{B} \times \mathbf{C}) = (\mathbf{A} \cdot \mathbf{C})\mathbf{B} - (\mathbf{A} \cdot \mathbf{B})\mathbf{C} \quad (\text{II-37})$$

$$\begin{aligned} (\mathbf{A} \times \mathbf{B}) \cdot (\mathbf{C} \times \mathbf{D}) &= \mathbf{A} \cdot \mathbf{B} \times (\mathbf{C} \times \mathbf{D}) \\ &= \mathbf{A} \cdot (\mathbf{B} \cdot \mathbf{D}\mathbf{C} - \mathbf{B} \cdot \mathbf{C}\mathbf{D}) \\ &= (\mathbf{A} \cdot \mathbf{C})(\mathbf{B} \cdot \mathbf{D}) - (\mathbf{A} \cdot \mathbf{D})(\mathbf{B} \cdot \mathbf{C}) \end{aligned} \quad (\text{II-38})$$

$$(\mathbf{A} \times \mathbf{B}) \times (\mathbf{C} \times \mathbf{D}) = (\mathbf{A} \times \mathbf{B} \cdot \mathbf{D})\mathbf{C} - (\mathbf{A} \times \mathbf{B} \cdot \mathbf{C})\mathbf{D} \quad (\text{II-39})$$

II.3.2 Differentiation

$$\nabla \cdot (\nabla \times \mathbf{A}) = 0 \quad (\text{II-40})$$

$$\nabla \times \nabla \psi = 0 \quad (\text{II-41})$$

$$\nabla(\phi + \psi) = \nabla\phi + \nabla\psi \quad (\text{II-42})$$

$$\nabla(\phi\psi) = \phi\nabla\psi + \psi\nabla\phi \quad (\text{II-43})$$

$$\nabla \cdot (\mathbf{A} + \mathbf{B}) = \nabla \cdot \mathbf{A} + \nabla \cdot \mathbf{B} \quad (\text{II-44})$$

$$\nabla \times (\mathbf{A} + \mathbf{B}) = \nabla \times \mathbf{A} + \nabla \times \mathbf{B} \quad (\text{II-45})$$

$$\nabla \cdot (\psi\mathbf{A}) = \mathbf{A} \cdot \nabla\psi + \psi\nabla \cdot \mathbf{A} \quad (\text{II-46})$$

$$\nabla \times (\psi\mathbf{A}) = \nabla\psi \times \mathbf{A} + \psi\nabla \times \mathbf{A} \quad (\text{II-47})$$

$$\nabla(\mathbf{A} \cdot \mathbf{B}) = (\mathbf{A} \cdot \nabla)\mathbf{B} + (\mathbf{B} \cdot \nabla)\mathbf{A} + \mathbf{A} \times (\nabla \times \mathbf{B}) + \mathbf{B} \times (\nabla \times \mathbf{A}) \quad (\text{II-48})$$

$$\nabla \cdot (\mathbf{A} \times \mathbf{B}) = \mathbf{B} \cdot \nabla \times \mathbf{A} - \mathbf{A} \cdot \nabla \times \mathbf{B} \quad (\text{II-49})$$

$$\nabla \times (\mathbf{A} \times \mathbf{B}) = \mathbf{A}\nabla \cdot \mathbf{B} - \mathbf{B}\nabla \cdot \mathbf{A} + (\mathbf{B} \cdot \nabla)\mathbf{A} - (\mathbf{A} \cdot \nabla)\mathbf{B} \quad (\text{II-50})$$

$$\nabla \times \nabla \times \mathbf{A} = \nabla(\nabla \cdot \mathbf{A}) - \nabla^2 \mathbf{A} \quad (\text{II-51})$$

II.3.3 Integration

$$\oint_C \mathbf{A} \cdot d\mathbf{l} = \iint_S (\nabla \times \mathbf{A}) \cdot d\mathbf{s} \quad \text{Stokes' theorem} \quad (\text{II-52})$$

$$\oiint_S \mathbf{A} \cdot d\mathbf{s} = \iiint_V (\nabla \cdot \mathbf{A}) \, dv \quad \text{divergence theorem} \quad (\text{II-53})$$

$$\oiint_S (\hat{\mathbf{n}} \times \mathbf{A}) \, ds = \iiint_V (\nabla \times \mathbf{A}) \, dv \quad (\text{II-54})$$

$$\oiint_S \psi \, ds = \iiint_V \nabla \psi \, dv \quad (\text{II-55})$$

$$\oint_C \psi \, dl = \iint_S \hat{\mathbf{n}} \times \nabla \psi \, ds \quad (\text{II-56})$$

APPENDIX III



Fresnel Integrals

$$C_0(x) = \int_0^x \frac{\cos(\tau)}{\sqrt{2\pi\tau}} d\tau \quad (\text{III-1})$$

$$S_0(x) = \int_0^x \frac{\sin(\tau)}{\sqrt{2\pi\tau}} d\tau \quad (\text{III-2})$$

$$C(x) = \int_0^x \cos\left(\frac{\pi}{2}\tau^2\right) d\tau \quad (\text{III-3})$$

$$S(x) = \int_0^x \sin\left(\frac{\pi}{2}\tau^2\right) d\tau \quad (\text{III-4})$$

$$C_1(x) = \int_x^\infty \cos(\tau^2) d\tau \quad (\text{III-5})$$

$$S_1(x) = \int_x^\infty \sin(\tau^2) d\tau \quad (\text{III-6})$$

$$C(x) - jS(x) = \int_0^x e^{-j(\pi/2)\tau^2} d\tau = \int_0^{(\pi/2)x^2} \frac{e^{-j\tau}}{\sqrt{2\pi\tau}} d\tau$$

$$C(x) - jS(x) = C_0\left(\frac{\pi}{2}x^2\right) - jS_0\left(\frac{\pi}{2}x^2\right) \quad (\text{III-7})$$

$$C_1(x) - jS_1(x) = \int_x^\infty e^{-j\tau^2} d\tau = \sqrt{\frac{\pi}{2}} \int_{x^2}^\infty \frac{e^{-j\tau}}{\sqrt{2\pi\tau}} d\tau$$

$$C_1(x) - jS_1(x) = \sqrt{\frac{\pi}{2}} \left\{ \int_0^\infty \frac{e^{-j\tau}}{\sqrt{2\pi\tau}} d\tau - \int_0^{x^2} \frac{e^{-j\tau}}{\sqrt{2\pi\tau}} d\tau \right\}$$

$$C_1(x) - jS_1(x) = \sqrt{\frac{\pi}{2}} \left\{ \left[\frac{1}{2} - j\frac{1}{2} \right] - [C_0(x^2) - jS_0(x^2)] \right\}$$

$$C_1(x) - jS_1(x) = \sqrt{\frac{\pi}{2}} \left\{ \left[\frac{1}{2} - C_0(x^2) \right] - j \left[\frac{1}{2} - S_0(x^2) \right] \right\} \quad (\text{III-8})$$

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x	$C_1(x)$	$S_1(x)$	$C(x)$	$S(x)$
0.0	0.62666	0.62666	0.0	0.0
0.1	0.52666	0.62632	0.10000	0.00052
0.2	0.42669	0.62399	0.19992	0.00419
0.3	0.32690	0.61766	0.29940	0.01412
0.4	0.22768	0.60536	0.39748	0.03336
0.5	0.12977	0.58518	0.49234	0.06473
0.6	0.03439	0.55532	0.58110	0.11054
0.7	-0.05672	0.51427	0.65965	0.17214
0.8	-0.14119	0.46092	0.72284	0.24934
0.9	-0.21606	0.39481	0.76482	0.33978
1.0	-0.27787	0.31639	0.77989	0.43826
1.1	-0.32285	0.22728	0.76381	0.53650
1.2	-0.34729	0.13054	0.71544	0.62340
1.3	-0.34803	0.03081	0.63855	0.68633
1.4	-0.32312	-0.06573	0.54310	0.71353
1.5	-0.27253	-0.15158	0.44526	0.69751
1.6	-0.19886	-0.21861	0.36546	0.63889
1.7	-0.10790	-0.25905	0.32383	0.54920
1.8	-0.00871	-0.26682	0.33363	0.45094
1.9	0.08680	-0.23918	0.39447	0.37335
2.0	0.16520	-0.17812	0.48825	0.34342
2.1	0.21359	-0.09141	0.58156	0.37427
2.2	0.22242	0.00743	0.63629	0.45570
2.3	0.18833	0.10054	0.62656	0.55315
2.4	0.11650	0.16879	0.55496	0.61969
2.5	0.02135	0.19614	0.45742	0.61918
2.6	-0.07518	0.17454	0.38894	0.54999
2.7	-0.14816	0.10789	0.39249	0.45292
2.8	-0.17646	0.01329	0.46749	0.39153
2.9	-0.15021	-0.08181	0.56237	0.41014
3.0	-0.07621	-0.14690	0.60572	0.49631
3.1	0.02152	-0.15883	0.56160	0.58181
3.2	0.10791	-0.11181	0.46632	0.59335
3.3	0.14907	-0.02260	0.40570	0.51929
3.4	0.12691	0.07301	0.43849	0.42965
3.5	0.04965	0.13335	0.53257	0.41525
3.6	-0.04819	0.12973	0.58795	0.49231
3.7	-0.11929	0.06258	0.54195	0.57498
3.8	-0.12649	-0.03483	0.44810	0.56562
3.9	-0.06469	-0.11030	0.42233	0.47521
4.0	0.03219	-0.12048	0.49842	0.42052
4.1	0.10690	-0.05815	0.57369	0.47580
4.2	0.11228	0.03885	0.54172	0.56320
4.3	0.04374	0.10751	0.44944	0.55400
4.4	-0.05287	0.10038	0.43833	0.46227
4.5	-0.10884	0.02149	0.52602	0.43427
4.6	-0.08188	-0.07126	0.56724	0.51619
4.7	0.00810	-0.10594	0.49143	0.56715
4.8	0.08905	-0.05381	0.43380	0.49675
4.9	0.09277	0.04224	0.50016	0.43507
5.0	0.01519	0.09874	0.56363	0.49919
5.1	-0.07411	0.06405	0.49979	0.56239

x	$C_1(x)$	$S_1(x)$	$C(x)$	$S(x)$
5.2	-0.09125	-0.03004	0.43889	0.49688
5.3	-0.01892	-0.09235	0.50778	0.44047
5.4	0.07063	-0.05976	0.55723	0.51403
5.5	0.08408	0.03440	0.47843	0.55369
5.6	0.00641	0.08900	0.45171	0.47004
5.7	-0.07642	0.04296	0.53846	0.45953
5.8	-0.06919	-0.05135	0.52984	0.54604
5.9	0.01998	-0.08231	0.44859	0.51633
6.0	0.08245	-0.01181	0.49953	0.44696
6.1	0.03946	0.07180	0.54950	0.51647
6.2	-0.05363	0.06018	0.46761	0.53982
6.3	-0.07284	-0.03144	0.47600	0.45555
6.4	0.00835	-0.07765	0.54960	0.49649
6.5	0.07574	-0.01326	0.48161	0.54538
6.6	0.03183	0.06872	0.46899	0.46307
6.7	-0.05828	0.04658	0.54674	0.49150
6.8	-0.05734	-0.04600	0.48307	0.54364
6.9	0.03317	-0.06440	0.47322	0.46244
7.0	0.06832	0.02077	0.54547	0.49970
7.1	-0.00944	0.06977	0.47332	0.53602
7.2	-0.06943	0.00041	0.48874	0.45725
7.3	-0.00864	-0.06793	0.53927	0.51894
7.4	0.06582	-0.01521	0.46010	0.51607
7.5	0.02018	0.06353	0.51601	0.46070
7.6	-0.06137	0.02367	0.51564	0.53885
7.7	-0.02580	-0.05958	0.46278	0.48202
7.8	0.05828	-0.02668	0.53947	0.48964
7.9	0.02638	0.05752	0.47598	0.53235
8.0	-0.05730	0.02494	0.49980	0.46021
8.1	-0.02238	-0.05752	0.52275	0.53204
8.2	0.05803	-0.01870	0.46384	0.48589
8.3	0.01387	0.05861	0.53775	0.49323
8.4	-0.05899	0.00789	0.47092	0.52429
8.5	-0.00080	-0.05881	0.51417	0.46534
8.6	0.05767	0.00729	0.50249	0.53693
8.7	-0.01616	0.05515	0.48274	0.46774
8.8	-0.05079	-0.02545	0.52797	0.52294
8.9	0.03461	-0.04425	0.46612	0.48856
9.0	0.03526	0.04293	0.53537	0.49985
9.1	-0.04951	0.02381	0.46661	0.51042
9.2	-0.01021	-0.05338	0.52914	0.48135
9.3	0.05354	0.00485	0.47628	0.52467
9.4	-0.02020	0.04920	0.51803	0.47134
9.5	-0.03995	-0.03426	0.48729	0.53100
9.6	0.04513	-0.02599	0.50813	0.46786
9.7	0.00837	0.05086	0.49549	0.53250
9.8	-0.04983	-0.01094	0.50192	0.46758
9.9	0.02916	-0.04124	0.49961	0.53215
10.0	0.02554	0.04298	0.49989	0.46817
10.1	-0.04927	0.00478	0.49961	0.53151
10.2	0.01738	-0.04583	0.50186	0.46885
10.3	0.03233	0.03621	0.49575	0.53061
10.4	-0.04681	0.01094	0.50751	0.47033

x	$C_1(x)$	$S_1(x)$	$C(x)$	$S(x)$
10.5	0.01360	−0.04563	0.48849	0.52804
10.6	0.03187	0.03477	0.51601	0.47460
10.7	−0.04595	0.00848	0.47936	0.52143
10.8	0.01789	−0.04270	0.52484	0.48413
10.9	0.02494	0.03850	0.47211	0.50867
11.0	−0.04541	−0.00202	0.52894	0.49991
11.1	0.02845	−0.03492	0.47284	0.49079
11.2	0.01008	0.04349	0.52195	0.51805
11.3	−0.03981	−0.01930	0.48675	0.47514
11.4	0.04005	−0.01789	0.50183	0.52786
11.5	−0.01282	0.04155	0.51052	0.47440
11.6	−0.02188	−0.03714	0.47890	0.51755
11.7	0.04164	0.00962	0.52679	0.49525
11.8	−0.03580	0.02267	0.47489	0.49013
11.9	0.00977	−0.04086	0.51544	0.52184
12.0	0.02059	0.03622	0.49993	0.47347
12.1	−0.03919	−0.01309	0.48426	0.52108
12.2	0.03792	−0.01555	0.52525	0.49345
12.3	−0.01914	0.03586	0.47673	0.48867
12.4	−0.00728	−0.03966	0.50951	0.52384
12.5	0.02960	0.02691	0.50969	0.47645
12.6	−0.03946	−0.00421	0.47653	0.50936
12.7	0.03445	−0.01906	0.52253	0.51097
12.8	−0.01783	0.03475	0.49376	0.47593
12.9	−0.00377	−0.03857	0.48523	0.51977
13.0	0.02325	0.03064	0.52449	0.49994
13.1	−0.03530	−0.01452	0.48598	0.48015
13.2	0.03760	−0.00459	0.49117	0.52244
13.3	−0.03075	0.02163	0.52357	0.49583
13.4	0.01744	−0.03299	0.48482	0.48173
13.5	−0.00129	0.03701	0.49103	0.52180
13.6	−0.01421	−0.03391	0.52336	0.49848
13.7	0.02639	0.02521	0.48908	0.47949
13.8	−0.03377	−0.01313	0.48534	0.51781
13.9	0.03597	−0.00002	0.52168	0.50737
14.0	−0.03352	0.01232	0.49996	0.47726
14.1	0.02749	−0.02240	0.47844	0.50668
14.2	−0.01916	0.02954	0.51205	0.51890
14.3	0.00979	−0.03357	0.51546	0.48398
14.4	−0.00043	0.03472	0.48131	0.48819
14.5	−0.00817	−0.03350	0.49164	0.52030
14.6	0.01553	0.03052	0.52113	0.50538
14.7	−0.02145	−0.02640	0.50301	0.47856
14.8	0.02591	0.02168	0.47853	0.49869
14.9	−0.02903	−0.01683	0.49971	0.52136
15.0	0.03103	0.01217	0.52122	0.49926

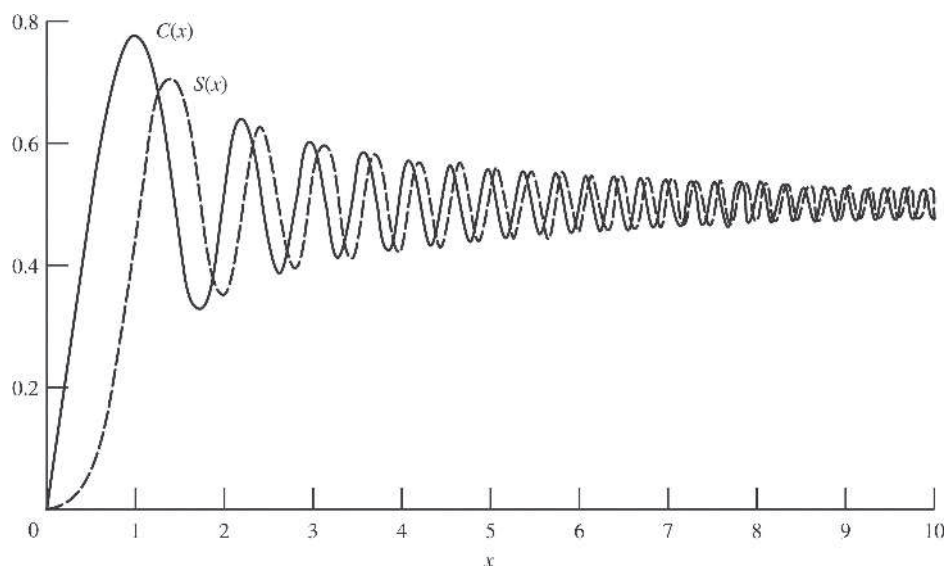


Figure III-1 Plots of $C(x)$ and $S(x)$ Fresnel integrals. (Source: C. A. Balanis, *Antenna Theory: Analysis and Design*, Third Edition, copyright © 2005, John Wiley & Sons, Inc. Reprinted by permission of John Wiley & Sons, Inc.)

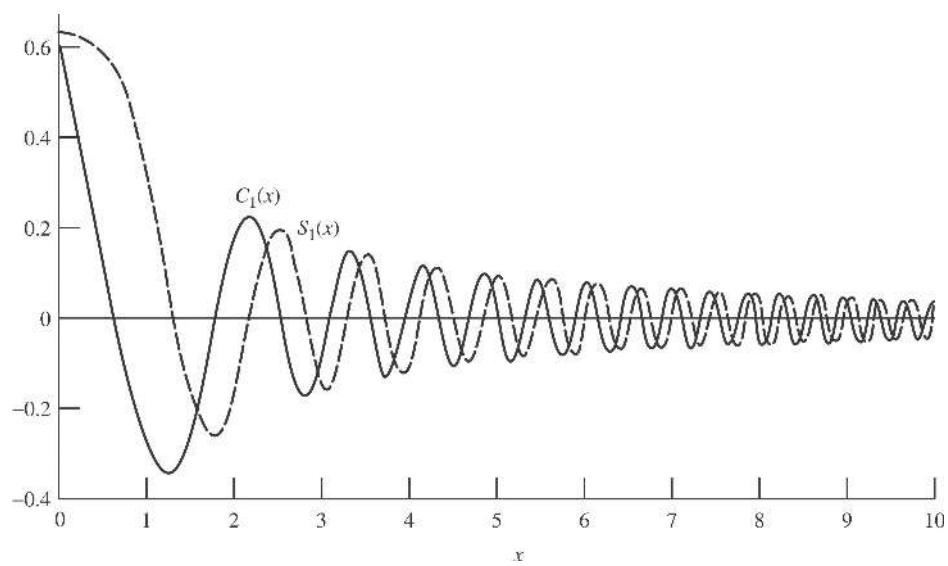


Figure III-2 Plots of $C_1(x)$ and $S_1(x)$ Fresnel integrals. (Source: C. A. Balanis, *Antenna Theory: Analysis and Design*, Third Edition, copyright © 2005, John Wiley & Sons, Inc. Reprinted by permission of John Wiley & Sons, Inc.)

APPENDIX IV



Bessel Functions

IV.1 BESSEL AND HANKEL FUNCTIONS

Bessel's equation can be written as

$$x^2 \frac{d^2 y}{dx^2} + x \frac{dy}{dx} + (x^2 - p^2)y = 0 \quad (\text{IV-1})$$

Using the method of Frobenius, we can write its solutions as

$$y(x) = A_1 J_p(x) + B_1 J_{-p}(x) \quad p \text{ not an integer} \quad (\text{IV-2})$$

or

$$y(x) = A_2 J_n(x) + B_2 Y_n(x) \quad n \text{ an integer} \quad (\text{IV-3})$$

where

$$J_p(x) = \sum_{m=0}^{\infty} \frac{(-1)^m (x/2)^{2m+p}}{m!(m+p)!} \quad (\text{IV-4})$$

$$J_{-p}(x) = \sum_{m=0}^{\infty} \frac{(-1)^m (x/2)^{2m-p}}{m!(m-p)!} \quad (\text{IV-5})$$

$$Y_p(x) = \frac{J_p(x) \cos(p\pi) - J_{-p}(x)}{\sin(p\pi)} \quad (\text{IV-6})$$

$$m! = \Gamma(m+1) \quad (\text{IV-7})$$

$J_p(x)$ is referred to as the Bessel function of the first kind of order p , $Y_p(x)$ as the Bessel function of the second kind of order p , and $\Gamma(x)$ as the gamma function.

When $p = n = \text{integer}$, using (IV-5) and (IV-7), it can be shown that

$$J_{-n}(x) = (-1)^n J_n(x) \quad (\text{IV-8})$$

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and no longer are the two Bessel functions independent of each other. Therefore, a second solution is required, and it is given by (IV-3). It can also be shown that

$$Y_n(x) = \lim_{p \rightarrow n} Y_p(x) = \lim_{p \rightarrow n} \frac{J_p(x) \cos(p\pi) - J_{-p}(x)}{\sin(p\pi)} \quad (\text{IV-9})$$

When the argument of the Bessel function is negative and $p = n$, using (IV-4) leads to

$$J_n(-x) = (-1)^n J_n(x) \quad (\text{IV-10})$$

In many applications, Bessel functions of small and large arguments are required. Using asymptotic methods, it can be shown that

$$\left. \begin{aligned} J_0(x) &\simeq 1 \\ Y_0(x) &\simeq \frac{2}{\pi} \ln \left(\frac{\gamma x}{2} \right) \\ \gamma &= 1.781 \end{aligned} \right\} \quad x \rightarrow 0 \quad (\text{IV-11})$$

$$\left. \begin{aligned} J_p(x) &\simeq \frac{1}{p!} \left(\frac{x}{2} \right)^p \\ Y_p(x) &\simeq -\frac{(p-1)!}{\pi} \left(\frac{2}{x} \right)^p \end{aligned} \right\} \begin{array}{l} x \rightarrow 0 \\ p > 0 \end{array} \quad (\text{IV-12})$$

and

$$\left. \begin{aligned} J_p(x) &\simeq \sqrt{\frac{2}{\pi x}} \cos \left(x - \frac{\pi}{4} - \frac{p\pi}{2} \right) \\ Y_p(x) &\simeq \sqrt{\frac{2}{\pi x}} \sin \left(x - \frac{\pi}{4} - \frac{p\pi}{2} \right) \end{aligned} \right\} \quad x \rightarrow \infty \quad (\text{IV-13})$$

For wave propagation, it is often convenient to introduce Hankel functions defined as

$$H_p^{(1)}(x) = J_p(x) + jY_p(x) \quad (\text{IV-14})$$

$$H_p^{(2)}(x) = J_p(x) - jY_p(x) \quad (\text{IV-15})$$

where $H_p^{(1)}(x)$ is the Hankel function of the first kind of order p and $H_p^{(2)}(x)$ is the Hankel function of the second kind of order p . For large arguments

$$H_p^{(1)}(x) \simeq \sqrt{\frac{2}{\pi x}} e^{j[x - p(\pi/2) - \pi/4]} \quad x \rightarrow \infty \quad (\text{IV-16})$$

$$H_p^{(2)}(x) \simeq \sqrt{\frac{2}{\pi x}} e^{-j[x - p(\pi/2) - \pi/4]} \quad x \rightarrow \infty \quad (\text{IV-17})$$

A derivative can be taken using either

$$\frac{d}{dx}[Z_p(\alpha x)] = \alpha Z_{p-1}(\alpha x) - \frac{p}{x} Z_p(\alpha x) \quad (\text{IV-18})$$

or

$$\frac{d}{dx}[Z_p(\alpha x)] = -\alpha Z_{p+1}(\alpha x) + \frac{p}{x} Z_p(\alpha x) \quad (\text{IV-19})$$

where Z_p can be a Bessel function (J_p, Y_p) or a Hankel function ($H_p^{(1)}$ or $H_p^{(2)}$).

A useful identity, relating Bessel functions and their derivatives, is given by

$$J_p(x)Y'_p(x) - Y_p(x)J'_p(x) = \frac{2}{\pi x} \quad (\text{IV-20})$$

and it is referred to as the Wronskian. The prime ($'$) indicates a derivative. Also

$$J_p(x)J'_{-p}(x) - J_{-p}(x)J'_p(x) = -\frac{2}{\pi x} \sin(p\pi) \quad (\text{IV-21})$$

Some useful integrals of Bessel functions are

$$\int x^{p+1} J_p(\alpha x) dx = \frac{1}{\alpha} x^{p+1} J_{p+1}(\alpha x) + C \quad (\text{IV-22})$$

$$\int x^{1-p} J_p(\alpha x) dx = -\frac{1}{\alpha} x^{1-p} J_{p-1}(\alpha x) + C \quad (\text{IV-23})$$

$$\int x^3 J_0(x) dx = x^3 J_1(x) - 2x^2 J_2(x) + C \quad (\text{IV-24})$$

$$\int x^6 J_1(x) dx = x^6 J_2(x) - 4x^5 J_3(x) + 8x^4 J_4(x) + C \quad (\text{IV-25})$$

$$\int J_3(x) dx = -J_2(x) - \frac{2}{x} J_1(x) + C \quad (\text{IV-26})$$

$$\int x J_1(x) dx = -x J_0(x) + \int J_0(x) dx + C \quad (\text{IV-27})$$

$$\int x^{-1} J_1(x) dx = -J_1(x) + \int J_0(x) dx + C \quad (\text{IV-28})$$

$$\int J_2(x) dx = -2J_1(x) + \int J_0(x) dx + C \quad (\text{IV-29})$$

$$\int x^m J_n(x) dx = x^m J_{n+1}(x) - (m-n-1) \int x^{m-1} J_{n+1}(x) dx \quad (\text{IV-30})$$

$$\int x^m J_n(x) dx = -x^m J_{n-1}(x) + (m+n-1) \int x^{m-1} J_{n-1}(x) dx \quad (\text{IV-31})$$

$$J_1(x) = \frac{2}{\pi} \int_0^{\pi/2} \sin(x \sin \theta) \sin \theta \, d\theta \quad (\text{IV-32})$$

$$\frac{1}{x} J_1(x) = \frac{2}{\pi} \int_0^{\pi/2} \cos(x \sin \theta) \cos^2 \theta \, d\theta \quad (\text{IV-33})$$

$$J_2(x) = \frac{2}{\pi} \int_0^{\pi/2} \cos(x \sin \theta) \cos 2\theta \, d\theta \quad (\text{IV-34})$$

$$J_n(x) = \frac{j^{-n}}{2\pi} \int_0^{2\pi} e^{jx \cos \phi} e^{jn\phi} \, d\phi \quad (\text{IV-35})$$

$$J_n(x) = \frac{j^{-n}}{\pi} \int_0^{\pi} \cos(n\phi) e^{jx \cos \phi} \, d\phi \quad (\text{IV-36})$$

$$J_n(x) = \frac{1}{\pi} \int_0^{\pi} \cos(x \sin \phi - n\phi) \, d\phi \quad (\text{IV-37})$$

$$J_{2n}(x) = \frac{2}{\pi} \int_0^{\pi/2} \cos(x \sin \phi) \cos(2n\phi) \, d\phi \quad (\text{IV-38})$$

$$J_{2n}(x) = (-1)^n \frac{2}{\pi} \int_0^{\pi/2} \cos(x \cos \phi) \cos(2n\phi) \, d\phi \quad (\text{IV-39})$$

The integrals

$$\int_0^x J_0(\tau) \, d\tau \quad \text{and} \quad \int_0^x Y_0(\tau) \, d\tau \quad (\text{IV-40})$$

often appear in solutions of problems but cannot be integrated in closed form. Graphs and tables for each, obtained using numerical techniques, are included.

IV.2 MODIFIED BESSEL FUNCTIONS

In addition to the regular cylindrical Bessel functions of the first and second kind, there exists another set of cylindrical Bessel functions that are referred to as the *modified* Bessel functions of the first and second kind, denoted respectively by $I_p(x)$ and $K_p(x)$. These modified cylindrical Bessel functions exhibit ascending and descending variations for increasing argument as shown, respectively, in Figures IV-5 and IV-6. For real values of the argument, the modified Bessel functions exhibit real values.

The modified Bessel functions are related to the regular Bessel and Hankel functions by

$$I_p(x) = j^{-p} J_p(jx) = j^p J_{-p}(jx) = j^p J_p(-jx) \quad (\text{IV-41})$$

$$K_p(x) = \frac{\pi}{2} j^{p+1} H_p^{(1)}(jx) = \frac{\pi}{2} (-j)^{p+1} H_p^{(2)}(-jx) \quad (\text{IV-42})$$

Some of the identities involving modified Bessel functions are

$$I_{-p}(x) = j^p J_{-p}(jx) \quad (\text{IV-43})$$

$$I_{-n}(x) = I_n(x) \quad n = 0, 1, 2, 3, \dots \quad (\text{IV-44})$$

$$K_{-n}(x) = K_n(x) \quad n = 0, 1, 2, 3, \dots \quad (\text{IV-45})$$

For large arguments, the modified Bessel functions can be computed using the asymptotic formulas

$$\left. \begin{aligned} I_p(x) &\simeq \frac{e^x}{\sqrt{2\pi x}} \\ K_p(x) &\simeq \sqrt{\frac{\pi}{2x}} e^{-x} \end{aligned} \right\} x \rightarrow \infty \quad \text{(IV-46a)}$$

$$\text{(IV-46b)}$$

Derivatives of both modified Bessel functions can be found using the same expressions, (IV-18) and (IV-19), as for the regular Bessel functions.

IV.3 SPHERICAL BESSEL AND HANKEL FUNCTIONS

There is another set of Bessel and Hankel functions, which are usually referred to as the *spherical* Bessel and Hankel functions. These spherical Bessel and Hankel functions of order n are related, respectively, to the regular cylindrical Bessel and Hankel of order $n + 1/2$ by

$$j_n(x) = \sqrt{\frac{\pi}{2x}} J_{n+1/2}(x) \quad \text{(IV-47a)}$$

$$y_n(x) = \sqrt{\frac{\pi}{2x}} Y_{n+1/2}(x) \quad \text{(IV-47b)}$$

$$h_n^{(1)}(x) = \sqrt{\frac{\pi}{2x}} H_{n+1/2}^{(1)}(x) \quad \text{(IV-47c)}$$

$$h_n^{(2)}(x) = \sqrt{\frac{\pi}{2x}} H_{n+1/2}^{(2)}(x) \quad \text{(IV-47d)}$$

where j_n , y_n , $h_n^{(1)}$, and $h_n^{(2)}$ are the spherical Bessel and Hankel functions. These spherical Bessel and Hankel functions are used as solutions to electromagnetic problems solved using spherical coordinates.

For small arguments

$$\left. \begin{aligned} j_n(x) &\simeq \frac{x^n}{1 \cdot 3 \cdot 5 \cdots (2n+1)} \\ y_n(x) &\simeq -1 \cdot 3 \cdot 5 \cdots (2n-1) x^{-(n+1)} \end{aligned} \right\} \begin{aligned} n &= 0, 1, 2, \dots \\ x &\rightarrow 0 \end{aligned} \quad \text{(IV-48a)}$$

$$\text{(IV-48b)}$$

Another set of spherical Bessel and Hankel functions, which appear in solutions of electromagnetic problems, is that denoted by $\hat{B}_n(x)$ where $\hat{B}_n$ can be used to represent $\hat{J}_n$, $\hat{Y}_n$, $\hat{H}_n^{(1)}$, or $\hat{H}_n^{(2)}$. These are related to the preceding spherical Bessel and Hankel functions [denoted by b_n to represent j_n , y_n , $h_n^{(1)}$, or $h_n^{(2)}$] and to the regular cylindrical Bessel and Hankel functions [denoted by $B_{n+1/2}$ to represent $J_{n+1/2}$, $Y_{n+1/2}$, $H_{n+1/2}^{(1)}$, or $H_{n+1/2}^{(2)}$] by

$$\hat{B}_n(x) = x b_n(x) = \sqrt{\frac{\pi x}{2}} B_{n+1/2}(x) \quad \text{(IV-49)}$$

x	$J_0(x)$	$J_1(x)$	$Y_0(x)$	$Y_1(x)$
0.0	1.00000	0.00000	$-\infty$	$-\infty$
0.1	0.99750	0.04994	-1.53424	-6.45895
0.2	0.99003	0.09950	-1.08110	-3.32382
0.3	0.97763	0.14832	-0.80727	-2.29310
0.4	0.96040	0.19603	-0.60602	-1.78087
0.5	0.93847	0.24227	-0.44452	-1.47147
0.6	0.91201	0.28670	-0.30851	-1.26039
0.7	0.88120	0.32900	-0.19066	-1.10325
0.8	0.84629	0.36884	-0.08680	-0.97814
0.9	0.80752	0.40595	0.00563	-0.87313
1.0	0.76520	0.44005	0.08826	-0.78121
1.1	0.71962	0.47090	0.16216	-0.69812
1.2	0.67113	0.49829	0.22808	-0.62114
1.3	0.62009	0.52202	0.28654	-0.54852
1.4	0.56686	0.54195	0.33789	-0.47915
1.5	0.51183	0.55794	0.38245	-0.41231
1.6	0.45540	0.56990	0.42043	-0.34758
1.7	0.39799	0.57777	0.45203	-0.28473
1.8	0.33999	0.58152	0.47743	-0.22366
1.9	0.28182	0.58116	0.49682	-0.16441
2.0	0.22389	0.57673	0.51038	-0.10703
2.1	0.16661	0.56829	0.51829	-0.05168
2.2	0.11036	0.55596	0.52078	0.00149
2.3	0.05554	0.53987	0.51807	0.05228
2.4	0.00251	0.52019	0.51041	0.10049
2.5	-0.04838	0.49710	0.49807	0.14592
2.6	-0.09681	0.47082	0.48133	0.18836
2.7	-0.14245	0.44161	0.46050	0.22763
2.8	-0.18504	0.40972	0.43592	0.26354
2.9	-0.22432	0.37544	0.40791	0.29594
3.0	-0.26005	0.33906	0.37686	0.32467
3.1	-0.29206	0.30092	0.34310	0.34963
3.2	-0.32019	0.26134	0.30705	0.37071
3.3	-0.34430	0.22066	0.26909	0.38785
3.4	-0.36430	0.17923	0.22962	0.40101
3.5	-0.38013	0.13738	0.18902	0.41019
3.6	-0.39177	0.09547	0.14771	0.41539
3.7	-0.39923	0.05383	0.10607	0.41667
3.8	-0.40256	0.01282	0.06450	0.41411
3.9	-0.40183	-0.02724	0.02338	0.40782
4.0	-0.39715	-0.06604	-0.01694	0.39793
4.1	-0.38868	-0.10328	-0.05609	0.38459
4.2	-0.37657	-0.13865	-0.09375	0.36801
4.3	-0.36102	-0.17190	-0.12960	0.34839
4.4	-0.34226	-0.20278	-0.16334	0.32597
4.5	-0.32054	-0.23106	-0.19471	0.30100
4.6	-0.29614	-0.25655	-0.22346	0.27375
4.7	-0.26933	-0.27908	-0.24939	0.24450
4.8	-0.24043	-0.29850	-0.27230	0.21356
4.9	-0.20974	-0.31470	-0.29205	0.18125
5.0	-0.17760	-0.32758	-0.30852	0.14786
5.1	-0.14434	-0.33710	-0.32160	0.11374
5.2	-0.11029	-0.34322	-0.33125	0.07919

x	$J_0(x)$	$J_1(x)$	$Y_0(x)$	$Y_1(x)$
5.3	−0.07580	−0.34596	−0.33744	0.04455
5.4	−0.04121	−0.34534	−0.34017	0.01013
5.5	−0.00684	−0.34144	−0.33948	−0.02376
5.6	0.02697	−0.33433	−0.33544	−0.05681
5.7	0.05992	−0.32415	−0.32816	−0.08872
5.8	0.09170	−0.31103	−0.31775	−0.11923
5.9	0.12203	−0.29514	−0.30437	−0.14808
6.0	0.15065	−0.27668	−0.28819	−0.17501
6.1	0.17729	−0.25587	−0.26943	−0.19981
6.2	0.20175	−0.23292	−0.24831	−0.22228
6.3	0.22381	−0.20809	−0.22506	−0.24225
6.4	0.24331	−0.18164	−0.19995	−0.25956
6.5	0.26009	−0.15384	−0.17324	−0.27409
6.6	0.27404	−0.12498	−0.14523	−0.28575
6.7	0.28506	−0.09534	−0.11619	−0.29446
6.8	0.29310	−0.06522	−0.08643	−0.30019
6.9	0.29810	−0.03490	−0.05625	−0.30292
7.0	0.30008	−0.00468	−0.02595	−0.30267
7.1	0.29905	0.02515	0.00418	−0.29948
7.2	0.29507	0.05433	0.03385	−0.29342
7.3	0.28822	0.08257	0.06277	−0.28459
7.4	0.27860	0.10962	0.09068	−0.27311
7.5	0.26634	0.13525	0.11731	−0.25913
7.6	0.25160	0.15921	0.14243	−0.24280
7.7	0.23456	0.18131	0.16580	−0.22432
7.8	0.21541	0.20136	0.18723	−0.20388
7.9	0.19436	0.21918	0.20652	−0.18172
8.0	0.17165	0.23464	0.22352	−0.15806
8.1	0.14752	0.24761	0.23809	−0.13315
8.2	0.12222	0.25800	0.25012	−0.10724
8.3	0.09601	0.26574	0.25951	−0.08060
8.4	0.06916	0.27079	0.26622	−0.05348
8.5	0.04194	0.27312	0.27021	−0.02617
8.6	0.01462	0.27276	0.27146	0.00108
8.7	−0.01252	0.26972	0.27000	0.02801
8.8	−0.03923	0.26407	0.26587	0.05436
8.9	−0.06525	0.25590	0.25916	0.07987
9.0	−0.09033	0.24531	0.24994	0.10431
9.1	−0.11424	0.23243	0.23834	0.12747
9.2	−0.13675	0.21741	0.22449	0.14911
9.3	−0.15765	0.20041	0.20857	0.16906
9.4	−0.17677	0.18163	0.19074	0.18714
9.5	−0.19393	0.16126	0.17121	0.20318
9.6	−0.20898	0.13952	0.15018	0.21706
9.7	−0.22180	0.11664	0.12787	0.22866
9.8	−0.23228	0.09284	0.10453	0.23789
9.9	−0.24034	0.06837	0.08038	0.24469
10.0	−0.24594	0.04347	0.05567	0.24902
10.1	−0.24903	0.01840	0.03066	0.25084
10.2	−0.24962	−0.00662	0.00558	0.25019
10.3	−0.24772	−0.03132	−0.01930	0.24707
10.4	−0.24337	−0.05547	−0.04375	0.24155
10.5	−0.23665	−0.07885	−0.06753	0.23370

x	$J_0(x)$	$J_1(x)$	$Y_0(x)$	$Y_1(x)$
10.6	-0.22764	-0.10123	-0.09042	0.22363
10.7	-0.21644	-0.12240	-0.11219	0.21144
10.8	-0.20320	-0.14217	-0.13264	0.19729
10.9	-0.18806	-0.16035	-0.15158	0.18132
11.0	-0.17119	-0.17679	-0.16885	0.16371
11.1	-0.15277	-0.19133	-0.18428	0.14464
11.2	-0.13299	-0.20385	-0.19773	0.12431
11.3	-0.11207	-0.21426	-0.20910	0.10294
11.4	-0.09021	-0.22245	-0.21829	0.08074
11.5	-0.06765	-0.22838	-0.22523	0.05794
11.6	-0.04462	-0.23200	-0.22987	0.03477
11.7	-0.02133	-0.23330	-0.23218	0.01145
11.8	0.00197	-0.23229	-0.23216	-0.01179
11.9	0.02505	-0.22898	-0.22983	-0.03471
12.0	0.04769	-0.22345	-0.22524	-0.05710
12.1	0.06967	-0.21575	-0.21844	-0.07874
12.2	0.09077	-0.20598	-0.20952	-0.09942
12.3	0.11080	-0.19426	-0.19859	-0.11895
12.4	0.12956	-0.18071	-0.18578	-0.13714
12.5	0.14689	-0.16549	-0.17121	-0.15384
12.6	0.16261	-0.14874	-0.15506	-0.16888
12.7	0.17659	-0.13066	-0.13750	-0.18213
12.8	0.18870	-0.11143	-0.11870	-0.19347
12.9	0.19885	-0.09125	-0.09887	-0.20282
13.0	0.20693	-0.07032	-0.07821	-0.21008
13.1	0.21289	-0.04885	-0.05692	-0.21521
13.2	0.21669	-0.02707	-0.03524	-0.21817
13.3	0.21830	-0.00518	-0.01336	-0.21895
13.4	0.21773	0.01660	0.00848	-0.21756
13.5	0.21499	0.03805	0.03008	-0.21402
13.6	0.21013	0.05896	0.05122	-0.20839
13.7	0.20322	0.07914	0.07169	-0.20074
13.8	0.19434	0.09839	0.09130	-0.19116
13.9	0.18358	0.11653	0.10986	-0.17975
14.0	0.17108	0.13338	0.12719	-0.16664
14.1	0.15695	0.14879	0.14314	-0.15198
14.2	0.14137	0.16261	0.15754	-0.13592
14.3	0.12449	0.17473	0.17028	-0.11862
14.4	0.10649	0.18503	0.18123	-0.10026
14.5	0.08755	0.19343	0.19030	-0.08104
14.6	0.06787	0.19986	0.19742	-0.06115
14.7	0.04764	0.20426	0.20252	-0.04079
14.8	0.02708	0.20660	0.20557	-0.02016
14.9	0.00639	0.20688	0.20655	0.00053
15.0	-0.01422	0.20511	0.20546	0.02107

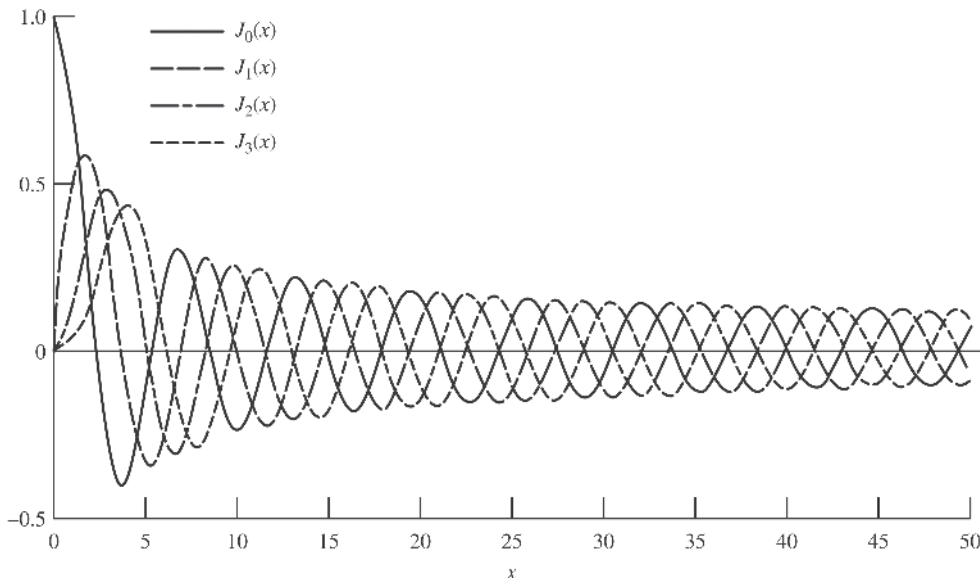


Figure IV-1 Bessel functions of the first kind [$J_0(x)$, $J_1(x)$, $J_2(x)$, and $J_3(x)$]. (Source: C. A. Balanis, *Antenna Theory: Analysis and Design*, Third Edition, copyright © 2005, John Wiley & Sons, Inc. Reprinted by permission of John Wiley & Sons, Inc.)

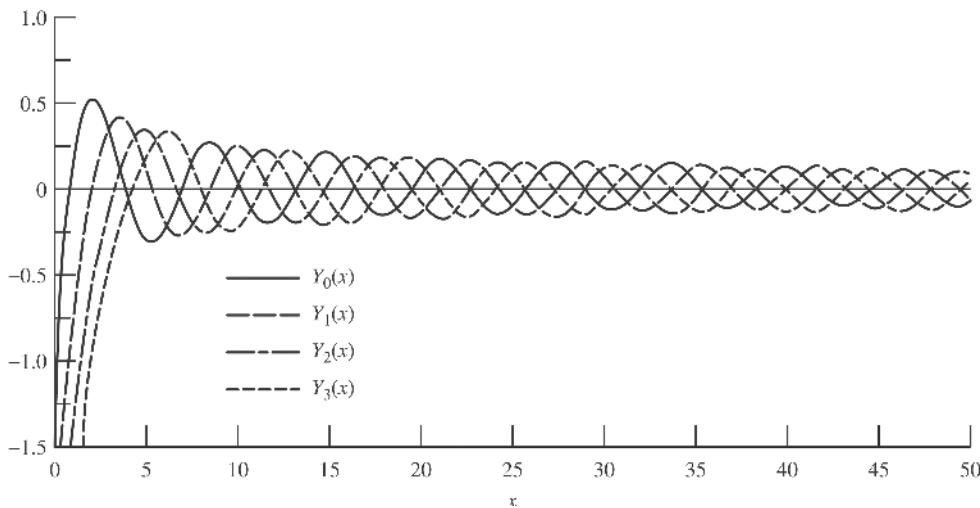


Figure IV-2 Bessel functions of the second kind [$Y_0(x)$, $Y_1(x)$, $Y_2(x)$, and $Y_3(x)$]. (Source: C. A. Balanis, *Antenna Theory: Analysis and Design*, Third Edition, copyright © 2005, John Wiley & Sons, Inc. Reprinted by permission of John Wiley & Sons, Inc.)

$J_1(x)/x$ function

x	$J_1(x)/x$	x	$J_1(x)/x$	x	$J_1(x)/x$
0.0	0.50000	5.0	-0.06552	10.0	0.00435
0.1	0.49938	5.1	-0.06610	10.1	0.00182
0.2	0.49750	5.2	-0.06600	10.2	-0.00065
0.3	0.49440	5.3	-0.06528	10.3	-0.00304
0.4	0.49007	5.4	-0.06395	10.4	-0.00533
0.5	0.48454	5.5	-0.06208	10.5	-0.00751
0.6	0.47783	5.6	-0.05970	10.6	-0.00955
0.7	0.46999	5.7	-0.05687	10.7	-0.01144
0.8	0.46105	5.8	-0.05363	10.8	-0.01316
0.9	0.45105	5.9	-0.05002	10.9	-0.01471
1.0	0.44005	6.0	-0.04611	11.0	-0.01607
1.1	0.42809	6.1	-0.04194	11.1	-0.01724
1.2	0.41524	6.2	-0.03757	11.2	-0.01820
1.3	0.40156	6.3	-0.03303	11.3	-0.01896
1.4	0.38710	6.4	-0.02838	11.4	-0.01951
1.5	0.37196	6.5	-0.02367	11.5	-0.01986
1.6	0.35618	6.6	-0.01894	11.6	-0.02000
1.7	0.33986	6.7	-0.01423	11.7	-0.01994
1.8	0.32306	6.8	-0.00959	11.8	-0.01969
1.9	0.30587	6.9	-0.00506	11.9	-0.01924
2.0	0.28836	7.0	-0.00067	12.0	-0.01862
2.1	0.27061	7.1	0.00354	12.1	-0.01783
2.2	0.25271	7.2	0.00755	12.2	-0.01688
2.3	0.23473	7.3	0.01131	12.3	-0.01579
2.4	0.21674	7.4	0.01481	12.4	-0.01457
2.5	0.19884	7.5	0.01803	12.5	-0.01324
2.6	0.18108	7.6	0.02095	12.6	-0.01180
2.7	0.16356	7.7	0.02355	12.7	-0.01029
2.8	0.14633	7.8	0.02582	12.8	-0.00871
2.9	0.12946	7.9	0.02774	12.9	-0.00707
3.0	0.11302	8.0	0.02933	13.0	-0.00541
3.1	0.09707	8.1	0.03057	13.1	-0.00373
3.2	0.08167	8.2	0.03146	13.2	-0.00205
3.3	0.06687	8.3	0.03202	13.3	-0.00039
3.4	0.05271	8.4	0.03224	13.4	0.00124
3.5	0.03925	8.5	0.03213	13.5	0.00282
3.6	0.02652	8.6	0.03172	13.6	0.00434
3.7	0.01455	8.7	0.03100	13.7	0.00578
3.8	0.00337	8.8	0.03001	13.8	0.00713
3.9	-0.00699	8.9	0.02875	13.9	0.00838
4.0	-0.01651	9.0	0.02726	14.0	0.00953
4.1	-0.02519	9.1	0.02554	14.1	0.01055
4.2	-0.03301	9.2	0.02363	14.2	0.01145
4.3	-0.03998	9.3	0.02155	14.3	0.01222
4.4	-0.04609	9.4	0.01932	14.4	0.01285
4.5	-0.05135	9.5	0.01697	14.5	0.01334
4.6	-0.05578	9.6	0.01453	14.6	0.01369
4.7	-0.05938	9.7	0.01202	14.7	0.01389
4.8	-0.06219	9.8	0.00947	14.8	0.01396
4.9	-0.06423	9.9	0.00691	14.9	0.01388
				15.0	0.01367

$\int_0^x J_0(\tau)d\tau$ and $\int_0^x Y_0(\tau)d\tau$ functions

x	$\int_0^x J_0(\tau)d\tau$	$\int_0^x Y_0(\tau)d\tau$	x	$\int_0^x J_0(\tau)d\tau$	$\int_0^x Y_0(\tau)d\tau$
0.0	0.00000	0.00000	5.0	0.71531	0.19971
0.1	0.09991	-0.21743	5.1	0.69920	0.16818
0.2	0.19933	-0.34570	5.2	0.68647	0.13551
0.3	0.29775	-0.43928	5.3	0.67716	0.10205
0.4	0.39469	-0.50952	5.4	0.67131	0.06814
0.5	0.48968	-0.56179	5.5	0.66891	0.03413
0.6	0.58224	-0.59927	5.6	0.66992	0.00035
0.7	0.67193	-0.62409	5.7	0.67427	-0.03284
0.8	0.75834	-0.63786	5.8	0.68187	-0.06517
0.9	0.84106	-0.64184	5.9	0.69257	-0.09630
1.0	0.91973	-0.63706	6.0	0.70622	-0.12595
1.1	0.99399	-0.62447	6.1	0.72263	-0.15385
1.2	1.06355	-0.60490	6.2	0.74160	-0.17975
1.3	1.12813	-0.57911	6.3	0.76290	-0.20344
1.4	1.18750	-0.54783	6.4	0.78628	-0.22470
1.5	1.24144	-0.51175	6.5	0.81147	-0.24338
1.6	1.28982	-0.47156	6.6	0.83820	-0.25931
1.7	1.33249	-0.42788	6.7	0.86618	-0.27239
1.8	1.36939	-0.38136	6.8	0.89512	-0.28252
1.9	1.40048	-0.33260	6.9	0.92470	-0.28966
2.0	1.42577	-0.28219	7.0	0.95464	-0.29377
2.1	1.44528	-0.23071	7.1	0.98462	-0.29486
2.2	1.45912	-0.17871	7.2	1.01435	-0.29295
2.3	1.46740	-0.12672	7.3	1.04354	-0.28811
2.4	1.47029	-0.07526	7.4	1.07190	-0.28043
2.5	1.46798	-0.02480	7.5	1.09917	-0.27002
2.6	1.46069	0.02420	7.6	1.12508	-0.25702
2.7	1.44871	0.07132	7.7	1.14941	-0.24159
2.8	1.43231	0.11617	7.8	1.17192	-0.22392
2.9	1.41181	0.15839	7.9	1.19243	-0.20421
3.0	1.38756	0.19765	8.0	1.21074	-0.18269
3.1	1.35992	0.23367	8.1	1.22671	-0.15959
3.2	1.32928	0.26620	8.2	1.24021	-0.13516
3.3	1.29602	0.29502	8.3	1.25112	-0.10966
3.4	1.26056	0.31996	8.4	1.25939	-0.08335
3.5	1.22330	0.34090	8.5	1.26494	-0.05650
3.6	1.18467	0.35775	8.6	1.26777	-0.02940
3.7	1.14509	0.37044	8.7	1.26787	-0.00230
3.8	1.10496	0.37896	8.8	1.26528	0.02451
3.9	1.06471	0.38335	8.9	1.26005	0.05078
4.0	1.02473	0.38366	9.0	1.25226	0.07625
4.1	0.98541	0.38000	9.1	1.24202	0.10069
4.2	0.94712	0.37250	9.2	1.22946	0.12385
4.3	0.91021	0.36131	9.3	1.21473	0.14552
4.4	0.87502	0.34665	9.4	1.19799	0.16550
4.5	0.84186	0.32872	9.5	1.17944	0.18361
4.6	0.81100	0.30779	9.6	1.15927	0.19969
4.7	0.78271	0.28413	9.7	1.13772	0.21360
4.8	0.75721	0.25802	9.8	1.11499	0.22523
4.9	0.73468	0.22977	9.9	1.09134	0.23448
			10.0	1.06701	0.24129

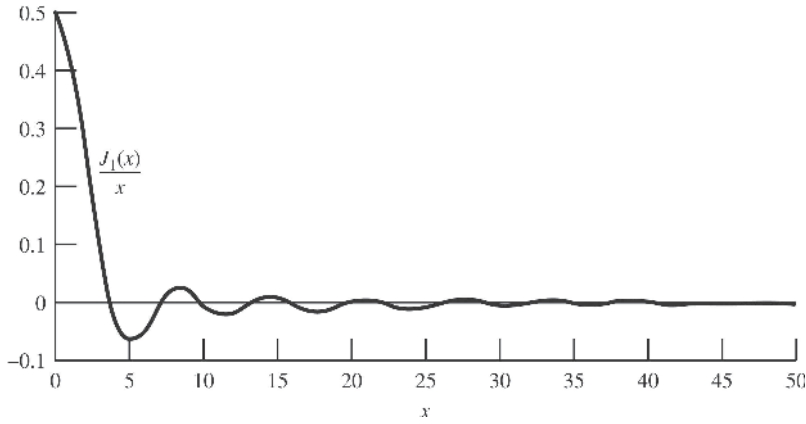


Figure IV-3 Plot of $J_1(x)/x$ function. (Source: C. A. Balanis, *Antenna Theory: Analysis and Design*, Third Edition, copyright © 2005, John Wiley & Sons, Inc. Reprinted by permission of John Wiley & Sons, Inc.)

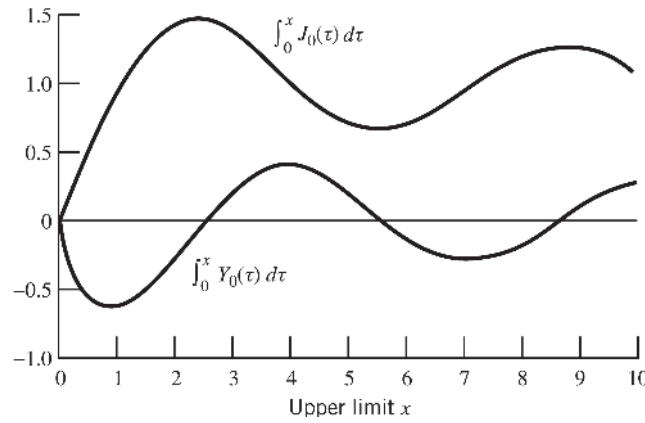


Figure IV-4 Plots of $\int_0^x J_0(\tau) d\tau$ and $\int_0^x Y_0(\tau) d\tau$ functions. (Source: C. A. Balanis, *Antenna Theory: Analysis and Design*, Third Edition, copyright © 2005, John Wiley & Sons, Inc. Reprinted by permission of John Wiley & Sons, Inc.)

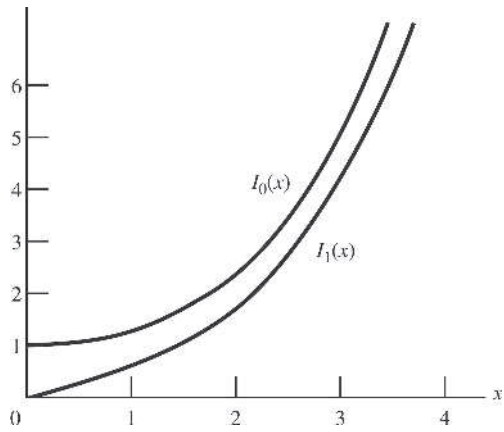


Figure IV-5 Modified Bessel functions of the first kind $[I_0(x)$ and $I_1(x)]$.

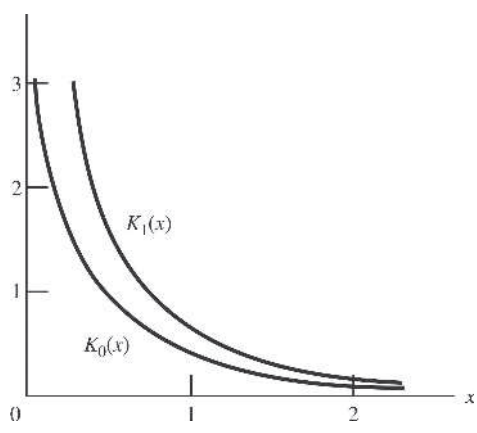


Figure IV-6 Modified Bessel functions of the second kind [$K_0(x)$ and $K_1(x)$].

APPENDIX V



Legendre Polynomials and Functions

V.1 LEGENDRE POLYNOMIALS AND FUNCTIONS

The *ordinary* Legendre differential equation can be written as

$$(1-x^2)\frac{d^2y}{dx^2}-2x\frac{dy}{dx}+p(p+1)y=0 \quad (\text{V-1})$$

Its solution can be written as

$$y(x)=A_1P_p(x)+B_1P_p(-x) \quad p \text{ not an integer} \quad (\text{V-2})$$

where $P_p(x)$ is referred to as the *Legendre function of the first kind*. If p is an integer ($p=n$), then $P_n(x)$ and $P_n(-x)$ are not two independent solutions because

$$P_n(-x)=(-1)^n P_n(x) \quad (\text{V-3})$$

Therefore, two independent solutions to (V-1) for $p=n$, an integer, are

$$y(x)=A_2P_n(x)+B_2Q_n(x) \quad (\text{V-4})$$

where $Q_n(x)$ is referred to as the *Legendre function of the second kind*.

When $p=n$, an integer, $P_n(x)$ are also referred to as the *Legendre polynomials* of order n , and are defined by

$$P_n(x)=\sum_{m=0}^M \frac{(-1)^m (2n-2m)!(x)^{n-2m}}{2^n m!(n-m)!(n-2m)!} \quad (\text{V-5})$$

where $M=n/2$ or $(n-1)/2$, whichever is an integer.

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The Legendre functions $Q_n(x)$ of the second kind are defined by

$$Q_n(x) = \lim_{p \rightarrow n} Q_p(x) = \lim_{p \rightarrow n} \frac{\pi}{2} \frac{P_p(x) \cos(p\pi) - P_p(-x)}{\sin(p\pi)} \quad (\text{V-6})$$

The Legendre polynomials (or Legendre functions of the first kind) $P_n(x)$ can also be obtained more conveniently using *Rodrigues' formula*

$$P_n(x) = \frac{1}{2^n n!} \frac{d^n}{dx^n} (x^2 - 1)^n \quad (\text{V-7})$$

which when expanded leads (for $n = 0, 1, 2, \dots, 7$) to

$$\begin{aligned} P_0(x) &= 1 \\ P_1(x) &= x \\ P_2(x) &= \frac{1}{2}(3x^2 - 1) \\ P_3(x) &= \frac{1}{2}(5x^3 - 3x) \\ P_4(x) &= \frac{1}{8}(35x^4 - 30x^2 + 3) \\ P_5(x) &= \frac{1}{8}(63x^5 - 70x^3 + 15x) \\ P_6(x) &= \frac{1}{16}(231x^6 - 315x^4 + 105x^2 - 5) \\ P_7(x) &= \frac{1}{16}(429x^7 - 693x^5 + 315x^3 - 35x) \end{aligned} \quad (\text{V-8})$$

If $x = \cos \theta$, the Legendre polynomials (or Legendre functions of the first kind) of (V-8) can be written as

$$\begin{aligned} P_0(\cos \theta) &= 1 \\ P_1(\cos \theta) &= \cos \theta \\ P_2(\cos \theta) &= \frac{1}{4}(3 \cos 2\theta + 1) \\ P_3(\cos \theta) &= \frac{1}{8}(5 \cos 3\theta + 3 \cos \theta) \\ P_4(\cos \theta) &= \frac{1}{64}(35 \cos 4\theta + 20 \cos 2\theta + 9) \\ P_5(\cos \theta) &= \frac{1}{128}(63 \cos 5\theta + 35 \cos 3\theta + 30 \cos \theta) \\ P_6(\cos \theta) &= \frac{1}{512}(231 \cos 6\theta + 126 \cos 4\theta + 105 \cos 2\theta + 50) \\ P_7(\cos \theta) &= \frac{1}{1024}(429 \cos 7\theta + 231 \cos 5\theta + 189 \cos 3\theta + 175 \cos \theta) \end{aligned} \quad (\text{V-9})$$

The Legendre functions $Q_n(x)$ of the second kind exhibit singularities at $x = \pm 1$ or $\theta = 0, \pi$ and can be obtained from the Legendre functions $P_n(x)$ of the first kind using the formula

$$Q_n(x) = P_n(x) \left\{ \frac{1}{2} \ln \left(\frac{1+x}{1-x} \right) - \psi(n) \right\} + \sum_{m=1}^n \frac{(-1)^m (n+m)!}{(m!)^2 (n-m)!} \psi(m) \left(\frac{1-x}{2} \right)^m \quad (\text{V-10})$$

where

$$\psi(n) = 1 + \frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{n} \quad (\text{V-10a})$$

When (V-10) is expanded, it leads (for $n = 0, 1, 2, 3$) to

$$\begin{aligned} Q_0(x) &= \frac{1}{2} \ln \left(\frac{1+x}{1-x} \right) \\ Q_1(x) &= \frac{x}{2} \ln \left(\frac{1+x}{1-x} \right) - 1 \\ Q_2(x) &= \frac{3x^2-1}{4} \ln \left(\frac{1+x}{1-x} \right) - \frac{3x}{2} \\ Q_3(x) &= \frac{5x^3-3x}{4} \ln \left(\frac{1+x}{1-x} \right) - \frac{5x^2}{2} + \frac{2}{3} \end{aligned} \quad (\text{V-11})$$

or for $x = \cos \theta$ to

$$\begin{aligned} Q_0(\cos \theta) &= \ln \left(\cot \frac{\theta}{2} \right) \\ Q_1(\cos \theta) &= \cos \theta \ln \left(\cot \frac{\theta}{2} \right) - 1 \\ Q_2(\cos \theta) &= \frac{1}{4} (1 + 3 \cos 2\theta) \ln \left(\cot \frac{\theta}{2} \right) - \frac{3}{2} \cos \theta \\ Q_3(\cos \theta) &= \frac{1}{8} (3 \cos \theta + 5 \cos 3\theta) \ln \left(\cot \frac{\theta}{2} \right) - \frac{5}{4} \cos 2\theta - \frac{7}{12} \end{aligned} \quad (\text{V-12})$$

The Legendre functions of the first $P_n(x)$ and second $Q_n(x)$ kind obey the following recurrence relations:

$$(n+1)R_{n+1}(x) - (2n+1)xR_n(x) + nR_{n-1}(x) = 0 \quad (\text{V-13a})$$

$$\frac{dR_{n+1}(x)}{dx} - x \frac{dR_n(x)}{dx} = (n+1)R_n(x) \quad (\text{V-13b})$$

$$x \frac{dR_n(x)}{dx} - \frac{dR_{n-1}(x)}{dx} = nR_n(x) \quad (\text{V-13c})$$

$$\frac{dR_{n+1}(x)}{dx} - \frac{dR_{n-1}(x)}{dx} = (2n+1)R_n(x) \quad (\text{V-13d})$$

$$(x^2-1) \frac{dR_n(x)}{dx} = nxR_n(x) - nR_{n-1}(x) = -(n+1)(xR_n - R_{n+1}) \quad (\text{V-13e})$$

where $R_n(x)$ can be either $P_n(x)$ or $Q_n(x)$.

Some other useful formulas involving Legendre polynomials $P_n(x)$ and $Q_n(x)$ are

$$\int_{-1}^1 P_m(x)P_n(x) dx = 0 \quad m \neq n \quad (\text{V-14a})$$

$$\int_{-1}^1 [P_n(x)]^2 dx = \frac{2}{2n+1} \quad (\text{V-14b})$$

$$\int_0^1 [Q_n(x)]^2 dx = \frac{1}{2n+1} \left[\frac{\pi^2}{4} - \frac{1}{(n+1)^2} - \frac{1}{(n+2)^2} - \dots \right] \quad (\text{V-14c})$$

which indicate that the Legendre polynomials are orthogonal in the range $-1 \leq x \leq 1$. Also

$$P_n(-x) = (-1)^n P_n(x) \quad (\text{V-15a})$$

$$P_n(x) = P_{-n-1}(x) \quad (\text{V-15b})$$

$$Q_n(-x) = (-1)^{n+1} Q_n(x) \quad (\text{V-15c})$$

$$P_n(0) = \begin{cases} 0 & n = \text{odd} \\ (-1)^{n/2} \frac{1 \cdot 3 \cdot 5 \cdots (n-1)}{2 \cdot 4 \cdot 6 \cdots n} & n = \text{even} \end{cases} \quad (\text{V-15d})$$

$$P_n(1) = 1 \quad (\text{V-15e})$$

$$P_n(-1) = \begin{cases} 1 & n = \text{even} \\ -1 & n = \text{odd} \end{cases} \quad (\text{V-15f})$$

$$Q_n(1) = +\infty \quad (\text{V-15g})$$

$$Q_n(-1) = \begin{cases} -\infty & n = \text{even} \\ +\infty & n = \text{odd} \end{cases} \quad (\text{V-15h})$$

$$P_n(x) = \frac{1}{\pi} \int_0^\pi (x + \sqrt{x^2 - 1} \cos \psi)^n d\psi \quad (\text{V-15i})$$

$$\int P_n(x) dx = \frac{P_{n+1}(x) - P_{n-1}(x)}{(2n+1)} \quad (\text{V-15j})$$

Plots of $P_n(x)$ and $Q_n(x)$ for $n = 0, 1, 2, 3$ in the range $-1 \leq x \leq 1$ are shown in Figures V-1 and V-2.

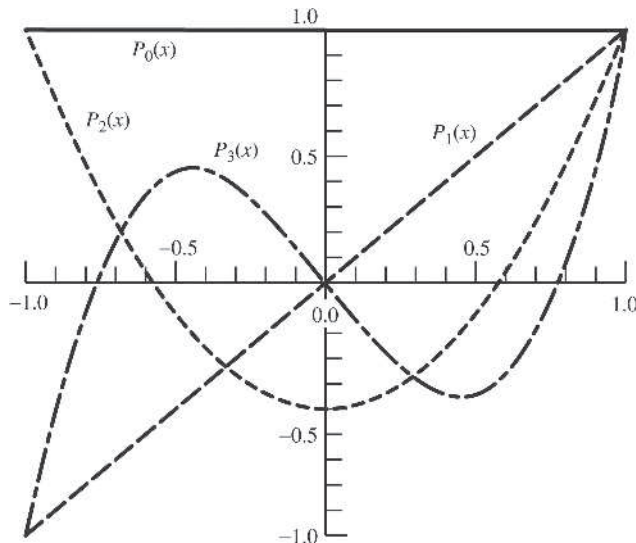


Figure V-1 Legendre functions of the first kind [$P_0(x)$, $P_1(x)$, $P_2(x)$, and $P_3(x)$].

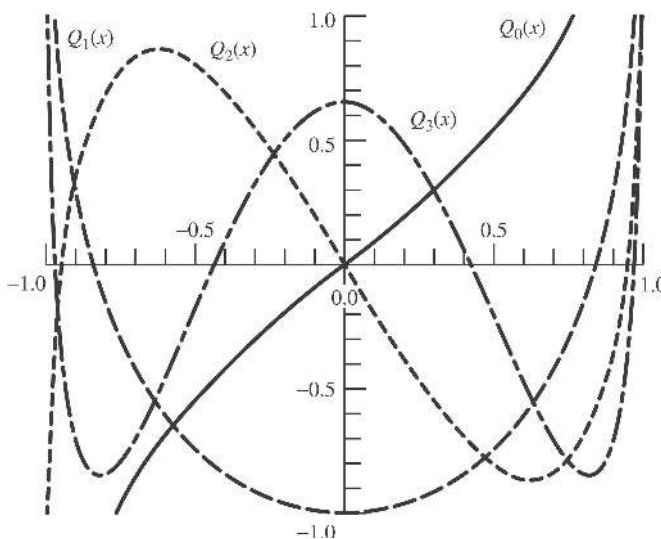


Figure V-2 Legendre functions of the second kind [$Q_0(x)$, $Q_1(x)$, $Q_2(x)$, and $Q_3(x)$].

V.2 ASSOCIATED LEGENDRE FUNCTIONS

In addition to the *ordinary* Legendre differential equation V-1, there also exists the *associated* Legendre differential equation

$$(1-x^2)\frac{d^2y}{dx^2} - 2x\frac{dy}{dx} + \left[n(n+1) - \frac{m^2}{1-x^2}\right]y = 0 \quad (\text{V-16})$$

whose solution, for nonnegative integer values of n and m , takes the form

$$y(x) = A_1 P_n^m(x) + B_1 Q_n^m(x) \quad (\text{V-17})$$

where $P_n^m(x)$ and $Q_n^m(x)$ are referred to, respectively, as the *associated Legendre functions of the first and second kind*.

The associated Legendre functions $P_n^m(x)$ and $Q_n^m(x)$ of the first and second kind are related, respectively, to the Legendre functions $P_n(x)$ and $Q_n(x)$ of the first and second kind by

$$\begin{aligned} P_n^m(x) &= (-1)^m (1-x^2)^{m/2} \frac{d^m P_n(x)}{dx^m} \\ &= (-1)^m \frac{(1-x^2)^{m/2}}{2^n n!} \frac{d^{m+n}(x^2-1)^n}{dx^{m+n}} \end{aligned} \quad (\text{V-18a})$$

$$Q_n^m(x) = (-1)^m (1-x^2)^{m/2} \frac{d^m Q_n(x)}{dx^m} \quad (\text{V-18b})$$

The associated Legendre functions $Q_n^m(x)$ of the second kind are singular at $x = \pm 1$, as are the Legendre functions $Q_n(x)$.

When (V-18a) and (V-18b) are expanded, we can write, using (V-8) and (V-11), the first few orders of $P_n^m(x)$ and $Q_n^m(x)$ as

$$\begin{aligned}
 P_0^0(x) &= P_0(x) = 1 & P_0^2(x) &= 0 \\
 P_1^0(x) &= P_1(x) = x & P_1^2(x) &= 0 \\
 P_2^0(x) &= P_2(x) = \frac{1}{2}(3x^2 - 1) & P_2^2(x) &= 3(1 - x^2) \\
 P_3^0(x) &= P_3(x) = \frac{1}{2}(5x^3 - 3x) & P_3^2(x) &= 15x(1 - x^2) \\
 &\vdots & &\vdots \\
 P_0^1(x) &= 0 & P_0^3(x) &= 0 \\
 P_1^1(x) &= -(1 - x^2)^{1/2} & P_1^3(x) &= 0 \\
 P_2^1(x) &= -3x(1 - x^2)^{1/2} & P_2^3(x) &= 0 \\
 P_3^1(x) &= -\frac{3}{2}(5x^2 - 1)(1 - x^2)^{1/2} & P_3^3(x) &= -15(1 - x^2)^{3/2} \\
 &\vdots & &\vdots
 \end{aligned} \tag{V-19a}$$

$$\begin{aligned}
 Q_0^0(x) &= Q_0(x) = \frac{1}{2} \ln \left(\frac{1+x}{1-x} \right) \\
 Q_1^0(x) &= Q_1(x) = \frac{x}{2} \ln \left(\frac{1+x}{1-x} \right) - 1 \\
 Q_2^0(x) &= Q_2(x) = \frac{3x^2 - 1}{4} \ln \left(\frac{1+x}{1-x} \right) - \frac{3x}{2} \\
 &\vdots \\
 Q_0^1(x) &= 0 \\
 Q_1^1(x) &= -(1 - x^2)^{1/2} \left[\frac{1}{2} \ln \left(\frac{1+x}{1-x} \right) + \frac{x}{1 - x^2} \right] \\
 Q_2^1(x) &= -(1 - x^2)^{1/2} \left[\frac{3x}{2} \ln \left(\frac{1+x}{1-x} \right) + \frac{3x^2 - 2}{1 - x^2} \right] \\
 &\vdots \\
 Q_0^2(x) &= 0 \\
 Q_1^2(x) &= 0 \\
 Q_2^2(x) &= (1 - x^2)^{1/2} \left[\frac{3}{2} \ln \left(\frac{1+x}{1-x} \right) + \frac{5x - 3x^2}{(1 - x^2)^2} \right] \\
 &\vdots
 \end{aligned} \tag{V-19b}$$

It should be noted that

$$P_n^0(x) = P_n(x) \tag{V-20a}$$

$$Q_n^0(x) = Q_n(x) \tag{V-20b}$$

$$P_n^m(x) = 0 \quad m > n \tag{V-20c}$$

$$Q_n^m(x) = 0 \quad m > n \quad (\text{V-20d})$$

$$P_n^m(-x) = (-1)^{n-m} P_n^m(x) \quad (\text{V-20e})$$

$$P_n^m(x) = P_{-n-1}^m(x) \quad (\text{V-20f})$$

$$Q_n^m(-x) = (-1)^{n+m+1} Q_n^m(x) \quad (\text{V-20g})$$

$$P_n^m(1) = \begin{cases} 1 & m = 0 \\ 0 & m > 0 \end{cases} \quad (\text{V-20h})$$

$$P_n^m(0) = \begin{cases} (-1)^{(n+m)/2} \frac{1 \cdot 3 \cdot 5 \cdots (n+m-1)}{2 \cdot 4 \cdot 6 \cdots (n-m)} & n+m = \text{even} \\ 0 & n+m = \text{odd} \end{cases} \quad (\text{V-20i})$$

$$Q_n^m(0) = \begin{cases} 0 & n+m = \text{even} \\ (-1)^{(n+m+1)/2} \frac{2 \cdot 4 \cdot 6 \cdots (n+m-1)}{1 \cdot 3 \cdot 5 \cdots (n-m)} & n+m = \text{odd} \end{cases} \quad (\text{V-20j})$$

$$\left. \frac{d^q P_n^m(x)}{dx^q} \right|_{x=0} = (-1)^q P_n^{m+q}(0) \quad (\text{V-20k})$$

$$\left. \frac{d^q Q_n^m(x)}{dx^q} \right|_{x=0} = (-1)^q Q_n^{m+q}(0) \quad (\text{V-20l})$$

Orthogonality relations of $P_n^m(x)$ in the range of $-1 \leq x \leq 1$ are

$$\int_{-1}^1 P_n^m(x) P_l^m(x) dx = 0 \quad n \neq l \quad (\text{V-21a})$$

$$\int_{-1}^1 [P_n^m(x)]^2 dx = \frac{2}{2n+1} \frac{(n+m)!}{(n-m)!} \quad (\text{V-21b})$$

$$\int_{-1}^1 \left[\frac{dP_n(x)}{dx} \right]^2 dx = n(n+1) \quad (\text{V-21c})$$

and useful recurrence formulas are

$$(n+1-m)R_{n+1}^m(x) - (2n+1)xR_n^m(x) + (n+m)R_{n-1}^m(x) = 0 \quad (\text{V-22a})$$

$$R_n^{m+2}(x) + \frac{2(m+1)x}{(1-x^2)^{1/2}} R_n^{m+1}(x) + (n-m)(n+m+1)R_n^m(x) = 0 \quad (\text{V-22b})$$

where $R_n^m(x)$ can be either $P_n^m(x)$ or $Q_n^m(x)$.

When m is not an integer in the associated Legendre differential equation V-16, the solutions become more complex and can be expressed in terms of *hypergeometric functions*. These solutions are beyond the scope of this book, and the reader is referred to the literature.

x	$P_0(x)$	$P_1(x)$	$P_2(x)$	$P_3(x)$
-1.00	1.00000	-1.00000	1.00000	-1.00000
-0.99	1.00000	-0.99000	0.97015	-0.94075
-0.98	1.00000	-0.98000	0.94060	-0.88298
-0.97	1.00000	-0.97000	0.91135	-0.82668
-0.96	1.00000	-0.96000	0.88240	-0.77184
-0.95	1.00000	-0.95000	0.85375	-0.71844
-0.94	1.00000	-0.94000	0.82540	-0.66646
-0.93	1.00000	-0.93000	0.79735	-0.61589
-0.92	1.00000	-0.92000	0.76960	-0.56672
-0.91	1.00000	-0.91000	0.74215	-0.51893
-0.90	1.00000	-0.90000	0.71500	-0.47250
-0.89	1.00000	-0.89000	0.68815	-0.42742
-0.88	1.00000	-0.88000	0.66160	-0.38368
-0.87	1.00000	-0.87000	0.63535	-0.34126
-0.86	1.00000	-0.86000	0.60940	-0.30014
-0.85	1.00000	-0.85000	0.58375	-0.26031
-0.84	1.00000	-0.84000	0.55840	-0.22176
-0.83	1.00000	-0.83000	0.53335	-0.18447
-0.82	1.00000	-0.82000	0.50860	-0.14842
-0.81	1.00000	-0.81000	0.48415	-0.11360
-0.80	1.00000	-0.80000	0.46000	-0.08000
-0.79	1.00000	-0.79000	0.43615	-0.04760
-0.78	1.00000	-0.78000	0.41260	-0.01638
-0.77	1.00000	-0.77000	0.38935	0.01367
-0.76	1.00000	-0.76000	0.36640	0.04256
-0.75	1.00000	-0.75000	0.34375	0.07031
-0.74	1.00000	-0.74000	0.32140	0.09694
-0.73	1.00000	-0.73000	0.29935	0.12246
-0.72	1.00000	-0.72000	0.27760	0.14688
-0.71	1.00000	-0.71000	0.25615	0.17022
-0.70	1.00000	-0.70000	0.23500	0.19250
-0.69	1.00000	-0.69000	0.21415	0.21373
-0.68	1.00000	-0.68000	0.19360	0.23392
-0.67	1.00000	-0.67000	0.17335	0.25309
-0.66	1.00000	-0.66000	0.15340	0.27126
-0.65	1.00000	-0.65000	0.13375	0.28844
-0.64	1.00000	-0.64000	0.11440	0.30464
-0.63	1.00000	-0.63000	0.09535	0.31988
-0.62	1.00000	-0.62000	0.07660	0.33418
-0.61	1.00000	-0.61000	0.05815	0.34755
-0.60	1.00000	-0.60000	0.04000	0.36000
-0.59	1.00000	-0.59000	0.02215	0.37155
-0.58	1.00000	-0.58000	0.00460	0.38222
-0.57	1.00000	-0.57000	-0.01265	0.39202
-0.56	1.00000	-0.56000	-0.02960	0.40096
-0.55	1.00000	-0.55000	-0.04625	0.40906
-0.54	1.00000	-0.54000	-0.06260	0.41634
-0.53	1.00000	-0.53000	-0.07865	0.42281
-0.52	1.00000	-0.52000	-0.09440	0.42848
-0.51	1.00000	-0.51000	-0.10985	0.43337

x	$P_0(x)$	$P_1(x)$	$P_2(x)$	$P_3(x)$
-0.50	1.00000	-0.50000	-0.12500	0.43750
-0.49	1.00000	-0.49000	-0.13985	0.44088
-0.48	1.00000	-0.48000	-0.15440	0.44352
-0.47	1.00000	-0.47000	-0.16865	0.44544
-0.46	1.00000	-0.46000	-0.18260	0.44666
-0.45	1.00000	-0.45000	-0.19625	0.44719
-0.44	1.00000	-0.44000	-0.20960	0.44704
-0.43	1.00000	-0.43000	-0.22265	0.44623
-0.42	1.00000	-0.42000	-0.23540	0.44478
-0.41	1.00000	-0.41000	-0.24785	0.44270
-0.40	1.00000	-0.40000	-0.26000	0.44000
-0.39	1.00000	-0.39000	-0.27185	0.43670
-0.38	1.00000	-0.38000	-0.28340	0.43282
-0.37	1.00000	-0.37000	-0.29465	0.42837
-0.36	1.00000	-0.36000	-0.30560	0.42336
-0.35	1.00000	-0.35000	-0.31625	0.41781
-0.34	1.00000	-0.34000	-0.32660	0.41174
-0.33	1.00000	-0.33000	-0.33665	0.40516
-0.32	1.00000	-0.32000	-0.34640	0.39808
-0.31	1.00000	-0.31000	-0.35585	0.39052
-0.30	1.00000	-0.30000	-0.36500	0.38250
-0.29	1.00000	-0.29000	-0.37385	0.37403
-0.28	1.00000	-0.28000	-0.38240	0.36512
-0.27	1.00000	-0.27000	-0.39065	0.35579
-0.26	1.00000	-0.26000	-0.39860	0.34606
-0.25	1.00000	-0.25000	-0.40625	0.33594
-0.24	1.00000	-0.24000	-0.41360	0.32544
-0.23	1.00000	-0.23000	-0.42065	0.31458
-0.22	1.00000	-0.22000	-0.42740	0.30338
-0.21	1.00000	-0.21000	-0.43385	0.29185
-0.20	1.00000	-0.20000	-0.44000	0.28000
-0.19	1.00000	-0.19000	-0.44585	0.26785
-0.18	1.00000	-0.18000	-0.45140	0.25542
-0.17	1.00000	-0.17000	-0.45665	0.24272
-0.16	1.00000	-0.16000	-0.46160	0.22976
-0.15	1.00000	-0.15000	-0.46625	0.21656
-0.14	1.00000	-0.14000	-0.47060	0.20314
-0.13	1.00000	-0.13000	-0.47465	0.18951
-0.12	1.00000	-0.12000	-0.47840	0.17568
-0.11	1.00000	-0.11000	-0.48185	0.16167
-0.10	1.00000	-0.10000	-0.48500	0.14750
-0.09	1.00000	-0.09000	-0.48785	0.13318
-0.08	1.00000	-0.08000	-0.49040	0.11872
-0.07	1.00000	-0.07000	-0.49265	0.10414
-0.06	1.00000	-0.06000	-0.49460	0.08946
-0.05	1.00000	-0.05000	-0.49625	0.07469
-0.04	1.00000	-0.04000	-0.49760	0.05984
-0.03	1.00000	-0.03000	-0.49865	0.04493
-0.02	1.00000	-0.02000	-0.49940	0.02998
-0.01	1.00000	-0.01000	-0.49985	0.01500
0.00	1.00000	0.00000	-0.50000	0.00000

x	$P_0(x)$	$P_1(x)$	$P_2(x)$	$P_3(x)$
0.01	1.00000	0.01000	-0.49985	-0.01500
0.02	1.00000	0.02000	-0.49940	-0.02998
0.03	1.00000	0.03000	-0.49865	-0.04493
0.04	1.00000	0.04000	-0.49760	-0.05984
0.05	1.00000	0.05000	-0.49625	-0.07469
0.06	1.00000	0.06000	-0.49460	-0.08946
0.07	1.00000	0.07000	-0.49265	-0.10414
0.08	1.00000	0.08000	-0.49040	-0.11872
0.09	1.00000	0.09000	-0.48785	-0.13318
0.10	1.00000	0.10000	-0.48500	-0.14750
0.11	1.00000	0.11000	-0.48185	-0.16167
0.12	1.00000	0.12000	-0.47840	-0.17568
0.13	1.00000	0.13000	-0.47465	-0.18951
0.14	1.00000	0.14000	-0.47060	-0.20314
0.15	1.00000	0.15000	-0.46625	-0.21656
0.16	1.00000	0.16000	-0.46160	-0.22976
0.17	1.00000	0.17000	-0.45665	-0.24272
0.18	1.00000	0.18000	-0.45140	-0.25542
0.19	1.00000	0.19000	-0.44585	-0.26785
0.20	1.00000	0.20000	-0.44000	-0.28000
0.21	1.00000	0.21000	-0.43385	-0.29185
0.22	1.00000	0.22000	-0.42740	-0.30338
0.23	1.00000	0.23000	-0.42065	-0.31458
0.24	1.00000	0.24000	-0.41360	-0.32544
0.25	1.00000	0.25000	-0.40625	-0.33594
0.26	1.00000	0.26000	-0.39860	-0.34606
0.27	1.00000	0.27000	-0.39065	-0.35579
0.28	1.00000	0.28000	-0.38240	-0.36512
0.29	1.00000	0.29000	-0.37385	-0.37403
0.30	1.00000	0.30000	-0.36500	-0.38250
0.31	1.00000	0.31000	-0.35585	-0.39052
0.32	1.00000	0.32000	-0.34640	-0.39808
0.33	1.00000	0.33000	-0.33665	-0.40516
0.34	1.00000	0.34000	-0.32660	-0.41174
0.35	1.00000	0.35000	-0.31625	-0.41781
0.36	1.00000	0.36000	-0.30560	-0.42336
0.37	1.00000	0.37000	-0.29465	-0.42837
0.38	1.00000	0.38000	-0.28340	-0.43282
0.39	1.00000	0.39000	-0.27185	-0.43670
0.40	1.00000	0.40000	-0.26000	-0.44000
0.41	1.00000	0.41000	-0.24785	-0.44270
0.42	1.00000	0.42000	-0.23540	-0.44478
0.43	1.00000	0.43000	-0.22265	-0.44623
0.44	1.00000	0.44000	-0.20960	-0.44704
0.45	1.00000	0.45000	-0.19625	-0.44719
0.46	1.00000	0.46000	-0.18260	-0.44666
0.47	1.00000	0.47000	-0.16865	-0.44544
0.48	1.00000	0.48000	-0.15440	-0.44352
0.49	1.00000	0.49000	-0.13985	-0.44088
0.50	1.00000	0.50000	-0.12500	-0.43750
0.51	1.00000	0.51000	-0.10985	-0.43337

x	$P_0(x)$	$P_1(x)$	$P_2(x)$	$P_3(x)$
0.52	1.00000	0.52000	-0.09440	-0.42848
0.53	1.00000	0.53000	-0.07865	-0.42281
0.54	1.00000	0.54000	-0.06260	-0.41634
0.55	1.00000	0.55000	-0.04625	-0.40906
0.56	1.00000	0.56000	-0.02960	-0.40096
0.57	1.00000	0.57000	-0.01265	-0.39202
0.58	1.00000	0.58000	0.00460	-0.38222
0.59	1.00000	0.59000	0.02215	-0.37155
0.60	1.00000	0.60000	0.04000	-0.36000
0.61	1.00000	0.61000	0.05815	-0.34755
0.62	1.00000	0.62000	0.07660	-0.33418
0.63	1.00000	0.63000	0.09535	-0.31988
0.64	1.00000	0.64000	0.11440	-0.30464
0.65	1.00000	0.65000	0.13375	-0.28844
0.66	1.00000	0.66000	0.15340	-0.27126
0.67	1.00000	0.67000	0.17335	-0.25309
0.68	1.00000	0.68000	0.19360	-0.23392
0.69	1.00000	0.69000	0.21415	-0.21373
0.70	1.00000	0.70000	0.23500	-0.19250
0.71	1.00000	0.71000	0.25615	-0.17022
0.72	1.00000	0.72000	0.27760	-0.14688
0.73	1.00000	0.73000	0.29935	-0.12246
0.74	1.00000	0.74000	0.32140	-0.09694
0.75	1.00000	0.75000	0.34375	-0.07031
0.76	1.00000	0.76000	0.36640	-0.04256
0.77	1.00000	0.77000	0.38935	-0.01367
0.78	1.00000	0.78000	0.41260	0.01638
0.79	1.00000	0.79000	0.43615	0.04760
0.80	1.00000	0.80000	0.46000	0.08000
0.81	1.00000	0.81000	0.48415	0.11360
0.82	1.00000	0.82000	0.50860	0.14842
0.83	1.00000	0.83000	0.53335	0.18447
0.84	1.00000	0.84000	0.55840	0.22176
0.85	1.00000	0.85000	0.58375	0.26031
0.86	1.00000	0.86000	0.60940	0.30014
0.87	1.00000	0.87000	0.63535	0.34126
0.88	1.00000	0.88000	0.66160	0.38368
0.89	1.00000	0.89000	0.68815	0.42742
0.90	1.00000	0.90000	0.71500	0.47250
0.91	1.00000	0.91000	0.74215	0.51893
0.92	1.00000	0.92000	0.76960	0.56672
0.93	1.00000	0.93000	0.79735	0.61589
0.94	1.00000	0.94000	0.82540	0.66646
0.95	1.00000	0.95000	0.85375	0.71844
0.96	1.00000	0.96000	0.88240	0.77184
0.97	1.00000	0.97000	0.91135	0.82668
0.98	1.00000	0.98000	0.94060	0.88298
0.99	1.00000	0.99000	0.97015	0.94075
1.00	1.00000	1.00000	1.00000	1.00000

x	$Q_0(x)$	$Q_1(x)$	$Q_2(x)$	$Q_3(x)$
-1.00	$-\infty$	$+\infty$	$-\infty$	$+\infty$
-0.99	-2.64665	1.62019	-1.08265	0.70625
-0.98	-2.29756	1.25161	-0.69109	0.29437
-0.97	-2.09230	1.02953	-0.45181	0.04408
-0.96	-1.94591	0.86807	-0.27707	-0.13540
-0.95	-1.83178	0.74019	-0.13888	-0.27356
-0.94	-1.73805	0.63377	-0.02459	-0.38399
-0.93	-1.65839	0.54230	0.07268	-0.47419
-0.92	-1.58903	0.46190	0.15708	-0.54880
-0.91	-1.52752	0.39005	0.23135	-0.61091
-0.90	-1.47222	0.32500	0.29736	-0.66271
-0.89	-1.42193	0.26551	0.35650	-0.70582
-0.88	-1.37577	0.21068	0.40979	-0.74148
-0.87	-1.33308	0.15978	0.45803	-0.77066
-0.86	-1.29334	0.11228	0.50184	-0.79415
-0.85	-1.25615	0.06773	0.54172	-0.81259
-0.84	-1.22117	0.02579	0.57810	-0.82653
-0.83	-1.18814	-0.01385	0.61131	-0.83641
-0.82	-1.15682	-0.05141	0.64164	-0.84264
-0.81	-1.12703	-0.08711	0.66935	-0.84555
-0.80	-1.09861	-0.12111	0.69464	-0.84544
-0.79	-1.07143	-0.15357	0.71770	-0.84259
-0.78	-1.04537	-0.18461	0.73868	-0.83721
-0.77	-1.02033	-0.21435	0.75774	-0.82953
-0.76	-0.99622	-0.24288	0.77499	-0.81973
-0.75	-0.97296	-0.27028	0.79055	-0.80799
-0.74	-0.95048	-0.29665	0.80452	-0.79447
-0.73	-0.92873	-0.32203	0.81699	-0.77931
-0.72	-0.90764	-0.34650	0.82804	-0.76265
-0.71	-0.88718	-0.37010	0.83775	-0.74460
-0.70	-0.86730	-0.39289	0.84618	-0.72529
-0.69	-0.84796	-0.41491	0.85341	-0.70481
-0.68	-0.82911	-0.43620	0.85948	-0.68328
-0.67	-0.81074	-0.45680	0.86446	-0.66078
-0.66	-0.79281	-0.47674	0.86838	-0.63739
-0.65	-0.77530	-0.49606	0.87130	-0.61321
-0.64	-0.75817	-0.51477	0.87326	-0.58830
-0.63	-0.74142	-0.53291	0.87431	-0.56275
-0.62	-0.72500	-0.55050	0.87446	-0.53662
-0.61	-0.70892	-0.56756	0.87378	-0.50997
-0.60	-0.69315	-0.58411	0.87227	-0.48287
-0.59	-0.67767	-0.60018	0.86999	-0.45537
-0.58	-0.66246	-0.61577	0.86695	-0.42754
-0.57	-0.64752	-0.63091	0.86319	-0.39942
-0.56	-0.63283	-0.64561	0.85873	-0.37107
-0.55	-0.61838	-0.65989	0.85360	-0.34254
-0.54	-0.60416	-0.67376	0.84782	-0.31387
-0.53	-0.59015	-0.68722	0.84141	-0.28510
-0.52	-0.57634	-0.70030	0.83441	-0.25628
-0.51	-0.56273	-0.71301	0.82682	-0.22745

x	$Q_0(x)$	$Q_1(x)$	$Q_2(x)$	$Q_3(x)$
-0.50	-0.54931	-0.72535	0.81866	-0.19865
-0.49	-0.53606	-0.73733	0.80997	-0.16992
-0.48	-0.52298	-0.74897	0.80075	-0.14129
-0.47	-0.51007	-0.76027	0.79102	-0.11279
-0.46	-0.49731	-0.77124	0.78081	-0.08446
-0.45	-0.48470	-0.78188	0.77012	-0.05634
-0.44	-0.47223	-0.79222	0.75898	-0.02844
-0.43	-0.45990	-0.80224	0.74740	-0.00080
-0.42	-0.44769	-0.81197	0.73539	0.02654
-0.41	-0.43561	-0.82140	0.72297	0.05357
-0.40	-0.42365	-0.83054	0.71015	0.08026
-0.39	-0.41180	-0.83940	0.69695	0.10658
-0.38	-0.40006	-0.84798	0.68338	0.13251
-0.37	-0.38842	-0.85628	0.66945	0.15803
-0.36	-0.37689	-0.86432	0.65518	0.18311
-0.35	-0.36544	-0.87209	0.64057	0.20773
-0.34	-0.35409	-0.87961	0.62565	0.23187
-0.33	-0.34283	-0.88687	0.61041	0.25552
-0.32	-0.33165	-0.89387	0.59488	0.27864
-0.31	-0.32055	-0.90063	0.57907	0.30124
-0.30	-0.30952	-0.90714	0.56297	0.32328
-0.29	-0.29857	-0.91342	0.54662	0.34474
-0.28	-0.28768	-0.91945	0.53001	0.36563
-0.27	-0.27686	-0.92525	0.51316	0.38591
-0.26	-0.26611	-0.93081	0.49607	0.40558
-0.25	-0.25541	-0.93615	0.47876	0.42461
-0.24	-0.24477	-0.94125	0.46124	0.44301
-0.23	-0.23419	-0.94614	0.44351	0.46074
-0.22	-0.22366	-0.95080	0.42559	0.47781
-0.21	-0.21317	-0.95523	0.40748	0.49420
-0.20	-0.20273	-0.95945	0.38920	0.50990
-0.19	-0.19234	-0.96346	0.37075	0.52490
-0.18	-0.18198	-0.96724	0.35215	0.53918
-0.17	-0.17167	-0.97082	0.33339	0.55275
-0.16	-0.16139	-0.97418	0.31450	0.56559
-0.15	-0.15114	-0.97733	0.29547	0.57769
-0.14	-0.14093	-0.98027	0.27632	0.58904
-0.13	-0.13074	-0.98300	0.25706	0.59964
-0.12	-0.12058	-0.98553	0.23769	0.60948
-0.11	-0.11045	-0.98785	0.21822	0.61856
-0.10	-0.10034	-0.98997	0.19866	0.62687
-0.09	-0.09024	-0.99188	0.17903	0.63440
-0.08	-0.08017	-0.99359	0.15932	0.64115
-0.07	-0.07011	-0.99509	0.13954	0.64711
-0.06	-0.06007	-0.99640	0.11971	0.65229
-0.05	-0.05004	-0.99750	0.09983	0.65668
-0.04	-0.04002	-0.99840	0.07991	0.66027
-0.03	-0.03001	-0.99910	0.05996	0.66307
-0.02	-0.02000	-0.99960	0.03999	0.66507
-0.01	-0.01000	-0.99990	0.02000	0.66627

x	$Q_0(x)$	$Q_1(x)$	$Q_2(x)$	$Q_3(x)$
0.00	0.00000	-1.00000	0.00000	0.66667
0.01	0.01000	-0.99990	-0.02000	0.66627
0.02	0.02000	-0.99960	-0.03999	0.66507
0.03	0.03001	-0.99910	-0.05996	0.66307
0.04	0.04002	-0.99840	-0.07991	0.66027
0.05	0.05004	-0.99750	-0.09983	0.65668
0.06	0.06007	-0.99640	-0.11971	0.65229
0.07	0.07011	-0.99509	-0.13954	0.64711
0.08	0.08017	-0.99359	-0.15932	0.64115
0.09	0.09024	-0.99188	-0.17903	0.63440
0.10	0.10033	-0.98997	-0.19866	0.62687
0.11	0.11045	-0.98785	-0.21822	0.61856
0.12	0.12058	-0.98553	-0.23769	0.60948
0.13	0.13074	-0.98300	-0.25706	0.59964
0.14	0.14092	-0.98027	-0.27632	0.58904
0.15	0.15114	-0.97733	-0.29547	0.57769
0.16	0.16139	-0.97418	-0.31450	0.56559
0.17	0.17167	-0.97082	-0.33339	0.55275
0.18	0.18198	-0.96724	-0.35215	0.53919
0.19	0.19234	-0.96346	-0.37075	0.52490
0.20	0.20273	-0.95945	-0.38920	0.50990
0.21	0.21317	-0.95523	-0.40748	0.49420
0.22	0.22366	-0.95080	-0.42559	0.47782
0.23	0.23419	-0.94614	-0.44351	0.46075
0.24	0.24477	-0.94125	-0.46124	0.44301
0.25	0.25541	-0.93615	-0.47876	0.42461
0.26	0.26611	-0.93081	-0.49607	0.40558
0.27	0.27686	-0.92525	-0.51316	0.38591
0.28	0.28768	-0.91945	-0.53001	0.36563
0.29	0.29857	-0.91342	-0.54662	0.34474
0.30	0.30952	-0.90714	-0.56297	0.32328
0.31	0.32054	-0.90063	-0.57907	0.30124
0.32	0.33165	-0.89387	-0.59488	0.27865
0.33	0.34283	-0.88687	-0.61041	0.25552
0.34	0.35409	-0.87961	-0.62565	0.23187
0.35	0.36544	-0.87210	-0.64057	0.20773
0.36	0.37689	-0.86432	-0.65518	0.18311
0.37	0.38842	-0.85628	-0.66945	0.15803
0.38	0.40006	-0.84798	-0.68338	0.13251
0.39	0.41180	-0.83940	-0.69695	0.10658
0.40	0.42365	-0.83054	-0.71015	0.08026
0.41	0.43561	-0.82140	-0.72297	0.05357
0.42	0.44769	-0.81197	-0.73539	0.02654
0.43	0.45990	-0.80225	-0.74740	-0.00080
0.44	0.47223	-0.79222	-0.75898	-0.02844
0.45	0.48470	-0.78189	-0.77012	-0.05633
0.46	0.49731	-0.77124	-0.78081	-0.08446
0.47	0.51007	-0.76027	-0.79102	-0.11279
0.48	0.52298	-0.74897	-0.80075	-0.14129
0.49	0.53606	-0.73733	-0.80997	-0.16992
0.50	0.54931	-0.72535	-0.81866	-0.19865
0.51	0.56273	-0.71301	-0.82682	-0.22746